

**NIST Interagency Report  
NISTIR 8525**

# **Face Analysis Technology Evaluation: Age Estimation and Verification**

Kayee Hanaoka  
Mei Ngan  
Joyce Yang  
George W. Quinn  
Austin Hom  
Patrick Grother

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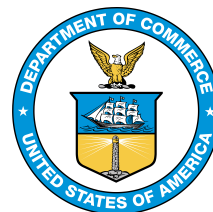
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## Abstract

The report documents technical performance of prototype age estimation algorithms submitted to the Age Estimation and Verification (AEV) track of the Face Analysis Technology Evaluation (FATE) executed by the National Institute of Standards and Technology (NIST).

## Acknowledgements

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## Other Related Reports

Results from the Face Recognition Technology Evaluation (FRTE) and Face Analysis Technology Evaluation (FATE) activities appear in the series of NIST Interagency Reports tabulated below. From 1999 to May 2024, FRTE and FATE were collectively known as FRVT.

| DATE       | PROG. | NISTIR                  | TITLE                                                                                             |
|------------|-------|-------------------------|---------------------------------------------------------------------------------------------------|
| 2014-03-20 | FATE  | <a href="#">7995</a>    | Performance of Automated Age Estimation Algorithms                                                |
| 2015-04-20 | FATE  | <a href="#">8052</a>    | Performance of Automated Gender Classification Algorithms                                         |
| 2014-05-21 | FRTE  | <a href="#">8009</a>    | Performance of Face Identification Algorithms                                                     |
| 2017-03-07 | FRTE  | <a href="#">8173</a>    | FIVE - Face In Video Evaluation: Face Recognition of Non-Cooperative Subjects                     |
| 2017-11-23 | FRTE  | <a href="#">8197</a>    | FRPC - The 2017 IARPA Face Recognition Prize Challenge                                            |
| 2020-01-03 | FRTE  | <a href="#">Draft</a>   | Part 1: Verification                                                                              |
| 2019-09-11 | FRTE  | <a href="#">8271</a>    | Part 2: Identification                                                                            |
| 2019-12-11 | FRTE  | <a href="#">8280</a>    | Part 3: Demographic Effects                                                                       |
| 2020-03-04 | FATE  | <a href="#">8292</a>    | Part 4: MORPH - Performance of Automated Face Morph Detection                                     |
| 2020-03-06 | FATE  | <a href="#">Draft</a>   | Part 5: Face Image Quality Assessment                                                             |
| 2020-07-24 | FRTE  | <a href="#">8311</a>    | Part 6A: Face Recognition Accuracy with Face Masks using Pre-COVID-19 Algorithms                  |
| 2022-01-20 | FRTE  | <a href="#">8331</a>    | Part 6B: Face Recognition Accuracy with Face Masks using Post-COVID-19 Algorithms                 |
| 2022-07-13 | FRTE  | <a href="#">8381</a>    | Part 7: Identification for Paperless Travel and Immigration                                       |
| 2022-09-30 | FRTE  | <a href="#">8429</a>    | Part 8: Summarizing Demographic Differentials                                                     |
| 2022-09-30 | FRTE  | <a href="#">8439</a>    | Part 9A: Face Recognition Verification Accuracy on Distinguishing Twins                           |
| 2023-09-20 | FATE  | <a href="#">8491</a>    | Part 10: Performance of Passive Software-based Presentation Attack Detection (PAD) Algorithms     |
| 2023-09-20 | FATE  | <a href="#">8485</a>    | Part 11: Face Image Quality Vector Assessment: Specific Image Defect Detection                    |
| 2024-05-29 | FATE  | <a href="#">8525</a>    | Part 12: Face Analysis Technology Evaluation: Age Estimation and Verification                     |
| 2025-08-18 | FATE  | <a href="#">8584</a>    | Part 4B: Considerations for Implementing Morph Detection in Operations                            |
| 2025-09-25 | FRTE  | <a href="#">500-343</a> | Face Recognition Technology Evaluation (FRTE): Preparation of Compact Face Images for 2D Barcodes |

Details appear on pages linked from <https://www.nist.gov/programs-projects/face-projects>.

## Key words

Face Analysis Technology Evaluation; FATE; AEV; Age estimation; Age Verification; Age Assurance

## Release notes

**2026-02-13:** The AEV track of the FATE remains open.

- ▷ This document is the twelfth update to [NIST Interagency Report 8525](#). It contains results for two developers: ROC, and Shufti.

**2026-01-23:** The AEV track of the FATE remains open.

- ▷ This document is the twelfth update to [NIST Interagency Report 8525](#). It contains results for two developers: Yoti, and Privately.

**2025-12-18:** The AEV track of the FATE remains open.

- ▷ This document is the twelfth update to [NIST Interagency Report 8525](#). It contains results for two developers: Innovatrics, and Daon.

**2025-08-28:** The AEV track of the FATE remains open.

- ▷ This document is the twelfth update to [NIST Interagency Report 8525](#). It contains results for three developers: Cyberlink Corp, Regula Forensics, and Youverse.

**2025-07-21:** The AEV track of the FATE remains open.

- ▷ This document is the eleventh update to [NIST Interagency Report 8525](#). It contains results for four developers: Cognitec Systems GmbH, Idemia, Innovatrics, and Yoti.

**2025-05-21:** The AEV track of the FATE remains open.

- ▷ This document is the tenth update to [NIST Interagency Report 8525](#). It contains results for one developers: Incode Technologies Inc.

**2025-05-14:** The AEV track of the FATE remains open.

- ▷ This document is the ninth update to [NIST Interagency Report 8525](#). It contains results for two developers: Czech Technical University in Prague, and Daon.

**2025-04-11:** The AEV track of the FATE remains open.

- ▷ This document is the eighth update to [NIST Interagency Report 8525](#). It contains results for three developers: Hangzhao Ailu Network Information Technology, Nominder, and Ondato UAB.
- ▷ The FATE [AEV homepage](#) includes age estimation noise results from the frontal video frames and the yaw variation video frames.

**2025-02-28:** The AEV track of the FATE remains open.

- ▷ This document is the seventh update to [NIST Interagency Report 8525](#). It contains results for one developers: Innovatrics.
- ▷ The FATE [AEV homepage](#) includes age estimation noise results from the frontal video frames and the yaw variation video frames.

**2025-01-28:** The AEV track of the FATE remains open.

- ▷ This document is the sixth update to [NIST Interagency Report 8525](#). It contains results for one developers: Dermalog.

**2024-12-20:** The AEV track of the FATE remains open.

- ▷ This document is the fifth update to [NIST Interagency Report 8525](#). It contains results for three developers: Nominder, Ondato UAB, and Viante.AI.

**2024-11-08:** The AEV track of the FATE remains open.

- ▷ This document is the fourth update to [NIST Interagency Report 8525](#). It contains results for two developers: ROC and Czech Technical University in Prague.

**2024-10-07:** The AEV track of the FATE remains open.

- ▷ This document is the third update to [NIST Interagency Report 8525](#). It contains results for two developers: Yoti and the Brno University of Technology.
- ▷ The FATE [AEV homepage](#) has been revamped to include MAE by demographic group and MAE by image type.

**2024-09-16:** The AEV track of the FATE remains open.

- ▷ This document is the second update to [NIST Interagency Report 8525](#). It contains results for two new developers: Idemia and Jumio.

**2024-08-23:** The AEV track of the FATE remains open.

- ▷ This document is the first update to [NIST Interagency Report 8525](#). It contains results for two new developers: Nominder and the Czech Technical University in Prague.
- ▷ The document also includes results for algorithms for one returning developer: Neurotechnology.

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## Executive Summary

### Motivation

Age assurance is the subject of recent legislation both inside and outside the United States. This has driven the need for age assurance methods to support applications such as verification that a person is above age 18, 21, or other key ages (e.g., for sale of alcohol or online access). While the mechanism for estimating age may not be specified in legislation, software-based face analysis is one potential approach supported by ubiquitous inexpensive cameras. Age estimation can operate statelessly with no requirement for persistent storage of a photo or biometric data derived from it.

### Overview

Age assurance applications can employ either an age verification (AV) algorithm that produces a yes/no guess at whether someone is above an age threshold, or an age estimation (AE) algorithm that emits a numeric age estimate that can be compared with the age threshold i.e. AV can be implemented with AE, or it can be computed without explicit AE.

This report quantifies the capability of six software-based AE and AV prototypes applied to around eleven million photos drawn from four operational repositories: immigration visas, arrest mugshots, border crossings, and immigration office photos. The report includes statements of age estimation (AE) accuracy globally and across demographic groups. It uses AE with thresholds to measure performance in age verification tasks and also reports the dependence on image quality. The report excludes performance measured in interactive sessions, in which a person can cooperatively present and re-present to a camera. It does not measure accuracy effects related to disguises, cosmetics, or other presentation attacks. It does not address policy nor recommend AV thresholds as these differ across applications and jurisdictions.

### Audience

This report is of primary utility to actual and prospective deployers of the technology faced with selection of particular AE implementations. It is additionally intended to inform policymakers interested in assessing absolute capability and whether that is sufficient for particular use-cases. The report will be useful to developers in exposing factor affecting performance and in documenting capability relative to other prototypes. This report is a snapshot; the AEV benchmark remain opens for new and returning developers as a means to track technological gains.

### Results

Age estimation accuracy has improved since we first measured it in 2014 - see [NISTIR 7995](#). However, for both age estimation (e.g., MAE) and metrics appropriate to “challenge” that require alternative means of age assurance if age is not above some limit, accuracy is strongly influenced by algorithm, sex, image quality, region-of-birth, age itself, and interactions between those factors. There is no uniformly superior algorithm, and algorithm rankings differ across those factors. Accuracy also depends on the metric: age estimation error vs. challenge-25-like false positive rates. Given variation across demographic groups, we anticipate developers improving capability over time. Future FATE AEV reports will address online safety applications (for 13-16 year olds), report on new datasets, and include extended analyses. Detailed results follow in the Technical Summary.

## Technical Summary

**Algorithms:** FATE AEV is an ongoing technology evaluation of age estimation implementations. This report is a snapshot of results documenting the application of six commercially-developed age estimation algorithms to four archived datasets. The algorithms are prototypes - so not necessarily available as mature integrable products. See sec. 4.

**Datasets:** We used four large operational sets of photographs. The sets are accompanied by subject metadata including age and, in some cases, sex and country-of-birth information. The four datasets are domestic *mugshots* collected in the United States after arrest, *application* photographs taken in immigration offices from a global population, *visa* photographs collected in consular offices in Mexico, and *border* crossing photographs of travelers entering the United States. See sec. 3.

**Accuracy gains since 2014:** This report includes results for a dataset used in our 2014 evaluation of age estimation algorithms. With a decade of research and development, and significant advances in deep neural network technology, we see five of six algorithms outperform the most accurate algorithm submitted to NIST in 2014. Using a common *visa* database, best mean absolute error (MAE) has reduced from 4.3 to 3.1 years. See sec. 5.1.

**Age estimation accuracy:** The AEV program aims to measure age estimation accuracy, and to quantify gains over time. This is done by running empirical trials using large datasets. Table 1 shows standard statements of age estimation error for all algorithms on the 18-24 subset of the four databases.

Notable from the table is that the main summary measure of how close age estimates are to ground truth - mean absolute error (MAE) - varies between algorithms operating on any one dataset, and between datasets using any one algorithm. This is a function of algorithm sensitivities to image quality and demographics, as discussed below. Within one dataset, aggregate MAE hides variation across sex and ethnicity. Moreover, MAE also does not specifically quantify how many age estimates are large overestimates, so it is not an appropriate metric for age verification (AV) tasks. See metrics in sec. 2.

**Table 1.** Mean, median and standard deviation of absolute error (MAE, MDAE, SD) and failure to process rate (FTP) for subjects between the age 18 to 24 year old. Num. Img shows how many images processed successfully by the algorithm. Lower values are better.

| Algorithm     | Dataset     | Num. Img | MAE | MDAE | SD  | FTP   |
|---------------|-------------|----------|-----|------|-----|-------|
| cognitec-000  | Mugshot     | 376773   | 2.4 | 1.8  | 2.1 | 0.000 |
| cognitec-000  | Border      | 236052   | 3.1 | 2.4  | 2.6 | 0.001 |
| cognitec-000  | Application | 111937   | 2.5 | 2.0  | 2.2 | 0.000 |
| cognitec-000  | Visa        | 857763   | 2.6 | 2.1  | 2.1 | 0.000 |
| cvut-001      | Mugshot     | 368979   | 3.5 | 3.0  | 3.1 | 0.021 |
| cvut-001      | Border      | 235829   | 3.3 | 2.5  | 3.5 | 0.002 |
| cvut-001      | Application | 111938   | 3.2 | 2.5  | 2.8 | 0.000 |
| cvut-001      | Visa        | 857761   | 4.5 | 3.7  | 3.7 | 0.000 |
| cvut-002      | Mugshot     | 368982   | 2.4 | 2.0  | 2.3 | 0.021 |
| cvut-002      | Border      | 236216   | 3.6 | 3.1  | 2.8 | 0.000 |
| cvut-002      | Application | 111938   | 2.5 | 2.0  | 2.4 | 0.000 |
| cvut-002      | Visa        | 857761   | 3.4 | 2.8  | 2.6 | 0.000 |
| cvut-003      | Mugshot     | 368895   | 2.7 | 2.2  | 2.4 | 0.021 |
| cvut-003      | Border      | 236216   | 3.4 | 3.0  | 2.6 | 0.000 |
| cvut-003      | Application | 111938   | 2.4 | 1.8  | 2.2 | 0.000 |
| cvut-003      | Visa        | 856824   | 4.6 | 4.2  | 2.9 | 0.001 |
| cyberlink-000 | Mugshot     | 376849   | 3.6 | 3.0  | 2.9 | 0.000 |
| cyberlink-000 | Border      | 236227   | 3.2 | 2.7  | 2.7 | 0.000 |
| cyberlink-000 | Application | 111938   | 3.5 | 3.1  | 2.7 | 0.000 |
| cyberlink-000 | Visa        | 857764   | 3.8 | 3.4  | 2.7 | 0.000 |
| cyberlink-001 | Mugshot     | 376849   | 2.8 | 2.0  | 2.4 | 0.000 |
| cyberlink-001 | Border      | 236227   | 2.8 | 2.3  | 2.3 | 0.000 |



**Table 1.** Mean, median and standard deviation of absolute error (MAE, MDAE, SD) and failure to process rate (FTP) for subjects between the age 18 to 24 year old. Num. Img shows how many images processed successfully by the algorithm. Lower values are better.  
(continued)

| Algorithm           | Dataset     | Num. Img | MAE  | MDAE | SD  | FTP   |
|---------------------|-------------|----------|------|------|-----|-------|
| cyberlink-001       | Application | 111938   | 2.6  | 2.0  | 2.3 | 0.000 |
| cyberlink-001       | Visa        | 857765   | 3.3  | 2.8  | 2.5 | 0.000 |
| daon-000            | Mugshot     | 376841   | 4.3  | 3.7  | 3.3 | 0.000 |
| daon-000            | Border      | 236231   | 4.8  | 4.2  | 3.6 | 0.000 |
| daon-000            | Application | 111938   | 4.1  | 3.4  | 3.2 | 0.000 |
| daon-000            | Visa        | 857763   | 4.7  | 4.3  | 3.1 | 0.000 |
| daon-001            | Mugshot     | 376835   | 3.5  | 3.0  | 2.8 | 0.000 |
| daon-001            | Border      | 236220   | 4.2  | 3.8  | 3.1 | 0.000 |
| daon-001            | Application | 111938   | 3.7  | 3.1  | 2.9 | 0.000 |
| daon-001            | Visa        | 857764   | 4.3  | 4.1  | 2.7 | 0.000 |
| dermalog-001        | Mugshot     | 376560   | 3.8  | 3.0  | 3.2 | 0.001 |
| dermalog-001        | Border      | 236251   | 5.1  | 4.6  | 3.6 | 0.000 |
| dermalog-001        | Application | 111938   | 3.3  | 2.6  | 2.8 | 0.000 |
| dermalog-001        | Visa        | 857766   | 4.7  | 4.2  | 3.2 | 0.000 |
| dermalog-002        | Mugshot     | 376560   | 3.9  | 3.0  | 3.4 | 0.001 |
| dermalog-002        | Border      | 236251   | 5.1  | 4.4  | 3.8 | 0.000 |
| dermalog-002        | Application | 111938   | 3.6  | 2.9  | 2.9 | 0.000 |
| dermalog-002        | Visa        | 857766   | 4.4  | 3.8  | 3.3 | 0.000 |
| hzailu-001          | Mugshot     | 376699   | 7.5  | 7.0  | 4.4 | 0.000 |
| hzailu-001          | Border      | 236236   | 5.7  | 5.1  | 3.9 | 0.000 |
| hzailu-001          | Application | 111938   | 5.9  | 5.5  | 3.8 | 0.000 |
| hzailu-001          | Visa        | 857763   | 7.8  | 7.6  | 3.9 | 0.000 |
| idemia-001          | Mugshot     | 376850   | 2.1  | 1.7  | 1.9 | 0.000 |
| idemia-001          | Border      | 236232   | 2.5  | 2.0  | 2.1 | 0.000 |
| idemia-001          | Application | 111938   | 2.6  | 2.1  | 2.3 | 0.000 |
| idemia-001          | Visa        | 857764   | 2.8  | 2.3  | 2.2 | 0.000 |
| idemia-002          | Mugshot     | 376850   | 2.4  | 1.9  | 2.1 | 0.000 |
| idemia-002          | Border      | 236232   | 2.7  | 2.2  | 2.2 | 0.000 |
| idemia-002          | Application | 111938   | 2.6  | 2.1  | 2.3 | 0.000 |
| idemia-002          | Visa        | 857764   | 3.5  | 2.9  | 2.7 | 0.000 |
| inbut-000           | Mugshot     | 376822   | 2.8  | 1.9  | 2.7 | 0.000 |
| inbut-000           | Border      | 236224   | 3.2  | 2.8  | 2.5 | 0.000 |
| inbut-000           | Application | 111938   | 3.7  | 2.9  | 3.3 | 0.000 |
| inbut-000           | Visa        | 857764   | 10.5 | 10.9 | 4.2 | 0.000 |
| incode-000          | Mugshot     | 376710   | 2.7  | 2.2  | 2.5 | 0.000 |
| incode-000          | Border      | 236181   | 4.5  | 3.6  | 3.9 | 0.000 |
| incode-000          | Application | 111938   | 3.0  | 2.4  | 2.5 | 0.000 |
| incode-000          | Visa        | 857762   | 3.5  | 2.9  | 2.7 | 0.000 |
| incode-001          | Mugshot     | 376606   | 2.5  | 1.9  | 2.5 | 0.001 |
| incode-001          | Border      | 236171   | 3.9  | 2.9  | 3.7 | 0.000 |
| incode-001          | Application | 111936   | 3.1  | 2.4  | 2.7 | 0.000 |
| incode-001          | Visa        | 857759   | 3.7  | 2.9  | 3.1 | 0.000 |
| innovatrics-000     | Mugshot     | 376794   | 2.0  | 1.5  | 1.8 | 0.000 |
| innovatrics-000     | Border      | 236171   | 2.5  | 2.1  | 2.1 | 0.000 |
| innovatrics-000     | Application | 111938   | 2.5  | 2.0  | 2.2 | 0.000 |
| innovatrics-000     | Visa        | 857764   | 3.2  | 2.9  | 2.3 | 0.000 |
| innovatrics-001     | Mugshot     | 376794   | 1.9  | 1.5  | 1.8 | 0.000 |
| innovatrics-001     | Border      | 236171   | 2.4  | 1.9  | 2.1 | 0.000 |
| innovatrics-001     | Application | 111938   | 2.3  | 1.8  | 2.1 | 0.000 |
| innovatrics-001     | Visa        | 857764   | 3.0  | 2.6  | 2.2 | 0.000 |
| innovatrics-002     | Mugshot     | 376823   | 2.3  | 1.8  | 2.0 | 0.000 |
| innovatrics-002     | Border      | 236208   | 2.7  | 2.3  | 2.2 | 0.000 |
| innovatrics-002     | Application | 111938   | 2.8  | 2.2  | 2.3 | 0.000 |
| innovatrics-002     | Visa        | 857765   | 3.2  | 2.6  | 2.4 | 0.000 |
| jumio-000           | Mugshot     | 376344   | 2.9  | 2.2  | 2.8 | 0.001 |
| jumio-000           | Border      | 226033   | 5.6  | 4.5  | 4.6 | 0.043 |
| jumio-000           | Application | 111759   | 3.4  | 2.7  | 2.9 | 0.002 |
| jumio-000           | Visa        | 835881   | 4.3  | 3.9  | 3.0 | 0.026 |
| neurotechnology-000 | Mugshot     | 376764   | 3.6  | 3.0  | 3.0 | 0.000 |
| neurotechnology-000 | Border      | 236169   | 3.4  | 2.8  | 2.8 | 0.000 |
| neurotechnology-000 | Application | 111938   | 3.7  | 3.1  | 2.9 | 0.000 |
| neurotechnology-000 | Visa        | 857764   | 3.2  | 2.5  | 2.7 | 0.000 |
| neurotechnology-001 | Mugshot     | 376726   | 2.9  | 2.0  | 2.7 | 0.000 |
| neurotechnology-001 | Border      | 236107   | 3.2  | 2.6  | 2.7 | 0.001 |
| neurotechnology-001 | Application | 111938   | 3.2  | 2.6  | 2.6 | 0.000 |
| neurotechnology-001 | Visa        | 857762   | 3.1  | 2.5  | 2.6 | 0.000 |
| nominder-000        | Mugshot     | 376720   | 2.8  | 2.2  | 2.5 | 0.000 |
| nominder-000        | Border      | 236203   | 3.2  | 2.7  | 2.6 | 0.000 |
| nominder-000        | Application | 111938   | 3.1  | 2.5  | 2.6 | 0.000 |
| nominder-000        | Visa        | 857765   | 3.5  | 2.9  | 2.7 | 0.000 |
| nominder-001        | Mugshot     | 376683   | 2.3  | 1.7  | 2.1 | 0.000 |

**Table 1.** Mean, median and standard deviation of absolute error (MAE, MDAE, SD) and failure to process rate (FTP) for subjects between the age 18 to 24 year old. Num. Img shows how many images processed successfully by the algorithm. Lower values are better. (continued)

| Algorithm     | Dataset     | Num. Img | MAE | MDAE | SD  | FTP   |
|---------------|-------------|----------|-----|------|-----|-------|
| nominder-001  | Border      | 236204   | 2.6 | 2.1  | 2.3 | 0.000 |
| nominder-001  | Application | 111938   | 2.4 | 1.9  | 2.2 | 0.000 |
| nominder-001  | Visa        | 857765   | 3.7 | 3.0  | 2.9 | 0.000 |
| ondato-000    | Mugshot     | 376835   | 2.8 | 2.2  | 2.6 | 0.000 |
| ondato-000    | Border      | 236209   | 2.8 | 2.3  | 2.4 | 0.000 |
| ondato-000    | Application | 111938   | 3.1 | 2.4  | 2.7 | 0.000 |
| ondato-000    | Visa        | 857764   | 3.1 | 2.6  | 2.5 | 0.000 |
| ondato-001    | Mugshot     | 376835   | 2.7 | 2.1  | 2.5 | 0.000 |
| ondato-001    | Border      | 236209   | 2.7 | 2.2  | 2.4 | 0.000 |
| ondato-001    | Application | 111938   | 2.9 | 2.3  | 2.5 | 0.000 |
| ondato-001    | Visa        | 857764   | 3.2 | 2.7  | 2.4 | 0.000 |
| privately-000 | Mugshot     | 376867   | 2.9 | 2.3  | 2.5 | 0.000 |
| privately-000 | Border      | 236272   | 3.9 | 3.3  | 3.1 | 0.000 |
| privately-000 | Application | 111938   | 2.9 | 2.3  | 2.5 | 0.000 |
| privately-000 | Visa        | 857774   | 3.4 | 2.8  | 2.6 | 0.000 |
| regula-000    | Mugshot     | 376840   | 2.0 | 1.5  | 1.9 | 0.000 |
| regula-000    | Border      | 236175   | 2.6 | 2.0  | 2.3 | 0.000 |
| regula-000    | Application | 111938   | 2.2 | 1.7  | 2.1 | 0.000 |
| regula-000    | Visa        | 857764   | 2.6 | 2.1  | 2.2 | 0.000 |
| roc-000       | Mugshot     | 376809   | 2.3 | 1.7  | 2.3 | 0.000 |
| roc-000       | Border      | 236191   | 4.0 | 3.4  | 3.1 | 0.000 |
| roc-000       | Application | 111938   | 3.5 | 2.9  | 2.8 | 0.000 |
| roc-000       | Visa        | 857764   | 3.7 | 2.8  | 3.1 | 0.000 |
| roc-001       | Mugshot     | 376818   | 2.2 | 1.6  | 2.1 | 0.000 |
| roc-001       | Border      | 236198   | 3.5 | 3.1  | 2.6 | 0.000 |
| roc-001       | Application | 111938   | 3.2 | 2.8  | 2.6 | 0.000 |
| roc-001       | Visa        | 857764   | 3.9 | 3.5  | 2.8 | 0.000 |
| roc-002       | Mugshot     | 376850   | 1.8 | 1.4  | 1.6 | 0.000 |
| roc-002       | Border      | 236199   | 3.2 | 2.7  | 2.5 | 0.000 |
| roc-002       | Application | 111938   | 2.5 | 2.0  | 2.2 | 0.000 |
| roc-002       | Visa        | 857764   | 2.4 | 1.9  | 2.0 | 0.000 |
| shufti-000    | Mugshot     | 376392   | 3.1 | 2.2  | 3.0 | 0.001 |
| shufti-000    | Border      | 234897   | 4.2 | 3.2  | 3.8 | 0.006 |
| shufti-000    | Application | 111932   | 4.0 | 3.2  | 3.4 | 0.000 |
| shufti-000    | Visa        | 857762   | 3.5 | 2.7  | 3.0 | 0.000 |
| unisse-001    | Mugshot     | 376776   | 3.5 | 3.0  | 3.1 | 0.000 |
| unisse-001    | Border      | 236182   | 3.4 | 3.0  | 2.8 | 0.000 |
| unisse-001    | Application | 111938   | 4.0 | 3.5  | 3.1 | 0.000 |
| unisse-001    | Visa        | 857764   | 4.1 | 3.6  | 3.0 | 0.000 |
| viant-000     | Mugshot     | 376674   | 4.8 | 5.0  | 2.7 | 0.001 |
| viant-000     | Border      | 236210   | 4.1 | 3.8  | 2.8 | 0.000 |
| viant-000     | Application | 111938   | 5.0 | 4.7  | 2.7 | 0.000 |
| viant-000     | Visa        | 857759   | 6.2 | 6.1  | 3.1 | 0.000 |
| yoti-001      | Mugshot     | 376274   | 2.7 | 2.1  | 2.5 | 0.002 |
| yoti-001      | Border      | 235988   | 5.1 | 4.5  | 3.8 | 0.001 |
| yoti-001      | Application | 111936   | 4.1 | 3.4  | 3.3 | 0.000 |
| yoti-001      | Visa        | 857713   | 3.8 | 3.1  | 3.0 | 0.000 |
| yoti-002      | Mugshot     | 376867   | 2.5 | 2.1  | 2.2 | 0.000 |
| yoti-002      | Border      | 236272   | 4.4 | 4.0  | 3.1 | 0.000 |
| yoti-002      | Application | 111938   | 3.5 | 2.9  | 2.7 | 0.000 |
| yoti-002      | Visa        | 857774   | 3.1 | 2.6  | 2.4 | 0.000 |
| yoti-003      | Mugshot     | 376867   | 2.4 | 1.9  | 2.1 | 0.000 |
| yoti-003      | Border      | 236272   | 4.3 | 3.8  | 3.1 | 0.000 |
| yoti-003      | Application | 111938   | 3.3 | 2.7  | 2.7 | 0.000 |
| yoti-003      | Visa        | 857774   | 3.1 | 2.6  | 2.4 | 0.000 |
| yoti-004      | Mugshot     | 376867   | 1.9 | 1.5  | 1.8 | 0.000 |
| yoti-004      | Border      | 236272   | 3.8 | 3.3  | 2.8 | 0.000 |
| yoti-004      | Application | 111938   | 2.6 | 2.1  | 2.3 | 0.000 |
| yoti-004      | Visa        | 857774   | 2.5 | 2.1  | 2.0 | 0.000 |
| yoverse-000   | Mugshot     | 376691   | 3.3 | 2.4  | 3.0 | 0.000 |
| yoverse-000   | Border      | 236217   | 4.7 | 4.0  | 3.6 | 0.000 |
| yoverse-000   | Application | 111938   | 3.1 | 2.5  | 2.6 | 0.000 |
| yoverse-000   | Visa        | 857689   | 3.1 | 2.5  | 2.5 | 0.000 |

**Impact of image quality:** Table 1 shows mugshots are easiest for three of six algorithms using the MAE metric, and border crossing photos are the hardest for four of six algorithms. Mugshots and application photos are collected with standardized photographic setups. The border photos are

collected with a webcam and exhibit more variation in head orientation away from frontal, and lower contrast. On such images, algorithms perform less well than on office-collected application photos of the same people (i.e. the same demographics). We have not conducted a detailed cause-and-effect analysis to relate age estimation error to explicit measurements of standardized quality metrics such as sharpness, exposure and head orientation. The datasets are detailed in 3. See sec. 5.6.

**Impact of eyeglasses:** Eyeglasses have an effect on age estimation error. Four of six algorithms show higher age estimation error (over-estimation or under-estimation) in both male and female when wearing glasses vs. not wearing glasses. See sec. 5.6.1.

**Age verification accuracy:** For restricted-age applications such as alcohol purchase, a Challenge-T policy accepts people with age estimated at or above T but requires additional age assurance checks on anyone assessed to have age below T. Table 2 shows false positive rates (FPR) for people below a legal limit (18 in the first four columns, and 21 in the second four columns) but who are estimated to be at or above the challenge age T. FPR, which should be low, decreases with higher Challenge-T values. The use of higher T values will cause more people at or above legal age to be subject to additional age assurance. The tables show that for four of six algorithms, FPR is markedly lower in men than women, the opposite is true for another, and parity is almost achieved for one more.

**Table 2.** For Application images, false positive rates (FPR) for 14-17 year olds, 14-20 years olds, at challenge ages, 25, 28 and 31. Values are lower the better.

| Algorithm           | Male<br>14-17<br>T=25 | Female<br>14-17<br>T=25 | Male<br>14-17<br>T=28 | Female<br>14-17<br>T=28 | Male<br>14-20<br>T=28 | Female<br>14-20<br>T=28 | Male<br>14-20<br>T=31 | Female<br>14-20<br>T=31 |
|---------------------|-----------------------|-------------------------|-----------------------|-------------------------|-----------------------|-------------------------|-----------------------|-------------------------|
| cognitec-000        | 0.005                 | 0.014                   | 0.001                 | 0.003                   | 0.007                 | 0.013                   | 0.002                 | 0.004                   |
| cvut-001            | 0.039                 | 0.100                   | 0.006                 | 0.025                   | 0.029                 | 0.052                   | 0.006                 | 0.018                   |
| cvut-002            | 0.011                 | 0.032                   | 0.002                 | 0.005                   | 0.012                 | 0.015                   | 0.004                 | 0.005                   |
| cvut-003            | 0.008                 | 0.029                   | 0.002                 | 0.005                   | 0.009                 | 0.017                   | 0.003                 | 0.006                   |
| cyberlink-000       | 0.054                 | 0.091                   | 0.012                 | 0.020                   | 0.051                 | 0.050                   | 0.008                 | 0.011                   |
| cyberlink-001       | 0.013                 | 0.023                   | 0.002                 | 0.005                   | 0.014                 | 0.016                   | 0.003                 | 0.005                   |
| daon-000            | 0.094                 | 0.067                   | 0.027                 | 0.024                   | 0.066                 | 0.048                   | 0.020                 | 0.017                   |
| daon-001            | 0.045                 | 0.034                   | 0.009                 | 0.011                   | 0.037                 | 0.030                   | 0.010                 | 0.010                   |
| dermalog-001        | 0.051                 | 0.112                   | 0.012                 | 0.036                   | 0.034                 | 0.069                   | 0.010                 | 0.021                   |
| dermalog-002        | 0.034                 | 0.167                   | 0.009                 | 0.037                   | 0.031                 | 0.081                   | 0.010                 | 0.020                   |
| hzailu-001          | 0.341                 | 0.361                   | 0.127                 | 0.154                   | 0.246                 | 0.247                   | 0.091                 | 0.080                   |
| idemia-001          | 0.004                 | 0.012                   | 0.002                 | 0.003                   | 0.009                 | 0.012                   | 0.003                 | 0.003                   |
| idemia-002          | 0.002                 | 0.009                   | 0.002                 | 0.003                   | 0.006                 | 0.012                   | 0.002                 | 0.003                   |
| inbut-000           | 0.100                 | 0.114                   | 0.055                 | 0.050                   | 0.072                 | 0.075                   | 0.034                 | 0.030                   |
| incode-000          | 0.006                 | 0.033                   | 0.002                 | 0.010                   | 0.008                 | 0.027                   | 0.002                 | 0.008                   |
| incode-001          | 0.006                 | 0.034                   | 0.002                 | 0.010                   | 0.011                 | 0.038                   | 0.004                 | 0.010                   |
| innovatrics-000     | 0.007                 | 0.030                   | 0.002                 | 0.005                   | 0.008                 | 0.015                   | 0.002                 | 0.004                   |
| innovatrics-001     | 0.005                 | 0.024                   | 0.001                 | 0.004                   | 0.007                 | 0.013                   | 0.002                 | 0.003                   |
| innovatrics-002     | 0.008                 | 0.034                   | 0.002                 | 0.006                   | 0.010                 | 0.020                   | 0.003                 | 0.006                   |
| jumio-000           | 0.021                 | 0.088                   | 0.009                 | 0.026                   | 0.019                 | 0.067                   | 0.007                 | 0.020                   |
| neurotechnology-000 | 0.248                 | 0.178                   | 0.062                 | 0.035                   | 0.101                 | 0.057                   | 0.028                 | 0.014                   |
| neurotechnology-001 | 0.157                 | 0.125                   | 0.019                 | 0.014                   | 0.040                 | 0.027                   | 0.011                 | 0.007                   |
| nominder-000        | 0.020                 | 0.041                   | 0.004                 | 0.012                   | 0.025                 | 0.031                   | 0.007                 | 0.009                   |
| nominder-001        | 0.007                 | 0.021                   | 0.002                 | 0.004                   | 0.009                 | 0.015                   | 0.003                 | 0.004                   |
| ondato-000          | 0.027                 | 0.073                   | 0.006                 | 0.018                   | 0.022                 | 0.040                   | 0.006                 | 0.011                   |
| ondato-001          | 0.014                 | 0.051                   | 0.003                 | 0.010                   | 0.012                 | 0.030                   | 0.004                 | 0.007                   |
| privately-000       | 0.014                 | 0.065                   | 0.003                 | 0.012                   | 0.008                 | 0.029                   | 0.002                 | 0.006                   |
| regula-000          | 0.003                 | 0.012                   | 0.001                 | 0.003                   | 0.005                 | 0.012                   | 0.002                 | 0.004                   |
| roc-000             | 0.041                 | 0.073                   | 0.006                 | 0.016                   | 0.030                 | 0.049                   | 0.006                 | 0.011                   |
| roc-001             | 0.013                 | 0.040                   | 0.002                 | 0.006                   | 0.015                 | 0.030                   | 0.004                 | 0.007                   |

|                     |       |       |       |       |       |       |       |       |
|---------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>roc-002</b>      | 0.006 | 0.016 | 0.002 | 0.003 | 0.008 | 0.014 | 0.002 | 0.003 |
| <b>shufti-000</b>   | 0.029 | 0.067 | 0.009 | 0.022 | 0.044 | 0.066 | 0.015 | 0.023 |
| <b>unissey-001</b>  | 0.107 | 0.127 | 0.031 | 0.038 | 0.091 | 0.092 | 0.018 | 0.019 |
| <b>vianite-000</b>  | 0.138 | 0.253 | 0.023 | 0.049 | 0.111 | 0.123 | 0.017 | 0.020 |
| <b>yoti-001</b>     | 0.012 | 0.090 | 0.003 | 0.032 | 0.017 | 0.093 | 0.004 | 0.033 |
| <b>yoti-002</b>     | 0.009 | 0.063 | 0.002 | 0.014 | 0.010 | 0.046 | 0.003 | 0.011 |
| <b>yoti-003</b>     | 0.008 | 0.056 | 0.003 | 0.014 | 0.011 | 0.046 | 0.004 | 0.011 |
| <b>yoti-004</b>     | 0.005 | 0.026 | 0.001 | 0.004 | 0.006 | 0.017 | 0.002 | 0.004 |
| <b>youverse-000</b> | 0.026 | 0.051 | 0.008 | 0.015 | 0.030 | 0.031 | 0.008 | 0.009 |

Table 2 uses two samples sets (14-17 and 14-20), but these subsets are imbalanced by age and sex per the characteristic demographics of the operational population the set was drawn from. Imbalance matters, because 14 year olds have markedly lower FPR than do those aged 17: Table 3 shows that Challenge-25 false positive rates increase by an order-of-magnitude as subjects age from 14 through 20 - for example for the roc-000 algorithm FPR at age 20 (0.295) is almost fifteen times larger than at age 14 (0.02). The implication of imbalance in a set is that a test that employs that set without taking any steps to correct for that imbalance, will yield unrepresentative results - see the discussion in 2.6.3. A test of an operational system should typically proceed with operationally representative numbers of 14, 15, . . . year old males and females.

**Table 3.** For Application images, false positive rates for Challenge-25 by actual age. The FPR values are averages of 12 FPR estimates from two sexes and six regions-of-birth. Lower values are the better.

| Algorithm                  | Actual Age |       |       |       |       |       |       |
|----------------------------|------------|-------|-------|-------|-------|-------|-------|
|                            | 14         | 15    | 16    | 17    | 18    | 19    | 20    |
| <b>cognitec-000</b>        | 0.003      | 0.009 | 0.016 | 0.027 | 0.044 | 0.078 | 0.125 |
| <b>cvut-001</b>            | 0.055      | 0.075 | 0.106 | 0.148 | 0.173 | 0.241 | 0.304 |
| <b>cvut-002</b>            | 0.013      | 0.018 | 0.035 | 0.064 | 0.083 | 0.136 | 0.197 |
| <b>cvut-003</b>            | 0.013      | 0.018 | 0.030 | 0.051 | 0.068 | 0.111 | 0.162 |
| <b>cyberlink-000</b>       | 0.041      | 0.059 | 0.096 | 0.149 | 0.199 | 0.281 | 0.375 |
| <b>cyberlink-001</b>       | 0.006      | 0.013 | 0.026 | 0.047 | 0.067 | 0.120 | 0.183 |
| <b>daon-000</b>            | 0.031      | 0.059 | 0.096 | 0.143 | 0.168 | 0.234 | 0.299 |
| <b>daon-001</b>            | 0.014      | 0.024 | 0.045 | 0.087 | 0.114 | 0.178 | 0.257 |
| <b>dermalog-001</b>        | 0.039      | 0.060 | 0.097 | 0.134 | 0.151 | 0.226 | 0.284 |
| <b>dermalog-002</b>        | 0.046      | 0.074 | 0.106 | 0.152 | 0.189 | 0.251 | 0.322 |
| <b>hzailu-001</b>          | 0.246      | 0.336 | 0.444 | 0.522 | 0.590 | 0.663 | 0.717 |
| <b>idemia-001</b>          | 0.001      | 0.007 | 0.012 | 0.024 | 0.040 | 0.078 | 0.132 |
| <b>idemia-002</b>          | 0.001      | 0.005 | 0.008 | 0.019 | 0.030 | 0.057 | 0.102 |
| <b>inbut-000</b>           | 0.092      | 0.123 | 0.159 | 0.200 | 0.202 | 0.288 | 0.371 |
| <b>incode-000</b>          | 0.008      | 0.012 | 0.031 | 0.047 | 0.062 | 0.112 | 0.171 |
| <b>incode-001</b>          | 0.006      | 0.012 | 0.029 | 0.053 | 0.078 | 0.140 | 0.214 |
| <b>innovatrics-000</b>     | 0.008      | 0.014 | 0.028 | 0.055 | 0.074 | 0.130 | 0.193 |
| <b>innovatrics-001</b>     | 0.007      | 0.012 | 0.025 | 0.048 | 0.065 | 0.114 | 0.168 |
| <b>innovatrics-002</b>     | 0.010      | 0.017 | 0.038 | 0.069 | 0.090 | 0.155 | 0.219 |
| <b>jumio-000</b>           | 0.037      | 0.052 | 0.079 | 0.115 | 0.147 | 0.219 | 0.303 |
| <b>neurotechnology-000</b> | 0.200      | 0.231 | 0.293 | 0.353 | 0.380 | 0.426 | 0.465 |
| <b>neurotechnology-001</b> | 0.058      | 0.090 | 0.120 | 0.169 | 0.201 | 0.243 | 0.302 |
| <b>nominder-000</b>        | 0.017      | 0.025 | 0.041 | 0.073 | 0.101 | 0.161 | 0.227 |
| <b>nominder-001</b>        | 0.004      | 0.008 | 0.023 | 0.042 | 0.060 | 0.099 | 0.150 |
| <b>ondato-000</b>          | 0.029      | 0.044 | 0.068 | 0.108 | 0.129 | 0.200 | 0.273 |
| <b>ondato-001</b>          | 0.015      | 0.029 | 0.047 | 0.078 | 0.107 | 0.169 | 0.248 |
| <b>privately-000</b>       | 0.014      | 0.026 | 0.047 | 0.061 | 0.077 | 0.110 | 0.160 |
| <b>regula-000</b>          | 0.001      | 0.004 | 0.012 | 0.024 | 0.040 | 0.076 | 0.120 |
| <b>roc-000</b>             | 0.020      | 0.037 | 0.063 | 0.103 | 0.138 | 0.216 | 0.295 |
| <b>roc-001</b>             | 0.004      | 0.017 | 0.031 | 0.068 | 0.104 | 0.187 | 0.277 |
| <b>roc-002</b>             | 0.002      | 0.005 | 0.015 | 0.027 | 0.040 | 0.084 | 0.140 |

|                     |       |       |       |       |       |       |       |
|---------------------|-------|-------|-------|-------|-------|-------|-------|
| <b>shufti-000</b>   | 0.021 | 0.036 | 0.062 | 0.103 | 0.140 | 0.242 | 0.337 |
| <b>unissey-001</b>  | 0.047 | 0.082 | 0.128 | 0.191 | 0.252 | 0.332 | 0.424 |
| <b>viant-000</b>    | 0.091 | 0.148 | 0.260 | 0.373 | 0.452 | 0.579 | 0.690 |
| <b>yoti-001</b>     | 0.018 | 0.033 | 0.068 | 0.110 | 0.155 | 0.236 | 0.327 |
| <b>yoti-002</b>     | 0.012 | 0.021 | 0.046 | 0.072 | 0.098 | 0.173 | 0.246 |
| <b>yoti-003</b>     | 0.010 | 0.017 | 0.039 | 0.067 | 0.087 | 0.159 | 0.242 |
| <b>yoti-004</b>     | 0.003 | 0.008 | 0.018 | 0.037 | 0.053 | 0.105 | 0.164 |
| <b>youverse-000</b> | 0.021 | 0.040 | 0.057 | 0.096 | 0.130 | 0.175 | 0.231 |

**Table 4.** For Application images, false positive rates for Challenge-28 by actual age. The FPR values are averages of 12 FPR estimates from two sexes and six regions-of-birth. Lower values are the better.

| Algorithm                  | Actual Age |       |       |       |       |       |       |
|----------------------------|------------|-------|-------|-------|-------|-------|-------|
|                            | 14         | 15    | 16    | 17    | 18    | 19    | 20    |
| <b>cognitec-000</b>        | 0.000      | 0.002 | 0.004 | 0.006 | 0.011 | 0.022 | 0.032 |
| <b>cvut-001</b>            | 0.012      | 0.017 | 0.028 | 0.042 | 0.056 | 0.091 | 0.124 |
| <b>cvut-002</b>            | 0.001      | 0.003 | 0.007 | 0.011 | 0.016 | 0.029 | 0.050 |
| <b>cvut-003</b>            | 0.002      | 0.003 | 0.007 | 0.009 | 0.015 | 0.031 | 0.048 |
| <b>cyberlink-000</b>       | 0.006      | 0.013 | 0.024 | 0.043 | 0.052 | 0.088 | 0.136 |
| <b>cyberlink-001</b>       | 0.000      | 0.002 | 0.006 | 0.011 | 0.014 | 0.031 | 0.048 |
| <b>daon-000</b>            | 0.012      | 0.020 | 0.031 | 0.052 | 0.060 | 0.091 | 0.127 |
| <b>daon-001</b>            | 0.003      | 0.007 | 0.014 | 0.027 | 0.039 | 0.060 | 0.097 |
| <b>dermalog-001</b>        | 0.011      | 0.018 | 0.028 | 0.041 | 0.047 | 0.077 | 0.107 |
| <b>dermalog-002</b>        | 0.010      | 0.018 | 0.027 | 0.041 | 0.052 | 0.078 | 0.113 |
| <b>hzailu-001</b>          | 0.103      | 0.141 | 0.205 | 0.263 | 0.320 | 0.380 | 0.445 |
| <b>idemia-001</b>          | 0.000      | 0.001 | 0.005 | 0.007 | 0.009 | 0.021 | 0.039 |
| <b>idemia-002</b>          | 0.000      | 0.001 | 0.004 | 0.007 | 0.009 | 0.018 | 0.033 |
| <b>inbut-000</b>           | 0.046      | 0.062 | 0.082 | 0.095 | 0.091 | 0.124 | 0.150 |
| <b>incode-000</b>          | 0.002      | 0.004 | 0.012 | 0.015 | 0.019 | 0.037 | 0.052 |
| <b>incode-001</b>          | 0.002      | 0.005 | 0.010 | 0.017 | 0.024 | 0.046 | 0.072 |
| <b>innovatrics-000</b>     | 0.001      | 0.002 | 0.006 | 0.010 | 0.014 | 0.027 | 0.044 |
| <b>innovatrics-001</b>     | 0.000      | 0.002 | 0.005 | 0.009 | 0.011 | 0.024 | 0.038 |
| <b>innovatrics-002</b>     | 0.001      | 0.005 | 0.008 | 0.014 | 0.019 | 0.038 | 0.059 |
| <b>jumio-000</b>           | 0.013      | 0.018 | 0.028 | 0.042 | 0.050 | 0.082 | 0.113 |
| <b>neurotechnology-000</b> | 0.058      | 0.071 | 0.094 | 0.123 | 0.138 | 0.162 | 0.191 |
| <b>neurotechnology-001</b> | 0.008      | 0.012 | 0.019 | 0.027 | 0.030 | 0.040 | 0.065 |
| <b>nominder-000</b>        | 0.003      | 0.008 | 0.013 | 0.024 | 0.030 | 0.054 | 0.085 |
| <b>nominder-001</b>        | 0.000      | 0.002 | 0.005 | 0.010 | 0.014 | 0.026 | 0.042 |
| <b>ondato-000</b>          | 0.006      | 0.013 | 0.019 | 0.030 | 0.037 | 0.062 | 0.093 |
| <b>ondato-001</b>          | 0.003      | 0.006 | 0.010 | 0.018 | 0.021 | 0.043 | 0.068 |
| <b>privately-000</b>       | 0.003      | 0.004 | 0.011 | 0.012 | 0.015 | 0.022 | 0.041 |
| <b>regula-000</b>          | 0.000      | 0.001 | 0.004 | 0.005 | 0.008 | 0.019 | 0.031 |
| <b>roc-000</b>             | 0.003      | 0.008 | 0.014 | 0.023 | 0.037 | 0.066 | 0.100 |
| <b>roc-001</b>             | 0.000      | 0.003 | 0.008 | 0.013 | 0.021 | 0.043 | 0.074 |
| <b>roc-002</b>             | 0.001      | 0.002 | 0.003 | 0.006 | 0.008 | 0.020 | 0.036 |
| <b>shufti-000</b>          | 0.006      | 0.012 | 0.019 | 0.035 | 0.054 | 0.094 | 0.140 |
| <b>unissey-001</b>         | 0.012      | 0.024 | 0.046 | 0.064 | 0.090 | 0.139 | 0.194 |
| <b>viant-000</b>           | 0.017      | 0.028 | 0.053 | 0.095 | 0.132 | 0.206 | 0.281 |
| <b>yoti-001</b>            | 0.007      | 0.012 | 0.025 | 0.040 | 0.054 | 0.093 | 0.124 |
| <b>yoti-002</b>            | 0.002      | 0.005 | 0.013 | 0.018 | 0.022 | 0.046 | 0.072 |
| <b>yoti-003</b>            | 0.002      | 0.005 | 0.011 | 0.019 | 0.024 | 0.047 | 0.073 |
| <b>yoti-004</b>            | 0.000      | 0.002 | 0.003 | 0.007 | 0.010 | 0.021 | 0.037 |
| <b>youverse-000</b>        | 0.008      | 0.013 | 0.021 | 0.029 | 0.047 | 0.058 | 0.088 |

Table 4 shows the effect of increasing the challenge age to 28. FPR reduces universally (necessarily). Note that in both tables the large variation down the columns, showing that some algorithms are markedly better than others.

In the main results section, we include results for all ages and all thresholds. We also include results for other datasets, where FPR is increased with lower quality photos. See sec. 5.3.

**Age verification accuracy vs age estimation:** Our API includes age verification function which allow developers to implement a dedicated classifier indicating whether the subject is above an age limit. All algorithms show that age-verification classifier do not perform better when comparing to a regression-like estimator. See sec. 5.3.1

**Demographics - Region of birth:** In addition to age, we report performance for six regions-of-birth: E. Africa, W. Africa, E. Europe, S. Asia, S. E. Asia, and E. Asia. This forms a proxy for ethnicity, as explained in section 5.4. We do not attempt to recover skin tone information from images, because this is problematic for technical reasons and, more importantly, because face processing algorithms should be expected to be sensitive to skin tone *and* many other phenotypes present in a face such as square jaws, prominent eye brows, high cheekbones, aquiline noses etc.

**Table 5.** For Application images, Challenge-25 FPR for men aged 17 by region of birth. Gini summarizes variability of the six FPR values to its right; lower values are better.

| Algorithm           | Gini | E Africa      | E Asia        | E Europe      | S Asia        | S E Asia      | W Africa    |
|---------------------|------|---------------|---------------|---------------|---------------|---------------|-------------|
| cognitec-000        | 0.53 | 0.006 ± 0.006 | 0.010 ± 0.004 | 0.000 ± 0.000 | 0.014 ± 0.007 | 0.02 ± 0.01   | 0.04 ± 0.02 |
| cvut-001            | 0.50 | 0.17 ± 0.04   | 0.05 ± 0.01   | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.05 ± 0.02   | 0.27 ± 0.05 |
| cvut-002            | 0.59 | 0.05 ± 0.02   | 0.006 ± 0.003 | 0.008 ± 0.008 | 0.04 ± 0.01   | 0.01 ± 0.01   | 0.11 ± 0.03 |
| cvut-003            | 0.65 | 0.03 ± 0.02   | 0.004 ± 0.003 | 0.003 ± 0.003 | 0.029 ± 0.009 | 0.006 ± 0.006 | 0.08 ± 0.03 |
| cyberlink-000       | 0.30 | 0.20 ± 0.04   | 0.09 ± 0.01   | 0.04 ± 0.02   | 0.12 ± 0.02   | 0.11 ± 0.03   | 0.19 ± 0.05 |
| cyberlink-001       | 0.22 | 0.04 ± 0.02   | 0.020 ± 0.006 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.02   | 0.05 ± 0.02 |
| daon-000            | 0.27 | 0.23 ± 0.04   | 0.17 ± 0.02   | 0.03 ± 0.01   | 0.19 ± 0.02   | 0.16 ± 0.04   | 0.13 ± 0.04 |
| daon-001            | 0.25 | 0.13 ± 0.04   | 0.10 ± 0.01   | 0.008 ± 0.008 | 0.09 ± 0.02   | 0.10 ± 0.03   | 0.11 ± 0.03 |
| dermalog-001        | 0.21 | 0.10 ± 0.03   | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.12 ± 0.02   | 0.10 ± 0.03   | 0.09 ± 0.03 |
| dermalog-002        | 0.14 | 0.07 ± 0.03   | 0.054 ± 0.010 | 0.06 ± 0.02   | 0.09 ± 0.02   | 0.08 ± 0.03   | 0.05 ± 0.02 |
| hzailu-001          | 0.15 | 0.55 ± 0.06   | 0.34 ± 0.02   | 0.44 ± 0.05   | 0.68 ± 0.03   | 0.53 ± 0.05   | 0.67 ± 0.05 |
| idemia-001          | 0.58 | 0.01 ± 0.01   | 0.007 ± 0.004 | 0.000 ± 0.000 | 0.009 ± 0.005 | 0.006 ± 0.006 | 0.04 ± 0.02 |
| idemia-002          | 0.75 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.006 ± 0.004 | 0.006 ± 0.006 | 0.03 ± 0.02 |
| inbut-000           | 0.28 | 0.30 ± 0.05   | 0.10 ± 0.02   | 0.12 ± 0.03   | 0.13 ± 0.02   | 0.13 ± 0.03   | 0.27 ± 0.05 |
| incode-000          | 0.49 | 0.02 ± 0.01   | 0.006 ± 0.004 | 0.003 ± 0.003 | 0.017 ± 0.007 | 0.01 ± 0.01   | 0.05 ± 0.02 |
| incode-001          | 0.47 | 0.02 ± 0.01   | 0.011 ± 0.004 | 0.000 ± 0.000 | 0.011 ± 0.005 | 0.02 ± 0.01   | 0.04 ± 0.02 |
| innovatrics-000     | 0.68 | 0.03 ± 0.01   | 0.006 ± 0.003 | 0.000 ± 0.000 | 0.020 ± 0.007 | 0.003 ± 0.003 | 0.07 ± 0.03 |
| innovatrics-001     | 0.72 | 0.02 ± 0.01   | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.012 ± 0.006 | 0.003 ± 0.003 | 0.06 ± 0.02 |
| innovatrics-002     | 0.67 | 0.05 ± 0.02   | 0.008 ± 0.004 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.003 ± 0.003 | 0.09 ± 0.03 |
| jumio-000           | 0.44 | 0.04 ± 0.02   | 0.019 ± 0.006 | 0.11 ± 0.03   | 0.017 ± 0.007 | 0.03 ± 0.01   | 0.06 ± 0.03 |
| neurotechnology-000 | 0.45 | 0.76 ± 0.04   | 0.36 ± 0.02   | 0.06 ± 0.02   | 0.20 ± 0.02   | 0.28 ± 0.04   | 0.84 ± 0.04 |
| neurotechnology-001 | 0.38 | 0.12 ± 0.03   | 0.37 ± 0.02   | 0.06 ± 0.02   | 0.20 ± 0.02   | 0.26 ± 0.05   | 0.10 ± 0.03 |
| nominder-000        | 0.41 | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.008 ± 0.008 | 0.04 ± 0.01   | 0.03 ± 0.02   | 0.10 ± 0.03 |
| nominder-001        | 0.34 | 0.01 ± 0.01   | 0.016 ± 0.006 | 0.008 ± 0.008 | 0.016 ± 0.007 | 0.03 ± 0.01   | 0.04 ± 0.02 |
| ondato-000          | 0.44 | 0.10 ± 0.03   | 0.04 ± 0.01   | 0.011 ± 0.008 | 0.05 ± 0.01   | 0.05 ± 0.02   | 0.16 ± 0.04 |
| ondato-001          | 0.46 | 0.06 ± 0.02   | 0.018 ± 0.006 | 0.006 ± 0.006 | 0.024 ± 0.009 | 0.04 ± 0.02   | 0.09 ± 0.03 |
| privately-000       | 0.36 | 0.01 ± 0.01   | 0.037 ± 0.009 | 0.003 ± 0.003 | 0.015 ± 0.007 | 0.03 ± 0.01   | 0.03 ± 0.02 |
| regula-000          | 0.66 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.007 ± 0.005 | 0.011 ± 0.008 | 0.03 ± 0.02 |
| roc-000             | 0.36 | 0.07 ± 0.02   | 0.12 ± 0.01   | 0.000 ± 0.000 | 0.04 ± 0.01   | 0.11 ± 0.03   | 0.11 ± 0.03 |
| roc-001             | 0.45 | 0.05 ± 0.02   | 0.042 ± 0.009 | 0.003 ± 0.003 | 0.019 ± 0.008 | 0.03 ± 0.02   | 0.08 ± 0.03 |
| roc-002             | 0.50 | 0.012 ± 0.009 | 0.014 ± 0.005 | 0.000 ± 0.000 | 0.010 ± 0.006 | 0.02 ± 0.01   | 0.04 ± 0.02 |
| shufti-000          | 0.31 | 0.11 ± 0.03   | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.06 ± 0.02   | 0.09 ± 0.03 |
| unissey-001         | 0.28 | 0.20 ± 0.04   | 0.26 ± 0.02   | 0.04 ± 0.02   | 0.15 ± 0.02   | 0.23 ± 0.04   | 0.14 ± 0.04 |
| viant-000           | 0.30 | 0.43 ± 0.05   | 0.25 ± 0.02   | 0.02 ± 0.01   | 0.35 ± 0.03   | 0.37 ± 0.05   | 0.47 ± 0.06 |
| yoti-001            | 0.48 | 0.02 ± 0.01   | 0.036 ± 0.009 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.02 ± 0.01   | 0.08 ± 0.03 |
| yoti-002            | 0.38 | 0.012 ± 0.009 | 0.028 ± 0.008 | 0.006 ± 0.006 | 0.018 ± 0.007 | 0.011 ± 0.008 | 0.04 ± 0.02 |
| yoti-003            | 0.39 | 0.012 ± 0.009 | 0.020 ± 0.006 | 0.006 ± 0.006 | 0.016 ± 0.007 | 0.03 ± 0.02   | 0.04 ± 0.02 |
| yoti-004            | 0.38 | 0.012 ± 0.009 | 0.011 ± 0.005 | 0.003 ± 0.003 | 0.011 ± 0.006 | 0.011 ± 0.008 | 0.03 ± 0.02 |

|                     |      |             |               |             |             |             |             |
|---------------------|------|-------------|---------------|-------------|-------------|-------------|-------------|
| <b>youverse-000</b> | 0.50 | 0.12 ± 0.03 | 0.021 ± 0.007 | 0.03 ± 0.02 | 0.07 ± 0.02 | 0.02 ± 0.01 | 0.17 ± 0.04 |
|---------------------|------|-------------|---------------|-------------|-------------|-------------|-------------|

Table 5 shows false positive rates for a Challenge-25 policy for 17 year old males. The lowest FPR values are in East Europeans though this varies for women and for men of other ages; the highest values occur in various other regions. See sec. 5.4

**Demographics - Sex:** Another obvious demographic variable is sex. Table 6 shows false positive rates, again for just 17 year olds, but with both sexes and three regions-of-birth. The main observation is generally higher FPR in women than men, but with exceptions for some algorithms operating on some ethnicities. The MAE values are also generally higher in women.

**Table 6.** For Application images, Challenge-25 FPR for subjects aged 17 by sex and region of birth. Lower values are the better.

| Algorithm                  | Gini | Female<br>E Africa | Female<br>E Asia | Female<br>E Europe | Male<br>E Africa | Male<br>E Asia | Male<br>E Europe |
|----------------------------|------|--------------------|------------------|--------------------|------------------|----------------|------------------|
| <b>cognitec-000</b>        | 0.74 | 0.08 ± 0.03        | 0.015 ± 0.006    | 0.006 ± 0.006      | 0.006 ± 0.006    | 0.010 ± 0.004  | 0.000 ± 0.000    |
| <b>cvut-001</b>            | 0.61 | 0.46 ± 0.05        | 0.12 ± 0.01      | 0.04 ± 0.02        | 0.17 ± 0.04      | 0.05 ± 0.01    | 0.03 ± 0.01      |
| <b>cvut-002</b>            | 0.65 | 0.17 ± 0.04        | 0.024 ± 0.007    | 0.03 ± 0.02        | 0.05 ± 0.02      | 0.006 ± 0.003  | 0.008 ± 0.008    |
| <b>cvut-003</b>            | 0.70 | 0.15 ± 0.04        | 0.022 ± 0.007    | 0.02 ± 0.01        | 0.03 ± 0.02      | 0.004 ± 0.003  | 0.003 ± 0.003    |
| <b>cyberlink-000</b>       | 0.41 | 0.28 ± 0.04        | 0.14 ± 0.02      | 0.06 ± 0.02        | 0.20 ± 0.04      | 0.09 ± 0.01    | 0.04 ± 0.02      |
| <b>cyberlink-001</b>       | 0.38 | 0.10 ± 0.03        | 0.034 ± 0.009    | 0.03 ± 0.02        | 0.04 ± 0.02      | 0.020 ± 0.006  | 0.02 ± 0.01      |
| <b>daon-000</b>            | 0.42 | 0.21 ± 0.04        | 0.10 ± 0.02      | 0.03 ± 0.02        | 0.23 ± 0.04      | 0.17 ± 0.02    | 0.03 ± 0.01      |
| <b>daon-001</b>            | 0.46 | 0.14 ± 0.04        | 0.06 ± 0.01      | 0.02 ± 0.01        | 0.13 ± 0.04      | 0.10 ± 0.01    | 0.008 ± 0.008    |
| <b>dermalog-001</b>        | 0.41 | 0.13 ± 0.03        | 0.14 ± 0.02      | 0.31 ± 0.05        | 0.10 ± 0.03      | 0.07 ± 0.01    | 0.04 ± 0.02      |
| <b>dermalog-002</b>        | 0.45 | 0.11 ± 0.03        | 0.29 ± 0.02      | 0.31 ± 0.05        | 0.07 ± 0.03      | 0.054 ± 0.010  | 0.06 ± 0.02      |
| <b>hzailu-001</b>          | 0.15 | 0.39 ± 0.05        | 0.40 ± 0.02      | 0.66 ± 0.05        | 0.55 ± 0.06      | 0.34 ± 0.02    | 0.44 ± 0.05      |
| <b>idemia-001</b>          | 0.59 | 0.04 ± 0.02        | 0.023 ± 0.007    | 0.003 ± 0.003      | 0.01 ± 0.01      | 0.007 ± 0.004  | 0.000 ± 0.000    |
| <b>idemia-002</b>          | 0.65 | 0.03 ± 0.02        | 0.013 ± 0.005    | 0.03 ± 0.02        | 0.000 ± 0.000    | 0.004 ± 0.003  | 0.000 ± 0.000    |
| <b>inbut-000</b>           | 0.37 | 0.39 ± 0.05        | 0.08 ± 0.01      | 0.22 ± 0.04        | 0.30 ± 0.05      | 0.10 ± 0.02    | 0.12 ± 0.03      |
| <b>incode-000</b>          | 0.59 | 0.11 ± 0.03        | 0.05 ± 0.01      | 0.07 ± 0.03        | 0.02 ± 0.01      | 0.006 ± 0.004  | 0.003 ± 0.003    |
| <b>incode-001</b>          | 0.55 | 0.11 ± 0.03        | 0.07 ± 0.01      | 0.06 ± 0.03        | 0.02 ± 0.01      | 0.011 ± 0.004  | 0.000 ± 0.000    |
| <b>innovatrics-000</b>     | 0.80 | 0.20 ± 0.04        | 0.020 ± 0.007    | 0.01 ± 0.01        | 0.03 ± 0.01      | 0.006 ± 0.003  | 0.000 ± 0.000    |
| <b>innovatrics-001</b>     | 0.81 | 0.17 ± 0.04        | 0.014 ± 0.005    | 0.02 ± 0.01        | 0.02 ± 0.01      | 0.004 ± 0.003  | 0.000 ± 0.000    |
| <b>innovatrics-002</b>     | 0.71 | 0.22 ± 0.04        | 0.024 ± 0.008    | 0.04 ± 0.02        | 0.05 ± 0.02      | 0.008 ± 0.004  | 0.003 ± 0.003    |
| <b>jumio-000</b>           | 0.45 | 0.25 ± 0.04        | 0.10 ± 0.01      | 0.16 ± 0.04        | 0.04 ± 0.02      | 0.019 ± 0.006  | 0.11 ± 0.03      |
| <b>neurotechnology-000</b> | 0.46 | 0.39 ± 0.05        | 0.24 ± 0.02      | 0.12 ± 0.03        | 0.76 ± 0.04      | 0.36 ± 0.02    | 0.06 ± 0.02      |
| <b>neurotechnology-001</b> | 0.41 | 0.08 ± 0.03        | 0.26 ± 0.02      | 0.14 ± 0.04        | 0.12 ± 0.03      | 0.37 ± 0.02    | 0.06 ± 0.02      |
| <b>nominder-000</b>        | 0.44 | 0.14 ± 0.04        | 0.06 ± 0.01      | 0.04 ± 0.02        | 0.04 ± 0.02      | 0.05 ± 0.01    | 0.008 ± 0.008    |
| <b>nominder-001</b>        | 0.48 | 0.08 ± 0.03        | 0.024 ± 0.007    | 0.03 ± 0.01        | 0.01 ± 0.01      | 0.016 ± 0.006  | 0.008 ± 0.008    |
| <b>ondato-000</b>          | 0.50 | 0.23 ± 0.04        | 0.11 ± 0.02      | 0.05 ± 0.02        | 0.10 ± 0.03      | 0.04 ± 0.01    | 0.011 ± 0.008    |
| <b>ondato-001</b>          | 0.55 | 0.16 ± 0.04        | 0.07 ± 0.01      | 0.04 ± 0.02        | 0.06 ± 0.02      | 0.018 ± 0.006  | 0.006 ± 0.006    |
| <b>privately-000</b>       | 0.54 | 0.08 ± 0.03        | 0.14 ± 0.02      | 0.07 ± 0.03        | 0.01 ± 0.01      | 0.037 ± 0.009  | 0.003 ± 0.003    |
| <b>regula-000</b>          | 0.76 | 0.07 ± 0.02        | 0.013 ± 0.005    | 0.02 ± 0.01        | 0.003 ± 0.003    | 0.002 ± 0.002  | 0.000 ± 0.000    |
| <b>roc-000</b>             | 0.50 | 0.18 ± 0.04        | 0.17 ± 0.02      | 0.02 ± 0.01        | 0.07 ± 0.02      | 0.12 ± 0.01    | 0.000 ± 0.000    |
| <b>roc-001</b>             | 0.59 | 0.14 ± 0.03        | 0.10 ± 0.01      | 0.003 ± 0.003      | 0.05 ± 0.02      | 0.042 ± 0.009  | 0.003 ± 0.003    |
| <b>roc-002</b>             | 0.55 | 0.05 ± 0.02        | 0.040 ± 0.009    | 0.008 ± 0.008      | 0.012 ± 0.009    | 0.014 ± 0.005  | 0.000 ± 0.000    |
| <b>shufti-000</b>          | 0.41 | 0.21 ± 0.04        | 0.09 ± 0.01      | 0.14 ± 0.04        | 0.11 ± 0.03      | 0.05 ± 0.01    | 0.02 ± 0.01      |
| <b>unissey-001</b>         | 0.25 | 0.17 ± 0.04        | 0.26 ± 0.02      | 0.19 ± 0.04        | 0.20 ± 0.04      | 0.26 ± 0.02    | 0.04 ± 0.02      |
| <b>viant-000</b>           | 0.45 | 0.60 ± 0.05        | 0.33 ± 0.02      | 0.12 ± 0.03        | 0.43 ± 0.05      | 0.25 ± 0.02    | 0.02 ± 0.01      |
| <b>yoti-001</b>            | 0.53 | 0.18 ± 0.04        | 0.14 ± 0.02      | 0.19 ± 0.04        | 0.02 ± 0.01      | 0.036 ± 0.009  | 0.003 ± 0.003    |
| <b>yoti-002</b>            | 0.52 | 0.10 ± 0.03        | 0.12 ± 0.02      | 0.06 ± 0.03        | 0.012 ± 0.009    | 0.028 ± 0.008  | 0.006 ± 0.006    |
| <b>yoti-003</b>            | 0.56 | 0.11 ± 0.03        | 0.12 ± 0.02      | 0.05 ± 0.02        | 0.012 ± 0.009    | 0.020 ± 0.006  | 0.006 ± 0.006    |
| <b>yoti-004</b>            | 0.54 | 0.06 ± 0.02        | 0.05 ± 0.01      | 0.011 ± 0.008      | 0.012 ± 0.009    | 0.011 ± 0.005  | 0.003 ± 0.003    |
| <b>youverse-000</b>        | 0.45 | 0.18 ± 0.04        | 0.037 ± 0.009    | 0.13 ± 0.03        | 0.12 ± 0.03      | 0.021 ± 0.007  | 0.03 ± 0.02      |

**Weighted Challenge-25 metrics:** In a Challenge-T application, the false positive rate varies by age and demographics. We therefore introduce an aggregated measure of ineffectiveness, which is a



population- and risk-weighted average of FPR. The population weights quantify the [age pyramid](#); the risk weights express how likely it is for a person of each age to attempt to use an AV system. The formula is presented and discussed in section 2.6.3; it is included not as a definitive model, rather to illustrate what an analyst might consider. For an 18 year legal limit application, Table 7 shows weighted Challenge-T FPR, aggregating both sexes and six regions-of-birth with equal weights, and 13, ..., 17 year olds with weights that de-emphasize the younger ages.

**Table 7.** For Application images in an  $L = 18$  age restriction application, **ineffectiveness** for four Challenge-T thresholds i.e. a population weighted aggregate of FPR. Lower values are better.

| Algorithm           | T=22          | T=25          | T=28          | T=31          |
|---------------------|---------------|---------------|---------------|---------------|
| cognitec-000        | 0.030 ± 0.012 | 0.006 ± 0.004 | 0.001 ± 0.001 | 0.001 ± 0.001 |
| cvut-001            | 0.143 ± 0.026 | 0.067 ± 0.018 | 0.016 ± 0.007 | 0.005 ± 0.003 |
| cvut-002            | 0.061 ± 0.017 | 0.018 ± 0.009 | 0.003 ± 0.002 | 0.001 ± 0.001 |
| cvut-003            | 0.048 ± 0.016 | 0.017 ± 0.008 | 0.003 ± 0.002 | 0.001 ± 0.001 |
| cyberlink-000       | 0.261 ± 0.035 | 0.054 ± 0.017 | 0.011 ± 0.006 | 0.002 ± 0.002 |
| cyberlink-001       | 0.068 ± 0.019 | 0.011 ± 0.007 | 0.002 ± 0.001 | 0.001 ± 0.001 |
| daon-000            | 0.132 ± 0.026 | 0.047 ± 0.015 | 0.017 ± 0.009 | 0.005 ± 0.004 |
| daon-001            | 0.069 ± 0.018 | 0.021 ± 0.010 | 0.006 ± 0.004 | 0.002 ± 0.002 |
| dermalog-001        | 0.183 ± 0.030 | 0.052 ± 0.016 | 0.015 ± 0.008 | 0.005 ± 0.004 |
| dermalog-002        | 0.194 ± 0.029 | 0.061 ± 0.016 | 0.014 ± 0.008 | 0.003 ± 0.003 |
| hzailu-001          | 0.501 ± 0.042 | 0.293 ± 0.038 | 0.126 ± 0.027 | 0.043 ± 0.016 |
| idemia-001          | 0.024 ± 0.010 | 0.004 ± 0.003 | 0.001 ± 0.001 | 0.000 ± 0.000 |
| idemia-002          | 0.015 ± 0.008 | 0.003 ± 0.002 | 0.001 ± 0.001 | 0.000 ± 0.000 |
| inbut-000           | 0.218 ± 0.033 | 0.108 ± 0.026 | 0.054 ± 0.018 | 0.026 ± 0.012 |
| incode-000          | 0.032 ± 0.013 | 0.012 ± 0.007 | 0.004 ± 0.003 | 0.001 ± 0.001 |
| incode-001          | 0.038 ± 0.013 | 0.011 ± 0.006 | 0.004 ± 0.003 | 0.001 ± 0.001 |
| innovatrics-000     | 0.070 ± 0.017 | 0.013 ± 0.006 | 0.002 ± 0.002 | 0.001 ± 0.001 |
| innovatrics-001     | 0.066 ± 0.017 | 0.011 ± 0.006 | 0.001 ± 0.001 | 0.001 ± 0.001 |
| innovatrics-002     | 0.067 ± 0.017 | 0.016 ± 0.007 | 0.003 ± 0.002 | 0.001 ± 0.001 |
| jumio-000           | 0.140 ± 0.027 | 0.047 ± 0.016 | 0.017 ± 0.009 | 0.006 ± 0.004 |
| neurotechnology-000 | 0.452 ± 0.035 | 0.220 ± 0.030 | 0.066 ± 0.018 | 0.016 ± 0.008 |
| neurotechnology-001 | 0.288 ± 0.035 | 0.075 ± 0.018 | 0.010 ± 0.007 | 0.004 ± 0.004 |
| nominder-000        | 0.111 ± 0.024 | 0.023 ± 0.011 | 0.006 ± 0.004 | 0.001 ± 0.001 |
| nominder-001        | 0.070 ± 0.019 | 0.008 ± 0.005 | 0.001 ± 0.001 | 0.001 ± 0.001 |
| ondato-000          | 0.144 ± 0.028 | 0.039 ± 0.014 | 0.010 ± 0.006 | 0.003 ± 0.002 |
| ondato-001          | 0.111 ± 0.024 | 0.023 ± 0.011 | 0.005 ± 0.004 | 0.001 ± 0.001 |
| privately-000       | 0.085 ± 0.020 | 0.021 ± 0.009 | 0.004 ± 0.003 | 0.001 ± 0.001 |
| regula-000          | 0.021 ± 0.009 | 0.003 ± 0.002 | 0.001 ± 0.001 | 0.001 ± 0.001 |
| roc-000             | 0.131 ± 0.024 | 0.030 ± 0.011 | 0.006 ± 0.004 | 0.001 ± 0.001 |
| roc-001             | 0.077 ± 0.017 | 0.012 ± 0.006 | 0.002 ± 0.002 | 0.001 ± 0.001 |
| roc-002             | 0.021 ± 0.009 | 0.005 ± 0.003 | 0.001 ± 0.001 | 0.001 ± 0.001 |
| shufti-000          | 0.095 ± 0.022 | 0.031 ± 0.012 | 0.009 ± 0.006 | 0.004 ± 0.003 |
| unissey-001         | 0.120 ± 0.025 | 0.067 ± 0.019 | 0.019 ± 0.010 | 0.004 ± 0.003 |
| vianite-000         | 0.370 ± 0.036 | 0.128 ± 0.025 | 0.025 ± 0.010 | 0.003 ± 0.002 |
| yoti-001            | 0.069 ± 0.018 | 0.029 ± 0.011 | 0.011 ± 0.006 | 0.003 ± 0.003 |
| yoti-002            | 0.049 ± 0.014 | 0.019 ± 0.008 | 0.004 ± 0.003 | 0.001 ± 0.001 |
| yoti-003            | 0.048 ± 0.014 | 0.016 ± 0.008 | 0.004 ± 0.003 | 0.001 ± 0.001 |
| yoti-004            | 0.046 ± 0.014 | 0.007 ± 0.004 | 0.001 ± 0.001 | 0.000 ± 0.000 |
| youverse-000        | 0.093 ± 0.023 | 0.031 ± 0.013 | 0.011 ± 0.007 | 0.003 ± 0.003 |

We additionally quantify inconvenience experienced by persons above the legal age limit whose age verification are delayed by a failed Challenge-T attempt. The aggregated measure is again a weighted average this time of false negative rates (FNR). The formula is presented and discussed in section 2.6.4. Table 8 shows Challenge-T FNR computed over all ages 18, ..., 71 with equal weights representing uniform likelihood that adults would engage with the system.



**Table 8.** For Application images in an  $L = 18$  age restriction application, **inconvenience** for four Challenge-T thresholds i.e. a population weighted aggregate of FNR. Lower values are better.

| Algorithm           | T=22          | T=25          | T=28          | T=31          |
|---------------------|---------------|---------------|---------------|---------------|
| cognitec-000        | 0.077 ± 0.005 | 0.139 ± 0.006 | 0.202 ± 0.006 | 0.264 ± 0.007 |
| cvut-001            | 0.069 ± 0.006 | 0.119 ± 0.007 | 0.188 ± 0.008 | 0.267 ± 0.009 |
| cvut-002            | 0.077 ± 0.007 | 0.138 ± 0.008 | 0.227 ± 0.009 | 0.282 ± 0.009 |
| cvut-003            | 0.077 ± 0.005 | 0.143 ± 0.006 | 0.222 ± 0.007 | 0.275 ± 0.007 |
| cyberlink-000       | 0.019 ± 0.003 | 0.090 ± 0.006 | 0.156 ± 0.007 | 0.254 ± 0.008 |
| cyberlink-001       | 0.057 ± 0.005 | 0.128 ± 0.006 | 0.203 ± 0.007 | 0.272 ± 0.008 |
| daon-000            | 0.068 ± 0.006 | 0.121 ± 0.007 | 0.185 ± 0.008 | 0.249 ± 0.009 |
| daon-001            | 0.065 ± 0.005 | 0.116 ± 0.006 | 0.176 ± 0.007 | 0.237 ± 0.007 |
| dermalog-001        | 0.051 ± 0.006 | 0.122 ± 0.008 | 0.193 ± 0.008 | 0.257 ± 0.009 |
| dermalog-002        | 0.056 ± 0.007 | 0.125 ± 0.009 | 0.210 ± 0.010 | 0.288 ± 0.010 |
| hzailu-001          | 0.022 ± 0.004 | 0.048 ± 0.006 | 0.107 ± 0.009 | 0.196 ± 0.010 |
| idemia-001          | 0.073 ± 0.005 | 0.130 ± 0.005 | 0.190 ± 0.006 | 0.250 ± 0.006 |
| idemia-002          | 0.085 ± 0.005 | 0.140 ± 0.005 | 0.194 ± 0.006 | 0.251 ± 0.006 |
| inbut-000           | 0.033 ± 0.003 | 0.077 ± 0.005 | 0.142 ± 0.006 | 0.213 ± 0.007 |
| incode-000          | 0.075 ± 0.005 | 0.120 ± 0.006 | 0.184 ± 0.006 | 0.247 ± 0.006 |
| incode-001          | 0.060 ± 0.004 | 0.100 ± 0.005 | 0.153 ± 0.005 | 0.210 ± 0.006 |
| innovatrics-000     | 0.048 ± 0.004 | 0.113 ± 0.005 | 0.190 ± 0.006 | 0.263 ± 0.006 |
| innovatrics-001     | 0.052 ± 0.004 | 0.121 ± 0.005 | 0.197 ± 0.006 | 0.273 ± 0.006 |
| innovatrics-002     | 0.055 ± 0.004 | 0.111 ± 0.005 | 0.185 ± 0.006 | 0.254 ± 0.006 |
| jumio-000           | 0.040 ± 0.004 | 0.089 ± 0.005 | 0.149 ± 0.006 | 0.214 ± 0.006 |
| neurotechnology-000 | 0.040 ± 0.005 | 0.107 ± 0.007 | 0.209 ± 0.009 | 0.308 ± 0.010 |
| neurotechnology-001 | 0.047 ± 0.005 | 0.140 ± 0.009 | 0.263 ± 0.010 | 0.343 ± 0.011 |
| nominder-000        | 0.049 ± 0.005 | 0.120 ± 0.006 | 0.182 ± 0.007 | 0.248 ± 0.007 |
| nominder-001        | 0.056 ± 0.004 | 0.129 ± 0.006 | 0.198 ± 0.006 | 0.263 ± 0.007 |
| ondato-000          | 0.044 ± 0.004 | 0.102 ± 0.006 | 0.174 ± 0.007 | 0.241 ± 0.007 |
| ondato-001          | 0.043 ± 0.004 | 0.103 ± 0.006 | 0.177 ± 0.006 | 0.247 ± 0.007 |
| privately-000       | 0.081 ± 0.006 | 0.165 ± 0.008 | 0.254 ± 0.009 | 0.331 ± 0.009 |
| regula-000          | 0.069 ± 0.004 | 0.126 ± 0.005 | 0.190 ± 0.005 | 0.255 ± 0.006 |
| roc-000             | 0.049 ± 0.004 | 0.099 ± 0.006 | 0.157 ± 0.006 | 0.221 ± 0.007 |
| roc-001             | 0.039 ± 0.003 | 0.088 ± 0.005 | 0.151 ± 0.005 | 0.216 ± 0.005 |
| roc-002             | 0.068 ± 0.004 | 0.119 ± 0.005 | 0.181 ± 0.005 | 0.248 ± 0.006 |
| shufti-000          | 0.042 ± 0.005 | 0.083 ± 0.006 | 0.132 ± 0.007 | 0.187 ± 0.007 |
| unissey-001         | 0.061 ± 0.006 | 0.091 ± 0.007 | 0.155 ± 0.008 | 0.246 ± 0.009 |
| vianite-000         | 0.006 ± 0.001 | 0.036 ± 0.003 | 0.105 ± 0.005 | 0.188 ± 0.006 |
| yoti-001            | 0.047 ± 0.004 | 0.082 ± 0.005 | 0.137 ± 0.006 | 0.199 ± 0.006 |
| yoti-002            | 0.056 ± 0.004 | 0.093 ± 0.005 | 0.156 ± 0.006 | 0.222 ± 0.006 |
| yoti-003            | 0.053 ± 0.004 | 0.092 ± 0.005 | 0.152 ± 0.005 | 0.216 ± 0.006 |
| yoti-004            | 0.048 ± 0.004 | 0.108 ± 0.005 | 0.181 ± 0.005 | 0.244 ± 0.006 |
| youverse-000        | 0.079 ± 0.006 | 0.144 ± 0.008 | 0.227 ± 0.009 | 0.312 ± 0.010 |

The two tables show the expected tradeoff between protecting young people and inconveniencing older subjects.

**Children:** We use a set of visa photos collected in one country, Mexico, to quantify age estimation accuracy in children between age 0 to 17. Accuracy is higher than in an adult population: For the most accurate algorithm mean absolute error in children under age 1 is 0.34 years (i.e. about four months) versus 3.5 years for a thirty year old. See sec. 5.5.

**Future work:** The FATE AEV evaluation remains open, so we will continue to evaluate and report on newly submitted prototypes. In future reports we will: evaluate performance of implementations that can exploit having a prior known-age reference photo of a subject (see our [API](#)); consider

whether video clips afford improved accuracy over still photographs; and extend demographic and quality analyses.

## 1. Introduction

### 1.1. Role of AEV Technology

Age assurance (AA) systems are fielded to support validation of a person's eligibility to access age-restricted online content or physical world products or spaces. As the [Future of Privacy Forum shows](#), an AA system may employ various methods depending on the purpose. These include human checks of a government-issued identification document, possession of a credit card or payment credential only issued to people of the required age, biometric verification against a photograph collected during a prior session in which age was verified, accessing high integrity age data from a trusted device, and use of face analysis to estimate age or verify age is above (or below) the limit.

This report is concerned only with evaluation of AE technologies used in an age verification process. It is intended to generically support applications including:

1. Verification that a person is above 18, 21 or other key ages, e.g. for sale of alcohol.
2. Verification that a person is below a certain age, e.g. entrance to a teen chat room.
3. Producing population age statistics for people visiting certain locations, e.g. movie theaters.
4. Digital advertising where a display might show an age-tailored advertisement.
5. Checks that a passport application photo is recent (not collected 10 years previously).
6. Age estimation for refugees, asylum seekers, and other undocumented individuals.

Age assurance is mandated in various jurisdictions for various purposes. In the United States, the Children's Online Privacy Protection Act (COPPA) of 1998[1], requires the Federal Trade Commission to prescribe regulations to protect the privacy of personal information collected from and about children on the internet, to provide greater parental control over the collection and use of that information, and for other purposes. Within the States, Arkansas[2], Mississippi[3], Montana[4], Louisiana[5], Utah[6], Virginia[7], and Texas[8] have passed age verification laws to require social media, adult websites, harmful to minors to use a reasonable age verification methods to verify the user's age eligibility.

There are new and emerging technical standards in this area. IEEE Standard 2089-2021 provides best practice for designing digital services that impact directly or indirectly on children[9]. ISO/IEC 27566 is a (recently renumbered) multipart standard for age assurance: [Part 1](#) provides a framework including vocabulary; [Part 2](#) guides architecture and interoperability; [Part 3](#) concerns benchmarking and analysis.

### 1.2. Age Estimation is Not Face Recognition

We distinguish facial age estimation (AE) from face recognition (FR). AE analyses one face to produce an estimate of age. FR is concerned with *who* is in an image. The two techniques employ different algorithmic machinery for these two purposes: AE is trained on photos accompanied by known-age values; FR is trained on pairs of photographs with identity labels. AE typically proceeds with *one* photograph or video - the image is passed to an algorithm which emits an age estimate, and the photo can be deleted. FR, on the other hand, compares identity information extracted from *two*

photos with the goal of determining if they are from the same person. The first sample is stored in a passport, or phone, or database. The second sample is collected for verification or identification. The Oxford English Dictionary is [explicit](#) in noting the “previous encounter”. The utility of FR is in binding a new photograph to an older one and its informative metadata, for example, that a person is allowed to enter a facility, or had previously applied for a passport under the same (or different) name, or had previously been deported from a country.

### 1.3. Scope

NIST published [Interagency Report 7995 Performance of Automated Age Estimation Algorithms](#) in 2014. The report showed that the most accurate algorithm can estimate the age of a person within five years of their actual age 67% of the time[10]. With the increase in human-computer interaction and as the motivation of needs to use age estimation systems continues to grow, NIST revamped the age estimation study. We restructured and launched the Face Analysis Technology Evaluation - Age Estimation and Verification track. This is an ongoing evaluation to benchmark software-based facial age estimation algorithms.

This evaluation is not a certification program nor an evaluation of a full age verification process. It does not evaluate mechanisms for detecting active attacks on AV systems. It does not establish minimum accuracy criteria, because different applications have different requirements.

### 1.4. The Technical Challenge

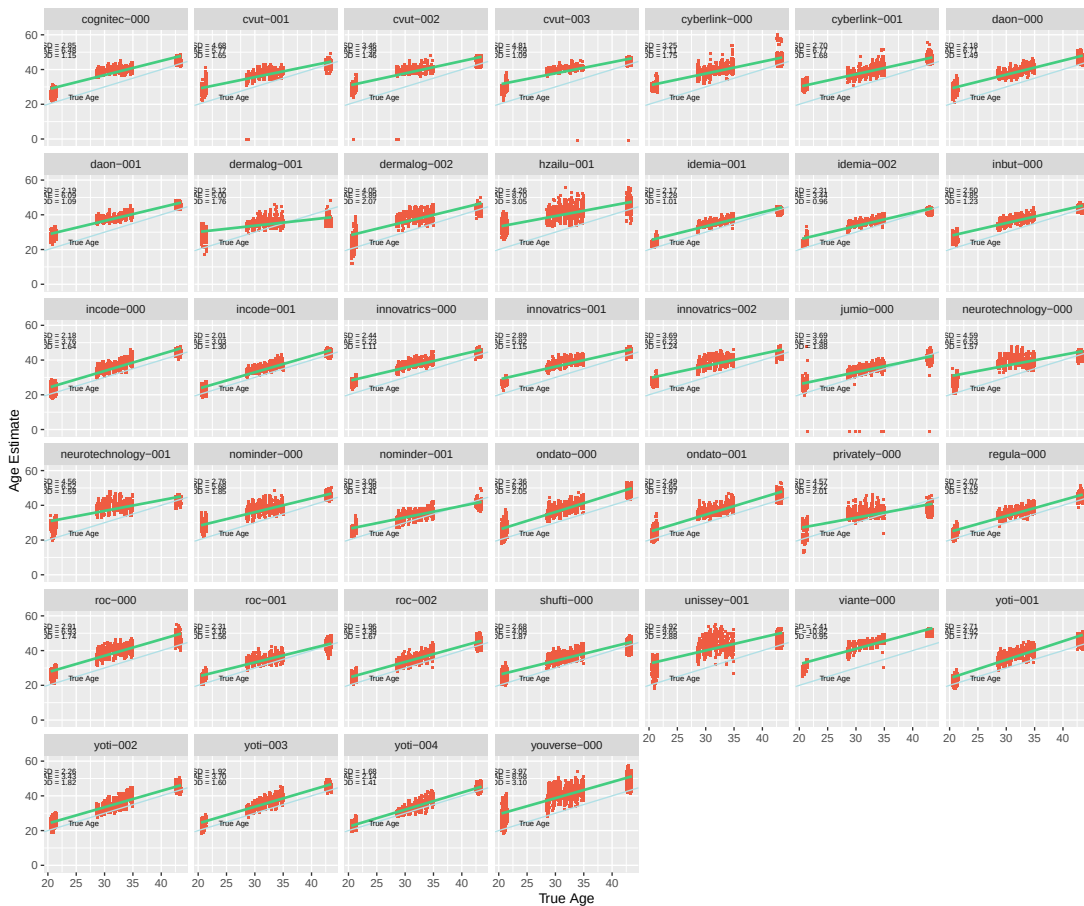
An AE algorithm is tasked with mapping pixel values in an image to an estimate of the subject’s age. The output estimate will generally be different to the actual age, and this estimate error should be low and bounded. An AE algorithm should tolerate at least four classes of variation. First are **photography-related** factors such as illumination, exposure, focus, and distortion. Second are **subject presentation** variations such as facial hair, eye glasses, dry vs. moisturized skin, and cosmetics. Third there are more **slowly varying** facial archetypes such as sun-damaged skin, creases, and jowls. Fourth there are **phenotypic** properties which includes skin tone but extends to a multitude of characteristics such as high cheek bones, bushy eye brows, dark eye brows, thin lips, square jaws, flat faces, and aquiline noses. This latter category are often properties of demographic groups associated with sex and ethnicity. Further more, should age estimation accuracy be constant across age, should it work as well in the old as the young? Ideally, yes, but operationally this is less important as many applications are focused on groups of people around a certain age, and algorithms could be tailored for that.

Figure 1 demonstrates some of the difficulty of the AE task. The figure shows the result of applying AE algorithms to self-portraits collected by artist Noah Kalina as part of his [Everyday](#) project. The images, collected daily since 2000, can seen via that link.

The figure reveals considerable **daily variance** in the age estimates. The daily increase should be 1/365 but the standard deviation of the absolute error is much larger - above 2 years in the best case. Why this occurs would require detailed developer investigation. Given the use of one camera model for the period 2009-2015 (age 27-34) and an impressively consistent two-arm frontal-selfie presentation to that camera, the sources of short-term variation are clothing, hair arrangement, ambient illumination which varies with the room, and sensor noise, none of which should affect

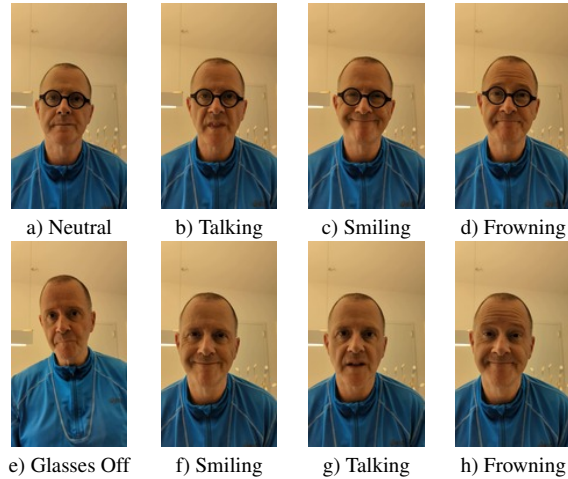
estimated age. In the longer-run the subject does adopt a heavier beard, but that too is elective not essentially part of ageing.

The figure also shows **systematic error** for all six algorithms: They overestimate Kalina’s age, with very few photographs ever giving an underestimate. The blue line  $y = x$  represents perfect estimation; the green line represents best fit linear regression. The separation of those lines is around five years, and this often persists for the entire interval. Why this occurs is not known, but could be investigated by developers who should have access to the photographs. An incorrect birthdate would yield consistent error, but date of birth has been reported. The images have good quality, so either some aspect of the subject’s appearance induces overestimation, or the algorithms are sensitive to information beyond the facial region. This concept - that some people look young, or old, for their age - is common in human experience, but whether Deep Neural Network (DNN) based algorithms can acquire information from photographs that humans cannot, or at least do not, is not clear.



**Fig. 1.** This figure shows age estimates plotted against ground truth age for six algorithms applied to 1991 photographs of the artist Noah Kalina taken daily throughout 2001, then between January and June in the years 2009 - 2015, then again throughout 2023. Note NIST purchased only photos for the first six months of each calendar year - the subject was born 1980-07-04. The inset values are standard deviation (SD) quantifying “noise”, mean absolute error (MAE) quantifying accuracy, and mean sequential day difference (MDD) which would ideally be 1/365. The lower blue line gives ground truth age. The upper line gives least squares best fit.

This kind of variation is evident in the very short term also. Figure 3 shows the results of applying AE algorithms to sequential frames in a 60 second video. The frames have size 1080x1920 and were extracted from an MP4 video captured by one of the authors using his mobile phone. The figure shows considerable inter-frame estimated age variation even in frames 0-180 and 900-1100 where the subject was largely motionless. In subsequent frames the subject talks, then smiles and frowns, and these induce larger variance. Age estimates tend to be higher with thick-rimmed glasses in the first half of the video, than without glasses in the second half.



**Fig. 2.** Image frames from a 60 second video when subject's facial expression changes, causing large variance.



**Fig. 3.** This figure shows the variation of age estimates for six algorithms applied to 1688 frames of Patrick Grother (figure 2) extracted from a 60 second video captured using an Android phone. The blue line is the ground truth age.

**Discussion:** Face recognition (FR) algorithms have become highly insensitive to nuisance variables such as pose, illumination and expression. This has been achieved over time using diverse in-the-wild datasets, and more recently data augmentation (synthetically changing images to diversify appearance while maintaining identity). FR algorithms are tasked with extracting features related to a person’s fixed identity; AE algorithms, on the other hand, are estimating a variable quantity, age. Whether AE can become insensitive to nuisance variables is not clear, and may actually be unnecessary. What is needed, at least in those applications involving determination that a person is younger or older than some limit, is to put bounds on AE measurement “noise” - i.e. to ensure that errors are inside some limit. In those Challenge-T applications, the appropriate task is to limit false positives.

## 2. Performance Metrics

This section details performance measures for age estimation algorithms. The metrics are statistics estimated by applying an AE algorithm to  $N$  images for which we have a known age,  $a_k$ , to get an

estimated age,  $\hat{a}_k$ . Unless stated otherwise, these are based on the absolute error

$$\epsilon_k = |\hat{a}_k - a_k| \quad (1)$$

### 2.1. Mean Absolute Error

**Mean Absolute Error (MAE)** is defined as the mean difference between the actual and estimated ages:

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^N \epsilon_k \quad (2)$$

This serves as our primary accuracy measure.

### 2.2. Median Absolute Error

**Median Absolute Error (MDAE)** is defined as the median of the absolute error values,  $\epsilon_k$ . The median is a robust statistic meaning that it is insensitive to outliers. Given  $N$  ordered error values,

$$\text{MDAE} = \begin{cases} \epsilon_{(N+1)/2}, & \text{if } N \text{ is odd} \\ (\epsilon_{N/2} + \epsilon_{N/2+1})/2 & \text{if } N \text{ is even} \end{cases} \quad (3)$$

### 2.3. Standard deviation

The **Standard Deviation (SD)** is used to quantify variation around the mean. It has been adopted in the AEV industry as a supplement MAE<sup>1</sup>.

$$\text{SD} = \frac{1}{N-1} \sum_{k=1}^N (\epsilon_k - \text{MAE})^2 \quad (4)$$

This statistic only quantifies variation around the mean, so an algorithm could have low SD yet be systematically underestimating or overestimating age, i.e. large MAE. For this reason we consider SD as a secondary performance indicator.

### 2.4. Accuracy

**Overall Accuracy (ACC)** is defined as the proportion of test images for which the absolute error is below a threshold,  $T$ ,

$$\text{ACC}(T) = 1 - \frac{1}{N} \sum_{k=1}^N H(\epsilon_k - T) \quad (5)$$

where  $H(x)$  is the step function

$$H(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1, & \text{if } x \geq 0 \end{cases} \quad (6)$$

<sup>1</sup>Yoti, [Age Estimation White Paper](#) (December 2023), 15



## 2.5. Inequities Across Demographic Groups

The **Gini coefficient**<sup>2</sup> has long been used in Economics to quantify earning or wealth inequality in a population. Here we adopt it to express variability in accuracy across  $M$  demographic groups. This has been done for biometric error rates, particularly in ISO/IEC 19795-10:2024.

$$G = \frac{\sum_{i=1}^M \sum_{j=1}^M |x_i - x_j|}{2M(M-1)\mu}, \quad (7)$$

where  $\mu$  is the arithmetic mean of  $x$ . This can be applied to MAE, MDAE or any other statistic.

## 2.6. Challenge-T

A main application of AEV systems is in verifying that a person is above a certain legally mandated age. For example, for alcohol purchases, this age is 21 years in most United States jurisdictions. Recognizing AEV algorithms naturally have age estimation errors, systems are typically configured to require another form of age verification if the estimated age is below a challenge age,  $T$ . The value of  $T$  depends on the legal restriction age,  $L$ , and the accuracy of the age estimator. For  $L = 18$ ,  $T = 25$  is common. For  $L = 21$ , a higher value of  $T$  would be required to achieve the same accuracy.

For an evaluation of AE algorithms we seek figures of merit for the effectiveness at the stated objective and for convenience for the legal population.

### 2.6.1. False Positive Rate

In Challenge- $T$  applications, a false positive occurs if the true age is less than the legal age limit,  $L$ , and the estimated age is greater than or equal to the challenge age,  $T$ . The false positive rate is estimated over a set of  $N$  samples for which  $a_k < L$ .

$$\text{FPR} = \frac{1}{N} \sum_i H(\hat{a}_k - T) \quad (8)$$

### 2.6.2. False Negative Rate

In Challenge- $T$  applications, a false negative occurs when someone above the legal age is estimated to have age below the challenge age,  $T$ . The false negative rate is estimated over a set of  $N$  samples for which  $a_k \geq L$ .

$$\text{FNR} = 1 - \frac{1}{N} \sum_i H(\hat{a}_k - T) \quad (9)$$

### 2.6.3. Ineffectiveness

The **Ineffectiveness** quantifies how well the system prevents people below the legal age limit ( $L$ ) from accessing the prohibited service or item. We define it in terms of how often it allows under-age persons to go unchallenged:

<sup>2</sup>See <https://mathworld.wolfram.com/GiniCoefficient.html>.

$$\text{INEFFECTIVENESS}(T) = \frac{\sum_{i=0}^{L-1} w_i p_i \text{FPR}_i}{\sum_{i=0}^{L-1} w_i p_i} \quad (10)$$

where  $w_i$  is the proportion of the population who have age  $i$ ,  $p_i$  is the likelihood that a person of age  $i$  would even try to defeat the system, and  $\text{FPR}_i$  is an empirical false positive rate for persons of age  $i$ . The denominator normalizes so that ineffectiveness represents a false positive rate residing on  $[0, 1]$ .  $\text{FPR}_i$  is estimated as the proportion of  $N_i$  individuals whose age is estimated at or above  $T$

$$\text{FPR}_i(T) = \frac{1}{N_i} \sum_{k=1}^{N_i} H(a_k - T) \quad (11)$$

The weights  $w_i$  should be estimated by an analyst from demographic tables. The probabilities  $p_i$  might be estimated via empirical observation or via social modelling. For the purpose of this report we assume an exponential model for this prior probability.

$$p_i = \alpha \frac{\exp(L - 1 - i)}{\exp(L - 1)} \quad (12)$$

For example, with  $L = 18$  and  $\alpha = 0.6$  the formula says that 60% of 17-year olds would try ( $p_{17} = 0.6$ ) but only 1.1% of 13 year olds would ( $p_{13} = 0.011$ ). This may be pessimistic for 17 year olds in many societies but for our purpose - comparison of AEV algorithms - we do not refine this further. To do so would require some sociological study. The model is ad hoc and certainly imprecise in real operations but is used here to illustrate what a local policy analyst should adapt or replace in real operations.

Note that if false positive rates vary across sex or ethnicity, the weights could be adapted for specific populations, for example it may be known that  $p_i$  for women is less than that for men. In that case equation 11 would include double sums over doubly subscripted variables ( $p_{is}$ ,  $\text{FPR}_{is}$ , etc., with  $s$  denoting sex). Extending that further we could incorporate region-of-birth measurements into the formulate via  $p_{isr}$ .

#### 2.6.4. Inconvenience

We additionally need a metric that quantifies how well the system does not inconvenience people above the legal age  $L$ . Inconvenience occurs when they are challenged and estimated to have age below  $T$ .

$$\text{INCONVENIENCE}(T) = \frac{\sum_{i=L}^{\text{MA}} w_i p_i \text{FNR}_i}{\sum_{i=L}^{\text{MA}} w_i p_i} \quad (13)$$

where  $T$  is the challenge age, MA is the maximum age of persons in the population,  $w_i$  is the proportion of the population who have age  $i$ ,  $p_i$  is the likelihood that a person of age  $i$  would use the system at all, and  $\text{FNR}_i$  is an empirical false negative rate for persons of age  $i$ . This is estimated as the proportion of  $N_i$  individuals whose age is estimated below  $T$ .

This formulation treats everyone at or above the legal age equally. One might argue that those of, or older than, the legal age but younger than the challenge age should expect to be inconvenienced

and that this should be discounted in the equation. We reject this, maintaining the existence of a challenge policy is needed because the age estimation technology is imperfect.

## 2.7. Variance in age estimates from frontal images

Given the fully frontal, minimal activity, face videos described in 3.6 we quantify noise in the age estimates as follows.

For the  $i$ -th video, the algorithm under test is applied to each of the  $j = 1 \dots n_i$  frames producing age estimates  $x_{ij}$  or a non-zero return code,  $r_{ij}$  indicating failure to compute or elective refusal to compute. For each video we compute an overall age estimate as the arithmetic mean,

$$\alpha_i = \frac{\sum_j^{n_i} x_{ij} \delta_{0r_{ij}}}{\sum_j^{n_i} \delta_{0r_{ij}}} \quad (14)$$

where the Kronecker  $\delta$  assigns zero weight to failed estimations. We then produce an overall noise statistic as the mean absolute error computed over all frames of all videos. The videos have the same length to within about 0.2%.

$$\eta = \sum_i \sum_j^{n_i} |x_{ij} - \alpha_i| \quad (15)$$

These values are tabulated on the variability tab of the [AEV results page](#).

## 2.8. Variance in age estimates with head yaw

Given the right-left head pose variation videos described in 3.7 we quantify angular dependence as described below using head pose yaw estimates produced by a leading algorithm submitted to the [FATE Quality](#) track.

For the  $i$ -th video, the age estimation algorithm is applied to each of the  $n_i$  frames producing age estimates  $x_{ij}$  or a non-zero return code,  $r_{ij}$  indicating failure, or elective refusal, to compute.

Using head yaw estimates  $\phi_{ij}$  for each frame, we compute an overall age estimate as the mean over frontal frames - those for which  $\phi_{ij} > 6$  deg.

$$\alpha_i = \frac{\sum_j^{n_i} x_{ij} \delta_{0r_{ij}} (1 - H(|\phi_{ij}| - 6))}{\sum_j^{n_i} \delta_{0r_{ij}} (1 - H(|\phi_{ij}| - 6))} \quad (16)$$

where the Kronecker  $\delta$  assigns zero weight to failed estimations, and the Heaviside function  $H(x)$  is 1 for  $x \geq 0$  and 0 otherwise.

We then recover an overall yaw sensitivity statistic  $u$  via simple linear regression.

$$|x_{ij} - \alpha_i| = u \sin |\phi_{ij}| + \epsilon_{ij} \quad (17)$$

where the sine is used to discount variations at small yaw angles. As  $\sin(\pi/2) = 1$  the coefficient  $u$  can be interpreted as the increase in age estimation error for 90 degrees of head rotation, smaller values connoting less pose sensitivity. This interpretation is limited because the the videos have yaw variations rarely extending beyond 60 degrees. The  $u$  values are tabulated on the variability tab of the [AEV results page](#).

### 3. Datasets

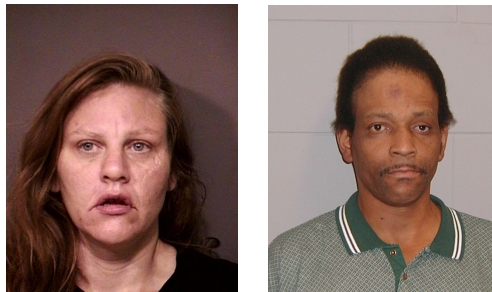
Table 9 shows the number of photographs and people in the datasets used in this report.

| Name                 | Visa                       | Mugshots                       | Application                  | Border               | Kalina Everyday            |
|----------------------|----------------------------|--------------------------------|------------------------------|----------------------|----------------------------|
| <b>Section</b>       | <a href="#">3.1</a>        | <a href="#">3.2</a>            | <a href="#">3.3</a>          | <a href="#">3.4</a>  | <a href="#">3.5</a>        |
| <b>DOB Precision</b> | YYYY-MM-DD                 | YYYY                           | YYYY-MM-DD                   | YYYY-MM-DD           | YYYY-MM-DD                 |
| <b>Age Precision</b> | YYYY-MM-DD                 | YYYY                           | YYYY-MM-DD                   | YYYY-MM-DD           | YYYY-MM-DD                 |
| <b>Age Range</b>     | 0-99                       | 17-99                          | 14-98                        | 14-91                | 20-21, 28-34, 42-43        |
| <b>Num. Images</b>   | 6249294                    | 1482667                        | 1054704                      | 2715230              | 1991                       |
| <b>Num. People</b>   | 5738091                    | 1482667                        | 802332                       | 632520               | 1                          |
| <b>Purpose</b>       | Exact repeat of 2014 study | AE accuracy on standard photos | Challenge-T and demographics | Effect of of quality | Longitudinal demonstration |

**Table 9.** Properties of the datasets.

#### 3.1. Visa Images

These images were used in the NIST 2014 Performance of Automated Age Estimation Algorithms[10] study. The images are of size 252x300 pixels. The mean interocular distance(IOD) is 69 pixels. The images are of subjects who applied for visas in United States consular facilities in Mexico. Many of the images are live capture. A minority of the images are photographs of paper photographs. When these images are passed to the algorithm, they are labeled with type “iso”.



**Fig. 4.** Examples of mugshot images used in the evaluation. Image source: NIST Special Database 32: Multiple Encounter Deceased Subjects (MEDS).

#### 3.2. Mugshot Images

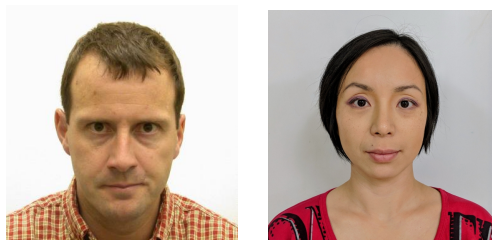
The images are collected in the United States in routine post-arrest booking processes. While the images are accompanied by one of five race labels specified in an EBTS standard<sup>3</sup> we don’t use them

<sup>3</sup>This standard is published by the Criminal Justice Information Services (CJIS) Division of the Federal Bureau of Investigation in the United States Department of Justice. It defines data records for transactions that (law enforce-

as they are unreliable. These images have reasonable compliance with the ANSI/NIST ITL 1-2011 Type 10 standard's subject acquisition profiles levels 10-20 for frontal images[11]. As shown in Figure 4 the images are of visually good quality, mostly sharp and well exposed per the standardized photographic equipment specified for their capture. The most common departure from standard requirements is the presence of mild pose variations around frontal. The images vary in size, with many being 480x600 pixels. They are JPEG compressed to produce filesizes of between 18 and 36 kilobytes. When these images are provided as input into the algorithm, they are labeled with the type "mugshot".

### 3.3. Application Images

The images are collected in an attended interview in immigration offices in the United States. The dataset is from a global population born in more than 100 countries, but for this study we use only those images from 34 countries grouped into six regions - see 5.4. The images are collected using dedicated capture equipment and lighting. They have a uniform, usually white background. As shown in Figure 5, the images have geometry in good conformance with the ISO/IEC19794-5 Full Frontal image type, pose is closely frontal, and eyeglasses are absent. The images, of size 300x300 pixels, are smaller than normally indicated by ISO. When these images are passed to an algorithm, they are labeled with the type "iso".



**Fig. 5.** The figure gives simulated samples of application type image used in the evaluation. Image sources: Authors.

### 3.4. Border Crossing Images

These images are taken with a webcam oriented by an immigration officer toward a cooperating subject - see the examples in Figure 6. The images are captured under time constraints, so roll, pitch and yaw often depart from frontal presentation, not exceeding  $\pm 45$  degrees. Images sometimes have moderate contrast because the background can be bright. This can lead to an under-exposed face. Resolution is limited by the use of inexpensive cameras, compression, and resizing to 240 x 300 pixels. The images have mean interocular distance of 38 pixels. We excluded images for which a leading face recognition engine found more than one face (typically from someone standing in the background). When these images are passed to an algorithm, they are labeled with the type "wild".

ment) agencies can make with the FBI. Version 10.0.7 of the standard was published on December 5, 2016. See <https://www.fbi/iospecs.cjis.gov>.



**Fig. 6.** The figure gives simulated samples of border type images used in the evaluation. The ages are 18yr 4mo, 41yr 3mo, and 52yr 2mo. Image sources: Author (right) and NIST Special Database 32 the Multiple Encounter Deceased Subjects dataset.

### 3.5. Kalina Everyday

The artist Noah Kalina has collected a self [portrait every day](#) since 2000. NIST purchased twelve months of photos for the years 2001 and 2023, and photos from January to June for each year 2009, 2010, 2011, 2012, 2013, 2014, and 2015. The images are attractive for longitudinal studies of age estimation (and face recognition), because they are collected with a fixed two-handed selfie technique. They therefore exhibit very little difference in head orientation, distance to camera, and margin around the head. The main sources of day-to-day variation are hairstyle and illumination. Also, the artist appears in various locations, so the backgrounds differ. All photos from 2001 have size 640x480, and all other years are 2816x2112. When these images are provided as input into the algorithm, they are labeled with the type “mugshot”.

### 3.6. Frontal videos

This dataset is comprised of frames extracted from close-up, fully-frontal, high quality videos of people looking at a camera. The subjects are not asked to do anything. While they occasionally blink, swallow, purse their lips, or move their head slightly, they are largely motionless.

There are 482 individuals. Each video has a 30 second duration. The video frame rate is 30 frames per second. Age estimation is done on each frame independently.

### 3.7. Yaw variation videos

This dataset is comprised of frames extracted from close-up high quality videos of people who are initially looking forward toward a camera and who then move their heads rightward, back to frontal, and then leftwards, and then back to frontal where they remain for more than 10 seconds. Subjects occasionally blink, swallow, or purse their lips. The maximum head rotation varies by subject but extends to from 30 degrees to 60 degrees.

There are 29 videos, one per individual. Each video has a 30 second duration. The video frame rate is 30 frames per second. Age estimation is done on each frame independently.

## 4. Algorithms

The FATE activity is open to participation worldwide, and the test will evaluate submissions on an ongoing basis. There is no charge to participate. The requirements to submit algorithms to NIST

are described in the [FATE Age Estimation And Verification Application Programming Interface \(API\)](#)[12] document. Participants provide their submissions in the form of libraries compiled on a specific Linux kernel, which are linked against NIST’s test harness to produce executables. NIST provides a validation package to participants to ensure that NIST’s execution of submitted libraries produces the expected output on NIST’s test machines.

**Table 10.** Participant [location](#), organization name, algorithm name, and submission date.

|    | Location | Developer Name                               | Algorithm           | Submission Date |
|----|----------|----------------------------------------------|---------------------|-----------------|
| 1  | DE       | Cognitec Systems GmbH                        | cognitec-000        | 2025-07-11      |
| 2  | CZ       | Czech Technical University in Prague         | cvut-001            | 2024-06-14      |
| 3  | CZ       | Czech Technical University in Prague         | cvut-002            | 2024-11-04      |
| 4  | CZ       | Czech Technical University in Prague         | cvut-003            | 2025-05-06      |
| 5  | TW       | Cyberlink Corp                               | cyberlink-000       | 2025-03-11      |
| 6  | TW       | Cyberlink Corp                               | cyberlink-001       | 2025-07-30      |
| 7  | IE       | Daon                                         | daon-000            | 2025-05-08      |
| 8  | IE       | Daon                                         | daon-001            | 2025-11-19      |
| 9  | DE       | Dermalog                                     | dermalog-001        | 2023-11-02      |
| 10 | DE       | Dermalog                                     | dermalog-002        | 2025-01-13      |
| 11 | CN       | Hangzhou Ailu Network Information Technology | hzailu-001          | 2025-03-28      |
| 12 | FR       | Idemia                                       | idemias-001         | 2024-08-26      |
| 13 | FR       | Idemia                                       | idemias-002         | 2025-06-23      |
| 14 | CZ       | Brno University of Technology-FIT DCGM       | inbut-000           | 2024-09-11      |
| 15 | US       | Incode Technologies Inc                      | incode-000          | 2023-09-05      |
| 16 | US       | Incode Technologies Inc                      | incode-001          | 2025-03-28      |
| 17 | SK       | Innovatrics                                  | innovatrics-000     | 2025-02-05      |
| 18 | SK       | Innovatrics                                  | innovatrics-001     | 2025-06-05      |
| 19 | SK       | Innovatrics                                  | innovatrics-002     | 2025-10-27      |
| 20 | US       | Jumio                                        | jumio-000           | 2024-08-30      |
| 21 | LT       | Neurotechnology                              | neurotechnology-000 | 2023-10-17      |
| 22 | LT       | Neurotechnology                              | neurotechnology-001 | 2024-07-19      |
| 23 | LT       | Nominder                                     | nominder-000        | 2024-07-22      |
| 24 | LT       | Nominder                                     | nominder-001        | 2024-11-25      |
| 25 | GB       | Ondato UAB                                   | ondato-000          | 2024-11-08      |
| 26 | GB       | Ondato UAB                                   | ondato-001          | 2025-03-07      |
| 27 | AU       | Privately SA                                 | privately-000       | 2026-01-14      |
| 28 | US       | Regula Forensics                             | regula-000          | 2025-07-23      |
| 29 | US       | ROC                                          | roc-000             | 2023-11-03      |
| 30 | US       | ROC                                          | roc-001             | 2024-10-29      |
| 31 | US       | ROC                                          | roc-002             | 2026-02-06      |
| 32 | GB       | Shufti                                       | shufti-000          | 2026-01-26      |
| 33 | FR       | Unissey                                      | unissey-001         | 2023-09-19      |
| 34 | AE       | Viante.AI                                    | viantes-000         | 2024-11-25      |
| 35 | GB       | Yoti                                         | yoti-001            | 2023-09-19      |
| 36 | GB       | Yoti                                         | yoti-002            | 2024-09-17      |
| 37 | GB       | Yoti                                         | yoti-003            | 2025-07-08      |
| 38 | GB       | Yoti                                         | yoti-004            | 2026-01-07      |
| 39 | PT       | Youverse                                     | youverse-000        | 2025-07-30      |

This report documents the results of algorithms submitted to FATE AEV for testing. Table 10 lists the algorithms that were tested.



**Table 11.** Algorithm resources: The size of the algorithm (the models + the libraries); the median duration (in milliseconds) for AE function calls. Timing was conducted over 1000 function invocations with an input image of 640x480 pixels.

| Algorithm           | MB      |         | Millisecs                |                        | Integer Age Estimates |
|---------------------|---------|---------|--------------------------|------------------------|-----------------------|
|                     | Config  | Lib     | Estimate Age (One Media) | Verify Age (One Media) |                       |
| cognitec-000        | 137.93  | 25.92   | 292.11                   | 293.47                 | No                    |
| cvut-001            | 2886.72 | 958.17  | 813.03                   | 828.61                 | Yes                   |
| cvut-002            | 353.46  | 943.60  | 812.03                   | 812.11                 | Yes                   |
| cvut-003            | 353.48  | 944.11  | 823.00                   | 815.05                 | No                    |
| cyberlink-000       | 117.50  | 130.30  | 419.67                   | 404.31                 | No                    |
| cyberlink-001       | 266.37  | 130.34  | 675.39                   | 689.25                 | No                    |
| daon-000            | 115.02  | 65.12   | 362.26                   | -                      | No                    |
| daon-001            | 565.14  | 65.20   | 1395.44                  | -                      | No                    |
| dermalog-001        | 0.00    | 187.52  | 26.70                    | 30.86                  | Yes                   |
| dermalog-002        | 0.00    | 168.32  | 27.41                    | 31.32                  | Yes                   |
| hzailu-001          | 185.49  | 280.69  | 100.55                   | 106.88                 | Yes                   |
| idemia-001          | 481.48  | 118.64  | 646.44                   | 642.18                 | No                    |
| idemia-002          | 1444.45 | 64.44   | 632.31                   | 644.35                 | No                    |
| inbut-000           | 0.00    | 791.00  | 711.81                   | 708.56                 | No                    |
| incode-000          | 457.10  | 40.46   | 235.54                   | 242.18                 | No                    |
| incode-001          | 910.32  | 40.46   | 868.99                   | 872.13                 | No                    |
| innovatrics-000     | 0.00    | 2491.85 | 975.28                   | 966.19                 | No                    |
| innovatrics-001     | 0.00    | 2474.18 | 981.35                   | 979.77                 | No                    |
| innovatrics-002     | 0.00    | 2367.96 | 1128.20                  | 1133.49                | No                    |
| jumio-000           | 0.00    | 626.68  | 256.02                   | 251.88                 | No                    |
| neurotechnology-000 | 113.42  | 95.91   | 153.18                   | 156.60                 | Yes                   |
| neurotechnology-001 | 53.42   | 119.96  | 96.70                    | 101.57                 | Yes                   |
| nominder-000        | 64.81   | 111.53  | 193.62                   | 191.53                 | No                    |
| nominder-001        | 118.72  | 111.54  | 249.76                   | 254.01                 | No                    |
| ondato-000          | 196.65  | 33.20   | 578.83                   | 570.83                 | No                    |
| ondato-001          | 196.65  | 33.20   | 562.06                   | 564.68                 | No                    |
| privately-000       | 293.60  | 27.33   | 63.59                    | -                      | No                    |
| regula-000          | 273.34  | 37.84   | 822.56                   | -                      | No                    |
| roc-000             | 0.00    | 473.82  | 134.15                   | 134.08                 | No                    |
| roc-001             | 0.00    | 565.40  | 683.73                   | 689.05                 | No                    |
| roc-002             | 0.00    | 737.29  | 1036.74                  | 1037.68                | No                    |
| shufti-000          | 194.02  | 52.54   | 612.70                   | 616.65                 | No                    |
| unissey-001         | 0.00    | 952.42  | 113.34                   | 117.91                 | Yes                   |
| viant-000           | 1224.51 | 16.15   | 1184.30                  | 1183.37                | Yes                   |
| yoti-001            | 226.95  | 428.53  | 267.86                   | 257.46                 | No                    |
| yoti-002            | 243.55  | 427.93  | 377.59                   | 379.94                 | No                    |
| yoti-003            | 810.14  | 934.89  | 734.52                   | 724.69                 | No                    |
| yoti-004            | 810.14  | 1028.74 | 1093.71                  | 1044.49                | No                    |
| youverse-000        | 84.39   | 315.24  | 311.41                   | 308.26                 | No                    |

Table 11 gives technical information about the algorithms. Column two shows the size of static read-only data provided with the algorithms. FATE AEV is a black-box evaluation, so we don't know and don't inquire as to what this data is. It typically includes small configuration files and much larger deep neural network (DNN) model files for face detection and age estimation. Some developers (e.g. roc) include such data in the libraries contained in the "lib" directory whose size is noted in column three. The fourth and fifth columns note execution duration for a call to the age estimation function and age verification function, that accepts in-memory image data and produced an estimated age value, and true/false decision. The timing is measured by calling high-resolution C++ std::chrono timers immediately before and after the function call. Timing is measured over 1000 images of size 640x480 pixels. Duration is measured on one core of a server-class Intel chip running



in a machine that is otherwise idle. The final column notes whether age estimates are continuous (fractional parts of a year) or discrete (integer number of years).

Integer age estimates will imply increased MAE for those datasets where fractional actual age information is available.

Note the duration varies by an order of magnitude; the total size of the config and lib directory varies by about a factor of five - which may have relevance to the use of these algorithms on embedded or low-powered devices.

## 5. Results

The following subsections address accuracy gains since 2014, comparative accuracy, Challenge-T performance, demographics, and the effect of quality.

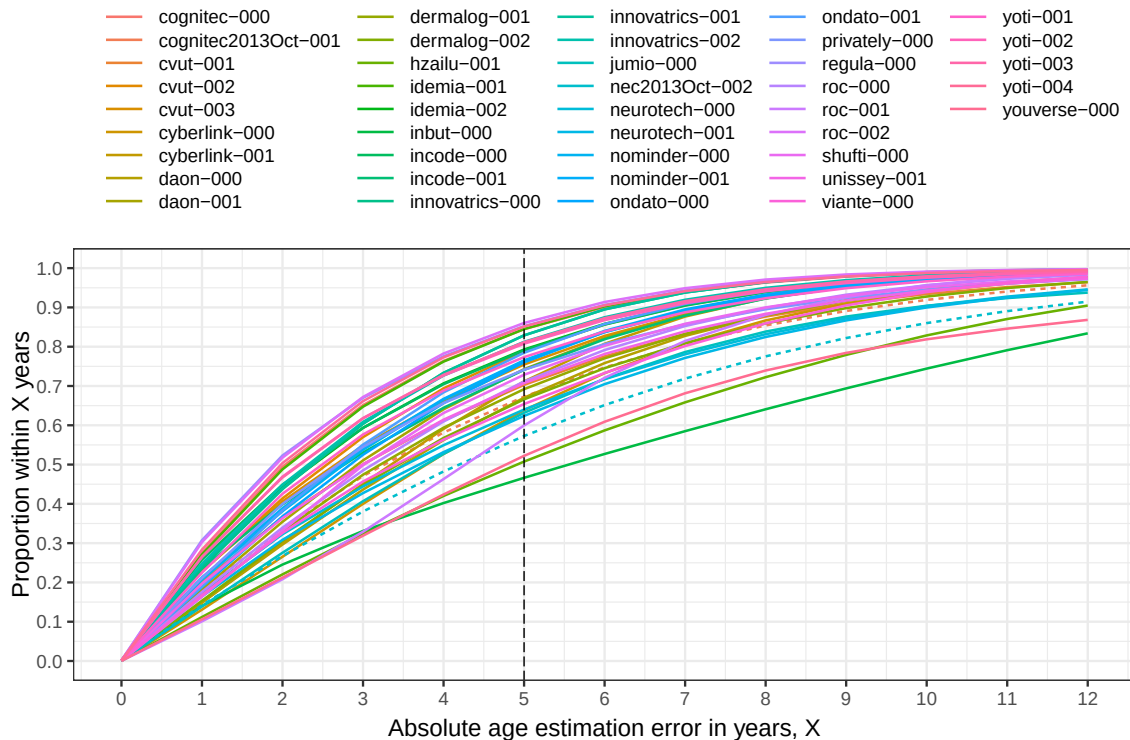
### 5.1. Age Estimation Improvements Since 2014

We measure algorithmic accuracy gains since 2014 by running the 2024 algorithms on the exact dataset used in 2014<sup>4</sup>. This set consists of 6.2 million *Visa* images with about 5.7 million people collected in Mexico with ages spanning 0 to 100.

Figure 7 shows the proportion of images estimated to have error less than the number of years given on the horizontal axis. The 2014 algorithms are shown with dotted lines, recent entries with solid lines. Table 12 shows the top-performing algorithm on this dataset, Incode-000, has a reduction of 1.19 years in mean absolute error (MAE) compared to the leader from 2014, Cognitec-001. Three other performance variables improve also. Only one algorithm is inferior.

Figure 8 shows the same gain in MAE and the proportion of images whose age estimate is better than the best 2014 algorithm. That proportion is never larger than 61%. This result implies a degree of randomness - an improved algorithm doesn't always give better estimates.

This publication is available free of charge from: <https://doi.org/10.6028/NIST.IR.8525>

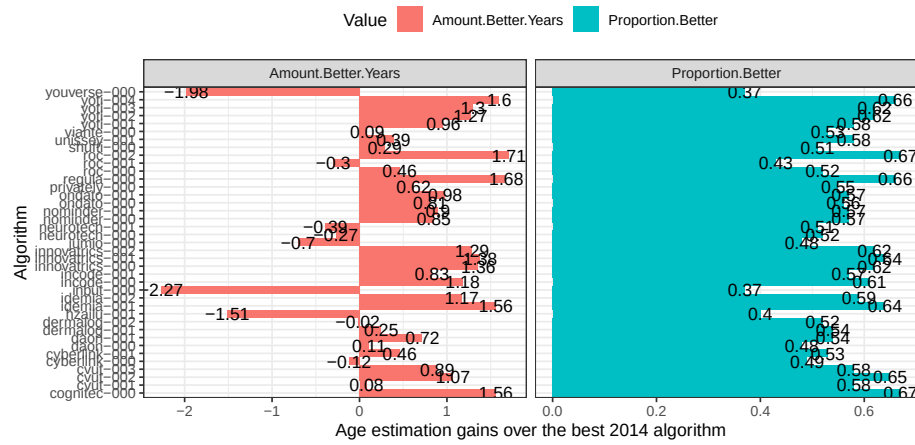


**Fig. 7.** This plot shows the accuracy of the 2024 algorithms compared against two top-performing 2014 algorithms. The lines plot cumulative accuracy at absolute age estimation error levels. Results were generated with the 6172425 images of the Mexican visa population.

<sup>4</sup>See [NIST Interagency Report 7995](#).

**Table 12.** Age estimation error statistics for the 2024 algorithms and the two most accurate algorithms from 2014 (in red). The values are mean and median absolute error (MAE, MDAE), standard deviation of the abs. error (SD), and mean error (ME). Dataset is the Mexican visa sample. MAE, MDAE, SD values are lower the better. ME values closer to zero the better, positive values are overestimate and negative values are underestimate.

| Algorithm           | MAE  | MDAE | ME    | SD   |
|---------------------|------|------|-------|------|
| regula-000          | 2.58 | 1.88 | 1.18  | 2.39 |
| roc-002             | 2.55 | 1.91 | 1.07  | 2.31 |
| yoti-004            | 2.66 | 1.98 | 1.06  | 2.44 |
| cognitec-000        | 2.70 | 2.04 | 0.39  | 2.42 |
| idemia-001          | 2.70 | 2.07 | 1.23  | 2.37 |
| yoti-002            | 2.99 | 2.19 | 1.43  | 2.79 |
| yoti-003            | 2.96 | 2.20 | 1.63  | 2.75 |
| innovatrics-002     | 2.97 | 2.30 | 1.02  | 2.59 |
| incode-000          | 3.08 | 2.33 | 1.48  | 2.74 |
| innovatrics-001     | 2.89 | 2.35 | 1.07  | 2.40 |
| idemia-002          | 3.09 | 2.36 | 2.04  | 2.67 |
| innovatrics-000     | 2.90 | 2.36 | 1.12  | 2.39 |
| yoti-001            | 3.30 | 2.46 | 1.74  | 2.95 |
| cvut-002            | 3.19 | 2.53 | 1.05  | 2.72 |
| cvut-003            | 3.37 | 2.64 | 1.70  | 2.89 |
| privately-000       | 3.65 | 2.67 | -0.16 | 3.37 |
| ondato-001          | 3.28 | 2.68 | 1.17  | 2.64 |
| nominder-001        | 3.36 | 2.71 | 1.65  | 2.79 |
| nominder-000        | 3.41 | 2.75 | 1.26  | 2.83 |
| incode-001          | 3.43 | 2.76 | 2.64  | 2.89 |
| ondato-000          | 3.46 | 2.84 | 1.06  | 2.82 |
| daon-001            | 3.54 | 2.93 | 2.22  | 2.78 |
| cvut-001            | 4.18 | 3.00 | 0.01  | 5.08 |
| unissey-001         | 3.87 | 3.00 | 0.87  | 3.35 |
| shufti-000          | 3.98 | 3.02 | 1.95  | 4.57 |
| roc-000             | 3.81 | 3.12 | 2.64  | 3.05 |
| dermalog-001        | 4.01 | 3.21 | 2.46  | 3.29 |
| cognitec2013Oct-001 | 4.27 | 3.25 | 0.30  | 3.86 |
| cyberlink-001       | 3.81 | 3.35 | 1.45  | 2.81 |
| viant-000           | 4.17 | 3.39 | 3.20  | 3.29 |
| dermalog-002        | 4.29 | 3.42 | 2.27  | 3.55 |
| daon-000            | 4.15 | 3.49 | 2.12  | 3.21 |
| neurotechnology-000 | 4.54 | 3.52 | -0.19 | 3.96 |
| neurotechnology-001 | 4.65 | 3.69 | -0.09 | 3.98 |
| jumio-000           | 4.98 | 3.77 | 3.61  | 5.05 |
| cyberlink-000       | 4.40 | 3.78 | 2.77  | 3.31 |
| nec2013Oct-002      | 5.32 | 4.19 | 0.31  | 4.61 |
| roc-001             | 4.55 | 4.27 | 4.00  | 2.88 |
| youverse-000        | 6.27 | 4.77 | -2.20 | 5.61 |
| hzailu-001          | 5.79 | 4.91 | 4.13  | 4.35 |
| inbut-000           | 6.49 | 5.55 | 5.82  | 5.03 |



**Fig. 8.** For 2024 algorithms, the right panel shows the fraction of images that have better accuracy estimates than the best-performing algorithm in our 2014 report (Cognitec-001 submitted to NIST 2013-10). The left panel shows the mean reduction in absolute age estimation error - negative values show worse performance now than in 2014. These results were generated over a consensus set of 6 166 298 images for which all algorithms returned an age estimate.

## 5.2. Comparing Algorithms

The prior section gives a ranking of algorithms on one dataset. We consider here three different datasets. Table 13 summarizes the mean age absolute error (MAE) for men and women, in three age groups, and in three datasets.

The table also includes the failure-to-process rate (FTP). FTP is uniformly very low, meaning the algorithms are able to process almost all photographs. Some algorithms may include a quality filter to electively decline to process images that the algorithm considers not suitable for AE. Low FTP is likely a product of face detection algorithms today that have very low false negative rates on cooperative datasets such as these. Note also it is a characteristic of deep neural networks to operate on *any* array of input pixels regardless of their semantic content.

**Table 13.** By sex, mean absolute error (MAE), and failure to process proportion (FTP) for three age bands. Lower values are better.

| Algorithm     | Dataset     | FTP   | Ages 18-30 |      | Ages 31-50 |      | Ages 51-80 |      |
|---------------|-------------|-------|------------|------|------------|------|------------|------|
|               |             |       | Female     | Male | Female     | Male | Female     | Male |
|               |             |       | MAE        | MAE  | MAE        | MAE  | MAE        | MAE  |
| cognitec-000  | Mugshot     | 0.000 | 2.8        | 2.4  | 3.4        | 2.9  | 4.3        | 3.9  |
| cognitec-000  | Application | 0.000 | 3.0        | 2.6  | 3.6        | 3.1  | 4.0        | 4.1  |
| cognitec-000  | Border      | 0.002 | 3.2        | 3.3  | 4.0        | 3.0  | 5.8        | 5.3  |
| cvut-001      | Mugshot     | 0.022 | 4.0        | 3.4  | 5.1        | 4.5  | 6.2        | 5.4  |
| cvut-001      | Application | 0.000 | 3.6        | 3.2  | 6.1        | 4.2  | 6.9        | 7.0  |
| cvut-001      | Border      | 0.002 | 3.8        | 3.4  | 7.8        | 6.1  | 7.4        | 7.5  |
| cvut-002      | Mugshot     | 0.022 | 2.9        | 2.6  | 4.3        | 3.6  | 5.1        | 4.9  |
| cvut-002      | Application | 0.000 | 3.4        | 2.9  | 6.0        | 4.6  | 6.5        | 5.9  |
| cvut-002      | Border      | 0.001 | 3.3        | 3.2  | 4.5        | 3.8  | 6.1        | 5.8  |
| cvut-003      | Mugshot     | 0.022 | 3.1        | 2.6  | 3.9        | 3.3  | 4.8        | 4.6  |
| cvut-003      | Application | 0.000 | 3.1        | 2.7  | 5.0        | 4.0  | 5.8        | 5.3  |
| cvut-003      | Border      | 0.001 | 3.1        | 2.9  | 4.4        | 3.8  | 5.8        | 5.5  |
| cyberlink-000 | Mugshot     | 0.000 | 3.4        | 3.5  | 4.4        | 3.8  | 5.4        | 4.8  |
| cyberlink-000 | Application | 0.000 | 3.1        | 3.3  | 4.8        | 3.6  | 5.1        | 4.8  |
| cyberlink-000 | Border      | 0.000 | 3.4        | 3.2  | 5.8        | 4.2  | 5.8        | 5.5  |
| cyberlink-001 | Mugshot     | 0.000 | 3.1        | 2.8  | 4.1        | 3.5  | 5.0        | 4.1  |
| cyberlink-001 | Application | 0.000 | 3.0        | 2.8  | 5.0        | 3.4  | 5.4        | 5.1  |
| cyberlink-001 | Border      | 0.000 | 3.6        | 2.9  | 7.0        | 4.4  | 7.4        | 6.6  |
| daon-000      | Mugshot     | 0.000 | 4.1        | 4.2  | 4.7        | 3.9  | 5.1        | 4.3  |
| daon-000      | Application | 0.000 | 4.3        | 3.8  | 5.0        | 3.8  | 5.3        | 4.5  |
| daon-000      | Border      | 0.000 | 5.0        | 4.4  | 6.5        | 4.2  | 8.3        | 5.7  |
| daon-001      | Mugshot     | 0.000 | 3.6        | 3.5  | 4.0        | 3.4  | 4.2        | 4.0  |
| daon-001      | Application | 0.000 | 3.8        | 3.5  | 4.1        | 3.2  | 4.2        | 3.7  |
| daon-001      | Border      | 0.001 | 4.4        | 3.7  | 5.3        | 3.6  | 6.7        | 4.8  |
| dermalog-001  | Mugshot     | 0.001 | 4.4        | 3.6  | 4.7        | 4.3  | 5.2        | 4.7  |
| dermalog-001  | Application | 0.000 | 3.6        | 3.1  | 4.5        | 4.3  | 5.6        | 4.8  |
| dermalog-001  | Border      | 0.000 | 4.3        | 4.3  | 4.1        | 3.9  | 5.4        | 4.5  |
| dermalog-002  | Mugshot     | 0.001 | 4.8        | 3.7  | 5.5        | 4.3  | 5.8        | 5.6  |
| dermalog-002  | Application | 0.000 | 3.7        | 3.3  | 5.2        | 5.2  | 6.2        | 6.6  |
| dermalog-002  | Border      | 0.000 | 4.4        | 4.5  | 5.2        | 4.8  | 6.3        | 5.6  |
| hzailu-001    | Mugshot     | 0.000 | 6.5        | 7.0  | 6.1        | 5.4  | 7.1        | 6.5  |
| hzailu-001    | Application | 0.000 | 4.4        | 5.2  | 5.3        | 4.5  | 6.9        | 5.8  |
| hzailu-001    | Border      | 0.000 | 4.3        | 5.2  | 5.6        | 4.7  | 7.4        | 6.4  |
| idemia-001    | Mugshot     | 0.000 | 2.5        | 2.2  | 3.1        | 2.8  | 3.8        | 3.3  |
| idemia-001    | Application | 0.000 | 2.9        | 2.7  | 3.5        | 2.8  | 4.0        | 3.6  |
| idemia-001    | Border      | 0.000 | 3.1        | 2.7  | 3.9        | 3.0  | 4.5        | 3.9  |
| idemia-002    | Mugshot     | 0.000 | 3.3        | 2.2  | 3.3        | 2.8  | 3.3        | 3.4  |
| idemia-002    | Application | 0.000 | 3.2        | 2.6  | 3.4        | 2.9  | 3.6        | 3.6  |
| idemia-002    | Border      | 0.000 | 3.5        | 2.7  | 3.8        | 3.1  | 3.9        | 3.9  |

**Table 13.** By sex, mean absolute error (MAE), and failure to process proportion (FTP) for three age bands. Lower values are better.  
(continued)

| Algorithm                  | Dataset     | FTP   | Ages 18-30 |      | Ages 31-50 |      | Ages 51-80 |      |
|----------------------------|-------------|-------|------------|------|------------|------|------------|------|
|                            |             |       | Female     | Male | Female     | Male | Female     | Male |
|                            |             |       | MAE        | MAE  | MAE        | MAE  | MAE        | MAE  |
| <b>inbut-000</b>           | Mugshot     | 0.000 | 2.9        | 2.9  | 3.4        | 3.1  | 4.5        | 3.8  |
| <b>inbut-000</b>           | Application | 0.000 | 3.2        | 3.6  | 3.4        | 3.4  | 4.0        | 3.5  |
| <b>inbut-000</b>           | Border      | 0.000 | 2.9        | 2.9  | 5.2        | 3.9  | 6.1        | 4.7  |
| <b>incode-000</b>          | Mugshot     | 0.000 | 3.6        | 2.6  | 3.5        | 3.1  | 3.7        | 3.9  |
| <b>incode-000</b>          | Application | 0.000 | 3.3        | 2.7  | 3.5        | 3.2  | 3.5        | 3.6  |
| <b>incode-000</b>          | Border      | 0.001 | 4.8        | 4.3  | 4.5        | 4.1  | 4.8        | 4.3  |
| <b>incode-001</b>          | Mugshot     | 0.001 | 3.5        | 2.4  | 3.2        | 2.9  | 3.1        | 3.4  |
| <b>incode-001</b>          | Application | 0.000 | 3.4        | 2.9  | 3.2        | 3.1  | 3.1        | 3.1  |
| <b>incode-001</b>          | Border      | 0.001 | 4.3        | 4.2  | 4.5        | 4.1  | 4.0        | 3.6  |
| <b>innovatrics-000</b>     | Mugshot     | 0.000 | 2.4        | 2.0  | 3.2        | 2.8  | 4.8        | 3.9  |
| <b>innovatrics-000</b>     | Application | 0.000 | 2.6        | 2.4  | 3.9        | 3.2  | 5.0        | 4.1  |
| <b>innovatrics-000</b>     | Border      | 0.001 | 2.8        | 2.5  | 4.9        | 3.6  | 6.0        | 4.9  |
| <b>innovatrics-001</b>     | Mugshot     | 0.000 | 2.4        | 2.0  | 3.3        | 2.9  | 4.8        | 3.8  |
| <b>innovatrics-001</b>     | Application | 0.000 | 2.5        | 2.3  | 4.4        | 3.4  | 4.8        | 4.0  |
| <b>innovatrics-001</b>     | Border      | 0.001 | 2.9        | 2.5  | 5.2        | 3.8  | 5.5        | 4.6  |
| <b>innovatrics-002</b>     | Mugshot     | 0.000 | 2.7        | 2.3  | 3.5        | 3.1  | 5.2        | 4.3  |
| <b>innovatrics-002</b>     | Application | 0.000 | 2.8        | 2.7  | 4.2        | 3.4  | 4.7        | 3.9  |
| <b>innovatrics-002</b>     | Border      | 0.001 | 3.5        | 2.8  | 5.6        | 3.9  | 5.6        | 4.6  |
| <b>jumio-000</b>           | Mugshot     | 0.002 | 4.2        | 2.5  | 3.3        | 3.0  | 3.7        | 3.7  |
| <b>jumio-000</b>           | Application | 0.002 | 3.6        | 2.8  | 3.2        | 3.0  | 3.7        | 3.5  |
| <b>jumio-000</b>           | Border      | 0.048 | 4.9        | 4.0  | 4.1        | 3.7  | 7.0        | 5.9  |
| <b>neurotechnology-000</b> | Mugshot     | 0.000 | 3.7        | 3.4  | 5.8        | 4.6  | 6.8        | 5.5  |
| <b>neurotechnology-000</b> | Application | 0.000 | 3.8        | 3.4  | 7.9        | 5.7  | 9.0        | 7.1  |
| <b>neurotechnology-000</b> | Border      | 0.001 | 4.3        | 3.4  | 8.9        | 6.3  | 10.6       | 7.5  |
| <b>neurotechnology-001</b> | Mugshot     | 0.000 | 3.4        | 3.0  | 6.3        | 5.0  | 6.9        | 5.5  |
| <b>neurotechnology-001</b> | Application | 0.000 | 3.8        | 3.2  | 8.0        | 5.9  | 9.0        | 7.1  |
| <b>neurotechnology-001</b> | Border      | 0.002 | 4.3        | 3.3  | 9.0        | 6.4  | 10.6       | 7.5  |
| <b>nominder-000</b>        | Mugshot     | 0.000 | 3.2        | 2.9  | 3.7        | 3.4  | 4.6        | 4.0  |
| <b>nominder-000</b>        | Application | 0.000 | 3.1        | 3.2  | 4.5        | 3.4  | 5.8        | 4.7  |
| <b>nominder-000</b>        | Border      | 0.001 | 3.3        | 3.1  | 5.3        | 4.1  | 6.9        | 5.9  |
| <b>nominder-001</b>        | Mugshot     | 0.001 | 2.7        | 2.2  | 3.2        | 2.8  | 4.0        | 3.4  |
| <b>nominder-001</b>        | Application | 0.000 | 2.8        | 2.6  | 4.3        | 3.3  | 4.9        | 4.4  |
| <b>nominder-001</b>        | Border      | 0.001 | 3.2        | 2.8  | 5.6        | 4.0  | 6.5        | 5.8  |
| <b>ondato-000</b>          | Mugshot     | 0.000 | 3.4        | 2.8  | 3.8        | 3.5  | 4.1        | 3.7  |
| <b>ondato-000</b>          | Application | 0.000 | 3.1        | 3.0  | 4.1        | 3.5  | 4.1        | 3.7  |
| <b>ondato-000</b>          | Border      | 0.001 | 3.5        | 3.0  | 7.2        | 4.8  | 8.9        | 6.7  |
| <b>ondato-001</b>          | Mugshot     | 0.000 | 3.4        | 2.6  | 3.5        | 3.3  | 4.1        | 3.8  |
| <b>ondato-001</b>          | Application | 0.000 | 2.9        | 2.6  | 3.7        | 3.2  | 4.3        | 3.8  |
| <b>ondato-001</b>          | Border      | 0.001 | 3.4        | 2.8  | 7.0        | 4.7  | 8.2        | 6.6  |
| <b>privately-000</b>       | Mugshot     | 0.000 | 3.9        | 2.8  | 6.1        | 4.9  | 8.0        | 6.1  |
| <b>privately-000</b>       | Application | 0.000 | 3.5        | 3.0  | 5.9        | 5.7  | 7.5        | 7.0  |
| <b>privately-000</b>       | Border      | 0.000 | 4.0        | 3.5  | 6.2        | 5.3  | 7.9        | 6.6  |
| <b>regula-000</b>          | Mugshot     | 0.000 | 2.9        | 2.0  | 3.4        | 3.2  | 3.5        | 3.7  |
| <b>regula-000</b>          | Application | 0.000 | 2.6        | 2.3  | 3.1        | 3.0  | 3.2        | 3.2  |
| <b>regula-000</b>          | Border      | 0.001 | 3.1        | 2.6  | 3.8        | 3.7  | 4.0        | 3.9  |
| <b>roc-000</b>             | Mugshot     | 0.000 | 3.0        | 2.5  | 3.9        | 3.2  | 4.4        | 3.8  |
| <b>roc-000</b>             | Application | 0.000 | 3.6        | 3.4  | 3.8        | 3.2  | 4.2        | 3.9  |
| <b>roc-000</b>             | Border      | 0.001 | 3.9        | 3.8  | 4.2        | 3.4  | 4.8        | 4.3  |
| <b>roc-001</b>             | Mugshot     | 0.000 | 2.6        | 2.3  | 3.3        | 3.0  | 3.6        | 3.4  |
| <b>roc-001</b>             | Application | 0.000 | 3.1        | 2.9  | 3.1        | 3.0  | 3.4        | 3.4  |
| <b>roc-001</b>             | Border      | 0.001 | 3.3        | 3.2  | 3.2        | 3.0  | 3.7        | 3.5  |
| <b>roc-002</b>             | Mugshot     | 0.000 | 2.4        | 2.0  | 3.2        | 2.9  | 3.7        | 3.7  |
| <b>roc-002</b>             | Application | 0.000 | 2.7        | 2.4  | 3.1        | 2.8  | 3.5        | 3.5  |
| <b>roc-002</b>             | Border      | 0.001 | 3.1        | 2.9  | 3.1        | 2.8  | 3.7        | 3.5  |

**Table 13.** By sex, mean absolute error (MAE), and failure to process proportion (FTP) for three age bands. Lower values are better. (continued)

| Algorithm    | Dataset     | FTP   | Ages 18-30 |      | Ages 31-50 |      | Ages 51-80 |      |
|--------------|-------------|-------|------------|------|------------|------|------------|------|
|              |             |       | Female     | Male | Female     | Male | Female     | Male |
|              |             |       | MAE        | MAE  | MAE        | MAE  | MAE        | MAE  |
| shufti-000   | Mugshot     | 0.002 | 4.2        | 2.9  | 4.0        | 3.5  | 4.1        | 3.9  |
| shufti-000   | Application | 0.000 | 4.1        | 4.3  | 3.8        | 3.8  | 4.5        | 4.0  |
| shufti-000   | Border      | 0.009 | 4.8        | 4.2  | 5.5        | 4.1  | 7.6        | 5.2  |
| unissey-001  | Mugshot     | 0.000 | 4.3        | 3.6  | 6.3        | 4.8  | 6.0        | 5.3  |
| unissey-001  | Application | 0.000 | 3.9        | 3.7  | 5.0        | 4.2  | 6.1        | 5.8  |
| unissey-001  | Border      | 0.001 | 4.1        | 3.4  | 7.1        | 5.8  | 8.4        | 7.2  |
| viant-000    | Mugshot     | 0.001 | 4.0        | 4.4  | 3.8        | 3.8  | 4.0        | 3.5  |
| viant-000    | Application | 0.000 | 3.5        | 4.3  | 3.3        | 3.4  | 3.7        | 3.4  |
| viant-000    | Border      | 0.001 | 3.1        | 3.5  | 3.8        | 3.3  | 5.2        | 4.1  |
| yoti-001     | Mugshot     | 0.002 | 3.8        | 2.4  | 3.9        | 3.4  | 3.7        | 3.7  |
| yoti-001     | Application | 0.000 | 4.7        | 3.2  | 3.8        | 3.2  | 3.8        | 3.5  |
| yoti-001     | Border      | 0.003 | 5.2        | 4.1  | 4.8        | 4.0  | 5.1        | 4.2  |
| yoti-002     | Mugshot     | 0.000 | 3.2        | 2.4  | 3.5        | 3.1  | 3.6        | 3.6  |
| yoti-002     | Application | 0.000 | 3.5        | 2.8  | 3.1        | 2.9  | 3.7        | 3.3  |
| yoti-002     | Border      | 0.000 | 4.1        | 3.5  | 4.2        | 3.7  | 4.4        | 3.8  |
| yoti-003     | Mugshot     | 0.000 | 3.1        | 2.3  | 3.4        | 3.1  | 3.4        | 3.5  |
| yoti-003     | Application | 0.000 | 3.4        | 2.8  | 3.0        | 2.9  | 3.3        | 3.2  |
| yoti-003     | Border      | 0.000 | 3.9        | 3.5  | 3.7        | 3.3  | 4.4        | 3.8  |
| yoti-004     | Mugshot     | 0.000 | 2.6        | 1.9  | 3.4        | 3.1  | 3.4        | 3.5  |
| yoti-004     | Application | 0.000 | 2.7        | 2.3  | 2.9        | 2.8  | 3.4        | 3.1  |
| yoti-004     | Border      | 0.000 | 3.5        | 3.0  | 3.9        | 3.4  | 4.8        | 3.8  |
| youverse-000 | Mugshot     | 0.001 | 3.8        | 3.3  | 5.8        | 4.8  | 9.6        | 9.2  |
| youverse-000 | Application | 0.000 | 3.7        | 3.3  | 7.3        | 4.8  | 12.3       | 9.7  |
| youverse-000 | Border      | 0.001 | 5.1        | 4.6  | 9.2        | 5.6  | 16.0       | 11.0 |

The table shows MAE trending higher with age group, reflecting wider variation in the amount of ageing evidence in human faces. This effect is much larger in some algorithms, perhaps reflecting commercial focus on AE estimation in younger subjects, and is absent for one algorithm (yoti-001) operating on women. MAE is often highest in the lower quality *Border* image dataset, and lowest in the *Application* photos. There are exceptions and these observations are overlaid by a higher MAE in women than in men that applies to all algorithms, all datasets and both age groups. The lowest MAE values are 2.4 and 2.5 years from yoti-001 and roc-000 operating on *Mugshot* photos of men, age 18-30. The incode-001 algorithm achieves 2.7 years on *Application* photos.

### 5.3. Challenge Age Analysis

A primary application of AEV technology is in preventing underage subjects from accessing alcohol, cigarettes, and adult content. Given some legal restriction age  $L^5$ , AE is applied with a threshold age  $T$ , which is necessarily larger than  $L$ , with a supplemental challenge process being applied to anyone whose age estimate is below  $T$ . The difference  $T - L$  is sometimes termed the buffer size. For  $L = 18$  a seven year buffer is conventional, so  $T = 25$  is typical. A false positive occurs if a person of age below  $L$  is estimated to have age at or above  $T$ . A false negative occurs if a person of age at or above  $L$  is estimated to have age below  $T$ . This is formalized in section 2. Smaller  $T$  values are desirable, because they imply lower false negative rates (FNR). Smaller  $T$  values are sustainable only if false positive rates (FPR) remain low enough to meet some policy-driven criterion.

To mimic a Challenge-25 application, Table 14 shows FPR for persons aged 14-17 who are estimated to have age at or above 25. The sample is from the global traveler population. Likewise Table 15 shows FPR for persons aged 14-20 who are estimated to have age at or above 28. The *Mugshot* entries are estimated from persons aged 18-20.

**Table 14.** False positive rate (FPR) and mean absolute error (MAE) for persons 14-17 estimated at 25 or more. Border photos have inferior quality to the application photos. Lower values are better.

| Algorithm     | Data        | Male  |     | Female |     |
|---------------|-------------|-------|-----|--------|-----|
|               |             | FPR   | MAE | FPR    | MAE |
| cognitec-000  | Border      | 0.051 | 3.4 | 0.050  | 3.3 |
| cognitec-000  | Application | 0.005 | 1.9 | 0.014  | 2.2 |
| cvut-001      | Border      | 0.043 | 3.2 | 0.117  | 4.4 |
| cvut-001      | Application | 0.039 | 2.8 | 0.100  | 3.7 |
| cvut-002      | Border      | 0.077 | 4.2 | 0.131  | 5.1 |
| cvut-002      | Application | 0.011 | 2.4 | 0.032  | 3.3 |
| cvut-003      | Border      | 0.049 | 3.8 | 0.115  | 4.9 |
| cvut-003      | Application | 0.008 | 2.3 | 0.029  | 3.4 |
| cyberlink-000 | Border      | 0.081 | 3.8 | 0.106  | 4.2 |
| cyberlink-000 | Application | 0.054 | 4.2 | 0.091  | 5.2 |
| cyberlink-001 | Border      | 0.031 | 2.9 | 0.027  | 3.0 |
| cyberlink-001 | Application | 0.013 | 3.1 | 0.023  | 3.7 |
| daon-000      | Border      | 0.202 | 5.4 | 0.110  | 4.3 |
| daon-000      | Application | 0.094 | 4.3 | 0.067  | 3.5 |
| daon-001      | Border      | 0.115 | 4.1 | 0.077  | 3.3 |
| daon-001      | Application | 0.045 | 3.3 | 0.034  | 3.0 |
| dermalog-001  | Border      | 0.166 | 4.3 | 0.283  | 5.9 |
| dermalog-001  | Application | 0.051 | 3.6 | 0.112  | 4.7 |
| dermalog-002  | Border      | 0.190 | 5.1 | 0.344  | 6.8 |
| dermalog-002  | Application | 0.034 | 3.2 | 0.167  | 5.2 |
| hzailu-001    | Border      | 0.423 | 7.6 | 0.393  | 7.3 |
| hzailu-001    | Application | 0.341 | 6.9 | 0.361  | 6.9 |
| idemia-001    | Border      | 0.008 | 2.0 | 0.015  | 2.4 |
| idemia-001    | Application | 0.004 | 2.0 | 0.012  | 2.5 |
| idemia-002    | Border      | 0.005 | 1.7 | 0.013  | 2.1 |
| idemia-002    | Application | 0.002 | 1.7 | 0.009  | 1.9 |
| inbut-000     | Border      | 0.062 | 3.4 | 0.090  | 4.3 |
| inbut-000     | Application | 0.100 | 4.1 | 0.114  | 4.6 |
| incode-000    | Border      | 0.065 | 2.9 | 0.177  | 4.8 |
| incode-000    | Application | 0.006 | 1.8 | 0.033  | 2.6 |
| incode-001    | Border      | 0.046 | 3.2 | 0.134  | 4.8 |

<sup>5</sup>For example,  $L = 18$  for alcohol and cigarettes in the United Kingdom, and  $L = 21$  in [most jurisdictions](#) in the United States.



**Table 14.** False positive rate (FPR) and mean absolute error (MAE) for persons 14-17 estimated at 25 or more. Border photos have inferior quality to the application photos. Lower values are better. (continued)

| Algorithm           | Data        | Male  |     | Female |     |
|---------------------|-------------|-------|-----|--------|-----|
|                     |             | FPR   | MAE | FPR    | MAE |
| incode-001          | Application | 0.006 | 1.8 | 0.034  | 3.0 |
| innovatrics-000     | Border      | 0.017 | 2.5 | 0.039  | 3.4 |
| innovatrics-000     | Application | 0.007 | 2.5 | 0.030  | 3.5 |
| innovatrics-001     | Border      | 0.013 | 2.5 | 0.032  | 3.1 |
| innovatrics-001     | Application | 0.005 | 2.4 | 0.024  | 3.4 |
| innovatrics-002     | Border      | 0.009 | 1.4 | 0.018  | 2.0 |
| innovatrics-002     | Application | 0.008 | 1.6 | 0.034  | 2.7 |
| jumio-000           | Border      | 0.229 | 7.0 | 0.429  | 9.4 |
| jumio-000           | Application | 0.021 | 3.0 | 0.088  | 4.8 |
| neurotechnology-000 | Border      | 0.182 | 5.7 | 0.128  | 4.7 |
| neurotechnology-000 | Application | 0.248 | 6.3 | 0.178  | 5.4 |
| neurotechnology-001 | Border      | 0.148 | 5.4 | 0.103  | 4.5 |
| neurotechnology-001 | Application | 0.157 | 5.6 | 0.125  | 4.9 |
| nominder-000        | Border      | 0.075 | 4.5 | 0.074  | 4.2 |
| nominder-000        | Application | 0.020 | 2.7 | 0.041  | 3.6 |
| nominder-001        | Border      | 0.026 | 3.7 | 0.036  | 3.8 |
| nominder-001        | Application | 0.007 | 2.8 | 0.021  | 3.9 |
| ondato-000          | Border      | 0.056 | 3.7 | 0.060  | 4.0 |
| ondato-000          | Application | 0.027 | 2.9 | 0.073  | 4.6 |
| ondato-001          | Border      | 0.040 | 3.4 | 0.046  | 3.8 |
| ondato-001          | Application | 0.014 | 3.0 | 0.051  | 4.5 |
| privately-000       | Border      | 0.091 | 4.0 | 0.137  | 4.8 |
| privately-000       | Application | 0.014 | 2.9 | 0.065  | 3.7 |
| regula-000          | Border      | 0.011 | 1.9 | 0.039  | 2.9 |
| regula-000          | Application | 0.003 | 1.5 | 0.012  | 2.4 |
| roc-000             | Border      | 0.089 | 4.9 | 0.144  | 5.3 |
| roc-000             | Application | 0.041 | 4.2 | 0.073  | 4.6 |
| roc-001             | Border      | 0.029 | 4.1 | 0.062  | 4.5 |
| roc-001             | Application | 0.013 | 3.6 | 0.040  | 4.4 |
| roc-002             | Border      | 0.024 | 3.3 | 0.047  | 3.6 |
| roc-002             | Application | 0.006 | 2.2 | 0.016  | 2.8 |
| shufti-000          | Border      | 0.062 | 3.5 | 0.085  | 4.2 |
| shufti-000          | Application | 0.029 | 3.0 | 0.067  | 4.1 |
| unissey-001         | Border      | 0.080 | 3.1 | 0.106  | 3.5 |
| unissey-001         | Application | 0.107 | 3.4 | 0.127  | 3.7 |
| viant-000           | Border      | 0.113 | 3.8 | 0.216  | 5.1 |
| viant-000           | Application | 0.138 | 4.9 | 0.253  | 6.2 |
| yoti-001            | Border      | 0.103 | 3.4 | 0.196  | 5.0 |
| yoti-001            | Application | 0.012 | 1.7 | 0.090  | 3.4 |
| yoti-002            | Border      | 0.067 | 2.7 | 0.196  | 4.8 |
| yoti-002            | Application | 0.009 | 1.7 | 0.063  | 3.2 |
| yoti-003            | Border      | 0.077 | 3.0 | 0.197  | 5.0 |
| yoti-003            | Application | 0.008 | 1.6 | 0.056  | 3.1 |
| yoti-004            | Border      | 0.052 | 3.0 | 0.166  | 4.9 |
| yoti-004            | Application | 0.005 | 1.7 | 0.026  | 3.2 |
| youverse-000        | Border      | 0.248 | 5.7 | 0.143  | 4.5 |
| youverse-000        | Application | 0.026 | 2.6 | 0.051  | 3.3 |

**Table 15.** False positive rate (FPR) and mean absolute error (MAE) for persons 14-20 estimated at 28 or more. Border photos have inferior quality to the mugshot and application photos. The mugshots are from people 18-20. Lower values are better.

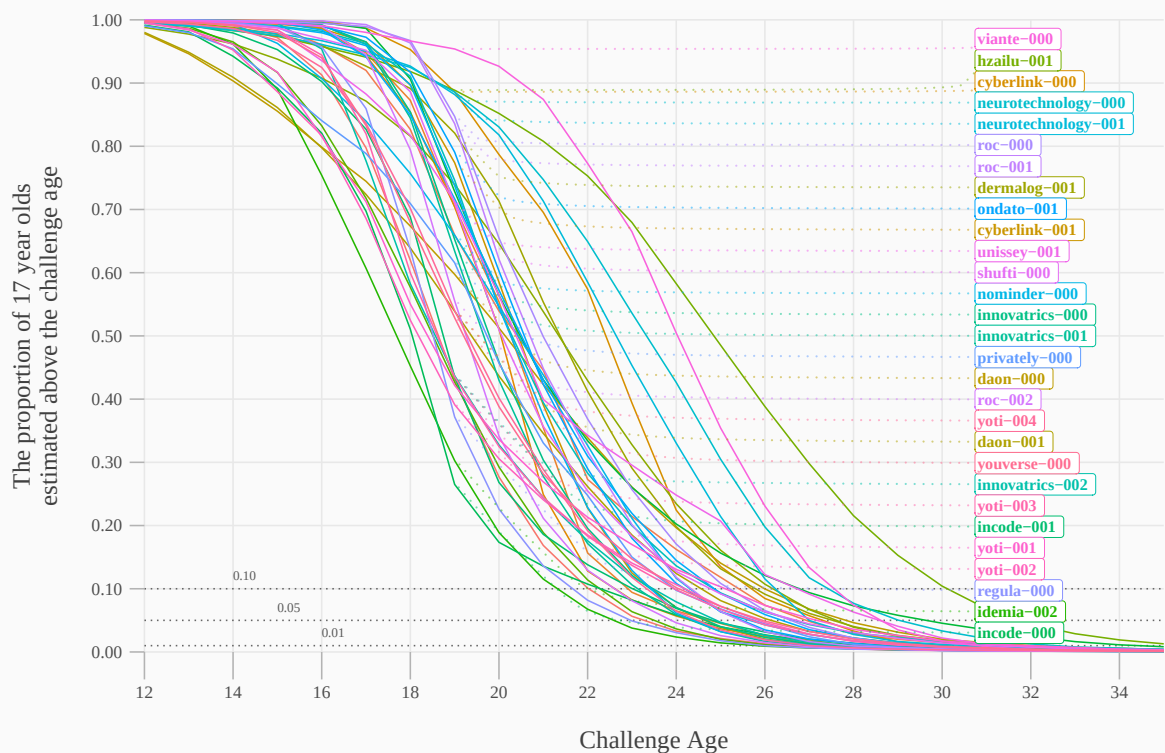
| Algorithm       | Data        | Male  |     | Female |     |
|-----------------|-------------|-------|-----|--------|-----|
|                 |             | FPR   | MAE | FPR    | MAE |
| cognitec-000    | Mugshot     | 0.009 | 2.2 | 0.020  | 2.8 |
| cognitec-000    | Border      | 0.041 | 3.4 | 0.032  | 3.1 |
| cognitec-000    | Application | 0.007 | 2.0 | 0.013  | 2.4 |
| cvut-001        | Mugshot     | 0.045 | 3.4 | 0.070  | 3.4 |
| cvut-001        | Border      | 0.019 | 3.2 | 0.042  | 4.1 |
| cvut-001        | Application | 0.029 | 2.8 | 0.052  | 3.4 |
| cvut-002        | Mugshot     | 0.012 | 2.3 | 0.021  | 2.6 |
| cvut-002        | Border      | 0.060 | 4.2 | 0.078  | 4.5 |
| cvut-002        | Application | 0.012 | 2.3 | 0.015  | 2.7 |
| cvut-003        | Mugshot     | 0.011 | 2.6 | 0.034  | 3.6 |
| cvut-003        | Border      | 0.026 | 3.8 | 0.047  | 4.4 |
| cvut-003        | Application | 0.009 | 2.3 | 0.017  | 2.8 |
| cyberlink-000   | Mugshot     | 0.053 | 3.8 | 0.074  | 4.1 |
| cyberlink-000   | Border      | 0.055 | 3.7 | 0.040  | 3.7 |
| cyberlink-000   | Application | 0.051 | 4.2 | 0.050  | 4.6 |
| cyberlink-001   | Mugshot     | 0.018 | 2.7 | 0.032  | 3.1 |
| cyberlink-001   | Border      | 0.025 | 2.8 | 0.017  | 2.6 |
| cyberlink-001   | Application | 0.014 | 2.9 | 0.016  | 3.1 |
| daon-000        | Mugshot     | 0.083 | 4.4 | 0.078  | 4.1 |
| daon-000        | Border      | 0.136 | 5.3 | 0.072  | 4.5 |
| daon-000        | Application | 0.066 | 4.2 | 0.048  | 3.8 |
| daon-001        | Mugshot     | 0.038 | 3.5 | 0.052  | 3.4 |
| daon-001        | Border      | 0.076 | 4.3 | 0.054  | 3.8 |
| daon-001        | Application | 0.037 | 3.5 | 0.030  | 3.4 |
| dermalog-001    | Mugshot     | 0.069 | 3.7 | 0.123  | 4.5 |
| dermalog-001    | Border      | 0.147 | 4.8 | 0.211  | 5.9 |
| dermalog-001    | Application | 0.034 | 3.4 | 0.069  | 4.2 |
| dermalog-002    | Mugshot     | 0.067 | 3.5 | 0.144  | 5.2 |
| dermalog-002    | Border      | 0.146 | 5.1 | 0.222  | 6.4 |
| dermalog-002    | Application | 0.031 | 3.1 | 0.081  | 5.0 |
| hzailu-001      | Mugshot     | 0.406 | 7.9 | 0.403  | 7.9 |
| hzailu-001      | Border      | 0.269 | 7.2 | 0.240  | 6.6 |
| hzailu-001      | Application | 0.246 | 6.9 | 0.247  | 6.6 |
| idemias-001     | Mugshot     | 0.004 | 1.9 | 0.009  | 2.7 |
| idemias-001     | Border      | 0.011 | 2.1 | 0.011  | 2.4 |
| idemias-001     | Application | 0.009 | 2.2 | 0.012  | 2.6 |
| idemias-002     | Mugshot     | 0.004 | 2.0 | 0.033  | 3.2 |
| idemias-002     | Border      | 0.007 | 1.8 | 0.014  | 2.5 |
| idemias-002     | Application | 0.006 | 1.8 | 0.012  | 2.4 |
| inbut-000       | Mugshot     | 0.041 | 2.7 | 0.044  | 3.3 |
| inbut-000       | Border      | 0.031 | 3.6 | 0.038  | 3.8 |
| inbut-000       | Application | 0.072 | 3.8 | 0.075  | 4.2 |
| incode-000      | Mugshot     | 0.010 | 1.9 | 0.048  | 3.1 |
| incode-000      | Border      | 0.061 | 3.1 | 0.138  | 4.8 |
| incode-000      | Application | 0.008 | 2.0 | 0.027  | 2.9 |
| incode-001      | Mugshot     | 0.009 | 1.6 | 0.055  | 3.4 |
| incode-001      | Border      | 0.052 | 3.0 | 0.110  | 4.3 |
| incode-001      | Application | 0.011 | 1.9 | 0.038  | 3.1 |
| innovatrics-000 | Mugshot     | 0.005 | 1.8 | 0.011  | 2.9 |
| innovatrics-000 | Border      | 0.013 | 2.6 | 0.017  | 3.0 |
| innovatrics-000 | Application | 0.008 | 2.4 | 0.015  | 3.1 |
| innovatrics-001 | Mugshot     | 0.004 | 1.8 | 0.010  | 2.8 |
| innovatrics-001 | Border      | 0.010 | 2.4 | 0.014  | 2.7 |
| innovatrics-001 | Application | 0.007 | 2.3 | 0.013  | 3.0 |
| innovatrics-002 | Mugshot     | 0.006 | 1.7 | 0.018  | 2.7 |
| innovatrics-002 | Border      | 0.011 | 1.9 | 0.013  | 2.3 |

**Table 15.** False positive rate (FPR) and mean absolute error (MAE) for persons 14-20 estimated at 28 or more. Border photos have inferior quality to the mugshot and application photos. The mugshots are from people 18-20. Lower values are better. (continued)

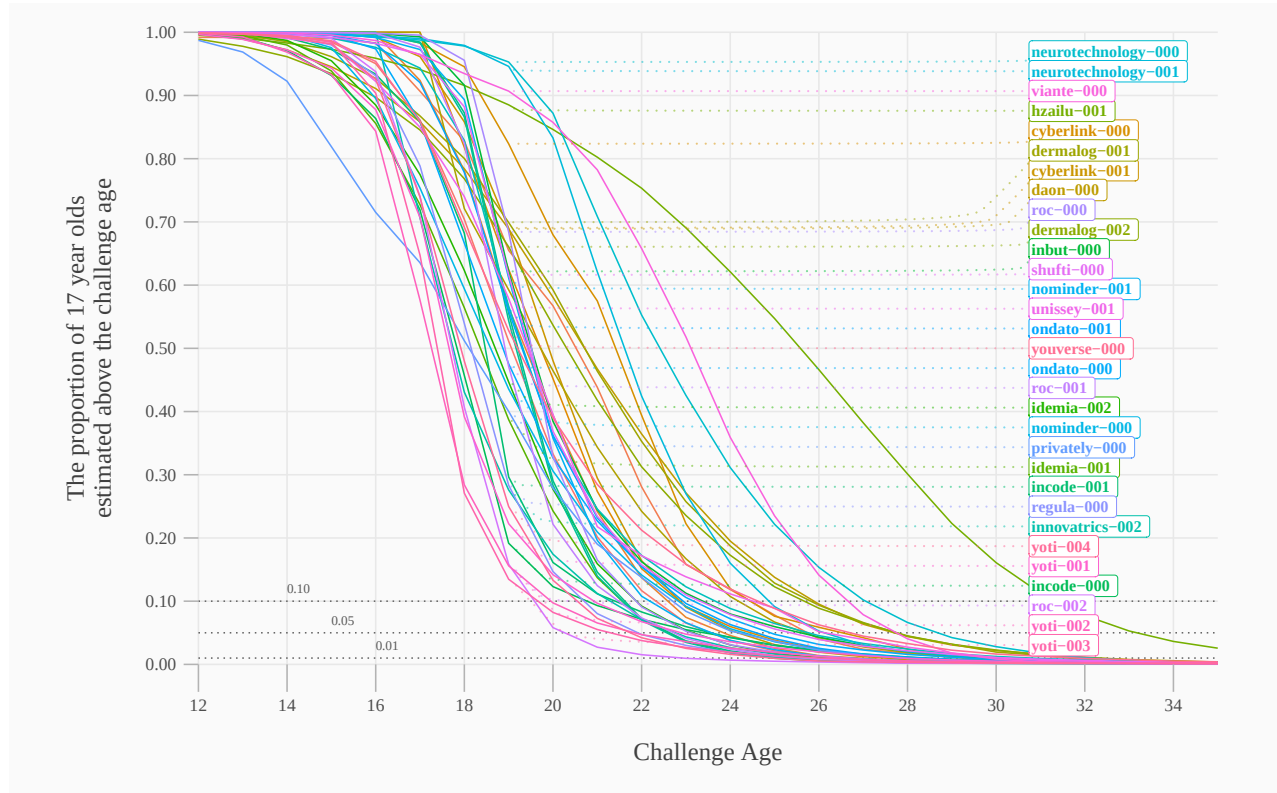
| Algorithm           | Data        | Male  |     | Female |     |
|---------------------|-------------|-------|-----|--------|-----|
|                     |             | FPR   | MAE | FPR    | MAE |
| innovatrics-002     | Application | 0.010 | 2.1 | 0.020  | 2.8 |
| jumio-000           | Mugshot     | 0.014 | 2.2 | 0.109  | 4.6 |
| jumio-000           | Border      | 0.156 | 6.0 | 0.298  | 8.1 |
| jumio-000           | Application | 0.019 | 2.7 | 0.067  | 4.4 |
| neurotechnology-000 | Mugshot     | 0.083 | 4.3 | 0.106  | 4.7 |
| neurotechnology-000 | Border      | 0.070 | 4.9 | 0.047  | 4.0 |
| neurotechnology-000 | Application | 0.101 | 5.6 | 0.057  | 4.6 |
| neurotechnology-001 | Mugshot     | 0.036 | 3.6 | 0.049  | 3.8 |
| neurotechnology-001 | Border      | 0.043 | 4.6 | 0.034  | 3.8 |
| neurotechnology-001 | Application | 0.040 | 4.9 | 0.027  | 4.1 |
| nominder-000        | Mugshot     | 0.019 | 2.5 | 0.038  | 3.2 |
| nominder-000        | Border      | 0.036 | 4.2 | 0.041  | 3.7 |
| nominder-000        | Application | 0.025 | 3.0 | 0.031  | 3.5 |
| nominder-001        | Mugshot     | 0.008 | 2.1 | 0.024  | 3.3 |
| nominder-001        | Border      | 0.017 | 3.4 | 0.020  | 3.1 |
| nominder-001        | Application | 0.009 | 2.6 | 0.015  | 3.3 |
| ondato-000          | Mugshot     | 0.018 | 2.4 | 0.054  | 3.6 |
| ondato-000          | Border      | 0.031 | 3.4 | 0.023  | 3.3 |
| ondato-000          | Application | 0.022 | 2.9 | 0.040  | 4.1 |
| ondato-001          | Mugshot     | 0.013 | 2.3 | 0.050  | 3.7 |
| ondato-001          | Border      | 0.023 | 3.2 | 0.021  | 3.2 |
| ondato-001          | Application | 0.012 | 2.8 | 0.030  | 4.0 |
| privately-000       | Mugshot     | 0.011 | 2.6 | 0.035  | 3.7 |
| privately-000       | Border      | 0.056 | 4.0 | 0.062  | 4.5 |
| privately-000       | Application | 0.008 | 2.6 | 0.029  | 3.7 |
| regula-000          | Mugshot     | 0.004 | 1.4 | 0.020  | 2.6 |
| regula-000          | Border      | 0.010 | 1.9 | 0.032  | 2.8 |
| regula-000          | Application | 0.005 | 1.6 | 0.012  | 2.3 |
| roc-000             | Mugshot     | 0.010 | 2.0 | 0.026  | 2.7 |
| roc-000             | Border      | 0.058 | 4.4 | 0.093  | 4.7 |
| roc-000             | Application | 0.030 | 3.8 | 0.049  | 4.1 |
| roc-001             | Mugshot     | 0.007 | 1.9 | 0.014  | 2.5 |
| roc-001             | Border      | 0.025 | 3.7 | 0.046  | 4.1 |
| roc-001             | Application | 0.015 | 3.3 | 0.030  | 4.0 |
| roc-002             | Mugshot     | 0.003 | 1.2 | 0.004  | 1.6 |
| roc-002             | Border      | 0.023 | 3.1 | 0.039  | 3.4 |
| roc-002             | Application | 0.008 | 2.1 | 0.014  | 2.6 |
| shufti-000          | Mugshot     | 0.019 | 2.4 | 0.099  | 4.2 |
| shufti-000          | Border      | 0.063 | 3.6 | 0.078  | 4.2 |
| shufti-000          | Application | 0.044 | 3.2 | 0.066  | 3.9 |
| unissey-001         | Mugshot     | 0.064 | 2.9 | 0.082  | 3.2 |
| unissey-001         | Border      | 0.050 | 3.1 | 0.060  | 3.4 |
| unissey-001         | Application | 0.091 | 3.7 | 0.092  | 3.8 |
| viant-000           | Mugshot     | 0.098 | 5.1 | 0.116  | 5.7 |
| viant-000           | Border      | 0.099 | 4.2 | 0.126  | 4.9 |
| viant-000           | Application | 0.111 | 5.3 | 0.123  | 6.0 |
| yoti-001            | Mugshot     | 0.007 | 1.8 | 0.057  | 3.2 |
| yoti-001            | Border      | 0.081 | 4.0 | 0.168  | 5.4 |
| yoti-001            | Application | 0.017 | 2.3 | 0.093  | 4.1 |
| yoti-002            | Mugshot     | 0.005 | 1.8 | 0.026  | 2.7 |
| yoti-002            | Border      | 0.053 | 3.4 | 0.129  | 5.1 |
| yoti-002            | Application | 0.010 | 2.2 | 0.046  | 3.5 |
| yoti-003            | Mugshot     | 0.005 | 1.5 | 0.025  | 2.4 |
| yoti-003            | Border      | 0.052 | 3.5 | 0.121  | 5.1 |
| yoti-003            | Application | 0.011 | 2.0 | 0.046  | 3.4 |
| yoti-004            | Mugshot     | 0.004 | 1.5 | 0.011  | 2.3 |

**Table 15.** False positive rate (FPR) and mean absolute error (MAE) for persons 14-20 estimated at 28 or more. Border photos have inferior quality to the mugshot and application photos. The mugshots are from people 18-20. Lower values are better. (continued)

| Algorithm    | Data        | Male  |     | Female |     |
|--------------|-------------|-------|-----|--------|-----|
|              |             | FPR   | MAE | FPR    | MAE |
| yoti-004     | Border      | 0.033 | 3.3 | 0.080  | 4.8 |
| yoti-004     | Application | 0.006 | 1.9 | 0.017  | 3.3 |
| youverse-000 | Mugshot     | 0.051 | 2.9 | 0.071  | 3.8 |
| youverse-000 | Border      | 0.181 | 5.6 | 0.083  | 4.3 |
| youverse-000 | Application | 0.030 | 2.7 | 0.031  | 3.1 |



**Fig. 9.** The proportion of 17 year olds whose age is estimated as at or above the given challenge age. The Application images are of subjects from the set of countries used in the demographics analysis.



**Fig. 10.** The proportion of 17 year olds whose age is estimated as at or above the given challenge age. The Mugshot images are of subjects from the set of countries used in the demographics analysis.

The seven-year buffer has been adopted operationally ( $L = 18$ ,  $T = 25$ ). It would usually be useful, if FPR objectives can be met, to reduce the  $T$  value to give improved false negative rates. We consider this in two ways. First in Figures 9 and 10 we plot for 17 year olds FPR against challenge age,  $T$ . The curves vary by algorithm, and decline monotonically with challenge age  $T$ . We also show, in Table 16, the  $T$  value needed to achieve three particular FPR objectives, 10%, 5% and 1% for 17 year olds. The results improve for younger persons. The *Application* images are from a global population so the results are dependent on the underlying demographic-specific FPR values documented in the next section. The *Mugshot* images are from a domestic population, 72% of which are men.

While a rough model of this change is shift invariant  $FPR(A, T) \approx FPR(A+k, T+k)$  meaning we'd expect similar FPR for Age-18-Challenge-25 as we would for Age-21-Challenge-28, it only holds approximately so empirical measurements should be preferred.

**Table 16.** For *Application* images, challenge ages need to attain FPR below that specified, by actual age. Lower values are better

| Algorithm     | FPR = 0.10 |    |    |    |    |    | FPR = 0.05 |    |    |    |    |    | FPR = 0.01 |    |    |    |    |    |
|---------------|------------|----|----|----|----|----|------------|----|----|----|----|----|------------|----|----|----|----|----|
|               | 15         | 16 | 17 | 18 | 19 | 20 | 15         | 16 | 17 | 18 | 19 | 20 | 15         | 16 | 17 | 18 | 19 | 20 |
| cognitec-000  | 20         | 21 | 23 | 23 | 25 | 26 | 21         | 23 | 24 | 25 | 26 | 27 | 24         | 26 | 27 | 28 | 29 | 31 |
| cvut-001      | 23         | 25 | 26 | 26 | 27 | 28 | 26         | 26 | 27 | 28 | 29 | 30 | 28         | 30 | 31 | 31 | 33 | 34 |
| cvut-002      | 22         | 22 | 23 | 24 | 26 | 26 | 22         | 24 | 25 | 26 | 27 | 28 | 26         | 27 | 28 | 29 | 30 | 32 |
| cvut-003      | 21         | 22 | 23 | 24 | 25 | 26 | 22         | 23 | 25 | 25 | 27 | 28 | 26         | 27 | 28 | 28 | 31 | 32 |
| cyberlink-000 | 24         | 25 | 26 | 27 | 28 | 29 | 25         | 26 | 28 | 28 | 29 | 30 | 28         | 29 | 31 | 30 | 32 | 33 |
| cyberlink-001 | 22         | 23 | 24 | 25 | 26 | 27 | 23         | 24 | 25 | 26 | 27 | 28 | 25         | 27 | 28 | 29 | 30 | 31 |

**Table 16.** For Application images, challenge ages need to attain FPR below that specified, by actual age. Lower values are better  
(continued)

| Algorithm           | FPR = 0.10 |    |    |    |    |    | FPR = 0.05 |    |    |    |    |    | FPR = 0.01 |    |    |    |    |    |
|---------------------|------------|----|----|----|----|----|------------|----|----|----|----|----|------------|----|----|----|----|----|
|                     | 15         | 16 | 17 | 18 | 19 | 20 | 15         | 16 | 17 | 18 | 19 | 20 | 15         | 16 | 17 | 18 | 19 | 20 |
| daon-000            | 24         | 25 | 26 | 27 | 28 | 29 | 26         | 27 | 28 | 29 | 30 | 31 | 30         | 31 | 33 | 33 | 34 | 36 |
| daon-001            | 22         | 24 | 25 | 26 | 27 | 28 | 24         | 25 | 27 | 27 | 28 | 30 | 27         | 29 | 30 | 31 | 32 | 34 |
| dermalog-001        | 25         | 25 | 26 | 27 | 28 | 29 | 26         | 27 | 28 | 28 | 30 | 30 | 30         | 31 | 32 | 32 | 33 | 34 |
| dermalog-002        | 25         | 26 | 27 | 27 | 28 | 29 | 27         | 27 | 28 | 29 | 30 | 30 | 30         | 30 | 32 | 32 | 33 | 34 |
| hzailu-001          | 29         | 30 | 31 | 31 | 32 | 33 | 30         | 31 | 32 | 33 | 34 | 34 | 35         | 35 | 36 | NA | NA | NA |
| idemia-001          | 20         | 22 | 23 | 24 | 25 | 26 | 21         | 23 | 24 | 25 | 26 | 27 | 24         | 25 | 27 | 28 | 29 | 31 |
| idemia-002          | 19         | 21 | 22 | 23 | 24 | 25 | 20         | 22 | 23 | 24 | 25 | 27 | 23         | 25 | 26 | 27 | 29 | 31 |
| inbut-000           | 25         | 26 | 27 | 27 | 28 | 29 | 28         | 29 | 30 | 30 | 31 | 31 | 34         | 35 | 35 | 34 | 36 | NA |
| incode-000          | 19         | 21 | 23 | 24 | 25 | 27 | 21         | 23 | 25 | 26 | 27 | 28 | 25         | 28 | 29 | 29 | 31 | 32 |
| incode-001          | 20         | 22 | 24 | 25 | 26 | 27 | 22         | 24 | 25 | 26 | 28 | 29 | 25         | 28 | 29 | 30 | 32 | 33 |
| innovatrics-000     | 22         | 23 | 24 | 24 | 26 | 26 | 23         | 24 | 25 | 26 | 27 | 28 | 25         | 27 | 28 | 28 | 29 | 31 |
| innovatrics-001     | 21         | 22 | 24 | 24 | 25 | 26 | 23         | 24 | 25 | 25 | 26 | 27 | 25         | 26 | 27 | 28 | 29 | 30 |
| innovatrics-002     | 21         | 22 | 24 | 25 | 26 | 27 | 23         | 24 | 25 | 26 | 27 | 28 | 25         | 27 | 28 | 29 | 30 | 32 |
| jumio-000           | 23         | 24 | 25 | 26 | 27 | 28 | 25         | 26 | 27 | 28 | 29 | 30 | 29         | 30 | 32 | 32 | 33 | 34 |
| neurotechnology-000 | 27         | 27 | 28 | 28 | 29 | 29 | 28         | 29 | 30 | 30 | 30 | 31 | 32         | 32 | 33 | 33 | 33 | 34 |
| neurotechnology-001 | 26         | 26 | 27 | 27 | 27 | 28 | 27         | 27 | 27 | 28 | 28 | 29 | 29         | 30 | 31 | 31 | 32 | 34 |
| nominder-000        | 23         | 24 | 25 | 25 | 26 | 28 | 24         | 25 | 26 | 27 | 28 | 29 | 27         | 28 | 30 | 30 | 32 | 33 |
| nominder-001        | 22         | 23 | 24 | 24 | 25 | 26 | 23         | 24 | 25 | 25 | 26 | 27 | 25         | 26 | 28 | 28 | 30 | 31 |
| ondato-000          | 23         | 24 | 25 | 26 | 27 | 28 | 25         | 26 | 27 | 27 | 28 | 29 | 28         | 29 | 30 | 31 | 32 | 34 |
| ondato-001          | 23         | 24 | 25 | 25 | 26 | 27 | 24         | 25 | 26 | 27 | 28 | 29 | 27         | 28 | 29 | 29 | 31 | 32 |
| privately-000       | 23         | 24 | 24 | 25 | 26 | 27 | 24         | 25 | 26 | 27 | 27 | 28 | 27         | 29 | 29 | 29 | 30 | 31 |
| regula-000          | 20         | 21 | 22 | 23 | 24 | 25 | 21         | 22 | 23 | 24 | 26 | 27 | 24         | 25 | 26 | 28 | 29 | 30 |
| roc-000             | 24         | 25 | 26 | 26 | 27 | 28 | 25         | 26 | 27 | 28 | 29 | 30 | 28         | 29 | 30 | 31 | 32 | 33 |
| roc-001             | 22         | 24 | 25 | 26 | 27 | 28 | 24         | 25 | 26 | 27 | 28 | 29 | 26         | 27 | 28 | 29 | 31 | 33 |
| roc-002             | 21         | 22 | 23 | 24 | 25 | 26 | 22         | 23 | 24 | 25 | 26 | 28 | 24         | 26 | 27 | 28 | 29 | 31 |
| shufti-000          | 23         | 24 | 25 | 26 | 28 | 29 | 24         | 26 | 27 | 28 | 30 | 31 | 29         | 30 | 32 | 33 | 35 | 36 |
| unissey-001         | 25         | 26 | 27 | 28 | 29 | 30 | 26         | 28 | 29 | 30 | 30 | 31 | 30         | 31 | 32 | 32 | 33 | 35 |
| viant-000           | 26         | 27 | 28 | 29 | 29 | 30 | 27         | 28 | 29 | 30 | 31 | 31 | 30         | 30 | 31 | 32 | 33 | 34 |
| yoti-001            | 21         | 24 | 26 | 27 | 28 | 29 | 24         | 26 | 28 | 28 | 30 | 31 | 29         | 30 | 32 | 33 | 35 | 36 |
| yoti-002            | 21         | 23 | 25 | 26 | 27 | 28 | 23         | 25 | 26 | 27 | 28 | 29 | 27         | 29 | 30 | 30 | 32 | 33 |
| yoti-003            | 21         | 23 | 24 | 25 | 27 | 28 | 23         | 25 | 26 | 27 | 28 | 29 | 27         | 28 | 30 | 30 | 32 | 33 |
| yoti-004            | 21         | 23 | 24 | 25 | 26 | 26 | 23         | 24 | 25 | 26 | 27 | 28 | 25         | 26 | 28 | 28 | 30 | 31 |
| youverse-000        | 22         | 23 | 25 | 25 | 26 | 27 | 24         | 25 | 26 | 27 | 28 | 29 | 28         | 29 | 30 | 31 | 32 | 33 |

### 5.3.1. Age Verification vs. Age Estimation

Our API includes an age estimation function returning a continuous valued age *and* an age verification function returning a boolean value indicating whether the subject is above an age limit. This latter function is included to allow developers to build and submit a dedicated classifier for particular age thresholds.

Table 17 shows that five developers implemented the function by simply calling the general purpose age estimation function and return TRUE if the age is at or above the input threshold. One developer, Unissey, implemented the age verification function returning true if the age is above (not at-or-above) the challenge age; this matters because their AE algorithm only returns integer age estimates. So that the FPR value shown for their AV function is actually that for challenge-26 not challenge-25.

In conclusion, we do not have any evidence (yet) that an age-verification classifier can outperform a regression-like estimator on the same task.

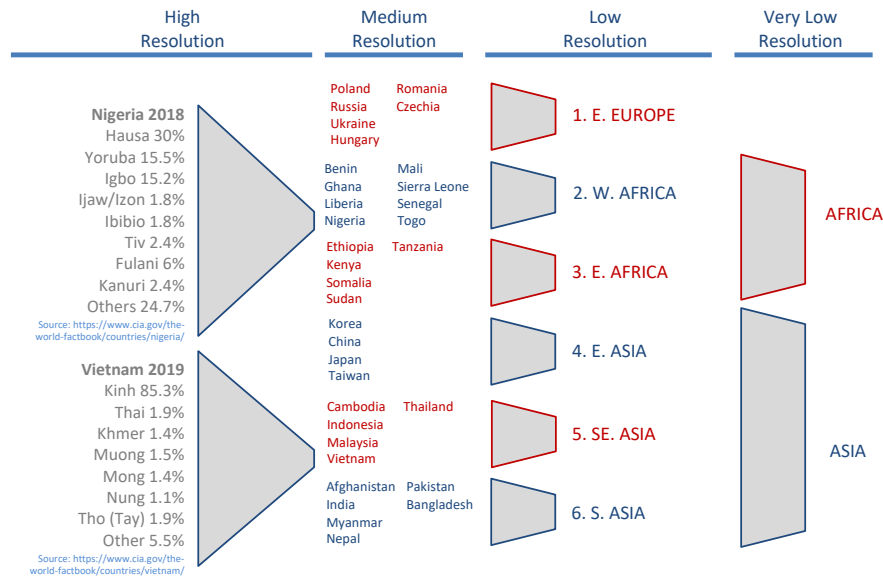
**Table 17.** Compare false positive rate (FPR) for AE and AV for persons 14-17 estimated at 25 or more. Border photos have inferior quality to the application photos. Lower values are better.

| Algorithm           | Data        | AE-FPR |        | AV-FPR |        |
|---------------------|-------------|--------|--------|--------|--------|
|                     |             | Male   | Female | Male   | Female |
| cvut-001            | Border      | 0.043  | 0.117  | 0.043  | 0.117  |
| cvut-001            | Application | 0.039  | 0.100  | 0.038  | 0.099  |
| cvut-002            | Border      | 0.077  | 0.131  | 0.077  | 0.131  |
| cvut-002            | Application | 0.011  | 0.032  | 0.011  | 0.032  |
| cyberlink-000       | Border      | 0.081  | 0.106  | 0.081  | 0.106  |
| cyberlink-000       | Application | 0.054  | 0.091  | 0.054  | 0.091  |
| dermalog-001        | Border      | 0.166  | 0.283  | 0.166  | 0.283  |
| dermalog-001        | Application | 0.051  | 0.112  | 0.051  | 0.112  |
| dermalog-002        | Border      | 0.190  | 0.344  | 0.190  | 0.344  |
| dermalog-002        | Application | 0.034  | 0.167  | 0.034  | 0.167  |
| hzailu-001          | Border      | 0.423  | 0.393  | 0.323  | 0.305  |
| hzailu-001          | Application | 0.341  | 0.361  | 0.259  | 0.282  |
| idemia-001          | Border      | 0.008  | 0.015  | 0.008  | 0.015  |
| idemia-001          | Application | 0.004  | 0.012  | 0.004  | 0.012  |
| inbut-000           | Border      | 0.062  | 0.090  | 0.062  | 0.090  |
| inbut-000           | Application | 0.100  | 0.114  | 0.100  | 0.113  |
| incode-000          | Border      | 0.065  | 0.177  | 0.065  | 0.177  |
| incode-000          | Application | 0.006  | 0.033  | 0.006  | 0.033  |
| incode-001          | Border      | 0.046  | 0.134  | 0.046  | 0.133  |
| incode-001          | Application | 0.006  | 0.034  | 0.006  | 0.034  |
| innovatrics-000     | Border      | 0.017  | 0.039  | 0.016  | 0.039  |
| innovatrics-000     | Application | 0.007  | 0.030  | 0.007  | 0.029  |
| jumio-000           | Border      | 0.229  | 0.429  | 0.229  | 0.428  |
| jumio-000           | Application | 0.021  | 0.088  | 0.021  | 0.088  |
| neurotechnology-000 | Border      | 0.182  | 0.128  | 0.182  | 0.128  |
| neurotechnology-000 | Application | 0.248  | 0.178  | 0.248  | 0.178  |
| neurotechnology-001 | Border      | 0.148  | 0.103  | 0.148  | 0.103  |
| neurotechnology-001 | Application | 0.157  | 0.125  | 0.157  | 0.125  |
| nominder-000        | Border      | 0.075  | 0.074  | 0.074  | 0.074  |
| nominder-000        | Application | 0.020  | 0.041  | 0.019  | 0.041  |
| nominder-001        | Border      | 0.026  | 0.036  | 0.026  | 0.036  |
| nominder-001        | Application | 0.007  | 0.021  | 0.007  | 0.021  |
| ondato-000          | Border      | 0.056  | 0.060  | 0.055  | 0.060  |
| ondato-000          | Application | 0.027  | 0.073  | 0.027  | 0.073  |
| ondato-001          | Border      | 0.040  | 0.046  | 0.040  | 0.046  |
| ondato-001          | Application | 0.014  | 0.051  | 0.014  | 0.051  |
| roc-000             | Border      | 0.089  | 0.144  | 0.089  | 0.144  |
| roc-000             | Application | 0.041  | 0.073  | 0.041  | 0.073  |
| roc-001             | Border      | 0.029  | 0.062  | 0.029  | 0.062  |
| roc-001             | Application | 0.013  | 0.040  | 0.013  | 0.039  |
| unissey-001         | Border      | 0.080  | 0.106  | 0.044  | 0.060  |
| unissey-001         | Application | 0.107  | 0.127  | 0.061  | 0.074  |
| viente-000          | Border      | 0.113  | 0.216  | 0.080  | 0.160  |
| viente-000          | Application | 0.138  | 0.253  | 0.083  | 0.163  |
| yoti-001            | Border      | 0.103  | 0.196  | 0.103  | 0.196  |
| yoti-001            | Application | 0.012  | 0.090  | 0.012  | 0.089  |
| yoti-002            | Border      | 0.067  | 0.196  | 0.067  | 0.196  |
| yoti-002            | Application | 0.009  | 0.063  | 0.009  | 0.063  |

#### 5.4. Demographic Analysis

The demographic analysis of race in this report is enabled by using country-of-birth as a proxy for ethnicity. We employ only those countries for which birth is a more or less reliable indication of





**Fig. 11.** Countries and regions used in quantifying demographic dependence on race. The fine-grained or local ethnicities shown at left are not available to us. The rightmost grouping is possible but not useful.

ethnicity as discussed below. To increase sample sizes, and reduce the verbosity and complexity of our results tables, we group countries into regions as follows:

- ▷ **East Africa:** Ethiopia, Kenya, Somalia, Sudan, Tanzania
- ▷ **West Africa:** Nigeria, Liberia, Sierra Leone, Benin, Ghana, Mali, Senegal, Togo
- ▷ **East Europe:** Poland, Ukraine, Russia, Hungary, Romania, Czechia
- ▷ **East Asia:** Korea, China, Japan, Taiwan
- ▷ **South East Asia:** Cambodia, Indonesia, Malaysia, Thailand, Vietnam
- ▷ **South Asia:** Afghanistan, India, Myanmar, Nepal, Pakistan, Bangladesh

Our use of proxy is imperfect in two ways. First, it ignores local ethnic variations: As discussed in Figure 11 our country labels are a lower “resolution” indication of ethnicity than the local ethnicities shown for Nigeria and Vietnam as examples. Our metadata does not include local information, only country-of-birth. Our region labels are, in-turn, an even lower resolution indication of ethnicity. If we find that age estimation accuracy shows region-of-birth dependence, then it would suggest country-level analysis is warranted. Second, some part of a population will have trans-national ancestry. This is probably not problematic if that proportion is low relative to inter-country variability in AEV accuracy. For this reason, we ignore countries where there has been considerable transcontinental immigration, such as the United Kingdom, France, and the United States.

Country-of-birth information has obvious operational relevance to deployment. But a higher goal would be to understand the effect of phenotypes which are a large set of features evident in a face such as high cheek bones, aquiline nose, prominent eye brows, and skin tone. Skin tone, which is just one of those phenotypes, is sometimes considered synonymous with race, has been shown to be



related to recognition false negative rates due to photography difficulties. However, for many face recognition algorithms, false positive rates vary with country-of-birth independent of skin tone e.g., they are higher in West Africa *and* China. This is considered to be an effect of under-representation of certain ethnicities in the algorithm training data.

#### 5.4.1. Fitzpatrick Skin Type

We do not use the Fitzpatrick skin type (FST)[13]. It was designed as dermatological statement of photosensitivity (sun-burn) but in recent years has been appropriated by AI researchers as a ready means of quantifying racial sensitivity of algorithms. This is far from sufficient and problematic for three reasons. First, Fitzpatrick types are not an accurate measure of skin pigmentation[14]. Second, FST was originally to be determined by a dermatological examination - recovery from a digital image is imprecise because the spectrum of the illuminant is usually unknown, and the settings of the camera and display device matter[15]. Finally, and most importantly, skin tone is but one phenotype accessible by AI algorithms extracting information from photographs of skin [16]. Deep neural networks for age estimation and other operations should be expected to find information that minimizes the cost function specified by the staff training the algorithm. A researcher would want to know whether hard tissue (e.g. high cheek bones, square jaws, aquiline noses) and soft tissue phenotypes (lips, eyelids, freckles) impact the cost function. For a summary of the issues see this [presentation](#) [17], and note slide 14 which shows intra-person variability of skin color in photographs.

We did not attempt to recover FST from our test photographs.

**Table 18.** For Application images, Challenge-25 FPR, by region of birth and sex. The final columns indicate the groups that give the highest and lowest FPR and the Gini summary of variability. Gini is computed over the 12 FPR values to its left; lower values are better.

| Algorithm       | Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----------------|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| cognitec-000    | 16  | 0.05 ± 0.02   | 0.008 ± 0.005 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.01 ± 0.01   | 0.06 ± 0.03   | 0.02 ± 0.01   | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.01 ± 0.01   | W Africa F    | E Europe F    | 0.66 |
| cognitec-000    | 17  | 0.08 ± 0.03   | 0.015 ± 0.006 | 0.006 ± 0.006 | 0.03 ± 0.01   | 0.02 ± 0.01   | 0.09 ± 0.03   | 0.006 ± 0.006 | 0.010 ± 0.004 | 0.000 ± 0.000 | 0.014 ± 0.007 | 0.02 ± 0.01   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.55 |
| cognitec-000    | 18  | 0.11 ± 0.03   | 0.027 ± 0.007 | 0.02 ± 0.01   | 0.030 ± 0.007 | 0.04 ± 0.01   | 0.14 ± 0.03   | 0.04 ± 0.01   | 0.018 ± 0.005 | 0.004 ± 0.004 | 0.029 ± 0.008 | 0.03 ± 0.01   | 0.05 ± 0.02   | W Africa F    | E Europe M    | 0.45 |
| cvut-001        | 16  | 0.38 ± 0.05   | 0.08 ± 0.01   | 0.03 ± 0.02   | 0.07 ± 0.02   | 0.06 ± 0.02   | 0.29 ± 0.05   | 0.11 ± 0.03   | 0.024 ± 0.007 | 0.009 ± 0.009 | 0.03 ± 0.01   | 0.01 ± 0.01   | 0.16 ± 0.05   | E Africa F    | E Europe M    | 0.58 |
| cvut-001        | 17  | 0.46 ± 0.05   | 0.12 ± 0.01   | 0.04 ± 0.02   | 0.11 ± 0.02   | 0.07 ± 0.02   | 0.34 ± 0.05   | 0.17 ± 0.04   | 0.05 ± 0.01   | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.05 ± 0.02   | 0.27 ± 0.05   | E Africa F    | E Europe M    | 0.50 |
| cvut-001        | 18  | 0.50 ± 0.04   | 0.13 ± 0.01   | 0.05 ± 0.02   | 0.14 ± 0.02   | 0.09 ± 0.02   | 0.37 ± 0.05   | 0.25 ± 0.03   | 0.056 ± 0.009 | 0.06 ± 0.02   | 0.10 ± 0.01   | 0.03 ± 0.01   | 0.28 ± 0.04   | E Africa F    | S E Asia M    | 0.47 |
| cvut-002        | 16  | 0.12 ± 0.03   | 0.015 ± 0.006 | 0.02 ± 0.01   | 0.019 ± 0.009 | 0.02 ± 0.01   | 0.14 ± 0.04   | 0.04 ± 0.02   | 0.003 ± 0.002 | 0.009 ± 0.009 | 0.013 ± 0.007 | 0.003 ± 0.003 | 0.03 ± 0.02   | W Africa F    | E Asia M      | 0.62 |
| cvut-002        | 17  | 0.17 ± 0.04   | 0.024 ± 0.007 | 0.03 ± 0.02   | 0.05 ± 0.01   | 0.04 ± 0.02   | 0.22 ± 0.04   | 0.05 ± 0.02   | 0.006 ± 0.003 | 0.008 ± 0.008 | 0.04 ± 0.01   | 0.01 ± 0.01   | 0.11 ± 0.03   | W Africa F    | E Asia M      | 0.55 |
| cvut-002        | 18  | 0.19 ± 0.03   | 0.028 ± 0.007 | 0.05 ± 0.02   | 0.08 ± 0.01   | 0.04 ± 0.01   | 0.25 ± 0.04   | 0.09 ± 0.02   | 0.014 ± 0.004 | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.008 ± 0.006 | 0.15 ± 0.04   | W Africa F    | S E Asia M    | 0.52 |
| cvut-003        | 16  | 0.10 ± 0.03   | 0.011 ± 0.005 | 0.02 ± 0.01   | 0.011 ± 0.008 | 0.02 ± 0.01   | 0.15 ± 0.04   | 0.04 ± 0.02   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.02 ± 0.01   | W Africa F    | S E Asia M    | 0.70 |
| cvut-003        | 17  | 0.15 ± 0.04   | 0.022 ± 0.007 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.02   | 0.20 ± 0.05   | 0.03 ± 0.02   | 0.004 ± 0.003 | 0.003 ± 0.003 | 0.029 ± 0.009 | 0.006 ± 0.006 | 0.08 ± 0.03   | W Africa F    | E Europe M    | 0.60 |
| cvut-003        | 18  | 0.14 ± 0.03   | 0.026 ± 0.006 | 0.03 ± 0.01   | 0.053 ± 0.010 | 0.04 ± 0.01   | 0.25 ± 0.04   | 0.10 ± 0.02   | 0.011 ± 0.004 | 0.015 ± 0.009 | 0.030 ± 0.008 | 0.006 ± 0.004 | 0.12 ± 0.03   | W Africa F    | S E Asia M    | 0.56 |
| cyberlink-000   | 16  | 0.18 ± 0.04   | 0.10 ± 0.02   | 0.05 ± 0.02   | 0.08 ± 0.02   | 0.10 ± 0.03   | 0.22 ± 0.05   | 0.14 ± 0.04   | 0.033 ± 0.009 | 0.01 ± 0.01   | 0.05 ± 0.02   | 0.08 ± 0.03   | 0.09 ± 0.04   | W Africa F    | E Europe M    | 0.36 |
| cyberlink-000   | 17  | 0.28 ± 0.04   | 0.14 ± 0.02   | 0.06 ± 0.02   | 0.12 ± 0.02   | 0.17 ± 0.04   | 0.28 ± 0.05   | 0.20 ± 0.04   | 0.09 ± 0.01   | 0.04 ± 0.02   | 0.12 ± 0.02   | 0.11 ± 0.03   | 0.19 ± 0.05   | W Africa F    | E Europe M    | 0.30 |
| cyberlink-000   | 18  | 0.26 ± 0.04   | 0.20 ± 0.02   | 0.11 ± 0.03   | 0.15 ± 0.02   | 0.22 ± 0.03   | 0.32 ± 0.04   | 0.31 ± 0.04   | 0.13 ± 0.01   | 0.06 ± 0.02   | 0.18 ± 0.02   | 0.21 ± 0.03   | 0.24 ± 0.04   | W Africa F    | E Europe M    | 0.23 |
| cyberlink-001   | 16  | 0.05 ± 0.02   | 0.021 ± 0.007 | 0.02 ± 0.01   | 0.02 ± 0.01   | 0.03 ± 0.02   | 0.08 ± 0.03   | 0.04 ± 0.02   | 0.010 ± 0.005 | 0.009 ± 0.009 | 0.015 ± 0.008 | 0.011 ± 0.008 | 0.02 ± 0.02   | W Africa F    | E Europe M    | 0.39 |
| cyberlink-001   | 17  | 0.10 ± 0.03   | 0.034 ± 0.009 | 0.03 ± 0.02   | 0.03 ± 0.01   | 0.05 ± 0.02   | 0.12 ± 0.03   | 0.04 ± 0.02   | 0.020 ± 0.006 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.02   | 0.05 ± 0.02   | W Africa F    | E Europe M    | 0.34 |
| cyberlink-001   | 18  | 0.12 ± 0.03   | 0.054 ± 0.009 | 0.04 ± 0.01   | 0.06 ± 0.01   | 0.06 ± 0.02   | 0.14 ± 0.03   | 0.09 ± 0.02   | 0.041 ± 0.008 | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.04 ± 0.01   | 0.09 ± 0.03   | W Africa F    | E Europe M    | 0.30 |
| daon-000        | 16  | 0.12 ± 0.03   | 0.08 ± 0.01   | 0.01 ± 0.01   | 0.06 ± 0.02   | 0.08 ± 0.03   | 0.16 ± 0.04   | 0.15 ± 0.04   | 0.12 ± 0.02   | 0.03 ± 0.01   | 0.12 ± 0.02   | 0.13 ± 0.03   | 0.07 ± 0.03   | W Africa F    | E Europe F    | 0.29 |
| daon-000        | 17  | 0.21 ± 0.04   | 0.10 ± 0.02   | 0.03 ± 0.02   | 0.09 ± 0.02   | 0.13 ± 0.03   | 0.25 ± 0.05   | 0.23 ± 0.04   | 0.17 ± 0.02   | 0.03 ± 0.01   | 0.19 ± 0.02   | 0.16 ± 0.04   | 0.13 ± 0.04   | W Africa F    | E Europe M    | 0.30 |
| daon-000        | 18  | 0.18 ± 0.03   | 0.11 ± 0.01   | 0.04 ± 0.01   | 0.09 ± 0.01   | 0.11 ± 0.02   | 0.27 ± 0.04   | 0.31 ± 0.04   | 0.19 ± 0.02   | 0.05 ± 0.02   | 0.26 ± 0.02   | 0.19 ± 0.03   | 0.23 ± 0.04   | E Africa M    | E Europe F    | 0.31 |
| daon-001        | 16  | 0.07 ± 0.02   | 0.03 ± 0.01   | 0.008 ± 0.008 | 0.023 ± 0.009 | 0.03 ± 0.02   | 0.11 ± 0.04   | 0.07 ± 0.02   | 0.007 ± 0.001 | 0.003 ± 0.003 | 0.04 ± 0.01   | 0.04 ± 0.02   | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.39 |
| daon-001        | 17  | 0.14 ± 0.04   | 0.06 ± 0.01   | 0.02 ± 0.01   | 0.04 ± 0.01   | 0.06 ± 0.02   | 0.18 ± 0.04   | 0.13 ± 0.04   | 0.10 ± 0.01   | 0.008 ± 0.008 | 0.09 ± 0.02   | 0.10 ± 0.03   | 0.11 ± 0.03   | W Africa F    | E Europe M    | 0.35 |
| daon-001        | 18  | 0.12 ± 0.03   | 0.08 ± 0.01   | 0.03 ± 0.01   | 0.05 ± 0.01   | 0.06 ± 0.02   | 0.23 ± 0.04   | 0.21 ± 0.03   | 0.12 ± 0.01   | 0.03 ± 0.01   | 0.17 ± 0.02   | 0.11 ± 0.02   | 0.15 ± 0.04   | W Africa F    | E Europe F    | 0.34 |
| dermalog-001    | 16  | 0.10 ± 0.03   | 0.10 ± 0.02   | 0.21 ± 0.04   | 0.16 ± 0.03   | 0.16 ± 0.04   | 0.06 ± 0.03   | 0.05 ± 0.02   | 0.05 ± 0.01   | 0.04 ± 0.02   | 0.10 ± 0.02   | 0.08 ± 0.03   | 0.04 ± 0.02   | E Europe F    | W Africa M    | 0.32 |
| dermalog-001    | 17  | 0.13 ± 0.03   | 0.14 ± 0.02   | 0.31 ± 0.05   | 0.24 ± 0.03   | 0.20 ± 0.04   | 0.07 ± 0.03   | 0.10 ± 0.03   | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.12 ± 0.02   | 0.10 ± 0.03   | 0.09 ± 0.03   | E Europe F    | E Europe M    | 0.33 |
| dermalog-001    | 18  | 0.13 ± 0.03   | 0.16 ± 0.02   | 0.27 ± 0.04   | 0.25 ± 0.02   | 0.20 ± 0.03   | 0.12 ± 0.03   | 0.13 ± 0.03   | 0.07 ± 0.01   | 0.05 ± 0.02   | 0.18 ± 0.02   | 0.10 ± 0.02   | 0.14 ± 0.03   | E Europe F    | E Europe M    | 0.26 |
| dermalog-002    | 16  | 0.07 ± 0.03   | 0.23 ± 0.02   | 0.24 ± 0.04   | 0.19 ± 0.03   | 0.21 ± 0.04   | 0.10 ± 0.04   | 0.04 ± 0.02   | 0.038 ± 0.009 | 0.04 ± 0.02   | 0.03 ± 0.01   | 0.04 ± 0.02   | 0.03 ± 0.02   | E Europe F    | W Africa M    | 0.45 |
| dermalog-002    | 17  | 0.11 ± 0.03   | 0.29 ± 0.02   | 0.31 ± 0.05   | 0.26 ± 0.03   | 0.30 ± 0.05   | 0.14 ± 0.03   | 0.07 ± 0.03   | 0.054 ± 0.010 | 0.06 ± 0.02   | 0.09 ± 0.02   | 0.08 ± 0.03   | 0.05 ± 0.02   | E Europe F    | W Africa M    | 0.39 |
| dermalog-002    | 18  | 0.13 ± 0.03   | 0.35 ± 0.02   | 0.39 ± 0.04   | 0.29 ± 0.02   | 0.32 ± 0.03   | 0.14 ± 0.03   | 0.12 ± 0.03   | 0.08 ± 0.01   | 0.11 ± 0.03   | 0.13 ± 0.02   | 0.11 ± 0.02   | 0.11 ± 0.03   | E Europe F    | E Asia M      | 0.33 |
| hzailu-001      | 16  | 0.33 ± 0.05   | 0.33 ± 0.02   | 0.56 ± 0.05   | 0.43 ± 0.03   | 0.43 ± 0.05   | 0.52 ± 0.06   | 0.51 ± 0.05   | 0.25 ± 0.02   | 0.38 ± 0.05   | 0.58 ± 0.03   | 0.46 ± 0.05   | 0.56 ± 0.06   | S Asia M      | E Asia M      | 0.14 |
| hzailu-001      | 17  | 0.39 ± 0.05   | 0.40 ± 0.02   | 0.66 ± 0.05   | 0.51 ± 0.03   | 0.48 ± 0.06   | 0.61 ± 0.05   | 0.55 ± 0.06   | 0.34 ± 0.02   | 0.44 ± 0.05   | 0.68 ± 0.03   | 0.53 ± 0.05   | 0.67 ± 0.05   | S Asia M      | E Asia M      | 0.13 |
| hzailu-001      | 18  | 0.46 ± 0.04   | 0.47 ± 0.02   | 0.73 ± 0.04   | 0.55 ± 0.02   | 0.51 ± 0.03   | 0.63 ± 0.04   | 0.70 ± 0.04   | 0.40 ± 0.02   | 0.55 ± 0.04   | 0.77 ± 0.02   | 0.57 ± 0.03   | 0.74 ± 0.04   | S Asia M      | E Asia M      | 0.12 |
| idemia-001      | 16  | 0.02 ± 0.01   | 0.007 ± 0.004 | 0.000 ± 0.000 | 0.004 ± 0.004 | 0.03 ± 0.02   | 0.05 ± 0.03   | 0.02 ± 0.01   | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.005 ± 0.005 | W Africa F    | E Europe F    | 0.67 |
| idemia-001      | 17  | 0.04 ± 0.02   | 0.023 ± 0.007 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.05 ± 0.02   | 0.09 ± 0.03   | 0.01 ± 0.01   | 0.007 ± 0.004 | 0.000 ± 0.000 | 0.009 ± 0.005 | 0.006 ± 0.006 | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.55 |
| idemia-001      | 18  | 0.08 ± 0.02   | 0.030 ± 0.007 | 0.014 ± 0.009 | 0.031 ± 0.009 | 0.04 ± 0.01   | 0.11 ± 0.03   | 0.04 ± 0.02   | 0.019 ± 0.005 | 0.000 ± 0.000 | 0.031 ± 0.008 | 0.02 ± 0.01   | 0.06 ± 0.02   | W Africa F    | E Europe M    | 0.42 |
| idemia-002      | 16  | 0.03 ± 0.02   | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.01 ± 0.01   | 0.03 ± 0.02   | 0.01 ± 0.01   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | W Africa F    | E Europe F    | 0.71 |
| idemia-002      | 17  | 0.03 ± 0.02   | 0.013 ± 0.005 | 0.03 ± 0.02   | 0.018 ± 0.008 | 0.02 ± 0.02   | 0.07 ± 0.02   | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.006 ± 0.004 | 0.006 ± 0.006 | 0.03 ± 0.02   | W Africa F    | E Africa M    | 0.55 |
| idemia-002      | 18  | 0.07 ± 0.02   | 0.015 ± 0.005 | 0.04 ± 0.01   | 0.025 ± 0.007 | 0.026 ± 0.009 | 0.09 ± 0.03   | 0.02 ± 0.01   | 0.012 ± 0.004 | 0.000 ± 0.000 | 0.012 ± 0.005 | 0.011 ± 0.007 | 0.03 ± 0.01   | W Africa F    | E Europe M    | 0.48 |
| inbut-000       | 16  | 0.26 ± 0.04   | 0.06 ± 0.01   | 0.19 ± 0.04   | 0.09 ± 0.02   | 0.09 ± 0.03   | 0.35 ± 0.06   | 0.23 ± 0.04   | 0.07 ± 0.01   | 0.12 ± 0.03   | 0.11 ± 0.02   | 0.12 ± 0.03   | 0.22 ± 0.05   | W Africa F    | E Asia F      | 0.33 |
| inbut-000       | 17  | 0.39 ± 0.05   | 0.08 ± 0.01   | 0.22 ± 0.04   | 0.16 ± 0.02   | 0.11 ± 0.03   | 0.40 ± 0.05   | 0.30 ± 0.05   | 0.10 ± 0.02   | 0.12 ± 0.03   | 0.13 ± 0.02   | 0.13 ± 0.03   | 0.27 ± 0.05   | W Africa F    | E Asia F      | 0.32 |
| inbut-000       | 18  | 0.40 ± 0.04   | 0.10 ± 0.01   | 0.23 ± 0.04   | 0.15 ± 0.02   | 0.11 ± 0.02   | 0.45 ± 0.05   | 0.27 ± 0.04   | 0.10 ± 0.01   | 0.15 ± 0.03   | 0.14 ± 0.02   | 0.08 ± 0.02   | 0.24 ± 0.04   | W Africa F    | S E Asia M    | 0.33 |
| incode-000      | 16  | 0.08 ± 0.02   | 0.026 ± 0.009 | 0.04 ± 0.02   | 0.03 ± 0.01   | 0.05 ± 0.02   | 0.09 ± 0.03   | 0.03 ± 0.02   | 0.002 ± 0.002 | 0.006 ± 0.006 | 0.004 ± 0.003 | 0.006 ± 0.006 | 0.01 ± 0.01   | W Africa F    | E Asia M      | 0.52 |
| incode-000      | 17  | 0.11 ± 0.03   | 0.05 ± 0.01   | 0.07 ± 0.03   | 0.06 ± 0.01   | 0.06 ± 0.02   | 0.11 ± 0.03   | 0.02 ± 0.01   | 0.006 ± 0.004 | 0.003 ± 0.003 | 0.017 ± 0.007 | 0.01 ± 0.01   | 0.05 ± 0.02   | E Africa F    | E Europe M    | 0.46 |
| incode-000      | 18  | 0.15 ± 0.03   | 0.07 ± 0.01   | 0.08 ± 0.02   | 0.08 ± 0.01   | 0.09 ± 0.02   | 0.13 ± 0.03   | 0.04 ± 0.01   | 0.013 ± 0.004 | 0.013 ± 0.009 | 0.023 ± 0.007 | 0.015 ± 0.008 | 0.06 ± 0.02   | E Africa F    | E Europe M    | 0.42 |
| incode-001      | 16  | 0.07 ± 0.02   | 0.023 ± 0.008 | 0.04 ± 0.02   | 0.03 ± 0.01   | 0.06 ± 0.02   | 0.08 ± 0.03   | 0.02 ± 0.01   | 0.005 ± 0.003 | 0.003 ± 0.003 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.005 | W Africa F    | S E Asia M    | 0.57 |
| incode-001      | 17  | 0.11 ± 0.03   | 0.07 ± 0.01   | 0.06 ± 0.03   | 0.06 ± 0.02   | 0.06 ± 0.02   | 0.16 ± 0.04   | 0.02 ± 0.01   | 0.011 ± 0.004 | 0.000 ± 0.000 | 0.011 ± 0.005 | 0.02 ± 0.01   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.49 |
| incode-001      | 18  | 0.13 ± 0.03   | 0.10 ± 0.01   | 0.08 ± 0.02   | 0.10 ± 0.01   | 0.15 ± 0.02   | 0.19 ± 0.03   | 0.04 ± 0.01   | 0.022 ± 0.007 | 0.009 ± 0.007 | 0.025 ± 0.007 | 0.03 ± 0.01   | 0.06 ± 0.02   | W Africa F    | E Europe M    | 0.43 |
| innovatrics-000 | 16  | 0.11 ± 0.03   | 0.015 ± 0.006 | 0.005 ± 0.005 | 0.015 ± 0.008 | 0.02 ± 0.01   | 0.12 ± 0.04   | 0.03 ± 0.02   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.02 ± 0.01   | W Africa F    | S E Asia M    | 0.70 |
| innovatrics-000 | 17  | 0.20 ± 0.04   | 0.020 ± 0.007 | 0.01 ± 0.01   | 0.07 ± 0.02   | 0.01 ± 0.01   | 0.22 ± 0.04   | 0.03 ± 0.01   | 0.006 ± 0.003 | 0.000 ± 0.000 | 0.020 ± 0.001 | 0.003 ± 0.003 | 0.07 ± 0.03   | W Africa F    | E Europe M    | 0.68 |
| innovatrics-000 | 18  | 0.23 ± 0.03   | 0.029 ± 0.007 | 0.02 ± 0.01   | 0.08 ± 0.01   | 0.027 ± 0.009 | 0.25 ± 0.04   | 0.07 ± 0.02   | 0.015 ± 0.005 | 0.011 ± 0.007 | 0.04 ± 0.01   | 0.010 ± 0.007 | 0.10 ± 0.03   | W Africa F    | S E Asia M    | 0.58 |
| innovatrics-001 | 16  | 0.09 ± 0.03   | 0.008 ± 0.004 | 0.005 ± 0.005 | 0.011 ± 0.006 | 0.005 ± 0.005 | 0.13 ± 0      |               |               |               |               |               |               |               |               |      |

|                     |    |             |               |               |               |             |             |               |               |               |               |               |               |              |            |      |
|---------------------|----|-------------|---------------|---------------|---------------|-------------|-------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|------------|------|
| innovatrics-002     | 17 | 0.22 ± 0.04 | 0.024 ± 0.008 | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.02 ± 0.01 | 0.31 ± 0.05 | 0.05 ± 0.02   | 0.008 ± 0.004 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.003 ± 0.003 | 0.09 ± 0.03   | W Africa F   | S E Asia M | 0.68 |
| innovatrics-002     | 18 | 0.26 ± 0.04 | 0.032 ± 0.008 | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.04 ± 0.01 | 0.33 ± 0.04 | 0.09 ± 0.02   | 0.018 ± 0.006 | 0.015 ± 0.009 | 0.044 ± 0.009 | 0.010 ± 0.007 | 0.13 ± 0.03   | W Africa F   | S E Asia M | 0.59 |
| jumio-000           | 16 | 0.18 ± 0.04 | 0.06 ± 0.01   | 0.11 ± 0.03   | 0.10 ± 0.02   | 0.12 ± 0.03 | 0.22 ± 0.05 | 0.04 ± 0.02   | 0.009 ± 0.005 | 0.08 ± 0.03   | 0.006 ± 0.004 | 0.011 ± 0.008 | 0.02 ± 0.01   | W Africa F   | S Asia M   | 0.50 |
| jumio-000           | 17 | 0.25 ± 0.04 | 0.10 ± 0.01   | 0.16 ± 0.04   | 0.16 ± 0.02   | 0.15 ± 0.04 | 0.29 ± 0.04 | 0.04 ± 0.02   | 0.019 ± 0.006 | 0.11 ± 0.03   | 0.017 ± 0.007 | 0.03 ± 0.01   | 0.06 ± 0.03   | W Africa F   | S Asia M   | 0.45 |
| jumio-000           | 18 | 0.28 ± 0.04 | 0.15 ± 0.01   | 0.18 ± 0.03   | 0.18 ± 0.02   | 0.20 ± 0.02 | 0.37 ± 0.04 | 0.07 ± 0.02   | 0.039 ± 0.007 | 0.13 ± 0.03   | 0.041 ± 0.009 | 0.03 ± 0.01   | 0.10 ± 0.03   | W Africa F   | S E Asia M | 0.40 |
| neurotechnology-000 | 16 | 0.31 ± 0.05 | 0.20 ± 0.02   | 0.08 ± 0.03   | 0.10 ± 0.02   | 0.16 ± 0.03 | 0.59 ± 0.06 | 0.66 ± 0.05   | 0.29 ± 0.02   | 0.03 ± 0.01   | 0.14 ± 0.02   | 0.19 ± 0.04   | 0.76 ± 0.06   | W Africa M   | E Europe M | 0.47 |
| neurotechnology-000 | 17 | 0.39 ± 0.05 | 0.24 ± 0.02   | 0.12 ± 0.03   | 0.13 ± 0.02   | 0.21 ± 0.04 | 0.64 ± 0.05 | 0.76 ± 0.04   | 0.36 ± 0.02   | 0.06 ± 0.02   | 0.20 ± 0.02   | 0.28 ± 0.04   | 0.84 ± 0.04   | W Africa M   | E Europe M | 0.41 |
| neurotechnology-000 | 18 | 0.40 ± 0.04 | 0.26 ± 0.02   | 0.13 ± 0.03   | 0.14 ± 0.02   | 0.23 ± 0.03 | 0.66 ± 0.05 | 0.77 ± 0.04   | 0.41 ± 0.02   | 0.10 ± 0.03   | 0.27 ± 0.02   | 0.34 ± 0.03   | 0.85 ± 0.03   | W Africa M   | E Europe M | 0.38 |
| neurotechnology-001 | 16 | 0.07 ± 0.02 | 0.21 ± 0.02   | 0.08 ± 0.03   | 0.10 ± 0.02   | 0.17 ± 0.04 | 0.06 ± 0.03 | 0.05 ± 0.02   | 0.30 ± 0.02   | 0.03 ± 0.02   | 0.12 ± 0.02   | 0.21 ± 0.04   | 0.03 ± 0.02   | E Asia M     | E Europe M | 0.41 |
| neurotechnology-001 | 17 | 0.08 ± 0.03 | 0.26 ± 0.02   | 0.14 ± 0.04   | 0.13 ± 0.02   | 0.23 ± 0.04 | 0.09 ± 0.03 | 0.12 ± 0.03   | 0.37 ± 0.02   | 0.06 ± 0.02   | 0.20 ± 0.02   | 0.26 ± 0.05   | 0.10 ± 0.03   | E Asia M     | E Europe M | 0.32 |
| neurotechnology-001 | 18 | 0.12 ± 0.03 | 0.26 ± 0.02   | 0.13 ± 0.03   | 0.14 ± 0.02   | 0.23 ± 0.03 | 0.11 ± 0.03 | 0.16 ± 0.03   | 0.41 ± 0.02   | 0.10 ± 0.02   | 0.25 ± 0.02   | 0.35 ± 0.04   | 0.15 ± 0.03   | E Asia M     | E Europe M | 0.28 |
| nominder-000        | 16 | 0.08 ± 0.03 | 0.028 ± 0.009 | 0.01 ± 0.01   | 0.03 ± 0.01   | 0.05 ± 0.02 | 0.18 ± 0.05 | 0.03 ± 0.02   | 0.026 ± 0.008 | 0.009 ± 0.009 | 0.013 ± 0.007 | 0.01 ± 0.01   | 0.02 ± 0.02   | W Africa F   | E Europe M | 0.52 |
| nominder-000        | 17 | 0.14 ± 0.04 | 0.06 ± 0.01   | 0.04 ± 0.02   | 0.04 ± 0.01   | 0.06 ± 0.02 | 0.27 ± 0.05 | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.008 ± 0.008 | 0.04 ± 0.01   | 0.03 ± 0.02   | 0.10 ± 0.03   | W Africa F   | E Europe M | 0.47 |
| nominder-000        | 18 | 0.17 ± 0.03 | 0.10 ± 0.01   | 0.05 ± 0.02   | 0.05 ± 0.01   | 0.07 ± 0.02 | 0.31 ± 0.04 | 0.09 ± 0.02   | 0.07 ± 0.01   | 0.03 ± 0.01   | 0.08 ± 0.01   | 0.06 ± 0.02   | 0.12 ± 0.03   | W Africa F   | E Europe M | 0.37 |
| nominder-001        | 16 | 0.04 ± 0.02 | 0.015 ± 0.006 | 0.02 ± 0.01   | 0.010 ± 0.006 | 0.03 ± 0.02 | 0.12 ± 0.04 | 0.01 ± 0.01   | 0.009 ± 0.004 | 0.003 ± 0.003 | 0.004 ± 0.003 | 0.008 ± 0.008 | 0.005 ± 0.005 | W Africa F   | E Europe M | 0.61 |
| nominder-001        | 17 | 0.08 ± 0.03 | 0.024 ± 0.007 | 0.03 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.01 | 0.18 ± 0.04 | 0.01 ± 0.01   | 0.016 ± 0.006 | 0.008 ± 0.008 | 0.016 ± 0.007 | 0.03 ± 0.01   | 0.04 ± 0.02   | W Africa F   | E Europe M | 0.51 |
| nominder-001        | 18 | 0.10 ± 0.02 | 0.033 ± 0.008 | 0.04 ± 0.02   | 0.033 ± 0.009 | 0.05 ± 0.01 | 0.24 ± 0.04 | 0.04 ± 0.02   | 0.039 ± 0.008 | 0.016 ± 0.009 | 0.031 ± 0.008 | 0.03 ± 0.01   | 0.07 ± 0.02   | W Africa F   | E Europe M | 0.43 |
| ondato-000          | 16 | 0.14 ± 0.03 | 0.06 ± 0.01   | 0.05 ± 0.02   | 0.05 ± 0.01   | 0.08 ± 0.03 | 0.23 ± 0.05 | 0.07 ± 0.02   | 0.026 ± 0.008 | 0.009 ± 0.009 | 0.017 ± 0.008 | 0.03 ± 0.02   | 0.05 ± 0.03   | W Africa F   | E Europe M | 0.47 |
| ondato-000          | 17 | 0.23 ± 0.04 | 0.11 ± 0.02   | 0.05 ± 0.02   | 0.10 ± 0.02   | 0.10 ± 0.03 | 0.30 ± 0.05 | 0.10 ± 0.03   | 0.04 ± 0.01   | 0.011 ± 0.008 | 0.05 ± 0.01   | 0.05 ± 0.02   | 0.16 ± 0.04   | W Africa F   | E Europe M | 0.43 |
| ondato-000          | 18 | 0.24 ± 0.03 | 0.14 ± 0.01   | 0.07 ± 0.02   | 0.12 ± 0.02   | 0.13 ± 0.02 | 0.34 ± 0.04 | 0.14 ± 0.03   | 0.06 ± 0.01   | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.06 ± 0.02   | 0.16 ± 0.04   | W Africa F   | E Europe M | 0.38 |
| ondato-001          | 16 | 0.10 ± 0.03 | 0.04 ± 0.01   | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.05 ± 0.02 | 0.16 ± 0.04 | 0.05 ± 0.02   | 0.014 ± 0.006 | 0.003 ± 0.003 | 0.007 ± 0.004 | 0.008 ± 0.008 | 0.04 ± 0.02   | W Africa F   | E Europe M | 0.50 |
| ondato-001          | 17 | 0.16 ± 0.04 | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.10 ± 0.02   | 0.11 ± 0.03 | 0.22 ± 0.04 | 0.06 ± 0.02   | 0.018 ± 0.006 | 0.006 ± 0.006 | 0.024 ± 0.009 | 0.04 ± 0.02   | 0.09 ± 0.03   | W Africa F   | E Europe M | 0.46 |
| ondato-001          | 18 | 0.20 ± 0.03 | 0.12 ± 0.01   | 0.07 ± 0.02   | 0.12 ± 0.02   | 0.14 ± 0.02 | 0.26 ± 0.04 | 0.11 ± 0.03   | 0.036 ± 0.008 | 0.013 ± 0.007 | 0.05 ± 0.01   | 0.04 ± 0.01   | 0.12 ± 0.03   | W Africa F   | E Europe M | 0.39 |
| privately-000       | 16 | 0.06 ± 0.02 | 0.11 ± 0.02   | 0.03 ± 0.02   | 0.06 ± 0.02   | 0.13 ± 0.03 | 0.08 ± 0.03 | 0.02 ± 0.01   | 0.026 ± 0.008 | 0.012 ± 0.009 | 0.006 ± 0.004 | 0.03 ± 0.01   | 0.005 ± 0.005 | S E Asia F   | W Africa M | 0.49 |
| privately-000       | 17 | 0.08 ± 0.03 | 0.14 ± 0.02   | 0.07 ± 0.03   | 0.08 ± 0.02   | 0.12 ± 0.03 | 0.11 ± 0.03 | 0.01 ± 0.01   | 0.037 ± 0.009 | 0.003 ± 0.003 | 0.015 ± 0.007 | 0.03 ± 0.01   | 0.03 ± 0.02   | E Asia F     | E Europe M | 0.46 |
| privately-000       | 18 | 0.10 ± 0.02 | 0.19 ± 0.02   | 0.08 ± 0.02   | 0.10 ± 0.01   | 0.18 ± 0.03 | 0.11 ± 0.03 | 0.05 ± 0.02   | 0.043 ± 0.008 | 0.007 ± 0.005 | 0.010 ± 0.004 | 0.03 ± 0.01   | 0.03 ± 0.02   | E Asia F     | E Europe M | 0.45 |
| regula-000          | 16 | 0.03 ± 0.02 | 0.003 ± 0.002 | 0.003 ± 0.003 | 0.004 ± 0.004 | 0.02 ± 0.01 | 0.05 ± 0.02 | 0.02 ± 0.01   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.009 ± 0.009 | W Africa F   | E Asia M   | 0.62 |
| regula-000          | 17 | 0.07 ± 0.02 | 0.013 ± 0.005 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.02 ± 0.01 | 0.09 ± 0.03 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.007 ± 0.005 | 0.011 ± 0.008 | 0.03 ± 0.02   | W Africa F   | E Europe M | 0.59 |
| regula-000          | 18 | 0.10 ± 0.03 | 0.015 ± 0.005 | 0.03 ± 0.01   | 0.034 ± 0.009 | 0.04 ± 0.01 | 0.14 ± 0.03 | 0.015 ± 0.009 | 0.009 ± 0.004 | 0.002 ± 0.002 | 0.014 ± 0.005 | 0.011 ± 0.007 | 0.06 ± 0.02   | W Africa F   | E Europe M | 0.56 |
| roc-000             | 16 | 0.11 ± 0.03 | 0.14 ± 0.02   | 0.005 ± 0.005 | 0.03 ± 0.01   | 0.11 ± 0.03 | 0.14 ± 0.04 | 0.05 ± 0.02   | 0.06 ± 0.01   | 0.003 ± 0.003 | 0.018 ± 0.008 | 0.05 ± 0.02   | 0.05 ± 0.03   | W Africa F   | E Europe M | 0.45 |
| roc-000             | 17 | 0.18 ± 0.04 | 0.17 ± 0.02   | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.13 ± 0.03 | 0.22 ± 0.04 | 0.07 ± 0.02   | 0.12 ± 0.01   | 0.000 ± 0.000 | 0.04 ± 0.01   | 0.11 ± 0.03   | 0.11 ± 0.03   | W Africa F   | E Europe M | 0.38 |
| roc-000             | 18 | 0.19 ± 0.03 | 0.22 ± 0.02   | 0.05 ± 0.02   | 0.07 ± 0.01   | 0.20 ± 0.03 | 0.26 ± 0.04 | 0.13 ± 0.03   | 0.17 ± 0.02   | 0.007 ± 0.005 | 0.07 ± 0.01   | 0.14 ± 0.03   | 0.15 ± 0.03   | W Africa F   | E Europe M | 0.33 |
| roc-001             | 16 | 0.09 ± 0.03 | 0.04 ± 0.01   | 0.000 ± 0.000 | 0.02 ± 0.01   | 0.05 ± 0.02 | 0.09 ± 0.03 | 0.03 ± 0.01   | 0.014 ± 0.005 | 0.003 ± 0.003 | 0.004 ± 0.003 | 0.01 ± 0.01   | 0.01 ± 0.01   | E Africa F   | E Europe F | 0.57 |
| roc-001             | 17 | 0.14 ± 0.03 | 0.10 ± 0.01   | 0.003 ± 0.003 | 0.07 ± 0.01   | 0.10 ± 0.03 | 0.18 ± 0.04 | 0.05 ± 0.02   | 0.042 ± 0.009 | 0.003 ± 0.003 | 0.019 ± 0.008 | 0.03 ± 0.02   | 0.08 ± 0.03   | W Africa F   | E Europe F | 0.47 |
| roc-001             | 18 | 0.19 ± 0.03 | 0.15 ± 0.02   | 0.02 ± 0.01   | 0.10 ± 0.01   | 0.14 ± 0.02 | 0.23 ± 0.04 | 0.09 ± 0.02   | 0.09 ± 0.01   | 0.002 ± 0.002 | 0.04 ± 0.01   | 0.06 ± 0.02   | 0.13 ± 0.03   | W Africa F   | E Europe M | 0.39 |
| roc-002             | 16 | 0.03 ± 0.02 | 0.019 ± 0.007 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.04 ± 0.02 | 0.04 ± 0.02 | 0.02 ± 0.01   | 0.004 ± 0.003 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.008 ± 0.008 | 0.005 ± 0.005 | W Africa F   | E Europe F | 0.57 |
| roc-002             | 17 | 0.05 ± 0.02 | 0.040 ± 0.009 | 0.008 ± 0.008 | 0.025 ± 0.009 | 0.06 ± 0.02 | 0.05 ± 0.02 | 0.012 ± 0.009 | 0.014 ± 0.005 | 0.000 ± 0.000 | 0.010 ± 0.006 | 0.02 ± 0.01   | 0.04 ± 0.02   | S E Asia F   | E Europe M | 0.42 |
| roc-002             | 18 | 0.08 ± 0.02 | 0.07 ± 0.01   | 0.007 ± 0.005 | 0.030 ± 0.008 | 0.07 ± 0.02 | 0.06 ± 0.02 | 0.03 ± 0.01   | 0.037 ± 0.007 | 0.002 ± 0.002 | 0.023 ± 0.006 | 0.022 ± 0.010 | 0.05 ± 0.02   | E Africa F   | E Europe M | 0.39 |
| shufti-000          | 16 | 0.13 ± 0.03 | 0.05 ± 0.01   | 0.11 ± 0.03   | 0.06 ± 0.02   | 0.07 ± 0.03 | 0.12 ± 0.04 | 0.10 ± 0.03   | 0.027 ± 0.007 | 0.02 ± 0.01   | 0.022 ± 0.009 | 0.02 ± 0.01   | 0.03 ± 0.02   | E Africa F   | E Europe M | 0.38 |
| shufti-000          | 17 | 0.21 ± 0.04 | 0.09 ± 0.01   | 0.14 ± 0.04   | 0.14 ± 0.02   | 0.11 ± 0.03 | 0.17 ± 0.04 | 0.11 ± 0.03   | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.06 ± 0.02   | 0.09 ± 0.03   | E Africa F   | E Europe M | 0.32 |
| shufti-000          | 18 | 0.24 ± 0.03 | 0.14 ± 0.01   | 0.17 ± 0.03   | 0.18 ± 0.02   | 0.15 ± 0.02 | 0.20 ± 0.04 | 0.16 ± 0.03   | 0.09 ± 0.01   | 0.03 ± 0.01   | 0.12 ± 0.01   | 0.09 ± 0.02   | 0.11 ± 0.03   | E Africa F   | E Europe M | 0.22 |
| unissey-001         | 16 | 0.12 ± 0.03 | 0.19 ± 0.02   | 0.10 ± 0.03   | 0.10 ± 0.02   | 0.16 ± 0.04 | 0.19 ± 0.05 | 0.11 ± 0.03   | 0.17 ± 0.02   | 0.05 ± 0.02   | 0.10 ± 0.02   | 0.17 ± 0.04   | 0.08 ± 0.03   | E Asia F     | E Europe M | 0.21 |
| unissey-001         | 17 | 0.17 ± 0.04 | 0.26 ± 0.02   | 0.19 ± 0.04   | 0.15 ± 0.03   | 0.21 ± 0.04 | 0.29 ± 0.05 | 0.20 ± 0.04   | 0.26 ± 0.02   | 0.04 ± 0.02   | 0.15 ± 0.02   | 0.23 ± 0.04   | 0.14 ± 0.04   | W Africa F   | E Europe M | 0.20 |
| unissey-001         | 18 | 0.22 ± 0.03 | 0.32 ± 0.02   | 0.22 ± 0.03   | 0.18 ± 0.02   | 0.24 ± 0.03 | 0.32 ± 0.04 | 0.27 ± 0.04   | 0.32 ± 0.02   | 0.12 ± 0.03   | 0.26 ± 0.02   | 0.28 ± 0.03   | 0.25 ± 0.04   | E Asia F     | E Europe M | 0.14 |
| viant-000           | 16 | 0.49 ± 0.05 | 0.21 ± 0.02   | 0.03 ± 0.02   | 0.41 ± 0.04   | 0.39 ± 0.05 | 0.43 ± 0.06 | 0.28 ± 0.05   | 0.14 ± 0.02   | 0.01 ± 0.01   | 0.21 ± 0.03   | 0.23 ± 0.04   | 0.28 ± 0.06   | E Africa F   | E Europe M | 0.34 |
| viant-000           | 17 | 0.60 ± 0.05 | 0.33 ± 0.02   | 0.12 ± 0.03   | 0.51 ± 0.03   | 0.46 ± 0.05 | 0.57 ± 0.05 | 0.43 ± 0.05   | 0.25 ± 0.02   | 0.02 ± 0.01   | 0.35 ± 0.03   | 0.37 ± 0.05   | 0.47 ± 0.06   | E Africa F   | E Europe M | 0.27 |
| viant-000           | 18 | 0.65 ± 0.04 | 0.42 ± 0.02   | 0.18 ± 0.03   | 0.56 ± 0.02   | 0.56 ± 0.03 | 0.61 ± 0.05 | 0.54 ± 0.04   | 0.36 ± 0.02   | 0.06 ± 0.02   | 0.48 ± 0.02   | 0.47 ± 0.04   | 0.55 ± 0.05   | E Africa F   | E Europe M | 0.21 |
| yoti-001            | 16 | 0.13 ± 0.04 | 0.08 ± 0.01   | 0.11 ± 0.03   | 0.12 ± 0.02   | 0.15 ± 0.03 | 0.17 ± 0.05 | 0.02 ± 0.02   | 0.010 ± 0.005 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.003 ± 0.003 | 0.02 ± 0.02   | W Africa F   | S E Asia M | 0.54 |
| yoti-001            | 17 | 0.18 ± 0.04 | 0.14 ± 0.02   | 0.19 ± 0.04   | 0.21 ± 0.03   | 0.18 ± 0.03 | 0.24 ± 0.04 | 0.02 ± 0.01   | 0.036 ± 0.009 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.02 ± 0.01   | 0.08 ± 0.03   | W Africa F   | E Europe M | 0.46 |
| yoti-001            | 18 | 0.23 ± 0.04 | 0.19 ± 0.02   | 0.26 ± 0.03   | 0.25 ± 0.02   | 0.26 ± 0.03 | 0.32 ± 0.04 | 0.07 ± 0.02   | 0.059 ± 0.009 | 0.03 ± 0.01   | 0.043 ± 0.009 | 0.04 ± 0.02   | 0.10 ± 0.03   | W Africa F   | E Europe M | 0.40 |
| yoti-002            | 16 | 0.07 ± 0.03 | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.07 ± 0.02   | 0.13 ± 0.03 | 0.11 ± 0.04 | 0.02 ± 0.01   | 0.011 ± 0.005 | 0.003 ± 0.003 | 0.008 ± 0.006 | 0.011 ± 0.008 | 0.009 ± 0.009 | S E Asia F   | E Europe M | 0.54 |
| yoti-002            | 17 | 0.10 ± 0.03 | 0.12 ± 0.02   | 0.06 ± 0.03   | 0.14 ± 0.02   | 0.16 ± 0.03 | 0.18 ± 0.04 | 0.012 ± 0.009 | 0.028 ± 0.008 | 0.006 ± 0.006 | 0.018 ± 0.007 | 0.011 ± 0.008 | 0.04 ± 0.02   | W Africa F   | E Europe M | 0.51 |
| yoti-002            | 18 | 0.12 ± 0.03 | 0.16 ± 0.02   | 0.09 ± 0.02   | 0.18 ± 0.02   | 0.22 ± 0.03 | 0.19 ± 0.04 | 0.04 ± 0.02   | 0.043 ± 0.008 | 0.005 ± 0.005 | 0.043 ± 0.009 | 0.03 ± 0.01   | 0.07 ± 0.02   | S E Asia F</ |            |      |

|              |    |                 |                   |                 |                 |                 |                 |                 |                   |                 |                 |                 |                 |            |          |      |
|--------------|----|-----------------|-------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-------------------|-----------------|-----------------|-----------------|-----------------|------------|----------|------|
| youverse-000 | 18 | $0.20 \pm 0.03$ | $0.046 \pm 0.009$ | $0.16 \pm 0.03$ | $0.13 \pm 0.02$ | $0.05 \pm 0.01$ | $0.24 \pm 0.04$ | $0.22 \pm 0.03$ | $0.031 \pm 0.007$ | $0.08 \pm 0.02$ | $0.11 \pm 0.02$ | $0.04 \pm 0.01$ | $0.25 \pm 0.04$ | W Africa M | E Asia M | 0.37 |
|--------------|----|-----------------|-------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-------------------|-----------------|-----------------|-----------------|-----------------|------------|----------|------|

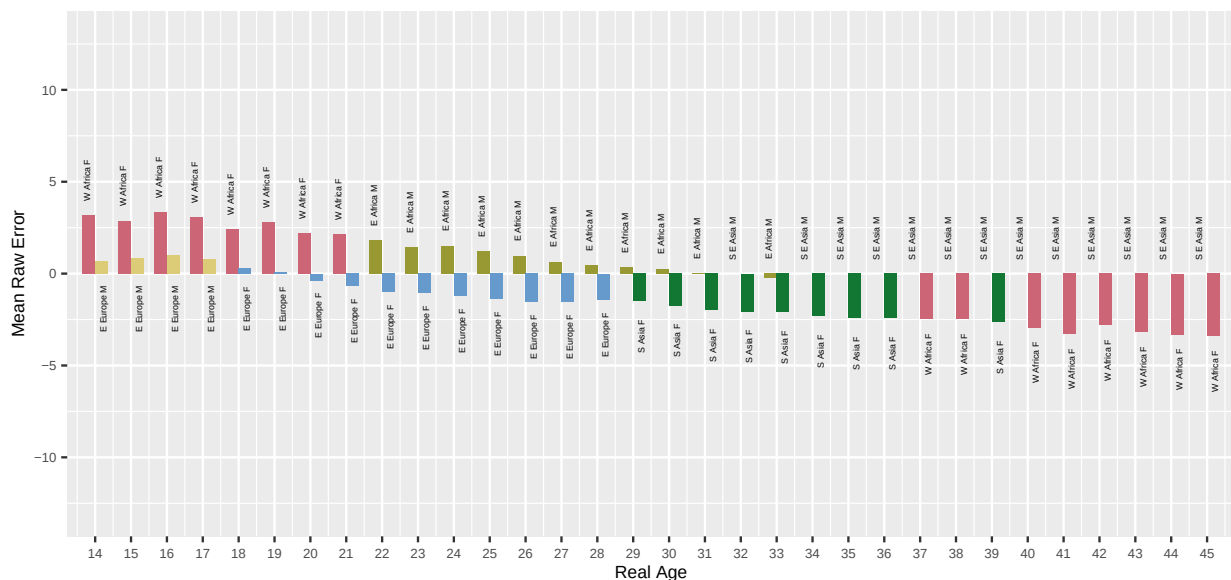
### 5.4.2. Empirical results

Using sex and region-of-birth metadata, Table 18 shows dependence of Challenge-25 false positive rates on twelve demographic groups. The images are higher quality office application portraits. Expanded information is given in Appendix A.3 which includes Tables 113 to 180 showing Challenge-25 false positive rates for ages 14-34, both sexes, six regions of birth and all algorithms.

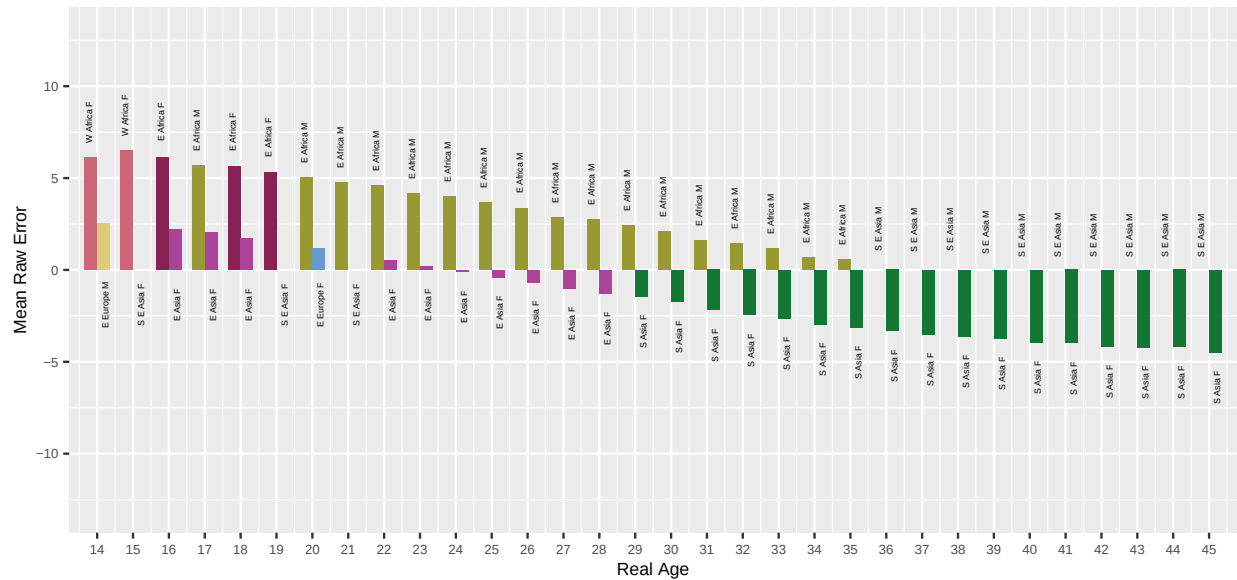
Each tabulated data point is computed from hundreds or thousands of images. For 17 year olds, the largest group is East Asian men with 1563 images and the smallest group is 316 West African men. For 18 year olds those numbers rise to 2269 and 420. The uncertainty estimates reflect this; they are computed by bootstrapping 1000 times and computing the interval that gives 95% coverage of the FPR values i.e. quantile (0.025,0.975).

For 16-18 year olds, general accuracy trends are few but we see FPR varies substantially with algorithm, sex, and region of birth. Generally, FPR is higher in women, but the degree is quite strongly algorithm dependent, and for unissey-001 men usually have higher FPR. We expect performance variations to be algorithm-specific, because 1) AI-based models are widely held to underperform on demographic groups that are under-represented in the training data and 2) algorithms are usually trained on image sets curated or owned by the developer.

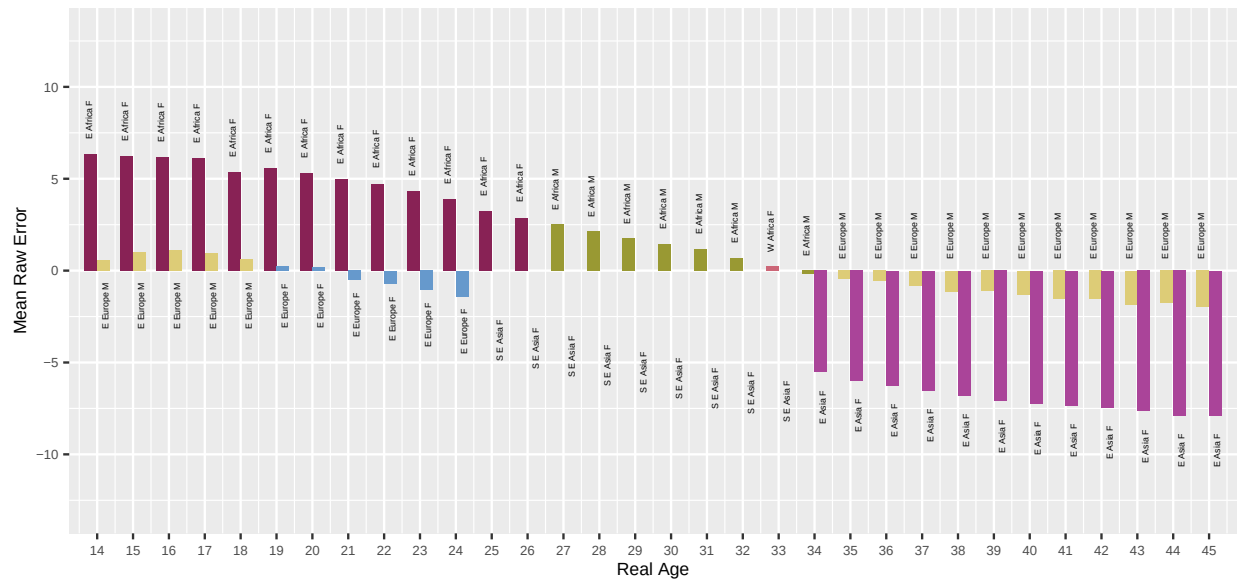
Figures 18- 87 give one summary of those tables by showing the groups that have worst- and best-case overall mean error for all true age values from 14 to 45. Figures are included for *Application* and *Border* images, both on one page. The figures include both positive and negative errors, positive corresponding to overestimation of age by an algorithm, and negative stating underestimation. A smaller magnitude of error is better. The images are application-type images.



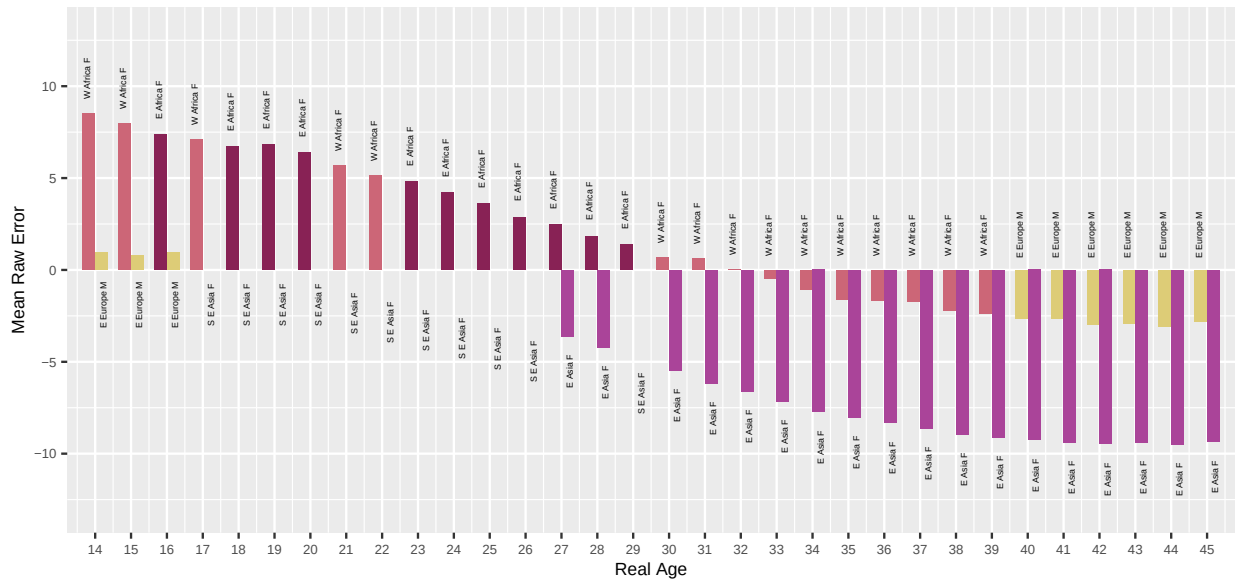
**Fig. 12.** For algorithm cognitec-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



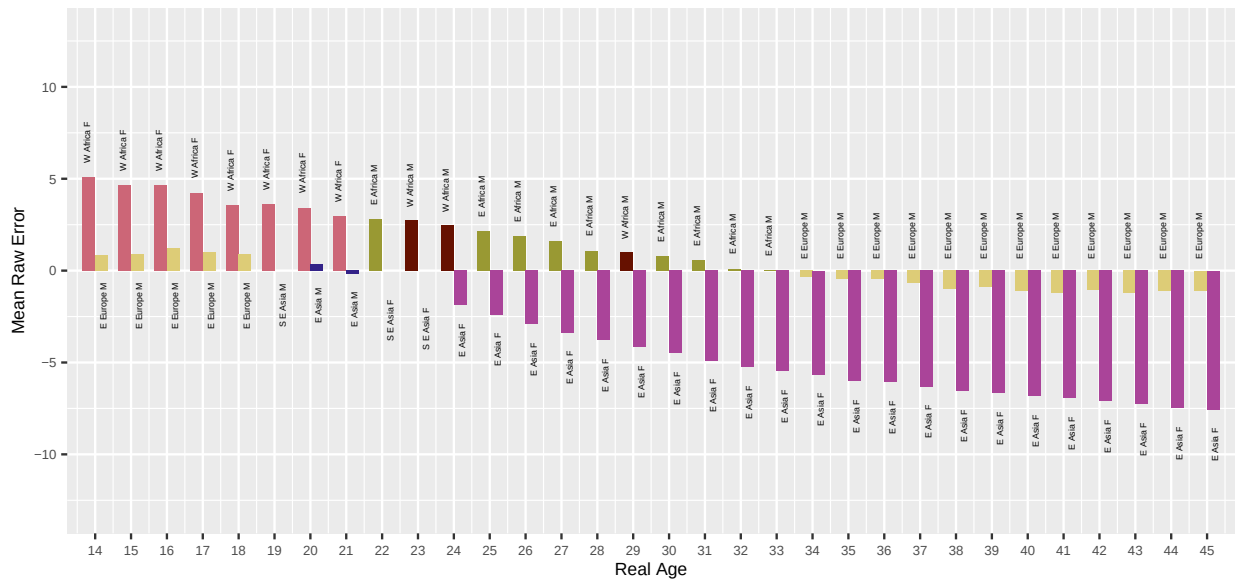
**Fig. 13.** For algorithm cognitec-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



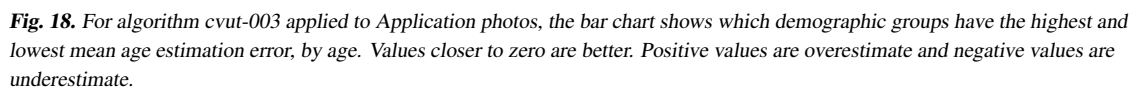
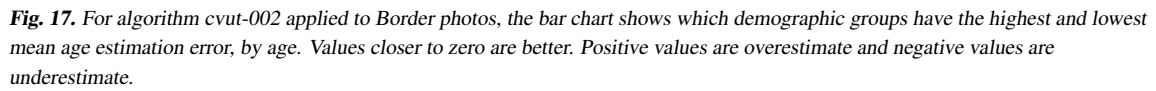
**Fig. 14.** For algorithm cvut-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



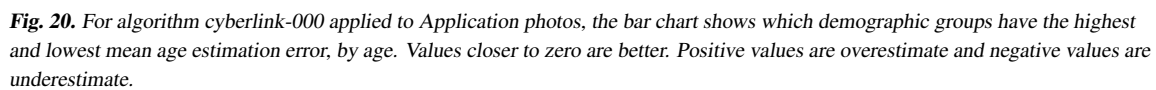
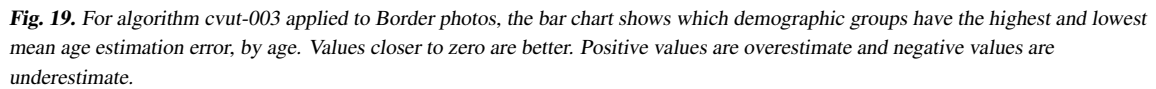
**Fig. 15.** For algorithm cvut-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

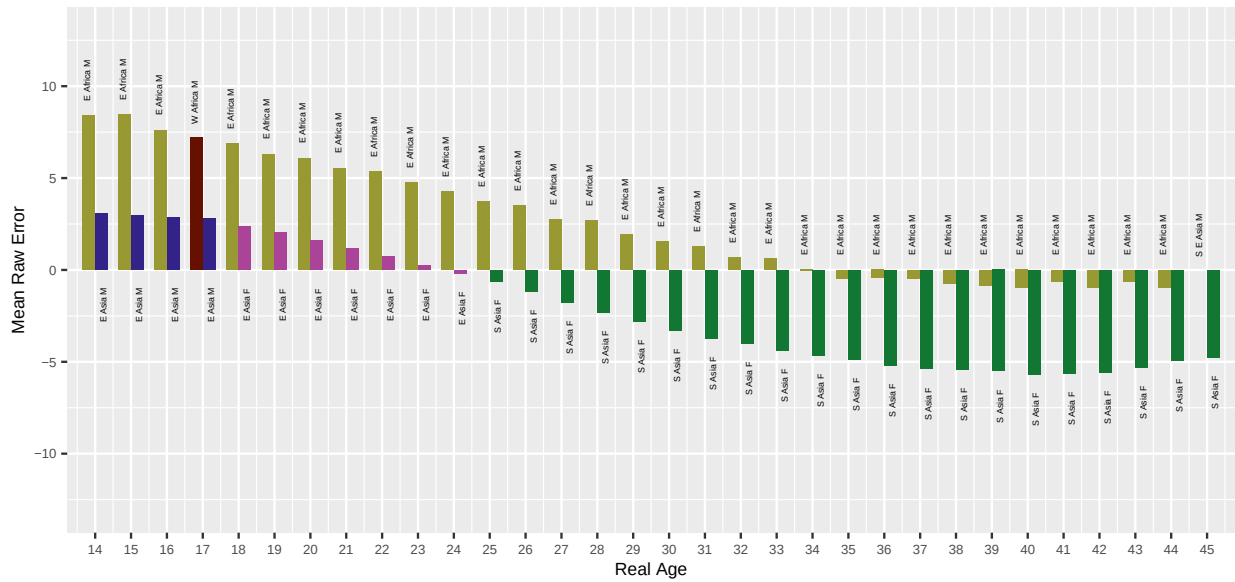


**Fig. 16.** For algorithm cvut-002 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

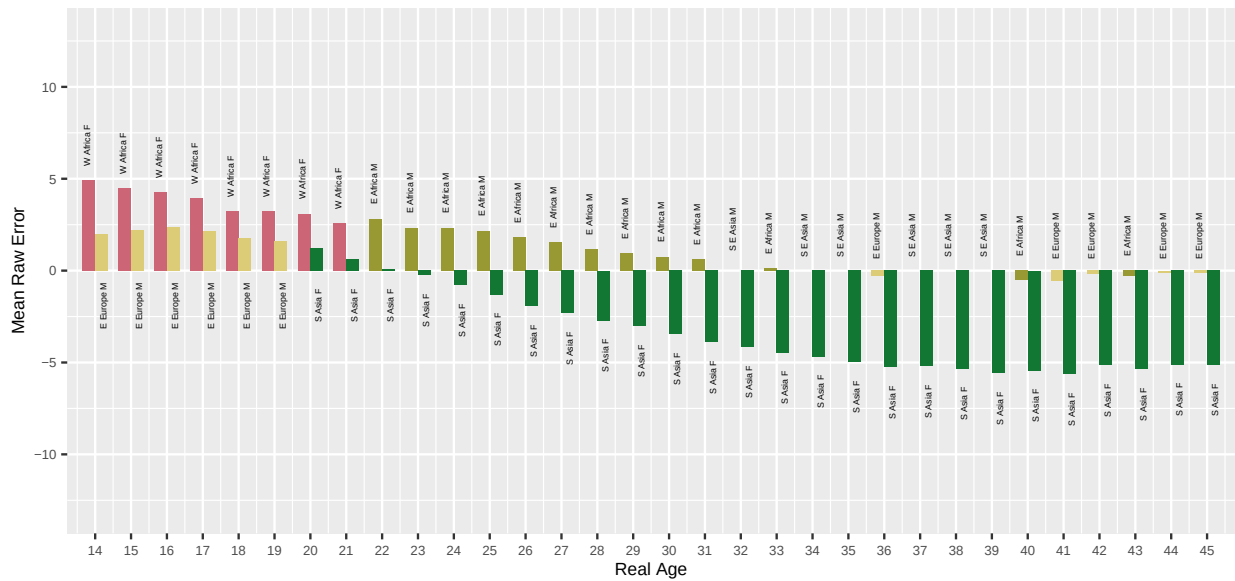




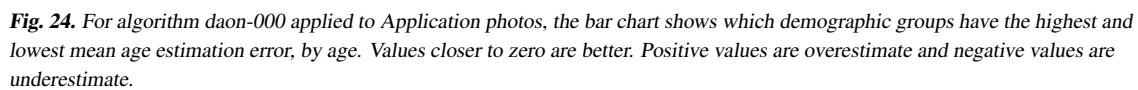
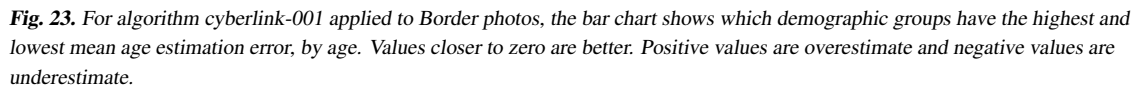


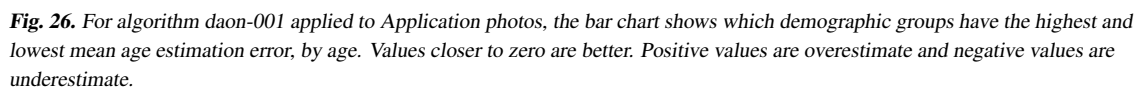
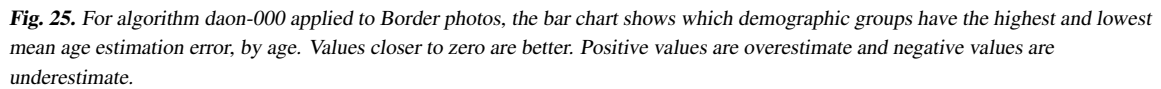


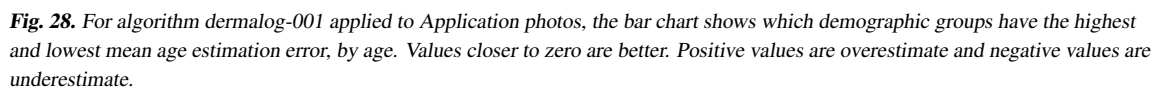
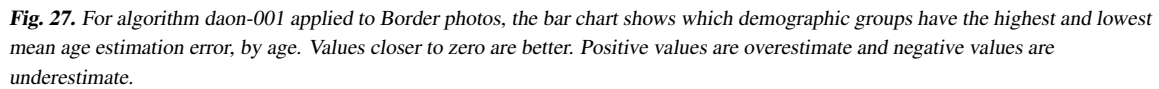
**Fig. 21.** For algorithm cyberlink-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

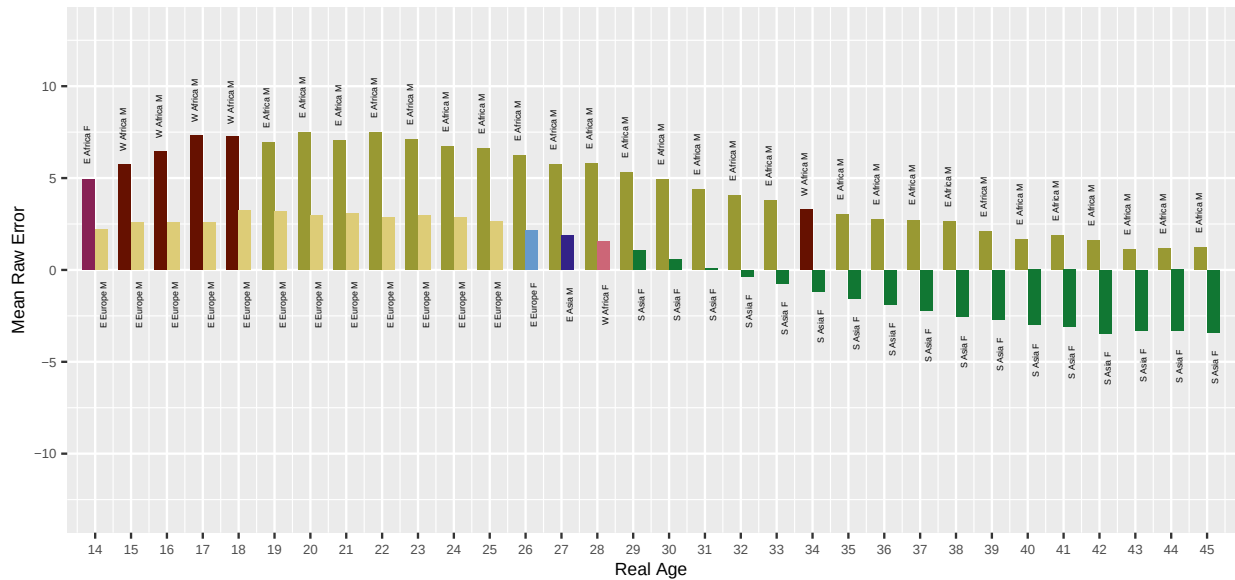


**Fig. 22.** For algorithm cyberlink-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

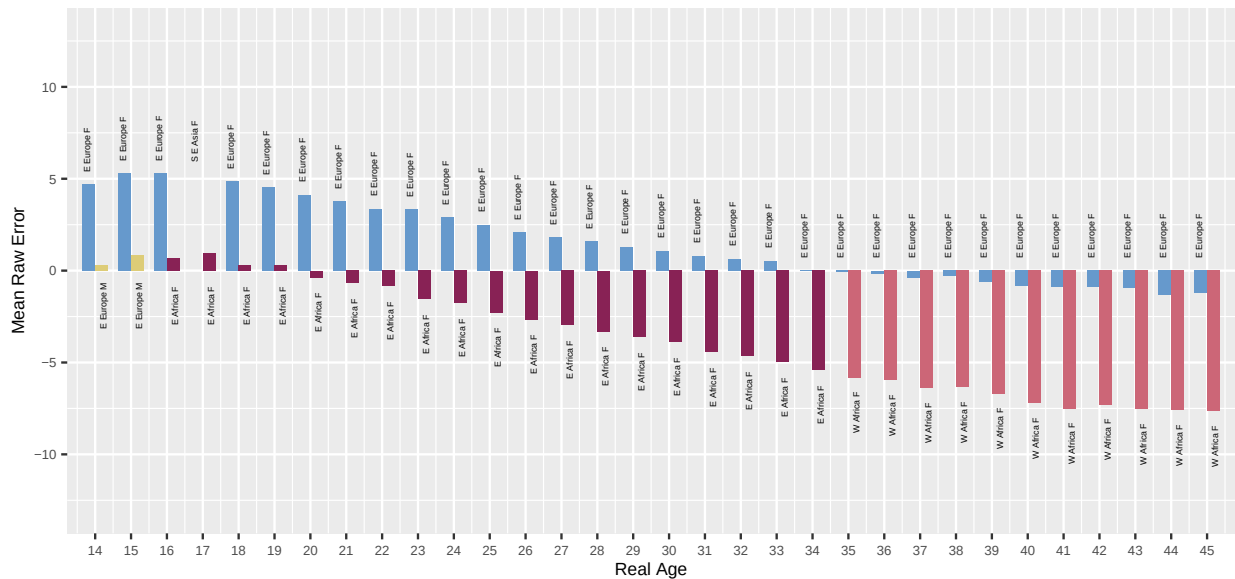




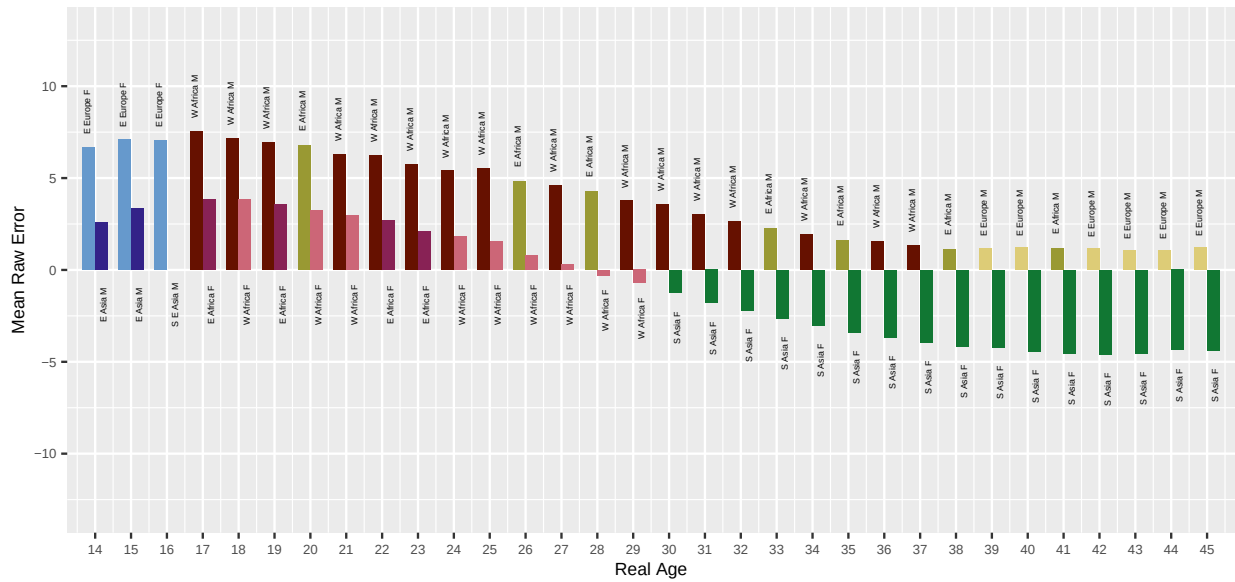




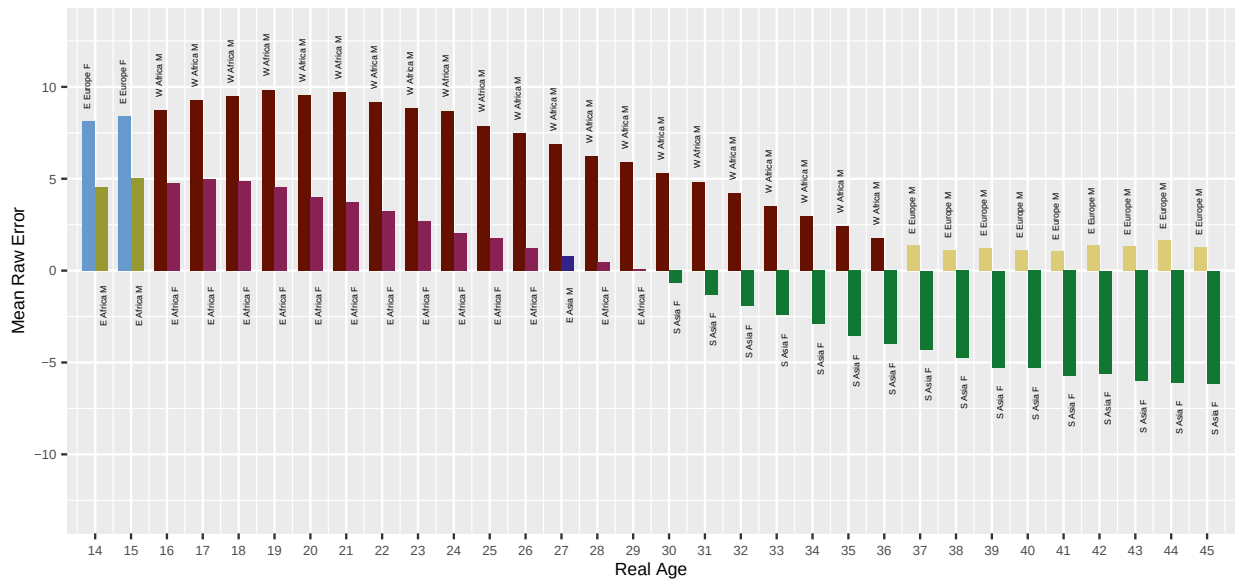
**Fig. 29.** For algorithm dermalog-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



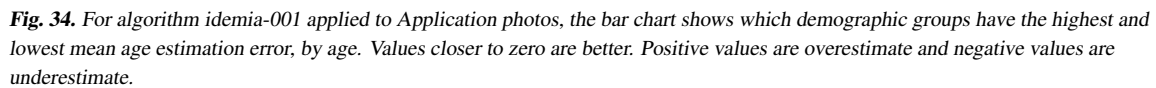
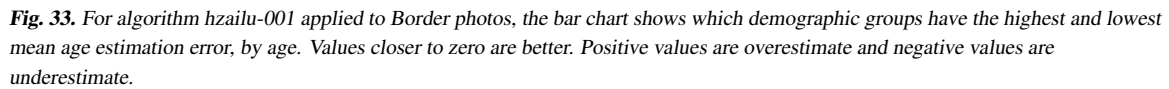
**Fig. 30.** For algorithm dermalog-002 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



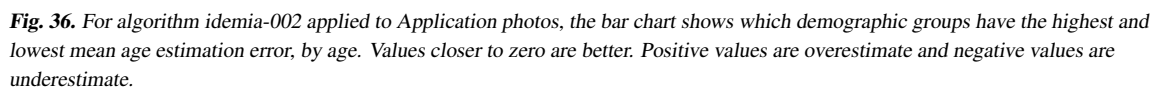
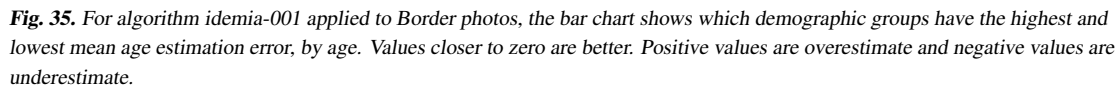
**Fig. 31.** For algorithm dermalog-002 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

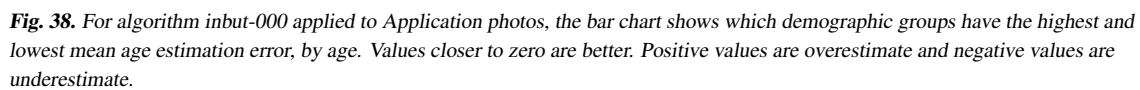
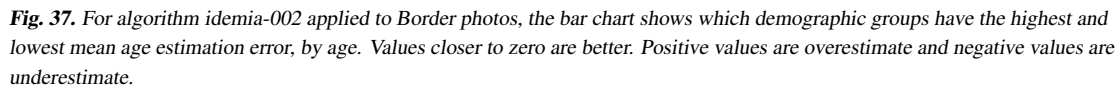


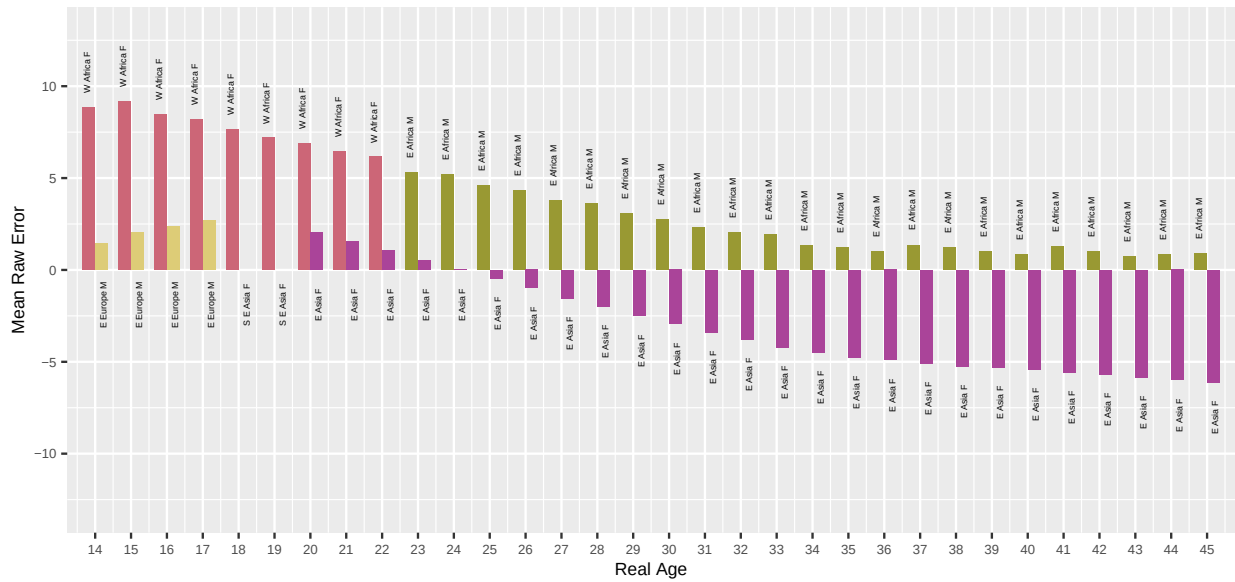
**Fig. 32.** For algorithm hzailu-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



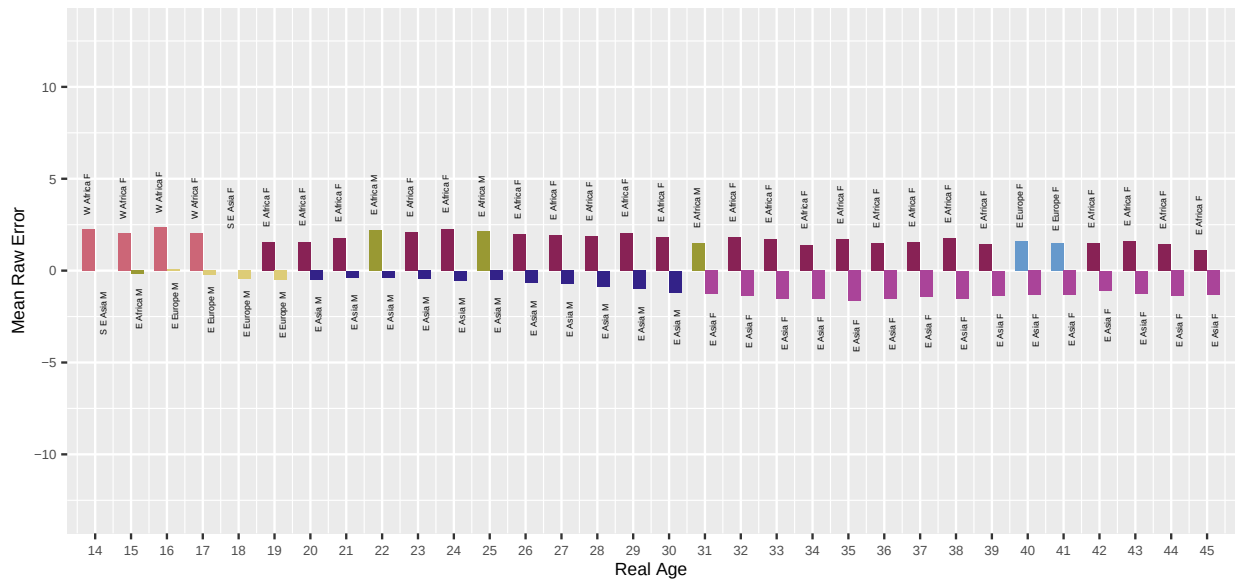




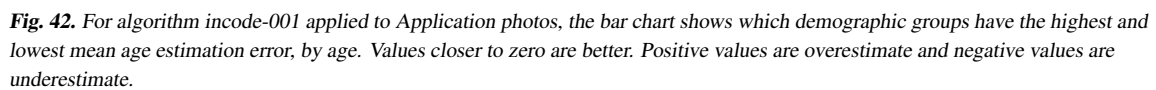
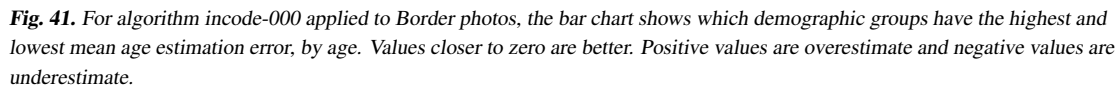


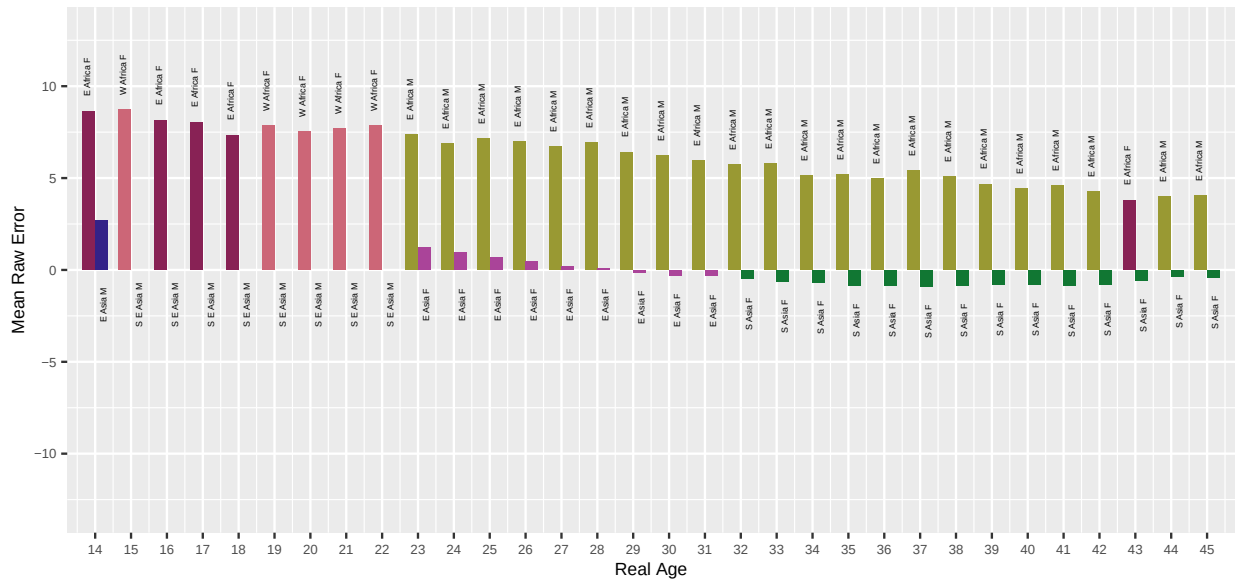


**Fig. 39.** For algorithm inbut-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

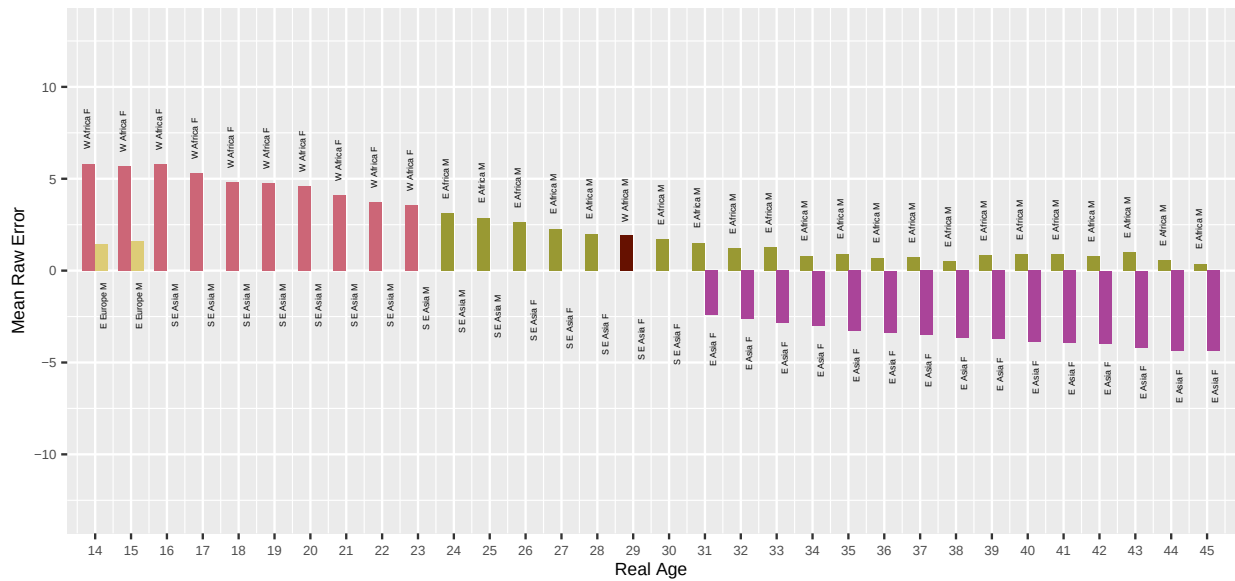


**Fig. 40.** For algorithm incode-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

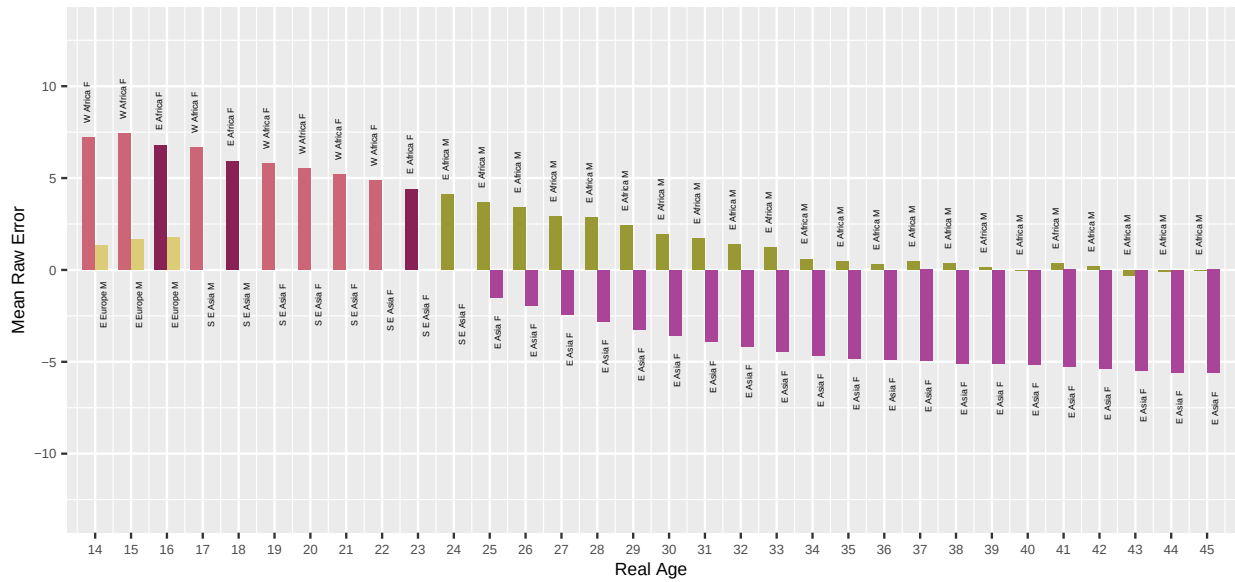




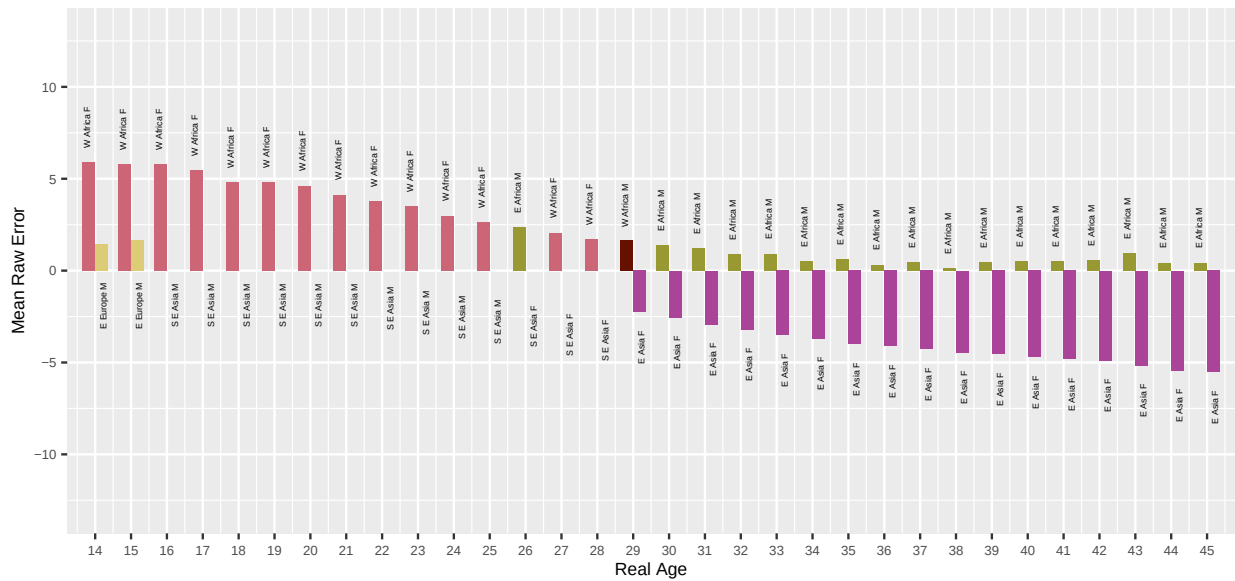
**Fig. 43.** For algorithm inc001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



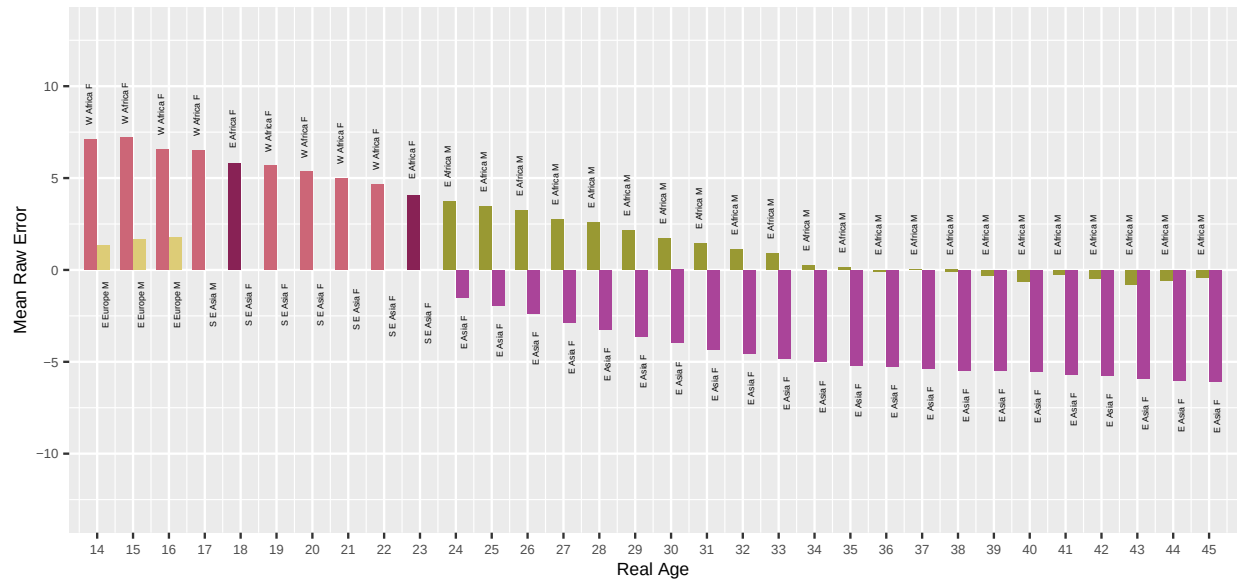
**Fig. 44.** For algorithm innovatrics-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



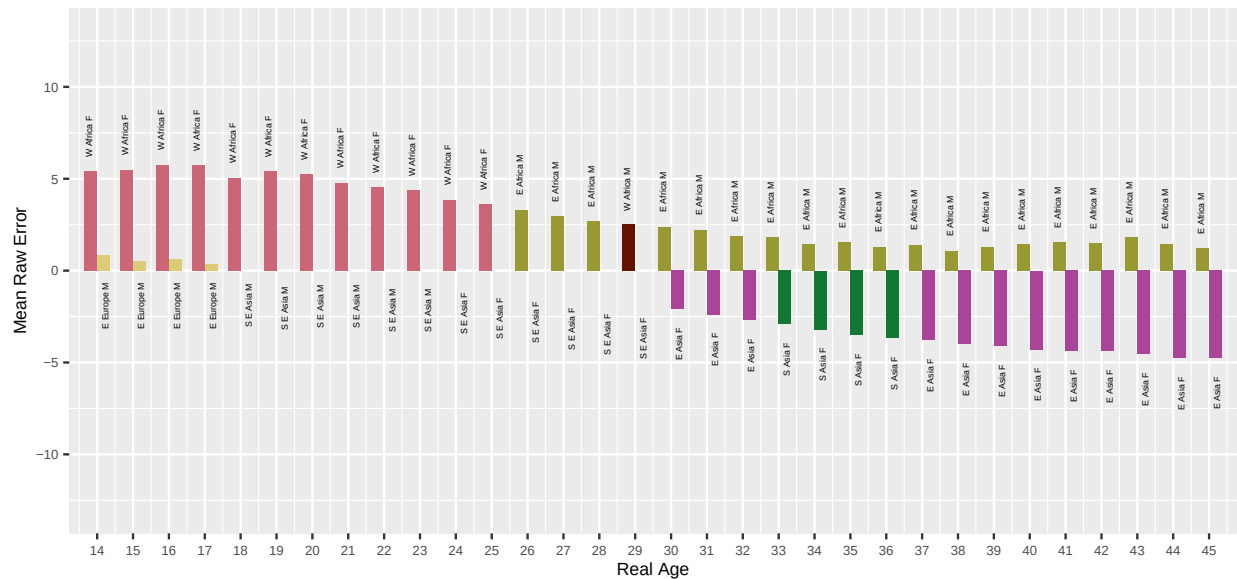
**Fig. 45.** For algorithm innovatrics-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



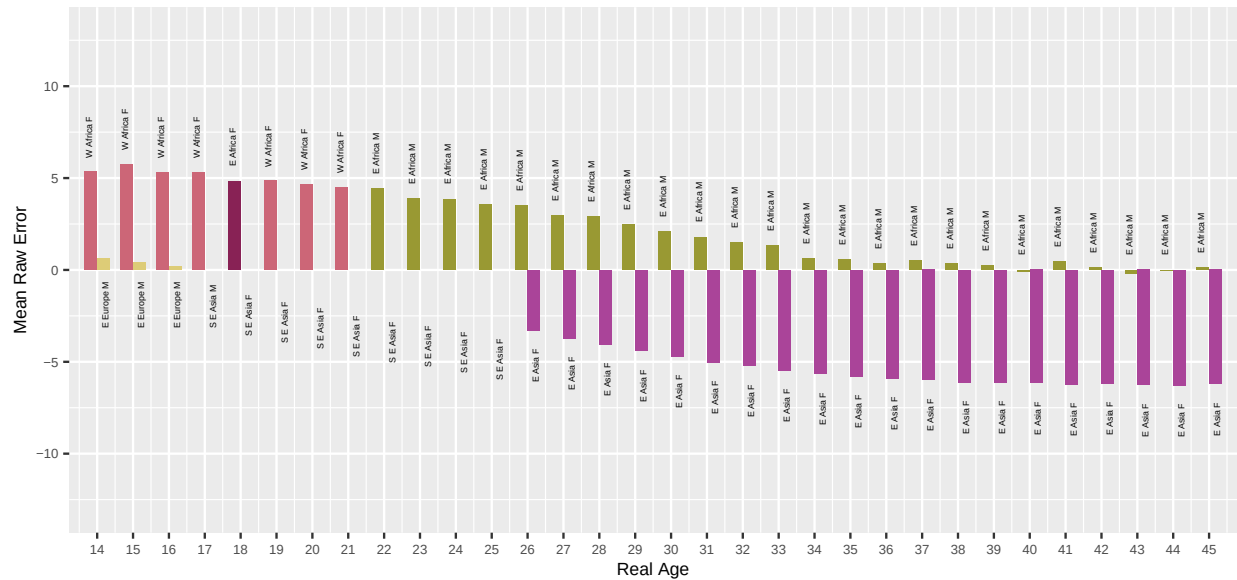
**Fig. 46.** For algorithm innovatrics-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



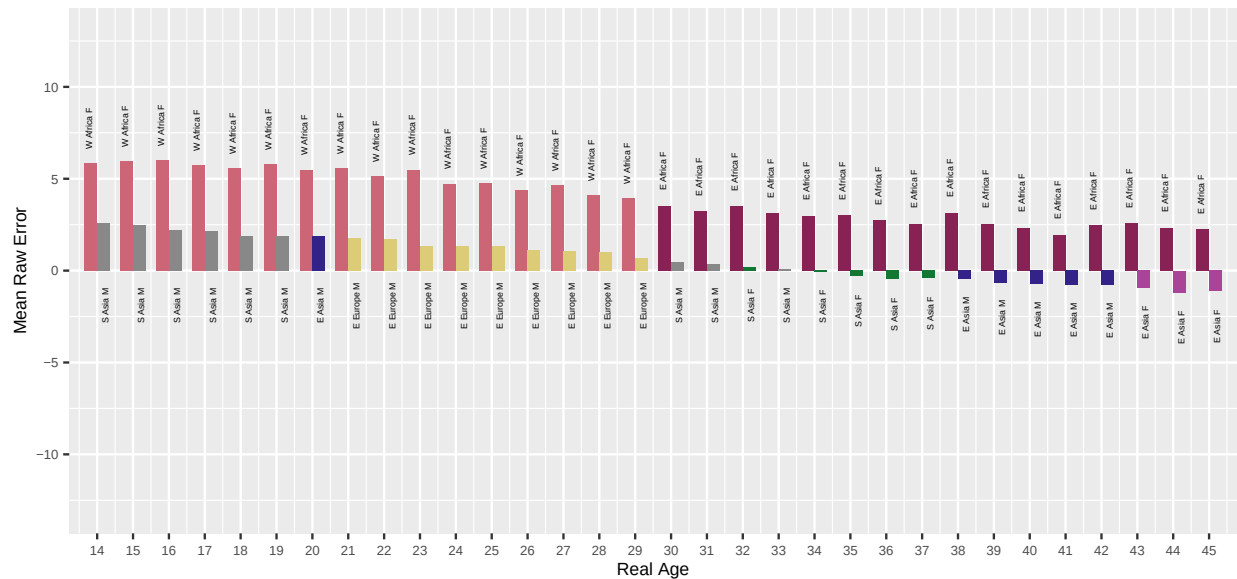
**Fig. 47.** For algorithm innovatrics-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



**Fig. 48.** For algorithm innovatrics-002 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

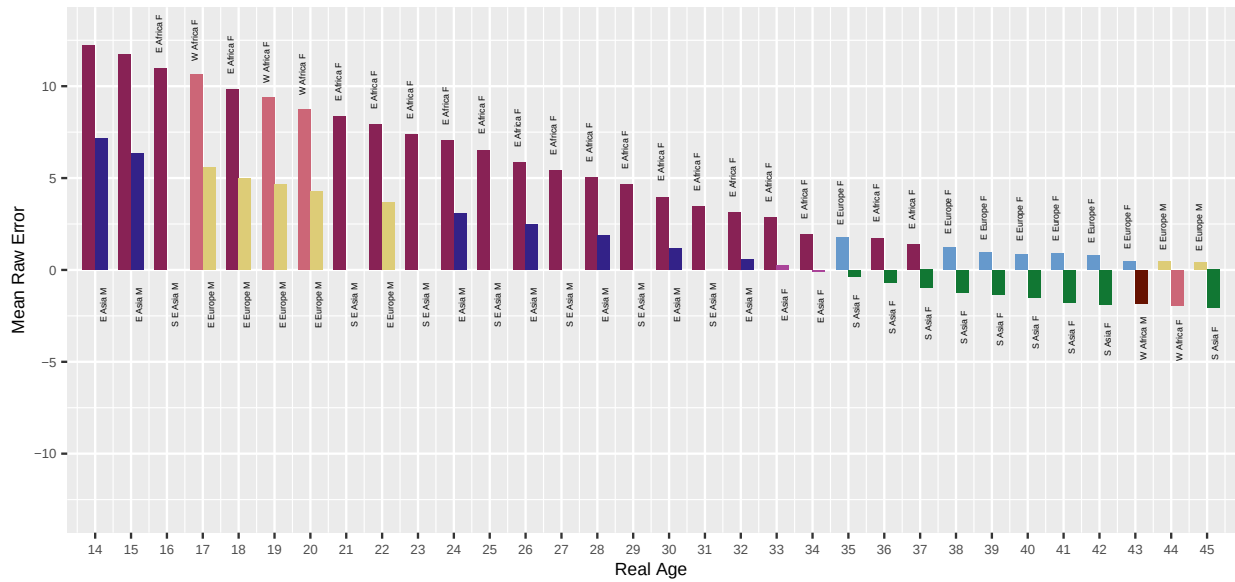


**Fig. 49.** For algorithm *innovatrics-002* applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

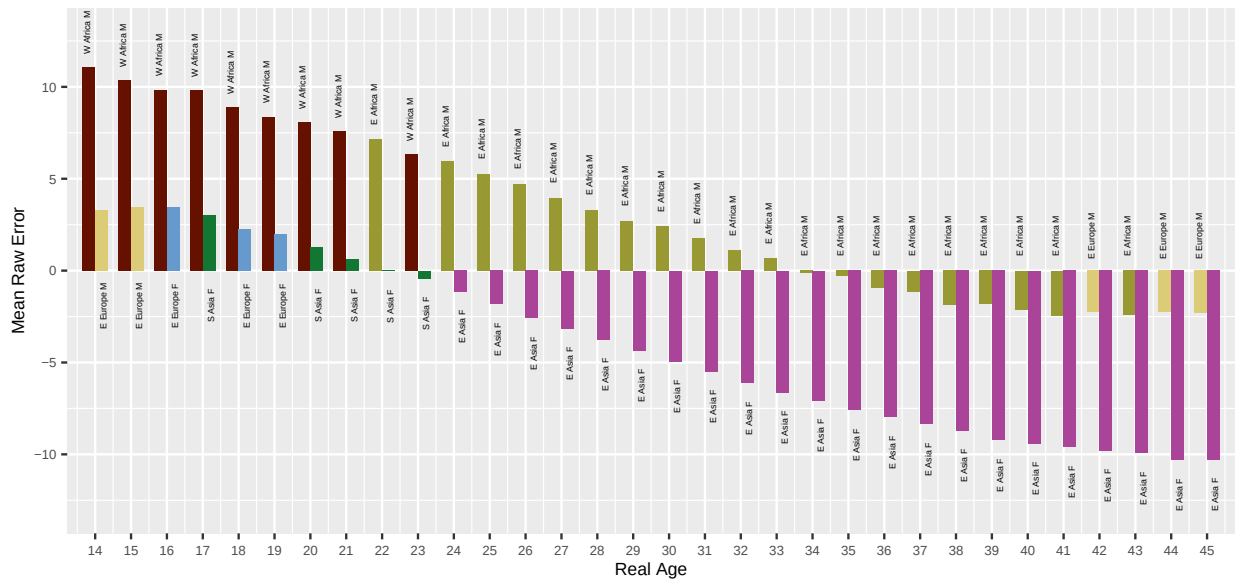


**Fig. 50.** For algorithm *jumio-000* applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

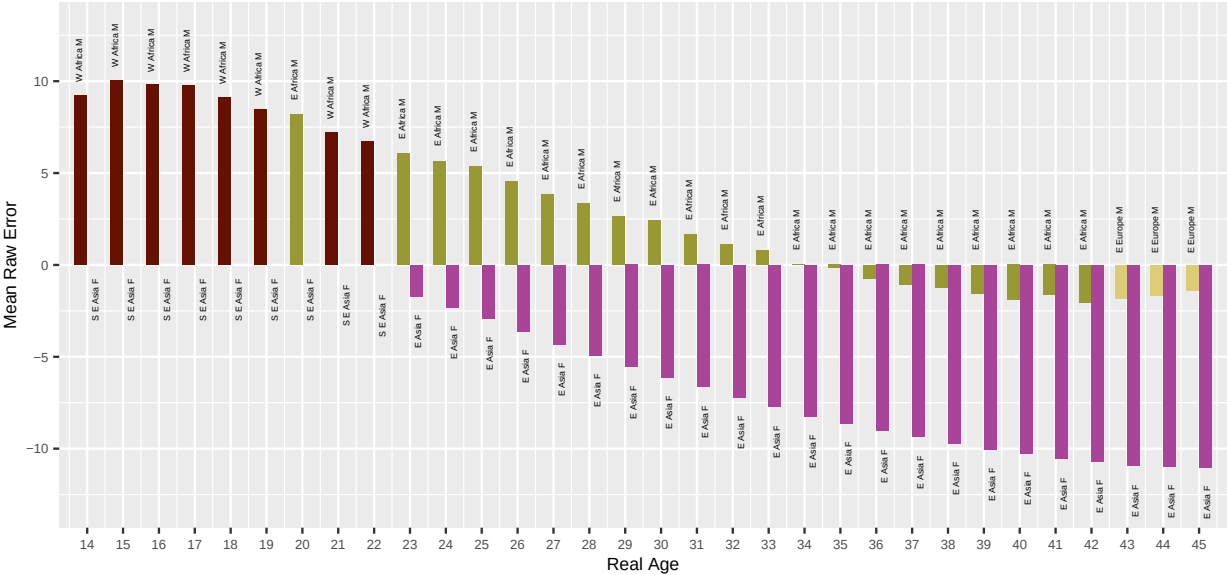




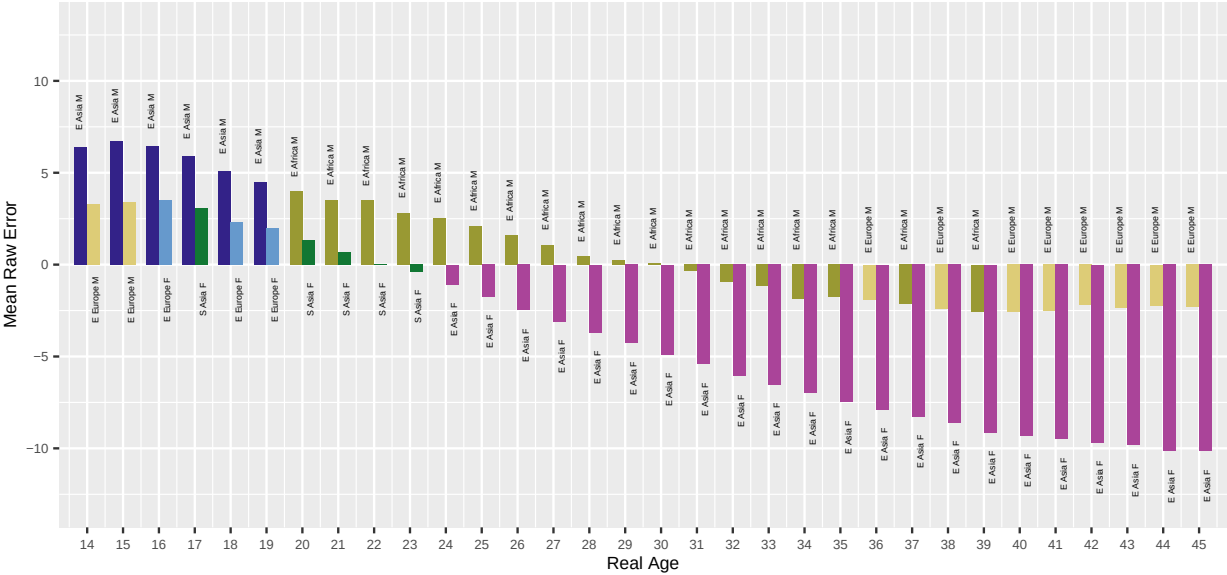
**Fig. 51.** For algorithm jumio-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



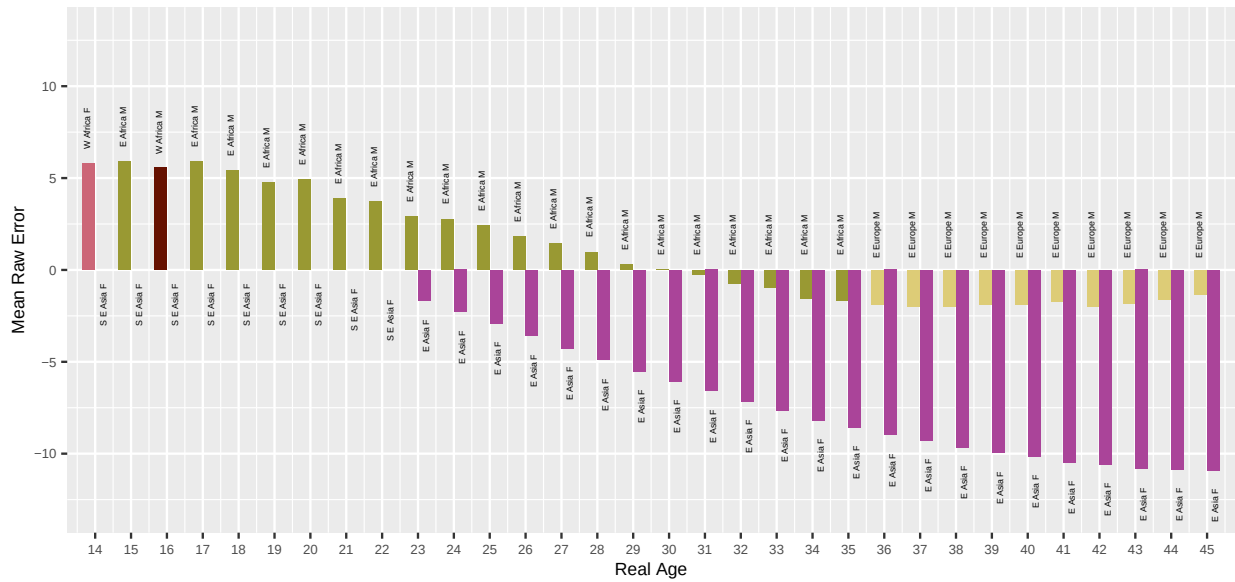
**Fig. 52.** For algorithm neurotechnology-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



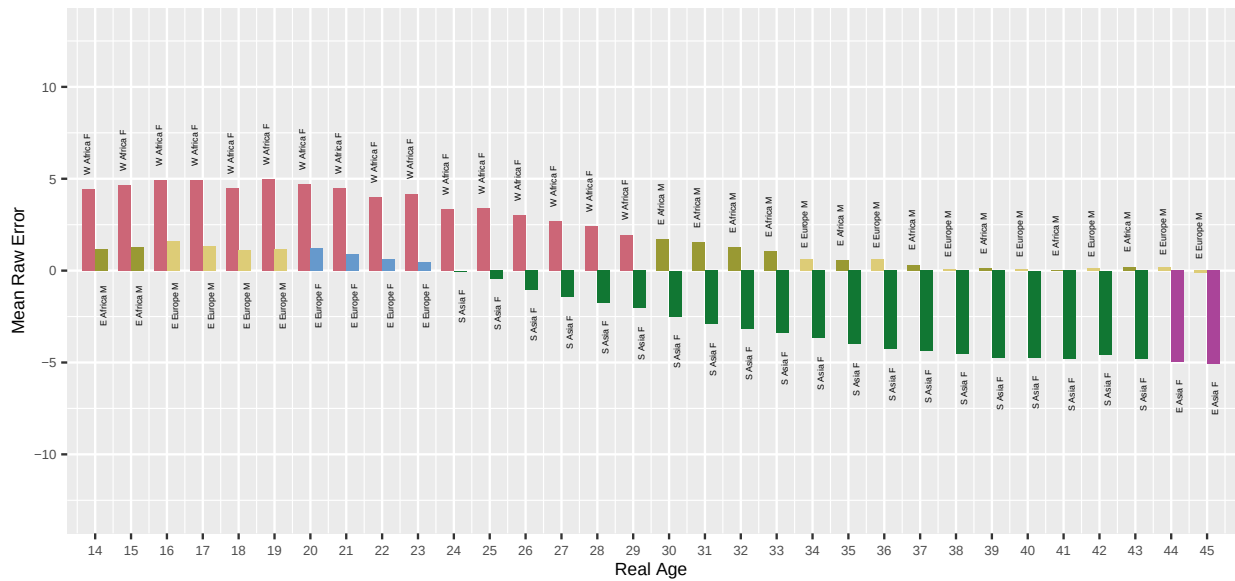
**Fig. 53.** For algorithm neurotechnology-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



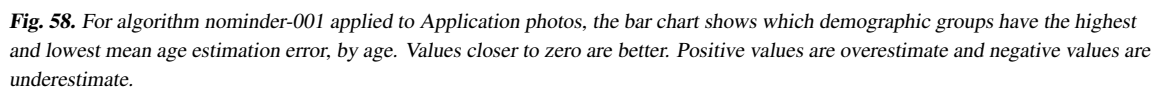
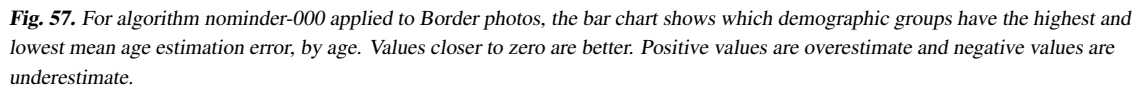
**Fig. 54.** For algorithm neurotechnology-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

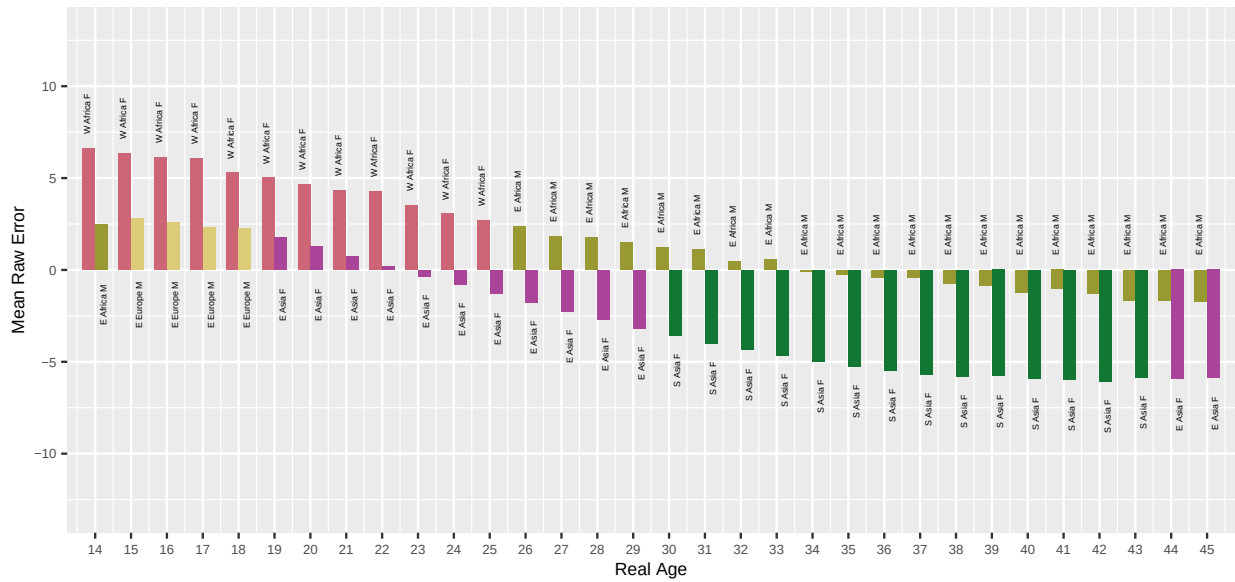


**Fig. 55.** For algorithm neurotechnology-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

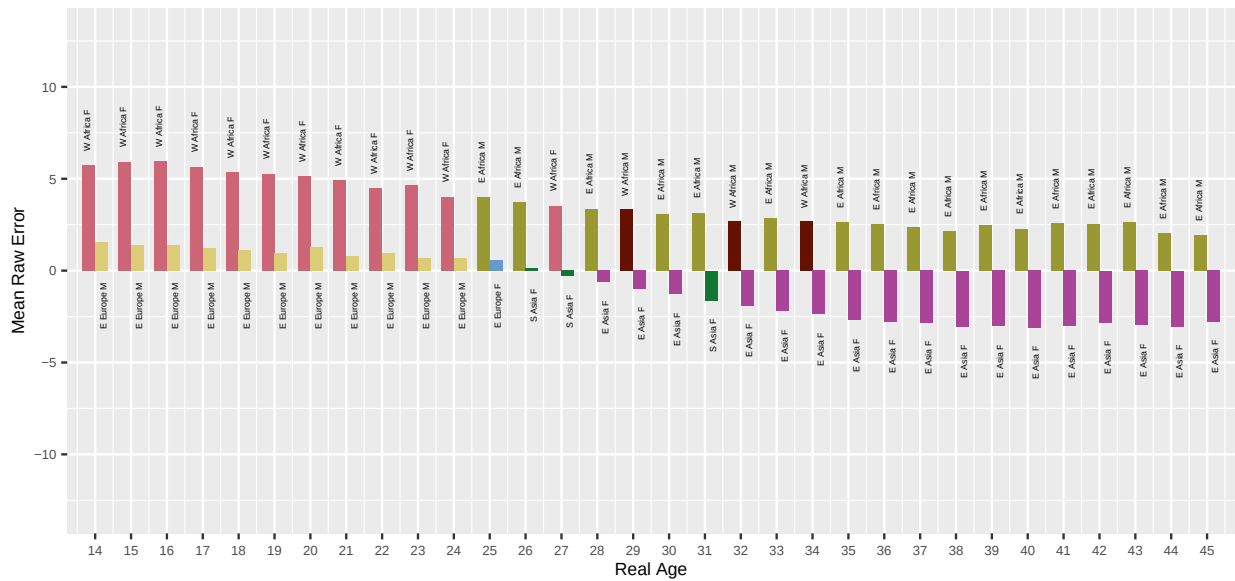


**Fig. 56.** For algorithm nominder-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

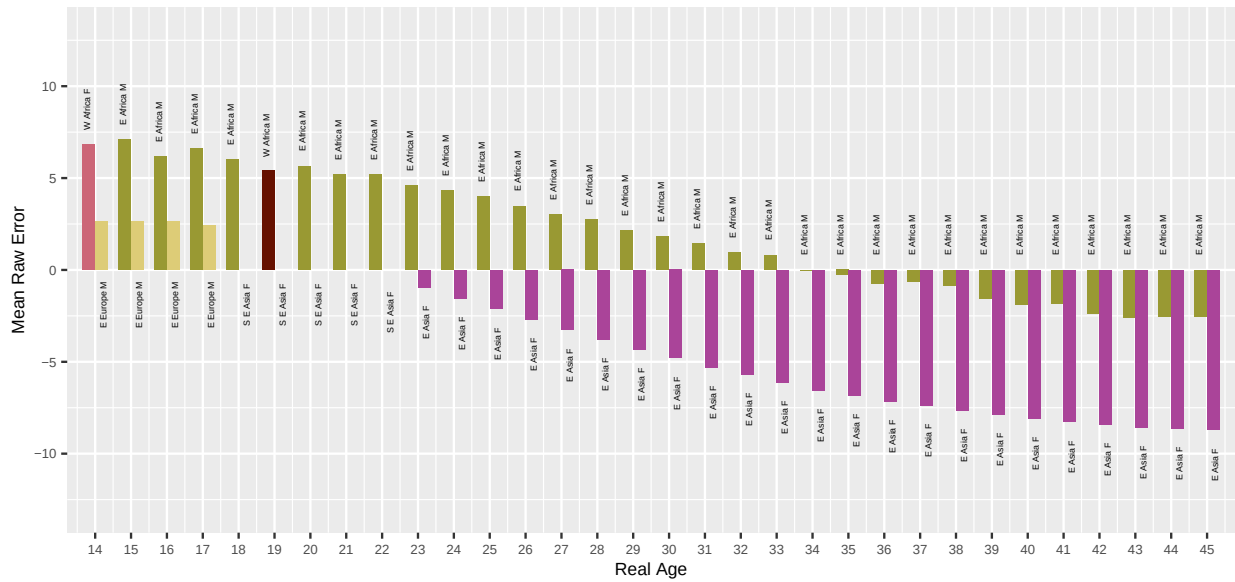




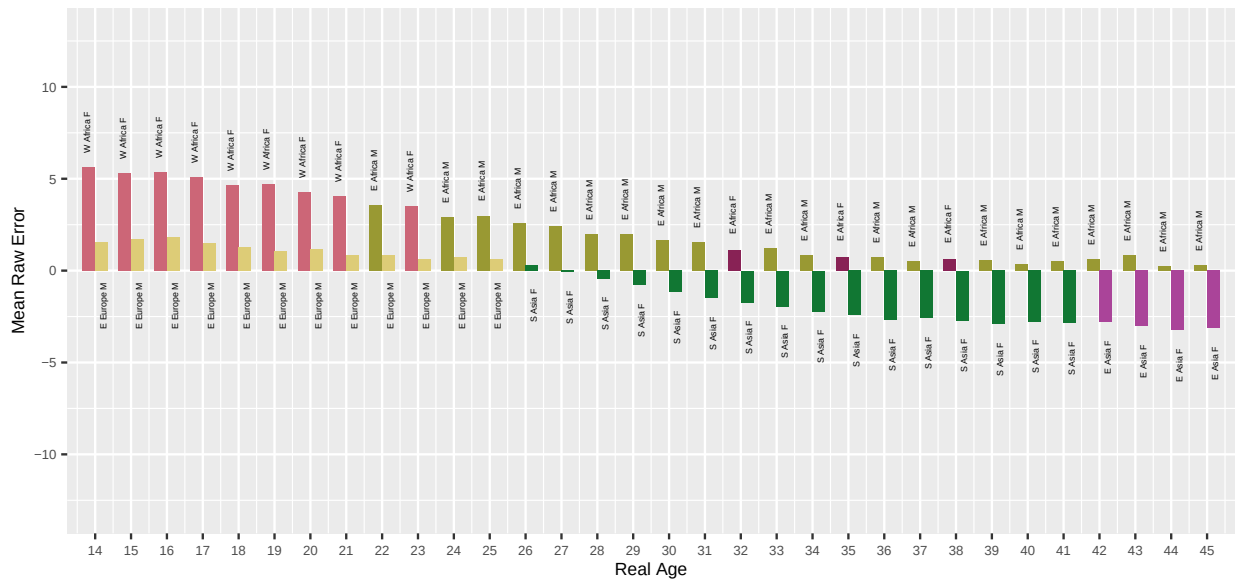
**Fig. 59.** For algorithm nominder-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



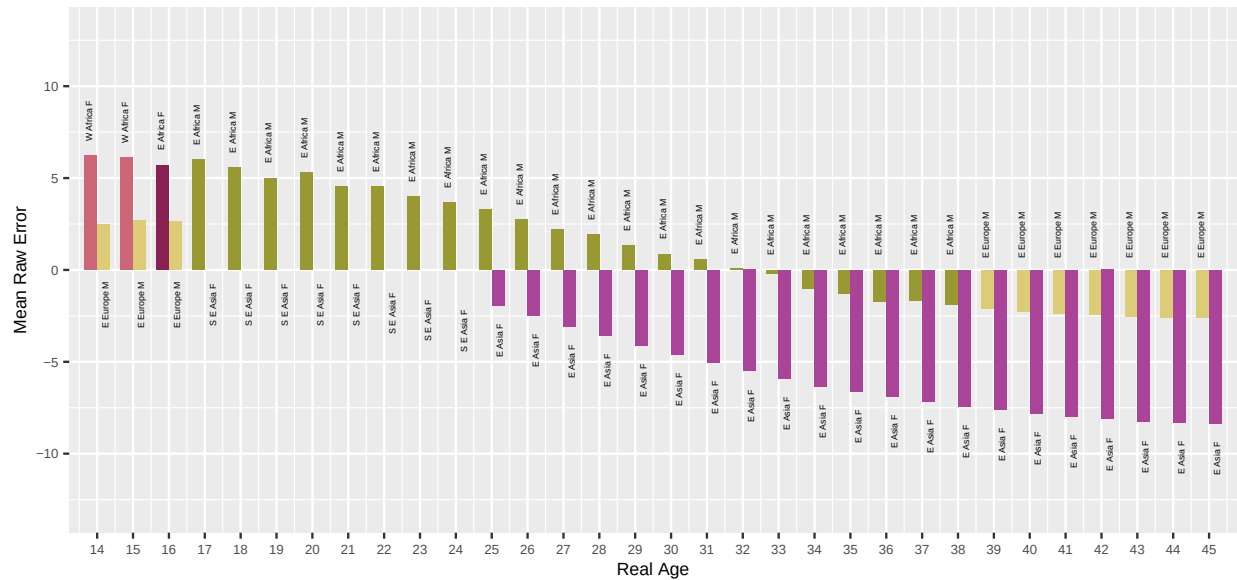
**Fig. 60.** For algorithm ondato-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



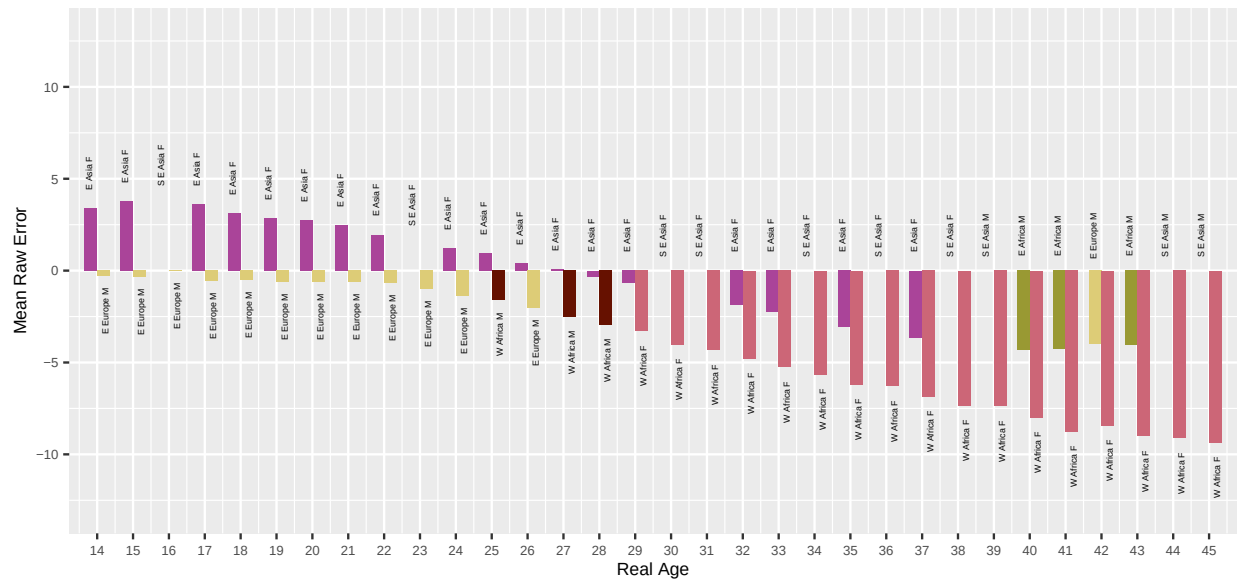
**Fig. 61.** For algorithm ondato-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



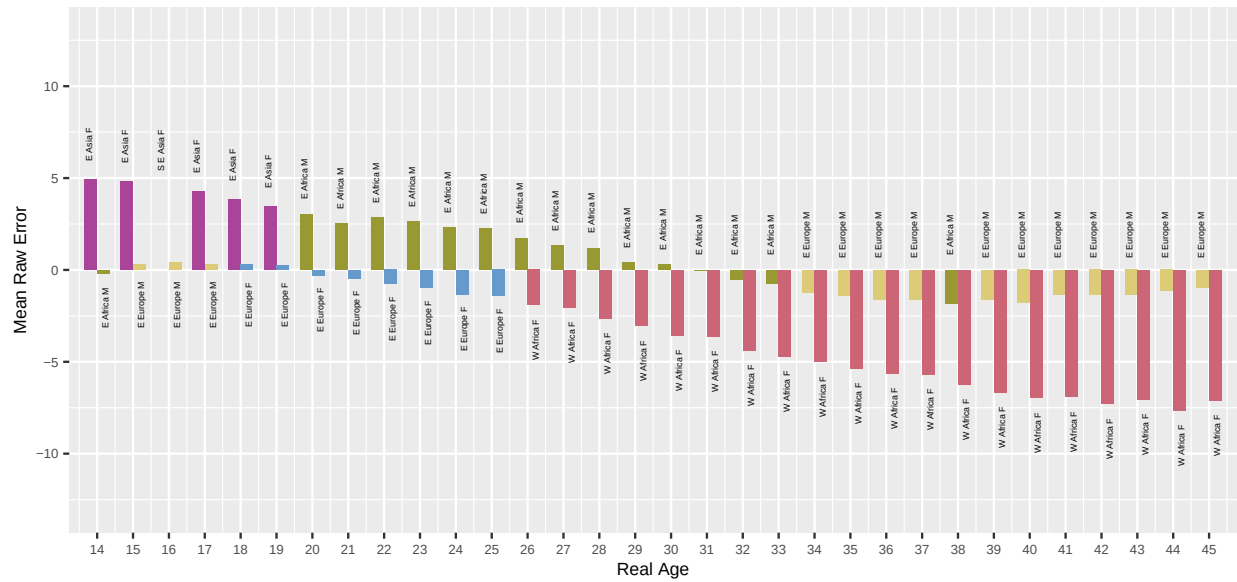
**Fig. 62.** For algorithm ondato-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



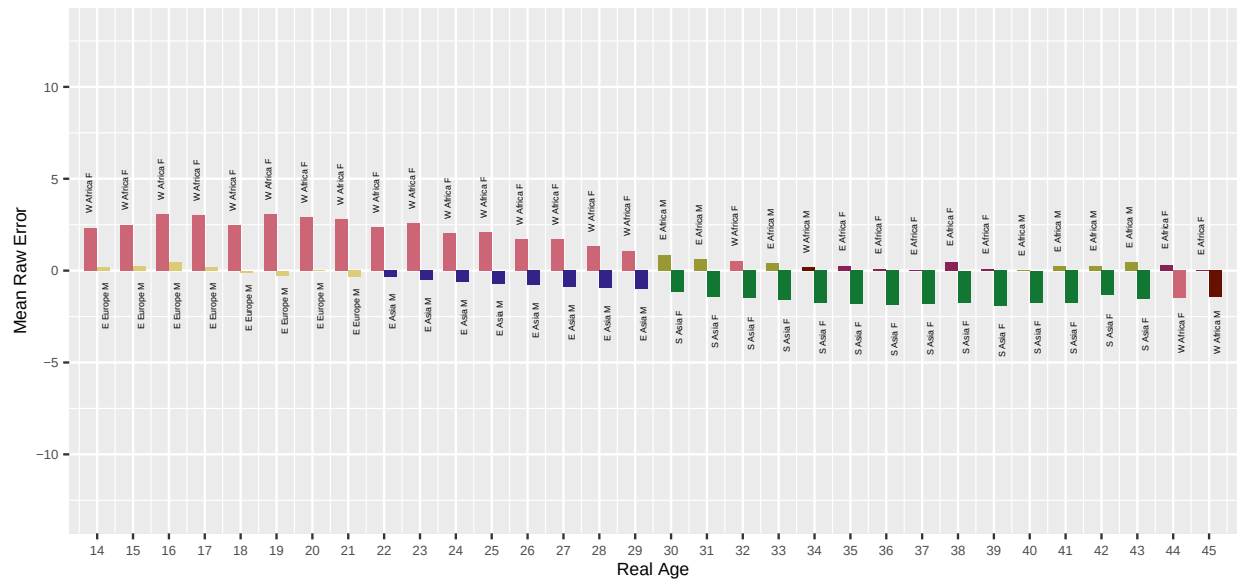
**Fig. 63.** For algorithm ondato-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



**Fig. 64.** For algorithm privately-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

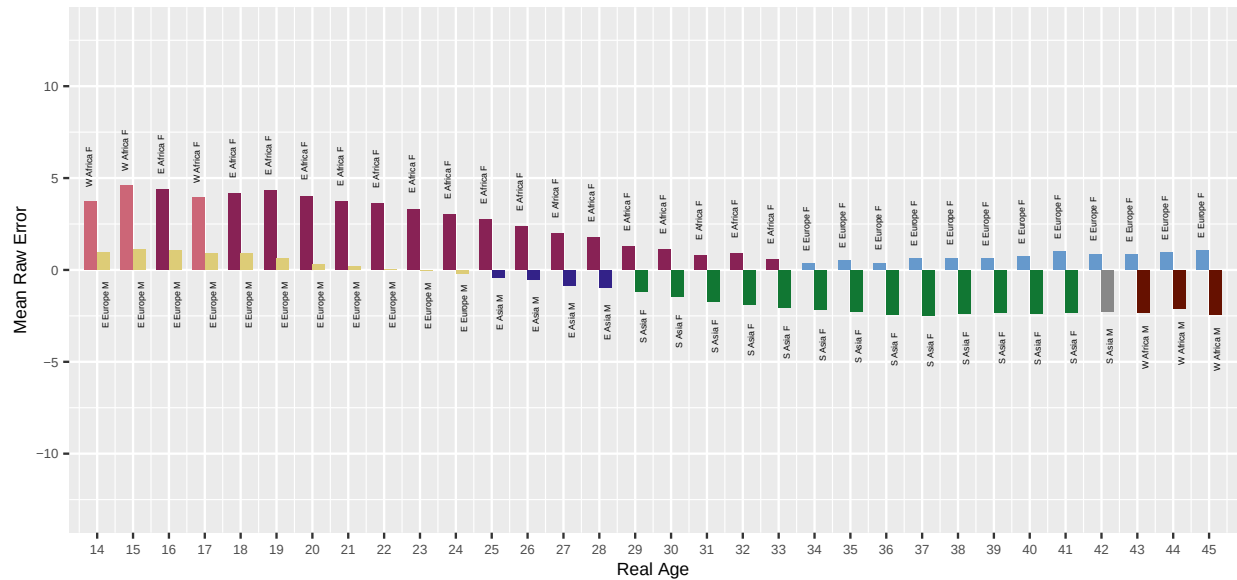


**Fig. 65.** For algorithm *privately-000* applied to *Border* photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

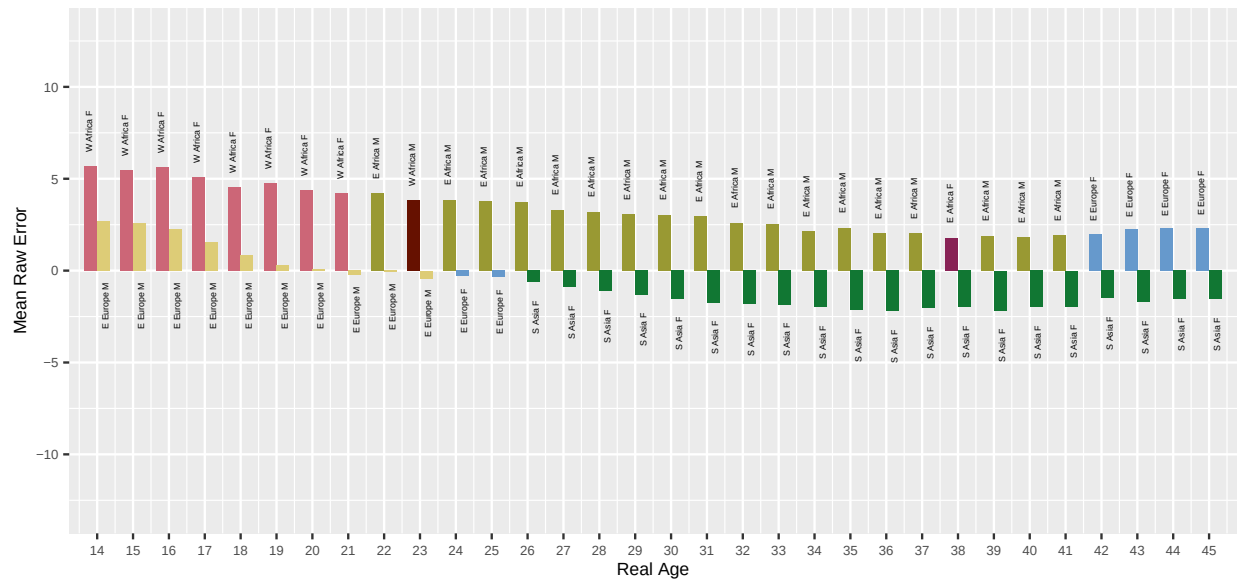


**Fig. 66.** For algorithm *regula-000* applied to *Application* photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

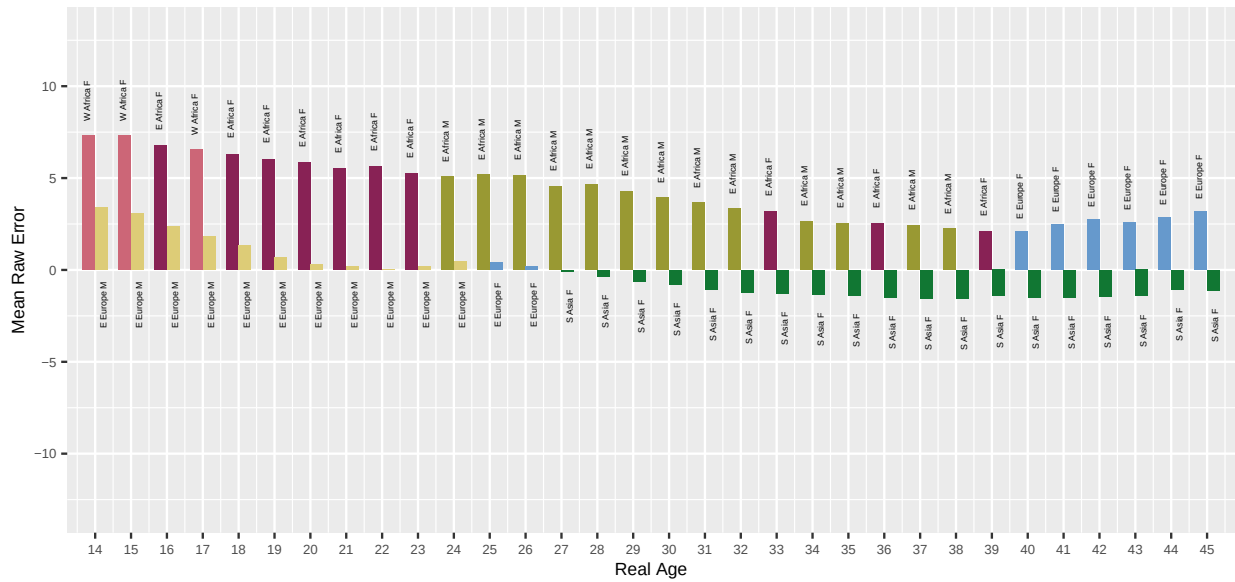




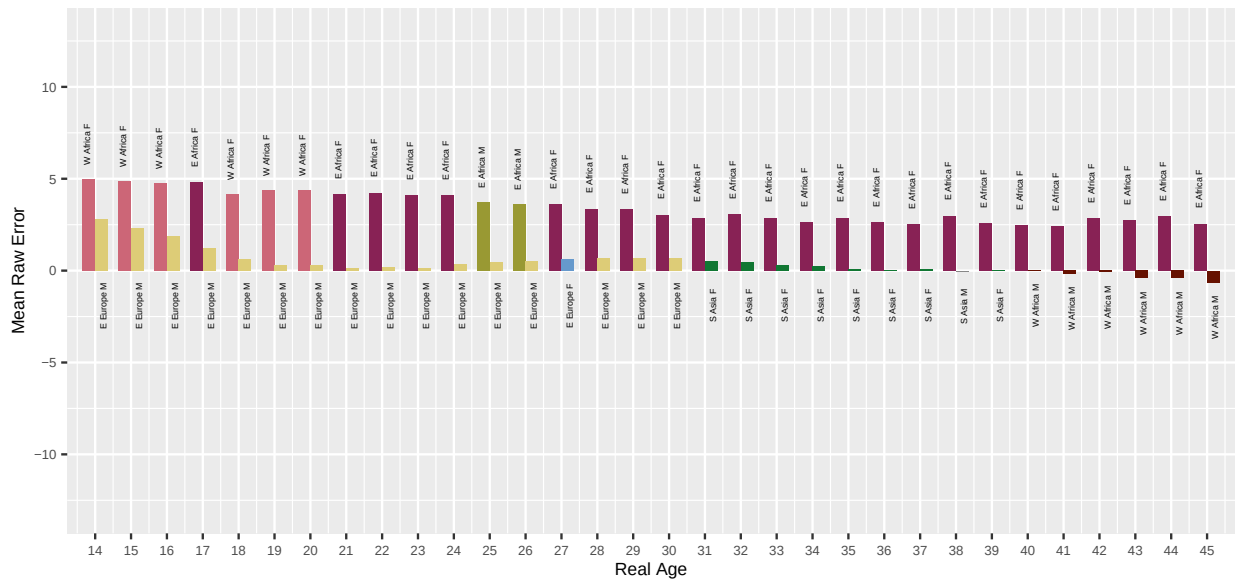
**Fig. 67.** For algorithm regula-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



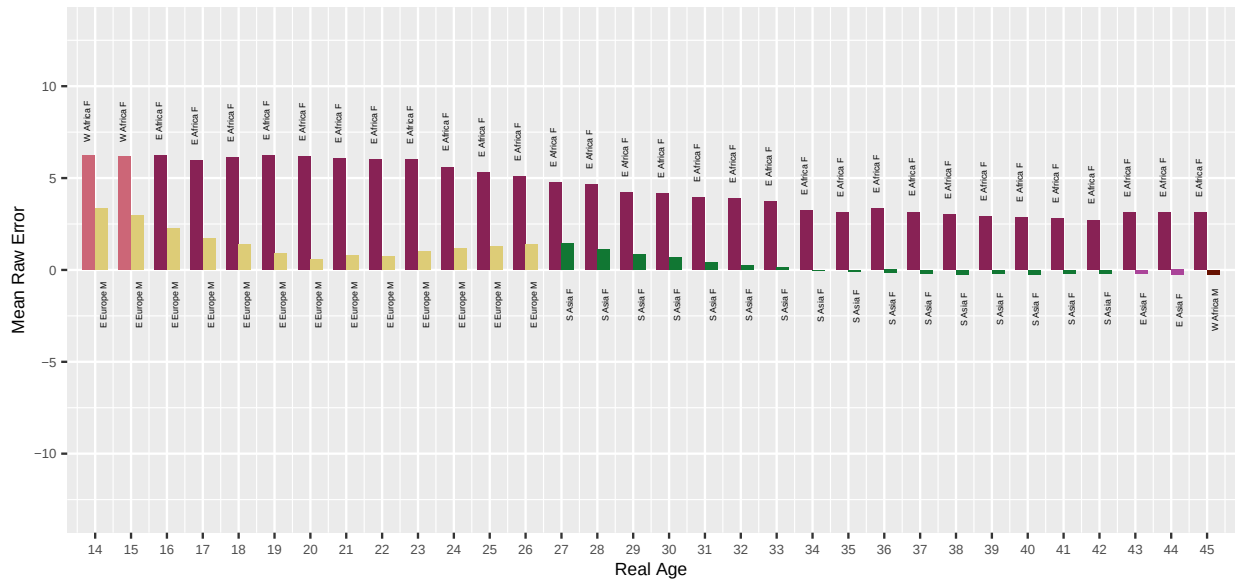
**Fig. 68.** For algorithm roc-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



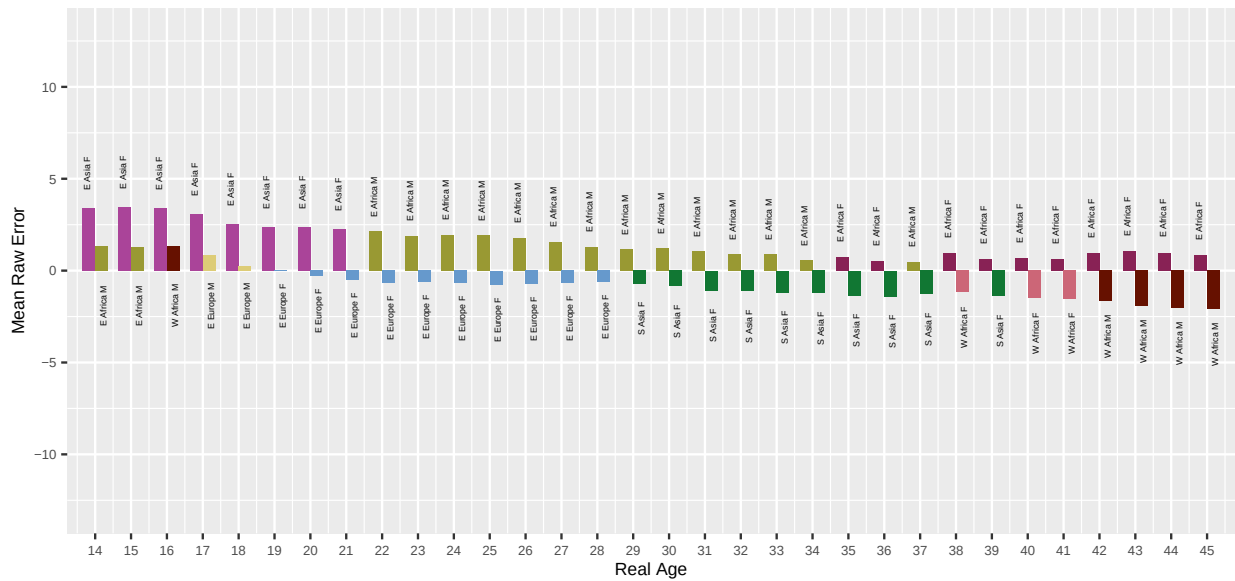
**Fig. 69.** For algorithm roc-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



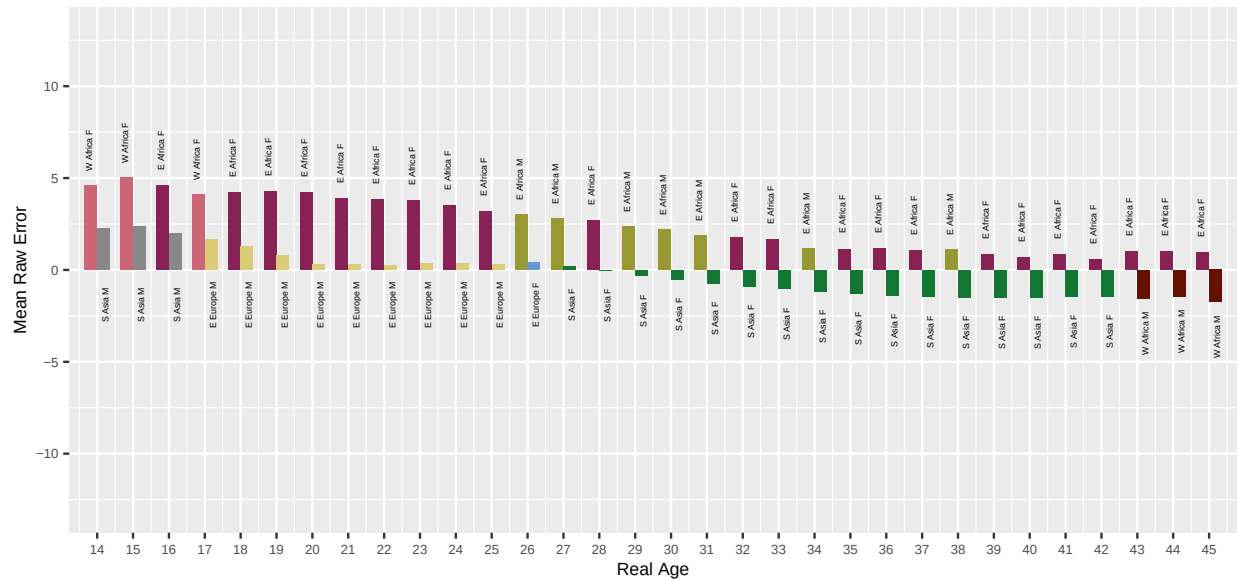
**Fig. 70.** For algorithm roc-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



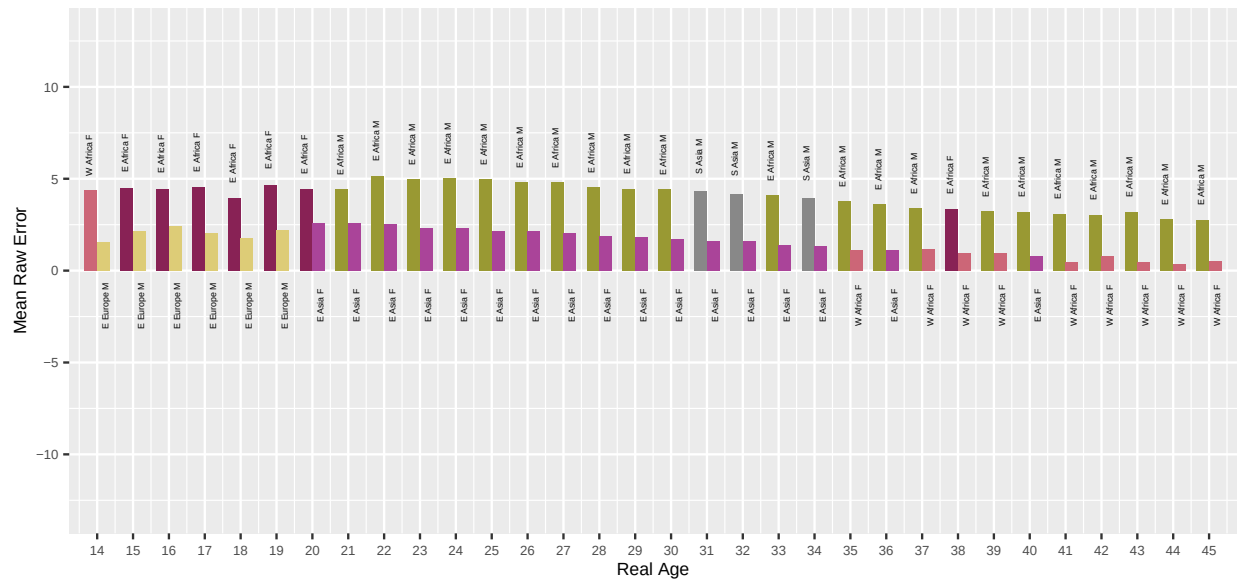
**Fig. 71.** For algorithm roc-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



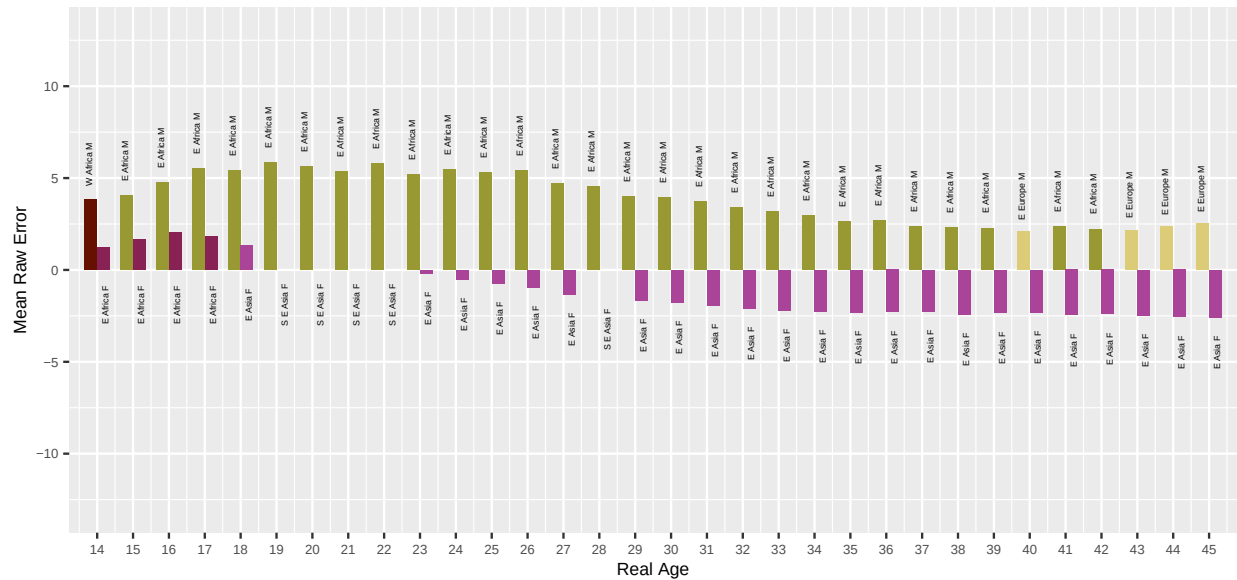
**Fig. 72.** For algorithm roc-002 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



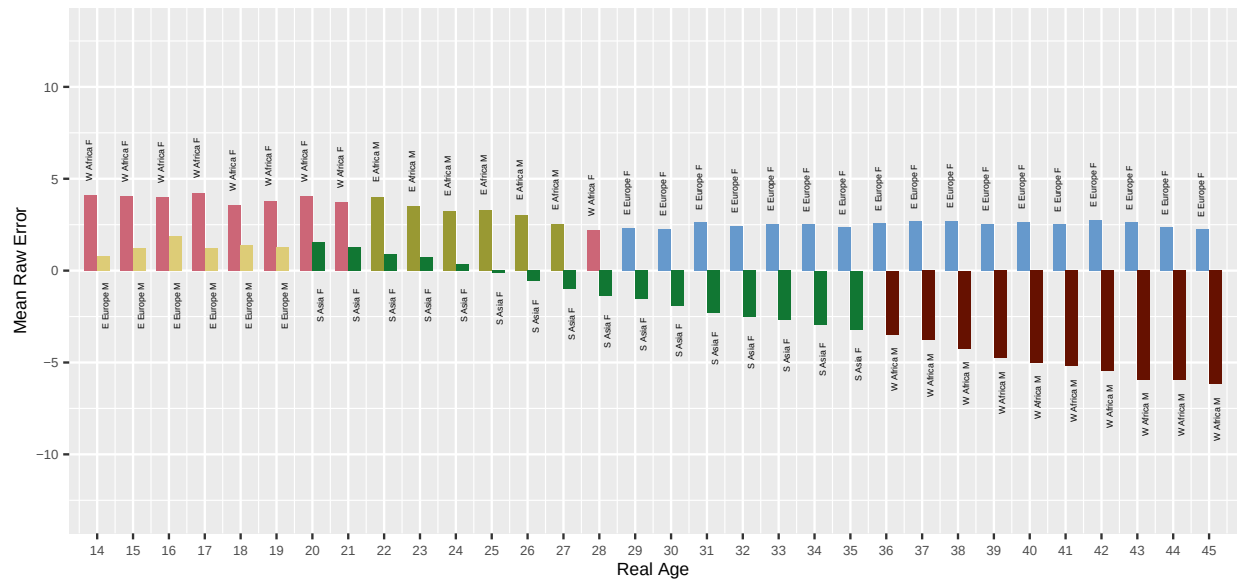
**Fig. 73.** For algorithm roc-002 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



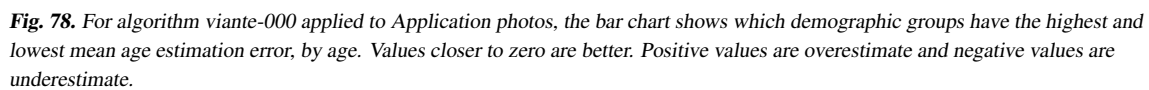
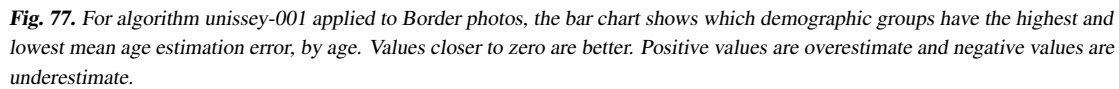
**Fig. 74.** For algorithm shufti-000 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

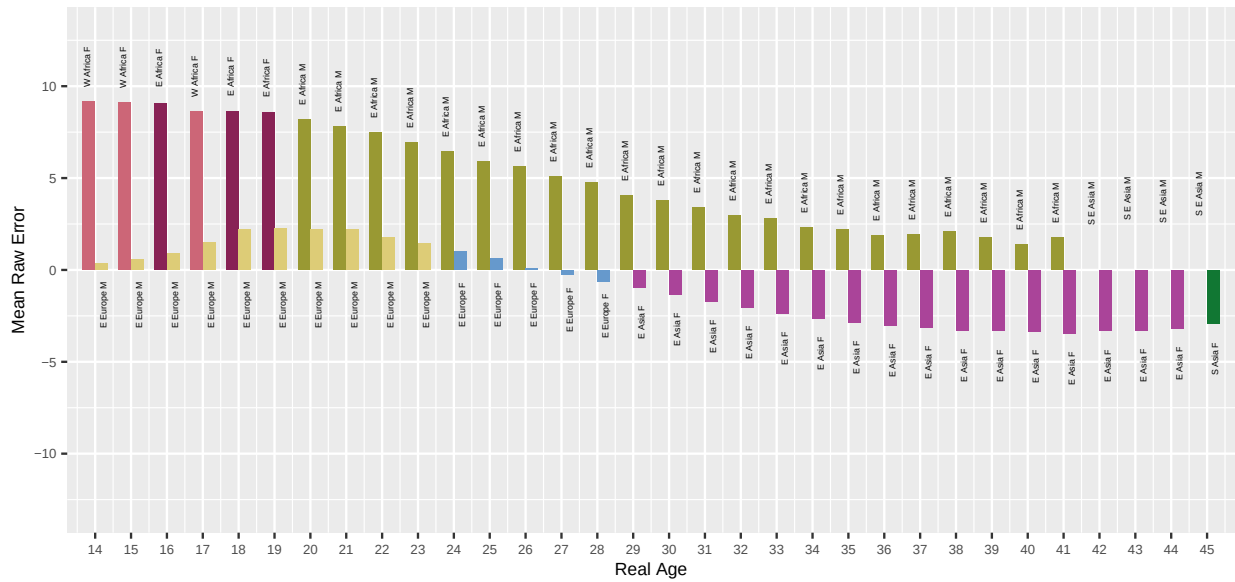


**Fig. 75.** For algorithm shufti-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

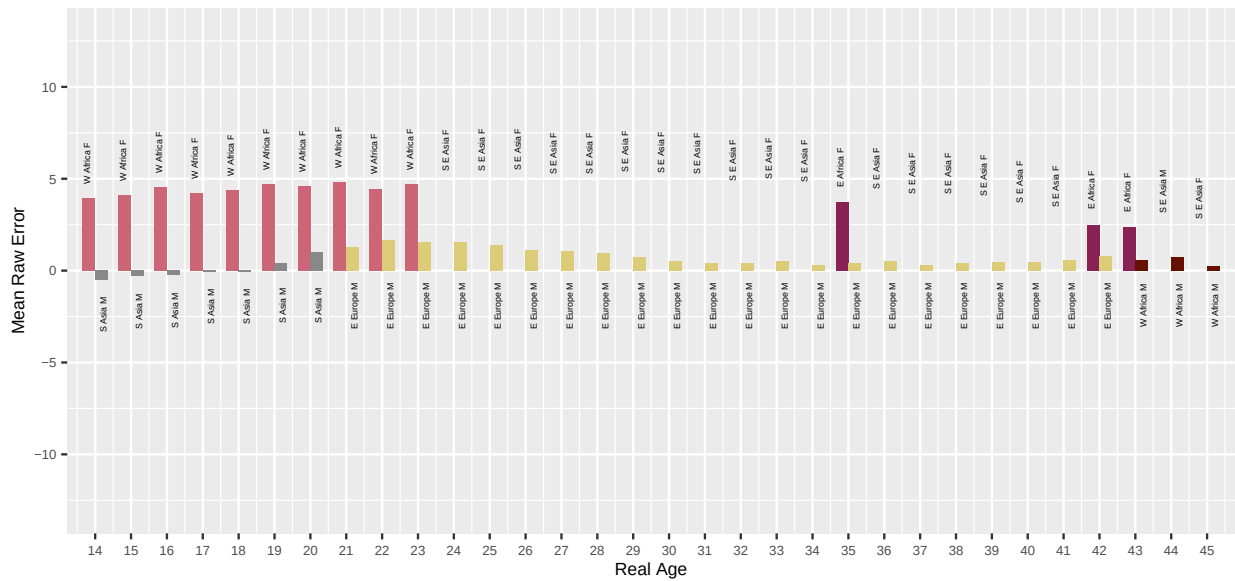


**Fig. 76.** For algorithm unissey-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

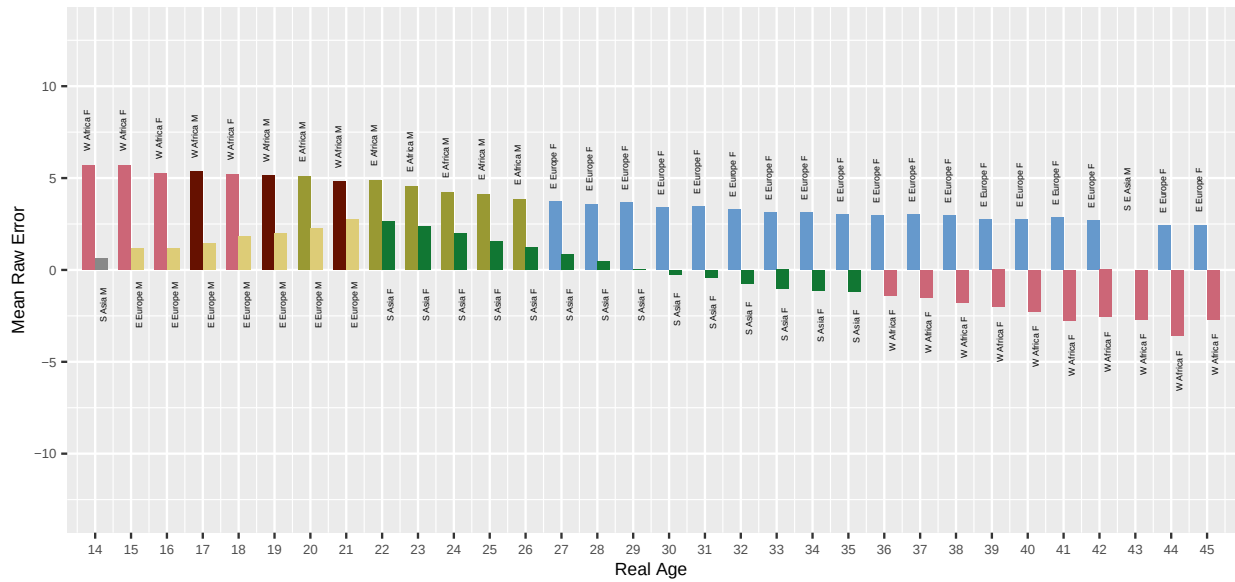




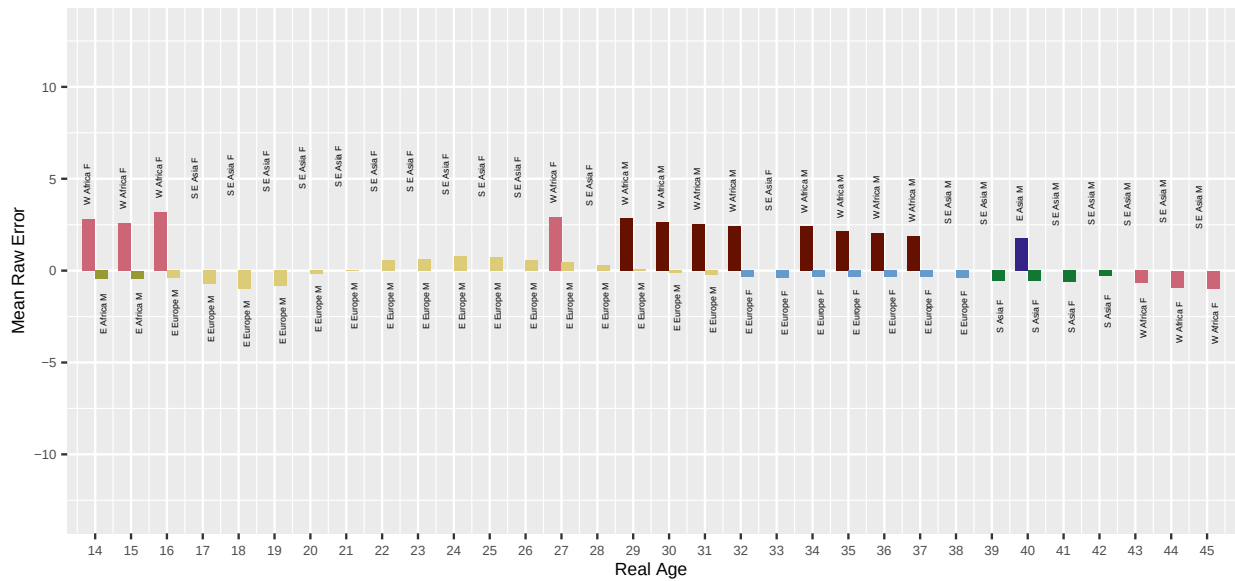
**Fig. 79.** For algorithm viante-000 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



**Fig. 80.** For algorithm yoti-001 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.

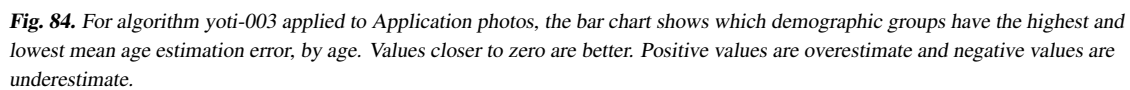
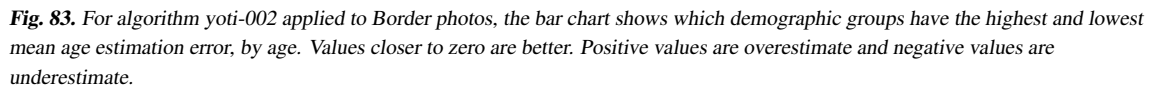


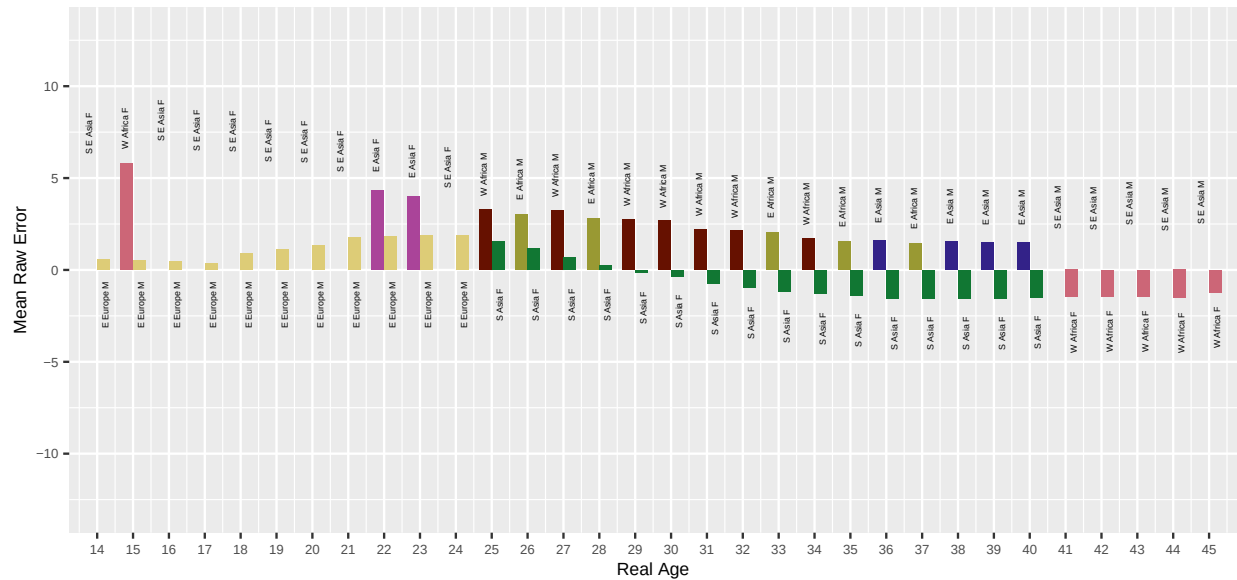
**Fig. 81.** For algorithm yoti-001 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



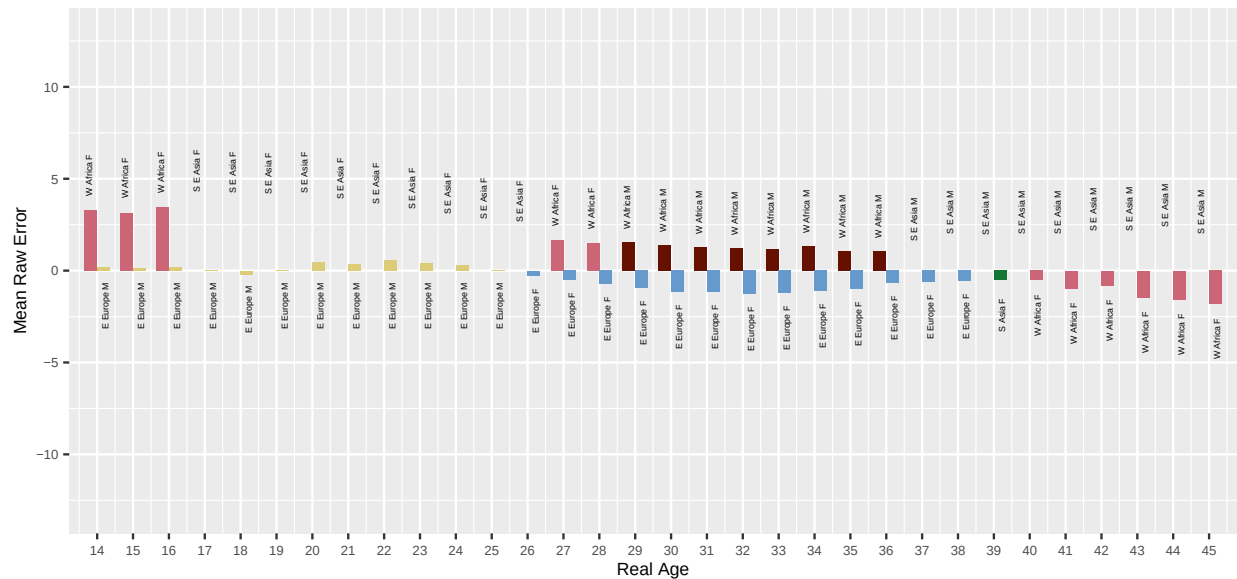
**Fig. 82.** For algorithm yoti-002 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



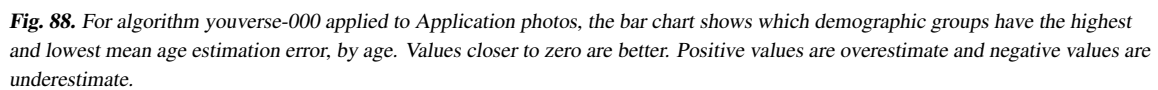
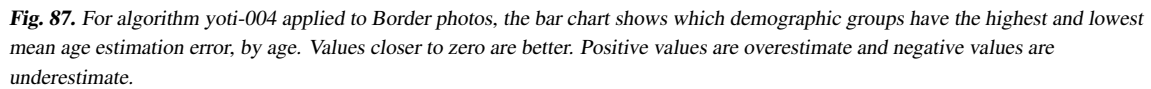


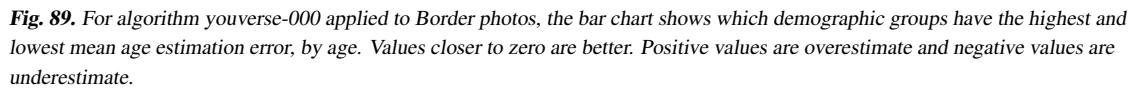


**Fig. 85.** For algorithm yoti-003 applied to Border photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.



**Fig. 86.** For algorithm yoti-004 applied to Application photos, the bar chart shows which demographic groups have the highest and lowest mean age estimation error, by age. Values closer to zero are better. Positive values are overestimate and negative values are underestimate.





## 5.5. Children

**Table 19.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample.

| Age | Num. Images | cognitec-000 |      |     | cvut-001 |      |     | cvut-002 |      |     | cvut-003 |      |       | cyberlink-000 |      |     | cyberlink-001 |      |     |
|-----|-------------|--------------|------|-----|----------|------|-----|----------|------|-----|----------|------|-------|---------------|------|-----|---------------|------|-----|
|     |             | Acc(3)       | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP   | Acc(3)        | MAE  | FTP | Acc(3)        | MAE  | FTP |
| 0   | 133387      | 0.688        | 0.87 | 0   | 0.868    | 0.58 | 0   | 0.915    | 0.53 | 0   | 0.975    | 0.41 | 0.001 | 0.039         | 5.54 | 0   | 0.160         | 3.45 | 0   |
| 1   | 94475       | 0.598        | 0.98 | 0   | 0.610    | 0.95 | 0   | 0.767    | 0.78 | 0   | 0.788    | 0.70 | 0.001 | 0.086         | 4.36 | 0   | 0.181         | 2.80 | 0   |
| 2   | 82596       | 0.516        | 1.11 | 0   | 0.543    | 1.18 | 0   | 0.438    | 1.13 | 0   | 0.505    | 1.04 | 0.001 | 0.073         | 4.78 | 0   | 0.109         | 3.68 | 0   |
| 3   | 82473       | 0.484        | 1.19 | 0   | 0.499    | 1.41 | 0   | 0.595    | 1.16 | 0   | 0.585    | 1.08 | 0.001 | 0.096         | 5.08 | 0   | 0.079         | 4.22 | 0   |
| 4   | 84055       | 0.503        | 1.16 | 0   | 0.397    | 1.62 | 0   | 0.437    | 1.32 | 0   | 0.509    | 1.17 | 0.001 | 0.149         | 5.25 | 0   | 0.069         | 4.51 | 0   |
| 5   | 92521       | 0.523        | 1.14 | 0   | 0.335    | 1.71 | 0   | 0.474    | 1.45 | 0   | 0.510    | 1.29 | 0.001 | 0.154         | 5.53 | 0   | 0.066         | 4.67 | 0   |
| 6   | 91639       | 0.521        | 1.18 | 0   | 0.416    | 1.78 | 0   | 0.423    | 1.71 | 0   | 0.473    | 1.46 | 0.001 | 0.112         | 5.96 | 0   | 0.057         | 4.85 | 0   |
| 7   | 89993       | 0.478        | 1.30 | 0   | 0.382    | 2.01 | 0   | 0.259    | 2.14 | 0   | 0.335    | 1.78 | 0.001 | 0.071         | 6.54 | 0   | 0.043         | 5.10 | 0   |
| 8   | 89428       | 0.471        | 1.34 | 0   | 0.252    | 2.26 | 0   | 0.170    | 2.38 | 0   | 0.245    | 1.94 | 0.001 | 0.061         | 6.46 | 0   | 0.050         | 5.03 | 0   |
| 9   | 89644       | 0.480        | 1.35 | 0   | 0.300    | 2.36 | 0   | 0.215    | 2.43 | 0   | 0.302    | 1.92 | 0.001 | 0.037         | 6.18 | 0   | 0.056         | 4.87 | 0   |
| 10  | 98337       | 0.475        | 1.40 | 0   | 0.212    | 2.38 | 0   | 0.156    | 2.43 | 0   | 0.292    | 1.91 | 0.001 | 0.020         | 5.80 | 0   | 0.050         | 4.70 | 0   |
| 11  | 97415       | 0.439        | 1.53 | 0   | 0.233    | 2.41 | 0   | 0.207    | 2.48 | 0   | 0.309    | 2.02 | 0.001 | 0.016         | 5.42 | 0   | 0.041         | 4.54 | 0   |
| 12  | 94326       | 0.390        | 1.74 | 0   | 0.258    | 2.53 | 0   | 0.220    | 2.60 | 0   | 0.328    | 2.24 | 0.001 | 0.028         | 5.11 | 0   | 0.035         | 4.41 | 0   |
| 13  | 94047       | 0.331        | 2.01 | 0   | 0.202    | 2.86 | 0   | 0.172    | 2.75 | 0   | 0.260    | 2.65 | 0.001 | 0.022         | 4.95 | 0   | 0.030         | 4.38 | 0   |
| 14  | 96915       | 0.296        | 2.23 | 0   | 0.240    | 3.34 | 0   | 0.092    | 2.85 | 0   | 0.186    | 3.11 | 0.001 | 0.011         | 4.84 | 0   | 0.033         | 4.34 | 0   |
| 15  | 97599       | 0.280        | 2.38 | 0   | 0.209    | 3.94 | 0   | 0.293    | 2.99 | 0   | 0.254    | 3.60 | 0.001 | 0.021         | 4.78 | 0   | 0.037         | 4.29 | 0   |
| 16  | 96388       | 0.273        | 2.47 | 0   | 0.133    | 4.36 | 0   | 0.270    | 3.27 | 0   | 0.238    | 4.11 | 0.001 | 0.040         | 4.63 | 0   | 0.054         | 4.13 | 0   |
| 17  | 99462       | 0.268        | 2.48 | 0   | 0.139    | 4.53 | 0   | 0.176    | 3.46 | 0   | 0.145    | 4.50 | 0.001 | 0.064         | 4.43 | 0   | 0.077         | 3.89 | 0   |

**Table 20.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample.

| Age | Num. Images | daon-000 |      |     | daon-001 |      |     | dermalog-001 |      |       | dermalog-002 |      |       | hzailu-001 |      |     | idemia-001 |      |     |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|------|-------|--------------|------|-------|------------|------|-----|------------|------|-----|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE  | FTP   | Acc(3)       | MAE  | FTP   | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP |
| 0   | 133387      | 0.004    | 4.70 | 0   | 0.008    | 3.49 | 0   | 0.183        | 2.05 | 0.001 | 0.047        | 3.00 | 0.001 | 0.397      | 1.85 | 0   | 0.759      | 1.00 | 0   |
| 1   | 94475       | 0.070    | 3.94 | 0   | 0.047    | 2.94 | 0   | 0.298        | 1.84 | 0.000 | 0.085        | 2.97 | 0.000 | 0.294      | 2.17 | 0   | 0.437      | 1.35 | 0   |
| 2   | 82596       | 0.106    | 3.75 | 0   | 0.089    | 2.78 | 0   | 0.400        | 1.89 | 0.000 | 0.123        | 2.97 | 0.000 | 0.215      | 2.68 | 0   | 0.366      | 1.49 | 0   |
| 3   | 82473       | 0.113    | 3.53 | 0   | 0.149    | 2.52 | 0   | 0.395        | 2.12 | 0.000 | 0.175        | 2.80 | 0.000 | 0.183      | 3.08 | 0   | 0.419      | 1.35 | 0   |
| 4   | 84055       | 0.132    | 3.19 | 0   | 0.220    | 2.17 | 0   | 0.353        | 2.37 | 0.000 | 0.223        | 2.62 | 0.000 | 0.176      | 3.30 | 0   | 0.549      | 1.09 | 0   |
| 5   | 92521       | 0.165    | 2.85 | 0   | 0.289    | 1.88 | 0   | 0.266        | 2.64 | 0.000 | 0.268        | 2.49 | 0.000 | 0.174      | 3.42 | 0   | 0.639      | 0.94 | 0   |
| 6   | 91639       | 0.183    | 2.68 | 0   | 0.326    | 1.72 | 0   | 0.174        | 2.98 | 0.000 | 0.273        | 2.59 | 0.000 | 0.164      | 3.74 | 0   | 0.635      | 0.94 | 0   |
| 7   | 89993       | 0.166    | 2.69 | 0   | 0.333    | 1.62 | 0   | 0.108        | 3.34 | 0.000 | 0.243        | 2.91 | 0.000 | 0.126      | 4.36 | 0   | 0.556      | 1.06 | 0   |
| 8   | 89428       | 0.191    | 2.46 | 0   | 0.410    | 1.41 | 0   | 0.116        | 3.27 | 0.000 | 0.255        | 2.94 | 0.000 | 0.130      | 4.56 | 0   | 0.482      | 1.22 | 0   |

**Table 20.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample. (continued)

| Age | Num. Images | daon-000 |      |     | daon-001 |      |     | dermalog-001 |      |       | dermalog-002 |      |       | hzailu-001 |      |     | idemia-001 |      |     |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|------|-------|--------------|------|-------|------------|------|-----|------------|------|-----|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE  | FTP   | Acc(3)       | MAE  | FTP   | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP |
| 9   | 89644       | 0.239    | 2.25 | 0   | 0.505    | 1.25 | 0   | 0.123        | 3.10 | 0.000 | 0.280        | 2.96 | 0.000 | 0.138      | 4.77 | 0   | 0.430      | 1.36 | 0   |
| 10  | 98337       | 0.305    | 2.11 | 0   | 0.550    | 1.21 | 0   | 0.168        | 2.84 | 0.000 | 0.298        | 3.00 | 0.000 | 0.147      | 5.04 | 0   | 0.392      | 1.50 | 0   |
| 11  | 97415       | 0.361    | 2.21 | 0   | 0.509    | 1.41 | 0   | 0.233        | 2.65 | 0.000 | 0.292        | 3.09 | 0.000 | 0.144      | 5.59 | 0   | 0.377      | 1.61 | 0   |
| 12  | 94326       | 0.350    | 2.66 | 0   | 0.411    | 1.86 | 0   | 0.290        | 2.60 | 0.000 | 0.259        | 3.28 | 0.000 | 0.122      | 6.44 | 0   | 0.369      | 1.72 | 0   |
| 13  | 94047       | 0.273    | 3.48 | 0   | 0.276    | 2.65 | 0   | 0.324        | 2.87 | 0.000 | 0.209        | 3.66 | 0.000 | 0.088      | 7.61 | 0   | 0.354      | 1.87 | 0   |
| 14  | 96915       | 0.177    | 4.35 | 0   | 0.189    | 3.50 | 0   | 0.274        | 3.43 | 0.000 | 0.164        | 4.10 | 0.000 | 0.055      | 8.66 | 0   | 0.331      | 2.06 | 0   |
| 15  | 97599       | 0.108    | 5.01 | 0   | 0.136    | 4.20 | 0   | 0.208        | 4.06 | 0.000 | 0.128        | 4.45 | 0.000 | 0.034      | 9.38 | 0   | 0.302      | 2.28 | 0   |
| 16  | 96388       | 0.084    | 5.27 | 0   | 0.104    | 4.58 | 0   | 0.158        | 4.50 | 0.000 | 0.111        | 4.60 | 0.000 | 0.022      | 9.59 | 0   | 0.274      | 2.46 | 0   |
| 17  | 99462       | 0.078    | 5.26 | 0   | 0.091    | 4.69 | 0   | 0.122        | 4.69 | 0.000 | 0.113        | 4.58 | 0.000 | 0.017      | 9.45 | 0   | 0.263      | 2.52 | 0   |

**Table 21.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample.

| Age | Num. Images | idemia-002 |      |     | inbut-000 |       |     | incode-000 |      |     | incode-001 |      |       | innovatrics-000 |      |     | innovatrics-001 |      |     |
|-----|-------------|------------|------|-----|-----------|-------|-----|------------|------|-----|------------|------|-------|-----------------|------|-----|-----------------|------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)    | MAE   | FTP | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP   | Acc(3)          | MAE  | FTP | Acc(3)          | MAE  | FTP |
| 0   | 133387      | 0.572      | 1.27 | 0   | 0.974     | 0.38  | 0   | 0.983      | 0.34 | 0   | 0.983      | 0.32 | 0.001 | 0.920           | 0.74 | 0   | 0.989           | 0.36 | 0   |
| 1   | 94475       | 0.379      | 1.59 | 0   | 0.700     | 0.85  | 0   | 0.895      | 0.63 | 0   | 0.905      | 0.57 | 0.000 | 0.713           | 0.79 | 0   | 0.823           | 0.61 | 0   |
| 2   | 82596       | 0.289      | 1.83 | 0   | 0.381     | 1.41  | 0   | 0.614      | 0.92 | 0   | 0.735      | 0.77 | 0.000 | 0.415           | 1.31 | 0   | 0.522           | 1.12 | 0   |
| 3   | 82473       | 0.280      | 1.78 | 0   | 0.300     | 1.55  | 0   | 0.538      | 1.10 | 0   | 0.762      | 0.75 | 0.000 | 0.324           | 1.48 | 0   | 0.382           | 1.34 | 0   |
| 4   | 84055       | 0.359      | 1.56 | 0   | 0.353     | 1.45  | 0   | 0.492      | 1.24 | 0   | 0.720      | 0.83 | 0.000 | 0.357           | 1.41 | 0   | 0.425           | 1.29 | 0   |
| 5   | 92521       | 0.461      | 1.34 | 0   | 0.470     | 1.35  | 0   | 0.478      | 1.34 | 0   | 0.576      | 1.07 | 0.000 | 0.451           | 1.32 | 0   | 0.501           | 1.24 | 0   |
| 6   | 91639       | 0.526      | 1.23 | 0   | 0.487     | 1.41  | 0   | 0.489      | 1.36 | 0   | 0.497      | 1.27 | 0.000 | 0.483           | 1.36 | 0   | 0.492           | 1.35 | 0   |
| 7   | 89993       | 0.527      | 1.22 | 0   | 0.415     | 1.71  | 0   | 0.508      | 1.27 | 0   | 0.484      | 1.38 | 0.000 | 0.404           | 1.62 | 0   | 0.402           | 1.64 | 0   |
| 8   | 89428       | 0.477      | 1.30 | 0   | 0.342     | 2.13  | 0   | 0.449      | 1.42 | 0   | 0.413      | 1.77 | 0.000 | 0.329           | 1.86 | 0   | 0.317           | 1.91 | 0   |
| 9   | 89644       | 0.440      | 1.34 | 0   | 0.280     | 2.68  | 0   | 0.406      | 1.56 | 0   | 0.348      | 2.29 | 0.000 | 0.290           | 2.03 | 0   | 0.276           | 2.09 | 0   |
| 10  | 98337       | 0.455      | 1.33 | 0   | 0.225     | 3.54  | 0   | 0.373      | 1.71 | 0   | 0.296      | 2.84 | 0.000 | 0.268           | 2.19 | 0   | 0.262           | 2.26 | 0   |
| 11  | 97415       | 0.471      | 1.36 | 0   | 0.182     | 4.94  | 0   | 0.351      | 1.87 | 0   | 0.251      | 3.36 | 0.000 | 0.250           | 2.45 | 0   | 0.241           | 2.55 | 0   |
| 12  | 94326       | 0.462      | 1.46 | 0   | 0.129     | 6.83  | 0   | 0.399      | 2.03 | 0   | 0.218      | 3.73 | 0.000 | 0.210           | 2.79 | 0   | 0.189           | 2.93 | 0   |
| 13  | 94047       | 0.425      | 1.71 | 0   | 0.073     | 8.89  | 0   | 0.345      | 2.31 | 0   | 0.146      | 3.90 | 0.000 | 0.151           | 3.16 | 0   | 0.124           | 3.32 | 0   |
| 14  | 96915       | 0.358      | 2.07 | 0   | 0.032     | 10.41 | 0   | 0.220      | 2.58 | 0   | 0.068      | 3.81 | 0.000 | 0.102           | 3.42 | 0   | 0.077           | 3.56 | 0   |
| 15  | 97599       | 0.301      | 2.47 | 0   | 0.014     | 11.34 | 0   | 0.171      | 2.71 | 0   | 0.031      | 3.52 | 0.000 | 0.082           | 3.54 | 0   | 0.059           | 3.65 | 0   |
| 16  | 96388       | 0.255      | 2.80 | 0   | 0.011     | 11.74 | 0   | 0.207      | 2.73 | 0   | 0.019      | 3.18 | 0.000 | 0.084           | 3.55 | 0   | 0.067           | 3.59 | 0   |
| 17  | 99462       | 0.227      | 2.98 | 0   | 0.014     | 11.74 | 0   | 0.303      | 2.69 | 0   | 0.141      | 2.87 | 0.000 | 0.112           | 3.48 | 0   | 0.104           | 3.44 | 0   |

**Table 22.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample.

| Age | Num. Images | innovatrics-002 |      |     | jumio-000 |       |       | neuro-000 |      |     | neuro-001 |      |     | nominder-000 |      |     | nominder-001 |      |     |
|-----|-------------|-----------------|------|-----|-----------|-------|-------|-----------|------|-----|-----------|------|-----|--------------|------|-----|--------------|------|-----|
|     |             | Acc(3)          | MAE  | FTP | Acc(3)    | MAE   | FTP   | Acc(3)    | MAE  | FTP | Acc(3)    | MAE  | FTP | Acc(3)       | MAE  | FTP | Acc(3)       | MAE  | FTP |
| 0   | 133387      | 0.963           | 0.41 | 0   | 0.000     | 18.43 | 0.018 | 0.469     | 1.23 | 0   | 0.429     | 1.87 | 0   | 0.906        | 0.68 | 0   | 0.915        | 0.66 | 0   |
| 1   | 94475       | 0.759           | 0.74 | 0   | 0.000     | 15.25 | 0.009 | 0.582     | 1.11 | 0   | 0.508     | 1.82 | 0   | 0.644        | 1.07 | 0   | 0.661        | 1.04 | 0   |
| 2   | 82596       | 0.512           | 1.18 | 0   | 0.000     | 13.27 | 0.011 | 0.559     | 1.25 | 0   | 0.498     | 1.80 | 0   | 0.321        | 1.80 | 0   | 0.294        | 1.87 | 0   |
| 3   | 82473       | 0.345           | 1.48 | 0   | 0.001     | 10.99 | 0.011 | 0.467     | 1.50 | 0   | 0.425     | 1.95 | 0   | 0.221        | 1.90 | 0   | 0.195        | 1.89 | 0   |
| 4   | 84055       | 0.340           | 1.48 | 0   | 0.006     | 9.37  | 0.015 | 0.412     | 1.64 | 0   | 0.376     | 2.09 | 0   | 0.268        | 1.75 | 0   | 0.272        | 1.57 | 0   |
| 5   | 92521       | 0.420           | 1.41 | 0   | 0.020     | 8.06  | 0.021 | 0.389     | 1.76 | 0   | 0.352     | 2.22 | 0   | 0.496        | 1.74 | 0   | 0.616        | 1.38 | 0   |
| 6   | 91639       | 0.444           | 1.49 | 0   | 0.044     | 7.29  | 0.028 | 0.365     | 1.96 | 0   | 0.325     | 2.49 | 0   | 0.408        | 2.12 | 0   | 0.533        | 1.65 | 0   |
| 7   | 89993       | 0.360           | 1.77 | 0   | 0.077     | 6.84  | 0.038 | 0.329     | 2.39 | 0   | 0.293     | 3.00 | 0   | 0.202        | 2.69 | 0   | 0.294        | 2.08 | 0   |
| 8   | 89428       | 0.281           | 1.98 | 0   | 0.095     | 6.83  | 0.032 | 0.302     | 2.83 | 0   | 0.269     | 3.42 | 0   | 0.149        | 2.91 | 0   | 0.248        | 2.46 | 0   |
| 9   | 89644       | 0.252           | 2.05 | 0   | 0.100     | 7.00  | 0.028 | 0.261     | 3.25 | 0   | 0.232     | 3.79 | 0   | 0.128        | 2.94 | 0   | 0.220        | 2.72 | 0   |
| 10  | 98337       | 0.256           | 2.02 | 0   | 0.085     | 7.10  | 0.030 | 0.206     | 3.68 | 0   | 0.183     | 4.15 | 0   | 0.117        | 2.91 | 0   | 0.195        | 2.96 | 0   |
| 11  | 97415       | 0.260           | 2.01 | 0   | 0.063     | 7.04  | 0.032 | 0.165     | 4.16 | 0   | 0.142     | 4.56 | 0   | 0.122        | 2.84 | 0   | 0.180        | 3.22 | 0   |
| 12  | 94326       | 0.235           | 1.99 | 0   | 0.043     | 6.68  | 0.038 | 0.122     | 4.68 | 0   | 0.105     | 4.99 | 0   | 0.137        | 2.78 | 0   | 0.164        | 3.54 | 0   |
| 13  | 94047       | 0.226           | 1.96 | 0   | 0.032     | 6.13  | 0.041 | 0.084     | 5.17 | 0   | 0.071     | 5.37 | 0   | 0.182        | 2.82 | 0   | 0.145        | 3.92 | 0   |
| 14  | 96915       | 0.321           | 1.91 | 0   | 0.024     | 5.62  | 0.041 | 0.051     | 5.34 | 0   | 0.044     | 5.46 | 0   | 0.233        | 2.91 | 0   | 0.114        | 4.17 | 0   |
| 15  | 97599       | 0.425           | 1.95 | 0   | 0.022     | 5.24  | 0.039 | 0.035     | 5.29 | 0   | 0.030     | 5.36 | 0   | 0.236        | 3.11 | 0   | 0.079        | 4.31 | 0   |
| 16  | 96388       | 0.445           | 2.10 | 0   | 0.026     | 4.94  | 0.035 | 0.035     | 4.96 | 0   | 0.031     | 5.01 | 0   | 0.217        | 3.28 | 0   | 0.066        | 4.28 | 0   |
| 17  | 99462       | 0.368           | 2.35 | 0   | 0.072     | 4.62  | 0.033 | 0.047     | 4.46 | 0   | 0.043     | 4.50 | 0   | 0.207        | 3.36 | 0   | 0.084        | 4.12 | 0   |

**Table 23.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample.

| Age | Num. Images | ondato-000 |      |     | ondato-001 |      |     | privately-000 |      |     | regula-000 |      |     | roc-000 |      |     | roc-001 |      |     |
|-----|-------------|------------|------|-----|------------|------|-----|---------------|------|-----|------------|------|-----|---------|------|-----|---------|------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)        | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)  | MAE  | FTP | Acc(3)  | MAE  | FTP |
| 0   | 133387      | 0.328      | 2.29 | 0   | 0.281      | 1.66 | 0   | 0.352         | 1.40 | 0   | 0.413      | 1.28 | 0   | 0.270   | 2.28 | 0   | 0.000   | 7.75 | 0   |
| 1   | 94475       | 0.439      | 1.68 | 0   | 0.275      | 1.76 | 0   | 0.497         | 1.30 | 0   | 0.934      | 0.46 | 0   | 0.326   | 1.80 | 0   | 0.000   | 7.52 | 0   |
| 2   | 82596       | 0.340      | 2.13 | 0   | 0.193      | 2.04 | 0   | 0.421         | 1.81 | 0   | 0.862      | 0.56 | 0   | 0.335   | 1.78 | 0   | 0.000   | 7.17 | 0   |
| 3   | 82473       | 0.281      | 2.52 | 0   | 0.216      | 2.18 | 0   | 0.265         | 2.25 | 0   | 0.796      | 0.66 | 0   | 0.342   | 1.75 | 0   | 0.000   | 6.87 | 0   |
| 4   | 84055       | 0.251      | 2.77 | 0   | 0.272      | 2.22 | 0   | 0.155         | 2.28 | 0   | 0.776      | 0.68 | 0   | 0.370   | 1.64 | 0   | 0.001   | 6.58 | 0   |
| 5   | 92521       | 0.234      | 2.94 | 0   | 0.279      | 2.34 | 0   | 0.183         | 2.05 | 0   | 0.737      | 0.74 | 0   | 0.396   | 1.59 | 0   | 0.001   | 6.32 | 0   |
| 6   | 91639       | 0.198      | 3.25 | 0   | 0.245      | 2.53 | 0   | 0.319         | 1.74 | 0   | 0.670      | 0.85 | 0   | 0.412   | 1.60 | 0   | 0.001   | 5.92 | 0   |
| 7   | 89993       | 0.157      | 3.65 | 0   | 0.204      | 2.82 | 0   | 0.449         | 1.52 | 0   | 0.622      | 0.91 | 0   | 0.416   | 1.66 | 0   | 0.002   | 5.22 | 0   |
| 8   | 89428       | 0.138      | 3.88 | 0   | 0.192      | 3.01 | 0   | 0.483         | 1.48 | 0   | 0.593      | 0.97 | 0   | 0.375   | 1.88 | 0   | 0.002   | 5.04 | 0   |
| 9   | 89644       | 0.126      | 4.00 | 0   | 0.193      | 3.14 | 0   | 0.486         | 1.52 | 0   | 0.595      | 1.00 | 0   | 0.312   | 2.19 | 0   | 0.002   | 4.81 | 0   |
| 10  | 98337       | 0.119      | 3.99 | 0   | 0.190      | 3.23 | 0   | 0.452         | 1.64 | 0   | 0.575      | 1.06 | 0   | 0.243   | 2.64 | 0   | 0.003   | 4.59 | 0   |

**Table 23.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample. (continued)

| Age | Num. Images | ondato-000 |      |     | ondato-001 |      |     | privately-000 |      |     | regula-000 |      |     | roc-000 |      |     | roc-001 |      |     |
|-----|-------------|------------|------|-----|------------|------|-----|---------------|------|-----|------------|------|-----|---------|------|-----|---------|------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)        | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)  | MAE  | FTP | Acc(3)  | MAE  | FTP |
| 11  | 97415       | 0.110      | 3.97 | 0   | 0.170      | 3.40 | 0   | 0.394         | 1.84 | 0   | 0.514      | 1.23 | 0   | 0.185   | 3.18 | 0   | 0.004   | 4.39 | 0   |
| 12  | 94326       | 0.096      | 3.85 | 0   | 0.140      | 3.55 | 0   | 0.369         | 2.00 | 0   | 0.437      | 1.46 | 0   | 0.124   | 3.76 | 0   | 0.005   | 4.18 | 0   |
| 13  | 94047       | 0.082      | 3.70 | 0   | 0.108      | 3.63 | 0   | 0.380         | 2.18 | 0   | 0.370      | 1.73 | 0   | 0.072   | 4.18 | 0   | 0.007   | 4.00 | 0   |
| 14  | 96915       | 0.085      | 3.50 | 0   | 0.089      | 3.60 | 0   | 0.368         | 2.41 | 0   | 0.314      | 1.98 | 0   | 0.037   | 4.20 | 0   | 0.009   | 3.81 | 0   |
| 15  | 97599       | 0.118      | 3.36 | 0   | 0.096      | 3.52 | 0   | 0.272         | 2.76 | 0   | 0.291      | 2.13 | 0   | 0.019   | 4.00 | 0   | 0.016   | 3.63 | 0   |
| 16  | 96388       | 0.173      | 3.25 | 0   | 0.120      | 3.44 | 0   | 0.194         | 3.05 | 0   | 0.289      | 2.21 | 0   | 0.023   | 3.65 | 0   | 0.044   | 3.47 | 0   |
| 17  | 99462       | 0.217      | 3.16 | 0   | 0.160      | 3.32 | 0   | 0.180         | 3.16 | 0   | 0.323      | 2.20 | 0   | 0.104   | 3.29 | 0   | 0.129   | 3.35 | 0   |

**Table 24.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample.

| Age | Num. Images | roc-002 |      |     | shufti-000 |      |       | unissey-001 |      |     | viante-000 |      |     | yoti-001 |      |       | yoti-002 |      |     |
|-----|-------------|---------|------|-----|------------|------|-------|-------------|------|-----|------------|------|-----|----------|------|-------|----------|------|-----|
|     |             | Acc(3)  | MAE  | FTP | Acc(3)     | MAE  | FTP   | Acc(3)      | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)   | MAE  | FTP   | Acc(3)   | MAE  | FTP |
| 0   | 133387      | 0.859   | 0.61 | 0   | 0.013      | 3.01 | 0.001 | 0.962       | 0.60 | 0   | 0.687      | 0.82 | 0   | 0.418    | 1.39 | 0.012 | 0.690    | 1.11 | 0   |
| 1   | 94475       | 0.872   | 0.56 | 0   | 0.014      | 3.06 | 0.000 | 0.229       | 1.31 | 0   | 0.461      | 1.16 | 0   | 0.396    | 1.46 | 0.005 | 0.759    | 0.96 | 0   |
| 2   | 82596       | 0.780   | 0.69 | 0   | 0.019      | 3.17 | 0.000 | 0.414       | 1.32 | 0   | 0.445      | 1.38 | 0   | 0.303    | 1.73 | 0.004 | 0.509    | 1.26 | 0   |
| 3   | 82473       | 0.696   | 0.83 | 0   | 0.044      | 3.11 | 0.000 | 0.577       | 1.24 | 0   | 0.248      | 1.66 | 0   | 0.249    | 1.82 | 0.003 | 0.392    | 1.45 | 0   |
| 4   | 84055       | 0.649   | 0.91 | 0   | 0.109      | 2.88 | 0.000 | 0.477       | 1.36 | 0   | 0.247      | 1.65 | 0   | 0.289    | 1.67 | 0.003 | 0.382    | 1.41 | 0   |
| 5   | 92521       | 0.610   | 1.00 | 0   | 0.218      | 2.60 | 0.000 | 0.502       | 1.60 | 0   | 0.352      | 1.68 | 0   | 0.367    | 1.51 | 0.002 | 0.477    | 1.28 | 0   |
| 6   | 91639       | 0.577   | 1.08 | 0   | 0.331      | 2.49 | 0.000 | 0.332       | 2.02 | 0   | 0.365      | 1.93 | 0   | 0.475    | 1.34 | 0.002 | 0.545    | 1.22 | 0   |
| 7   | 89993       | 0.563   | 1.10 | 0   | 0.343      | 2.61 | 0.000 | 0.214       | 2.49 | 0   | 0.252      | 2.43 | 0   | 0.529    | 1.30 | 0.001 | 0.520    | 1.34 | 0   |
| 8   | 89428       | 0.507   | 1.22 | 0   | 0.323      | 2.76 | 0.000 | 0.251       | 2.71 | 0   | 0.161      | 2.57 | 0   | 0.490    | 1.42 | 0.000 | 0.463    | 1.47 | 0   |
| 9   | 89644       | 0.477   | 1.30 | 0   | 0.270      | 2.94 | 0.000 | 0.226       | 2.81 | 0   | 0.106      | 2.35 | 0   | 0.447    | 1.55 | 0.000 | 0.437    | 1.49 | 0   |
| 10  | 98337       | 0.439   | 1.43 | 0   | 0.217      | 3.04 | 0.000 | 0.162       | 2.80 | 0   | 0.083      | 1.97 | 0   | 0.415    | 1.68 | 0.000 | 0.442    | 1.49 | 0   |
| 11  | 97415       | 0.394   | 1.60 | 0   | 0.166      | 3.19 | 0.000 | 0.107       | 2.79 | 0   | 0.266      | 1.62 | 0   | 0.391    | 1.80 | 0.000 | 0.450    | 1.49 | 0   |
| 12  | 94326       | 0.340   | 1.78 | 0   | 0.127      | 3.26 | 0.000 | 0.261       | 2.81 | 0   | 0.453      | 1.70 | 0   | 0.340    | 1.97 | 0.000 | 0.434    | 1.56 | 0   |
| 13  | 94047       | 0.279   | 1.92 | 0   | 0.105      | 3.28 | 0.000 | 0.198       | 3.07 | 0   | 0.493      | 2.57 | 0   | 0.298    | 2.17 | 0.000 | 0.386    | 1.69 | 0   |
| 14  | 96915       | 0.256   | 1.93 | 0   | 0.114      | 3.22 | 0.000 | 0.222       | 3.36 | 0   | 0.294      | 3.99 | 0   | 0.299    | 2.35 | 0.000 | 0.349    | 1.82 | 0   |
| 15  | 97599       | 0.290   | 1.82 | 0   | 0.115      | 3.11 | 0.000 | 0.200       | 3.67 | 0   | 0.164      | 5.38 | 0   | 0.294    | 2.61 | 0.000 | 0.359    | 1.94 | 0   |
| 16  | 96388       | 0.386   | 1.65 | 0   | 0.165      | 3.00 | 0.000 | 0.152       | 3.86 | 0   | 0.100      | 6.19 | 0   | 0.292    | 2.85 | 0.000 | 0.389    | 2.06 | 0   |
| 17  | 99462       | 0.478   | 1.53 | 0   | 0.283      | 2.85 | 0.000 | 0.166       | 3.85 | 0   | 0.052      | 6.54 | 0   | 0.287    | 3.06 | 0.000 | 0.382    | 2.22 | 0   |



**Table 25.** AE error statistics in children age 0 to 17. Acc(1) is the proportion of age estimates within 1 years (higher value is better), MAE is mean absolute error (lower value is better), and FTP is the proportion of images the algorithm failed to process (lower is better). Dataset is the Mexican visa sample.

| Age | Num. Images | yoti-003 |      |     | yoti-004 |      |     | youverse-000 |      |       |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|------|-------|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE  | FTP   |
| 0   | 133387      | 0.859    | 1.42 | 0   | 0.812    | 1.14 | 0   | 0.211        | 3.31 | 0.026 |
| 1   | 94475       | 0.729    | 1.27 | 0   | 0.633    | 1.09 | 0   | 0.131        | 3.88 | 0.006 |
| 2   | 82596       | 0.473    | 1.45 | 0   | 0.488    | 1.23 | 0   | 0.085        | 4.64 | 0.003 |
| 3   | 82473       | 0.345    | 1.56 | 0   | 0.445    | 1.28 | 0   | 0.063        | 5.04 | 0.003 |
| 4   | 84055       | 0.372    | 1.42 | 0   | 0.493    | 1.18 | 0   | 0.054        | 5.31 | 0.002 |
| 5   | 92521       | 0.495    | 1.22 | 0   | 0.566    | 1.08 | 0   | 0.050        | 5.47 | 0.002 |
| 6   | 91639       | 0.577    | 1.14 | 0   | 0.584    | 1.09 | 0   | 0.041        | 5.75 | 0.001 |
| 7   | 89993       | 0.546    | 1.28 | 0   | 0.522    | 1.26 | 0   | 0.030        | 6.18 | 0.001 |
| 8   | 89428       | 0.480    | 1.44 | 0   | 0.458    | 1.41 | 0   | 0.024        | 6.27 | 0.001 |
| 9   | 89644       | 0.429    | 1.55 | 0   | 0.420    | 1.49 | 0   | 0.022        | 6.13 | 0.000 |
| 10  | 98337       | 0.377    | 1.65 | 0   | 0.402    | 1.56 | 0   | 0.021        | 5.86 | 0.000 |
| 11  | 97415       | 0.333    | 1.73 | 0   | 0.385    | 1.65 | 0   | 0.017        | 5.53 | 0.000 |
| 12  | 94326       | 0.322    | 1.78 | 0   | 0.351    | 1.77 | 0   | 0.012        | 5.13 | 0.000 |
| 13  | 94047       | 0.359    | 1.77 | 0   | 0.331    | 1.90 | 0   | 0.011        | 4.69 | 0.000 |
| 14  | 96915       | 0.401    | 1.76 | 0   | 0.335    | 1.99 | 0   | 0.013        | 4.32 | 0.000 |
| 15  | 97599       | 0.422    | 1.85 | 0   | 0.347    | 2.11 | 0   | 0.034        | 3.99 | 0.000 |
| 16  | 96388       | 0.415    | 2.02 | 0   | 0.334    | 2.27 | 0   | 0.114        | 3.67 | 0.000 |
| 17  | 99462       | 0.377    | 2.23 | 0   | 0.308    | 2.39 | 0   | 0.207        | 3.39 | 0.000 |

Table 25 shows MAE and two other performance statistics for a large dataset of *Visa* images of children. The most accurate algorithm, Incode-000, is markedly more accurate than others on babies and infants. It estimates the age of children less than 1 year old with MAE 0.34, i.e. about four months. This rises almost monotonically through 17 years. All other algorithms exhibit a similar trend.

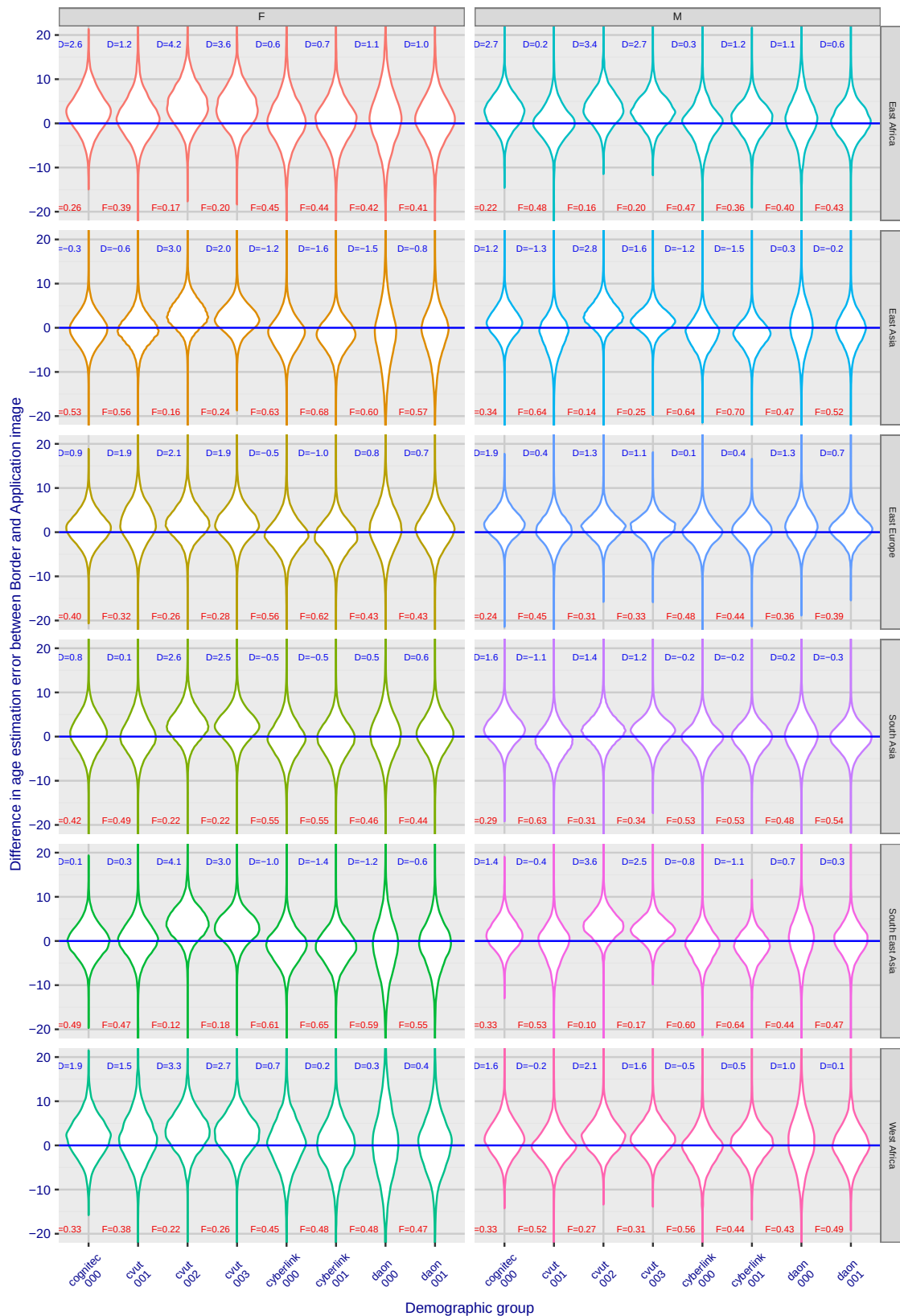
## 5.6. Image Quality

Clearly age estimation accuracy will depend on the quality of the images. If the image is out of focus, or underexposed, or has some other properties that sufficiently depart from standardized portrait format, then either the age estimation algorithms will elect to not process the image, or it will emit a degraded age estimate.

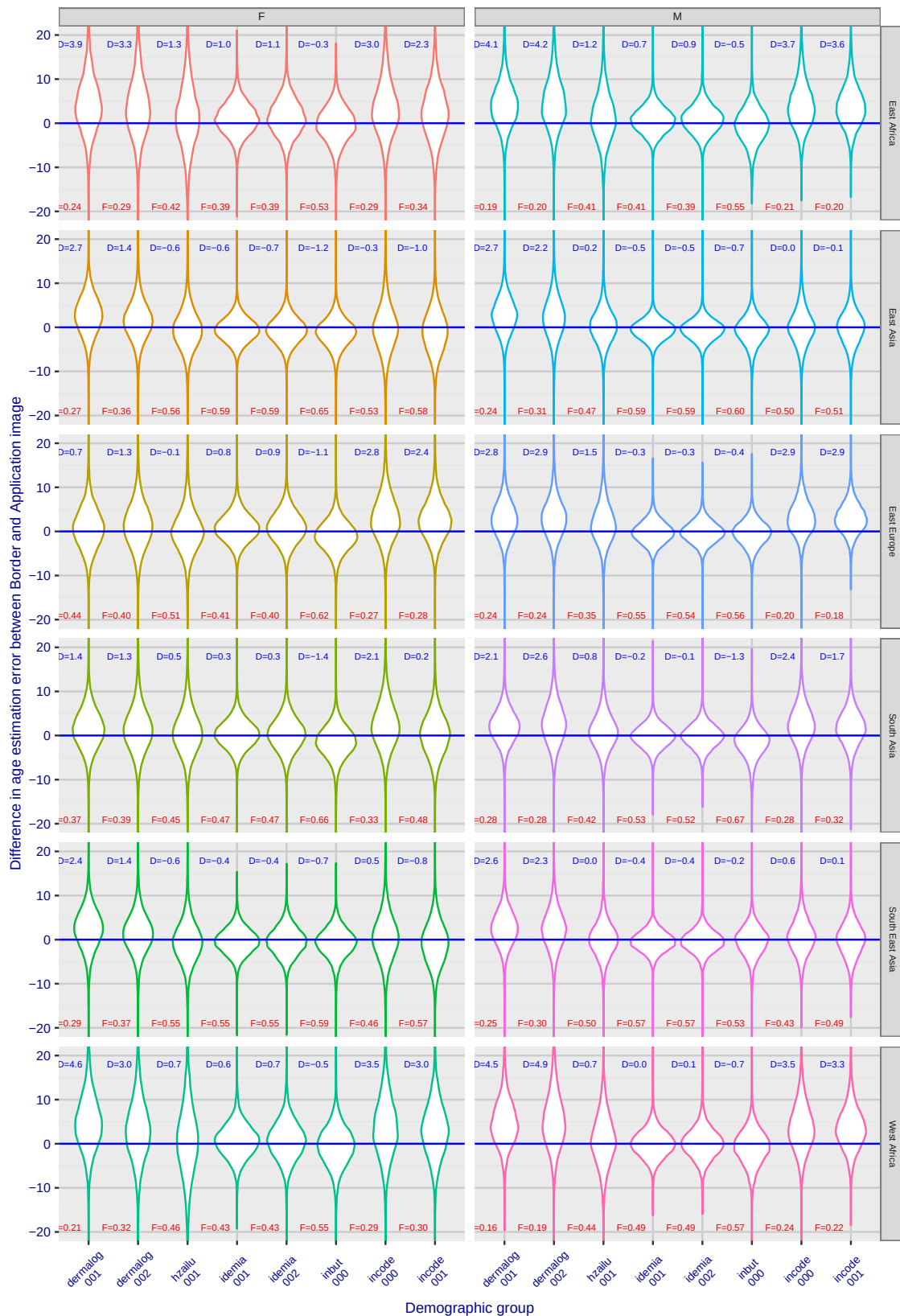
We do not have the resources to conduct an exhaustive exploration of the effect of quality variables (resolution, exposure, blur, etc.) on AE error. Instead, we note that *Border* images give generally degraded age estimation metrics, relative to *Application* images. As described in section 3, the *Border* images are collected under time constraints using inexpensive webcams, with mild departures from frontal head orientation.

Given collection condition 1 (*Border*), person  $i$  of age  $a_{i1}$  and estimated age  $e_{i1}$  the age estimation error is  $\epsilon_{i1} = e_{i1} - a_{i1}$ . Likewise for the second (*Application*) photo of the same person  $\epsilon_{i2} = e_{i2} - a_{i2}$ . We use only those people aged 14 to 36, for which the photos are collected within two years of each other  $|a_{i1} - a_{i2}| \leq 2$ . Figure 94 shows the distribution of the change in age estimation error between two collection conditions,  $\epsilon_{i1} - \epsilon_{i2}$ . Positive values indicate larger error in *Border* images. Pairwise analysis gives more powerful insight than simply comparing the two error distributions. The figure breaks out this change by sex and region-of-birth. The figure includes two summary statistics: First is median change in error,  $D$ ; second is the proportion of images with a decrease in error,  $F$ . For example, for dermalog-001 operating on E. African women, 24% of persons have a better age estimate using *Border* images than *Application* images, but the median change in error is 3.9 years.

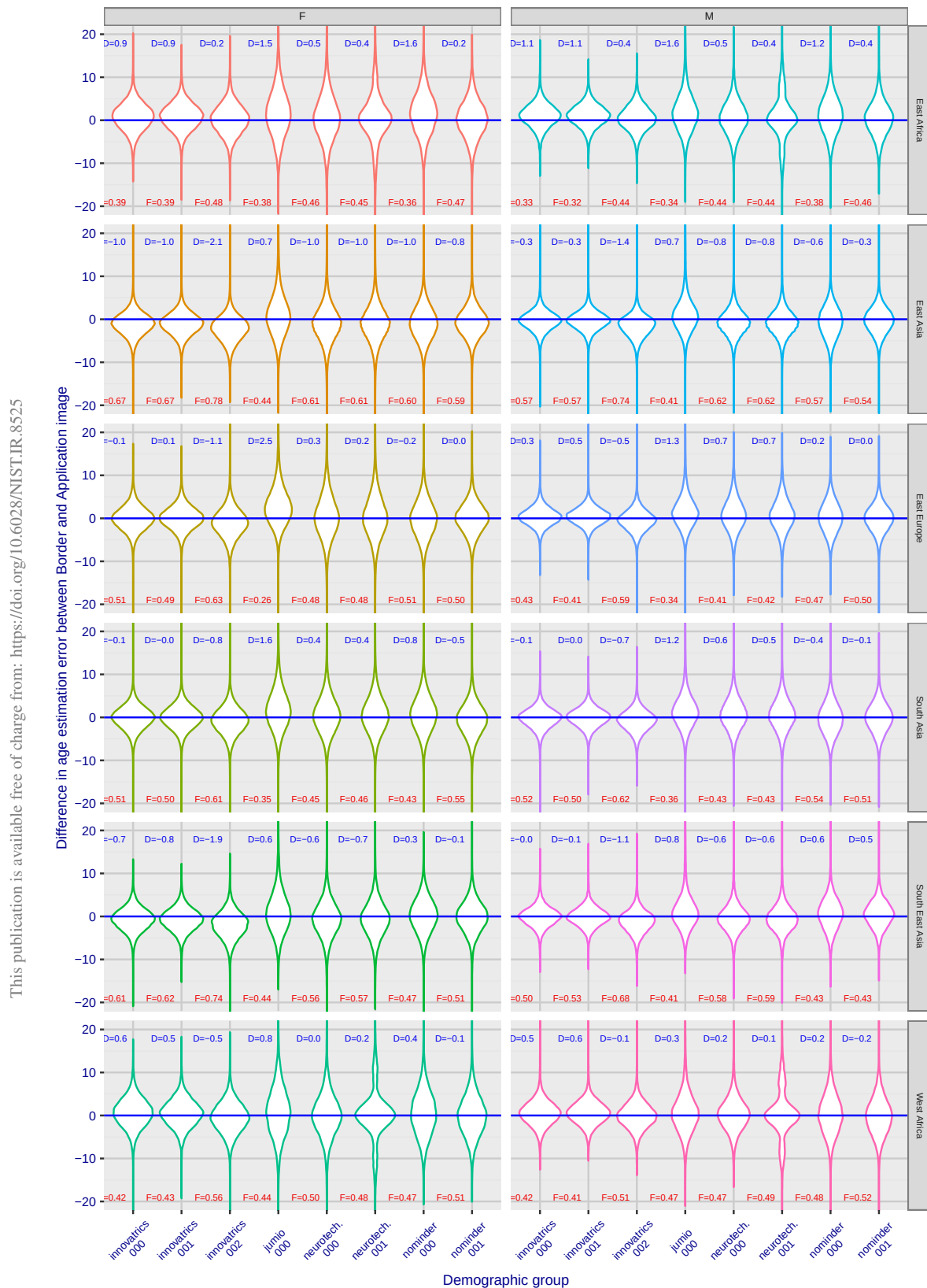
The ideal form of this plot is for the violins to be wide and not tall, showing little change in AE error between imaging conditions. Neurotechnology-000, roc-000 and yoti-001 give generally smaller changes in error. We note most algorithms give worse age estimates on *Border* images, with the exception of unissey-001 which has generally lower error.



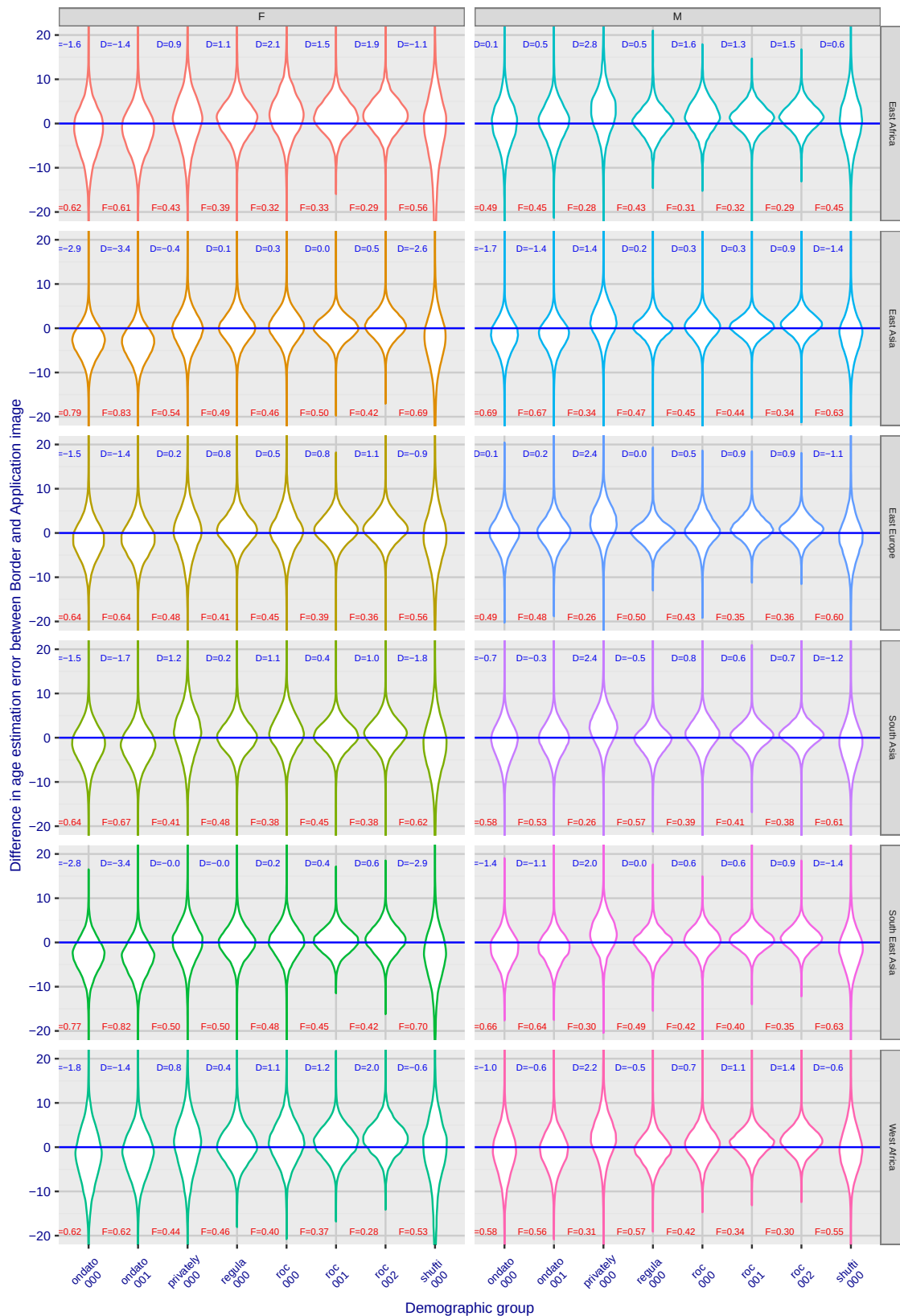
**Fig. 90.** Distribution of the pairwise age estimation error difference for Border and Application images. Positive values indicate larger error on Border images.



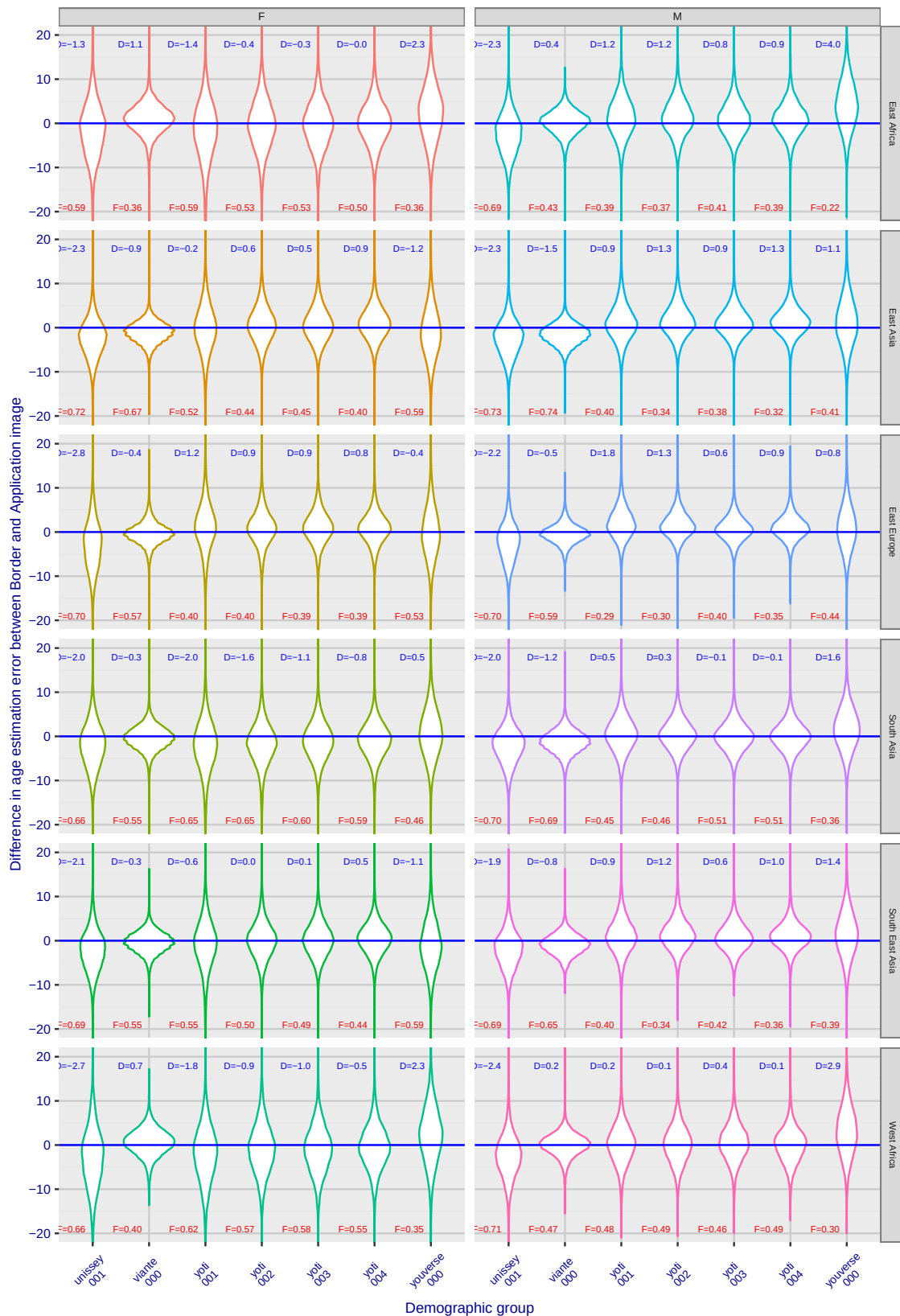
**Fig. 91.** Distribution of the pairwise age estimation error difference for Border and Application images. Positive values indicate larger error on Border images.



**Fig. 92.** Distribution of the pairwise age estimation error difference for Border and Application images. Positive values indicate larger error on Border images.



**Fig. 93.** Distribution of the pairwise age estimation error difference for Border and Application images. Positive values indicate larger error on Border images.



**Fig. 94.** Distribution of the pairwise age estimation error difference for Border and Application images. Positive values indicate larger error on Border images.

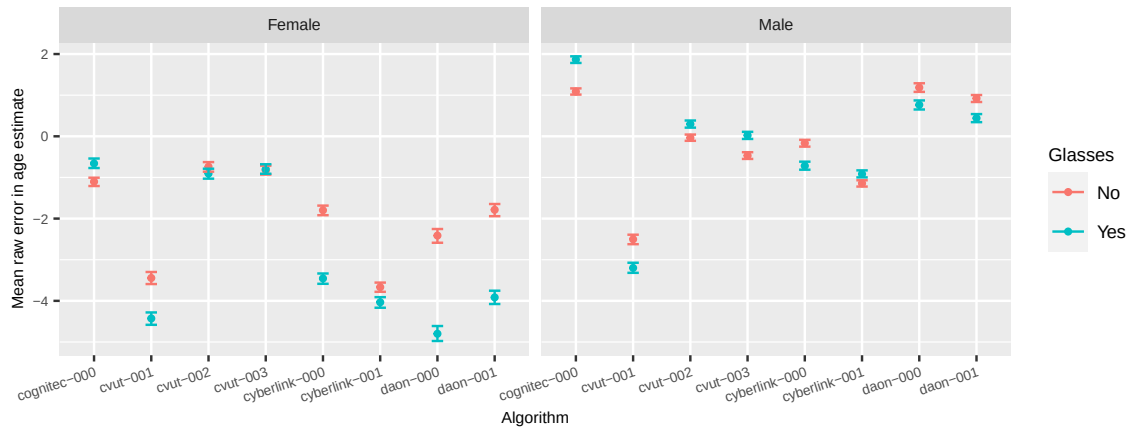
### 5.6.1. Effect of Eyeglasses

Figure 99 shows age estimation error results for subjects photographed with and without glasses. The photographs are a subset of the border images described in section 3.4. The population consists of 9181 subjects for whom we have paired photos - one with glasses and one without taken within three months years of each other. There are 16677 pairs. All subjects are between 14 and 36 years old.

Ground truth for presence of eyeglasses is established automatically using a [Specific Image Defect Detection](#) (SIDD) algorithm to determine the presence of glasses, setting a threshold of 0.5 on the quality component. The SIDD algorithm is chosen based on both accuracy and speed. More information regarding SIDD algorithms' performance can be found in the [FATE SIDD report](#) which is a regularly updated version of NIST Interagency Report 8485.

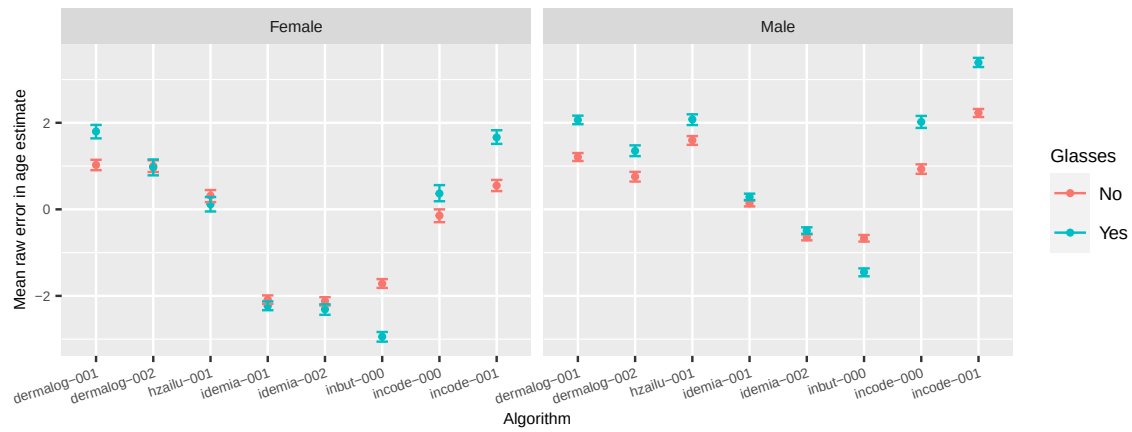
Error bars are computed by bootstrapping the calculation of mean error 1000 times and taking the interval that gives 95% coverage of the mean. Note that the error bars are not a measure of the deviation of the actual error values.

The figure shows that dermalog\_001 and incode\_000 overestimate age of both men and women more with glasses than without. Yoti\_001 does the same but only in women. The roc\_000 algorithm overestimates age of both sexes without glasses more than with glasses. In contrast, neurotechnology\_000 and unissey\_001 have large mean age estimate error for underestimate the age of subjects wearing glasses.

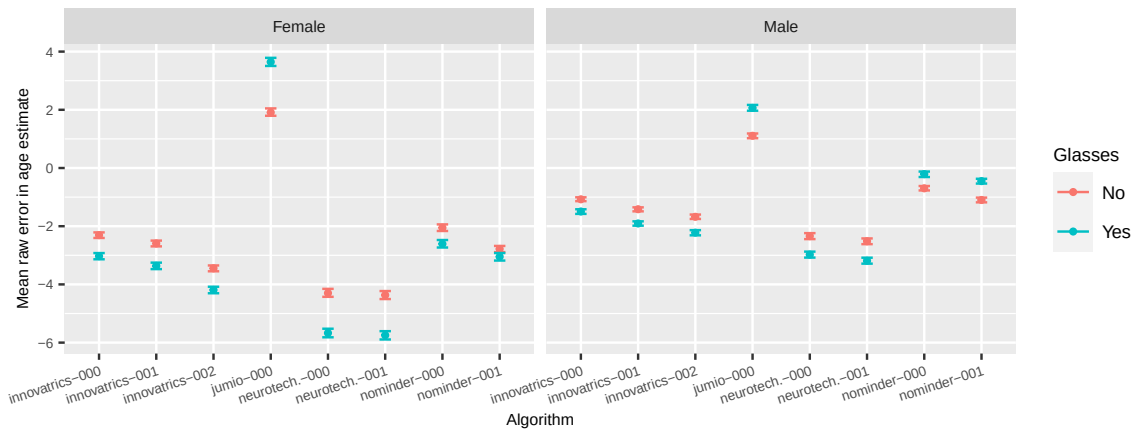


**Fig. 95.** Mean raw age estimation error for border crossing subjects for whom we have images with and without eyeglasses.

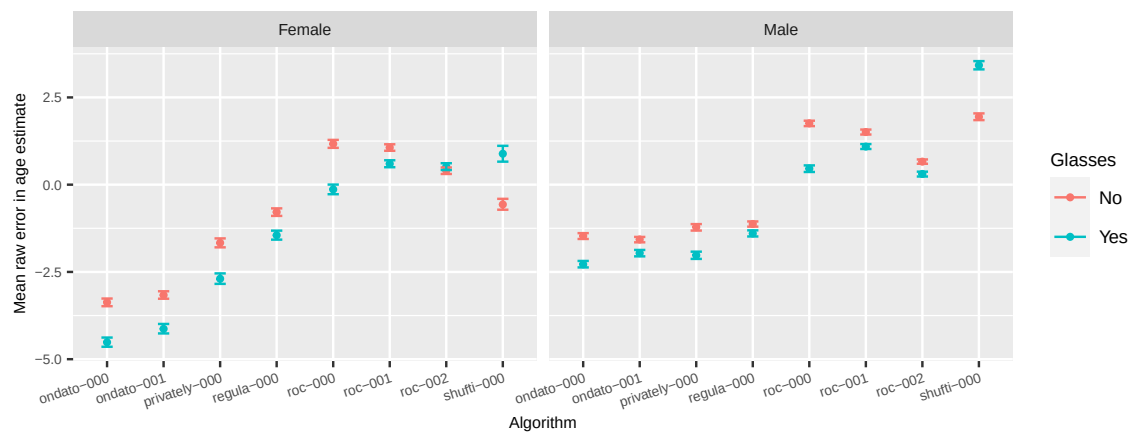




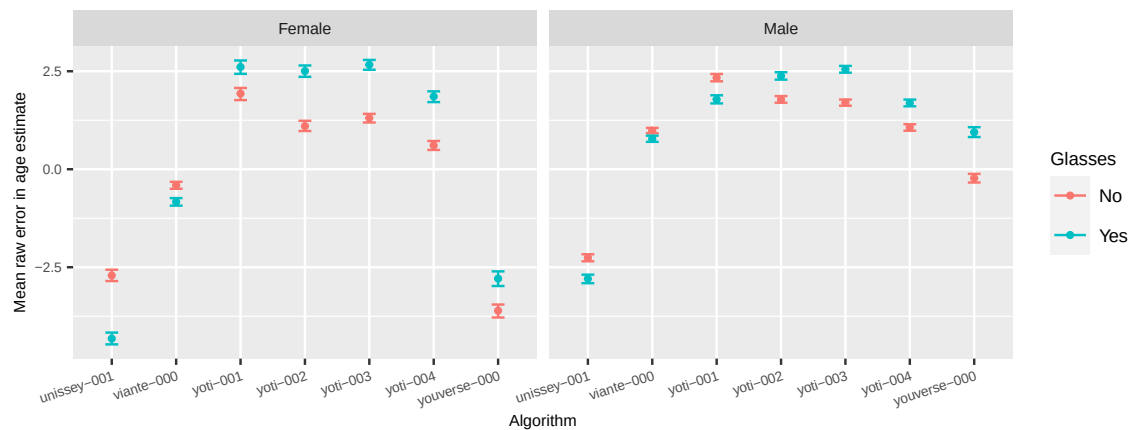
**Fig. 96.** Mean raw age estimation error for border crossing subjects for whom we have images with and without eyeglasses.



**Fig. 97.** Mean raw age estimation error for border crossing subjects for whom we have images with and without eyeglasses.

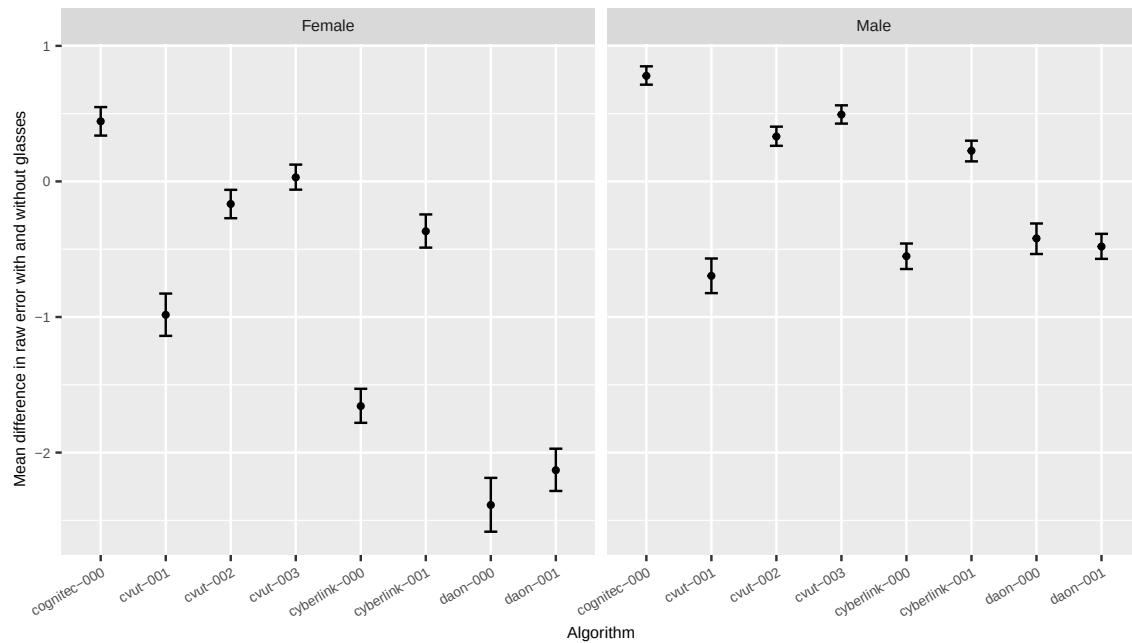


**Fig. 98.** Mean raw age estimation error for border crossing subjects for whom we have images with and without eyeglasses.

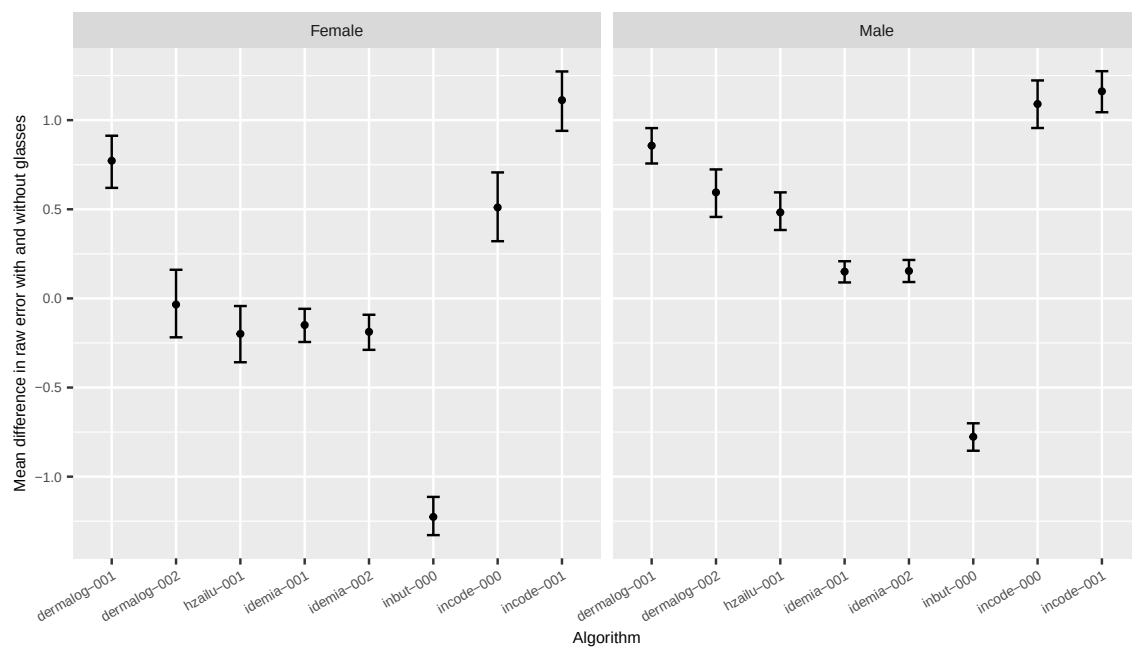


**Fig. 99.** Mean raw age estimation error for border crossing subjects for whom we have images with and without eyeglasses.

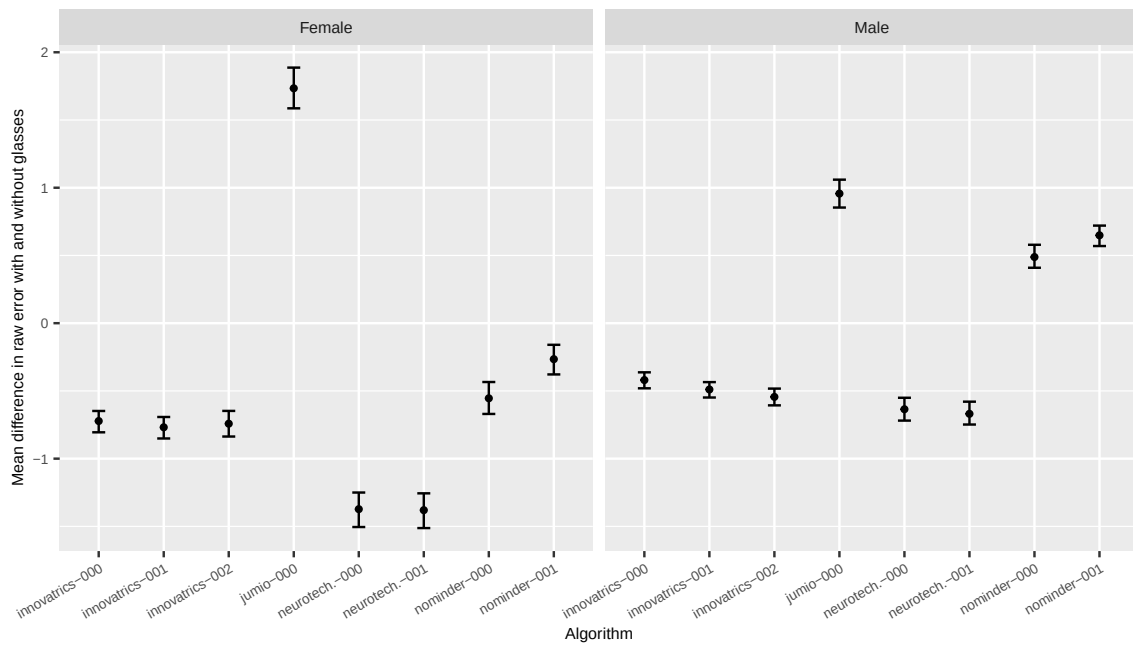
In Figure 104, the difference-in-error with and without glasses is calculated for the same images as in the previous plot, pairwise by subject. The error bars are computed by calculating mean difference-in-error 1000 times and taking the interval that gives 95% coverage of the mean. The results are consistent with those in the previous figure.



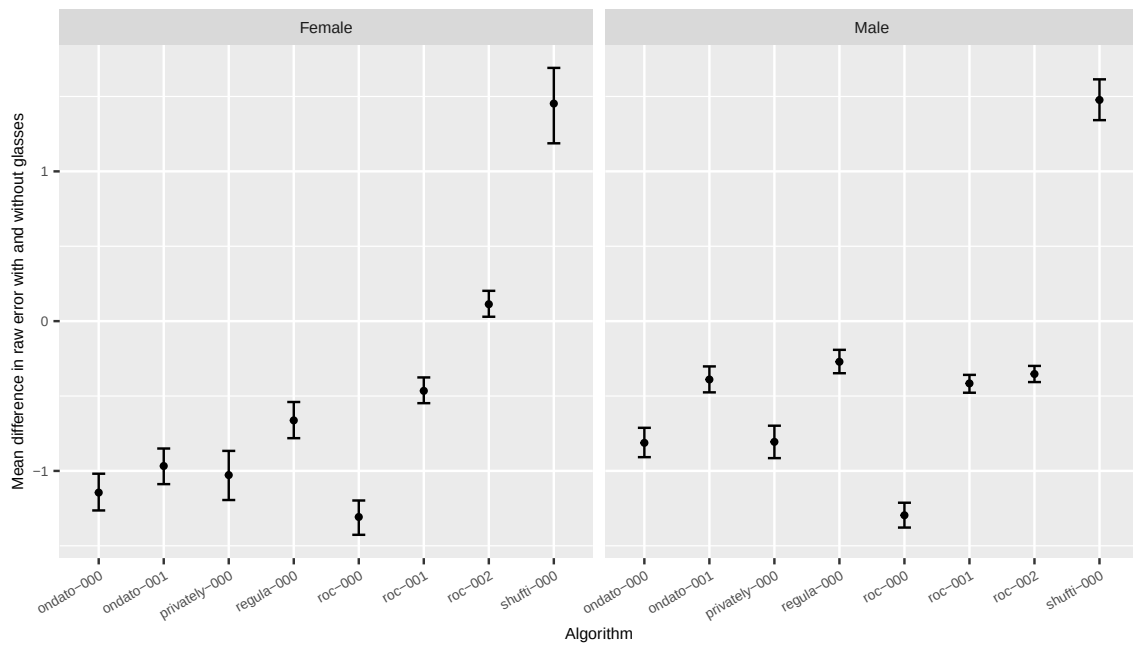
**Fig. 100.** Difference in age estimation error for border crossing subjects with and without eyeglasses.



**Fig. 101.** Difference in age estimation error for border crossing subjects with and without eyeglasses.

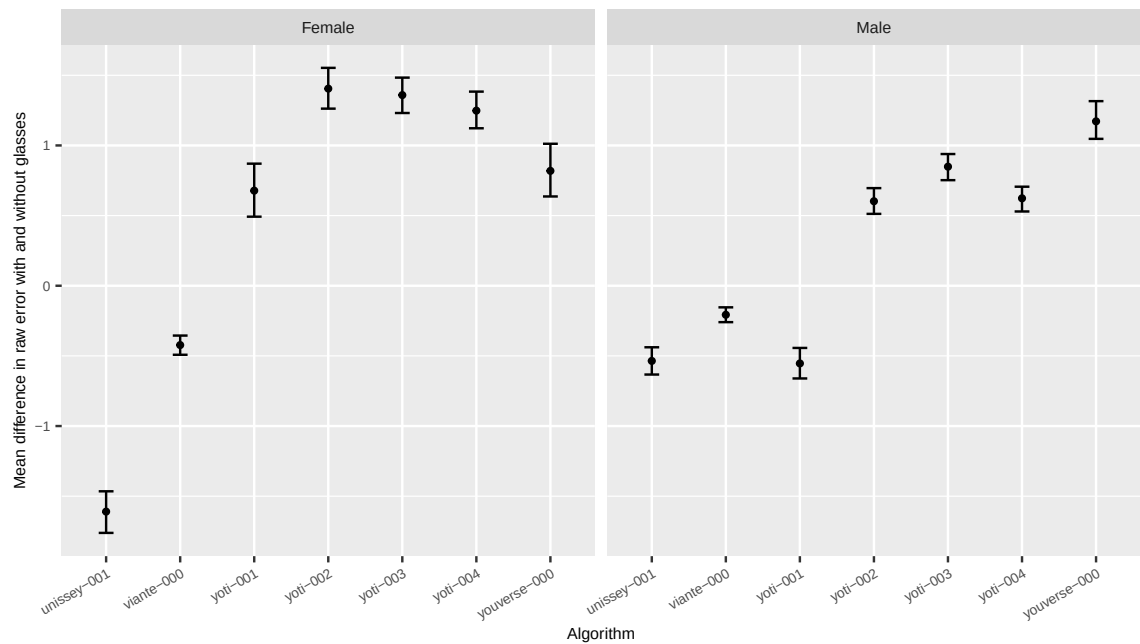


**Fig. 102.** Difference in age estimation error for border crossing subjects with and without eyeglasses.



**Fig. 103.** Difference in age estimation error for border crossing subjects with and without eyeglasses.

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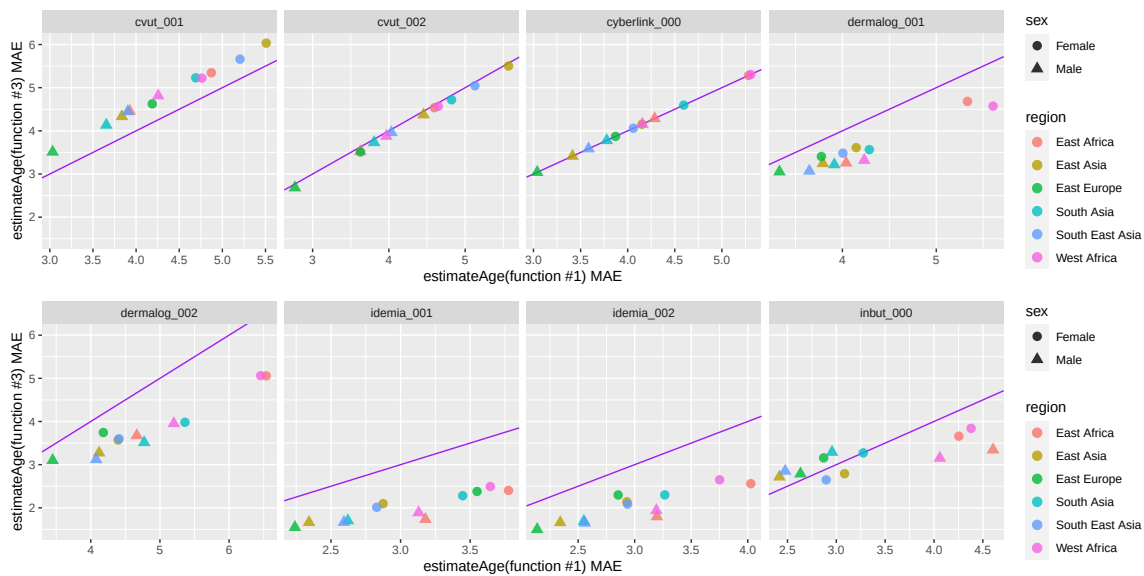
**Fig. 104.** Difference in age estimation error for border crossing subjects with and without eyeglasses.

### 5.7. Age Estimation given a prior image of known age

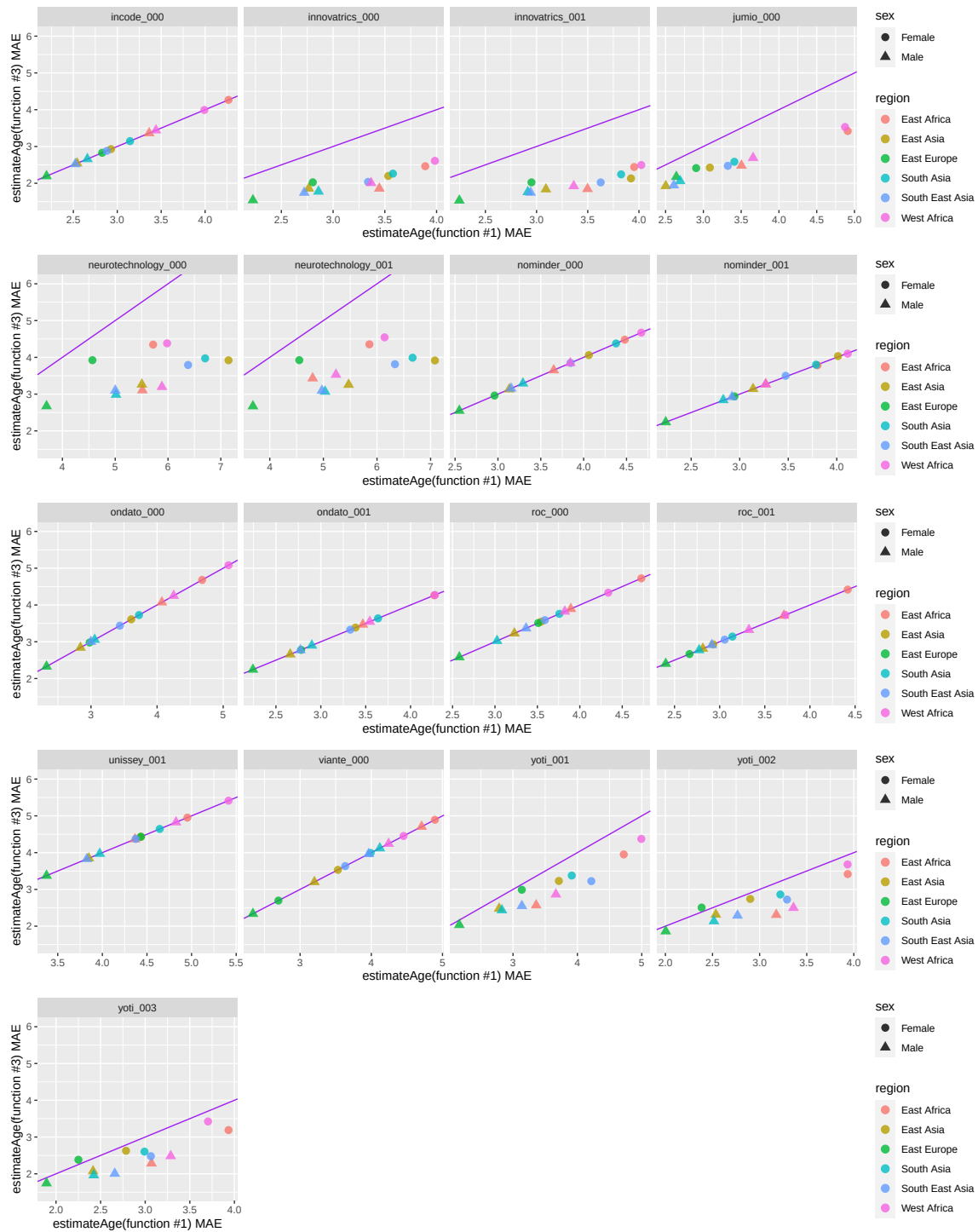
The API includes two age estimation functions: The first returns a continuous-valued age estimate from a single image; the second returns an age estimate also but is additionally provided with a prior image and a known age for that image.

Figure 106 shows MAE computed over a subset of the Application image dataset. This subset consists of 209,450 images of individuals aged between 15 and 50. Each query image is paired with a prior image of the same person and their known age, and then processed through both age estimation function calls. Accuracy gains are evident when the points fall below the  $y = x$  line e.g. for algorithms from Dermalog, Innovatricks, Jumio, Neurotechnology and Yoti. Most other algorithms give identical age estimates - which would be consistent with ignoring the known-age photo.

While this analysis serves as an existence proof that some implementations can produce better accuracy given a prior photo and age, this result is academic because most operational use-cases proceed without storage or presentation of prior images and known ages. There are instances where it could be meaningful. For example, if Alice submits a photo for a passport at time  $T$ , then re-submits a photo at time  $T + 10$  years for passport renewal, then an age estimation algorithm could be used to check the new photo has age around 10 years larger than the old one. This would be effective at preventing people from “recycling” old photos.



**Fig. 105.** In this figure, some algorithms show an improved MAE when given a prior image of known age to the second age estimate function. The x-axis represents the MAE for the first age estimation function, which returns the age estimate of a single image. In contrast, the y-axis represents the MAE for the second age estimation function, which returns the age estimate of an image when both the prior image and the known age of the same individual were provided.



**Fig. 106.** In this figure, some algorithms show an improved MAE when given a prior image of known age to the second age estimate function. The x-axis represents the MAE for the first age estimation function, which returns the age estimate of a single image. In contrast, the y-axis represents the MAE for the second age estimation function, which returns the age estimate of an image when both the prior image and the known age of the same individual were provided.

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## **Appendix A. Detailed Algorithm Results**

### **Appendix A.1. Age Estimation Performance by Age**

**Table 26.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better).

| Age | Num. Images | cognitec-000 |      |     | cvut-001 |      |     | cvut-002 |      |     | cvut-003 |      |       | cyberlink-000 |      |     | cyberlink-001 |      |     |
|-----|-------------|--------------|------|-----|----------|------|-----|----------|------|-----|----------|------|-------|---------------|------|-----|---------------|------|-----|
|     |             | Acc(3)       | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP   | Acc(3)        | MAE  | FTP | Acc(3)        | MAE  | FTP |
| 0   | 133387      | 0.99         | 0.87 | 0   | 1.00     | 0.58 | 0   | 1.00     | 0.53 | 0   | 1.00     | 0.41 | 0.001 | 0.41          | 5.54 | 0   | 0.66          | 3.45 | 0   |
| 1   | 94475       | 0.99         | 0.98 | 0   | 0.99     | 0.95 | 0   | 1.00     | 0.78 | 0   | 1.00     | 0.70 | 0.001 | 0.47          | 4.36 | 0   | 0.64          | 2.80 | 0   |
| 2   | 82596       | 0.98         | 1.11 | 0   | 0.95     | 1.18 | 0   | 0.98     | 1.13 | 0   | 0.99     | 1.04 | 0.001 | 0.44          | 4.78 | 0   | 0.43          | 3.68 | 0   |
| 3   | 82473       | 0.97         | 1.19 | 0   | 0.87     | 1.41 | 0   | 0.96     | 1.16 | 0   | 0.97     | 1.08 | 0.001 | 0.41          | 5.08 | 0   | 0.35          | 4.22 | 0   |
| 4   | 84055       | 0.97         | 1.16 | 0   | 0.88     | 1.62 | 0   | 0.93     | 1.32 | 0   | 0.96     | 1.17 | 0.001 | 0.39          | 5.25 | 0   | 0.30          | 4.51 | 0   |
| 5   | 92521       | 0.97         | 1.14 | 0   | 0.86     | 1.71 | 0   | 0.89     | 1.45 | 0   | 0.92     | 1.29 | 0.001 | 0.34          | 5.53 | 0   | 0.27          | 4.67 | 0   |
| 6   | 91639       | 0.96         | 1.18 | 0   | 0.79     | 1.78 | 0   | 0.80     | 1.71 | 0   | 0.87     | 1.46 | 0.001 | 0.30          | 5.96 | 0   | 0.25          | 4.85 | 0   |
| 7   | 89993       | 0.93         | 1.30 | 0   | 0.80     | 2.01 | 0   | 0.77     | 2.14 | 0   | 0.85     | 1.78 | 0.001 | 0.18          | 6.54 | 0   | 0.21          | 5.10 | 0   |
| 8   | 89428       | 0.92         | 1.34 | 0   | 0.67     | 2.26 | 0   | 0.65     | 2.38 | 0   | 0.81     | 1.94 | 0.001 | 0.13          | 6.46 | 0   | 0.21          | 5.03 | 0   |
| 9   | 89644       | 0.92         | 1.35 | 0   | 0.70     | 2.36 | 0   | 0.70     | 2.43 | 0   | 0.83     | 1.92 | 0.001 | 0.11          | 6.18 | 0   | 0.21          | 4.87 | 0   |
| 10  | 98337       | 0.90         | 1.40 | 0   | 0.71     | 2.38 | 0   | 0.72     | 2.43 | 0   | 0.84     | 1.91 | 0.001 | 0.13          | 5.80 | 0   | 0.20          | 4.70 | 0   |
| 11  | 97415       | 0.88         | 1.53 | 0   | 0.69     | 2.41 | 0   | 0.68     | 2.48 | 0   | 0.79     | 2.02 | 0.001 | 0.13          | 5.42 | 0   | 0.21          | 4.54 | 0   |
| 12  | 94326       | 0.84         | 1.74 | 0   | 0.72     | 2.53 | 0   | 0.55     | 2.60 | 0   | 0.72     | 2.24 | 0.001 | 0.10          | 5.11 | 0   | 0.23          | 4.41 | 0   |
| 13  | 94047       | 0.78         | 2.01 | 0   | 0.69     | 2.86 | 0   | 0.72     | 2.75 | 0   | 0.71     | 2.65 | 0.001 | 0.14          | 4.95 | 0   | 0.26          | 4.38 | 0   |
| 14  | 96915       | 0.73         | 2.23 | 0   | 0.56     | 3.34 | 0   | 0.67     | 2.85 | 0   | 0.65     | 3.11 | 0.001 | 0.23          | 4.84 | 0   | 0.27          | 4.34 | 0   |
| 15  | 97599       | 0.70         | 2.38 | 0   | 0.49     | 3.94 | 0   | 0.58     | 2.99 | 0   | 0.53     | 3.60 | 0.001 | 0.26          | 4.78 | 0   | 0.28          | 4.29 | 0   |
| 16  | 96388       | 0.67         | 2.47 | 0   | 0.40     | 4.36 | 0   | 0.50     | 3.27 | 0   | 0.41     | 4.11 | 0.001 | 0.28          | 4.63 | 0   | 0.32          | 4.13 | 0   |
| 17  | 99462       | 0.67         | 2.48 | 0   | 0.38     | 4.53 | 0   | 0.46     | 3.46 | 0   | 0.35     | 4.50 | 0.001 | 0.29          | 4.43 | 0   | 0.39          | 3.89 | 0   |
| 18  | 98765       | 0.68         | 2.48 | 0   | 0.39     | 4.61 | 0   | 0.45     | 3.59 | 0   | 0.31     | 4.88 | 0.001 | 0.33          | 4.30 | 0   | 0.44          | 3.72 | 0   |
| 19  | 105717      | 0.68         | 2.48 | 0   | 0.40     | 4.60 | 0   | 0.47     | 3.54 | 0   | 0.30     | 4.93 | 0.001 | 0.38          | 4.15 | 0   | 0.49          | 3.53 | 0   |
| 20  | 118643      | 0.68         | 2.49 | 0   | 0.40     | 4.52 | 0   | 0.49     | 3.43 | 0   | 0.30     | 4.80 | 0.001 | 0.46          | 3.98 | 0   | 0.53          | 3.37 | 0   |
| 21  | 124060      | 0.68         | 2.50 | 0   | 0.42     | 4.47 | 0   | 0.57     | 3.31 | 0   | 0.33     | 4.65 | 0.001 | 0.49          | 3.82 | 0   | 0.55          | 3.26 | 0   |
| 22  | 128864      | 0.67         | 2.55 | 0   | 0.45     | 4.43 | 0   | 0.58     | 3.25 | 0   | 0.36     | 4.51 | 0.001 | 0.47          | 3.69 | 0   | 0.56          | 3.18 | 0   |
| 23  | 133507      | 0.65         | 2.63 | 0   | 0.43     | 4.41 | 0   | 0.55     | 3.25 | 0   | 0.36     | 4.36 | 0.001 | 0.44          | 3.60 | 0   | 0.56          | 3.16 | 0   |
| 24  | 138094      | 0.64         | 2.73 | 0   | 0.42     | 4.38 | 0   | 0.54     | 3.29 | 0   | 0.38     | 4.18 | 0.001 | 0.45          | 3.55 | 0   | 0.56          | 3.13 | 0   |
| 25  | 139920      | 0.61         | 2.86 | 0   | 0.40     | 4.38 | 0   | 0.54     | 3.43 | 0   | 0.41     | 4.07 | 0.001 | 0.53          | 3.51 | 0   | 0.55          | 3.13 | 0   |
| 26  | 141741      | 0.58         | 2.96 | 0   | 0.41     | 4.40 | 0   | 0.54     | 3.52 | 0   | 0.45     | 3.93 | 0.001 | 0.57          | 3.40 | 0   | 0.55          | 3.12 | 0   |
| 27  | 138491      | 0.56         | 3.08 | 0   | 0.44     | 4.38 | 0   | 0.51     | 3.57 | 0   | 0.48     | 3.81 | 0.001 | 0.54          | 3.29 | 0   | 0.56          | 3.12 | 0   |
| 28  | 133633      | 0.54         | 3.16 | 0   | 0.46     | 4.32 | 0   | 0.48     | 3.63 | 0   | 0.50     | 3.76 | 0.001 | 0.52          | 3.29 | 0   | 0.56          | 3.13 | 0   |
| 29  | 129054      | 0.53         | 3.22 | 0   | 0.45     | 4.28 | 0   | 0.48     | 3.71 | 0   | 0.51     | 3.76 | 0.001 | 0.53          | 3.38 | 0   | 0.56          | 3.15 | 0   |
| 30  | 123437      | 0.53         | 3.25 | 0   | 0.41     | 4.25 | 0   | 0.46     | 3.78 | 0   | 0.49     | 3.80 | 0.001 | 0.54          | 3.48 | 0   | 0.56          | 3.19 | 0   |
| 31  | 118174      | 0.53         | 3.26 | 0   | 0.43     | 4.22 | 0   | 0.44     | 3.80 | 0   | 0.47     | 3.87 | 0.001 | 0.51          | 3.55 | 0   | 0.56          | 3.22 | 0   |
| 32  | 113131      | 0.53         | 3.26 | 0   | 0.48     | 4.20 | 0   | 0.45     | 3.81 | 0   | 0.47     | 3.95 | 0.001 | 0.49          | 3.61 | 0   | 0.55          | 3.31 | 0   |
| 33  | 109146      | 0.54         | 3.23 | 0   | 0.54     | 4.10 | 0   | 0.45     | 3.76 | 0   | 0.45     | 4.00 | 0.001 | 0.53          | 3.67 | 0   | 0.53          | 3.38 | 0   |
| 34  | 105908      | 0.55         | 3.21 | 0   | 0.51     | 3.97 | 0   | 0.48     | 3.69 | 0   | 0.43     | 4.05 | 0.001 | 0.55          | 3.78 | 0   | 0.51          | 3.48 | 0   |
| 35  | 103149      | 0.56         | 3.13 | 0   | 0.49     | 3.84 | 0   | 0.48     | 3.60 | 0   | 0.40     | 4.04 | 0.001 | 0.52          | 3.89 | 0   | 0.51          | 3.55 | 0   |
| 36  | 99901       | 0.56         | 3.10 | 0   | 0.55     | 3.90 | 0   | 0.52     | 3.53 | 0   | 0.40     | 4.01 | 0.001 | 0.47          | 4.08 | 0   | 0.50          | 3.63 | 0   |

**Table 26.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | cognitec-000 |      |     | cvut-001 |      |     | cvut-002 |      |     | cvut-003 |      |       | cyberlink-000 |      |     | cyberlink-001 |      |     |
|-----|-------------|--------------|------|-----|----------|------|-----|----------|------|-----|----------|------|-------|---------------|------|-----|---------------|------|-----|
|     |             | Acc(3)       | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP   | Acc(3)        | MAE  | FTP | Acc(3)        | MAE  | FTP |
| 37  | 97516       | 0.58         | 3.05 | 0   | 0.57     | 4.01 | 0   | 0.53     | 3.45 | 0   | 0.41     | 3.92 | 0.001 | 0.43          | 4.33 | 0   | 0.49          | 3.67 | 0   |
| 38  | 93939       | 0.58         | 3.03 | 0   | 0.49     | 4.09 | 0   | 0.52     | 3.41 | 0   | 0.42     | 3.79 | 0.001 | 0.39          | 4.55 | 0   | 0.48          | 3.73 | 0   |
| 39  | 91461       | 0.59         | 2.98 | 0   | 0.44     | 4.10 | 0   | 0.54     | 3.34 | 0   | 0.46     | 3.60 | 0.001 | 0.33          | 4.71 | 0   | 0.47          | 3.80 | 0   |
| 40  | 94056       | 0.58         | 2.98 | 0   | 0.47     | 4.14 | 0   | 0.54     | 3.33 | 0   | 0.51     | 3.42 | 0.001 | 0.31          | 4.78 | 0   | 0.46          | 3.87 | 0   |
| 41  | 90173       | 0.59         | 2.96 | 0   | 0.49     | 4.27 | 0   | 0.55     | 3.29 | 0   | 0.55     | 3.24 | 0.001 | 0.33          | 4.77 | 0   | 0.45          | 3.92 | 0   |
| 42  | 86423       | 0.59         | 2.97 | 0   | 0.46     | 4.30 | 0   | 0.53     | 3.24 | 0   | 0.59     | 3.08 | 0.001 | 0.39          | 4.71 | 0   | 0.43          | 3.93 | 0   |
| 43  | 83252       | 0.59         | 2.98 | 0   | 0.43     | 4.37 | 0   | 0.54     | 3.24 | 0   | 0.59     | 2.98 | 0.001 | 0.43          | 4.66 | 0   | 0.43          | 3.93 | 0   |
| 44  | 80602       | 0.59         | 2.99 | 0   | 0.45     | 4.47 | 0   | 0.55     | 3.21 | 0   | 0.60     | 2.94 | 0.001 | 0.44          | 4.61 | 0   | 0.43          | 3.87 | 0   |
| 45  | 77502       | 0.58         | 3.01 | 0   | 0.42     | 4.44 | 0   | 0.55     | 3.19 | 0   | 0.60     | 2.96 | 0.001 | 0.43          | 4.56 | 0   | 0.45          | 3.79 | 0   |
| 46  | 74182       | 0.58         | 3.02 | 0   | 0.44     | 4.50 | 0   | 0.55     | 3.15 | 0   | 0.58     | 2.97 | 0.001 | 0.40          | 4.57 | 0   | 0.48          | 3.66 | 0   |
| 47  | 70698       | 0.57         | 3.02 | 0   | 0.50     | 4.47 | 0   | 0.58     | 3.10 | 0   | 0.57     | 2.97 | 0.001 | 0.37          | 4.57 | 0   | 0.51          | 3.51 | 0   |
| 48  | 67856       | 0.57         | 3.02 | 0   | 0.50     | 4.46 | 0   | 0.58     | 3.06 | 0   | 0.58     | 2.95 | 0.001 | 0.35          | 4.53 | 0   | 0.54          | 3.35 | 0   |
| 49  | 65674       | 0.58         | 3.04 | 0   | 0.50     | 4.47 | 0   | 0.59     | 3.06 | 0   | 0.60     | 2.97 | 0.001 | 0.36          | 4.43 | 0   | 0.56          | 3.24 | 0   |
| 50  | 64896       | 0.58         | 3.03 | 0   | 0.49     | 4.58 | 0   | 0.58     | 3.06 | 0   | 0.60     | 2.95 | 0.001 | 0.38          | 4.23 | 0   | 0.58          | 3.13 | 0   |
| 51  | 63410       | 0.58         | 3.04 | 0   | 0.49     | 4.84 | 0   | 0.57     | 3.14 | 0   | 0.58     | 2.99 | 0.001 | 0.42          | 4.01 | 0   | 0.60          | 3.08 | 0   |
| 52  | 60991       | 0.57         | 3.07 | 0   | 0.41     | 5.12 | 0   | 0.56     | 3.23 | 0   | 0.57     | 3.08 | 0.001 | 0.48          | 3.83 | 0   | 0.60          | 3.05 | 0   |
| 53  | 59550       | 0.57         | 3.08 | 0   | 0.31     | 5.42 | 0   | 0.52     | 3.34 | 0   | 0.56     | 3.20 | 0.001 | 0.55          | 3.71 | 0   | 0.60          | 3.04 | 0   |
| 54  | 56895       | 0.57         | 3.09 | 0   | 0.29     | 5.54 | 0   | 0.52     | 3.42 | 0   | 0.54     | 3.30 | 0.001 | 0.56          | 3.67 | 0   | 0.59          | 3.07 | 0   |
| 55  | 55297       | 0.56         | 3.14 | 0   | 0.27     | 5.87 | 0   | 0.50     | 3.49 | 0   | 0.51     | 3.36 | 0.001 | 0.53          | 3.69 | 0   | 0.58          | 3.15 | 0   |
| 56  | 52574       | 0.56         | 3.16 | 0   | 0.30     | 6.09 | 0   | 0.51     | 3.55 | 0   | 0.51     | 3.39 | 0.001 | 0.52          | 3.76 | 0   | 0.56          | 3.25 | 0   |
| 57  | 49823       | 0.55         | 3.22 | 0   | 0.28     | 6.31 | 0   | 0.49     | 3.64 | 0   | 0.51     | 3.44 | 0.001 | 0.47          | 3.91 | 0   | 0.52          | 3.46 | 0   |
| 58  | 48570       | 0.54         | 3.29 | 0   | 0.25     | 6.42 | 0   | 0.48     | 3.75 | 0   | 0.51     | 3.48 | 0.001 | 0.44          | 4.07 | 0   | 0.47          | 3.69 | 0   |
| 59  | 46040       | 0.52         | 3.37 | 0   | 0.27     | 6.70 | 0   | 0.47     | 3.88 | 0   | 0.50     | 3.55 | 0.001 | 0.39          | 4.22 | 0   | 0.44          | 3.94 | 0   |
| 60  | 46037       | 0.51         | 3.45 | 0   | 0.28     | 6.88 | 0   | 0.44     | 4.01 | 0   | 0.49     | 3.62 | 0.001 | 0.39          | 4.32 | 0   | 0.39          | 4.20 | 0   |
| 61  | 43860       | 0.49         | 3.56 | 0   | 0.26     | 7.14 | 0   | 0.41     | 4.14 | 0   | 0.48     | 3.65 | 0.001 | 0.44          | 4.37 | 0   | 0.36          | 4.44 | 0   |
| 62  | 41407       | 0.49         | 3.65 | 0   | 0.21     | 7.33 | 0   | 0.40     | 4.26 | 0   | 0.49     | 3.72 | 0.001 | 0.47          | 4.38 | 0   | 0.33          | 4.68 | 0   |
| 63  | 39444       | 0.47         | 3.73 | 0   | 0.19     | 7.48 | 0   | 0.40     | 4.37 | 0   | 0.47     | 3.78 | 0.001 | 0.50          | 4.38 | 0   | 0.33          | 4.87 | 0   |
| 64  | 37425       | 0.46         | 3.86 | 0   | 0.21     | 7.68 | 0   | 0.39     | 4.40 | 0   | 0.47     | 3.79 | 0.001 | 0.49          | 4.46 | 0   | 0.32          | 5.04 | 0   |
| 65  | 34834       | 0.45         | 3.96 | 0   | 0.20     | 7.75 | 0   | 0.40     | 4.47 | 0   | 0.46     | 3.87 | 0.001 | 0.47          | 4.70 | 0   | 0.31          | 5.21 | 0   |
| 66  | 32020       | 0.44         | 4.04 | 0   | 0.20     | 7.99 | 0   | 0.41     | 4.52 | 0   | 0.46     | 3.99 | 0.001 | 0.42          | 4.94 | 0   | 0.32          | 5.31 | 0   |
| 67  | 28898       | 0.42         | 4.16 | 0   | 0.22     | 8.03 | 0   | 0.42     | 4.63 | 0   | 0.45     | 4.13 | 0.001 | 0.32          | 5.24 | 0   | 0.32          | 5.44 | 0   |
| 68  | 26630       | 0.40         | 4.35 | 0   | 0.16     | 8.17 | 0   | 0.37     | 4.75 | 0   | 0.39     | 4.30 | 0.001 | 0.25          | 5.49 | 0   | 0.30          | 5.59 | 0   |
| 69  | 24864       | 0.39         | 4.43 | 0   | 0.21     | 8.14 | 0   | 0.37     | 4.88 | 0   | 0.37     | 4.45 | 0.001 | 0.23          | 5.54 | 0   | 0.31          | 5.64 | 0   |
| 70  | 23407       | 0.39         | 4.51 | 0   | 0.21     | 8.32 | 0   | 0.35     | 4.98 | 0   | 0.36     | 4.60 | 0.001 | 0.23          | 5.54 | 0   | 0.30          | 5.71 | 0   |
| 71  | 20736       | 0.38         | 4.68 | 0   | 0.22     | 8.12 | 0   | 0.30     | 5.16 | 0   | 0.34     | 4.80 | 0.001 | 0.25          | 5.43 | 0   | 0.30          | 5.77 | 0   |
| 72  | 18423       | 0.37         | 4.79 | 0   | 0.27     | 8.14 | 0   | 0.29     | 5.32 | 0   | 0.32     | 5.03 | 0.001 | 0.28          | 5.33 | 0   | 0.31          | 5.81 | 0   |
| 73  | 16215       | 0.36         | 4.93 | 0   | 0.29     | 7.99 | 0   | 0.28     | 5.44 | 0   | 0.30     | 5.18 | 0.001 | 0.31          | 5.14 | 0   | 0.31          | 5.85 | 0   |
| 74  | 14848       | 0.35         | 4.99 | 0   | 0.31     | 7.94 | 0   | 0.29     | 5.48 | 0   | 0.32     | 5.31 | 0.001 | 0.35          | 4.87 | 0   | 0.32          | 5.76 | 0   |

**Table 26.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | cognitec-000 |       |     | cvut-001 |       |     | cvut-002 |      |     | cvut-003 |       |       | cyberlink-000 |      |     | cyberlink-001 |      |     |
|-----|-------------|--------------|-------|-----|----------|-------|-----|----------|------|-----|----------|-------|-------|---------------|------|-----|---------------|------|-----|
|     |             | Acc(3)       | MAE   | FTP | Acc(3)   | MAE   | FTP | Acc(3)   | MAE  | FTP | Acc(3)   | MAE   | FTP   | Acc(3)        | MAE  | FTP | Acc(3)        | MAE  | FTP |
| 75  | 13197       | 0.33         | 5.29  | 0   | 0.31     | 7.92  | 0   | 0.30     | 5.72 | 0   | 0.31     | 5.59  | 0.001 | 0.41          | 4.72 | 0   | 0.32          | 5.89 | 0   |
| 76  | 11886       | 0.32         | 5.43  | 0   | 0.28     | 8.00  | 0   | 0.32     | 5.75 | 0   | 0.32     | 5.78  | 0.001 | 0.46          | 4.45 | 0   | 0.31          | 5.95 | 0   |
| 77  | 10839       | 0.30         | 5.68  | 0   | 0.30     | 8.07  | 0   | 0.32     | 5.94 | 0   | 0.28     | 6.17  | 0.002 | 0.52          | 4.16 | 0   | 0.31          | 5.99 | 0   |
| 78  | 9361        | 0.29         | 5.87  | 0   | 0.25     | 8.38  | 0   | 0.30     | 6.00 | 0   | 0.25     | 6.46  | 0.001 | 0.54          | 4.06 | 0   | 0.31          | 6.05 | 0   |
| 79  | 7540        | 0.26         | 6.06  | 0   | 0.19     | 8.36  | 0   | 0.31     | 6.07 | 0   | 0.21     | 6.75  | 0.001 | 0.58          | 3.88 | 0   | 0.30          | 5.97 | 0   |
| 80  | 6267        | 0.26         | 6.28  | 0   | 0.20     | 8.25  | 0   | 0.29     | 6.24 | 0   | 0.18     | 7.20  | 0.001 | 0.55          | 4.02 | 0   | 0.28          | 6.08 | 0   |
| 81  | 5221        | 0.24         | 6.44  | 0   | 0.15     | 8.30  | 0   | 0.28     | 6.14 | 0   | 0.12     | 7.48  | 0.000 | 0.48          | 4.08 | 0   | 0.30          | 5.95 | 0   |
| 82  | 3962        | 0.22         | 6.79  | 0   | 0.27     | 8.45  | 0   | 0.31     | 6.25 | 0   | 0.10     | 7.91  | 0.001 | 0.44          | 4.42 | 0   | 0.29          | 6.11 | 0   |
| 83  | 3277        | 0.19         | 7.19  | 0   | 0.27     | 8.35  | 0   | 0.27     | 6.47 | 0   | 0.06     | 8.66  | 0.001 | 0.35          | 4.77 | 0   | 0.30          | 6.12 | 0   |
| 84  | 2615        | 0.18         | 7.45  | 0   | 0.27     | 8.32  | 0   | 0.29     | 6.39 | 0   | 0.04     | 8.99  | 0.000 | 0.28          | 5.17 | 0   | 0.30          | 6.23 | 0   |
| 85  | 2206        | 0.16         | 7.69  | 0   | 0.29     | 8.42  | 0   | 0.29     | 6.58 | 0   | 0.02     | 9.66  | 0.001 | 0.24          | 5.57 | 0   | 0.31          | 6.17 | 0   |
| 86  | 1831        | 0.14         | 8.15  | 0   | 0.28     | 8.82  | 0   | 0.31     | 6.66 | 0   | 0.01     | 10.42 | 0.002 | 0.21          | 6.08 | 0   | 0.31          | 6.36 | 0   |
| 87  | 1402        | 0.13         | 8.78  | 0   | 0.27     | 9.22  | 0   | 0.30     | 7.12 | 0   | 0.00     | 11.03 | 0.002 | 0.20          | 6.67 | 0   | 0.30          | 6.69 | 0   |
| 88  | 1254        | 0.09         | 9.24  | 0   | 0.10     | 10.37 | 0   | 0.23     | 7.31 | 0   | 0.00     | 11.74 | 0.000 | 0.14          | 7.18 | 0   | 0.27          | 6.99 | 0   |
| 89  | 907         | 0.09         | 9.26  | 0   | 0.11     | 10.05 | 0   | 0.23     | 7.31 | 0   | 0.00     | 12.33 | 0.000 | 0.15          | 7.46 | 0   | 0.33          | 6.61 | 0   |
| 90  | 670         | 0.06         | 10.30 | 0   | 0.09     | 10.44 | 0   | 0.21     | 8.29 | 0   | 0.00     | 13.42 | 0.001 | 0.15          | 8.24 | 0   | 0.27          | 7.54 | 0   |

**Table 27.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better).

| Age | Num. Images | daon-000 |      |     | daon-001 |      |     | dermalog-001 |      |       | dermalog-002 |      |       | hzailu-001 |      |     | idemia-001 |      |     |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|------|-------|--------------|------|-------|------------|------|-----|------------|------|-----|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE  | FTP   | Acc(3)       | MAE  | FTP   | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP |
| 0   | 133387      | 0.26     | 4.70 | 0   | 0.43     | 3.49 | 0   | 0.90         | 2.05 | 0.001 | 0.60         | 3.00 | 0.001 | 0.83       | 1.85 | 0   | 0.94       | 1.00 | 0   |
| 1   | 94475       | 0.35     | 3.94 | 0   | 0.57     | 2.94 | 0   | 0.87         | 1.84 | 0.000 | 0.60         | 2.97 | 0.000 | 0.76       | 2.17 | 0   | 0.93       | 1.35 | 0   |
| 2   | 82596       | 0.38     | 3.75 | 0   | 0.59     | 2.78 | 0   | 0.84         | 1.89 | 0.000 | 0.58         | 2.97 | 0.000 | 0.64       | 2.68 | 0   | 0.93       | 1.49 | 0   |
| 3   | 82473       | 0.41     | 3.53 | 0   | 0.65     | 2.52 | 0   | 0.76         | 2.12 | 0.000 | 0.61         | 2.80 | 0.000 | 0.56       | 3.08 | 0   | 0.95       | 1.35 | 0   |
| 4   | 84055       | 0.48     | 3.19 | 0   | 0.74     | 2.17 | 0   | 0.69         | 2.37 | 0.000 | 0.66         | 2.62 | 0.000 | 0.52       | 3.30 | 0   | 0.97       | 1.09 | 0   |
| 5   | 92521       | 0.56     | 2.85 | 0   | 0.81     | 1.88 | 0   | 0.63         | 2.64 | 0.000 | 0.70         | 2.49 | 0.000 | 0.52       | 3.42 | 0   | 0.98       | 0.94 | 0   |
| 6   | 91639       | 0.59     | 2.68 | 0   | 0.86     | 1.72 | 0   | 0.57         | 2.98 | 0.000 | 0.70         | 2.59 | 0.000 | 0.49       | 3.74 | 0   | 0.98       | 0.94 | 0   |
| 7   | 89993       | 0.59     | 2.69 | 0   | 0.90     | 1.62 | 0   | 0.46         | 3.34 | 0.000 | 0.66         | 2.91 | 0.000 | 0.43       | 4.36 | 0   | 0.97       | 1.06 | 0   |
| 8   | 89428       | 0.68     | 2.46 | 0   | 0.93     | 1.41 | 0   | 0.50         | 3.27 | 0.000 | 0.67         | 2.94 | 0.000 | 0.43       | 4.56 | 0   | 0.96       | 1.22 | 0   |
| 9   | 89644       | 0.77     | 2.25 | 0   | 0.94     | 1.25 | 0   | 0.59         | 3.10 | 0.000 | 0.68         | 2.96 | 0.000 | 0.43       | 4.77 | 0   | 0.94       | 1.36 | 0   |
| 10  | 98337       | 0.81     | 2.11 | 0   | 0.94     | 1.21 | 0   | 0.68         | 2.84 | 0.000 | 0.67         | 3.00 | 0.000 | 0.43       | 5.04 | 0   | 0.91       | 1.50 | 0   |
| 11  | 97415       | 0.79     | 2.21 | 0   | 0.90     | 1.41 | 0   | 0.74         | 2.65 | 0.000 | 0.65         | 3.09 | 0.000 | 0.40       | 5.59 | 0   | 0.88       | 1.61 | 0   |
| 12  | 94326       | 0.71     | 2.66 | 0   | 0.83     | 1.86 | 0   | 0.74         | 2.60 | 0.000 | 0.61         | 3.28 | 0.000 | 0.33       | 6.44 | 0   | 0.84       | 1.72 | 0   |
| 13  | 94047       | 0.58     | 3.48 | 0   | 0.69     | 2.65 | 0   | 0.68         | 2.87 | 0.000 | 0.53         | 3.66 | 0.000 | 0.24       | 7.61 | 0   | 0.81       | 1.87 | 0   |

**Table 27.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | daon-000 |      |     | daon-001 |      |     | dermalog-001 |      |       | dermalog-002 |      |       | hzailu-001 |      |     | idemias-001 |      |     |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|------|-------|--------------|------|-------|------------|------|-----|-------------|------|-----|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE  | FTP   | Acc(3)       | MAE  | FTP   | Acc(3)     | MAE  | FTP | Acc(3)      | MAE  | FTP |
| 14  | 96915       | 0.44     | 4.35 | 0   | 0.53     | 3.50 | 0   | 0.59         | 3.43 | 0.000 | 0.45         | 4.10 | 0.000 | 0.16       | 8.66 | 0   | 0.77        | 2.06 | 0   |
| 15  | 97599       | 0.33     | 5.01 | 0   | 0.40     | 4.20 | 0   | 0.48         | 4.06 | 0.000 | 0.39         | 4.45 | 0.000 | 0.10       | 9.38 | 0   | 0.72        | 2.28 | 0   |
| 16  | 96388       | 0.28     | 5.27 | 0   | 0.33     | 4.58 | 0   | 0.41         | 4.50 | 0.000 | 0.36         | 4.60 | 0.000 | 0.07       | 9.59 | 0   | 0.69        | 2.46 | 0   |
| 17  | 99462       | 0.27     | 5.26 | 0   | 0.31     | 4.69 | 0   | 0.37         | 4.69 | 0.000 | 0.36         | 4.58 | 0.000 | 0.05       | 9.45 | 0   | 0.67        | 2.52 | 0   |
| 18  | 98765       | 0.28     | 5.16 | 0   | 0.30     | 4.70 | 0   | 0.33         | 4.89 | 0.000 | 0.36         | 4.62 | 0.000 | 0.05       | 9.12 | 0   | 0.63        | 2.73 | 0   |
| 19  | 105717      | 0.29     | 5.07 | 0   | 0.30     | 4.67 | 0   | 0.32         | 4.92 | 0.000 | 0.37         | 4.57 | 0.000 | 0.06       | 8.72 | 0   | 0.61        | 2.81 | 0   |
| 20  | 118643      | 0.31     | 4.90 | 0   | 0.31     | 4.55 | 0   | 0.34         | 4.81 | 0.000 | 0.39         | 4.49 | 0.000 | 0.06       | 8.34 | 0   | 0.62        | 2.77 | 0   |
| 21  | 124060      | 0.33     | 4.71 | 0   | 0.34     | 4.39 | 0   | 0.36         | 4.71 | 0.000 | 0.41         | 4.40 | 0.000 | 0.08       | 7.89 | 0   | 0.62        | 2.78 | 0   |
| 22  | 128864      | 0.36     | 4.53 | 0   | 0.37     | 4.22 | 0   | 0.37         | 4.59 | 0.000 | 0.43         | 4.36 | 0.000 | 0.10       | 7.47 | 0   | 0.61        | 2.80 | 0   |
| 23  | 133507      | 0.39     | 4.41 | 0   | 0.39     | 4.08 | 0   | 0.39         | 4.51 | 0.000 | 0.44         | 4.33 | 0.000 | 0.13       | 7.07 | 0   | 0.61        | 2.82 | 0   |
| 24  | 138094      | 0.41     | 4.26 | 0   | 0.42     | 3.93 | 0   | 0.41         | 4.42 | 0.000 | 0.44         | 4.31 | 0.000 | 0.17       | 6.66 | 0   | 0.60        | 2.85 | 0   |
| 25  | 139920      | 0.43     | 4.22 | 0   | 0.44     | 3.85 | 0   | 0.40         | 4.36 | 0.000 | 0.43         | 4.36 | 0.000 | 0.21       | 6.36 | 0   | 0.59        | 2.91 | 0   |
| 26  | 141741      | 0.44     | 4.15 | 0   | 0.46     | 3.76 | 0   | 0.42         | 4.30 | 0.000 | 0.43         | 4.38 | 0.000 | 0.26       | 6.08 | 0   | 0.59        | 2.93 | 0   |
| 27  | 138491      | 0.45     | 4.14 | 0   | 0.48     | 3.71 | 0   | 0.41         | 4.27 | 0.000 | 0.42         | 4.44 | 0.000 | 0.30       | 5.83 | 0   | 0.58        | 2.97 | 0   |
| 28  | 133633      | 0.45     | 4.15 | 0   | 0.49     | 3.69 | 0   | 0.43         | 4.27 | 0.000 | 0.41         | 4.51 | 0.000 | 0.34       | 5.66 | 0   | 0.57        | 3.02 | 0   |
| 29  | 129054      | 0.44     | 4.21 | 0   | 0.49     | 3.71 | 0   | 0.42         | 4.26 | 0.000 | 0.41         | 4.56 | 0.000 | 0.37       | 5.53 | 0   | 0.57        | 3.05 | 0   |
| 30  | 123437      | 0.43     | 4.24 | 0   | 0.48     | 3.74 | 0   | 0.44         | 4.25 | 0.000 | 0.40         | 4.60 | 0.000 | 0.38       | 5.47 | 0   | 0.56        | 3.10 | 0   |
| 31  | 118174      | 0.43     | 4.28 | 0   | 0.48     | 3.77 | 0   | 0.45         | 4.28 | 0.000 | 0.39         | 4.66 | 0.000 | 0.40       | 5.41 | 0   | 0.56        | 3.12 | 0   |
| 32  | 113131      | 0.42     | 4.32 | 0   | 0.48     | 3.81 | 0   | 0.46         | 4.32 | 0.000 | 0.39         | 4.73 | 0.000 | 0.39       | 5.47 | 0   | 0.56        | 3.14 | 0   |
| 33  | 109146      | 0.42     | 4.37 | 0   | 0.47     | 3.86 | 0   | 0.46         | 4.38 | 0.000 | 0.39         | 4.73 | 0.000 | 0.37       | 5.48 | 0   | 0.55        | 3.15 | 0   |
| 34  | 105908      | 0.41     | 4.42 | 0   | 0.46     | 3.92 | 0   | 0.44         | 4.45 | 0.000 | 0.39         | 4.79 | 0.000 | 0.35       | 5.57 | 0   | 0.55        | 3.20 | 0   |
| 35  | 103149      | 0.41     | 4.44 | 0   | 0.45     | 3.95 | 0   | 0.44         | 4.54 | 0.000 | 0.39         | 4.79 | 0.000 | 0.32       | 5.63 | 0   | 0.55        | 3.19 | 0   |
| 36  | 99901       | 0.40     | 4.48 | 0   | 0.44     | 4.01 | 0   | 0.41         | 4.65 | 0.000 | 0.38         | 4.86 | 0.000 | 0.30       | 5.73 | 0   | 0.54        | 3.22 | 0   |
| 37  | 97516       | 0.40     | 4.51 | 0   | 0.44     | 4.03 | 0   | 0.39         | 4.73 | 0.000 | 0.38         | 4.88 | 0.000 | 0.29       | 5.72 | 0   | 0.54        | 3.21 | 0   |
| 38  | 93939       | 0.40     | 4.49 | 0   | 0.43     | 4.03 | 0   | 0.35         | 4.78 | 0.000 | 0.37         | 4.89 | 0.000 | 0.28       | 5.75 | 0   | 0.55        | 3.21 | 0   |
| 39  | 91461       | 0.40     | 4.47 | 0   | 0.43     | 4.04 | 0   | 0.36         | 4.80 | 0.000 | 0.37         | 4.89 | 0.000 | 0.29       | 5.71 | 0   | 0.55        | 3.20 | 0   |
| 40  | 94056       | 0.40     | 4.49 | 0   | 0.43     | 4.08 | 0   | 0.33         | 4.89 | 0.000 | 0.37         | 4.92 | 0.000 | 0.29       | 5.71 | 0   | 0.54        | 3.22 | 0   |
| 41  | 90173       | 0.40     | 4.45 | 0   | 0.43     | 4.05 | 0   | 0.34         | 4.81 | 0.000 | 0.36         | 4.90 | 0.000 | 0.30       | 5.66 | 0   | 0.54        | 3.22 | 0   |
| 42  | 86423       | 0.41     | 4.43 | 0   | 0.43     | 4.03 | 0   | 0.35         | 4.71 | 0.000 | 0.37         | 4.87 | 0.000 | 0.31       | 5.56 | 0   | 0.54        | 3.21 | 0   |
| 43  | 83252       | 0.41     | 4.40 | 0   | 0.44     | 4.01 | 0   | 0.36         | 4.63 | 0.000 | 0.37         | 4.86 | 0.000 | 0.32       | 5.53 | 0   | 0.55        | 3.21 | 0   |
| 44  | 80602       | 0.42     | 4.34 | 0   | 0.44     | 3.97 | 0   | 0.39         | 4.51 | 0.000 | 0.37         | 4.83 | 0.000 | 0.33       | 5.48 | 0   | 0.55        | 3.20 | 0   |
| 45  | 77502       | 0.43     | 4.32 | 0   | 0.45     | 3.94 | 0   | 0.40         | 4.41 | 0.000 | 0.38         | 4.79 | 0.000 | 0.34       | 5.39 | 0   | 0.54        | 3.19 | 0   |
| 46  | 74182       | 0.43     | 4.26 | 0   | 0.45     | 3.91 | 0   | 0.42         | 4.29 | 0.000 | 0.38         | 4.74 | 0.000 | 0.34       | 5.35 | 0   | 0.54        | 3.19 | 0   |
| 47  | 70698       | 0.43     | 4.23 | 0   | 0.45     | 3.85 | 0   | 0.44         | 4.18 | 0.000 | 0.39         | 4.68 | 0.000 | 0.35       | 5.28 | 0   | 0.55        | 3.19 | 0   |
| 48  | 67856       | 0.45     | 4.17 | 0   | 0.46     | 3.79 | 0   | 0.43         | 4.13 | 0.000 | 0.39         | 4.63 | 0.000 | 0.36       | 5.22 | 0   | 0.55        | 3.17 | 0   |
| 49  | 65674       | 0.45     | 4.13 | 0   | 0.47     | 3.74 | 0   | 0.45         | 4.08 | 0.000 | 0.40         | 4.59 | 0.000 | 0.36       | 5.21 | 0   | 0.55        | 3.16 | 0   |
| 50  | 64896       | 0.45     | 4.09 | 0   | 0.48     | 3.68 | 0   | 0.45         | 4.02 | 0.000 | 0.41         | 4.52 | 0.000 | 0.37       | 5.15 | 0   | 0.55        | 3.16 | 0   |
| 51  | 63410       | 0.46     | 4.08 | 0   | 0.49     | 3.63 | 0   | 0.46         | 3.99 | 0.000 | 0.42         | 4.46 | 0.000 | 0.38       | 5.09 | 0   | 0.56        | 3.14 | 0   |

**Table 27.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | daon-000 |       |     | daon-001 |      |     | dermalog-001 |       |       | dermalog-002 |       |       | hzailu-001 |      |     | idemia-001 |      |     |
|-----|-------------|----------|-------|-----|----------|------|-----|--------------|-------|-------|--------------|-------|-------|------------|------|-----|------------|------|-----|
|     |             | Acc(3)   | MAE   | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE   | FTP   | Acc(3)       | MAE   | FTP   | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP |
| 52  | 60991       | 0.45     | 4.07  | 0   | 0.50     | 3.58 | 0   | 0.46         | 3.94  | 0.000 | 0.43         | 4.46  | 0.000 | 0.38       | 5.08 | 0   | 0.56       | 3.15 | 0   |
| 53  | 59550       | 0.45     | 4.05  | 0   | 0.51     | 3.53 | 0   | 0.46         | 3.90  | 0.000 | 0.43         | 4.42  | 0.000 | 0.38       | 5.06 | 0   | 0.56       | 3.12 | 0   |
| 54  | 56895       | 0.45     | 4.05  | 0   | 0.52     | 3.48 | 0   | 0.47         | 3.89  | 0.000 | 0.43         | 4.43  | 0.000 | 0.38       | 5.06 | 0   | 0.57       | 3.14 | 0   |
| 55  | 55297       | 0.44     | 4.06  | 0   | 0.52     | 3.45 | 0   | 0.46         | 3.87  | 0.000 | 0.43         | 4.41  | 0.000 | 0.38       | 5.03 | 0   | 0.56       | 3.16 | 0   |
| 56  | 52574       | 0.44     | 4.04  | 0   | 0.53     | 3.42 | 0   | 0.48         | 3.83  | 0.000 | 0.43         | 4.43  | 0.000 | 0.39       | 5.01 | 0   | 0.56       | 3.18 | 0   |
| 57  | 49823       | 0.44     | 4.07  | 0   | 0.54     | 3.43 | 0   | 0.48         | 3.84  | 0.000 | 0.43         | 4.48  | 0.000 | 0.38       | 5.10 | 0   | 0.56       | 3.24 | 0   |
| 58  | 48570       | 0.45     | 4.02  | 0   | 0.52     | 3.42 | 0   | 0.49         | 3.84  | 0.000 | 0.43         | 4.49  | 0.000 | 0.38       | 5.11 | 0   | 0.55       | 3.28 | 0   |
| 59  | 46040       | 0.45     | 4.06  | 0   | 0.51     | 3.47 | 0   | 0.49         | 3.85  | 0.000 | 0.42         | 4.61  | 0.000 | 0.38       | 5.11 | 0   | 0.54       | 3.37 | 0   |
| 60  | 46037       | 0.45     | 4.07  | 0   | 0.51     | 3.51 | 0   | 0.47         | 3.90  | 0.000 | 0.41         | 4.68  | 0.000 | 0.37       | 5.21 | 0   | 0.53       | 3.45 | 0   |
| 61  | 43860       | 0.46     | 4.06  | 0   | 0.50     | 3.54 | 0   | 0.47         | 3.95  | 0.000 | 0.40         | 4.77  | 0.000 | 0.37       | 5.23 | 0   | 0.51       | 3.56 | 0   |
| 62  | 41407       | 0.45     | 4.10  | 0   | 0.50     | 3.54 | 0   | 0.47         | 4.03  | 0.000 | 0.40         | 4.87  | 0.000 | 0.35       | 5.33 | 0   | 0.50       | 3.61 | 0   |
| 63  | 39444       | 0.45     | 4.16  | 0   | 0.50     | 3.53 | 0   | 0.45         | 4.12  | 0.000 | 0.38         | 4.98  | 0.000 | 0.35       | 5.37 | 0   | 0.49       | 3.68 | 0   |
| 64  | 37425       | 0.44     | 4.21  | 0   | 0.50     | 3.54 | 0   | 0.43         | 4.17  | 0.000 | 0.36         | 5.11  | 0.000 | 0.35       | 5.38 | 0   | 0.48       | 3.74 | 0   |
| 65  | 34834       | 0.43     | 4.27  | 0   | 0.51     | 3.54 | 0   | 0.42         | 4.27  | 0.000 | 0.35         | 5.17  | 0.000 | 0.34       | 5.46 | 0   | 0.47       | 3.79 | 0   |
| 66  | 32020       | 0.43     | 4.31  | 0   | 0.51     | 3.52 | 0   | 0.41         | 4.31  | 0.000 | 0.33         | 5.25  | 0.000 | 0.34       | 5.45 | 0   | 0.47       | 3.83 | 0   |
| 67  | 28898       | 0.41     | 4.44  | 0   | 0.50     | 3.58 | 0   | 0.40         | 4.36  | 0.000 | 0.31         | 5.29  | 0.000 | 0.33       | 5.50 | 0   | 0.46       | 3.89 | 0   |
| 68  | 26630       | 0.40     | 4.59  | 0   | 0.49     | 3.64 | 0   | 0.37         | 4.44  | 0.000 | 0.30         | 5.33  | 0.000 | 0.32       | 5.54 | 0   | 0.46       | 3.92 | 0   |
| 69  | 24864       | 0.39     | 4.65  | 0   | 0.49     | 3.66 | 0   | 0.38         | 4.43  | 0.000 | 0.30         | 5.32  | 0.000 | 0.32       | 5.55 | 0   | 0.45       | 3.91 | 0   |
| 70  | 23407       | 0.39     | 4.71  | 0   | 0.49     | 3.67 | 0   | 0.39         | 4.32  | 0.000 | 0.31         | 5.17  | 0.000 | 0.32       | 5.54 | 0   | 0.46       | 3.88 | 0   |
| 71  | 20736       | 0.38     | 4.81  | 0   | 0.48     | 3.71 | 0   | 0.39         | 4.29  | 0.000 | 0.33         | 5.09  | 0.000 | 0.33       | 5.51 | 0   | 0.46       | 3.87 | 0   |
| 72  | 18423       | 0.37     | 4.95  | 0   | 0.47     | 3.82 | 0   | 0.43         | 4.22  | 0.000 | 0.36         | 4.94  | 0.000 | 0.33       | 5.52 | 0   | 0.46       | 3.86 | 0   |
| 73  | 16215       | 0.36     | 5.02  | 0   | 0.47     | 3.81 | 0   | 0.47         | 4.11  | 0.000 | 0.40         | 4.78  | 0.000 | 0.33       | 5.55 | 0   | 0.47       | 3.78 | 0   |
| 74  | 14848       | 0.37     | 5.02  | 0   | 0.48     | 3.82 | 0   | 0.51         | 3.93  | 0.000 | 0.45         | 4.58  | 0.000 | 0.34       | 5.47 | 0   | 0.47       | 3.72 | 0   |
| 75  | 13197       | 0.35     | 5.26  | 0   | 0.46     | 4.00 | 0   | 0.52         | 3.93  | 0.000 | 0.49         | 4.55  | 0.000 | 0.33       | 5.60 | 0   | 0.48       | 3.69 | 0   |
| 76  | 11886       | 0.35     | 5.35  | 0   | 0.45     | 4.03 | 0   | 0.56         | 3.86  | 0.000 | 0.53         | 4.39  | 0.000 | 0.34       | 5.52 | 0   | 0.49       | 3.60 | 0   |
| 77  | 10839       | 0.35     | 5.47  | 0   | 0.44     | 4.10 | 0   | 0.54         | 4.10  | 0.000 | 0.53         | 4.49  | 0.000 | 0.34       | 5.67 | 0   | 0.50       | 3.52 | 0   |
| 78  | 9361        | 0.33     | 5.73  | 0   | 0.44     | 4.24 | 0   | 0.50         | 4.27  | 0.000 | 0.51         | 4.64  | 0.000 | 0.34       | 5.73 | 0   | 0.52       | 3.46 | 0   |
| 79  | 7540        | 0.32     | 5.86  | 0   | 0.43     | 4.30 | 0   | 0.42         | 4.57  | 0.000 | 0.46         | 4.83  | 0.000 | 0.34       | 5.81 | 0   | 0.54       | 3.35 | 0   |
| 80  | 6267        | 0.30     | 6.10  | 0   | 0.42     | 4.47 | 0   | 0.35         | 4.98  | 0.000 | 0.40         | 5.27  | 0.000 | 0.33       | 6.04 | 0   | 0.55       | 3.31 | 0   |
| 81  | 5221        | 0.29     | 6.29  | 0   | 0.41     | 4.54 | 0   | 0.28         | 5.34  | 0.000 | 0.33         | 5.62  | 0.000 | 0.32       | 6.09 | 0   | 0.58       | 3.22 | 0   |
| 82  | 3962        | 0.26     | 6.68  | 0   | 0.39     | 4.79 | 0   | 0.18         | 5.95  | 0.000 | 0.22         | 6.12  | 0.000 | 0.32       | 6.23 | 0   | 0.57       | 3.29 | 0   |
| 83  | 3277        | 0.21     | 7.12  | 0   | 0.35     | 5.03 | 0   | 0.12         | 6.50  | 0.000 | 0.13         | 6.67  | 0.000 | 0.31       | 6.44 | 0   | 0.57       | 3.28 | 0   |
| 84  | 2615        | 0.18     | 7.58  | 0   | 0.32     | 5.34 | 0   | 0.06         | 7.19  | 0.000 | 0.07         | 7.40  | 0.000 | 0.27       | 6.87 | 0   | 0.56       | 3.44 | 0   |
| 85  | 2206        | 0.15     | 7.97  | 0   | 0.29     | 5.53 | 0   | 0.03         | 7.81  | 0.000 | 0.04         | 7.97  | 0.000 | 0.28       | 7.11 | 0   | 0.52       | 3.60 | 0   |
| 86  | 1831        | 0.09     | 8.75  | 0   | 0.27     | 5.95 | 0   | 0.01         | 8.42  | 0.000 | 0.02         | 8.67  | 0.000 | 0.24       | 7.44 | 0   | 0.48       | 3.95 | 0   |
| 87  | 1402        | 0.07     | 9.20  | 0   | 0.23     | 6.41 | 0   | 0.00         | 9.20  | 0.001 | 0.01         | 9.35  | 0.001 | 0.21       | 8.07 | 0   | 0.39       | 4.41 | 0   |
| 88  | 1254        | 0.03     | 10.01 | 0   | 0.17     | 6.92 | 0   | 0.00         | 10.04 | 0.000 | 0.00         | 10.32 | 0.000 | 0.17       | 8.59 | 0   | 0.30       | 5.05 | 0   |
| 89  | 907         | 0.02     | 10.51 | 0   | 0.14     | 7.20 | 0   | 0.00         | 10.57 | 0.000 | 0.00         | 10.72 | 0.000 | 0.14       | 8.79 | 0   | 0.23       | 5.29 | 0   |

**Table 27.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | daon-000 |       |     | daon-001 |      |     | dermalog-001 |       |       | dermalog-002 |       |       | hzailu-001 |      |     | idemia-001 |      |     |
|-----|-------------|----------|-------|-----|----------|------|-----|--------------|-------|-------|--------------|-------|-------|------------|------|-----|------------|------|-----|
|     |             | Acc(3)   | MAE   | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE   | FTP   | Acc(3)       | MAE   | FTP   | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP |
| 90  | 670         | 0.01     | 11.31 | 0   | 0.10     | 7.94 | 0   | 0.00         | 11.48 | 0.000 | 0.00         | 11.81 | 0.000 | 0.09       | 9.81 | 0   | 0.14       | 6.34 | 0   |

**Table 28.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better).

| Age | Num. Images | idemia-002 |      |     | inbut-000 |       |     | incode-000 |      |     | incode-001 |      |       | innovatrics-000 |      |     | innovatrics-001 |      |     |
|-----|-------------|------------|------|-----|-----------|-------|-----|------------|------|-----|------------|------|-------|-----------------|------|-----|-----------------|------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)    | MAE   | FTP | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP   | Acc(3)          | MAE  | FTP | Acc(3)          | MAE  | FTP |
| 0   | 133387      | 0.93       | 1.27 | 0   | 1.00      | 0.38  | 0   | 1.00       | 0.34 | 0   | 0.99       | 0.32 | 0.001 | 1.00            | 0.74 | 0   | 1.00            | 0.36 | 0   |
| 1   | 94475       | 0.89       | 1.59 | 0   | 0.99      | 0.85  | 0   | 1.00       | 0.63 | 0   | 0.99       | 0.57 | 0.000 | 0.99            | 0.79 | 0   | 1.00            | 0.61 | 0   |
| 2   | 82596       | 0.85       | 1.83 | 0   | 0.96      | 1.41  | 0   | 0.99       | 0.92 | 0   | 0.99       | 0.77 | 0.000 | 0.97            | 1.31 | 0   | 0.98            | 1.12 | 0   |
| 3   | 82473       | 0.87       | 1.78 | 0   | 0.95      | 1.55  | 0   | 0.98       | 1.10 | 0   | 0.99       | 0.75 | 0.000 | 0.96            | 1.48 | 0   | 0.97            | 1.34 | 0   |
| 4   | 84055       | 0.90       | 1.56 | 0   | 0.94      | 1.45  | 0   | 0.96       | 1.24 | 0   | 0.98       | 0.83 | 0.000 | 0.95            | 1.41 | 0   | 0.96            | 1.29 | 0   |
| 5   | 92521       | 0.93       | 1.34 | 0   | 0.93      | 1.35  | 0   | 0.93       | 1.34 | 0   | 0.97       | 1.07 | 0.000 | 0.94            | 1.32 | 0   | 0.94            | 1.24 | 0   |
| 6   | 91639       | 0.93       | 1.23 | 0   | 0.90      | 1.41  | 0   | 0.91       | 1.36 | 0   | 0.94       | 1.27 | 0.000 | 0.91            | 1.36 | 0   | 0.90            | 1.35 | 0   |
| 7   | 89993       | 0.93       | 1.22 | 0   | 0.84      | 1.71  | 0   | 0.93       | 1.27 | 0   | 0.92       | 1.38 | 0.000 | 0.85            | 1.62 | 0   | 0.85            | 1.64 | 0   |
| 8   | 89428       | 0.93       | 1.30 | 0   | 0.77      | 2.13  | 0   | 0.91       | 1.42 | 0   | 0.85       | 1.77 | 0.000 | 0.80            | 1.86 | 0   | 0.80            | 1.91 | 0   |
| 9   | 89644       | 0.94       | 1.34 | 0   | 0.70      | 2.68  | 0   | 0.89       | 1.56 | 0   | 0.77       | 2.29 | 0.000 | 0.77            | 2.03 | 0   | 0.77            | 2.09 | 0   |
| 10  | 98337       | 0.93       | 1.33 | 0   | 0.63      | 3.54  | 0   | 0.88       | 1.71 | 0   | 0.67       | 2.84 | 0.000 | 0.74            | 2.19 | 0   | 0.72            | 2.26 | 0   |
| 11  | 97415       | 0.92       | 1.36 | 0   | 0.52      | 4.94  | 0   | 0.83       | 1.87 | 0   | 0.55       | 3.36 | 0.000 | 0.68            | 2.45 | 0   | 0.64            | 2.55 | 0   |
| 12  | 94326       | 0.89       | 1.46 | 0   | 0.35      | 6.83  | 0   | 0.76       | 2.03 | 0   | 0.43       | 3.73 | 0.000 | 0.59            | 2.79 | 0   | 0.55            | 2.93 | 0   |
| 13  | 94047       | 0.84       | 1.71 | 0   | 0.20      | 8.89  | 0   | 0.68       | 2.31 | 0   | 0.30       | 3.90 | 0.000 | 0.51            | 3.16 | 0   | 0.46            | 3.32 | 0   |
| 14  | 96915       | 0.78       | 2.07 | 0   | 0.12      | 10.41 | 0   | 0.64       | 2.58 | 0   | 0.20       | 3.81 | 0.000 | 0.46            | 3.42 | 0   | 0.43            | 3.56 | 0   |
| 15  | 97599       | 0.70       | 2.47 | 0   | 0.08      | 11.34 | 0   | 0.68       | 2.71 | 0   | 0.40       | 3.52 | 0.000 | 0.45            | 3.54 | 0   | 0.42            | 3.65 | 0   |
| 16  | 96388       | 0.64       | 2.80 | 0   | 0.07      | 11.74 | 0   | 0.73       | 2.73 | 0   | 0.71       | 3.18 | 0.000 | 0.46            | 3.55 | 0   | 0.45            | 3.59 | 0   |
| 17  | 99462       | 0.61       | 2.98 | 0   | 0.06      | 11.74 | 0   | 0.70       | 2.69 | 0   | 0.72       | 2.87 | 0.000 | 0.47            | 3.48 | 0   | 0.48            | 3.44 | 0   |
| 18  | 98765       | 0.56       | 3.32 | 0   | 0.06      | 11.47 | 0   | 0.62       | 2.98 | 0   | 0.64       | 2.98 | 0.000 | 0.46            | 3.48 | 0   | 0.47            | 3.38 | 0   |
| 19  | 105717      | 0.53       | 3.48 | 0   | 0.05      | 11.24 | 0   | 0.56       | 3.28 | 0   | 0.58       | 3.25 | 0.000 | 0.48            | 3.42 | 0   | 0.50            | 3.26 | 0   |
| 20  | 118643      | 0.53       | 3.47 | 0   | 0.05      | 10.99 | 0   | 0.53       | 3.50 | 0   | 0.55       | 3.52 | 0.000 | 0.50            | 3.32 | 0   | 0.54            | 3.11 | 0   |
| 21  | 124060      | 0.52       | 3.52 | 0   | 0.05      | 10.62 | 0   | 0.49       | 3.63 | 0   | 0.52       | 3.79 | 0.000 | 0.53            | 3.23 | 0   | 0.57            | 2.99 | 0   |
| 22  | 128864      | 0.51       | 3.57 | 0   | 0.05      | 10.26 | 0   | 0.46       | 3.65 | 0   | 0.46       | 3.94 | 0.000 | 0.54            | 3.16 | 0   | 0.59            | 2.91 | 0   |
| 23  | 133507      | 0.50       | 3.60 | 0   | 0.06      | 9.91  | 0   | 0.47       | 3.59 | 0   | 0.43       | 4.01 | 0.000 | 0.55            | 3.12 | 0   | 0.60            | 2.86 | 0   |
| 24  | 138094      | 0.49       | 3.62 | 0   | 0.06      | 9.57  | 0   | 0.49       | 3.50 | 0   | 0.43       | 4.01 | 0.000 | 0.56            | 3.09 | 0   | 0.60            | 2.83 | 0   |
| 25  | 139920      | 0.48       | 3.65 | 0   | 0.07      | 9.30  | 0   | 0.51       | 3.44 | 0   | 0.43       | 3.99 | 0.000 | 0.56            | 3.10 | 0   | 0.60            | 2.86 | 0   |
| 26  | 141741      | 0.48       | 3.64 | 0   | 0.08      | 9.02  | 0   | 0.53       | 3.36 | 0   | 0.44       | 3.95 | 0.000 | 0.56            | 3.09 | 0   | 0.60            | 2.87 | 0   |
| 27  | 138491      | 0.49       | 3.64 | 0   | 0.09      | 8.75  | 0   | 0.54       | 3.36 | 0   | 0.44       | 3.91 | 0.000 | 0.56            | 3.10 | 0   | 0.60            | 2.90 | 0   |
| 28  | 133633      | 0.48       | 3.66 | 0   | 0.10      | 8.56  | 0   | 0.54       | 3.39 | 0   | 0.45       | 3.90 | 0.000 | 0.55            | 3.12 | 0   | 0.59            | 2.92 | 0   |

**Table 28.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | idemia-002 |      |     | inbut-000 |      |     | incode-000 |      |     | incode-001 |      |       | innovatrics-000 |      |     | innovatrics-001 |      |     |
|-----|-------------|------------|------|-----|-----------|------|-----|------------|------|-----|------------|------|-------|-----------------|------|-----|-----------------|------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)    | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP   | Acc(3)          | MAE  | FTP | Acc(3)          | MAE  | FTP |
| 29  | 129054      | 0.49       | 3.67 | 0   | 0.12      | 8.39 | 0   | 0.53       | 3.44 | 0   | 0.45       | 3.90 | 0.000 | 0.55            | 3.14 | 0   | 0.58            | 2.96 | 0   |
| 30  | 123437      | 0.48       | 3.69 | 0   | 0.13      | 8.24 | 0   | 0.52       | 3.52 | 0   | 0.45       | 3.92 | 0.000 | 0.54            | 3.16 | 0   | 0.57            | 3.01 | 0   |
| 31  | 118174      | 0.48       | 3.70 | 0   | 0.14      | 8.07 | 0   | 0.51       | 3.63 | 0   | 0.44       | 3.97 | 0.000 | 0.54            | 3.14 | 0   | 0.56            | 3.02 | 0   |
| 32  | 113131      | 0.48       | 3.72 | 0   | 0.15      | 7.91 | 0   | 0.49       | 3.74 | 0   | 0.44       | 4.01 | 0.000 | 0.54            | 3.11 | 0   | 0.56            | 3.03 | 0   |
| 33  | 109146      | 0.48       | 3.70 | 0   | 0.17      | 7.80 | 0   | 0.46       | 3.84 | 0   | 0.42       | 4.08 | 0.000 | 0.55            | 3.07 | 0   | 0.57            | 3.00 | 0   |
| 34  | 105908      | 0.47       | 3.75 | 0   | 0.18      | 7.65 | 0   | 0.44       | 3.95 | 0   | 0.41       | 4.16 | 0.000 | 0.57            | 3.03 | 0   | 0.58            | 2.96 | 0   |
| 35  | 103149      | 0.47       | 3.74 | 0   | 0.19      | 7.46 | 0   | 0.42       | 4.00 | 0   | 0.40       | 4.22 | 0.000 | 0.58            | 2.96 | 0   | 0.60            | 2.89 | 0   |
| 36  | 99901       | 0.47       | 3.75 | 0   | 0.21      | 7.31 | 0   | 0.41       | 4.02 | 0   | 0.39       | 4.26 | 0.000 | 0.59            | 2.91 | 0   | 0.61            | 2.86 | 0   |
| 37  | 97516       | 0.47       | 3.72 | 0   | 0.22      | 7.13 | 0   | 0.41       | 3.99 | 0   | 0.38       | 4.31 | 0.000 | 0.60            | 2.86 | 0   | 0.61            | 2.83 | 0   |
| 38  | 93939       | 0.48       | 3.71 | 0   | 0.24      | 6.91 | 0   | 0.42       | 3.92 | 0   | 0.39       | 4.31 | 0.000 | 0.60            | 2.85 | 0   | 0.60            | 2.86 | 0   |
| 39  | 91461       | 0.48       | 3.69 | 0   | 0.25      | 6.74 | 0   | 0.45       | 3.80 | 0   | 0.39       | 4.30 | 0.000 | 0.61            | 2.81 | 0   | 0.60            | 2.88 | 0   |
| 40  | 94056       | 0.49       | 3.65 | 0   | 0.25      | 6.76 | 0   | 0.47       | 3.72 | 0   | 0.38       | 4.32 | 0.000 | 0.61            | 2.81 | 0   | 0.60            | 2.90 | 0   |
| 41  | 90173       | 0.48       | 3.65 | 0   | 0.26      | 6.51 | 0   | 0.49       | 3.62 | 0   | 0.39       | 4.29 | 0.000 | 0.61            | 2.80 | 0   | 0.59            | 2.92 | 0   |
| 42  | 86423       | 0.48       | 3.63 | 0   | 0.26      | 6.31 | 0   | 0.51       | 3.56 | 0   | 0.39       | 4.28 | 0.000 | 0.62            | 2.77 | 0   | 0.59            | 2.95 | 0   |
| 43  | 83252       | 0.49       | 3.62 | 0   | 0.27      | 6.15 | 0   | 0.52       | 3.52 | 0   | 0.40       | 4.25 | 0.000 | 0.62            | 2.77 | 0   | 0.58            | 3.01 | 0   |
| 44  | 80602       | 0.49       | 3.58 | 0   | 0.27      | 5.94 | 0   | 0.51       | 3.52 | 0   | 0.40       | 4.23 | 0.000 | 0.62            | 2.77 | 0   | 0.58            | 3.09 | 0   |
| 45  | 77502       | 0.49       | 3.57 | 0   | 0.28      | 5.73 | 0   | 0.51       | 3.54 | 0   | 0.41       | 4.22 | 0.000 | 0.62            | 2.79 | 0   | 0.56            | 3.18 | 0   |
| 46  | 74182       | 0.49       | 3.56 | 0   | 0.29      | 5.52 | 0   | 0.50       | 3.59 | 0   | 0.41       | 4.23 | 0.000 | 0.61            | 2.81 | 0   | 0.54            | 3.28 | 0   |
| 47  | 70698       | 0.49       | 3.55 | 0   | 0.30      | 5.35 | 0   | 0.50       | 3.65 | 0   | 0.41       | 4.24 | 0.000 | 0.61            | 2.83 | 0   | 0.53            | 3.37 | 0   |
| 48  | 67856       | 0.50       | 3.52 | 0   | 0.32      | 5.16 | 0   | 0.50       | 3.72 | 0   | 0.42       | 4.28 | 0.000 | 0.61            | 2.84 | 0   | 0.51            | 3.44 | 0   |
| 49  | 65674       | 0.50       | 3.51 | 0   | 0.34      | 4.98 | 0   | 0.51       | 3.79 | 0   | 0.42       | 4.29 | 0.000 | 0.60            | 2.86 | 0   | 0.50            | 3.52 | 0   |
| 50  | 64896       | 0.50       | 3.49 | 0   | 0.35      | 4.80 | 0   | 0.51       | 3.89 | 0   | 0.41       | 4.32 | 0.000 | 0.60            | 2.85 | 0   | 0.49            | 3.57 | 0   |
| 51  | 63410       | 0.51       | 3.47 | 0   | 0.37      | 4.66 | 0   | 0.50       | 4.03 | 0   | 0.40       | 4.36 | 0.000 | 0.60            | 2.87 | 0   | 0.48            | 3.64 | 0   |
| 52  | 60991       | 0.51       | 3.46 | 0   | 0.38      | 4.51 | 0   | 0.48       | 4.16 | 0   | 0.39       | 4.37 | 0.000 | 0.60            | 2.88 | 0   | 0.48            | 3.67 | 0   |
| 53  | 59550       | 0.52       | 3.42 | 0   | 0.39      | 4.40 | 0   | 0.44       | 4.28 | 0   | 0.37       | 4.34 | 0.000 | 0.60            | 2.87 | 0   | 0.48            | 3.65 | 0   |
| 54  | 56895       | 0.52       | 3.43 | 0   | 0.40      | 4.29 | 0   | 0.38       | 4.36 | 0   | 0.35       | 4.27 | 0.000 | 0.60            | 2.87 | 0   | 0.48            | 3.62 | 0   |
| 55  | 55297       | 0.53       | 3.41 | 0   | 0.41      | 4.17 | 0   | 0.32       | 4.38 | 0   | 0.35       | 4.15 | 0.000 | 0.60            | 2.91 | 0   | 0.49            | 3.57 | 0   |
| 56  | 52574       | 0.53       | 3.42 | 0   | 0.42      | 4.08 | 0   | 0.28       | 4.30 | 0   | 0.37       | 4.01 | 0.000 | 0.60            | 2.91 | 0   | 0.51            | 3.46 | 0   |
| 57  | 49823       | 0.53       | 3.45 | 0   | 0.44      | 3.97 | 0   | 0.31       | 4.10 | 0   | 0.43       | 3.80 | 0.000 | 0.58            | 2.97 | 0   | 0.52            | 3.41 | 0   |
| 58  | 48570       | 0.53       | 3.47 | 0   | 0.45      | 3.85 | 0   | 0.41       | 3.78 | 0   | 0.48       | 3.57 | 0.000 | 0.57            | 3.03 | 0   | 0.54            | 3.29 | 0   |
| 59  | 46040       | 0.52       | 3.53 | 0   | 0.47      | 3.80 | 0   | 0.51       | 3.39 | 0   | 0.52       | 3.39 | 0.000 | 0.56            | 3.11 | 0   | 0.55            | 3.23 | 0   |
| 60  | 46037       | 0.51       | 3.58 | 0   | 0.48      | 3.70 | 0   | 0.58       | 2.99 | 0   | 0.55       | 3.23 | 0.000 | 0.55            | 3.18 | 0   | 0.57            | 3.13 | 0   |
| 61  | 43860       | 0.50       | 3.67 | 0   | 0.49      | 3.65 | 0   | 0.64       | 2.70 | 0   | 0.57       | 3.15 | 0.000 | 0.53            | 3.28 | 0   | 0.58            | 3.09 | 0   |
| 62  | 41407       | 0.49       | 3.70 | 0   | 0.50      | 3.56 | 0   | 0.70       | 2.52 | 0   | 0.58       | 3.06 | 0.000 | 0.52            | 3.36 | 0   | 0.58            | 3.06 | 0   |
| 63  | 39444       | 0.48       | 3.77 | 0   | 0.52      | 3.50 | 0   | 0.71       | 2.47 | 0   | 0.59       | 3.05 | 0.000 | 0.50            | 3.47 | 0   | 0.57            | 3.10 | 0   |
| 64  | 37425       | 0.47       | 3.81 | 0   | 0.52      | 3.50 | 0   | 0.70       | 2.50 | 0   | 0.58       | 3.06 | 0.000 | 0.49            | 3.58 | 0   | 0.56            | 3.13 | 0   |
| 65  | 34834       | 0.47       | 3.85 | 0   | 0.52      | 3.48 | 0   | 0.66       | 2.57 | 0   | 0.58       | 3.05 | 0.000 | 0.47            | 3.69 | 0   | 0.55            | 3.21 | 0   |
| 66  | 32020       | 0.46       | 3.88 | 0   | 0.52      | 3.50 | 0   | 0.63       | 2.68 | 0   | 0.58       | 3.04 | 0.000 | 0.46            | 3.79 | 0   | 0.53            | 3.33 | 0   |



**Table 28.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | idemia-002 |      |     | inbut-000 |      |     | incode-000 |      |     | incode-001 |      |       | innovatrics-000 |       |     | innovatrics-001 |       |     |
|-----|-------------|------------|------|-----|-----------|------|-----|------------|------|-----|------------|------|-------|-----------------|-------|-----|-----------------|-------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)    | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP   | Acc(3)          | MAE   | FTP | Acc(3)          | MAE   | FTP |
| 67  | 28898       | 0.46       | 3.94 | 0   | 0.50      | 3.55 | 0   | 0.59       | 2.84 | 0   | 0.57       | 3.08 | 0.000 | 0.44            | 3.95  | 0   | 0.51            | 3.46  | 0   |
| 68  | 26630       | 0.45       | 3.96 | 0   | 0.49      | 3.61 | 0   | 0.56       | 3.03 | 0   | 0.57       | 3.07 | 0.000 | 0.41            | 4.15  | 0   | 0.47            | 3.68  | 0   |
| 69  | 24864       | 0.45       | 3.94 | 0   | 0.49      | 3.63 | 0   | 0.53       | 3.17 | 0   | 0.57       | 3.08 | 0.000 | 0.40            | 4.28  | 0   | 0.45            | 3.84  | 0   |
| 70  | 23407       | 0.46       | 3.91 | 0   | 0.48      | 3.68 | 0   | 0.52       | 3.29 | 0   | 0.56       | 3.09 | 0.000 | 0.38            | 4.45  | 0   | 0.44            | 3.99  | 0   |
| 71  | 20736       | 0.45       | 3.89 | 0   | 0.48      | 3.74 | 0   | 0.49       | 3.45 | 0   | 0.56       | 3.11 | 0.000 | 0.36            | 4.63  | 0   | 0.42            | 4.16  | 0   |
| 72  | 18423       | 0.46       | 3.87 | 0   | 0.47      | 3.76 | 0   | 0.48       | 3.63 | 0   | 0.55       | 3.15 | 0.000 | 0.33            | 4.89  | 0   | 0.39            | 4.43  | 0   |
| 73  | 16215       | 0.47       | 3.78 | 0   | 0.46      | 3.85 | 0   | 0.46       | 3.78 | 0   | 0.55       | 3.14 | 0.000 | 0.30            | 5.14  | 0   | 0.37            | 4.64  | 0   |
| 74  | 14848       | 0.47       | 3.72 | 0   | 0.47      | 3.82 | 0   | 0.45       | 3.91 | 0   | 0.56       | 3.13 | 0.000 | 0.27            | 5.38  | 0   | 0.35            | 4.80  | 0   |
| 75  | 13197       | 0.48       | 3.65 | 0   | 0.46      | 3.97 | 0   | 0.43       | 4.15 | 0   | 0.55       | 3.18 | 0.000 | 0.23            | 5.82  | 0   | 0.32            | 5.14  | 0   |
| 76  | 11886       | 0.49       | 3.59 | 0   | 0.46      | 3.95 | 0   | 0.41       | 4.33 | 0   | 0.55       | 3.18 | 0.000 | 0.19            | 6.13  | 0   | 0.31            | 5.27  | 0   |
| 77  | 10839       | 0.50       | 3.49 | 0   | 0.46      | 3.99 | 0   | 0.37       | 4.61 | 0   | 0.55       | 3.19 | 0.000 | 0.14            | 6.60  | 0   | 0.28            | 5.55  | 0   |
| 78  | 9361        | 0.52       | 3.41 | 0   | 0.46      | 4.11 | 0   | 0.35       | 4.86 | 0   | 0.56       | 3.25 | 0.000 | 0.10            | 7.03  | 0   | 0.27            | 5.83  | 0   |
| 79  | 7540        | 0.55       | 3.26 | 0   | 0.46      | 4.21 | 0   | 0.32       | 5.05 | 0   | 0.54       | 3.28 | 0.000 | 0.07            | 7.46  | 0   | 0.24            | 6.09  | 0   |
| 80  | 6267        | 0.56       | 3.23 | 0   | 0.45      | 4.38 | 0   | 0.30       | 5.33 | 0   | 0.53       | 3.38 | 0.000 | 0.04            | 7.94  | 0   | 0.22            | 6.34  | 0   |
| 81  | 5221        | 0.58       | 3.12 | 0   | 0.45      | 4.41 | 0   | 0.26       | 5.63 | 0   | 0.52       | 3.41 | 0.000 | 0.02            | 8.42  | 0   | 0.20            | 6.57  | 0   |
| 82  | 3962        | 0.58       | 3.16 | 0   | 0.42      | 4.63 | 0   | 0.23       | 5.98 | 0   | 0.51       | 3.58 | 0.000 | 0.00            | 8.93  | 0   | 0.17            | 6.93  | 0   |
| 83  | 3277        | 0.59       | 3.11 | 0   | 0.36      | 5.12 | 0   | 0.20       | 6.48 | 0   | 0.48       | 3.78 | 0.000 | 0.00            | 9.54  | 0   | 0.14            | 7.37  | 0   |
| 84  | 2615        | 0.58       | 3.26 | 0   | 0.31      | 5.50 | 0   | 0.18       | 6.80 | 0   | 0.47       | 3.88 | 0.000 | 0.00            | 10.17 | 0   | 0.13            | 7.68  | 0   |
| 85  | 2206        | 0.55       | 3.39 | 0   | 0.25      | 5.84 | 0   | 0.17       | 7.17 | 0   | 0.44       | 4.04 | 0.000 | 0.00            | 10.69 | 0   | 0.11            | 8.01  | 0   |
| 86  | 1831        | 0.51       | 3.74 | 0   | 0.18      | 6.41 | 0   | 0.14       | 7.56 | 0   | 0.42       | 4.30 | 0.000 | 0.00            | 11.35 | 0   | 0.07            | 8.65  | 0   |
| 87  | 1402        | 0.43       | 4.18 | 0   | 0.13      | 6.87 | 0   | 0.11       | 8.16 | 0   | 0.39       | 4.52 | 0.000 | 0.00            | 12.13 | 0   | 0.06            | 9.07  | 0   |
| 88  | 1254        | 0.33       | 4.83 | 0   | 0.08      | 7.64 | 0   | 0.09       | 8.59 | 0   | 0.39       | 4.76 | 0.000 | 0.00            | 12.87 | 0   | 0.05            | 9.71  | 0   |
| 89  | 907         | 0.25       | 5.08 | 0   | 0.06      | 7.81 | 0   | 0.06       | 8.85 | 0   | 0.36       | 4.75 | 0.000 | 0.00            | 13.33 | 0   | 0.03            | 9.92  | 0   |
| 90  | 670         | 0.16       | 6.12 | 0   | 0.03      | 8.80 | 0   | 0.02       | 9.77 | 0   | 0.31       | 5.58 | 0.000 | 0.00            | 14.39 | 0   | 0.02            | 10.93 | 0   |

**Table 29.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better).

| Age | Num. Images | innovatrics-002 |      |     | jumio-000 |       |       | neuro-000 |      |     | neuro-001 |      |       | nominder-000 |      |     | nominder-001 |      |     |
|-----|-------------|-----------------|------|-----|-----------|-------|-------|-----------|------|-----|-----------|------|-------|--------------|------|-----|--------------|------|-----|
|     |             | Acc(3)          | MAE  | FTP | Acc(3)    | MAE   | FTP   | Acc(3)    | MAE  | FTP | Acc(3)    | MAE  | FTP   | Acc(3)       | MAE  | FTP | Acc(3)       | MAE  | FTP |
| 0   | 133387      | 1.00            | 0.41 | 0   | 0.00      | 18.43 | 0.018 | 0.97      | 1.23 | 0   | 0.85      | 1.87 | 0.000 | 0.99         | 0.68 | 0   | 0.99         | 0.66 | 0   |
| 1   | 94475       | 0.99            | 0.74 | 0   | 0.01      | 15.25 | 0.009 | 0.96      | 1.11 | 0   | 0.84      | 1.82 | 0.000 | 0.92         | 1.07 | 0   | 0.92         | 1.04 | 0   |
| 2   | 82596       | 0.97            | 1.18 | 0   | 0.02      | 13.27 | 0.011 | 0.92      | 1.25 | 0   | 0.82      | 1.80 | 0.000 | 0.84         | 1.80 | 0   | 0.83         | 1.87 | 0   |
| 3   | 82473       | 0.95            | 1.48 | 0   | 0.06      | 10.99 | 0.011 | 0.88      | 1.50 | 0   | 0.80      | 1.95 | 0.000 | 0.91         | 1.90 | 0   | 0.95         | 1.89 | 0   |
| 4   | 84055       | 0.94            | 1.48 | 0   | 0.11      | 9.37  | 0.015 | 0.86      | 1.64 | 0   | 0.79      | 2.09 | 0.000 | 0.87         | 1.75 | 0   | 0.92         | 1.57 | 0   |
| 5   | 92521       | 0.92            | 1.41 | 0   | 0.19      | 8.06  | 0.021 | 0.84      | 1.76 | 0   | 0.78      | 2.22 | 0.000 | 0.80         | 1.74 | 0   | 0.87         | 1.38 | 0   |

**Table 29.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | innovatrics-002 |      |     | jumio-000 |      |       | neuro-000 |      |     | neuro-001 |      |       | nominder-000 |      |     | nominder-001 |      |     |
|-----|-------------|-----------------|------|-----|-----------|------|-------|-----------|------|-----|-----------|------|-------|--------------|------|-----|--------------|------|-----|
|     |             | Acc(3)          | MAE  | FTP | Acc(3)    | MAE  | FTP   | Acc(3)    | MAE  | FTP | Acc(3)    | MAE  | FTP   | Acc(3)       | MAE  | FTP | Acc(3)       | MAE  | FTP |
| 6   | 91639       | 0.88            | 1.49 | 0   | 0.26      | 7.29 | 0.028 | 0.82      | 1.96 | 0   | 0.75      | 2.49 | 0.000 | 0.71         | 2.12 | 0   | 0.82         | 1.65 | 0   |
| 7   | 89993       | 0.82            | 1.77 | 0   | 0.30      | 6.84 | 0.038 | 0.75      | 2.39 | 0   | 0.67      | 3.00 | 0.000 | 0.59         | 2.69 | 0   | 0.77         | 2.08 | 0   |
| 8   | 89428       | 0.78            | 1.98 | 0   | 0.30      | 6.83 | 0.032 | 0.67      | 2.83 | 0   | 0.60      | 3.42 | 0.000 | 0.54         | 2.91 | 0   | 0.70         | 2.46 | 0   |
| 9   | 89644       | 0.77            | 2.05 | 0   | 0.27      | 7.00 | 0.028 | 0.60      | 3.25 | 0   | 0.54      | 3.79 | 0.000 | 0.53         | 2.94 | 0   | 0.63         | 2.72 | 0   |
| 10  | 98337       | 0.77            | 2.02 | 0   | 0.23      | 7.10 | 0.030 | 0.53      | 3.68 | 0   | 0.47      | 4.15 | 0.000 | 0.57         | 2.91 | 0   | 0.61         | 2.96 | 0   |
| 11  | 97415       | 0.77            | 2.01 | 0   | 0.18      | 7.04 | 0.032 | 0.45      | 4.16 | 0   | 0.40      | 4.56 | 0.000 | 0.63         | 2.84 | 0   | 0.57         | 3.22 | 0   |
| 12  | 94326       | 0.83            | 1.99 | 0   | 0.14      | 6.68 | 0.038 | 0.35      | 4.68 | 0   | 0.31      | 4.99 | 0.000 | 0.66         | 2.78 | 0   | 0.51         | 3.54 | 0   |
| 13  | 94047       | 0.85            | 1.96 | 0   | 0.12      | 6.13 | 0.041 | 0.25      | 5.17 | 0   | 0.21      | 5.37 | 0.000 | 0.64         | 2.82 | 0   | 0.41         | 3.92 | 0   |
| 14  | 96915       | 0.83            | 1.91 | 0   | 0.13      | 5.62 | 0.041 | 0.18      | 5.34 | 0   | 0.16      | 5.46 | 0.000 | 0.61         | 2.91 | 0   | 0.34         | 4.17 | 0   |
| 15  | 97599       | 0.79            | 1.95 | 0   | 0.21      | 5.24 | 0.039 | 0.16      | 5.29 | 0   | 0.14      | 5.36 | 0.000 | 0.57         | 3.11 | 0   | 0.32         | 4.31 | 0   |
| 16  | 96388       | 0.75            | 2.10 | 0   | 0.32      | 4.94 | 0.035 | 0.20      | 4.96 | 0   | 0.19      | 5.01 | 0.000 | 0.54         | 3.28 | 0   | 0.35         | 4.28 | 0   |
| 17  | 99462       | 0.71            | 2.35 | 0   | 0.39      | 4.62 | 0.033 | 0.31      | 4.46 | 0   | 0.30      | 4.50 | 0.000 | 0.53         | 3.36 | 0   | 0.40         | 4.12 | 0   |
| 18  | 98765       | 0.65            | 2.74 | 0   | 0.37      | 4.73 | 0.032 | 0.44      | 3.93 | 0   | 0.42      | 3.94 | 0.000 | 0.51         | 3.46 | 0   | 0.42         | 4.06 | 0   |
| 19  | 105717      | 0.60            | 2.99 | 0   | 0.36      | 4.67 | 0.029 | 0.52      | 3.47 | 0   | 0.51      | 3.45 | 0.000 | 0.50         | 3.52 | 0   | 0.45         | 3.94 | 0   |
| 20  | 118643      | 0.56            | 3.14 | 0   | 0.38      | 4.46 | 0.027 | 0.59      | 3.15 | 0   | 0.58      | 3.11 | 0.000 | 0.51         | 3.51 | 0   | 0.48         | 3.74 | 0   |
| 21  | 124060      | 0.54            | 3.24 | 0   | 0.39      | 4.37 | 0.026 | 0.64      | 2.91 | 0   | 0.64      | 2.85 | 0.000 | 0.51         | 3.50 | 0   | 0.51         | 3.62 | 0   |
| 22  | 128864      | 0.54            | 3.31 | 0   | 0.40      | 4.23 | 0.023 | 0.65      | 2.88 | 0   | 0.66      | 2.79 | 0.000 | 0.51         | 3.49 | 0   | 0.52         | 3.53 | 0   |
| 23  | 133507      | 0.52            | 3.33 | 0   | 0.41      | 4.09 | 0.023 | 0.63      | 2.97 | 0   | 0.65      | 2.85 | 0.000 | 0.51         | 3.50 | 0   | 0.53         | 3.49 | 0   |
| 24  | 138094      | 0.50            | 3.36 | 0   | 0.43      | 3.91 | 0.022 | 0.58      | 3.12 | 0   | 0.61      | 3.00 | 0.000 | 0.51         | 3.51 | 0   | 0.53         | 3.45 | 0   |
| 25  | 139920      | 0.50            | 3.42 | 0   | 0.45      | 3.77 | 0.021 | 0.53      | 3.35 | 0   | 0.56      | 3.24 | 0.000 | 0.50         | 3.56 | 0   | 0.52         | 3.46 | 0   |
| 26  | 141741      | 0.50            | 3.44 | 0   | 0.47      | 3.61 | 0.019 | 0.49      | 3.60 | 0   | 0.50      | 3.52 | 0.000 | 0.50         | 3.57 | 0   | 0.52         | 3.43 | 0   |
| 27  | 138491      | 0.50            | 3.42 | 0   | 0.50      | 3.47 | 0.018 | 0.44      | 3.90 | 0   | 0.45      | 3.86 | 0.000 | 0.49         | 3.59 | 0   | 0.51         | 3.46 | 0   |
| 28  | 133633      | 0.50            | 3.39 | 0   | 0.51      | 3.38 | 0.018 | 0.41      | 4.18 | 0   | 0.40      | 4.20 | 0.000 | 0.49         | 3.63 | 0   | 0.51         | 3.49 | 0   |
| 29  | 129054      | 0.52            | 3.36 | 0   | 0.53      | 3.29 | 0.017 | 0.38      | 4.46 | 0   | 0.36      | 4.52 | 0.000 | 0.49         | 3.64 | 0   | 0.51         | 3.53 | 0   |
| 30  | 123437      | 0.53            | 3.34 | 0   | 0.54      | 3.24 | 0.015 | 0.36      | 4.71 | 0   | 0.33      | 4.81 | 0.000 | 0.50         | 3.67 | 0   | 0.50         | 3.58 | 0   |
| 31  | 118174      | 0.54            | 3.30 | 0   | 0.55      | 3.21 | 0.015 | 0.35      | 4.93 | 0   | 0.32      | 5.06 | 0.000 | 0.50         | 3.69 | 0   | 0.50         | 3.60 | 0   |
| 32  | 113131      | 0.55            | 3.30 | 0   | 0.55      | 3.21 | 0.015 | 0.33      | 5.17 | 0   | 0.31      | 5.32 | 0.000 | 0.49         | 3.72 | 0   | 0.49         | 3.66 | 0   |
| 33  | 109146      | 0.54            | 3.30 | 0   | 0.56      | 3.21 | 0.014 | 0.33      | 5.36 | 0   | 0.31      | 5.50 | 0.000 | 0.49         | 3.74 | 0   | 0.48         | 3.70 | 0   |
| 34  | 105908      | 0.54            | 3.31 | 0   | 0.55      | 3.26 | 0.015 | 0.32      | 5.52 | 0   | 0.31      | 5.65 | 0.000 | 0.48         | 3.78 | 0   | 0.48         | 3.75 | 0   |
| 35  | 103149      | 0.53            | 3.32 | 0   | 0.55      | 3.29 | 0.014 | 0.31      | 5.69 | 0   | 0.30      | 5.81 | 0.000 | 0.48         | 3.79 | 0   | 0.48         | 3.75 | 0   |
| 36  | 99901       | 0.52            | 3.33 | 0   | 0.54      | 3.33 | 0.014 | 0.31      | 5.79 | 0   | 0.31      | 5.89 | 0.000 | 0.47         | 3.82 | 0   | 0.47         | 3.77 | 0   |
| 37  | 97516       | 0.52            | 3.31 | 0   | 0.53      | 3.37 | 0.014 | 0.30      | 5.91 | 0   | 0.30      | 6.00 | 0.000 | 0.47         | 3.81 | 0   | 0.48         | 3.76 | 0   |
| 38  | 93939       | 0.53            | 3.28 | 0   | 0.53      | 3.40 | 0.014 | 0.29      | 5.98 | 0   | 0.30      | 6.06 | 0.000 | 0.47         | 3.82 | 0   | 0.48         | 3.74 | 0   |
| 39  | 91461       | 0.55            | 3.21 | 0   | 0.52      | 3.41 | 0.013 | 0.30      | 6.00 | 0   | 0.30      | 6.06 | 0.000 | 0.48         | 3.82 | 0   | 0.49         | 3.71 | 0   |
| 40  | 94056       | 0.56            | 3.18 | 0   | 0.52      | 3.48 | 0.013 | 0.29      | 6.06 | 0   | 0.29      | 6.11 | 0.000 | 0.47         | 3.87 | 0   | 0.49         | 3.70 | 0   |
| 41  | 90173       | 0.57            | 3.14 | 0   | 0.51      | 3.48 | 0.014 | 0.29      | 6.07 | 0   | 0.29      | 6.10 | 0.000 | 0.47         | 3.87 | 0   | 0.49         | 3.67 | 0   |
| 42  | 86423       | 0.58            | 3.12 | 0   | 0.51      | 3.50 | 0.013 | 0.29      | 6.09 | 0   | 0.29      | 6.12 | 0.000 | 0.46         | 3.89 | 0   | 0.49         | 3.64 | 0   |
| 43  | 83252       | 0.58            | 3.15 | 0   | 0.51      | 3.51 | 0.013 | 0.30      | 6.04 | 0   | 0.30      | 6.05 | 0.000 | 0.45         | 3.92 | 0   | 0.49         | 3.62 | 0   |

**Table 29.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | innovatrics-002 |      |     | jumio-000 |      |       | neuro-000 |       |     | neuro-001 |       |       | nominder-000 |      |     | nominder-001 |      |     |
|-----|-------------|-----------------|------|-----|-----------|------|-------|-----------|-------|-----|-----------|-------|-------|--------------|------|-----|--------------|------|-----|
|     |             | Acc(3)          | MAE  | FTP | Acc(3)    | MAE  | FTP   | Acc(3)    | MAE   | FTP | Acc(3)    | MAE   | FTP   | Acc(3)       | MAE  | FTP | Acc(3)       | MAE  | FTP |
| 44  | 80602       | 0.57            | 3.19 | 0   | 0.51      | 3.51 | 0.014 | 0.30      | 6.03  | 0   | 0.30      | 6.04  | 0.000 | 0.44         | 3.95 | 0   | 0.49         | 3.60 | 0   |
| 45  | 77502       | 0.57            | 3.25 | 0   | 0.51      | 3.48 | 0.013 | 0.30      | 5.95  | 0   | 0.30      | 5.95  | 0.000 | 0.44         | 3.93 | 0   | 0.50         | 3.54 | 0   |
| 46  | 74182       | 0.56            | 3.32 | 0   | 0.51      | 3.48 | 0.014 | 0.31      | 5.83  | 0   | 0.31      | 5.83  | 0.000 | 0.44         | 3.89 | 0   | 0.50         | 3.49 | 0   |
| 47  | 70698       | 0.54            | 3.43 | 0   | 0.51      | 3.45 | 0.014 | 0.32      | 5.72  | 0   | 0.32      | 5.71  | 0.000 | 0.46         | 3.81 | 0   | 0.50         | 3.43 | 0   |
| 48  | 67856       | 0.52            | 3.55 | 0   | 0.51      | 3.42 | 0.015 | 0.33      | 5.57  | 0   | 0.34      | 5.57  | 0.000 | 0.47         | 3.71 | 0   | 0.52         | 3.34 | 0   |
| 49  | 65674       | 0.49            | 3.70 | 0   | 0.52      | 3.38 | 0.015 | 0.35      | 5.46  | 0   | 0.35      | 5.44  | 0.000 | 0.49         | 3.63 | 0   | 0.54         | 3.24 | 0   |
| 50  | 64896       | 0.46            | 3.81 | 0   | 0.53      | 3.32 | 0.015 | 0.37      | 5.27  | 0   | 0.37      | 5.25  | 0.000 | 0.51         | 3.53 | 0   | 0.55         | 3.15 | 0   |
| 51  | 63410       | 0.44            | 3.94 | 0   | 0.54      | 3.26 | 0.015 | 0.39      | 5.13  | 0   | 0.39      | 5.11  | 0.000 | 0.53         | 3.46 | 0   | 0.57         | 3.09 | 0   |
| 52  | 60991       | 0.42            | 4.00 | 0   | 0.55      | 3.21 | 0.017 | 0.40      | 5.04  | 0   | 0.40      | 5.01  | 0.000 | 0.53         | 3.42 | 0   | 0.58         | 3.03 | 0   |
| 53  | 59550       | 0.43            | 4.02 | 0   | 0.56      | 3.15 | 0.016 | 0.41      | 4.94  | 0   | 0.41      | 4.91  | 0.000 | 0.54         | 3.38 | 0   | 0.59         | 2.99 | 0   |
| 54  | 56895       | 0.44            | 4.01 | 0   | 0.56      | 3.10 | 0.018 | 0.41      | 4.88  | 0   | 0.41      | 4.86  | 0.000 | 0.53         | 3.41 | 0   | 0.59         | 2.99 | 0   |
| 55  | 55297       | 0.45            | 3.98 | 0   | 0.57      | 3.09 | 0.018 | 0.41      | 4.80  | 0   | 0.42      | 4.78  | 0.000 | 0.53         | 3.42 | 0   | 0.58         | 3.02 | 0   |
| 56  | 52574       | 0.47            | 3.95 | 0   | 0.57      | 3.07 | 0.019 | 0.42      | 4.72  | 0   | 0.42      | 4.70  | 0.000 | 0.52         | 3.43 | 0   | 0.58         | 3.05 | 0   |
| 57  | 49823       | 0.47            | 3.99 | 0   | 0.57      | 3.06 | 0.020 | 0.41      | 4.74  | 0   | 0.41      | 4.72  | 0.000 | 0.51         | 3.51 | 0   | 0.56         | 3.15 | 0   |
| 58  | 48570       | 0.48            | 3.99 | 0   | 0.57      | 3.05 | 0.023 | 0.41      | 4.74  | 0   | 0.41      | 4.73  | 0.000 | 0.51         | 3.54 | 0   | 0.55         | 3.19 | 0   |
| 59  | 46040       | 0.46            | 4.06 | 0   | 0.57      | 3.08 | 0.023 | 0.41      | 4.78  | 0   | 0.41      | 4.78  | 0.000 | 0.51         | 3.61 | 0   | 0.54         | 3.31 | 0   |
| 60  | 46037       | 0.45            | 4.07 | 0   | 0.57      | 3.09 | 0.024 | 0.40      | 4.82  | 0   | 0.40      | 4.82  | 0.000 | 0.49         | 3.70 | 0   | 0.52         | 3.41 | 0   |
| 61  | 43860       | 0.43            | 4.13 | 0   | 0.56      | 3.12 | 0.025 | 0.40      | 4.88  | 0   | 0.40      | 4.88  | 0.000 | 0.48         | 3.77 | 0   | 0.51         | 3.50 | 0   |
| 62  | 41407       | 0.41            | 4.16 | 0   | 0.55      | 3.15 | 0.027 | 0.39      | 4.98  | 0   | 0.39      | 4.99  | 0.000 | 0.47         | 3.87 | 0   | 0.50         | 3.61 | 0   |
| 63  | 39444       | 0.41            | 4.17 | 0   | 0.55      | 3.23 | 0.028 | 0.38      | 5.20  | 0   | 0.38      | 5.22  | 0.000 | 0.45         | 4.02 | 0   | 0.47         | 3.78 | 0   |
| 64  | 37425       | 0.40            | 4.14 | 0   | 0.53      | 3.29 | 0.030 | 0.38      | 5.39  | 0   | 0.38      | 5.40  | 0.000 | 0.44         | 4.14 | 0   | 0.46         | 3.89 | 0   |
| 65  | 34834       | 0.41            | 4.07 | 0   | 0.52      | 3.37 | 0.032 | 0.35      | 5.64  | 0   | 0.35      | 5.66  | 0.000 | 0.42         | 4.26 | 0   | 0.43         | 4.06 | 0   |
| 66  | 32020       | 0.43            | 3.96 | 0   | 0.52      | 3.42 | 0.032 | 0.31      | 5.91  | 0   | 0.31      | 5.93  | 0.000 | 0.40         | 4.38 | 0   | 0.43         | 4.13 | 0   |
| 67  | 28898       | 0.46            | 3.82 | 0   | 0.51      | 3.53 | 0.036 | 0.26      | 6.39  | 0   | 0.26      | 6.42  | 0.000 | 0.39         | 4.57 | 0   | 0.41         | 4.30 | 0   |
| 68  | 26630       | 0.51            | 3.70 | 0   | 0.48      | 3.70 | 0.037 | 0.20      | 6.94  | 0   | 0.19      | 6.96  | 0.000 | 0.36         | 4.79 | 0   | 0.39         | 4.47 | 0   |
| 69  | 24864       | 0.54            | 3.54 | 0   | 0.47      | 3.82 | 0.040 | 0.13      | 7.46  | 0   | 0.13      | 7.48  | 0.000 | 0.34         | 4.93 | 0   | 0.38         | 4.56 | 0   |
| 70  | 23407       | 0.56            | 3.44 | 0   | 0.45      | 3.96 | 0.043 | 0.06      | 8.08  | 0   | 0.06      | 8.11  | 0.000 | 0.33         | 5.08 | 0   | 0.37         | 4.64 | 0   |
| 71  | 20736       | 0.57            | 3.39 | 0   | 0.43      | 4.13 | 0.042 | 0.01      | 8.66  | 0   | 0.01      | 8.69  | 0.000 | 0.31         | 5.26 | 0   | 0.36         | 4.73 | 0   |
| 72  | 18423       | 0.56            | 3.41 | 0   | 0.40      | 4.37 | 0.048 | 0.00      | 9.36  | 0   | 0.00      | 9.37  | 0.000 | 0.30         | 5.49 | 0   | 0.35         | 4.86 | 0   |
| 73  | 16215       | 0.54            | 3.54 | 0   | 0.38      | 4.57 | 0.048 | 0.00      | 9.98  | 0   | 0.00      | 10.00 | 0.000 | 0.29         | 5.62 | 0   | 0.36         | 4.89 | 0   |
| 74  | 14848       | 0.51            | 3.63 | 0   | 0.37      | 4.72 | 0.050 | 0.00      | 10.65 | 0   | 0.00      | 10.67 | 0.000 | 0.29         | 5.75 | 0   | 0.37         | 4.87 | 0   |
| 75  | 13197       | 0.46            | 3.89 | 0   | 0.31      | 5.14 | 0.049 | 0.00      | 11.39 | 0   | 0.00      | 11.41 | 0.000 | 0.28         | 6.00 | 0   | 0.37         | 5.00 | 0   |
| 76  | 11886       | 0.42            | 4.07 | 0   | 0.28      | 5.46 | 0.055 | 0.00      | 12.02 | 0   | 0.00      | 12.04 | 0.000 | 0.28         | 6.14 | 0   | 0.37         | 5.04 | 0   |
| 77  | 10839       | 0.38            | 4.35 | 0   | 0.25      | 5.77 | 0.055 | 0.00      | 12.85 | 0   | 0.00      | 12.86 | 0.000 | 0.26         | 6.40 | 0   | 0.38         | 5.08 | 0   |
| 78  | 9361        | 0.35            | 4.67 | 0   | 0.21      | 6.21 | 0.061 | 0.00      | 13.55 | 0   | 0.00      | 13.57 | 0.000 | 0.26         | 6.58 | 0   | 0.37         | 5.23 | 0   |
| 79  | 7540        | 0.32            | 4.89 | 0   | 0.17      | 6.52 | 0.068 | 0.00      | 14.40 | 0   | 0.00      | 14.41 | 0.000 | 0.26         | 6.65 | 0   | 0.37         | 5.19 | 0   |
| 80  | 6267        | 0.30            | 5.22 | 0   | 0.14      | 7.03 | 0.066 | 0.00      | 15.18 | 0   | 0.00      | 15.20 | 0.000 | 0.25         | 6.90 | 0   | 0.36         | 5.37 | 0   |
| 81  | 5221        | 0.28            | 5.41 | 0   | 0.11      | 7.38 | 0.072 | 0.00      | 16.00 | 0   | 0.00      | 16.01 | 0.000 | 0.24         | 7.01 | 0   | 0.34         | 5.51 | 0   |

**Table 29.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | innovatrics-002 |       |     | jumio-000 |       |       | neuro-000 |       |     | neuro-001 |       |       | nominder-000 |       |     | nominder-001 |      |     |
|-----|-------------|-----------------|-------|-----|-----------|-------|-------|-----------|-------|-----|-----------|-------|-------|--------------|-------|-----|--------------|------|-----|
|     |             | Acc(3)          | MAE   | FTP | Acc(3)    | MAE   | FTP   | Acc(3)    | MAE   | FTP | Acc(3)    | MAE   | FTP   | Acc(3)       | MAE   | FTP | Acc(3)       | MAE  | FTP |
| 82  | 3962        | 0.25            | 5.81  | 0   | 0.08      | 7.94  | 0.076 | 0.00      | 16.78 | 0   | 0.00      | 16.80 | 0.000 | 0.23         | 7.36  | 0   | 0.33         | 5.69 | 0   |
| 83  | 3277        | 0.21            | 6.31  | 0   | 0.06      | 8.48  | 0.070 | 0.00      | 17.66 | 0   | 0.00      | 17.68 | 0.000 | 0.19         | 7.65  | 0   | 0.27         | 6.11 | 0   |
| 84  | 2615        | 0.16            | 6.76  | 0   | 0.04      | 9.03  | 0.081 | 0.00      | 18.45 | 0   | 0.00      | 18.49 | 0.000 | 0.16         | 8.02  | 0   | 0.26         | 6.39 | 0   |
| 85  | 2206        | 0.14            | 7.19  | 0   | 0.03      | 9.62  | 0.079 | 0.00      | 19.29 | 0   | 0.00      | 19.32 | 0.000 | 0.15         | 8.22  | 0   | 0.22         | 6.63 | 0   |
| 86  | 1831        | 0.11            | 7.64  | 0   | 0.01      | 10.17 | 0.082 | 0.00      | 20.24 | 0   | 0.00      | 20.25 | 0.000 | 0.14         | 8.61  | 0   | 0.20         | 7.06 | 0   |
| 87  | 1402        | 0.07            | 8.21  | 0   | 0.01      | 10.81 | 0.086 | 0.00      | 21.17 | 0   | 0.00      | 21.15 | 0.000 | 0.11         | 9.11  | 0   | 0.15         | 7.49 | 0   |
| 88  | 1254        | 0.04            | 9.07  | 0   | 0.01      | 11.54 | 0.093 | 0.00      | 22.10 | 0   | 0.00      | 22.14 | 0.000 | 0.11         | 9.40  | 0   | 0.12         | 8.13 | 0   |
| 89  | 907         | 0.02            | 9.24  | 0   | 0.01      | 12.06 | 0.097 | 0.00      | 22.86 | 0   | 0.00      | 22.89 | 0.001 | 0.10         | 9.57  | 0   | 0.11         | 8.32 | 0   |
| 90  | 670         | 0.01            | 10.21 | 0   | 0.00      | 13.01 | 0.099 | 0.00      | 24.05 | 0   | 0.00      | 24.06 | 0.000 | 0.09         | 10.64 | 0   | 0.07         | 9.27 | 0   |

**Table 30.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better).

| Age | Num. Images | ondato-000 |      |     | ondato-001 |      |     | privately-000 |      |     | regula-000 |      |     | roc-000 |      |     | roc-001 |      |     |
|-----|-------------|------------|------|-----|------------|------|-----|---------------|------|-----|------------|------|-----|---------|------|-----|---------|------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)        | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)  | MAE  | FTP | Acc(3)  | MAE  | FTP |
| 0   | 133387      | 0.85       | 2.29 | 0   | 0.93       | 1.66 | 0   | 0.98          | 1.40 | 0   | 0.98       | 1.28 | 0   | 0.72    | 2.28 | 0   | 0.00    | 7.75 | 0   |
| 1   | 94475       | 0.86       | 1.68 | 0   | 0.90       | 1.76 | 0   | 0.93          | 1.30 | 0   | 1.00       | 0.46 | 0   | 0.83    | 1.80 | 0   | 0.01    | 7.52 | 0   |
| 2   | 82596       | 0.77       | 2.13 | 0   | 0.84       | 2.04 | 0   | 0.75          | 1.81 | 0   | 1.00       | 0.56 | 0   | 0.83    | 1.78 | 0   | 0.02    | 7.17 | 0   |
| 3   | 82473       | 0.69       | 2.52 | 0   | 0.78       | 2.18 | 0   | 0.67          | 2.25 | 0   | 1.00       | 0.66 | 0   | 0.84    | 1.75 | 0   | 0.02    | 6.87 | 0   |
| 4   | 84055       | 0.64       | 2.77 | 0   | 0.74       | 2.22 | 0   | 0.74          | 2.28 | 0   | 1.00       | 0.68 | 0   | 0.87    | 1.64 | 0   | 0.02    | 6.58 | 0   |
| 5   | 92521       | 0.61       | 2.94 | 0   | 0.71       | 2.34 | 0   | 0.82          | 2.05 | 0   | 1.00       | 0.74 | 0   | 0.88    | 1.59 | 0   | 0.03    | 6.32 | 0   |
| 6   | 91639       | 0.55       | 3.25 | 0   | 0.68       | 2.53 | 0   | 0.87          | 1.74 | 0   | 0.99       | 0.85 | 0   | 0.87    | 1.60 | 0   | 0.03    | 5.92 | 0   |
| 7   | 89993       | 0.48       | 3.65 | 0   | 0.63       | 2.82 | 0   | 0.90          | 1.52 | 0   | 0.99       | 0.91 | 0   | 0.85    | 1.66 | 0   | 0.05    | 5.22 | 0   |
| 8   | 89428       | 0.44       | 3.88 | 0   | 0.60       | 3.01 | 0   | 0.90          | 1.48 | 0   | 0.99       | 0.97 | 0   | 0.80    | 1.88 | 0   | 0.06    | 5.04 | 0   |
| 9   | 89644       | 0.41       | 4.00 | 0   | 0.57       | 3.14 | 0   | 0.90          | 1.52 | 0   | 0.98       | 1.00 | 0   | 0.73    | 2.19 | 0   | 0.08    | 4.81 | 0   |
| 10  | 98337       | 0.39       | 3.99 | 0   | 0.54       | 3.23 | 0   | 0.88          | 1.64 | 0   | 0.97       | 1.06 | 0   | 0.64    | 2.64 | 0   | 0.10    | 4.59 | 0   |
| 11  | 97415       | 0.38       | 3.97 | 0   | 0.50       | 3.40 | 0   | 0.85          | 1.84 | 0   | 0.94       | 1.23 | 0   | 0.53    | 3.18 | 0   | 0.13    | 4.39 | 0   |
| 12  | 94326       | 0.38       | 3.85 | 0   | 0.45       | 3.55 | 0   | 0.82          | 2.00 | 0   | 0.90       | 1.46 | 0   | 0.39    | 3.76 | 0   | 0.16    | 4.18 | 0   |
| 13  | 94047       | 0.41       | 3.70 | 0   | 0.42       | 3.63 | 0   | 0.76          | 2.18 | 0   | 0.84       | 1.73 | 0   | 0.26    | 4.18 | 0   | 0.20    | 4.00 | 0   |
| 14  | 96915       | 0.47       | 3.50 | 0   | 0.42       | 3.60 | 0   | 0.70          | 2.41 | 0   | 0.79       | 1.98 | 0   | 0.19    | 4.20 | 0   | 0.29    | 3.81 | 0   |
| 15  | 97599       | 0.52       | 3.36 | 0   | 0.46       | 3.52 | 0   | 0.65          | 2.76 | 0   | 0.76       | 2.13 | 0   | 0.30    | 4.00 | 0   | 0.42    | 3.63 | 0   |
| 16  | 96388       | 0.54       | 3.25 | 0   | 0.50       | 3.44 | 0   | 0.62          | 3.05 | 0   | 0.75       | 2.21 | 0   | 0.52    | 3.65 | 0   | 0.54    | 3.47 | 0   |
| 17  | 99462       | 0.56       | 3.16 | 0   | 0.53       | 3.32 | 0   | 0.60          | 3.16 | 0   | 0.75       | 2.20 | 0   | 0.62    | 3.29 | 0   | 0.55    | 3.35 | 0   |
| 18  | 98765       | 0.55       | 3.21 | 0   | 0.53       | 3.33 | 0   | 0.52          | 3.34 | 0   | 0.71       | 2.36 | 0   | 0.60    | 3.25 | 0   | 0.49    | 3.53 | 0   |
| 19  | 105717      | 0.56       | 3.21 | 0   | 0.53       | 3.32 | 0   | 0.50          | 3.37 | 0   | 0.69       | 2.47 | 0   | 0.59    | 3.28 | 0   | 0.46    | 3.69 | 0   |
| 20  | 118643      | 0.56       | 3.18 | 0   | 0.53       | 3.28 | 0   | 0.53          | 3.34 | 0   | 0.68       | 2.55 | 0   | 0.58    | 3.38 | 0   | 0.46    | 3.79 | 0   |

**Table 30.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | ondato-000 |      |     | ondato-001 |      |     | privately-000 |      |     | regula-000 |      |     | roc-000 |      |     | roc-001 |      |     |
|-----|-------------|------------|------|-----|------------|------|-----|---------------|------|-----|------------|------|-----|---------|------|-----|---------|------|-----|
|     |             | Acc(3)     | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)        | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)  | MAE  | FTP | Acc(3)  | MAE  | FTP |
| 21  | 124060      | 0.56       | 3.16 | 0   | 0.54       | 3.24 | 0   | 0.54          | 3.35 | 0   | 0.66       | 2.64 | 0   | 0.55    | 3.60 | 0   | 0.44    | 3.93 | 0   |
| 22  | 128864      | 0.56       | 3.12 | 0   | 0.54       | 3.19 | 0   | 0.54          | 3.34 | 0   | 0.63       | 2.72 | 0   | 0.51    | 3.81 | 0   | 0.42    | 4.03 | 0   |
| 23  | 133507      | 0.57       | 3.09 | 0   | 0.55       | 3.14 | 0   | 0.53          | 3.38 | 0   | 0.62       | 2.76 | 0   | 0.46    | 3.99 | 0   | 0.41    | 4.13 | 0   |
| 24  | 138094      | 0.58       | 3.04 | 0   | 0.57       | 3.07 | 0   | 0.55          | 3.38 | 0   | 0.62       | 2.75 | 0   | 0.43    | 4.14 | 0   | 0.40    | 4.21 | 0   |
| 25  | 139920      | 0.58       | 3.03 | 0   | 0.57       | 3.04 | 0   | 0.54          | 3.38 | 0   | 0.62       | 2.74 | 0   | 0.41    | 4.28 | 0   | 0.38    | 4.32 | 0   |
| 26  | 141741      | 0.58       | 3.02 | 0   | 0.58       | 2.99 | 0   | 0.53          | 3.37 | 0   | 0.63       | 2.72 | 0   | 0.40    | 4.37 | 0   | 0.37    | 4.37 | 0   |
| 27  | 138491      | 0.58       | 3.04 | 0   | 0.58       | 2.99 | 0   | 0.54          | 3.42 | 0   | 0.63       | 2.73 | 0   | 0.39    | 4.49 | 0   | 0.36    | 4.44 | 0   |
| 28  | 133633      | 0.56       | 3.10 | 0   | 0.57       | 3.02 | 0   | 0.54          | 3.46 | 0   | 0.63       | 2.74 | 0   | 0.37    | 4.58 | 0   | 0.36    | 4.50 | 0   |
| 29  | 129054      | 0.55       | 3.16 | 0   | 0.57       | 3.05 | 0   | 0.54          | 3.52 | 0   | 0.62       | 2.80 | 0   | 0.37    | 4.65 | 0   | 0.35    | 4.57 | 0   |
| 30  | 123437      | 0.54       | 3.21 | 0   | 0.56       | 3.09 | 0   | 0.53          | 3.53 | 0   | 0.61       | 2.88 | 0   | 0.36    | 4.73 | 0   | 0.34    | 4.60 | 0   |
| 31  | 118174      | 0.54       | 3.28 | 0   | 0.56       | 3.11 | 0   | 0.52          | 3.59 | 0   | 0.59       | 2.96 | 0   | 0.35    | 4.75 | 0   | 0.34    | 4.61 | 0   |
| 32  | 113131      | 0.53       | 3.34 | 0   | 0.55       | 3.15 | 0   | 0.52          | 3.67 | 0   | 0.58       | 3.04 | 0   | 0.35    | 4.77 | 0   | 0.34    | 4.60 | 0   |
| 33  | 109146      | 0.53       | 3.35 | 0   | 0.56       | 3.15 | 0   | 0.51          | 3.76 | 0   | 0.56       | 3.11 | 0   | 0.35    | 4.77 | 0   | 0.34    | 4.60 | 0   |
| 34  | 105908      | 0.53       | 3.38 | 0   | 0.56       | 3.17 | 0   | 0.49          | 3.88 | 0   | 0.55       | 3.20 | 0   | 0.34    | 4.78 | 0   | 0.34    | 4.63 | 0   |
| 35  | 103149      | 0.53       | 3.43 | 0   | 0.56       | 3.19 | 0   | 0.47          | 4.03 | 0   | 0.55       | 3.23 | 0   | 0.35    | 4.73 | 0   | 0.35    | 4.60 | 0   |
| 36  | 99901       | 0.53       | 3.47 | 0   | 0.56       | 3.22 | 0   | 0.44          | 4.21 | 0   | 0.54       | 3.28 | 0   | 0.35    | 4.73 | 0   | 0.35    | 4.59 | 0   |
| 37  | 97516       | 0.52       | 3.52 | 0   | 0.56       | 3.25 | 0   | 0.41          | 4.36 | 0   | 0.53       | 3.31 | 0   | 0.35    | 4.67 | 0   | 0.36    | 4.58 | 0   |
| 38  | 93939       | 0.51       | 3.61 | 0   | 0.55       | 3.32 | 0   | 0.38          | 4.53 | 0   | 0.53       | 3.34 | 0   | 0.37    | 4.60 | 0   | 0.36    | 4.56 | 0   |
| 39  | 91461       | 0.50       | 3.66 | 0   | 0.54       | 3.38 | 0   | 0.36          | 4.69 | 0   | 0.53       | 3.35 | 0   | 0.37    | 4.55 | 0   | 0.37    | 4.55 | 0   |
| 40  | 94056       | 0.49       | 3.73 | 0   | 0.53       | 3.48 | 0   | 0.34          | 4.82 | 0   | 0.52       | 3.41 | 0   | 0.38    | 4.50 | 0   | 0.37    | 4.60 | 0   |
| 41  | 90173       | 0.48       | 3.79 | 0   | 0.51       | 3.57 | 0   | 0.33          | 4.91 | 0   | 0.51       | 3.44 | 0   | 0.39    | 4.43 | 0   | 0.37    | 4.60 | 0   |
| 42  | 86423       | 0.47       | 3.84 | 0   | 0.49       | 3.66 | 0   | 0.32          | 4.95 | 0   | 0.51       | 3.46 | 0   | 0.40    | 4.36 | 0   | 0.37    | 4.61 | 0   |
| 43  | 83252       | 0.46       | 3.90 | 0   | 0.48       | 3.74 | 0   | 0.32          | 4.95 | 0   | 0.51       | 3.48 | 0   | 0.41    | 4.33 | 0   | 0.37    | 4.62 | 0   |
| 44  | 80602       | 0.46       | 3.90 | 0   | 0.47       | 3.80 | 0   | 0.34          | 4.91 | 0   | 0.50       | 3.49 | 0   | 0.41    | 4.30 | 0   | 0.37    | 4.65 | 0   |
| 45  | 77502       | 0.46       | 3.90 | 0   | 0.46       | 3.83 | 0   | 0.37          | 4.79 | 0   | 0.50       | 3.51 | 0   | 0.42    | 4.27 | 0   | 0.36    | 4.66 | 0   |
| 46  | 74182       | 0.46       | 3.88 | 0   | 0.46       | 3.86 | 0   | 0.41          | 4.64 | 0   | 0.50       | 3.50 | 0   | 0.42    | 4.27 | 0   | 0.36    | 4.67 | 0   |
| 47  | 70698       | 0.46       | 3.86 | 0   | 0.45       | 3.85 | 0   | 0.44          | 4.49 | 0   | 0.51       | 3.50 | 0   | 0.43    | 4.25 | 0   | 0.35    | 4.70 | 0   |
| 48  | 67856       | 0.47       | 3.80 | 0   | 0.46       | 3.81 | 0   | 0.46          | 4.33 | 0   | 0.51       | 3.49 | 0   | 0.43    | 4.25 | 0   | 0.35    | 4.69 | 0   |
| 49  | 65674       | 0.49       | 3.74 | 0   | 0.46       | 3.77 | 0   | 0.48          | 4.23 | 0   | 0.52       | 3.44 | 0   | 0.43    | 4.24 | 0   | 0.35    | 4.67 | 0   |
| 50  | 64896       | 0.50       | 3.69 | 0   | 0.48       | 3.69 | 0   | 0.49          | 4.16 | 0   | 0.52       | 3.42 | 0   | 0.44    | 4.23 | 0   | 0.35    | 4.63 | 0   |
| 51  | 63410       | 0.50       | 3.66 | 0   | 0.49       | 3.62 | 0   | 0.49          | 4.18 | 0   | 0.52       | 3.42 | 0   | 0.43    | 4.26 | 0   | 0.36    | 4.58 | 0   |
| 52  | 60991       | 0.51       | 3.61 | 0   | 0.50       | 3.54 | 0   | 0.47          | 4.20 | 0   | 0.52       | 3.40 | 0   | 0.43    | 4.24 | 0   | 0.36    | 4.51 | 0   |
| 53  | 59550       | 0.51       | 3.58 | 0   | 0.52       | 3.44 | 0   | 0.45          | 4.23 | 0   | 0.52       | 3.38 | 0   | 0.44    | 4.18 | 0   | 0.38    | 4.41 | 0   |
| 54  | 56895       | 0.52       | 3.55 | 0   | 0.54       | 3.36 | 0   | 0.45          | 4.29 | 0   | 0.52       | 3.36 | 0   | 0.44    | 4.18 | 0   | 0.39    | 4.32 | 0   |
| 55  | 55297       | 0.51       | 3.57 | 0   | 0.55       | 3.31 | 0   | 0.44          | 4.39 | 0   | 0.53       | 3.35 | 0   | 0.44    | 4.13 | 0   | 0.40    | 4.21 | 0   |
| 56  | 52574       | 0.50       | 3.56 | 0   | 0.55       | 3.26 | 0   | 0.42          | 4.49 | 0   | 0.53       | 3.31 | 0   | 0.44    | 4.08 | 0   | 0.42    | 4.08 | 0   |
| 57  | 49823       | 0.50       | 3.59 | 0   | 0.56       | 3.25 | 0   | 0.40          | 4.67 | 0   | 0.54       | 3.29 | 0   | 0.45    | 4.02 | 0   | 0.43    | 3.97 | 0   |
| 58  | 48570       | 0.50       | 3.60 | 0   | 0.56       | 3.25 | 0   | 0.38          | 4.83 | 0   | 0.55       | 3.23 | 0   | 0.45    | 3.97 | 0   | 0.45    | 3.82 | 0   |

**Table 30.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | ondato-000 |       |     | ondato-001 |       |     | privately-000 |      |     | regula-000 |      |     | roc-000 |       |     | roc-001 |      |     |
|-----|-------------|------------|-------|-----|------------|-------|-----|---------------|------|-----|------------|------|-----|---------|-------|-----|---------|------|-----|
|     |             | Acc(3)     | MAE   | FTP | Acc(3)     | MAE   | FTP | Acc(3)        | MAE  | FTP | Acc(3)     | MAE  | FTP | Acc(3)  | MAE   | FTP | Acc(3)  | MAE  | FTP |
| 59  | 46040       | 0.49       | 3.64  | 0   | 0.55       | 3.29  | 0   | 0.37          | 4.98 | 0   | 0.55       | 3.23 | 0   | 0.46    | 3.90  | 0   | 0.47    | 3.70 | 0   |
| 60  | 46037       | 0.49       | 3.68  | 0   | 0.54       | 3.35  | 0   | 0.35          | 5.19 | 0   | 0.55       | 3.21 | 0   | 0.46    | 3.85  | 0   | 0.49    | 3.59 | 0   |
| 61  | 43860       | 0.48       | 3.70  | 0   | 0.53       | 3.41  | 0   | 0.33          | 5.38 | 0   | 0.55       | 3.20 | 0   | 0.47    | 3.77  | 0   | 0.51    | 3.49 | 0   |
| 62  | 41407       | 0.48       | 3.73  | 0   | 0.51       | 3.52  | 0   | 0.32          | 5.54 | 0   | 0.56       | 3.16 | 0   | 0.49    | 3.69  | 0   | 0.53    | 3.35 | 0   |
| 63  | 39444       | 0.48       | 3.78  | 0   | 0.49       | 3.63  | 0   | 0.30          | 5.79 | 0   | 0.56       | 3.17 | 0   | 0.49    | 3.64  | 0   | 0.54    | 3.27 | 0   |
| 64  | 37425       | 0.47       | 3.84  | 0   | 0.48       | 3.73  | 0   | 0.30          | 5.97 | 0   | 0.55       | 3.16 | 0   | 0.50    | 3.59  | 0   | 0.55    | 3.21 | 0   |
| 65  | 34834       | 0.46       | 3.92  | 0   | 0.46       | 3.84  | 0   | 0.30          | 6.17 | 0   | 0.55       | 3.16 | 0   | 0.50    | 3.54  | 0   | 0.56    | 3.14 | 0   |
| 66  | 32020       | 0.46       | 3.95  | 0   | 0.45       | 3.94  | 0   | 0.31          | 6.33 | 0   | 0.56       | 3.13 | 0   | 0.51    | 3.52  | 0   | 0.57    | 3.10 | 0   |
| 67  | 28898       | 0.45       | 4.04  | 0   | 0.44       | 4.02  | 0   | 0.31          | 6.59 | 0   | 0.56       | 3.13 | 0   | 0.50    | 3.54  | 0   | 0.56    | 3.14 | 0   |
| 68  | 26630       | 0.44       | 4.16  | 0   | 0.42       | 4.20  | 0   | 0.28          | 6.84 | 0   | 0.56       | 3.14 | 0   | 0.51    | 3.55  | 0   | 0.56    | 3.14 | 0   |
| 69  | 24864       | 0.44       | 4.22  | 0   | 0.42       | 4.28  | 0   | 0.26          | 7.05 | 0   | 0.56       | 3.14 | 0   | 0.50    | 3.56  | 0   | 0.56    | 3.16 | 0   |
| 70  | 23407       | 0.44       | 4.34  | 0   | 0.41       | 4.41  | 0   | 0.23          | 7.27 | 0   | 0.56       | 3.13 | 0   | 0.49    | 3.64  | 0   | 0.55    | 3.19 | 0   |
| 71  | 20736       | 0.43       | 4.46  | 0   | 0.41       | 4.52  | 0   | 0.19          | 7.59 | 0   | 0.56       | 3.14 | 0   | 0.48    | 3.72  | 0   | 0.54    | 3.27 | 0   |
| 72  | 18423       | 0.42       | 4.69  | 0   | 0.40       | 4.73  | 0   | 0.15          | 7.72 | 0   | 0.55       | 3.17 | 0   | 0.48    | 3.79  | 0   | 0.53    | 3.35 | 0   |
| 73  | 16215       | 0.40       | 4.85  | 0   | 0.40       | 4.83  | 0   | 0.13          | 7.72 | 0   | 0.55       | 3.19 | 0   | 0.45    | 3.97  | 0   | 0.51    | 3.49 | 0   |
| 74  | 14848       | 0.37       | 5.06  | 0   | 0.39       | 4.95  | 0   | 0.15          | 7.79 | 0   | 0.55       | 3.18 | 0   | 0.45    | 4.06  | 0   | 0.50    | 3.54 | 0   |
| 75  | 13197       | 0.31       | 5.57  | 0   | 0.34       | 5.37  | 0   | 0.16          | 7.85 | 0   | 0.54       | 3.31 | 0   | 0.41    | 4.35  | 0   | 0.48    | 3.73 | 0   |
| 76  | 11886       | 0.25       | 5.95  | 0   | 0.30       | 5.66  | 0   | 0.17          | 7.77 | 0   | 0.53       | 3.37 | 0   | 0.39    | 4.51  | 0   | 0.45    | 3.91 | 0   |
| 77  | 10839       | 0.17       | 6.49  | 0   | 0.23       | 6.17  | 0   | 0.19          | 7.87 | 0   | 0.52       | 3.45 | 0   | 0.36    | 4.84  | 0   | 0.43    | 4.13 | 0   |
| 78  | 9361        | 0.10       | 7.11  | 0   | 0.16       | 6.72  | 0   | 0.21          | 7.66 | 0   | 0.51       | 3.56 | 0   | 0.33    | 5.21  | 0   | 0.42    | 4.30 | 0   |
| 79  | 7540        | 0.04       | 7.65  | 0   | 0.09       | 7.18  | 0   | 0.23          | 7.48 | 0   | 0.50       | 3.63 | 0   | 0.30    | 5.45  | 0   | 0.40    | 4.41 | 0   |
| 80  | 6267        | 0.01       | 8.31  | 0   | 0.02       | 7.78  | 0   | 0.26          | 7.27 | 0   | 0.48       | 3.82 | 0   | 0.29    | 5.75  | 0   | 0.38    | 4.64 | 0   |
| 81  | 5221        | 0.00       | 8.91  | 0   | 0.00       | 8.34  | 0   | 0.29          | 7.15 | 0   | 0.47       | 3.83 | 0   | 0.25    | 6.09  | 0   | 0.36    | 4.83 | 0   |
| 82  | 3962        | 0.00       | 9.61  | 0   | 0.00       | 8.98  | 0   | 0.33          | 6.92 | 0   | 0.44       | 4.11 | 0   | 0.23    | 6.46  | 0   | 0.34    | 5.05 | 0   |
| 83  | 3277        | 0.00       | 10.40 | 0   | 0.00       | 9.71  | 0   | 0.36          | 6.88 | 0   | 0.40       | 4.36 | 0   | 0.19    | 7.06  | 0   | 0.31    | 5.34 | 0   |
| 84  | 2615        | 0.00       | 11.06 | 0   | 0.00       | 10.41 | 0   | 0.38          | 6.88 | 0   | 0.40       | 4.47 | 0   | 0.17    | 7.45  | 0   | 0.29    | 5.57 | 0   |
| 85  | 2206        | 0.00       | 11.76 | 0   | 0.00       | 11.07 | 0   | 0.39          | 6.93 | 0   | 0.37       | 4.69 | 0   | 0.14    | 7.88  | 0   | 0.28    | 5.80 | 0   |
| 86  | 1831        | 0.00       | 12.56 | 0   | 0.00       | 11.87 | 0   | 0.37          | 7.03 | 0   | 0.33       | 5.02 | 0   | 0.12    | 8.26  | 0   | 0.28    | 6.01 | 0   |
| 87  | 1402        | 0.00       | 13.37 | 0   | 0.00       | 12.71 | 0   | 0.33          | 7.58 | 0   | 0.29       | 5.42 | 0   | 0.09    | 9.02  | 0   | 0.24    | 6.45 | 0   |
| 88  | 1254        | 0.00       | 14.28 | 0   | 0.00       | 13.61 | 0   | 0.28          | 8.40 | 0   | 0.25       | 5.79 | 0   | 0.07    | 9.67  | 0   | 0.22    | 6.89 | 0   |
| 89  | 907         | 0.00       | 14.92 | 0   | 0.00       | 14.22 | 0   | 0.24          | 8.60 | 0   | 0.22       | 5.97 | 0   | 0.07    | 9.85  | 0   | 0.22    | 6.96 | 0   |
| 90  | 670         | 0.00       | 16.05 | 0   | 0.00       | 15.34 | 0   | 0.18          | 9.09 | 0   | 0.17       | 6.84 | 0   | 0.06    | 10.89 | 0   | 0.19    | 7.68 | 0   |

**Table 31.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better).

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| Age | Num. Images | roc-002 |      |     | shufti-000 |      |       | unissey-001 |      |     | viant-000 |      |     | yoti-001 |      |       | yoti-002 |      |     |
|-----|-------------|---------|------|-----|------------|------|-------|-------------|------|-----|-----------|------|-----|----------|------|-------|----------|------|-----|
|     |             | Acc(3)  | MAE  | FTP | Acc(3)     | MAE  | FTP   | Acc(3)      | MAE  | FTP | Acc(3)    | MAE  | FTP | Acc(3)   | MAE  | FTP   | Acc(3)   | MAE  | FTP |
| 0   | 133387      | 1.00    | 0.61 | 0   | 0.61       | 3.01 | 0.001 | 1.00        | 0.60 | 0   | 1.00      | 0.82 | 0   | 0.96     | 1.39 | 0.012 | 0.97     | 1.11 | 0   |
| 1   | 94475       | 1.00    | 0.56 | 0   | 0.58       | 3.06 | 0.000 | 0.99        | 1.31 | 0   | 0.98      | 1.16 | 0   | 0.93     | 1.46 | 0.005 | 0.98     | 0.96 | 0   |
| 2   | 82596       | 1.00    | 0.69 | 0   | 0.58       | 3.17 | 0.000 | 0.95        | 1.32 | 0   | 0.95      | 1.38 | 0   | 0.88     | 1.73 | 0.004 | 0.95     | 1.26 | 0   |
| 3   | 82473       | 0.99    | 0.83 | 0   | 0.64       | 3.11 | 0.000 | 0.95        | 1.24 | 0   | 0.94      | 1.66 | 0   | 0.88     | 1.82 | 0.003 | 0.95     | 1.45 | 0   |
| 4   | 84055       | 0.98    | 0.91 | 0   | 0.72       | 2.88 | 0.000 | 0.87        | 1.36 | 0   | 0.92      | 1.65 | 0   | 0.91     | 1.67 | 0.003 | 0.95     | 1.41 | 0   |
| 5   | 92521       | 0.97    | 1.00 | 0   | 0.77       | 2.60 | 0.000 | 0.85        | 1.60 | 0   | 0.87      | 1.68 | 0   | 0.92     | 1.51 | 0.002 | 0.95     | 1.28 | 0   |
| 6   | 91639       | 0.96    | 1.08 | 0   | 0.76       | 2.49 | 0.000 | 0.79        | 2.02 | 0   | 0.76      | 1.93 | 0   | 0.93     | 1.34 | 0.002 | 0.93     | 1.22 | 0   |
| 7   | 89993       | 0.96    | 1.10 | 0   | 0.69       | 2.61 | 0.000 | 0.70        | 2.49 | 0   | 0.60      | 2.43 | 0   | 0.91     | 1.30 | 0.001 | 0.91     | 1.34 | 0   |
| 8   | 89428       | 0.94    | 1.22 | 0   | 0.64       | 2.76 | 0.000 | 0.59        | 2.71 | 0   | 0.52      | 2.57 | 0   | 0.89     | 1.42 | 0.000 | 0.89     | 1.47 | 0   |
| 9   | 89644       | 0.93    | 1.30 | 0   | 0.60       | 2.94 | 0.000 | 0.49        | 2.81 | 0   | 0.79      | 2.35 | 0   | 0.86     | 1.55 | 0.000 | 0.88     | 1.49 | 0   |
| 10  | 98337       | 0.90    | 1.43 | 0   | 0.57       | 3.04 | 0.000 | 0.68        | 2.80 | 0   | 0.93      | 1.97 | 0   | 0.84     | 1.68 | 0.000 | 0.89     | 1.49 | 0   |
| 11  | 97415       | 0.86    | 1.60 | 0   | 0.53       | 3.19 | 0.000 | 0.61        | 2.79 | 0   | 0.95      | 1.62 | 0   | 0.83     | 1.80 | 0.000 | 0.89     | 1.49 | 0   |
| 12  | 94326       | 0.82    | 1.78 | 0   | 0.53       | 3.26 | 0.000 | 0.67        | 2.81 | 0   | 0.88      | 1.70 | 0   | 0.80     | 1.97 | 0.000 | 0.87     | 1.56 | 0   |
| 13  | 94047       | 0.81    | 1.92 | 0   | 0.51       | 3.28 | 0.000 | 0.59        | 3.07 | 0   | 0.74      | 2.57 | 0   | 0.77     | 2.17 | 0.000 | 0.85     | 1.69 | 0   |
| 14  | 96915       | 0.82    | 1.93 | 0   | 0.51       | 3.22 | 0.000 | 0.53        | 3.36 | 0   | 0.55      | 3.99 | 0   | 0.74     | 2.35 | 0.000 | 0.84     | 1.82 | 0   |
| 15  | 97599       | 0.85    | 1.82 | 0   | 0.60       | 3.11 | 0.000 | 0.49        | 3.67 | 0   | 0.36      | 5.38 | 0   | 0.70     | 2.61 | 0.000 | 0.82     | 1.94 | 0   |
| 16  | 96388       | 0.87    | 1.65 | 0   | 0.65       | 3.00 | 0.000 | 0.49        | 3.86 | 0   | 0.23      | 6.19 | 0   | 0.66     | 2.85 | 0.000 | 0.78     | 2.06 | 0   |
| 17  | 99462       | 0.87    | 1.53 | 0   | 0.66       | 2.85 | 0.000 | 0.57        | 3.85 | 0   | 0.16      | 6.54 | 0   | 0.62     | 3.06 | 0.000 | 0.75     | 2.22 | 0   |
| 18  | 98765       | 0.84    | 1.65 | 0   | 0.61       | 3.11 | 0.000 | 0.53        | 3.86 | 0   | 0.12      | 6.86 | 0   | 0.56     | 3.46 | 0.000 | 0.68     | 2.63 | 0   |
| 19  | 105717      | 0.80    | 1.89 | 0   | 0.58       | 3.29 | 0.000 | 0.48        | 3.93 | 0   | 0.09      | 6.86 | 0   | 0.52     | 3.70 | 0.000 | 0.62     | 2.93 | 0   |
| 20  | 118643      | 0.76    | 2.15 | 0   | 0.58       | 3.38 | 0.000 | 0.44        | 4.03 | 0   | 0.10      | 6.60 | 0   | 0.48     | 3.79 | 0.000 | 0.57     | 3.11 | 0   |
| 21  | 124060      | 0.70    | 2.41 | 0   | 0.55       | 3.50 | 0.000 | 0.40        | 4.16 | 0   | 0.12      | 6.34 | 0   | 0.46     | 3.83 | 0.000 | 0.53     | 3.23 | 0   |
| 22  | 128864      | 0.65    | 2.60 | 0   | 0.53       | 3.61 | 0.000 | 0.43        | 4.23 | 0   | 0.16      | 6.07 | 0   | 0.45     | 3.85 | 0.000 | 0.51     | 3.29 | 0   |
| 23  | 133507      | 0.61    | 2.73 | 0   | 0.51       | 3.69 | 0.000 | 0.37        | 4.23 | 0   | 0.21      | 5.83 | 0   | 0.46     | 3.83 | 0.000 | 0.52     | 3.28 | 0   |
| 24  | 138094      | 0.59    | 2.83 | 0   | 0.50       | 3.75 | 0.000 | 0.37        | 4.15 | 0   | 0.24      | 5.61 | 0   | 0.48     | 3.76 | 0.000 | 0.55     | 3.21 | 0   |
| 25  | 139920      | 0.59    | 2.88 | 0   | 0.49       | 3.81 | 0.000 | 0.40        | 4.14 | 0   | 0.26      | 5.47 | 0   | 0.50     | 3.69 | 0.000 | 0.58     | 3.12 | 0   |
| 26  | 141741      | 0.59    | 2.92 | 0   | 0.48       | 3.85 | 0.000 | 0.45        | 4.14 | 0   | 0.28      | 5.35 | 0   | 0.52     | 3.61 | 0.000 | 0.61     | 3.03 | 0   |
| 27  | 138491      | 0.58    | 2.97 | 0   | 0.47       | 3.91 | 0.000 | 0.49        | 4.11 | 0   | 0.29      | 5.30 | 0   | 0.53     | 3.59 | 0.000 | 0.61     | 3.03 | 0   |
| 28  | 133633      | 0.57    | 3.03 | 0   | 0.46       | 4.00 | 0.000 | 0.46        | 4.10 | 0   | 0.29      | 5.31 | 0   | 0.53     | 3.60 | 0.000 | 0.61     | 3.07 | 0   |
| 29  | 129054      | 0.56    | 3.09 | 0   | 0.45       | 4.05 | 0.000 | 0.52        | 4.13 | 0   | 0.28      | 5.31 | 0   | 0.53     | 3.64 | 0.000 | 0.60     | 3.18 | 0   |
| 30  | 123437      | 0.55    | 3.16 | 0   | 0.44       | 4.14 | 0.000 | 0.49        | 4.21 | 0   | 0.27      | 5.32 | 0   | 0.51     | 3.70 | 0.000 | 0.57     | 3.32 | 0   |
| 31  | 118174      | 0.54    | 3.20 | 0   | 0.43       | 4.21 | 0.000 | 0.45        | 4.31 | 0   | 0.29      | 5.27 | 0   | 0.50     | 3.78 | 0.000 | 0.54     | 3.47 | 0   |
| 32  | 113131      | 0.54    | 3.21 | 0   | 0.42       | 4.25 | 0.000 | 0.40        | 4.43 | 0   | 0.29      | 5.23 | 0   | 0.48     | 3.84 | 0.000 | 0.51     | 3.60 | 0   |
| 33  | 109146      | 0.53    | 3.23 | 0   | 0.41       | 4.31 | 0.000 | 0.35        | 4.59 | 0   | 0.35      | 5.18 | 0   | 0.46     | 3.90 | 0.000 | 0.49     | 3.71 | 0   |
| 34  | 105908      | 0.54    | 3.22 | 0   | 0.40       | 4.36 | 0.000 | 0.33        | 4.65 | 0   | 0.36      | 5.16 | 0   | 0.45     | 3.94 | 0.000 | 0.47     | 3.81 | 0   |
| 35  | 103149      | 0.54    | 3.20 | 0   | 0.40       | 4.40 | 0.000 | 0.36        | 4.61 | 0   | 0.40      | 5.08 | 0   | 0.45     | 3.95 | 0.000 | 0.46     | 3.86 | 0   |
| 36  | 99901       | 0.55    | 3.18 | 0   | 0.40       | 4.42 | 0.000 | 0.40        | 4.50 | 0   | 0.39      | 5.09 | 0   | 0.45     | 3.95 | 0.000 | 0.46     | 3.91 | 0   |

**Table 31.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | roc-002 |      |     | shufti-000 |       |       | unissey-001 |      |     | viant-000 |      |     | yoti-001 |      |       | yoti-002 |      |     |
|-----|-------------|---------|------|-----|------------|-------|-------|-------------|------|-----|-----------|------|-----|----------|------|-------|----------|------|-----|
|     |             | Acc(3)  | MAE  | FTP | Acc(3)     | MAE   | FTP   | Acc(3)      | MAE  | FTP | Acc(3)    | MAE  | FTP | Acc(3)   | MAE  | FTP   | Acc(3)   | MAE  | FTP |
| 37  | 97516       | 0.55    | 3.15 | 0   | 0.39       | 4.42  | 0.000 | 0.44        | 4.32 | 0   | 0.36      | 5.07 | 0   | 0.45     | 3.92 | 0.000 | 0.45     | 3.91 | 0   |
| 38  | 93939       | 0.55    | 3.13 | 0   | 0.40       | 4.41  | 0.000 | 0.47        | 4.24 | 0   | 0.34      | 5.05 | 0   | 0.46     | 3.86 | 0.000 | 0.45     | 3.93 | 0   |
| 39  | 91461       | 0.56    | 3.10 | 0   | 0.40       | 4.39  | 0.000 | 0.48        | 4.21 | 0   | 0.30      | 5.06 | 0   | 0.47     | 3.84 | 0.000 | 0.46     | 3.91 | 0   |
| 40  | 94056       | 0.56    | 3.11 | 0   | 0.40       | 4.39  | 0.000 | 0.52        | 4.22 | 0   | 0.26      | 5.05 | 0   | 0.47     | 3.83 | 0.000 | 0.46     | 3.89 | 0   |
| 41  | 90173       | 0.56    | 3.10 | 0   | 0.40       | 4.37  | 0.000 | 0.45        | 4.25 | 0   | 0.25      | 4.87 | 0   | 0.48     | 3.80 | 0.000 | 0.46     | 3.89 | 0   |
| 42  | 86423       | 0.57    | 3.08 | 0   | 0.40       | 4.34  | 0.000 | 0.41        | 4.29 | 0   | 0.28      | 4.62 | 0   | 0.48     | 3.80 | 0.000 | 0.46     | 3.88 | 0   |
| 43  | 83252       | 0.57    | 3.09 | 0   | 0.41       | 4.31  | 0.000 | 0.39        | 4.31 | 0   | 0.32      | 4.31 | 0   | 0.48     | 3.80 | 0.000 | 0.46     | 3.90 | 0   |
| 44  | 80602       | 0.57    | 3.10 | 0   | 0.42       | 4.25  | 0.000 | 0.37        | 4.35 | 0   | 0.33      | 3.99 | 0   | 0.47     | 3.83 | 0.000 | 0.45     | 3.90 | 0   |
| 45  | 77502       | 0.56    | 3.12 | 0   | 0.42       | 4.19  | 0.000 | 0.42        | 4.33 | 0   | 0.43      | 3.67 | 0   | 0.46     | 3.85 | 0.000 | 0.45     | 3.90 | 0   |
| 46  | 74182       | 0.56    | 3.14 | 0   | 0.42       | 4.19  | 0.000 | 0.41        | 4.24 | 0   | 0.50      | 3.38 | 0   | 0.46     | 3.86 | 0.000 | 0.45     | 3.88 | 0   |
| 47  | 70698       | 0.55    | 3.16 | 0   | 0.43       | 4.16  | 0.000 | 0.44        | 4.13 | 0   | 0.57      | 3.11 | 0   | 0.46     | 3.89 | 0.000 | 0.45     | 3.85 | 0   |
| 48  | 67856       | 0.56    | 3.15 | 0   | 0.43       | 4.12  | 0.000 | 0.46        | 4.06 | 0   | 0.63      | 2.89 | 0   | 0.46     | 3.90 | 0.000 | 0.46     | 3.84 | 0   |
| 49  | 65674       | 0.55    | 3.16 | 0   | 0.44       | 4.07  | 0.000 | 0.48        | 4.04 | 0   | 0.63      | 2.76 | 0   | 0.47     | 3.89 | 0.000 | 0.48     | 3.78 | 0   |
| 50  | 64896       | 0.55    | 3.16 | 0   | 0.44       | 4.03  | 0.000 | 0.51        | 4.01 | 0   | 0.64      | 2.73 | 0   | 0.47     | 3.90 | 0.000 | 0.49     | 3.75 | 0   |
| 51  | 63410       | 0.55    | 3.17 | 0   | 0.45       | 3.96  | 0.000 | 0.47        | 4.04 | 0   | 0.64      | 2.80 | 0   | 0.47     | 3.90 | 0.000 | 0.49     | 3.72 | 0   |
| 52  | 60991       | 0.56    | 3.18 | 0   | 0.45       | 3.88  | 0.000 | 0.45        | 4.07 | 0   | 0.64      | 2.89 | 0   | 0.47     | 3.92 | 0.000 | 0.49     | 3.71 | 0   |
| 53  | 59550       | 0.56    | 3.15 | 0   | 0.46       | 3.78  | 0.000 | 0.44        | 4.13 | 0   | 0.60      | 2.97 | 0   | 0.46     | 3.91 | 0.000 | 0.49     | 3.70 | 0   |
| 54  | 56895       | 0.55    | 3.16 | 0   | 0.47       | 3.68  | 0.000 | 0.43        | 4.20 | 0   | 0.55      | 3.06 | 0   | 0.45     | 3.93 | 0.000 | 0.48     | 3.71 | 0   |
| 55  | 55297       | 0.56    | 3.14 | 0   | 0.47       | 3.57  | 0.000 | 0.42        | 4.29 | 0   | 0.54      | 3.14 | 0   | 0.44     | 3.89 | 0.000 | 0.48     | 3.69 | 0   |
| 56  | 52574       | 0.56    | 3.13 | 0   | 0.49       | 3.42  | 0.000 | 0.40        | 4.30 | 0   | 0.53      | 3.19 | 0   | 0.44     | 3.85 | 0.000 | 0.47     | 3.68 | 0   |
| 57  | 49823       | 0.57    | 3.13 | 0   | 0.51       | 3.23  | 0.000 | 0.44        | 4.40 | 0   | 0.52      | 3.21 | 0   | 0.44     | 3.77 | 0.000 | 0.48     | 3.63 | 0   |
| 58  | 48570       | 0.56    | 3.13 | 0   | 0.53       | 3.06  | 0.000 | 0.44        | 4.46 | 0   | 0.53      | 3.16 | 0   | 0.46     | 3.64 | 0.000 | 0.49     | 3.56 | 0   |
| 59  | 46040       | 0.56    | 3.14 | 0   | 0.57       | 2.86  | 0.000 | 0.44        | 4.52 | 0   | 0.54      | 3.10 | 0   | 0.48     | 3.52 | 0.000 | 0.50     | 3.53 | 0   |
| 60  | 46037       | 0.55    | 3.19 | 0   | 0.64       | 2.65  | 0.000 | 0.42        | 4.69 | 0   | 0.55      | 2.99 | 0   | 0.50     | 3.38 | 0.000 | 0.51     | 3.46 | 0   |
| 61  | 43860       | 0.55    | 3.21 | 0   | 0.74       | 2.43  | 0.000 | 0.41        | 4.86 | 0   | 0.64      | 2.86 | 0   | 0.53     | 3.25 | 0.000 | 0.51     | 3.43 | 0   |
| 62  | 41407       | 0.55    | 3.23 | 0   | 0.76       | 2.28  | 0.000 | 0.39        | 5.12 | 0   | 0.64      | 2.75 | 0   | 0.57     | 3.07 | 0.000 | 0.53     | 3.32 | 0   |
| 63  | 39444       | 0.54    | 3.27 | 0   | 0.74       | 2.28  | 0.000 | 0.30        | 5.35 | 0   | 0.65      | 2.65 | 0   | 0.60     | 2.96 | 0.001 | 0.54     | 3.28 | 0   |
| 64  | 37425       | 0.54    | 3.30 | 0   | 0.70       | 2.57  | 0.000 | 0.29        | 5.56 | 0   | 0.64      | 2.66 | 0   | 0.63     | 2.83 | 0.001 | 0.55     | 3.23 | 0   |
| 65  | 34834       | 0.53    | 3.35 | 0   | 0.63       | 3.18  | 0.000 | 0.28        | 5.71 | 0   | 0.65      | 2.76 | 0   | 0.65     | 2.81 | 0.001 | 0.56     | 3.21 | 0   |
| 66  | 32020       | 0.52    | 3.40 | 0   | 0.48       | 3.85  | 0.000 | 0.28        | 5.87 | 0   | 0.65      | 2.86 | 0   | 0.65     | 2.82 | 0.001 | 0.56     | 3.16 | 0   |
| 67  | 28898       | 0.52    | 3.46 | 0   | 0.25       | 4.61  | 0.000 | 0.26        | 6.07 | 0   | 0.54      | 3.01 | 0   | 0.63     | 2.91 | 0.001 | 0.57     | 3.14 | 0   |
| 68  | 26630       | 0.50    | 3.54 | 0   | 0.14       | 5.59  | 0.000 | 0.27        | 6.19 | 0   | 0.52      | 3.23 | 0   | 0.60     | 3.12 | 0.001 | 0.57     | 3.18 | 0   |
| 69  | 24864       | 0.50    | 3.59 | 0   | 0.10       | 6.44  | 0.000 | 0.27        | 6.33 | 0   | 0.49      | 3.46 | 0   | 0.55     | 3.38 | 0.002 | 0.56     | 3.20 | 0   |
| 70  | 23407       | 0.48    | 3.68 | 0   | 0.09       | 7.50  | 0.000 | 0.27        | 6.36 | 0   | 0.46      | 3.73 | 0   | 0.49     | 3.69 | 0.002 | 0.55     | 3.26 | 0   |
| 71  | 20736       | 0.47    | 3.76 | 0   | 0.08       | 8.59  | 0.000 | 0.27        | 6.42 | 0   | 0.39      | 4.03 | 0   | 0.43     | 4.07 | 0.001 | 0.56     | 3.30 | 0   |
| 72  | 18423       | 0.47    | 3.81 | 0   | 0.08       | 9.75  | 0.000 | 0.27        | 6.49 | 0   | 0.36      | 4.33 | 0   | 0.36     | 4.52 | 0.002 | 0.57     | 3.40 | 0   |
| 73  | 16215       | 0.45    | 3.93 | 0   | 0.08       | 11.08 | 0.000 | 0.28        | 6.49 | 0   | 0.31      | 4.60 | 0   | 0.30     | 4.95 | 0.002 | 0.54     | 3.50 | 0   |
| 74  | 14848       | 0.45    | 3.95 | 0   | 0.08       | 12.42 | 0.000 | 0.28        | 6.43 | 0   | 0.32      | 4.82 | 0   | 0.24     | 5.42 | 0.002 | 0.51     | 3.69 | 0   |



**Table 31.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

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| Age | Num. Images | roc-002 |      |     | shufti-000 |       |       | unissey-001 |      |     | vianti-000 |       |     | yoti-001 |       |       | yoti-002 |       |     |
|-----|-------------|---------|------|-----|------------|-------|-------|-------------|------|-----|------------|-------|-----|----------|-------|-------|----------|-------|-----|
|     |             | Acc(3)  | MAE  | FTP | Acc(3)     | MAE   | FTP   | Acc(3)      | MAE  | FTP | Acc(3)     | MAE   | FTP | Acc(3)   | MAE   | FTP   | Acc(3)   | MAE   | FTP |
| 75  | 13197       | 0.44    | 4.13 | 0   | 0.08       | 13.73 | 0.000 | 0.28        | 6.53 | 0   | 0.29       | 5.20  | 0   | 0.17     | 6.09  | 0.003 | 0.45     | 4.15  | 0   |
| 76  | 11886       | 0.43    | 4.20 | 0   | 0.08       | 15.25 | 0.000 | 0.29        | 6.46 | 0   | 0.27       | 5.51  | 0   | 0.12     | 6.68  | 0.002 | 0.38     | 4.64  | 0   |
| 77  | 10839       | 0.41    | 4.39 | 0   | 0.08       | 16.94 | 0.000 | 0.29        | 6.50 | 0   | 0.25       | 5.99  | 0   | 0.08     | 7.33  | 0.003 | 0.27     | 5.22  | 0   |
| 78  | 9361        | 0.40    | 4.51 | 0   | 0.09       | 18.64 | 0.000 | 0.30        | 6.58 | 0   | 0.14       | 6.41  | 0   | 0.04     | 7.97  | 0.004 | 0.08     | 5.90  | 0   |
| 79  | 7540        | 0.39    | 4.56 | 0   | 0.10       | 20.50 | 0.000 | 0.32        | 6.46 | 0   | 0.10       | 6.80  | 0   | 0.01     | 8.56  | 0.003 | 0.00     | 6.52  | 0   |
| 80  | 6267        | 0.38    | 4.69 | 0   | 0.11       | 22.58 | 0.000 | 0.32        | 6.62 | 0   | 0.07       | 7.28  | 0   | 0.00     | 9.37  | 0.007 | 0.00     | 7.24  | 0   |
| 81  | 5221        | 0.38    | 4.75 | 0   | 0.11       | 25.46 | 0.000 | 0.34        | 6.48 | 0   | 0.00       | 7.75  | 0   | 0.00     | 10.02 | 0.005 | 0.00     | 7.91  | 0   |
| 82  | 3962        | 0.38    | 4.87 | 0   | 0.10       | 26.77 | 0.000 | 0.34        | 6.56 | 0   | 0.00       | 8.32  | 0   | 0.00     | 10.72 | 0.007 | 0.00     | 8.74  | 0   |
| 83  | 3277        | 0.34    | 5.13 | 0   | 0.09       | 29.02 | 0.000 | 0.35        | 6.63 | 0   | 0.00       | 8.90  | 0   | 0.00     | 11.61 | 0.007 | 0.00     | 9.57  | 0   |
| 84  | 2615        | 0.32    | 5.34 | 0   | 0.08       | 31.06 | 0.000 | 0.34        | 6.85 | 0   | 0.00       | 9.55  | 0   | 0.00     | 12.26 | 0.007 | 0.00     | 10.35 | 0   |
| 85  | 2206        | 0.31    | 5.41 | 0   | 0.06       | 33.71 | 0.000 | 0.33        | 6.98 | 0   | 0.00       | 10.07 | 0   | 0.00     | 13.07 | 0.008 | 0.00     | 11.20 | 0   |
| 86  | 1831        | 0.31    | 5.55 | 0   | 0.02       | 37.96 | 0.000 | 0.29        | 7.18 | 0   | 0.00       | 10.77 | 0   | 0.00     | 13.86 | 0.011 | 0.00     | 12.17 | 0   |
| 87  | 1402        | 0.29    | 5.87 | 0   | 0.00       | 38.35 | 0.000 | 0.23        | 7.61 | 0   | 0.00       | 11.57 | 0   | 0.00     | 14.69 | 0.011 | 0.00     | 13.08 | 0   |
| 88  | 1254        | 0.23    | 6.43 | 0   | 0.00       | 39.91 | 0.000 | 0.19        | 8.13 | 0   | 0.00       | 12.34 | 0   | 0.00     | 15.78 | 0.014 | 0.00     | 14.03 | 0   |
| 89  | 907         | 0.26    | 6.29 | 0   | 0.00       | 41.86 | 0.000 | 0.15        | 7.91 | 0   | 0.00       | 12.91 | 0   | 0.00     | 16.57 | 0.012 | 0.00     | 14.93 | 0   |
| 90  | 670         | 0.21    | 6.96 | 0   | 0.00       | 44.54 | 0.000 | 0.10        | 8.24 | 0   | 0.00       | 13.99 | 0   | 0.00     | 17.55 | 0.015 | 0.00     | 16.11 | 0   |

**Table 32.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better).

| Age | Num. Images | yoti-003 |      |     | yoti-004 |      |     | youverse-000 |      |       |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|------|-------|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE  | FTP   |
| 0   | 133387      | 0.94     | 1.42 | 0   | 0.96     | 1.14 | 0   | 0.60         | 3.31 | 0.026 |
| 1   | 94475       | 0.95     | 1.27 | 0   | 0.98     | 1.09 | 0   | 0.42         | 3.88 | 0.006 |
| 2   | 82596       | 0.93     | 1.45 | 0   | 0.97     | 1.23 | 0   | 0.29         | 4.64 | 0.003 |
| 3   | 82473       | 0.94     | 1.56 | 0   | 0.97     | 1.28 | 0   | 0.25         | 5.04 | 0.003 |
| 4   | 84055       | 0.95     | 1.42 | 0   | 0.97     | 1.18 | 0   | 0.22         | 5.31 | 0.002 |
| 5   | 92521       | 0.96     | 1.22 | 0   | 0.97     | 1.08 | 0   | 0.20         | 5.47 | 0.002 |
| 6   | 91639       | 0.94     | 1.14 | 0   | 0.95     | 1.09 | 0   | 0.17         | 5.75 | 0.001 |
| 7   | 89993       | 0.90     | 1.28 | 0   | 0.92     | 1.26 | 0   | 0.13         | 6.18 | 0.001 |
| 8   | 89428       | 0.87     | 1.44 | 0   | 0.90     | 1.41 | 0   | 0.11         | 6.27 | 0.001 |
| 9   | 89644       | 0.86     | 1.55 | 0   | 0.89     | 1.49 | 0   | 0.10         | 6.13 | 0.000 |
| 10  | 98337       | 0.87     | 1.65 | 0   | 0.88     | 1.56 | 0   | 0.10         | 5.86 | 0.000 |
| 11  | 97415       | 0.86     | 1.73 | 0   | 0.86     | 1.65 | 0   | 0.09         | 5.53 | 0.000 |
| 12  | 94326       | 0.85     | 1.78 | 0   | 0.84     | 1.77 | 0   | 0.09         | 5.13 | 0.000 |
| 13  | 94047       | 0.84     | 1.77 | 0   | 0.81     | 1.90 | 0   | 0.14         | 4.69 | 0.000 |

**Table 32.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | yoti-003 |      |     | yoti-004 |      |     | youverse-000 |      |       |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|------|-------|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE  | FTP   |
| 14  | 96915       | 0.83     | 1.76 | 0   | 0.79     | 1.99 | 0   | 0.29         | 4.32 | 0.000 |
| 15  | 97599       | 0.82     | 1.85 | 0   | 0.75     | 2.11 | 0   | 0.43         | 3.99 | 0.000 |
| 16  | 96388       | 0.78     | 2.02 | 0   | 0.72     | 2.27 | 0   | 0.49         | 3.67 | 0.000 |
| 17  | 99462       | 0.75     | 2.23 | 0   | 0.69     | 2.39 | 0   | 0.52         | 3.39 | 0.000 |
| 18  | 98765       | 0.68     | 2.65 | 0   | 0.64     | 2.63 | 0   | 0.54         | 3.29 | 0.000 |
| 19  | 105717      | 0.61     | 2.94 | 0   | 0.62     | 2.72 | 0   | 0.57         | 3.17 | 0.000 |
| 20  | 118643      | 0.56     | 3.11 | 0   | 0.63     | 2.68 | 0   | 0.58         | 3.11 | 0.000 |
| 21  | 124060      | 0.53     | 3.23 | 0   | 0.65     | 2.60 | 0   | 0.59         | 3.05 | 0.000 |
| 22  | 128864      | 0.51     | 3.28 | 0   | 0.67     | 2.49 | 0   | 0.60         | 3.02 | 0.000 |
| 23  | 133507      | 0.52     | 3.28 | 0   | 0.70     | 2.39 | 0   | 0.59         | 3.01 | 0.000 |
| 24  | 138094      | 0.54     | 3.24 | 0   | 0.73     | 2.30 | 0   | 0.58         | 3.05 | 0.000 |
| 25  | 139920      | 0.56     | 3.19 | 0   | 0.74     | 2.27 | 0   | 0.57         | 3.13 | 0.000 |
| 26  | 141741      | 0.58     | 3.11 | 0   | 0.73     | 2.29 | 0   | 0.55         | 3.22 | 0.000 |
| 27  | 138491      | 0.59     | 3.11 | 0   | 0.72     | 2.40 | 0   | 0.52         | 3.33 | 0.000 |
| 28  | 133633      | 0.59     | 3.14 | 0   | 0.69     | 2.57 | 0   | 0.50         | 3.44 | 0.000 |
| 29  | 129054      | 0.58     | 3.21 | 0   | 0.65     | 2.76 | 0   | 0.49         | 3.54 | 0.000 |
| 30  | 123437      | 0.56     | 3.33 | 0   | 0.60     | 2.96 | 0   | 0.48         | 3.65 | 0.000 |
| 31  | 118174      | 0.54     | 3.44 | 0   | 0.56     | 3.15 | 0   | 0.47         | 3.77 | 0.000 |
| 32  | 113131      | 0.51     | 3.56 | 0   | 0.53     | 3.34 | 0   | 0.46         | 3.89 | 0.000 |
| 33  | 109146      | 0.49     | 3.67 | 0   | 0.50     | 3.48 | 0   | 0.46         | 3.99 | 0.000 |
| 34  | 105908      | 0.47     | 3.78 | 0   | 0.48     | 3.59 | 0   | 0.45         | 4.12 | 0.000 |
| 35  | 103149      | 0.46     | 3.85 | 0   | 0.47     | 3.65 | 0   | 0.43         | 4.31 | 0.000 |
| 36  | 99901       | 0.45     | 3.90 | 0   | 0.47     | 3.67 | 0   | 0.42         | 4.44 | 0.000 |
| 37  | 97516       | 0.45     | 3.91 | 0   | 0.48     | 3.63 | 0   | 0.40         | 4.65 | 0.000 |
| 38  | 93939       | 0.45     | 3.94 | 0   | 0.48     | 3.60 | 0   | 0.37         | 4.90 | 0.000 |
| 39  | 91461       | 0.46     | 3.91 | 0   | 0.50     | 3.53 | 0   | 0.34         | 5.12 | 0.000 |
| 40  | 94056       | 0.45     | 3.92 | 0   | 0.50     | 3.51 | 0   | 0.32         | 5.29 | 0.000 |
| 41  | 90173       | 0.45     | 3.92 | 0   | 0.51     | 3.46 | 0   | 0.30         | 5.57 | 0.000 |
| 42  | 86423       | 0.45     | 3.88 | 0   | 0.52     | 3.41 | 0   | 0.27         | 5.87 | 0.000 |
| 43  | 83252       | 0.46     | 3.86 | 0   | 0.52     | 3.37 | 0   | 0.24         | 6.19 | 0.000 |
| 44  | 80602       | 0.45     | 3.85 | 0   | 0.53     | 3.34 | 0   | 0.21         | 6.54 | 0.000 |
| 45  | 77502       | 0.45     | 3.81 | 0   | 0.53     | 3.31 | 0   | 0.19         | 6.89 | 0.000 |
| 46  | 74182       | 0.46     | 3.77 | 0   | 0.53     | 3.28 | 0   | 0.17         | 7.26 | 0.000 |
| 47  | 70698       | 0.47     | 3.71 | 0   | 0.54     | 3.25 | 0   | 0.15         | 7.61 | 0.000 |
| 48  | 67856       | 0.48     | 3.66 | 0   | 0.55     | 3.23 | 0   | 0.13         | 8.01 | 0.000 |
| 49  | 65674       | 0.50     | 3.55 | 0   | 0.56     | 3.18 | 0   | 0.11         | 8.43 | 0.000 |
| 50  | 64896       | 0.52     | 3.48 | 0   | 0.56     | 3.15 | 0   | 0.10         | 8.84 | 0.000 |
| 51  | 63410       | 0.53     | 3.42 | 0   | 0.57     | 3.13 | 0   | 0.08         | 9.31 | 0.000 |

**Table 32.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | yoti-003 |      |     | yoti-004 |      |     | youverse-000 |       |       |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|-------|-------|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE   | FTP   |
| 52  | 60991       | 0.54     | 3.40 | 0   | 0.57     | 3.12 | 0   | 0.07         | 9.74  | 0.000 |
| 53  | 59550       | 0.53     | 3.40 | 0   | 0.57     | 3.10 | 0   | 0.06         | 10.18 | 0.000 |
| 54  | 56895       | 0.53     | 3.41 | 0   | 0.57     | 3.10 | 0   | 0.05         | 10.64 | 0.000 |
| 55  | 55297       | 0.52     | 3.44 | 0   | 0.56     | 3.10 | 0   | 0.04         | 11.12 | 0.000 |
| 56  | 52574       | 0.51     | 3.46 | 0   | 0.56     | 3.11 | 0   | 0.04         | 11.60 | 0.000 |
| 57  | 49823       | 0.50     | 3.44 | 0   | 0.57     | 3.08 | 0   | 0.03         | 12.13 | 0.000 |
| 58  | 48570       | 0.51     | 3.41 | 0   | 0.57     | 3.06 | 0   | 0.03         | 12.64 | 0.000 |
| 59  | 46040       | 0.51     | 3.41 | 0   | 0.57     | 3.08 | 0   | 0.02         | 13.14 | 0.000 |
| 60  | 46037       | 0.52     | 3.37 | 0   | 0.57     | 3.09 | 0   | 0.02         | 13.63 | 0.000 |
| 61  | 43860       | 0.53     | 3.35 | 0   | 0.57     | 3.11 | 0   | 0.02         | 14.14 | 0.000 |
| 62  | 41407       | 0.54     | 3.29 | 0   | 0.57     | 3.13 | 0   | 0.01         | 14.70 | 0.000 |
| 63  | 39444       | 0.54     | 3.28 | 0   | 0.56     | 3.17 | 0   | 0.01         | 15.24 | 0.000 |
| 64  | 37425       | 0.55     | 3.27 | 0   | 0.55     | 3.24 | 0   | 0.01         | 15.80 | 0.000 |
| 65  | 34834       | 0.55     | 3.26 | 0   | 0.54     | 3.29 | 0   | 0.01         | 16.35 | 0.000 |
| 66  | 32020       | 0.55     | 3.23 | 0   | 0.53     | 3.35 | 0   | 0.01         | 16.96 | 0.000 |
| 67  | 28898       | 0.55     | 3.25 | 0   | 0.51     | 3.45 | 0   | 0.00         | 17.58 | 0.000 |
| 68  | 26630       | 0.54     | 3.30 | 0   | 0.50     | 3.55 | 0   | 0.00         | 18.20 | 0.000 |
| 69  | 24864       | 0.54     | 3.33 | 0   | 0.50     | 3.59 | 0   | 0.00         | 18.83 | 0.000 |
| 70  | 23407       | 0.54     | 3.33 | 0   | 0.49     | 3.63 | 0   | 0.00         | 19.45 | 0.000 |
| 71  | 20736       | 0.52     | 3.42 | 0   | 0.48     | 3.72 | 0   | 0.00         | 20.09 | 0.000 |
| 72  | 18423       | 0.51     | 3.52 | 0   | 0.47     | 3.83 | 0   | 0.00         | 20.82 | 0.000 |
| 73  | 16215       | 0.50     | 3.60 | 0   | 0.47     | 3.89 | 0   | 0.00         | 21.48 | 0.000 |
| 74  | 14848       | 0.48     | 3.71 | 0   | 0.46     | 3.94 | 0   | 0.00         | 22.08 | 0.000 |
| 75  | 13197       | 0.46     | 3.94 | 0   | 0.44     | 4.12 | 0   | 0.00         | 22.93 | 0.000 |
| 76  | 11886       | 0.44     | 4.04 | 0   | 0.44     | 4.20 | 0   | 0.00         | 23.52 | 0.000 |
| 77  | 10839       | 0.42     | 4.21 | 0   | 0.42     | 4.35 | 0   | 0.00         | 24.26 | 0.000 |
| 78  | 9361        | 0.40     | 4.46 | 0   | 0.41     | 4.51 | 0   | 0.00         | 24.95 | 0.000 |
| 79  | 7540        | 0.38     | 4.58 | 0   | 0.39     | 4.64 | 0   | 0.00         | 25.67 | 0.000 |
| 80  | 6267        | 0.36     | 4.79 | 0   | 0.37     | 4.80 | 0   | 0.00         | 26.54 | 0.000 |
| 81  | 5221        | 0.35     | 4.91 | 0   | 0.36     | 4.92 | 0   | 0.00         | 27.29 | 0.000 |
| 82  | 3962        | 0.32     | 5.25 | 0   | 0.33     | 5.23 | 0   | 0.00         | 28.08 | 0.000 |
| 83  | 3277        | 0.30     | 5.45 | 0   | 0.30     | 5.47 | 0   | 0.00         | 28.84 | 0.000 |
| 84  | 2615        | 0.29     | 5.71 | 0   | 0.30     | 5.65 | 0   | 0.00         | 29.72 | 0.000 |
| 85  | 2206        | 0.27     | 5.93 | 0   | 0.28     | 5.72 | 0   | 0.00         | 30.26 | 0.000 |
| 86  | 1831        | 0.25     | 6.28 | 0   | 0.28     | 6.08 | 0   | 0.00         | 31.21 | 0.000 |
| 87  | 1402        | 0.21     | 6.85 | 0   | 0.24     | 6.43 | 0   | 0.00         | 32.31 | 0.000 |
| 88  | 1254        | 0.18     | 7.36 | 0   | 0.22     | 6.87 | 0   | 0.00         | 33.22 | 0.000 |
| 89  | 907         | 0.15     | 7.46 | 0   | 0.21     | 6.93 | 0   | 0.00         | 33.91 | 0.000 |

**Table 32.** For visa images AE error statistics by age, 0 to 90. Acc(3) is proportion of age estimates within 3 years (higher value is better), MAE is mean absolute error (MAE, lower is better), and the failure to process proportion (FTP, lower is better). (continued)

| Age | Num. Images | yoti-003 |      |     | yoti-004 |      |     | youverse-000 |       |       |
|-----|-------------|----------|------|-----|----------|------|-----|--------------|-------|-------|
|     |             | Acc(3)   | MAE  | FTP | Acc(3)   | MAE  | FTP | Acc(3)       | MAE   | FTP   |
| 90  | 670         | 0.09     | 8.43 | 0   | 0.21     | 7.69 | 0   | 0.00         | 34.86 | 0.000 |

## **Appendix A.2. Age Estimation Performance Across Demographics**

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**Table 33.** Mean raw error for Application-type images for algorithm *cognitec-000*. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 2.3 ± 0.3     | 1.5 ± 0.1    | 1.4 ± 0.2     | 1.1 ± 0.1   | 1.1 ± 0.2    | 3.2 ± 0.3     | 1.2 ± 0.2     | 1.43 ± 0.10 | 0.7 ± 0.1     | 1.0 ± 0.1    | 1.2 ± 0.2    | 1.5 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 2.4 ± 0.4     | 1.5 ± 0.1    | 1.4 ± 0.2     | 1.4 ± 0.2   | 1.3 ± 0.3    | 2.9 ± 0.4     | 1.1 ± 0.3     | 1.55 ± 0.09 | 0.9 ± 0.2     | 1.4 ± 0.1    | 1.5 ± 0.2    | 1.6 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 2.1 ± 0.4     | 1.5 ± 0.1    | 1.1 ± 0.2     | 1.0 ± 0.2   | 1.5 ± 0.3    | 3.3 ± 0.5     | 1.7 ± 0.3     | 1.44 ± 0.09 | 1.0 ± 0.2     | 1.5 ± 0.1    | 1.5 ± 0.2    | 1.8 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 2.4 ± 0.4     | 1.3 ± 0.1    | 0.9 ± 0.2     | 1.0 ± 0.2   | 1.4 ± 0.3    | 3.1 ± 0.3     | 1.5 ± 0.3     | 1.3 ± 0.1   | 0.8 ± 0.2     | 1.5 ± 0.1    | 1.6 ± 0.2    | 2.0 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 1.7 ± 0.3     | 0.8 ± 0.1    | 0.3 ± 0.2     | 0.6 ± 0.1   | 1.0 ± 0.2    | 2.4 ± 0.4     | 1.2 ± 0.2     | 0.84 ± 0.09 | 0.4 ± 0.2     | 1.1 ± 0.1    | 1.3 ± 0.2    | 1.2 ± 0.3     | W Africa F   | E Europe F   |
| 19  | 1.8 ± 0.3     | 0.6 ± 0.1    | 0.1 ± 0.2     | 0.7 ± 0.1   | 1.0 ± 0.2    | 2.8 ± 0.3     | 1.5 ± 0.3     | 0.8 ± 0.1   | 0.1 ± 0.2     | 1.3 ± 0.1    | 1.1 ± 0.2    | 1.4 ± 0.3     | W Africa F   | E Europe F   |
| 20  | 1.9 ± 0.3     | 0.6 ± 0.1    | -0.4 ± 0.2    | 0.4 ± 0.2   | 0.9 ± 0.2    | 2.2 ± 0.3     | 1.5 ± 0.3     | 0.6 ± 0.1   | 0.3 ± 0.2     | 1.3 ± 0.1    | 1.1 ± 0.2    | 1.2 ± 0.3     | W Africa F   | E Europe F   |
| 21  | 1.4 ± 0.3     | 0.6 ± 0.1    | -0.7 ± 0.2    | 0.3 ± 0.1   | 0.6 ± 0.2    | 2.2 ± 0.3     | 1.4 ± 0.2     | 0.5 ± 0.1   | -0.1 ± 0.2    | 1.2 ± 0.1    | 0.9 ± 0.2    | 1.0 ± 0.3     | W Africa F   | E Europe F   |
| 22  | 1.2 ± 0.3     | 0.3 ± 0.1    | -1.0 ± 0.1    | 0.0 ± 0.1   | 0.5 ± 0.2    | 1.6 ± 0.3     | 1.8 ± 0.3     | 0.3 ± 0.1   | 0.0 ± 0.2     | 1.1 ± 0.1    | 0.8 ± 0.2    | 0.7 ± 0.3     | E Africa M   | E Europe F   |
| 23  | 1.1 ± 0.3     | 0.2 ± 0.1    | -1.1 ± 0.1    | -0.1 ± 0.1  | 0.4 ± 0.1    | 1.3 ± 0.3     | 1.4 ± 0.3     | 0.2 ± 0.1   | -0.2 ± 0.2    | 1.0 ± 0.1    | 0.6 ± 0.2    | 0.8 ± 0.2     | E Africa M   | E Europe F   |
| 24  | 0.9 ± 0.3     | -0.1 ± 0.1   | -1.2 ± 0.1    | -0.3 ± 0.1  | 0.1 ± 0.1    | 1.0 ± 0.3     | 1.5 ± 0.3     | -0.0 ± 0.1  | -0.2 ± 0.2    | 1.1 ± 0.1    | 0.6 ± 0.2    | 0.5 ± 0.2     | E Africa M   | E Europe F   |
| 25  | 0.2 ± 0.3     | -0.2 ± 0.1   | -1.4 ± 0.1    | -0.7 ± 0.1  | -0.1 ± 0.1   | 0.8 ± 0.3     | 1.2 ± 0.3     | 0.0 ± 0.1   | -0.3 ± 0.2    | 0.9 ± 0.1    | 0.5 ± 0.2    | 0.3 ± 0.2     | E Africa M   | E Europe F   |
| 26  | -0.2 ± 0.3    | -0.35 ± 0.10 | -1.5 ± 0.1    | -0.9 ± 0.1  | -0.1 ± 0.1   | 0.4 ± 0.2     | 1.0 ± 0.2     | 0.1 ± 0.1   | -0.4 ± 0.2    | 0.8 ± 0.1    | 0.4 ± 0.2    | -0.0 ± 0.2    | E Africa M   | E Europe F   |
| 27  | -0.3 ± 0.3    | -0.50 ± 0.10 | -1.5 ± 0.1    | -1.2 ± 0.1  | -0.5 ± 0.1   | 0.3 ± 0.2     | 0.7 ± 0.2     | -0.1 ± 0.1  | -0.4 ± 0.1    | 0.6 ± 0.1    | 0.5 ± 0.2    | -0.3 ± 0.2    | E Africa M   | E Europe F   |
| 28  | -0.6 ± 0.2    | -0.60 ± 0.09 | -1.4 ± 0.1    | -1.3 ± 0.1  | -0.6 ± 0.1   | -0.2 ± 0.2    | 0.4 ± 0.2     | -0.0 ± 0.1  | -0.4 ± 0.1    | 0.46 ± 0.10  | 0.2 ± 0.2    | -0.6 ± 0.2    | S Asia M     | E Europe F   |
| 29  | -0.6 ± 0.3    | -0.83 ± 0.09 | -1.4 ± 0.1    | -1.5 ± 0.1  | -0.8 ± 0.1   | -0.5 ± 0.2    | 0.2 ± 0.2     | -0.1 ± 0.1  | -0.5 ± 0.2    | 0.33 ± 0.09  | -0.0 ± 0.2   | -0.5 ± 0.2    | S Asia M     | S Asia F     |
| 30  | -1.0 ± 0.3    | -0.94 ± 0.09 | -1.5 ± 0.1    | -1.8 ± 0.1  | -0.8 ± 0.1   | -0.8 ± 0.2    | 0.3 ± 0.2     | -0.3 ± 0.1  | -0.7 ± 0.2    | 0.14 ± 0.09  | -0.1 ± 0.2   | -0.8 ± 0.2    | E Africa M   | S Asia F     |
| 31  | -1.3 ± 0.3    | -1.05 ± 0.09 | -1.5 ± 0.1    | -2.0 ± 0.1  | -1.1 ± 0.1   | -1.0 ± 0.2    | 0.0 ± 0.2     | -0.4 ± 0.1  | -0.7 ± 0.2    | 0.01 ± 0.09  | -0.1 ± 0.2   | -0.9 ± 0.2    | E Africa M   | S Asia F     |
| 32  | -1.1 ± 0.3    | -1.13 ± 0.10 | -1.4 ± 0.1    | -2.1 ± 0.1  | -1.1 ± 0.2   | -1.1 ± 0.3    | -0.4 ± 0.2    | -0.5 ± 0.1  | -0.7 ± 0.2    | -0.06 ± 0.09 | -0.1 ± 0.2   | -1.2 ± 0.2    | S Asia M     | S Asia F     |
| 33  | -1.5 ± 0.3    | -1.27 ± 0.10 | -1.6 ± 0.1    | -2.1 ± 0.1  | -1.3 ± 0.2   | -1.5 ± 0.3    | -0.2 ± 0.2    | -0.5 ± 0.1  | -0.6 ± 0.2    | -0.30 ± 0.09 | -0.4 ± 0.2   | -1.3 ± 0.2    | E Africa M   | S Asia F     |
| 34  | -1.8 ± 0.3    | -1.29 ± 0.10 | -1.5 ± 0.2    | -2.3 ± 0.1  | -1.3 ± 0.2   | -1.7 ± 0.3    | -0.8 ± 0.2    | -0.7 ± 0.1  | -0.9 ± 0.2    | -0.41 ± 0.10 | -0.4 ± 0.2   | -1.5 ± 0.2    | S E Asia M   | S Asia F     |
| 35  | -1.6 ± 0.3    | -1.5 ± 0.1   | -1.6 ± 0.2    | -2.4 ± 0.1  | -1.6 ± 0.2   | -2.2 ± 0.3    | -0.6 ± 0.3    | -0.8 ± 0.1  | -0.8 ± 0.2    | -0.49 ± 0.10 | -0.5 ± 0.2   | -1.6 ± 0.2    | S Asia M     | S Asia F     |
| 36  | -1.8 ± 0.3    | -1.6 ± 0.1   | -1.5 ± 0.2    | -2.4 ± 0.1  | -1.3 ± 0.2   | -2.1 ± 0.3    | -0.8 ± 0.3    | -0.8 ± 0.1  | -0.7 ± 0.2    | -0.71 ± 0.10 | -0.3 ± 0.2   | -1.7 ± 0.3    | S E Asia M   | S Asia F     |
| 37  | -1.9 ± 0.3    | -1.5 ± 0.1   | -1.5 ± 0.2    | -2.4 ± 0.1  | -1.5 ± 0.2   | -2.5 ± 0.3    | -0.8 ± 0.3    | -0.9 ± 0.1  | -0.9 ± 0.2    | -0.9 ± 0.1   | -0.6 ± 0.2   | -1.9 ± 0.2    | S E Asia M   | W Africa F   |
| 38  | -1.4 ± 0.4    | -1.7 ± 0.1   | -1.5 ± 0.2    | -2.4 ± 0.1  | -1.4 ± 0.2   | -2.5 ± 0.3    | -1.2 ± 0.3    | -0.9 ± 0.1  | -0.9 ± 0.2    | -1.0 ± 0.1   | -0.3 ± 0.2   | -1.9 ± 0.3    | S E Asia M   | W Africa F   |
| 39  | -1.9 ± 0.4    | -1.7 ± 0.1   | -1.6 ± 0.2    | -2.6 ± 0.1  | -1.4 ± 0.2   | -2.2 ± 0.3    | -0.8 ± 0.3    | -1.1 ± 0.1  | -0.9 ± 0.2    | -1.1 ± 0.1   | -0.5 ± 0.2   | -2.1 ± 0.3    | S E Asia M   | S Asia F     |
| 40  | -2.0 ± 0.4    | -1.7 ± 0.1   | -1.4 ± 0.2    | -2.5 ± 0.2  | -1.4 ± 0.2   | -2.9 ± 0.3    | -0.9 ± 0.3    | -1.1 ± 0.1  | -0.8 ± 0.2    | -1.2 ± 0.1   | -0.7 ± 0.2   | -2.1 ± 0.3    | S E Asia M   | W Africa F   |
| 41  | -2.2 ± 0.4    | -1.8 ± 0.1   | -1.5 ± 0.2    | -2.4 ± 0.2  | -1.6 ± 0.2   | -3.3 ± 0.3    | -1.0 ± 0.3    | -1.1 ± 0.1  | -0.9 ± 0.2    | -1.3 ± 0.1   | -0.8 ± 0.2   | -2.3 ± 0.3    | S E Asia M   | W Africa F   |
| 42  | -1.8 ± 0.4    | -1.7 ± 0.1   | -1.4 ± 0.2    | -2.2 ± 0.2  | -1.6 ± 0.2   | -2.8 ± 0.3    | -1.0 ± 0.3    | -1.2 ± 0.1  | -0.7 ± 0.2    | -1.5 ± 0.1   | -0.5 ± 0.2   | -2.3 ± 0.3    | S E Asia M   | W Africa F   |
| 43  | -1.7 ± 0.4    | -1.8 ± 0.1   | -1.6 ± 0.2    | -2.4 ± 0.2  | -1.6 ± 0.2   | -3.2 ± 0.3    | -1.0 ± 0.3    | -1.2 ± 0.1  | -0.9 ± 0.2    | -1.5 ± 0.1   | -0.7 ± 0.2   | -2.7 ± 0.3    | S E Asia M   | W Africa F   |
| 44  | -1.6 ± 0.5    | -1.9 ± 0.1   | -1.8 ± 0.2    | -2.4 ± 0.2  | -1.3 ± 0.2   | -3.3 ± 0.4    | -1.5 ± 0.3    | -1.2 ± 0.1  | -0.5 ± 0.2    | -1.7 ± 0.1   | -0.5 ± 0.2   | -2.6 ± 0.3    | S E Asia M   | W Africa F   |
| 45  | -2.0 ± 0.5    | -1.9 ± 0.1   | -1.7 ± 0.2    | -2.3 ± 0.2  | -1.4 ± 0.2   | -3.4 ± 0.4    | -1.5 ± 0.4    | -1.3 ± 0.1  | -0.7 ± 0.2    | -1.6 ± 0.1   | -0.6 ± 0.2   | -2.9 ± 0.3    | S E Asia M   | W Africa F   |
| 46  | -1.3 ± 0.5    | -2.0 ± 0.1   | -1.7 ± 0.2    | -2.2 ± 0.2  | -1.6 ± 0.2   | -3.3 ± 0.4    | -1.6 ± 0.4    | -1.4 ± 0.1  | -0.5 ± 0.2    | -1.6 ± 0.2   | -0.6 ± 0.3   | -2.9 ± 0.3    | E Europe M   | W Africa F   |
| 47  | -1.5 ± 0.5    | -1.9 ± 0.1   | -1.7 ± 0.2    | -2.3 ± 0.2  | -1.4 ± 0.2   | -3.5 ± 0.4    | -1.5 ± 0.4    | -1.4 ± 0.1  | -0.6 ± 0.2    | -1.6 ± 0.2   | -0.4 ± 0.3   | -2.8 ± 0.3    | S E Asia M   | W Africa F   |
| 48  | -1.9 ± 0.5    | -2.0 ± 0.1   | -1.7 ± 0.2    | -2.1 ± 0.2  | -1.5 ± 0.2   | -3.5 ± 0.4    | -1.8 ± 0.4    | -1.5 ± 0.1  | -0.6 ± 0.3    | -1.5 ± 0.2   | -0.4 ± 0.3   | -3.3 ± 0.3    | S E Asia M   | W Africa F   |
| 49  | -1.5 ± 0.6    | -2.1 ± 0.1   | -1.8 ± 0.2    | -2.1 ± 0.2  | -1.3 ± 0.2   | -3.8 ± 0.5    | -2.3 ± 0.4    | -1.5 ± 0.1  | -0.8 ± 0.2    | -1.8 ± 0.2   | -0.2 ± 0.3   | -3.1 ± 0.3    | S E Asia M   | W Africa F   |
| 50  | -2.2 ± 0.6    | -2.1 ± 0.1   | -1.6 ± 0.2    | -2.2 ± 0.2  | -1.4 ± 0.2   | -3.8 ± 0.4    | -2.0 ± 0.4    | -1.5 ± 0.1  | -1.0 ± 0.2    | -1.8 ± 0.2   | -0.1 ± 0.3   | -3.6 ± 0.3    | S E Asia M   | W Africa F   |
| 51  | -1.6 ± 0.5    | -2.1 ± 0.1   | -1.8 ± 0.2    | -2.1 ± 0.2  | -1.5 ± 0.2   | -3.7 ± 0.5    | -2.5 ± 0.5    | -1.6 ± 0.1  | -0.9 ± 0.2    | -1.8 ± 0.2   | -0.1 ± 0.3   | -3.8 ± 0.4    | S E Asia M   | W Africa M   |
| 52  | -2.1 ± 0.6    | -2.2 ± 0.1   | -1.6 ± 0.2    | -2.2 ± 0.2  | -1.2 ± 0.2   | -4.2 ± 0.5    | -2.5 ± 0.5    | -1.7 ± 0.1  | -1.0 ± 0.3    | -2.0 ± 0.2   | -0.5 ± 0.3   | -3.7 ± 0.4    | S E Asia M   | W Africa F   |

**Table 34.** Mean raw error for Application-type images for algorithm cvut-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 6.3 ± 0.4     | 3.1 ± 0.1    | 2.4 ± 0.2     | 2.9 ± 0.2   | 2.5 ± 0.3    | 6.3 ± 0.5     | 2.6 ± 0.4     | 1.9 ± 0.1   | 0.6 ± 0.2     | 1.1 ± 0.1   | 1.6 ± 0.3    | 3.8 ± 0.5     | E Africa F   | E Europe M   |
| 15  | 6.2 ± 0.5     | 3.2 ± 0.2    | 2.3 ± 0.2     | 3.1 ± 0.2   | 2.6 ± 0.3    | 5.3 ± 0.5     | 2.6 ± 0.4     | 2.5 ± 0.1   | 1.0 ± 0.2     | 1.5 ± 0.2   | 2.2 ± 0.3    | 4.3 ± 0.4     | E Africa F   | E Europe M   |
| 16  | 6.2 ± 0.4     | 3.1 ± 0.2    | 2.1 ± 0.2     | 2.7 ± 0.2   | 2.7 ± 0.3    | 5.4 ± 0.6     | 3.5 ± 0.3     | 2.6 ± 0.1   | 1.1 ± 0.3     | 1.8 ± 0.2   | 2.4 ± 0.3    | 4.1 ± 0.4     | E Africa F   | E Europe M   |
| 17  | 6.1 ± 0.5     | 2.8 ± 0.2    | 1.6 ± 0.2     | 2.6 ± 0.2   | 2.3 ± 0.3    | 5.2 ± 0.5     | 3.5 ± 0.4     | 2.5 ± 0.1   | 0.9 ± 0.2     | 2.3 ± 0.2   | 2.5 ± 0.3    | 4.2 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 5.3 ± 0.4     | 2.0 ± 0.1    | 0.7 ± 0.3     | 1.9 ± 0.2   | 1.7 ± 0.2    | 4.2 ± 0.5     | 3.3 ± 0.3     | 1.8 ± 0.1   | 0.6 ± 0.3     | 1.9 ± 0.2   | 1.8 ± 0.2    | 3.5 ± 0.4     | E Africa F   | E Europe M   |
| 19  | 5.6 ± 0.4     | 1.7 ± 0.1    | 0.3 ± 0.3     | 1.9 ± 0.2   | 1.2 ± 0.2    | 4.8 ± 0.4     | 3.3 ± 0.3     | 1.6 ± 0.1   | 0.5 ± 0.3     | 2.1 ± 0.2   | 1.3 ± 0.2    | 3.7 ± 0.4     | E Africa F   | E Europe F   |
| 20  | 5.3 ± 0.4     | 1.4 ± 0.1    | 0.2 ± 0.2     | 1.4 ± 0.2   | 0.8 ± 0.2    | 4.5 ± 0.4     | 3.8 ± 0.3     | 1.2 ± 0.1   | 1.0 ± 0.3     | 2.2 ± 0.2   | 1.0 ± 0.2    | 3.6 ± 0.4     | E Africa F   | E Europe F   |
| 21  | 5.0 ± 0.3     | 1.0 ± 0.1    | -0.5 ± 0.2    | 1.2 ± 0.2   | 0.3 ± 0.2    | 4.2 ± 0.4     | 3.8 ± 0.3     | 0.6 ± 0.1   | 0.8 ± 0.3     | 2.1 ± 0.2   | 0.6 ± 0.2    | 3.4 ± 0.3     | E Africa F   | E Europe F   |
| 22  | 4.7 ± 0.3     | 0.4 ± 0.1    | -0.7 ± 0.2    | 0.8 ± 0.2   | -0.2 ± 0.2   | 4.0 ± 0.4     | 4.0 ± 0.3     | 0.2 ± 0.1   | 1.2 ± 0.3     | 2.0 ± 0.2   | 0.1 ± 0.2    | 3.5 ± 0.3     | E Africa F   | E Europe F   |
| 23  | 4.3 ± 0.3     | -0.0 ± 0.1   | -1.0 ± 0.2    | 0.6 ± 0.1   | -0.7 ± 0.1   | 3.6 ± 0.4     | 3.6 ± 0.3     | -0.2 ± 0.1  | 0.9 ± 0.3     | 2.0 ± 0.2   | -0.3 ± 0.2   | 3.4 ± 0.3     | E Africa F   | E Europe F   |
| 24  | 3.9 ± 0.3     | -0.7 ± 0.1   | -1.4 ± 0.2    | 0.3 ± 0.1   | -1.2 ± 0.1   | 3.5 ± 0.3     | 3.5 ± 0.3     | -0.5 ± 0.1  | 1.2 ± 0.2     | 1.9 ± 0.2   | -0.8 ± 0.2   | 3.1 ± 0.3     | E Africa F   | E Europe F   |
| 25  | 3.2 ± 0.3     | -1.16 ± 0.10 | -1.8 ± 0.2    | -0.3 ± 0.1  | -1.9 ± 0.1   | 3.0 ± 0.3     | 3.2 ± 0.3     | -0.6 ± 0.2  | 1.1 ± 0.2     | 1.8 ± 0.1   | -0.8 ± 0.2   | 2.6 ± 0.2     | E Africa F   | S E Asia F   |
| 26  | 2.9 ± 0.2     | -1.70 ± 0.09 | -2.0 ± 0.2    | -0.7 ± 0.1  | -2.3 ± 0.1   | 2.6 ± 0.3     | 2.9 ± 0.2     | -0.8 ± 0.1  | 0.9 ± 0.2     | 1.5 ± 0.1   | -0.9 ± 0.3   | 2.1 ± 0.2     | E Africa F   | S E Asia F   |
| 27  | 2.4 ± 0.3     | -2.26 ± 0.08 | -2.2 ± 0.2    | -1.1 ± 0.1  | -2.8 ± 0.1   | 2.1 ± 0.3     | 2.5 ± 0.2     | -1.1 ± 0.2  | 1.2 ± 0.2     | 1.3 ± 0.1   | -1.0 ± 0.3   | 1.7 ± 0.2     | E Africa M   | S E Asia F   |
| 28  | 2.0 ± 0.3     | -2.73 ± 0.08 | -2.4 ± 0.2    | -1.4 ± 0.1  | -3.2 ± 0.1   | 1.9 ± 0.3     | 2.1 ± 0.2     | -1.1 ± 0.2  | 1.2 ± 0.2     | 0.9 ± 0.1   | -1.2 ± 0.3   | 1.2 ± 0.2     | E Africa M   | S E Asia F   |
| 29  | 1.5 ± 0.3     | -3.25 ± 0.08 | -2.5 ± 0.2    | -1.8 ± 0.1  | -3.6 ± 0.1   | 1.5 ± 0.3     | 1.7 ± 0.2     | -1.0 ± 0.2  | 0.7 ± 0.2     | 0.6 ± 0.1   | -1.2 ± 0.3   | 1.1 ± 0.2     | E Africa M   | S E Asia F   |
| 30  | 0.9 ± 0.3     | -3.72 ± 0.08 | -2.7 ± 0.2    | -2.3 ± 0.1  | -3.9 ± 0.2   | 1.0 ± 0.3     | 1.5 ± 0.2     | -1.3 ± 0.2  | 0.6 ± 0.2     | 0.1 ± 0.1   | -1.3 ± 0.3   | 0.4 ± 0.2     | E Africa M   | S E Asia F   |
| 31  | 0.5 ± 0.2     | -4.25 ± 0.08 | -2.6 ± 0.2    | -2.8 ± 0.1  | -4.5 ± 0.2   | 0.7 ± 0.3     | 1.2 ± 0.2     | -1.3 ± 0.2  | 0.2 ± 0.2     | -0.2 ± 0.1  | -1.5 ± 0.3   | 0.1 ± 0.2     | E Africa M   | S E Asia F   |
| 32  | 0.3 ± 0.2     | -4.71 ± 0.09 | -2.9 ± 0.2    | -3.0 ± 0.1  | -4.7 ± 0.2   | 0.3 ± 0.3     | 0.7 ± 0.2     | -1.3 ± 0.2  | 0.3 ± 0.2     | -0.6 ± 0.1  | -1.3 ± 0.3   | -0.4 ± 0.2    | E Africa M   | S E Asia F   |
| 33  | -0.4 ± 0.3    | -5.13 ± 0.09 | -3.0 ± 0.2    | -3.3 ± 0.1  | -5.2 ± 0.2   | 0.2 ± 0.3     | 0.2 ± 0.2     | -1.3 ± 0.2  | 0.3 ± 0.2     | -0.8 ± 0.1  | -1.4 ± 0.3   | -0.8 ± 0.2    | E Europe M   | S E Asia F   |
| 34  | -0.8 ± 0.3    | -5.5 ± 0.1   | -3.2 ± 0.2    | -3.5 ± 0.1  | -5.3 ± 0.2   | -0.4 ± 0.3    | -0.2 ± 0.2    | -1.6 ± 0.2  | -0.2 ± 0.2    | -1.2 ± 0.1  | -1.5 ± 0.3   | -1.4 ± 0.2    | E Africa M   | E Asia F     |
| 35  | -1.1 ± 0.3    | -6.0 ± 0.1   | -3.4 ± 0.2    | -4.0 ± 0.1  | -5.6 ± 0.2   | -0.6 ± 0.3    | -0.5 ± 0.2    | -1.6 ± 0.2  | -0.5 ± 0.2    | -1.6 ± 0.1  | -1.6 ± 0.3   | -1.6 ± 0.2    | E Europe M   | E Asia F     |
| 36  | -1.1 ± 0.3    | -6.3 ± 0.1   | -3.4 ± 0.2    | -4.2 ± 0.1  | -5.5 ± 0.2   | -1.1 ± 0.3    | -1.0 ± 0.3    | -1.7 ± 0.2  | -0.6 ± 0.2    | -1.8 ± 0.1  | -1.6 ± 0.3   | -2.1 ± 0.2    | E Europe M   | E Asia F     |
| 37  | -1.8 ± 0.4    | -6.5 ± 0.1   | -3.6 ± 0.2    | -4.3 ± 0.2  | -5.9 ± 0.2   | -1.6 ± 0.3    | -1.1 ± 0.3    | -1.9 ± 0.2  | -0.8 ± 0.2    | -2.1 ± 0.1  | -1.7 ± 0.3   | -2.5 ± 0.2    | E Europe M   | E Asia F     |
| 38  | -1.8 ± 0.3    | -6.8 ± 0.1   | -3.6 ± 0.3    | -4.4 ± 0.2  | -5.6 ± 0.3   | -1.9 ± 0.3    | -1.8 ± 0.3    | -2.0 ± 0.2  | -1.1 ± 0.3    | -2.4 ± 0.1  | -1.5 ± 0.3   | -3.0 ± 0.2    | E Europe M   | E Asia F     |
| 39  | -2.1 ± 0.4    | -7.1 ± 0.1   | -3.6 ± 0.3    | -4.7 ± 0.2  | -5.8 ± 0.3   | -2.1 ± 0.3    | -1.6 ± 0.3    | -2.3 ± 0.1  | -1.1 ± 0.2    | -2.7 ± 0.1  | -1.8 ± 0.3   | -3.5 ± 0.2    | E Europe M   | E Asia F     |
| 40  | -2.5 ± 0.4    | -7.2 ± 0.1   | -3.5 ± 0.3    | -4.7 ± 0.2  | -5.8 ± 0.3   | -2.6 ± 0.3    | -1.8 ± 0.3    | -2.6 ± 0.1  | -1.3 ± 0.2    | -2.8 ± 0.1  | -2.1 ± 0.3   | -3.8 ± 0.2    | E Europe M   | E Asia F     |
| 41  | -3.4 ± 0.4    | -7.3 ± 0.1   | -3.6 ± 0.3    | -4.8 ± 0.2  | -5.8 ± 0.3   | -3.1 ± 0.3    | -2.1 ± 0.3    | -3.0 ± 0.1  | -1.6 ± 0.2    | -3.0 ± 0.2  | -2.5 ± 0.3   | -3.9 ± 0.3    | E Europe M   | E Asia F     |
| 42  | -2.9 ± 0.4    | -7.5 ± 0.1   | -3.5 ± 0.3    | -4.4 ± 0.2  | -5.9 ± 0.3   | -3.3 ± 0.4    | -2.5 ± 0.4    | -3.4 ± 0.1  | -1.6 ± 0.3    | -3.1 ± 0.2  | -2.5 ± 0.3   | -4.3 ± 0.3    | E Europe M   | E Asia F     |
| 43  | -3.1 ± 0.5    | -7.7 ± 0.1   | -3.5 ± 0.3    | -4.8 ± 0.3  | -5.8 ± 0.3   | -3.7 ± 0.3    | -2.2 ± 0.4    | -3.7 ± 0.1  | -1.9 ± 0.3    | -3.3 ± 0.2  | -2.7 ± 0.3   | -5.1 ± 0.3    | E Europe M   | E Asia F     |
| 44  | -3.2 ± 0.5    | -7.9 ± 0.1   | -3.7 ± 0.3    | -4.6 ± 0.3  | -5.2 ± 0.3   | -4.1 ± 0.4    | -2.9 ± 0.4    | -4.1 ± 0.1  | -1.8 ± 0.3    | -3.3 ± 0.2  | -2.7 ± 0.3   | -5.0 ± 0.3    | E Europe M   | E Asia F     |
| 45  | -3.9 ± 0.5    | -7.9 ± 0.2   | -3.5 ± 0.3    | -4.5 ± 0.3  | -5.5 ± 0.3   | -4.2 ± 0.4    | -3.2 ± 0.4    | -4.6 ± 0.1  | -2.0 ± 0.3    | -3.2 ± 0.2  | -2.9 ± 0.4   | -5.3 ± 0.3    | E Europe M   | E Asia F     |
| 46  | -3.6 ± 0.6    | -8.1 ± 0.2   | -3.5 ± 0.3    | -4.4 ± 0.3  | -5.3 ± 0.3   | -4.6 ± 0.5    | -3.3 ± 0.5    | -5.0 ± 0.1  | -1.9 ± 0.3    | -3.2 ± 0.2  | -3.1 ± 0.4   | -5.5 ± 0.3    | E Europe M   | E Asia F     |
| 47  | -3.6 ± 0.7    | -8.1 ± 0.2   | -3.5 ± 0.4    | -4.5 ± 0.3  | -5.5 ± 0.3   | -4.7 ± 0.4    | -3.0 ± 0.4    | -5.4 ± 0.1  | -2.3 ± 0.4    | -3.3 ± 0.2  | -3.3 ± 0.4   | -5.5 ± 0.4    | E Europe M   | E Asia F     |
| 48  | -4.5 ± 0.6    | -8.2 ± 0.2   | -3.7 ± 0.3    | -4.3 ± 0.3  | -5.5 ± 0.3   | -5.1 ± 0.4    | -3.6 ± 0.5    | -5.8 ± 0.2  | -2.3 ± 0.4    | -3.5 ± 0.2  | -2.9 ± 0.4   | -6.1 ± 0.4    | E Europe M   | E Asia F     |
| 49  | -3.9 ± 0.7    | -8.3 ± 0.2   | -3.7 ± 0.3    | -4.2 ± 0.3  | -5.3 ± 0.3   | -5.5 ± 0.5    | -3.7 ± 0.6    | -6.2 ± 0.2  | -2.1 ± 0.4    | -3.4 ± 0.2  | -3.4 ± 0.4   | -6.1 ± 0.4    | E Europe M   | E Asia F     |
| 50  | -4.6 ± 0.7    | -8.3 ± 0.2   | -3.5 ± 0.3    | -4.6 ± 0.3  | -4.9 ± 0.4   | -5.4 ± 0.5    | -4.0 ± 0.6    | -6.3 ± 0.2  | -2.4 ± 0.4    | -3.5 ± 0.3  | -2.8 ± 0.5   | -6.2 ± 0.4    | E Europe M   | E Asia F     |
| 51  | -4.1 ± 0.6    | -8.4 ± 0.2   | -3.7 ± 0.3    | -4.3 ± 0.3  | -5.1 ± 0.4   | -6.0 ± 0.5    | -4.2 ± 0.6    | -6.3 ± 0.2  | -1.7 ± 0.4    | -3.8 ± 0.3  | -2.4 ± 0.5   | -6.7 ± 0.5    | E Europe M   | E Asia F     |
| 52  | -4.4 ± 0.7    | -8.2 ± 0.2   | -3.7 ± 0.3    | -4.7 ± 0.3  | -4.8 ± 0.4   | -6.1 ± 0.6    | -3.5 ± 0.7    | -6.6 ± 0.2  | -2.0 ± 0.3    | -4.0 ± 0.2  | -2.9 ± 0.5   | -6.8 ± 0.5    | E Europe M   | E Asia F     |

**Table 35.** Mean raw error for Application-type images for algorithm cvut-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|--------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.3 ± 0.3     | 3.4 ± 0.1    | 2.7 ± 0.2     | 2.6 ± 0.1    | 2.8 ± 0.2    | 5.1 ± 0.3     | 1.6 ± 0.2     | 2.2 ± 0.1    | 0.9 ± 0.2     | 1.2 ± 0.1   | 1.5 ± 0.2    | 2.0 ± 0.4     | W Africa F   | E Europe M   |
| 15  | 4.0 ± 0.4     | 3.3 ± 0.1    | 2.7 ± 0.2     | 2.9 ± 0.2    | 2.8 ± 0.3    | 4.7 ± 0.3     | 1.5 ± 0.3     | 2.4 ± 0.1    | 0.9 ± 0.3     | 1.5 ± 0.1   | 1.9 ± 0.3    | 2.4 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 3.8 ± 0.3     | 3.0 ± 0.1    | 2.4 ± 0.2     | 2.5 ± 0.2    | 2.7 ± 0.3    | 4.7 ± 0.4     | 2.0 ± 0.3     | 2.24 ± 0.10  | 1.2 ± 0.2     | 1.8 ± 0.1   | 2.1 ± 0.2    | 2.3 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 3.7 ± 0.3     | 2.47 ± 0.10  | 2.0 ± 0.3     | 2.3 ± 0.2    | 2.0 ± 0.3    | 4.2 ± 0.3     | 2.1 ± 0.4     | 2.0 ± 0.1    | 1.0 ± 0.2     | 2.2 ± 0.2   | 2.0 ± 0.2    | 2.7 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 2.9 ± 0.3     | 1.57 ± 0.10  | 1.4 ± 0.2     | 1.7 ± 0.1    | 1.4 ± 0.2    | 3.6 ± 0.3     | 2.3 ± 0.3     | 1.27 ± 0.10  | 0.9 ± 0.2     | 1.9 ± 0.2   | 1.3 ± 0.2    | 2.4 ± 0.4     | W Africa F   | E Europe M   |
| 19  | 2.8 ± 0.3     | 1.0 ± 0.1    | 1.0 ± 0.2     | 1.5 ± 0.2    | 1.1 ± 0.2    | 3.6 ± 0.3     | 2.4 ± 0.3     | 0.8 ± 0.1    | 0.8 ± 0.2     | 2.0 ± 0.1   | 0.8 ± 0.2    | 2.6 ± 0.4     | W Africa F   | S E Asia M   |
| 20  | 2.5 ± 0.3     | 0.5 ± 0.1    | 0.7 ± 0.2     | 1.1 ± 0.2    | 0.5 ± 0.1    | 3.4 ± 0.3     | 2.6 ± 0.3     | 0.37 ± 0.10  | 1.2 ± 0.2     | 2.0 ± 0.1   | 0.5 ± 0.2    | 2.9 ± 0.3     | W Africa F   | E Asia M     |
| 21  | 2.2 ± 0.3     | -0.0 ± 0.1   | 0.3 ± 0.2     | 0.8 ± 0.1    | -0.0 ± 0.1   | 2.9 ± 0.3     | 2.4 ± 0.3     | -0.18 ± 0.09 | 0.9 ± 0.2     | 1.9 ± 0.1   | 0.2 ± 0.1    | 2.7 ± 0.3     | W Africa F   | E Asia M     |
| 22  | 1.9 ± 0.3     | -0.81 ± 0.09 | -0.1 ± 0.2    | 0.3 ± 0.1    | -0.8 ± 0.1   | 2.6 ± 0.3     | 2.8 ± 0.3     | -0.70 ± 0.09 | 1.1 ± 0.2     | 1.6 ± 0.2   | -0.4 ± 0.1   | 2.6 ± 0.3     | E Africa M   | S E Asia F   |
| 23  | 1.3 ± 0.3     | -1.23 ± 0.10 | -0.4 ± 0.2    | 0.1 ± 0.1    | -1.3 ± 0.1   | 2.3 ± 0.3     | 2.6 ± 0.3     | -1.20 ± 0.10 | 0.7 ± 0.2     | 1.5 ± 0.2   | -0.9 ± 0.2   | 2.7 ± 0.3     | W Africa M   | S E Asia F   |
| 24  | 1.0 ± 0.3     | -1.88 ± 0.09 | -0.8 ± 0.1    | -0.4 ± 0.1   | -1.7 ± 0.1   | 1.8 ± 0.2     | 2.3 ± 0.3     | -1.7 ± 0.1   | 0.6 ± 0.2     | 1.3 ± 0.1   | -1.4 ± 0.2   | 2.5 ± 0.2     | W Africa M   | E Asia F     |
| 25  | 0.4 ± 0.3     | -2.40 ± 0.10 | -1.1 ± 0.1    | -1.0 ± 0.1   | -2.3 ± 0.1   | 1.4 ± 0.3     | 2.2 ± 0.3     | -2.1 ± 0.1   | 0.5 ± 0.2     | 1.1 ± 0.1   | -1.6 ± 0.2   | 2.1 ± 0.3     | E Africa M   | E Asia F     |
| 26  | -0.0 ± 0.3    | -2.92 ± 0.09 | -1.5 ± 0.1    | -1.50 ± 0.10 | -2.7 ± 0.1   | 0.8 ± 0.3     | 1.8 ± 0.3     | -2.4 ± 0.1   | 0.1 ± 0.2     | 0.7 ± 0.1   | -2.0 ± 0.2   | 1.9 ± 0.2     | W Africa M   | E Asia F     |
| 27  | -0.3 ± 0.3    | -3.38 ± 0.09 | -1.7 ± 0.1    | -2.0 ± 0.1   | -3.2 ± 0.1   | 0.6 ± 0.2     | 1.6 ± 0.2     | -2.8 ± 0.1   | 0.4 ± 0.2     | 0.4 ± 0.1   | -2.0 ± 0.2   | 1.4 ± 0.2     | E Africa M   | E Asia F     |
| 28  | -0.7 ± 0.3    | -3.78 ± 0.09 | -1.9 ± 0.1    | -2.41 ± 0.10 | -3.6 ± 0.1   | 0.2 ± 0.2     | 1.1 ± 0.2     | -3.1 ± 0.1   | 0.5 ± 0.2     | -0.1 ± 0.1  | -2.4 ± 0.2   | 1.0 ± 0.2     | E Africa M   | E Asia F     |
| 29  | -0.9 ± 0.3    | -4.15 ± 0.09 | -2.0 ± 0.2    | -2.9 ± 0.1   | -3.8 ± 0.1   | -0.2 ± 0.2    | 0.8 ± 0.3     | -3.3 ± 0.1   | 0.1 ± 0.2     | -0.5 ± 0.1  | -2.9 ± 0.2   | 1.0 ± 0.2     | W Africa M   | E Asia F     |
| 30  | -1.3 ± 0.3    | -4.50 ± 0.09 | -2.3 ± 0.2    | -3.3 ± 0.1   | -4.1 ± 0.2   | -0.7 ± 0.2    | 0.8 ± 0.2     | -3.8 ± 0.1   | 0.1 ± 0.2     | -1.0 ± 0.1  | -3.1 ± 0.2   | 0.5 ± 0.2     | E Africa M   | E Asia F     |
| 31  | -1.7 ± 0.3    | -4.91 ± 0.10 | -2.2 ± 0.2    | -3.8 ± 0.1   | -4.6 ± 0.2   | -1.1 ± 0.3    | 0.6 ± 0.3     | -4.0 ± 0.1   | -0.2 ± 0.2    | -1.3 ± 0.1  | -3.3 ± 0.2   | 0.1 ± 0.2     | E Africa M   | E Asia F     |
| 32  | -1.6 ± 0.3    | -5.24 ± 0.10 | -2.4 ± 0.2    | -4.1 ± 0.1   | -4.8 ± 0.2   | -1.4 ± 0.3    | 0.1 ± 0.2     | -4.2 ± 0.1   | -0.1 ± 0.2    | -1.8 ± 0.1  | -3.4 ± 0.3   | -0.3 ± 0.2    | E Africa M   | E Asia F     |
| 33  | -2.1 ± 0.3    | -5.5 ± 0.1   | -2.5 ± 0.2    | -4.4 ± 0.1   | -5.1 ± 0.2   | -1.6 ± 0.3    | -0.1 ± 0.3    | -4.4 ± 0.1   | -0.0 ± 0.2    | -2.1 ± 0.1  | -3.6 ± 0.2   | -0.5 ± 0.2    | E Europe M   | E Asia F     |
| 34  | -2.5 ± 0.3    | -5.7 ± 0.1   | -2.7 ± 0.2    | -4.6 ± 0.1   | -5.1 ± 0.2   | -2.1 ± 0.3    | -0.5 ± 0.2    | -4.7 ± 0.1   | -0.3 ± 0.2    | -2.4 ± 0.1  | -4.0 ± 0.3   | -1.0 ± 0.2    | E Europe M   | E Asia F     |
| 35  | -2.5 ± 0.3    | -6.0 ± 0.1   | -2.7 ± 0.2    | -5.0 ± 0.1   | -5.5 ± 0.2   | -2.4 ± 0.3    | -0.7 ± 0.3    | -4.8 ± 0.1   | -0.5 ± 0.2    | -2.6 ± 0.1  | -4.0 ± 0.3   | -1.5 ± 0.2    | E Europe M   | E Asia F     |
| 36  | -2.7 ± 0.4    | -6.1 ± 0.1   | -2.7 ± 0.2    | -5.3 ± 0.1   | -5.5 ± 0.2   | -2.5 ± 0.3    | -1.1 ± 0.3    | -4.9 ± 0.1   | -0.5 ± 0.2    | -2.9 ± 0.1  | -4.1 ± 0.3   | -1.9 ± 0.2    | E Europe M   | E Asia F     |
| 37  | -2.7 ± 0.4    | -6.3 ± 0.1   | -2.9 ± 0.2    | -5.4 ± 0.1   | -5.9 ± 0.2   | -3.1 ± 0.3    | -1.1 ± 0.3    | -5.1 ± 0.1   | -0.7 ± 0.2    | -3.2 ± 0.1  | -4.4 ± 0.3   | -2.3 ± 0.2    | E Europe M   | E Asia F     |
| 38  | -2.7 ± 0.4    | -6.5 ± 0.1   | -2.8 ± 0.2    | -5.5 ± 0.2   | -5.6 ± 0.2   | -3.3 ± 0.3    | -1.6 ± 0.3    | -5.1 ± 0.1   | -1.0 ± 0.2    | -3.3 ± 0.1  | -4.1 ± 0.3   | -2.6 ± 0.2    | E Europe M   | E Asia F     |
| 39  | -2.8 ± 0.4    | -6.7 ± 0.1   | -3.0 ± 0.3    | -5.7 ± 0.2   | -5.7 ± 0.2   | -3.5 ± 0.3    | -1.4 ± 0.3    | -5.3 ± 0.1   | -0.9 ± 0.2    | -3.6 ± 0.1  | -4.1 ± 0.3   | -3.0 ± 0.2    | E Europe M   | E Asia F     |
| 40  | -3.1 ± 0.4    | -6.8 ± 0.1   | -2.6 ± 0.3    | -5.6 ± 0.2   | -6.0 ± 0.2   | -4.0 ± 0.3    | -1.6 ± 0.3    | -5.5 ± 0.2   | -1.1 ± 0.3    | -3.6 ± 0.1  | -4.4 ± 0.3   | -3.4 ± 0.3    | E Europe M   | E Asia F     |
| 41  | -3.8 ± 0.4    | -6.9 ± 0.1   | -3.1 ± 0.3    | -5.8 ± 0.2   | -5.9 ± 0.2   | -4.3 ± 0.3    | -2.0 ± 0.3    | -5.5 ± 0.1   | -1.2 ± 0.3    | -3.8 ± 0.2  | -4.3 ± 0.3   | -3.8 ± 0.3    | E Europe M   | E Asia F     |
| 42  | -3.3 ± 0.4    | -7.1 ± 0.1   | -3.0 ± 0.3    | -5.4 ± 0.2   | -6.2 ± 0.2   | -4.3 ± 0.3    | -2.3 ± 0.4    | -5.6 ± 0.2   | -1.0 ± 0.3    | -3.9 ± 0.2  | -4.2 ± 0.3   | -3.9 ± 0.2    | E Europe M   | E Asia F     |
| 43  | -3.2 ± 0.4    | -7.3 ± 0.1   | -2.8 ± 0.3    | -5.7 ± 0.2   | -6.4 ± 0.3   | -4.8 ± 0.4    | -2.0 ± 0.4    | -5.7 ± 0.2   | -1.2 ± 0.3    | -3.9 ± 0.2  | -4.3 ± 0.3   | -4.7 ± 0.3    | E Europe M   | E Asia F     |
| 44  | -3.3 ± 0.5    | -7.5 ± 0.1   | -3.0 ± 0.3    | -5.6 ± 0.3   | -5.8 ± 0.2   | -5.3 ± 0.4    | -2.5 ± 0.3    | -5.8 ± 0.2   | -1.1 ± 0.3    | -4.1 ± 0.2  | -4.2 ± 0.3   | -4.8 ± 0.3    | E Europe M   | E Asia F     |
| 45  | -3.8 ± 0.5    | -7.6 ± 0.1   | -3.0 ± 0.3    | -5.3 ± 0.2   | -6.1 ± 0.3   | -5.4 ± 0.4    | -3.0 ± 0.5    | -5.8 ± 0.1   | -1.1 ± 0.3    | -3.8 ± 0.2  | -4.2 ± 0.3   | -5.1 ± 0.3    | E Europe M   | E Asia F     |
| 46  | -3.1 ± 0.5    | -7.7 ± 0.1   | -3.1 ± 0.3    | -5.1 ± 0.3   | -6.3 ± 0.3   | -5.6 ± 0.4    | -2.9 ± 0.5    | -5.9 ± 0.2   | -1.3 ± 0.4    | -3.9 ± 0.2  | -4.1 ± 0.4   | -5.4 ± 0.3    | E Europe M   | E Asia F     |
| 47  | -3.5 ± 0.6    | -7.7 ± 0.1   | -3.0 ± 0.3    | -5.1 ± 0.3   | -6.2 ± 0.3   | -5.8 ± 0.4    | -2.7 ± 0.5    | -6.0 ± 0.2   | -1.2 ± 0.3    | -3.9 ± 0.2  | -3.8 ± 0.3   | -5.3 ± 0.3    | E Europe M   | E Asia F     |
| 48  | -4.0 ± 0.6    | -7.8 ± 0.1   | -3.0 ± 0.3    | -4.9 ± 0.3   | -6.2 ± 0.3   | -5.9 ± 0.4    | -3.5 ± 0.6    | -5.8 ± 0.2   | -1.1 ± 0.4    | -3.9 ± 0.2  | -3.8 ± 0.4   | -6.0 ± 0.4    | E Europe M   | E Asia F     |
| 49  | -3.4 ± 0.6    | -7.9 ± 0.2   | -3.3 ± 0.4    | -4.9 ± 0.3   | -6.3 ± 0.3   | -6.3 ± 0.4    | -3.4 ± 0.5    | -5.8 ± 0.2   | -1.1 ± 0.3    | -3.9 ± 0.3  | -3.7 ± 0.4   | -6.1 ± 0.3    | E Europe M   | E Asia F     |
| 50  | -4.4 ± 0.6    | -7.9 ± 0.2   | -2.8 ± 0.4    | -4.8 ± 0.3   | -6.2 ± 0.3   | -6.3 ± 0.5    | -3.6 ± 0.5    | -5.8 ± 0.2   | -1.3 ± 0.4    | -3.9 ± 0.3  | -3.2 ± 0.4   | -6.3 ± 0.4    | E Europe M   | E Asia F     |
| 51  | -3.9 ± 0.6    | -7.9 ± 0.2   | -2.7 ± 0.3    | -4.6 ± 0.3   | -6.3 ± 0.3   | -6.7 ± 0.4    | -3.9 ± 0.6    | -5.7 ± 0.2   | -0.8 ± 0.3    | -4.1 ± 0.3  | -2.9 ± 0.4   | -6.4 ± 0.4    | E Europe M   | E Asia F     |
| 52  | -4.2 ± 0.8    | -8.0 ± 0.2   | -2.7 ± 0.3    | -4.7 ± 0.3   | -5.7 ± 0.3   | -7.0 ± 0.5    | -3.9 ± 0.7    | -5.7 ± 0.2   | -0.9 ± 0.3    | -4.0 ± 0.3  | -3.6 ± 0.4   | -7.0 ± 0.5    | E Europe M   | E Asia F     |



**Table 36.** Mean raw error for Application-type images for algorithm cvut-003. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 4.8 ± 0.2     | 3.73 ± 0.08  | 3.1 ± 0.1     | 3.4 ± 0.1    | 3.5 ± 0.2    | 5.5 ± 0.3     | 2.6 ± 0.2     | 2.46 ± 0.08  | 1.4 ± 0.1     | 1.89 ± 0.09  | 2.0 ± 0.2    | 2.8 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 4.3 ± 0.3     | 3.56 ± 0.09  | 3.0 ± 0.2     | 3.3 ± 0.2    | 3.3 ± 0.2    | 5.0 ± 0.3     | 2.2 ± 0.2     | 2.50 ± 0.09  | 1.3 ± 0.2     | 1.9 ± 0.1    | 2.2 ± 0.2    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.1 ± 0.3     | 3.21 ± 0.10  | 2.7 ± 0.2     | 2.9 ± 0.1    | 3.2 ± 0.2    | 5.0 ± 0.4     | 2.6 ± 0.3     | 2.33 ± 0.08  | 1.5 ± 0.2     | 2.2 ± 0.1    | 2.3 ± 0.2    | 2.5 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 4.0 ± 0.3     | 2.75 ± 0.09  | 2.3 ± 0.2     | 2.7 ± 0.1    | 2.7 ± 0.2    | 4.7 ± 0.3     | 2.5 ± 0.2     | 2.13 ± 0.09  | 1.3 ± 0.2     | 2.5 ± 0.1    | 2.2 ± 0.2    | 2.9 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 3.0 ± 0.3     | 1.97 ± 0.08  | 1.7 ± 0.2     | 2.1 ± 0.1    | 1.9 ± 0.1    | 4.2 ± 0.3     | 2.5 ± 0.2     | 1.56 ± 0.07  | 1.3 ± 0.2     | 2.2 ± 0.1    | 1.7 ± 0.2    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 3.0 ± 0.3     | 1.45 ± 0.09  | 1.3 ± 0.2     | 1.9 ± 0.1    | 1.6 ± 0.1    | 4.3 ± 0.3     | 2.5 ± 0.3     | 1.20 ± 0.07  | 1.0 ± 0.2     | 2.3 ± 0.1    | 1.2 ± 0.1    | 2.8 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 2.6 ± 0.3     | 1.0 ± 0.1    | 1.0 ± 0.2     | 1.4 ± 0.1    | 1.1 ± 0.1    | 4.1 ± 0.3     | 2.8 ± 0.3     | 0.75 ± 0.08  | 1.2 ± 0.2     | 2.2 ± 0.1    | 0.9 ± 0.1    | 2.8 ± 0.3     | W Africa F   | E Asia M     |
| 21  | 2.3 ± 0.3     | 0.53 ± 0.10  | 0.5 ± 0.2     | 1.1 ± 0.1    | 0.6 ± 0.1    | 3.6 ± 0.3     | 2.7 ± 0.3     | 0.24 ± 0.07  | 1.0 ± 0.2     | 1.9 ± 0.1    | 0.5 ± 0.1    | 2.7 ± 0.3     | W Africa F   | E Asia M     |
| 22  | 2.3 ± 0.3     | -0.18 ± 0.09 | 0.2 ± 0.2     | 0.7 ± 0.1    | -0.1 ± 0.1   | 3.2 ± 0.3     | 3.0 ± 0.3     | -0.23 ± 0.08 | 0.9 ± 0.2     | 1.6 ± 0.1    | -0.1 ± 0.1   | 2.6 ± 0.3     | W Africa F   | E Asia M     |
| 23  | 1.7 ± 0.3     | -0.50 ± 0.09 | -0.2 ± 0.2    | 0.4 ± 0.1    | -0.4 ± 0.1   | 3.1 ± 0.3     | 2.7 ± 0.3     | -0.64 ± 0.09 | 0.4 ± 0.2     | 1.4 ± 0.1    | -0.5 ± 0.1   | 2.7 ± 0.3     | W Africa F   | E Asia M     |
| 24  | 1.4 ± 0.3     | -1.11 ± 0.09 | -0.6 ± 0.1    | -0.1 ± 0.1   | -0.8 ± 0.1   | 2.5 ± 0.3     | 2.6 ± 0.3     | -1.03 ± 0.10 | 0.4 ± 0.2     | 1.2 ± 0.1    | -0.8 ± 0.1   | 2.5 ± 0.2     | W Africa F   | E Asia F     |
| 25  | 0.9 ± 0.3     | -1.54 ± 0.08 | -0.8 ± 0.1    | -0.58 ± 0.10 | -1.3 ± 0.1   | 2.2 ± 0.3     | 2.5 ± 0.3     | -1.3 ± 0.1   | 0.2 ± 0.2     | 0.9 ± 0.1    | -1.1 ± 0.2   | 2.2 ± 0.2     | E Africa M   | E Asia F     |
| 26  | 0.5 ± 0.3     | -2.01 ± 0.09 | -1.1 ± 0.1    | -1.15 ± 0.09 | -1.6 ± 0.1   | 1.8 ± 0.2     | 2.2 ± 0.2     | -1.6 ± 0.1   | -0.2 ± 0.2    | 0.6 ± 0.1    | -1.3 ± 0.2   | 2.0 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 0.4 ± 0.3     | -2.43 ± 0.08 | -1.3 ± 0.1    | -1.57 ± 0.09 | -2.0 ± 0.1   | 1.5 ± 0.2     | 1.9 ± 0.2     | -2.0 ± 0.1   | -0.1 ± 0.2    | 0.3 ± 0.1    | -1.3 ± 0.2   | 1.7 ± 0.2     | E Africa M   | E Asia F     |
| 28  | 0.1 ± 0.3     | -2.77 ± 0.09 | -1.4 ± 0.1    | -1.96 ± 0.09 | -2.3 ± 0.1   | 1.1 ± 0.2     | 1.5 ± 0.2     | -2.1 ± 0.1   | -0.0 ± 0.2    | -0.1 ± 0.1   | -1.6 ± 0.2   | 1.3 ± 0.2     | E Africa M   | E Asia F     |
| 29  | -0.1 ± 0.3    | -3.03 ± 0.09 | -1.5 ± 0.1    | -2.34 ± 0.10 | -2.6 ± 0.1   | 0.9 ± 0.2     | 1.3 ± 0.2     | -2.2 ± 0.1   | -0.3 ± 0.2    | -0.51 ± 0.10 | -2.0 ± 0.2   | 1.3 ± 0.2     | E Africa M   | E Asia F     |
| 30  | -0.4 ± 0.3    | -3.39 ± 0.09 | -1.7 ± 0.2    | -2.7 ± 0.1   | -2.7 ± 0.2   | 0.4 ± 0.2     | 1.2 ± 0.2     | -2.5 ± 0.1   | -0.3 ± 0.2    | -0.90 ± 0.09 | -2.1 ± 0.2   | 0.9 ± 0.2     | E Africa M   | E Asia F     |
| 31  | -0.7 ± 0.3    | -3.66 ± 0.09 | -1.5 ± 0.2    | -3.2 ± 0.1   | -3.1 ± 0.2   | 0.1 ± 0.2     | 1.0 ± 0.2     | -2.6 ± 0.1   | -0.5 ± 0.2    | -1.22 ± 0.10 | -2.3 ± 0.2   | 0.6 ± 0.2     | E Africa M   | E Asia F     |
| 32  | -0.6 ± 0.3    | -3.94 ± 0.10 | -1.7 ± 0.2    | -3.4 ± 0.1   | -3.2 ± 0.2   | -0.3 ± 0.2    | 0.6 ± 0.2     | -2.8 ± 0.1   | -0.3 ± 0.2    | -1.57 ± 0.10 | -2.1 ± 0.2   | 0.2 ± 0.2     | E Africa M   | E Asia F     |
| 33  | -1.2 ± 0.3    | -4.0 ± 0.1   | -1.8 ± 0.2    | -3.7 ± 0.1   | -3.5 ± 0.2   | -0.4 ± 0.2    | 0.4 ± 0.2     | -2.8 ± 0.1   | -0.3 ± 0.2    | -1.8 ± 0.1   | -2.4 ± 0.2   | -0.2 ± 0.2    | E Africa M   | E Asia F     |
| 34  | -1.4 ± 0.3    | -4.2 ± 0.1   | -1.9 ± 0.2    | -3.9 ± 0.1   | -3.5 ± 0.2   | -1.0 ± 0.2    | -0.0 ± 0.2    | -3.0 ± 0.1   | -0.6 ± 0.2    | -2.1 ± 0.1   | -2.5 ± 0.3   | -0.5 ± 0.2    | E Africa M   | E Asia F     |
| 35  | -1.5 ± 0.3    | -4.5 ± 0.1   | -1.9 ± 0.2    | -4.2 ± 0.1   | -3.7 ± 0.2   | -1.3 ± 0.2    | -0.2 ± 0.2    | -3.1 ± 0.1   | -0.7 ± 0.2    | -2.4 ± 0.1   | -2.7 ± 0.3   | -1.0 ± 0.2    | E Africa M   | E Asia F     |
| 36  | -1.7 ± 0.3    | -4.4 ± 0.1   | -1.8 ± 0.2    | -4.4 ± 0.1   | -3.6 ± 0.2   | -1.5 ± 0.2    | -0.6 ± 0.2    | -3.2 ± 0.1   | -0.6 ± 0.2    | -2.6 ± 0.1   | -2.6 ± 0.2   | -1.4 ± 0.2    | E Africa M   | S Asia F     |
| 37  | -1.8 ± 0.3    | -4.6 ± 0.1   | -2.0 ± 0.2    | -4.5 ± 0.1   | -4.1 ± 0.2   | -2.0 ± 0.3    | -0.6 ± 0.2    | -3.4 ± 0.1   | -0.9 ± 0.2    | -2.9 ± 0.1   | -2.8 ± 0.2   | -1.8 ± 0.2    | E Africa M   | E Asia F     |
| 38  | -1.8 ± 0.3    | -4.7 ± 0.1   | -1.9 ± 0.2    | -4.6 ± 0.2   | -3.8 ± 0.2   | -2.3 ± 0.3    | -1.1 ± 0.3    | -3.5 ± 0.1   | -1.2 ± 0.2    | -3.1 ± 0.1   | -2.7 ± 0.2   | -2.2 ± 0.2    | E Africa M   | E Asia F     |
| 39  | -2.0 ± 0.3    | -4.9 ± 0.1   | -2.2 ± 0.2    | -4.8 ± 0.2   | -4.0 ± 0.2   | -2.6 ± 0.3    | -1.1 ± 0.3    | -3.6 ± 0.1   | -1.1 ± 0.2    | -3.3 ± 0.1   | -2.8 ± 0.2   | -2.6 ± 0.2    | E Africa M   | E Asia F     |
| 40  | -2.3 ± 0.4    | -5.0 ± 0.1   | -1.9 ± 0.2    | -4.7 ± 0.2   | -4.2 ± 0.2   | -3.1 ± 0.3    | -1.2 ± 0.3    | -3.9 ± 0.1   | -1.1 ± 0.2    | -3.3 ± 0.1   | -3.0 ± 0.2   | -3.0 ± 0.2    | E Europe M   | E Asia F     |
| 41  | -3.1 ± 0.4    | -5.1 ± 0.1   | -2.2 ± 0.2    | -4.8 ± 0.2   | -4.3 ± 0.2   | -3.5 ± 0.3    | -1.6 ± 0.3    | -4.0 ± 0.1   | -1.3 ± 0.2    | -3.5 ± 0.1   | -3.2 ± 0.2   | -3.3 ± 0.2    | E Europe M   | E Asia F     |
| 42  | -2.6 ± 0.4    | -5.2 ± 0.1   | -2.2 ± 0.2    | -4.5 ± 0.2   | -4.5 ± 0.2   | -3.6 ± 0.3    | -1.8 ± 0.3    | -4.2 ± 0.1   | -1.0 ± 0.2    | -3.6 ± 0.2   | -3.1 ± 0.2   | -3.6 ± 0.2    | E Europe M   | E Asia F     |
| 43  | -2.6 ± 0.4    | -5.5 ± 0.1   | -2.2 ± 0.2    | -4.8 ± 0.2   | -4.7 ± 0.2   | -4.0 ± 0.3    | -1.5 ± 0.3    | -4.2 ± 0.1   | -1.3 ± 0.3    | -3.6 ± 0.2   | -3.2 ± 0.3   | -4.2 ± 0.2    | E Europe M   | E Asia F     |
| 44  | -2.8 ± 0.4    | -5.7 ± 0.1   | -2.5 ± 0.3    | -4.7 ± 0.2   | -4.3 ± 0.2   | -4.3 ± 0.3    | -2.2 ± 0.3    | -4.4 ± 0.1   | -1.0 ± 0.3    | -3.7 ± 0.2   | -3.1 ± 0.3   | -4.4 ± 0.2    | E Europe M   | E Asia F     |
| 45  | -3.2 ± 0.4    | -5.8 ± 0.1   | -2.5 ± 0.3    | -4.6 ± 0.2   | -4.7 ± 0.2   | -4.5 ± 0.3    | -2.4 ± 0.3    | -4.6 ± 0.1   | -1.1 ± 0.3    | -3.5 ± 0.2   | -3.2 ± 0.3   | -4.7 ± 0.3    | E Europe M   | E Asia F     |
| 46  | -2.6 ± 0.5    | -6.0 ± 0.1   | -2.6 ± 0.3    | -4.3 ± 0.2   | -4.9 ± 0.2   | -4.9 ± 0.3    | -2.4 ± 0.4    | -4.6 ± 0.1   | -1.0 ± 0.3    | -3.3 ± 0.2   | -3.2 ± 0.3   | -5.1 ± 0.3    | E Europe M   | E Asia F     |
| 47  | -2.8 ± 0.5    | -6.1 ± 0.1   | -2.5 ± 0.3    | -4.3 ± 0.2   | -4.8 ± 0.2   | -5.0 ± 0.3    | -2.2 ± 0.4    | -4.7 ± 0.1   | -1.1 ± 0.3    | -3.4 ± 0.2   | -3.0 ± 0.3   | -5.0 ± 0.3    | E Europe M   | E Asia F     |
| 48  | -3.4 ± 0.6    | -6.2 ± 0.1   | -2.5 ± 0.3    | -4.1 ± 0.3   | -5.0 ± 0.2   | -5.3 ± 0.4    | -2.9 ± 0.4    | -4.7 ± 0.1   | -0.9 ± 0.3    | -3.3 ± 0.2   | -3.0 ± 0.3   | -5.8 ± 0.3    | E Europe M   | E Asia F     |
| 49  | -3.0 ± 0.6    | -6.3 ± 0.1   | -2.7 ± 0.3    | -4.0 ± 0.3   | -4.9 ± 0.2   | -5.7 ± 0.4    | -3.1 ± 0.5    | -4.8 ± 0.2   | -0.9 ± 0.3    | -3.4 ± 0.2   | -2.9 ± 0.3   | -6.0 ± 0.3    | E Europe M   | E Asia F     |
| 50  | -3.9 ± 0.6    | -6.4 ± 0.1   | -2.4 ± 0.3    | -4.0 ± 0.2   | -5.0 ± 0.3   | -5.6 ± 0.4    | -3.3 ± 0.5    | -4.7 ± 0.2   | -1.1 ± 0.3    | -3.4 ± 0.2   | -2.5 ± 0.3   | -6.1 ± 0.3    | E Europe M   | E Asia F     |
| 51  | -3.3 ± 0.6    | -6.4 ± 0.1   | -2.5 ± 0.3    | -3.9 ± 0.3   | -5.1 ± 0.2   | -6.1 ± 0.4    | -3.5 ± 0.5    | -4.7 ± 0.2   | -0.7 ± 0.3    | -3.4 ± 0.3   | -2.4 ± 0.3   | -6.4 ± 0.3    | E Europe M   | W Africa M   |
| 52  | -3.4 ± 0.7    | -6.4 ± 0.1   | -2.3 ± 0.3    | -4.0 ± 0.3   | -4.7 ± 0.2   | -6.5 ± 0.4    | -3.3 ± 0.5    | -4.7 ± 0.2   | -0.8 ± 0.3    | -3.4 ± 0.3   | -2.8 ± 0.4   | -6.6 ± 0.4    | E Europe M   | W Africa M   |

**Table 37.** Mean raw error for Application-type images for algorithm cyberlink-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 5.8 ± 0.3     | 5.3 ± 0.1    | 4.7 ± 0.2     | 4.9 ± 0.2    | 5.5 ± 0.3    | 6.6 ± 0.3     | 5.0 ± 0.3     | 4.0 ± 0.1   | 2.8 ± 0.2     | 3.4 ± 0.1    | 4.0 ± 0.3    | 5.1 ± 0.4     | W Africa F   | E Europe M   |
| 15  | 5.5 ± 0.3     | 5.2 ± 0.1    | 4.7 ± 0.2     | 4.9 ± 0.2    | 5.6 ± 0.3    | 6.3 ± 0.4     | 4.6 ± 0.3     | 4.3 ± 0.1   | 3.2 ± 0.2     | 3.8 ± 0.2    | 4.4 ± 0.3    | 4.8 ± 0.4     | W Africa F   | E Europe M   |
| 16  | 5.5 ± 0.3     | 5.2 ± 0.1    | 4.9 ± 0.2     | 4.7 ± 0.2    | 5.3 ± 0.3    | 6.3 ± 0.5     | 5.0 ± 0.3     | 4.1 ± 0.1   | 3.5 ± 0.2     | 3.9 ± 0.2    | 4.4 ± 0.3    | 5.0 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 5.6 ± 0.4     | 4.8 ± 0.1    | 4.3 ± 0.2     | 4.4 ± 0.2    | 5.1 ± 0.2    | 5.8 ± 0.4     | 4.9 ± 0.3     | 4.2 ± 0.1   | 3.2 ± 0.2     | 4.3 ± 0.2    | 4.4 ± 0.3    | 5.2 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 4.7 ± 0.3     | 4.2 ± 0.1    | 3.8 ± 0.2     | 3.9 ± 0.1    | 4.6 ± 0.2    | 5.1 ± 0.3     | 4.7 ± 0.3     | 3.7 ± 0.1   | 3.0 ± 0.2     | 3.9 ± 0.1    | 4.2 ± 0.2    | 4.6 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.6 ± 0.3     | 3.8 ± 0.1    | 3.3 ± 0.2     | 3.5 ± 0.1    | 4.2 ± 0.2    | 5.2 ± 0.3     | 4.7 ± 0.3     | 3.8 ± 0.1   | 2.9 ± 0.2     | 4.1 ± 0.1    | 4.0 ± 0.2    | 4.7 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.3 ± 0.3     | 3.4 ± 0.1    | 2.8 ± 0.2     | 3.0 ± 0.2    | 3.8 ± 0.1    | 4.7 ± 0.3     | 4.8 ± 0.3     | 3.6 ± 0.1   | 2.8 ± 0.2     | 4.2 ± 0.1    | 4.1 ± 0.2    | 4.5 ± 0.3     | E Africa M   | E Europe M   |
| 21  | 4.0 ± 0.3     | 2.91 ± 0.10  | 2.3 ± 0.2     | 2.3 ± 0.1    | 3.1 ± 0.1    | 4.3 ± 0.3     | 4.6 ± 0.2     | 3.5 ± 0.1   | 2.4 ± 0.2     | 4.0 ± 0.1    | 3.9 ± 0.2    | 4.2 ± 0.3     | E Africa M   | S Asia F     |
| 22  | 3.8 ± 0.3     | 2.42 ± 0.09  | 2.0 ± 0.2     | 1.7 ± 0.1    | 2.7 ± 0.1    | 4.0 ± 0.3     | 4.8 ± 0.3     | 3.3 ± 0.1   | 2.3 ± 0.2     | 3.7 ± 0.1    | 3.5 ± 0.2    | 3.9 ± 0.3     | E Africa M   | S Asia F     |
| 23  | 3.0 ± 0.3     | 2.00 ± 0.09  | 1.6 ± 0.1     | 1.2 ± 0.1    | 2.2 ± 0.1    | 3.6 ± 0.3     | 4.3 ± 0.3     | 3.1 ± 0.1   | 2.0 ± 0.2     | 3.6 ± 0.1    | 3.2 ± 0.2    | 3.9 ± 0.2     | E Africa M   | S Asia F     |
| 24  | 2.7 ± 0.3     | 1.35 ± 0.08  | 1.2 ± 0.1     | 0.6 ± 0.1    | 1.8 ± 0.1    | 2.9 ± 0.3     | 4.2 ± 0.2     | 2.9 ± 0.1   | 1.8 ± 0.2     | 3.3 ± 0.1    | 3.0 ± 0.2    | 3.5 ± 0.2     | E Africa M   | S Asia F     |
| 25  | 2.1 ± 0.3     | 0.89 ± 0.08  | 0.7 ± 0.1     | 0.05 ± 0.10  | 1.2 ± 0.1    | 2.8 ± 0.3     | 4.0 ± 0.2     | 2.5 ± 0.1   | 1.6 ± 0.1     | 2.9 ± 0.1    | 2.7 ± 0.2    | 3.2 ± 0.2     | E Africa M   | S Asia F     |
| 26  | 1.6 ± 0.3     | 0.36 ± 0.08  | 0.3 ± 0.1     | -0.60 ± 0.10 | 0.8 ± 0.1    | 2.3 ± 0.3     | 3.6 ± 0.2     | 2.2 ± 0.1   | 1.2 ± 0.1     | 2.5 ± 0.1    | 2.4 ± 0.2    | 2.9 ± 0.2     | E Africa M   | S Asia F     |
| 27  | 1.5 ± 0.3     | -0.17 ± 0.08 | 0.0 ± 0.1     | -1.08 ± 0.10 | 0.3 ± 0.1    | 2.1 ± 0.3     | 3.3 ± 0.2     | 1.9 ± 0.1   | 1.0 ± 0.1     | 2.1 ± 0.1    | 2.3 ± 0.2    | 2.5 ± 0.2     | E Africa M   | S Asia F     |
| 28  | 1.0 ± 0.3     | -0.62 ± 0.08 | -0.3 ± 0.1    | -1.4 ± 0.1   | -0.1 ± 0.1   | 1.6 ± 0.3     | 2.8 ± 0.2     | 1.6 ± 0.1   | 0.8 ± 0.1     | 1.61 ± 0.09  | 1.9 ± 0.2    | 2.0 ± 0.2     | E Africa M   | S Asia F     |
| 29  | 0.8 ± 0.3     | -1.03 ± 0.08 | -0.6 ± 0.1    | -1.8 ± 0.1   | -0.5 ± 0.1   | 1.1 ± 0.3     | 2.6 ± 0.2     | 1.24 ± 0.10 | 0.4 ± 0.1     | 1.17 ± 0.09  | 1.4 ± 0.2    | 2.0 ± 0.2     | E Africa M   | S Asia F     |
| 30  | 0.3 ± 0.3     | -1.45 ± 0.08 | -1.0 ± 0.1    | -2.2 ± 0.1   | -0.7 ± 0.1   | 0.5 ± 0.3     | 2.5 ± 0.2     | 0.88 ± 0.10 | 0.1 ± 0.2     | 0.73 ± 0.09  | 1.2 ± 0.2    | 1.5 ± 0.2     | E Africa M   | S Asia F     |
| 31  | -0.0 ± 0.3    | -1.77 ± 0.08 | -1.2 ± 0.2    | -2.6 ± 0.1   | -1.1 ± 0.1   | 0.3 ± 0.3     | 2.4 ± 0.2     | 0.6 ± 0.1   | -0.0 ± 0.2    | 0.43 ± 0.09  | 0.8 ± 0.2    | 1.2 ± 0.2     | E Africa M   | S Asia F     |
| 32  | -0.2 ± 0.3    | -2.22 ± 0.09 | -1.4 ± 0.2    | -2.9 ± 0.1   | -1.3 ± 0.2   | -0.0 ± 0.3    | 1.9 ± 0.2     | 0.1 ± 0.1   | -0.1 ± 0.2    | 0.11 ± 0.09  | 0.7 ± 0.2    | 0.8 ± 0.2     | E Africa M   | S Asia F     |
| 33  | -0.3 ± 0.3    | -2.47 ± 0.09 | -1.9 ± 0.2    | -3.2 ± 0.1   | -1.7 ± 0.2   | -0.1 ± 0.3    | 1.8 ± 0.3     | -0.2 ± 0.1  | 0.0 ± 0.2     | -0.21 ± 0.10 | 0.5 ± 0.2    | 0.5 ± 0.2     | E Africa M   | S Asia F     |
| 34  | -0.8 ± 0.3    | -2.68 ± 0.10 | -1.9 ± 0.2    | -3.4 ± 0.1   | -1.9 ± 0.2   | -0.6 ± 0.3    | 1.3 ± 0.3     | -0.6 ± 0.1  | -0.3 ± 0.2    | -0.5 ± 0.1   | 0.2 ± 0.2    | 0.2 ± 0.2     | E Africa M   | S Asia F     |
| 35  | -0.8 ± 0.4    | -3.1 ± 0.1   | -1.9 ± 0.2    | -3.7 ± 0.1   | -2.1 ± 0.2   | -0.9 ± 0.3    | 1.2 ± 0.3     | -0.8 ± 0.1  | -0.3 ± 0.2    | -0.7 ± 0.1   | -0.1 ± 0.2   | -0.1 ± 0.2    | E Africa M   | S Asia F     |
| 36  | -0.6 ± 0.4    | -3.1 ± 0.1   | -2.0 ± 0.2    | -3.9 ± 0.1   | -2.0 ± 0.2   | -1.0 ± 0.4    | 0.9 ± 0.3     | -1.1 ± 0.1  | -0.2 ± 0.2    | -1.1 ± 0.1   | -0.2 ± 0.2   | -0.4 ± 0.2    | E Africa M   | S Asia F     |
| 37  | -0.8 ± 0.4    | -3.2 ± 0.1   | -2.1 ± 0.3    | -3.8 ± 0.2   | -2.3 ± 0.2   | -1.4 ± 0.4    | 0.9 ± 0.3     | -1.3 ± 0.1  | -0.5 ± 0.2    | -1.2 ± 0.1   | -0.5 ± 0.3   | -0.7 ± 0.2    | E Africa M   | S Asia F     |
| 38  | -0.5 ± 0.5    | -3.5 ± 0.1   | -2.0 ± 0.3    | -4.0 ± 0.2   | -2.0 ± 0.2   | -1.4 ± 0.4    | 0.7 ± 0.3     | -1.4 ± 0.1  | -0.5 ± 0.2    | -1.6 ± 0.1   | -0.1 ± 0.3   | -0.9 ± 0.3    | E Africa M   | S Asia F     |
| 39  | -0.6 ± 0.5    | -3.5 ± 0.1   | -1.9 ± 0.3    | -4.0 ± 0.2   | -1.9 ± 0.2   | -1.5 ± 0.4    | 1.0 ± 0.3     | -1.6 ± 0.1  | -0.6 ± 0.2    | -1.7 ± 0.1   | -0.1 ± 0.3   | -1.0 ± 0.3    | E Africa M   | S Asia F     |
| 40  | -0.8 ± 0.5    | -3.6 ± 0.1   | -1.3 ± 0.3    | -4.0 ± 0.2   | -2.0 ± 0.3   | -2.1 ± 0.4    | 0.7 ± 0.4     | -1.7 ± 0.1  | -0.4 ± 0.2    | -1.9 ± 0.1   | -0.4 ± 0.3   | -1.4 ± 0.3    | E Africa M   | S Asia F     |
| 41  | -1.1 ± 0.6    | -3.5 ± 0.1   | -1.5 ± 0.3    | -4.0 ± 0.2   | -1.9 ± 0.3   | -2.2 ± 0.5    | 0.5 ± 0.4     | -1.6 ± 0.1  | -0.6 ± 0.3    | -1.9 ± 0.1   | -0.4 ± 0.3   | -1.3 ± 0.3    | E Africa M   | S Asia F     |
| 42  | -0.5 ± 0.6    | -3.3 ± 0.1   | -1.2 ± 0.3    | -3.5 ± 0.2   | -1.8 ± 0.3   | -1.6 ± 0.5    | 0.5 ± 0.4     | -1.7 ± 0.2  | -0.4 ± 0.3    | -2.1 ± 0.2   | -0.1 ± 0.3   | -1.6 ± 0.3    | E Africa M   | S Asia F     |
| 43  | -0.6 ± 0.6    | -3.3 ± 0.2   | -1.2 ± 0.3    | -3.6 ± 0.3   | -1.8 ± 0.3   | -1.9 ± 0.5    | 0.9 ± 0.4     | -1.7 ± 0.2  | -0.6 ± 0.3    | -2.1 ± 0.2   | -0.3 ± 0.3   | -2.0 ± 0.3    | E Africa M   | S Asia F     |
| 44  | -0.1 ± 0.7    | -3.2 ± 0.2   | -1.1 ± 0.3    | -3.3 ± 0.3   | -1.2 ± 0.3   | -2.2 ± 0.5    | 0.2 ± 0.4     | -1.6 ± 0.2  | -0.3 ± 0.3    | -2.2 ± 0.2   | -0.1 ± 0.3   | -2.0 ± 0.3    | E Africa M   | S Asia F     |
| 45  | -0.9 ± 0.6    | -3.0 ± 0.2   | -0.8 ± 0.3    | -3.3 ± 0.3   | -1.3 ± 0.3   | -2.0 ± 0.5    | 0.1 ± 0.4     | -1.8 ± 0.2  | -0.3 ± 0.3    | -2.2 ± 0.2   | -0.1 ± 0.3   | -2.4 ± 0.4    | E Africa M   | S Asia F     |
| 46  | 0.1 ± 0.7     | -3.0 ± 0.2   | -0.7 ± 0.3    | -2.7 ± 0.3   | -1.4 ± 0.3   | -2.0 ± 0.5    | 0.0 ± 0.5     | -1.8 ± 0.2  | -0.0 ± 0.3    | -1.9 ± 0.2   | -0.1 ± 0.3   | -2.3 ± 0.4    | E Africa F   | E Asia F     |
| 47  | 0.2 ± 0.7     | -2.6 ± 0.2   | -0.5 ± 0.4    | -2.7 ± 0.3   | -1.1 ± 0.3   | -1.7 ± 0.6    | 0.6 ± 0.5     | -1.9 ± 0.2  | -0.1 ± 0.3    | -1.9 ± 0.2   | 0.2 ± 0.3    | -2.2 ± 0.4    | E Africa M   | S Asia F     |
| 48  | 0.3 ± 0.7     | -2.7 ± 0.2   | -0.6 ± 0.3    | -2.3 ± 0.3   | -1.0 ± 0.3   | -1.8 ± 0.6    | -0.1 ± 0.5    | -1.8 ± 0.2  | 0.1 ± 0.4     | -1.9 ± 0.2   | -0.2 ± 0.3   | -2.6 ± 0.4    | E Africa F   | E Asia F     |
| 49  | 0.7 ± 0.7     | -2.5 ± 0.2   | -0.4 ± 0.3    | -2.3 ± 0.3   | -0.8 ± 0.3   | -2.0 ± 0.6    | -0.4 ± 0.5    | -2.0 ± 0.2  | 0.1 ± 0.3     | -1.8 ± 0.2   | 0.1 ± 0.3    | -2.6 ± 0.4    | E Africa F   | W Africa M   |
| 50  | -0.5 ± 0.8    | -2.4 ± 0.2   | -0.1 ± 0.3    | -2.2 ± 0.3   | -0.9 ± 0.3   | -1.8 ± 0.6    | -0.3 ± 0.6    | -2.0 ± 0.2  | -0.1 ± 0.3    | -1.9 ± 0.3   | 0.2 ± 0.3    | -2.9 ± 0.4    | S E Asia M   | W Africa M   |
| 51  | 0.3 ± 0.7     | -2.4 ± 0.2   | -0.5 ± 0.3    | -2.1 ± 0.3   | -1.0 ± 0.3   | -1.6 ± 0.6    | -0.9 ± 0.6    | -2.0 ± 0.2  | 0.4 ± 0.3     | -2.0 ± 0.3   | 0.0 ± 0.3    | -3.0 ± 0.4    | E Europe M   | W Africa M   |
| 52  | 0.3 ± 0.8     | -2.4 ± 0.2   | 0.2 ± 0.4     | -2.3 ± 0.3   | -0.8 ± 0.3   | -2.5 ± 0.6    | -0.2 ± 0.6    | -2.3 ± 0.2  | 0.2 ± 0.4     | -2.0 ± 0.3   | -0.4 ± 0.3   | -3.3 ± 0.5    | E Africa F   | W Africa M   |

**Table 38.** Mean raw error for Application-type images for algorithm cyberlink-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 4.5 ± 0.2     | 3.9 ± 0.1    | 3.1 ± 0.2     | 3.3 ± 0.1    | 3.7 ± 0.2    | 4.9 ± 0.3     | 3.5 ± 0.2     | 3.3 ± 0.1   | 2.0 ± 0.2     | 2.6 ± 0.1    | 2.9 ± 0.2    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 4.2 ± 0.2     | 3.9 ± 0.1    | 3.2 ± 0.2     | 3.3 ± 0.2    | 3.7 ± 0.3    | 4.5 ± 0.3     | 3.0 ± 0.2     | 3.29 ± 0.09 | 2.2 ± 0.2     | 2.7 ± 0.1    | 3.1 ± 0.2    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 3.6 ± 0.3     | 3.6 ± 0.1    | 3.1 ± 0.2     | 2.9 ± 0.2    | 3.5 ± 0.3    | 4.3 ± 0.4     | 3.3 ± 0.2     | 2.99 ± 0.09 | 2.4 ± 0.2     | 2.7 ± 0.1    | 2.9 ± 0.2    | 3.4 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 3.5 ± 0.3     | 3.1 ± 0.1    | 2.8 ± 0.2     | 2.7 ± 0.2    | 3.2 ± 0.3    | 3.9 ± 0.3     | 2.9 ± 0.2     | 2.8 ± 0.1   | 2.1 ± 0.2     | 2.7 ± 0.1    | 2.9 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 2.7 ± 0.3     | 2.5 ± 0.1    | 2.2 ± 0.2     | 2.1 ± 0.1    | 2.6 ± 0.2    | 3.2 ± 0.3     | 2.6 ± 0.2     | 2.25 ± 0.10 | 1.8 ± 0.2     | 2.3 ± 0.1    | 2.4 ± 0.2    | 2.7 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 2.5 ± 0.3     | 2.1 ± 0.1    | 1.7 ± 0.2     | 1.7 ± 0.1    | 2.3 ± 0.2    | 3.2 ± 0.3     | 2.5 ± 0.2     | 2.1 ± 0.1   | 1.6 ± 0.2     | 2.4 ± 0.1    | 2.1 ± 0.2    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 2.2 ± 0.3     | 1.8 ± 0.1    | 1.3 ± 0.2     | 1.2 ± 0.2    | 2.0 ± 0.1    | 3.1 ± 0.3     | 2.5 ± 0.3     | 1.9 ± 0.1   | 1.6 ± 0.2     | 2.5 ± 0.1    | 2.1 ± 0.2    | 2.4 ± 0.3     | W Africa F   | S Asia F     |
| 21  | 1.7 ± 0.3     | 1.4 ± 0.1    | 0.9 ± 0.2     | 0.6 ± 0.1    | 1.3 ± 0.1    | 2.6 ± 0.3     | 2.5 ± 0.2     | 1.7 ± 0.1   | 1.3 ± 0.2     | 2.3 ± 0.1    | 2.0 ± 0.2    | 2.1 ± 0.3     | W Africa F   | S Asia F     |
| 22  | 1.5 ± 0.3     | 0.86 ± 0.09  | 0.6 ± 0.1     | 0.1 ± 0.1    | 0.9 ± 0.1    | 2.2 ± 0.3     | 2.8 ± 0.3     | 1.5 ± 0.1   | 1.3 ± 0.2     | 2.2 ± 0.1    | 1.6 ± 0.2    | 1.9 ± 0.3     | E Africa M   | S Asia F     |
| 23  | 0.8 ± 0.2     | 0.48 ± 0.09  | 0.2 ± 0.1     | -0.2 ± 0.1   | 0.5 ± 0.1    | 1.8 ± 0.3     | 2.3 ± 0.3     | 1.3 ± 0.1   | 1.0 ± 0.2     | 2.2 ± 0.1    | 1.5 ± 0.2    | 1.9 ± 0.2     | E Africa M   | S Asia F     |
| 24  | 0.4 ± 0.3     | -0.10 ± 0.09 | -0.1 ± 0.1    | -0.8 ± 0.1   | 0.1 ± 0.1    | 1.4 ± 0.3     | 2.3 ± 0.2     | 1.2 ± 0.1   | 1.0 ± 0.2     | 2.1 ± 0.1    | 1.3 ± 0.2    | 1.7 ± 0.2     | E Africa M   | S Asia F     |
| 25  | 0.1 ± 0.3     | -0.44 ± 0.08 | -0.5 ± 0.1    | -1.32 ± 0.10 | -0.4 ± 0.1   | 1.1 ± 0.3     | 2.2 ± 0.2     | 1.1 ± 0.1   | 0.8 ± 0.2     | 1.9 ± 0.1    | 1.3 ± 0.2    | 1.4 ± 0.2     | E Africa M   | S Asia F     |
| 26  | -0.4 ± 0.2    | -0.93 ± 0.08 | -0.8 ± 0.1    | -1.92 ± 0.10 | -0.7 ± 0.1   | 0.6 ± 0.3     | 1.8 ± 0.2     | 0.9 ± 0.1   | 0.6 ± 0.2     | 1.7 ± 0.1    | 1.3 ± 0.2    | 1.0 ± 0.2     | E Africa M   | S Asia F     |
| 27  | -0.6 ± 0.2    | -1.39 ± 0.08 | -1.0 ± 0.1    | -2.32 ± 0.10 | -1.1 ± 0.1   | 0.5 ± 0.3     | 1.5 ± 0.2     | 0.7 ± 0.1   | 0.7 ± 0.2     | 1.4 ± 0.1    | 1.3 ± 0.2    | 0.8 ± 0.2     | E Africa M   | S Asia F     |
| 28  | -1.0 ± 0.2    | -1.70 ± 0.08 | -1.4 ± 0.1    | -2.71 ± 0.10 | -1.4 ± 0.1   | -0.0 ± 0.3    | 1.2 ± 0.2     | 0.7 ± 0.1   | 0.6 ± 0.2     | 1.1 ± 0.1    | 1.0 ± 0.2    | 0.3 ± 0.2     | E Africa M   | S Asia F     |
| 29  | -1.2 ± 0.3    | -2.07 ± 0.08 | -1.6 ± 0.1    | -3.01 ± 0.10 | -1.7 ± 0.1   | -0.3 ± 0.3    | 0.9 ± 0.2     | 0.5 ± 0.1   | 0.3 ± 0.2     | 0.83 ± 0.09  | 0.6 ± 0.2    | 0.3 ± 0.2     | E Africa M   | S Asia F     |
| 30  | -1.9 ± 0.3    | -2.37 ± 0.09 | -1.9 ± 0.1    | -3.4 ± 0.1   | -1.9 ± 0.1   | -0.9 ± 0.3    | 0.7 ± 0.2     | 0.3 ± 0.1   | 0.1 ± 0.2     | 0.52 ± 0.09  | 0.6 ± 0.2    | -0.2 ± 0.2    | E Africa M   | S Asia F     |
| 31  | -2.1 ± 0.3    | -2.58 ± 0.09 | -2.2 ± 0.1    | -3.9 ± 0.1   | -2.3 ± 0.1   | -0.9 ± 0.3    | 0.6 ± 0.2     | 0.1 ± 0.1   | -0.1 ± 0.2    | 0.32 ± 0.09  | 0.4 ± 0.2    | -0.5 ± 0.2    | E Africa M   | S Asia F     |
| 32  | -2.1 ± 0.3    | -2.91 ± 0.10 | -2.4 ± 0.1    | -4.2 ± 0.1   | -2.4 ± 0.2   | -1.3 ± 0.3    | 0.2 ± 0.2     | -0.2 ± 0.1  | -0.1 ± 0.2    | 0.10 ± 0.09  | 0.3 ± 0.2    | -0.8 ± 0.2    | S E Asia M   | S Asia F     |
| 33  | -2.5 ± 0.3    | -3.0 ± 0.1   | -2.8 ± 0.1    | -4.5 ± 0.1   | -2.8 ± 0.2   | -1.3 ± 0.3    | 0.1 ± 0.2     | -0.3 ± 0.1  | -0.1 ± 0.2    | -0.20 ± 0.09 | 0.1 ± 0.2    | -1.1 ± 0.2    | S E Asia M   | S Asia F     |
| 34  | -2.9 ± 0.3    | -3.1 ± 0.1   | -3.0 ± 0.2    | -4.7 ± 0.1   | -2.9 ± 0.2   | -1.9 ± 0.3    | -0.3 ± 0.3    | -0.5 ± 0.1  | -0.3 ± 0.2    | -0.3 ± 0.1   | -0.0 ± 0.2   | -1.3 ± 0.2    | S E Asia M   | S Asia F     |
| 35  | -2.8 ± 0.3    | -3.5 ± 0.1   | -3.1 ± 0.2    | -5.0 ± 0.1   | -3.1 ± 0.2   | -2.1 ± 0.3    | -0.5 ± 0.3    | -0.7 ± 0.1  | -0.4 ± 0.2    | -0.6 ± 0.1   | -0.2 ± 0.2   | -1.7 ± 0.2    | S E Asia M   | S Asia F     |
| 36  | -2.8 ± 0.4    | -3.5 ± 0.1   | -3.1 ± 0.2    | -5.3 ± 0.1   | -3.0 ± 0.2   | -2.2 ± 0.3    | -0.5 ± 0.3    | -0.8 ± 0.1  | -0.3 ± 0.2    | -0.8 ± 0.1   | -0.4 ± 0.2   | -2.0 ± 0.2    | E Europe M   | S Asia F     |
| 37  | -3.1 ± 0.4    | -3.6 ± 0.1   | -3.3 ± 0.2    | -5.2 ± 0.1   | -3.3 ± 0.2   | -2.6 ± 0.4    | -0.6 ± 0.3    | -1.1 ± 0.1  | -0.6 ± 0.2    | -0.9 ± 0.1   | -0.5 ± 0.3   | -2.3 ± 0.2    | S E Asia M   | S Asia F     |
| 38  | -2.9 ± 0.4    | -3.8 ± 0.1   | -3.4 ± 0.2    | -5.3 ± 0.2   | -3.1 ± 0.2   | -2.7 ± 0.4    | -0.9 ± 0.3    | -1.2 ± 0.1  | -0.7 ± 0.2    | -1.1 ± 0.1   | -0.3 ± 0.2   | -2.7 ± 0.2    | S E Asia M   | S Asia F     |
| 39  | -3.0 ± 0.5    | -3.9 ± 0.1   | -3.4 ± 0.2    | -5.6 ± 0.2   | -3.2 ± 0.2   | -2.6 ± 0.4    | -0.5 ± 0.3    | -1.3 ± 0.1  | -0.6 ± 0.2    | -1.2 ± 0.1   | -0.4 ± 0.2   | -2.9 ± 0.3    | S E Asia M   | S Asia F     |
| 40  | -3.2 ± 0.5    | -4.0 ± 0.1   | -3.0 ± 0.2    | -5.4 ± 0.2   | -3.5 ± 0.2   | -3.3 ± 0.4    | -0.6 ± 0.4    | -1.5 ± 0.1  | -0.5 ± 0.2    | -1.4 ± 0.1   | -0.6 ± 0.2   | -3.0 ± 0.3    | E Europe M   | S Asia F     |
| 41  | -3.5 ± 0.5    | -4.0 ± 0.1   | -3.1 ± 0.2    | -5.6 ± 0.2   | -3.4 ± 0.2   | -3.6 ± 0.4    | -0.7 ± 0.4    | -1.5 ± 0.1  | -0.6 ± 0.2    | -1.5 ± 0.1   | -0.8 ± 0.2   | -3.2 ± 0.3    | E Europe M   | S Asia F     |
| 42  | -3.1 ± 0.5    | -3.9 ± 0.1   | -2.8 ± 0.3    | -5.1 ± 0.2   | -3.5 ± 0.2   | -3.1 ± 0.5    | -0.7 ± 0.4    | -1.6 ± 0.1  | -0.2 ± 0.3    | -1.7 ± 0.1   | -0.5 ± 0.2   | -3.4 ± 0.3    | E Europe M   | S Asia F     |
| 43  | -3.1 ± 0.6    | -4.1 ± 0.1   | -2.8 ± 0.3    | -5.4 ± 0.2   | -3.5 ± 0.2   | -3.5 ± 0.5    | -0.4 ± 0.4    | -1.6 ± 0.1  | -0.3 ± 0.3    | -1.6 ± 0.2   | -0.8 ± 0.3   | -3.7 ± 0.3    | E Europe M   | S Asia F     |
| 44  | -2.8 ± 0.6    | -4.2 ± 0.1   | -3.0 ± 0.3    | -5.1 ± 0.3   | -3.1 ± 0.3   | -3.5 ± 0.5    | -1.1 ± 0.4    | -1.7 ± 0.1  | -0.1 ± 0.3    | -1.8 ± 0.2   | -0.6 ± 0.3   | -3.5 ± 0.3    | E Europe M   | S Asia F     |
| 45  | -3.1 ± 0.6    | -4.1 ± 0.1   | -2.8 ± 0.3    | -5.1 ± 0.3   | -3.2 ± 0.3   | -3.4 ± 0.5    | -1.0 ± 0.4    | -1.9 ± 0.1  | -0.1 ± 0.3    | -1.8 ± 0.2   | -0.8 ± 0.3   | -3.9 ± 0.4    | E Europe M   | S Asia F     |
| 46  | -2.0 ± 0.7    | -4.2 ± 0.1   | -2.6 ± 0.3    | -4.6 ± 0.3   | -3.3 ± 0.3   | -3.4 ± 0.5    | -0.9 ± 0.5    | -2.0 ± 0.1  | 0.1 ± 0.3     | -1.6 ± 0.2   | -0.7 ± 0.3   | -4.1 ± 0.4    | E Europe M   | S Asia F     |
| 47  | -1.7 ± 0.7    | -4.0 ± 0.1   | -2.4 ± 0.3    | -4.6 ± 0.3   | -2.9 ± 0.3   | -3.1 ± 0.5    | -0.6 ± 0.4    | -2.0 ± 0.1  | 0.1 ± 0.3     | -1.7 ± 0.2   | -0.6 ± 0.3   | -3.7 ± 0.4    | E Europe M   | S Asia F     |
| 48  | -2.4 ± 0.7    | -3.9 ± 0.2   | -2.2 ± 0.3    | -4.3 ± 0.3   | -2.9 ± 0.3   | -3.2 ± 0.5    | -1.3 ± 0.5    | -2.1 ± 0.1  | 0.1 ± 0.3     | -1.8 ± 0.2   | -1.0 ± 0.3   | -4.2 ± 0.4    | E Europe M   | S Asia F     |
| 49  | -1.7 ± 0.7    | -4.0 ± 0.2   | -2.4 ± 0.3    | -4.2 ± 0.3   | -2.6 ± 0.3   | -3.6 ± 0.6    | -1.5 ± 0.5    | -2.4 ± 0.1  | -0.0 ± 0.3    | -1.8 ± 0.2   | -1.0 ± 0.3   | -4.3 ± 0.4    | E Europe M   | W Africa M   |
| 50  | -2.5 ± 0.8    | -3.9 ± 0.2   | -1.8 ± 0.3    | -4.1 ± 0.3   | -2.7 ± 0.3   | -3.3 ± 0.6    | -1.4 ± 0.5    | -2.5 ± 0.1  | -0.2 ± 0.3    | -2.0 ± 0.2   | -0.9 ± 0.3   | -4.4 ± 0.4    | E Europe M   | W Africa M   |
| 51  | -2.1 ± 0.7    | -3.9 ± 0.2   | -2.1 ± 0.3    | -4.0 ± 0.3   | -2.8 ± 0.3   | -3.2 ± 0.6    | -1.7 ± 0.6    | -2.7 ± 0.1  | -0.1 ± 0.3    | -2.1 ± 0.2   | -1.2 ± 0.3   | -4.4 ± 0.4    | E Europe M   | W Africa M   |
| 52  | -2.0 ± 0.8    | -3.8 ± 0.2   | -1.5 ± 0.3    | -4.0 ± 0.3   | -2.3 ± 0.3   | -3.9 ± 0.6    | -1.5 ± 0.5    | -2.9 ± 0.1  | -0.3 ± 0.3    | -2.3 ± 0.2   | -1.6 ± 0.3   | -4.7 ± 0.5    | E Europe M   | W Africa M   |

**Table 39.** Mean raw error for Application-type images for algorithm daon-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 1.7 ± 0.4     | 0.4 ± 0.2   | 0.8 ± 0.2     | 1.2 ± 0.2   | 0.3 ± 0.4    | 1.1 ± 0.6     | 2.8 ± 0.4     | 3.6 ± 0.2   | 1.5 ± 0.2     | 3.7 ± 0.2   | 3.1 ± 0.4    | 1.6 ± 0.4     | S Asia M     | S E Asia F   |
| 15  | 1.9 ± 0.5     | 0.7 ± 0.2   | 0.7 ± 0.2     | 1.8 ± 0.3   | 0.3 ± 0.5    | 1.4 ± 0.6     | 2.9 ± 0.4     | 4.3 ± 0.2   | 1.7 ± 0.3     | 4.2 ± 0.2   | 4.2 ± 0.4    | 2.3 ± 0.5     | E Asia M     | S E Asia F   |
| 16  | 2.2 ± 0.5     | 0.7 ± 0.2   | 1.1 ± 0.3     | 1.3 ± 0.3   | 0.8 ± 0.5    | 1.9 ± 0.8     | 4.3 ± 0.4     | 4.2 ± 0.2   | 2.1 ± 0.3     | 4.7 ± 0.2   | 4.5 ± 0.4    | 2.3 ± 0.5     | S Asia M     | E Asia F     |
| 17  | 2.8 ± 0.6     | 0.6 ± 0.2   | 1.0 ± 0.3     | 1.4 ± 0.3   | 0.8 ± 0.5    | 2.6 ± 0.7     | 4.9 ± 0.4     | 4.1 ± 0.2   | 1.8 ± 0.3     | 4.9 ± 0.2   | 4.1 ± 0.4    | 3.3 ± 0.5     | S Asia M     | E Asia F     |
| 18  | 1.8 ± 0.4     | -0.0 ± 0.2  | 0.4 ± 0.3     | 0.9 ± 0.2   | -0.1 ± 0.3   | 1.9 ± 0.6     | 4.5 ± 0.4     | 3.3 ± 0.2   | 1.4 ± 0.2     | 4.4 ± 0.2   | 3.5 ± 0.3    | 3.2 ± 0.4     | E Africa M   | S E Asia F   |
| 19  | 2.7 ± 0.5     | -0.1 ± 0.2  | -0.0 ± 0.3    | 1.0 ± 0.2   | -0.0 ± 0.3   | 2.5 ± 0.6     | 5.0 ± 0.4     | 3.1 ± 0.2   | 1.4 ± 0.2     | 4.6 ± 0.2   | 3.2 ± 0.3    | 3.4 ± 0.4     | E Africa M   | E Asia F     |
| 20  | 2.6 ± 0.5     | -0.1 ± 0.2  | -0.1 ± 0.2    | 0.6 ± 0.2   | -0.3 ± 0.3   | 2.9 ± 0.6     | 5.1 ± 0.4     | 2.8 ± 0.2   | 1.4 ± 0.3     | 4.6 ± 0.2   | 2.7 ± 0.3    | 3.6 ± 0.4     | E Africa M   | S E Asia F   |
| 21  | 2.3 ± 0.4     | -0.4 ± 0.2  | -0.3 ± 0.2    | 0.2 ± 0.2   | -0.3 ± 0.3   | 3.1 ± 0.6     | 5.2 ± 0.3     | 2.4 ± 0.2   | 1.2 ± 0.2     | 4.1 ± 0.2   | 2.6 ± 0.3    | 3.4 ± 0.4     | E Africa M   | E Asia F     |
| 22  | 2.3 ± 0.4     | -0.6 ± 0.2  | -0.4 ± 0.2    | -0.3 ± 0.2  | -0.3 ± 0.3   | 2.5 ± 0.5     | 5.4 ± 0.3     | 2.1 ± 0.2   | 1.2 ± 0.2     | 4.0 ± 0.2   | 1.9 ± 0.2    | 3.4 ± 0.4     | E Africa M   | E Asia F     |
| 23  | 1.8 ± 0.4     | -0.7 ± 0.2  | -0.5 ± 0.2    | -0.5 ± 0.2  | -0.5 ± 0.2   | 2.6 ± 0.5     | 5.1 ± 0.3     | 1.8 ± 0.2   | 0.9 ± 0.2     | 3.7 ± 0.2   | 1.7 ± 0.3    | 2.9 ± 0.3     | E Africa M   | E Asia F     |
| 24  | 1.9 ± 0.4     | -1.0 ± 0.2  | -0.7 ± 0.2    | -0.9 ± 0.1  | -0.6 ± 0.2   | 2.3 ± 0.5     | 4.8 ± 0.3     | 1.5 ± 0.2   | 1.0 ± 0.2     | 3.5 ± 0.1   | 1.4 ± 0.2    | 2.8 ± 0.3     | E Africa M   | E Asia F     |
| 25  | 1.6 ± 0.3     | -1.1 ± 0.2  | -0.8 ± 0.2    | -1.4 ± 0.1  | -0.7 ± 0.2   | 2.0 ± 0.5     | 4.7 ± 0.3     | 1.3 ± 0.2   | 0.9 ± 0.2     | 3.2 ± 0.1   | 1.5 ± 0.2    | 2.7 ± 0.3     | E Africa M   | S Asia F     |
| 26  | 1.2 ± 0.3     | -1.3 ± 0.2  | -0.8 ± 0.2    | -1.9 ± 0.1  | -0.7 ± 0.2   | 2.1 ± 0.4     | 4.3 ± 0.3     | 1.0 ± 0.2   | 0.7 ± 0.2     | 2.9 ± 0.1   | 1.2 ± 0.2    | 2.4 ± 0.3     | E Africa M   | S Asia F     |
| 27  | 1.0 ± 0.3     | -1.5 ± 0.2  | -0.8 ± 0.2    | -2.3 ± 0.1  | -0.9 ± 0.2   | 1.8 ± 0.4     | 3.9 ± 0.3     | 0.9 ± 0.2   | 0.8 ± 0.2     | 2.5 ± 0.1   | 1.3 ± 0.2    | 2.1 ± 0.3     | E Africa M   | S Asia F     |
| 28  | 0.3 ± 0.3     | -1.6 ± 0.2  | -1.0 ± 0.2    | -2.8 ± 0.1  | -1.1 ± 0.2   | 1.4 ± 0.4     | 3.8 ± 0.3     | 0.8 ± 0.2   | 0.8 ± 0.2     | 2.1 ± 0.1   | 1.1 ± 0.2    | 1.8 ± 0.3     | E Africa M   | S Asia F     |
| 29  | 0.3 ± 0.3     | -1.5 ± 0.1  | -0.9 ± 0.2    | -3.0 ± 0.1  | -1.1 ± 0.2   | 1.0 ± 0.4     | 3.6 ± 0.3     | 0.7 ± 0.2   | 0.7 ± 0.2     | 1.7 ± 0.1   | 1.0 ± 0.3    | 2.0 ± 0.3     | E Africa M   | S Asia F     |
| 30  | -0.1 ± 0.3    | -1.8 ± 0.1  | -1.0 ± 0.2    | -3.4 ± 0.1  | -1.0 ± 0.2   | 0.7 ± 0.4     | 3.3 ± 0.3     | 0.4 ± 0.2   | 0.5 ± 0.2     | 1.3 ± 0.1   | 0.9 ± 0.3    | 1.5 ± 0.3     | E Africa M   | S Asia F     |
| 31  | -0.4 ± 0.3    | -1.9 ± 0.1  | -0.9 ± 0.2    | -3.9 ± 0.1  | -1.4 ± 0.2   | 0.3 ± 0.4     | 3.2 ± 0.3     | 0.4 ± 0.2   | 0.6 ± 0.2     | 1.0 ± 0.1   | 0.9 ± 0.3    | 1.3 ± 0.3     | E Africa M   | S Asia F     |
| 32  | -0.4 ± 0.3    | -1.8 ± 0.1  | -1.1 ± 0.2    | -4.1 ± 0.1  | -1.3 ± 0.2   | 0.3 ± 0.4     | 2.5 ± 0.3     | 0.5 ± 0.2   | 0.6 ± 0.2     | 0.8 ± 0.1   | 1.1 ± 0.3    | 1.1 ± 0.3     | E Africa M   | S Asia F     |
| 33  | -0.8 ± 0.4    | -1.9 ± 0.2  | -1.0 ± 0.2    | -4.3 ± 0.1  | -1.6 ± 0.2   | -0.1 ± 0.4    | 2.5 ± 0.3     | 0.2 ± 0.2   | 0.8 ± 0.2     | 0.4 ± 0.1   | 0.8 ± 0.3    | 0.9 ± 0.3     | E Africa M   | S Asia F     |
| 34  | -1.2 ± 0.4    | -1.8 ± 0.1  | -1.0 ± 0.2    | -4.5 ± 0.1  | -1.7 ± 0.2   | -0.6 ± 0.4    | 1.9 ± 0.3     | 0.2 ± 0.2   | 0.6 ± 0.2     | 0.3 ± 0.1   | 0.7 ± 0.3    | 0.6 ± 0.3     | E Africa M   | S Asia F     |
| 35  | -1.1 ± 0.4    | -2.1 ± 0.2  | -1.0 ± 0.2    | -4.9 ± 0.2  | -1.8 ± 0.2   | -0.7 ± 0.4    | 2.0 ± 0.3     | 0.3 ± 0.2   | 0.7 ± 0.2     | -0.0 ± 0.1  | 1.0 ± 0.3    | 0.5 ± 0.3     | E Africa M   | S Asia F     |
| 36  | -1.3 ± 0.4    | -2.1 ± 0.2  | -0.9 ± 0.2    | -5.1 ± 0.2  | -1.5 ± 0.2   | -0.8 ± 0.4    | 1.6 ± 0.3     | 0.3 ± 0.2   | 0.8 ± 0.2     | -0.2 ± 0.1  | 0.8 ± 0.3    | 0.1 ± 0.3     | E Africa M   | S Asia F     |
| 37  | -1.9 ± 0.4    | -2.0 ± 0.2  | -0.9 ± 0.2    | -5.1 ± 0.2  | -1.7 ± 0.2   | -0.8 ± 0.4    | 1.3 ± 0.3     | 0.1 ± 0.2   | 0.5 ± 0.2     | -0.4 ± 0.1  | 0.6 ± 0.3    | -0.1 ± 0.3    | E Africa M   | S Asia F     |
| 38  | -1.4 ± 0.4    | -2.2 ± 0.2  | -0.6 ± 0.3    | -5.2 ± 0.2  | -1.7 ± 0.3   | -1.6 ± 0.5    | 1.1 ± 0.3     | 0.1 ± 0.2   | 0.6 ± 0.2     | -0.7 ± 0.1  | 1.0 ± 0.3    | -0.3 ± 0.3    | E Africa M   | S Asia F     |
| 39  | -1.7 ± 0.4    | -2.2 ± 0.2  | -0.8 ± 0.2    | -5.5 ± 0.2  | -1.5 ± 0.3   | -1.3 ± 0.4    | 1.4 ± 0.3     | 0.1 ± 0.2   | 0.7 ± 0.2     | -0.6 ± 0.1  | 1.1 ± 0.3    | -0.4 ± 0.3    | E Africa M   | S Asia F     |
| 40  | -2.1 ± 0.5    | -2.3 ± 0.2  | -0.6 ± 0.3    | -5.4 ± 0.2  | -1.7 ± 0.2   | -2.0 ± 0.5    | 0.9 ± 0.3     | -0.1 ± 0.2  | 0.8 ± 0.2     | -0.9 ± 0.1  | 1.3 ± 0.3    | -0.9 ± 0.3    | S E Asia M   | S Asia F     |
| 41  | -2.4 ± 0.5    | -2.3 ± 0.2  | -0.6 ± 0.3    | -5.4 ± 0.2  | -2.0 ± 0.3   | -2.2 ± 0.5    | 1.2 ± 0.4     | 0.1 ± 0.2   | 0.8 ± 0.2     | -0.9 ± 0.1  | 1.1 ± 0.3    | -1.1 ± 0.3    | E Africa M   | S Asia F     |
| 42  | -2.4 ± 0.5    | -2.2 ± 0.1  | -0.2 ± 0.3    | -4.8 ± 0.2  | -1.7 ± 0.3   | -2.3 ± 0.5    | 0.8 ± 0.4     | 0.2 ± 0.2   | 1.0 ± 0.2     | -1.1 ± 0.2  | 1.3 ± 0.3    | -1.3 ± 0.3    | S E Asia M   | S Asia F     |
| 43  | -2.1 ± 0.6    | -2.4 ± 0.1  | -0.3 ± 0.3    | -5.2 ± 0.3  | -2.0 ± 0.3   | -2.8 ± 0.5    | 0.9 ± 0.4     | 0.1 ± 0.2   | 0.8 ± 0.3     | -0.9 ± 0.2  | 1.2 ± 0.3    | -1.8 ± 0.3    | S E Asia M   | S Asia F     |
| 44  | -2.1 ± 0.6    | -2.6 ± 0.1  | -0.5 ± 0.3    | -4.9 ± 0.3  | -1.8 ± 0.3   | -2.9 ± 0.6    | 0.1 ± 0.4     | 0.1 ± 0.2   | 1.2 ± 0.3     | -1.1 ± 0.2  | 1.2 ± 0.3    | -1.6 ± 0.3    | S E Asia M   | S Asia F     |
| 45  | -2.7 ± 0.6    | -2.7 ± 0.2  | -0.6 ± 0.3    | -4.8 ± 0.3  | -2.0 ± 0.3   | -2.6 ± 0.5    | -0.2 ± 0.4    | -0.1 ± 0.2  | 0.6 ± 0.3     | -0.9 ± 0.2  | 1.2 ± 0.3    | -2.1 ± 0.4    | S E Asia M   | S Asia F     |
| 46  | -1.9 ± 0.6    | -3.0 ± 0.1  | -0.5 ± 0.3    | -4.4 ± 0.3  | -2.3 ± 0.3   | -2.8 ± 0.5    | 0.0 ± 0.4     | -0.3 ± 0.2  | 0.8 ± 0.3     | -0.8 ± 0.2  | 1.2 ± 0.3    | -2.4 ± 0.3    | S E Asia M   | S Asia F     |
| 47  | -1.8 ± 0.6    | -2.8 ± 0.2  | -0.5 ± 0.3    | -4.4 ± 0.3  | -2.3 ± 0.3   | -3.0 ± 0.6    | 0.2 ± 0.4     | -0.3 ± 0.2  | 0.9 ± 0.3     | -0.7 ± 0.2  | 1.1 ± 0.3    | -2.1 ± 0.4    | S E Asia M   | S Asia F     |
| 48  | -2.4 ± 0.7    | -3.0 ± 0.2  | -0.5 ± 0.3    | -4.1 ± 0.3  | -2.3 ± 0.3   | -3.2 ± 0.5    | -0.7 ± 0.5    | -0.4 ± 0.2  | 0.9 ± 0.3     | -0.7 ± 0.2  | 0.9 ± 0.3    | -3.0 ± 0.4    | E Europe M   | S Asia F     |
| 49  | -2.0 ± 0.7    | -3.1 ± 0.2  | -0.6 ± 0.3    | -4.0 ± 0.3  | -2.2 ± 0.3   | -3.4 ± 0.6    | -1.0 ± 0.5    | -0.5 ± 0.2  | 0.7 ± 0.3     | -0.8 ± 0.2  | 1.0 ± 0.3    | -2.8 ± 0.4    | S E Asia M   | S Asia F     |
| 50  | -2.7 ± 0.8    | -3.2 ± 0.2  | -0.5 ± 0.3    | -3.4 ± 0.3  | -2.6 ± 0.3   | -3.3 ± 0.5    | -1.3 ± 0.5    | -0.7 ± 0.2  | 0.6 ± 0.3     | -0.8 ± 0.2  | 1.0 ± 0.3    | -3.5 ± 0.4    | S E Asia M   | W Africa M   |
| 51  | -2.3 ± 0.6    | -3.2 ± 0.2  | -0.6 ± 0.3    | -3.8 ± 0.3  | -2.6 ± 0.3   | -3.5 ± 0.6    | -1.6 ± 0.6    | -0.9 ± 0.2  | 0.8 ± 0.3     | -0.9 ± 0.2  | 0.7 ± 0.3    | -3.6 ± 0.4    | E Europe M   | S Asia F     |
| 52  | -2.7 ± 0.7    | -3.4 ± 0.2  | -0.4 ± 0.3    | -3.6 ± 0.3  | -2.4 ± 0.3   | -4.0 ± 0.6    | -1.5 ± 0.5    | -1.2 ± 0.2  | 0.7 ± 0.3     | -0.9 ± 0.2  | 0.3 ± 0.3    | -3.7 ± 0.4    | E Europe M   | W Africa F   |

**Table 40.** Mean raw error for Application-type images for algorithm daon-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 0.8 ± 0.3     | -0.3 ± 0.2  | 0.3 ± 0.2     | 0.1 ± 0.2   | -0.8 ± 0.3   | 0.4 ± 0.4     | 1.4 ± 0.3     | 2.3 ± 0.2   | 0.7 ± 0.2     | 1.8 ± 0.2   | 1.6 ± 0.3    | 0.8 ± 0.3     | E Asia M     | S E Asia F   |
| 15  | 0.9 ± 0.4     | 0.1 ± 0.2   | 0.3 ± 0.2     | 0.7 ± 0.2   | -0.4 ± 0.4   | 0.7 ± 0.5     | 1.4 ± 0.4     | 3.1 ± 0.2   | 1.1 ± 0.2     | 2.7 ± 0.2   | 2.7 ± 0.3    | 1.5 ± 0.4     | E Asia M     | S E Asia F   |
| 16  | 1.3 ± 0.5     | 0.1 ± 0.2   | 0.7 ± 0.3     | 0.5 ± 0.2   | 0.1 ± 0.4    | 1.4 ± 0.7     | 2.9 ± 0.4     | 3.3 ± 0.2   | 1.7 ± 0.3     | 3.4 ± 0.2   | 3.1 ± 0.3    | 1.9 ± 0.4     | S Asia M     | S E Asia F   |
| 17  | 1.9 ± 0.5     | 0.2 ± 0.2   | 0.5 ± 0.3     | 0.6 ± 0.2   | 0.2 ± 0.4    | 2.0 ± 0.6     | 3.6 ± 0.4     | 3.4 ± 0.2   | 1.5 ± 0.2     | 3.8 ± 0.2   | 3.4 ± 0.3    | 2.9 ± 0.5     | S Asia M     | S E Asia F   |
| 18  | 1.3 ± 0.4     | -0.4 ± 0.2  | 0.0 ± 0.3     | 0.2 ± 0.2   | -0.5 ± 0.3   | 1.6 ± 0.6     | 3.4 ± 0.3     | 2.8 ± 0.1   | 1.3 ± 0.2     | 3.6 ± 0.2   | 2.7 ± 0.3    | 2.6 ± 0.4     | S Asia M     | S E Asia F   |
| 19  | 2.0 ± 0.4     | -0.3 ± 0.2  | -0.3 ± 0.3    | 0.3 ± 0.2   | -0.5 ± 0.3   | 2.3 ± 0.6     | 4.1 ± 0.3     | 2.7 ± 0.1   | 1.2 ± 0.2     | 3.9 ± 0.2   | 2.5 ± 0.2    | 3.1 ± 0.4     | E Africa M   | S E Asia F   |
| 20  | 1.9 ± 0.4     | -0.3 ± 0.2  | -0.2 ± 0.2    | 0.1 ± 0.2   | -0.6 ± 0.3   | 2.7 ± 0.5     | 4.7 ± 0.3     | 2.5 ± 0.1   | 1.5 ± 0.2     | 4.0 ± 0.1   | 2.3 ± 0.2    | 3.6 ± 0.4     | E Africa M   | S E Asia F   |
| 21  | 1.9 ± 0.4     | -0.4 ± 0.2  | -0.4 ± 0.2    | -0.2 ± 0.2  | -0.6 ± 0.2   | 2.8 ± 0.5     | 4.8 ± 0.3     | 2.2 ± 0.1   | 1.1 ± 0.2     | 3.8 ± 0.1   | 2.2 ± 0.2    | 3.4 ± 0.4     | E Africa M   | S E Asia F   |
| 22  | 2.1 ± 0.4     | -0.6 ± 0.2  | -0.4 ± 0.2    | -0.6 ± 0.2  | -0.6 ± 0.2   | 2.5 ± 0.5     | 5.2 ± 0.3     | 2.0 ± 0.1   | 1.2 ± 0.2     | 3.7 ± 0.1   | 1.7 ± 0.2    | 3.5 ± 0.3     | E Africa M   | E Asia F     |
| 23  | 1.8 ± 0.3     | -0.7 ± 0.2  | -0.4 ± 0.2    | -0.8 ± 0.1  | -0.6 ± 0.2   | 2.7 ± 0.4     | 5.0 ± 0.3     | 1.8 ± 0.1   | 1.0 ± 0.2     | 3.6 ± 0.1   | 1.6 ± 0.2    | 3.3 ± 0.3     | E Africa M   | S Asia F     |
| 24  | 1.8 ± 0.3     | -0.9 ± 0.2  | -0.5 ± 0.2    | -1.1 ± 0.1  | -0.8 ± 0.2   | 2.3 ± 0.4     | 5.0 ± 0.3     | 1.5 ± 0.1   | 1.1 ± 0.2     | 3.4 ± 0.1   | 1.4 ± 0.2    | 3.2 ± 0.3     | E Africa M   | S Asia F     |
| 25  | 1.6 ± 0.3     | -0.9 ± 0.1  | -0.6 ± 0.2    | -1.5 ± 0.1  | -0.9 ± 0.2   | 2.4 ± 0.4     | 4.8 ± 0.3     | 1.4 ± 0.1   | 1.0 ± 0.2     | 3.2 ± 0.1   | 1.4 ± 0.2    | 3.1 ± 0.3     | E Africa M   | S Asia F     |
| 26  | 1.4 ± 0.3     | -1.1 ± 0.1  | -0.5 ± 0.1    | -1.9 ± 0.1  | -0.9 ± 0.2   | 2.4 ± 0.3     | 4.5 ± 0.2     | 1.1 ± 0.1   | 0.8 ± 0.2     | 2.9 ± 0.1   | 1.3 ± 0.2    | 2.9 ± 0.3     | E Africa M   | S Asia F     |
| 27  | 1.3 ± 0.3     | -1.1 ± 0.1  | -0.5 ± 0.1    | -2.3 ± 0.1  | -1.0 ± 0.2   | 2.2 ± 0.3     | 4.3 ± 0.2     | 1.1 ± 0.1   | 1.0 ± 0.2     | 2.6 ± 0.1   | 1.4 ± 0.2    | 2.7 ± 0.2     | E Africa M   | S Asia F     |
| 28  | 0.9 ± 0.3     | -1.3 ± 0.1  | -0.6 ± 0.1    | -2.6 ± 0.1  | -1.1 ± 0.2   | 1.9 ± 0.3     | 4.1 ± 0.2     | 1.0 ± 0.1   | 1.0 ± 0.2     | 2.3 ± 0.1   | 1.3 ± 0.2    | 2.4 ± 0.2     | E Africa M   | S Asia F     |
| 29  | 1.0 ± 0.3     | -1.3 ± 0.1  | -0.4 ± 0.1    | -2.8 ± 0.1  | -1.1 ± 0.2   | 1.6 ± 0.3     | 4.0 ± 0.2     | 0.9 ± 0.1   | 0.8 ± 0.2     | 1.93 ± 0.10 | 1.2 ± 0.2    | 2.6 ± 0.2     | E Africa M   | S Asia F     |
| 30  | 0.7 ± 0.3     | -1.3 ± 0.1  | -0.5 ± 0.2    | -3.1 ± 0.1  | -1.1 ± 0.2   | 1.3 ± 0.3     | 3.8 ± 0.2     | 0.8 ± 0.1   | 0.7 ± 0.2     | 1.63 ± 0.09 | 1.2 ± 0.2    | 2.2 ± 0.2     | E Africa M   | S Asia F     |
| 31  | 0.4 ± 0.3     | -1.4 ± 0.1  | -0.3 ± 0.2    | -3.5 ± 0.1  | -1.3 ± 0.2   | 1.1 ± 0.3     | 3.7 ± 0.2     | 0.8 ± 0.1   | 0.8 ± 0.2     | 1.42 ± 0.10 | 1.1 ± 0.2    | 2.0 ± 0.2     | E Africa M   | S Asia F     |
| 32  | 0.6 ± 0.3     | -1.4 ± 0.1  | -0.4 ± 0.2    | -3.6 ± 0.1  | -1.4 ± 0.2   | 1.0 ± 0.3     | 3.3 ± 0.2     | 0.9 ± 0.1   | 0.8 ± 0.2     | 1.24 ± 0.10 | 1.3 ± 0.2    | 1.8 ± 0.2     | E Africa M   | S Asia F     |
| 33  | 0.2 ± 0.3     | -1.5 ± 0.1  | -0.4 ± 0.2    | -3.8 ± 0.1  | -1.5 ± 0.2   | 0.8 ± 0.3     | 3.2 ± 0.2     | 0.6 ± 0.1   | 1.0 ± 0.2     | 0.95 ± 0.09 | 1.0 ± 0.2    | 1.6 ± 0.2     | E Africa M   | S Asia F     |
| 34  | -0.1 ± 0.3    | -1.4 ± 0.1  | -0.3 ± 0.2    | -3.9 ± 0.1  | -1.6 ± 0.2   | 0.5 ± 0.3     | 2.7 ± 0.3     | 0.6 ± 0.1   | 0.8 ± 0.2     | 0.8 ± 0.1   | 1.0 ± 0.3    | 1.4 ± 0.2     | E Africa M   | S Asia F     |
| 35  | 0.0 ± 0.3     | -1.6 ± 0.1  | -0.2 ± 0.2    | -4.2 ± 0.1  | -1.7 ± 0.2   | 0.3 ± 0.3     | 2.7 ± 0.2     | 0.7 ± 0.1   | 0.9 ± 0.2     | 0.6 ± 0.1   | 1.0 ± 0.3    | 1.2 ± 0.2     | E Africa M   | S Asia F     |
| 36  | -0.0 ± 0.4    | -1.6 ± 0.1  | 0.0 ± 0.2     | -4.3 ± 0.1  | -1.5 ± 0.2   | 0.3 ± 0.3     | 2.4 ± 0.3     | 0.6 ± 0.1   | 1.0 ± 0.2     | 0.4 ± 0.1   | 1.0 ± 0.3    | 0.8 ± 0.2     | E Africa M   | S Asia F     |
| 37  | -0.3 ± 0.4    | -1.5 ± 0.1  | 0.0 ± 0.2     | -4.2 ± 0.2  | -1.6 ± 0.2   | 0.3 ± 0.4     | 2.2 ± 0.3     | 0.5 ± 0.1   | 0.8 ± 0.2     | 0.4 ± 0.1   | 0.8 ± 0.2    | 0.5 ± 0.2     | E Africa M   | S Asia F     |
| 38  | 0.2 ± 0.4     | -1.7 ± 0.1  | 0.2 ± 0.2     | -4.2 ± 0.2  | -1.5 ± 0.2   | -0.3 ± 0.4    | 1.8 ± 0.3     | 0.5 ± 0.1   | 0.8 ± 0.2     | 0.1 ± 0.1   | 1.3 ± 0.3    | 0.3 ± 0.2     | E Africa M   | S Asia F     |
| 39  | -0.1 ± 0.4    | -1.6 ± 0.1  | 0.2 ± 0.2     | -4.4 ± 0.2  | -1.3 ± 0.2   | -0.0 ± 0.4    | 2.1 ± 0.3     | 0.5 ± 0.1   | 0.9 ± 0.2     | 0.1 ± 0.1   | 1.2 ± 0.2    | 0.1 ± 0.2     | E Africa M   | S Asia F     |
| 40  | -0.4 ± 0.4    | -1.7 ± 0.1  | 0.4 ± 0.2     | -4.2 ± 0.2  | -1.5 ± 0.2   | -0.5 ± 0.4    | 1.7 ± 0.3     | 0.4 ± 0.1   | 1.0 ± 0.2     | -0.1 ± 0.1  | 1.2 ± 0.2    | -0.2 ± 0.3    | E Africa M   | S Asia F     |
| 41  | -0.7 ± 0.4    | -1.6 ± 0.1  | 0.5 ± 0.2     | -4.1 ± 0.2  | -1.6 ± 0.2   | -1.0 ± 0.4    | 1.9 ± 0.3     | 0.4 ± 0.1   | 1.0 ± 0.2     | -0.1 ± 0.1  | 1.0 ± 0.2    | -0.4 ± 0.3    | E Africa M   | S Asia F     |
| 42  | -0.5 ± 0.4    | -1.5 ± 0.1  | 0.7 ± 0.2     | -3.5 ± 0.2  | -1.4 ± 0.2   | -0.8 ± 0.4    | 1.5 ± 0.3     | 0.5 ± 0.1   | 1.1 ± 0.2     | -0.3 ± 0.1  | 1.3 ± 0.2    | -0.6 ± 0.3    | E Africa M   | S Asia F     |
| 43  | -0.1 ± 0.5    | -1.7 ± 0.1  | 0.7 ± 0.2     | -3.7 ± 0.2  | -1.6 ± 0.2   | -1.1 ± 0.4    | 1.5 ± 0.3     | 0.3 ± 0.1   | 0.9 ± 0.2     | -0.2 ± 0.1  | 1.2 ± 0.2    | -1.1 ± 0.3    | E Africa M   | S Asia F     |
| 44  | -0.4 ± 0.5    | -1.8 ± 0.1  | 0.4 ± 0.2     | -3.5 ± 0.2  | -1.3 ± 0.2   | -1.3 ± 0.4    | 0.9 ± 0.3     | 0.3 ± 0.1   | 1.3 ± 0.2     | -0.3 ± 0.2  | 1.2 ± 0.3    | -1.0 ± 0.3    | E Europe M   | S Asia F     |
| 45  | -0.7 ± 0.5    | -1.8 ± 0.1  | 0.4 ± 0.2     | -3.3 ± 0.2  | -1.5 ± 0.2   | -1.0 ± 0.4    | 0.6 ± 0.4     | 0.1 ± 0.1   | 0.9 ± 0.2     | -0.2 ± 0.2  | 1.1 ± 0.2    | -1.5 ± 0.3    | S E Asia M   | S Asia F     |
| 46  | 0.2 ± 0.5     | -2.1 ± 0.1  | 0.5 ± 0.3     | -2.9 ± 0.2  | -1.7 ± 0.2   | -1.1 ± 0.4    | 0.8 ± 0.4     | -0.0 ± 0.1  | 1.0 ± 0.3     | -0.1 ± 0.2  | 1.0 ± 0.2    | -1.7 ± 0.3    | S E Asia M   | S Asia F     |
| 47  | 0.0 ± 0.5     | -2.0 ± 0.1  | 0.5 ± 0.3     | -2.9 ± 0.3  | -1.7 ± 0.2   | -1.4 ± 0.4    | 1.0 ± 0.4     | -0.2 ± 0.1  | 1.1 ± 0.3     | -0.0 ± 0.2  | 1.0 ± 0.2    | -1.6 ± 0.3    | E Europe M   | S Asia F     |
| 48  | -0.5 ± 0.6    | -2.1 ± 0.1  | 0.6 ± 0.2     | -2.6 ± 0.3  | -1.7 ± 0.2   | -1.5 ± 0.4    | 0.2 ± 0.4     | -0.2 ± 0.1  | 1.1 ± 0.3     | -0.1 ± 0.2  | 0.9 ± 0.3    | -2.3 ± 0.3    | E Europe M   | S Asia F     |
| 49  | -0.0 ± 0.6    | -2.3 ± 0.1  | 0.4 ± 0.2     | -2.4 ± 0.3  | -1.7 ± 0.2   | -2.1 ± 0.5    | -0.0 ± 0.4    | -0.3 ± 0.1  | 0.9 ± 0.3     | -0.2 ± 0.2  | 0.9 ± 0.2    | -2.2 ± 0.3    | E Europe M   | S Asia F     |
| 50  | -0.6 ± 0.7    | -2.3 ± 0.1  | 0.6 ± 0.3     | -2.2 ± 0.2  | -1.8 ± 0.2   | -1.6 ± 0.4    | -0.2 ± 0.4    | -0.5 ± 0.1  | 0.9 ± 0.3     | -0.2 ± 0.2  | 1.0 ± 0.3    | -2.7 ± 0.3    | S E Asia M   | W Africa M   |
| 51  | -0.3 ± 0.5    | -2.4 ± 0.1  | 0.5 ± 0.3     | -2.3 ± 0.2  | -1.9 ± 0.2   | -1.8 ± 0.5    | -0.8 ± 0.5    | -0.6 ± 0.1  | 1.1 ± 0.3     | -0.2 ± 0.2  | 0.7 ± 0.3    | -3.0 ± 0.3    | E Europe M   | W Africa M   |
| 52  | -0.7 ± 0.6    | -2.5 ± 0.1  | 0.7 ± 0.3     | -2.4 ± 0.2  | -1.7 ± 0.2   | -2.4 ± 0.5    | -0.5 ± 0.5    | -0.9 ± 0.1  | 1.0 ± 0.3     | -0.3 ± 0.2  | 0.4 ± 0.3    | -3.1 ± 0.4    | E Europe M   | W Africa M   |

**Table 41.** Mean raw error for Application-type images for algorithm dermalog-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 2.2 ± 0.4     | 4.4 ± 0.2   | 5.0 ± 0.3     | 4.1 ± 0.2   | 4.9 ± 0.4    | 3.0 ± 0.4     | 1.8 ± 0.3     | 3.3 ± 0.1   | 2.4 ± 0.3     | 2.6 ± 0.2    | 3.1 ± 0.3    | 2.4 ± 0.3     | E Europe F   | E Africa M   |
| 15  | 2.7 ± 0.5     | 4.4 ± 0.2   | 5.3 ± 0.3     | 5.0 ± 0.3   | 4.9 ± 0.4    | 2.7 ± 0.4     | 1.3 ± 0.3     | 3.6 ± 0.1   | 2.6 ± 0.3     | 3.5 ± 0.2    | 3.6 ± 0.3    | 2.7 ± 0.3     | E Europe F   | E Africa M   |
| 16  | 2.0 ± 0.5     | 4.3 ± 0.2   | 5.2 ± 0.3     | 4.7 ± 0.3   | 5.1 ± 0.4    | 2.6 ± 0.5     | 2.2 ± 0.3     | 3.6 ± 0.1   | 2.9 ± 0.3     | 3.9 ± 0.2    | 4.1 ± 0.3    | 3.0 ± 0.4     | E Europe F   | E Africa F   |
| 17  | 2.1 ± 0.5     | 3.9 ± 0.2   | 5.4 ± 0.4     | 4.8 ± 0.3   | 4.6 ± 0.3    | 2.2 ± 0.4     | 2.2 ± 0.4     | 3.2 ± 0.1   | 2.4 ± 0.3     | 3.9 ± 0.2    | 3.6 ± 0.3    | 3.2 ± 0.4     | E Europe F   | E Africa F   |
| 18  | 1.0 ± 0.4     | 3.2 ± 0.1   | 4.3 ± 0.3     | 4.0 ± 0.2   | 3.8 ± 0.2    | 1.5 ± 0.4     | 2.3 ± 0.3     | 2.4 ± 0.1   | 1.9 ± 0.2     | 3.5 ± 0.2    | 3.0 ± 0.2    | 2.8 ± 0.3     | E Europe F   | E Africa F   |
| 19  | 1.5 ± 0.4     | 2.7 ± 0.1   | 4.1 ± 0.3     | 3.8 ± 0.2   | 3.3 ± 0.2    | 1.5 ± 0.3     | 2.8 ± 0.3     | 2.0 ± 0.1   | 1.7 ± 0.2     | 3.5 ± 0.2    | 2.5 ± 0.2    | 2.8 ± 0.3     | E Europe F   | E Africa F   |
| 20  | 1.0 ± 0.4     | 2.4 ± 0.2   | 3.8 ± 0.3     | 3.5 ± 0.2   | 2.9 ± 0.2    | 1.0 ± 0.3     | 2.8 ± 0.4     | 1.7 ± 0.1   | 1.6 ± 0.2     | 3.5 ± 0.1    | 2.2 ± 0.2    | 2.7 ± 0.3     | E Europe F   | E Africa F   |
| 21  | 0.6 ± 0.4     | 1.9 ± 0.1   | 3.3 ± 0.2     | 3.2 ± 0.2   | 2.5 ± 0.2    | 0.7 ± 0.4     | 2.7 ± 0.3     | 1.1 ± 0.1   | 1.0 ± 0.2     | 3.1 ± 0.1    | 1.8 ± 0.2    | 2.6 ± 0.3     | E Europe F   | E Africa F   |
| 22  | 0.7 ± 0.4     | 1.2 ± 0.1   | 3.1 ± 0.2     | 2.8 ± 0.1   | 1.8 ± 0.2    | 0.0 ± 0.3     | 3.2 ± 0.3     | 0.7 ± 0.1   | 1.0 ± 0.2     | 2.8 ± 0.1    | 1.4 ± 0.2    | 2.3 ± 0.3     | E Africa M   | W Africa F   |
| 23  | 0.2 ± 0.3     | 0.9 ± 0.1   | 2.7 ± 0.2     | 2.6 ± 0.1   | 1.5 ± 0.2    | -0.5 ± 0.3    | 2.9 ± 0.3     | 0.1 ± 0.1   | 0.6 ± 0.2     | 2.5 ± 0.1    | 0.9 ± 0.2    | 2.3 ± 0.3     | E Africa M   | W Africa F   |
| 24  | 0.0 ± 0.3     | 0.4 ± 0.1   | 2.3 ± 0.2     | 2.2 ± 0.1   | 1.1 ± 0.1    | -1.0 ± 0.3    | 2.7 ± 0.3     | -0.2 ± 0.1  | 0.4 ± 0.2     | 2.3 ± 0.1    | 0.4 ± 0.2    | 2.0 ± 0.3     | E Africa M   | W Africa F   |
| 25  | -0.7 ± 0.3    | -0.0 ± 0.1  | 1.9 ± 0.2     | 1.9 ± 0.1   | 0.5 ± 0.1    | -1.4 ± 0.3    | 2.8 ± 0.3     | -0.7 ± 0.1  | -0.1 ± 0.2    | 1.9 ± 0.1    | 0.4 ± 0.2    | 1.5 ± 0.3     | E Africa M   | W Africa F   |
| 26  | -0.7 ± 0.3    | -0.4 ± 0.1  | 1.7 ± 0.2     | 1.3 ± 0.1   | 0.4 ± 0.1    | -1.7 ± 0.3    | 2.3 ± 0.3     | -1.0 ± 0.1  | -0.2 ± 0.2    | 1.6 ± 0.1    | -0.1 ± 0.2   | 1.1 ± 0.2     | E Africa M   | W Africa F   |
| 27  | -1.0 ± 0.3    | -0.7 ± 0.1  | 1.4 ± 0.1     | 0.8 ± 0.1   | -0.1 ± 0.1   | -2.1 ± 0.3    | 2.1 ± 0.3     | -1.3 ± 0.1  | -0.2 ± 0.2    | 1.2 ± 0.1    | -0.1 ± 0.2   | 0.8 ± 0.2     | E Africa M   | W Africa F   |
| 28  | -1.2 ± 0.3    | -1.1 ± 0.1  | 1.2 ± 0.1     | 0.5 ± 0.1   | -0.5 ± 0.2   | -2.4 ± 0.3    | 1.6 ± 0.3     | -1.6 ± 0.1  | -0.4 ± 0.2    | 0.8 ± 0.1    | -0.5 ± 0.2   | 0.4 ± 0.2     | E Africa M   | W Africa F   |
| 29  | -1.7 ± 0.3    | -1.4 ± 0.1  | 0.9 ± 0.2     | 0.2 ± 0.1   | -0.8 ± 0.2   | -2.7 ± 0.3    | 1.5 ± 0.3     | -1.8 ± 0.1  | -0.8 ± 0.2    | 0.46 ± 0.10  | -0.9 ± 0.2   | 0.2 ± 0.2     | E Africa M   | W Africa F   |
| 30  | -1.9 ± 0.3    | -1.6 ± 0.1  | 0.4 ± 0.2     | -0.3 ± 0.1  | -1.0 ± 0.2   | -3.1 ± 0.3    | 1.4 ± 0.3     | -2.1 ± 0.1  | -1.0 ± 0.2    | 0.04 ± 0.10  | -1.1 ± 0.2   | -0.1 ± 0.2    | E Africa M   | W Africa F   |
| 31  | -2.4 ± 0.3    | -1.9 ± 0.1  | 0.3 ± 0.2     | -0.7 ± 0.1  | -1.4 ± 0.2   | -3.5 ± 0.3    | 0.9 ± 0.3     | -2.2 ± 0.1  | -1.2 ± 0.2    | -0.31 ± 0.10 | -1.4 ± 0.2   | -0.4 ± 0.2    | E Africa M   | W Africa F   |
| 32  | -2.4 ± 0.4    | -2.1 ± 0.1  | 0.1 ± 0.2     | -1.1 ± 0.1  | -1.6 ± 0.2   | -3.5 ± 0.3    | 0.5 ± 0.3     | -2.4 ± 0.1  | -1.4 ± 0.2    | -0.5 ± 0.1   | -1.4 ± 0.2   | -0.8 ± 0.2    | E Africa M   | W Africa F   |
| 33  | -3.0 ± 0.4    | -2.3 ± 0.1  | -0.0 ± 0.2    | -1.4 ± 0.1  | -1.8 ± 0.2   | -3.9 ± 0.3    | 0.7 ± 0.3     | -2.7 ± 0.1  | -1.3 ± 0.2    | -1.0 ± 0.1   | -1.6 ± 0.3   | -0.8 ± 0.2    | E Africa M   | W Africa F   |
| 34  | -2.8 ± 0.4    | -2.4 ± 0.1  | -0.4 ± 0.2    | -1.7 ± 0.1  | -1.9 ± 0.2   | -4.2 ± 0.3    | 0.0 ± 0.3     | -2.8 ± 0.1  | -1.7 ± 0.2    | -1.3 ± 0.1   | -1.6 ± 0.3   | -1.2 ± 0.2    | E Africa M   | W Africa F   |
| 35  | -3.0 ± 0.4    | -2.6 ± 0.1  | -0.7 ± 0.2    | -2.1 ± 0.1  | -2.3 ± 0.2   | -4.3 ± 0.3    | 0.1 ± 0.3     | -3.0 ± 0.2  | -1.8 ± 0.2    | -1.6 ± 0.1   | -2.1 ± 0.3   | -1.4 ± 0.3    | E Africa M   | W Africa F   |
| 36  | -2.9 ± 0.4    | -2.7 ± 0.1  | -0.6 ± 0.2    | -2.4 ± 0.1  | -2.4 ± 0.2   | -4.4 ± 0.4    | -0.3 ± 0.3    | -3.2 ± 0.1  | -1.7 ± 0.3    | -2.0 ± 0.1   | -2.0 ± 0.3   | -1.6 ± 0.3    | E Africa M   | W Africa F   |
| 37  | -3.4 ± 0.4    | -2.9 ± 0.1  | -0.9 ± 0.2    | -2.7 ± 0.1  | -2.5 ± 0.2   | -4.5 ± 0.4    | -0.1 ± 0.3    | -3.1 ± 0.2  | -1.9 ± 0.3    | -2.3 ± 0.1   | -2.3 ± 0.3   | -1.8 ± 0.3    | E Africa M   | W Africa F   |
| 38  | -3.1 ± 0.5    | -3.1 ± 0.1  | -1.0 ± 0.2    | -2.9 ± 0.1  | -2.4 ± 0.2   | -4.7 ± 0.4    | -0.5 ± 0.4    | -3.3 ± 0.2  | -1.9 ± 0.3    | -2.6 ± 0.1   | -1.5 ± 0.3   | -1.7 ± 0.3    | E Africa M   | W Africa F   |
| 39  | -3.3 ± 0.5    | -3.3 ± 0.1  | -1.3 ± 0.2    | -3.2 ± 0.2  | -2.6 ± 0.2   | -4.5 ± 0.4    | -0.4 ± 0.4    | -3.4 ± 0.2  | -1.6 ± 0.3    | -2.7 ± 0.1   | -1.8 ± 0.3   | -2.2 ± 0.3    | E Africa M   | W Africa F   |
| 40  | -3.1 ± 0.5    | -3.4 ± 0.1  | -1.2 ± 0.2    | -3.2 ± 0.2  | -2.8 ± 0.2   | -5.0 ± 0.4    | -0.4 ± 0.4    | -3.5 ± 0.2  | -1.9 ± 0.3    | -2.9 ± 0.1   | -1.8 ± 0.3   | -2.5 ± 0.3    | E Africa M   | W Africa F   |
| 41  | -3.8 ± 0.5    | -3.5 ± 0.1  | -1.4 ± 0.2    | -3.5 ± 0.2  | -2.9 ± 0.2   | -5.2 ± 0.5    | -0.6 ± 0.4    | -3.3 ± 0.2  | -1.6 ± 0.3    | -2.9 ± 0.2   | -1.7 ± 0.3   | -2.2 ± 0.4    | E Africa M   | W Africa F   |
| 42  | -2.9 ± 0.5    | -3.5 ± 0.1  | -1.4 ± 0.3    | -3.3 ± 0.2  | -3.0 ± 0.2   | -5.0 ± 0.5    | -0.5 ± 0.4    | -3.3 ± 0.2  | -1.3 ± 0.3    | -3.1 ± 0.2   | -1.4 ± 0.3   | -2.3 ± 0.3    | E Africa M   | W Africa F   |
| 43  | -3.2 ± 0.5    | -3.7 ± 0.1  | -1.3 ± 0.3    | -3.8 ± 0.2  | -3.2 ± 0.2   | -5.2 ± 0.5    | -0.3 ± 0.4    | -3.0 ± 0.2  | -1.5 ± 0.3    | -3.1 ± 0.2   | -1.2 ± 0.3   | -2.8 ± 0.4    | E Africa M   | W Africa F   |
| 44  | -2.9 ± 0.6    | -4.0 ± 0.1  | -1.8 ± 0.3    | -3.6 ± 0.2  | -2.7 ± 0.2   | -5.2 ± 0.5    | -1.1 ± 0.4    | -2.9 ± 0.2  | -1.0 ± 0.3    | -3.2 ± 0.2   | -1.0 ± 0.3   | -2.6 ± 0.4    | E Europe M   | W Africa F   |
| 45  | -3.6 ± 0.6    | -3.9 ± 0.1  | -1.9 ± 0.3    | -3.8 ± 0.2  | -3.0 ± 0.3   | -5.4 ± 0.5    | -1.3 ± 0.5    | -2.9 ± 0.2  | -0.9 ± 0.4    | -2.9 ± 0.2   | -1.1 ± 0.3   | -2.9 ± 0.4    | E Europe M   | W Africa F   |
| 46  | -2.8 ± 0.6    | -4.1 ± 0.1  | -2.0 ± 0.3    | -3.6 ± 0.2  | -3.3 ± 0.3   | -5.0 ± 0.5    | -1.2 ± 0.5    | -2.8 ± 0.2  | -0.8 ± 0.4    | -2.8 ± 0.2   | -1.1 ± 0.3   | -3.2 ± 0.4    | E Europe M   | W Africa F   |
| 47  | -2.7 ± 0.7    | -4.0 ± 0.1  | -1.9 ± 0.3    | -3.6 ± 0.2  | -3.1 ± 0.3   | -5.1 ± 0.5    | -0.7 ± 0.5    | -2.4 ± 0.2  | -0.8 ± 0.3    | -2.9 ± 0.2   | -0.6 ± 0.3   | -2.7 ± 0.4    | S E Asia M   | W Africa F   |
| 48  | -3.0 ± 0.7    | -4.1 ± 0.1  | -2.0 ± 0.3    | -3.7 ± 0.3  | -3.2 ± 0.3   | -5.0 ± 0.5    | -1.2 ± 0.5    | -2.5 ± 0.2  | -0.8 ± 0.4    | -2.9 ± 0.2   | -0.9 ± 0.3   | -3.1 ± 0.5    | E Europe M   | W Africa F   |
| 49  | -3.1 ± 0.7    | -4.2 ± 0.2  | -2.1 ± 0.3    | -3.7 ± 0.2  | -3.0 ± 0.3   | -5.4 ± 0.6    | -1.6 ± 0.6    | -2.4 ± 0.2  | -0.7 ± 0.3    | -2.8 ± 0.2   | -1.0 ± 0.3   | -3.4 ± 0.4    | E Europe M   | W Africa F   |
| 50  | -3.5 ± 0.8    | -4.3 ± 0.2  | -2.1 ± 0.3    | -3.7 ± 0.2  | -3.3 ± 0.3   | -5.0 ± 0.6    | -1.9 ± 0.5    | -2.2 ± 0.2  | -0.9 ± 0.4    | -2.9 ± 0.3   | -0.5 ± 0.3   | -3.2 ± 0.4    | S E Asia M   | W Africa F   |
| 51  | -2.9 ± 0.6    | -4.5 ± 0.2  | -2.2 ± 0.3    | -3.9 ± 0.2  | -3.3 ± 0.3   | -5.2 ± 0.6    | -1.7 ± 0.6    | -2.2 ± 0.2  | -0.8 ± 0.3    | -2.8 ± 0.3   | -0.8 ± 0.3   | -3.4 ± 0.5    | S E Asia M   | W Africa F   |
| 52  | -3.0 ± 0.7    | -4.4 ± 0.2  | -2.2 ± 0.3    | -4.3 ± 0.2  | -3.1 ± 0.3   | -5.8 ± 0.6    | -1.8 ± 0.6    | -2.2 ± 0.2  | -1.1 ± 0.4    | -2.9 ± 0.3   | -1.1 ± 0.3   | -3.6 ± 0.5    | S E Asia M   | W Africa F   |

**Table 42.** Mean raw error for Application-type images for algorithm dermalog-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 1.2 ± 0.4     | 4.7 ± 0.2    | 4.5 ± 0.4     | 3.2 ± 0.3   | 4.3 ± 0.4    | 2.7 ± 0.5     | 1.1 ± 0.3     | 1.6 ± 0.2   | 0.3 ± 0.3     | 1.2 ± 0.2   | 1.7 ± 0.3    | 2.3 ± 0.4     | E Asia F     | E Europe M   |
| 15  | 1.0 ± 0.5     | 5.1 ± 0.2    | 5.3 ± 0.4     | 3.8 ± 0.3   | 4.9 ± 0.5    | 2.6 ± 0.6     | 1.2 ± 0.3     | 2.2 ± 0.2   | 0.8 ± 0.4     | 2.0 ± 0.2   | 2.1 ± 0.3    | 2.5 ± 0.4     | E Europe F   | E Europe M   |
| 16  | 0.7 ± 0.5     | 5.3 ± 0.2    | 5.3 ± 0.4     | 4.0 ± 0.3   | 5.2 ± 0.4    | 2.6 ± 0.5     | 2.2 ± 0.3     | 2.4 ± 0.2   | 1.5 ± 0.4     | 2.4 ± 0.2   | 2.8 ± 0.3    | 2.7 ± 0.4     | E Asia F     | E Africa F   |
| 17  | 0.9 ± 0.5     | 5.1 ± 0.2    | 4.9 ± 0.5     | 4.1 ± 0.3   | 5.1 ± 0.4    | 2.4 ± 0.5     | 2.0 ± 0.3     | 2.4 ± 0.2   | 1.7 ± 0.3     | 2.7 ± 0.2   | 2.8 ± 0.3    | 2.7 ± 0.4     | E Asia F     | E Africa F   |
| 18  | 0.3 ± 0.4     | 4.6 ± 0.2    | 4.8 ± 0.4     | 3.4 ± 0.2   | 4.5 ± 0.3    | 1.7 ± 0.4     | 2.0 ± 0.3     | 1.8 ± 0.1   | 1.7 ± 0.3     | 2.4 ± 0.2   | 2.3 ± 0.3    | 2.4 ± 0.3     | E Europe F   | E Africa F   |
| 19  | 0.3 ± 0.4     | 4.1 ± 0.2    | 4.5 ± 0.3     | 3.5 ± 0.2   | 4.3 ± 0.2    | 1.5 ± 0.4     | 2.2 ± 0.3     | 1.7 ± 0.1   | 1.5 ± 0.3     | 2.5 ± 0.2   | 2.0 ± 0.2    | 2.3 ± 0.3     | E Europe F   | E Africa F   |
| 20  | -0.4 ± 0.4    | 4.0 ± 0.1    | 4.1 ± 0.3     | 2.9 ± 0.2   | 4.1 ± 0.2    | 1.5 ± 0.4     | 2.2 ± 0.3     | 1.4 ± 0.1   | 1.7 ± 0.3     | 2.5 ± 0.2   | 1.9 ± 0.2    | 2.0 ± 0.3     | E Europe F   | E Africa F   |
| 21  | -0.7 ± 0.4    | 3.5 ± 0.1    | 3.8 ± 0.3     | 2.5 ± 0.2   | 3.5 ± 0.2    | 0.9 ± 0.4     | 1.8 ± 0.3     | 1.1 ± 0.1   | 1.6 ± 0.3     | 2.4 ± 0.2   | 1.5 ± 0.2    | 2.0 ± 0.3     | E Europe F   | E Africa F   |
| 22  | -0.8 ± 0.4    | 3.0 ± 0.1    | 3.4 ± 0.2     | 1.7 ± 0.2   | 3.0 ± 0.2    | 0.3 ± 0.4     | 2.0 ± 0.3     | 0.6 ± 0.1   | 1.8 ± 0.3     | 1.9 ± 0.2   | 1.0 ± 0.2    | 1.5 ± 0.3     | E Europe F   | E Africa F   |
| 23  | -1.6 ± 0.4    | 2.5 ± 0.1    | 3.3 ± 0.2     | 1.7 ± 0.2   | 2.6 ± 0.2    | -0.2 ± 0.3    | 1.4 ± 0.3     | 0.3 ± 0.1   | 1.3 ± 0.3     | 1.7 ± 0.2   | 0.5 ± 0.2    | 1.3 ± 0.3     | E Europe F   | E Africa F   |
| 24  | -1.8 ± 0.4    | 2.1 ± 0.1    | 2.9 ± 0.2     | 1.1 ± 0.1   | 2.1 ± 0.1    | -0.9 ± 0.3    | 1.2 ± 0.3     | -0.1 ± 0.1  | 1.2 ± 0.2     | 1.5 ± 0.2   | 0.2 ± 0.2    | 1.0 ± 0.3     | E Europe F   | E Africa F   |
| 25  | -2.3 ± 0.3    | 1.57 ± 0.10  | 2.5 ± 0.2     | 0.5 ± 0.1   | 1.5 ± 0.1    | -1.5 ± 0.3    | 0.9 ± 0.3     | -0.4 ± 0.1  | 1.0 ± 0.2     | 1.1 ± 0.1   | -0.1 ± 0.2   | 0.4 ± 0.3     | E Europe F   | E Africa F   |
| 26  | -2.7 ± 0.3    | 1.05 ± 0.09  | 2.1 ± 0.2     | -0.1 ± 0.1  | 1.1 ± 0.1    | -1.9 ± 0.3    | 0.6 ± 0.3     | -0.8 ± 0.1  | 0.6 ± 0.2     | 0.5 ± 0.1   | -0.6 ± 0.2   | -0.1 ± 0.2    | E Europe F   | E Africa F   |
| 27  | -2.9 ± 0.3    | 0.60 ± 0.09  | 1.8 ± 0.2     | -0.7 ± 0.1  | 0.6 ± 0.1    | -2.3 ± 0.3    | 0.1 ± 0.3     | -0.9 ± 0.1  | 0.8 ± 0.2     | 0.1 ± 0.1   | -0.4 ± 0.2   | -0.5 ± 0.2    | E Europe F   | E Africa F   |
| 28  | -3.3 ± 0.3    | 0.11 ± 0.09  | 1.6 ± 0.2     | -1.2 ± 0.1  | 0.2 ± 0.1    | -2.8 ± 0.3    | -0.4 ± 0.2    | -1.3 ± 0.2  | 0.6 ± 0.2     | -0.4 ± 0.1  | -0.8 ± 0.2   | -0.9 ± 0.2    | E Europe F   | E Africa F   |
| 29  | -3.6 ± 0.3    | -0.27 ± 0.09 | 1.3 ± 0.2     | -1.7 ± 0.1  | -0.2 ± 0.1   | -3.0 ± 0.3    | -0.8 ± 0.3    | -1.5 ± 0.1  | 0.2 ± 0.2     | -1.0 ± 0.1  | -1.2 ± 0.2   | -1.2 ± 0.2    | E Europe F   | E Africa F   |
| 30  | -3.9 ± 0.3    | -0.67 ± 0.09 | 1.1 ± 0.2     | -2.2 ± 0.1  | -0.6 ± 0.2   | -3.8 ± 0.3    | -1.1 ± 0.3    | -1.9 ± 0.1  | 0.0 ± 0.2     | -1.4 ± 0.1  | -1.5 ± 0.2   | -1.7 ± 0.2    | E Europe F   | E Africa F   |
| 31  | -4.4 ± 0.3    | -1.1 ± 0.1   | 0.8 ± 0.2     | -2.7 ± 0.1  | -0.9 ± 0.2   | -4.4 ± 0.3    | -1.5 ± 0.3    | -2.0 ± 0.1  | -0.4 ± 0.2    | -1.8 ± 0.1  | -2.0 ± 0.3   | -2.1 ± 0.2    | E Europe F   | E Africa F   |
| 32  | -4.6 ± 0.4    | -1.4 ± 0.1   | 0.6 ± 0.2     | -3.1 ± 0.1  | -1.3 ± 0.2   | -4.5 ± 0.3    | -2.0 ± 0.3    | -2.4 ± 0.1  | -0.4 ± 0.2    | -2.2 ± 0.1  | -1.8 ± 0.3   | -2.6 ± 0.3    | E Europe F   | E Africa F   |
| 33  | -5.0 ± 0.4    | -1.8 ± 0.1   | 0.5 ± 0.2     | -3.5 ± 0.1  | -1.7 ± 0.2   | -4.8 ± 0.3    | -2.1 ± 0.3    | -2.8 ± 0.2  | -0.3 ± 0.2    | -2.8 ± 0.1  | -2.0 ± 0.3   | -2.8 ± 0.3    | E Europe F   | E Africa F   |
| 34  | -5.4 ± 0.4    | -1.9 ± 0.1   | 0.0 ± 0.2     | -3.8 ± 0.1  | -1.6 ± 0.2   | -5.4 ± 0.3    | -2.7 ± 0.3    | -2.9 ± 0.2  | -0.8 ± 0.2    | -3.1 ± 0.1  | -2.5 ± 0.3   | -3.2 ± 0.3    | E Europe F   | W Africa F   |
| 35  | -5.3 ± 0.4    | -2.3 ± 0.1   | -0.1 ± 0.2    | -4.2 ± 0.1  | -2.1 ± 0.2   | -5.8 ± 0.4    | -2.9 ± 0.3    | -3.1 ± 0.2  | -1.0 ± 0.2    | -3.5 ± 0.1  | -2.7 ± 0.3   | -3.7 ± 0.3    | E Europe F   | W Africa F   |
| 36  | -5.7 ± 0.4    | -2.3 ± 0.1   | -0.2 ± 0.2    | -4.7 ± 0.1  | -2.2 ± 0.2   | -6.0 ± 0.4    | -3.3 ± 0.4    | -3.4 ± 0.2  | -0.9 ± 0.2    | -3.9 ± 0.1  | -2.9 ± 0.3   | -4.1 ± 0.3    | E Europe F   | W Africa F   |
| 37  | -6.1 ± 0.5    | -2.7 ± 0.1   | -0.4 ± 0.2    | -4.9 ± 0.2  | -2.5 ± 0.2   | -6.4 ± 0.4    | -3.4 ± 0.4    | -3.4 ± 0.2  | -1.3 ± 0.3    | -4.3 ± 0.1  | -3.1 ± 0.3   | -4.2 ± 0.3    | E Europe F   | W Africa F   |
| 38  | -5.6 ± 0.5    | -2.9 ± 0.1   | -0.3 ± 0.2    | -5.2 ± 0.2  | -2.4 ± 0.2   | -6.4 ± 0.4    | -3.7 ± 0.4    | -3.7 ± 0.2  | -1.3 ± 0.2    | -4.6 ± 0.1  | -2.7 ± 0.3   | -4.5 ± 0.3    | E Europe F   | W Africa F   |
| 39  | -6.0 ± 0.5    | -3.0 ± 0.1   | -0.6 ± 0.3    | -5.5 ± 0.2  | -2.5 ± 0.2   | -6.7 ± 0.4    | -3.8 ± 0.4    | -3.8 ± 0.2  | -1.4 ± 0.3    | -4.8 ± 0.2  | -2.9 ± 0.3   | -4.9 ± 0.3    | E Europe F   | W Africa F   |
| 40  | -6.5 ± 0.6    | -3.1 ± 0.1   | -0.8 ± 0.3    | -5.5 ± 0.2  | -2.6 ± 0.3   | -7.2 ± 0.4    | -3.8 ± 0.4    | -3.9 ± 0.2  | -1.6 ± 0.3    | -5.1 ± 0.2  | -3.1 ± 0.3   | -5.4 ± 0.3    | E Europe F   | W Africa F   |
| 41  | -6.6 ± 0.6    | -3.2 ± 0.1   | -0.9 ± 0.3    | -5.6 ± 0.2  | -3.0 ± 0.3   | -7.5 ± 0.5    | -4.5 ± 0.4    | -3.9 ± 0.2  | -1.9 ± 0.3    | -5.3 ± 0.2  | -3.3 ± 0.3   | -5.3 ± 0.4    | E Europe F   | W Africa F   |
| 42  | -5.9 ± 0.6    | -3.2 ± 0.1   | -0.9 ± 0.3    | -5.5 ± 0.2  | -2.7 ± 0.3   | -7.3 ± 0.5    | -4.3 ± 0.5    | -3.9 ± 0.2  | -1.7 ± 0.3    | -5.5 ± 0.2  | -3.0 ± 0.3   | -5.6 ± 0.4    | E Europe F   | W Africa F   |
| 43  | -6.4 ± 0.6    | -3.4 ± 0.1   | -0.9 ± 0.3    | -5.8 ± 0.3  | -3.0 ± 0.3   | -7.5 ± 0.5    | -4.1 ± 0.5    | -3.7 ± 0.2  | -2.0 ± 0.3    | -5.5 ± 0.2  | -3.2 ± 0.3   | -6.1 ± 0.4    | E Europe F   | W Africa F   |
| 44  | -6.4 ± 0.7    | -3.6 ± 0.1   | -1.3 ± 0.3    | -5.6 ± 0.3  | -2.4 ± 0.3   | -7.6 ± 0.5    | -4.9 ± 0.5    | -3.8 ± 0.2  | -1.7 ± 0.3    | -5.7 ± 0.2  | -3.0 ± 0.3   | -5.9 ± 0.4    | E Europe F   | W Africa F   |
| 45  | -6.6 ± 0.7    | -3.5 ± 0.2   | -1.2 ± 0.3    | -5.6 ± 0.3  | -2.8 ± 0.3   | -7.7 ± 0.6    | -4.8 ± 0.5    | -4.0 ± 0.2  | -1.9 ± 0.3    | -5.7 ± 0.2  | -3.0 ± 0.4   | -6.3 ± 0.4    | E Europe F   | W Africa F   |
| 46  | -5.6 ± 0.7    | -3.7 ± 0.2   | -1.7 ± 0.3    | -5.4 ± 0.3  | -3.2 ± 0.3   | -7.9 ± 0.6    | -4.8 ± 0.5    | -4.0 ± 0.2  | -2.0 ± 0.3    | -5.3 ± 0.2  | -2.8 ± 0.4   | -6.5 ± 0.5    | E Europe F   | W Africa F   |
| 47  | -5.6 ± 0.8    | -3.5 ± 0.2   | -1.4 ± 0.4    | -5.4 ± 0.3  | -2.8 ± 0.3   | -7.6 ± 0.6    | -4.4 ± 0.5    | -3.6 ± 0.2  | -2.3 ± 0.3    | -5.5 ± 0.2  | -2.9 ± 0.4   | -6.3 ± 0.5    | E Europe F   | W Africa F   |
| 48  | -6.0 ± 0.9    | -3.5 ± 0.2   | -1.5 ± 0.3    | -5.4 ± 0.3  | -3.0 ± 0.3   | -7.5 ± 0.6    | -5.5 ± 0.6    | -3.9 ± 0.2  | -2.2 ± 0.4    | -5.5 ± 0.3  | -2.8 ± 0.4   | -6.8 ± 0.5    | E Europe F   | W Africa F   |
| 49  | -6.0 ± 0.9    | -3.7 ± 0.2   | -1.5 ± 0.3    | -5.2 ± 0.3  | -2.8 ± 0.3   | -7.8 ± 0.7    | -5.1 ± 0.6    | -3.7 ± 0.2  | -2.2 ± 0.4    | -5.5 ± 0.3  | -2.8 ± 0.4   | -6.8 ± 0.5    | E Europe F   | W Africa F   |
| 50  | -6.1 ± 0.9    | -3.7 ± 0.2   | -1.5 ± 0.3    | -5.3 ± 0.3  | -3.1 ± 0.3   | -7.5 ± 0.6    | -6.0 ± 0.7    | -3.8 ± 0.2  | -2.5 ± 0.4    | -5.8 ± 0.3  | -2.3 ± 0.4   | -7.1 ± 0.5    | E Europe F   | W Africa F   |
| 51  | -5.8 ± 0.8    | -3.8 ± 0.2   | -1.7 ± 0.3    | -5.2 ± 0.3  | -3.1 ± 0.3   | -8.0 ± 0.7    | -5.9 ± 0.7    | -3.6 ± 0.2  | -2.3 ± 0.4    | -6.0 ± 0.3  | -2.3 ± 0.4   | -6.5 ± 0.6    | E Europe F   | W Africa F   |
| 52  | -5.9 ± 0.9    | -3.9 ± 0.2   | -1.7 ± 0.3    | -5.6 ± 0.3  | -2.6 ± 0.3   | -8.4 ± 0.7    | -5.7 ± 0.7    | -3.6 ± 0.2  | -2.7 ± 0.4    | -6.0 ± 0.3  | -2.9 ± 0.4   | -7.6 ± 0.6    | E Europe F   | W Africa F   |

**Table 43.** Mean raw error for Application-type images for algorithm hzailu-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.1 ± 0.5     | 5.4 ± 0.3    | 8.1 ± 0.5     | 6.2 ± 0.3   | 5.5 ± 0.5    | 7.9 ± 0.7     | 4.5 ± 0.6     | 5.5 ± 0.2   | 4.8 ± 0.5     | 7.1 ± 0.2   | 5.6 ± 0.4    | 6.3 ± 1.0     | E Europe F   | E Africa M   |
| 15  | 5.1 ± 0.7     | 5.8 ± 0.3    | 8.4 ± 0.5     | 6.8 ± 0.3   | 5.8 ± 0.5    | 7.8 ± 0.8     | 5.0 ± 0.6     | 5.9 ± 0.2   | 5.9 ± 0.5     | 8.0 ± 0.3   | 6.5 ± 0.4    | 7.3 ± 0.9     | E Europe F   | E Africa M   |
| 16  | 4.8 ± 0.6     | 5.8 ± 0.3    | 8.2 ± 0.5     | 6.5 ± 0.3   | 6.5 ± 0.5    | 7.8 ± 0.8     | 7.6 ± 0.6     | 5.8 ± 0.2   | 6.5 ± 0.5     | 8.2 ± 0.2   | 7.2 ± 0.4    | 8.7 ± 0.9     | W Africa M   | E Africa F   |
| 17  | 5.0 ± 0.6     | 5.7 ± 0.2    | 8.3 ± 0.5     | 6.6 ± 0.3   | 6.1 ± 0.5    | 8.0 ± 0.7     | 7.8 ± 0.6     | 5.7 ± 0.2   | 6.4 ± 0.4     | 8.4 ± 0.2   | 7.3 ± 0.4    | 9.3 ± 0.7     | W Africa M   | E Africa F   |
| 18  | 4.9 ± 0.5     | 5.2 ± 0.2    | 7.8 ± 0.4     | 5.8 ± 0.2   | 5.7 ± 0.3    | 7.2 ± 0.6     | 8.2 ± 0.5     | 5.0 ± 0.1   | 6.0 ± 0.4     | 8.2 ± 0.2   | 6.4 ± 0.3    | 9.5 ± 0.6     | W Africa M   | E Africa F   |
| 19  | 4.6 ± 0.5     | 4.8 ± 0.2    | 7.4 ± 0.3     | 6.0 ± 0.2   | 5.6 ± 0.3    | 7.0 ± 0.6     | 8.6 ± 0.4     | 4.7 ± 0.1   | 5.9 ± 0.3     | 8.1 ± 0.2   | 5.8 ± 0.3    | 9.8 ± 0.6     | W Africa M   | E Africa F   |
| 20  | 4.0 ± 0.5     | 4.6 ± 0.2    | 7.1 ± 0.3     | 5.3 ± 0.2   | 5.1 ± 0.3    | 6.9 ± 0.5     | 8.8 ± 0.4     | 4.2 ± 0.1   | 6.2 ± 0.3     | 7.8 ± 0.2   | 5.8 ± 0.2    | 9.5 ± 0.5     | W Africa M   | E Africa F   |
| 21  | 3.7 ± 0.5     | 4.3 ± 0.2    | 6.7 ± 0.2     | 4.9 ± 0.2   | 4.9 ± 0.2    | 6.6 ± 0.5     | 8.4 ± 0.4     | 3.8 ± 0.1   | 6.2 ± 0.3     | 7.5 ± 0.2   | 5.3 ± 0.2    | 9.7 ± 0.5     | W Africa M   | E Africa F   |
| 22  | 3.4 ± 0.4     | 3.7 ± 0.2    | 6.3 ± 0.2     | 4.2 ± 0.2   | 4.3 ± 0.2    | 5.9 ± 0.5     | 8.6 ± 0.4     | 3.2 ± 0.1   | 6.0 ± 0.3     | 7.0 ± 0.2   | 4.6 ± 0.2    | 9.2 ± 0.4     | W Africa M   | E Asia M     |
| 23  | 2.7 ± 0.4     | 3.5 ± 0.1    | 5.9 ± 0.2     | 3.9 ± 0.2   | 4.0 ± 0.2    | 5.6 ± 0.4     | 7.7 ± 0.3     | 2.7 ± 0.1   | 5.6 ± 0.3     | 6.5 ± 0.1   | 4.1 ± 0.2    | 8.9 ± 0.4     | W Africa M   | E Asia M     |
| 24  | 2.0 ± 0.4     | 2.8 ± 0.1    | 5.4 ± 0.2     | 3.3 ± 0.1   | 3.5 ± 0.2    | 4.9 ± 0.4     | 7.4 ± 0.3     | 2.2 ± 0.1   | 5.3 ± 0.2     | 6.2 ± 0.1   | 3.5 ± 0.2    | 8.7 ± 0.3     | W Africa M   | E Africa F   |
| 25  | 1.9 ± 0.3     | 2.4 ± 0.1    | 4.8 ± 0.2     | 2.6 ± 0.1   | 2.9 ± 0.2    | 4.5 ± 0.4     | 7.2 ± 0.3     | 1.8 ± 0.1   | 5.0 ± 0.2     | 5.6 ± 0.1   | 3.1 ± 0.2    | 7.9 ± 0.3     | W Africa M   | E Asia M     |
| 26  | 1.3 ± 0.3     | 1.9 ± 0.1    | 4.3 ± 0.1     | 1.9 ± 0.1   | 2.5 ± 0.1    | 3.9 ± 0.4     | 6.8 ± 0.3     | 1.3 ± 0.1   | 4.6 ± 0.2     | 5.1 ± 0.1   | 2.5 ± 0.2    | 7.5 ± 0.3     | W Africa M   | E Asia M     |
| 27  | 1.2 ± 0.3     | 1.39 ± 0.10  | 3.8 ± 0.1     | 1.3 ± 0.1   | 2.0 ± 0.1    | 3.4 ± 0.3     | 6.2 ± 0.3     | 0.8 ± 0.1   | 4.5 ± 0.2     | 4.4 ± 0.1   | 2.5 ± 0.2    | 6.9 ± 0.3     | W Africa M   | E Asia M     |
| 28  | 0.5 ± 0.3     | 0.9 ± 0.1    | 3.2 ± 0.1     | 0.5 ± 0.1   | 1.5 ± 0.1    | 3.0 ± 0.4     | 5.5 ± 0.3     | 0.4 ± 0.1   | 4.1 ± 0.2     | 3.8 ± 0.1   | 1.8 ± 0.2    | 6.2 ± 0.3     | W Africa M   | E Asia M     |
| 29  | 0.3 ± 0.3     | 0.4 ± 0.1    | 2.7 ± 0.1     | 0.1 ± 0.1   | 1.0 ± 0.2    | 2.5 ± 0.3     | 5.3 ± 0.3     | 0.1 ± 0.1   | 3.5 ± 0.2     | 3.1 ± 0.1   | 1.3 ± 0.2    | 5.9 ± 0.2     | W Africa M   | S Asia F     |
| 30  | -0.1 ± 0.3    | -0.08 ± 0.10 | 2.1 ± 0.2     | -0.7 ± 0.1  | 0.7 ± 0.2    | 1.9 ± 0.3     | 4.7 ± 0.3     | -0.5 ± 0.1  | 3.1 ± 0.2     | 2.41 ± 0.10 | 1.1 ± 0.2    | 5.3 ± 0.2     | W Africa M   | S Asia F     |
| 31  | -0.7 ± 0.3    | -0.5 ± 0.1   | 1.8 ± 0.2     | -1.3 ± 0.1  | 0.1 ± 0.2    | 1.3 ± 0.3     | 4.1 ± 0.3     | -0.8 ± 0.1  | 2.6 ± 0.2     | 1.79 ± 0.10 | 0.4 ± 0.3    | 4.8 ± 0.2     | W Africa M   | S Asia F     |
| 32  | -1.0 ± 0.4    | -1.0 ± 0.1   | 1.2 ± 0.2     | -1.9 ± 0.1  | -0.5 ± 0.2   | 0.9 ± 0.3     | 3.4 ± 0.3     | -1.2 ± 0.1  | 2.8 ± 0.2     | 1.4 ± 0.1   | 0.5 ± 0.3    | 4.2 ± 0.2     | W Africa M   | S Asia F     |
| 33  | -1.8 ± 0.4    | -1.3 ± 0.1   | 0.9 ± 0.2     | -2.4 ± 0.1  | -0.7 ± 0.2   | 0.5 ± 0.4     | 3.4 ± 0.3     | -1.5 ± 0.2  | 2.3 ± 0.2     | 0.9 ± 0.1   | 0.2 ± 0.3    | 3.5 ± 0.2     | W Africa M   | S Asia F     |
| 34  | -1.9 ± 0.4    | -1.6 ± 0.1   | 0.3 ± 0.2     | -2.9 ± 0.1  | -1.2 ± 0.2   | -0.1 ± 0.4    | 2.3 ± 0.3     | -1.7 ± 0.2  | 1.7 ± 0.3     | 0.4 ± 0.1   | -0.2 ± 0.3   | 3.0 ± 0.3     | W Africa M   | S Asia F     |
| 35  | -2.2 ± 0.4    | -2.2 ± 0.1   | -0.1 ± 0.2    | -3.6 ± 0.1  | -1.8 ± 0.2   | -0.8 ± 0.4    | 2.1 ± 0.3     | -2.0 ± 0.2  | 1.5 ± 0.3     | -0.0 ± 0.1  | -0.4 ± 0.3   | 2.4 ± 0.3     | W Africa M   | S Asia F     |
| 36  | -2.5 ± 0.4    | -2.4 ± 0.1   | -0.4 ± 0.3    | -4.0 ± 0.2  | -1.9 ± 0.2   | -0.9 ± 0.4    | 1.5 ± 0.3     | -2.0 ± 0.2  | 1.5 ± 0.3     | -0.4 ± 0.1  | -0.4 ± 0.3   | 1.8 ± 0.3     | W Africa M   | S Asia F     |
| 37  | -3.1 ± 0.5    | -2.7 ± 0.1   | -0.7 ± 0.3    | -4.3 ± 0.2  | -2.4 ± 0.2   | -1.5 ± 0.4    | 1.3 ± 0.3     | -2.0 ± 0.2  | 1.4 ± 0.3     | -0.8 ± 0.1  | -0.7 ± 0.4   | 1.3 ± 0.3     | E Europe M   | S Asia F     |
| 38  | -2.8 ± 0.5    | -3.1 ± 0.1   | -0.7 ± 0.3    | -4.7 ± 0.2  | -2.5 ± 0.2   | -2.0 ± 0.4    | 0.6 ± 0.3     | -2.2 ± 0.2  | 1.1 ± 0.3     | -1.2 ± 0.1  | -0.4 ± 0.3   | 1.0 ± 0.3     | E Europe M   | S Asia F     |
| 39  | -3.3 ± 0.5    | -3.4 ± 0.1   | -1.5 ± 0.3    | -5.3 ± 0.2  | -2.9 ± 0.2   | -2.2 ± 0.4    | 0.6 ± 0.4     | -2.3 ± 0.2  | 1.3 ± 0.3     | -1.4 ± 0.1  | -0.4 ± 0.3   | 0.3 ± 0.3     | E Europe M   | S Asia F     |
| 40  | -3.8 ± 0.6    | -3.6 ± 0.1   | -1.1 ± 0.3    | -5.3 ± 0.2  | -3.2 ± 0.3   | -2.7 ± 0.4    | 0.0 ± 0.4     | -2.3 ± 0.2  | 1.1 ± 0.4     | -1.8 ± 0.2  | -0.7 ± 0.3   | -0.1 ± 0.3    | E Europe M   | S Asia F     |
| 41  | -4.4 ± 0.6    | -3.9 ± 0.1   | -1.7 ± 0.3    | -5.7 ± 0.2  | -3.3 ± 0.3   | -3.4 ± 0.5    | -0.1 ± 0.4    | -2.2 ± 0.2  | 1.1 ± 0.4     | -1.9 ± 0.2  | -0.4 ± 0.4   | -0.7 ± 0.3    | E Europe M   | S Asia F     |
| 42  | -3.6 ± 0.6    | -4.0 ± 0.1   | -1.4 ± 0.4    | -5.6 ± 0.3  | -3.6 ± 0.3   | -3.5 ± 0.5    | -0.8 ± 0.4    | -2.2 ± 0.2  | 1.4 ± 0.4     | -2.3 ± 0.2  | -0.3 ± 0.4   | -1.0 ± 0.3    | E Europe M   | S Asia F     |
| 43  | -4.7 ± 0.7    | -4.2 ± 0.2   | -1.3 ± 0.4    | -6.0 ± 0.3  | -4.0 ± 0.3   | -4.0 ± 0.5    | -0.6 ± 0.4    | -2.0 ± 0.2  | 1.3 ± 0.4     | -2.4 ± 0.2  | -0.6 ± 0.4   | -1.9 ± 0.3    | E Europe M   | S Asia F     |
| 44  | -4.3 ± 0.7    | -4.5 ± 0.2   | -1.9 ± 0.4    | -6.1 ± 0.3  | -3.7 ± 0.3   | -4.1 ± 0.5    | -0.8 ± 0.5    | -2.0 ± 0.2  | 1.6 ± 0.4     | -2.6 ± 0.2  | -0.4 ± 0.4   | -1.9 ± 0.4    | E Europe M   | S Asia F     |
| 45  | -5.2 ± 0.7    | -4.6 ± 0.2   | -1.8 ± 0.4    | -6.1 ± 0.3  | -4.2 ± 0.3   | -4.4 ± 0.5    | -1.9 ± 0.5    | -2.0 ± 0.2  | 1.3 ± 0.4     | -2.4 ± 0.2  | -0.5 ± 0.4   | -2.9 ± 0.4    | E Europe M   | S Asia F     |
| 46  | -4.0 ± 0.8    | -4.8 ± 0.2   | -2.0 ± 0.4    | -6.1 ± 0.3  | -4.7 ± 0.3   | -5.0 ± 0.5    | -1.9 ± 0.5    | -2.0 ± 0.2  | 1.7 ± 0.4     | -2.3 ± 0.3  | -0.4 ± 0.4   | -2.9 ± 0.4    | E Europe M   | S Asia F     |
| 47  | -3.7 ± 0.8    | -4.9 ± 0.2   | -1.3 ± 0.5    | -6.1 ± 0.3  | -4.5 ± 0.3   | -5.0 ± 0.5    | -1.8 ± 0.5    | -1.7 ± 0.2  | 1.2 ± 0.4     | -2.5 ± 0.3  | 0.1 ± 0.4    | -3.2 ± 0.4    | E Europe M   | S Asia F     |
| 48  | -5.1 ± 0.8    | -4.9 ± 0.2   | -1.5 ± 0.4    | -5.7 ± 0.4  | -4.5 ± 0.3   | -5.8 ± 0.6    | -2.3 ± 0.6    | -1.8 ± 0.2  | 1.4 ± 0.5     | -2.4 ± 0.3  | -0.0 ± 0.4   | -3.5 ± 0.4    | E Europe M   | W Africa F   |
| 49  | -4.9 ± 0.9    | -5.1 ± 0.2   | -1.6 ± 0.5    | -5.9 ± 0.4  | -4.5 ± 0.3   | -5.9 ± 0.6    | -2.5 ± 0.7    | -1.9 ± 0.2  | 1.8 ± 0.4     | -2.5 ± 0.3  | -0.2 ± 0.4   | -4.1 ± 0.4    | E Europe M   | W Africa F   |
| 50  | -5.5 ± 0.9    | -5.2 ± 0.2   | -1.1 ± 0.4    | -6.0 ± 0.4  | -4.8 ± 0.3   | -5.8 ± 0.6    | -2.8 ± 0.7    | -1.7 ± 0.2  | 1.3 ± 0.5     | -2.7 ± 0.3  | 0.1 ± 0.4    | -4.6 ± 0.4    | E Europe M   | S Asia F     |
| 51  | -4.6 ± 0.8    | -5.3 ± 0.2   | -1.3 ± 0.5    | -5.8 ± 0.4  | -4.8 ± 0.3   | -6.4 ± 0.6    | -3.0 ± 0.8    | -1.6 ± 0.2  | 1.5 ± 0.4     | -2.6 ± 0.3  | 0.4 ± 0.4    | -4.6 ± 0.5    | E Europe M   | W Africa F   |
| 52  | -4.5 ± 1.0    | -5.5 ± 0.2   | -1.2 ± 0.4    | -5.7 ± 0.4  | -4.7 ± 0.3   | -7.4 ± 0.6    | -2.4 ± 0.8    | -1.7 ± 0.2  | 1.4 ± 0.5     | -2.5 ± 0.3  | -0.3 ± 0.4   | -5.3 ± 0.5    | E Europe M   | W Africa F   |



**Table 44.** Mean raw error for Application-type images for algorithm idemia-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 0.8 ± 0.3     | 1.6 ± 0.1    | 0.3 ± 0.2     | 1.0 ± 0.2    | 1.3 ± 0.3    | 2.3 ± 0.3     | 0.1 ± 0.3     | 0.5 ± 0.1   | -0.7 ± 0.2    | -0.5 ± 0.1  | -0.3 ± 0.2   | 0.6 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 1.0 ± 0.3     | 1.9 ± 0.1    | 0.1 ± 0.2     | 1.3 ± 0.2    | 1.7 ± 0.3    | 2.4 ± 0.4     | 0.0 ± 0.3     | 0.9 ± 0.1   | -0.6 ± 0.2    | -0.2 ± 0.1  | 0.4 ± 0.2    | 0.8 ± 0.4     | W Africa F   | E Europe M   |
| 16  | 1.2 ± 0.3     | 1.9 ± 0.1    | 0.2 ± 0.2     | 1.3 ± 0.2    | 2.1 ± 0.3    | 2.7 ± 0.5     | 0.8 ± 0.3     | 1.1 ± 0.1   | -0.7 ± 0.2    | 0.1 ± 0.2   | 0.6 ± 0.2    | 0.8 ± 0.4     | W Africa F   | E Europe M   |
| 17  | 1.6 ± 0.3     | 1.7 ± 0.1    | -0.0 ± 0.3    | 1.4 ± 0.2    | 2.1 ± 0.3    | 2.6 ± 0.4     | 1.0 ± 0.3     | 1.1 ± 0.1   | -0.8 ± 0.2    | 0.4 ± 0.2   | 1.0 ± 0.2    | 1.4 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 1.0 ± 0.3     | 1.2 ± 0.1    | -0.8 ± 0.3    | 0.6 ± 0.1    | 1.7 ± 0.2    | 2.2 ± 0.4     | 1.0 ± 0.3     | 0.8 ± 0.1   | -1.1 ± 0.2    | 0.4 ± 0.1   | 1.0 ± 0.2    | 0.8 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 1.2 ± 0.3     | 1.1 ± 0.1    | -1.1 ± 0.2    | 0.7 ± 0.2    | 1.7 ± 0.2    | 2.3 ± 0.3     | 1.6 ± 0.3     | 1.1 ± 0.1   | -1.0 ± 0.2    | 1.0 ± 0.1   | 0.9 ± 0.2    | 1.2 ± 0.4     | W Africa F   | E Europe F   |
| 20  | 1.1 ± 0.3     | 1.0 ± 0.1    | -1.3 ± 0.2    | 0.6 ± 0.2    | 1.6 ± 0.2    | 2.2 ± 0.3     | 1.8 ± 0.3     | 1.0 ± 0.1   | -0.6 ± 0.2    | 1.3 ± 0.1   | 1.2 ± 0.2    | 1.6 ± 0.4     | W Africa F   | E Europe F   |
| 21  | 0.9 ± 0.3     | 0.9 ± 0.1    | -1.6 ± 0.2    | 0.3 ± 0.1    | 1.4 ± 0.2    | 1.7 ± 0.3     | 2.0 ± 0.3     | 1.0 ± 0.1   | -0.7 ± 0.2    | 1.6 ± 0.1   | 1.3 ± 0.2    | 1.4 ± 0.3     | E Africa M   | E Europe F   |
| 22  | 0.9 ± 0.3     | 0.5 ± 0.1    | -1.9 ± 0.2    | 0.0 ± 0.1    | 1.3 ± 0.2    | 1.4 ± 0.3     | 2.6 ± 0.3     | 0.9 ± 0.1   | -0.5 ± 0.2    | 1.6 ± 0.1   | 1.2 ± 0.2    | 1.3 ± 0.3     | E Africa M   | E Europe F   |
| 23  | 0.8 ± 0.3     | 0.4 ± 0.1    | -2.2 ± 0.2    | -0.0 ± 0.1   | 1.1 ± 0.1    | 1.3 ± 0.3     | 2.5 ± 0.3     | 0.8 ± 0.1   | -0.6 ± 0.2    | 1.8 ± 0.1   | 1.1 ± 0.2    | 1.7 ± 0.3     | E Africa M   | E Europe F   |
| 24  | 0.6 ± 0.3     | 0.09 ± 0.10  | -2.6 ± 0.1    | -0.3 ± 0.1   | 0.9 ± 0.1    | 1.0 ± 0.3     | 2.6 ± 0.3     | 0.7 ± 0.1   | -0.4 ± 0.2    | 2.0 ± 0.1   | 1.1 ± 0.2    | 1.5 ± 0.3     | E Africa M   | E Europe F   |
| 25  | 0.3 ± 0.2     | -0.13 ± 0.09 | -2.7 ± 0.1    | -0.63 ± 0.10 | 0.6 ± 0.1    | 0.8 ± 0.3     | 2.6 ± 0.3     | 0.7 ± 0.1   | -0.4 ± 0.2    | 2.0 ± 0.1   | 1.1 ± 0.2    | 1.6 ± 0.3     | E Africa M   | E Europe F   |
| 26  | 0.0 ± 0.2     | -0.37 ± 0.09 | -3.0 ± 0.1    | -0.91 ± 0.10 | 0.6 ± 0.1    | 0.5 ± 0.2     | 2.4 ± 0.2     | 0.6 ± 0.1   | -0.4 ± 0.2    | 2.0 ± 0.1   | 1.1 ± 0.2    | 1.3 ± 0.2     | E Africa M   | E Europe F   |
| 27  | 0.0 ± 0.2     | -0.57 ± 0.08 | -3.1 ± 0.1    | -1.18 ± 0.09 | 0.3 ± 0.1    | 0.4 ± 0.2     | 2.2 ± 0.2     | 0.5 ± 0.1   | -0.2 ± 0.2    | 1.9 ± 0.1   | 1.2 ± 0.2    | 1.2 ± 0.2     | E Africa M   | E Europe F   |
| 28  | -0.2 ± 0.2    | -0.75 ± 0.08 | -3.3 ± 0.1    | -1.34 ± 0.10 | 0.1 ± 0.1    | 0.1 ± 0.2     | 2.1 ± 0.2     | 0.5 ± 0.1   | -0.2 ± 0.2    | 1.86 ± 0.10 | 1.1 ± 0.2    | 1.1 ± 0.2     | E Africa M   | E Europe F   |
| 29  | -0.1 ± 0.2    | -0.88 ± 0.08 | -3.4 ± 0.1    | -1.51 ± 0.09 | -0.0 ± 0.1   | -0.1 ± 0.2    | 2.1 ± 0.2     | 0.5 ± 0.1   | -0.2 ± 0.2    | 1.78 ± 0.09 | 0.8 ± 0.2    | 1.4 ± 0.2     | E Africa M   | E Europe F   |
| 30  | -0.5 ± 0.2    | -0.97 ± 0.08 | -3.4 ± 0.1    | -1.76 ± 0.10 | -0.1 ± 0.1   | -0.5 ± 0.2    | 2.1 ± 0.2     | 0.3 ± 0.1   | -0.1 ± 0.2    | 1.66 ± 0.08 | 0.9 ± 0.2    | 1.1 ± 0.2     | E Africa M   | E Europe F   |
| 31  | -0.7 ± 0.2    | -1.13 ± 0.08 | -3.4 ± 0.1    | -2.0 ± 0.1   | -0.3 ± 0.1   | -0.6 ± 0.2    | 2.0 ± 0.2     | 0.2 ± 0.1   | -0.1 ± 0.2    | 1.53 ± 0.09 | 0.8 ± 0.2    | 1.0 ± 0.2     | E Africa M   | E Europe F   |
| 32  | -0.4 ± 0.3    | -1.19 ± 0.09 | -3.3 ± 0.1    | -2.2 ± 0.1   | -0.4 ± 0.1   | -0.7 ± 0.2    | 1.8 ± 0.2     | 0.0 ± 0.1   | -0.0 ± 0.2    | 1.54 ± 0.09 | 0.7 ± 0.2    | 1.1 ± 0.2     | E Africa M   | E Europe F   |
| 33  | -0.8 ± 0.3    | -1.27 ± 0.09 | -3.3 ± 0.2    | -2.2 ± 0.1   | -0.5 ± 0.1   | -0.9 ± 0.2    | 1.8 ± 0.2     | -0.1 ± 0.1  | 0.2 ± 0.2     | 1.42 ± 0.09 | 0.6 ± 0.2    | 0.8 ± 0.2     | E Africa M   | E Europe F   |
| 34  | -0.9 ± 0.3    | -1.30 ± 0.09 | -3.3 ± 0.2    | -2.3 ± 0.1   | -0.6 ± 0.1   | -1.0 ± 0.2    | 1.4 ± 0.2     | -0.2 ± 0.1  | 0.1 ± 0.2     | 1.34 ± 0.09 | 0.6 ± 0.2    | 0.9 ± 0.2     | E Africa M   | E Europe F   |
| 35  | -0.8 ± 0.3    | -1.48 ± 0.09 | -3.3 ± 0.2    | -2.5 ± 0.1   | -0.8 ± 0.2   | -1.3 ± 0.3    | 1.5 ± 0.2     | -0.2 ± 0.1  | 0.2 ± 0.2     | 1.26 ± 0.09 | 0.4 ± 0.2    | 0.7 ± 0.2     | E Africa M   | E Europe F   |
| 36  | -0.9 ± 0.3    | -1.50 ± 0.09 | -3.0 ± 0.2    | -2.7 ± 0.1   | -0.7 ± 0.2   | -1.2 ± 0.3    | 1.4 ± 0.3     | -0.2 ± 0.1  | 0.4 ± 0.2     | 1.08 ± 0.09 | 0.6 ± 0.2    | 0.6 ± 0.2     | E Africa M   | E Europe F   |
| 37  | -1.0 ± 0.3    | -1.56 ± 0.09 | -3.0 ± 0.2    | -2.6 ± 0.1   | -0.9 ± 0.2   | -1.4 ± 0.3    | 1.4 ± 0.3     | -0.3 ± 0.1  | 0.3 ± 0.2     | 0.99 ± 0.10 | 0.4 ± 0.2    | 0.5 ± 0.2     | E Africa M   | E Europe F   |
| 38  | -0.6 ± 0.3    | -1.66 ± 0.09 | -2.8 ± 0.2    | -2.6 ± 0.1   | -0.7 ± 0.2   | -1.5 ± 0.3    | 1.1 ± 0.3     | -0.4 ± 0.1  | 0.3 ± 0.2     | 0.84 ± 0.10 | 0.7 ± 0.2    | 0.4 ± 0.2     | E Africa M   | E Europe F   |
| 39  | -1.1 ± 0.3    | -1.76 ± 0.09 | -2.8 ± 0.2    | -2.7 ± 0.1   | -0.8 ± 0.2   | -1.1 ± 0.3    | 1.3 ± 0.3     | -0.5 ± 0.1  | 0.3 ± 0.2     | 0.81 ± 0.10 | 0.7 ± 0.2    | 0.2 ± 0.2     | E Africa M   | E Europe F   |
| 40  | -1.1 ± 0.4    | -1.90 ± 0.09 | -2.3 ± 0.2    | -2.5 ± 0.1   | -1.0 ± 0.2   | -1.7 ± 0.3    | 1.2 ± 0.3     | -0.5 ± 0.1  | 0.4 ± 0.2     | 0.8 ± 0.1   | 0.6 ± 0.2    | -0.0 ± 0.2    | E Africa M   | S Asia F     |
| 41  | -1.3 ± 0.4    | -1.90 ± 0.09 | -2.3 ± 0.2    | -2.6 ± 0.2   | -0.9 ± 0.2   | -2.1 ± 0.3    | 1.2 ± 0.3     | -0.6 ± 0.1  | 0.4 ± 0.2     | 0.7 ± 0.1   | 0.5 ± 0.2    | -0.1 ± 0.2    | E Africa M   | S Asia F     |
| 42  | -0.9 ± 0.4    | -1.95 ± 0.09 | -2.2 ± 0.2    | -2.2 ± 0.2   | -1.0 ± 0.2   | -1.8 ± 0.3    | 1.2 ± 0.3     | -0.6 ± 0.1  | 0.4 ± 0.2     | 0.5 ± 0.1   | 0.9 ± 0.2    | -0.2 ± 0.2    | E Africa M   | E Europe F   |
| 43  | -0.9 ± 0.4    | -2.18 ± 0.09 | -2.2 ± 0.2    | -2.3 ± 0.2   | -1.1 ± 0.2   | -2.1 ± 0.3    | 1.3 ± 0.3     | -0.7 ± 0.1  | 0.3 ± 0.2     | 0.6 ± 0.1   | 0.8 ± 0.2    | -0.5 ± 0.3    | E Africa M   | S Asia F     |
| 44  | -0.9 ± 0.5    | -2.32 ± 0.09 | -2.4 ± 0.2    | -2.2 ± 0.2   | -0.7 ± 0.2   | -2.1 ± 0.3    | 0.8 ± 0.3     | -0.8 ± 0.1  | 0.6 ± 0.2     | 0.5 ± 0.1   | 0.9 ± 0.2    | -0.5 ± 0.3    | S E Asia M   | E Europe F   |
| 45  | -1.2 ± 0.4    | -2.36 ± 0.09 | -2.1 ± 0.2    | -2.2 ± 0.2   | -1.1 ± 0.2   | -2.3 ± 0.3    | 0.6 ± 0.3     | -0.9 ± 0.1  | 0.3 ± 0.2     | 0.5 ± 0.1   | 0.7 ± 0.2    | -0.7 ± 0.3    | S E Asia M   | E Asia F     |
| 46  | -0.4 ± 0.5    | -2.53 ± 0.10 | -2.0 ± 0.2    | -1.9 ± 0.2   | -1.2 ± 0.2   | -2.2 ± 0.3    | 0.5 ± 0.4     | -1.0 ± 0.1  | 0.6 ± 0.2     | 0.6 ± 0.2   | 0.8 ± 0.2    | -1.0 ± 0.3    | S E Asia M   | E Asia F     |
| 47  | -0.5 ± 0.5    | -2.5 ± 0.1   | -1.9 ± 0.3    | -2.0 ± 0.2   | -1.1 ± 0.2   | -2.3 ± 0.3    | 0.7 ± 0.4     | -0.9 ± 0.1  | 0.4 ± 0.2     | 0.6 ± 0.2   | 1.0 ± 0.3    | -0.9 ± 0.3    | S E Asia M   | E Asia F     |
| 48  | -0.9 ± 0.5    | -2.6 ± 0.1   | -1.9 ± 0.2    | -1.7 ± 0.2   | -1.2 ± 0.2   | -2.5 ± 0.4    | 0.1 ± 0.4     | -0.9 ± 0.1  | 0.5 ± 0.3     | 0.6 ± 0.2   | 0.8 ± 0.3    | -1.4 ± 0.3    | S E Asia M   | E Asia F     |
| 49  | -0.6 ± 0.5    | -2.7 ± 0.1   | -2.0 ± 0.2    | -1.6 ± 0.2   | -0.9 ± 0.2   | -2.8 ± 0.4    | -0.0 ± 0.4    | -1.0 ± 0.1  | 0.4 ± 0.2     | 0.5 ± 0.2   | 0.9 ± 0.3    | -1.4 ± 0.3    | S E Asia M   | W Africa F   |
| 50  | -1.0 ± 0.6    | -2.8 ± 0.1   | -1.5 ± 0.2    | -1.5 ± 0.2   | -1.3 ± 0.2   | -2.7 ± 0.4    | -0.0 ± 0.4    | -0.9 ± 0.1  | 0.2 ± 0.3     | 0.5 ± 0.2   | 1.1 ± 0.3    | -1.9 ± 0.3    | S E Asia M   | E Asia F     |
| 51  | -0.9 ± 0.5    | -2.8 ± 0.1   | -1.8 ± 0.3    | -1.5 ± 0.2   | -1.3 ± 0.2   | -2.8 ± 0.4    | -0.5 ± 0.5    | -1.0 ± 0.1  | 0.4 ± 0.2     | 0.5 ± 0.2   | 1.1 ± 0.3    | -2.0 ± 0.3    | S E Asia M   | W Africa F   |
| 52  | -1.0 ± 0.6    | -2.9 ± 0.1   | -1.5 ± 0.2    | -1.6 ± 0.2   | -1.1 ± 0.2   | -3.3 ± 0.4    | -0.3 ± 0.5    | -1.0 ± 0.1  | 0.3 ± 0.3     | 0.3 ± 0.2   | 0.8 ± 0.3    | -2.1 ± 0.4    | S E Asia M   | W Africa F   |

**Table 45.** Mean raw error for Application-type images for algorithm idemia-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 0.2 ± 0.2     | 0.53 ± 0.09  | 0.4 ± 0.2     | 0.4 ± 0.1   | 0.4 ± 0.2    | 1.1 ± 0.3     | 0.7 ± 0.2     | 0.4 ± 0.1   | 0.0 ± 0.2     | -0.04 ± 0.10 | -0.2 ± 0.2   | 1.1 ± 0.3     | W Africa F   | S E Asia M   |
| 15  | 0.1 ± 0.3     | 0.5 ± 0.1    | 0.2 ± 0.2     | 0.5 ± 0.2   | 0.5 ± 0.3    | 1.2 ± 0.4     | 0.5 ± 0.2     | 0.6 ± 0.1   | 0.1 ± 0.2     | -0.0 ± 0.1   | 0.2 ± 0.2    | 1.1 ± 0.3     | W Africa F   | S Asia M     |
| 16  | 0.3 ± 0.3     | 0.4 ± 0.1    | 0.5 ± 0.3     | 0.4 ± 0.2   | 0.7 ± 0.3    | 1.4 ± 0.5     | 1.0 ± 0.3     | 0.64 ± 0.10 | 0.2 ± 0.2     | 0.1 ± 0.1    | 0.3 ± 0.2    | 0.9 ± 0.3     | W Africa F   | S Asia M     |
| 17  | 0.4 ± 0.3     | 0.2 ± 0.1    | 0.4 ± 0.3     | 0.3 ± 0.2   | 0.6 ± 0.3    | 1.4 ± 0.4     | 1.0 ± 0.2     | 0.5 ± 0.1   | -0.0 ± 0.2    | 0.2 ± 0.1    | 0.5 ± 0.2    | 1.1 ± 0.4     | W Africa F   | E Europe M   |
| 18  | -0.1 ± 0.3    | -0.3 ± 0.1   | -0.3 ± 0.3    | -0.3 ± 0.2  | 0.3 ± 0.2    | 1.0 ± 0.4     | 0.7 ± 0.2     | 0.2 ± 0.1   | -0.2 ± 0.2    | 0.0 ± 0.1    | 0.4 ± 0.2    | 0.4 ± 0.3     | W Africa F   | E Asia F     |
| 19  | 0.2 ± 0.3     | -0.5 ± 0.1   | -0.5 ± 0.3    | -0.2 ± 0.2  | 0.2 ± 0.2    | 1.1 ± 0.3     | 1.0 ± 0.2     | 0.3 ± 0.1   | -0.1 ± 0.2    | 0.3 ± 0.1    | 0.3 ± 0.2    | 0.5 ± 0.3     | W Africa F   | E Europe F   |
| 20  | 0.2 ± 0.4     | -0.5 ± 0.1   | -0.5 ± 0.3    | -0.2 ± 0.2  | 0.3 ± 0.2    | 1.1 ± 0.4     | 1.1 ± 0.3     | 0.2 ± 0.1   | 0.2 ± 0.2     | 0.4 ± 0.1    | 0.4 ± 0.2    | 0.6 ± 0.3     | W Africa F   | E Europe F   |
| 21  | 0.2 ± 0.3     | -0.5 ± 0.1   | -0.6 ± 0.2    | -0.3 ± 0.2  | 0.1 ± 0.2    | 0.7 ± 0.4     | 1.1 ± 0.3     | 0.2 ± 0.1   | 0.1 ± 0.2     | 0.5 ± 0.1    | 0.5 ± 0.2    | 0.4 ± 0.3     | E Africa M   | E Europe F   |
| 22  | 0.3 ± 0.3     | -0.8 ± 0.1   | -0.6 ± 0.2    | -0.5 ± 0.2  | 0.1 ± 0.2    | 0.5 ± 0.3     | 1.5 ± 0.3     | 0.1 ± 0.1   | 0.2 ± 0.2     | 0.4 ± 0.1    | 0.4 ± 0.2    | 0.1 ± 0.3     | E Africa M   | E Asia F     |
| 23  | 0.3 ± 0.3     | -0.9 ± 0.1   | -0.7 ± 0.2    | -0.3 ± 0.1  | -0.0 ± 0.1   | 0.5 ± 0.3     | 1.3 ± 0.3     | 0.0 ± 0.1   | 0.1 ± 0.2     | 0.5 ± 0.1    | 0.4 ± 0.2    | 0.3 ± 0.3     | E Africa M   | E Asia F     |
| 24  | 0.2 ± 0.3     | -1.0 ± 0.1   | -0.8 ± 0.2    | -0.4 ± 0.1  | -0.1 ± 0.1   | 0.3 ± 0.3     | 1.3 ± 0.3     | -0.1 ± 0.1  | 0.3 ± 0.2     | 0.6 ± 0.1    | 0.3 ± 0.2    | 0.1 ± 0.3     | E Africa M   | E Asia F     |
| 25  | -0.1 ± 0.3    | -1.2 ± 0.1   | -0.8 ± 0.1    | -0.6 ± 0.1  | -0.3 ± 0.1   | 0.3 ± 0.3     | 1.4 ± 0.3     | -0.0 ± 0.1  | 0.2 ± 0.2     | 0.5 ± 0.1    | 0.4 ± 0.2    | 0.2 ± 0.3     | E Africa M   | E Asia F     |
| 26  | -0.2 ± 0.3    | -1.3 ± 0.1   | -0.9 ± 0.1    | -0.7 ± 0.1  | -0.1 ± 0.1   | 0.1 ± 0.3     | 1.2 ± 0.2     | -0.0 ± 0.1  | 0.2 ± 0.2     | 0.6 ± 0.1    | 0.4 ± 0.2    | -0.2 ± 0.2    | E Africa M   | E Asia F     |
| 27  | 0.0 ± 0.3     | -1.40 ± 0.09 | -0.9 ± 0.1    | -0.8 ± 0.1  | -0.3 ± 0.1   | 0.1 ± 0.3     | 1.0 ± 0.3     | -0.1 ± 0.1  | 0.3 ± 0.1     | 0.5 ± 0.1    | 0.6 ± 0.2    | -0.2 ± 0.2    | E Africa M   | E Asia F     |
| 28  | -0.1 ± 0.3    | -1.47 ± 0.09 | -0.8 ± 0.1    | -0.7 ± 0.1  | -0.4 ± 0.1   | -0.0 ± 0.3    | 1.0 ± 0.2     | -0.0 ± 0.1  | 0.2 ± 0.1     | 0.5 ± 0.1    | 0.5 ± 0.2    | -0.3 ± 0.2    | E Africa M   | E Asia F     |
| 29  | 0.1 ± 0.3     | -1.48 ± 0.09 | -0.8 ± 0.1    | -0.7 ± 0.1  | -0.4 ± 0.1   | -0.2 ± 0.3    | 1.0 ± 0.2     | -0.1 ± 0.1  | 0.2 ± 0.1     | 0.45 ± 0.10  | 0.3 ± 0.2    | 0.1 ± 0.2     | E Africa M   | E Asia F     |
| 30  | -0.2 ± 0.3    | -1.46 ± 0.09 | -0.8 ± 0.1    | -0.8 ± 0.1  | -0.4 ± 0.2   | -0.4 ± 0.3    | 1.1 ± 0.2     | -0.2 ± 0.1  | 0.2 ± 0.1     | 0.37 ± 0.09  | 0.4 ± 0.2    | -0.1 ± 0.2    | E Africa M   | E Asia F     |
| 31  | -0.2 ± 0.3    | -1.51 ± 0.09 | -0.7 ± 0.2    | -0.9 ± 0.1  | -0.5 ± 0.1   | -0.4 ± 0.3    | 1.1 ± 0.2     | -0.2 ± 0.1  | 0.2 ± 0.1     | 0.27 ± 0.09  | 0.4 ± 0.2    | -0.1 ± 0.2    | E Africa M   | E Asia F     |
| 32  | 0.2 ± 0.3     | -1.44 ± 0.10 | -0.6 ± 0.2    | -0.9 ± 0.1  | -0.5 ± 0.2   | -0.2 ± 0.3    | 0.9 ± 0.2     | -0.3 ± 0.1  | 0.2 ± 0.2     | 0.36 ± 0.09  | 0.3 ± 0.2    | 0.0 ± 0.2     | E Africa M   | E Asia F     |
| 33  | -0.1 ± 0.3    | -1.41 ± 0.10 | -0.6 ± 0.2    | -0.7 ± 0.1  | -0.4 ± 0.2   | -0.4 ± 0.3    | 1.0 ± 0.3     | -0.4 ± 0.1  | 0.3 ± 0.2     | 0.29 ± 0.09  | 0.3 ± 0.2    | -0.1 ± 0.2    | E Africa M   | E Asia F     |
| 34  | 0.0 ± 0.3     | -1.30 ± 0.10 | -0.6 ± 0.2    | -0.7 ± 0.1  | -0.4 ± 0.2   | -0.3 ± 0.3    | 0.7 ± 0.3     | -0.4 ± 0.1  | 0.2 ± 0.2     | 0.27 ± 0.10  | 0.4 ± 0.2    | 0.0 ± 0.2     | E Africa M   | E Asia F     |
| 35  | 0.2 ± 0.3     | -1.4 ± 0.1   | -0.6 ± 0.2    | -0.8 ± 0.1  | -0.5 ± 0.2   | -0.5 ± 0.3    | 0.9 ± 0.3     | -0.3 ± 0.1  | 0.2 ± 0.2     | 0.26 ± 0.10  | 0.2 ± 0.2    | -0.0 ± 0.2    | E Africa M   | E Asia F     |
| 36  | 0.2 ± 0.4     | -1.3 ± 0.1   | -0.4 ± 0.2    | -0.8 ± 0.1  | -0.3 ± 0.2   | -0.3 ± 0.3    | 0.8 ± 0.3     | -0.4 ± 0.1  | 0.4 ± 0.2     | 0.13 ± 0.10  | 0.4 ± 0.2    | 0.0 ± 0.2     | E Africa M   | E Asia F     |
| 37  | 0.2 ± 0.4     | -1.21 ± 0.10 | -0.4 ± 0.2    | -0.5 ± 0.1  | -0.3 ± 0.2   | -0.4 ± 0.3    | 0.9 ± 0.3     | -0.4 ± 0.1  | 0.2 ± 0.2     | 0.1 ± 0.1    | 0.2 ± 0.2    | -0.1 ± 0.2    | E Africa M   | E Asia F     |
| 38  | 0.8 ± 0.4     | -1.2 ± 0.1   | -0.3 ± 0.2    | -0.5 ± 0.1  | 0.0 ± 0.2    | -0.4 ± 0.4    | 0.8 ± 0.3     | -0.4 ± 0.1  | 0.1 ± 0.2     | -0.0 ± 0.1   | 0.6 ± 0.2    | -0.0 ± 0.2    | E Africa F   | E Asia F     |
| 39  | 0.3 ± 0.4     | -1.2 ± 0.1   | -0.3 ± 0.2    | -0.5 ± 0.1  | 0.0 ± 0.2    | 0.2 ± 0.3     | 1.0 ± 0.3     | -0.5 ± 0.1  | 0.1 ± 0.2     | 0.0 ± 0.1    | 0.6 ± 0.2    | -0.1 ± 0.2    | E Africa M   | E Asia F     |
| 40  | 0.4 ± 0.4     | -1.2 ± 0.1   | 0.1 ± 0.2     | -0.3 ± 0.2  | -0.0 ± 0.2   | -0.3 ± 0.3    | 1.0 ± 0.3     | -0.6 ± 0.1  | 0.1 ± 0.2     | 0.0 ± 0.1    | 0.6 ± 0.2    | -0.3 ± 0.2    | E Africa M   | E Asia F     |
| 41  | 0.3 ± 0.4     | -1.1 ± 0.1   | 0.1 ± 0.2     | -0.4 ± 0.2  | 0.1 ± 0.2    | -0.6 ± 0.3    | 1.1 ± 0.3     | -0.6 ± 0.1  | 0.1 ± 0.2     | -0.0 ± 0.1   | 0.5 ± 0.2    | -0.2 ± 0.3    | E Africa M   | E Asia F     |
| 42  | 0.8 ± 0.4     | -1.0 ± 0.1   | 0.2 ± 0.2     | 0.0 ± 0.2   | 0.2 ± 0.2    | -0.3 ± 0.3    | 1.2 ± 0.3     | -0.6 ± 0.1  | 0.1 ± 0.2     | -0.1 ± 0.1   | 0.8 ± 0.2    | -0.3 ± 0.3    | E Africa M   | E Asia F     |
| 43  | 0.9 ± 0.4     | -1.1 ± 0.1   | 0.2 ± 0.2     | -0.1 ± 0.2  | 0.2 ± 0.2    | -0.5 ± 0.3    | 1.4 ± 0.3     | -0.7 ± 0.1  | -0.1 ± 0.2    | -0.0 ± 0.1   | 0.8 ± 0.2    | -0.5 ± 0.3    | E Africa M   | E Asia F     |
| 44  | 0.9 ± 0.5     | -1.2 ± 0.1   | -0.1 ± 0.2    | -0.1 ± 0.2  | 0.6 ± 0.2    | -0.4 ± 0.4    | 1.0 ± 0.4     | -0.8 ± 0.1  | 0.2 ± 0.2     | -0.1 ± 0.1   | 0.8 ± 0.2    | -0.4 ± 0.3    | E Africa M   | E Asia F     |
| 45  | 0.6 ± 0.5     | -1.1 ± 0.1   | 0.1 ± 0.2     | -0.1 ± 0.2  | 0.3 ± 0.2    | -0.6 ± 0.4    | 0.9 ± 0.4     | -0.9 ± 0.1  | -0.1 ± 0.2    | 0.0 ± 0.1    | 0.7 ± 0.2    | -0.6 ± 0.3    | E Africa M   | E Asia F     |
| 46  | 1.5 ± 0.5     | -1.2 ± 0.1   | 0.2 ± 0.2     | 0.2 ± 0.2   | 0.3 ± 0.2    | -0.4 ± 0.4    | 0.9 ± 0.4     | -0.9 ± 0.1  | 0.2 ± 0.2     | 0.1 ± 0.2    | 0.8 ± 0.2    | -0.7 ± 0.3    | E Africa F   | E Asia F     |
| 47  | 1.4 ± 0.5     | -1.1 ± 0.1   | 0.3 ± 0.3     | 0.1 ± 0.2   | 0.5 ± 0.2    | -0.5 ± 0.4    | 1.2 ± 0.4     | -0.9 ± 0.1  | -0.0 ± 0.2    | 0.1 ± 0.2    | 1.0 ± 0.3    | -0.6 ± 0.3    | E Africa F   | E Asia F     |
| 48  | 1.0 ± 0.5     | -1.1 ± 0.1   | 0.3 ± 0.2     | 0.2 ± 0.2   | 0.4 ± 0.2    | -0.7 ± 0.4    | 0.6 ± 0.5     | -0.9 ± 0.1  | 0.1 ± 0.3     | 0.2 ± 0.2    | 0.8 ± 0.3    | -0.9 ± 0.3    | E Africa F   | E Asia F     |
| 49  | 1.3 ± 0.6     | -1.2 ± 0.1   | 0.1 ± 0.2     | 0.3 ± 0.2   | 0.7 ± 0.2    | -0.9 ± 0.4    | 0.5 ± 0.5     | -1.0 ± 0.1  | -0.0 ± 0.2    | 0.0 ± 0.2    | 0.9 ± 0.3    | -0.9 ± 0.3    | E Africa F   | E Asia F     |
| 50  | 0.9 ± 0.6     | -1.2 ± 0.1   | 0.5 ± 0.2     | 0.3 ± 0.2   | 0.4 ± 0.2    | -0.8 ± 0.4    | 0.6 ± 0.4     | -0.9 ± 0.1  | -0.2 ± 0.3    | 0.1 ± 0.2    | 1.1 ± 0.3    | -1.3 ± 0.3    | S E Asia M   | W Africa M   |
| 51  | 1.1 ± 0.5     | -1.1 ± 0.1   | 0.2 ± 0.2     | 0.3 ± 0.2   | 0.3 ± 0.2    | -0.9 ± 0.4    | 0.1 ± 0.5     | -1.0 ± 0.1  | 0.0 ± 0.2     | 0.1 ± 0.2    | 1.1 ± 0.3    | -1.4 ± 0.4    | E Africa F   | W Africa M   |
| 52  | 1.0 ± 0.6     | -1.3 ± 0.1   | 0.4 ± 0.2     | 0.1 ± 0.2   | 0.6 ± 0.2    | -1.4 ± 0.4    | 0.4 ± 0.5     | -1.0 ± 0.1  | -0.1 ± 0.3    | -0.1 ± 0.2   | 0.8 ± 0.3    | -1.4 ± 0.4    | E Africa F   | W Africa M   |

**Table 46.** Mean raw error for Application-type images for algorithm inbut-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 6.7 ± 0.4     | 4.0 ± 0.1    | 5.1 ± 0.4     | 4.5 ± 0.2    | 4.0 ± 0.2    | 7.7 ± 0.5     | 6.5 ± 0.5     | 3.3 ± 0.1   | 2.9 ± 0.3     | 3.7 ± 0.2   | 3.5 ± 0.3    | 5.9 ± 0.5     | W Africa F   | E Europe M   |
| 15  | 7.1 ± 0.4     | 3.7 ± 0.1    | 5.4 ± 0.4     | 4.6 ± 0.2    | 3.9 ± 0.3    | 7.2 ± 0.5     | 6.2 ± 0.5     | 3.5 ± 0.2   | 3.7 ± 0.3     | 3.9 ± 0.2   | 3.5 ± 0.4    | 6.2 ± 0.5     | W Africa F   | E Asia M     |
| 16  | 6.7 ± 0.4     | 3.4 ± 0.1    | 5.6 ± 0.4     | 4.2 ± 0.2    | 3.7 ± 0.3    | 7.7 ± 0.6     | 6.4 ± 0.4     | 3.3 ± 0.2   | 4.1 ± 0.4     | 3.9 ± 0.2   | 3.8 ± 0.4    | 6.0 ± 0.5     | W Africa F   | E Asia M     |
| 17  | 6.9 ± 0.5     | 3.1 ± 0.1    | 5.2 ± 0.4     | 4.2 ± 0.2    | 3.5 ± 0.3    | 7.1 ± 0.4     | 6.0 ± 0.4     | 3.1 ± 0.2   | 3.8 ± 0.3     | 3.7 ± 0.2   | 3.5 ± 0.4    | 6.0 ± 0.5     | W Africa F   | E Asia F     |
| 18  | 6.1 ± 0.3     | 2.4 ± 0.1    | 4.3 ± 0.4     | 3.5 ± 0.2    | 2.5 ± 0.2    | 6.6 ± 0.4     | 5.3 ± 0.3     | 2.6 ± 0.1   | 3.5 ± 0.3     | 3.2 ± 0.2   | 2.3 ± 0.3    | 4.7 ± 0.4     | W Africa F   | S E Asia M   |
| 19  | 6.2 ± 0.4     | 2.2 ± 0.1    | 4.0 ± 0.3     | 3.5 ± 0.2    | 2.4 ± 0.2    | 6.5 ± 0.4     | 5.6 ± 0.4     | 2.7 ± 0.1   | 3.6 ± 0.3     | 3.6 ± 0.2   | 2.3 ± 0.3    | 5.4 ± 0.4     | W Africa F   | E Asia F     |
| 20  | 5.9 ± 0.3     | 2.1 ± 0.1    | 3.8 ± 0.3     | 3.4 ± 0.2    | 2.3 ± 0.2    | 6.6 ± 0.4     | 5.4 ± 0.4     | 2.5 ± 0.1   | 3.7 ± 0.3     | 3.4 ± 0.2   | 2.1 ± 0.2    | 5.1 ± 0.4     | W Africa F   | S E Asia M   |
| 21  | 5.7 ± 0.3     | 2.0 ± 0.1    | 3.7 ± 0.2     | 3.3 ± 0.1    | 2.2 ± 0.2    | 6.0 ± 0.3     | 5.7 ± 0.4     | 2.4 ± 0.1   | 3.2 ± 0.3     | 3.4 ± 0.2   | 2.3 ± 0.2    | 4.9 ± 0.3     | W Africa F   | E Asia F     |
| 22  | 5.7 ± 0.3     | 1.6 ± 0.1    | 3.2 ± 0.2     | 3.1 ± 0.1    | 1.8 ± 0.2    | 5.7 ± 0.3     | 6.4 ± 0.3     | 2.2 ± 0.1   | 3.5 ± 0.2     | 3.5 ± 0.2   | 2.1 ± 0.2    | 4.6 ± 0.3     | E Africa M   | E Asia F     |
| 23  | 5.4 ± 0.3     | 1.5 ± 0.1    | 3.0 ± 0.2     | 2.9 ± 0.1    | 1.6 ± 0.1    | 5.7 ± 0.3     | 5.9 ± 0.3     | 1.9 ± 0.1   | 3.3 ± 0.2     | 3.5 ± 0.2   | 1.9 ± 0.2    | 5.0 ± 0.3     | E Africa M   | E Asia F     |
| 24  | 5.0 ± 0.3     | 1.07 ± 0.10  | 2.8 ± 0.1     | 2.8 ± 0.1    | 1.2 ± 0.1    | 5.3 ± 0.3     | 5.8 ± 0.3     | 1.7 ± 0.1   | 3.2 ± 0.2     | 3.5 ± 0.1   | 1.8 ± 0.2    | 4.7 ± 0.3     | E Africa M   | E Asia F     |
| 25  | 4.4 ± 0.2     | 0.85 ± 0.08  | 2.3 ± 0.1     | 2.5 ± 0.1    | 0.9 ± 0.1    | 4.9 ± 0.3     | 5.5 ± 0.3     | 1.4 ± 0.1   | 3.2 ± 0.2     | 3.4 ± 0.1   | 1.7 ± 0.2    | 4.5 ± 0.3     | E Africa M   | E Asia F     |
| 26  | 4.0 ± 0.2     | 0.42 ± 0.08  | 2.0 ± 0.1     | 2.13 ± 0.10  | 0.7 ± 0.1    | 4.6 ± 0.2     | 5.3 ± 0.3     | 1.3 ± 0.1   | 2.7 ± 0.2     | 3.2 ± 0.1   | 1.7 ± 0.2    | 4.0 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 3.9 ± 0.2     | 0.12 ± 0.07  | 1.7 ± 0.1     | 1.78 ± 0.10  | 0.4 ± 0.1    | 4.3 ± 0.2     | 5.1 ± 0.3     | 1.0 ± 0.1   | 2.8 ± 0.2     | 3.0 ± 0.1   | 1.7 ± 0.2    | 4.1 ± 0.2     | E Africa M   | E Asia F     |
| 28  | 3.8 ± 0.2     | -0.28 ± 0.08 | 1.4 ± 0.1     | 1.48 ± 0.10  | -0.0 ± 0.1   | 4.0 ± 0.2     | 4.8 ± 0.2     | 0.8 ± 0.1   | 2.7 ± 0.2     | 2.8 ± 0.1   | 1.5 ± 0.2    | 3.8 ± 0.2     | E Africa M   | E Asia F     |
| 29  | 3.4 ± 0.2     | -0.53 ± 0.07 | 1.1 ± 0.1     | 1.08 ± 0.09  | -0.3 ± 0.1   | 3.7 ± 0.2     | 4.7 ± 0.2     | 0.7 ± 0.1   | 2.7 ± 0.2     | 2.6 ± 0.1   | 1.1 ± 0.2    | 3.9 ± 0.2     | E Africa M   | E Asia F     |
| 30  | 3.0 ± 0.2     | -0.85 ± 0.07 | 0.8 ± 0.1     | 0.58 ± 0.09  | -0.6 ± 0.1   | 3.2 ± 0.2     | 4.9 ± 0.3     | 0.4 ± 0.1   | 2.4 ± 0.2     | 2.4 ± 0.1   | 1.1 ± 0.2    | 3.7 ± 0.2     | E Africa M   | E Asia F     |
| 31  | 2.6 ± 0.2     | -1.12 ± 0.08 | 0.7 ± 0.1     | 0.15 ± 0.10  | -0.9 ± 0.1   | 3.2 ± 0.2     | 4.6 ± 0.2     | 0.4 ± 0.1   | 2.3 ± 0.2     | 2.2 ± 0.1   | 1.0 ± 0.2    | 3.4 ± 0.2     | E Africa M   | E Asia F     |
| 32  | 2.6 ± 0.2     | -1.34 ± 0.08 | 0.4 ± 0.1     | -0.20 ± 0.10 | -1.0 ± 0.1   | 3.0 ± 0.2     | 4.3 ± 0.3     | 0.3 ± 0.1   | 2.3 ± 0.2     | 1.9 ± 0.1   | 0.8 ± 0.2    | 3.3 ± 0.2     | E Africa M   | E Asia F     |
| 33  | 2.2 ± 0.2     | -1.53 ± 0.09 | 0.2 ± 0.1     | -0.51 ± 0.10 | -1.3 ± 0.1   | 2.5 ± 0.2     | 4.5 ± 0.3     | 0.2 ± 0.1   | 2.4 ± 0.2     | 1.7 ± 0.1   | 0.5 ± 0.2    | 3.3 ± 0.2     | E Africa M   | E Asia F     |
| 34  | 1.9 ± 0.3     | -1.66 ± 0.09 | -0.0 ± 0.2    | -0.9 ± 0.1   | -1.5 ± 0.1   | 2.2 ± 0.2     | 4.1 ± 0.3     | 0.3 ± 0.1   | 2.0 ± 0.2     | 1.4 ± 0.1   | 0.7 ± 0.2    | 3.3 ± 0.2     | E Africa M   | E Asia F     |
| 35  | 2.0 ± 0.3     | -1.92 ± 0.09 | -0.2 ± 0.2    | -1.3 ± 0.1   | -1.6 ± 0.2   | 1.8 ± 0.2     | 4.2 ± 0.3     | 0.2 ± 0.1   | 2.0 ± 0.2     | 1.2 ± 0.1   | 0.4 ± 0.2    | 3.1 ± 0.2     | E Africa M   | E Asia F     |
| 36  | 1.7 ± 0.3     | -1.95 ± 0.10 | -0.4 ± 0.2    | -1.5 ± 0.1   | -1.7 ± 0.2   | 1.9 ± 0.3     | 4.1 ± 0.3     | 0.2 ± 0.1   | 1.9 ± 0.2     | 0.9 ± 0.1   | 0.5 ± 0.2    | 3.2 ± 0.2     | E Africa M   | E Asia F     |
| 37  | 1.4 ± 0.3     | -2.11 ± 0.09 | -0.6 ± 0.2    | -1.6 ± 0.1   | -1.9 ± 0.2   | 1.5 ± 0.3     | 4.2 ± 0.3     | 0.1 ± 0.1   | 1.7 ± 0.2     | 0.6 ± 0.1   | 0.2 ± 0.2    | 2.9 ± 0.2     | E Africa M   | E Asia F     |
| 38  | 1.8 ± 0.3     | -2.30 ± 0.09 | -0.7 ± 0.2    | -1.7 ± 0.1   | -1.9 ± 0.2   | 1.2 ± 0.3     | 3.7 ± 0.3     | 0.0 ± 0.1   | 1.6 ± 0.2     | 0.4 ± 0.1   | 0.3 ± 0.2    | 2.9 ± 0.2     | E Africa M   | E Asia F     |
| 39  | 1.2 ± 0.3     | -2.29 ± 0.09 | -0.8 ± 0.2    | -1.9 ± 0.1   | -1.9 ± 0.2   | 1.2 ± 0.3     | 3.9 ± 0.3     | -0.2 ± 0.1  | 1.7 ± 0.2     | 0.3 ± 0.1   | 0.1 ± 0.2    | 2.9 ± 0.2     | E Africa M   | E Asia F     |
| 40  | 1.1 ± 0.4     | -2.48 ± 0.10 | -0.6 ± 0.2    | -1.8 ± 0.1   | -2.0 ± 0.2   | 0.7 ± 0.3     | 4.0 ± 0.3     | -0.3 ± 0.1  | 1.5 ± 0.2     | 0.3 ± 0.1   | -0.1 ± 0.2   | 2.6 ± 0.3     | E Africa M   | E Asia F     |
| 41  | 0.7 ± 0.4     | -2.60 ± 0.10 | -0.8 ± 0.2    | -1.9 ± 0.2   | -2.1 ± 0.2   | 0.4 ± 0.3     | 3.9 ± 0.3     | -0.5 ± 0.1  | 1.6 ± 0.2     | 0.2 ± 0.1   | -0.4 ± 0.2   | 2.7 ± 0.3     | E Africa M   | E Asia F     |
| 42  | 0.9 ± 0.4     | -2.68 ± 0.09 | -0.8 ± 0.2    | -1.6 ± 0.2   | -2.2 ± 0.2   | 0.3 ± 0.3     | 3.9 ± 0.4     | -0.7 ± 0.1  | 1.8 ± 0.2     | 0.2 ± 0.1   | -0.3 ± 0.2   | 2.4 ± 0.3     | E Africa M   | E Asia F     |
| 43  | 1.2 ± 0.4     | -2.81 ± 0.10 | -0.9 ± 0.2    | -1.7 ± 0.2   | -2.2 ± 0.2   | 0.1 ± 0.3     | 4.2 ± 0.4     | -0.8 ± 0.1  | 1.6 ± 0.2     | 0.3 ± 0.1   | -0.3 ± 0.2   | 2.2 ± 0.3     | E Africa M   | E Asia F     |
| 44  | 1.0 ± 0.5     | -3.09 ± 0.10 | -1.2 ± 0.2    | -1.5 ± 0.2   | -2.0 ± 0.2   | -0.3 ± 0.4    | 3.7 ± 0.4     | -0.9 ± 0.1  | 1.7 ± 0.2     | 0.1 ± 0.2   | -0.5 ± 0.2   | 2.3 ± 0.3     | E Africa M   | E Asia F     |
| 45  | 0.8 ± 0.5     | -3.14 ± 0.10 | -1.1 ± 0.2    | -1.5 ± 0.2   | -2.2 ± 0.2   | -0.2 ± 0.4    | 3.5 ± 0.4     | -1.1 ± 0.1  | 1.7 ± 0.3     | 0.2 ± 0.2   | -0.7 ± 0.2   | 2.1 ± 0.3     | E Africa M   | E Asia F     |
| 46  | 1.7 ± 0.5     | -3.37 ± 0.10 | -1.0 ± 0.3    | -1.4 ± 0.2   | -2.3 ± 0.2   | -0.4 ± 0.4    | 3.5 ± 0.4     | -1.2 ± 0.1  | 1.7 ± 0.3     | 0.3 ± 0.2   | -0.6 ± 0.2   | 1.7 ± 0.3     | E Africa M   | E Asia F     |
| 47  | 1.2 ± 0.5     | -3.3 ± 0.1   | -1.1 ± 0.3    | -1.3 ± 0.2   | -2.3 ± 0.2   | -0.4 ± 0.4    | 4.1 ± 0.4     | -1.4 ± 0.1  | 1.5 ± 0.3     | 0.3 ± 0.2   | -0.5 ± 0.2   | 1.9 ± 0.3     | E Africa M   | E Asia F     |
| 48  | 0.9 ± 0.6     | -3.4 ± 0.1   | -0.9 ± 0.3    | -1.1 ± 0.2   | -2.4 ± 0.2   | -0.7 ± 0.4    | 3.5 ± 0.5     | -1.3 ± 0.1  | 1.5 ± 0.3     | 0.4 ± 0.2   | -0.5 ± 0.3   | 1.6 ± 0.3     | E Africa M   | E Asia F     |
| 49  | 1.7 ± 0.6     | -3.5 ± 0.1   | -1.4 ± 0.3    | -0.9 ± 0.2   | -2.2 ± 0.2   | -0.8 ± 0.4    | 3.5 ± 0.5     | -1.4 ± 0.1  | 1.3 ± 0.3     | 0.3 ± 0.2   | -0.7 ± 0.2   | 1.7 ± 0.3     | E Africa M   | E Asia F     |
| 50  | 0.7 ± 0.6     | -3.5 ± 0.1   | -0.9 ± 0.2    | -1.1 ± 0.2   | -2.4 ± 0.2   | -0.8 ± 0.4    | 3.2 ± 0.4     | -1.5 ± 0.1  | 1.0 ± 0.3     | 0.1 ± 0.2   | -0.3 ± 0.3   | 1.2 ± 0.3     | E Africa M   | E Asia F     |
| 51  | 1.0 ± 0.5     | -3.5 ± 0.1   | -1.2 ± 0.3    | -1.0 ± 0.2   | -2.5 ± 0.2   | -0.8 ± 0.4    | 2.9 ± 0.5     | -1.6 ± 0.1  | 1.0 ± 0.2     | 0.1 ± 0.2   | -0.5 ± 0.2   | 1.2 ± 0.4     | E Africa M   | E Asia F     |
| 52  | 0.7 ± 0.6     | -3.6 ± 0.1   | -1.0 ± 0.3    | -1.0 ± 0.2   | -2.1 ± 0.2   | -1.0 ± 0.5    | 2.9 ± 0.5     | -1.5 ± 0.1  | 1.1 ± 0.3     | -0.0 ± 0.2  | -0.8 ± 0.3   | 1.2 ± 0.4     | E Africa M   | E Asia F     |

**Table 47.** Mean raw error for Application-type images for algorithm incode-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 1.2 ± 0.3     | 0.7 ± 0.1    | 1.3 ± 0.3     | 0.9 ± 0.2   | 1.0 ± 0.3    | 2.3 ± 0.3     | 0.1 ± 0.3     | -0.2 ± 0.1  | -0.2 ± 0.2    | -0.4 ± 0.1  | -0.4 ± 0.2   | 0.4 ± 0.3     | W Africa F   | S Asia M     |
| 15  | 1.4 ± 0.4     | 0.9 ± 0.2    | 1.3 ± 0.3     | 1.3 ± 0.2   | 1.4 ± 0.4    | 2.0 ± 0.4     | -0.2 ± 0.3    | 0.0 ± 0.1   | -0.1 ± 0.2    | -0.1 ± 0.1  | 0.2 ± 0.2    | 0.8 ± 0.3     | W Africa F   | E Africa M   |
| 16  | 1.3 ± 0.4     | 0.9 ± 0.2    | 1.8 ± 0.3     | 1.4 ± 0.2   | 1.6 ± 0.4    | 2.4 ± 0.5     | 0.7 ± 0.3     | 0.1 ± 0.1   | 0.1 ± 0.2     | 0.2 ± 0.1   | 0.3 ± 0.2    | 0.7 ± 0.3     | W Africa F   | E Asia M     |
| 17  | 1.8 ± 0.5     | 0.9 ± 0.2    | 1.6 ± 0.3     | 1.3 ± 0.2   | 1.4 ± 0.3    | 2.0 ± 0.4     | 0.3 ± 0.3     | 0.1 ± 0.1   | -0.2 ± 0.2    | 0.5 ± 0.1   | 0.5 ± 0.2    | 1.0 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 1.0 ± 0.4     | 0.5 ± 0.2    | 0.9 ± 0.3     | 0.9 ± 0.2   | 1.5 ± 0.2    | 1.3 ± 0.4     | 0.2 ± 0.3     | -0.3 ± 0.1  | -0.5 ± 0.2    | 0.2 ± 0.1   | 0.3 ± 0.2    | 0.3 ± 0.3     | S E Asia F   | E Europe M   |
| 19  | 1.6 ± 0.4     | 0.6 ± 0.2    | 1.0 ± 0.3     | 1.0 ± 0.2   | 1.4 ± 0.2    | 1.5 ± 0.4     | 0.8 ± 0.3     | -0.3 ± 0.1  | -0.5 ± 0.2    | 0.5 ± 0.1   | 0.1 ± 0.2    | 0.7 ± 0.4     | E Africa F   | E Europe M   |
| 20  | 1.6 ± 0.4     | 0.7 ± 0.2    | 1.0 ± 0.3     | 1.0 ± 0.2   | 1.3 ± 0.2    | 1.4 ± 0.4     | 1.0 ± 0.3     | -0.5 ± 0.1  | -0.2 ± 0.2    | 0.7 ± 0.1   | 0.3 ± 0.2    | 0.9 ± 0.4     | E Africa F   | E Asia M     |
| 21  | 1.8 ± 0.4     | 0.7 ± 0.2    | 1.0 ± 0.3     | 0.8 ± 0.2   | 1.4 ± 0.2    | 1.7 ± 0.4     | 1.4 ± 0.3     | -0.4 ± 0.1  | -0.3 ± 0.2    | 0.9 ± 0.1   | 0.4 ± 0.2    | 0.8 ± 0.4     | E Africa F   | E Asia M     |
| 22  | 1.8 ± 0.4     | 0.5 ± 0.1    | 1.1 ± 0.2     | 0.7 ± 0.2   | 1.1 ± 0.2    | 1.1 ± 0.4     | 2.2 ± 0.3     | -0.4 ± 0.1  | 0.2 ± 0.2     | 1.0 ± 0.2   | 0.1 ± 0.2    | 0.6 ± 0.3     | E Africa M   | E Asia M     |
| 23  | 2.1 ± 0.3     | 0.3 ± 0.1    | 1.2 ± 0.2     | 0.7 ± 0.1   | 1.0 ± 0.2    | 1.3 ± 0.3     | 2.0 ± 0.3     | -0.5 ± 0.1  | 0.2 ± 0.2     | 1.1 ± 0.1   | 0.2 ± 0.2    | 1.2 ± 0.3     | E Africa F   | E Asia M     |
| 24  | 2.3 ± 0.4     | 0.0 ± 0.1    | 1.3 ± 0.2     | 0.5 ± 0.1   | 0.8 ± 0.1    | 1.0 ± 0.3     | 2.0 ± 0.3     | -0.5 ± 0.1  | 0.4 ± 0.2     | 1.1 ± 0.1   | 0.1 ± 0.2    | 1.1 ± 0.3     | E Africa F   | E Asia M     |
| 25  | 1.8 ± 0.3     | 0.0 ± 0.1    | 1.0 ± 0.2     | 0.3 ± 0.1   | 0.6 ± 0.1    | 1.2 ± 0.3     | 2.1 ± 0.3     | -0.5 ± 0.1  | 0.4 ± 0.2     | 1.2 ± 0.1   | 0.1 ± 0.2    | 0.9 ± 0.3     | E Africa M   | E Asia M     |
| 26  | 2.0 ± 0.3     | -0.3 ± 0.1   | 1.1 ± 0.1     | 0.1 ± 0.1   | 0.5 ± 0.1    | 1.1 ± 0.3     | 1.9 ± 0.3     | -0.7 ± 0.1  | 0.3 ± 0.2     | 1.1 ± 0.1   | 0.1 ± 0.2    | 0.7 ± 0.2     | E Africa F   | E Asia M     |
| 27  | 1.9 ± 0.3     | -0.54 ± 0.10 | 1.1 ± 0.2     | -0.1 ± 0.1  | 0.3 ± 0.1    | 1.0 ± 0.3     | 1.8 ± 0.3     | -0.7 ± 0.1  | 0.4 ± 0.2     | 0.8 ± 0.1   | 0.1 ± 0.2    | 0.5 ± 0.2     | E Africa F   | E Asia M     |
| 28  | 1.9 ± 0.3     | -0.80 ± 0.09 | 1.0 ± 0.1     | -0.3 ± 0.1  | 0.0 ± 0.1    | 0.9 ± 0.3     | 1.6 ± 0.2     | -0.9 ± 0.1  | 0.5 ± 0.2     | 0.8 ± 0.1   | -0.2 ± 0.2   | 0.2 ± 0.2     | E Africa F   | E Asia M     |
| 29  | 2.1 ± 0.3     | -0.99 ± 0.09 | 1.0 ± 0.1     | -0.5 ± 0.1  | 0.0 ± 0.1    | 0.7 ± 0.3     | 1.6 ± 0.2     | -1.0 ± 0.1  | 0.3 ± 0.2     | 0.53 ± 0.10 | -0.3 ± 0.2   | 0.6 ± 0.2     | E Africa F   | E Asia M     |
| 30  | 1.8 ± 0.3     | -1.11 ± 0.09 | 0.9 ± 0.2     | -0.7 ± 0.1  | -0.1 ± 0.1   | 0.4 ± 0.3     | 1.5 ± 0.2     | -1.2 ± 0.1  | 0.3 ± 0.2     | 0.38 ± 0.10 | -0.2 ± 0.2   | 0.4 ± 0.2     | E Africa F   | E Asia M     |
| 31  | 1.5 ± 0.3     | -1.27 ± 0.10 | 1.1 ± 0.2     | -0.9 ± 0.1  | -0.2 ± 0.1   | 0.3 ± 0.3     | 1.5 ± 0.3     | -1.2 ± 0.1  | 0.3 ± 0.2     | 0.30 ± 0.10 | -0.4 ± 0.2   | 0.4 ± 0.2     | E Africa M   | E Asia F     |
| 32  | 1.8 ± 0.3     | -1.38 ± 0.10 | 1.0 ± 0.2     | -0.9 ± 0.1  | -0.3 ± 0.2   | 0.4 ± 0.3     | 1.2 ± 0.3     | -1.3 ± 0.1  | 0.4 ± 0.2     | 0.2 ± 0.1   | -0.3 ± 0.2   | 0.3 ± 0.2     | E Africa F   | E Asia F     |
| 33  | 1.7 ± 0.3     | -1.5 ± 0.1   | 1.0 ± 0.2     | -0.9 ± 0.1  | -0.4 ± 0.2   | 0.5 ± 0.3     | 1.5 ± 0.3     | -1.4 ± 0.1  | 0.7 ± 0.2     | 0.1 ± 0.1   | -0.5 ± 0.2   | 0.2 ± 0.2     | E Africa F   | E Asia F     |
| 34  | 1.4 ± 0.3     | -1.5 ± 0.1   | 1.1 ± 0.2     | -1.0 ± 0.1  | -0.3 ± 0.2   | 0.2 ± 0.3     | 1.0 ± 0.3     | -1.4 ± 0.1  | 0.5 ± 0.2     | 0.1 ± 0.1   | -0.2 ± 0.2   | 0.3 ± 0.2     | E Africa F   | E Asia F     |
| 35  | 1.7 ± 0.3     | -1.6 ± 0.1   | 1.1 ± 0.2     | -1.1 ± 0.1  | -0.4 ± 0.2   | -0.0 ± 0.3    | 1.3 ± 0.3     | -1.2 ± 0.1  | 0.6 ± 0.2     | 0.1 ± 0.1   | -0.4 ± 0.2   | 0.2 ± 0.3     | E Africa F   | E Asia F     |
| 36  | 1.5 ± 0.3     | -1.5 ± 0.1   | 1.2 ± 0.2     | -1.1 ± 0.1  | -0.3 ± 0.2   | 0.3 ± 0.3     | 1.2 ± 0.3     | -1.1 ± 0.1  | 0.8 ± 0.2     | 0.1 ± 0.1   | -0.1 ± 0.3   | 0.2 ± 0.3     | E Africa F   | E Asia F     |
| 37  | 1.6 ± 0.3     | -1.4 ± 0.1   | 1.4 ± 0.2     | -0.9 ± 0.1  | -0.2 ± 0.2   | 0.0 ± 0.4     | 1.1 ± 0.3     | -1.1 ± 0.1  | 0.7 ± 0.2     | 0.1 ± 0.1   | -0.3 ± 0.2   | 0.0 ± 0.2     | E Africa F   | E Asia F     |
| 38  | 1.7 ± 0.4     | -1.5 ± 0.1   | 1.4 ± 0.2     | -0.8 ± 0.1  | -0.3 ± 0.2   | 0.1 ± 0.4     | 0.8 ± 0.3     | -0.9 ± 0.1  | 0.8 ± 0.2     | 0.1 ± 0.1   | 0.2 ± 0.2    | 0.2 ± 0.3     | E Africa F   | E Asia F     |
| 39  | 1.4 ± 0.4     | -1.4 ± 0.1   | 1.3 ± 0.2     | -0.8 ± 0.1  | -0.1 ± 0.2   | 0.2 ± 0.3     | 1.0 ± 0.3     | -0.8 ± 0.1  | 1.0 ± 0.2     | 0.2 ± 0.1   | 0.1 ± 0.2    | 0.1 ± 0.3     | E Africa F   | E Asia F     |
| 40  | 1.4 ± 0.4     | -1.3 ± 0.1   | 1.6 ± 0.2     | -0.6 ± 0.1  | -0.2 ± 0.2   | -0.3 ± 0.4    | 0.7 ± 0.3     | -0.7 ± 0.1  | 1.0 ± 0.2     | 0.1 ± 0.1   | -0.0 ± 0.2   | -0.2 ± 0.3    | E Europe F   | E Asia F     |
| 41  | 1.0 ± 0.4     | -1.3 ± 0.1   | 1.5 ± 0.2     | -0.7 ± 0.2  | -0.3 ± 0.2   | -0.6 ± 0.4    | 1.0 ± 0.3     | -0.7 ± 0.1  | 1.0 ± 0.2     | 0.1 ± 0.1   | -0.1 ± 0.2   | -0.2 ± 0.3    | E Europe F   | E Asia F     |
| 42  | 1.4 ± 0.4     | -1.1 ± 0.1   | 1.5 ± 0.2     | -0.2 ± 0.2  | -0.2 ± 0.2   | -0.2 ± 0.4    | 0.9 ± 0.3     | -0.6 ± 0.1  | 1.1 ± 0.2     | 0.0 ± 0.1   | 0.1 ± 0.2    | -0.3 ± 0.3    | E Europe F   | E Asia F     |
| 43  | 1.6 ± 0.4     | -1.2 ± 0.1   | 1.3 ± 0.2     | -0.6 ± 0.2  | -0.3 ± 0.2   | -0.5 ± 0.4    | 0.8 ± 0.3     | -0.5 ± 0.1  | 0.8 ± 0.2     | 0.0 ± 0.1   | 0.1 ± 0.2    | -0.7 ± 0.3    | E Africa F   | E Asia F     |
| 44  | 1.4 ± 0.5     | -1.4 ± 0.1   | 1.0 ± 0.2     | -0.5 ± 0.2  | -0.3 ± 0.2   | -0.6 ± 0.4    | 0.4 ± 0.3     | -0.6 ± 0.1  | 1.1 ± 0.2     | 0.0 ± 0.1   | 0.1 ± 0.2    | -0.4 ± 0.3    | E Africa F   | E Asia F     |
| 45  | 1.1 ± 0.5     | -1.3 ± 0.1   | 1.0 ± 0.2     | -0.6 ± 0.2  | -0.3 ± 0.2   | -0.6 ± 0.4    | 0.3 ± 0.4     | -0.7 ± 0.1  | 0.8 ± 0.2     | 0.1 ± 0.1   | -0.2 ± 0.2   | -0.8 ± 0.3    | E Africa F   | E Asia F     |
| 46  | 2.2 ± 0.5     | -1.5 ± 0.1   | 0.9 ± 0.2     | -0.5 ± 0.2  | -0.6 ± 0.2   | -0.5 ± 0.4    | 0.3 ± 0.4     | -0.6 ± 0.1  | 0.8 ± 0.2     | 0.2 ± 0.2   | -0.1 ± 0.2   | -0.8 ± 0.3    | E Africa F   | E Asia F     |
| 47  | 1.9 ± 0.5     | -1.4 ± 0.1   | 0.9 ± 0.2     | -0.5 ± 0.2  | -0.5 ± 0.2   | -0.6 ± 0.4    | 0.8 ± 0.4     | -0.7 ± 0.1  | 0.9 ± 0.2     | 0.2 ± 0.2   | 0.1 ± 0.3    | -0.8 ± 0.3    | E Africa F   | E Asia F     |
| 48  | 1.1 ± 0.6     | -1.4 ± 0.1   | 1.1 ± 0.2     | -0.4 ± 0.2  | -0.5 ± 0.2   | -0.6 ± 0.4    | 0.3 ± 0.5     | -0.7 ± 0.1  | 1.0 ± 0.3     | 0.4 ± 0.2   | -0.0 ± 0.3   | -1.1 ± 0.4    | E Africa F   | E Asia F     |
| 49  | 2.0 ± 0.6     | -1.6 ± 0.1   | 0.8 ± 0.2     | -0.4 ± 0.2  | -0.3 ± 0.2   | -0.5 ± 0.5    | 0.1 ± 0.5     | -0.5 ± 0.1  | 0.9 ± 0.3     | 0.2 ± 0.2   | 0.1 ± 0.3    | -0.8 ± 0.4    | E Africa F   | E Asia F     |
| 50  | 1.5 ± 0.7     | -1.4 ± 0.1   | 1.0 ± 0.2     | -0.4 ± 0.2  | -0.3 ± 0.2   | -0.5 ± 0.5    | 0.1 ± 0.5     | -0.6 ± 0.1  | 0.9 ± 0.3     | 0.2 ± 0.2   | 0.4 ± 0.3    | -1.2 ± 0.4    | E Africa F   | E Asia F     |
| 51  | 1.8 ± 0.6     | -1.4 ± 0.1   | 0.7 ± 0.3     | -0.2 ± 0.2  | -0.2 ± 0.2   | -0.2 ± 0.5    | -0.3 ± 0.6    | -0.5 ± 0.2  | 1.2 ± 0.3     | 0.5 ± 0.2   | 0.3 ± 0.3    | -1.2 ± 0.4    | E Africa F   | E Asia F     |
| 52  | 1.7 ± 0.7     | -1.3 ± 0.1   | 1.1 ± 0.3     | -0.3 ± 0.2  | 0.2 ± 0.2    | -0.8 ± 0.5    | 0.1 ± 0.6     | -0.5 ± 0.2  | 1.2 ± 0.3     | 0.6 ± 0.2   | 0.4 ± 0.3    | -1.0 ± 0.5    | E Africa F   | E Asia F     |

**Table 48.** Mean raw error for Application-type images for algorithm incode-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.7 ± 0.3     | 1.7 ± 0.1   | 1.4 ± 0.3     | 1.5 ± 0.2   | 1.6 ± 0.3    | 3.4 ± 0.3     | 1.0 ± 0.3     | 0.7 ± 0.1   | -0.0 ± 0.2    | 0.1 ± 0.1   | 0.5 ± 0.2    | 1.2 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 2.9 ± 0.3     | 1.9 ± 0.2   | 1.7 ± 0.3     | 2.1 ± 0.2   | 2.2 ± 0.4    | 3.1 ± 0.4     | 0.8 ± 0.3     | 1.1 ± 0.1   | 0.2 ± 0.2     | 0.5 ± 0.1   | 1.0 ± 0.2    | 1.5 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 2.7 ± 0.4     | 2.0 ± 0.2   | 2.1 ± 0.3     | 2.1 ± 0.2   | 2.4 ± 0.4    | 3.4 ± 0.5     | 1.5 ± 0.3     | 1.14 ± 0.10 | 0.4 ± 0.2     | 0.7 ± 0.1   | 1.0 ± 0.2    | 1.2 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 3.1 ± 0.4     | 2.2 ± 0.2   | 1.9 ± 0.3     | 2.2 ± 0.2   | 2.3 ± 0.3    | 3.5 ± 0.4     | 1.2 ± 0.3     | 1.1 ± 0.1   | 0.1 ± 0.2     | 0.8 ± 0.1   | 1.0 ± 0.2    | 1.6 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 2.3 ± 0.3     | 1.7 ± 0.2   | 1.3 ± 0.3     | 1.9 ± 0.2   | 2.4 ± 0.2    | 2.9 ± 0.4     | 0.9 ± 0.2     | 0.8 ± 0.1   | -0.1 ± 0.2    | 0.5 ± 0.1   | 0.8 ± 0.2    | 0.9 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 3.1 ± 0.4     | 2.1 ± 0.2   | 1.5 ± 0.3     | 2.0 ± 0.2   | 2.5 ± 0.2    | 3.2 ± 0.4     | 1.4 ± 0.3     | 1.1 ± 0.1   | -0.3 ± 0.2    | 0.7 ± 0.1   | 0.9 ± 0.2    | 1.4 ± 0.4     | W Africa F   | E Europe M   |
| 20  | 3.2 ± 0.4     | 2.1 ± 0.2   | 1.5 ± 0.2     | 2.0 ± 0.2   | 2.5 ± 0.2    | 3.3 ± 0.4     | 1.9 ± 0.3     | 1.1 ± 0.1   | 0.2 ± 0.2     | 1.0 ± 0.1   | 1.1 ± 0.2    | 1.6 ± 0.4     | W Africa F   | E Europe M   |
| 21  | 3.4 ± 0.3     | 2.3 ± 0.2   | 1.5 ± 0.2     | 1.8 ± 0.1   | 2.6 ± 0.2    | 3.3 ± 0.4     | 2.4 ± 0.3     | 1.4 ± 0.1   | 0.0 ± 0.2     | 1.3 ± 0.1   | 1.3 ± 0.2    | 1.7 ± 0.4     | E Africa F   | E Europe M   |
| 22  | 3.6 ± 0.4     | 2.2 ± 0.1   | 1.4 ± 0.2     | 1.6 ± 0.1   | 2.5 ± 0.2    | 3.0 ± 0.4     | 3.3 ± 0.3     | 1.7 ± 0.1   | 0.5 ± 0.2     | 1.5 ± 0.2   | 1.3 ± 0.2    | 1.7 ± 0.4     | E Africa F   | E Europe M   |
| 23  | 3.7 ± 0.3     | 2.1 ± 0.1   | 1.6 ± 0.2     | 1.6 ± 0.1   | 2.5 ± 0.2    | 3.5 ± 0.3     | 3.3 ± 0.3     | 1.8 ± 0.1   | 0.5 ± 0.2     | 1.7 ± 0.1   | 1.6 ± 0.2    | 2.3 ± 0.3     | E Africa F   | E Europe M   |
| 24  | 3.9 ± 0.3     | 1.9 ± 0.1   | 1.5 ± 0.2     | 1.3 ± 0.1   | 2.4 ± 0.1    | 3.0 ± 0.3     | 3.5 ± 0.3     | 1.9 ± 0.1   | 0.7 ± 0.2     | 1.8 ± 0.1   | 1.8 ± 0.2    | 2.4 ± 0.3     | E Africa F   | E Europe M   |
| 25  | 3.6 ± 0.3     | 1.9 ± 0.1   | 1.4 ± 0.1     | 1.1 ± 0.1   | 2.3 ± 0.1    | 3.3 ± 0.3     | 3.6 ± 0.3     | 2.0 ± 0.1   | 0.8 ± 0.2     | 2.0 ± 0.1   | 1.9 ± 0.2    | 2.4 ± 0.3     | E Africa M   | E Europe M   |
| 26  | 3.7 ± 0.3     | 1.8 ± 0.1   | 1.4 ± 0.1     | 0.9 ± 0.1   | 2.3 ± 0.1    | 3.3 ± 0.3     | 3.5 ± 0.3     | 1.9 ± 0.1   | 0.8 ± 0.2     | 1.9 ± 0.1   | 2.0 ± 0.2    | 2.3 ± 0.2     | E Africa F   | E Europe M   |
| 27  | 3.8 ± 0.3     | 1.65 ± 0.09 | 1.4 ± 0.1     | 0.7 ± 0.1   | 2.1 ± 0.1    | 3.4 ± 0.3     | 3.4 ± 0.3     | 2.0 ± 0.1   | 0.9 ± 0.2     | 1.8 ± 0.1   | 2.1 ± 0.2    | 2.4 ± 0.2     | E Africa F   | S Asia F     |
| 28  | 3.7 ± 0.3     | 1.50 ± 0.09 | 1.3 ± 0.1     | 0.6 ± 0.1   | 2.0 ± 0.1    | 3.2 ± 0.3     | 3.3 ± 0.2     | 2.0 ± 0.1   | 1.0 ± 0.2     | 1.72 ± 0.10 | 2.0 ± 0.2    | 2.1 ± 0.2     | E Africa F   | S Asia F     |
| 29  | 4.0 ± 0.3     | 1.26 ± 0.09 | 1.3 ± 0.1     | 0.4 ± 0.1   | 2.0 ± 0.1    | 3.2 ± 0.3     | 3.3 ± 0.2     | 1.9 ± 0.1   | 0.8 ± 0.1     | 1.57 ± 0.09 | 1.7 ± 0.2    | 2.5 ± 0.2     | E Africa F   | S Asia F     |
| 30  | 4.0 ± 0.3     | 1.25 ± 0.09 | 1.1 ± 0.1     | 0.3 ± 0.1   | 1.9 ± 0.1    | 2.9 ± 0.3     | 3.2 ± 0.2     | 1.8 ± 0.1   | 0.9 ± 0.1     | 1.48 ± 0.09 | 1.9 ± 0.2    | 2.3 ± 0.2     | E Africa F   | S Asia F     |
| 31  | 3.7 ± 0.3     | 1.07 ± 0.09 | 1.2 ± 0.1     | 0.2 ± 0.1   | 1.8 ± 0.1    | 2.9 ± 0.3     | 3.3 ± 0.2     | 1.7 ± 0.1   | 0.9 ± 0.1     | 1.46 ± 0.09 | 1.7 ± 0.2    | 2.4 ± 0.2     | E Africa F   | S Asia F     |
| 32  | 4.0 ± 0.3     | 1.00 ± 0.09 | 1.1 ± 0.1     | 0.2 ± 0.1   | 1.7 ± 0.2    | 3.2 ± 0.3     | 3.0 ± 0.2     | 1.7 ± 0.1   | 0.9 ± 0.2     | 1.41 ± 0.09 | 1.7 ± 0.2    | 2.3 ± 0.2     | E Africa F   | S Asia F     |
| 33  | 3.9 ± 0.3     | 0.83 ± 0.09 | 1.0 ± 0.1     | 0.1 ± 0.1   | 1.7 ± 0.2    | 3.0 ± 0.3     | 3.1 ± 0.2     | 1.6 ± 0.1   | 1.1 ± 0.2     | 1.27 ± 0.09 | 1.5 ± 0.2    | 2.3 ± 0.2     | E Africa F   | S Asia F     |
| 34  | 3.6 ± 0.3     | 0.83 ± 0.09 | 1.1 ± 0.1     | 0.0 ± 0.1   | 1.6 ± 0.2    | 2.9 ± 0.3     | 2.7 ± 0.2     | 1.5 ± 0.1   | 0.9 ± 0.2     | 1.29 ± 0.10 | 1.6 ± 0.2    | 2.3 ± 0.2     | E Africa F   | S Asia F     |
| 35  | 4.0 ± 0.3     | 0.60 ± 0.10 | 1.0 ± 0.2     | -0.0 ± 0.1  | 1.5 ± 0.2    | 2.6 ± 0.3     | 3.0 ± 0.2     | 1.6 ± 0.1   | 1.0 ± 0.2     | 1.27 ± 0.10 | 1.5 ± 0.2    | 2.0 ± 0.2     | E Africa F   | S Asia F     |
| 36  | 3.8 ± 0.3     | 0.63 ± 0.10 | 1.1 ± 0.2     | -0.0 ± 0.1  | 1.5 ± 0.2    | 2.9 ± 0.3     | 2.8 ± 0.3     | 1.5 ± 0.1   | 1.2 ± 0.2     | 1.20 ± 0.10 | 1.5 ± 0.2    | 2.1 ± 0.2     | E Africa F   | S Asia F     |
| 37  | 3.9 ± 0.3     | 0.64 ± 0.10 | 1.2 ± 0.2     | 0.1 ± 0.1   | 1.5 ± 0.2    | 2.8 ± 0.3     | 2.8 ± 0.3     | 1.5 ± 0.1   | 1.0 ± 0.2     | 1.24 ± 0.10 | 1.3 ± 0.2    | 2.0 ± 0.2     | E Africa F   | S Asia F     |
| 38  | 4.2 ± 0.4     | 0.58 ± 0.10 | 1.2 ± 0.2     | 0.2 ± 0.1   | 1.4 ± 0.2    | 2.7 ± 0.3     | 2.6 ± 0.3     | 1.5 ± 0.1   | 1.1 ± 0.2     | 1.2 ± 0.1   | 1.6 ± 0.2    | 2.0 ± 0.2     | E Africa F   | S Asia F     |
| 39  | 3.9 ± 0.3     | 0.7 ± 0.1   | 1.2 ± 0.2     | 0.1 ± 0.1   | 1.5 ± 0.2    | 2.8 ± 0.3     | 2.8 ± 0.3     | 1.5 ± 0.1   | 1.3 ± 0.2     | 1.2 ± 0.1   | 1.6 ± 0.2    | 1.8 ± 0.2     | E Africa F   | S Asia F     |
| 40  | 3.7 ± 0.4     | 0.6 ± 0.1   | 1.6 ± 0.2     | 0.3 ± 0.1   | 1.4 ± 0.2    | 2.5 ± 0.3     | 2.7 ± 0.3     | 1.5 ± 0.1   | 1.2 ± 0.2     | 1.2 ± 0.1   | 1.4 ± 0.2    | 1.7 ± 0.2     | E Africa F   | S Asia F     |
| 41  | 3.5 ± 0.4     | 0.6 ± 0.1   | 1.5 ± 0.2     | 0.2 ± 0.1   | 1.3 ± 0.2    | 2.3 ± 0.3     | 2.9 ± 0.3     | 1.5 ± 0.1   | 1.3 ± 0.2     | 1.2 ± 0.1   | 1.4 ± 0.2    | 1.7 ± 0.3     | E Africa F   | S Asia F     |
| 42  | 3.9 ± 0.4     | 0.81 ± 0.10 | 1.6 ± 0.2     | 0.6 ± 0.2   | 1.3 ± 0.2    | 2.4 ± 0.3     | 2.7 ± 0.3     | 1.6 ± 0.1   | 1.4 ± 0.2     | 1.1 ± 0.1   | 1.6 ± 0.2    | 1.6 ± 0.3     | E Africa F   | S Asia F     |
| 43  | 4.1 ± 0.4     | 0.7 ± 0.1   | 1.4 ± 0.2     | 0.3 ± 0.2   | 1.3 ± 0.2    | 2.2 ± 0.3     | 3.1 ± 0.3     | 1.6 ± 0.1   | 1.3 ± 0.2     | 1.1 ± 0.1   | 1.6 ± 0.2    | 1.3 ± 0.3     | E Africa F   | S Asia F     |
| 44  | 3.9 ± 0.5     | 0.6 ± 0.1   | 1.3 ± 0.2     | 0.5 ± 0.2   | 1.4 ± 0.2    | 2.2 ± 0.4     | 2.5 ± 0.4     | 1.6 ± 0.1   | 1.6 ± 0.2     | 1.0 ± 0.1   | 1.7 ± 0.2    | 1.5 ± 0.3     | E Africa F   | S Asia F     |
| 45  | 3.9 ± 0.4     | 0.7 ± 0.1   | 1.4 ± 0.2     | 0.4 ± 0.2   | 1.3 ± 0.2    | 2.0 ± 0.3     | 2.6 ± 0.4     | 1.6 ± 0.1   | 1.4 ± 0.2     | 1.2 ± 0.1   | 1.5 ± 0.2    | 1.3 ± 0.3     | E Africa F   | S Asia F     |
| 46  | 4.7 ± 0.5     | 0.6 ± 0.1   | 1.4 ± 0.2     | 0.6 ± 0.2   | 1.2 ± 0.2    | 2.1 ± 0.4     | 2.7 ± 0.4     | 1.7 ± 0.1   | 1.5 ± 0.2     | 1.3 ± 0.2   | 1.7 ± 0.2    | 1.1 ± 0.3     | E Africa F   | S Asia F     |
| 47  | 4.5 ± 0.5     | 0.8 ± 0.1   | 1.5 ± 0.2     | 0.5 ± 0.2   | 1.2 ± 0.2    | 2.0 ± 0.4     | 3.3 ± 0.4     | 1.7 ± 0.1   | 1.6 ± 0.2     | 1.4 ± 0.2   | 1.8 ± 0.3    | 1.3 ± 0.3     | E Africa F   | S Asia F     |
| 48  | 3.8 ± 0.6     | 0.7 ± 0.1   | 1.6 ± 0.2     | 0.8 ± 0.2   | 1.2 ± 0.2    | 2.1 ± 0.4     | 2.8 ± 0.5     | 1.6 ± 0.1   | 1.6 ± 0.3     | 1.5 ± 0.2   | 1.9 ± 0.3    | 1.1 ± 0.3     | E Africa F   | E Asia F     |
| 49  | 4.5 ± 0.5     | 0.7 ± 0.1   | 1.5 ± 0.2     | 0.9 ± 0.2   | 1.5 ± 0.2    | 2.1 ± 0.4     | 2.7 ± 0.5     | 1.8 ± 0.1   | 1.6 ± 0.2     | 1.5 ± 0.2   | 1.9 ± 0.3    | 1.2 ± 0.3     | E Africa F   | E Asia F     |
| 50  | 3.9 ± 0.5     | 0.8 ± 0.1   | 1.7 ± 0.2     | 0.8 ± 0.2   | 1.4 ± 0.2    | 1.9 ± 0.4     | 2.9 ± 0.5     | 1.8 ± 0.1   | 1.5 ± 0.3     | 1.5 ± 0.2   | 2.3 ± 0.3    | 1.0 ± 0.3     | E Africa F   | E Asia F     |
| 51  | 4.2 ± 0.5     | 0.9 ± 0.1   | 1.6 ± 0.2     | 1.1 ± 0.2   | 1.5 ± 0.2    | 2.1 ± 0.4     | 2.5 ± 0.5     | 1.8 ± 0.1   | 1.9 ± 0.3     | 1.7 ± 0.2   | 2.2 ± 0.3    | 1.0 ± 0.4     | E Africa F   | E Asia F     |
| 52  | 4.0 ± 0.5     | 0.9 ± 0.1   | 1.9 ± 0.2     | 1.1 ± 0.2   | 1.8 ± 0.2    | 1.7 ± 0.4     | 2.8 ± 0.5     | 1.8 ± 0.1   | 2.0 ± 0.3     | 1.7 ± 0.2   | 2.1 ± 0.3    | 1.1 ± 0.4     | E Africa F   | E Asia F     |

**Table 49.** Mean raw error for Application-type images for algorithm innovatrics-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 5.3 ± 0.2     | 3.13 ± 0.09  | 2.8 ± 0.2     | 3.7 ± 0.1    | 2.9 ± 0.2    | 5.8 ± 0.3     | 3.4 ± 0.2     | 2.06 ± 0.09  | 1.5 ± 0.2     | 2.41 ± 0.10  | 1.9 ± 0.2    | 3.4 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 5.2 ± 0.3     | 2.99 ± 0.10  | 2.8 ± 0.2     | 3.8 ± 0.2    | 2.7 ± 0.2    | 5.7 ± 0.3     | 3.3 ± 0.2     | 2.15 ± 0.08  | 1.6 ± 0.2     | 2.5 ± 0.1    | 1.9 ± 0.1    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 5.0 ± 0.3     | 2.8 ± 0.1    | 2.8 ± 0.2     | 3.5 ± 0.2    | 2.6 ± 0.2    | 5.8 ± 0.4     | 3.7 ± 0.2     | 2.02 ± 0.08  | 1.9 ± 0.2     | 2.4 ± 0.1    | 1.7 ± 0.1    | 3.4 ± 0.3     | W Africa F   | S E Asia M   |
| 17  | 5.3 ± 0.3     | 2.5 ± 0.1    | 2.6 ± 0.2     | 3.3 ± 0.2    | 2.2 ± 0.2    | 5.3 ± 0.3     | 3.4 ± 0.2     | 1.88 ± 0.09  | 1.7 ± 0.2     | 2.4 ± 0.1    | 1.6 ± 0.2    | 3.8 ± 0.4     | W Africa F   | S E Asia M   |
| 18  | 4.5 ± 0.2     | 1.9 ± 0.1    | 1.9 ± 0.2     | 2.9 ± 0.1    | 1.7 ± 0.1    | 4.8 ± 0.3     | 3.2 ± 0.2     | 1.37 ± 0.09  | 1.4 ± 0.2     | 2.3 ± 0.1    | 1.1 ± 0.2    | 3.3 ± 0.3     | W Africa F   | S E Asia M   |
| 19  | 4.5 ± 0.3     | 1.6 ± 0.1    | 1.7 ± 0.2     | 2.8 ± 0.1    | 1.4 ± 0.1    | 4.8 ± 0.2     | 3.4 ± 0.2     | 1.30 ± 0.10  | 1.4 ± 0.2     | 2.3 ± 0.1    | 0.7 ± 0.2    | 3.4 ± 0.3     | W Africa F   | S E Asia M   |
| 20  | 4.3 ± 0.2     | 1.4 ± 0.1    | 1.4 ± 0.2     | 2.4 ± 0.1    | 1.1 ± 0.1    | 4.6 ± 0.2     | 3.3 ± 0.2     | 1.06 ± 0.10  | 1.5 ± 0.2     | 2.4 ± 0.1    | 0.6 ± 0.1    | 3.3 ± 0.3     | W Africa F   | S E Asia M   |
| 21  | 3.9 ± 0.2     | 1.1 ± 0.1    | 1.0 ± 0.2     | 2.2 ± 0.1    | 0.9 ± 0.1    | 4.1 ± 0.2     | 3.2 ± 0.2     | 0.83 ± 0.09  | 1.2 ± 0.2     | 2.2 ± 0.1    | 0.4 ± 0.2    | 3.3 ± 0.3     | W Africa F   | S E Asia M   |
| 22  | 3.7 ± 0.2     | 0.7 ± 0.1    | 0.6 ± 0.1     | 1.8 ± 0.1    | 0.4 ± 0.1    | 3.7 ± 0.2     | 3.6 ± 0.2     | 0.53 ± 0.09  | 1.3 ± 0.2     | 2.1 ± 0.1    | 0.1 ± 0.1    | 3.0 ± 0.3     | W Africa F   | S E Asia M   |
| 23  | 3.3 ± 0.2     | 0.40 ± 0.09  | 0.4 ± 0.1     | 1.60 ± 0.10  | 0.1 ± 0.1    | 3.5 ± 0.2     | 3.2 ± 0.2     | 0.30 ± 0.10  | 0.9 ± 0.2     | 2.1 ± 0.1    | -0.1 ± 0.1   | 3.2 ± 0.2     | W Africa F   | S E Asia M   |
| 24  | 3.0 ± 0.2     | -0.04 ± 0.08 | 0.3 ± 0.1     | 1.16 ± 0.09  | -0.2 ± 0.1   | 3.0 ± 0.2     | 3.1 ± 0.2     | -0.07 ± 0.10 | 0.9 ± 0.1     | 2.0 ± 0.1    | -0.3 ± 0.1   | 2.9 ± 0.2     | W Africa M   | S E Asia M   |
| 25  | 2.6 ± 0.2     | -0.33 ± 0.08 | -0.1 ± 0.1    | 0.71 ± 0.08  | -0.6 ± 0.1   | 2.8 ± 0.2     | 2.9 ± 0.2     | -0.26 ± 0.10 | 0.7 ± 0.1     | 1.77 ± 0.10  | -0.6 ± 0.1   | 2.7 ± 0.2     | E Africa M   | S E Asia F   |
| 26  | 2.1 ± 0.2     | -0.77 ± 0.07 | -0.3 ± 0.1    | 0.24 ± 0.07  | -0.8 ± 0.1   | 2.5 ± 0.2     | 2.6 ± 0.2     | -0.53 ± 0.09 | 0.4 ± 0.1     | 1.51 ± 0.09  | -0.7 ± 0.1   | 2.3 ± 0.2     | E Africa M   | S E Asia F   |
| 27  | 1.9 ± 0.2     | -1.09 ± 0.07 | -0.5 ± 0.1    | -0.21 ± 0.08 | -1.2 ± 0.1   | 2.2 ± 0.2     | 2.3 ± 0.2     | -0.88 ± 0.10 | 0.4 ± 0.1     | 1.20 ± 0.09  | -0.8 ± 0.1   | 2.1 ± 0.2     | E Africa M   | S E Asia F   |
| 28  | 1.6 ± 0.2     | -1.50 ± 0.07 | -0.7 ± 0.1    | -0.56 ± 0.08 | -1.6 ± 0.1   | 1.8 ± 0.2     | 2.0 ± 0.2     | -1.1 ± 0.1   | 0.2 ± 0.1     | 0.86 ± 0.08  | -1.1 ± 0.2   | 1.8 ± 0.2     | E Africa M   | S E Asia F   |
| 29  | 1.5 ± 0.2     | -1.77 ± 0.07 | -0.8 ± 0.1    | -0.96 ± 0.08 | -1.9 ± 0.1   | 1.5 ± 0.2     | 1.8 ± 0.2     | -1.30 ± 0.09 | -0.0 ± 0.1    | 0.50 ± 0.08  | -1.4 ± 0.2   | 1.9 ± 0.2     | W Africa M   | S E Asia F   |
| 30  | 1.1 ± 0.2     | -2.09 ± 0.07 | -1.1 ± 0.1    | -1.40 ± 0.08 | -2.1 ± 0.1   | 1.0 ± 0.2     | 1.7 ± 0.2     | -1.61 ± 0.09 | -0.2 ± 0.1    | 0.16 ± 0.08  | -1.6 ± 0.2   | 1.5 ± 0.2     | E Africa M   | E Asia F     |
| 31  | 0.8 ± 0.2     | -2.41 ± 0.07 | -1.2 ± 0.1    | -1.80 ± 0.08 | -2.3 ± 0.1   | 0.8 ± 0.2     | 1.5 ± 0.2     | -1.9 ± 0.1   | -0.4 ± 0.1    | -0.10 ± 0.08 | -1.7 ± 0.2   | 1.3 ± 0.2     | E Africa M   | E Asia F     |
| 32  | 0.7 ± 0.2     | -2.62 ± 0.08 | -1.3 ± 0.1    | -2.08 ± 0.09 | -2.5 ± 0.1   | 0.7 ± 0.2     | 1.2 ± 0.2     | -2.0 ± 0.1   | -0.4 ± 0.1    | -0.37 ± 0.08 | -1.8 ± 0.2   | 1.1 ± 0.2     | E Africa M   | E Asia F     |
| 33  | 0.5 ± 0.2     | -2.84 ± 0.08 | -1.6 ± 0.1    | -2.34 ± 0.09 | -2.7 ± 0.1   | 0.2 ± 0.2     | 1.3 ± 0.2     | -2.2 ± 0.1   | -0.5 ± 0.1    | -0.66 ± 0.08 | -2.0 ± 0.2   | 0.9 ± 0.2     | E Africa M   | E Asia F     |
| 34  | 0.1 ± 0.3     | -2.99 ± 0.08 | -1.7 ± 0.1    | -2.59 ± 0.10 | -2.9 ± 0.1   | 0.0 ± 0.2     | 0.8 ± 0.2     | -2.3 ± 0.1   | -0.7 ± 0.2    | -0.83 ± 0.09 | -2.0 ± 0.2   | 0.7 ± 0.2     | E Africa M   | E Asia F     |
| 35  | 0.3 ± 0.3     | -3.27 ± 0.09 | -1.9 ± 0.1    | -2.9 ± 0.1   | -3.1 ± 0.1   | -0.5 ± 0.2    | 0.9 ± 0.2     | -2.4 ± 0.1   | -0.7 ± 0.2    | -1.01 ± 0.09 | -2.2 ± 0.2   | 0.4 ± 0.2     | E Africa M   | E Asia F     |
| 36  | -0.1 ± 0.3    | -3.37 ± 0.09 | -1.9 ± 0.1    | -3.1 ± 0.1   | -3.1 ± 0.2   | -0.5 ± 0.2    | 0.7 ± 0.3     | -2.4 ± 0.1   | -0.6 ± 0.2    | -1.21 ± 0.09 | -2.0 ± 0.2   | 0.3 ± 0.2     | E Africa M   | E Asia F     |
| 37  | -0.3 ± 0.3    | -3.51 ± 0.09 | -2.0 ± 0.2    | -3.1 ± 0.1   | -3.2 ± 0.2   | -1.0 ± 0.3    | 0.7 ± 0.3     | -2.4 ± 0.1   | -0.8 ± 0.2    | -1.3 ± 0.1   | -2.2 ± 0.2   | 0.0 ± 0.2     | E Africa M   | E Asia F     |
| 38  | 0.0 ± 0.3     | -3.67 ± 0.09 | -2.0 ± 0.2    | -3.2 ± 0.1   | -3.2 ± 0.2   | -1.3 ± 0.3    | 0.5 ± 0.3     | -2.5 ± 0.1   | -0.9 ± 0.2    | -1.48 ± 0.10 | -2.0 ± 0.2   | -0.1 ± 0.2    | E Africa M   | E Asia F     |
| 39  | -0.5 ± 0.3    | -3.73 ± 0.09 | -2.2 ± 0.2    | -3.4 ± 0.1   | -3.2 ± 0.2   | -1.3 ± 0.3    | 0.8 ± 0.3     | -2.5 ± 0.1   | -0.9 ± 0.2    | -1.5 ± 0.1   | -1.9 ± 0.2   | -0.2 ± 0.2    | E Africa M   | E Asia F     |
| 40  | -0.6 ± 0.3    | -3.90 ± 0.09 | -2.0 ± 0.2    | -3.4 ± 0.1   | -3.4 ± 0.2   | -1.8 ± 0.3    | 0.9 ± 0.3     | -2.7 ± 0.1   | -0.9 ± 0.2    | -1.5 ± 0.1   | -1.9 ± 0.2   | -0.3 ± 0.2    | E Africa M   | E Asia F     |
| 41  | -1.1 ± 0.4    | -3.95 ± 0.09 | -2.2 ± 0.2    | -3.4 ± 0.2   | -3.5 ± 0.2   | -2.1 ± 0.3    | 0.9 ± 0.3     | -2.6 ± 0.1   | -0.9 ± 0.2    | -1.6 ± 0.1   | -2.1 ± 0.2   | -0.3 ± 0.3    | E Africa M   | E Asia F     |
| 42  | -0.7 ± 0.4    | -3.97 ± 0.09 | -2.2 ± 0.2    | -3.1 ± 0.2   | -3.6 ± 0.2   | -2.1 ± 0.3    | 0.8 ± 0.3     | -2.5 ± 0.1   | -0.8 ± 0.2    | -1.7 ± 0.1   | -1.7 ± 0.2   | -0.4 ± 0.3    | E Africa M   | E Asia F     |
| 43  | -0.6 ± 0.4    | -4.18 ± 0.09 | -2.3 ± 0.2    | -3.3 ± 0.2   | -3.5 ± 0.2   | -2.3 ± 0.3    | 1.0 ± 0.4     | -2.6 ± 0.1   | -0.9 ± 0.2    | -1.6 ± 0.1   | -1.8 ± 0.2   | -0.8 ± 0.3    | E Africa M   | E Asia F     |
| 44  | -0.8 ± 0.4    | -4.39 ± 0.10 | -2.6 ± 0.2    | -3.1 ± 0.2   | -3.3 ± 0.2   | -2.6 ± 0.3    | 0.6 ± 0.4     | -2.6 ± 0.1   | -0.8 ± 0.2    | -1.7 ± 0.1   | -1.7 ± 0.2   | -0.5 ± 0.3    | E Africa M   | E Asia F     |
| 45  | -1.0 ± 0.4    | -4.39 ± 0.10 | -2.6 ± 0.2    | -3.1 ± 0.2   | -3.5 ± 0.2   | -2.7 ± 0.3    | 0.4 ± 0.4     | -2.7 ± 0.1   | -1.0 ± 0.2    | -1.5 ± 0.1   | -1.8 ± 0.2   | -0.8 ± 0.3    | E Africa M   | E Asia F     |
| 46  | -0.4 ± 0.5    | -4.6 ± 0.1   | -2.5 ± 0.2    | -2.8 ± 0.2   | -3.7 ± 0.2   | -2.6 ± 0.3    | 0.5 ± 0.4     | -2.7 ± 0.1   | -0.8 ± 0.2    | -1.5 ± 0.2   | -1.6 ± 0.3   | -1.0 ± 0.3    | E Africa M   | E Asia F     |
| 47  | -0.9 ± 0.5    | -4.5 ± 0.1   | -2.7 ± 0.2    | -2.9 ± 0.2   | -3.6 ± 0.2   | -2.7 ± 0.3    | 0.9 ± 0.4     | -2.7 ± 0.1   | -0.9 ± 0.2    | -1.5 ± 0.2   | -1.4 ± 0.3   | -0.7 ± 0.3    | E Africa M   | E Asia F     |
| 48  | -1.2 ± 0.5    | -4.6 ± 0.1   | -2.6 ± 0.2    | -2.6 ± 0.2   | -3.6 ± 0.2   | -2.8 ± 0.4    | 0.4 ± 0.4     | -2.7 ± 0.1   | -0.9 ± 0.3    | -1.5 ± 0.2   | -1.5 ± 0.3   | -1.0 ± 0.4    | E Africa M   | E Asia F     |
| 49  | -0.8 ± 0.5    | -4.7 ± 0.1   | -3.0 ± 0.2    | -2.4 ± 0.2   | -3.3 ± 0.2   | -3.3 ± 0.4    | 0.0 ± 0.5     | -2.6 ± 0.1   | -1.0 ± 0.2    | -1.7 ± 0.2   | -1.5 ± 0.2   | -0.9 ± 0.3    | E Africa M   | E Asia F     |
| 50  | -1.4 ± 0.5    | -4.6 ± 0.1   | -2.7 ± 0.2    | -2.6 ± 0.2   | -3.5 ± 0.2   | -3.1 ± 0.4    | 0.1 ± 0.4     | -2.6 ± 0.1   | -1.2 ± 0.2    | -1.7 ± 0.2   | -1.2 ± 0.2   | -1.3 ± 0.3    | E Africa M   | E Asia F     |
| 51  | -1.3 ± 0.5    | -4.6 ± 0.1   | -3.1 ± 0.2    | -2.5 ± 0.2   | -3.5 ± 0.2   | -3.2 ± 0.4    | -0.3 ± 0.5    | -2.6 ± 0.1   | -1.1 ± 0.2    | -1.6 ± 0.2   | -1.3 ± 0.3   | -1.4 ± 0.4    | E Africa M   | E Asia F     |
| 52  | -1.6 ± 0.5    | -4.6 ± 0.1   | -2.8 ± 0.2    | -2.7 ± 0.2   | -3.2 ± 0.2   | -3.5 ± 0.4    | -0.2 ± 0.5    | -2.7 ± 0.1   | -1.2 ± 0.3    | -1.8 ± 0.2   | -1.6 ± 0.3   | -1.3 ± 0.4    | E Africa M   | E Asia F     |

**Table 50.** Mean raw error for Application-type images for algorithm innovatrics-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 5.2 ± 0.2     | 2.99 ± 0.08  | 2.9 ± 0.2     | 3.5 ± 0.1    | 2.9 ± 0.1    | 5.9 ± 0.3     | 3.5 ± 0.2     | 2.08 ± 0.09  | 1.5 ± 0.2     | 2.38 ± 0.09  | 1.9 ± 0.1    | 3.4 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 5.1 ± 0.3     | 2.80 ± 0.09  | 2.8 ± 0.2     | 3.5 ± 0.1    | 2.6 ± 0.2    | 5.8 ± 0.3     | 3.3 ± 0.2     | 2.10 ± 0.08  | 1.6 ± 0.2     | 2.4 ± 0.1    | 1.9 ± 0.1    | 3.6 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.8 ± 0.3     | 2.5 ± 0.1    | 2.8 ± 0.2     | 3.2 ± 0.1    | 2.5 ± 0.2    | 5.8 ± 0.4     | 3.7 ± 0.2     | 1.93 ± 0.07  | 1.8 ± 0.2     | 2.3 ± 0.1    | 1.7 ± 0.1    | 3.4 ± 0.3     | W Africa F   | S E Asia M   |
| 17  | 5.0 ± 0.3     | 2.2 ± 0.1    | 2.6 ± 0.2     | 3.0 ± 0.2    | 2.1 ± 0.2    | 5.4 ± 0.3     | 3.4 ± 0.2     | 1.70 ± 0.09  | 1.6 ± 0.2     | 2.3 ± 0.1    | 1.5 ± 0.1    | 3.8 ± 0.4     | W Africa F   | S E Asia M   |
| 18  | 4.4 ± 0.2     | 1.65 ± 0.10  | 1.9 ± 0.2     | 2.5 ± 0.1    | 1.5 ± 0.1    | 4.8 ± 0.3     | 3.2 ± 0.2     | 1.13 ± 0.08  | 1.2 ± 0.2     | 2.0 ± 0.1    | 1.0 ± 0.2    | 3.3 ± 0.3     | W Africa F   | S E Asia M   |
| 19  | 4.2 ± 0.2     | 1.4 ± 0.1    | 1.6 ± 0.2     | 2.4 ± 0.1    | 1.2 ± 0.1    | 4.8 ± 0.2     | 3.4 ± 0.2     | 1.02 ± 0.09  | 1.1 ± 0.2     | 2.1 ± 0.1    | 0.6 ± 0.1    | 3.4 ± 0.3     | W Africa F   | S E Asia M   |
| 20  | 4.0 ± 0.2     | 1.1 ± 0.1    | 1.2 ± 0.2     | 2.1 ± 0.1    | 0.9 ± 0.1    | 4.6 ± 0.2     | 3.2 ± 0.2     | 0.71 ± 0.09  | 1.2 ± 0.2     | 2.1 ± 0.1    | 0.3 ± 0.1    | 3.3 ± 0.3     | W Africa F   | S E Asia M   |
| 21  | 3.6 ± 0.2     | 0.8 ± 0.1    | 0.9 ± 0.1     | 1.8 ± 0.1    | 0.6 ± 0.1    | 4.1 ± 0.2     | 3.1 ± 0.2     | 0.48 ± 0.09  | 0.9 ± 0.2     | 1.9 ± 0.1    | 0.1 ± 0.1    | 3.2 ± 0.3     | W Africa F   | S E Asia M   |
| 22  | 3.4 ± 0.2     | 0.35 ± 0.09  | 0.5 ± 0.1     | 1.34 ± 0.10  | 0.2 ± 0.1    | 3.8 ± 0.2     | 3.4 ± 0.2     | 0.17 ± 0.09  | 0.9 ± 0.2     | 1.8 ± 0.1    | -0.2 ± 0.1   | 2.9 ± 0.2     | W Africa F   | S E Asia M   |
| 23  | 2.9 ± 0.2     | 0.07 ± 0.08  | 0.2 ± 0.1     | 1.17 ± 0.09  | -0.1 ± 0.1   | 3.5 ± 0.2     | 3.0 ± 0.2     | -0.12 ± 0.09 | 0.6 ± 0.2     | 1.8 ± 0.1    | -0.4 ± 0.1   | 3.0 ± 0.2     | W Africa F   | S E Asia M   |
| 24  | 2.7 ± 0.2     | -0.43 ± 0.08 | -0.0 ± 0.1    | 0.73 ± 0.09  | -0.53 ± 0.10 | 3.0 ± 0.2     | 2.9 ± 0.2     | -0.41 ± 0.10 | 0.5 ± 0.1     | 1.6 ± 0.1    | -0.6 ± 0.1   | 2.7 ± 0.2     | W Africa F   | S E Asia M   |
| 25  | 2.2 ± 0.2     | -0.70 ± 0.07 | -0.36 ± 0.10  | 0.28 ± 0.08  | -0.9 ± 0.1   | 2.6 ± 0.2     | 2.6 ± 0.2     | -0.61 ± 0.09 | 0.4 ± 0.1     | 1.46 ± 0.09  | -0.9 ± 0.1   | 2.5 ± 0.2     | W Africa F   | S E Asia M   |
| 26  | 1.7 ± 0.2     | -1.15 ± 0.07 | -0.61 ± 0.09  | -0.18 ± 0.07 | -1.15 ± 0.10 | 2.3 ± 0.2     | 2.3 ± 0.2     | -0.92 ± 0.09 | 0.1 ± 0.1     | 1.19 ± 0.09  | -1.0 ± 0.1   | 2.1 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 1.6 ± 0.2     | -1.51 ± 0.07 | -0.89 ± 0.10  | -0.67 ± 0.07 | -1.60 ± 0.10 | 2.1 ± 0.2     | 2.0 ± 0.2     | -1.19 ± 0.09 | 0.0 ± 0.1     | 0.86 ± 0.08  | -1.2 ± 0.1   | 1.9 ± 0.2     | W Africa F   | S E Asia F   |
| 28  | 1.2 ± 0.2     | -1.89 ± 0.07 | -1.09 ± 0.10  | -1.03 ± 0.08 | -1.9 ± 0.1   | 1.7 ± 0.2     | 1.7 ± 0.2     | -1.43 ± 0.10 | 0.1 ± 0.1     | 0.56 ± 0.08  | -1.5 ± 0.1   | 1.5 ± 0.2     | W Africa F   | S E Asia F   |
| 29  | 1.1 ± 0.2     | -2.26 ± 0.07 | -1.2 ± 0.1    | -1.42 ± 0.08 | -2.2 ± 0.1   | 1.3 ± 0.2     | 1.5 ± 0.2     | -1.74 ± 0.08 | -0.4 ± 0.1    | 0.22 ± 0.08  | -1.9 ± 0.1   | 1.6 ± 0.2     | W Africa M   | E Asia F     |
| 30  | 0.7 ± 0.2     | -2.56 ± 0.07 | -1.5 ± 0.1    | -1.86 ± 0.08 | -2.4 ± 0.1   | 0.9 ± 0.2     | 1.4 ± 0.2     | -2.12 ± 0.09 | -0.6 ± 0.1    | -0.19 ± 0.07 | -2.0 ± 0.2   | 1.2 ± 0.2     | E Africa M   | E Asia F     |
| 31  | 0.3 ± 0.2     | -2.93 ± 0.07 | -1.6 ± 0.1    | -2.29 ± 0.08 | -2.7 ± 0.1   | 0.6 ± 0.2     | 1.2 ± 0.2     | -2.36 ± 0.09 | -0.9 ± 0.1    | -0.44 ± 0.08 | -2.1 ± 0.2   | 1.0 ± 0.2     | E Africa M   | E Asia F     |
| 32  | 0.4 ± 0.2     | -3.21 ± 0.07 | -1.8 ± 0.1    | -2.58 ± 0.09 | -3.0 ± 0.1   | 0.4 ± 0.2     | 0.9 ± 0.2     | -2.61 ± 0.09 | -0.9 ± 0.1    | -0.73 ± 0.08 | -2.4 ± 0.2   | 0.8 ± 0.2     | E Africa M   | E Asia F     |
| 33  | -0.0 ± 0.2    | -3.51 ± 0.07 | -2.0 ± 0.1    | -2.84 ± 0.09 | -3.2 ± 0.1   | 0.0 ± 0.2     | 0.9 ± 0.2     | -2.91 ± 0.10 | -0.9 ± 0.1    | -1.04 ± 0.08 | -2.6 ± 0.2   | 0.6 ± 0.2     | E Africa M   | E Asia F     |
| 34  | -0.4 ± 0.3    | -3.71 ± 0.08 | -2.2 ± 0.1    | -3.13 ± 0.09 | -3.3 ± 0.1   | -0.3 ± 0.2    | 0.5 ± 0.2     | -3.12 ± 0.10 | -1.2 ± 0.1    | -1.24 ± 0.09 | -2.6 ± 0.2   | 0.4 ± 0.2     | E Africa M   | E Asia F     |
| 35  | -0.2 ± 0.3    | -4.01 ± 0.08 | -2.4 ± 0.1    | -3.41 ± 0.10 | -3.6 ± 0.1   | -0.8 ± 0.2    | 0.6 ± 0.3     | -3.2 ± 0.1   | -1.2 ± 0.2    | -1.47 ± 0.09 | -2.8 ± 0.2   | 0.1 ± 0.2     | E Africa M   | E Asia F     |
| 36  | -0.6 ± 0.3    | -4.11 ± 0.09 | -2.4 ± 0.1    | -3.68 ± 0.10 | -3.6 ± 0.2   | -0.8 ± 0.2    | 0.3 ± 0.3     | -3.3 ± 0.1   | -1.2 ± 0.2    | -1.70 ± 0.09 | -2.7 ± 0.2   | 0.1 ± 0.2     | E Africa M   | E Asia F     |
| 37  | -0.8 ± 0.3    | -4.28 ± 0.09 | -2.5 ± 0.2    | -3.8 ± 0.1   | -3.7 ± 0.2   | -1.3 ± 0.2    | 0.5 ± 0.3     | -3.4 ± 0.1   | -1.3 ± 0.2    | -1.85 ± 0.10 | -2.9 ± 0.2   | -0.2 ± 0.2    | E Africa M   | E Asia F     |
| 38  | -0.5 ± 0.3    | -4.46 ± 0.09 | -2.6 ± 0.2    | -3.9 ± 0.1   | -3.7 ± 0.2   | -1.6 ± 0.3    | 0.1 ± 0.3     | -3.4 ± 0.1   | -1.4 ± 0.2    | -2.02 ± 0.10 | -2.6 ± 0.2   | -0.3 ± 0.2    | E Africa M   | E Asia F     |
| 39  | -1.0 ± 0.3    | -4.55 ± 0.09 | -2.7 ± 0.2    | -4.1 ± 0.1   | -3.8 ± 0.2   | -1.7 ± 0.3    | 0.4 ± 0.3     | -3.5 ± 0.1   | -1.3 ± 0.2    | -2.1 ± 0.1   | -2.6 ± 0.2   | -0.4 ± 0.2    | E Africa M   | E Asia F     |
| 40  | -1.2 ± 0.3    | -4.71 ± 0.09 | -2.5 ± 0.2    | -4.1 ± 0.1   | -4.0 ± 0.2   | -2.2 ± 0.3    | 0.5 ± 0.3     | -3.6 ± 0.1   | -1.4 ± 0.2    | -2.1 ± 0.1   | -2.6 ± 0.2   | -0.6 ± 0.2    | E Africa M   | E Asia F     |
| 41  | -1.6 ± 0.4    | -4.83 ± 0.09 | -2.7 ± 0.2    | -4.3 ± 0.1   | -4.1 ± 0.2   | -2.6 ± 0.3    | 0.5 ± 0.3     | -3.6 ± 0.1   | -1.4 ± 0.2    | -2.2 ± 0.1   | -2.7 ± 0.2   | -0.6 ± 0.3    | E Africa M   | E Asia F     |
| 42  | -1.3 ± 0.4    | -4.89 ± 0.09 | -2.7 ± 0.2    | -3.9 ± 0.2   | -4.3 ± 0.2   | -2.6 ± 0.3    | 0.6 ± 0.4     | -3.6 ± 0.1   | -1.2 ± 0.2    | -2.3 ± 0.1   | -2.5 ± 0.2   | -0.8 ± 0.3    | E Africa M   | E Asia F     |
| 43  | -1.2 ± 0.4    | -5.16 ± 0.09 | -2.8 ± 0.2    | -4.2 ± 0.2   | -4.3 ± 0.2   | -2.8 ± 0.3    | 0.9 ± 0.4     | -3.6 ± 0.1   | -1.4 ± 0.2    | -2.2 ± 0.1   | -2.6 ± 0.2   | -1.1 ± 0.3    | E Africa M   | E Asia F     |
| 44  | -1.2 ± 0.5    | -5.44 ± 0.09 | -3.2 ± 0.2    | -4.1 ± 0.2   | -4.2 ± 0.2   | -3.2 ± 0.3    | 0.4 ± 0.4     | -3.6 ± 0.1   | -1.2 ± 0.2    | -2.3 ± 0.1   | -2.4 ± 0.2   | -0.9 ± 0.3    | E Africa M   | E Asia F     |
| 45  | -1.4 ± 0.5    | -5.52 ± 0.09 | -3.1 ± 0.2    | -4.0 ± 0.2   | -4.4 ± 0.2   | -3.2 ± 0.3    | 0.4 ± 0.4     | -3.7 ± 0.1   | -1.4 ± 0.2    | -2.2 ± 0.2   | -2.5 ± 0.2   | -1.0 ± 0.3    | E Africa M   | E Asia F     |
| 46  | -0.5 ± 0.6    | -5.80 ± 0.10 | -3.1 ± 0.2    | -3.7 ± 0.2   | -4.7 ± 0.2   | -3.3 ± 0.4    | 0.7 ± 0.5     | -3.8 ± 0.1   | -1.2 ± 0.2    | -2.0 ± 0.2   | -2.4 ± 0.3   | -1.3 ± 0.3    | E Africa M   | E Asia F     |
| 47  | -0.7 ± 0.6    | -5.7 ± 0.1   | -3.2 ± 0.2    | -3.7 ± 0.2   | -4.5 ± 0.2   | -3.3 ± 0.4    | 1.5 ± 0.5     | -3.7 ± 0.1   | -1.3 ± 0.2    | -1.9 ± 0.2   | -2.2 ± 0.3   | -0.8 ± 0.4    | E Africa M   | E Asia F     |
| 48  | -1.2 ± 0.6    | -5.8 ± 0.1   | -3.1 ± 0.2    | -3.4 ± 0.2   | -4.6 ± 0.2   | -3.4 ± 0.4    | 0.9 ± 0.5     | -3.7 ± 0.1   | -1.2 ± 0.3    | -1.8 ± 0.2   | -2.2 ± 0.3   | -1.2 ± 0.4    | E Africa M   | E Asia F     |
| 49  | -0.3 ± 0.7    | -6.0 ± 0.1   | -3.4 ± 0.2    | -3.2 ± 0.2   | -4.4 ± 0.2   | -3.7 ± 0.4    | 0.7 ± 0.5     | -3.6 ± 0.1   | -1.3 ± 0.3    | -1.9 ± 0.2   | -2.1 ± 0.3   | -1.0 ± 0.3    | E Africa M   | E Asia F     |
| 50  | -1.1 ± 0.7    | -5.9 ± 0.1   | -3.1 ± 0.2    | -3.3 ± 0.2   | -4.5 ± 0.2   | -3.6 ± 0.4    | 0.9 ± 0.5     | -3.5 ± 0.1   | -1.4 ± 0.3    | -1.9 ± 0.2   | -1.7 ± 0.3   | -1.2 ± 0.4    | E Africa M   | E Asia F     |
| 51  | -0.3 ± 0.7    | -5.9 ± 0.1   | -3.4 ± 0.2    | -3.0 ± 0.2   | -4.4 ± 0.2   | -3.5 ± 0.5    | 0.6 ± 0.6     | -3.4 ± 0.2   | -1.2 ± 0.3    | -1.7 ± 0.2   | -1.6 ± 0.3   | -1.2 ± 0.4    | E Africa M   | E Asia F     |
| 52  | -0.5 ± 0.7    | -5.8 ± 0.1   | -3.0 ± 0.3    | -3.1 ± 0.2   | -3.9 ± 0.3   | -3.7 ± 0.5    | 1.1 ± 0.6     | -3.4 ± 0.2   | -1.2 ± 0.3    | -1.7 ± 0.2   | -1.7 ± 0.3   | -0.9 ± 0.4    | E Africa M   | E Asia F     |

**Table 51.** Mean raw error for Application-type images for algorithm innovatrics-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 4.2 ± 0.3     | 2.08 ± 0.07  | 1.9 ± 0.2     | 2.5 ± 0.1    | 1.9 ± 0.1    | 5.4 ± 0.4     | 2.3 ± 0.2     | 1.36 ± 0.07 | 0.82 ± 0.10   | 1.42 ± 0.07  | 1.2 ± 0.1    | 2.4 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 4.3 ± 0.3     | 1.9 ± 0.1    | 1.8 ± 0.2     | 2.4 ± 0.2    | 1.6 ± 0.2    | 5.5 ± 0.4     | 2.0 ± 0.2     | 1.18 ± 0.08 | 0.5 ± 0.1     | 1.20 ± 0.10  | 0.9 ± 0.1    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.3 ± 0.4     | 1.7 ± 0.1    | 1.9 ± 0.2     | 2.2 ± 0.2    | 1.5 ± 0.2    | 5.8 ± 0.5     | 2.8 ± 0.3     | 0.96 ± 0.09 | 0.6 ± 0.2     | 1.1 ± 0.1    | 0.6 ± 0.1    | 2.7 ± 0.3     | W Africa F   | S E Asia M   |
| 17  | 4.8 ± 0.3     | 1.5 ± 0.1    | 1.7 ± 0.3     | 2.1 ± 0.2    | 1.3 ± 0.2    | 5.7 ± 0.4     | 2.8 ± 0.3     | 0.9 ± 0.1   | 0.3 ± 0.2     | 1.3 ± 0.2    | 0.5 ± 0.2    | 3.5 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.2 ± 0.3     | 1.1 ± 0.1    | 1.3 ± 0.3     | 1.8 ± 0.1    | 0.8 ± 0.2    | 5.1 ± 0.3     | 2.8 ± 0.2     | 0.51 ± 0.10 | 0.4 ± 0.2     | 1.4 ± 0.1    | 0.0 ± 0.2    | 3.1 ± 0.3     | W Africa F   | S E Asia M   |
| 19  | 4.2 ± 0.3     | 0.9 ± 0.1    | 1.0 ± 0.3     | 1.8 ± 0.1    | 0.7 ± 0.2    | 5.4 ± 0.3     | 3.3 ± 0.3     | 0.7 ± 0.1   | 0.5 ± 0.2     | 1.7 ± 0.1    | -0.1 ± 0.2   | 3.6 ± 0.4     | W Africa F   | S E Asia M   |
| 20  | 4.1 ± 0.3     | 0.8 ± 0.1    | 0.8 ± 0.2     | 1.6 ± 0.2    | 0.5 ± 0.2    | 5.3 ± 0.3     | 3.4 ± 0.3     | 0.6 ± 0.1   | 0.8 ± 0.2     | 2.0 ± 0.1    | -0.0 ± 0.2   | 3.6 ± 0.3     | W Africa F   | S E Asia M   |
| 21  | 3.8 ± 0.3     | 0.7 ± 0.1    | 0.7 ± 0.2     | 1.5 ± 0.1    | 0.4 ± 0.2    | 4.8 ± 0.3     | 3.5 ± 0.3     | 0.5 ± 0.1   | 0.7 ± 0.2     | 2.0 ± 0.1    | -0.0 ± 0.2   | 3.6 ± 0.3     | W Africa F   | S E Asia M   |
| 22  | 3.8 ± 0.3     | 0.3 ± 0.1    | 0.4 ± 0.2     | 1.1 ± 0.1    | 0.0 ± 0.1    | 4.5 ± 0.3     | 4.0 ± 0.3     | 0.3 ± 0.1   | 1.1 ± 0.2     | 2.0 ± 0.1    | -0.2 ± 0.2   | 3.5 ± 0.3     | W Africa F   | S E Asia M   |
| 23  | 3.3 ± 0.2     | 0.2 ± 0.1    | 0.3 ± 0.2     | 1.0 ± 0.1    | -0.1 ± 0.1   | 4.4 ± 0.2     | 3.6 ± 0.3     | 0.3 ± 0.1   | 0.8 ± 0.2     | 2.1 ± 0.1    | -0.3 ± 0.2   | 3.8 ± 0.3     | W Africa F   | S E Asia M   |
| 24  | 3.1 ± 0.2     | -0.20 ± 0.10 | 0.2 ± 0.1     | 0.60 ± 0.10  | -0.4 ± 0.1   | 3.9 ± 0.2     | 3.6 ± 0.3     | 0.0 ± 0.1   | 0.8 ± 0.2     | 2.0 ± 0.1    | -0.3 ± 0.2   | 3.5 ± 0.2     | W Africa F   | S E Asia F   |
| 25  | 2.8 ± 0.2     | -0.48 ± 0.09 | 0.0 ± 0.1     | 0.11 ± 0.09  | -0.7 ± 0.1   | 3.6 ± 0.2     | 3.4 ± 0.2     | -0.1 ± 0.1  | 0.8 ± 0.2     | 1.7 ± 0.1    | -0.5 ± 0.2   | 3.3 ± 0.2     | W Africa F   | S E Asia F   |
| 26  | 2.2 ± 0.2     | -0.90 ± 0.08 | -0.2 ± 0.1    | -0.32 ± 0.08 | -0.9 ± 0.1   | 3.2 ± 0.2     | 3.3 ± 0.2     | -0.2 ± 0.1  | 0.5 ± 0.2     | 1.54 ± 0.10  | -0.5 ± 0.2   | 3.0 ± 0.2     | E Africa M   | S E Asia F   |
| 27  | 2.1 ± 0.2     | -1.17 ± 0.08 | -0.4 ± 0.1    | -0.80 ± 0.08 | -1.3 ± 0.1   | 2.9 ± 0.2     | 2.9 ± 0.2     | -0.5 ± 0.1  | 0.5 ± 0.1     | 1.27 ± 0.09  | -0.5 ± 0.2   | 2.8 ± 0.2     | W Africa F   | S E Asia F   |
| 28  | 1.8 ± 0.2     | -1.51 ± 0.08 | -0.5 ± 0.1    | -1.12 ± 0.08 | -1.6 ± 0.1   | 2.6 ± 0.2     | 2.7 ± 0.2     | -0.6 ± 0.1  | 0.5 ± 0.1     | 0.91 ± 0.09  | -0.7 ± 0.2   | 2.4 ± 0.2     | E Africa M   | S E Asia F   |
| 29  | 1.6 ± 0.2     | -1.79 ± 0.08 | -0.6 ± 0.1    | -1.47 ± 0.09 | -1.8 ± 0.1   | 2.1 ± 0.2     | 2.4 ± 0.2     | -0.8 ± 0.1  | 0.1 ± 0.1     | 0.61 ± 0.08  | -1.1 ± 0.2   | 2.5 ± 0.2     | W Africa M   | E Asia F     |
| 30  | 1.2 ± 0.2     | -2.09 ± 0.08 | -0.8 ± 0.1    | -1.93 ± 0.09 | -1.9 ± 0.1   | 1.7 ± 0.2     | 2.4 ± 0.2     | -1.0 ± 0.1  | 0.0 ± 0.1     | 0.23 ± 0.08  | -1.1 ± 0.2   | 2.1 ± 0.2     | E Africa M   | E Asia F     |
| 31  | 0.8 ± 0.2     | -2.40 ± 0.08 | -0.9 ± 0.1    | -2.36 ± 0.09 | -2.3 ± 0.1   | 1.4 ± 0.2     | 2.2 ± 0.2     | -1.2 ± 0.1  | -0.2 ± 0.1    | -0.02 ± 0.08 | -1.3 ± 0.2   | 1.8 ± 0.2     | E Africa M   | E Asia F     |
| 32  | 0.7 ± 0.2     | -2.70 ± 0.08 | -1.0 ± 0.1    | -2.65 ± 0.10 | -2.5 ± 0.1   | 1.1 ± 0.2     | 1.9 ± 0.2     | -1.4 ± 0.1  | -0.2 ± 0.2    | -0.25 ± 0.08 | -1.4 ± 0.2   | 1.6 ± 0.2     | E Africa M   | E Asia F     |
| 33  | 0.4 ± 0.2     | -2.91 ± 0.08 | -1.3 ± 0.1    | -2.9 ± 0.1   | -2.7 ± 0.1   | 0.8 ± 0.2     | 1.8 ± 0.2     | -1.6 ± 0.1  | -0.2 ± 0.2    | -0.58 ± 0.09 | -1.5 ± 0.2   | 1.4 ± 0.2     | E Africa M   | E Asia F     |
| 34  | -0.0 ± 0.3    | -3.11 ± 0.08 | -1.5 ± 0.2    | -3.2 ± 0.1   | -2.8 ± 0.1   | 0.4 ± 0.2     | 1.4 ± 0.3     | -1.8 ± 0.1  | -0.5 ± 0.2    | -0.76 ± 0.09 | -1.6 ± 0.2   | 1.2 ± 0.2     | E Africa M   | S Asia F     |
| 35  | 0.1 ± 0.3     | -3.44 ± 0.09 | -1.6 ± 0.2    | -3.5 ± 0.1   | -3.2 ± 0.2   | -0.1 ± 0.2    | 1.5 ± 0.3     | -1.9 ± 0.1  | -0.5 ± 0.2    | -0.97 ± 0.10 | -1.9 ± 0.2   | 0.9 ± 0.2     | E Africa M   | S Asia F     |
| 36  | -0.2 ± 0.3    | -3.53 ± 0.09 | -1.6 ± 0.2    | -3.7 ± 0.1   | -3.1 ± 0.2   | -0.3 ± 0.2    | 1.3 ± 0.3     | -2.0 ± 0.1  | -0.3 ± 0.2    | -1.1 ± 0.1   | -1.6 ± 0.2   | 0.8 ± 0.2     | E Africa M   | S Asia F     |
| 37  | -0.5 ± 0.3    | -3.79 ± 0.09 | -1.6 ± 0.2    | -3.7 ± 0.1   | -3.3 ± 0.2   | -0.7 ± 0.3    | 1.4 ± 0.3     | -2.1 ± 0.1  | -0.5 ± 0.2    | -1.3 ± 0.1   | -1.8 ± 0.2   | 0.5 ± 0.2     | E Africa M   | E Asia F     |
| 38  | -0.1 ± 0.3    | -3.99 ± 0.09 | -1.7 ± 0.2    | -3.8 ± 0.1   | -3.2 ± 0.2   | -0.9 ± 0.3    | 1.1 ± 0.3     | -2.2 ± 0.1  | -0.7 ± 0.2    | -1.4 ± 0.1   | -1.6 ± 0.2   | 0.4 ± 0.2     | E Africa M   | E Asia F     |
| 39  | -0.8 ± 0.3    | -4.10 ± 0.10 | -1.8 ± 0.2    | -4.0 ± 0.1   | -3.4 ± 0.2   | -1.0 ± 0.3    | 1.3 ± 0.3     | -2.3 ± 0.1  | -0.6 ± 0.2    | -1.4 ± 0.1   | -1.6 ± 0.2   | 0.3 ± 0.2     | E Africa M   | E Asia F     |
| 40  | -0.9 ± 0.4    | -4.29 ± 0.10 | -1.7 ± 0.2    | -3.8 ± 0.2   | -3.5 ± 0.2   | -1.6 ± 0.3    | 1.4 ± 0.4     | -2.3 ± 0.1  | -0.6 ± 0.2    | -1.5 ± 0.1   | -1.7 ± 0.2   | 0.2 ± 0.3     | E Africa M   | E Asia F     |
| 41  | -1.3 ± 0.4    | -4.37 ± 0.10 | -1.8 ± 0.2    | -3.9 ± 0.2   | -3.6 ± 0.2   | -2.0 ± 0.3    | 1.6 ± 0.4     | -2.3 ± 0.1  | -0.6 ± 0.2    | -1.6 ± 0.1   | -1.8 ± 0.2   | 0.2 ± 0.3     | E Africa M   | E Asia F     |
| 42  | -0.9 ± 0.4    | -4.4 ± 0.1   | -1.7 ± 0.2    | -3.5 ± 0.2   | -3.6 ± 0.2   | -2.0 ± 0.3    | 1.5 ± 0.4     | -2.3 ± 0.1  | -0.3 ± 0.2    | -1.7 ± 0.1   | -1.5 ± 0.3   | 0.2 ± 0.3     | E Africa M   | E Asia F     |
| 43  | -0.9 ± 0.4    | -4.5 ± 0.1   | -1.9 ± 0.2    | -3.8 ± 0.2   | -3.8 ± 0.2   | -2.2 ± 0.3    | 1.8 ± 0.4     | -2.3 ± 0.1  | -0.5 ± 0.2    | -1.5 ± 0.1   | -1.6 ± 0.3   | -0.3 ± 0.3    | E Africa M   | E Asia F     |
| 44  | -1.1 ± 0.5    | -4.8 ± 0.1   | -2.0 ± 0.2    | -3.5 ± 0.2   | -3.5 ± 0.2   | -2.5 ± 0.4    | 1.5 ± 0.4     | -2.3 ± 0.1  | -0.3 ± 0.2    | -1.6 ± 0.2   | -1.4 ± 0.3   | -0.0 ± 0.3    | E Africa M   | E Asia F     |
| 45  | -1.4 ± 0.5    | -4.8 ± 0.1   | -2.2 ± 0.2    | -3.4 ± 0.2   | -3.7 ± 0.2   | -2.6 ± 0.4    | 1.2 ± 0.5     | -2.4 ± 0.1  | -0.4 ± 0.2    | -1.4 ± 0.2   | -1.5 ± 0.3   | -0.0 ± 0.4    | E Africa M   | E Asia F     |
| 46  | -0.6 ± 0.6    | -5.0 ± 0.1   | -2.3 ± 0.2    | -3.1 ± 0.2   | -3.9 ± 0.2   | -2.7 ± 0.3    | 1.7 ± 0.5     | -2.3 ± 0.1  | -0.1 ± 0.3    | -1.3 ± 0.2   | -1.2 ± 0.3   | -0.3 ± 0.4    | E Africa M   | E Asia F     |
| 47  | -0.8 ± 0.6    | -4.8 ± 0.1   | -2.3 ± 0.3    | -3.1 ± 0.2   | -3.7 ± 0.2   | -2.8 ± 0.4    | 2.3 ± 0.5     | -2.2 ± 0.2  | -0.4 ± 0.3    | -1.2 ± 0.2   | -0.9 ± 0.3   | 0.1 ± 0.4     | E Africa M   | E Asia F     |
| 48  | -1.0 ± 0.6    | -4.9 ± 0.1   | -2.4 ± 0.2    | -2.8 ± 0.3   | -3.8 ± 0.2   | -2.9 ± 0.4    | 1.8 ± 0.5     | -2.1 ± 0.2  | -0.2 ± 0.3    | -1.0 ± 0.2   | -1.0 ± 0.3   | -0.3 ± 0.4    | E Africa M   | E Asia F     |
| 49  | -0.5 ± 0.6    | -5.0 ± 0.1   | -2.8 ± 0.3    | -2.5 ± 0.3   | -3.4 ± 0.3   | -3.4 ± 0.4    | 1.5 ± 0.6     | -1.9 ± 0.2  | -0.3 ± 0.3    | -1.1 ± 0.2   | -0.9 ± 0.3   | -0.2 ± 0.4    | E Africa M   | E Asia F     |
| 50  | -1.0 ± 0.7    | -4.8 ± 0.1   | -2.3 ± 0.3    | -2.6 ± 0.2   | -3.7 ± 0.2   | -3.1 ± 0.4    | 1.8 ± 0.6     | -1.8 ± 0.2  | -0.1 ± 0.3    | -1.0 ± 0.2   | -0.3 ± 0.3   | -0.4 ± 0.4    | E Africa M   | E Asia F     |
| 51  | -0.4 ± 0.7    | -4.8 ± 0.1   | -2.7 ± 0.3    | -2.3 ± 0.3   | -3.6 ± 0.3   | -3.1 ± 0.5    | 1.5 ± 0.7     | -1.6 ± 0.2  | 0.0 ± 0.3     | -0.8 ± 0.3   | -0.4 ± 0.3   | -0.3 ± 0.5    | E Africa M   | E Asia F     |
| 52  | -0.7 ± 0.7    | -4.7 ± 0.2   | -2.1 ± 0.3    | -2.3 ± 0.3   | -3.0 ± 0.3   | -3.5 ± 0.4    | 2.3 ± 0.6     | -1.5 ± 0.2  | -0.1 ± 0.3    | -0.8 ± 0.2   | -0.5 ± 0.4   | -0.0 ± 0.5    | E Africa M   | E Asia F     |



**Table 52.** Mean raw error for Application-type images for algorithm jumio-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 5.4 ± 0.3     | 4.3 ± 0.1    | 4.2 ± 0.4     | 4.5 ± 0.2   | 4.7 ± 0.3    | 5.8 ± 0.4     | 3.8 ± 0.3     | 2.8 ± 0.1   | 3.2 ± 0.5     | 2.6 ± 0.1    | 2.7 ± 0.2    | 3.6 ± 0.3     | W Africa F   | S Asia M     |
| 15  | 5.7 ± 0.4     | 4.3 ± 0.2    | 4.4 ± 0.4     | 4.7 ± 0.2   | 4.7 ± 0.3    | 6.0 ± 0.4     | 3.1 ± 0.3     | 2.95 ± 0.10 | 3.6 ± 0.5     | 2.5 ± 0.1    | 2.9 ± 0.2    | 3.2 ± 0.3     | W Africa F   | S Asia M     |
| 16  | 5.2 ± 0.4     | 4.1 ± 0.1    | 4.5 ± 0.4     | 4.6 ± 0.2   | 4.7 ± 0.3    | 6.0 ± 0.5     | 2.9 ± 0.3     | 2.62 ± 0.09 | 3.6 ± 0.5     | 2.2 ± 0.1    | 2.6 ± 0.2    | 3.0 ± 0.3     | W Africa F   | S Asia M     |
| 17  | 5.3 ± 0.4     | 3.9 ± 0.2    | 4.3 ± 0.4     | 4.5 ± 0.2   | 4.6 ± 0.3    | 5.8 ± 0.4     | 2.4 ± 0.3     | 2.4 ± 0.1   | 3.2 ± 0.5     | 2.1 ± 0.1    | 2.5 ± 0.2    | 3.0 ± 0.4     | W Africa F   | S Asia M     |
| 18  | 4.7 ± 0.3     | 3.5 ± 0.1    | 3.6 ± 0.3     | 3.9 ± 0.1   | 4.1 ± 0.2    | 5.6 ± 0.4     | 2.3 ± 0.2     | 2.04 ± 0.09 | 2.8 ± 0.4     | 1.9 ± 0.1    | 2.0 ± 0.2    | 2.6 ± 0.3     | W Africa F   | S Asia M     |
| 19  | 5.0 ± 0.4     | 3.4 ± 0.1    | 3.4 ± 0.3     | 3.9 ± 0.2   | 4.0 ± 0.2    | 5.8 ± 0.4     | 2.5 ± 0.3     | 2.0 ± 0.1   | 2.2 ± 0.3     | 1.9 ± 0.1    | 2.0 ± 0.2    | 2.9 ± 0.3     | W Africa F   | S Asia M     |
| 20  | 5.1 ± 0.3     | 3.2 ± 0.1    | 3.1 ± 0.2     | 3.6 ± 0.2   | 3.9 ± 0.2    | 5.5 ± 0.4     | 2.8 ± 0.3     | 1.9 ± 0.1   | 2.2 ± 0.3     | 2.0 ± 0.1    | 2.0 ± 0.2    | 3.0 ± 0.3     | W Africa F   | E Asia M     |
| 21  | 4.8 ± 0.3     | 3.0 ± 0.1    | 2.8 ± 0.2     | 3.3 ± 0.1   | 3.7 ± 0.2    | 5.6 ± 0.4     | 2.9 ± 0.3     | 1.9 ± 0.1   | 1.7 ± 0.3     | 2.0 ± 0.1    | 2.0 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.9 ± 0.3     | 2.7 ± 0.1    | 2.5 ± 0.2     | 3.0 ± 0.1   | 3.5 ± 0.2    | 5.1 ± 0.4     | 3.5 ± 0.3     | 1.9 ± 0.1   | 1.7 ± 0.2     | 1.9 ± 0.1    | 1.9 ± 0.2    | 2.9 ± 0.3     | W Africa F   | E Europe M   |
| 23  | 4.7 ± 0.3     | 2.5 ± 0.1    | 2.3 ± 0.2     | 2.9 ± 0.1   | 3.2 ± 0.1    | 5.5 ± 0.3     | 3.0 ± 0.3     | 1.7 ± 0.1   | 1.3 ± 0.2     | 1.9 ± 0.1    | 1.8 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 24  | 4.7 ± 0.3     | 2.2 ± 0.1    | 2.2 ± 0.2     | 2.5 ± 0.1   | 3.0 ± 0.1    | 4.7 ± 0.3     | 3.2 ± 0.3     | 1.7 ± 0.1   | 1.3 ± 0.2     | 1.9 ± 0.1    | 1.7 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 25  | 4.3 ± 0.3     | 2.10 ± 0.10  | 1.9 ± 0.1     | 2.2 ± 0.1   | 2.9 ± 0.1    | 4.7 ± 0.3     | 3.1 ± 0.2     | 1.6 ± 0.1   | 1.4 ± 0.2     | 1.7 ± 0.1    | 1.7 ± 0.2    | 3.0 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 4.3 ± 0.3     | 1.84 ± 0.10  | 1.7 ± 0.1     | 1.80 ± 0.10 | 2.8 ± 0.1    | 4.4 ± 0.3     | 3.0 ± 0.2     | 1.3 ± 0.1   | 1.1 ± 0.2     | 1.5 ± 0.1    | 1.6 ± 0.2    | 2.7 ± 0.2     | W Africa F   | E Europe M   |
| 27  | 4.1 ± 0.3     | 1.63 ± 0.10  | 1.6 ± 0.1     | 1.49 ± 0.10 | 2.5 ± 0.1    | 4.6 ± 0.3     | 2.7 ± 0.2     | 1.2 ± 0.1   | 1.1 ± 0.1     | 1.3 ± 0.1    | 1.6 ± 0.2    | 2.6 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 3.8 ± 0.3     | 1.36 ± 0.09  | 1.4 ± 0.1     | 1.28 ± 0.09 | 2.3 ± 0.1    | 4.1 ± 0.3     | 2.3 ± 0.2     | 1.1 ± 0.1   | 1.0 ± 0.1     | 0.99 ± 0.10  | 1.4 ± 0.2    | 2.3 ± 0.2     | W Africa F   | S Asia M     |
| 29  | 3.9 ± 0.3     | 1.18 ± 0.09  | 1.3 ± 0.1     | 1.00 ± 0.10 | 2.2 ± 0.1    | 3.9 ± 0.2     | 2.4 ± 0.2     | 0.9 ± 0.1   | 0.7 ± 0.1     | 0.72 ± 0.09  | 1.1 ± 0.2    | 2.4 ± 0.2     | W Africa F   | E Europe M   |
| 30  | 3.5 ± 0.3     | 1.05 ± 0.09  | 1.1 ± 0.1     | 0.73 ± 0.10 | 2.1 ± 0.1    | 3.5 ± 0.3     | 2.2 ± 0.2     | 0.7 ± 0.1   | 0.6 ± 0.1     | 0.49 ± 0.09  | 1.1 ± 0.2    | 2.0 ± 0.2     | E Africa F   | S Asia M     |
| 31  | 3.2 ± 0.3     | 0.87 ± 0.09  | 1.0 ± 0.1     | 0.4 ± 0.1   | 1.9 ± 0.1    | 3.1 ± 0.2     | 2.0 ± 0.2     | 0.5 ± 0.1   | 0.5 ± 0.1     | 0.37 ± 0.09  | 0.9 ± 0.2    | 1.8 ± 0.2     | E Africa F   | S Asia M     |
| 32  | 3.5 ± 0.3     | 0.70 ± 0.09  | 0.8 ± 0.1     | 0.2 ± 0.1   | 1.7 ± 0.2    | 3.2 ± 0.3     | 1.6 ± 0.2     | 0.3 ± 0.1   | 0.3 ± 0.1     | 0.22 ± 0.09  | 0.8 ± 0.2    | 1.6 ± 0.2     | E Africa F   | S Asia F     |
| 33  | 3.1 ± 0.3     | 0.49 ± 0.09  | 0.7 ± 0.1     | 0.1 ± 0.1   | 1.7 ± 0.2    | 2.9 ± 0.3     | 1.7 ± 0.2     | 0.1 ± 0.1   | 0.4 ± 0.1     | 0.06 ± 0.09  | 0.6 ± 0.2    | 1.4 ± 0.2     | E Africa F   | S Asia M     |
| 34  | 3.0 ± 0.3     | 0.34 ± 0.09  | 0.5 ± 0.1     | -0.1 ± 0.1  | 1.4 ± 0.1    | 2.7 ± 0.3     | 1.2 ± 0.2     | -0.1 ± 0.1  | 0.1 ± 0.2     | -0.06 ± 0.10 | 0.5 ± 0.2    | 1.5 ± 0.2     | E Africa F   | S Asia M     |
| 35  | 3.0 ± 0.3     | 0.14 ± 0.10  | 0.4 ± 0.2     | -0.3 ± 0.1  | 1.0 ± 0.2    | 2.3 ± 0.3     | 1.4 ± 0.3     | -0.2 ± 0.1  | 0.1 ± 0.2     | -0.1 ± 0.1   | 0.3 ± 0.2    | 1.2 ± 0.2     | E Africa F   | S Asia F     |
| 36  | 2.7 ± 0.3     | 0.01 ± 0.10  | 0.4 ± 0.2     | -0.5 ± 0.1  | 1.0 ± 0.2    | 2.4 ± 0.3     | 1.2 ± 0.3     | -0.3 ± 0.1  | 0.1 ± 0.2     | -0.2 ± 0.1   | 0.2 ± 0.2    | 1.1 ± 0.2     | E Africa F   | S Asia F     |
| 37  | 2.6 ± 0.3     | -0.12 ± 0.09 | 0.3 ± 0.2     | -0.4 ± 0.1  | 0.8 ± 0.2    | 2.1 ± 0.3     | 1.2 ± 0.3     | -0.4 ± 0.1  | 0.0 ± 0.2     | -0.3 ± 0.1   | -0.0 ± 0.2   | 1.0 ± 0.2     | E Africa F   | S Asia F     |
| 38  | 3.1 ± 0.4     | -0.36 ± 0.10 | 0.1 ± 0.2     | -0.4 ± 0.1  | 0.7 ± 0.2    | 1.9 ± 0.3     | 0.9 ± 0.3     | -0.5 ± 0.1  | -0.1 ± 0.2    | -0.4 ± 0.1   | 0.3 ± 0.2    | 0.8 ± 0.3     | E Africa F   | E Asia M     |
| 39  | 2.6 ± 0.3     | -0.37 ± 0.10 | 0.0 ± 0.2     | -0.5 ± 0.1  | 0.5 ± 0.2    | 1.7 ± 0.3     | 1.3 ± 0.3     | -0.7 ± 0.1  | 0.1 ± 0.2     | -0.3 ± 0.1   | 0.2 ± 0.2    | 0.7 ± 0.3     | E Africa F   | E Asia M     |
| 40  | 2.3 ± 0.4     | -0.55 ± 0.10 | 0.5 ± 0.2     | -0.4 ± 0.1  | 0.4 ± 0.2    | 1.4 ± 0.3     | 1.1 ± 0.3     | -0.7 ± 0.1  | 0.0 ± 0.2     | -0.3 ± 0.1   | -0.1 ± 0.2   | 0.7 ± 0.3     | E Africa F   | E Asia M     |
| 41  | 1.9 ± 0.4     | -0.69 ± 0.10 | 0.2 ± 0.2     | -0.5 ± 0.2  | 0.2 ± 0.2    | 1.0 ± 0.3     | 1.3 ± 0.3     | -0.8 ± 0.1  | 0.0 ± 0.2     | -0.3 ± 0.1   | -0.2 ± 0.2   | 0.8 ± 0.3     | E Africa F   | E Asia M     |
| 42  | 2.5 ± 0.4     | -0.73 ± 0.10 | 0.3 ± 0.2     | -0.1 ± 0.2  | 0.1 ± 0.2    | 1.3 ± 0.4     | 1.4 ± 0.4     | -0.8 ± 0.1  | 0.2 ± 0.2     | -0.3 ± 0.1   | -0.1 ± 0.2   | 0.6 ± 0.3     | E Africa F   | E Asia M     |
| 43  | 2.6 ± 0.4     | -0.9 ± 0.1   | 0.3 ± 0.2     | -0.3 ± 0.2  | -0.1 ± 0.2   | 1.1 ± 0.3     | 1.6 ± 0.4     | -0.8 ± 0.1  | 0.0 ± 0.2     | -0.1 ± 0.1   | -0.0 ± 0.2   | 0.4 ± 0.3     | E Africa F   | E Asia F     |
| 44  | 2.3 ± 0.5     | -1.2 ± 0.1   | 0.1 ± 0.2     | -0.2 ± 0.2  | -0.1 ± 0.2   | 0.6 ± 0.4     | 1.2 ± 0.4     | -0.8 ± 0.1  | 0.4 ± 0.2     | -0.2 ± 0.2   | -0.0 ± 0.2   | 0.6 ± 0.3     | E Africa F   | E Asia F     |
| 45  | 2.3 ± 0.5     | -1.1 ± 0.1   | -0.0 ± 0.2    | -0.2 ± 0.2  | -0.2 ± 0.2   | 0.7 ± 0.4     | 1.2 ± 0.4     | -0.9 ± 0.1  | 0.2 ± 0.2     | 0.1 ± 0.2    | -0.2 ± 0.2   | 0.3 ± 0.3     | E Africa F   | E Asia F     |
| 46  | 3.0 ± 0.5     | -1.4 ± 0.1   | 0.1 ± 0.2     | -0.0 ± 0.2  | -0.3 ± 0.2   | 1.0 ± 0.4     | 1.2 ± 0.4     | -0.9 ± 0.1  | 0.3 ± 0.3     | 0.2 ± 0.2    | -0.2 ± 0.3   | 0.4 ± 0.4     | E Africa F   | E Asia F     |
| 47  | 2.5 ± 0.5     | -1.3 ± 0.1   | 0.2 ± 0.3     | 0.0 ± 0.2   | -0.3 ± 0.2   | 0.7 ± 0.4     | 1.7 ± 0.4     | -1.0 ± 0.1  | 0.3 ± 0.2     | 0.4 ± 0.2    | -0.1 ± 0.3   | 0.5 ± 0.4     | E Africa F   | E Asia F     |
| 48  | 2.2 ± 0.5     | -1.4 ± 0.1   | 0.3 ± 0.2     | 0.2 ± 0.2   | -0.5 ± 0.2   | 0.4 ± 0.4     | 1.1 ± 0.5     | -0.9 ± 0.1  | 0.4 ± 0.3     | 0.4 ± 0.2    | 0.1 ± 0.3    | 0.4 ± 0.4     | E Africa F   | E Asia F     |
| 49  | 2.7 ± 0.5     | -1.6 ± 0.1   | 0.1 ± 0.3     | 0.2 ± 0.2   | -0.1 ± 0.2   | 0.4 ± 0.4     | 1.3 ± 0.5     | -0.9 ± 0.1  | 0.4 ± 0.3     | 0.4 ± 0.2    | 0.0 ± 0.3    | 0.4 ± 0.4     | E Africa F   | E Asia F     |
| 50  | 2.3 ± 0.6     | -1.5 ± 0.1   | 0.2 ± 0.2     | 0.2 ± 0.2   | -0.4 ± 0.2   | 0.4 ± 0.4     | 1.1 ± 0.5     | -0.8 ± 0.1  | 0.2 ± 0.3     | 0.4 ± 0.2    | 0.3 ± 0.3    | 0.2 ± 0.4     | E Africa F   | E Asia F     |
| 51  | 2.0 ± 0.5     | -1.5 ± 0.1   | 0.1 ± 0.2     | 0.2 ± 0.2   | -0.4 ± 0.2   | 0.1 ± 0.4     | 0.9 ± 0.6     | -0.9 ± 0.1  | 0.4 ± 0.2     | 0.5 ± 0.2    | 0.3 ± 0.3    | 0.2 ± 0.4     | E Africa F   | E Asia F     |
| 52  | 1.9 ± 0.6     | -1.6 ± 0.1   | 0.2 ± 0.2     | 0.1 ± 0.2   | -0.1 ± 0.2   | -0.0 ± 0.4    | 0.8 ± 0.6     | -1.0 ± 0.1  | 0.2 ± 0.3     | 0.4 ± 0.2    | -0.1 ± 0.3   | 0.3 ± 0.4     | E Africa F   | E Asia F     |

**Table 53.** Mean raw error for Application-type images for algorithm neurotechnology-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 5.7 ± 0.5     | 5.4 ± 0.2    | 3.8 ± 0.3     | 3.8 ± 0.2   | 3.9 ± 0.4    | 9.3 ± 0.5     | 8.6 ± 0.4     | 6.3 ± 0.2    | 3.3 ± 0.3     | 4.7 ± 0.2    | 5.1 ± 0.4    | 11.1 ± 0.5    | W Africa M   | E Europe M   |
| 15  | 4.9 ± 0.5     | 5.7 ± 0.2    | 3.4 ± 0.3     | 3.9 ± 0.2   | 4.1 ± 0.4    | 8.4 ± 0.5     | 8.7 ± 0.5     | 6.7 ± 0.2    | 3.4 ± 0.3     | 5.1 ± 0.2    | 5.7 ± 0.4    | 10.4 ± 0.5    | W Africa M   | E Europe M   |
| 16  | 5.0 ± 0.5     | 5.0 ± 0.2    | 3.5 ± 0.3     | 3.4 ± 0.3   | 4.4 ± 0.4    | 8.3 ± 0.5     | 9.3 ± 0.3     | 6.4 ± 0.1    | 3.7 ± 0.3     | 5.2 ± 0.2    | 5.8 ± 0.3    | 9.8 ± 0.4     | W Africa M   | S Asia F     |
| 17  | 5.3 ± 0.5     | 4.5 ± 0.2    | 3.2 ± 0.4     | 3.0 ± 0.2   | 3.9 ± 0.4    | 8.2 ± 0.4     | 9.2 ± 0.3     | 5.9 ± 0.1    | 3.3 ± 0.2     | 5.0 ± 0.2    | 5.4 ± 0.3    | 9.8 ± 0.4     | W Africa M   | S Asia F     |
| 18  | 4.2 ± 0.4     | 3.4 ± 0.2    | 2.4 ± 0.3     | 2.3 ± 0.2   | 3.0 ± 0.3    | 7.1 ± 0.4     | 8.5 ± 0.3     | 5.1 ± 0.1    | 2.8 ± 0.2     | 4.4 ± 0.1    | 4.9 ± 0.2    | 8.9 ± 0.3     | W Africa M   | S Asia F     |
| 19  | 3.9 ± 0.4     | 2.6 ± 0.2    | 2.0 ± 0.3     | 2.0 ± 0.2   | 2.2 ± 0.2    | 6.8 ± 0.4     | 8.3 ± 0.3     | 4.5 ± 0.1    | 2.1 ± 0.2     | 4.3 ± 0.1    | 4.1 ± 0.2    | 8.4 ± 0.3     | W Africa M   | S Asia F     |
| 20  | 3.2 ± 0.4     | 1.9 ± 0.2    | 1.6 ± 0.2     | 1.3 ± 0.2   | 1.5 ± 0.2    | 6.1 ± 0.3     | 7.9 ± 0.3     | 3.8 ± 0.1    | 2.0 ± 0.2     | 4.0 ± 0.1    | 3.4 ± 0.2    | 8.1 ± 0.3     | W Africa M   | S Asia F     |
| 21  | 2.7 ± 0.4     | 1.1 ± 0.2    | 1.1 ± 0.2     | 0.6 ± 0.1   | 1.1 ± 0.2    | 5.6 ± 0.3     | 7.3 ± 0.3     | 3.02 ± 0.10  | 1.7 ± 0.2     | 3.4 ± 0.1    | 2.9 ± 0.2    | 7.6 ± 0.3     | W Africa M   | S Asia F     |
| 22  | 2.6 ± 0.3     | 0.2 ± 0.1    | 0.7 ± 0.2     | -0.0 ± 0.1  | 0.2 ± 0.2    | 5.1 ± 0.3     | 7.2 ± 0.2     | 2.30 ± 0.10  | 1.3 ± 0.2     | 2.8 ± 0.1    | 2.2 ± 0.2    | 7.0 ± 0.3     | E Africa M   | S Asia F     |
| 23  | 1.9 ± 0.3     | -0.3 ± 0.1   | 0.4 ± 0.2     | -0.4 ± 0.1  | -0.3 ± 0.2   | 4.3 ± 0.3     | 6.3 ± 0.2     | 1.7 ± 0.1    | 0.8 ± 0.2     | 2.4 ± 0.1    | 1.4 ± 0.2    | 6.3 ± 0.2     | W Africa M   | S Asia F     |
| 24  | 1.3 ± 0.3     | -1.2 ± 0.1   | -0.1 ± 0.2    | -1.1 ± 0.1  | -0.8 ± 0.1   | 3.7 ± 0.3     | 6.0 ± 0.2     | 1.0 ± 0.1    | 0.5 ± 0.2     | 1.9 ± 0.1    | 0.9 ± 0.1    | 5.7 ± 0.2     | W Africa M   | E Asia F     |
| 25  | 0.9 ± 0.3     | -1.8 ± 0.1   | -0.4 ± 0.2    | -1.7 ± 0.1  | -1.7 ± 0.1   | 3.0 ± 0.3     | 5.3 ± 0.2     | 0.3 ± 0.1    | 0.2 ± 0.2     | 1.3 ± 0.1    | 0.2 ± 0.1    | 5.2 ± 0.2     | E Africa M   | E Asia F     |
| 26  | 0.3 ± 0.3     | -2.6 ± 0.1   | -0.8 ± 0.2    | -2.4 ± 0.1  | -2.2 ± 0.1   | 2.4 ± 0.2     | 4.7 ± 0.2     | -0.37 ± 0.10 | -0.2 ± 0.2    | 0.7 ± 0.1    | -0.3 ± 0.2   | 4.4 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 0.1 ± 0.3     | -3.2 ± 0.1   | -1.2 ± 0.2    | -3.0 ± 0.1  | -2.7 ± 0.1   | 1.7 ± 0.2     | 4.0 ± 0.2     | -0.99 ± 0.10 | -0.4 ± 0.2    | 0.1 ± 0.1    | -0.8 ± 0.2   | 3.7 ± 0.2     | E Africa M   | E Asia F     |
| 28  | -0.5 ± 0.3    | -3.77 ± 0.09 | -1.6 ± 0.2    | -3.6 ± 0.1  | -3.3 ± 0.1   | 1.2 ± 0.2     | 3.3 ± 0.2     | -1.6 ± 0.1   | -0.5 ± 0.2    | -0.6 ± 0.1   | -1.5 ± 0.2   | 3.0 ± 0.2     | E Africa M   | E Asia F     |
| 29  | -0.9 ± 0.3    | -4.35 ± 0.09 | -1.8 ± 0.2    | -4.2 ± 0.1  | -4.0 ± 0.1   | 0.5 ± 0.2     | 2.7 ± 0.2     | -2.15 ± 0.09 | -1.1 ± 0.2    | -1.19 ± 0.10 | -1.9 ± 0.2   | 2.6 ± 0.2     | E Africa M   | E Asia F     |
| 30  | -1.5 ± 0.3    | -4.94 ± 0.09 | -2.2 ± 0.2    | -4.8 ± 0.1  | -4.4 ± 0.1   | -0.1 ± 0.2    | 2.4 ± 0.2     | -2.84 ± 0.09 | -1.3 ± 0.2    | -1.72 ± 0.10 | -2.6 ± 0.2   | 1.9 ± 0.2     | E Africa M   | E Asia F     |
| 31  | -1.9 ± 0.3    | -5.49 ± 0.09 | -2.4 ± 0.2    | -5.4 ± 0.1  | -4.9 ± 0.2   | -0.8 ± 0.2    | 1.8 ± 0.2     | -3.3 ± 0.1   | -1.6 ± 0.2    | -2.26 ± 0.10 | -3.2 ± 0.2   | 1.1 ± 0.2     | E Africa M   | E Asia F     |
| 32  | -1.9 ± 0.3    | -6.12 ± 0.09 | -2.6 ± 0.2    | -6.0 ± 0.1  | -5.4 ± 0.2   | -1.3 ± 0.3    | 1.1 ± 0.2     | -3.9 ± 0.1   | -1.6 ± 0.2    | -2.73 ± 0.10 | -3.6 ± 0.2   | 0.5 ± 0.2     | E Africa M   | E Asia F     |
| 33  | -2.7 ± 0.3    | -6.63 ± 0.10 | -2.9 ± 0.2    | -6.4 ± 0.1  | -6.0 ± 0.2   | -1.7 ± 0.3    | 0.7 ± 0.2     | -4.6 ± 0.1   | -1.6 ± 0.2    | -3.2 ± 0.1   | -3.8 ± 0.2   | -0.0 ± 0.2    | E Africa M   | E Asia F     |
| 34  | -3.1 ± 0.3    | -7.1 ± 0.1   | -3.3 ± 0.2    | -6.9 ± 0.1  | -6.4 ± 0.2   | -2.4 ± 0.3    | -0.1 ± 0.2    | -5.0 ± 0.1   | -2.0 ± 0.2    | -3.6 ± 0.1   | -4.2 ± 0.3   | -0.7 ± 0.2    | E Africa M   | E Asia F     |
| 35  | -3.3 ± 0.3    | -7.6 ± 0.1   | -3.5 ± 0.2    | -7.4 ± 0.1  | -6.8 ± 0.2   | -2.8 ± 0.3    | -0.3 ± 0.2    | -5.4 ± 0.1   | -2.2 ± 0.2    | -4.0 ± 0.1   | -4.7 ± 0.3   | -1.1 ± 0.2    | E Africa M   | E Asia F     |
| 36  | -3.6 ± 0.4    | -8.0 ± 0.1   | -3.6 ± 0.3    | -7.8 ± 0.1  | -7.0 ± 0.2   | -3.2 ± 0.3    | -0.9 ± 0.3    | -5.8 ± 0.1   | -2.0 ± 0.3    | -4.5 ± 0.1   | -4.8 ± 0.3   | -1.9 ± 0.2    | E Africa M   | E Asia F     |
| 37  | -3.9 ± 0.4    | -8.4 ± 0.1   | -3.7 ± 0.3    | -8.0 ± 0.2  | -7.6 ± 0.2   | -3.5 ± 0.3    | -1.1 ± 0.3    | -6.2 ± 0.1   | -2.5 ± 0.3    | -4.7 ± 0.1   | -5.3 ± 0.3   | -2.4 ± 0.2    | E Africa M   | E Asia F     |
| 38  | -3.9 ± 0.4    | -8.7 ± 0.1   | -3.7 ± 0.3    | -8.3 ± 0.2  | -7.7 ± 0.2   | -4.2 ± 0.4    | -1.9 ± 0.3    | -6.5 ± 0.1   | -2.5 ± 0.3    | -5.1 ± 0.1   | -4.8 ± 0.3   | -2.8 ± 0.2    | E Africa M   | E Asia F     |
| 39  | -4.1 ± 0.4    | -9.2 ± 0.1   | -3.8 ± 0.3    | -8.8 ± 0.2  | -7.6 ± 0.2   | -4.1 ± 0.4    | -1.8 ± 0.3    | -6.8 ± 0.2   | -2.6 ± 0.3    | -5.3 ± 0.1   | -5.0 ± 0.3   | -3.3 ± 0.2    | E Africa M   | E Asia F     |
| 40  | -4.7 ± 0.5    | -9.4 ± 0.1   | -3.8 ± 0.3    | -8.8 ± 0.2  | -8.0 ± 0.3   | -4.8 ± 0.4    | -2.1 ± 0.3    | -7.0 ± 0.2   | -2.6 ± 0.3    | -5.4 ± 0.2   | -5.2 ± 0.3   | -3.9 ± 0.3    | E Africa M   | E Asia F     |
| 41  | -5.3 ± 0.5    | -9.6 ± 0.1   | -4.0 ± 0.3    | -9.0 ± 0.2  | -8.2 ± 0.3   | -5.6 ± 0.4    | -2.4 ± 0.4    | -7.0 ± 0.2   | -2.6 ± 0.3    | -5.5 ± 0.2   | -5.2 ± 0.4   | -4.1 ± 0.3    | E Africa M   | E Asia F     |
| 42  | -4.5 ± 0.5    | -9.8 ± 0.1   | -3.4 ± 0.4    | -8.8 ± 0.3  | -8.0 ± 0.3   | -5.3 ± 0.4    | -2.6 ± 0.4    | -7.0 ± 0.2   | -2.2 ± 0.3    | -5.9 ± 0.2   | -4.8 ± 0.4   | -4.3 ± 0.3    | E Europe M   | E Asia F     |
| 43  | -5.3 ± 0.6    | -9.9 ± 0.2   | -3.3 ± 0.4    | -9.1 ± 0.3  | -8.3 ± 0.3   | -5.9 ± 0.5    | -2.4 ± 0.4    | -6.9 ± 0.2   | -2.4 ± 0.4    | -5.9 ± 0.2   | -5.0 ± 0.4   | -5.0 ± 0.3    | E Europe M   | E Asia F     |
| 44  | -5.0 ± 0.6    | -10.3 ± 0.2  | -3.7 ± 0.4    | -9.1 ± 0.3  | -7.7 ± 0.3   | -5.8 ± 0.5    | -3.0 ± 0.5    | -6.9 ± 0.2   | -2.2 ± 0.4    | -5.7 ± 0.2   | -4.6 ± 0.4   | -4.8 ± 0.3    | E Europe M   | E Asia F     |
| 45  | -5.7 ± 0.6    | -10.3 ± 0.2  | -3.7 ± 0.4    | -9.0 ± 0.3  | -8.1 ± 0.3   | -5.9 ± 0.5    | -3.2 ± 0.5    | -7.0 ± 0.2   | -2.3 ± 0.4    | -5.8 ± 0.2   | -4.4 ± 0.4   | -5.2 ± 0.4    | E Europe M   | E Asia F     |
| 46  | -4.8 ± 0.7    | -10.6 ± 0.2  | -3.6 ± 0.4    | -8.8 ± 0.3  | -8.4 ± 0.4   | -6.4 ± 0.5    | -3.1 ± 0.5    | -6.8 ± 0.2   | -1.9 ± 0.4    | -5.5 ± 0.2   | -4.0 ± 0.4   | -5.5 ± 0.4    | E Europe M   | E Asia F     |
| 47  | -4.4 ± 0.7    | -10.2 ± 0.2  | -3.2 ± 0.4    | -8.8 ± 0.3  | -7.9 ± 0.4   | -6.2 ± 0.5    | -2.7 ± 0.5    | -6.5 ± 0.2   | -2.4 ± 0.4    | -5.2 ± 0.3   | -3.9 ± 0.4   | -4.9 ± 0.5    | E Europe M   | E Asia F     |
| 48  | -5.4 ± 0.8    | -10.1 ± 0.2  | -3.3 ± 0.4    | -8.6 ± 0.4  | -7.8 ± 0.4   | -6.5 ± 0.6    | -3.7 ± 0.6    | -6.5 ± 0.3   | -2.1 ± 0.4    | -5.2 ± 0.3   | -3.7 ± 0.5   | -6.1 ± 0.5    | E Europe M   | E Asia F     |
| 49  | -5.1 ± 0.8    | -10.3 ± 0.2  | -3.5 ± 0.4    | -8.5 ± 0.4  | -7.4 ± 0.4   | -6.8 ± 0.6    | -3.6 ± 0.6    | -6.2 ± 0.3   | -2.1 ± 0.4    | -5.1 ± 0.3   | -3.4 ± 0.5   | -5.8 ± 0.5    | E Europe M   | E Asia F     |
| 50  | -5.6 ± 0.9    | -9.9 ± 0.2   | -3.1 ± 0.4    | -8.5 ± 0.3  | -7.7 ± 0.4   | -6.4 ± 0.6    | -3.8 ± 0.7    | -6.0 ± 0.3   | -2.2 ± 0.4    | -5.2 ± 0.3   | -3.2 ± 0.5   | -5.7 ± 0.5    | E Europe M   | E Asia F     |
| 51  | -5.0 ± 0.7    | -9.9 ± 0.3   | -3.6 ± 0.4    | -8.3 ± 0.4  | -7.3 ± 0.4   | -6.6 ± 0.6    | -4.0 ± 0.7    | -5.8 ± 0.3   | -1.8 ± 0.4    | -5.1 ± 0.3   | -3.0 ± 0.5   | -6.1 ± 0.6    | E Europe M   | E Asia F     |
| 52  | -5.3 ± 0.8    | -9.9 ± 0.3   | -3.3 ± 0.4    | -8.6 ± 0.3  | -6.5 ± 0.4   | -7.0 ± 0.6    | -3.2 ± 0.7    | -5.6 ± 0.3   | -2.0 ± 0.4    | -5.1 ± 0.3   | -3.4 ± 0.5   | -6.0 ± 0.6    | E Europe M   | E Asia F     |

**Table 54.** Mean raw error for Application-type images for algorithm neurotechnology-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 4.4 ± 0.3     | 5.6 ± 0.2    | 3.8 ± 0.3     | 3.9 ± 0.2   | 4.0 ± 0.4    | 5.7 ± 0.3     | 4.6 ± 0.3     | 6.4 ± 0.2    | 3.3 ± 0.3     | 4.7 ± 0.2    | 5.1 ± 0.4    | 4.7 ± 0.3     | E Asia M     | E Europe M   |
| 15  | 3.9 ± 0.3     | 5.8 ± 0.2    | 3.5 ± 0.3     | 4.0 ± 0.2   | 4.1 ± 0.5    | 4.9 ± 0.3     | 4.4 ± 0.3     | 6.7 ± 0.2    | 3.4 ± 0.3     | 5.1 ± 0.2    | 5.7 ± 0.4    | 4.9 ± 0.4     | E Asia M     | E Europe M   |
| 16  | 3.8 ± 0.3     | 5.1 ± 0.2    | 3.6 ± 0.3     | 3.5 ± 0.3   | 4.4 ± 0.4    | 4.9 ± 0.4     | 4.3 ± 0.2     | 6.4 ± 0.1    | 3.8 ± 0.3     | 5.1 ± 0.2    | 5.9 ± 0.3    | 4.5 ± 0.4     | E Asia M     | S Asia F     |
| 17  | 3.2 ± 0.3     | 4.6 ± 0.2    | 3.2 ± 0.4     | 3.1 ± 0.2   | 4.0 ± 0.4    | 4.4 ± 0.3     | 4.5 ± 0.3     | 5.9 ± 0.1    | 3.3 ± 0.2     | 4.9 ± 0.2    | 5.5 ± 0.3    | 4.9 ± 0.4     | E Asia M     | S Asia F     |
| 18  | 2.7 ± 0.2     | 3.5 ± 0.2    | 2.4 ± 0.3     | 2.3 ± 0.2   | 3.0 ± 0.3    | 3.4 ± 0.3     | 4.1 ± 0.3     | 5.1 ± 0.1    | 2.7 ± 0.2     | 4.4 ± 0.1    | 4.9 ± 0.2    | 4.0 ± 0.3     | E Asia M     | S Asia F     |
| 19  | 2.3 ± 0.3     | 2.7 ± 0.2    | 2.1 ± 0.3     | 2.0 ± 0.2   | 2.2 ± 0.2    | 3.1 ± 0.3     | 4.0 ± 0.3     | 4.5 ± 0.1    | 2.1 ± 0.2     | 4.1 ± 0.1    | 4.1 ± 0.2    | 3.6 ± 0.3     | E Asia M     | S Asia F     |
| 20  | 1.8 ± 0.3     | 2.0 ± 0.2    | 1.7 ± 0.2     | 1.3 ± 0.2   | 1.6 ± 0.2    | 2.4 ± 0.3     | 4.0 ± 0.3     | 3.8 ± 0.1    | 2.0 ± 0.2     | 3.8 ± 0.1    | 3.5 ± 0.2    | 3.5 ± 0.3     | E Africa M   | S Asia F     |
| 21  | 0.9 ± 0.3     | 1.2 ± 0.2    | 1.1 ± 0.2     | 0.7 ± 0.1   | 1.1 ± 0.2    | 1.9 ± 0.3     | 3.5 ± 0.3     | 3.1 ± 0.1    | 1.8 ± 0.2     | 3.3 ± 0.1    | 2.9 ± 0.2    | 3.0 ± 0.3     | E Africa M   | S Asia F     |
| 22  | 0.8 ± 0.3     | 0.3 ± 0.1    | 0.8 ± 0.2     | 0.0 ± 0.1   | 0.3 ± 0.2    | 1.5 ± 0.3     | 3.5 ± 0.3     | 2.36 ± 0.10  | 1.3 ± 0.2     | 2.7 ± 0.1    | 2.2 ± 0.2    | 2.8 ± 0.3     | E Africa M   | S Asia F     |
| 23  | 0.2 ± 0.2     | -0.2 ± 0.1   | 0.4 ± 0.2     | -0.4 ± 0.1  | -0.2 ± 0.2   | 0.8 ± 0.3     | 2.8 ± 0.3     | 1.7 ± 0.1    | 0.9 ± 0.2     | 2.2 ± 0.1    | 1.4 ± 0.2    | 2.3 ± 0.3     | E Africa M   | S Asia F     |
| 24  | -0.5 ± 0.2    | -1.1 ± 0.1   | 0.0 ± 0.2     | -1.0 ± 0.1  | -0.8 ± 0.1   | 0.2 ± 0.3     | 2.5 ± 0.3     | 1.1 ± 0.1    | 0.5 ± 0.2     | 1.8 ± 0.1    | 0.9 ± 0.1    | 1.6 ± 0.2     | E Africa M   | E Asia F     |
| 25  | -1.0 ± 0.2    | -1.7 ± 0.1   | -0.4 ± 0.2    | -1.7 ± 0.1  | -1.6 ± 0.1   | -0.3 ± 0.3    | 2.1 ± 0.3     | 0.4 ± 0.1    | 0.3 ± 0.2     | 1.2 ± 0.1    | 0.2 ± 0.1    | 1.3 ± 0.2     | E Africa M   | E Asia F     |
| 26  | -1.6 ± 0.2    | -2.5 ± 0.1   | -0.8 ± 0.2    | -2.3 ± 0.1  | -2.1 ± 0.1   | -1.0 ± 0.3    | 1.6 ± 0.3     | -0.31 ± 0.10 | -0.1 ± 0.2    | 0.6 ± 0.1    | -0.3 ± 0.2   | 0.7 ± 0.2     | E Africa M   | E Asia F     |
| 27  | -1.8 ± 0.3    | -3.09 ± 0.10 | -1.2 ± 0.2    | -3.0 ± 0.1  | -2.6 ± 0.1   | -1.5 ± 0.3    | 1.1 ± 0.3     | -0.94 ± 0.10 | -0.4 ± 0.2    | -0.0 ± 0.1   | -0.8 ± 0.2   | 0.4 ± 0.2     | E Africa M   | E Asia F     |
| 28  | -2.6 ± 0.3    | -3.70 ± 0.09 | -1.5 ± 0.2    | -3.6 ± 0.1  | -3.3 ± 0.1   | -2.1 ± 0.3    | 0.5 ± 0.3     | -1.6 ± 0.1   | -0.5 ± 0.2    | -0.7 ± 0.1   | -1.5 ± 0.2   | -0.3 ± 0.2    | E Africa M   | E Asia F     |
| 29  | -2.8 ± 0.3    | -4.27 ± 0.10 | -1.7 ± 0.2    | -4.1 ± 0.1  | -4.0 ± 0.1   | -2.6 ± 0.3    | 0.3 ± 0.3     | -2.08 ± 0.09 | -1.1 ± 0.2    | -1.30 ± 0.10 | -1.9 ± 0.2   | -0.5 ± 0.2    | E Africa M   | E Asia F     |
| 30  | -3.2 ± 0.3    | -4.90 ± 0.09 | -2.1 ± 0.2    | -4.8 ± 0.1  | -4.3 ± 0.1   | -3.0 ± 0.3    | 0.1 ± 0.3     | -2.77 ± 0.09 | -1.3 ± 0.2    | -1.84 ± 0.10 | -2.6 ± 0.2   | -1.0 ± 0.2    | E Africa M   | E Asia F     |
| 31  | -3.9 ± 0.3    | -5.41 ± 0.09 | -2.3 ± 0.2    | -5.3 ± 0.1  | -4.9 ± 0.2   | -3.7 ± 0.3    | -0.4 ± 0.3    | -3.3 ± 0.1   | -1.6 ± 0.2    | -2.38 ± 0.10 | -3.2 ± 0.2   | -1.7 ± 0.2    | E Africa M   | E Asia F     |
| 32  | -3.8 ± 0.3    | -6.06 ± 0.10 | -2.5 ± 0.2    | -5.9 ± 0.1  | -5.3 ± 0.2   | -3.8 ± 0.3    | -1.0 ± 0.3    | -3.8 ± 0.1   | -1.6 ± 0.2    | -2.9 ± 0.1   | -3.6 ± 0.2   | -2.1 ± 0.3    | E Africa M   | E Asia F     |
| 33  | -4.3 ± 0.3    | -6.54 ± 0.10 | -2.8 ± 0.2    | -6.4 ± 0.1  | -6.0 ± 0.2   | -4.3 ± 0.3    | -1.2 ± 0.3    | -4.5 ± 0.1   | -1.5 ± 0.2    | -3.3 ± 0.1   | -3.8 ± 0.2   | -2.3 ± 0.3    | E Africa M   | E Asia F     |
| 34  | -4.5 ± 0.4    | -7.0 ± 0.1   | -3.2 ± 0.2    | -6.8 ± 0.1  | -6.3 ± 0.2   | -4.8 ± 0.3    | -1.9 ± 0.3    | -4.9 ± 0.1   | -2.0 ± 0.2    | -3.7 ± 0.1   | -4.2 ± 0.3   | -2.8 ± 0.3    | E Africa M   | E Asia F     |
| 35  | -4.7 ± 0.4    | -7.5 ± 0.1   | -3.3 ± 0.2    | -7.4 ± 0.1  | -6.8 ± 0.2   | -5.1 ± 0.4    | -1.7 ± 0.3    | -5.4 ± 0.1   | -2.1 ± 0.2    | -4.1 ± 0.1   | -4.6 ± 0.3   | -3.0 ± 0.3    | E Africa M   | E Asia F     |
| 36  | -5.0 ± 0.4    | -7.9 ± 0.1   | -3.4 ± 0.3    | -7.8 ± 0.1  | -6.8 ± 0.2   | -5.3 ± 0.4    | -2.4 ± 0.4    | -5.8 ± 0.1   | -1.9 ± 0.3    | -4.6 ± 0.1   | -4.8 ± 0.3   | -3.7 ± 0.3    | E Europe M   | E Asia F     |
| 37  | -5.3 ± 0.4    | -8.3 ± 0.1   | -3.6 ± 0.3    | -8.0 ± 0.2  | -7.5 ± 0.2   | -5.5 ± 0.4    | -2.2 ± 0.4    | -6.1 ± 0.1   | -2.5 ± 0.2    | -4.8 ± 0.1   | -5.2 ± 0.3   | -3.9 ± 0.3    | E Africa M   | E Asia F     |
| 38  | -5.1 ± 0.5    | -8.6 ± 0.1   | -3.5 ± 0.3    | -8.3 ± 0.2  | -7.5 ± 0.2   | -5.8 ± 0.5    | -2.8 ± 0.4    | -6.4 ± 0.1   | -2.4 ± 0.3    | -5.3 ± 0.1   | -4.8 ± 0.3   | -4.1 ± 0.3    | E Europe M   | E Asia F     |
| 39  | -5.0 ± 0.5    | -9.1 ± 0.1   | -3.6 ± 0.3    | -8.7 ± 0.2  | -7.5 ± 0.3   | -5.8 ± 0.5    | -2.6 ± 0.4    | -6.7 ± 0.2   | -2.5 ± 0.3    | -5.5 ± 0.1   | -4.9 ± 0.3   | -4.6 ± 0.3    | E Europe M   | E Asia F     |
| 40  | -5.5 ± 0.5    | -9.3 ± 0.1   | -3.6 ± 0.3    | -8.7 ± 0.2  | -7.9 ± 0.3   | -6.2 ± 0.5    | -2.8 ± 0.4    | -6.9 ± 0.2   | -2.6 ± 0.3    | -5.5 ± 0.2   | -5.1 ± 0.3   | -5.0 ± 0.3    | E Europe M   | E Asia F     |
| 41  | -6.2 ± 0.6    | -9.5 ± 0.1   | -3.8 ± 0.3    | -9.0 ± 0.2  | -8.1 ± 0.3   | -7.1 ± 0.5    | -3.0 ± 0.4    | -6.9 ± 0.2   | -2.5 ± 0.3    | -5.6 ± 0.2   | -5.2 ± 0.4   | -5.1 ± 0.4    | E Europe M   | E Asia F     |
| 42  | -5.2 ± 0.6    | -9.7 ± 0.2   | -3.3 ± 0.4    | -8.7 ± 0.2  | -7.9 ± 0.3   | -6.4 ± 0.5    | -3.0 ± 0.5    | -6.9 ± 0.2   | -2.2 ± 0.3    | -6.0 ± 0.2   | -4.7 ± 0.4   | -5.0 ± 0.4    | E Europe M   | E Asia F     |
| 43  | -5.8 ± 0.6    | -9.8 ± 0.2   | -3.2 ± 0.4    | -9.0 ± 0.3  | -8.1 ± 0.3   | -6.8 ± 0.5    | -2.7 ± 0.5    | -6.8 ± 0.2   | -2.4 ± 0.4    | -5.9 ± 0.2   | -4.9 ± 0.4   | -5.7 ± 0.4    | E Europe M   | E Asia F     |
| 44  | -5.7 ± 0.7    | -10.2 ± 0.2  | -3.6 ± 0.4    | -9.0 ± 0.3  | -7.5 ± 0.3   | -6.5 ± 0.6    | -3.2 ± 0.5    | -6.7 ± 0.2   | -2.2 ± 0.4    | -5.7 ± 0.2   | -4.5 ± 0.4   | -5.3 ± 0.4    | E Europe M   | E Asia F     |
| 45  | -6.1 ± 0.7    | -10.1 ± 0.2  | -3.6 ± 0.4    | -8.9 ± 0.3  | -8.0 ± 0.3   | -6.4 ± 0.6    | -3.4 ± 0.5    | -6.9 ± 0.2   | -2.3 ± 0.3    | -5.8 ± 0.2   | -4.3 ± 0.4   | -5.7 ± 0.4    | E Europe M   | E Asia F     |
| 46  | -5.2 ± 0.7    | -10.5 ± 0.2  | -3.4 ± 0.4    | -8.6 ± 0.3  | -8.2 ± 0.4   | -6.9 ± 0.6    | -3.3 ± 0.5    | -6.7 ± 0.2   | -1.9 ± 0.4    | -5.5 ± 0.3   | -3.9 ± 0.4   | -5.8 ± 0.4    | E Europe M   | E Asia F     |
| 47  | -4.5 ± 0.8    | -10.1 ± 0.2  | -3.1 ± 0.4    | -8.8 ± 0.3  | -7.7 ± 0.4   | -6.6 ± 0.6    | -2.8 ± 0.5    | -6.3 ± 0.3   | -2.3 ± 0.4    | -5.2 ± 0.3   | -3.8 ± 0.4   | -5.2 ± 0.5    | E Europe M   | E Asia F     |
| 48  | -5.6 ± 0.8    | -10.0 ± 0.2  | -3.2 ± 0.4    | -8.5 ± 0.3  | -7.6 ± 0.4   | -6.8 ± 0.6    | -3.8 ± 0.6    | -6.3 ± 0.3   | -2.1 ± 0.4    | -5.2 ± 0.3   | -3.6 ± 0.5   | -6.3 ± 0.5    | E Europe M   | E Asia F     |
| 49  | -5.4 ± 0.8    | -10.1 ± 0.2  | -3.4 ± 0.4    | -8.4 ± 0.4  | -7.2 ± 0.4   | -7.2 ± 0.7    | -3.7 ± 0.6    | -6.1 ± 0.3   | -2.0 ± 0.4    | -5.1 ± 0.3   | -3.3 ± 0.5   | -6.0 ± 0.5    | E Europe M   | E Asia F     |
| 50  | -5.7 ± 0.9    | -9.8 ± 0.2   | -3.0 ± 0.4    | -8.4 ± 0.3  | -7.5 ± 0.4   | -6.7 ± 0.7    | -3.8 ± 0.7    | -5.9 ± 0.3   | -2.2 ± 0.4    | -5.2 ± 0.3   | -3.1 ± 0.5   | -5.8 ± 0.5    | E Europe M   | E Asia F     |
| 51  | -5.1 ± 0.8    | -9.8 ± 0.3   | -3.5 ± 0.4    | -8.2 ± 0.3  | -7.2 ± 0.4   | -6.9 ± 0.7    | -4.1 ± 0.7    | -5.6 ± 0.3   | -1.7 ± 0.4    | -5.1 ± 0.3   | -2.9 ± 0.5   | -6.3 ± 0.6    | E Europe M   | E Asia F     |
| 52  | -5.3 ± 0.8    | -9.8 ± 0.3   | -3.2 ± 0.4    | -8.5 ± 0.3  | -6.4 ± 0.4   | -7.2 ± 0.7    | -3.2 ± 0.7    | -5.4 ± 0.3   | -1.9 ± 0.4    | -5.1 ± 0.3   | -3.3 ± 0.5   | -6.1 ± 0.6    | E Europe M   | E Asia F     |

**Table 55.** Mean raw error for Application-type images for algorithm nominder-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.5 ± 0.3     | 3.5 ± 0.1    | 2.7 ± 0.2     | 2.2 ± 0.2    | 2.8 ± 0.3    | 4.4 ± 0.5     | 1.1 ± 0.3     | 2.2 ± 0.1   | 1.4 ± 0.2     | 1.2 ± 0.1   | 1.4 ± 0.2    | 2.0 ± 0.4     | W Africa F   | E Africa M   |
| 15  | 3.5 ± 0.4     | 3.8 ± 0.2    | 2.6 ± 0.3     | 2.5 ± 0.2    | 3.1 ± 0.4    | 4.6 ± 0.5     | 1.3 ± 0.3     | 2.6 ± 0.2   | 1.5 ± 0.2     | 1.6 ± 0.2   | 2.2 ± 0.3    | 2.5 ± 0.4     | W Africa F   | E Africa M   |
| 16  | 3.4 ± 0.4     | 3.8 ± 0.2    | 2.5 ± 0.3     | 2.4 ± 0.2    | 3.1 ± 0.4    | 4.9 ± 0.5     | 1.9 ± 0.3     | 2.7 ± 0.2   | 1.6 ± 0.2     | 2.0 ± 0.2   | 2.4 ± 0.3    | 2.7 ± 0.4     | W Africa F   | E Europe M   |
| 17  | 3.7 ± 0.4     | 3.6 ± 0.2    | 2.4 ± 0.3     | 2.4 ± 0.2    | 3.0 ± 0.3    | 4.9 ± 0.5     | 2.2 ± 0.3     | 2.9 ± 0.2   | 1.3 ± 0.2     | 2.6 ± 0.2   | 2.3 ± 0.3    | 3.3 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 3.1 ± 0.3     | 2.9 ± 0.1    | 1.7 ± 0.3     | 1.7 ± 0.2    | 2.4 ± 0.2    | 4.5 ± 0.4     | 2.2 ± 0.3     | 2.4 ± 0.1   | 1.1 ± 0.2     | 2.5 ± 0.2   | 2.2 ± 0.3    | 2.9 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 3.2 ± 0.3     | 2.6 ± 0.1    | 1.6 ± 0.3     | 1.7 ± 0.2    | 2.1 ± 0.2    | 5.0 ± 0.4     | 2.7 ± 0.3     | 2.5 ± 0.1   | 1.2 ± 0.2     | 2.8 ± 0.2   | 1.9 ± 0.2    | 3.3 ± 0.4     | W Africa F   | E Europe M   |
| 20  | 2.7 ± 0.3     | 2.4 ± 0.1    | 1.2 ± 0.2     | 1.5 ± 0.2    | 1.9 ± 0.2    | 4.7 ± 0.4     | 3.0 ± 0.3     | 2.3 ± 0.1   | 1.3 ± 0.2     | 3.0 ± 0.1   | 1.9 ± 0.2    | 3.4 ± 0.3     | W Africa F   | E Europe F   |
| 21  | 2.6 ± 0.3     | 2.0 ± 0.1    | 0.9 ± 0.2     | 1.0 ± 0.1    | 1.5 ± 0.2    | 4.5 ± 0.4     | 3.0 ± 0.3     | 2.1 ± 0.1   | 1.0 ± 0.2     | 2.9 ± 0.1   | 1.8 ± 0.2    | 3.4 ± 0.3     | W Africa F   | E Europe F   |
| 22  | 2.5 ± 0.3     | 1.4 ± 0.1    | 0.6 ± 0.2     | 0.6 ± 0.1    | 1.0 ± 0.2    | 4.0 ± 0.3     | 3.4 ± 0.3     | 1.8 ± 0.1   | 1.3 ± 0.2     | 2.8 ± 0.2   | 1.4 ± 0.2    | 3.1 ± 0.3     | W Africa F   | S Asia F     |
| 23  | 2.1 ± 0.3     | 1.1 ± 0.1    | 0.5 ± 0.2     | 0.5 ± 0.1    | 0.8 ± 0.1    | 4.2 ± 0.3     | 2.9 ± 0.3     | 1.5 ± 0.1   | 1.0 ± 0.2     | 2.7 ± 0.2   | 1.2 ± 0.2    | 3.1 ± 0.3     | W Africa F   | E Europe F   |
| 24  | 2.1 ± 0.3     | 0.58 ± 0.10  | 0.3 ± 0.1     | -0.0 ± 0.1   | 0.4 ± 0.1    | 3.3 ± 0.3     | 2.9 ± 0.3     | 1.2 ± 0.1   | 1.1 ± 0.2     | 2.7 ± 0.1   | 0.9 ± 0.2    | 2.9 ± 0.3     | W Africa F   | S Asia F     |
| 25  | 1.5 ± 0.3     | 0.20 ± 0.09  | 0.0 ± 0.1     | -0.4 ± 0.1   | 0.0 ± 0.1    | 3.4 ± 0.3     | 2.8 ± 0.3     | 1.1 ± 0.1   | 1.2 ± 0.2     | 2.5 ± 0.1   | 0.8 ± 0.2    | 2.7 ± 0.3     | W Africa F   | S Asia F     |
| 26  | 1.2 ± 0.3     | -0.27 ± 0.09 | -0.0 ± 0.1    | -1.04 ± 0.10 | -0.2 ± 0.1   | 3.0 ± 0.3     | 2.5 ± 0.2     | 0.8 ± 0.1   | 1.1 ± 0.2     | 2.4 ± 0.1   | 0.8 ± 0.2    | 2.5 ± 0.2     | W Africa F   | S Asia F     |
| 27  | 0.9 ± 0.3     | -0.61 ± 0.09 | -0.1 ± 0.1    | -1.44 ± 0.10 | -0.6 ± 0.1   | 2.7 ± 0.2     | 2.3 ± 0.2     | 0.7 ± 0.1   | 1.3 ± 0.2     | 2.1 ± 0.1   | 0.7 ± 0.2    | 2.1 ± 0.2     | W Africa F   | S Asia F     |
| 28  | 0.5 ± 0.3     | -0.98 ± 0.09 | -0.2 ± 0.1    | -1.75 ± 0.10 | -0.9 ± 0.1   | 2.4 ± 0.2     | 2.1 ± 0.2     | 0.4 ± 0.1   | 1.3 ± 0.2     | 1.7 ± 0.1   | 0.3 ± 0.2    | 1.8 ± 0.2     | W Africa F   | S Asia F     |
| 29  | 0.5 ± 0.3     | -1.33 ± 0.09 | -0.2 ± 0.1    | -2.0 ± 0.1   | -1.2 ± 0.1   | 2.0 ± 0.2     | 1.8 ± 0.2     | 0.2 ± 0.1   | 1.1 ± 0.2     | 1.4 ± 0.1   | 0.0 ± 0.2    | 1.9 ± 0.2     | W Africa F   | S Asia F     |
| 30  | -0.1 ± 0.2    | -1.69 ± 0.09 | -0.3 ± 0.1    | -2.5 ± 0.1   | -1.3 ± 0.1   | 1.4 ± 0.2     | 1.7 ± 0.2     | -0.0 ± 0.1  | 1.0 ± 0.2     | 1.01 ± 0.09 | -0.1 ± 0.2   | 1.5 ± 0.2     | E Africa M   | S Asia F     |
| 31  | -0.4 ± 0.3    | -1.92 ± 0.09 | -0.4 ± 0.2    | -2.9 ± 0.1   | -1.6 ± 0.1   | 1.2 ± 0.2     | 1.5 ± 0.2     | -0.2 ± 0.1  | 0.8 ± 0.2     | 0.73 ± 0.10 | -0.3 ± 0.2   | 1.3 ± 0.2     | E Africa M   | S Asia F     |
| 32  | -0.4 ± 0.3    | -2.19 ± 0.09 | -0.4 ± 0.2    | -3.2 ± 0.1   | -1.8 ± 0.2   | 0.8 ± 0.2     | 1.3 ± 0.2     | -0.4 ± 0.1  | 0.8 ± 0.2     | 0.53 ± 0.09 | -0.3 ± 0.2   | 0.9 ± 0.2     | E Africa M   | S Asia F     |
| 33  | -0.9 ± 0.3    | -2.49 ± 0.09 | -0.5 ± 0.2    | -3.4 ± 0.1   | -2.3 ± 0.2   | 0.4 ± 0.2     | 1.0 ± 0.2     | -0.8 ± 0.1  | 0.9 ± 0.2     | 0.28 ± 0.10 | -0.5 ± 0.2   | 0.6 ± 0.2     | E Africa M   | S Asia F     |
| 34  | -1.0 ± 0.3    | -2.70 ± 0.09 | -0.6 ± 0.2    | -3.7 ± 0.1   | -2.5 ± 0.2   | -0.2 ± 0.2    | 0.5 ± 0.3     | -1.0 ± 0.1  | 0.6 ± 0.2     | 0.0 ± 0.1   | -0.7 ± 0.2   | 0.4 ± 0.2     | E Europe M   | S Asia F     |
| 35  | -1.1 ± 0.3    | -3.14 ± 0.09 | -0.6 ± 0.2    | -4.0 ± 0.1   | -2.8 ± 0.2   | -0.5 ± 0.3    | 0.6 ± 0.3     | -1.1 ± 0.1  | 0.5 ± 0.2     | -0.2 ± 0.1  | -0.9 ± 0.3   | 0.2 ± 0.2     | E Africa M   | S Asia F     |
| 36  | -1.4 ± 0.3    | -3.3 ± 0.1   | -0.4 ± 0.2    | -4.3 ± 0.1   | -2.9 ± 0.2   | -0.8 ± 0.3    | 0.3 ± 0.3     | -1.3 ± 0.1  | 0.6 ± 0.2     | -0.4 ± 0.1  | -0.9 ± 0.3   | -0.4 ± 0.2    | E Europe M   | S Asia F     |
| 37  | -1.7 ± 0.3    | -3.56 ± 0.10 | -0.5 ± 0.2    | -4.4 ± 0.1   | -3.1 ± 0.2   | -1.4 ± 0.3    | 0.3 ± 0.3     | -1.5 ± 0.1  | 0.2 ± 0.2     | -0.5 ± 0.1  | -1.1 ± 0.3   | -0.6 ± 0.2    | E Africa M   | S Asia F     |
| 38  | -1.6 ± 0.3    | -3.8 ± 0.1   | -0.5 ± 0.2    | -4.5 ± 0.1   | -3.2 ± 0.2   | -1.6 ± 0.3    | -0.1 ± 0.3    | -1.7 ± 0.1  | 0.1 ± 0.2     | -0.7 ± 0.1  | -0.9 ± 0.3   | -0.8 ± 0.3    | E Europe M   | S Asia F     |
| 39  | -1.8 ± 0.3    | -4.0 ± 0.1   | -0.5 ± 0.2    | -4.8 ± 0.2   | -3.3 ± 0.2   | -1.7 ± 0.3    | 0.1 ± 0.3     | -1.8 ± 0.1  | 0.0 ± 0.2     | -0.8 ± 0.1  | -0.8 ± 0.3   | -1.2 ± 0.3    | E Africa M   | S Asia F     |
| 40  | -2.2 ± 0.4    | -4.2 ± 0.1   | -0.2 ± 0.2    | -4.7 ± 0.2   | -3.6 ± 0.2   | -2.3 ± 0.3    | -0.1 ± 0.3    | -1.8 ± 0.1  | 0.1 ± 0.2     | -0.8 ± 0.1  | -0.8 ± 0.3   | -1.3 ± 0.3    | E Europe M   | S Asia F     |
| 41  | -2.5 ± 0.4    | -4.3 ± 0.1   | -0.5 ± 0.2    | -4.8 ± 0.2   | -3.7 ± 0.2   | -3.1 ± 0.3    | -0.1 ± 0.4    | -1.9 ± 0.1  | -0.0 ± 0.2    | -1.0 ± 0.1  | -1.0 ± 0.3   | -1.5 ± 0.3    | E Europe M   | S Asia F     |
| 42  | -2.4 ± 0.4    | -4.5 ± 0.1   | -0.1 ± 0.3    | -4.6 ± 0.2   | -3.9 ± 0.2   | -2.8 ± 0.3    | -0.2 ± 0.4    | -2.0 ± 0.1  | 0.1 ± 0.2     | -1.1 ± 0.1  | -0.8 ± 0.3   | -1.9 ± 0.3    | E Europe M   | S Asia F     |
| 43  | -2.2 ± 0.4    | -4.7 ± 0.1   | -0.4 ± 0.2    | -4.8 ± 0.2   | -4.0 ± 0.2   | -3.4 ± 0.4    | 0.2 ± 0.4     | -2.1 ± 0.1  | -0.0 ± 0.3    | -1.1 ± 0.2  | -0.8 ± 0.3   | -2.1 ± 0.3    | E Africa M   | S Asia F     |
| 44  | -2.6 ± 0.5    | -5.0 ± 0.1   | -0.6 ± 0.3    | -4.7 ± 0.2   | -3.8 ± 0.2   | -3.8 ± 0.4    | -0.3 ± 0.4    | -2.2 ± 0.1  | 0.2 ± 0.3     | -1.1 ± 0.2  | -0.7 ± 0.3   | -2.2 ± 0.3    | E Europe M   | E Asia F     |
| 45  | -2.5 ± 0.5    | -5.1 ± 0.1   | -0.6 ± 0.3    | -4.7 ± 0.2   | -4.3 ± 0.2   | -3.8 ± 0.4    | -0.5 ± 0.4    | -2.4 ± 0.1  | -0.1 ± 0.3    | -1.1 ± 0.2  | -0.9 ± 0.3   | -2.4 ± 0.3    | E Europe M   | E Asia F     |
| 46  | -2.1 ± 0.6    | -5.4 ± 0.1   | -0.4 ± 0.3    | -4.6 ± 0.2   | -4.5 ± 0.2   | -3.9 ± 0.4    | -0.3 ± 0.5    | -2.4 ± 0.1  | 0.5 ± 0.3     | -0.8 ± 0.2  | -0.8 ± 0.3   | -2.8 ± 0.3    | E Europe M   | E Asia F     |
| 47  | -2.4 ± 0.6    | -5.3 ± 0.1   | -0.6 ± 0.3    | -4.5 ± 0.2   | -4.3 ± 0.2   | -4.2 ± 0.4    | -0.2 ± 0.4    | -2.5 ± 0.1  | 0.1 ± 0.3     | -1.0 ± 0.2  | -0.6 ± 0.3   | -2.9 ± 0.3    | E Europe M   | E Asia F     |
| 48  | -3.1 ± 0.6    | -5.4 ± 0.1   | -0.5 ± 0.3    | -4.3 ± 0.3   | -4.4 ± 0.2   | -4.3 ± 0.4    | -1.2 ± 0.5    | -2.5 ± 0.1  | 0.2 ± 0.3     | -0.9 ± 0.2  | -0.7 ± 0.3   | -3.3 ± 0.4    | E Europe M   | E Asia F     |
| 49  | -2.3 ± 0.6    | -5.6 ± 0.1   | -0.8 ± 0.3    | -4.3 ± 0.3   | -4.1 ± 0.3   | -4.7 ± 0.5    | -1.1 ± 0.5    | -2.5 ± 0.2  | 0.0 ± 0.3     | -1.0 ± 0.2  | -0.6 ± 0.3   | -3.4 ± 0.4    | E Europe M   | E Asia F     |
| 50  | -3.2 ± 0.7    | -5.5 ± 0.1   | -0.5 ± 0.3    | -4.3 ± 0.2   | -4.7 ± 0.3   | -5.0 ± 0.5    | -1.5 ± 0.5    | -2.5 ± 0.2  | 0.0 ± 0.3     | -1.2 ± 0.2  | -0.4 ± 0.3   | -3.7 ± 0.4    | E Europe M   | E Asia F     |
| 51  | -2.3 ± 0.6    | -5.5 ± 0.2   | -1.0 ± 0.3    | -4.2 ± 0.3   | -4.7 ± 0.3   | -5.1 ± 0.5    | -1.8 ± 0.6    | -2.5 ± 0.2  | 0.1 ± 0.3     | -1.3 ± 0.2  | -0.4 ± 0.3   | -4.1 ± 0.4    | E Europe M   | E Asia F     |
| 52  | -2.9 ± 0.7    | -5.7 ± 0.2   | -0.7 ± 0.3    | -4.5 ± 0.3   | -4.3 ± 0.3   | -5.5 ± 0.5    | -1.6 ± 0.5    | -2.8 ± 0.2  | -0.0 ± 0.3    | -1.5 ± 0.2  | -0.9 ± 0.3   | -3.8 ± 0.4    | E Europe M   | E Asia F     |

**Table 56.** Mean raw error for Application-type images for algorithm nominder-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 4.0 ± 0.2     | 3.9 ± 0.1    | 3.7 ± 0.2     | 3.4 ± 0.1    | 3.5 ± 0.3    | 5.4 ± 0.3     | 2.4 ± 0.2     | 2.8 ± 0.1   | 2.0 ± 0.2     | 1.7 ± 0.1    | 2.1 ± 0.2    | 2.7 ± 0.3     | W Africa F   | S Asia M     |
| 15  | 3.9 ± 0.2     | 4.0 ± 0.1    | 3.3 ± 0.2     | 3.6 ± 0.2    | 3.7 ± 0.3    | 5.1 ± 0.3     | 2.2 ± 0.2     | 3.3 ± 0.1   | 2.1 ± 0.2     | 2.0 ± 0.1    | 2.9 ± 0.2    | 2.9 ± 0.3     | W Africa F   | S Asia M     |
| 16  | 3.7 ± 0.3     | 3.8 ± 0.1    | 3.3 ± 0.2     | 3.3 ± 0.2    | 3.6 ± 0.3    | 5.3 ± 0.4     | 2.6 ± 0.2     | 3.2 ± 0.1   | 1.9 ± 0.2     | 2.2 ± 0.1    | 2.7 ± 0.2    | 2.5 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.8 ± 0.3     | 3.4 ± 0.1    | 2.7 ± 0.2     | 3.2 ± 0.2    | 3.1 ± 0.3    | 4.9 ± 0.3     | 2.0 ± 0.2     | 3.0 ± 0.1   | 1.5 ± 0.2     | 2.2 ± 0.1    | 2.6 ± 0.2    | 2.8 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.0 ± 0.2     | 2.7 ± 0.1    | 1.9 ± 0.3     | 2.4 ± 0.1    | 2.7 ± 0.2    | 4.4 ± 0.3     | 1.8 ± 0.2     | 2.4 ± 0.1   | 0.9 ± 0.2     | 1.7 ± 0.1    | 2.2 ± 0.2    | 2.1 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 3.0 ± 0.3     | 2.26 ± 0.10  | 1.6 ± 0.2     | 2.2 ± 0.1    | 2.2 ± 0.2    | 4.5 ± 0.3     | 2.0 ± 0.2     | 2.2 ± 0.1   | 0.8 ± 0.2     | 1.7 ± 0.1    | 1.7 ± 0.2    | 2.2 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 2.7 ± 0.2     | 2.0 ± 0.1    | 1.2 ± 0.2     | 1.8 ± 0.1    | 1.9 ± 0.1    | 4.2 ± 0.3     | 2.1 ± 0.3     | 1.8 ± 0.1   | 0.7 ± 0.2     | 1.7 ± 0.1    | 1.5 ± 0.2    | 2.2 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 2.5 ± 0.2     | 1.52 ± 0.09  | 0.8 ± 0.2     | 1.3 ± 0.1    | 1.4 ± 0.1    | 3.7 ± 0.3     | 2.0 ± 0.2     | 1.58 ± 0.09 | 0.4 ± 0.2     | 1.5 ± 0.1    | 1.3 ± 0.2    | 2.1 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 2.4 ± 0.2     | 0.83 ± 0.09  | 0.4 ± 0.2     | 0.9 ± 0.1    | 0.9 ± 0.1    | 3.3 ± 0.3     | 2.4 ± 0.3     | 1.3 ± 0.1   | 0.5 ± 0.2     | 1.4 ± 0.1    | 0.9 ± 0.1    | 1.8 ± 0.3     | W Africa F   | E Europe F   |
| 23  | 1.9 ± 0.2     | 0.44 ± 0.08  | 0.2 ± 0.1     | 0.51 ± 0.10  | 0.5 ± 0.1    | 3.4 ± 0.3     | 2.2 ± 0.3     | 0.9 ± 0.1   | 0.2 ± 0.2     | 1.4 ± 0.1    | 0.7 ± 0.2    | 2.0 ± 0.3     | W Africa F   | E Europe F   |
| 24  | 1.7 ± 0.2     | -0.15 ± 0.08 | -0.2 ± 0.1    | 0.04 ± 0.09  | 0.1 ± 0.1    | 2.8 ± 0.3     | 2.2 ± 0.3     | 0.6 ± 0.1   | 0.3 ± 0.2     | 1.4 ± 0.1    | 0.2 ± 0.1    | 1.9 ± 0.3     | W Africa F   | E Europe F   |
| 25  | 1.3 ± 0.2     | -0.49 ± 0.08 | -0.6 ± 0.1    | -0.38 ± 0.09 | -0.3 ± 0.1   | 2.5 ± 0.2     | 2.1 ± 0.3     | 0.3 ± 0.1   | 0.3 ± 0.2     | 1.3 ± 0.1    | 0.1 ± 0.1    | 1.9 ± 0.2     | W Africa F   | E Europe F   |
| 26  | 0.9 ± 0.2     | -1.03 ± 0.07 | -0.7 ± 0.1    | -0.90 ± 0.08 | -0.5 ± 0.1   | 2.4 ± 0.2     | 1.9 ± 0.2     | -0.0 ± 0.1  | 0.0 ± 0.2     | 1.2 ± 0.1    | -0.0 ± 0.2   | 1.7 ± 0.2     | W Africa F   | E Asia F     |
| 27  | 0.7 ± 0.2     | -1.44 ± 0.07 | -0.9 ± 0.1    | -1.30 ± 0.08 | -0.9 ± 0.1   | 2.2 ± 0.2     | 1.8 ± 0.2     | -0.2 ± 0.1  | 0.1 ± 0.2     | 0.9 ± 0.1    | -0.2 ± 0.2   | 1.5 ± 0.2     | W Africa F   | E Asia F     |
| 28  | 0.5 ± 0.2     | -1.81 ± 0.07 | -1.1 ± 0.1    | -1.60 ± 0.09 | -1.3 ± 0.1   | 1.8 ± 0.2     | 1.6 ± 0.2     | -0.5 ± 0.1  | 0.1 ± 0.2     | 0.59 ± 0.10  | -0.4 ± 0.2   | 1.2 ± 0.2     | W Africa F   | E Asia F     |
| 29  | 0.5 ± 0.2     | -2.25 ± 0.07 | -1.2 ± 0.1    | -1.95 ± 0.09 | -1.4 ± 0.1   | 1.5 ± 0.2     | 1.6 ± 0.2     | -0.7 ± 0.1  | -0.2 ± 0.2    | 0.35 ± 0.09  | -0.9 ± 0.2   | 1.5 ± 0.2     | E Africa M   | E Asia F     |
| 30  | 0.1 ± 0.2     | -2.55 ± 0.08 | -1.4 ± 0.1    | -2.33 ± 0.09 | -1.7 ± 0.1   | 1.1 ± 0.2     | 1.5 ± 0.2     | -1.0 ± 0.1  | -0.3 ± 0.2    | 0.07 ± 0.09  | -0.9 ± 0.2   | 1.2 ± 0.2     | E Africa M   | E Asia F     |
| 31  | -0.2 ± 0.2    | -2.87 ± 0.08 | -1.5 ± 0.1    | -2.64 ± 0.10 | -1.9 ± 0.1   | 0.9 ± 0.2     | 1.5 ± 0.2     | -1.3 ± 0.1  | -0.4 ± 0.2    | -0.12 ± 0.09 | -1.2 ± 0.2   | 1.1 ± 0.2     | E Africa M   | E Asia F     |
| 32  | -0.1 ± 0.2    | -3.11 ± 0.08 | -1.6 ± 0.1    | -2.9 ± 0.1   | -2.0 ± 0.1   | 0.6 ± 0.2     | 1.2 ± 0.2     | -1.5 ± 0.1  | -0.4 ± 0.2    | -0.29 ± 0.09 | -1.3 ± 0.2   | 0.9 ± 0.2     | E Africa M   | E Asia F     |
| 33  | -0.4 ± 0.3    | -3.49 ± 0.09 | -1.8 ± 0.2    | -3.0 ± 0.1   | -2.3 ± 0.2   | 0.2 ± 0.2     | 1.2 ± 0.2     | -1.8 ± 0.1  | -0.4 ± 0.2    | -0.47 ± 0.09 | -1.5 ± 0.2   | 0.8 ± 0.2     | E Africa M   | E Asia F     |
| 34  | -0.6 ± 0.3    | -3.64 ± 0.09 | -2.0 ± 0.2    | -3.2 ± 0.1   | -2.5 ± 0.2   | -0.1 ± 0.2    | 0.8 ± 0.2     | -2.1 ± 0.1  | -0.6 ± 0.2    | -0.71 ± 0.09 | -1.6 ± 0.2   | 0.8 ± 0.2     | E Africa M   | E Asia F     |
| 35  | -0.5 ± 0.3    | -4.00 ± 0.09 | -1.9 ± 0.2    | -3.4 ± 0.1   | -2.8 ± 0.2   | -0.3 ± 0.3    | 1.0 ± 0.2     | -2.1 ± 0.1  | -0.8 ± 0.2    | -0.83 ± 0.09 | -2.0 ± 0.2   | 0.6 ± 0.2     | E Africa M   | E Asia F     |
| 36  | -0.7 ± 0.3    | -4.06 ± 0.10 | -2.0 ± 0.2    | -3.5 ± 0.1   | -2.7 ± 0.2   | -0.4 ± 0.3    | 0.9 ± 0.3     | -2.4 ± 0.1  | -0.7 ± 0.2    | -1.00 ± 0.10 | -1.9 ± 0.2   | 0.5 ± 0.2     | E Africa M   | E Asia F     |
| 37  | -1.1 ± 0.3    | -4.27 ± 0.10 | -2.1 ± 0.2    | -3.6 ± 0.1   | -2.9 ± 0.2   | -0.6 ± 0.3    | 1.0 ± 0.3     | -2.7 ± 0.1  | -0.9 ± 0.2    | -1.1 ± 0.1   | -2.2 ± 0.2   | 0.2 ± 0.2     | E Africa M   | E Asia F     |
| 38  | -0.7 ± 0.3    | -4.5 ± 0.1   | -2.0 ± 0.2    | -3.7 ± 0.1   | -2.9 ± 0.2   | -0.9 ± 0.3    | 0.5 ± 0.3     | -2.7 ± 0.1  | -1.0 ± 0.2    | -1.4 ± 0.1   | -1.9 ± 0.2   | 0.2 ± 0.2     | E Africa M   | E Asia F     |
| 39  | -1.0 ± 0.3    | -4.6 ± 0.1   | -2.2 ± 0.2    | -3.9 ± 0.1   | -3.0 ± 0.2   | -1.0 ± 0.3    | 0.9 ± 0.3     | -3.0 ± 0.1  | -1.0 ± 0.2    | -1.4 ± 0.1   | -1.9 ± 0.2   | 0.1 ± 0.2     | E Africa M   | E Asia F     |
| 40  | -1.2 ± 0.4    | -4.7 ± 0.1   | -1.8 ± 0.2    | -3.7 ± 0.1   | -3.0 ± 0.2   | -1.3 ± 0.3    | 0.8 ± 0.3     | -3.0 ± 0.1  | -1.1 ± 0.2    | -1.5 ± 0.1   | -2.1 ± 0.2   | 0.1 ± 0.3     | E Africa M   | E Asia F     |
| 41  | -1.6 ± 0.4    | -4.7 ± 0.1   | -2.0 ± 0.2    | -3.9 ± 0.2   | -3.1 ± 0.2   | -1.9 ± 0.3    | 0.8 ± 0.3     | -3.1 ± 0.1  | -1.2 ± 0.2    | -1.7 ± 0.1   | -2.2 ± 0.2   | -0.0 ± 0.3    | E Africa M   | E Asia F     |
| 42  | -1.3 ± 0.4    | -4.7 ± 0.1   | -1.8 ± 0.2    | -3.6 ± 0.2   | -3.1 ± 0.2   | -1.5 ± 0.3    | 0.9 ± 0.3     | -3.1 ± 0.1  | -1.0 ± 0.2    | -1.7 ± 0.1   | -2.1 ± 0.2   | -0.2 ± 0.3    | E Africa M   | E Asia F     |
| 43  | -1.3 ± 0.4    | -4.9 ± 0.1   | -1.9 ± 0.2    | -3.9 ± 0.2   | -3.2 ± 0.2   | -1.9 ± 0.3    | 1.2 ± 0.3     | -3.3 ± 0.1  | -1.3 ± 0.2    | -1.7 ± 0.1   | -2.2 ± 0.2   | -0.4 ± 0.3    | E Africa M   | E Asia F     |
| 44  | -1.5 ± 0.4    | -5.0 ± 0.1   | -2.0 ± 0.2    | -3.8 ± 0.2   | -3.1 ± 0.2   | -2.2 ± 0.4    | 0.7 ± 0.4     | -3.3 ± 0.1  | -1.0 ± 0.2    | -1.9 ± 0.1   | -2.0 ± 0.3   | -0.3 ± 0.3    | E Africa M   | E Asia F     |
| 45  | -1.6 ± 0.4    | -5.0 ± 0.1   | -2.0 ± 0.2    | -3.8 ± 0.2   | -3.2 ± 0.2   | -2.1 ± 0.4    | 0.6 ± 0.4     | -3.5 ± 0.1  | -1.2 ± 0.2    | -1.7 ± 0.2   | -2.1 ± 0.3   | -0.6 ± 0.3    | E Africa M   | E Asia F     |
| 46  | -1.0 ± 0.5    | -5.2 ± 0.1   | -1.6 ± 0.3    | -3.6 ± 0.2   | -3.5 ± 0.2   | -2.1 ± 0.4    | 0.6 ± 0.4     | -3.5 ± 0.1  | -0.8 ± 0.2    | -1.6 ± 0.2   | -2.0 ± 0.3   | -0.7 ± 0.3    | E Africa M   | E Asia F     |
| 47  | -1.3 ± 0.5    | -5.0 ± 0.1   | -1.8 ± 0.3    | -3.6 ± 0.2   | -3.2 ± 0.2   | -2.1 ± 0.4    | 1.2 ± 0.4     | -3.6 ± 0.1  | -1.1 ± 0.2    | -1.7 ± 0.2   | -1.9 ± 0.3   | -0.5 ± 0.3    | E Africa M   | E Asia F     |
| 48  | -1.6 ± 0.6    | -5.0 ± 0.1   | -1.7 ± 0.3    | -3.4 ± 0.2   | -3.2 ± 0.2   | -2.2 ± 0.4    | 0.6 ± 0.5     | -3.6 ± 0.1  | -1.1 ± 0.3    | -1.6 ± 0.2   | -1.9 ± 0.3   | -0.9 ± 0.4    | E Africa M   | E Asia F     |
| 49  | -1.1 ± 0.6    | -5.1 ± 0.1   | -1.8 ± 0.3    | -3.4 ± 0.2   | -2.9 ± 0.3   | -2.4 ± 0.5    | 0.2 ± 0.4     | -3.6 ± 0.1  | -1.2 ± 0.3    | -1.7 ± 0.2   | -1.9 ± 0.3   | -0.7 ± 0.3    | E Africa M   | E Asia F     |
| 50  | -1.6 ± 0.6    | -4.9 ± 0.1   | -1.6 ± 0.3    | -3.4 ± 0.2   | -3.1 ± 0.3   | -2.2 ± 0.5    | 0.1 ± 0.4     | -3.6 ± 0.1  | -1.2 ± 0.3    | -1.6 ± 0.2   | -1.5 ± 0.3   | -1.0 ± 0.3    | E Africa M   | E Asia F     |
| 51  | -1.3 ± 0.5    | -4.8 ± 0.1   | -2.0 ± 0.3    | -3.2 ± 0.2   | -3.1 ± 0.3   | -2.3 ± 0.5    | -0.4 ± 0.5    | -3.7 ± 0.1  | -1.2 ± 0.2    | -1.7 ± 0.2   | -1.6 ± 0.3   | -1.2 ± 0.4    | E Africa M   | E Asia F     |
| 52  | -1.5 ± 0.6    | -4.8 ± 0.2   | -1.5 ± 0.3    | -3.5 ± 0.2   | -2.6 ± 0.3   | -2.5 ± 0.5    | -0.2 ± 0.5    | -3.7 ± 0.1  | -1.4 ± 0.3    | -1.9 ± 0.2   | -2.0 ± 0.3   | -1.2 ± 0.4    | E Africa M   | E Asia F     |

**Table 57.** Mean raw error for Application-type images for algorithm ondato-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.4 ± 0.3     | 4.9 ± 0.1    | 3.3 ± 0.3     | 4.4 ± 0.2    | 4.7 ± 0.3    | 5.8 ± 0.4     | 4.0 ± 0.3     | 3.2 ± 0.1   | 1.5 ± 0.2     | 2.5 ± 0.1   | 2.7 ± 0.2    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 5.3 ± 0.4     | 4.9 ± 0.1    | 3.1 ± 0.3     | 4.3 ± 0.2    | 4.6 ± 0.3    | 5.9 ± 0.5     | 3.3 ± 0.3     | 3.0 ± 0.1   | 1.4 ± 0.2     | 2.3 ± 0.1   | 2.8 ± 0.2    | 3.5 ± 0.4     | W Africa F   | E Europe M   |
| 16  | 4.8 ± 0.3     | 4.6 ± 0.1    | 3.4 ± 0.3     | 4.0 ± 0.2    | 4.6 ± 0.3    | 6.0 ± 0.5     | 3.0 ± 0.3     | 2.8 ± 0.1   | 1.4 ± 0.2     | 2.3 ± 0.1   | 2.7 ± 0.2    | 3.3 ± 0.4     | W Africa F   | E Europe M   |
| 17  | 5.0 ± 0.4     | 4.2 ± 0.1    | 2.9 ± 0.3     | 3.8 ± 0.2    | 4.0 ± 0.3    | 5.7 ± 0.4     | 2.9 ± 0.3     | 2.8 ± 0.1   | 1.2 ± 0.2     | 2.6 ± 0.2   | 2.7 ± 0.3    | 3.5 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.3 ± 0.3     | 3.6 ± 0.1    | 2.3 ± 0.3     | 3.2 ± 0.1    | 3.6 ± 0.2    | 5.4 ± 0.4     | 2.8 ± 0.3     | 2.4 ± 0.1   | 1.1 ± 0.2     | 2.3 ± 0.1   | 2.6 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.3 ± 0.3     | 3.2 ± 0.1    | 2.2 ± 0.3     | 2.9 ± 0.1    | 3.0 ± 0.2    | 5.3 ± 0.4     | 3.3 ± 0.3     | 2.3 ± 0.1   | 1.0 ± 0.2     | 2.5 ± 0.1   | 2.4 ± 0.2    | 3.4 ± 0.4     | W Africa F   | E Europe M   |
| 20  | 4.0 ± 0.3     | 2.8 ± 0.1    | 2.0 ± 0.2     | 2.5 ± 0.2    | 2.8 ± 0.2    | 5.1 ± 0.4     | 3.4 ± 0.3     | 2.1 ± 0.1   | 1.3 ± 0.2     | 2.6 ± 0.1   | 2.4 ± 0.2    | 3.8 ± 0.4     | W Africa F   | E Europe M   |
| 21  | 3.7 ± 0.3     | 2.5 ± 0.1    | 1.5 ± 0.2     | 2.1 ± 0.1    | 2.5 ± 0.1    | 4.9 ± 0.4     | 3.7 ± 0.3     | 2.0 ± 0.1   | 0.8 ± 0.2     | 2.5 ± 0.1   | 2.3 ± 0.2    | 3.8 ± 0.4     | W Africa F   | E Europe M   |
| 22  | 3.7 ± 0.3     | 2.0 ± 0.1    | 1.4 ± 0.2     | 1.7 ± 0.1    | 2.1 ± 0.1    | 4.5 ± 0.4     | 4.2 ± 0.3     | 1.7 ± 0.1   | 1.0 ± 0.2     | 2.4 ± 0.1   | 2.0 ± 0.2    | 3.6 ± 0.3     | W Africa F   | E Europe M   |
| 23  | 3.6 ± 0.3     | 1.62 ± 0.09  | 1.1 ± 0.2     | 1.4 ± 0.1    | 1.6 ± 0.1    | 4.6 ± 0.4     | 3.9 ± 0.3     | 1.5 ± 0.1   | 0.7 ± 0.2     | 2.4 ± 0.1   | 1.8 ± 0.2    | 3.8 ± 0.3     | W Africa F   | E Europe M   |
| 24  | 3.3 ± 0.3     | 1.04 ± 0.09  | 0.9 ± 0.1     | 0.9 ± 0.1    | 1.3 ± 0.1    | 4.0 ± 0.3     | 3.8 ± 0.3     | 1.2 ± 0.1   | 0.7 ± 0.2     | 2.3 ± 0.1   | 1.6 ± 0.2    | 3.7 ± 0.3     | W Africa F   | E Europe M   |
| 25  | 3.0 ± 0.3     | 0.75 ± 0.08  | 0.6 ± 0.1     | 0.6 ± 0.1    | 0.9 ± 0.1    | 3.9 ± 0.3     | 4.0 ± 0.3     | 1.1 ± 0.1   | 0.7 ± 0.2     | 2.2 ± 0.1   | 1.3 ± 0.2    | 3.7 ± 0.3     | E Africa M   | E Europe F   |
| 26  | 2.9 ± 0.3     | 0.25 ± 0.08  | 0.4 ± 0.1     | 0.11 ± 0.10  | 0.7 ± 0.1    | 3.6 ± 0.3     | 3.7 ± 0.3     | 0.8 ± 0.1   | 0.3 ± 0.2     | 2.0 ± 0.1   | 1.1 ± 0.2    | 3.5 ± 0.3     | E Africa M   | S Asia F     |
| 27  | 2.9 ± 0.3     | -0.21 ± 0.08 | 0.2 ± 0.1     | -0.30 ± 0.09 | 0.2 ± 0.1    | 3.5 ± 0.3     | 3.5 ± 0.3     | 0.6 ± 0.1   | 0.5 ± 0.2     | 1.7 ± 0.1   | 1.2 ± 0.2    | 3.2 ± 0.3     | W Africa F   | S Asia F     |
| 28  | 2.3 ± 0.3     | -0.63 ± 0.08 | -0.0 ± 0.1    | -0.59 ± 0.10 | -0.1 ± 0.1   | 3.1 ± 0.3     | 3.4 ± 0.2     | 0.5 ± 0.1   | 0.4 ± 0.2     | 1.5 ± 0.1   | 1.0 ± 0.2    | 2.9 ± 0.2     | E Africa M   | E Asia F     |
| 29  | 2.4 ± 0.3     | -0.99 ± 0.08 | -0.2 ± 0.1    | -0.9 ± 0.1   | -0.3 ± 0.1   | 2.8 ± 0.3     | 3.2 ± 0.3     | 0.2 ± 0.1   | 0.2 ± 0.2     | 1.27 ± 0.10 | 0.8 ± 0.2    | 3.4 ± 0.2     | W Africa M   | E Asia F     |
| 30  | 2.3 ± 0.3     | -1.29 ± 0.08 | -0.4 ± 0.1    | -1.3 ± 0.1   | -0.6 ± 0.1   | 2.5 ± 0.3     | 3.1 ± 0.3     | 0.0 ± 0.1   | 0.1 ± 0.2     | 1.07 ± 0.10 | 0.7 ± 0.2    | 2.9 ± 0.2     | E Africa M   | E Asia F     |
| 31  | 1.8 ± 0.3     | -1.59 ± 0.08 | -0.4 ± 0.1    | -1.6 ± 0.1   | -0.8 ± 0.1   | 2.2 ± 0.3     | 3.1 ± 0.3     | -0.1 ± 0.1  | -0.1 ± 0.2    | 0.9 ± 0.1   | 0.6 ± 0.2    | 2.8 ± 0.2     | E Africa M   | S Asia F     |
| 32  | 2.1 ± 0.3     | -1.91 ± 0.09 | -0.7 ± 0.2    | -1.7 ± 0.1   | -1.0 ± 0.2   | 2.2 ± 0.3     | 2.6 ± 0.3     | -0.2 ± 0.1  | -0.1 ± 0.2    | 0.77 ± 0.10 | 0.5 ± 0.2    | 2.7 ± 0.2     | W Africa M   | E Asia F     |
| 33  | 1.9 ± 0.3     | -2.19 ± 0.09 | -0.8 ± 0.2    | -2.0 ± 0.1   | -1.2 ± 0.2   | 2.0 ± 0.3     | 2.8 ± 0.3     | -0.4 ± 0.1  | 0.3 ± 0.2     | 0.5 ± 0.1   | 0.3 ± 0.2    | 2.5 ± 0.3     | E Africa M   | E Asia F     |
| 34  | 1.7 ± 0.3     | -2.37 ± 0.10 | -0.9 ± 0.2    | -2.2 ± 0.1   | -1.3 ± 0.2   | 1.8 ± 0.3     | 2.5 ± 0.3     | -0.5 ± 0.1  | -0.1 ± 0.2    | 0.5 ± 0.1   | 0.2 ± 0.2    | 2.7 ± 0.3     | W Africa M   | E Asia F     |
| 35  | 1.9 ± 0.4     | -2.7 ± 0.1   | -0.8 ± 0.2    | -2.4 ± 0.1   | -1.6 ± 0.2   | 1.6 ± 0.3     | 2.6 ± 0.3     | -0.5 ± 0.1  | -0.0 ± 0.2    | 0.4 ± 0.1   | 0.2 ± 0.3    | 2.2 ± 0.3     | E Africa M   | E Asia F     |
| 36  | 1.8 ± 0.4     | -2.8 ± 0.1   | -0.8 ± 0.2    | -2.5 ± 0.1   | -1.4 ± 0.2   | 1.9 ± 0.3     | 2.5 ± 0.3     | -0.5 ± 0.1  | 0.1 ± 0.2     | 0.2 ± 0.1   | 0.4 ± 0.3    | 2.2 ± 0.3     | E Africa M   | E Asia F     |
| 37  | 1.6 ± 0.4     | -2.8 ± 0.1   | -0.8 ± 0.2    | -2.3 ± 0.1   | -1.5 ± 0.2   | 1.5 ± 0.4     | 2.4 ± 0.3     | -0.6 ± 0.1  | -0.1 ± 0.2    | 0.2 ± 0.1   | 0.1 ± 0.3    | 2.0 ± 0.3     | E Africa M   | E Asia F     |
| 38  | 2.0 ± 0.4     | -3.0 ± 0.1   | -0.8 ± 0.2    | -2.4 ± 0.2   | -1.5 ± 0.2   | 1.4 ± 0.4     | 2.2 ± 0.3     | -0.5 ± 0.1  | -0.1 ± 0.2    | 0.1 ± 0.1   | 0.4 ± 0.3    | 2.1 ± 0.3     | E Africa M   | E Asia F     |
| 39  | 1.8 ± 0.4     | -3.0 ± 0.1   | -0.8 ± 0.2    | -2.4 ± 0.2   | -1.5 ± 0.2   | 1.2 ± 0.4     | 2.5 ± 0.3     | -0.6 ± 0.1  | 0.1 ± 0.2     | 0.1 ± 0.1   | 0.5 ± 0.3    | 1.8 ± 0.3     | E Africa M   | E Asia F     |
| 40  | 1.6 ± 0.4     | -3.1 ± 0.1   | -0.4 ± 0.2    | -2.2 ± 0.2   | -1.6 ± 0.2   | 1.1 ± 0.4     | 2.3 ± 0.3     | -0.6 ± 0.1  | 0.1 ± 0.2     | 0.2 ± 0.1   | 0.5 ± 0.2    | 1.9 ± 0.3     | E Africa M   | E Asia F     |
| 41  | 1.1 ± 0.4     | -3.0 ± 0.1   | -0.5 ± 0.2    | -2.2 ± 0.2   | -1.8 ± 0.2   | 0.4 ± 0.4     | 2.6 ± 0.4     | -0.5 ± 0.1  | 0.1 ± 0.2     | 0.1 ± 0.1   | 0.3 ± 0.3    | 1.8 ± 0.3     | E Africa M   | E Asia F     |
| 42  | 1.4 ± 0.5     | -2.8 ± 0.1   | -0.2 ± 0.3    | -1.7 ± 0.2   | -1.5 ± 0.2   | 0.9 ± 0.4     | 2.5 ± 0.4     | -0.3 ± 0.1  | 0.4 ± 0.2     | 0.0 ± 0.1   | 0.7 ± 0.3    | 1.6 ± 0.3     | E Africa M   | E Asia F     |
| 43  | 1.7 ± 0.5     | -3.0 ± 0.1   | -0.3 ± 0.3    | -1.9 ± 0.2   | -1.6 ± 0.2   | 0.6 ± 0.4     | 2.6 ± 0.4     | -0.3 ± 0.1  | 0.1 ± 0.2     | 0.1 ± 0.2   | 0.7 ± 0.3    | 1.3 ± 0.3     | E Africa M   | E Asia F     |
| 44  | 1.4 ± 0.5     | -3.0 ± 0.1   | -0.4 ± 0.3    | -1.7 ± 0.2   | -1.4 ± 0.2   | 0.2 ± 0.4     | 2.0 ± 0.4     | -0.2 ± 0.2  | 0.7 ± 0.3     | 0.1 ± 0.2   | 0.9 ± 0.3    | 1.4 ± 0.3     | E Africa M   | E Asia F     |
| 45  | 1.3 ± 0.5     | -2.8 ± 0.1   | -0.4 ± 0.3    | -1.6 ± 0.2   | -1.4 ± 0.2   | 0.2 ± 0.4     | 1.9 ± 0.4     | -0.3 ± 0.2  | 0.3 ± 0.3     | 0.3 ± 0.2   | 0.8 ± 0.3    | 0.9 ± 0.3     | E Africa M   | E Asia F     |
| 46  | 1.7 ± 0.5     | -2.9 ± 0.1   | -0.3 ± 0.3    | -1.4 ± 0.2   | -1.7 ± 0.2   | 0.4 ± 0.4     | 2.1 ± 0.4     | -0.2 ± 0.2  | 0.5 ± 0.3     | 0.4 ± 0.2   | 0.8 ± 0.3    | 0.8 ± 0.3     | E Africa M   | E Asia F     |
| 47  | 1.5 ± 0.5     | -2.5 ± 0.1   | -0.2 ± 0.3    | -1.4 ± 0.2   | -1.5 ± 0.3   | 0.1 ± 0.4     | 2.2 ± 0.4     | -0.1 ± 0.2  | 0.6 ± 0.3     | 0.5 ± 0.2   | 0.9 ± 0.3    | 0.9 ± 0.4     | E Africa M   | E Asia F     |
| 48  | 0.9 ± 0.6     | -2.5 ± 0.1   | -0.1 ± 0.3    | -1.1 ± 0.2   | -1.4 ± 0.2   | 0.3 ± 0.4     | 1.6 ± 0.5     | -0.2 ± 0.1  | 0.6 ± 0.3     | 0.5 ± 0.2   | 0.8 ± 0.3    | 0.6 ± 0.4     | E Africa M   | E Asia F     |
| 49  | 1.6 ± 0.5     | -2.6 ± 0.1   | -0.3 ± 0.3    | -1.2 ± 0.2   | -1.2 ± 0.3   | -0.3 ± 0.4    | 1.5 ± 0.5     | -0.1 ± 0.2  | 0.5 ± 0.3     | 0.2 ± 0.2   | 0.9 ± 0.3    | 0.5 ± 0.4     | E Africa F   | E Asia F     |
| 50  | 1.1 ± 0.6     | -2.4 ± 0.2   | -0.1 ± 0.3    | -1.2 ± 0.2   | -1.4 ± 0.2   | -0.1 ± 0.4    | 1.1 ± 0.5     | -0.2 ± 0.1  | 0.4 ± 0.3     | 0.3 ± 0.2   | 1.0 ± 0.3    | 0.4 ± 0.3     | E Africa F   | E Asia F     |
| 51  | 1.0 ± 0.5     | -2.4 ± 0.1   | -0.5 ± 0.3    | -1.2 ± 0.2   | -1.4 ± 0.2   | -0.4 ± 0.5    | 0.9 ± 0.6     | -0.3 ± 0.1  | 0.7 ± 0.3     | 0.3 ± 0.2   | 1.1 ± 0.3    | -0.0 ± 0.4    | S E Asia M   | E Asia F     |
| 52  | 0.8 ± 0.6     | -2.4 ± 0.1   | -0.3 ± 0.3    | -1.5 ± 0.2   | -1.2 ± 0.2   | -0.7 ± 0.4    | 1.2 ± 0.5     | -0.4 ± 0.2  | 0.3 ± 0.3     | 0.1 ± 0.2   | 0.6 ± 0.3    | 0.0 ± 0.4     | E Africa M   | E Asia F     |

**Table 58.** Mean raw error for Application-type images for algorithm ondato-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 4.8 ± 0.3     | 4.9 ± 0.1    | 3.5 ± 0.2     | 4.1 ± 0.2    | 4.5 ± 0.3    | 5.6 ± 0.4     | 3.7 ± 0.3     | 3.4 ± 0.1   | 1.5 ± 0.2     | 2.2 ± 0.1    | 2.7 ± 0.2    | 3.7 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 4.9 ± 0.4     | 4.9 ± 0.1    | 3.5 ± 0.2     | 4.3 ± 0.2    | 4.6 ± 0.3    | 5.3 ± 0.4     | 3.1 ± 0.3     | 3.34 ± 0.10 | 1.7 ± 0.2     | 2.3 ± 0.1    | 3.0 ± 0.2    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.6 ± 0.3     | 4.5 ± 0.1    | 3.4 ± 0.2     | 4.0 ± 0.2    | 4.5 ± 0.3    | 5.4 ± 0.4     | 3.2 ± 0.3     | 3.08 ± 0.09 | 1.8 ± 0.2     | 2.4 ± 0.1    | 2.9 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 4.5 ± 0.3     | 4.2 ± 0.1    | 3.1 ± 0.2     | 3.9 ± 0.2    | 4.2 ± 0.3    | 5.1 ± 0.3     | 3.1 ± 0.3     | 2.86 ± 0.10 | 1.5 ± 0.2     | 2.6 ± 0.1    | 2.8 ± 0.2    | 3.3 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.0 ± 0.3     | 3.6 ± 0.1    | 2.5 ± 0.3     | 3.4 ± 0.1    | 3.8 ± 0.2    | 4.6 ± 0.3     | 2.7 ± 0.2     | 2.32 ± 0.09 | 1.3 ± 0.2     | 2.3 ± 0.1    | 2.4 ± 0.2    | 2.8 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.0 ± 0.3     | 3.2 ± 0.1    | 2.4 ± 0.2     | 3.2 ± 0.1    | 3.4 ± 0.2    | 4.7 ± 0.3     | 3.0 ± 0.3     | 2.11 ± 0.10 | 1.1 ± 0.2     | 2.3 ± 0.1    | 2.2 ± 0.2    | 3.1 ± 0.4     | W Africa F   | E Europe M   |
| 20  | 3.7 ± 0.3     | 3.0 ± 0.1    | 2.2 ± 0.2     | 2.9 ± 0.2    | 3.2 ± 0.2    | 4.3 ± 0.3     | 3.1 ± 0.3     | 1.89 ± 0.10 | 1.2 ± 0.2     | 2.4 ± 0.1    | 2.1 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 3.3 ± 0.3     | 2.8 ± 0.1    | 1.7 ± 0.2     | 2.4 ± 0.1    | 3.0 ± 0.1    | 4.1 ± 0.3     | 3.2 ± 0.3     | 1.69 ± 0.09 | 0.9 ± 0.2     | 2.3 ± 0.1    | 2.0 ± 0.2    | 2.9 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 3.3 ± 0.3     | 2.3 ± 0.1    | 1.5 ± 0.2     | 1.9 ± 0.1    | 2.5 ± 0.1    | 3.4 ± 0.3     | 3.6 ± 0.3     | 1.4 ± 0.1   | 0.9 ± 0.2     | 2.1 ± 0.1    | 1.6 ± 0.1    | 2.8 ± 0.3     | E Africa M   | E Europe M   |
| 23  | 3.1 ± 0.2     | 2.00 ± 0.09  | 1.3 ± 0.1     | 1.7 ± 0.1    | 2.1 ± 0.1    | 3.5 ± 0.3     | 3.1 ± 0.3     | 1.2 ± 0.1   | 0.7 ± 0.2     | 2.0 ± 0.1    | 1.5 ± 0.2    | 2.8 ± 0.3     | W Africa F   | E Europe M   |
| 24  | 2.8 ± 0.3     | 1.49 ± 0.09  | 1.1 ± 0.1     | 1.17 ± 0.10  | 1.8 ± 0.1    | 2.8 ± 0.3     | 2.9 ± 0.2     | 0.9 ± 0.1   | 0.7 ± 0.2     | 1.9 ± 0.1    | 1.3 ± 0.2    | 2.8 ± 0.2     | E Africa M   | E Europe M   |
| 25  | 2.4 ± 0.2     | 1.23 ± 0.08  | 0.9 ± 0.1     | 0.76 ± 0.09  | 1.5 ± 0.1    | 2.7 ± 0.3     | 3.0 ± 0.2     | 0.8 ± 0.1   | 0.7 ± 0.1     | 1.7 ± 0.1    | 1.1 ± 0.1    | 2.4 ± 0.2     | E Africa M   | E Europe M   |
| 26  | 2.3 ± 0.2     | 0.82 ± 0.08  | 0.7 ± 0.1     | 0.29 ± 0.09  | 1.3 ± 0.1    | 2.3 ± 0.2     | 2.6 ± 0.2     | 0.5 ± 0.1   | 0.4 ± 0.1     | 1.4 ± 0.1    | 1.0 ± 0.2    | 2.1 ± 0.2     | E Africa M   | S Asia F     |
| 27  | 2.1 ± 0.3     | 0.38 ± 0.08  | 0.5 ± 0.1     | -0.08 ± 0.09 | 0.8 ± 0.1    | 2.1 ± 0.2     | 2.4 ± 0.2     | 0.3 ± 0.1   | 0.5 ± 0.1     | 1.14 ± 0.10  | 0.9 ± 0.2    | 1.8 ± 0.2     | E Africa M   | S Asia F     |
| 28  | 1.7 ± 0.2     | 0.01 ± 0.08  | 0.4 ± 0.1     | -0.44 ± 0.09 | 0.5 ± 0.1    | 1.8 ± 0.2     | 2.0 ± 0.2     | 0.2 ± 0.1   | 0.3 ± 0.1     | 0.93 ± 0.09  | 0.7 ± 0.2    | 1.4 ± 0.2     | E Africa M   | S Asia F     |
| 29  | 1.7 ± 0.3     | -0.31 ± 0.08 | 0.1 ± 0.1     | -0.79 ± 0.09 | 0.3 ± 0.1    | 1.4 ± 0.2     | 2.0 ± 0.2     | -0.0 ± 0.1  | 0.2 ± 0.1     | 0.62 ± 0.09  | 0.3 ± 0.2    | 1.6 ± 0.2     | E Africa M   | S Asia F     |
| 30  | 1.5 ± 0.3     | -0.59 ± 0.08 | -0.0 ± 0.1    | -1.18 ± 0.09 | 0.1 ± 0.1    | 1.0 ± 0.2     | 1.7 ± 0.2     | -0.2 ± 0.1  | 0.1 ± 0.2     | 0.38 ± 0.09  | 0.4 ± 0.2    | 1.2 ± 0.2     | E Africa M   | S Asia F     |
| 31  | 0.9 ± 0.3     | -0.90 ± 0.08 | -0.0 ± 0.1    | -1.51 ± 0.10 | -0.2 ± 0.1   | 0.6 ± 0.2     | 1.5 ± 0.2     | -0.4 ± 0.1  | -0.1 ± 0.2    | 0.11 ± 0.09  | 0.2 ± 0.2    | 1.0 ± 0.2     | E Africa M   | S Asia F     |
| 32  | 1.1 ± 0.3     | -1.15 ± 0.09 | -0.3 ± 0.1    | -1.7 ± 0.1   | -0.4 ± 0.1   | 0.6 ± 0.3     | 1.0 ± 0.2     | -0.6 ± 0.1  | -0.1 ± 0.2    | -0.04 ± 0.09 | 0.1 ± 0.2    | 0.7 ± 0.2     | E Africa F   | S Asia F     |
| 33  | 0.9 ± 0.3     | -1.41 ± 0.09 | -0.5 ± 0.1    | -2.0 ± 0.1   | -0.6 ± 0.2   | 0.3 ± 0.3     | 1.2 ± 0.2     | -0.8 ± 0.1  | 0.1 ± 0.2     | -0.36 ± 0.09 | -0.1 ± 0.2   | 0.5 ± 0.2     | E Africa M   | S Asia F     |
| 34  | 0.7 ± 0.3     | -1.69 ± 0.09 | -0.7 ± 0.2    | -2.2 ± 0.1   | -0.9 ± 0.1   | 0.0 ± 0.3     | 0.8 ± 0.3     | -1.0 ± 0.1  | -0.2 ± 0.2    | -0.47 ± 0.10 | -0.3 ± 0.2   | 0.7 ± 0.2     | E Africa M   | S Asia F     |
| 35  | 0.8 ± 0.3     | -1.98 ± 0.10 | -0.6 ± 0.2    | -2.4 ± 0.1   | -1.1 ± 0.2   | -0.3 ± 0.3    | 0.7 ± 0.3     | -1.0 ± 0.1  | -0.3 ± 0.2    | -0.7 ± 0.1   | -0.4 ± 0.2   | 0.1 ± 0.2     | E Africa F   | S Asia F     |
| 36  | 0.5 ± 0.3     | -2.11 ± 0.10 | -0.6 ± 0.2    | -2.7 ± 0.1   | -1.1 ± 0.2   | -0.2 ± 0.3    | 0.7 ± 0.3     | -1.2 ± 0.1  | -0.2 ± 0.2    | -0.8 ± 0.1   | -0.3 ± 0.2   | 0.0 ± 0.2     | E Africa M   | S Asia F     |
| 37  | 0.2 ± 0.3     | -2.28 ± 0.10 | -0.8 ± 0.2    | -2.6 ± 0.1   | -1.2 ± 0.2   | -0.6 ± 0.3    | 0.5 ± 0.3     | -1.3 ± 0.1  | -0.5 ± 0.2    | -1.0 ± 0.1   | -0.6 ± 0.2   | -0.2 ± 0.2    | E Africa M   | S Asia F     |
| 38  | 0.6 ± 0.4     | -2.5 ± 0.1   | -0.9 ± 0.2    | -2.7 ± 0.1   | -1.4 ± 0.2   | -0.6 ± 0.3    | 0.2 ± 0.3     | -1.3 ± 0.1  | -0.6 ± 0.2    | -1.1 ± 0.1   | -0.4 ± 0.2   | -0.3 ± 0.2    | E Africa F   | S Asia F     |
| 39  | 0.2 ± 0.4     | -2.5 ± 0.1   | -1.0 ± 0.2    | -2.9 ± 0.1   | -1.5 ± 0.2   | -0.8 ± 0.3    | 0.6 ± 0.3     | -1.4 ± 0.1  | -0.5 ± 0.2    | -1.1 ± 0.1   | -0.3 ± 0.2   | -0.5 ± 0.3    | E Africa M   | S Asia F     |
| 40  | 0.2 ± 0.4     | -2.8 ± 0.1   | -0.9 ± 0.2    | -2.8 ± 0.2   | -1.8 ± 0.2   | -1.2 ± 0.3    | 0.4 ± 0.3     | -1.5 ± 0.1  | -0.7 ± 0.2    | -1.1 ± 0.1   | -0.6 ± 0.2   | -0.6 ± 0.3    | E Africa M   | S Asia F     |
| 41  | -0.3 ± 0.4    | -2.8 ± 0.1   | -0.9 ± 0.2    | -2.9 ± 0.2   | -1.9 ± 0.2   | -1.6 ± 0.4    | 0.5 ± 0.3     | -1.5 ± 0.1  | -0.7 ± 0.2    | -1.2 ± 0.1   | -0.7 ± 0.2   | -0.6 ± 0.3    | E Africa M   | S Asia F     |
| 42  | 0.1 ± 0.4     | -2.8 ± 0.1   | -0.7 ± 0.2    | -2.5 ± 0.2   | -2.0 ± 0.2   | -1.2 ± 0.4    | 0.6 ± 0.3     | -1.5 ± 0.1  | -0.4 ± 0.2    | -1.3 ± 0.1   | -0.5 ± 0.2   | -0.8 ± 0.3    | E Africa M   | E Asia F     |
| 43  | 0.4 ± 0.4     | -3.0 ± 0.1   | -0.9 ± 0.2    | -2.8 ± 0.2   | -2.1 ± 0.2   | -1.5 ± 0.4    | 0.8 ± 0.4     | -1.5 ± 0.1  | -0.8 ± 0.2    | -1.3 ± 0.1   | -0.5 ± 0.3   | -1.2 ± 0.3    | E Africa M   | E Asia F     |
| 44  | 0.0 ± 0.5     | -3.2 ± 0.1   | -1.1 ± 0.2    | -2.6 ± 0.2   | -2.1 ± 0.2   | -2.0 ± 0.4    | 0.2 ± 0.4     | -1.5 ± 0.1  | -0.4 ± 0.2    | -1.2 ± 0.2   | -0.4 ± 0.3   | -0.9 ± 0.3    | E Africa M   | E Asia F     |
| 45  | -0.0 ± 0.5    | -3.1 ± 0.1   | -1.0 ± 0.2    | -2.6 ± 0.2   | -2.2 ± 0.2   | -1.9 ± 0.4    | 0.3 ± 0.4     | -1.6 ± 0.1  | -0.6 ± 0.2    | -1.0 ± 0.2   | -0.4 ± 0.3   | -1.5 ± 0.3    | E Africa M   | E Asia F     |
| 46  | 0.5 ± 0.5     | -3.4 ± 0.1   | -1.0 ± 0.2    | -2.4 ± 0.2   | -2.4 ± 0.2   | -1.6 ± 0.4    | 0.3 ± 0.5     | -1.6 ± 0.1  | -0.3 ± 0.3    | -0.7 ± 0.2   | -0.5 ± 0.3   | -1.6 ± 0.3    | E Africa F   | E Asia F     |
| 47  | 0.2 ± 0.6     | -3.2 ± 0.1   | -0.8 ± 0.3    | -2.3 ± 0.2   | -2.3 ± 0.2   | -2.0 ± 0.4    | 0.6 ± 0.4     | -1.5 ± 0.1  | -0.1 ± 0.3    | -0.7 ± 0.2   | -0.2 ± 0.3   | -1.4 ± 0.4    | E Africa M   | E Asia F     |
| 48  | -0.2 ± 0.6    | -3.1 ± 0.1   | -0.6 ± 0.3    | -2.1 ± 0.2   | -2.2 ± 0.2   | -1.9 ± 0.4    | -0.0 ± 0.5    | -1.5 ± 0.1  | -0.1 ± 0.3    | -0.5 ± 0.2   | -0.3 ± 0.3   | -1.6 ± 0.4    | E Africa M   | E Asia F     |
| 49  | 0.3 ± 0.6     | -3.3 ± 0.1   | -0.7 ± 0.3    | -2.0 ± 0.2   | -2.1 ± 0.2   | -2.5 ± 0.4    | 0.1 ± 0.5     | -1.4 ± 0.2  | -0.1 ± 0.3    | -0.7 ± 0.2   | -0.1 ± 0.3   | -1.6 ± 0.4    | E Africa F   | E Asia F     |
| 50  | -0.2 ± 0.6    | -3.1 ± 0.1   | -0.5 ± 0.3    | -2.0 ± 0.2   | -2.2 ± 0.2   | -2.2 ± 0.4    | -0.3 ± 0.5    | -1.3 ± 0.2  | -0.1 ± 0.3    | -0.5 ± 0.2   | 0.1 ± 0.3    | -1.7 ± 0.4    | S E Asia M   | E Asia F     |
| 51  | 0.1 ± 0.6     | -3.1 ± 0.1   | -0.9 ± 0.3    | -1.9 ± 0.2   | -2.2 ± 0.2   | -2.5 ± 0.5    | -0.4 ± 0.6    | -1.3 ± 0.2  | 0.2 ± 0.3     | -0.5 ± 0.2   | 0.2 ± 0.3    | -2.2 ± 0.4    | S E Asia M   | E Asia F     |
| 52  | -0.2 ± 0.6    | -3.1 ± 0.1   | -0.5 ± 0.3    | -2.1 ± 0.2   | -1.8 ± 0.2   | -2.8 ± 0.4    | -0.2 ± 0.6    | -1.4 ± 0.2  | 0.0 ± 0.3     | -0.6 ± 0.2   | -0.1 ± 0.3   | -1.9 ± 0.4    | E Europe M   | E Asia F     |

**Table 59.** Mean raw error for Application-type images for algorithm privately-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 1.2 ± 0.4     | 3.4 ± 0.2    | 0.7 ± 0.3     | 1.6 ± 0.2   | 2.8 ± 0.4    | 2.0 ± 0.4     | 1.1 ± 0.3     | 3.2 ± 0.1   | -0.3 ± 0.2    | 1.2 ± 0.1    | 2.0 ± 0.3    | 1.7 ± 0.3     | E Asia F     | E Europe M   |
| 15  | 1.3 ± 0.4     | 3.8 ± 0.2    | 0.7 ± 0.3     | 2.1 ± 0.3   | 3.3 ± 0.4    | 1.9 ± 0.5     | 0.9 ± 0.3     | 3.4 ± 0.1   | -0.3 ± 0.2    | 1.4 ± 0.2    | 2.3 ± 0.3    | 1.9 ± 0.4     | E Asia F     | E Europe M   |
| 16  | 1.2 ± 0.4     | 3.8 ± 0.2    | 0.9 ± 0.4     | 1.9 ± 0.3   | 3.7 ± 0.4    | 1.9 ± 0.6     | 1.4 ± 0.3     | 3.1 ± 0.1   | -0.0 ± 0.3    | 1.4 ± 0.2    | 2.5 ± 0.3    | 1.3 ± 0.3     | E Asia F     | E Europe M   |
| 17  | 1.5 ± 0.4     | 3.6 ± 0.2    | 1.0 ± 0.4     | 1.8 ± 0.3   | 3.2 ± 0.4    | 1.6 ± 0.4     | 1.2 ± 0.3     | 2.9 ± 0.1   | -0.5 ± 0.3    | 1.5 ± 0.2    | 2.1 ± 0.3    | 1.3 ± 0.4     | E Asia F     | E Europe M   |
| 18  | 1.1 ± 0.3     | 3.1 ± 0.2    | 0.8 ± 0.4     | 1.2 ± 0.2   | 3.0 ± 0.3    | 0.8 ± 0.4     | 0.9 ± 0.3     | 2.1 ± 0.1   | -0.5 ± 0.3    | 0.8 ± 0.1    | 1.7 ± 0.2    | 0.7 ± 0.3     | E Asia F     | E Europe M   |
| 19  | 1.1 ± 0.4     | 2.8 ± 0.2    | 0.6 ± 0.4     | 1.2 ± 0.2   | 2.7 ± 0.2    | 0.9 ± 0.4     | 0.8 ± 0.3     | 1.8 ± 0.1   | -0.6 ± 0.3    | 0.9 ± 0.1    | 1.2 ± 0.2    | 0.5 ± 0.3     | E Asia F     | E Europe M   |
| 20  | 0.7 ± 0.4     | 2.8 ± 0.2    | 0.4 ± 0.3     | 0.9 ± 0.2   | 2.5 ± 0.2    | 0.9 ± 0.4     | 0.8 ± 0.3     | 1.3 ± 0.1   | -0.6 ± 0.3    | 0.7 ± 0.1    | 0.9 ± 0.2    | 0.4 ± 0.3     | E Asia F     | E Europe M   |
| 21  | 0.4 ± 0.3     | 2.5 ± 0.1    | -0.0 ± 0.3    | 0.4 ± 0.2   | 2.3 ± 0.2    | 0.3 ± 0.4     | 0.5 ± 0.3     | 0.9 ± 0.1   | -0.6 ± 0.2    | 0.4 ± 0.1    | 0.5 ± 0.2    | -0.3 ± 0.3    | E Asia F     | E Europe M   |
| 22  | 0.2 ± 0.3     | 2.0 ± 0.1    | -0.1 ± 0.2    | -0.2 ± 0.2  | 1.8 ± 0.2    | 0.0 ± 0.4     | 0.8 ± 0.3     | 0.4 ± 0.1   | -0.7 ± 0.2    | -0.0 ± 0.1   | -0.2 ± 0.2   | -0.5 ± 0.3    | E Asia F     | E Europe M   |
| 23  | -0.3 ± 0.3    | 1.6 ± 0.1    | -0.4 ± 0.2    | -0.5 ± 0.1  | 1.6 ± 0.2    | -0.3 ± 0.3    | 0.1 ± 0.3     | -0.2 ± 0.1  | -1.0 ± 0.2    | -0.3 ± 0.1   | -0.7 ± 0.2   | -0.8 ± 0.2    | E Asia F     | E Europe M   |
| 24  | -0.7 ± 0.3    | 1.2 ± 0.1    | -0.7 ± 0.2    | -0.9 ± 0.1  | 1.1 ± 0.1    | -0.9 ± 0.3    | -0.1 ± 0.3    | -0.6 ± 0.1  | -1.4 ± 0.2    | -0.6 ± 0.1   | -1.1 ± 0.2   | -1.3 ± 0.2    | E Asia F     | E Europe M   |
| 25  | -1.0 ± 0.3    | 0.9 ± 0.1    | -1.1 ± 0.2    | -1.5 ± 0.1  | 0.6 ± 0.1    | -1.3 ± 0.3    | -0.2 ± 0.3    | -0.9 ± 0.1  | -1.6 ± 0.2    | -1.1 ± 0.1   | -1.3 ± 0.2   | -1.6 ± 0.2    | E Asia F     | W Africa M   |
| 26  | -1.4 ± 0.2    | 0.4 ± 0.1    | -1.4 ± 0.1    | -2.0 ± 0.1  | 0.3 ± 0.1    | -1.9 ± 0.3    | -0.7 ± 0.2    | -1.4 ± 0.1  | -2.1 ± 0.2    | -1.5 ± 0.1   | -1.8 ± 0.2   | -2.0 ± 0.2    | E Asia F     | E Europe M   |
| 27  | -1.5 ± 0.3    | 0.08 ± 0.09  | -1.7 ± 0.1    | -2.4 ± 0.1  | -0.1 ± 0.1   | -2.3 ± 0.2    | -1.0 ± 0.2    | -1.7 ± 0.1  | -2.2 ± 0.2    | -1.9 ± 0.1   | -1.8 ± 0.2   | -2.5 ± 0.2    | E Asia F     | W Africa M   |
| 28  | -2.2 ± 0.3    | -0.31 ± 0.09 | -2.1 ± 0.1    | -2.9 ± 0.1  | -0.4 ± 0.1   | -2.9 ± 0.2    | -1.4 ± 0.2    | -2.0 ± 0.1  | -2.2 ± 0.2    | -2.32 ± 0.10 | -2.3 ± 0.2   | -3.0 ± 0.2    | E Asia F     | W Africa M   |
| 29  | -2.5 ± 0.2    | -0.68 ± 0.09 | -2.3 ± 0.1    | -3.3 ± 0.1  | -0.8 ± 0.1   | -3.3 ± 0.2    | -1.7 ± 0.2    | -2.3 ± 0.1  | -2.6 ± 0.2    | -2.81 ± 0.10 | -2.6 ± 0.2   | -3.1 ± 0.2    | E Asia F     | S Asia F     |
| 30  | -3.1 ± 0.2    | -1.10 ± 0.09 | -2.7 ± 0.1    | -3.8 ± 0.1  | -1.0 ± 0.1   | -4.0 ± 0.2    | -2.0 ± 0.3    | -2.7 ± 0.1  | -2.8 ± 0.2    | -3.27 ± 0.09 | -2.9 ± 0.2   | -3.6 ± 0.2    | S E Asia F   | W Africa F   |
| 31  | -3.5 ± 0.3    | -1.52 ± 0.09 | -2.9 ± 0.2    | -4.3 ± 0.1  | -1.5 ± 0.1   | -4.3 ± 0.2    | -2.2 ± 0.3    | -2.9 ± 0.1  | -3.2 ± 0.2    | -3.63 ± 0.10 | -3.2 ± 0.2   | -4.1 ± 0.2    | E Asia F     | S Asia F     |
| 32  | -3.8 ± 0.3    | -1.87 ± 0.09 | -3.3 ± 0.2    | -4.7 ± 0.1  | -2.0 ± 0.2   | -4.8 ± 0.3    | -2.6 ± 0.3    | -3.1 ± 0.1  | -3.1 ± 0.2    | -3.88 ± 0.10 | -3.3 ± 0.2   | -4.4 ± 0.2    | E Asia F     | W Africa F   |
| 33  | -4.4 ± 0.3    | -2.27 ± 0.10 | -3.7 ± 0.2    | -5.1 ± 0.1  | -2.4 ± 0.2   | -5.2 ± 0.3    | -3.0 ± 0.3    | -3.4 ± 0.1  | -3.2 ± 0.2    | -4.40 ± 0.10 | -3.6 ± 0.2   | -4.7 ± 0.2    | E Asia F     | W Africa F   |
| 34  | -4.6 ± 0.3    | -2.59 ± 0.10 | -4.1 ± 0.2    | -5.4 ± 0.1  | -2.7 ± 0.2   | -5.7 ± 0.3    | -3.5 ± 0.3    | -3.7 ± 0.1  | -3.5 ± 0.2    | -4.6 ± 0.1   | -3.7 ± 0.3   | -5.0 ± 0.2    | E Asia F     | W Africa F   |
| 35  | -4.7 ± 0.3    | -3.1 ± 0.1   | -4.3 ± 0.2    | -5.8 ± 0.1  | -3.2 ± 0.2   | -6.2 ± 0.3    | -3.4 ± 0.3    | -3.8 ± 0.1  | -3.8 ± 0.2    | -5.1 ± 0.1   | -4.1 ± 0.3   | -5.4 ± 0.2    | E Asia F     | W Africa F   |
| 36  | -5.0 ± 0.3    | -3.3 ± 0.1   | -4.4 ± 0.2    | -6.2 ± 0.1  | -3.3 ± 0.2   | -6.3 ± 0.3    | -4.1 ± 0.3    | -4.1 ± 0.1  | -3.7 ± 0.2    | -5.3 ± 0.1   | -4.1 ± 0.3   | -5.7 ± 0.3    | E Asia F     | W Africa F   |
| 37  | -5.6 ± 0.3    | -3.6 ± 0.1   | -4.8 ± 0.2    | -6.3 ± 0.1  | -3.9 ± 0.2   | -6.9 ± 0.3    | -4.0 ± 0.3    | -4.2 ± 0.1  | -4.2 ± 0.2    | -5.6 ± 0.1   | -4.4 ± 0.3   | -6.0 ± 0.3    | E Asia F     | W Africa F   |
| 38  | -5.4 ± 0.4    | -4.0 ± 0.1   | -4.9 ± 0.2    | -6.6 ± 0.2  | -3.8 ± 0.2   | -7.3 ± 0.4    | -4.5 ± 0.3    | -4.6 ± 0.1  | -4.3 ± 0.2    | -5.9 ± 0.1   | -4.0 ± 0.3   | -6.2 ± 0.3    | S E Asia F   | W Africa F   |
| 39  | -5.8 ± 0.4    | -4.2 ± 0.1   | -5.2 ± 0.2    | -7.1 ± 0.2  | -4.2 ± 0.2   | -7.4 ± 0.4    | -4.2 ± 0.4    | -4.8 ± 0.1  | -4.2 ± 0.2    | -6.1 ± 0.1   | -4.0 ± 0.3   | -6.5 ± 0.3    | S E Asia M   | W Africa F   |
| 40  | -6.1 ± 0.4    | -4.5 ± 0.1   | -5.0 ± 0.3    | -7.0 ± 0.2  | -4.6 ± 0.2   | -8.0 ± 0.3    | -4.3 ± 0.4    | -5.0 ± 0.1  | -4.5 ± 0.2    | -6.2 ± 0.1   | -4.4 ± 0.3   | -6.7 ± 0.3    | E Africa M   | W Africa F   |
| 41  | -6.5 ± 0.4    | -4.7 ± 0.1   | -5.5 ± 0.3    | -7.4 ± 0.2  | -4.7 ± 0.2   | -8.8 ± 0.4    | -4.3 ± 0.4    | -5.0 ± 0.2  | -4.3 ± 0.3    | -6.4 ± 0.2   | -4.4 ± 0.3   | -6.9 ± 0.4    | E Europe M   | W Africa F   |
| 42  | -6.2 ± 0.5    | -4.7 ± 0.1   | -5.1 ± 0.3    | -7.2 ± 0.2  | -5.1 ± 0.2   | -8.4 ± 0.4    | -4.3 ± 0.4    | -5.0 ± 0.2  | -4.0 ± 0.3    | -6.4 ± 0.2   | -4.1 ± 0.3   | -7.0 ± 0.3    | E Europe M   | W Africa F   |
| 43  | -6.5 ± 0.5    | -5.0 ± 0.1   | -5.4 ± 0.3    | -7.6 ± 0.2  | -5.4 ± 0.3   | -9.0 ± 0.4    | -4.1 ± 0.5    | -5.0 ± 0.2  | -4.4 ± 0.3    | -6.4 ± 0.2   | -4.1 ± 0.3   | -7.3 ± 0.4    | E Africa M   | W Africa F   |
| 44  | -6.3 ± 0.6    | -5.2 ± 0.1   | -5.5 ± 0.3    | -7.5 ± 0.3  | -5.1 ± 0.3   | -9.1 ± 0.5    | -4.6 ± 0.5    | -4.9 ± 0.2  | -4.2 ± 0.3    | -6.4 ± 0.2   | -3.8 ± 0.3   | -6.9 ± 0.4    | S E Asia M   | W Africa F   |
| 45  | -6.7 ± 0.6    | -5.2 ± 0.2   | -5.5 ± 0.3    | -7.5 ± 0.3  | -5.4 ± 0.3   | -9.4 ± 0.5    | -4.6 ± 0.5    | -5.1 ± 0.2  | -4.2 ± 0.4    | -6.3 ± 0.2   | -4.0 ± 0.3   | -7.4 ± 0.4    | S E Asia M   | W Africa F   |
| 46  | -6.3 ± 0.6    | -5.4 ± 0.2   | -5.7 ± 0.4    | -7.2 ± 0.3  | -5.8 ± 0.3   | -9.5 ± 0.5    | -4.4 ± 0.5    | -5.0 ± 0.2  | -4.0 ± 0.4    | -6.0 ± 0.2   | -3.7 ± 0.4   | -7.5 ± 0.4    | S E Asia M   | W Africa F   |
| 47  | -5.7 ± 0.7    | -5.3 ± 0.2   | -5.4 ± 0.4    | -7.3 ± 0.3  | -5.4 ± 0.3   | -9.5 ± 0.5    | -3.8 ± 0.6    | -4.8 ± 0.2  | -4.0 ± 0.4    | -6.0 ± 0.3   | -3.5 ± 0.3   | -7.2 ± 0.5    | S E Asia M   | W Africa F   |
| 48  | -6.5 ± 0.7    | -5.3 ± 0.2   | -5.2 ± 0.4    | -7.2 ± 0.3  | -5.6 ± 0.3   | -9.6 ± 0.5    | -4.7 ± 0.6    | -4.9 ± 0.2  | -3.7 ± 0.4    | -5.9 ± 0.3   | -3.5 ± 0.4   | -7.6 ± 0.5    | S E Asia M   | W Africa F   |
| 49  | -6.5 ± 0.7    | -5.4 ± 0.2   | -5.5 ± 0.4    | -7.0 ± 0.3  | -5.3 ± 0.3   | -9.7 ± 0.6    | -4.9 ± 0.6    | -4.7 ± 0.2  | -3.9 ± 0.4    | -5.8 ± 0.3   | -3.5 ± 0.4   | -7.8 ± 0.5    | S E Asia M   | W Africa F   |
| 50  | -6.5 ± 0.8    | -5.5 ± 0.2   | -5.1 ± 0.4    | -7.0 ± 0.3  | -5.7 ± 0.3   | -9.6 ± 0.6    | -4.4 ± 0.6    | -4.7 ± 0.2  | -4.0 ± 0.4    | -5.9 ± 0.3   | -3.3 ± 0.4   | -7.4 ± 0.5    | S E Asia M   | W Africa F   |
| 51  | -5.9 ± 0.7    | -5.5 ± 0.2   | -5.1 ± 0.4    | -7.0 ± 0.3  | -5.9 ± 0.3   | -10.1 ± 0.6   | -5.0 ± 0.7    | -4.7 ± 0.2  | -3.8 ± 0.4    | -5.8 ± 0.3   | -3.4 ± 0.3   | -7.9 ± 0.5    | S E Asia M   | W Africa F   |
| 52  | -6.4 ± 0.8    | -5.7 ± 0.2   | -4.7 ± 0.4    | -7.1 ± 0.3  | -5.3 ± 0.3   | -10.7 ± 0.6   | -4.7 ± 0.7    | -4.7 ± 0.2  | -3.9 ± 0.4    | -5.8 ± 0.3   | -3.9 ± 0.4   | -8.0 ± 0.5    | E Europe M   | W Africa F   |



**Table 60.** Mean raw error for Application-type images for algorithm regula-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 2.1 ± 0.3     | 1.98 ± 0.09  | 1.2 ± 0.2     | 2.1 ± 0.1    | 1.9 ± 0.2    | 2.3 ± 0.3     | 0.4 ± 0.2     | 1.38 ± 0.08  | 0.2 ± 0.1     | 0.7 ± 0.1    | 0.9 ± 0.2    | 0.4 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 2.5 ± 0.3     | 1.81 ± 0.09  | 1.3 ± 0.2     | 2.5 ± 0.2    | 2.1 ± 0.3    | 2.4 ± 0.4     | 0.4 ± 0.2     | 1.35 ± 0.07  | 0.3 ± 0.1     | 1.0 ± 0.1    | 1.1 ± 0.1    | 0.9 ± 0.3     | S Asia F     | E Europe M   |
| 16  | 2.4 ± 0.3     | 1.59 ± 0.09  | 1.5 ± 0.2     | 2.2 ± 0.2    | 2.1 ± 0.3    | 3.1 ± 0.5     | 1.0 ± 0.2     | 1.12 ± 0.06  | 0.5 ± 0.2     | 1.0 ± 0.1    | 1.0 ± 0.1    | 0.8 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 2.6 ± 0.3     | 1.3 ± 0.1    | 1.5 ± 0.2     | 2.3 ± 0.2    | 1.9 ± 0.2    | 3.0 ± 0.3     | 0.8 ± 0.2     | 0.88 ± 0.08  | 0.2 ± 0.1     | 1.0 ± 0.1    | 0.9 ± 0.2    | 1.4 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 2.2 ± 0.3     | 0.74 ± 0.10  | 0.8 ± 0.3     | 1.7 ± 0.1    | 1.6 ± 0.2    | 2.5 ± 0.4     | 0.7 ± 0.2     | 0.32 ± 0.08  | -0.1 ± 0.2    | 0.6 ± 0.1    | 0.5 ± 0.2    | 1.1 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 2.4 ± 0.3     | 0.5 ± 0.1    | 0.8 ± 0.2     | 1.7 ± 0.1    | 1.5 ± 0.2    | 3.1 ± 0.3     | 1.0 ± 0.2     | 0.21 ± 0.09  | -0.3 ± 0.2    | 0.7 ± 0.1    | 0.3 ± 0.2    | 1.4 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 2.3 ± 0.3     | 0.5 ± 0.1    | 0.6 ± 0.2     | 1.6 ± 0.1    | 1.4 ± 0.2    | 2.9 ± 0.3     | 1.1 ± 0.3     | -0.02 ± 0.09 | -0.0 ± 0.2    | 0.8 ± 0.1    | 0.3 ± 0.2    | 1.6 ± 0.3     | W Africa F   | E Asia M     |
| 21  | 2.2 ± 0.2     | 0.4 ± 0.1    | 0.5 ± 0.2     | 1.3 ± 0.1    | 1.3 ± 0.2    | 2.8 ± 0.3     | 1.3 ± 0.2     | -0.13 ± 0.09 | -0.3 ± 0.2    | 0.7 ± 0.1    | 0.2 ± 0.2    | 1.6 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 2.2 ± 0.2     | 0.1 ± 0.1    | 0.4 ± 0.2     | 1.0 ± 0.1    | 1.1 ± 0.1    | 2.4 ± 0.3     | 1.8 ± 0.3     | -0.34 ± 0.09 | -0.1 ± 0.2    | 0.6 ± 0.1    | 0.1 ± 0.2    | 1.5 ± 0.3     | W Africa F   | E Asia M     |
| 23  | 2.0 ± 0.2     | 0.1 ± 0.1    | 0.3 ± 0.1     | 0.8 ± 0.1    | 0.9 ± 0.1    | 2.6 ± 0.3     | 1.5 ± 0.3     | -0.5 ± 0.1   | -0.2 ± 0.2    | 0.6 ± 0.1    | 0.1 ± 0.2    | 1.8 ± 0.2     | W Africa F   | E Asia M     |
| 24  | 1.9 ± 0.2     | -0.15 ± 0.10 | 0.3 ± 0.1     | 0.52 ± 0.10  | 0.7 ± 0.1    | 2.0 ± 0.2     | 1.6 ± 0.2     | -0.6 ± 0.1   | -0.2 ± 0.1    | 0.6 ± 0.1    | 0.0 ± 0.2    | 1.6 ± 0.2     | W Africa F   | E Asia M     |
| 25  | 1.4 ± 0.2     | -0.13 ± 0.10 | 0.1 ± 0.1     | 0.15 ± 0.09  | 0.6 ± 0.1    | 2.1 ± 0.2     | 1.5 ± 0.2     | -0.7 ± 0.1   | -0.2 ± 0.1    | 0.42 ± 0.10  | -0.0 ± 0.2   | 1.5 ± 0.2     | W Africa F   | E Asia M     |
| 26  | 1.2 ± 0.2     | -0.34 ± 0.09 | -0.0 ± 0.1    | -0.18 ± 0.09 | 0.6 ± 0.1    | 1.7 ± 0.2     | 1.4 ± 0.2     | -0.8 ± 0.1   | -0.3 ± 0.1    | 0.29 ± 0.09  | -0.2 ± 0.2   | 1.3 ± 0.2     | W Africa F   | E Asia M     |
| 27  | 1.2 ± 0.2     | -0.43 ± 0.09 | -0.1 ± 0.1    | -0.53 ± 0.08 | 0.3 ± 0.1    | 1.7 ± 0.2     | 1.2 ± 0.2     | -0.9 ± 0.1   | -0.4 ± 0.1    | 0.10 ± 0.09  | 0.0 ± 0.2    | 1.2 ± 0.2     | W Africa F   | E Asia M     |
| 28  | 0.9 ± 0.2     | -0.57 ± 0.09 | -0.1 ± 0.1    | -0.68 ± 0.09 | 0.1 ± 0.1    | 1.3 ± 0.2     | 0.8 ± 0.2     | -0.9 ± 0.1   | -0.5 ± 0.1    | -0.05 ± 0.09 | -0.2 ± 0.2   | 0.9 ± 0.2     | W Africa F   | E Asia M     |
| 29  | 0.9 ± 0.2     | -0.74 ± 0.08 | -0.3 ± 0.1    | -0.93 ± 0.09 | 0.1 ± 0.1    | 1.1 ± 0.2     | 0.8 ± 0.2     | -1.0 ± 0.1   | -0.6 ± 0.1    | -0.25 ± 0.09 | -0.5 ± 0.2   | 1.1 ± 0.2     | W Africa F   | E Asia M     |
| 30  | 0.7 ± 0.2     | -0.73 ± 0.09 | -0.4 ± 0.1    | -1.17 ± 0.09 | -0.0 ± 0.1   | 0.7 ± 0.2     | 0.9 ± 0.2     | -1.1 ± 0.1   | -0.8 ± 0.1    | -0.38 ± 0.09 | -0.4 ± 0.2   | 0.7 ± 0.2     | E Africa M   | S Asia F     |
| 31  | 0.4 ± 0.2     | -0.87 ± 0.09 | -0.3 ± 0.1    | -1.40 ± 0.10 | -0.2 ± 0.1   | 0.5 ± 0.2     | 0.6 ± 0.2     | -1.2 ± 0.1   | -0.9 ± 0.1    | -0.48 ± 0.09 | -0.4 ± 0.2   | 0.5 ± 0.2     | E Africa M   | S Asia F     |
| 32  | 0.4 ± 0.2     | -0.88 ± 0.09 | -0.4 ± 0.1    | -1.5 ± 0.1   | -0.3 ± 0.1   | 0.5 ± 0.2     | 0.4 ± 0.2     | -1.2 ± 0.1   | -0.9 ± 0.1    | -0.57 ± 0.09 | -0.5 ± 0.2   | 0.4 ± 0.2     | W Africa F   | S Asia F     |
| 33  | 0.2 ± 0.3     | -0.98 ± 0.09 | -0.5 ± 0.1    | -1.6 ± 0.1   | -0.4 ± 0.1   | 0.3 ± 0.2     | 0.4 ± 0.2     | -1.3 ± 0.1   | -0.8 ± 0.1    | -0.68 ± 0.09 | -0.8 ± 0.2   | 0.1 ± 0.2     | E Africa M   | S Asia F     |
| 34  | 0.0 ± 0.3     | -0.95 ± 0.09 | -0.5 ± 0.1    | -1.7 ± 0.1   | -0.5 ± 0.1   | 0.1 ± 0.2     | 0.0 ± 0.2     | -1.3 ± 0.1   | -1.0 ± 0.1    | -0.71 ± 0.10 | -0.7 ± 0.2   | 0.2 ± 0.2     | W Africa M   | S Asia F     |
| 35  | 0.2 ± 0.3     | -1.11 ± 0.09 | -0.6 ± 0.1    | -1.8 ± 0.1   | -0.6 ± 0.2   | -0.3 ± 0.2    | 0.2 ± 0.3     | -1.3 ± 0.1   | -0.9 ± 0.2    | -0.70 ± 0.10 | -0.9 ± 0.2   | -0.1 ± 0.2    | E Africa F   | S Asia F     |
| 36  | 0.1 ± 0.3     | -1.04 ± 0.10 | -0.5 ± 0.1    | -1.9 ± 0.1   | -0.5 ± 0.2   | -0.3 ± 0.2    | -0.0 ± 0.3    | -1.4 ± 0.1   | -0.8 ± 0.2    | -0.81 ± 0.10 | -0.6 ± 0.2   | -0.2 ± 0.2    | E Africa F   | S Asia F     |
| 37  | 0.0 ± 0.3     | -1.13 ± 0.09 | -0.5 ± 0.2    | -1.8 ± 0.1   | -0.7 ± 0.2   | -0.4 ± 0.3    | 0.0 ± 0.3     | -1.3 ± 0.1   | -0.9 ± 0.2    | -0.9 ± 0.1   | -0.8 ± 0.2   | -0.4 ± 0.2    | E Africa F   | S Asia F     |
| 38  | 0.5 ± 0.3     | -1.22 ± 0.09 | -0.5 ± 0.2    | -1.7 ± 0.1   | -0.6 ± 0.2   | -0.6 ± 0.3    | -0.2 ± 0.3    | -1.3 ± 0.1   | -1.0 ± 0.2    | -0.9 ± 0.1   | -0.4 ± 0.2   | -0.5 ± 0.2    | E Africa F   | S Asia F     |
| 39  | 0.1 ± 0.3     | -1.22 ± 0.09 | -0.5 ± 0.2    | -1.9 ± 0.1   | -0.5 ± 0.2   | -0.6 ± 0.3    | 0.0 ± 0.3     | -1.3 ± 0.1   | -0.8 ± 0.2    | -0.8 ± 0.1   | -0.6 ± 0.2   | -0.7 ± 0.2    | E Africa F   | S Asia F     |
| 40  | -0.1 ± 0.4    | -1.24 ± 0.09 | -0.3 ± 0.2    | -1.7 ± 0.1   | -0.7 ± 0.2   | -1.0 ± 0.3    | 0.0 ± 0.3     | -1.3 ± 0.1   | -0.7 ± 0.2    | -0.8 ± 0.1   | -0.6 ± 0.2   | -0.8 ± 0.2    | E Africa M   | S Asia F     |
| 41  | -0.4 ± 0.4    | -1.29 ± 0.09 | -0.4 ± 0.2    | -1.8 ± 0.2   | -0.7 ± 0.2   | -1.3 ± 0.3    | 0.2 ± 0.3     | -1.3 ± 0.1   | -0.7 ± 0.2    | -0.8 ± 0.1   | -0.5 ± 0.2   | -0.8 ± 0.3    | E Africa M   | S Asia F     |
| 42  | -0.1 ± 0.4    | -1.28 ± 0.09 | -0.3 ± 0.2    | -1.3 ± 0.2   | -0.8 ± 0.2   | -1.1 ± 0.3    | 0.3 ± 0.3     | -1.2 ± 0.1   | -0.4 ± 0.2    | -0.8 ± 0.1   | -0.3 ± 0.2   | -1.0 ± 0.3    | E Africa M   | S Asia F     |
| 43  | 0.2 ± 0.4     | -1.41 ± 0.09 | -0.3 ± 0.2    | -1.5 ± 0.2   | -0.7 ± 0.2   | -1.2 ± 0.3    | 0.5 ± 0.4     | -1.2 ± 0.1   | -0.6 ± 0.2    | -0.8 ± 0.1   | -0.2 ± 0.2   | -1.2 ± 0.3    | E Africa M   | S Asia F     |
| 44  | 0.3 ± 0.4     | -1.48 ± 0.10 | -0.6 ± 0.2    | -1.4 ± 0.2   | -0.5 ± 0.2   | -1.3 ± 0.4    | 0.0 ± 0.4     | -1.1 ± 0.1   | -0.2 ± 0.2    | -0.8 ± 0.1   | 0.1 ± 0.3    | -1.1 ± 0.3    | E Africa F   | E Asia F     |
| 45  | -0.0 ± 0.5    | -1.45 ± 0.10 | -0.5 ± 0.2    | -1.3 ± 0.2   | -0.6 ± 0.2   | -1.2 ± 0.3    | -0.1 ± 0.4    | -1.0 ± 0.1   | -0.3 ± 0.2    | -0.6 ± 0.2   | -0.0 ± 0.3   | -1.3 ± 0.3    | S E Asia M   | E Asia F     |
| 46  | 0.9 ± 0.5     | -1.6 ± 0.1   | -0.4 ± 0.2    | -1.1 ± 0.2   | -0.8 ± 0.2   | -1.2 ± 0.4    | 0.2 ± 0.4     | -1.0 ± 0.1   | -0.2 ± 0.2    | -0.4 ± 0.2   | 0.3 ± 0.3    | -1.4 ± 0.3    | E Africa F   | E Asia F     |
| 47  | 0.7 ± 0.5     | -1.4 ± 0.1   | -0.4 ± 0.2    | -1.2 ± 0.2   | -0.4 ± 0.2   | -1.3 ± 0.4    | 0.8 ± 0.4     | -0.9 ± 0.1   | -0.1 ± 0.2    | -0.3 ± 0.2   | 0.5 ± 0.3    | -1.3 ± 0.3    | E Africa M   | E Asia F     |
| 48  | 0.5 ± 0.5     | -1.4 ± 0.1   | -0.3 ± 0.2    | -0.9 ± 0.2   | -0.5 ± 0.2   | -1.0 ± 0.4    | 0.3 ± 0.5     | -0.8 ± 0.1   | 0.1 ± 0.3     | -0.2 ± 0.2   | 0.5 ± 0.3    | -1.4 ± 0.4    | S E Asia M   | E Asia F     |
| 49  | 1.0 ± 0.6     | -1.5 ± 0.1   | -0.5 ± 0.2    | -0.7 ± 0.2   | -0.0 ± 0.2   | -1.2 ± 0.4    | 0.2 ± 0.5     | -0.7 ± 0.1   | 0.0 ± 0.2     | -0.2 ± 0.2   | 0.7 ± 0.3    | -1.3 ± 0.3    | E Africa F   | E Asia F     |
| 50  | 0.6 ± 0.6     | -1.4 ± 0.1   | -0.1 ± 0.2    | -0.8 ± 0.2   | -0.2 ± 0.2   | -1.2 ± 0.4    | 0.3 ± 0.5     | -0.6 ± 0.1   | -0.2 ± 0.3    | -0.2 ± 0.2   | 1.1 ± 0.3    | -1.6 ± 0.3    | S E Asia M   | W Africa M   |
| 51  | 0.8 ± 0.5     | -1.3 ± 0.1   | -0.4 ± 0.2    | -0.5 ± 0.2   | -0.0 ± 0.2   | -1.0 ± 0.4    | 0.1 ± 0.6     | -0.6 ± 0.1   | 0.2 ± 0.2     | 0.1 ± 0.2    | 1.0 ± 0.3    | -1.7 ± 0.4    | S E Asia M   | W Africa M   |
| 52  | 0.6 ± 0.6     | -1.3 ± 0.1   | -0.1 ± 0.2    | -0.5 ± 0.2   | 0.2 ± 0.2    | -1.2 ± 0.4    | 0.4 ± 0.6     | -0.5 ± 0.1   | 0.2 ± 0.3     | 0.1 ± 0.2    | 0.9 ± 0.3    | -1.5 ± 0.4    | S E Asia M   | W Africa M   |

**Table 61.** Mean raw error for Application-type images for algorithm roc-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.1 ± 0.3     | 5.4 ± 0.1   | 3.7 ± 0.2     | 3.7 ± 0.1   | 4.5 ± 0.3    | 5.7 ± 0.3     | 4.2 ± 0.2     | 5.4 ± 0.1   | 2.7 ± 0.2     | 3.2 ± 0.1   | 4.5 ± 0.3    | 4.6 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 4.8 ± 0.3     | 5.5 ± 0.1   | 3.2 ± 0.2     | 3.8 ± 0.2   | 4.7 ± 0.3    | 5.4 ± 0.4     | 4.1 ± 0.3     | 5.4 ± 0.1   | 2.6 ± 0.1     | 3.2 ± 0.1   | 4.6 ± 0.2    | 4.6 ± 0.3     | E Asia F     | E Europe M   |
| 16  | 4.7 ± 0.3     | 5.4 ± 0.2   | 2.8 ± 0.1     | 3.4 ± 0.2   | 4.9 ± 0.3    | 5.6 ± 0.4     | 4.2 ± 0.2     | 4.9 ± 0.1   | 2.3 ± 0.2     | 3.0 ± 0.1   | 4.4 ± 0.2    | 4.2 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 4.8 ± 0.3     | 5.0 ± 0.1   | 2.3 ± 0.2     | 3.1 ± 0.2   | 4.5 ± 0.3    | 5.1 ± 0.3     | 3.7 ± 0.2     | 4.7 ± 0.1   | 1.58 ± 0.10   | 2.8 ± 0.1   | 4.2 ± 0.2    | 4.4 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 3.9 ± 0.3     | 4.3 ± 0.1   | 1.4 ± 0.2     | 2.2 ± 0.1   | 4.2 ± 0.2    | 4.5 ± 0.3     | 3.4 ± 0.2     | 4.1 ± 0.1   | 0.8 ± 0.2     | 2.1 ± 0.1   | 3.8 ± 0.2    | 3.8 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 3.9 ± 0.3     | 4.1 ± 0.1   | 1.1 ± 0.2     | 2.0 ± 0.1   | 4.0 ± 0.2    | 4.7 ± 0.3     | 3.7 ± 0.3     | 4.0 ± 0.1   | 0.3 ± 0.2     | 2.1 ± 0.1   | 3.5 ± 0.2    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 3.6 ± 0.3     | 4.0 ± 0.1   | 0.5 ± 0.2     | 1.7 ± 0.2   | 3.7 ± 0.2    | 4.4 ± 0.3     | 3.5 ± 0.3     | 3.8 ± 0.1   | 0.1 ± 0.2     | 2.1 ± 0.1   | 3.6 ± 0.2    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 3.2 ± 0.3     | 3.8 ± 0.1   | 0.2 ± 0.2     | 1.2 ± 0.1   | 3.5 ± 0.2    | 4.2 ± 0.3     | 3.6 ± 0.3     | 3.7 ± 0.1   | -0.2 ± 0.2    | 2.1 ± 0.1   | 3.6 ± 0.2    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 3.2 ± 0.3     | 3.3 ± 0.1   | -0.0 ± 0.2    | 0.8 ± 0.1   | 3.4 ± 0.2    | 3.7 ± 0.3     | 4.2 ± 0.3     | 3.5 ± 0.1   | -0.1 ± 0.2    | 2.1 ± 0.2   | 3.2 ± 0.2    | 3.7 ± 0.3     | E Africa M   | E Europe M   |
| 23  | 3.0 ± 0.3     | 3.2 ± 0.1   | -0.2 ± 0.2    | 0.5 ± 0.1   | 3.2 ± 0.2    | 3.7 ± 0.3     | 3.8 ± 0.3     | 3.3 ± 0.1   | -0.4 ± 0.2    | 2.1 ± 0.2   | 3.2 ± 0.2    | 3.9 ± 0.3     | W Africa M   | E Europe M   |
| 24  | 2.8 ± 0.3     | 2.8 ± 0.1   | -0.3 ± 0.2    | 0.1 ± 0.1   | 2.9 ± 0.1    | 3.3 ± 0.3     | 3.9 ± 0.3     | 3.1 ± 0.1   | -0.1 ± 0.2    | 2.1 ± 0.1   | 3.0 ± 0.2    | 3.7 ± 0.3     | E Africa M   | E Europe F   |
| 25  | 2.4 ± 0.3     | 2.69 ± 0.10 | -0.3 ± 0.2    | -0.3 ± 0.1  | 2.7 ± 0.1    | 3.2 ± 0.3     | 3.8 ± 0.3     | 2.9 ± 0.1   | 0.0 ± 0.2     | 1.9 ± 0.1   | 2.9 ± 0.2    | 3.6 ± 0.2     | E Africa M   | E Europe F   |
| 26  | 2.1 ± 0.3     | 2.35 ± 0.10 | -0.3 ± 0.2    | -0.6 ± 0.1  | 2.6 ± 0.1    | 2.9 ± 0.3     | 3.7 ± 0.2     | 2.7 ± 0.1   | -0.1 ± 0.2    | 1.8 ± 0.1   | 2.7 ± 0.2    | 3.2 ± 0.2     | E Africa M   | S Asia F     |
| 27  | 2.3 ± 0.3     | 2.06 ± 0.09 | -0.3 ± 0.2    | -0.9 ± 0.1  | 2.2 ± 0.1    | 2.7 ± 0.2     | 3.3 ± 0.2     | 2.5 ± 0.1   | 0.2 ± 0.2     | 1.6 ± 0.1   | 2.7 ± 0.2    | 3.0 ± 0.2     | E Africa M   | S Asia F     |
| 28  | 1.9 ± 0.3     | 1.78 ± 0.09 | -0.2 ± 0.2    | -1.1 ± 0.1  | 2.0 ± 0.1    | 2.4 ± 0.3     | 3.2 ± 0.2     | 2.3 ± 0.1   | 0.4 ± 0.2     | 1.4 ± 0.1   | 2.3 ± 0.2    | 2.5 ± 0.2     | E Africa M   | S Asia F     |
| 29  | 2.0 ± 0.3     | 1.63 ± 0.09 | -0.0 ± 0.2    | -1.3 ± 0.1  | 1.8 ± 0.1    | 2.1 ± 0.2     | 3.1 ± 0.2     | 2.1 ± 0.1   | 0.5 ± 0.2     | 1.3 ± 0.1   | 2.1 ± 0.2    | 2.7 ± 0.2     | E Africa M   | S Asia F     |
| 30  | 1.6 ± 0.3     | 1.45 ± 0.09 | -0.0 ± 0.2    | -1.5 ± 0.1  | 1.8 ± 0.1    | 1.8 ± 0.3     | 3.0 ± 0.2     | 1.8 ± 0.1   | 0.5 ± 0.2     | 1.02 ± 0.10 | 2.0 ± 0.2    | 2.3 ± 0.2     | E Africa M   | S Asia F     |
| 31  | 1.6 ± 0.3     | 1.19 ± 0.09 | 0.1 ± 0.2     | -1.7 ± 0.1  | 1.5 ± 0.1    | 1.6 ± 0.2     | 3.0 ± 0.2     | 1.6 ± 0.1   | 0.4 ± 0.2     | 0.84 ± 0.10 | 1.7 ± 0.2    | 2.2 ± 0.2     | E Africa M   | S Asia F     |
| 32  | 1.7 ± 0.3     | 1.00 ± 0.09 | 0.3 ± 0.2     | -1.8 ± 0.1  | 1.3 ± 0.2    | 1.6 ± 0.3     | 2.6 ± 0.2     | 1.4 ± 0.1   | 0.5 ± 0.2     | 0.75 ± 0.10 | 1.6 ± 0.2    | 1.9 ± 0.2     | E Africa M   | S Asia F     |
| 33  | 1.4 ± 0.3     | 0.80 ± 0.09 | 0.3 ± 0.2     | -1.9 ± 0.1  | 1.1 ± 0.2    | 1.2 ± 0.3     | 2.6 ± 0.3     | 1.1 ± 0.1   | 1.0 ± 0.2     | 0.51 ± 0.10 | 1.2 ± 0.2    | 1.7 ± 0.2     | E Africa M   | S Asia F     |
| 34  | 1.5 ± 0.3     | 0.66 ± 0.10 | 0.5 ± 0.2     | -2.0 ± 0.1  | 1.0 ± 0.2    | 1.2 ± 0.3     | 2.2 ± 0.3     | 0.9 ± 0.1   | 0.7 ± 0.2     | 0.4 ± 0.1   | 1.2 ± 0.2    | 1.6 ± 0.2     | E Africa M   | S Asia F     |
| 35  | 1.6 ± 0.4     | 0.38 ± 0.10 | 0.4 ± 0.2     | -2.1 ± 0.1  | 0.6 ± 0.2    | 0.7 ± 0.3     | 2.3 ± 0.3     | 0.8 ± 0.1   | 0.7 ± 0.2     | 0.3 ± 0.1   | 0.9 ± 0.2    | 1.2 ± 0.2     | E Africa M   | S Asia F     |
| 36  | 1.4 ± 0.4     | 0.4 ± 0.1   | 0.9 ± 0.2     | -2.2 ± 0.1  | 0.6 ± 0.2    | 0.8 ± 0.3     | 2.0 ± 0.3     | 0.6 ± 0.1   | 0.9 ± 0.2     | 0.0 ± 0.1   | 1.1 ± 0.2    | 1.1 ± 0.2     | E Africa M   | S Asia F     |
| 37  | 1.4 ± 0.4     | 0.2 ± 0.1   | 1.0 ± 0.2     | -2.0 ± 0.2  | 0.5 ± 0.2    | 0.8 ± 0.3     | 2.0 ± 0.3     | 0.5 ± 0.1   | 0.6 ± 0.2     | -0.1 ± 0.1  | 0.7 ± 0.2    | 0.8 ± 0.2     | E Africa M   | S Asia F     |
| 38  | 1.8 ± 0.4     | 0.0 ± 0.1   | 1.3 ± 0.2     | -2.0 ± 0.2  | 0.6 ± 0.2    | 0.4 ± 0.3     | 1.6 ± 0.3     | 0.4 ± 0.1   | 0.7 ± 0.2     | -0.3 ± 0.1  | 1.0 ± 0.2    | 0.7 ± 0.2     | E Africa F   | S Asia F     |
| 39  | 1.5 ± 0.4     | -0.0 ± 0.1  | 1.2 ± 0.2     | -2.2 ± 0.2  | 0.5 ± 0.2    | 0.6 ± 0.3     | 1.9 ± 0.3     | 0.2 ± 0.1   | 0.7 ± 0.2     | -0.3 ± 0.1  | 1.0 ± 0.2    | 0.6 ± 0.2     | E Africa M   | S Asia F     |
| 40  | 1.5 ± 0.4     | -0.1 ± 0.1  | 1.7 ± 0.3     | -2.0 ± 0.2  | 0.3 ± 0.2    | 0.1 ± 0.3     | 1.8 ± 0.3     | 0.1 ± 0.1   | 0.6 ± 0.2     | -0.4 ± 0.1  | 0.8 ± 0.2    | 0.4 ± 0.2     | E Africa M   | S Asia F     |
| 41  | 1.1 ± 0.4     | -0.2 ± 0.1  | 1.7 ± 0.3     | -2.0 ± 0.2  | 0.3 ± 0.2    | -0.1 ± 0.4    | 1.9 ± 0.3     | 0.1 ± 0.1   | 0.8 ± 0.2     | -0.5 ± 0.1  | 0.6 ± 0.2    | 0.3 ± 0.3     | E Africa M   | S Asia F     |
| 42  | 1.6 ± 0.5     | -0.1 ± 0.1  | 2.0 ± 0.3     | -1.5 ± 0.2  | 0.4 ± 0.2    | 0.0 ± 0.4     | 1.8 ± 0.3     | 0.1 ± 0.1   | 1.0 ± 0.2     | -0.7 ± 0.1  | 0.9 ± 0.2    | 0.2 ± 0.3     | E Europe F   | S Asia F     |
| 43  | 1.6 ± 0.5     | -0.3 ± 0.1  | 2.2 ± 0.3     | -1.7 ± 0.2  | 0.3 ± 0.2    | -0.0 ± 0.4    | 1.9 ± 0.3     | 0.0 ± 0.1   | 0.8 ± 0.2     | -0.6 ± 0.1  | 0.8 ± 0.3    | -0.2 ± 0.3    | E Europe F   | S Asia F     |
| 44  | 1.5 ± 0.5     | -0.4 ± 0.1  | 2.3 ± 0.3     | -1.6 ± 0.2  | 0.6 ± 0.2    | -0.2 ± 0.4    | 1.3 ± 0.3     | -0.1 ± 0.1  | 1.0 ± 0.2     | -0.7 ± 0.2  | 1.0 ± 0.3    | -0.3 ± 0.3    | E Europe F   | S Asia F     |
| 45  | 1.3 ± 0.5     | -0.3 ± 0.1  | 2.3 ± 0.3     | -1.6 ± 0.2  | 0.4 ± 0.2    | -0.1 ± 0.4    | 1.2 ± 0.4     | -0.1 ± 0.1  | 0.9 ± 0.2     | -0.7 ± 0.2  | 0.8 ± 0.3    | -0.6 ± 0.3    | E Europe F   | S Asia F     |
| 46  | 2.3 ± 0.6     | -0.5 ± 0.1  | 2.5 ± 0.3     | -1.2 ± 0.2  | 0.3 ± 0.2    | 0.1 ± 0.4     | 1.2 ± 0.4     | -0.2 ± 0.1  | 1.1 ± 0.3     | -0.6 ± 0.2  | 0.8 ± 0.3    | -0.7 ± 0.3    | E Europe F   | S Asia F     |
| 47  | 2.0 ± 0.6     | -0.3 ± 0.1  | 2.9 ± 0.3     | -1.2 ± 0.2  | 0.4 ± 0.3    | -0.1 ± 0.4    | 1.5 ± 0.4     | -0.1 ± 0.2  | 1.0 ± 0.3     | -0.6 ± 0.2  | 1.0 ± 0.3    | -0.5 ± 0.3    | E Europe F   | S Asia F     |
| 48  | 1.8 ± 0.6     | -0.3 ± 0.1  | 3.1 ± 0.3     | -1.0 ± 0.3  | 0.6 ± 0.3    | -0.2 ± 0.5    | 0.9 ± 0.5     | -0.2 ± 0.1  | 1.3 ± 0.3     | -0.5 ± 0.2  | 0.9 ± 0.3    | -1.0 ± 0.3    | E Europe F   | S Asia F     |
| 49  | 2.1 ± 0.6     | -0.3 ± 0.2  | 3.0 ± 0.3     | -0.9 ± 0.3  | 0.6 ± 0.3    | -0.5 ± 0.5    | 0.6 ± 0.5     | -0.1 ± 0.2  | 1.0 ± 0.3     | -0.7 ± 0.2  | 1.0 ± 0.3    | -0.9 ± 0.3    | E Europe F   | W Africa M   |
| 50  | 1.5 ± 0.6     | -0.2 ± 0.2  | 3.5 ± 0.3     | -0.8 ± 0.2  | 0.5 ± 0.3    | -0.3 ± 0.5    | 0.5 ± 0.4     | -0.0 ± 0.2  | 1.1 ± 0.3     | -0.6 ± 0.2  | 1.3 ± 0.3    | -1.3 ± 0.3    | E Europe F   | W Africa M   |
| 51  | 2.1 ± 0.6     | -0.0 ± 0.2  | 3.4 ± 0.3     | -0.8 ± 0.3  | 0.6 ± 0.3    | -0.0 ± 0.5    | 0.2 ± 0.5     | 0.0 ± 0.2   | 1.3 ± 0.3     | -0.6 ± 0.2  | 1.4 ± 0.3    | -1.3 ± 0.4    | E Europe F   | W Africa M   |
| 52  | 1.5 ± 0.7     | -0.0 ± 0.2  | 3.8 ± 0.3     | -0.9 ± 0.3  | 0.9 ± 0.3    | -0.5 ± 0.5    | 0.3 ± 0.5     | 0.0 ± 0.2   | 1.2 ± 0.3     | -0.8 ± 0.2  | 0.9 ± 0.3    | -1.4 ± 0.4    | E Europe F   | W Africa M   |

**Table 62.** Mean raw error for Application-type images for algorithm roc-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 4.8 ± 0.2     | 4.98 ± 0.09  | 3.4 ± 0.1     | 4.2 ± 0.1   | 4.8 ± 0.2    | 4.9 ± 0.2     | 3.7 ± 0.1     | 4.72 ± 0.08 | 2.79 ± 0.09   | 3.54 ± 0.08  | 4.3 ± 0.1    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 15  | 4.8 ± 0.2     | 4.8 ± 0.1    | 2.8 ± 0.1     | 4.1 ± 0.1   | 4.7 ± 0.2    | 4.8 ± 0.3     | 3.4 ± 0.2     | 4.43 ± 0.07 | 2.33 ± 0.09   | 3.23 ± 0.08  | 4.0 ± 0.1    | 3.4 ± 0.2     | E Asia F     | E Europe M   |
| 16  | 4.6 ± 0.3     | 4.8 ± 0.1    | 2.4 ± 0.1     | 3.8 ± 0.1   | 4.5 ± 0.3    | 4.7 ± 0.4     | 3.4 ± 0.2     | 4.05 ± 0.08 | 1.9 ± 0.1     | 2.82 ± 0.09  | 3.7 ± 0.2    | 3.1 ± 0.3     | E Asia F     | E Europe M   |
| 17  | 4.8 ± 0.3     | 4.5 ± 0.1    | 2.0 ± 0.1     | 3.7 ± 0.2   | 4.4 ± 0.2    | 4.6 ± 0.3     | 3.1 ± 0.2     | 3.9 ± 0.1   | 1.2 ± 0.1     | 2.5 ± 0.1    | 3.5 ± 0.2    | 3.6 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 3.9 ± 0.2     | 4.0 ± 0.1    | 1.4 ± 0.2     | 3.1 ± 0.1   | 4.0 ± 0.2    | 4.1 ± 0.3     | 2.9 ± 0.2     | 3.55 ± 0.09 | 0.6 ± 0.1     | 2.0 ± 0.1    | 3.2 ± 0.2    | 3.1 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.2 ± 0.3     | 3.8 ± 0.1    | 0.9 ± 0.2     | 3.1 ± 0.1   | 3.9 ± 0.2    | 4.4 ± 0.3     | 3.3 ± 0.2     | 3.5 ± 0.1   | 0.3 ± 0.1     | 2.1 ± 0.1    | 3.0 ± 0.2    | 3.3 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.3 ± 0.3     | 3.6 ± 0.1    | 0.7 ± 0.2     | 2.8 ± 0.2   | 3.7 ± 0.1    | 4.4 ± 0.3     | 3.3 ± 0.3     | 3.5 ± 0.1   | 0.3 ± 0.2     | 2.1 ± 0.1    | 3.0 ± 0.2    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 4.1 ± 0.3     | 3.5 ± 0.1    | 0.5 ± 0.2     | 2.6 ± 0.1   | 3.4 ± 0.1    | 4.0 ± 0.3     | 3.4 ± 0.3     | 3.36 ± 0.09 | 0.1 ± 0.2     | 2.1 ± 0.1    | 3.0 ± 0.2    | 3.4 ± 0.3     | E Africa F   | E Europe M   |
| 22  | 4.2 ± 0.3     | 3.21 ± 0.10  | 0.5 ± 0.2     | 2.5 ± 0.1   | 3.2 ± 0.1    | 3.7 ± 0.3     | 4.0 ± 0.3     | 3.19 ± 0.10 | 0.2 ± 0.2     | 2.1 ± 0.1    | 2.8 ± 0.1    | 3.3 ± 0.3     | E Africa F   | E Europe M   |
| 23  | 4.1 ± 0.3     | 2.94 ± 0.09  | 0.5 ± 0.2     | 2.3 ± 0.1   | 3.0 ± 0.1    | 3.6 ± 0.3     | 3.8 ± 0.3     | 3.01 ± 0.09 | 0.2 ± 0.2     | 2.2 ± 0.1    | 2.7 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 4.1 ± 0.3     | 2.66 ± 0.08  | 0.5 ± 0.1     | 2.0 ± 0.1   | 2.7 ± 0.1    | 3.3 ± 0.2     | 3.8 ± 0.3     | 2.76 ± 0.09 | 0.4 ± 0.2     | 2.1 ± 0.1    | 2.5 ± 0.1    | 3.3 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 3.6 ± 0.3     | 2.41 ± 0.08  | 0.5 ± 0.1     | 1.77 ± 0.09 | 2.4 ± 0.1    | 3.4 ± 0.2     | 3.7 ± 0.2     | 2.58 ± 0.09 | 0.5 ± 0.2     | 1.9 ± 0.1    | 2.3 ± 0.1    | 3.2 ± 0.2     | E Africa M   | E Europe M   |
| 26  | 3.4 ± 0.2     | 2.12 ± 0.08  | 0.7 ± 0.1     | 1.50 ± 0.09 | 2.2 ± 0.1    | 2.9 ± 0.2     | 3.6 ± 0.2     | 2.29 ± 0.09 | 0.5 ± 0.2     | 1.9 ± 0.1    | 2.1 ± 0.1    | 2.9 ± 0.2     | E Africa M   | E Europe M   |
| 27  | 3.6 ± 0.2     | 1.88 ± 0.07  | 0.6 ± 0.1     | 1.26 ± 0.09 | 1.9 ± 0.1    | 2.8 ± 0.2     | 3.4 ± 0.2     | 2.11 ± 0.09 | 0.7 ± 0.1     | 1.62 ± 0.09  | 2.0 ± 0.1    | 2.7 ± 0.2     | E Africa F   | E Europe F   |
| 28  | 3.4 ± 0.2     | 1.60 ± 0.07  | 0.8 ± 0.1     | 1.16 ± 0.09 | 1.7 ± 0.1    | 2.5 ± 0.2     | 3.1 ± 0.2     | 1.84 ± 0.10 | 0.7 ± 0.1     | 1.47 ± 0.09  | 1.8 ± 0.1    | 2.4 ± 0.2     | E Africa F   | E Europe M   |
| 29  | 3.3 ± 0.3     | 1.41 ± 0.07  | 0.8 ± 0.1     | 0.96 ± 0.09 | 1.5 ± 0.1    | 2.2 ± 0.2     | 3.0 ± 0.2     | 1.68 ± 0.09 | 0.7 ± 0.1     | 1.18 ± 0.09  | 1.5 ± 0.1    | 2.5 ± 0.2     | E Africa F   | E Europe M   |
| 30  | 3.0 ± 0.2     | 1.25 ± 0.07  | 0.8 ± 0.1     | 0.73 ± 0.09 | 1.4 ± 0.1    | 2.0 ± 0.2     | 2.9 ± 0.2     | 1.38 ± 0.09 | 0.7 ± 0.1     | 0.95 ± 0.08  | 1.3 ± 0.2    | 2.1 ± 0.2     | E Africa F   | E Europe M   |
| 31  | 2.9 ± 0.3     | 1.02 ± 0.07  | 0.9 ± 0.1     | 0.51 ± 0.10 | 1.2 ± 0.1    | 1.9 ± 0.2     | 2.8 ± 0.2     | 1.25 ± 0.09 | 0.7 ± 0.2     | 0.81 ± 0.08  | 1.2 ± 0.2    | 1.9 ± 0.2     | E Africa F   | S Asia F     |
| 32  | 3.1 ± 0.3     | 0.91 ± 0.08  | 1.0 ± 0.1     | 0.45 ± 0.10 | 1.0 ± 0.1    | 1.9 ± 0.2     | 2.7 ± 0.2     | 1.08 ± 0.09 | 0.8 ± 0.2     | 0.65 ± 0.08  | 1.1 ± 0.2    | 1.7 ± 0.2     | E Africa F   | S Asia F     |
| 33  | 2.9 ± 0.3     | 0.71 ± 0.08  | 1.0 ± 0.1     | 0.3 ± 0.1   | 0.9 ± 0.1    | 1.8 ± 0.2     | 2.7 ± 0.2     | 0.87 ± 0.10 | 1.0 ± 0.2     | 0.47 ± 0.08  | 0.8 ± 0.2    | 1.4 ± 0.2     | E Africa F   | S Asia F     |
| 34  | 2.6 ± 0.3     | 0.60 ± 0.08  | 1.1 ± 0.1     | 0.3 ± 0.1   | 0.8 ± 0.1    | 1.5 ± 0.2     | 2.4 ± 0.2     | 0.76 ± 0.10 | 0.9 ± 0.2     | 0.36 ± 0.09  | 0.8 ± 0.2    | 1.2 ± 0.2     | E Africa F   | S Asia F     |
| 35  | 2.9 ± 0.3     | 0.46 ± 0.09  | 1.0 ± 0.2     | 0.1 ± 0.1   | 0.5 ± 0.1    | 1.2 ± 0.2     | 2.5 ± 0.2     | 0.8 ± 0.1   | 0.8 ± 0.2     | 0.28 ± 0.09  | 0.7 ± 0.2    | 1.0 ± 0.2     | E Africa F   | S Asia F     |
| 36  | 2.6 ± 0.3     | 0.44 ± 0.09  | 1.3 ± 0.2     | -0.0 ± 0.1  | 0.5 ± 0.2    | 1.4 ± 0.2     | 2.4 ± 0.3     | 0.6 ± 0.1   | 1.1 ± 0.2     | 0.14 ± 0.09  | 0.8 ± 0.2    | 0.8 ± 0.2     | E Africa F   | S Asia F     |
| 37  | 2.5 ± 0.3     | 0.27 ± 0.09  | 1.3 ± 0.2     | 0.1 ± 0.1   | 0.5 ± 0.2    | 1.3 ± 0.3     | 2.4 ± 0.3     | 0.6 ± 0.1   | 1.0 ± 0.2     | 0.08 ± 0.10  | 0.6 ± 0.2    | 0.6 ± 0.2     | E Africa F   | S Asia F     |
| 38  | 3.0 ± 0.4     | 0.20 ± 0.09  | 1.5 ± 0.2     | 0.0 ± 0.1   | 0.6 ± 0.2    | 1.1 ± 0.3     | 2.1 ± 0.3     | 0.5 ± 0.1   | 1.1 ± 0.2     | -0.05 ± 0.10 | 0.8 ± 0.2    | 0.5 ± 0.2     | E Africa F   | S Asia M     |
| 39  | 2.6 ± 0.4     | 0.19 ± 0.09  | 1.5 ± 0.2     | 0.0 ± 0.1   | 0.4 ± 0.2    | 1.0 ± 0.3     | 2.3 ± 0.3     | 0.4 ± 0.1   | 1.2 ± 0.2     | 0.0 ± 0.1    | 0.7 ± 0.2    | 0.3 ± 0.2     | E Africa F   | S Asia F     |
| 40  | 2.5 ± 0.4     | 0.10 ± 0.09  | 1.8 ± 0.2     | 0.1 ± 0.1   | 0.4 ± 0.2    | 0.6 ± 0.3     | 2.3 ± 0.3     | 0.4 ± 0.1   | 1.4 ± 0.2     | 0.0 ± 0.1    | 0.7 ± 0.2    | 0.1 ± 0.2     | E Africa F   | S Asia M     |
| 41  | 2.4 ± 0.4     | 0.09 ± 0.10  | 1.9 ± 0.2     | 0.1 ± 0.2   | 0.4 ± 0.2    | 0.5 ± 0.3     | 2.3 ± 0.3     | 0.3 ± 0.1   | 1.4 ± 0.2     | 0.0 ± 0.1    | 0.7 ± 0.2    | -0.2 ± 0.2    | E Africa F   | W Africa M   |
| 42  | 2.9 ± 0.4     | 0.06 ± 0.10  | 2.0 ± 0.2     | 0.5 ± 0.2   | 0.5 ± 0.2    | 0.7 ± 0.3     | 2.3 ± 0.3     | 0.5 ± 0.1   | 1.7 ± 0.2     | 0.0 ± 0.1    | 1.1 ± 0.2    | -0.1 ± 0.2    | E Africa F   | W Africa M   |
| 43  | 2.7 ± 0.4     | -0.10 ± 0.10 | 2.1 ± 0.2     | 0.4 ± 0.2   | 0.6 ± 0.2    | 0.4 ± 0.3     | 2.4 ± 0.4     | 0.4 ± 0.1   | 1.7 ± 0.2     | 0.1 ± 0.1    | 1.0 ± 0.3    | -0.4 ± 0.3    | E Africa F   | W Africa M   |
| 44  | 3.0 ± 0.5     | -0.1 ± 0.1   | 2.0 ± 0.2     | 0.5 ± 0.2   | 0.8 ± 0.2    | 0.4 ± 0.3     | 2.0 ± 0.4     | 0.6 ± 0.1   | 2.1 ± 0.3     | 0.1 ± 0.2    | 1.3 ± 0.3    | -0.4 ± 0.3    | E Africa F   | W Africa M   |
| 45  | 2.5 ± 0.5     | -0.1 ± 0.1   | 2.0 ± 0.2     | 0.7 ± 0.2   | 0.6 ± 0.2    | 0.1 ± 0.3     | 1.8 ± 0.4     | 0.5 ± 0.1   | 2.1 ± 0.3     | 0.4 ± 0.2    | 1.2 ± 0.3    | -0.6 ± 0.3    | E Africa F   | W Africa M   |
| 46  | 3.8 ± 0.5     | -0.2 ± 0.1   | 2.5 ± 0.2     | 0.9 ± 0.2   | 0.6 ± 0.2    | 0.2 ± 0.3     | 1.8 ± 0.4     | 0.6 ± 0.1   | 2.6 ± 0.3     | 0.6 ± 0.2    | 1.4 ± 0.3    | -1.0 ± 0.3    | E Africa F   | W Africa M   |
| 47  | 3.3 ± 0.6     | -0.0 ± 0.1   | 2.5 ± 0.3     | 0.9 ± 0.2   | 0.7 ± 0.2    | 0.1 ± 0.4     | 2.1 ± 0.4     | 0.8 ± 0.1   | 2.7 ± 0.3     | 0.6 ± 0.2    | 1.7 ± 0.3    | -0.7 ± 0.3    | E Africa F   | W Africa M   |
| 48  | 2.9 ± 0.6     | -0.0 ± 0.1   | 2.6 ± 0.2     | 1.3 ± 0.2   | 0.7 ± 0.2    | -0.0 ± 0.4    | 1.5 ± 0.4     | 0.8 ± 0.1   | 2.9 ± 0.3     | 0.9 ± 0.2    | 1.6 ± 0.3    | -1.1 ± 0.3    | E Europe M   | W Africa M   |
| 49  | 3.4 ± 0.6     | -0.1 ± 0.1   | 2.5 ± 0.2     | 1.4 ± 0.2   | 1.0 ± 0.2    | -0.2 ± 0.4    | 1.6 ± 0.5     | 0.8 ± 0.1   | 2.7 ± 0.3     | 0.8 ± 0.2    | 1.7 ± 0.3    | -1.1 ± 0.3    | E Africa F   | W Africa M   |
| 50  | 3.1 ± 0.6     | 0.1 ± 0.1    | 2.8 ± 0.2     | 1.4 ± 0.2   | 0.8 ± 0.2    | -0.2 ± 0.4    | 1.5 ± 0.4     | 0.9 ± 0.1   | 2.7 ± 0.3     | 1.0 ± 0.2    | 2.2 ± 0.3    | -1.5 ± 0.3    | E Africa F   | W Africa M   |
| 51  | 3.4 ± 0.6     | 0.2 ± 0.1    | 2.5 ± 0.2     | 1.6 ± 0.2   | 1.0 ± 0.2    | -0.1 ± 0.4    | 1.2 ± 0.5     | 1.1 ± 0.1   | 3.1 ± 0.3     | 1.1 ± 0.2    | 2.0 ± 0.3    | -1.7 ± 0.4    | E Africa F   | W Africa M   |
| 52  | 3.3 ± 0.6     | 0.3 ± 0.1    | 2.8 ± 0.2     | 1.6 ± 0.2   | 1.3 ± 0.2    | -0.3 ± 0.4    | 1.4 ± 0.5     | 1.0 ± 0.1   | 2.8 ± 0.3     | 1.0 ± 0.2    | 1.9 ± 0.3    | -1.5 ± 0.4    | E Africa F   | W Africa M   |

**Table 63.** Mean raw error for Application-type images for algorithm roc-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia  | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 2.6 ± 0.2     | 3.39 ± 0.10  | 2.5 ± 0.1     | 2.3 ± 0.1    | 2.8 ± 0.2    | 2.8 ± 0.2     | 1.3 ± 0.2     | 2.7 ± 0.1    | 1.5 ± 0.1     | 1.69 ± 0.10  | 1.8 ± 0.2    | 1.5 ± 0.2     | E Asia F     | E Africa M   |
| 15  | 2.6 ± 0.2     | 3.5 ± 0.1    | 2.0 ± 0.1     | 2.3 ± 0.1    | 2.8 ± 0.3    | 2.6 ± 0.3     | 1.3 ± 0.2     | 2.85 ± 0.09  | 1.5 ± 0.1     | 1.7 ± 0.1    | 2.1 ± 0.2    | 1.6 ± 0.3     | E Asia F     | E Africa M   |
| 16  | 2.4 ± 0.2     | 3.4 ± 0.1    | 1.7 ± 0.1     | 1.9 ± 0.1    | 2.9 ± 0.3    | 2.7 ± 0.4     | 1.5 ± 0.2     | 2.77 ± 0.09  | 1.3 ± 0.2     | 1.59 ± 0.09  | 2.0 ± 0.2    | 1.3 ± 0.2     | E Asia F     | E Europe M   |
| 17  | 2.5 ± 0.3     | 3.1 ± 0.1    | 1.3 ± 0.1     | 1.7 ± 0.1    | 2.8 ± 0.3    | 2.3 ± 0.3     | 1.2 ± 0.2     | 2.6 ± 0.1    | 0.8 ± 0.1     | 1.3 ± 0.1    | 1.9 ± 0.2    | 1.8 ± 0.4     | E Asia F     | E Europe M   |
| 18  | 1.7 ± 0.2     | 2.5 ± 0.1    | 0.5 ± 0.2     | 1.0 ± 0.1    | 2.3 ± 0.2    | 1.8 ± 0.3     | 0.8 ± 0.2     | 2.18 ± 0.10  | 0.2 ± 0.1     | 0.8 ± 0.1    | 1.6 ± 0.2    | 1.3 ± 0.3     | E Asia F     | E Europe M   |
| 19  | 2.1 ± 0.3     | 2.4 ± 0.1    | -0.0 ± 0.2    | 0.9 ± 0.1    | 2.1 ± 0.2    | 2.0 ± 0.3     | 1.2 ± 0.3     | 2.1 ± 0.1    | -0.0 ± 0.2    | 0.8 ± 0.1    | 1.3 ± 0.2    | 1.5 ± 0.3     | E Asia F     | E Europe F   |
| 20  | 2.0 ± 0.3     | 2.3 ± 0.1    | -0.3 ± 0.2    | 0.7 ± 0.1    | 2.0 ± 0.2    | 1.9 ± 0.3     | 1.3 ± 0.3     | 2.0 ± 0.1    | -0.1 ± 0.2    | 0.8 ± 0.1    | 1.3 ± 0.2    | 1.5 ± 0.3     | E Asia F     | E Europe F   |
| 21  | 1.9 ± 0.3     | 2.3 ± 0.1    | -0.5 ± 0.2    | 0.5 ± 0.1    | 1.9 ± 0.2    | 1.5 ± 0.3     | 1.7 ± 0.3     | 1.9 ± 0.1    | -0.4 ± 0.2    | 0.8 ± 0.1    | 1.2 ± 0.2    | 1.5 ± 0.3     | E Asia F     | E Europe F   |
| 22  | 2.0 ± 0.3     | 2.1 ± 0.1    | -0.7 ± 0.1    | 0.3 ± 0.1    | 1.7 ± 0.1    | 1.1 ± 0.3     | 2.1 ± 0.3     | 1.7 ± 0.1    | -0.2 ± 0.2    | 0.8 ± 0.1    | 1.0 ± 0.2    | 1.3 ± 0.3     | E Africa M   | E Europe F   |
| 23  | 1.9 ± 0.3     | 1.85 ± 0.10  | -0.6 ± 0.1    | 0.3 ± 0.1    | 1.6 ± 0.1    | 1.1 ± 0.3     | 1.9 ± 0.3     | 1.5 ± 0.1    | -0.4 ± 0.2    | 1.0 ± 0.1    | 0.9 ± 0.2    | 1.6 ± 0.3     | E Africa M   | E Europe F   |
| 24  | 1.7 ± 0.3     | 1.71 ± 0.09  | -0.7 ± 0.1    | 0.0 ± 0.1    | 1.3 ± 0.1    | 1.0 ± 0.2     | 1.9 ± 0.3     | 1.4 ± 0.1    | -0.2 ± 0.2    | 1.0 ± 0.1    | 0.9 ± 0.2    | 1.4 ± 0.2     | E Africa M   | E Europe F   |
| 25  | 1.3 ± 0.2     | 1.48 ± 0.08  | -0.8 ± 0.1    | -0.08 ± 0.10 | 1.2 ± 0.1    | 0.9 ± 0.2     | 1.9 ± 0.2     | 1.3 ± 0.1    | -0.3 ± 0.2    | 1.0 ± 0.1    | 0.9 ± 0.2    | 1.3 ± 0.2     | E Africa M   | E Europe F   |
| 26  | 1.2 ± 0.2     | 1.23 ± 0.08  | -0.7 ± 0.1    | -0.32 ± 0.10 | 1.1 ± 0.1    | 0.6 ± 0.2     | 1.8 ± 0.2     | 1.13 ± 0.10  | -0.2 ± 0.1    | 0.9 ± 0.1    | 0.9 ± 0.2    | 0.9 ± 0.2     | E Africa M   | E Europe F   |
| 27  | 1.2 ± 0.2     | 1.00 ± 0.08  | -0.7 ± 0.1    | -0.50 ± 0.09 | 0.8 ± 0.1    | 0.4 ± 0.2     | 1.6 ± 0.2     | 1.0 ± 0.1    | -0.2 ± 0.1    | 0.70 ± 0.09  | 0.9 ± 0.2    | 0.8 ± 0.2     | E Africa M   | E Europe F   |
| 28  | 1.0 ± 0.2     | 0.74 ± 0.07  | -0.6 ± 0.1    | -0.56 ± 0.10 | 0.7 ± 0.1    | 0.1 ± 0.2     | 1.3 ± 0.2     | 0.8 ± 0.1    | -0.2 ± 0.1    | 0.59 ± 0.09  | 0.9 ± 0.2    | 0.5 ± 0.2     | E Africa M   | E Europe F   |
| 29  | 1.1 ± 0.3     | 0.51 ± 0.07  | -0.7 ± 0.1    | -0.71 ± 0.10 | 0.5 ± 0.1    | -0.0 ± 0.2    | 1.2 ± 0.2     | 0.62 ± 0.09  | -0.3 ± 0.1    | 0.34 ± 0.08  | 0.5 ± 0.2    | 0.6 ± 0.2     | E Africa M   | S Asia F     |
| 30  | 0.8 ± 0.2     | 0.32 ± 0.07  | -0.7 ± 0.1    | -0.85 ± 0.10 | 0.3 ± 0.1    | -0.4 ± 0.2    | 1.2 ± 0.2     | 0.43 ± 0.10  | -0.4 ± 0.1    | 0.09 ± 0.08  | 0.6 ± 0.2    | 0.3 ± 0.2     | E Africa M   | S Asia F     |
| 31  | 0.6 ± 0.3     | 0.05 ± 0.08  | -0.6 ± 0.1    | -1.08 ± 0.10 | 0.2 ± 0.1    | -0.4 ± 0.2    | 1.0 ± 0.2     | 0.33 ± 0.10  | -0.5 ± 0.1    | -0.10 ± 0.08 | 0.3 ± 0.2    | -0.0 ± 0.2    | E Africa M   | S Asia F     |
| 32  | 0.7 ± 0.3     | 0.01 ± 0.08  | -0.6 ± 0.1    | -1.1 ± 0.1   | 0.0 ± 0.1    | -0.3 ± 0.2    | 0.9 ± 0.2     | 0.14 ± 0.10  | -0.5 ± 0.1    | -0.26 ± 0.08 | 0.2 ± 0.2    | -0.1 ± 0.2    | E Africa M   | S Asia F     |
| 33  | 0.6 ± 0.3     | -0.18 ± 0.08 | -0.7 ± 0.1    | -1.2 ± 0.1   | -0.1 ± 0.1   | -0.6 ± 0.2    | 0.9 ± 0.2     | -0.1 ± 0.1   | -0.4 ± 0.1    | -0.50 ± 0.08 | -0.0 ± 0.2   | -0.4 ± 0.2    | E Africa M   | S Asia F     |
| 34  | 0.4 ± 0.3     | -0.24 ± 0.08 | -0.6 ± 0.1    | -1.2 ± 0.1   | -0.1 ± 0.1   | -0.8 ± 0.2    | 0.5 ± 0.2     | -0.18 ± 0.10 | -0.6 ± 0.1    | -0.62 ± 0.09 | 0.0 ± 0.2    | -0.3 ± 0.2    | E Africa M   | S Asia F     |
| 35  | 0.7 ± 0.3     | -0.46 ± 0.09 | -0.8 ± 0.1    | -1.4 ± 0.1   | -0.4 ± 0.1   | -1.0 ± 0.3    | 0.6 ± 0.2     | -0.2 ± 0.1   | -0.6 ± 0.2    | -0.70 ± 0.09 | -0.1 ± 0.2   | -0.6 ± 0.2    | E Africa F   | S Asia F     |
| 36  | 0.5 ± 0.3     | -0.47 ± 0.09 | -0.6 ± 0.2    | -1.4 ± 0.1   | -0.4 ± 0.1   | -0.9 ± 0.2    | 0.5 ± 0.3     | -0.4 ± 0.1   | -0.4 ± 0.2    | -0.92 ± 0.09 | 0.0 ± 0.2    | -0.8 ± 0.2    | E Africa M   | S Asia F     |
| 37  | 0.4 ± 0.3     | -0.61 ± 0.09 | -0.5 ± 0.2    | -1.3 ± 0.1   | -0.5 ± 0.1   | -1.0 ± 0.3    | 0.5 ± 0.3     | -0.5 ± 0.1   | -0.7 ± 0.2    | -1.01 ± 0.10 | -0.2 ± 0.2   | -1.0 ± 0.2    | E Africa M   | S Asia F     |
| 38  | 0.9 ± 0.4     | -0.68 ± 0.09 | -0.5 ± 0.2    | -1.2 ± 0.1   | -0.4 ± 0.2   | -1.1 ± 0.3    | 0.2 ± 0.3     | -0.5 ± 0.1   | -0.6 ± 0.2    | -1.14 ± 0.10 | -0.1 ± 0.2   | -1.1 ± 0.2    | E Africa F   | S Asia F     |
| 39  | 0.6 ± 0.4     | -0.77 ± 0.09 | -0.6 ± 0.2    | -1.4 ± 0.1   | -0.6 ± 0.2   | -1.1 ± 0.3    | 0.5 ± 0.3     | -0.6 ± 0.1   | -0.5 ± 0.2    | -1.15 ± 0.10 | -0.3 ± 0.2   | -1.2 ± 0.2    | E Africa F   | S Asia F     |
| 40  | 0.7 ± 0.4     | -0.81 ± 0.09 | -0.4 ± 0.2    | -1.1 ± 0.1   | -0.7 ± 0.2   | -1.5 ± 0.3    | 0.5 ± 0.3     | -0.6 ± 0.1   | -0.6 ± 0.2    | -1.2 ± 0.1   | -0.3 ± 0.2   | -1.5 ± 0.2    | E Africa F   | W Africa F   |
| 41  | 0.6 ± 0.4     | -0.94 ± 0.09 | -0.5 ± 0.2    | -1.1 ± 0.2   | -0.8 ± 0.2   | -1.5 ± 0.3    | 0.5 ± 0.3     | -0.8 ± 0.1   | -0.6 ± 0.2    | -1.2 ± 0.1   | -0.5 ± 0.2   | -1.5 ± 0.2    | E Africa F   | W Africa M   |
| 42  | 1.0 ± 0.4     | -0.89 ± 0.09 | -0.3 ± 0.2    | -0.7 ± 0.2   | -0.7 ± 0.2   | -1.4 ± 0.3    | 0.4 ± 0.3     | -0.7 ± 0.1   | -0.5 ± 0.2    | -1.1 ± 0.1   | -0.3 ± 0.2   | -1.7 ± 0.2    | E Africa F   | W Africa M   |
| 43  | 1.1 ± 0.4     | -1.07 ± 0.09 | -0.3 ± 0.2    | -0.8 ± 0.2   | -0.7 ± 0.2   | -1.6 ± 0.3    | 0.6 ± 0.3     | -0.8 ± 0.1   | -0.6 ± 0.2    | -1.1 ± 0.1   | -0.4 ± 0.2   | -1.9 ± 0.3    | E Africa F   | W Africa M   |
| 44  | 1.0 ± 0.5     | -1.07 ± 0.09 | -0.5 ± 0.2    | -0.7 ± 0.2   | -0.6 ± 0.2   | -1.5 ± 0.3    | 0.1 ± 0.3     | -0.8 ± 0.1   | -0.3 ± 0.2    | -1.1 ± 0.1   | -0.2 ± 0.2   | -2.1 ± 0.3    | E Africa F   | W Africa M   |
| 45  | 0.8 ± 0.5     | -1.12 ± 0.10 | -0.4 ± 0.2    | -0.6 ± 0.2   | -0.7 ± 0.2   | -1.6 ± 0.3    | -0.0 ± 0.3    | -0.9 ± 0.1   | -0.5 ± 0.2    | -0.9 ± 0.2   | -0.5 ± 0.2   | -2.1 ± 0.3    | E Africa F   | W Africa M   |
| 46  | 1.7 ± 0.5     | -1.22 ± 0.10 | 0.0 ± 0.2     | -0.3 ± 0.2   | -0.9 ± 0.2   | -1.5 ± 0.4    | -0.1 ± 0.4    | -0.8 ± 0.1   | -0.2 ± 0.2    | -0.7 ± 0.2   | -0.4 ± 0.2   | -2.4 ± 0.3    | E Africa F   | W Africa M   |
| 47  | 1.6 ± 0.5     | -1.1 ± 0.1   | -0.2 ± 0.3    | -0.4 ± 0.2   | -0.8 ± 0.2   | -1.7 ± 0.4    | 0.3 ± 0.4     | -0.9 ± 0.1   | -0.3 ± 0.2    | -0.6 ± 0.2   | -0.1 ± 0.3   | -2.3 ± 0.3    | E Africa F   | W Africa M   |
| 48  | 1.1 ± 0.6     | -1.2 ± 0.1   | 0.1 ± 0.3     | -0.1 ± 0.2   | -0.9 ± 0.2   | -1.5 ± 0.4    | -0.2 ± 0.4    | -0.9 ± 0.1   | -0.2 ± 0.3    | -0.4 ± 0.2   | -0.1 ± 0.3   | -2.6 ± 0.3    | E Africa F   | W Africa M   |
| 49  | 1.5 ± 0.6     | -1.3 ± 0.1   | -0.1 ± 0.3    | 0.0 ± 0.2    | -0.6 ± 0.2   | -1.7 ± 0.4    | -0.2 ± 0.5    | -0.9 ± 0.1   | -0.2 ± 0.3    | -0.4 ± 0.2   | -0.1 ± 0.3   | -2.7 ± 0.3    | E Africa F   | W Africa M   |
| 50  | 1.3 ± 0.6     | -1.3 ± 0.1   | 0.5 ± 0.3     | 0.0 ± 0.2    | -0.8 ± 0.2   | -1.8 ± 0.4    | -0.3 ± 0.4    | -0.9 ± 0.1   | -0.3 ± 0.3    | -0.3 ± 0.2   | 0.2 ± 0.3    | -2.9 ± 0.3    | E Africa F   | W Africa M   |
| 51  | 1.4 ± 0.6     | -1.3 ± 0.1   | 0.3 ± 0.3     | 0.2 ± 0.2    | -0.7 ± 0.2   | -1.5 ± 0.4    | -0.7 ± 0.5    | -0.8 ± 0.1   | -0.3 ± 0.2    | -0.1 ± 0.2   | -0.0 ± 0.3   | -3.1 ± 0.4    | E Africa F   | W Africa M   |
| 52  | 1.5 ± 0.6     | -1.2 ± 0.1   | 0.6 ± 0.3     | 0.1 ± 0.2    | -0.4 ± 0.2   | -1.8 ± 0.4    | -0.6 ± 0.5    | -0.9 ± 0.1   | -0.4 ± 0.3    | -0.1 ± 0.2   | -0.1 ± 0.3   | -2.8 ± 0.4    | E Africa F   | W Africa M   |

**Table 64.** Mean raw error for Application-type images for algorithm shufti-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.4 ± 0.4     | 3.6 ± 0.1   | 3.1 ± 0.3     | 3.9 ± 0.2   | 3.4 ± 0.3    | 4.4 ± 0.4     | 2.9 ± 0.3     | 2.7 ± 0.1   | 1.6 ± 0.2     | 2.4 ± 0.1   | 2.3 ± 0.2    | 3.4 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 4.5 ± 0.4     | 3.7 ± 0.2   | 3.6 ± 0.3     | 4.2 ± 0.2   | 3.5 ± 0.3    | 4.4 ± 0.5     | 2.7 ± 0.4     | 2.9 ± 0.1   | 2.1 ± 0.2     | 2.6 ± 0.1   | 2.6 ± 0.2    | 3.2 ± 0.4     | E Africa F   | E Europe M   |
| 16  | 4.5 ± 0.4     | 3.4 ± 0.2   | 3.8 ± 0.4     | 3.9 ± 0.2   | 3.7 ± 0.3    | 4.2 ± 0.5     | 3.5 ± 0.4     | 2.8 ± 0.1   | 2.4 ± 0.2     | 2.9 ± 0.2   | 2.7 ± 0.2    | 3.1 ± 0.4     | E Africa F   | E Europe M   |
| 17  | 4.5 ± 0.4     | 3.2 ± 0.2   | 3.7 ± 0.4     | 4.0 ± 0.2   | 3.2 ± 0.3    | 4.1 ± 0.4     | 3.4 ± 0.3     | 2.8 ± 0.1   | 2.0 ± 0.2     | 2.8 ± 0.2   | 2.8 ± 0.2    | 3.5 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 3.9 ± 0.4     | 2.7 ± 0.1   | 3.0 ± 0.4     | 3.6 ± 0.2   | 3.1 ± 0.2    | 3.5 ± 0.4     | 3.2 ± 0.3     | 2.5 ± 0.1   | 1.8 ± 0.2     | 3.1 ± 0.2   | 2.7 ± 0.2    | 2.7 ± 0.3     | E Africa F   | E Europe M   |
| 19  | 4.7 ± 0.4     | 2.6 ± 0.2   | 3.4 ± 0.4     | 3.8 ± 0.2   | 3.2 ± 0.2    | 3.9 ± 0.4     | 3.5 ± 0.3     | 2.9 ± 0.1   | 2.2 ± 0.3     | 3.4 ± 0.2   | 2.9 ± 0.2    | 3.4 ± 0.4     | E Africa F   | E Europe M   |
| 20  | 4.4 ± 0.4     | 2.6 ± 0.2   | 3.4 ± 0.3     | 3.6 ± 0.2   | 3.2 ± 0.2    | 3.5 ± 0.3     | 3.6 ± 0.3     | 2.9 ± 0.1   | 2.7 ± 0.3     | 3.8 ± 0.2   | 3.0 ± 0.2    | 3.4 ± 0.4     | E Africa F   | E Asia F     |
| 21  | 4.3 ± 0.3     | 2.6 ± 0.2   | 3.4 ± 0.3     | 3.6 ± 0.2   | 3.0 ± 0.2    | 3.9 ± 0.4     | 4.4 ± 0.3     | 3.3 ± 0.1   | 2.9 ± 0.3     | 4.0 ± 0.2   | 3.4 ± 0.2    | 3.6 ± 0.4     | E Africa M   | E Asia F     |
| 22  | 4.2 ± 0.4     | 2.5 ± 0.2   | 3.3 ± 0.2     | 3.4 ± 0.2   | 2.9 ± 0.2    | 3.0 ± 0.4     | 5.2 ± 0.4     | 3.5 ± 0.1   | 3.4 ± 0.3     | 4.2 ± 0.2   | 3.3 ± 0.2    | 3.5 ± 0.3     | E Africa M   | E Asia F     |
| 23  | 4.3 ± 0.3     | 2.3 ± 0.1   | 3.3 ± 0.2     | 3.3 ± 0.1   | 2.9 ± 0.2    | 3.5 ± 0.3     | 5.0 ± 0.3     | 3.5 ± 0.1   | 3.6 ± 0.3     | 4.4 ± 0.2   | 3.4 ± 0.2    | 4.1 ± 0.3     | E Africa M   | E Asia F     |
| 24  | 4.4 ± 0.3     | 2.3 ± 0.1   | 3.2 ± 0.2     | 3.2 ± 0.1   | 2.8 ± 0.2    | 3.1 ± 0.3     | 5.0 ± 0.3     | 3.6 ± 0.1   | 3.8 ± 0.3     | 4.4 ± 0.2   | 3.3 ± 0.2    | 4.1 ± 0.3     | E Africa M   | E Asia F     |
| 25  | 3.9 ± 0.3     | 2.1 ± 0.1   | 3.2 ± 0.2     | 3.0 ± 0.1   | 2.8 ± 0.2    | 2.7 ± 0.3     | 5.0 ± 0.3     | 3.4 ± 0.1   | 3.8 ± 0.2     | 4.7 ± 0.1   | 3.4 ± 0.2    | 3.9 ± 0.3     | E Africa M   | E Asia F     |
| 26  | 4.1 ± 0.3     | 2.2 ± 0.1   | 3.1 ± 0.2     | 2.9 ± 0.1   | 2.9 ± 0.2    | 2.8 ± 0.3     | 4.8 ± 0.3     | 3.3 ± 0.1   | 3.7 ± 0.2     | 4.7 ± 0.1   | 3.4 ± 0.2    | 4.1 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 4.0 ± 0.3     | 2.1 ± 0.1   | 3.3 ± 0.2     | 2.7 ± 0.1   | 2.5 ± 0.2    | 2.7 ± 0.3     | 4.8 ± 0.3     | 3.3 ± 0.1   | 3.8 ± 0.2     | 4.6 ± 0.1   | 3.3 ± 0.2    | 3.8 ± 0.2     | E Africa M   | E Asia F     |
| 28  | 3.6 ± 0.3     | 1.9 ± 0.1   | 3.1 ± 0.2     | 2.6 ± 0.1   | 2.6 ± 0.2    | 2.5 ± 0.3     | 4.5 ± 0.2     | 3.2 ± 0.1   | 4.1 ± 0.2     | 4.5 ± 0.1   | 3.2 ± 0.2    | 3.6 ± 0.2     | S Asia M     | E Asia F     |
| 29  | 3.8 ± 0.3     | 1.8 ± 0.1   | 2.9 ± 0.2     | 2.4 ± 0.1   | 2.5 ± 0.2    | 2.3 ± 0.3     | 4.5 ± 0.2     | 3.1 ± 0.1   | 3.7 ± 0.2     | 4.5 ± 0.1   | 2.8 ± 0.2    | 3.9 ± 0.2     | S Asia M     | E Asia F     |
| 30  | 3.7 ± 0.3     | 1.7 ± 0.1   | 2.8 ± 0.2     | 2.3 ± 0.1   | 2.5 ± 0.2    | 2.1 ± 0.3     | 4.3 ± 0.2     | 3.0 ± 0.1   | 3.7 ± 0.2     | 4.4 ± 0.1   | 3.1 ± 0.2    | 3.5 ± 0.2     | S Asia M     | E Asia F     |
| 31  | 3.4 ± 0.3     | 1.6 ± 0.1   | 2.8 ± 0.2     | 2.2 ± 0.1   | 2.3 ± 0.2    | 1.8 ± 0.3     | 4.1 ± 0.2     | 2.8 ± 0.1   | 3.6 ± 0.2     | 4.31 ± 0.10 | 3.1 ± 0.2    | 3.3 ± 0.2     | S Asia M     | E Asia F     |
| 32  | 3.4 ± 0.3     | 1.6 ± 0.1   | 2.7 ± 0.2     | 2.1 ± 0.1   | 2.3 ± 0.2    | 2.0 ± 0.3     | 3.8 ± 0.3     | 2.7 ± 0.1   | 3.4 ± 0.2     | 4.19 ± 0.10 | 3.0 ± 0.2    | 3.3 ± 0.2     | S Asia M     | E Asia F     |
| 33  | 3.3 ± 0.3     | 1.4 ± 0.1   | 2.5 ± 0.2     | 2.1 ± 0.1   | 2.4 ± 0.2    | 1.7 ± 0.3     | 4.1 ± 0.3     | 2.7 ± 0.1   | 3.4 ± 0.2     | 4.0 ± 0.1   | 2.6 ± 0.2    | 3.0 ± 0.2     | E Africa M   | E Asia F     |
| 34  | 3.1 ± 0.3     | 1.4 ± 0.1   | 2.4 ± 0.2     | 2.0 ± 0.1   | 2.2 ± 0.2    | 1.6 ± 0.3     | 3.4 ± 0.3     | 2.5 ± 0.1   | 3.3 ± 0.2     | 3.9 ± 0.1   | 2.8 ± 0.2    | 3.0 ± 0.2     | S Asia M     | E Asia F     |
| 35  | 3.3 ± 0.3     | 1.1 ± 0.1   | 2.3 ± 0.2     | 1.9 ± 0.1   | 1.9 ± 0.2    | 1.3 ± 0.3     | 3.7 ± 0.3     | 2.4 ± 0.1   | 3.0 ± 0.2     | 3.8 ± 0.1   | 2.4 ± 0.2    | 2.6 ± 0.2     | S Asia M     | E Asia F     |
| 36  | 2.9 ± 0.3     | 1.1 ± 0.1   | 2.4 ± 0.2     | 1.9 ± 0.1   | 2.0 ± 0.2    | 1.4 ± 0.3     | 3.6 ± 0.3     | 2.3 ± 0.1   | 3.3 ± 0.2     | 3.61 ± 0.10 | 2.6 ± 0.2    | 2.4 ± 0.2     | S Asia M     | E Asia F     |
| 37  | 3.0 ± 0.3     | 1.2 ± 0.1   | 2.2 ± 0.2     | 1.9 ± 0.1   | 1.8 ± 0.2    | 1.2 ± 0.3     | 3.3 ± 0.3     | 2.2 ± 0.1   | 2.9 ± 0.2     | 3.4 ± 0.1   | 2.4 ± 0.2    | 2.3 ± 0.2     | S Asia M     | E Asia F     |
| 38  | 3.3 ± 0.4     | 1.0 ± 0.1   | 2.2 ± 0.2     | 2.0 ± 0.1   | 1.8 ± 0.2    | 1.0 ± 0.3     | 3.0 ± 0.3     | 2.0 ± 0.1   | 2.8 ± 0.2     | 3.3 ± 0.1   | 2.5 ± 0.2    | 2.0 ± 0.2     | E Africa F   | E Asia F     |
| 39  | 2.8 ± 0.4     | 1.0 ± 0.1   | 2.0 ± 0.2     | 1.9 ± 0.1   | 1.7 ± 0.2    | 1.1 ± 0.3     | 3.2 ± 0.3     | 1.8 ± 0.1   | 2.9 ± 0.2     | 3.1 ± 0.1   | 2.3 ± 0.2    | 2.0 ± 0.2     | E Africa M   | E Asia F     |
| 40  | 2.6 ± 0.4     | 0.8 ± 0.1   | 2.3 ± 0.2     | 1.9 ± 0.2   | 1.5 ± 0.2    | 1.1 ± 0.3     | 3.2 ± 0.3     | 1.8 ± 0.1   | 2.6 ± 0.2     | 3.0 ± 0.1   | 2.1 ± 0.2    | 1.7 ± 0.3     | E Africa M   | E Asia F     |
| 41  | 2.4 ± 0.4     | 0.8 ± 0.1   | 2.1 ± 0.2     | 1.8 ± 0.2   | 1.4 ± 0.2    | 0.5 ± 0.4     | 3.1 ± 0.3     | 1.7 ± 0.1   | 2.7 ± 0.2     | 2.8 ± 0.1   | 2.0 ± 0.2    | 1.7 ± 0.3     | E Africa M   | W Africa F   |
| 42  | 2.6 ± 0.4     | 0.9 ± 0.1   | 2.1 ± 0.2     | 1.9 ± 0.2   | 1.5 ± 0.2    | 0.8 ± 0.4     | 3.0 ± 0.3     | 1.6 ± 0.1   | 2.8 ± 0.2     | 2.7 ± 0.1   | 2.0 ± 0.2    | 1.3 ± 0.3     | E Africa M   | W Africa F   |
| 43  | 2.6 ± 0.4     | 0.6 ± 0.1   | 2.1 ± 0.3     | 1.8 ± 0.2   | 1.4 ± 0.2    | 0.5 ± 0.4     | 3.2 ± 0.4     | 1.5 ± 0.1   | 2.6 ± 0.2     | 2.7 ± 0.1   | 2.0 ± 0.2    | 1.2 ± 0.3     | E Africa M   | W Africa F   |
| 44  | 2.4 ± 0.5     | 0.5 ± 0.1   | 1.8 ± 0.3     | 1.8 ± 0.2   | 1.5 ± 0.2    | 0.3 ± 0.4     | 2.7 ± 0.4     | 1.3 ± 0.1   | 2.8 ± 0.3     | 2.3 ± 0.1   | 2.0 ± 0.2    | 1.3 ± 0.3     | E Europe M   | W Africa F   |
| 45  | 2.5 ± 0.5     | 0.5 ± 0.1   | 2.0 ± 0.2     | 1.7 ± 0.2   | 1.4 ± 0.2    | 0.5 ± 0.4     | 2.8 ± 0.4     | 1.2 ± 0.1   | 2.6 ± 0.3     | 2.4 ± 0.2   | 1.7 ± 0.2    | 0.9 ± 0.3     | E Africa M   | W Africa F   |
| 46  | 3.0 ± 0.5     | 0.3 ± 0.1   | 2.1 ± 0.3     | 1.8 ± 0.2   | 1.3 ± 0.2    | 0.2 ± 0.5     | 3.1 ± 0.4     | 1.0 ± 0.1   | 2.8 ± 0.3     | 2.2 ± 0.2   | 1.8 ± 0.3    | 0.9 ± 0.3     | E Africa M   | W Africa F   |
| 47  | 2.7 ± 0.5     | 0.4 ± 0.1   | 2.0 ± 0.4     | 1.8 ± 0.2   | 1.4 ± 0.2    | 0.4 ± 0.4     | 3.0 ± 0.4     | 1.0 ± 0.1   | 2.5 ± 0.3     | 2.3 ± 0.2   | 1.7 ± 0.3    | 0.9 ± 0.3     | E Africa M   | E Asia F     |
| 48  | 2.2 ± 0.6     | 0.3 ± 0.1   | 2.3 ± 0.3     | 1.9 ± 0.2   | 1.2 ± 0.2    | 0.3 ± 0.4     | 2.4 ± 0.5     | 0.8 ± 0.1   | 2.6 ± 0.3     | 2.3 ± 0.2   | 1.9 ± 0.3    | 0.8 ± 0.4     | E Europe M   | E Asia F     |
| 49  | 2.9 ± 0.6     | 0.1 ± 0.1   | 2.2 ± 0.3     | 2.0 ± 0.2   | 1.5 ± 0.2    | 0.4 ± 0.4     | 2.2 ± 0.5     | 0.8 ± 0.1   | 2.5 ± 0.3     | 2.1 ± 0.2   | 1.7 ± 0.3    | 0.7 ± 0.4     | E Africa F   | E Asia F     |
| 50  | 2.4 ± 0.6     | 0.1 ± 0.1   | 2.4 ± 0.3     | 2.0 ± 0.2   | 1.2 ± 0.2    | 0.6 ± 0.5     | 2.3 ± 0.5     | 0.7 ± 0.1   | 2.3 ± 0.3     | 2.1 ± 0.2   | 2.1 ± 0.3    | 0.3 ± 0.4     | E Africa F   | E Asia F     |
| 51  | 2.3 ± 0.5     | 0.1 ± 0.1   | 2.3 ± 0.3     | 1.9 ± 0.2   | 1.3 ± 0.2    | 0.4 ± 0.5     | 1.7 ± 0.6     | 0.5 ± 0.1   | 2.3 ± 0.3     | 2.0 ± 0.2   | 2.0 ± 0.3    | 0.6 ± 0.4     | E Europe F   | E Asia F     |
| 52  | 1.9 ± 0.6     | 0.1 ± 0.1   | 2.3 ± 0.3     | 1.8 ± 0.2   | 1.6 ± 0.3    | 0.2 ± 0.5     | 2.1 ± 0.6     | 0.4 ± 0.1   | 2.3 ± 0.3     | 1.9 ± 0.2   | 1.7 ± 0.3    | 0.5 ± 0.5     | E Europe M   | E Asia F     |

**Table 65.** Mean raw error for Application-type images for algorithm unissey-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.6 ± 0.3     | 3.8 ± 0.2    | 2.3 ± 0.3     | 2.2 ± 0.2   | 2.8 ± 0.3    | 4.1 ± 0.5     | 2.1 ± 0.3     | 3.0 ± 0.2   | 0.8 ± 0.3     | 1.7 ± 0.2   | 2.3 ± 0.3    | 2.6 ± 0.4     | W Africa F   | E Europe M   |
| 15  | 2.5 ± 0.4     | 4.1 ± 0.2    | 2.6 ± 0.3     | 2.5 ± 0.2   | 3.0 ± 0.4    | 3.9 ± 0.6     | 2.0 ± 0.3     | 3.6 ± 0.2   | 1.2 ± 0.3     | 2.4 ± 0.2   | 3.0 ± 0.4    | 2.6 ± 0.4     | E Asia F     | E Europe M   |
| 16  | 2.3 ± 0.4     | 4.0 ± 0.2    | 2.6 ± 0.3     | 2.4 ± 0.2   | 3.4 ± 0.4    | 4.0 ± 0.6     | 2.9 ± 0.4     | 3.8 ± 0.2   | 1.9 ± 0.3     | 2.8 ± 0.2   | 3.6 ± 0.4    | 2.5 ± 0.4     | E Asia F     | E Europe M   |
| 17  | 2.6 ± 0.4     | 3.9 ± 0.2    | 2.9 ± 0.4     | 2.3 ± 0.2   | 3.0 ± 0.4    | 4.2 ± 0.5     | 3.1 ± 0.4     | 3.9 ± 0.2   | 1.2 ± 0.3     | 3.0 ± 0.2   | 3.4 ± 0.4    | 3.1 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 2.1 ± 0.4     | 3.4 ± 0.2    | 2.4 ± 0.4     | 1.8 ± 0.2   | 2.6 ± 0.3    | 3.6 ± 0.5     | 3.1 ± 0.4     | 3.4 ± 0.2   | 1.4 ± 0.3     | 3.0 ± 0.2   | 3.2 ± 0.3    | 2.8 ± 0.4     | W Africa F   | E Europe M   |
| 19  | 2.2 ± 0.4     | 3.2 ± 0.2    | 2.0 ± 0.4     | 2.0 ± 0.2   | 2.5 ± 0.3    | 3.8 ± 0.5     | 3.7 ± 0.4     | 3.3 ± 0.2   | 1.3 ± 0.3     | 3.3 ± 0.2   | 2.9 ± 0.3    | 2.6 ± 0.4     | W Africa F   | E Europe M   |
| 20  | 2.1 ± 0.4     | 3.1 ± 0.2    | 2.3 ± 0.3     | 1.5 ± 0.2   | 2.2 ± 0.2    | 4.0 ± 0.5     | 3.9 ± 0.4     | 3.2 ± 0.2   | 1.7 ± 0.3     | 3.4 ± 0.2   | 3.1 ± 0.3    | 2.8 ± 0.4     | W Africa F   | S Asia F     |
| 21  | 1.8 ± 0.4     | 2.8 ± 0.2    | 2.0 ± 0.3     | 1.3 ± 0.2   | 2.0 ± 0.2    | 3.6 ± 0.5     | 3.7 ± 0.4     | 3.1 ± 0.2   | 1.7 ± 0.3     | 3.2 ± 0.2   | 3.0 ± 0.3    | 2.6 ± 0.4     | E Africa M   | S Asia F     |
| 22  | 1.5 ± 0.4     | 2.3 ± 0.2    | 2.2 ± 0.3     | 0.9 ± 0.2   | 1.6 ± 0.2    | 3.5 ± 0.5     | 4.0 ± 0.3     | 2.8 ± 0.2   | 2.0 ± 0.3     | 3.1 ± 0.2   | 2.5 ± 0.3    | 2.1 ± 0.3     | E Africa M   | S Asia F     |
| 23  | 1.3 ± 0.3     | 2.1 ± 0.1    | 2.2 ± 0.3     | 0.7 ± 0.2   | 1.5 ± 0.2    | 3.1 ± 0.4     | 3.5 ± 0.3     | 2.8 ± 0.2   | 1.7 ± 0.3     | 2.9 ± 0.2   | 2.4 ± 0.3    | 1.8 ± 0.3     | E Africa M   | S Asia F     |
| 24  | 1.1 ± 0.3     | 1.6 ± 0.1    | 2.2 ± 0.2     | 0.4 ± 0.1   | 1.2 ± 0.2    | 2.7 ± 0.4     | 3.3 ± 0.3     | 2.4 ± 0.2   | 2.0 ± 0.3     | 2.7 ± 0.1   | 2.1 ± 0.2    | 1.6 ± 0.3     | E Africa M   | S Asia F     |
| 25  | 1.0 ± 0.3     | 1.3 ± 0.1    | 2.2 ± 0.2     | -0.1 ± 0.1  | 0.6 ± 0.2    | 2.8 ± 0.4     | 3.3 ± 0.3     | 2.2 ± 0.2   | 2.1 ± 0.2     | 2.4 ± 0.1   | 2.0 ± 0.2    | 1.3 ± 0.3     | E Africa M   | S Asia F     |
| 26  | 1.1 ± 0.3     | 0.8 ± 0.1    | 2.3 ± 0.2     | -0.6 ± 0.1  | 0.4 ± 0.2    | 2.3 ± 0.3     | 3.0 ± 0.3     | 1.8 ± 0.1   | 1.9 ± 0.2     | 2.0 ± 0.1   | 1.8 ± 0.2    | 0.7 ± 0.2     | E Africa M   | S Asia F     |
| 27  | 0.7 ± 0.3     | 0.48 ± 0.10  | 2.2 ± 0.2     | -1.0 ± 0.1  | 0.0 ± 0.1    | 2.0 ± 0.3     | 2.5 ± 0.3     | 1.6 ± 0.1   | 2.1 ± 0.2     | 1.6 ± 0.1   | 1.6 ± 0.2    | 0.3 ± 0.2     | E Africa M   | S Asia F     |
| 28  | 0.2 ± 0.3     | 0.01 ± 0.10  | 2.2 ± 0.2     | -1.4 ± 0.1  | -0.3 ± 0.2   | 2.1 ± 0.3     | 2.0 ± 0.3     | 1.4 ± 0.1   | 2.1 ± 0.2     | 1.1 ± 0.1   | 1.4 ± 0.2    | -0.4 ± 0.2    | E Europe F   | S Asia F     |
| 29  | 0.3 ± 0.3     | -0.38 ± 0.10 | 2.3 ± 0.2     | -1.5 ± 0.1  | -0.7 ± 0.2   | 1.7 ± 0.3     | 1.9 ± 0.3     | 1.1 ± 0.1   | 1.8 ± 0.2     | 0.7 ± 0.1   | 0.9 ± 0.2    | -0.3 ± 0.2    | E Europe F   | S Asia F     |
| 30  | -0.1 ± 0.3    | -0.81 ± 0.09 | 2.3 ± 0.2     | -1.9 ± 0.1  | -0.9 ± 0.2   | 1.2 ± 0.3     | 1.6 ± 0.3     | 0.7 ± 0.1   | 1.8 ± 0.2     | 0.2 ± 0.1   | 0.7 ± 0.2    | -1.0 ± 0.2    | E Europe F   | S Asia F     |
| 31  | -0.3 ± 0.3    | -1.21 ± 0.10 | 2.6 ± 0.2     | -2.3 ± 0.1  | -1.3 ± 0.2   | 0.9 ± 0.3     | 1.3 ± 0.3     | 0.5 ± 0.1   | 1.7 ± 0.2     | -0.2 ± 0.1  | 0.4 ± 0.3    | -1.5 ± 0.2    | E Europe F   | S Asia F     |
| 32  | -0.2 ± 0.3    | -1.56 ± 0.10 | 2.4 ± 0.2     | -2.5 ± 0.1  | -1.6 ± 0.2   | 0.6 ± 0.3     | 0.9 ± 0.3     | 0.2 ± 0.1   | 1.8 ± 0.2     | -0.5 ± 0.1  | 0.2 ± 0.2    | -2.0 ± 0.2    | E Europe F   | S Asia F     |
| 33  | -0.7 ± 0.4    | -1.9 ± 0.1   | 2.5 ± 0.2     | -2.7 ± 0.1  | -2.0 ± 0.2   | 0.6 ± 0.4     | 0.5 ± 0.3     | -0.3 ± 0.1  | 2.1 ± 0.2     | -0.9 ± 0.1  | -0.1 ± 0.3   | -2.2 ± 0.2    | E Europe F   | S Asia F     |
| 34  | -0.8 ± 0.4    | -2.2 ± 0.1   | 2.5 ± 0.3     | -2.9 ± 0.1  | -2.3 ± 0.2   | -0.1 ± 0.4    | 0.0 ± 0.3     | -0.6 ± 0.1  | 1.9 ± 0.2     | -1.1 ± 0.1  | -0.6 ± 0.3   | -2.7 ± 0.2    | E Europe F   | S Asia F     |
| 35  | -0.9 ± 0.4    | -2.6 ± 0.1   | 2.4 ± 0.3     | -3.2 ± 0.2  | -2.7 ± 0.2   | -0.6 ± 0.4    | -0.2 ± 0.3    | -0.8 ± 0.1  | 1.5 ± 0.2     | -1.4 ± 0.1  | -0.7 ± 0.3   | -2.9 ± 0.3    | E Europe F   | S Asia F     |
| 36  | -1.2 ± 0.4    | -2.8 ± 0.1   | 2.6 ± 0.3     | -3.5 ± 0.2  | -2.7 ± 0.2   | -0.5 ± 0.4    | -0.7 ± 0.3    | -1.1 ± 0.1  | 1.6 ± 0.2     | -1.7 ± 0.1  | -1.0 ± 0.3   | -3.5 ± 0.3    | E Europe F   | W Africa M   |
| 37  | -1.2 ± 0.4    | -3.1 ± 0.1   | 2.7 ± 0.3     | -3.4 ± 0.2  | -3.0 ± 0.2   | -1.1 ± 0.4    | -0.6 ± 0.3    | -1.4 ± 0.1  | 1.3 ± 0.2     | -1.9 ± 0.1  | -1.3 ± 0.3   | -3.8 ± 0.3    | E Europe F   | W Africa M   |
| 38  | -1.2 ± 0.4    | -3.4 ± 0.1   | 2.7 ± 0.3     | -3.7 ± 0.2  | -3.1 ± 0.2   | -1.6 ± 0.4    | -1.1 ± 0.3    | -1.6 ± 0.1  | 1.2 ± 0.2     | -2.2 ± 0.1  | -0.9 ± 0.3   | -4.2 ± 0.3    | E Europe F   | W Africa M   |
| 39  | -1.3 ± 0.5    | -3.7 ± 0.1   | 2.5 ± 0.3     | -3.8 ± 0.2  | -3.1 ± 0.2   | -1.5 ± 0.4    | -1.2 ± 0.3    | -1.9 ± 0.1  | 1.0 ± 0.2     | -2.3 ± 0.1  | -1.1 ± 0.3   | -4.7 ± 0.3    | E Europe F   | W Africa M   |
| 40  | -1.8 ± 0.5    | -3.9 ± 0.1   | 2.7 ± 0.3     | -3.7 ± 0.2  | -3.4 ± 0.2   | -2.0 ± 0.4    | -1.4 ± 0.4    | -2.2 ± 0.1  | 1.1 ± 0.2     | -2.5 ± 0.1  | -1.3 ± 0.3   | -5.0 ± 0.3    | E Europe F   | W Africa M   |
| 41  | -2.4 ± 0.5    | -4.1 ± 0.1   | 2.5 ± 0.3     | -3.9 ± 0.2  | -3.6 ± 0.2   | -2.7 ± 0.4    | -1.5 ± 0.4    | -2.3 ± 0.1  | 0.8 ± 0.3     | -2.7 ± 0.2  | -1.5 ± 0.3   | -5.2 ± 0.3    | E Europe F   | W Africa M   |
| 42  | -2.1 ± 0.5    | -4.1 ± 0.1   | 2.8 ± 0.3     | -3.6 ± 0.2  | -3.7 ± 0.3   | -2.4 ± 0.4    | -1.6 ± 0.4    | -2.5 ± 0.1  | 0.8 ± 0.3     | -3.0 ± 0.2  | -1.4 ± 0.3   | -5.5 ± 0.3    | E Europe F   | W Africa M   |
| 43  | -2.0 ± 0.5    | -4.4 ± 0.1   | 2.6 ± 0.3     | -3.8 ± 0.3  | -3.8 ± 0.3   | -3.2 ± 0.5    | -1.6 ± 0.4    | -2.7 ± 0.1  | 0.6 ± 0.3     | -2.9 ± 0.2  | -1.7 ± 0.3   | -5.9 ± 0.4    | E Europe F   | W Africa M   |
| 44  | -1.9 ± 0.6    | -4.6 ± 0.1   | 2.4 ± 0.3     | -3.8 ± 0.3  | -3.5 ± 0.3   | -3.2 ± 0.5    | -2.2 ± 0.4    | -2.8 ± 0.2  | 0.8 ± 0.3     | -3.1 ± 0.2  | -1.6 ± 0.3   | -5.9 ± 0.4    | E Europe F   | W Africa M   |
| 45  | -2.4 ± 0.6    | -4.6 ± 0.1   | 2.2 ± 0.3     | -3.7 ± 0.3  | -3.8 ± 0.3   | -3.3 ± 0.5    | -2.7 ± 0.5    | -3.2 ± 0.2  | 0.6 ± 0.3     | -3.0 ± 0.2  | -1.9 ± 0.3   | -6.2 ± 0.4    | E Europe F   | W Africa M   |
| 46  | -1.6 ± 0.6    | -4.9 ± 0.1   | 2.2 ± 0.3     | -3.5 ± 0.3  | -4.1 ± 0.3   | -3.4 ± 0.5    | -2.6 ± 0.5    | -3.3 ± 0.2  | 0.7 ± 0.3     | -2.9 ± 0.2  | -1.9 ± 0.3   | -6.6 ± 0.4    | E Europe F   | W Africa M   |
| 47  | -1.6 ± 0.7    | -4.8 ± 0.2   | 2.2 ± 0.3     | -3.5 ± 0.3  | -3.8 ± 0.3   | -3.7 ± 0.5    | -2.3 ± 0.5    | -3.3 ± 0.2  | 0.3 ± 0.3     | -2.8 ± 0.2  | -1.8 ± 0.3   | -6.1 ± 0.4    | E Europe F   | W Africa M   |
| 48  | -2.5 ± 0.7    | -4.9 ± 0.2   | 1.8 ± 0.3     | -3.3 ± 0.3  | -4.1 ± 0.3   | -3.7 ± 0.5    | -3.0 ± 0.6    | -3.4 ± 0.2  | 0.4 ± 0.3     | -2.8 ± 0.2  | -2.2 ± 0.3   | -6.8 ± 0.4    | E Europe F   | W Africa M   |
| 49  | -2.3 ± 0.7    | -5.0 ± 0.2   | 1.8 ± 0.3     | -3.6 ± 0.3  | -3.7 ± 0.3   | -4.0 ± 0.5    | -3.2 ± 0.6    | -3.6 ± 0.2  | 0.3 ± 0.3     | -3.0 ± 0.2  | -2.0 ± 0.3   | -6.9 ± 0.5    | E Europe F   | W Africa M   |
| 50  | -2.9 ± 0.8    | -5.1 ± 0.2   | 2.0 ± 0.3     | -3.3 ± 0.3  | -4.0 ± 0.3   | -4.0 ± 0.5    | -3.2 ± 0.6    | -3.6 ± 0.2  | -0.0 ± 0.3    | -3.1 ± 0.3  | -1.8 ± 0.3   | -7.3 ± 0.5    | E Europe F   | W Africa M   |
| 51  | -2.3 ± 0.7    | -5.1 ± 0.2   | 1.6 ± 0.3     | -3.5 ± 0.3  | -4.3 ± 0.3   | -4.5 ± 0.6    | -3.5 ± 0.6    | -3.6 ± 0.2  | 0.2 ± 0.3     | -3.0 ± 0.3  | -2.0 ± 0.3   | -7.3 ± 0.5    | E Europe F   | W Africa M   |
| 52  | -2.9 ± 0.8    | -5.2 ± 0.2   | 1.5 ± 0.3     | -3.6 ± 0.3  | -3.7 ± 0.3   | -5.0 ± 0.6    | -2.9 ± 0.6    | -3.9 ± 0.2  | -0.1 ± 0.3    | -3.1 ± 0.3  | -2.3 ± 0.4   | -7.3 ± 0.5    | E Europe F   | W Africa M   |

**Table 66.** Mean raw error for Application-type images for algorithm viante-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.8 ± 0.5     | 5.3 ± 0.2    | 2.4 ± 0.3     | 5.8 ± 0.2   | 5.5 ± 0.4    | 7.1 ± 0.5     | 3.3 ± 0.4     | 3.6 ± 0.1   | 0.6 ± 0.1     | 3.2 ± 0.2   | 3.7 ± 0.3    | 4.8 ± 0.5     | W Africa F   | E Europe M   |
| 15  | 6.9 ± 0.5     | 5.9 ± 0.1    | 3.1 ± 0.3     | 6.8 ± 0.3   | 6.5 ± 0.4    | 7.5 ± 0.4     | 4.1 ± 0.4     | 4.7 ± 0.1   | 0.8 ± 0.2     | 4.7 ± 0.2   | 5.1 ± 0.3    | 6.2 ± 0.5     | W Africa F   | E Europe M   |
| 16  | 7.0 ± 0.4     | 6.0 ± 0.1    | 3.8 ± 0.3     | 7.1 ± 0.2   | 7.1 ± 0.3    | 7.4 ± 0.5     | 6.2 ± 0.4     | 5.3 ± 0.1   | 1.7 ± 0.3     | 5.7 ± 0.2   | 5.9 ± 0.3    | 6.4 ± 0.4     | W Africa F   | E Europe M   |
| 17  | 7.5 ± 0.4     | 5.9 ± 0.1    | 4.2 ± 0.3     | 7.1 ± 0.2   | 6.7 ± 0.3    | 7.3 ± 0.3     | 6.5 ± 0.3     | 5.4 ± 0.1   | 2.2 ± 0.3     | 6.2 ± 0.2   | 6.2 ± 0.3    | 7.2 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 6.8 ± 0.3     | 5.3 ± 0.1    | 3.8 ± 0.3     | 6.4 ± 0.2   | 6.2 ± 0.2    | 6.6 ± 0.4     | 6.2 ± 0.3     | 5.0 ± 0.1   | 2.4 ± 0.2     | 5.8 ± 0.2   | 5.9 ± 0.2    | 6.3 ± 0.3     | E Africa F   | E Europe M   |
| 19  | 6.9 ± 0.3     | 4.9 ± 0.1    | 3.6 ± 0.2     | 6.4 ± 0.1   | 5.9 ± 0.2    | 6.8 ± 0.3     | 6.7 ± 0.3     | 5.0 ± 0.1   | 2.6 ± 0.2     | 6.2 ± 0.1   | 5.7 ± 0.2    | 6.6 ± 0.3     | E Africa F   | E Europe M   |
| 20  | 6.5 ± 0.3     | 4.7 ± 0.1    | 3.5 ± 0.2     | 6.0 ± 0.2   | 5.5 ± 0.1    | 6.5 ± 0.2     | 6.7 ± 0.3     | 4.9 ± 0.1   | 2.8 ± 0.2     | 6.2 ± 0.1   | 5.6 ± 0.2    | 6.5 ± 0.3     | E Africa M   | E Europe M   |
| 21  | 6.2 ± 0.2     | 4.30 ± 0.09  | 2.9 ± 0.1     | 5.7 ± 0.1   | 5.1 ± 0.1    | 6.0 ± 0.2     | 6.6 ± 0.2     | 4.66 ± 0.09 | 2.5 ± 0.2     | 6.0 ± 0.1   | 5.4 ± 0.2    | 6.2 ± 0.3     | E Africa M   | E Europe M   |
| 22  | 6.0 ± 0.2     | 3.71 ± 0.08  | 2.5 ± 0.1     | 5.4 ± 0.1   | 4.6 ± 0.1    | 5.4 ± 0.2     | 6.9 ± 0.2     | 4.34 ± 0.09 | 2.3 ± 0.2     | 5.9 ± 0.1   | 5.0 ± 0.1    | 5.9 ± 0.2     | E Africa M   | E Europe M   |
| 23  | 5.5 ± 0.2     | 3.29 ± 0.07  | 2.04 ± 0.09   | 5.1 ± 0.1   | 4.20 ± 0.10  | 5.1 ± 0.2     | 6.4 ± 0.2     | 3.95 ± 0.09 | 1.9 ± 0.1     | 5.6 ± 0.1   | 4.8 ± 0.2    | 5.8 ± 0.2     | E Africa M   | E Europe M   |
| 24  | 5.1 ± 0.2     | 2.70 ± 0.06  | 1.55 ± 0.07   | 4.70 ± 0.09 | 3.67 ± 0.09  | 4.6 ± 0.2     | 6.2 ± 0.2     | 3.57 ± 0.09 | 1.6 ± 0.1     | 5.5 ± 0.1   | 4.5 ± 0.1    | 5.4 ± 0.2     | E Africa M   | E Europe F   |
| 25  | 4.5 ± 0.2     | 2.14 ± 0.06  | 1.01 ± 0.07   | 4.17 ± 0.09 | 3.10 ± 0.09  | 4.2 ± 0.2     | 5.8 ± 0.2     | 3.25 ± 0.09 | 1.3 ± 0.1     | 5.2 ± 0.1   | 4.2 ± 0.1    | 5.1 ± 0.2     | E Africa M   | E Europe F   |
| 26  | 3.9 ± 0.2     | 1.62 ± 0.06  | 0.59 ± 0.07   | 3.67 ± 0.08 | 2.78 ± 0.09  | 3.7 ± 0.2     | 5.5 ± 0.2     | 2.85 ± 0.09 | 1.0 ± 0.1     | 4.95 ± 0.10 | 3.9 ± 0.1    | 4.4 ± 0.2     | E Africa M   | E Europe F   |
| 27  | 3.7 ± 0.2     | 1.08 ± 0.06  | 0.17 ± 0.08   | 3.19 ± 0.08 | 2.21 ± 0.09  | 3.3 ± 0.2     | 5.1 ± 0.2     | 2.48 ± 0.08 | 0.8 ± 0.1     | 4.65 ± 0.09 | 3.8 ± 0.1    | 4.0 ± 0.2     | E Africa M   | E Europe F   |
| 28  | 3.2 ± 0.2     | 0.57 ± 0.06  | -0.15 ± 0.09  | 2.78 ± 0.08 | 1.83 ± 0.09  | 2.7 ± 0.2     | 4.7 ± 0.2     | 2.18 ± 0.09 | 0.7 ± 0.1     | 4.29 ± 0.09 | 3.3 ± 0.1    | 3.4 ± 0.2     | E Africa M   | E Europe F   |
| 29  | 3.0 ± 0.2     | 0.14 ± 0.06  | -0.3 ± 0.1    | 2.32 ± 0.08 | 1.38 ± 0.10  | 2.3 ± 0.2     | 4.4 ± 0.2     | 1.90 ± 0.09 | 0.6 ± 0.1     | 4.04 ± 0.08 | 3.0 ± 0.2    | 3.3 ± 0.2     | E Africa M   | E Europe F   |
| 30  | 2.5 ± 0.2     | -0.28 ± 0.06 | -0.5 ± 0.1    | 1.86 ± 0.08 | 1.1 ± 0.1    | 1.8 ± 0.2     | 4.3 ± 0.2     | 1.48 ± 0.09 | 0.5 ± 0.2     | 3.76 ± 0.08 | 2.9 ± 0.2    | 2.8 ± 0.2     | E Africa M   | E Europe F   |
| 31  | 2.2 ± 0.2     | -0.68 ± 0.06 | -0.5 ± 0.1    | 1.40 ± 0.09 | 0.7 ± 0.1    | 1.4 ± 0.2     | 4.1 ± 0.2     | 1.26 ± 0.10 | 0.5 ± 0.2     | 3.51 ± 0.09 | 2.6 ± 0.2    | 2.3 ± 0.2     | E Africa M   | E Asia F     |
| 32  | 2.2 ± 0.3     | -1.01 ± 0.07 | -0.5 ± 0.1    | 1.06 ± 0.09 | 0.3 ± 0.1    | 1.3 ± 0.2     | 3.7 ± 0.2     | 1.07 ± 0.10 | 0.7 ± 0.2     | 3.41 ± 0.09 | 2.5 ± 0.2    | 2.0 ± 0.2     | E Africa M   | E Asia F     |
| 33  | 1.7 ± 0.3     | -1.33 ± 0.07 | -0.5 ± 0.1    | 0.8 ± 0.1   | -0.0 ± 0.1   | 0.8 ± 0.2     | 3.7 ± 0.2     | 0.8 ± 0.1   | 0.9 ± 0.2     | 3.22 ± 0.09 | 2.3 ± 0.2    | 1.6 ± 0.2     | E Africa M   | E Asia F     |
| 34  | 1.6 ± 0.3     | -1.56 ± 0.07 | -0.6 ± 0.1    | 0.5 ± 0.1   | -0.2 ± 0.1   | 0.5 ± 0.2     | 3.2 ± 0.3     | 0.6 ± 0.1   | 0.6 ± 0.2     | 3.2 ± 0.1   | 2.3 ± 0.2    | 1.6 ± 0.2     | E Africa M   | E Asia F     |
| 35  | 1.6 ± 0.3     | -1.89 ± 0.08 | -0.7 ± 0.1    | 0.1 ± 0.1   | -0.6 ± 0.1   | 0.3 ± 0.3     | 3.4 ± 0.3     | 0.5 ± 0.1   | 0.6 ± 0.2     | 3.0 ± 0.1   | 2.2 ± 0.2    | 1.3 ± 0.2     | E Africa M   | E Asia F     |
| 36  | 1.3 ± 0.3     | -1.96 ± 0.09 | -0.7 ± 0.1    | -0.2 ± 0.1  | -0.5 ± 0.2   | 0.3 ± 0.3     | 3.0 ± 0.3     | 0.5 ± 0.1   | 0.6 ± 0.2     | 2.8 ± 0.1   | 2.2 ± 0.2    | 1.1 ± 0.2     | E Africa M   | E Asia F     |
| 37  | 1.3 ± 0.3     | -2.19 ± 0.09 | -0.9 ± 0.1    | -0.3 ± 0.1  | -0.8 ± 0.2   | -0.1 ± 0.3    | 3.2 ± 0.3     | 0.5 ± 0.1   | 0.2 ± 0.2     | 2.7 ± 0.1   | 2.0 ± 0.2    | 0.9 ± 0.2     | E Africa M   | E Asia F     |
| 38  | 1.8 ± 0.4     | -2.31 ± 0.09 | -0.9 ± 0.2    | -0.4 ± 0.1  | -0.6 ± 0.2   | -0.2 ± 0.3    | 2.7 ± 0.3     | 0.4 ± 0.1   | 0.2 ± 0.2     | 2.6 ± 0.1   | 2.5 ± 0.3    | 0.7 ± 0.2     | E Africa M   | E Asia F     |
| 39  | 1.4 ± 0.4     | -2.40 ± 0.10 | -1.0 ± 0.2    | -0.6 ± 0.2  | -0.6 ± 0.2   | 0.0 ± 0.3     | 3.0 ± 0.3     | 0.4 ± 0.1   | 0.3 ± 0.2     | 2.5 ± 0.1   | 2.5 ± 0.2    | 0.6 ± 0.2     | E Africa M   | E Asia F     |
| 40  | 1.4 ± 0.4     | -2.4 ± 0.1   | -0.7 ± 0.2    | -0.5 ± 0.2  | -0.7 ± 0.2   | -0.4 ± 0.3    | 2.9 ± 0.3     | 0.4 ± 0.1   | 0.4 ± 0.2     | 2.5 ± 0.1   | 2.5 ± 0.2    | 0.4 ± 0.3     | E Africa M   | E Asia F     |
| 41  | 0.9 ± 0.4     | -2.4 ± 0.1   | -0.8 ± 0.2    | -0.6 ± 0.2  | -0.6 ± 0.2   | -0.7 ± 0.4    | 2.8 ± 0.3     | 0.5 ± 0.1   | 0.5 ± 0.2     | 2.4 ± 0.1   | 2.4 ± 0.2    | 0.4 ± 0.3     | E Africa M   | E Asia F     |
| 42  | 1.5 ± 0.4     | -2.2 ± 0.1   | -0.4 ± 0.2    | -0.3 ± 0.2  | -0.7 ± 0.2   | -0.4 ± 0.4    | 2.6 ± 0.3     | 0.6 ± 0.1   | 0.7 ± 0.2     | 2.2 ± 0.1   | 2.6 ± 0.2    | 0.2 ± 0.3     | S E Asia M   | E Asia F     |
| 43  | 1.5 ± 0.4     | -2.2 ± 0.1   | -0.3 ± 0.2    | -0.6 ± 0.2  | -0.5 ± 0.2   | -0.7 ± 0.4    | 2.7 ± 0.3     | 0.7 ± 0.1   | 0.6 ± 0.2     | 2.1 ± 0.1   | 2.7 ± 0.2    | -0.1 ± 0.3    | E Africa M   | E Asia F     |
| 44  | 1.4 ± 0.5     | -2.2 ± 0.1   | -0.5 ± 0.2    | -0.6 ± 0.2  | -0.1 ± 0.2   | -0.8 ± 0.4    | 2.1 ± 0.3     | 0.7 ± 0.1   | 1.0 ± 0.2     | 1.9 ± 0.1   | 2.7 ± 0.3    | 0.1 ± 0.3     | S E Asia M   | E Asia F     |
| 45  | 1.2 ± 0.5     | -1.9 ± 0.1   | -0.2 ± 0.2    | -0.7 ± 0.2  | -0.3 ± 0.2   | -0.6 ± 0.4    | 1.7 ± 0.3     | 0.7 ± 0.1   | 0.8 ± 0.2     | 1.7 ± 0.1   | 2.7 ± 0.3    | -0.2 ± 0.3    | S E Asia M   | E Asia F     |
| 46  | 1.8 ± 0.5     | -1.9 ± 0.1   | -0.1 ± 0.2    | -0.7 ± 0.2  | -0.4 ± 0.2   | -0.7 ± 0.4    | 1.7 ± 0.4     | 0.7 ± 0.1   | 0.9 ± 0.2     | 1.7 ± 0.1   | 2.7 ± 0.2    | -0.4 ± 0.3    | S E Asia M   | E Asia F     |
| 47  | 1.5 ± 0.6     | -1.6 ± 0.1   | -0.1 ± 0.2    | -0.9 ± 0.2  | 0.0 ± 0.2    | -0.7 ± 0.4    | 1.9 ± 0.4     | 0.8 ± 0.1   | 0.6 ± 0.2     | 1.4 ± 0.2   | 2.7 ± 0.3    | -0.2 ± 0.3    | S E Asia M   | E Asia F     |
| 48  | 1.2 ± 0.5     | -1.4 ± 0.1   | -0.0 ± 0.2    | -0.9 ± 0.2  | -0.0 ± 0.2   | -1.0 ± 0.4    | 1.1 ± 0.4     | 0.8 ± 0.1   | 0.6 ± 0.2     | 1.1 ± 0.1   | 2.5 ± 0.2    | -0.7 ± 0.4    | S E Asia M   | E Asia F     |
| 49  | 1.4 ± 0.6     | -1.4 ± 0.1   | -0.4 ± 0.2    | -1.1 ± 0.2  | 0.2 ± 0.2    | -1.2 ± 0.4    | 0.8 ± 0.4     | 0.8 ± 0.1   | 0.3 ± 0.2     | 0.8 ± 0.2   | 2.4 ± 0.2    | -0.7 ± 0.3    | S E Asia M   | E Asia F     |
| 50  | 0.8 ± 0.6     | -1.3 ± 0.1   | -0.2 ± 0.2    | -1.3 ± 0.2  | -0.1 ± 0.2   | -1.1 ± 0.4    | 0.7 ± 0.4     | 0.7 ± 0.1   | -0.0 ± 0.2    | 0.6 ± 0.2   | 2.6 ± 0.2    | -1.1 ± 0.3    | S E Asia M   | E Asia F     |
| 51  | 1.4 ± 0.5     | -1.2 ± 0.1   | -0.5 ± 0.2    | -1.6 ± 0.2  | -0.1 ± 0.2   | -1.1 ± 0.5    | 0.3 ± 0.5     | 0.7 ± 0.1   | 0.0 ± 0.2     | 0.4 ± 0.2   | 2.3 ± 0.2    | -1.4 ± 0.3    | S E Asia M   | S Asia F     |
| 52  | 1.0 ± 0.6     | -1.2 ± 0.1   | -0.4 ± 0.2    | -1.7 ± 0.2  | 0.0 ± 0.2    | -1.6 ± 0.5    | 0.5 ± 0.5     | 0.5 ± 0.1   | -0.2 ± 0.3    | 0.1 ± 0.2   | 1.8 ± 0.2    | -1.4 ± 0.4    | S E Asia M   | S Asia F     |

**Table 67.** Mean raw error for Application-type images for algorithm yoti-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.1 ± 0.3     | 2.0 ± 0.1   | 1.7 ± 0.3     | 1.8 ± 0.2   | 2.1 ± 0.3    | 3.9 ± 0.4     | 0.2 ± 0.3     | 0.59 ± 0.09 | -0.0 ± 0.2    | -0.5 ± 0.1  | 0.5 ± 0.1    | 1.1 ± 0.3     | W Africa F   | S Asia M     |
| 15  | 3.2 ± 0.4     | 2.2 ± 0.2   | 2.5 ± 0.4     | 2.3 ± 0.3   | 2.3 ± 0.4    | 4.1 ± 0.5     | 0.1 ± 0.3     | 0.6 ± 0.1   | 0.1 ± 0.2     | -0.3 ± 0.1  | 0.4 ± 0.2    | 1.6 ± 0.4     | W Africa F   | S Asia M     |
| 16  | 3.3 ± 0.4     | 2.4 ± 0.2   | 2.7 ± 0.4     | 2.7 ± 0.3   | 3.1 ± 0.5    | 4.5 ± 0.6     | 1.2 ± 0.3     | 0.5 ± 0.1   | 0.4 ± 0.2     | -0.3 ± 0.2  | 0.4 ± 0.2    | 1.7 ± 0.4     | W Africa F   | S Asia M     |
| 17  | 3.3 ± 0.5     | 2.6 ± 0.2   | 3.5 ± 0.5     | 3.1 ± 0.3   | 3.1 ± 0.5    | 4.2 ± 0.5     | 0.9 ± 0.3     | 0.7 ± 0.1   | 0.3 ± 0.2     | -0.1 ± 0.2  | 0.3 ± 0.3    | 2.4 ± 0.4     | W Africa F   | S Asia M     |
| 18  | 3.1 ± 0.4     | 2.4 ± 0.2   | 3.2 ± 0.4     | 3.0 ± 0.2   | 3.6 ± 0.3    | 4.4 ± 0.5     | 1.1 ± 0.3     | 0.6 ± 0.1   | 0.2 ± 0.2     | -0.1 ± 0.2  | 0.6 ± 0.3    | 2.0 ± 0.3     | W Africa F   | S Asia M     |
| 19  | 3.7 ± 0.4     | 2.8 ± 0.2   | 3.5 ± 0.4     | 3.6 ± 0.2   | 3.9 ± 0.3    | 4.7 ± 0.5     | 1.5 ± 0.3     | 1.1 ± 0.2   | 0.6 ± 0.3     | 0.4 ± 0.2   | 0.8 ± 0.3    | 2.5 ± 0.4     | W Africa F   | S Asia M     |
| 20  | 3.4 ± 0.4     | 3.2 ± 0.2   | 3.5 ± 0.3     | 3.4 ± 0.2   | 4.2 ± 0.3    | 4.6 ± 0.4     | 2.0 ± 0.3     | 1.5 ± 0.2   | 1.2 ± 0.3     | 1.0 ± 0.2   | 1.4 ± 0.3    | 2.7 ± 0.4     | W Africa F   | S Asia M     |
| 21  | 3.8 ± 0.4     | 3.5 ± 0.2   | 3.6 ± 0.3     | 3.8 ± 0.2   | 4.4 ± 0.3    | 4.8 ± 0.4     | 2.2 ± 0.3     | 1.8 ± 0.2   | 1.3 ± 0.2     | 1.5 ± 0.2   | 1.7 ± 0.3    | 2.8 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 3.8 ± 0.4     | 3.6 ± 0.2   | 3.7 ± 0.2     | 3.7 ± 0.2   | 4.5 ± 0.2    | 4.5 ± 0.4     | 2.9 ± 0.3     | 2.0 ± 0.1   | 1.7 ± 0.2     | 1.8 ± 0.2   | 1.9 ± 0.2    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 23  | 3.6 ± 0.3     | 3.7 ± 0.2   | 3.5 ± 0.2     | 3.8 ± 0.2   | 4.5 ± 0.2    | 4.7 ± 0.4     | 2.6 ± 0.3     | 2.0 ± 0.1   | 1.5 ± 0.2     | 2.1 ± 0.2   | 2.2 ± 0.2    | 3.3 ± 0.3     | W Africa F   | E Europe M   |
| 24  | 3.8 ± 0.4     | 3.7 ± 0.1   | 3.5 ± 0.2     | 3.7 ± 0.2   | 4.6 ± 0.2    | 4.4 ± 0.3     | 2.7 ± 0.3     | 2.1 ± 0.1   | 1.6 ± 0.1     | 2.3 ± 0.1   | 2.4 ± 0.2    | 2.9 ± 0.3     | S E Asia F   | E Europe M   |
| 25  | 3.6 ± 0.3     | 3.9 ± 0.1   | 3.2 ± 0.1     | 3.6 ± 0.1   | 4.7 ± 0.2    | 4.3 ± 0.3     | 2.6 ± 0.3     | 2.2 ± 0.1   | 1.4 ± 0.1     | 2.3 ± 0.1   | 2.5 ± 0.2    | 2.9 ± 0.3     | S E Asia F   | E Europe M   |
| 26  | 3.7 ± 0.3     | 3.8 ± 0.1   | 3.0 ± 0.1     | 3.4 ± 0.1   | 4.7 ± 0.2    | 4.1 ± 0.3     | 2.6 ± 0.2     | 2.1 ± 0.1   | 1.1 ± 0.1     | 2.3 ± 0.1   | 2.6 ± 0.2    | 2.7 ± 0.2     | S E Asia F   | E Europe M   |
| 27  | 3.9 ± 0.3     | 3.9 ± 0.1   | 2.9 ± 0.1     | 3.3 ± 0.1   | 4.5 ± 0.2    | 4.2 ± 0.3     | 2.4 ± 0.2     | 2.1 ± 0.1   | 1.0 ± 0.1     | 2.2 ± 0.1   | 2.7 ± 0.2    | 2.7 ± 0.2     | S E Asia F   | E Europe M   |
| 28  | 3.5 ± 0.3     | 3.8 ± 0.1   | 2.6 ± 0.1     | 3.2 ± 0.1   | 4.5 ± 0.2    | 3.8 ± 0.3     | 2.1 ± 0.2     | 2.2 ± 0.1   | 1.0 ± 0.1     | 2.2 ± 0.1   | 2.6 ± 0.2    | 2.4 ± 0.2     | S E Asia F   | E Europe M   |
| 29  | 3.9 ± 0.3     | 3.8 ± 0.1   | 2.5 ± 0.1     | 3.1 ± 0.1   | 4.6 ± 0.2    | 3.8 ± 0.3     | 2.2 ± 0.2     | 2.1 ± 0.1   | 0.7 ± 0.1     | 1.98 ± 0.10 | 2.4 ± 0.2    | 2.7 ± 0.2     | S E Asia F   | E Europe M   |
| 30  | 3.6 ± 0.3     | 3.8 ± 0.1   | 2.3 ± 0.1     | 2.9 ± 0.1   | 4.5 ± 0.2    | 3.5 ± 0.3     | 2.1 ± 0.2     | 2.1 ± 0.1   | 0.5 ± 0.1     | 1.98 ± 0.10 | 2.7 ± 0.2    | 2.4 ± 0.2     | S E Asia F   | E Europe M   |
| 31  | 3.4 ± 0.3     | 3.7 ± 0.1   | 2.2 ± 0.1     | 2.8 ± 0.1   | 4.5 ± 0.2    | 3.1 ± 0.3     | 2.2 ± 0.2     | 2.0 ± 0.1   | 0.4 ± 0.1     | 1.87 ± 0.09 | 2.5 ± 0.2    | 2.2 ± 0.2     | S E Asia F   | E Europe M   |
| 32  | 3.6 ± 0.3     | 3.6 ± 0.1   | 2.0 ± 0.1     | 2.7 ± 0.1   | 4.1 ± 0.2    | 3.3 ± 0.3     | 1.9 ± 0.2     | 2.1 ± 0.1   | 0.4 ± 0.2     | 1.87 ± 0.09 | 2.7 ± 0.2    | 2.2 ± 0.2     | S E Asia F   | E Europe M   |
| 33  | 3.4 ± 0.3     | 3.5 ± 0.1   | 1.8 ± 0.2     | 2.6 ± 0.1   | 4.2 ± 0.2    | 3.1 ± 0.3     | 2.0 ± 0.3     | 2.1 ± 0.1   | 0.5 ± 0.2     | 1.69 ± 0.09 | 2.4 ± 0.2    | 2.1 ± 0.2     | S E Asia F   | E Europe M   |
| 34  | 3.2 ± 0.3     | 3.4 ± 0.1   | 1.7 ± 0.2     | 2.5 ± 0.1   | 4.1 ± 0.2    | 2.8 ± 0.3     | 1.7 ± 0.3     | 2.1 ± 0.1   | 0.3 ± 0.2     | 1.68 ± 0.10 | 2.5 ± 0.2    | 2.1 ± 0.2     | S E Asia F   | E Europe M   |
| 35  | 3.7 ± 0.3     | 3.2 ± 0.1   | 1.7 ± 0.2     | 2.3 ± 0.1   | 3.7 ± 0.2    | 2.5 ± 0.3     | 2.1 ± 0.3     | 2.2 ± 0.1   | 0.4 ± 0.2     | 1.67 ± 0.10 | 2.6 ± 0.2    | 1.8 ± 0.2     | S E Asia F   | E Europe M   |
| 36  | 3.0 ± 0.3     | 3.1 ± 0.1   | 1.6 ± 0.2     | 2.2 ± 0.1   | 3.7 ± 0.2    | 2.8 ± 0.3     | 1.9 ± 0.3     | 2.2 ± 0.1   | 0.5 ± 0.2     | 1.57 ± 0.10 | 2.6 ± 0.2    | 1.7 ± 0.2     | S E Asia F   | E Europe M   |
| 37  | 2.9 ± 0.4     | 2.97 ± 0.10 | 1.7 ± 0.2     | 2.1 ± 0.1   | 3.4 ± 0.2    | 2.5 ± 0.3     | 2.0 ± 0.3     | 2.3 ± 0.1   | 0.3 ± 0.2     | 1.5 ± 0.1   | 2.3 ± 0.2    | 1.6 ± 0.2     | S E Asia F   | E Europe M   |
| 38  | 3.1 ± 0.4     | 2.79 ± 0.10 | 1.5 ± 0.2     | 2.0 ± 0.1   | 3.4 ± 0.2    | 2.3 ± 0.4     | 1.5 ± 0.3     | 2.1 ± 0.1   | 0.4 ± 0.2     | 1.4 ± 0.1   | 2.6 ± 0.2    | 1.6 ± 0.3     | S E Asia F   | E Europe M   |
| 39  | 2.6 ± 0.3     | 2.63 ± 0.10 | 1.4 ± 0.2     | 1.8 ± 0.1   | 3.3 ± 0.2    | 2.1 ± 0.4     | 1.8 ± 0.3     | 2.1 ± 0.1   | 0.5 ± 0.2     | 1.4 ± 0.1   | 2.6 ± 0.2    | 1.5 ± 0.3     | S E Asia F   | E Europe M   |
| 40  | 2.4 ± 0.4     | 2.47 ± 0.10 | 1.6 ± 0.2     | 1.7 ± 0.1   | 2.9 ± 0.2    | 2.0 ± 0.3     | 1.7 ± 0.3     | 2.1 ± 0.1   | 0.5 ± 0.2     | 1.3 ± 0.1   | 2.5 ± 0.2    | 1.3 ± 0.3     | S E Asia F   | E Europe M   |
| 41  | 2.0 ± 0.4     | 2.18 ± 0.09 | 1.4 ± 0.2     | 1.5 ± 0.1   | 2.7 ± 0.2    | 1.1 ± 0.4     | 1.7 ± 0.3     | 2.0 ± 0.1   | 0.6 ± 0.2     | 1.3 ± 0.1   | 2.3 ± 0.2    | 1.3 ± 0.3     | S E Asia F   | E Europe M   |
| 42  | 2.4 ± 0.4     | 2.19 ± 0.09 | 1.5 ± 0.2     | 1.6 ± 0.2   | 2.5 ± 0.2    | 1.6 ± 0.4     | 1.7 ± 0.3     | 1.9 ± 0.1   | 0.8 ± 0.2     | 1.2 ± 0.1   | 2.4 ± 0.2    | 0.9 ± 0.3     | S E Asia F   | E Europe M   |
| 43  | 2.2 ± 0.4     | 1.91 ± 0.10 | 1.3 ± 0.2     | 1.2 ± 0.2   | 2.3 ± 0.2    | 0.9 ± 0.4     | 1.6 ± 0.3     | 1.8 ± 0.1   | 0.6 ± 0.2     | 1.2 ± 0.1   | 2.4 ± 0.2    | 0.6 ± 0.3     | S E Asia M   | E Europe M   |
| 44  | 1.9 ± 0.5     | 1.63 ± 0.10 | 1.0 ± 0.2     | 1.1 ± 0.2   | 2.3 ± 0.2    | 0.9 ± 0.4     | 1.2 ± 0.3     | 1.7 ± 0.1   | 0.9 ± 0.2     | 1.0 ± 0.1   | 2.4 ± 0.2    | 0.7 ± 0.3     | S E Asia M   | W Africa M   |
| 45  | 1.4 ± 0.4     | 1.61 ± 0.10 | 1.2 ± 0.2     | 0.9 ± 0.2   | 2.1 ± 0.2    | 0.7 ± 0.4     | 1.1 ± 0.4     | 1.6 ± 0.1   | 0.8 ± 0.2     | 1.1 ± 0.1   | 1.9 ± 0.2    | 0.2 ± 0.3     | S E Asia F   | W Africa M   |
| 46  | 1.9 ± 0.4     | 1.32 ± 0.10 | 1.2 ± 0.2     | 0.9 ± 0.2   | 1.8 ± 0.2    | 0.4 ± 0.4     | 1.0 ± 0.4     | 1.5 ± 0.1   | 0.9 ± 0.3     | 1.2 ± 0.2   | 2.0 ± 0.2    | -0.0 ± 0.3    | S E Asia M   | W Africa M   |
| 47  | 1.6 ± 0.5     | 1.3 ± 0.1   | 1.3 ± 0.3     | 0.8 ± 0.2   | 1.8 ± 0.2    | 0.3 ± 0.4     | 1.2 ± 0.4     | 1.3 ± 0.1   | 1.2 ± 0.3     | 1.2 ± 0.2   | 2.1 ± 0.2    | 0.2 ± 0.3     | S E Asia M   | W Africa M   |
| 48  | 0.6 ± 0.5     | 1.1 ± 0.1   | 1.3 ± 0.2     | 0.8 ± 0.2   | 1.7 ± 0.2    | 0.2 ± 0.4     | 0.6 ± 0.4     | 1.2 ± 0.1   | 1.2 ± 0.3     | 1.1 ± 0.2   | 1.8 ± 0.2    | -0.3 ± 0.3    | S E Asia M   | W Africa M   |
| 49  | 0.8 ± 0.5     | 0.9 ± 0.1   | 1.1 ± 0.2     | 0.5 ± 0.2   | 1.8 ± 0.2    | -0.0 ± 0.4    | 0.4 ± 0.5     | 1.0 ± 0.1   | 1.1 ± 0.3     | 1.1 ± 0.2   | 1.8 ± 0.2    | -0.3 ± 0.3    | S E Asia M   | W Africa M   |
| 50  | 0.9 ± 0.6     | 0.9 ± 0.1   | 1.4 ± 0.2     | 0.4 ± 0.2   | 1.5 ± 0.2    | -0.1 ± 0.5    | 0.4 ± 0.4     | 0.9 ± 0.1   | 0.9 ± 0.3     | 0.9 ± 0.2   | 1.9 ± 0.2    | -0.9 ± 0.3    | S E Asia M   | W Africa M   |
| 51  | 0.3 ± 0.5     | 0.6 ± 0.1   | 1.2 ± 0.2     | 0.2 ± 0.2   | 1.4 ± 0.2    | -0.4 ± 0.5    | -0.3 ± 0.5    | 0.7 ± 0.1   | 1.2 ± 0.3     | 0.9 ± 0.2   | 1.7 ± 0.2    | -1.1 ± 0.4    | S E Asia M   | W Africa M   |
| 52  | 0.5 ± 0.6     | 0.6 ± 0.1   | 1.3 ± 0.2     | -0.1 ± 0.2  | 1.5 ± 0.2    | -1.0 ± 0.5    | -0.1 ± 0.5    | 0.5 ± 0.1   | 1.1 ± 0.3     | 0.6 ± 0.2   | 1.4 ± 0.3    | -1.3 ± 0.4    | S E Asia F   | W Africa M   |



**Table 68.** Mean raw error for Application-type images for algorithm yoti-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 2.1 ± 0.3     | 2.2 ± 0.1   | 0.6 ± 0.2     | 1.6 ± 0.2   | 2.2 ± 0.3    | 2.8 ± 0.4     | -0.4 ± 0.2    | 0.29 ± 0.09 | -0.2 ± 0.1    | -0.44 ± 0.10 | 0.0 ± 0.1    | 0.3 ± 0.3     | W Africa F   | S Asia M     |
| 15  | 2.4 ± 0.4     | 2.5 ± 0.2   | 1.0 ± 0.3     | 2.3 ± 0.2   | 2.5 ± 0.4    | 2.6 ± 0.4     | -0.4 ± 0.3    | 0.3 ± 0.1   | -0.3 ± 0.1    | -0.4 ± 0.1   | -0.0 ± 0.2   | 1.0 ± 0.4     | W Africa F   | S Asia M     |
| 16  | 2.1 ± 0.4     | 2.8 ± 0.2   | 1.1 ± 0.3     | 2.6 ± 0.3   | 3.2 ± 0.4    | 3.2 ± 0.6     | 0.4 ± 0.3     | 0.2 ± 0.1   | -0.4 ± 0.2    | -0.3 ± 0.1   | -0.1 ± 0.2   | 0.9 ± 0.3     | S E Asia F   | E Europe M   |
| 17  | 2.1 ± 0.4     | 3.1 ± 0.2   | 1.0 ± 0.3     | 2.8 ± 0.3   | 3.1 ± 0.4    | 2.8 ± 0.4     | 0.3 ± 0.3     | 0.4 ± 0.1   | -0.7 ± 0.2    | -0.2 ± 0.2   | -0.3 ± 0.3   | 1.5 ± 0.4     | S E Asia F   | E Europe M   |
| 18  | 1.6 ± 0.3     | 2.7 ± 0.2   | 0.7 ± 0.3     | 2.6 ± 0.2   | 3.3 ± 0.3    | 2.3 ± 0.4     | 0.2 ± 0.3     | 0.1 ± 0.1   | -1.0 ± 0.2    | -0.2 ± 0.1   | -0.2 ± 0.3   | 0.8 ± 0.3     | S E Asia F   | E Europe M   |
| 19  | 2.2 ± 0.4     | 3.0 ± 0.2   | 1.0 ± 0.3     | 3.0 ± 0.2   | 3.5 ± 0.2    | 2.9 ± 0.4     | 0.7 ± 0.3     | 0.6 ± 0.2   | -0.8 ± 0.2    | 0.3 ± 0.2    | 0.1 ± 0.3    | 1.5 ± 0.4     | S E Asia F   | E Europe M   |
| 20  | 2.0 ± 0.3     | 3.2 ± 0.2   | 1.0 ± 0.3     | 3.0 ± 0.2   | 3.7 ± 0.2    | 2.9 ± 0.4     | 1.0 ± 0.3     | 0.9 ± 0.1   | -0.2 ± 0.2    | 0.8 ± 0.2    | 0.6 ± 0.3    | 1.6 ± 0.4     | S E Asia F   | E Europe M   |
| 21  | 2.1 ± 0.3     | 3.2 ± 0.1   | 1.1 ± 0.2     | 3.0 ± 0.2   | 3.8 ± 0.2    | 3.0 ± 0.4     | 1.5 ± 0.3     | 1.2 ± 0.1   | -0.0 ± 0.2    | 1.3 ± 0.2    | 1.0 ± 0.3    | 1.9 ± 0.4     | S E Asia F   | E Europe M   |
| 22  | 2.5 ± 0.3     | 3.0 ± 0.1   | 1.1 ± 0.2     | 3.0 ± 0.2   | 3.5 ± 0.2    | 2.7 ± 0.4     | 2.4 ± 0.3     | 1.5 ± 0.1   | 0.6 ± 0.2     | 1.5 ± 0.2    | 1.1 ± 0.2    | 1.7 ± 0.3     | S E Asia F   | E Europe M   |
| 23  | 2.2 ± 0.3     | 3.0 ± 0.1   | 1.1 ± 0.2     | 2.9 ± 0.1   | 3.5 ± 0.2    | 3.2 ± 0.3     | 2.2 ± 0.3     | 1.4 ± 0.1   | 0.7 ± 0.2     | 1.9 ± 0.1    | 1.4 ± 0.2    | 2.4 ± 0.3     | S E Asia F   | E Europe M   |
| 24  | 2.5 ± 0.3     | 2.8 ± 0.1   | 1.1 ± 0.1     | 2.7 ± 0.1   | 3.3 ± 0.1    | 2.7 ± 0.3     | 2.2 ± 0.3     | 1.5 ± 0.1   | 0.8 ± 0.2     | 2.0 ± 0.1    | 1.5 ± 0.2    | 2.5 ± 0.3     | S E Asia F   | E Europe M   |
| 25  | 2.2 ± 0.3     | 2.68 ± 0.10 | 0.9 ± 0.1     | 2.4 ± 0.1   | 3.2 ± 0.1    | 2.9 ± 0.3     | 2.3 ± 0.3     | 1.5 ± 0.1   | 0.7 ± 0.1     | 2.0 ± 0.1    | 1.5 ± 0.2    | 2.7 ± 0.3     | S E Asia F   | E Europe M   |
| 26  | 2.2 ± 0.3     | 2.47 ± 0.10 | 0.8 ± 0.1     | 2.04 ± 0.10 | 3.2 ± 0.1    | 2.6 ± 0.3     | 2.4 ± 0.2     | 1.4 ± 0.1   | 0.6 ± 0.1     | 1.92 ± 0.10  | 1.5 ± 0.2    | 2.6 ± 0.2     | S E Asia F   | E Europe M   |
| 27  | 2.2 ± 0.3     | 2.38 ± 0.09 | 0.6 ± 0.1     | 1.78 ± 0.10 | 2.9 ± 0.1    | 2.9 ± 0.3     | 2.3 ± 0.2     | 1.3 ± 0.1   | 0.5 ± 0.1     | 1.68 ± 0.09  | 1.6 ± 0.2    | 2.6 ± 0.2     | S E Asia F   | E Europe M   |
| 28  | 2.0 ± 0.3     | 2.13 ± 0.09 | 0.4 ± 0.1     | 1.56 ± 0.10 | 2.8 ± 0.1    | 2.5 ± 0.3     | 2.1 ± 0.2     | 1.3 ± 0.1   | 0.3 ± 0.1     | 1.54 ± 0.09  | 1.5 ± 0.2    | 2.3 ± 0.2     | S E Asia F   | E Europe M   |
| 29  | 2.4 ± 0.3     | 2.02 ± 0.09 | 0.2 ± 0.1     | 1.19 ± 0.10 | 2.9 ± 0.1    | 2.4 ± 0.2     | 2.0 ± 0.2     | 1.1 ± 0.1   | 0.1 ± 0.1     | 1.35 ± 0.08  | 1.3 ± 0.2    | 2.8 ± 0.2     | S E Asia F   | E Europe M   |
| 30  | 2.0 ± 0.3     | 1.95 ± 0.09 | -0.1 ± 0.1    | 0.96 ± 0.10 | 2.6 ± 0.1    | 2.0 ± 0.3     | 2.1 ± 0.2     | 1.1 ± 0.1   | -0.1 ± 0.1    | 1.20 ± 0.08  | 1.5 ± 0.2    | 2.6 ± 0.2     | S E Asia F   | E Europe M   |
| 31  | 1.5 ± 0.3     | 1.84 ± 0.09 | -0.1 ± 0.1    | 0.72 ± 0.10 | 2.5 ± 0.1    | 1.9 ± 0.3     | 2.1 ± 0.2     | 1.0 ± 0.1   | -0.2 ± 0.1    | 1.08 ± 0.08  | 1.4 ± 0.2    | 2.5 ± 0.2     | S E Asia F   | E Europe M   |
| 32  | 1.9 ± 0.3     | 1.78 ± 0.09 | -0.3 ± 0.1    | 0.52 ± 0.10 | 2.4 ± 0.2    | 1.9 ± 0.3     | 1.9 ± 0.2     | 1.2 ± 0.1   | -0.3 ± 0.1    | 1.02 ± 0.08  | 1.4 ± 0.2    | 2.4 ± 0.2     | W Africa M   | E Europe F   |
| 33  | 1.7 ± 0.3     | 1.72 ± 0.09 | -0.4 ± 0.1    | 0.3 ± 0.1   | 2.5 ± 0.1    | 1.7 ± 0.3     | 1.9 ± 0.2     | 1.2 ± 0.1   | -0.1 ± 0.2    | 0.83 ± 0.08  | 1.3 ± 0.2    | 2.3 ± 0.2     | S E Asia F   | E Europe F   |
| 34  | 1.4 ± 0.3     | 1.70 ± 0.09 | -0.4 ± 0.1    | 0.1 ± 0.1   | 2.2 ± 0.1    | 1.4 ± 0.3     | 1.7 ± 0.2     | 1.3 ± 0.1   | -0.3 ± 0.2    | 0.80 ± 0.09  | 1.5 ± 0.2    | 2.4 ± 0.2     | W Africa M   | E Europe F   |
| 35  | 1.6 ± 0.3     | 1.55 ± 0.09 | -0.4 ± 0.2    | -0.1 ± 0.1  | 2.0 ± 0.2    | 0.9 ± 0.3     | 1.8 ± 0.2     | 1.5 ± 0.1   | -0.1 ± 0.2    | 0.72 ± 0.09  | 1.6 ± 0.2    | 2.1 ± 0.2     | W Africa M   | E Europe F   |
| 36  | 1.2 ± 0.3     | 1.50 ± 0.09 | -0.3 ± 0.2    | -0.2 ± 0.1  | 2.0 ± 0.2    | 1.2 ± 0.3     | 1.6 ± 0.3     | 1.6 ± 0.1   | 0.1 ± 0.2     | 0.64 ± 0.09  | 1.6 ± 0.2    | 2.1 ± 0.2     | W Africa M   | E Europe F   |
| 37  | 1.2 ± 0.3     | 1.47 ± 0.09 | -0.4 ± 0.2    | -0.3 ± 0.1  | 1.9 ± 0.2    | 0.9 ± 0.3     | 1.7 ± 0.3     | 1.7 ± 0.1   | -0.1 ± 0.2    | 0.66 ± 0.10  | 1.5 ± 0.2    | 1.9 ± 0.2     | W Africa M   | E Europe F   |
| 38  | 1.5 ± 0.3     | 1.29 ± 0.09 | -0.4 ± 0.2    | -0.4 ± 0.1  | 1.8 ± 0.2    | 0.6 ± 0.3     | 1.4 ± 0.3     | 1.7 ± 0.1   | 0.0 ± 0.2     | 0.50 ± 0.10  | 1.8 ± 0.2    | 1.7 ± 0.2     | S E Asia M   | E Europe F   |
| 39  | 1.0 ± 0.3     | 1.20 ± 0.09 | -0.3 ± 0.2    | -0.6 ± 0.1  | 1.7 ± 0.1    | 0.5 ± 0.3     | 1.6 ± 0.3     | 1.7 ± 0.1   | 0.1 ± 0.2     | 0.56 ± 0.10  | 1.8 ± 0.2    | 1.6 ± 0.2     | S E Asia M   | S Asia F     |
| 40  | 0.9 ± 0.4     | 1.01 ± 0.09 | 0.0 ± 0.2     | -0.6 ± 0.1  | 1.5 ± 0.2    | 0.1 ± 0.3     | 1.5 ± 0.3     | 1.7 ± 0.1   | 0.1 ± 0.2     | 0.5 ± 0.1    | 1.6 ± 0.2    | 1.3 ± 0.2     | E Asia M     | S Asia F     |
| 41  | 0.4 ± 0.4     | 0.85 ± 0.09 | -0.2 ± 0.2    | -0.6 ± 0.1  | 1.3 ± 0.2    | -0.2 ± 0.3    | 1.5 ± 0.3     | 1.7 ± 0.1   | 0.3 ± 0.2     | 0.4 ± 0.1    | 1.7 ± 0.2    | 1.3 ± 0.2     | E Asia M     | S Asia F     |
| 42  | 0.9 ± 0.4     | 0.84 ± 0.09 | -0.1 ± 0.2    | -0.3 ± 0.2  | 1.2 ± 0.2    | 0.0 ± 0.3     | 1.2 ± 0.3     | 1.7 ± 0.1   | 0.3 ± 0.2     | 0.4 ± 0.1    | 1.8 ± 0.2    | 0.8 ± 0.2     | S E Asia M   | S Asia F     |
| 43  | 0.8 ± 0.4     | 0.57 ± 0.09 | -0.1 ± 0.2    | -0.6 ± 0.2  | 1.0 ± 0.2    | -0.7 ± 0.3    | 1.4 ± 0.3     | 1.7 ± 0.1   | 0.2 ± 0.2     | 0.4 ± 0.1    | 1.7 ± 0.2    | 0.6 ± 0.3     | S E Asia M   | W Africa F   |
| 44  | 0.9 ± 0.5     | 0.37 ± 0.09 | -0.4 ± 0.2    | -0.5 ± 0.2  | 1.3 ± 0.2    | -0.9 ± 0.4    | 0.9 ± 0.3     | 1.6 ± 0.1   | 0.5 ± 0.2     | 0.2 ± 0.1    | 1.9 ± 0.2    | 0.6 ± 0.2     | S E Asia M   | W Africa F   |
| 45  | 0.5 ± 0.5     | 0.40 ± 0.09 | -0.3 ± 0.2    | -0.6 ± 0.2  | 1.1 ± 0.2    | -1.0 ± 0.3    | 0.7 ± 0.4     | 1.6 ± 0.1   | 0.4 ± 0.2     | 0.4 ± 0.1    | 1.5 ± 0.2    | 0.2 ± 0.3     | E Asia M     | W Africa F   |
| 46  | 1.2 ± 0.5     | 0.18 ± 0.10 | -0.2 ± 0.2    | -0.4 ± 0.2  | 0.8 ± 0.2    | -1.1 ± 0.4    | 0.7 ± 0.4     | 1.4 ± 0.1   | 0.5 ± 0.2     | 0.5 ± 0.1    | 1.6 ± 0.2    | -0.2 ± 0.3    | S E Asia M   | W Africa F   |
| 47  | 0.7 ± 0.5     | 0.22 ± 0.10 | -0.1 ± 0.2    | -0.4 ± 0.2  | 1.0 ± 0.2    | -1.3 ± 0.4    | 1.1 ± 0.4     | 1.3 ± 0.1   | 0.6 ± 0.2     | 0.5 ± 0.2    | 1.9 ± 0.2    | -0.1 ± 0.3    | S E Asia M   | W Africa F   |
| 48  | 0.4 ± 0.6     | 0.1 ± 0.1   | 0.0 ± 0.2     | -0.4 ± 0.2  | 0.8 ± 0.2    | -1.5 ± 0.4    | 0.6 ± 0.5     | 1.2 ± 0.1   | 0.7 ± 0.3     | 0.5 ± 0.2    | 1.8 ± 0.2    | -0.5 ± 0.3    | S E Asia M   | W Africa F   |
| 49  | 0.5 ± 0.6     | -0.1 ± 0.1  | -0.1 ± 0.2    | -0.3 ± 0.2  | 1.1 ± 0.2    | -1.6 ± 0.5    | 0.2 ± 0.5     | 1.1 ± 0.1   | 0.6 ± 0.2     | 0.5 ± 0.2    | 1.6 ± 0.2    | -0.6 ± 0.3    | S E Asia M   | W Africa F   |
| 50  | 0.3 ± 0.6     | -0.1 ± 0.1  | 0.2 ± 0.2     | -0.2 ± 0.2  | 0.8 ± 0.2    | -2.2 ± 0.4    | 0.1 ± 0.4     | 1.0 ± 0.1   | 0.4 ± 0.3     | 0.5 ± 0.2    | 1.9 ± 0.2    | -1.2 ± 0.3    | S E Asia M   | W Africa F   |
| 51  | -0.1 ± 0.6    | -0.1 ± 0.1  | 0.1 ± 0.2     | -0.1 ± 0.2  | 0.8 ± 0.2    | -2.1 ± 0.5    | -0.3 ± 0.5    | 0.7 ± 0.1   | 0.8 ± 0.2     | 0.5 ± 0.2    | 1.6 ± 0.2    | -1.1 ± 0.3    | S E Asia M   | W Africa F   |
| 52  | -0.3 ± 0.7    | -0.1 ± 0.1  | 0.2 ± 0.2     | -0.4 ± 0.2  | 1.0 ± 0.2    | -2.2 ± 0.5    | -0.2 ± 0.5    | 0.5 ± 0.1   | 0.7 ± 0.3     | 0.4 ± 0.2    | 1.3 ± 0.2    | -1.3 ± 0.4    | S E Asia M   | W Africa F   |

**Table 69.** Mean raw error for Application-type images for algorithm yoti-003. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.7 ± 0.3     | 2.5 ± 0.1   | 1.0 ± 0.2     | 2.3 ± 0.2   | 2.8 ± 0.3    | 3.3 ± 0.3     | 0.6 ± 0.2     | 1.0 ± 0.1   | 0.1 ± 0.1     | 0.3 ± 0.1   | 0.7 ± 0.2    | 1.1 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 3.0 ± 0.3     | 2.8 ± 0.2   | 1.1 ± 0.2     | 2.8 ± 0.2   | 3.1 ± 0.4    | 3.0 ± 0.3     | 0.6 ± 0.2     | 1.2 ± 0.1   | 0.0 ± 0.1     | 0.5 ± 0.1   | 0.9 ± 0.2    | 1.5 ± 0.3     | S E Asia F   | E Europe M   |
| 16  | 2.9 ± 0.3     | 3.0 ± 0.2   | 1.1 ± 0.3     | 2.7 ± 0.2   | 3.4 ± 0.4    | 3.5 ± 0.5     | 1.1 ± 0.2     | 1.2 ± 0.1   | -0.1 ± 0.2    | 0.6 ± 0.1   | 1.1 ± 0.2    | 1.1 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 2.9 ± 0.4     | 3.2 ± 0.2   | 1.0 ± 0.3     | 3.0 ± 0.2   | 3.6 ± 0.4    | 3.1 ± 0.4     | 0.8 ± 0.2     | 1.3 ± 0.1   | -0.4 ± 0.1    | 0.8 ± 0.1   | 1.3 ± 0.2    | 1.5 ± 0.3     | S E Asia F   | E Europe M   |
| 18  | 2.0 ± 0.3     | 2.7 ± 0.2   | 0.5 ± 0.3     | 2.6 ± 0.2   | 3.4 ± 0.2    | 2.5 ± 0.4     | 0.6 ± 0.2     | 1.0 ± 0.1   | -0.7 ± 0.2    | 0.6 ± 0.1   | 1.1 ± 0.2    | 0.8 ± 0.3     | S E Asia F   | E Europe M   |
| 19  | 2.8 ± 0.4     | 3.0 ± 0.2   | 0.7 ± 0.3     | 3.0 ± 0.2   | 3.5 ± 0.2    | 2.9 ± 0.4     | 0.9 ± 0.3     | 1.4 ± 0.1   | -0.6 ± 0.2    | 1.0 ± 0.1   | 1.3 ± 0.2    | 1.3 ± 0.4     | S E Asia F   | E Europe M   |
| 20  | 2.7 ± 0.4     | 3.2 ± 0.1   | 0.7 ± 0.2     | 3.0 ± 0.2   | 3.9 ± 0.2    | 2.9 ± 0.4     | 1.2 ± 0.3     | 1.7 ± 0.1   | 0.1 ± 0.2     | 1.4 ± 0.1   | 1.7 ± 0.2    | 1.3 ± 0.4     | S E Asia F   | E Europe M   |
| 21  | 3.1 ± 0.3     | 3.3 ± 0.1   | 0.9 ± 0.2     | 3.1 ± 0.2   | 3.8 ± 0.2    | 3.1 ± 0.4     | 1.6 ± 0.3     | 2.0 ± 0.1   | 0.1 ± 0.2     | 1.8 ± 0.1   | 2.0 ± 0.2    | 1.5 ± 0.4     | S E Asia F   | E Europe M   |
| 22  | 3.1 ± 0.3     | 3.1 ± 0.1   | 0.9 ± 0.2     | 3.0 ± 0.1   | 3.7 ± 0.2    | 2.7 ± 0.3     | 2.3 ± 0.3     | 2.0 ± 0.1   | 0.7 ± 0.2     | 2.0 ± 0.1   | 2.0 ± 0.2    | 1.5 ± 0.3     | S E Asia F   | E Europe M   |
| 23  | 3.2 ± 0.3     | 3.1 ± 0.1   | 0.9 ± 0.2     | 2.9 ± 0.1   | 3.6 ± 0.1    | 3.0 ± 0.3     | 2.2 ± 0.3     | 1.9 ± 0.1   | 0.8 ± 0.2     | 2.2 ± 0.1   | 2.2 ± 0.2    | 2.3 ± 0.3     | S E Asia F   | E Europe M   |
| 24  | 3.3 ± 0.3     | 2.9 ± 0.1   | 1.0 ± 0.1     | 2.9 ± 0.1   | 3.4 ± 0.1    | 2.8 ± 0.3     | 2.4 ± 0.3     | 2.0 ± 0.1   | 0.9 ± 0.1     | 2.3 ± 0.1   | 2.2 ± 0.2    | 2.2 ± 0.3     | S E Asia F   | E Europe M   |
| 25  | 2.9 ± 0.3     | 2.83 ± 0.09 | 0.8 ± 0.1     | 2.53 ± 0.10 | 3.2 ± 0.1    | 2.9 ± 0.3     | 2.5 ± 0.3     | 1.9 ± 0.1   | 0.9 ± 0.1     | 2.3 ± 0.1   | 2.2 ± 0.2    | 2.5 ± 0.3     | S E Asia F   | E Europe F   |
| 26  | 2.9 ± 0.3     | 2.62 ± 0.09 | 0.7 ± 0.1     | 2.23 ± 0.09 | 3.2 ± 0.1    | 2.6 ± 0.3     | 2.4 ± 0.2     | 1.78 ± 0.10 | 0.6 ± 0.1     | 2.27 ± 0.10 | 2.2 ± 0.2    | 2.4 ± 0.2     | S E Asia F   | E Europe M   |
| 27  | 2.9 ± 0.3     | 2.46 ± 0.09 | 0.6 ± 0.1     | 2.06 ± 0.09 | 2.9 ± 0.1    | 2.8 ± 0.3     | 2.4 ± 0.2     | 1.7 ± 0.1   | 0.5 ± 0.1     | 2.11 ± 0.09 | 2.2 ± 0.2    | 2.5 ± 0.2     | S E Asia F   | E Europe M   |
| 28  | 2.6 ± 0.3     | 2.24 ± 0.08 | 0.36 ± 0.10   | 1.83 ± 0.09 | 2.7 ± 0.1    | 2.6 ± 0.3     | 2.1 ± 0.2     | 1.7 ± 0.1   | 0.4 ± 0.1     | 2.03 ± 0.09 | 2.2 ± 0.2    | 2.2 ± 0.2     | S E Asia F   | E Europe M   |
| 29  | 2.7 ± 0.3     | 2.07 ± 0.08 | 0.2 ± 0.1     | 1.54 ± 0.09 | 2.7 ± 0.1    | 2.4 ± 0.2     | 2.1 ± 0.2     | 1.5 ± 0.1   | 0.2 ± 0.1     | 1.83 ± 0.08 | 1.9 ± 0.2    | 2.7 ± 0.2     | E Africa F   | E Europe M   |
| 30  | 2.5 ± 0.3     | 1.94 ± 0.08 | -0.0 ± 0.1    | 1.37 ± 0.09 | 2.5 ± 0.1    | 2.1 ± 0.3     | 2.2 ± 0.2     | 1.41 ± 0.10 | 0.1 ± 0.1     | 1.75 ± 0.08 | 2.1 ± 0.2    | 2.5 ± 0.2     | E Africa F   | E Europe F   |
| 31  | 2.1 ± 0.3     | 1.76 ± 0.08 | 0.0 ± 0.1     | 1.13 ± 0.09 | 2.4 ± 0.1    | 2.1 ± 0.2     | 2.2 ± 0.2     | 1.4 ± 0.1   | 0.0 ± 0.1     | 1.65 ± 0.08 | 1.9 ± 0.2    | 2.3 ± 0.2     | S E Asia F   | E Europe M   |
| 32  | 2.3 ± 0.3     | 1.64 ± 0.09 | -0.2 ± 0.1    | 1.04 ± 0.09 | 2.2 ± 0.1    | 2.0 ± 0.3     | 1.9 ± 0.2     | 1.4 ± 0.1   | -0.1 ± 0.1    | 1.59 ± 0.08 | 1.8 ± 0.2    | 2.4 ± 0.2     | W Africa M   | E Europe F   |
| 33  | 2.1 ± 0.3     | 1.47 ± 0.09 | -0.2 ± 0.1    | 0.80 ± 0.10 | 2.2 ± 0.1    | 1.8 ± 0.3     | 1.9 ± 0.2     | 1.3 ± 0.1   | 0.1 ± 0.1     | 1.40 ± 0.08 | 1.7 ± 0.2    | 2.2 ± 0.2     | S E Asia F   | E Europe F   |
| 34  | 1.8 ± 0.3     | 1.40 ± 0.09 | -0.1 ± 0.1    | 0.73 ± 0.10 | 2.1 ± 0.1    | 1.8 ± 0.3     | 1.7 ± 0.2     | 1.4 ± 0.1   | -0.1 ± 0.1    | 1.33 ± 0.08 | 1.8 ± 0.2    | 2.3 ± 0.2     | W Africa M   | E Europe M   |
| 35  | 2.1 ± 0.3     | 1.22 ± 0.09 | -0.2 ± 0.1    | 0.45 ± 0.10 | 1.8 ± 0.1    | 1.4 ± 0.3     | 1.8 ± 0.2     | 1.5 ± 0.1   | 0.0 ± 0.2     | 1.24 ± 0.08 | 1.8 ± 0.2    | 2.0 ± 0.2     | E Africa F   | E Europe F   |
| 36  | 1.5 ± 0.3     | 1.16 ± 0.09 | -0.0 ± 0.1    | 0.39 ± 0.10 | 1.7 ± 0.2    | 1.6 ± 0.3     | 1.6 ± 0.2     | 1.5 ± 0.1   | 0.2 ± 0.2     | 1.09 ± 0.09 | 1.8 ± 0.2    | 2.1 ± 0.2     | W Africa M   | E Europe F   |
| 37  | 1.6 ± 0.3     | 1.05 ± 0.09 | -0.0 ± 0.2    | 0.3 ± 0.1   | 1.6 ± 0.1    | 1.3 ± 0.3     | 1.5 ± 0.3     | 1.6 ± 0.1   | 0.1 ± 0.2     | 1.04 ± 0.09 | 1.7 ± 0.2    | 1.8 ± 0.2     | W Africa M   | E Europe F   |
| 38  | 1.8 ± 0.3     | 0.86 ± 0.09 | -0.1 ± 0.2    | 0.3 ± 0.1   | 1.5 ± 0.1    | 1.3 ± 0.3     | 1.3 ± 0.3     | 1.6 ± 0.1   | 0.1 ± 0.2     | 0.86 ± 0.09 | 2.0 ± 0.2    | 1.7 ± 0.2     | S E Asia M   | E Europe F   |
| 39  | 1.4 ± 0.3     | 0.81 ± 0.09 | -0.0 ± 0.2    | 0.1 ± 0.1   | 1.4 ± 0.1    | 1.1 ± 0.3     | 1.6 ± 0.3     | 1.6 ± 0.1   | 0.2 ± 0.2     | 0.83 ± 0.09 | 1.8 ± 0.2    | 1.6 ± 0.2     | S E Asia M   | E Europe F   |
| 40  | 1.3 ± 0.3     | 0.67 ± 0.09 | 0.3 ± 0.2     | 0.1 ± 0.1   | 1.2 ± 0.2    | 1.0 ± 0.3     | 1.3 ± 0.3     | 1.7 ± 0.1   | 0.2 ± 0.2     | 0.76 ± 0.10 | 1.9 ± 0.2    | 1.3 ± 0.2     | S E Asia M   | S Asia F     |
| 41  | 1.0 ± 0.4     | 0.57 ± 0.09 | 0.1 ± 0.2     | -0.0 ± 0.1  | 1.1 ± 0.2    | 0.5 ± 0.3     | 1.4 ± 0.3     | 1.7 ± 0.1   | 0.3 ± 0.2     | 0.65 ± 0.10 | 1.9 ± 0.2    | 1.4 ± 0.2     | S E Asia M   | S Asia F     |
| 42  | 1.3 ± 0.4     | 0.62 ± 0.09 | 0.2 ± 0.2     | 0.1 ± 0.2   | 1.0 ± 0.2    | 0.6 ± 0.3     | 1.3 ± 0.3     | 1.7 ± 0.1   | 0.3 ± 0.2     | 0.6 ± 0.1   | 2.0 ± 0.2    | 0.9 ± 0.2     | S E Asia M   | S Asia F     |
| 43  | 1.1 ± 0.4     | 0.38 ± 0.09 | 0.2 ± 0.2     | -0.1 ± 0.2  | 0.9 ± 0.2    | 0.1 ± 0.3     | 1.3 ± 0.3     | 1.7 ± 0.1   | 0.3 ± 0.2     | 0.6 ± 0.1   | 1.9 ± 0.2    | 0.7 ± 0.3     | S E Asia M   | S Asia F     |
| 44  | 1.1 ± 0.4     | 0.26 ± 0.09 | -0.1 ± 0.2    | 0.0 ± 0.2   | 1.1 ± 0.2    | -0.0 ± 0.4    | 0.9 ± 0.3     | 1.7 ± 0.1   | 0.6 ± 0.2     | 0.4 ± 0.1   | 2.1 ± 0.2    | 0.8 ± 0.3     | S E Asia M   | E Europe F   |
| 45  | 1.0 ± 0.4     | 0.38 ± 0.10 | 0.0 ± 0.2     | -0.0 ± 0.2  | 1.0 ± 0.2    | -0.1 ± 0.3    | 0.7 ± 0.4     | 1.7 ± 0.1   | 0.4 ± 0.2     | 0.5 ± 0.1   | 1.7 ± 0.2    | 0.4 ± 0.3     | S E Asia M   | W Africa F   |
| 46  | 1.3 ± 0.5     | 0.17 ± 0.10 | 0.1 ± 0.2     | 0.0 ± 0.2   | 0.8 ± 0.2    | -0.4 ± 0.4    | 0.7 ± 0.4     | 1.6 ± 0.1   | 0.5 ± 0.2     | 0.5 ± 0.1   | 2.0 ± 0.2    | 0.2 ± 0.3     | S E Asia M   | W Africa F   |
| 47  | 1.1 ± 0.5     | 0.2 ± 0.1   | 0.2 ± 0.2     | -0.0 ± 0.2  | 0.9 ± 0.2    | -0.5 ± 0.4    | 1.1 ± 0.4     | 1.5 ± 0.1   | 0.7 ± 0.2     | 0.5 ± 0.2   | 2.1 ± 0.2    | 0.4 ± 0.3     | S E Asia M   | W Africa F   |
| 48  | 0.6 ± 0.5     | 0.14 ± 0.10 | 0.3 ± 0.2     | 0.1 ± 0.2   | 0.8 ± 0.2    | -0.4 ± 0.4    | 0.6 ± 0.4     | 1.4 ± 0.1   | 0.7 ± 0.3     | 0.5 ± 0.2   | 2.1 ± 0.2    | 0.1 ± 0.3     | S E Asia M   | W Africa F   |
| 49  | 0.9 ± 0.5     | 0.0 ± 0.1   | 0.3 ± 0.2     | 0.2 ± 0.2   | 1.0 ± 0.2    | -0.5 ± 0.4    | 0.1 ± 0.5     | 1.3 ± 0.1   | 0.7 ± 0.2     | 0.5 ± 0.2   | 1.8 ± 0.2    | 0.2 ± 0.3     | S E Asia M   | W Africa F   |
| 50  | 0.8 ± 0.6     | 0.0 ± 0.1   | 0.5 ± 0.2     | 0.1 ± 0.2   | 0.8 ± 0.2    | -1.1 ± 0.4    | 0.3 ± 0.4     | 1.2 ± 0.1   | 0.4 ± 0.2     | 0.4 ± 0.2   | 2.1 ± 0.2    | -0.5 ± 0.3    | S E Asia M   | W Africa F   |
| 51  | 0.4 ± 0.5     | -0.0 ± 0.1  | 0.4 ± 0.2     | 0.2 ± 0.2   | 0.8 ± 0.2    | -1.0 ± 0.4    | -0.1 ± 0.5    | 1.0 ± 0.1   | 0.9 ± 0.2     | 0.5 ± 0.2   | 1.9 ± 0.2    | -0.2 ± 0.3    | S E Asia M   | W Africa F   |
| 52  | 0.5 ± 0.6     | -0.1 ± 0.1  | 0.5 ± 0.2     | 0.0 ± 0.2   | 1.1 ± 0.2    | -1.2 ± 0.4    | -0.2 ± 0.5    | 0.8 ± 0.1   | 0.7 ± 0.3     | 0.4 ± 0.2   | 1.6 ± 0.2    | -0.5 ± 0.4    | S E Asia M   | W Africa F   |

**Table 70.** Mean raw error for Application-type images for algorithm yoti-004. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia  | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|--------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.6 ± 0.3     | 2.3 ± 0.1    | 1.2 ± 0.2     | 2.2 ± 0.2    | 2.5 ± 0.3    | 3.3 ± 0.3     | 1.0 ± 0.2     | 0.59 ± 0.09 | 0.2 ± 0.1     | 0.50 ± 0.10 | 0.7 ± 0.2    | 1.5 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 2.8 ± 0.3     | 2.7 ± 0.2    | 1.3 ± 0.2     | 2.9 ± 0.2    | 3.1 ± 0.3    | 3.2 ± 0.4     | 1.0 ± 0.3     | 0.75 ± 0.10 | 0.2 ± 0.1     | 0.7 ± 0.1   | 1.0 ± 0.2    | 1.9 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 2.6 ± 0.3     | 3.0 ± 0.2    | 1.4 ± 0.2     | 3.0 ± 0.2    | 3.4 ± 0.3    | 3.5 ± 0.4     | 1.7 ± 0.2     | 1.0 ± 0.1   | 0.2 ± 0.2     | 1.0 ± 0.1   | 1.2 ± 0.2    | 1.8 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 2.7 ± 0.3     | 3.2 ± 0.2    | 1.4 ± 0.2     | 3.1 ± 0.2    | 3.5 ± 0.3    | 3.1 ± 0.3     | 1.5 ± 0.2     | 1.2 ± 0.1   | 0.0 ± 0.2     | 1.1 ± 0.1   | 1.5 ± 0.2    | 2.2 ± 0.3     | S E Asia F   | E Europe M   |
| 18  | 2.2 ± 0.3     | 2.8 ± 0.1    | 1.0 ± 0.3     | 2.8 ± 0.1    | 3.5 ± 0.2    | 2.6 ± 0.3     | 1.3 ± 0.2     | 1.1 ± 0.1   | -0.2 ± 0.2    | 1.1 ± 0.1   | 1.5 ± 0.2    | 1.5 ± 0.3     | S E Asia F   | E Europe M   |
| 19  | 2.5 ± 0.3     | 2.9 ± 0.1    | 1.2 ± 0.3     | 2.9 ± 0.1    | 3.4 ± 0.2    | 2.9 ± 0.3     | 1.5 ± 0.2     | 1.4 ± 0.1   | -0.0 ± 0.2    | 1.4 ± 0.1   | 1.6 ± 0.2    | 1.9 ± 0.3     | S E Asia F   | E Europe M   |
| 20  | 2.2 ± 0.3     | 2.9 ± 0.1    | 1.0 ± 0.2     | 2.7 ± 0.2    | 3.4 ± 0.2    | 2.8 ± 0.3     | 1.7 ± 0.2     | 1.5 ± 0.1   | 0.5 ± 0.2     | 1.6 ± 0.1   | 1.6 ± 0.2    | 1.8 ± 0.3     | S E Asia F   | E Europe M   |
| 21  | 2.2 ± 0.3     | 2.8 ± 0.1    | 0.8 ± 0.2     | 2.5 ± 0.1    | 3.1 ± 0.1    | 2.7 ± 0.3     | 1.8 ± 0.2     | 1.48 ± 0.10 | 0.4 ± 0.2     | 1.6 ± 0.1   | 1.7 ± 0.2    | 1.8 ± 0.3     | S E Asia F   | E Europe M   |
| 22  | 2.2 ± 0.3     | 2.4 ± 0.1    | 0.7 ± 0.1     | 2.2 ± 0.1    | 2.9 ± 0.1    | 2.3 ± 0.3     | 2.1 ± 0.2     | 1.39 ± 0.09 | 0.5 ± 0.2     | 1.6 ± 0.1   | 1.5 ± 0.1    | 1.5 ± 0.3     | S E Asia F   | E Europe M   |
| 23  | 2.0 ± 0.2     | 2.18 ± 0.09  | 0.4 ± 0.1     | 2.0 ± 0.1    | 2.6 ± 0.1    | 2.3 ± 0.3     | 1.7 ± 0.3     | 1.13 ± 0.09 | 0.4 ± 0.1     | 1.5 ± 0.1   | 1.4 ± 0.1    | 1.9 ± 0.3     | S E Asia F   | E Europe M   |
| 24  | 1.9 ± 0.3     | 1.82 ± 0.08  | 0.3 ± 0.1     | 1.67 ± 0.10  | 2.3 ± 0.1    | 1.8 ± 0.3     | 1.7 ± 0.2     | 0.97 ± 0.09 | 0.3 ± 0.1     | 1.38 ± 0.10 | 1.3 ± 0.1    | 1.7 ± 0.2     | S E Asia F   | E Europe M   |
| 25  | 1.5 ± 0.2     | 1.59 ± 0.08  | 0.01 ± 0.09   | 1.31 ± 0.09  | 2.0 ± 0.1    | 1.9 ± 0.2     | 1.6 ± 0.2     | 0.77 ± 0.09 | 0.0 ± 0.1     | 1.23 ± 0.09 | 1.1 ± 0.1    | 1.7 ± 0.2     | S E Asia F   | E Europe F   |
| 26  | 1.3 ± 0.2     | 1.30 ± 0.08  | -0.27 ± 0.09  | 0.95 ± 0.09  | 1.9 ± 0.1    | 1.6 ± 0.2     | 1.4 ± 0.2     | 0.63 ± 0.09 | -0.2 ± 0.1    | 1.08 ± 0.09 | 1.0 ± 0.1    | 1.5 ± 0.2     | S E Asia F   | E Europe F   |
| 27  | 1.3 ± 0.2     | 1.11 ± 0.08  | -0.50 ± 0.09  | 0.71 ± 0.08  | 1.6 ± 0.1    | 1.7 ± 0.2     | 1.3 ± 0.2     | 0.44 ± 0.09 | -0.4 ± 0.1    | 0.82 ± 0.08 | 1.0 ± 0.1    | 1.4 ± 0.2     | W Africa F   | E Europe F   |
| 28  | 1.0 ± 0.2     | 0.87 ± 0.08  | -0.74 ± 0.09  | 0.49 ± 0.09  | 1.4 ± 0.1    | 1.5 ± 0.2     | 0.9 ± 0.2     | 0.4 ± 0.1   | -0.6 ± 0.1    | 0.73 ± 0.08 | 0.9 ± 0.2    | 1.2 ± 0.2     | W Africa F   | E Europe F   |
| 29  | 1.2 ± 0.2     | 0.69 ± 0.08  | -0.96 ± 0.10  | 0.23 ± 0.09  | 1.4 ± 0.1    | 1.2 ± 0.2     | 0.9 ± 0.2     | 0.3 ± 0.1   | -0.8 ± 0.1    | 0.56 ± 0.08 | 0.5 ± 0.2    | 1.6 ± 0.2     | W Africa M   | E Europe F   |
| 30  | 1.0 ± 0.2     | 0.60 ± 0.08  | -1.2 ± 0.1    | 0.06 ± 0.09  | 1.3 ± 0.1    | 1.0 ± 0.2     | 1.0 ± 0.2     | 0.2 ± 0.1   | -0.9 ± 0.1    | 0.55 ± 0.08 | 0.8 ± 0.2    | 1.4 ± 0.2     | W Africa M   | E Europe F   |
| 31  | 0.5 ± 0.2     | 0.46 ± 0.08  | -1.2 ± 0.1    | -0.10 ± 0.09 | 1.2 ± 0.1    | 0.9 ± 0.2     | 0.9 ± 0.2     | 0.2 ± 0.1   | -0.9 ± 0.1    | 0.48 ± 0.08 | 0.7 ± 0.2    | 1.3 ± 0.2     | W Africa M   | E Europe F   |
| 32  | 0.7 ± 0.3     | 0.41 ± 0.09  | -1.3 ± 0.1    | -0.13 ± 0.10 | 1.1 ± 0.1    | 0.8 ± 0.2     | 0.7 ± 0.2     | 0.3 ± 0.1   | -0.9 ± 0.2    | 0.49 ± 0.08 | 0.7 ± 0.2    | 1.2 ± 0.2     | W Africa M   | E Europe F   |
| 33  | 0.5 ± 0.3     | 0.35 ± 0.09  | -1.2 ± 0.1    | -0.3 ± 0.1   | 1.2 ± 0.1    | 0.7 ± 0.3     | 0.9 ± 0.2     | 0.3 ± 0.1   | -0.6 ± 0.2    | 0.41 ± 0.09 | 0.5 ± 0.2    | 1.2 ± 0.2     | W Africa M   | E Europe F   |
| 34  | 0.3 ± 0.3     | 0.37 ± 0.09  | -1.1 ± 0.2    | -0.2 ± 0.1   | 1.0 ± 0.1    | 0.6 ± 0.3     | 0.6 ± 0.2     | 0.5 ± 0.1   | -0.6 ± 0.2    | 0.48 ± 0.09 | 0.8 ± 0.2    | 1.3 ± 0.2     | W Africa M   | E Europe F   |
| 35  | 0.6 ± 0.3     | 0.25 ± 0.09  | -1.0 ± 0.2    | -0.4 ± 0.1   | 0.9 ± 0.2    | 0.0 ± 0.3     | 0.8 ± 0.2     | 0.7 ± 0.1   | -0.3 ± 0.2    | 0.51 ± 0.09 | 0.9 ± 0.2    | 1.1 ± 0.2     | W Africa M   | E Europe F   |
| 36  | 0.2 ± 0.3     | 0.26 ± 0.10  | -0.7 ± 0.2    | -0.4 ± 0.1   | 0.9 ± 0.2    | 0.3 ± 0.3     | 0.6 ± 0.3     | 0.7 ± 0.1   | -0.1 ± 0.2    | 0.46 ± 0.09 | 0.9 ± 0.2    | 1.1 ± 0.2     | W Africa M   | E Europe F   |
| 37  | 0.3 ± 0.3     | 0.36 ± 0.09  | -0.6 ± 0.2    | -0.4 ± 0.1   | 0.8 ± 0.2    | -0.0 ± 0.3    | 0.7 ± 0.3     | 0.9 ± 0.1   | -0.1 ± 0.2    | 0.54 ± 0.09 | 0.9 ± 0.2    | 0.9 ± 0.2     | S E Asia M   | E Europe F   |
| 38  | 0.4 ± 0.3     | 0.24 ± 0.09  | -0.6 ± 0.2    | -0.3 ± 0.1   | 0.9 ± 0.2    | -0.2 ± 0.3    | 0.4 ± 0.3     | 0.8 ± 0.1   | -0.0 ± 0.2    | 0.43 ± 0.09 | 1.3 ± 0.2    | 0.7 ± 0.2     | S E Asia M   | E Europe F   |
| 39  | -0.0 ± 0.3    | 0.29 ± 0.09  | -0.4 ± 0.2    | -0.5 ± 0.1   | 0.9 ± 0.2    | -0.2 ± 0.3    | 0.7 ± 0.3     | 0.9 ± 0.1   | 0.1 ± 0.2     | 0.47 ± 0.09 | 1.2 ± 0.2    | 0.6 ± 0.2     | S E Asia M   | S Asia F     |
| 40  | -0.3 ± 0.4    | 0.17 ± 0.09  | -0.1 ± 0.2    | -0.4 ± 0.1   | 0.7 ± 0.2    | -0.5 ± 0.3    | 0.6 ± 0.3     | 1.0 ± 0.1   | 0.2 ± 0.2     | 0.4 ± 0.1   | 1.3 ± 0.2    | 0.4 ± 0.2     | S E Asia M   | W Africa F   |
| 41  | -0.6 ± 0.4    | 0.12 ± 0.09  | -0.2 ± 0.2    | -0.5 ± 0.1   | 0.6 ± 0.2    | -1.0 ± 0.3    | 0.7 ± 0.3     | 1.0 ± 0.1   | 0.3 ± 0.2     | 0.4 ± 0.1   | 1.3 ± 0.2    | 0.4 ± 0.2     | S E Asia M   | W Africa F   |
| 42  | -0.3 ± 0.4    | 0.20 ± 0.09  | -0.0 ± 0.2    | -0.3 ± 0.2   | 0.5 ± 0.2    | -0.8 ± 0.3    | 0.5 ± 0.3     | 1.0 ± 0.1   | 0.5 ± 0.2     | 0.3 ± 0.1   | 1.4 ± 0.2    | 0.1 ± 0.2     | S E Asia M   | W Africa F   |
| 43  | -0.5 ± 0.4    | 0.00 ± 0.09  | -0.2 ± 0.2    | -0.5 ± 0.2   | 0.5 ± 0.2    | -1.5 ± 0.3    | 0.6 ± 0.3     | 1.1 ± 0.1   | 0.3 ± 0.2     | 0.3 ± 0.1   | 1.5 ± 0.2    | -0.3 ± 0.3    | S E Asia M   | W Africa F   |
| 44  | -0.3 ± 0.5    | -0.11 ± 0.09 | -0.4 ± 0.2    | -0.4 ± 0.2   | 0.7 ± 0.2    | -1.6 ± 0.4    | 0.2 ± 0.3     | 1.1 ± 0.1   | 0.6 ± 0.2     | 0.2 ± 0.1   | 1.7 ± 0.2    | -0.0 ± 0.3    | S E Asia M   | W Africa F   |
| 45  | -0.6 ± 0.5    | -0.05 ± 0.09 | -0.2 ± 0.2    | -0.6 ± 0.2   | 0.5 ± 0.2    | -1.8 ± 0.3    | 0.1 ± 0.3     | 1.1 ± 0.1   | 0.4 ± 0.2     | 0.3 ± 0.1   | 1.3 ± 0.2    | -0.5 ± 0.3    | S E Asia M   | W Africa F   |
| 46  | -0.1 ± 0.5    | -0.24 ± 0.09 | -0.2 ± 0.2    | -0.5 ± 0.2   | 0.3 ± 0.2    | -1.8 ± 0.4    | -0.0 ± 0.4    | 1.1 ± 0.1   | 0.4 ± 0.2     | 0.4 ± 0.1   | 1.4 ± 0.2    | -0.8 ± 0.3    | S E Asia M   | W Africa F   |
| 47  | -0.4 ± 0.6    | -0.19 ± 0.09 | -0.1 ± 0.2    | -0.6 ± 0.2   | 0.3 ± 0.2    | -2.1 ± 0.4    | 0.4 ± 0.4     | 1.0 ± 0.1   | 0.5 ± 0.2     | 0.4 ± 0.2   | 1.6 ± 0.2    | -0.5 ± 0.3    | S E Asia M   | W Africa F   |
| 48  | -0.8 ± 0.6    | -0.30 ± 0.09 | -0.0 ± 0.2    | -0.4 ± 0.2   | 0.3 ± 0.2    | -2.0 ± 0.4    | -0.1 ± 0.4    | 0.9 ± 0.1   | 0.6 ± 0.2     | 0.3 ± 0.2   | 1.6 ± 0.2    | -0.9 ± 0.3    | S E Asia M   | W Africa F   |
| 49  | -0.5 ± 0.6    | -0.5 ± 0.1   | -0.1 ± 0.2    | -0.4 ± 0.2   | 0.5 ± 0.2    | -2.1 ± 0.4    | -0.5 ± 0.4    | 0.8 ± 0.1   | 0.5 ± 0.2     | 0.3 ± 0.2   | 1.4 ± 0.2    | -0.9 ± 0.3    | S E Asia M   | W Africa F   |
| 50  | -0.9 ± 0.7    | -0.44 ± 0.10 | 0.1 ± 0.2     | -0.4 ± 0.2   | 0.3 ± 0.2    | -2.7 ± 0.4    | -0.5 ± 0.4    | 0.7 ± 0.1   | 0.2 ± 0.2     | 0.3 ± 0.2   | 1.5 ± 0.2    | -1.3 ± 0.3    | S E Asia M   | W Africa F   |
| 51  | -1.3 ± 0.6    | -0.5 ± 0.1   | 0.0 ± 0.2     | -0.3 ± 0.2   | 0.2 ± 0.2    | -2.8 ± 0.5    | -0.9 ± 0.5    | 0.5 ± 0.1   | 0.5 ± 0.2     | 0.4 ± 0.2   | 1.4 ± 0.2    | -1.4 ± 0.3    | S E Asia M   | W Africa F   |
| 52  | -1.0 ± 0.6    | -0.6 ± 0.1   | 0.1 ± 0.2     | -0.5 ± 0.2   | 0.5 ± 0.2    | -2.8 ± 0.5    | -0.7 ± 0.5    | 0.3 ± 0.1   | 0.5 ± 0.3     | 0.3 ± 0.2   | 1.2 ± 0.2    | -1.4 ± 0.4    | S E Asia M   | W Africa F   |

**Table 71.** Mean raw error for Application-type images for algorithm youverse-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia  | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia  | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|--------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|--------------|--------------|---------------|--------------|--------------|
| 14  | 3.5 ± 0.3     | 2.7 ± 0.1    | 3.9 ± 0.3     | 3.9 ± 0.2   | 2.4 ± 0.2    | 4.2 ± 0.5     | 2.7 ± 0.3     | 2.40 ± 0.08 | 2.4 ± 0.2     | 2.9 ± 0.1    | 2.3 ± 0.2    | 2.9 ± 0.4     | W Africa F   | S E Asia M   |
| 15  | 3.6 ± 0.4     | 2.5 ± 0.1    | 4.0 ± 0.3     | 3.9 ± 0.2   | 2.1 ± 0.3    | 4.5 ± 0.5     | 2.5 ± 0.3     | 2.04 ± 0.09 | 2.3 ± 0.2     | 2.7 ± 0.1    | 2.2 ± 0.2    | 3.4 ± 0.4     | W Africa F   | E Asia M     |
| 16  | 3.7 ± 0.4     | 2.0 ± 0.1    | 4.1 ± 0.3     | 3.3 ± 0.2   | 2.0 ± 0.3    | 3.9 ± 0.5     | 3.0 ± 0.3     | 1.6 ± 0.1   | 2.4 ± 0.3     | 2.7 ± 0.2    | 2.2 ± 0.2    | 3.1 ± 0.5     | E Europe F   | E Asia M     |
| 17  | 3.8 ± 0.5     | 1.5 ± 0.1    | 3.8 ± 0.3     | 2.9 ± 0.2   | 1.3 ± 0.3    | 4.0 ± 0.5     | 3.2 ± 0.4     | 1.3 ± 0.1   | 2.0 ± 0.3     | 2.9 ± 0.2    | 1.8 ± 0.3    | 3.5 ± 0.5     | W Africa F   | E Asia M     |
| 18  | 3.0 ± 0.3     | 1.0 ± 0.1    | 3.2 ± 0.3     | 2.7 ± 0.2   | 1.3 ± 0.2    | 3.2 ± 0.4     | 3.6 ± 0.4     | 0.7 ± 0.1   | 1.8 ± 0.2     | 2.7 ± 0.1    | 1.3 ± 0.2    | 3.6 ± 0.4     | E Africa M   | E Asia M     |
| 19  | 2.8 ± 0.4     | 0.4 ± 0.1    | 2.6 ± 0.3     | 2.3 ± 0.2   | 0.6 ± 0.2    | 3.3 ± 0.4     | 3.5 ± 0.3     | 0.5 ± 0.1   | 1.7 ± 0.3     | 2.9 ± 0.2    | 0.9 ± 0.2    | 3.4 ± 0.4     | E Africa M   | E Asia F     |
| 20  | 2.7 ± 0.4     | -0.0 ± 0.1   | 2.8 ± 0.2     | 1.8 ± 0.2   | -0.1 ± 0.2   | 3.1 ± 0.4     | 3.7 ± 0.4     | 0.1 ± 0.1   | 1.7 ± 0.3     | 2.8 ± 0.2    | 0.8 ± 0.2    | 3.7 ± 0.4     | E Africa M   | S E Asia F   |
| 21  | 2.1 ± 0.3     | -0.6 ± 0.1   | 2.1 ± 0.2     | 1.4 ± 0.1   | -0.7 ± 0.2   | 2.5 ± 0.4     | 3.7 ± 0.3     | -0.2 ± 0.1  | 1.5 ± 0.3     | 2.5 ± 0.2    | 0.6 ± 0.2    | 4.0 ± 0.4     | W Africa M   | S E Asia F   |
| 22  | 1.7 ± 0.3     | -1.1 ± 0.1   | 2.0 ± 0.2     | 0.9 ± 0.1   | -1.3 ± 0.2   | 2.1 ± 0.4     | 3.8 ± 0.3     | -0.5 ± 0.1  | 1.6 ± 0.3     | 2.5 ± 0.2    | 0.1 ± 0.2    | 3.5 ± 0.3     | E Africa M   | S E Asia F   |
| 23  | 1.2 ± 0.3     | -1.6 ± 0.1   | 1.6 ± 0.2     | 0.7 ± 0.1   | -1.8 ± 0.1   | 1.9 ± 0.3     | 3.6 ± 0.3     | -0.7 ± 0.1  | 1.5 ± 0.3     | 2.3 ± 0.2    | -0.2 ± 0.2   | 3.2 ± 0.3     | E Africa M   | S E Asia F   |
| 24  | 0.8 ± 0.3     | -2.3 ± 0.1   | 1.4 ± 0.2     | 0.2 ± 0.1   | -2.2 ± 0.1   | 1.1 ± 0.3     | 3.2 ± 0.3     | -1.0 ± 0.1  | 1.5 ± 0.2     | 2.1 ± 0.1    | -0.4 ± 0.2   | 3.1 ± 0.3     | E Africa M   | E Asia F     |
| 25  | 0.5 ± 0.3     | -2.8 ± 0.1   | 1.0 ± 0.1     | -0.2 ± 0.1  | -2.8 ± 0.1   | 0.8 ± 0.3     | 2.9 ± 0.3     | -1.2 ± 0.1  | 1.3 ± 0.2     | 1.9 ± 0.1    | -0.7 ± 0.2   | 2.6 ± 0.3     | E Africa M   | E Asia F     |
| 26  | -0.2 ± 0.3    | -3.4 ± 0.1   | 0.7 ± 0.1     | -0.7 ± 0.1  | -3.3 ± 0.1   | 0.0 ± 0.3     | 2.7 ± 0.3     | -1.7 ± 0.1  | 0.9 ± 0.2     | 1.5 ± 0.1    | -0.8 ± 0.2   | 2.5 ± 0.2     | E Africa M   | E Asia F     |
| 27  | -0.5 ± 0.3    | -3.93 ± 0.10 | 0.4 ± 0.1     | -1.3 ± 0.1  | -3.6 ± 0.1   | -0.5 ± 0.2    | 2.2 ± 0.2     | -1.9 ± 0.1  | 1.0 ± 0.2     | 1.2 ± 0.1    | -1.2 ± 0.2   | 2.2 ± 0.2     | W Africa M   | E Asia F     |
| 28  | -0.9 ± 0.3    | -4.41 ± 0.10 | 0.2 ± 0.1     | -1.7 ± 0.1  | -4.2 ± 0.1   | -0.9 ± 0.3    | 1.8 ± 0.2     | -2.1 ± 0.1  | 0.9 ± 0.2     | 0.6 ± 0.1    | -1.5 ± 0.2   | 1.5 ± 0.2     | E Africa M   | E Asia F     |
| 29  | -1.2 ± 0.3    | -4.82 ± 0.10 | -0.1 ± 0.1    | -2.1 ± 0.1  | -4.6 ± 0.1   | -1.6 ± 0.3    | 1.5 ± 0.2     | -2.4 ± 0.1  | 0.7 ± 0.2     | 0.2 ± 0.1    | -1.6 ± 0.2   | 1.3 ± 0.2     | E Africa M   | E Asia F     |
| 30  | -1.8 ± 0.3    | -5.36 ± 0.10 | -0.3 ± 0.2    | -2.6 ± 0.1  | -4.9 ± 0.1   | -2.1 ± 0.3    | 0.9 ± 0.2     | -2.7 ± 0.1  | 0.4 ± 0.2     | -0.27 ± 0.10 | -2.0 ± 0.2   | 1.0 ± 0.2     | W Africa M   | E Asia F     |
| 31  | -2.2 ± 0.3    | -5.7 ± 0.1   | -0.4 ± 0.2    | -3.1 ± 0.1  | -5.5 ± 0.2   | -2.6 ± 0.3    | 0.8 ± 0.2     | -2.9 ± 0.1  | 0.3 ± 0.2     | -0.6 ± 0.1   | -2.4 ± 0.3   | 0.5 ± 0.2     | E Africa M   | E Asia F     |
| 32  | -2.3 ± 0.3    | -6.1 ± 0.1   | -0.7 ± 0.2    | -3.4 ± 0.1  | -5.9 ± 0.2   | -3.1 ± 0.3    | 0.3 ± 0.2     | -3.2 ± 0.1  | 0.2 ± 0.2     | -1.0 ± 0.1   | -2.4 ± 0.3   | 0.2 ± 0.2     | E Africa M   | E Asia F     |
| 33  | -3.2 ± 0.3    | -6.4 ± 0.1   | -0.8 ± 0.2    | -3.7 ± 0.1  | -6.3 ± 0.2   | -3.4 ± 0.3    | -0.0 ± 0.3    | -3.5 ± 0.2  | 0.1 ± 0.2     | -1.45 ± 0.10 | -2.9 ± 0.3   | -0.2 ± 0.2    | E Europe M   | E Asia F     |
| 34  | -3.6 ± 0.3    | -6.8 ± 0.1   | -1.1 ± 0.2    | -4.0 ± 0.1  | -6.8 ± 0.2   | -4.0 ± 0.3    | -0.5 ± 0.3    | -3.9 ± 0.1  | -0.3 ± 0.2    | -1.8 ± 0.1   | -3.0 ± 0.3   | -0.6 ± 0.2    | E Europe M   | E Asia F     |
| 35  | -3.9 ± 0.3    | -7.3 ± 0.1   | -1.3 ± 0.2    | -4.5 ± 0.1  | -7.2 ± 0.2   | -4.6 ± 0.3    | -0.8 ± 0.3    | -4.0 ± 0.2  | -0.5 ± 0.2    | -2.1 ± 0.1   | -3.2 ± 0.3   | -1.1 ± 0.2    | E Europe M   | E Asia F     |
| 36  | -4.3 ± 0.3    | -7.7 ± 0.1   | -1.4 ± 0.2    | -4.8 ± 0.1  | -7.3 ± 0.2   | -5.0 ± 0.3    | -1.4 ± 0.3    | -4.2 ± 0.2  | -0.3 ± 0.2    | -2.4 ± 0.1   | -3.4 ± 0.3   | -1.8 ± 0.2    | E Europe M   | E Asia F     |
| 37  | -4.6 ± 0.3    | -8.0 ± 0.1   | -1.6 ± 0.2    | -5.0 ± 0.1  | -7.8 ± 0.2   | -5.5 ± 0.3    | -1.5 ± 0.3    | -4.5 ± 0.2  | -0.8 ± 0.2    | -2.6 ± 0.1   | -3.6 ± 0.3   | -2.3 ± 0.2    | E Europe M   | E Asia F     |
| 38  | -4.9 ± 0.4    | -8.3 ± 0.1   | -1.6 ± 0.3    | -5.3 ± 0.2  | -8.0 ± 0.2   | -5.8 ± 0.3    | -2.2 ± 0.3    | -4.6 ± 0.2  | -0.9 ± 0.2    | -3.0 ± 0.1   | -3.5 ± 0.3   | -2.4 ± 0.3    | E Europe M   | E Asia F     |
| 39  | -5.1 ± 0.4    | -8.6 ± 0.1   | -2.1 ± 0.3    | -5.8 ± 0.2  | -8.2 ± 0.2   | -6.2 ± 0.3    | -2.0 ± 0.3    | -4.9 ± 0.2  | -1.1 ± 0.2    | -3.3 ± 0.1   | -3.6 ± 0.3   | -3.1 ± 0.3    | E Europe M   | E Asia F     |
| 40  | -5.5 ± 0.4    | -8.9 ± 0.1   | -2.1 ± 0.3    | -5.9 ± 0.2  | -8.2 ± 0.2   | -6.9 ± 0.3    | -2.4 ± 0.3    | -5.1 ± 0.2  | -1.1 ± 0.3    | -3.6 ± 0.1   | -3.7 ± 0.3   | -3.5 ± 0.3    | E Europe M   | E Asia F     |
| 41  | -6.2 ± 0.4    | -9.2 ± 0.1   | -2.4 ± 0.3    | -6.2 ± 0.2  | -8.7 ± 0.2   | -7.8 ± 0.4    | -2.8 ± 0.3    | -5.2 ± 0.2  | -1.5 ± 0.3    | -3.9 ± 0.1   | -3.8 ± 0.3   | -3.7 ± 0.3    | E Europe M   | E Asia F     |
| 42  | -6.0 ± 0.5    | -9.4 ± 0.1   | -2.4 ± 0.3    | -6.2 ± 0.2  | -8.7 ± 0.3   | -8.0 ± 0.4    | -3.3 ± 0.3    | -5.4 ± 0.2  | -1.4 ± 0.3    | -4.2 ± 0.2   | -3.6 ± 0.3   | -4.2 ± 0.3    | E Europe M   | E Asia F     |
| 43  | -6.8 ± 0.5    | -9.8 ± 0.1   | -2.7 ± 0.3    | -6.6 ± 0.2  | -9.2 ± 0.3   | -8.7 ± 0.4    | -3.3 ± 0.4    | -5.6 ± 0.2  | -2.2 ± 0.3    | -4.3 ± 0.2   | -3.9 ± 0.3   | -5.2 ± 0.3    | E Europe M   | E Asia F     |
| 44  | -6.7 ± 0.5    | -10.2 ± 0.1  | -3.3 ± 0.3    | -6.8 ± 0.2  | -9.2 ± 0.3   | -8.4 ± 0.4    | -4.1 ± 0.4    | -5.8 ± 0.2  | -1.9 ± 0.3    | -4.7 ± 0.2   | -3.8 ± 0.3   | -5.3 ± 0.3    | E Europe M   | E Asia F     |
| 45  | -7.1 ± 0.5    | -10.3 ± 0.2  | -3.2 ± 0.3    | -6.8 ± 0.2  | -9.6 ± 0.3   | -9.0 ± 0.4    | -4.8 ± 0.4    | -6.0 ± 0.2  | -2.5 ± 0.3    | -4.7 ± 0.2   | -4.1 ± 0.3   | -6.0 ± 0.3    | E Europe M   | E Asia F     |
| 46  | -6.7 ± 0.6    | -10.7 ± 0.2  | -3.9 ± 0.3    | -6.9 ± 0.2  | -9.8 ± 0.3   | -9.6 ± 0.4    | -4.7 ± 0.4    | -6.3 ± 0.2  | -2.1 ± 0.3    | -4.9 ± 0.2   | -4.4 ± 0.3   | -6.1 ± 0.3    | E Europe M   | E Asia F     |
| 47  | -7.5 ± 0.6    | -10.9 ± 0.2  | -4.0 ± 0.4    | -7.3 ± 0.3  | -10.1 ± 0.3  | -9.9 ± 0.4    | -4.9 ± 0.4    | -6.5 ± 0.2  | -2.6 ± 0.3    | -5.1 ± 0.2   | -4.1 ± 0.3   | -6.5 ± 0.4    | E Europe M   | E Asia F     |
| 48  | -8.1 ± 0.6    | -11.2 ± 0.2  | -4.3 ± 0.3    | -7.4 ± 0.3  | -10.1 ± 0.3  | -10.4 ± 0.5   | -5.6 ± 0.5    | -6.8 ± 0.2  | -2.4 ± 0.4    | -5.2 ± 0.2   | -4.5 ± 0.4   | -6.9 ± 0.4    | E Europe M   | E Asia F     |
| 49  | -7.9 ± 0.6    | -11.6 ± 0.2  | -4.5 ± 0.3    | -7.6 ± 0.3  | -10.4 ± 0.3  | -10.7 ± 0.5   | -6.0 ± 0.5    | -7.0 ± 0.2  | -3.0 ± 0.3    | -5.5 ± 0.2   | -4.6 ± 0.4   | -7.6 ± 0.4    | E Europe M   | E Asia F     |
| 50  | -8.6 ± 0.7    | -11.7 ± 0.2  | -4.6 ± 0.4    | -7.6 ± 0.3  | -10.7 ± 0.3  | -11.0 ± 0.5   | -6.5 ± 0.5    | -7.1 ± 0.2  | -3.0 ± 0.4    | -5.7 ± 0.3   | -4.9 ± 0.4   | -7.8 ± 0.4    | E Europe M   | E Asia F     |
| 51  | -9.1 ± 0.7    | -12.0 ± 0.2  | -4.9 ± 0.4    | -8.1 ± 0.3  | -11.0 ± 0.3  | -11.5 ± 0.5   | -6.7 ± 0.6    | -7.3 ± 0.2  | -3.0 ± 0.4    | -5.8 ± 0.3   | -4.7 ± 0.4   | -8.8 ± 0.4    | E Europe M   | E Asia F     |
| 52  | -8.9 ± 0.7    | -12.2 ± 0.2  | -5.1 ± 0.4    | -8.4 ± 0.3  | -10.9 ± 0.3  | -11.8 ± 0.5   | -6.5 ± 0.5    | -7.7 ± 0.2  | -3.1 ± 0.4    | -6.2 ± 0.3   | -5.2 ± 0.4   | -8.8 ± 0.4    | E Europe M   | E Asia F     |

**Table 72.** Mean absolute error for Application-type images for algorithm *cognitec-000*. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.0 ± 0.2     | 2.01 ± 0.08 | 1.8 ± 0.1     | 1.93 ± 0.10 | 1.9 ± 0.2    | 3.4 ± 0.3     | 2.3 ± 0.2     | 1.86 ± 0.08 | 1.39 ± 0.10   | 1.92 ± 0.08 | 1.9 ± 0.1    | 2.1 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 3.2 ± 0.3     | 2.02 ± 0.10 | 1.8 ± 0.1     | 2.1 ± 0.1   | 2.1 ± 0.2    | 3.2 ± 0.3     | 2.2 ± 0.2     | 1.82 ± 0.08 | 1.4 ± 0.1     | 1.88 ± 0.09 | 2.0 ± 0.1    | 2.2 ± 0.2     | W Africa F   | E Europe M   |
| 16  | 3.2 ± 0.3     | 2.2 ± 0.1   | 1.6 ± 0.1     | 2.0 ± 0.1   | 2.4 ± 0.2    | 3.8 ± 0.4     | 2.3 ± 0.2     | 1.71 ± 0.07 | 1.5 ± 0.1     | 1.9 ± 0.1   | 1.9 ± 0.2    | 2.4 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.3 ± 0.3     | 2.15 ± 0.10 | 1.6 ± 0.2     | 2.1 ± 0.1   | 2.2 ± 0.2    | 3.6 ± 0.3     | 2.1 ± 0.2     | 1.72 ± 0.08 | 1.3 ± 0.1     | 1.9 ± 0.1   | 2.0 ± 0.2    | 2.5 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.2 ± 0.2     | 2.24 ± 0.08 | 1.9 ± 0.2     | 2.34 ± 0.09 | 2.3 ± 0.1    | 3.5 ± 0.3     | 2.3 ± 0.2     | 1.77 ± 0.07 | 1.6 ± 0.1     | 2.02 ± 0.09 | 2.0 ± 0.2    | 2.3 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 3.4 ± 0.2     | 2.23 ± 0.08 | 1.9 ± 0.2     | 2.42 ± 0.10 | 2.5 ± 0.1    | 3.5 ± 0.3     | 2.5 ± 0.2     | 1.93 ± 0.08 | 1.6 ± 0.1     | 2.10 ± 0.09 | 2.0 ± 0.1    | 2.4 ± 0.2     | W Africa F   | E Europe M   |
| 20  | 3.5 ± 0.2     | 2.37 ± 0.09 | 1.9 ± 0.1     | 2.5 ± 0.1   | 2.5 ± 0.1    | 3.3 ± 0.2     | 2.7 ± 0.2     | 1.98 ± 0.08 | 1.6 ± 0.1     | 2.22 ± 0.09 | 2.2 ± 0.1    | 2.5 ± 0.2     | E Africa F   | E Europe M   |
| 21  | 3.4 ± 0.2     | 2.43 ± 0.08 | 2.13 ± 0.10   | 2.63 ± 0.08 | 2.5 ± 0.1    | 3.5 ± 0.2     | 2.6 ± 0.2     | 2.06 ± 0.07 | 1.7 ± 0.1     | 2.34 ± 0.09 | 2.3 ± 0.1    | 2.5 ± 0.2     | W Africa F   | E Europe M   |
| 22  | 3.4 ± 0.2     | 2.57 ± 0.08 | 2.31 ± 0.09   | 2.69 ± 0.08 | 2.65 ± 0.10  | 3.4 ± 0.2     | 3.1 ± 0.2     | 2.27 ± 0.08 | 1.8 ± 0.1     | 2.42 ± 0.10 | 2.2 ± 0.1    | 2.5 ± 0.2     | E Africa F   | E Europe M   |
| 23  | 3.5 ± 0.2     | 2.59 ± 0.07 | 2.54 ± 0.09   | 2.84 ± 0.08 | 2.65 ± 0.09  | 3.3 ± 0.2     | 3.3 ± 0.2     | 2.42 ± 0.08 | 1.9 ± 0.1     | 2.52 ± 0.09 | 2.4 ± 0.1    | 2.7 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 3.7 ± 0.2     | 2.66 ± 0.07 | 2.63 ± 0.08   | 2.99 ± 0.07 | 2.70 ± 0.08  | 3.4 ± 0.2     | 3.2 ± 0.2     | 2.53 ± 0.08 | 2.05 ± 0.09   | 2.67 ± 0.08 | 2.5 ± 0.1    | 2.8 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 3.7 ± 0.2     | 2.77 ± 0.06 | 2.79 ± 0.08   | 3.09 ± 0.06 | 2.76 ± 0.08  | 3.4 ± 0.2     | 3.4 ± 0.2     | 2.66 ± 0.08 | 2.17 ± 0.10   | 2.64 ± 0.07 | 2.6 ± 0.1    | 3.0 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 3.7 ± 0.2     | 2.84 ± 0.06 | 2.92 ± 0.08   | 3.23 ± 0.06 | 2.81 ± 0.08  | 3.4 ± 0.2     | 3.4 ± 0.1     | 2.65 ± 0.08 | 2.27 ± 0.10   | 2.69 ± 0.07 | 2.7 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Europe M   |
| 27  | 3.9 ± 0.2     | 2.88 ± 0.06 | 3.08 ± 0.08   | 3.38 ± 0.06 | 2.91 ± 0.08  | 3.6 ± 0.1     | 3.4 ± 0.1     | 2.72 ± 0.07 | 2.30 ± 0.09   | 2.70 ± 0.07 | 2.7 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 28  | 3.9 ± 0.2     | 2.92 ± 0.06 | 3.00 ± 0.08   | 3.51 ± 0.06 | 3.00 ± 0.08  | 3.6 ± 0.1     | 3.3 ± 0.1     | 2.76 ± 0.08 | 2.36 ± 0.09   | 2.73 ± 0.06 | 2.7 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 4.1 ± 0.2     | 2.97 ± 0.06 | 3.02 ± 0.08   | 3.62 ± 0.07 | 3.13 ± 0.09  | 3.7 ± 0.1     | 3.3 ± 0.1     | 2.69 ± 0.07 | 2.44 ± 0.09   | 2.75 ± 0.06 | 2.7 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 4.1 ± 0.2     | 3.05 ± 0.06 | 3.09 ± 0.09   | 3.78 ± 0.07 | 3.10 ± 0.09  | 3.8 ± 0.1     | 3.5 ± 0.1     | 2.82 ± 0.07 | 2.43 ± 0.10   | 2.77 ± 0.06 | 2.8 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.1 ± 0.2     | 3.07 ± 0.06 | 3.02 ± 0.09   | 3.91 ± 0.07 | 3.21 ± 0.09  | 3.9 ± 0.1     | 3.5 ± 0.1     | 2.82 ± 0.08 | 2.51 ± 0.09   | 2.80 ± 0.06 | 3.0 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 4.2 ± 0.2     | 3.13 ± 0.06 | 2.99 ± 0.09   | 3.98 ± 0.07 | 3.4 ± 0.1    | 4.1 ± 0.2     | 3.6 ± 0.1     | 2.77 ± 0.08 | 2.40 ± 0.10   | 2.83 ± 0.05 | 2.8 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 4.5 ± 0.2     | 3.23 ± 0.06 | 3.1 ± 0.1     | 4.00 ± 0.08 | 3.3 ± 0.1    | 4.2 ± 0.2     | 3.7 ± 0.1     | 2.87 ± 0.08 | 2.30 ± 0.10   | 2.88 ± 0.06 | 2.9 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 4.5 ± 0.2     | 3.23 ± 0.06 | 3.0 ± 0.1     | 4.07 ± 0.08 | 3.4 ± 0.1    | 4.3 ± 0.2     | 3.6 ± 0.2     | 2.90 ± 0.07 | 2.5 ± 0.1     | 2.89 ± 0.06 | 3.0 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.6 ± 0.2     | 3.30 ± 0.07 | 3.1 ± 0.1     | 4.06 ± 0.08 | 3.5 ± 0.1    | 4.4 ± 0.2     | 3.6 ± 0.2     | 2.90 ± 0.08 | 2.5 ± 0.1     | 2.89 ± 0.06 | 3.1 ± 0.1    | 4.1 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 4.8 ± 0.2     | 3.36 ± 0.07 | 3.0 ± 0.1     | 4.06 ± 0.09 | 3.4 ± 0.1    | 4.4 ± 0.2     | 3.7 ± 0.2     | 2.91 ± 0.07 | 2.5 ± 0.1     | 2.93 ± 0.06 | 3.0 ± 0.1    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 37  | 4.6 ± 0.2     | 3.29 ± 0.07 | 3.0 ± 0.1     | 4.08 ± 0.09 | 3.4 ± 0.1    | 4.8 ± 0.2     | 3.8 ± 0.2     | 2.93 ± 0.08 | 2.6 ± 0.1     | 2.98 ± 0.06 | 2.9 ± 0.1    | 4.1 ± 0.2     | W Africa F   | E Europe M   |
| 38  | 4.5 ± 0.2     | 3.42 ± 0.07 | 2.9 ± 0.1     | 4.04 ± 0.09 | 3.4 ± 0.1    | 4.8 ± 0.2     | 3.7 ± 0.2     | 2.99 ± 0.08 | 2.5 ± 0.1     | 2.97 ± 0.06 | 3.0 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 39  | 4.6 ± 0.2     | 3.37 ± 0.07 | 3.0 ± 0.1     | 4.2 ± 0.1   | 3.4 ± 0.1    | 4.4 ± 0.2     | 3.6 ± 0.2     | 3.00 ± 0.08 | 2.6 ± 0.1     | 2.96 ± 0.07 | 3.1 ± 0.1    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 40  | 4.6 ± 0.2     | 3.32 ± 0.07 | 3.0 ± 0.1     | 4.0 ± 0.1   | 3.5 ± 0.1    | 4.8 ± 0.2     | 3.4 ± 0.2     | 2.92 ± 0.08 | 2.7 ± 0.1     | 3.09 ± 0.07 | 2.9 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 41  | 4.7 ± 0.3     | 3.40 ± 0.07 | 2.9 ± 0.1     | 4.0 ± 0.1   | 3.4 ± 0.1    | 4.9 ± 0.2     | 3.7 ± 0.2     | 2.91 ± 0.08 | 2.7 ± 0.1     | 3.09 ± 0.07 | 3.0 ± 0.1    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 42  | 4.6 ± 0.3     | 3.28 ± 0.07 | 3.0 ± 0.1     | 3.9 ± 0.1   | 3.4 ± 0.1    | 4.6 ± 0.2     | 3.5 ± 0.2     | 2.93 ± 0.08 | 2.6 ± 0.1     | 3.11 ± 0.08 | 2.9 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 43  | 4.5 ± 0.3     | 3.35 ± 0.07 | 3.1 ± 0.1     | 3.9 ± 0.1   | 3.4 ± 0.1    | 4.7 ± 0.2     | 3.5 ± 0.2     | 2.93 ± 0.08 | 2.8 ± 0.1     | 3.19 ± 0.08 | 3.1 ± 0.1    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 44  | 4.7 ± 0.3     | 3.38 ± 0.07 | 3.2 ± 0.1     | 4.1 ± 0.1   | 3.4 ± 0.1    | 4.9 ± 0.3     | 3.5 ± 0.2     | 2.93 ± 0.08 | 2.6 ± 0.1     | 3.26 ± 0.09 | 3.1 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 45  | 4.7 ± 0.3     | 3.38 ± 0.07 | 2.9 ± 0.1     | 4.0 ± 0.1   | 3.3 ± 0.1    | 4.8 ± 0.3     | 3.6 ± 0.2     | 3.00 ± 0.08 | 2.8 ± 0.1     | 3.23 ± 0.09 | 3.1 ± 0.2    | 4.5 ± 0.2     | W Africa F   | E Europe M   |
| 46  | 4.8 ± 0.3     | 3.34 ± 0.07 | 3.1 ± 0.1     | 3.8 ± 0.1   | 3.5 ± 0.1    | 4.8 ± 0.3     | 3.7 ± 0.3     | 3.06 ± 0.08 | 2.9 ± 0.1     | 3.2 ± 0.1   | 3.2 ± 0.2    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 47  | 4.7 ± 0.4     | 3.34 ± 0.08 | 3.2 ± 0.2     | 3.9 ± 0.1   | 3.3 ± 0.1    | 4.9 ± 0.3     | 3.8 ± 0.2     | 3.05 ± 0.09 | 2.7 ± 0.1     | 3.4 ± 0.1   | 3.2 ± 0.2    | 4.5 ± 0.2     | W Africa F   | E Europe M   |
| 48  | 4.6 ± 0.3     | 3.30 ± 0.08 | 3.2 ± 0.1     | 3.9 ± 0.1   | 3.3 ± 0.1    | 5.0 ± 0.3     | 4.0 ± 0.3     | 3.10 ± 0.08 | 2.7 ± 0.2     | 3.4 ± 0.1   | 3.2 ± 0.2    | 4.7 ± 0.3     | W Africa F   | E Europe M   |
| 49  | 4.6 ± 0.4     | 3.44 ± 0.08 | 3.2 ± 0.2     | 3.9 ± 0.1   | 3.4 ± 0.1    | 5.4 ± 0.3     | 4.0 ± 0.3     | 3.10 ± 0.09 | 2.7 ± 0.2     | 3.5 ± 0.1   | 3.2 ± 0.2    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 50  | 5.0 ± 0.4     | 3.40 ± 0.09 | 3.1 ± 0.2     | 3.9 ± 0.1   | 3.4 ± 0.1    | 5.2 ± 0.3     | 3.8 ± 0.3     | 3.15 ± 0.09 | 2.8 ± 0.1     | 3.5 ± 0.1   | 3.3 ± 0.2    | 4.7 ± 0.2     | W Africa F   | E Europe M   |
| 51  | 4.6 ± 0.3     | 3.46 ± 0.09 | 3.2 ± 0.2     | 3.9 ± 0.2   | 3.3 ± 0.1    | 5.2 ± 0.3     | 4.3 ± 0.4     | 3.20 ± 0.09 | 2.7 ± 0.1     | 3.6 ± 0.1   | 3.4 ± 0.2    | 4.9 ± 0.3     | W Africa F   | E Europe M   |
| 52  | 4.9 ± 0.4     | 3.44 ± 0.09 | 3.2 ± 0.2     | 4.0 ± 0.1   | 3.4 ± 0.1    | 5.4 ± 0.3     | 4.0 ± 0.3     | 3.25 ± 0.09 | 2.8 ± 0.2     | 3.6 ± 0.1   | 3.4 ± 0.2    | 4.9 ± 0.3     | W Africa F   | E Europe M   |

**Table 73.** Mean absolute error for Application-type images for algorithm cvut-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 6.6 ± 0.4     | 3.4 ± 0.1   | 2.7 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 6.3 ± 0.5     | 3.8 ± 0.3     | 2.5 ± 0.1   | 1.9 ± 0.1     | 1.8 ± 0.1   | 2.2 ± 0.2    | 4.2 ± 0.4     | E Africa F   | S Asia M     |
| 15  | 6.5 ± 0.4     | 3.5 ± 0.1   | 2.5 ± 0.2     | 3.4 ± 0.2   | 3.2 ± 0.3    | 5.4 ± 0.5     | 3.4 ± 0.3     | 2.9 ± 0.1   | 1.7 ± 0.2     | 2.2 ± 0.1   | 2.8 ± 0.2    | 4.5 ± 0.4     | E Africa F   | E Europe M   |
| 16  | 6.3 ± 0.4     | 3.5 ± 0.2   | 2.3 ± 0.2     | 3.2 ± 0.2   | 3.2 ± 0.3    | 5.5 ± 0.6     | 3.8 ± 0.3     | 3.0 ± 0.1   | 1.8 ± 0.2     | 2.6 ± 0.1   | 3.0 ± 0.2    | 4.3 ± 0.4     | E Africa F   | E Europe M   |
| 17  | 6.4 ± 0.4     | 3.2 ± 0.1   | 1.9 ± 0.2     | 3.1 ± 0.2   | 2.8 ± 0.2    | 5.3 ± 0.4     | 3.9 ± 0.3     | 2.9 ± 0.1   | 1.7 ± 0.2     | 2.9 ± 0.2   | 3.0 ± 0.2    | 4.4 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 5.8 ± 0.3     | 2.9 ± 0.1   | 1.9 ± 0.2     | 2.9 ± 0.1   | 2.6 ± 0.2    | 4.8 ± 0.4     | 4.0 ± 0.3     | 2.68 ± 0.08 | 2.1 ± 0.2     | 2.9 ± 0.1   | 2.5 ± 0.2    | 4.1 ± 0.3     | E Africa F   | E Europe F   |
| 19  | 6.2 ± 0.3     | 2.7 ± 0.1   | 2.3 ± 0.2     | 3.0 ± 0.1   | 2.5 ± 0.1    | 5.4 ± 0.4     | 4.2 ± 0.3     | 2.47 ± 0.09 | 2.6 ± 0.2     | 3.0 ± 0.1   | 2.3 ± 0.1    | 4.3 ± 0.3     | E Africa F   | S E Asia M   |
| 20  | 5.9 ± 0.3     | 2.7 ± 0.1   | 2.6 ± 0.1     | 3.1 ± 0.1   | 2.5 ± 0.1    | 5.3 ± 0.3     | 4.4 ± 0.3     | 2.07 ± 0.09 | 2.8 ± 0.2     | 3.1 ± 0.1   | 2.0 ± 0.2    | 4.4 ± 0.3     | E Africa F   | S E Asia M   |
| 21  | 5.7 ± 0.3     | 2.64 ± 0.09 | 2.9 ± 0.1     | 3.2 ± 0.1   | 2.5 ± 0.1    | 5.4 ± 0.3     | 4.4 ± 0.2     | 2.04 ± 0.10 | 3.1 ± 0.2     | 3.0 ± 0.1   | 1.9 ± 0.1    | 4.1 ± 0.2     | E Africa F   | S E Asia M   |
| 22  | 5.6 ± 0.2     | 2.46 ± 0.07 | 3.1 ± 0.1     | 3.32 ± 0.10 | 2.6 ± 0.1    | 5.5 ± 0.3     | 4.9 ± 0.2     | 2.15 ± 0.08 | 3.3 ± 0.2     | 3.2 ± 0.1   | 2.2 ± 0.1    | 4.2 ± 0.2     | E Africa F   | E Asia M     |
| 23  | 5.4 ± 0.2     | 2.52 ± 0.06 | 3.3 ± 0.1     | 3.37 ± 0.09 | 2.68 ± 0.09  | 5.5 ± 0.2     | 4.7 ± 0.2     | 2.41 ± 0.09 | 3.3 ± 0.2     | 3.4 ± 0.1   | 2.6 ± 0.1    | 4.3 ± 0.2     | W Africa F   | E Asia M     |
| 24  | 5.1 ± 0.2     | 2.50 ± 0.07 | 3.3 ± 0.1     | 3.35 ± 0.08 | 2.77 ± 0.08  | 5.4 ± 0.2     | 4.5 ± 0.2     | 2.74 ± 0.10 | 3.3 ± 0.2     | 3.4 ± 0.1   | 2.7 ± 0.1    | 4.0 ± 0.2     | W Africa F   | E Asia F     |
| 25  | 4.7 ± 0.2     | 2.55 ± 0.07 | 3.7 ± 0.1     | 3.45 ± 0.08 | 3.11 ± 0.09  | 5.0 ± 0.2     | 4.4 ± 0.2     | 3.21 ± 0.09 | 3.3 ± 0.1     | 3.39 ± 0.10 | 3.3 ± 0.1    | 3.7 ± 0.2     | W Africa F   | E Asia F     |
| 26  | 4.5 ± 0.2     | 2.59 ± 0.06 | 3.96 ± 0.10   | 3.28 ± 0.07 | 3.36 ± 0.09  | 4.9 ± 0.2     | 4.0 ± 0.2     | 3.30 ± 0.08 | 3.4 ± 0.1     | 3.30 ± 0.09 | 3.8 ± 0.1    | 3.5 ± 0.2     | W Africa F   | E Asia F     |
| 27  | 4.3 ± 0.2     | 2.84 ± 0.07 | 4.1 ± 0.1     | 3.37 ± 0.07 | 3.63 ± 0.10  | 4.7 ± 0.2     | 3.6 ± 0.2     | 3.62 ± 0.09 | 3.3 ± 0.1     | 3.23 ± 0.09 | 4.1 ± 0.1    | 3.2 ± 0.1     | W Africa F   | E Asia F     |
| 28  | 4.1 ± 0.2     | 3.26 ± 0.07 | 4.2 ± 0.1     | 3.55 ± 0.08 | 4.1 ± 0.1    | 4.7 ± 0.2     | 3.3 ± 0.1     | 3.9 ± 0.1   | 3.0 ± 0.1     | 3.05 ± 0.08 | 4.1 ± 0.1    | 2.9 ± 0.1     | W Africa F   | W Africa M   |
| 29  | 3.9 ± 0.2     | 3.69 ± 0.07 | 4.2 ± 0.1     | 3.80 ± 0.08 | 4.4 ± 0.1    | 4.3 ± 0.2     | 3.3 ± 0.2     | 3.8 ± 0.1   | 2.9 ± 0.1     | 3.01 ± 0.07 | 4.2 ± 0.2    | 3.0 ± 0.1     | S E Asia F   | E Europe M   |
| 30  | 3.8 ± 0.2     | 4.15 ± 0.06 | 4.3 ± 0.1     | 4.00 ± 0.08 | 4.8 ± 0.1    | 4.2 ± 0.2     | 3.3 ± 0.2     | 3.9 ± 0.1   | 2.7 ± 0.1     | 3.08 ± 0.07 | 4.3 ± 0.2    | 2.8 ± 0.1     | S E Asia F   | E Europe M   |
| 31  | 3.6 ± 0.2     | 4.66 ± 0.07 | 4.2 ± 0.1     | 4.38 ± 0.09 | 5.3 ± 0.1    | 4.0 ± 0.2     | 3.3 ± 0.1     | 4.2 ± 0.1   | 2.8 ± 0.1     | 3.12 ± 0.06 | 4.5 ± 0.2    | 2.9 ± 0.1     | S E Asia F   | E Europe M   |
| 32  | 3.4 ± 0.2     | 5.09 ± 0.07 | 4.3 ± 0.1     | 4.54 ± 0.09 | 5.7 ± 0.1    | 4.1 ± 0.2     | 3.2 ± 0.1     | 4.2 ± 0.1   | 3.0 ± 0.1     | 3.28 ± 0.06 | 4.6 ± 0.2    | 3.1 ± 0.1     | S E Asia F   | E Europe M   |
| 33  | 3.7 ± 0.2     | 5.55 ± 0.08 | 4.6 ± 0.1     | 4.89 ± 0.09 | 6.0 ± 0.1    | 3.9 ± 0.2     | 3.3 ± 0.1     | 4.3 ± 0.1   | 2.9 ± 0.1     | 3.30 ± 0.06 | 4.5 ± 0.2    | 3.1 ± 0.1     | S E Asia F   | E Europe M   |
| 34  | 3.6 ± 0.2     | 6.04 ± 0.08 | 4.6 ± 0.2     | 5.02 ± 0.10 | 6.2 ± 0.1    | 3.8 ± 0.2     | 3.1 ± 0.2     | 4.3 ± 0.1   | 2.9 ± 0.1     | 3.44 ± 0.08 | 4.4 ± 0.2    | 3.2 ± 0.1     | S E Asia F   | E Europe M   |
| 35  | 3.7 ± 0.2     | 6.47 ± 0.08 | 4.8 ± 0.2     | 5.3 ± 0.1   | 6.5 ± 0.2    | 3.6 ± 0.2     | 3.3 ± 0.2     | 4.2 ± 0.1   | 2.9 ± 0.1     | 3.57 ± 0.08 | 4.4 ± 0.2    | 3.3 ± 0.1     | S E Asia F   | E Europe M   |
| 36  | 3.9 ± 0.2     | 6.72 ± 0.09 | 4.6 ± 0.2     | 5.5 ± 0.1   | 6.7 ± 0.2    | 3.9 ± 0.2     | 3.5 ± 0.2     | 4.0 ± 0.1   | 2.9 ± 0.1     | 3.70 ± 0.08 | 4.3 ± 0.2    | 3.4 ± 0.1     | E Asia F     | E Europe M   |
| 37  | 4.3 ± 0.3     | 7.0 ± 0.1   | 4.9 ± 0.2     | 5.6 ± 0.1   | 6.9 ± 0.2    | 4.2 ± 0.2     | 3.6 ± 0.2     | 3.7 ± 0.1   | 2.9 ± 0.1     | 3.96 ± 0.08 | 3.9 ± 0.2    | 3.8 ± 0.1     | E Asia F     | E Europe M   |
| 38  | 4.2 ± 0.2     | 7.2 ± 0.1   | 5.0 ± 0.2     | 5.8 ± 0.1   | 6.8 ± 0.2    | 4.4 ± 0.2     | 3.9 ± 0.2     | 3.3 ± 0.1   | 3.1 ± 0.2     | 4.15 ± 0.09 | 3.5 ± 0.2    | 4.1 ± 0.2     | E Asia F     | E Europe M   |
| 39  | 4.4 ± 0.2     | 7.4 ± 0.1   | 5.0 ± 0.2     | 6.1 ± 0.1   | 6.7 ± 0.2    | 4.1 ± 0.2     | 3.9 ± 0.2     | 3.0 ± 0.1   | 2.9 ± 0.1     | 4.28 ± 0.09 | 3.3 ± 0.2    | 4.4 ± 0.2     | E Asia F     | E Europe M   |
| 40  | 4.7 ± 0.2     | 7.6 ± 0.1   | 5.0 ± 0.2     | 5.9 ± 0.1   | 7.0 ± 0.2    | 4.3 ± 0.2     | 4.1 ± 0.2     | 3.1 ± 0.1   | 3.0 ± 0.1     | 4.37 ± 0.10 | 3.4 ± 0.2    | 4.7 ± 0.2     | E Asia F     | E Europe M   |
| 41  | 5.3 ± 0.3     | 7.7 ± 0.1   | 5.2 ± 0.2     | 6.2 ± 0.2   | 7.1 ± 0.2    | 4.8 ± 0.2     | 4.1 ± 0.2     | 3.4 ± 0.1   | 3.2 ± 0.2     | 4.6 ± 0.1   | 3.8 ± 0.2    | 5.1 ± 0.2     | E Asia F     | E Europe M   |
| 42  | 5.0 ± 0.3     | 7.8 ± 0.1   | 5.1 ± 0.2     | 6.0 ± 0.2   | 7.1 ± 0.2    | 5.1 ± 0.2     | 4.3 ± 0.3     | 3.8 ± 0.1   | 3.3 ± 0.2     | 4.7 ± 0.1   | 3.9 ± 0.2    | 5.4 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 5.4 ± 0.3     | 8.0 ± 0.1   | 5.4 ± 0.2     | 6.5 ± 0.2   | 7.1 ± 0.2    | 5.4 ± 0.2     | 4.1 ± 0.3     | 4.1 ± 0.1   | 3.6 ± 0.2     | 4.7 ± 0.1   | 4.3 ± 0.2    | 6.1 ± 0.2     | E Asia F     | E Europe M   |
| 44  | 5.4 ± 0.3     | 8.2 ± 0.1   | 5.4 ± 0.2     | 6.3 ± 0.2   | 6.7 ± 0.2    | 5.6 ± 0.2     | 4.7 ± 0.3     | 4.6 ± 0.1   | 3.8 ± 0.2     | 4.7 ± 0.1   | 4.7 ± 0.2    | 6.0 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 5.8 ± 0.3     | 8.2 ± 0.1   | 5.3 ± 0.2     | 6.3 ± 0.2   | 6.7 ± 0.3    | 5.6 ± 0.3     | 4.9 ± 0.3     | 5.1 ± 0.1   | 4.0 ± 0.2     | 4.7 ± 0.1   | 5.0 ± 0.2    | 6.4 ± 0.2     | E Asia F     | E Europe M   |
| 46  | 5.9 ± 0.4     | 8.5 ± 0.1   | 5.3 ± 0.2     | 6.1 ± 0.2   | 6.7 ± 0.3    | 6.1 ± 0.3     | 5.1 ± 0.3     | 5.5 ± 0.1   | 4.2 ± 0.2     | 4.8 ± 0.2   | 5.3 ± 0.2    | 6.8 ± 0.2     | E Asia F     | E Europe M   |
| 47  | 6.2 ± 0.5     | 8.5 ± 0.1   | 5.1 ± 0.2     | 5.9 ± 0.2   | 6.7 ± 0.3    | 6.1 ± 0.3     | 5.0 ± 0.3     | 6.0 ± 0.1   | 4.5 ± 0.2     | 4.8 ± 0.2   | 5.5 ± 0.2    | 6.8 ± 0.3     | E Asia F     | E Europe M   |
| 48  | 6.3 ± 0.5     | 8.7 ± 0.1   | 5.0 ± 0.3     | 5.9 ± 0.2   | 6.6 ± 0.3    | 6.2 ± 0.3     | 5.5 ± 0.3     | 6.5 ± 0.1   | 4.5 ± 0.3     | 4.9 ± 0.2   | 5.9 ± 0.3    | 7.4 ± 0.3     | E Asia F     | E Europe M   |
| 49  | 5.9 ± 0.5     | 8.8 ± 0.1   | 4.8 ± 0.3     | 5.4 ± 0.2   | 6.5 ± 0.3    | 6.6 ± 0.4     | 5.7 ± 0.4     | 6.9 ± 0.1   | 4.3 ± 0.3     | 4.7 ± 0.2   | 5.9 ± 0.3    | 7.1 ± 0.3     | E Asia F     | E Europe M   |
| 50  | 6.6 ± 0.5     | 8.8 ± 0.2   | 4.4 ± 0.3     | 5.6 ± 0.2   | 6.5 ± 0.3    | 6.6 ± 0.4     | 5.7 ± 0.4     | 7.1 ± 0.1   | 4.3 ± 0.3     | 4.8 ± 0.2   | 6.1 ± 0.3    | 7.3 ± 0.3     | E Asia F     | E Europe M   |
| 51  | 6.1 ± 0.5     | 9.0 ± 0.2   | 4.5 ± 0.3     | 5.4 ± 0.2   | 6.7 ± 0.3    | 6.9 ± 0.4     | 5.8 ± 0.4     | 7.2 ± 0.1   | 4.0 ± 0.2     | 4.9 ± 0.2   | 6.2 ± 0.3    | 7.8 ± 0.4     | E Asia F     | E Europe M   |
| 52  | 6.7 ± 0.5     | 8.8 ± 0.2   | 4.7 ± 0.2     | 5.6 ± 0.2   | 6.4 ± 0.3    | 7.4 ± 0.4     | 5.7 ± 0.5     | 7.5 ± 0.2   | 3.8 ± 0.2     | 5.0 ± 0.2   | 6.3 ± 0.3    | 7.8 ± 0.4     | E Asia F     | E Europe M   |

**Table 74.** Mean absolute error for Application-type images for algorithm cvut-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.5 ± 0.2     | 3.69 ± 0.09 | 2.9 ± 0.1     | 3.1 ± 0.1   | 3.3 ± 0.2    | 5.1 ± 0.3     | 2.6 ± 0.2     | 2.69 ± 0.09 | 1.68 ± 0.10   | 1.99 ± 0.09 | 2.2 ± 0.2    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 4.4 ± 0.3     | 3.49 ± 0.10 | 2.9 ± 0.2     | 3.1 ± 0.2   | 3.3 ± 0.3    | 4.7 ± 0.3     | 2.3 ± 0.2     | 2.67 ± 0.09 | 1.7 ± 0.2     | 2.0 ± 0.1   | 2.4 ± 0.2    | 2.6 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.0 ± 0.3     | 3.2 ± 0.1   | 2.6 ± 0.2     | 2.8 ± 0.1   | 3.0 ± 0.2    | 4.7 ± 0.4     | 2.8 ± 0.3     | 2.44 ± 0.08 | 1.7 ± 0.2     | 2.2 ± 0.1   | 2.3 ± 0.2    | 2.6 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 3.9 ± 0.3     | 2.60 ± 0.09 | 2.4 ± 0.2     | 2.7 ± 0.1   | 2.5 ± 0.2    | 4.3 ± 0.3     | 2.8 ± 0.3     | 2.23 ± 0.09 | 1.6 ± 0.1     | 2.7 ± 0.1   | 2.4 ± 0.2    | 3.3 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 3.5 ± 0.2     | 2.14 ± 0.08 | 2.2 ± 0.2     | 2.5 ± 0.1   | 2.0 ± 0.1    | 4.0 ± 0.3     | 2.9 ± 0.2     | 2.01 ± 0.07 | 2.1 ± 0.2     | 2.8 ± 0.1   | 2.1 ± 0.2    | 3.4 ± 0.3     | W Africa F   | E Asia M     |
| 19  | 3.5 ± 0.3     | 1.70 ± 0.08 | 2.1 ± 0.2     | 2.4 ± 0.1   | 2.0 ± 0.1    | 4.0 ± 0.3     | 3.0 ± 0.2     | 1.70 ± 0.07 | 2.1 ± 0.2     | 2.8 ± 0.1   | 1.9 ± 0.1    | 3.4 ± 0.3     | W Africa F   | E Asia M     |
| 20  | 3.5 ± 0.3     | 1.53 ± 0.10 | 2.2 ± 0.2     | 2.2 ± 0.1   | 1.8 ± 0.1    | 3.8 ± 0.3     | 3.2 ± 0.3     | 1.35 ± 0.08 | 2.2 ± 0.2     | 2.7 ± 0.1   | 1.7 ± 0.1    | 3.6 ± 0.3     | W Africa F   | E Asia M     |
| 21  | 3.3 ± 0.2     | 1.64 ± 0.08 | 2.2 ± 0.1     | 2.36 ± 0.09 | 1.90 ± 0.10  | 3.8 ± 0.3     | 3.4 ± 0.3     | 1.33 ± 0.07 | 2.3 ± 0.2     | 2.6 ± 0.1   | 1.5 ± 0.1    | 3.5 ± 0.3     | W Africa F   | E Asia M     |
| 22  | 3.7 ± 0.2     | 2.00 ± 0.06 | 2.4 ± 0.1     | 2.48 ± 0.08 | 2.34 ± 0.09  | 3.6 ± 0.2     | 3.5 ± 0.2     | 1.73 ± 0.06 | 2.3 ± 0.2     | 2.7 ± 0.1   | 1.84 ± 0.10  | 3.5 ± 0.2     | E Africa F   | E Asia M     |
| 23  | 3.6 ± 0.2     | 2.47 ± 0.06 | 2.6 ± 0.1     | 2.61 ± 0.08 | 2.71 ± 0.09  | 3.6 ± 0.2     | 3.7 ± 0.2     | 2.17 ± 0.06 | 2.3 ± 0.2     | 2.8 ± 0.1   | 2.2 ± 0.1    | 3.6 ± 0.2     | E Africa M   | E Asia M     |
| 24  | 3.5 ± 0.2     | 2.91 ± 0.06 | 2.51 ± 0.10   | 2.70 ± 0.07 | 2.96 ± 0.07  | 3.4 ± 0.2     | 3.6 ± 0.2     | 2.56 ± 0.08 | 2.3 ± 0.1     | 2.7 ± 0.1   | 2.4 ± 0.1    | 3.4 ± 0.2     | E Africa M   | E Europe M   |
| 25  | 3.6 ± 0.2     | 3.26 ± 0.07 | 2.73 ± 0.09   | 2.88 ± 0.07 | 3.39 ± 0.09  | 3.5 ± 0.2     | 3.7 ± 0.2     | 2.96 ± 0.08 | 2.4 ± 0.1     | 2.73 ± 0.09 | 2.8 ± 0.1    | 3.5 ± 0.2     | E Africa M   | E Europe M   |
| 26  | 3.6 ± 0.2     | 3.63 ± 0.07 | 2.89 ± 0.10   | 3.00 ± 0.07 | 3.65 ± 0.09  | 3.3 ± 0.2     | 3.6 ± 0.2     | 3.09 ± 0.07 | 2.6 ± 0.2     | 2.73 ± 0.08 | 3.1 ± 0.1    | 3.3 ± 0.2     | S E Asia F   | E Europe M   |
| 27  | 3.8 ± 0.2     | 3.89 ± 0.07 | 3.09 ± 0.10   | 3.35 ± 0.07 | 4.00 ± 0.10  | 3.3 ± 0.1     | 3.6 ± 0.2     | 3.42 ± 0.10 | 2.8 ± 0.1     | 2.86 ± 0.08 | 3.2 ± 0.1    | 3.3 ± 0.2     | S E Asia F   | E Europe M   |
| 28  | 4.0 ± 0.2     | 4.30 ± 0.08 | 3.42 ± 0.10   | 3.71 ± 0.07 | 4.3 ± 0.1    | 3.5 ± 0.1     | 3.6 ± 0.1     | 3.69 ± 0.10 | 2.9 ± 0.1     | 2.99 ± 0.07 | 3.4 ± 0.1    | 3.1 ± 0.1     | S E Asia F   | E Europe M   |
| 29  | 4.2 ± 0.2     | 4.60 ± 0.08 | 3.67 ± 0.10   | 4.06 ± 0.07 | 4.6 ± 0.1    | 3.7 ± 0.2     | 3.6 ± 0.2     | 3.81 ± 0.10 | 3.0 ± 0.1     | 3.09 ± 0.07 | 3.8 ± 0.1    | 3.3 ± 0.1     | S E Asia F   | E Europe M   |
| 30  | 4.3 ± 0.2     | 4.94 ± 0.08 | 3.9 ± 0.1     | 4.38 ± 0.08 | 4.9 ± 0.1    | 3.8 ± 0.1     | 3.5 ± 0.2     | 4.3 ± 0.1   | 3.0 ± 0.1     | 3.26 ± 0.07 | 4.0 ± 0.2    | 3.0 ± 0.1     | E Asia F     | E Europe M   |
| 31  | 4.4 ± 0.2     | 5.30 ± 0.08 | 3.9 ± 0.1     | 4.74 ± 0.08 | 5.3 ± 0.1    | 3.8 ± 0.2     | 3.5 ± 0.2     | 4.5 ± 0.1   | 3.1 ± 0.1     | 3.40 ± 0.07 | 4.3 ± 0.2    | 2.9 ± 0.1     | E Asia F     | W Africa M   |
| 32  | 4.4 ± 0.2     | 5.57 ± 0.09 | 4.1 ± 0.1     | 5.06 ± 0.09 | 5.5 ± 0.1    | 4.0 ± 0.2     | 3.4 ± 0.2     | 4.7 ± 0.1   | 3.2 ± 0.2     | 3.64 ± 0.07 | 4.3 ± 0.2    | 2.9 ± 0.1     | E Asia F     | W Africa M   |
| 33  | 4.6 ± 0.2     | 5.82 ± 0.09 | 4.1 ± 0.1     | 5.29 ± 0.09 | 5.7 ± 0.1    | 4.0 ± 0.2     | 3.5 ± 0.2     | 4.9 ± 0.1   | 2.7 ± 0.1     | 3.78 ± 0.08 | 4.4 ± 0.2    | 2.8 ± 0.1     | E Asia F     | E Europe M   |
| 34  | 4.6 ± 0.2     | 6.00 ± 0.10 | 4.2 ± 0.1     | 5.4 ± 0.1   | 5.6 ± 0.2    | 4.1 ± 0.2     | 3.2 ± 0.2     | 5.2 ± 0.1   | 2.8 ± 0.1     | 3.91 ± 0.08 | 4.8 ± 0.2    | 3.0 ± 0.1     | E Asia F     | E Europe M   |
| 35  | 4.9 ± 0.2     | 6.3 ± 0.1   | 4.3 ± 0.1     | 5.7 ± 0.1   | 6.0 ± 0.2    | 4.1 ± 0.2     | 3.4 ± 0.2     | 5.2 ± 0.1   | 2.8 ± 0.1     | 3.99 ± 0.08 | 4.8 ± 0.2    | 3.2 ± 0.2     | E Asia F     | E Europe M   |
| 36  | 4.9 ± 0.2     | 6.4 ± 0.1   | 4.1 ± 0.2     | 6.0 ± 0.1   | 6.0 ± 0.2    | 4.1 ± 0.2     | 3.6 ± 0.2     | 5.2 ± 0.1   | 2.8 ± 0.2     | 4.15 ± 0.09 | 4.8 ± 0.2    | 3.2 ± 0.2     | E Asia F     | E Europe M   |
| 37  | 4.8 ± 0.2     | 6.5 ± 0.1   | 4.2 ± 0.2     | 6.1 ± 0.1   | 6.3 ± 0.2    | 4.6 ± 0.2     | 3.4 ± 0.2     | 5.3 ± 0.1   | 2.8 ± 0.1     | 4.39 ± 0.10 | 4.9 ± 0.2    | 3.4 ± 0.2     | E Asia F     | E Europe M   |
| 38  | 4.8 ± 0.3     | 6.7 ± 0.1   | 4.1 ± 0.2     | 6.2 ± 0.1   | 6.0 ± 0.2    | 4.8 ± 0.2     | 3.6 ± 0.2     | 5.3 ± 0.1   | 2.9 ± 0.2     | 4.42 ± 0.10 | 4.8 ± 0.2    | 3.6 ± 0.2     | E Asia F     | E Europe M   |
| 39  | 4.9 ± 0.3     | 6.9 ± 0.1   | 4.3 ± 0.2     | 6.3 ± 0.1   | 6.1 ± 0.2    | 4.7 ± 0.2     | 3.6 ± 0.2     | 5.5 ± 0.1   | 2.9 ± 0.1     | 4.6 ± 0.1   | 4.8 ± 0.2    | 3.9 ± 0.2     | E Asia F     | E Europe M   |
| 40  | 5.1 ± 0.3     | 7.0 ± 0.1   | 4.0 ± 0.2     | 6.4 ± 0.2   | 6.4 ± 0.2    | 5.0 ± 0.3     | 3.7 ± 0.2     | 5.7 ± 0.1   | 3.3 ± 0.2     | 4.7 ± 0.1   | 5.0 ± 0.2    | 4.4 ± 0.2     | E Asia F     | E Europe M   |
| 41  | 5.4 ± 0.3     | 7.1 ± 0.1   | 4.4 ± 0.2     | 6.5 ± 0.2   | 6.4 ± 0.2    | 5.3 ± 0.3     | 4.0 ± 0.2     | 5.7 ± 0.1   | 3.2 ± 0.2     | 4.9 ± 0.1   | 5.1 ± 0.2    | 4.7 ± 0.2     | E Asia F     | E Europe M   |
| 42  | 5.2 ± 0.3     | 7.3 ± 0.1   | 4.5 ± 0.2     | 6.3 ± 0.2   | 6.5 ± 0.2    | 5.3 ± 0.2     | 4.2 ± 0.2     | 5.9 ± 0.1   | 3.4 ± 0.2     | 5.0 ± 0.1   | 5.1 ± 0.2    | 4.7 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 5.3 ± 0.3     | 7.5 ± 0.1   | 4.6 ± 0.2     | 6.6 ± 0.2   | 6.8 ± 0.2    | 5.7 ± 0.3     | 4.2 ± 0.3     | 6.0 ± 0.1   | 3.4 ± 0.2     | 4.9 ± 0.1   | 5.4 ± 0.3    | 5.5 ± 0.2     | E Asia F     | E Europe M   |
| 44  | 5.6 ± 0.3     | 7.7 ± 0.1   | 4.6 ± 0.2     | 6.5 ± 0.2   | 6.4 ± 0.2    | 6.2 ± 0.4     | 4.1 ± 0.2     | 6.1 ± 0.1   | 3.4 ± 0.2     | 5.1 ± 0.2   | 5.4 ± 0.3    | 5.5 ± 0.3     | E Asia F     | E Europe M   |
| 45  | 5.6 ± 0.3     | 7.8 ± 0.1   | 4.5 ± 0.2     | 6.4 ± 0.2   | 6.6 ± 0.2    | 6.2 ± 0.3     | 4.5 ± 0.3     | 6.2 ± 0.1   | 3.3 ± 0.2     | 4.9 ± 0.1   | 5.4 ± 0.3    | 5.9 ± 0.3     | E Asia F     | E Europe M   |
| 46  | 5.5 ± 0.4     | 7.9 ± 0.1   | 4.5 ± 0.2     | 6.1 ± 0.2   | 6.8 ± 0.2    | 6.3 ± 0.3     | 4.4 ± 0.3     | 6.4 ± 0.2   | 3.8 ± 0.3     | 5.0 ± 0.2   | 5.4 ± 0.3    | 6.1 ± 0.3     | E Asia F     | E Europe M   |
| 47  | 5.5 ± 0.4     | 7.8 ± 0.1   | 4.5 ± 0.3     | 6.0 ± 0.2   | 6.6 ± 0.3    | 6.4 ± 0.4     | 4.6 ± 0.4     | 6.4 ± 0.2   | 3.5 ± 0.2     | 5.1 ± 0.2   | 5.2 ± 0.3    | 6.0 ± 0.3     | E Asia F     | E Europe M   |
| 48  | 5.6 ± 0.5     | 7.9 ± 0.1   | 4.2 ± 0.3     | 6.1 ± 0.2   | 6.6 ± 0.2    | 6.5 ± 0.4     | 5.0 ± 0.4     | 6.3 ± 0.2   | 3.6 ± 0.3     | 5.1 ± 0.2   | 5.2 ± 0.3    | 6.7 ± 0.3     | E Asia F     | E Europe M   |
| 49  | 5.4 ± 0.4     | 8.1 ± 0.2   | 4.6 ± 0.3     | 6.0 ± 0.3   | 6.7 ± 0.3    | 6.9 ± 0.4     | 4.9 ± 0.3     | 6.3 ± 0.2   | 3.5 ± 0.2     | 5.1 ± 0.2   | 5.1 ± 0.3    | 6.7 ± 0.3     | E Asia F     | E Europe M   |
| 50  | 5.9 ± 0.5     | 8.1 ± 0.2   | 4.3 ± 0.3     | 5.7 ± 0.2   | 6.7 ± 0.3    | 6.7 ± 0.4     | 5.1 ± 0.4     | 6.3 ± 0.2   | 3.7 ± 0.2     | 5.2 ± 0.2   | 5.0 ± 0.3    | 6.8 ± 0.3     | E Asia F     | E Europe M   |
| 51  | 5.7 ± 0.4     | 8.1 ± 0.2   | 4.2 ± 0.2     | 5.8 ± 0.2   | 6.7 ± 0.3    | 7.1 ± 0.4     | 5.4 ± 0.4     | 6.4 ± 0.2   | 3.5 ± 0.2     | 5.4 ± 0.2   | 4.9 ± 0.2    | 7.0 ± 0.3     | E Asia F     | E Europe M   |
| 52  | 6.0 ± 0.6     | 8.2 ± 0.2   | 4.3 ± 0.2     | 5.7 ± 0.2   | 6.2 ± 0.2    | 7.4 ± 0.4     | 5.2 ± 0.5     | 6.3 ± 0.2   | 3.5 ± 0.2     | 5.3 ± 0.2   | 5.3 ± 0.3    | 7.5 ± 0.4     | E Asia F     | E Europe M   |

**Table 75.** Mean absolute error for Application-type images for algorithm cvut-003. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.8 ± 0.2     | 3.73 ± 0.08 | 3.1 ± 0.1     | 3.4 ± 0.1   | 3.5 ± 0.2    | 5.5 ± 0.3     | 2.6 ± 0.2     | 2.46 ± 0.08 | 1.42 ± 0.10   | 1.90 ± 0.09 | 2.0 ± 0.2    | 2.8 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 4.3 ± 0.3     | 3.57 ± 0.09 | 3.0 ± 0.2     | 3.4 ± 0.1   | 3.4 ± 0.2    | 5.1 ± 0.3     | 2.3 ± 0.2     | 2.54 ± 0.08 | 1.5 ± 0.1     | 2.1 ± 0.1   | 2.3 ± 0.2    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.1 ± 0.3     | 3.24 ± 0.10 | 2.7 ± 0.2     | 2.9 ± 0.1   | 3.2 ± 0.2    | 5.0 ± 0.4     | 2.7 ± 0.2     | 2.44 ± 0.08 | 1.8 ± 0.2     | 2.4 ± 0.1   | 2.4 ± 0.2    | 2.6 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 4.0 ± 0.3     | 2.80 ± 0.09 | 2.4 ± 0.2     | 2.8 ± 0.1   | 2.8 ± 0.2    | 4.7 ± 0.3     | 2.7 ± 0.2     | 2.30 ± 0.08 | 1.8 ± 0.1     | 2.8 ± 0.1   | 2.5 ± 0.2    | 3.1 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.3 ± 0.2     | 2.24 ± 0.07 | 2.2 ± 0.2     | 2.42 ± 0.10 | 2.2 ± 0.1    | 4.3 ± 0.3     | 2.9 ± 0.2     | 1.97 ± 0.05 | 2.0 ± 0.1     | 2.74 ± 0.08 | 2.1 ± 0.1    | 3.1 ± 0.2     | W Africa F   | E Asia M     |
| 19  | 3.4 ± 0.3     | 1.81 ± 0.07 | 2.0 ± 0.2     | 2.4 ± 0.1   | 2.0 ± 0.1    | 4.4 ± 0.3     | 2.9 ± 0.2     | 1.64 ± 0.06 | 1.9 ± 0.1     | 2.63 ± 0.09 | 1.76 ± 0.10  | 3.2 ± 0.3     | W Africa F   | E Asia M     |
| 20  | 3.3 ± 0.3     | 1.57 ± 0.09 | 1.9 ± 0.1     | 2.2 ± 0.1   | 1.8 ± 0.1    | 4.3 ± 0.3     | 3.1 ± 0.3     | 1.34 ± 0.06 | 1.9 ± 0.2     | 2.47 ± 0.10 | 1.5 ± 0.1    | 3.3 ± 0.3     | W Africa F   | E Asia M     |
| 21  | 3.3 ± 0.2     | 1.63 ± 0.07 | 2.1 ± 0.1     | 2.26 ± 0.09 | 1.9 ± 0.1    | 4.1 ± 0.3     | 3.2 ± 0.2     | 1.24 ± 0.05 | 1.9 ± 0.1     | 2.32 ± 0.10 | 1.40 ± 0.10  | 3.4 ± 0.3     | W Africa F   | E Asia M     |
| 22  | 3.8 ± 0.2     | 1.82 ± 0.05 | 2.32 ± 0.10   | 2.38 ± 0.07 | 2.18 ± 0.09  | 4.0 ± 0.2     | 3.6 ± 0.2     | 1.56 ± 0.05 | 1.9 ± 0.1     | 2.3 ± 0.1   | 1.63 ± 0.09  | 3.4 ± 0.2     | W Africa F   | E Asia M     |
| 23  | 3.7 ± 0.2     | 2.16 ± 0.05 | 2.52 ± 0.09   | 2.42 ± 0.07 | 2.41 ± 0.08  | 4.0 ± 0.2     | 3.6 ± 0.2     | 1.84 ± 0.05 | 1.9 ± 0.1     | 2.4 ± 0.1   | 1.93 ± 0.09  | 3.6 ± 0.2     | W Africa F   | E Asia M     |
| 24  | 3.8 ± 0.2     | 2.51 ± 0.05 | 2.50 ± 0.08   | 2.55 ± 0.06 | 2.60 ± 0.07  | 3.9 ± 0.2     | 3.7 ± 0.2     | 2.13 ± 0.06 | 2.0 ± 0.1     | 2.48 ± 0.09 | 2.08 ± 0.09  | 3.5 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 3.8 ± 0.2     | 2.72 ± 0.05 | 2.74 ± 0.08   | 2.67 ± 0.06 | 2.95 ± 0.07  | 3.8 ± 0.2     | 3.8 ± 0.2     | 2.49 ± 0.07 | 2.1 ± 0.1     | 2.53 ± 0.08 | 2.48 ± 0.09  | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 3.9 ± 0.2     | 3.05 ± 0.06 | 2.89 ± 0.08   | 2.79 ± 0.06 | 3.16 ± 0.08  | 3.6 ± 0.2     | 3.7 ± 0.2     | 2.61 ± 0.07 | 2.3 ± 0.1     | 2.61 ± 0.07 | 2.8 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 27  | 4.1 ± 0.2     | 3.27 ± 0.06 | 3.08 ± 0.09   | 3.17 ± 0.06 | 3.43 ± 0.08  | 3.6 ± 0.1     | 3.7 ± 0.2     | 2.90 ± 0.07 | 2.7 ± 0.1     | 2.83 ± 0.07 | 3.0 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 28  | 4.2 ± 0.2     | 3.61 ± 0.06 | 3.36 ± 0.08   | 3.56 ± 0.06 | 3.68 ± 0.09  | 3.7 ± 0.1     | 3.6 ± 0.1     | 3.15 ± 0.08 | 2.82 ± 0.10   | 2.99 ± 0.06 | 3.2 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 4.3 ± 0.2     | 3.83 ± 0.06 | 3.59 ± 0.08   | 3.87 ± 0.06 | 3.95 ± 0.09  | 3.7 ± 0.1     | 3.6 ± 0.1     | 3.30 ± 0.08 | 2.93 ± 0.10   | 3.09 ± 0.05 | 3.6 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 4.3 ± 0.1     | 4.16 ± 0.06 | 3.74 ± 0.09   | 4.17 ± 0.06 | 4.1 ± 0.1    | 3.6 ± 0.1     | 3.5 ± 0.1     | 3.53 ± 0.08 | 2.9 ± 0.1     | 3.27 ± 0.05 | 3.6 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.2 ± 0.2     | 4.38 ± 0.07 | 3.81 ± 0.10   | 4.50 ± 0.07 | 4.4 ± 0.1    | 3.5 ± 0.1     | 3.4 ± 0.1     | 3.72 ± 0.09 | 3.0 ± 0.1     | 3.40 ± 0.06 | 3.9 ± 0.2    | 2.9 ± 0.1     | S Asia F     | W Africa M   |
| 32  | 4.1 ± 0.2     | 4.60 ± 0.07 | 3.8 ± 0.1     | 4.68 ± 0.08 | 4.5 ± 0.1    | 3.6 ± 0.1     | 3.2 ± 0.1     | 3.78 ± 0.09 | 2.9 ± 0.1     | 3.50 ± 0.06 | 3.8 ± 0.1    | 2.9 ± 0.1     | S Asia F     | W Africa M   |
| 33  | 4.3 ± 0.2     | 4.71 ± 0.08 | 3.9 ± 0.1     | 4.89 ± 0.08 | 4.5 ± 0.1    | 3.4 ± 0.1     | 3.3 ± 0.1     | 3.79 ± 0.09 | 2.6 ± 0.1     | 3.55 ± 0.07 | 3.8 ± 0.2    | 2.7 ± 0.1     | S Asia F     | E Europe M   |
| 34  | 4.0 ± 0.2     | 4.81 ± 0.08 | 3.8 ± 0.1     | 4.96 ± 0.09 | 4.4 ± 0.1    | 3.5 ± 0.1     | 3.1 ± 0.1     | 3.81 ± 0.10 | 2.6 ± 0.1     | 3.68 ± 0.07 | 3.8 ± 0.2    | 2.7 ± 0.1     | S Asia F     | E Europe M   |
| 35  | 4.2 ± 0.2     | 5.00 ± 0.09 | 3.8 ± 0.1     | 5.12 ± 0.10 | 4.7 ± 0.2    | 3.5 ± 0.2     | 3.1 ± 0.1     | 3.8 ± 0.1   | 2.7 ± 0.1     | 3.82 ± 0.08 | 3.8 ± 0.2    | 2.9 ± 0.1     | S Asia F     | E Europe M   |
| 36  | 4.2 ± 0.2     | 4.91 ± 0.10 | 3.5 ± 0.1     | 5.3 ± 0.1   | 4.6 ± 0.2    | 3.4 ± 0.2     | 3.2 ± 0.2     | 3.8 ± 0.1   | 2.7 ± 0.1     | 3.97 ± 0.08 | 3.7 ± 0.2    | 2.9 ± 0.1     | S Asia F     | E Europe M   |
| 37  | 4.1 ± 0.2     | 4.99 ± 0.09 | 3.6 ± 0.1     | 5.4 ± 0.1   | 4.8 ± 0.2    | 3.8 ± 0.2     | 3.1 ± 0.2     | 3.8 ± 0.1   | 2.8 ± 0.1     | 4.18 ± 0.09 | 3.7 ± 0.2    | 3.0 ± 0.1     | S Asia F     | E Europe M   |
| 38  | 4.2 ± 0.2     | 5.10 ± 0.10 | 3.5 ± 0.1     | 5.4 ± 0.1   | 4.5 ± 0.2    | 3.9 ± 0.2     | 3.4 ± 0.2     | 3.86 ± 0.10 | 2.9 ± 0.1     | 4.32 ± 0.09 | 3.6 ± 0.2    | 3.2 ± 0.1     | S Asia F     | E Europe M   |
| 39  | 4.3 ± 0.2     | 5.20 ± 0.09 | 3.5 ± 0.2     | 5.6 ± 0.1   | 4.6 ± 0.2    | 4.0 ± 0.2     | 3.4 ± 0.2     | 4.0 ± 0.1   | 2.9 ± 0.1     | 4.44 ± 0.09 | 3.7 ± 0.2    | 3.6 ± 0.1     | S Asia F     | E Europe M   |
| 40  | 4.4 ± 0.3     | 5.31 ± 0.09 | 3.4 ± 0.1     | 5.5 ± 0.1   | 4.8 ± 0.2    | 4.2 ± 0.2     | 3.4 ± 0.2     | 4.25 ± 0.10 | 3.2 ± 0.1     | 4.49 ± 0.10 | 3.8 ± 0.2    | 3.9 ± 0.1     | S Asia F     | E Europe M   |
| 41  | 4.9 ± 0.3     | 5.41 ± 0.10 | 3.6 ± 0.1     | 5.7 ± 0.1   | 4.9 ± 0.2    | 4.6 ± 0.2     | 3.8 ± 0.2     | 4.36 ± 0.09 | 3.2 ± 0.1     | 4.7 ± 0.1   | 4.2 ± 0.2    | 4.3 ± 0.2     | S Asia F     | E Europe M   |
| 42  | 4.7 ± 0.3     | 5.53 ± 0.09 | 3.9 ± 0.2     | 5.5 ± 0.1   | 5.1 ± 0.2    | 4.6 ± 0.2     | 3.9 ± 0.2     | 4.58 ± 0.09 | 3.3 ± 0.2     | 4.8 ± 0.1   | 4.2 ± 0.2    | 4.5 ± 0.2     | S Asia F     | E Europe M   |
| 43  | 4.9 ± 0.3     | 5.79 ± 0.09 | 4.1 ± 0.2     | 5.8 ± 0.2   | 5.3 ± 0.2    | 5.1 ± 0.2     | 3.8 ± 0.2     | 4.68 ± 0.10 | 3.4 ± 0.2     | 4.7 ± 0.1   | 4.4 ± 0.2    | 5.0 ± 0.2     | S Asia F     | E Europe M   |
| 44  | 5.2 ± 0.3     | 6.01 ± 0.09 | 4.2 ± 0.2     | 5.7 ± 0.2   | 5.1 ± 0.2    | 5.3 ± 0.2     | 4.0 ± 0.2     | 4.8 ± 0.1   | 3.3 ± 0.2     | 4.8 ± 0.1   | 4.5 ± 0.2    | 5.0 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 5.2 ± 0.3     | 6.14 ± 0.10 | 4.0 ± 0.2     | 5.8 ± 0.2   | 5.3 ± 0.2    | 5.3 ± 0.2     | 4.0 ± 0.2     | 5.0 ± 0.1   | 3.4 ± 0.2     | 4.6 ± 0.1   | 4.6 ± 0.2    | 5.4 ± 0.2     | E Asia F     | E Europe M   |
| 46  | 5.3 ± 0.3     | 6.3 ± 0.1   | 4.0 ± 0.2     | 5.5 ± 0.2   | 5.5 ± 0.2    | 5.7 ± 0.3     | 4.1 ± 0.3     | 5.1 ± 0.1   | 3.6 ± 0.2     | 4.6 ± 0.1   | 4.7 ± 0.2    | 5.7 ± 0.2     | E Asia F     | E Europe M   |
| 47  | 5.1 ± 0.4     | 6.3 ± 0.1   | 3.9 ± 0.2     | 5.5 ± 0.2   | 5.3 ± 0.2    | 5.7 ± 0.3     | 4.2 ± 0.3     | 5.2 ± 0.1   | 3.3 ± 0.2     | 4.7 ± 0.2   | 4.5 ± 0.2    | 5.7 ± 0.2     | E Asia F     | E Europe M   |
| 48  | 5.3 ± 0.4     | 6.4 ± 0.1   | 3.7 ± 0.2     | 5.5 ± 0.2   | 5.5 ± 0.2    | 5.9 ± 0.3     | 4.6 ± 0.3     | 5.2 ± 0.1   | 3.5 ± 0.2     | 4.7 ± 0.2   | 4.7 ± 0.2    | 6.5 ± 0.3     | W Africa M   | E Europe M   |
| 49  | 5.0 ± 0.4     | 6.5 ± 0.1   | 3.9 ± 0.2     | 5.4 ± 0.2   | 5.4 ± 0.2    | 6.4 ± 0.3     | 4.7 ± 0.3     | 5.4 ± 0.1   | 3.4 ± 0.2     | 4.8 ± 0.2   | 4.5 ± 0.2    | 6.4 ± 0.3     | E Asia F     | E Europe M   |
| 50  | 5.7 ± 0.5     | 6.6 ± 0.1   | 3.8 ± 0.2     | 5.3 ± 0.2   | 5.5 ± 0.2    | 6.1 ± 0.4     | 4.8 ± 0.3     | 5.4 ± 0.1   | 3.6 ± 0.2     | 4.9 ± 0.2   | 4.5 ± 0.2    | 6.6 ± 0.3     | E Asia F     | E Europe M   |
| 51  | 5.3 ± 0.4     | 6.6 ± 0.1   | 4.0 ± 0.2     | 5.4 ± 0.2   | 5.6 ± 0.2    | 6.6 ± 0.4     | 5.0 ± 0.4     | 5.4 ± 0.1   | 3.4 ± 0.2     | 5.0 ± 0.2   | 4.5 ± 0.2    | 6.9 ± 0.3     | W Africa M   | E Europe M   |
| 52  | 5.6 ± 0.5     | 6.7 ± 0.1   | 4.0 ± 0.2     | 5.4 ± 0.2   | 5.4 ± 0.2    | 6.8 ± 0.4     | 4.7 ± 0.4     | 5.4 ± 0.1   | 3.5 ± 0.2     | 5.0 ± 0.2   | 4.7 ± 0.2    | 7.1 ± 0.3     | W Africa M   | E Europe M   |



**Table 76.** Mean absolute error for Application-type images for algorithm cyberlink-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.9 ± 0.3     | 5.3 ± 0.1   | 4.7 ± 0.2     | 5.0 ± 0.1   | 5.6 ± 0.2    | 6.6 ± 0.3     | 5.1 ± 0.3     | 4.2 ± 0.1   | 3.2 ± 0.2     | 3.8 ± 0.1   | 4.1 ± 0.2    | 5.1 ± 0.4     | W Africa F   | E Europe M   |
| 15  | 5.6 ± 0.3     | 5.2 ± 0.1   | 4.7 ± 0.2     | 5.0 ± 0.2   | 5.7 ± 0.3    | 6.3 ± 0.4     | 4.7 ± 0.3     | 4.3 ± 0.1   | 3.3 ± 0.2     | 3.9 ± 0.1   | 4.5 ± 0.3    | 4.8 ± 0.4     | W Africa F   | E Europe M   |
| 16  | 5.5 ± 0.3     | 5.2 ± 0.1   | 4.9 ± 0.2     | 4.7 ± 0.2   | 5.3 ± 0.3    | 6.3 ± 0.5     | 5.0 ± 0.3     | 4.2 ± 0.1   | 3.6 ± 0.2     | 3.9 ± 0.2   | 4.5 ± 0.3    | 5.0 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 5.7 ± 0.4     | 4.8 ± 0.1   | 4.3 ± 0.2     | 4.5 ± 0.2   | 5.1 ± 0.2    | 5.8 ± 0.4     | 5.0 ± 0.3     | 4.3 ± 0.1   | 3.3 ± 0.2     | 4.3 ± 0.2   | 4.5 ± 0.3    | 5.2 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 4.9 ± 0.3     | 4.38 ± 0.10 | 4.0 ± 0.2     | 4.1 ± 0.1   | 4.7 ± 0.1    | 5.3 ± 0.3     | 4.9 ± 0.2     | 4.05 ± 0.10 | 3.3 ± 0.2     | 4.2 ± 0.1   | 4.4 ± 0.2    | 4.7 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 4.9 ± 0.3     | 4.08 ± 0.09 | 3.5 ± 0.2     | 3.8 ± 0.1   | 4.4 ± 0.1    | 5.4 ± 0.3     | 4.8 ± 0.3     | 4.0 ± 0.1   | 3.3 ± 0.2     | 4.3 ± 0.1   | 4.3 ± 0.2    | 4.9 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.7 ± 0.3     | 3.67 ± 0.10 | 3.1 ± 0.2     | 3.4 ± 0.1   | 3.9 ± 0.1    | 4.8 ± 0.3     | 4.9 ± 0.2     | 3.9 ± 0.1   | 3.0 ± 0.2     | 4.3 ± 0.1   | 4.3 ± 0.2    | 4.6 ± 0.3     | E Africa M   | E Europe M   |
| 21  | 4.5 ± 0.3     | 3.14 ± 0.09 | 2.8 ± 0.1     | 3.0 ± 0.1   | 3.3 ± 0.1    | 4.5 ± 0.3     | 4.7 ± 0.2     | 3.76 ± 0.10 | 2.8 ± 0.2     | 4.2 ± 0.1   | 4.1 ± 0.2    | 4.4 ± 0.3     | E Africa M   | E Europe F   |
| 22  | 4.4 ± 0.3     | 2.71 ± 0.08 | 2.5 ± 0.1     | 2.56 ± 0.09 | 3.0 ± 0.1    | 4.4 ± 0.3     | 5.0 ± 0.2     | 3.6 ± 0.1   | 2.8 ± 0.2     | 3.9 ± 0.1   | 3.7 ± 0.2    | 4.1 ± 0.2     | E Africa M   | E Europe F   |
| 23  | 3.9 ± 0.2     | 2.45 ± 0.07 | 2.3 ± 0.1     | 2.42 ± 0.09 | 2.7 ± 0.1    | 4.2 ± 0.3     | 4.5 ± 0.2     | 3.37 ± 0.10 | 2.5 ± 0.1     | 3.8 ± 0.1   | 3.5 ± 0.2    | 4.1 ± 0.2     | E Africa M   | E Europe F   |
| 24  | 3.9 ± 0.2     | 2.22 ± 0.06 | 2.41 ± 0.10   | 2.55 ± 0.07 | 2.53 ± 0.08  | 3.9 ± 0.2     | 4.5 ± 0.2     | 3.29 ± 0.09 | 2.5 ± 0.1     | 3.6 ± 0.1   | 3.4 ± 0.1    | 3.8 ± 0.2     | E Africa M   | E Asia F     |
| 25  | 3.8 ± 0.2     | 2.12 ± 0.05 | 2.58 ± 0.08   | 2.76 ± 0.06 | 2.41 ± 0.08  | 4.2 ± 0.2     | 4.4 ± 0.2     | 3.15 ± 0.09 | 2.46 ± 0.10   | 3.38 ± 0.09 | 3.3 ± 0.1    | 3.7 ± 0.2     | E Africa M   | E Asia F     |
| 26  | 3.8 ± 0.2     | 2.09 ± 0.05 | 2.54 ± 0.08   | 2.95 ± 0.06 | 2.44 ± 0.07  | 4.0 ± 0.2     | 4.1 ± 0.2     | 2.87 ± 0.08 | 2.2 ± 0.1     | 3.11 ± 0.08 | 3.1 ± 0.1    | 3.5 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 4.0 ± 0.2     | 2.12 ± 0.05 | 2.49 ± 0.09   | 3.14 ± 0.06 | 2.32 ± 0.08  | 4.1 ± 0.2     | 4.0 ± 0.2     | 2.63 ± 0.08 | 2.0 ± 0.1     | 2.76 ± 0.08 | 3.0 ± 0.1    | 3.3 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 4.0 ± 0.2     | 2.23 ± 0.05 | 2.58 ± 0.10   | 3.42 ± 0.07 | 2.43 ± 0.08  | 4.0 ± 0.2     | 3.7 ± 0.2     | 2.48 ± 0.08 | 2.1 ± 0.1     | 2.57 ± 0.08 | 2.7 ± 0.1    | 3.2 ± 0.2     | W Africa F   | E Europe M   |
| 29  | 4.2 ± 0.2     | 2.53 ± 0.05 | 2.66 ± 0.09   | 3.60 ± 0.07 | 2.66 ± 0.08  | 4.1 ± 0.2     | 3.7 ± 0.2     | 2.42 ± 0.07 | 2.19 ± 0.10   | 2.61 ± 0.07 | 2.7 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 4.1 ± 0.2     | 2.85 ± 0.05 | 2.97 ± 0.09   | 3.98 ± 0.07 | 3.04 ± 0.09  | 4.3 ± 0.2     | 3.9 ± 0.2     | 2.39 ± 0.06 | 2.4 ± 0.1     | 2.78 ± 0.06 | 2.7 ± 0.1    | 3.2 ± 0.1     | W Africa F   | E Asia M     |
| 31  | 4.3 ± 0.2     | 3.11 ± 0.05 | 3.3 ± 0.1     | 4.27 ± 0.07 | 3.22 ± 0.09  | 4.3 ± 0.2     | 4.0 ± 0.2     | 2.36 ± 0.07 | 2.7 ± 0.1     | 2.96 ± 0.06 | 2.8 ± 0.1    | 3.4 ± 0.1     | W Africa F   | E Asia M     |
| 32  | 4.5 ± 0.2     | 3.33 ± 0.06 | 3.6 ± 0.1     | 4.53 ± 0.08 | 3.5 ± 0.1    | 4.5 ± 0.2     | 3.9 ± 0.2     | 2.21 ± 0.07 | 2.9 ± 0.1     | 3.02 ± 0.05 | 2.7 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Asia M     |
| 33  | 4.8 ± 0.2     | 3.57 ± 0.07 | 3.9 ± 0.1     | 4.76 ± 0.08 | 3.6 ± 0.1    | 4.5 ± 0.2     | 4.1 ± 0.2     | 2.30 ± 0.07 | 2.9 ± 0.1     | 3.12 ± 0.06 | 2.7 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Asia M     |
| 34  | 4.7 ± 0.2     | 3.87 ± 0.07 | 4.2 ± 0.1     | 4.94 ± 0.09 | 3.8 ± 0.1    | 4.5 ± 0.2     | 3.9 ± 0.2     | 2.52 ± 0.07 | 3.1 ± 0.1     | 3.25 ± 0.06 | 2.8 ± 0.1    | 3.4 ± 0.1     | S Asia F     | E Asia M     |
| 35  | 5.1 ± 0.2     | 4.28 ± 0.07 | 4.5 ± 0.1     | 5.10 ± 0.09 | 4.1 ± 0.1    | 4.7 ± 0.2     | 3.6 ± 0.2     | 2.71 ± 0.07 | 3.1 ± 0.1     | 3.32 ± 0.07 | 3.1 ± 0.2    | 3.6 ± 0.2     | S Asia F     | E Asia M     |
| 36  | 5.5 ± 0.3     | 4.54 ± 0.08 | 4.4 ± 0.1     | 5.4 ± 0.1   | 4.4 ± 0.1    | 4.9 ± 0.2     | 3.8 ± 0.2     | 2.91 ± 0.08 | 2.9 ± 0.1     | 3.40 ± 0.07 | 3.1 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 37  | 5.5 ± 0.3     | 4.80 ± 0.07 | 4.8 ± 0.2     | 5.6 ± 0.1   | 4.7 ± 0.1    | 5.3 ± 0.2     | 3.9 ± 0.2     | 3.20 ± 0.08 | 3.1 ± 0.1     | 3.61 ± 0.08 | 3.3 ± 0.2    | 3.7 ± 0.2     | S Asia F     | E Europe M   |
| 38  | 5.7 ± 0.3     | 5.15 ± 0.08 | 4.8 ± 0.2     | 5.8 ± 0.1   | 4.9 ± 0.1    | 5.8 ± 0.2     | 4.1 ± 0.2     | 3.49 ± 0.08 | 3.1 ± 0.1     | 3.70 ± 0.08 | 3.6 ± 0.2    | 4.0 ± 0.2     | S Asia F     | E Europe M   |
| 39  | 5.8 ± 0.3     | 5.32 ± 0.08 | 4.9 ± 0.2     | 5.9 ± 0.1   | 5.1 ± 0.1    | 5.7 ± 0.2     | 4.3 ± 0.2     | 3.69 ± 0.08 | 3.1 ± 0.1     | 3.85 ± 0.08 | 3.8 ± 0.2    | 4.3 ± 0.2     | S Asia F     | E Europe M   |
| 40  | 6.1 ± 0.3     | 5.45 ± 0.08 | 4.9 ± 0.2     | 6.0 ± 0.1   | 5.2 ± 0.1    | 6.1 ± 0.2     | 4.5 ± 0.2     | 3.72 ± 0.08 | 3.2 ± 0.2     | 4.07 ± 0.09 | 3.5 ± 0.1    | 4.5 ± 0.2     | W Africa F   | E Europe M   |
| 41  | 6.5 ± 0.3     | 5.50 ± 0.09 | 5.0 ± 0.2     | 6.2 ± 0.1   | 5.1 ± 0.2    | 6.4 ± 0.3     | 4.6 ± 0.2     | 3.80 ± 0.08 | 3.4 ± 0.1     | 4.19 ± 0.09 | 3.8 ± 0.2    | 4.7 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 6.2 ± 0.3     | 5.42 ± 0.09 | 4.9 ± 0.2     | 5.9 ± 0.2   | 5.0 ± 0.2    | 6.0 ± 0.3     | 4.6 ± 0.2     | 3.78 ± 0.09 | 3.3 ± 0.2     | 4.25 ± 0.10 | 3.8 ± 0.2    | 4.6 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 6.2 ± 0.3     | 5.3 ± 0.1   | 4.8 ± 0.2     | 5.9 ± 0.2   | 5.1 ± 0.2    | 6.1 ± 0.3     | 4.4 ± 0.3     | 3.72 ± 0.10 | 3.4 ± 0.2     | 4.2 ± 0.1   | 3.7 ± 0.2    | 4.7 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 6.7 ± 0.4     | 5.2 ± 0.1   | 4.7 ± 0.2     | 5.7 ± 0.2   | 4.9 ± 0.2    | 6.2 ± 0.3     | 4.0 ± 0.3     | 3.5 ± 0.1   | 3.4 ± 0.2     | 4.3 ± 0.1   | 3.8 ± 0.2    | 4.3 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 6.3 ± 0.4     | 5.0 ± 0.1   | 4.4 ± 0.2     | 5.7 ± 0.2   | 4.5 ± 0.2    | 6.3 ± 0.3     | 4.2 ± 0.3     | 3.5 ± 0.1   | 3.4 ± 0.2     | 4.2 ± 0.1   | 3.8 ± 0.2    | 4.7 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 6.5 ± 0.4     | 5.0 ± 0.1   | 4.4 ± 0.2     | 5.3 ± 0.2   | 4.5 ± 0.2    | 6.1 ± 0.3     | 4.2 ± 0.3     | 3.6 ± 0.1   | 3.7 ± 0.2     | 4.3 ± 0.1   | 3.9 ± 0.2    | 4.8 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 6.2 ± 0.4     | 4.9 ± 0.1   | 4.4 ± 0.2     | 5.3 ± 0.2   | 4.5 ± 0.2    | 6.2 ± 0.3     | 4.6 ± 0.3     | 3.7 ± 0.1   | 3.8 ± 0.2     | 4.4 ± 0.1   | 3.9 ± 0.2    | 4.9 ± 0.3     | W Africa F   | E Asia M     |
| 48  | 6.1 ± 0.4     | 4.9 ± 0.1   | 4.2 ± 0.2     | 5.4 ± 0.2   | 4.4 ± 0.2    | 6.0 ± 0.3     | 4.7 ± 0.3     | 3.84 ± 0.09 | 4.0 ± 0.2     | 4.5 ± 0.1   | 4.0 ± 0.2    | 5.3 ± 0.3     | E Africa F   | E Asia M     |
| 49  | 6.0 ± 0.4     | 4.9 ± 0.1   | 4.1 ± 0.2     | 5.3 ± 0.2   | 4.4 ± 0.2    | 5.9 ± 0.4     | 4.6 ± 0.3     | 4.02 ± 0.09 | 4.1 ± 0.2     | 4.5 ± 0.1   | 3.8 ± 0.2    | 5.0 ± 0.2     | E Africa F   | S E Asia M   |
| 50  | 6.0 ± 0.5     | 4.8 ± 0.1   | 4.1 ± 0.2     | 5.0 ± 0.2   | 4.3 ± 0.2    | 5.7 ± 0.4     | 4.6 ± 0.3     | 3.93 ± 0.09 | 3.9 ± 0.2     | 4.6 ± 0.2   | 3.7 ± 0.2    | 5.4 ± 0.3     | E Africa F   | S E Asia M   |
| 51  | 5.5 ± 0.5     | 4.6 ± 0.1   | 4.0 ± 0.2     | 4.9 ± 0.2   | 4.1 ± 0.2    | 5.5 ± 0.4     | 4.7 ± 0.4     | 3.8 ± 0.1   | 3.6 ± 0.2     | 4.5 ± 0.2   | 3.5 ± 0.2    | 5.5 ± 0.3     | E Africa F   | S E Asia M   |
| 52  | 5.8 ± 0.6     | 4.2 ± 0.1   | 4.1 ± 0.3     | 4.8 ± 0.2   | 3.5 ± 0.2    | 5.6 ± 0.4     | 4.2 ± 0.4     | 3.6 ± 0.1   | 3.6 ± 0.2     | 4.3 ± 0.2   | 3.1 ± 0.2    | 5.6 ± 0.3     | E Africa F   | S E Asia M   |

**Table 77.** Mean absolute error for Application-type images for algorithm cyberlink-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.7 ± 0.2     | 4.04 ± 0.10 | 3.2 ± 0.2     | 3.6 ± 0.1   | 4.0 ± 0.2    | 5.0 ± 0.3     | 3.8 ± 0.2     | 3.48 ± 0.09 | 2.3 ± 0.1     | 3.01 ± 0.09 | 3.2 ± 0.2    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 4.2 ± 0.2     | 4.0 ± 0.1   | 3.3 ± 0.2     | 3.5 ± 0.2   | 3.8 ± 0.3    | 4.5 ± 0.3     | 3.2 ± 0.2     | 3.35 ± 0.09 | 2.4 ± 0.2     | 2.9 ± 0.1   | 3.3 ± 0.2    | 3.6 ± 0.2     | W Africa F   | E Europe M   |
| 16  | 3.7 ± 0.2     | 3.7 ± 0.1   | 3.2 ± 0.2     | 3.1 ± 0.1   | 3.7 ± 0.2    | 4.3 ± 0.4     | 3.4 ± 0.2     | 3.04 ± 0.08 | 2.6 ± 0.2     | 2.8 ± 0.1   | 3.0 ± 0.2    | 3.4 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.9 ± 0.3     | 3.3 ± 0.1   | 3.0 ± 0.2     | 2.9 ± 0.1   | 3.4 ± 0.2    | 4.0 ± 0.3     | 3.0 ± 0.2     | 2.92 ± 0.10 | 2.4 ± 0.2     | 2.8 ± 0.1   | 2.9 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.2 ± 0.2     | 2.96 ± 0.08 | 2.7 ± 0.2     | 2.6 ± 0.1   | 3.0 ± 0.1    | 3.5 ± 0.3     | 2.8 ± 0.2     | 2.64 ± 0.08 | 2.4 ± 0.1     | 2.79 ± 0.10 | 2.7 ± 0.2    | 2.9 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 3.1 ± 0.2     | 2.67 ± 0.08 | 2.4 ± 0.2     | 2.4 ± 0.1   | 2.7 ± 0.1    | 3.6 ± 0.3     | 2.9 ± 0.2     | 2.49 ± 0.08 | 2.3 ± 0.1     | 2.8 ± 0.1   | 2.6 ± 0.1    | 3.1 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 3.1 ± 0.2     | 2.37 ± 0.09 | 2.2 ± 0.1     | 2.5 ± 0.1   | 2.5 ± 0.1    | 3.5 ± 0.3     | 2.9 ± 0.2     | 2.39 ± 0.09 | 2.1 ± 0.2     | 2.9 ± 0.1   | 2.6 ± 0.1    | 2.9 ± 0.2     | W Africa F   | E Europe M   |
| 21  | 2.9 ± 0.2     | 2.12 ± 0.07 | 2.1 ± 0.1     | 2.36 ± 0.08 | 2.2 ± 0.1    | 3.5 ± 0.3     | 3.0 ± 0.2     | 2.30 ± 0.08 | 2.1 ± 0.1     | 2.9 ± 0.1   | 2.6 ± 0.2    | 2.8 ± 0.2     | W Africa F   | E Europe F   |
| 22  | 3.0 ± 0.2     | 1.99 ± 0.06 | 2.10 ± 0.09   | 2.42 ± 0.08 | 2.16 ± 0.09  | 3.5 ± 0.3     | 3.4 ± 0.2     | 2.37 ± 0.08 | 2.2 ± 0.1     | 2.9 ± 0.1   | 2.5 ± 0.1    | 2.8 ± 0.2     | W Africa F   | E Asia F     |
| 23  | 3.1 ± 0.2     | 2.04 ± 0.06 | 2.13 ± 0.09   | 2.60 ± 0.07 | 2.20 ± 0.08  | 3.6 ± 0.2     | 3.4 ± 0.2     | 2.35 ± 0.08 | 2.1 ± 0.1     | 3.0 ± 0.1   | 2.6 ± 0.1    | 3.0 ± 0.2     | W Africa F   | E Asia F     |
| 24  | 3.3 ± 0.2     | 2.07 ± 0.06 | 2.06 ± 0.07   | 2.83 ± 0.07 | 2.19 ± 0.07  | 3.6 ± 0.2     | 3.4 ± 0.2     | 2.46 ± 0.09 | 2.2 ± 0.1     | 3.04 ± 0.09 | 2.6 ± 0.1    | 3.0 ± 0.2     | W Africa F   | E Europe F   |
| 25  | 3.5 ± 0.2     | 2.13 ± 0.05 | 2.28 ± 0.07   | 3.03 ± 0.07 | 2.39 ± 0.08  | 4.0 ± 0.2     | 3.4 ± 0.2     | 2.61 ± 0.09 | 2.2 ± 0.1     | 2.98 ± 0.09 | 2.9 ± 0.1    | 3.1 ± 0.2     | W Africa F   | E Asia F     |
| 26  | 3.6 ± 0.1     | 2.38 ± 0.05 | 2.34 ± 0.07   | 3.33 ± 0.06 | 2.58 ± 0.08  | 3.9 ± 0.2     | 3.2 ± 0.2     | 2.52 ± 0.08 | 2.3 ± 0.1     | 2.92 ± 0.08 | 2.9 ± 0.1    | 3.0 ± 0.1     | W Africa F   | E Europe M   |
| 27  | 3.9 ± 0.1     | 2.66 ± 0.05 | 2.45 ± 0.07   | 3.59 ± 0.07 | 2.82 ± 0.08  | 4.2 ± 0.2     | 3.3 ± 0.2     | 2.58 ± 0.08 | 2.3 ± 0.1     | 2.84 ± 0.07 | 3.0 ± 0.1    | 3.0 ± 0.1     | W Africa F   | E Europe M   |
| 28  | 4.0 ± 0.2     | 3.01 ± 0.05 | 2.65 ± 0.07   | 3.93 ± 0.07 | 3.08 ± 0.08  | 4.3 ± 0.2     | 3.2 ± 0.1     | 2.73 ± 0.08 | 2.42 ± 0.10   | 2.81 ± 0.07 | 2.9 ± 0.1    | 2.9 ± 0.1     | W Africa F   | E Europe M   |
| 29  | 4.1 ± 0.2     | 3.30 ± 0.05 | 2.81 ± 0.08   | 4.12 ± 0.07 | 3.34 ± 0.09  | 4.4 ± 0.2     | 3.1 ± 0.1     | 2.70 ± 0.07 | 2.45 ± 0.10   | 2.77 ± 0.06 | 2.9 ± 0.1    | 3.0 ± 0.1     | W Africa F   | E Europe M   |
| 30  | 4.2 ± 0.2     | 3.56 ± 0.06 | 3.06 ± 0.08   | 4.38 ± 0.08 | 3.56 ± 0.09  | 4.5 ± 0.2     | 3.2 ± 0.1     | 2.68 ± 0.07 | 2.39 ± 0.10   | 2.82 ± 0.06 | 2.8 ± 0.1    | 3.0 ± 0.1     | W Africa F   | E Europe M   |
| 31  | 4.6 ± 0.2     | 3.78 ± 0.06 | 3.24 ± 0.09   | 4.77 ± 0.08 | 3.77 ± 0.09  | 4.5 ± 0.2     | 3.3 ± 0.2     | 2.73 ± 0.08 | 2.51 ± 0.10   | 2.87 ± 0.06 | 3.0 ± 0.1    | 3.1 ± 0.1     | S Asia F     | E Europe M   |
| 32  | 4.7 ± 0.2     | 4.00 ± 0.07 | 3.40 ± 0.10   | 5.01 ± 0.09 | 3.9 ± 0.1    | 4.8 ± 0.2     | 3.2 ± 0.1     | 2.71 ± 0.07 | 2.59 ± 0.10   | 2.94 ± 0.06 | 3.0 ± 0.1    | 3.2 ± 0.1     | S Asia F     | E Europe M   |
| 33  | 5.1 ± 0.2     | 4.12 ± 0.07 | 3.7 ± 0.1     | 5.28 ± 0.09 | 4.1 ± 0.1    | 4.8 ± 0.2     | 3.5 ± 0.2     | 2.76 ± 0.08 | 2.40 ± 0.10   | 2.97 ± 0.06 | 3.0 ± 0.1    | 3.2 ± 0.1     | S Asia F     | E Europe M   |
| 34  | 5.0 ± 0.2     | 4.28 ± 0.08 | 4.0 ± 0.1     | 5.46 ± 0.10 | 4.1 ± 0.1    | 4.9 ± 0.2     | 3.6 ± 0.2     | 2.85 ± 0.08 | 2.7 ± 0.1     | 3.12 ± 0.06 | 2.9 ± 0.1    | 3.4 ± 0.1     | S Asia F     | E Europe M   |
| 35  | 5.4 ± 0.2     | 4.49 ± 0.08 | 4.2 ± 0.1     | 5.65 ± 0.10 | 4.3 ± 0.1    | 4.9 ± 0.2     | 3.6 ± 0.2     | 2.90 ± 0.08 | 2.7 ± 0.1     | 3.20 ± 0.06 | 3.1 ± 0.1    | 3.8 ± 0.1     | S Asia F     | E Europe M   |
| 36  | 5.7 ± 0.2     | 4.53 ± 0.09 | 4.2 ± 0.1     | 6.0 ± 0.1   | 4.4 ± 0.1    | 5.1 ± 0.2     | 4.1 ± 0.2     | 2.91 ± 0.08 | 2.7 ± 0.1     | 3.24 ± 0.07 | 3.1 ± 0.1    | 3.9 ± 0.1     | S Asia F     | E Europe M   |
| 37  | 6.0 ± 0.3     | 4.58 ± 0.08 | 4.5 ± 0.1     | 6.0 ± 0.1   | 4.6 ± 0.1    | 5.6 ± 0.2     | 4.0 ± 0.2     | 2.99 ± 0.08 | 2.9 ± 0.1     | 3.36 ± 0.07 | 3.1 ± 0.2    | 4.1 ± 0.1     | S Asia F     | E Europe M   |
| 38  | 5.7 ± 0.3     | 4.81 ± 0.09 | 4.6 ± 0.1     | 6.2 ± 0.1   | 4.6 ± 0.1    | 5.8 ± 0.3     | 4.2 ± 0.2     | 3.05 ± 0.08 | 3.1 ± 0.1     | 3.33 ± 0.07 | 3.1 ± 0.2    | 4.4 ± 0.2     | S Asia F     | E Asia M     |
| 39  | 6.0 ± 0.3     | 4.92 ± 0.09 | 4.6 ± 0.1     | 6.3 ± 0.1   | 4.8 ± 0.1    | 5.8 ± 0.3     | 4.1 ± 0.2     | 3.09 ± 0.09 | 3.2 ± 0.1     | 3.40 ± 0.07 | 3.3 ± 0.1    | 4.6 ± 0.2     | S Asia F     | E Asia M     |
| 40  | 6.2 ± 0.3     | 5.03 ± 0.09 | 4.5 ± 0.2     | 6.4 ± 0.1   | 5.0 ± 0.2    | 6.0 ± 0.3     | 4.3 ± 0.2     | 3.14 ± 0.08 | 3.3 ± 0.1     | 3.52 ± 0.08 | 3.1 ± 0.1    | 5.0 ± 0.2     | S Asia F     | S E Asia M   |
| 41  | 6.4 ± 0.3     | 5.06 ± 0.09 | 4.6 ± 0.2     | 6.6 ± 0.1   | 5.0 ± 0.2    | 6.2 ± 0.3     | 4.5 ± 0.2     | 3.25 ± 0.08 | 3.3 ± 0.1     | 3.59 ± 0.08 | 3.4 ± 0.2    | 5.2 ± 0.2     | S Asia F     | E Asia M     |
| 42  | 6.3 ± 0.3     | 5.01 ± 0.09 | 4.8 ± 0.2     | 6.3 ± 0.2   | 4.9 ± 0.2    | 6.1 ± 0.3     | 4.5 ± 0.2     | 3.36 ± 0.08 | 3.3 ± 0.2     | 3.66 ± 0.09 | 3.3 ± 0.2    | 5.1 ± 0.2     | S Asia F     | E Europe M   |
| 43  | 6.4 ± 0.4     | 5.21 ± 0.10 | 4.6 ± 0.2     | 6.5 ± 0.2   | 5.1 ± 0.2    | 6.3 ± 0.3     | 4.4 ± 0.3     | 3.37 ± 0.08 | 3.3 ± 0.2     | 3.67 ± 0.09 | 3.4 ± 0.2    | 5.5 ± 0.2     | S Asia F     | E Europe M   |
| 44  | 6.6 ± 0.4     | 5.30 ± 0.10 | 4.8 ± 0.2     | 6.4 ± 0.2   | 4.9 ± 0.2    | 6.5 ± 0.3     | 4.4 ± 0.3     | 3.32 ± 0.09 | 3.1 ± 0.2     | 3.7 ± 0.1   | 3.5 ± 0.2    | 5.3 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 6.4 ± 0.4     | 5.2 ± 0.1   | 4.5 ± 0.2     | 6.4 ± 0.2   | 5.0 ± 0.2    | 6.4 ± 0.3     | 4.4 ± 0.3     | 3.32 ± 0.09 | 3.2 ± 0.2     | 3.7 ± 0.1   | 3.4 ± 0.2    | 5.7 ± 0.2     | S Asia F     | E Europe M   |
| 46  | 6.7 ± 0.4     | 5.3 ± 0.1   | 4.4 ± 0.2     | 6.0 ± 0.2   | 5.1 ± 0.2    | 6.2 ± 0.3     | 4.6 ± 0.3     | 3.24 ± 0.09 | 3.2 ± 0.2     | 3.6 ± 0.1   | 3.5 ± 0.2    | 5.6 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 5.9 ± 0.4     | 5.1 ± 0.1   | 4.5 ± 0.2     | 6.1 ± 0.2   | 4.8 ± 0.2    | 6.2 ± 0.3     | 4.3 ± 0.3     | 3.2 ± 0.1   | 3.1 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.2    | 5.4 ± 0.3     | W Africa F   | E Europe M   |
| 48  | 5.9 ± 0.5     | 5.0 ± 0.1   | 4.2 ± 0.2     | 5.9 ± 0.2   | 4.7 ± 0.2    | 6.0 ± 0.4     | 4.5 ± 0.3     | 3.29 ± 0.09 | 3.1 ± 0.2     | 3.7 ± 0.1   | 3.4 ± 0.2    | 5.8 ± 0.3     | W Africa F   | E Europe M   |
| 49  | 5.7 ± 0.5     | 5.1 ± 0.1   | 4.3 ± 0.2     | 5.6 ± 0.2   | 4.6 ± 0.2    | 6.2 ± 0.4     | 4.3 ± 0.3     | 3.5 ± 0.1   | 3.1 ± 0.2     | 3.6 ± 0.1   | 3.2 ± 0.2    | 5.5 ± 0.3     | W Africa F   | E Europe M   |
| 50  | 6.0 ± 0.5     | 5.0 ± 0.1   | 4.1 ± 0.2     | 5.4 ± 0.2   | 4.5 ± 0.2    | 6.0 ± 0.4     | 4.1 ± 0.3     | 3.49 ± 0.10 | 3.1 ± 0.2     | 3.6 ± 0.1   | 3.1 ± 0.2    | 5.6 ± 0.3     | W Africa F   | E Europe M   |
| 51  | 5.5 ± 0.5     | 4.8 ± 0.1   | 4.1 ± 0.2     | 5.4 ± 0.2   | 4.3 ± 0.2    | 6.1 ± 0.4     | 4.2 ± 0.4     | 3.56 ± 0.10 | 2.7 ± 0.1     | 3.7 ± 0.2   | 3.1 ± 0.2    | 5.6 ± 0.3     | W Africa F   | E Europe M   |
| 52  | 5.8 ± 0.6     | 4.7 ± 0.1   | 3.9 ± 0.2     | 5.3 ± 0.2   | 4.0 ± 0.2    | 6.2 ± 0.5     | 3.9 ± 0.4     | 3.6 ± 0.1   | 2.9 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.2    | 5.8 ± 0.4     | W Africa F   | E Europe M   |

**Table 78.** Mean absolute error for Application-type images for algorithm daon-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.3 ± 0.3     | 2.9 ± 0.1   | 1.9 ± 0.2     | 3.0 ± 0.1   | 3.1 ± 0.2    | 3.3 ± 0.4     | 3.7 ± 0.3     | 4.3 ± 0.2   | 2.2 ± 0.2     | 4.2 ± 0.2   | 3.9 ± 0.3    | 2.6 ± 0.4     | E Asia M     | E Europe F   |
| 15  | 3.7 ± 0.3     | 3.4 ± 0.2   | 2.0 ± 0.2     | 3.6 ± 0.2   | 3.6 ± 0.3    | 3.6 ± 0.4     | 4.0 ± 0.3     | 4.7 ± 0.2   | 2.3 ± 0.2     | 4.6 ± 0.2   | 4.7 ± 0.3    | 3.4 ± 0.4     | S E Asia M   | E Europe F   |
| 16  | 4.4 ± 0.4     | 3.7 ± 0.2   | 2.4 ± 0.2     | 3.4 ± 0.2   | 4.0 ± 0.3    | 5.0 ± 0.6     | 4.8 ± 0.3     | 4.6 ± 0.2   | 2.6 ± 0.2     | 5.0 ± 0.2   | 4.9 ± 0.3    | 3.4 ± 0.3     | W Africa F   | E Europe F   |
| 17  | 5.0 ± 0.4     | 4.0 ± 0.1   | 2.5 ± 0.2     | 3.6 ± 0.2   | 4.1 ± 0.3    | 5.5 ± 0.4     | 5.3 ± 0.4     | 4.5 ± 0.2   | 2.3 ± 0.2     | 5.0 ± 0.2   | 4.5 ± 0.3    | 4.3 ± 0.5     | W Africa F   | E Europe M   |
| 18  | 4.6 ± 0.3     | 4.2 ± 0.1   | 2.6 ± 0.2     | 3.6 ± 0.1   | 4.0 ± 0.2    | 5.8 ± 0.4     | 5.2 ± 0.3     | 4.1 ± 0.1   | 2.4 ± 0.2     | 4.9 ± 0.1   | 4.1 ± 0.3    | 4.5 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 5.2 ± 0.3     | 4.2 ± 0.1   | 2.8 ± 0.2     | 3.5 ± 0.1   | 4.2 ± 0.2    | 6.1 ± 0.4     | 5.5 ± 0.3     | 3.9 ± 0.1   | 2.5 ± 0.2     | 4.9 ± 0.1   | 3.9 ± 0.2    | 4.4 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 5.1 ± 0.3     | 4.4 ± 0.1   | 2.7 ± 0.2     | 3.7 ± 0.1   | 4.3 ± 0.2    | 6.3 ± 0.4     | 5.5 ± 0.3     | 3.6 ± 0.1   | 2.5 ± 0.2     | 4.9 ± 0.1   | 3.7 ± 0.2    | 4.6 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 4.8 ± 0.3     | 4.4 ± 0.1   | 2.8 ± 0.1     | 3.5 ± 0.1   | 4.3 ± 0.2    | 6.7 ± 0.4     | 5.6 ± 0.3     | 3.4 ± 0.1   | 2.5 ± 0.2     | 4.4 ± 0.1   | 3.7 ± 0.2    | 4.5 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 5.0 ± 0.2     | 4.3 ± 0.1   | 2.9 ± 0.1     | 3.6 ± 0.1   | 4.4 ± 0.1    | 6.4 ± 0.3     | 5.7 ± 0.3     | 3.4 ± 0.1   | 2.5 ± 0.2     | 4.3 ± 0.1   | 3.4 ± 0.2    | 4.4 ± 0.3     | W Africa F   | E Europe M   |
| 23  | 4.9 ± 0.2     | 4.5 ± 0.1   | 3.0 ± 0.1     | 3.62 ± 0.10 | 4.4 ± 0.1    | 6.6 ± 0.3     | 5.6 ± 0.3     | 3.4 ± 0.1   | 2.5 ± 0.1     | 4.2 ± 0.1   | 3.6 ± 0.2    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 5.0 ± 0.2     | 4.5 ± 0.1   | 3.0 ± 0.1     | 3.62 ± 0.09 | 4.3 ± 0.1    | 6.5 ± 0.3     | 5.4 ± 0.3     | 3.4 ± 0.1   | 2.6 ± 0.1     | 4.0 ± 0.1   | 3.3 ± 0.1    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 5.0 ± 0.2     | 4.5 ± 0.1   | 3.2 ± 0.1     | 3.74 ± 0.09 | 4.3 ± 0.1    | 6.5 ± 0.3     | 5.4 ± 0.3     | 3.5 ± 0.1   | 2.6 ± 0.1     | 3.9 ± 0.1   | 3.6 ± 0.2    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 4.7 ± 0.2     | 4.6 ± 0.1   | 3.25 ± 0.09   | 3.96 ± 0.08 | 4.4 ± 0.1    | 6.1 ± 0.2     | 5.1 ± 0.2     | 3.4 ± 0.1   | 2.7 ± 0.1     | 3.78 ± 0.10 | 3.6 ± 0.2    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 27  | 4.7 ± 0.2     | 4.6 ± 0.1   | 3.35 ± 0.10   | 4.10 ± 0.08 | 4.3 ± 0.1    | 6.1 ± 0.2     | 4.9 ± 0.2     | 3.5 ± 0.1   | 2.8 ± 0.1     | 3.60 ± 0.09 | 3.7 ± 0.2    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 4.6 ± 0.2     | 4.6 ± 0.1   | 3.46 ± 0.10   | 4.46 ± 0.09 | 4.4 ± 0.1    | 6.0 ± 0.2     | 4.9 ± 0.2     | 3.6 ± 0.1   | 2.9 ± 0.1     | 3.49 ± 0.08 | 3.7 ± 0.2    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 29  | 4.8 ± 0.2     | 4.54 ± 0.10 | 3.6 ± 0.1     | 4.64 ± 0.09 | 4.3 ± 0.1    | 6.0 ± 0.2     | 4.9 ± 0.2     | 3.7 ± 0.1   | 3.0 ± 0.1     | 3.37 ± 0.07 | 3.8 ± 0.2    | 4.6 ± 0.2     | W Africa F   | E Europe M   |
| 30  | 4.9 ± 0.2     | 4.69 ± 0.10 | 3.7 ± 0.1     | 4.83 ± 0.09 | 4.5 ± 0.1    | 6.0 ± 0.2     | 4.8 ± 0.2     | 3.69 ± 0.10 | 3.0 ± 0.1     | 3.39 ± 0.07 | 3.8 ± 0.2    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 31  | 5.0 ± 0.2     | 4.7 ± 0.1   | 3.8 ± 0.1     | 5.26 ± 0.09 | 4.5 ± 0.1    | 5.7 ± 0.2     | 4.8 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.1     | 3.37 ± 0.07 | 4.0 ± 0.2    | 4.6 ± 0.2     | W Africa F   | E Europe M   |
| 32  | 5.0 ± 0.2     | 4.7 ± 0.1   | 3.9 ± 0.1     | 5.45 ± 0.10 | 4.6 ± 0.2    | 5.9 ± 0.2     | 4.6 ± 0.2     | 3.71 ± 0.10 | 3.1 ± 0.1     | 3.41 ± 0.07 | 4.0 ± 0.2    | 4.5 ± 0.2     | W Africa F   | E Europe M   |
| 33  | 5.3 ± 0.2     | 4.6 ± 0.1   | 4.1 ± 0.1     | 5.7 ± 0.1   | 4.5 ± 0.2    | 5.6 ± 0.2     | 4.6 ± 0.2     | 3.69 ± 0.10 | 3.0 ± 0.1     | 3.38 ± 0.07 | 3.9 ± 0.2    | 4.4 ± 0.2     | S Asia F     | E Europe M   |
| 34  | 5.2 ± 0.2     | 4.65 ± 0.10 | 4.1 ± 0.1     | 5.8 ± 0.1   | 4.5 ± 0.2    | 6.0 ± 0.2     | 4.4 ± 0.2     | 3.74 ± 0.10 | 3.2 ± 0.1     | 3.47 ± 0.07 | 4.1 ± 0.2    | 4.5 ± 0.2     | W Africa F   | E Europe M   |
| 35  | 5.4 ± 0.2     | 4.7 ± 0.1   | 4.3 ± 0.1     | 6.1 ± 0.1   | 4.6 ± 0.2    | 5.5 ± 0.2     | 4.5 ± 0.2     | 3.8 ± 0.1   | 3.1 ± 0.1     | 3.53 ± 0.07 | 4.2 ± 0.2    | 4.7 ± 0.2     | S Asia F     | E Europe M   |
| 36  | 5.5 ± 0.3     | 4.7 ± 0.1   | 4.0 ± 0.1     | 6.3 ± 0.1   | 4.5 ± 0.2    | 5.7 ± 0.3     | 4.5 ± 0.2     | 3.7 ± 0.1   | 3.3 ± 0.1     | 3.58 ± 0.07 | 4.1 ± 0.2    | 4.7 ± 0.2     | S Asia F     | E Europe M   |
| 37  | 5.6 ± 0.3     | 4.6 ± 0.1   | 4.2 ± 0.2     | 6.4 ± 0.1   | 4.5 ± 0.2    | 5.5 ± 0.3     | 4.3 ± 0.2     | 3.74 ± 0.10 | 3.1 ± 0.1     | 3.71 ± 0.08 | 3.9 ± 0.2    | 4.4 ± 0.2     | S Asia F     | E Europe M   |
| 38  | 5.6 ± 0.3     | 4.6 ± 0.1   | 4.2 ± 0.2     | 6.5 ± 0.1   | 4.5 ± 0.2    | 6.2 ± 0.3     | 4.2 ± 0.2     | 3.8 ± 0.1   | 3.2 ± 0.1     | 3.71 ± 0.08 | 4.2 ± 0.2    | 4.4 ± 0.2     | S Asia F     | E Europe M   |
| 39  | 5.6 ± 0.3     | 4.5 ± 0.1   | 4.1 ± 0.2     | 6.8 ± 0.1   | 4.4 ± 0.2    | 5.7 ± 0.3     | 4.2 ± 0.2     | 3.68 ± 0.10 | 3.2 ± 0.1     | 3.76 ± 0.08 | 4.3 ± 0.2    | 4.6 ± 0.2     | S Asia F     | E Europe M   |
| 40  | 5.8 ± 0.3     | 4.5 ± 0.1   | 4.0 ± 0.2     | 6.8 ± 0.2   | 4.4 ± 0.2    | 5.7 ± 0.3     | 4.1 ± 0.2     | 3.70 ± 0.10 | 3.2 ± 0.1     | 3.87 ± 0.08 | 4.0 ± 0.2    | 4.8 ± 0.2     | S Asia F     | E Europe M   |
| 41  | 5.9 ± 0.3     | 4.5 ± 0.1   | 3.8 ± 0.2     | 6.8 ± 0.2   | 4.5 ± 0.2    | 5.8 ± 0.3     | 4.3 ± 0.2     | 3.68 ± 0.10 | 3.3 ± 0.2     | 3.97 ± 0.09 | 4.1 ± 0.2    | 4.8 ± 0.2     | S Asia F     | E Europe M   |
| 42  | 5.8 ± 0.3     | 4.3 ± 0.1   | 3.8 ± 0.2     | 6.5 ± 0.2   | 4.2 ± 0.2    | 5.9 ± 0.3     | 4.1 ± 0.2     | 3.64 ± 0.09 | 3.2 ± 0.1     | 3.91 ± 0.10 | 4.1 ± 0.2    | 4.5 ± 0.2     | S Asia F     | E Europe M   |
| 43  | 6.0 ± 0.4     | 4.3 ± 0.1   | 3.8 ± 0.2     | 6.8 ± 0.2   | 4.3 ± 0.2    | 6.0 ± 0.4     | 4.0 ± 0.2     | 3.53 ± 0.09 | 3.2 ± 0.2     | 4.0 ± 0.1   | 4.1 ± 0.2    | 4.7 ± 0.2     | S Asia F     | E Europe M   |
| 44  | 5.8 ± 0.4     | 4.4 ± 0.1   | 3.7 ± 0.2     | 6.7 ± 0.2   | 4.3 ± 0.2    | 6.2 ± 0.4     | 3.9 ± 0.2     | 3.56 ± 0.10 | 3.2 ± 0.2     | 3.9 ± 0.1   | 4.1 ± 0.2    | 4.5 ± 0.2     | S Asia F     | E Europe M   |
| 45  | 5.7 ± 0.4     | 4.4 ± 0.1   | 3.7 ± 0.2     | 6.7 ± 0.2   | 4.2 ± 0.2    | 5.9 ± 0.4     | 3.8 ± 0.2     | 3.51 ± 0.10 | 3.1 ± 0.2     | 3.9 ± 0.1   | 3.9 ± 0.2    | 4.9 ± 0.2     | S Asia F     | E Europe M   |
| 46  | 5.8 ± 0.4     | 4.6 ± 0.1   | 3.7 ± 0.2     | 6.2 ± 0.2   | 4.4 ± 0.2    | 5.9 ± 0.3     | 3.8 ± 0.3     | 3.39 ± 0.10 | 3.2 ± 0.2     | 3.9 ± 0.1   | 4.0 ± 0.2    | 4.6 ± 0.2     | S Asia F     | E Europe M   |
| 47  | 5.4 ± 0.4     | 4.4 ± 0.1   | 3.8 ± 0.2     | 6.2 ± 0.2   | 4.3 ± 0.2    | 5.9 ± 0.4     | 3.8 ± 0.3     | 3.24 ± 0.10 | 3.1 ± 0.2     | 4.0 ± 0.1   | 3.6 ± 0.2    | 4.5 ± 0.2     | S Asia F     | E Europe M   |
| 48  | 5.7 ± 0.5     | 4.4 ± 0.1   | 3.7 ± 0.2     | 6.2 ± 0.2   | 4.2 ± 0.2    | 5.7 ± 0.4     | 4.0 ± 0.3     | 3.20 ± 0.09 | 3.3 ± 0.2     | 3.8 ± 0.1   | 3.4 ± 0.2    | 4.8 ± 0.3     | S Asia F     | E Asia M     |
| 49  | 5.7 ± 0.5     | 4.5 ± 0.1   | 3.6 ± 0.2     | 5.9 ± 0.2   | 4.2 ± 0.2    | 5.8 ± 0.4     | 4.1 ± 0.3     | 3.23 ± 0.10 | 3.2 ± 0.2     | 3.8 ± 0.1   | 3.4 ± 0.2    | 4.6 ± 0.3     | S Asia F     | E Europe M   |
| 50  | 5.7 ± 0.6     | 4.5 ± 0.1   | 3.6 ± 0.2     | 5.5 ± 0.2   | 4.2 ± 0.2    | 5.4 ± 0.4     | 4.2 ± 0.3     | 3.15 ± 0.10 | 3.2 ± 0.2     | 3.9 ± 0.1   | 3.5 ± 0.2    | 4.7 ± 0.3     | E Africa F   | E Asia M     |
| 51  | 5.0 ± 0.4     | 4.5 ± 0.1   | 3.6 ± 0.2     | 5.7 ± 0.2   | 4.4 ± 0.2    | 5.7 ± 0.4     | 4.4 ± 0.4     | 3.24 ± 0.10 | 3.2 ± 0.2     | 3.8 ± 0.2   | 3.4 ± 0.2    | 4.9 ± 0.3     | S Asia F     | E Europe M   |
| 52  | 5.6 ± 0.5     | 4.5 ± 0.1   | 3.9 ± 0.2     | 5.5 ± 0.2   | 4.2 ± 0.2    | 5.9 ± 0.4     | 4.0 ± 0.4     | 3.18 ± 0.10 | 3.4 ± 0.2     | 3.8 ± 0.1   | 3.3 ± 0.2    | 5.0 ± 0.3     | W Africa F   | E Asia M     |

**Table 79.** Mean absolute error for Application-type images for algorithm daon-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.7 ± 0.2     | 2.6 ± 0.1   | 1.8 ± 0.1     | 2.5 ± 0.1   | 2.7 ± 0.2    | 2.6 ± 0.3     | 3.1 ± 0.2     | 3.3 ± 0.1   | 1.9 ± 0.1     | 3.1 ± 0.1   | 2.9 ± 0.2    | 2.2 ± 0.2     | E Asia M     | E Europe F   |
| 15  | 3.0 ± 0.3     | 2.9 ± 0.1   | 1.9 ± 0.1     | 2.9 ± 0.2   | 3.1 ± 0.3    | 3.1 ± 0.4     | 3.1 ± 0.2     | 3.6 ± 0.1   | 1.9 ± 0.2     | 3.5 ± 0.2   | 3.4 ± 0.2    | 2.8 ± 0.3     | E Asia M     | E Europe M   |
| 16  | 3.7 ± 0.3     | 3.2 ± 0.1   | 2.2 ± 0.2     | 2.8 ± 0.2   | 3.4 ± 0.3    | 4.2 ± 0.5     | 3.7 ± 0.3     | 3.7 ± 0.1   | 2.2 ± 0.2     | 3.8 ± 0.2   | 3.6 ± 0.3    | 2.8 ± 0.3     | W Africa F   | E Europe F   |
| 17  | 4.0 ± 0.3     | 3.5 ± 0.1   | 2.4 ± 0.2     | 3.0 ± 0.1   | 3.6 ± 0.3    | 4.8 ± 0.4     | 4.1 ± 0.3     | 3.7 ± 0.1   | 2.0 ± 0.2     | 4.0 ± 0.2   | 3.7 ± 0.3    | 3.6 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.0 ± 0.3     | 3.7 ± 0.1   | 2.6 ± 0.2     | 3.1 ± 0.1   | 3.5 ± 0.2    | 5.0 ± 0.4     | 4.2 ± 0.3     | 3.5 ± 0.1   | 2.3 ± 0.2     | 4.1 ± 0.1   | 3.3 ± 0.2    | 3.8 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.4 ± 0.3     | 3.8 ± 0.1   | 2.7 ± 0.2     | 3.1 ± 0.1   | 3.7 ± 0.1    | 5.2 ± 0.4     | 4.7 ± 0.3     | 3.4 ± 0.1   | 2.3 ± 0.2     | 4.3 ± 0.1   | 3.2 ± 0.2    | 3.9 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.5 ± 0.3     | 3.9 ± 0.1   | 2.7 ± 0.1     | 3.3 ± 0.1   | 3.8 ± 0.1    | 5.3 ± 0.3     | 5.1 ± 0.3     | 3.2 ± 0.1   | 2.3 ± 0.2     | 4.3 ± 0.1   | 3.2 ± 0.2    | 4.3 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 4.3 ± 0.2     | 3.9 ± 0.1   | 2.8 ± 0.1     | 3.2 ± 0.1   | 3.8 ± 0.1    | 5.8 ± 0.3     | 5.2 ± 0.3     | 2.99 ± 0.10 | 2.2 ± 0.1     | 4.0 ± 0.1   | 3.1 ± 0.2    | 4.3 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.5 ± 0.2     | 3.9 ± 0.1   | 2.9 ± 0.1     | 3.29 ± 0.09 | 3.9 ± 0.1    | 5.5 ± 0.3     | 5.5 ± 0.3     | 2.9 ± 0.1   | 2.3 ± 0.1     | 4.0 ± 0.1   | 2.8 ± 0.1    | 4.4 ± 0.3     | E Africa M   | E Europe M   |
| 23  | 4.5 ± 0.2     | 3.89 ± 0.10 | 2.9 ± 0.1     | 3.31 ± 0.09 | 3.8 ± 0.1    | 5.6 ± 0.3     | 5.5 ± 0.3     | 2.94 ± 0.10 | 2.3 ± 0.1     | 4.0 ± 0.1   | 3.1 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 4.6 ± 0.2     | 4.03 ± 0.10 | 2.86 ± 0.10   | 3.36 ± 0.08 | 3.7 ± 0.1    | 5.6 ± 0.2     | 5.4 ± 0.3     | 2.94 ± 0.10 | 2.3 ± 0.1     | 3.8 ± 0.1   | 2.8 ± 0.1    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 4.7 ± 0.2     | 3.94 ± 0.09 | 3.02 ± 0.09   | 3.49 ± 0.08 | 3.7 ± 0.1    | 5.7 ± 0.2     | 5.4 ± 0.2     | 2.99 ± 0.10 | 2.3 ± 0.1     | 3.74 ± 0.10 | 3.0 ± 0.1    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 4.4 ± 0.2     | 3.93 ± 0.09 | 3.00 ± 0.09   | 3.60 ± 0.07 | 3.8 ± 0.1    | 5.4 ± 0.2     | 5.2 ± 0.2     | 2.91 ± 0.09 | 2.4 ± 0.1     | 3.61 ± 0.09 | 3.1 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 27  | 4.6 ± 0.2     | 3.90 ± 0.09 | 3.05 ± 0.09   | 3.76 ± 0.08 | 3.7 ± 0.1    | 5.5 ± 0.2     | 4.9 ± 0.2     | 3.04 ± 0.09 | 2.5 ± 0.1     | 3.41 ± 0.08 | 3.3 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 4.4 ± 0.2     | 3.87 ± 0.08 | 3.10 ± 0.09   | 4.02 ± 0.08 | 3.7 ± 0.1    | 5.3 ± 0.2     | 4.9 ± 0.2     | 3.09 ± 0.10 | 2.7 ± 0.1     | 3.32 ± 0.08 | 3.2 ± 0.1    | 4.1 ± 0.2     | W Africa F   | E Europe M   |
| 29  | 4.5 ± 0.2     | 3.77 ± 0.08 | 3.14 ± 0.09   | 4.13 ± 0.08 | 3.6 ± 0.1    | 5.1 ± 0.2     | 4.8 ± 0.2     | 3.16 ± 0.09 | 2.7 ± 0.1     | 3.15 ± 0.07 | 3.2 ± 0.1    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 30  | 4.4 ± 0.2     | 3.83 ± 0.08 | 3.25 ± 0.10   | 4.32 ± 0.08 | 3.7 ± 0.1    | 5.3 ± 0.2     | 4.7 ± 0.2     | 3.12 ± 0.08 | 2.6 ± 0.1     | 3.17 ± 0.06 | 3.3 ± 0.1    | 4.1 ± 0.1     | W Africa F   | E Europe M   |
| 31  | 4.5 ± 0.2     | 3.83 ± 0.08 | 3.29 ± 0.10   | 4.66 ± 0.09 | 3.7 ± 0.1    | 4.9 ± 0.2     | 4.7 ± 0.2     | 3.16 ± 0.09 | 2.8 ± 0.1     | 3.15 ± 0.06 | 3.4 ± 0.2    | 4.1 ± 0.1     | W Africa F   | E Europe M   |
| 32  | 4.5 ± 0.2     | 3.82 ± 0.08 | 3.3 ± 0.1     | 4.75 ± 0.09 | 3.9 ± 0.1    | 5.0 ± 0.2     | 4.4 ± 0.2     | 3.19 ± 0.08 | 2.8 ± 0.1     | 3.14 ± 0.06 | 3.5 ± 0.1    | 4.0 ± 0.1     | W Africa F   | E Europe M   |
| 33  | 4.7 ± 0.2     | 3.80 ± 0.08 | 3.5 ± 0.1     | 4.96 ± 0.09 | 3.8 ± 0.1    | 4.8 ± 0.2     | 4.5 ± 0.2     | 3.12 ± 0.09 | 2.8 ± 0.1     | 3.06 ± 0.06 | 3.4 ± 0.2    | 3.9 ± 0.1     | S Asia F     | E Europe M   |
| 34  | 4.5 ± 0.2     | 3.80 ± 0.08 | 3.5 ± 0.1     | 5.09 ± 0.10 | 3.7 ± 0.1    | 4.8 ± 0.2     | 4.2 ± 0.2     | 3.17 ± 0.08 | 2.8 ± 0.1     | 3.17 ± 0.06 | 3.4 ± 0.2    | 3.8 ± 0.1     | S Asia F     | E Europe M   |
| 35  | 4.7 ± 0.2     | 3.84 ± 0.09 | 3.7 ± 0.1     | 5.26 ± 0.10 | 3.8 ± 0.1    | 4.7 ± 0.2     | 4.2 ± 0.2     | 3.21 ± 0.08 | 2.8 ± 0.1     | 3.14 ± 0.07 | 3.5 ± 0.2    | 3.9 ± 0.1     | S Asia F     | E Europe M   |
| 36  | 4.7 ± 0.2     | 3.79 ± 0.09 | 3.4 ± 0.1     | 5.4 ± 0.1   | 3.8 ± 0.1    | 4.6 ± 0.2     | 4.1 ± 0.2     | 3.15 ± 0.08 | 2.9 ± 0.1     | 3.14 ± 0.07 | 3.5 ± 0.2    | 3.8 ± 0.1     | S Asia F     | E Europe M   |
| 37  | 4.7 ± 0.2     | 3.66 ± 0.08 | 3.6 ± 0.1     | 5.5 ± 0.1   | 3.7 ± 0.1    | 4.6 ± 0.2     | 4.0 ± 0.2     | 3.12 ± 0.08 | 2.8 ± 0.1     | 3.21 ± 0.07 | 3.2 ± 0.2    | 3.5 ± 0.1     | S Asia F     | E Europe M   |
| 38  | 4.6 ± 0.2     | 3.74 ± 0.08 | 3.6 ± 0.1     | 5.6 ± 0.1   | 3.8 ± 0.1    | 4.9 ± 0.2     | 3.8 ± 0.2     | 3.18 ± 0.08 | 2.8 ± 0.1     | 3.15 ± 0.07 | 3.5 ± 0.2    | 3.5 ± 0.1     | S Asia F     | E Europe M   |
| 39  | 4.6 ± 0.2     | 3.65 ± 0.08 | 3.3 ± 0.1     | 5.6 ± 0.1   | 3.6 ± 0.1    | 4.6 ± 0.2     | 3.7 ± 0.2     | 3.05 ± 0.08 | 2.9 ± 0.1     | 3.19 ± 0.07 | 3.5 ± 0.2    | 3.5 ± 0.1     | S Asia F     | E Europe M   |
| 40  | 4.6 ± 0.3     | 3.67 ± 0.08 | 3.4 ± 0.1     | 5.5 ± 0.1   | 3.6 ± 0.1    | 4.5 ± 0.2     | 3.7 ± 0.2     | 3.00 ± 0.08 | 3.0 ± 0.1     | 3.24 ± 0.07 | 3.3 ± 0.1    | 3.8 ± 0.2     | S Asia F     | E Europe M   |
| 41  | 4.8 ± 0.3     | 3.58 ± 0.08 | 3.2 ± 0.1     | 5.5 ± 0.1   | 3.6 ± 0.1    | 4.5 ± 0.3     | 3.9 ± 0.2     | 2.95 ± 0.08 | 2.9 ± 0.1     | 3.25 ± 0.08 | 3.3 ± 0.1    | 3.8 ± 0.2     | S Asia F     | E Europe M   |
| 42  | 4.5 ± 0.3     | 3.40 ± 0.08 | 3.2 ± 0.1     | 5.2 ± 0.2   | 3.4 ± 0.1    | 4.8 ± 0.3     | 3.6 ± 0.2     | 2.94 ± 0.08 | 2.9 ± 0.1     | 3.17 ± 0.08 | 3.3 ± 0.2    | 3.6 ± 0.2     | S Asia F     | E Europe M   |
| 43  | 4.6 ± 0.3     | 3.47 ± 0.08 | 3.1 ± 0.1     | 5.3 ± 0.2   | 3.6 ± 0.1    | 4.5 ± 0.3     | 3.5 ± 0.2     | 2.84 ± 0.08 | 2.8 ± 0.2     | 3.23 ± 0.09 | 3.3 ± 0.2    | 3.7 ± 0.2     | S Asia F     | E Europe M   |
| 44  | 4.9 ± 0.3     | 3.42 ± 0.08 | 3.1 ± 0.2     | 5.2 ± 0.2   | 3.5 ± 0.1    | 4.8 ± 0.3     | 3.4 ± 0.2     | 2.83 ± 0.08 | 2.9 ± 0.2     | 3.24 ± 0.09 | 3.3 ± 0.2    | 3.7 ± 0.2     | S Asia F     | E Asia M     |
| 45  | 4.6 ± 0.3     | 3.35 ± 0.08 | 3.0 ± 0.2     | 5.1 ± 0.2   | 3.4 ± 0.1    | 4.4 ± 0.3     | 3.5 ± 0.2     | 2.81 ± 0.08 | 2.8 ± 0.1     | 3.22 ± 0.10 | 3.1 ± 0.1    | 3.9 ± 0.2     | S Asia F     | E Europe M   |
| 46  | 4.8 ± 0.3     | 3.47 ± 0.09 | 3.1 ± 0.2     | 4.7 ± 0.2   | 3.5 ± 0.1    | 4.2 ± 0.2     | 3.4 ± 0.2     | 2.71 ± 0.08 | 3.0 ± 0.2     | 3.2 ± 0.1   | 3.2 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 4.4 ± 0.3     | 3.32 ± 0.08 | 3.2 ± 0.2     | 4.6 ± 0.2   | 3.4 ± 0.2    | 4.5 ± 0.3     | 3.6 ± 0.2     | 2.66 ± 0.08 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.1 ± 0.2    | 3.8 ± 0.2     | S Asia F     | E Asia M     |
| 48  | 4.4 ± 0.4     | 3.34 ± 0.09 | 3.1 ± 0.2     | 4.6 ± 0.2   | 3.3 ± 0.2    | 4.4 ± 0.3     | 3.7 ± 0.3     | 2.64 ± 0.07 | 3.1 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 4.0 ± 0.2     | S Asia F     | E Asia M     |
| 49  | 4.5 ± 0.3     | 3.44 ± 0.09 | 3.1 ± 0.2     | 4.3 ± 0.2   | 3.4 ± 0.2    | 4.5 ± 0.3     | 3.6 ± 0.3     | 2.71 ± 0.08 | 3.0 ± 0.2     | 3.2 ± 0.1   | 2.9 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 50  | 4.6 ± 0.4     | 3.47 ± 0.09 | 3.1 ± 0.2     | 4.1 ± 0.2   | 3.3 ± 0.1    | 4.0 ± 0.3     | 3.3 ± 0.3     | 2.66 ± 0.08 | 2.9 ± 0.2     | 3.3 ± 0.1   | 2.9 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Asia M     |
| 51  | 4.1 ± 0.3     | 3.48 ± 0.09 | 3.1 ± 0.2     | 4.2 ± 0.2   | 3.4 ± 0.2    | 4.4 ± 0.3     | 3.8 ± 0.3     | 2.74 ± 0.08 | 3.0 ± 0.2     | 3.3 ± 0.1   | 2.9 ± 0.2    | 4.3 ± 0.2     | W Africa F   | E Asia M     |
| 52  | 4.6 ± 0.4     | 3.46 ± 0.09 | 3.3 ± 0.2     | 4.1 ± 0.2   | 3.3 ± 0.1    | 4.5 ± 0.3     | 3.6 ± 0.3     | 2.72 ± 0.08 | 3.2 ± 0.2     | 3.3 ± 0.1   | 2.9 ± 0.2    | 4.4 ± 0.3     | E Africa F   | E Asia M     |

**Table 80.** Mean absolute error for Application-type images for algorithm dermalog-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.4 ± 0.3     | 4.7 ± 0.1   | 5.1 ± 0.3     | 4.8 ± 0.2   | 5.3 ± 0.3    | 3.4 ± 0.3     | 2.4 ± 0.2     | 3.7 ± 0.1   | 2.9 ± 0.3     | 3.6 ± 0.2   | 3.6 ± 0.2    | 2.7 ± 0.3     | S E Asia F   | E Africa M   |
| 15  | 4.0 ± 0.3     | 4.6 ± 0.2   | 5.4 ± 0.3     | 5.4 ± 0.2   | 5.2 ± 0.4    | 3.4 ± 0.3     | 2.4 ± 0.2     | 3.9 ± 0.1   | 3.0 ± 0.3     | 4.1 ± 0.2   | 4.0 ± 0.2    | 3.0 ± 0.3     | S Asia F     | E Africa M   |
| 16  | 4.0 ± 0.3     | 4.6 ± 0.1   | 5.3 ± 0.3     | 5.2 ± 0.2   | 5.3 ± 0.3    | 3.4 ± 0.4     | 3.0 ± 0.3     | 3.8 ± 0.1   | 3.3 ± 0.3     | 4.3 ± 0.2   | 4.2 ± 0.3    | 3.3 ± 0.3     | E Europe F   | E Africa M   |
| 17  | 4.2 ± 0.3     | 4.2 ± 0.1   | 5.6 ± 0.3     | 5.1 ± 0.2   | 4.9 ± 0.3    | 3.2 ± 0.3     | 3.1 ± 0.3     | 3.4 ± 0.1   | 2.8 ± 0.2     | 4.2 ± 0.2   | 3.8 ± 0.3    | 3.6 ± 0.4     | E Europe F   | E Europe M   |
| 18  | 3.8 ± 0.2     | 3.7 ± 0.1   | 4.8 ± 0.3     | 4.5 ± 0.1   | 4.1 ± 0.2    | 3.1 ± 0.2     | 3.4 ± 0.2     | 2.88 ± 0.09 | 2.7 ± 0.2     | 4.1 ± 0.1   | 3.3 ± 0.2    | 3.4 ± 0.3     | E Europe F   | E Europe M   |
| 19  | 4.2 ± 0.3     | 3.3 ± 0.1   | 4.6 ± 0.3     | 4.4 ± 0.1   | 3.7 ± 0.2    | 3.2 ± 0.2     | 3.8 ± 0.3     | 2.52 ± 0.09 | 2.6 ± 0.2     | 3.9 ± 0.1   | 3.0 ± 0.2    | 3.6 ± 0.3     | E Europe F   | E Asia M     |
| 20  | 4.4 ± 0.3     | 3.1 ± 0.1   | 4.3 ± 0.2     | 4.0 ± 0.2   | 3.5 ± 0.2    | 3.0 ± 0.2     | 3.8 ± 0.3     | 2.32 ± 0.09 | 2.5 ± 0.2     | 3.8 ± 0.1   | 2.8 ± 0.2    | 3.3 ± 0.3     | E Africa F   | E Asia M     |
| 21  | 4.3 ± 0.2     | 2.8 ± 0.1   | 4.1 ± 0.2     | 3.9 ± 0.1   | 3.3 ± 0.1    | 3.3 ± 0.3     | 3.8 ± 0.3     | 2.06 ± 0.08 | 2.3 ± 0.2     | 3.5 ± 0.1   | 2.6 ± 0.2    | 3.5 ± 0.3     | E Africa F   | E Asia M     |
| 22  | 4.3 ± 0.2     | 2.59 ± 0.09 | 3.8 ± 0.2     | 3.7 ± 0.1   | 2.9 ± 0.1    | 3.3 ± 0.2     | 4.5 ± 0.3     | 2.14 ± 0.08 | 2.5 ± 0.2     | 3.4 ± 0.1   | 2.3 ± 0.1    | 3.4 ± 0.2     | E Africa M   | E Asia M     |
| 23  | 4.2 ± 0.2     | 2.68 ± 0.08 | 3.7 ± 0.1     | 3.6 ± 0.1   | 2.9 ± 0.1    | 3.3 ± 0.2     | 4.4 ± 0.3     | 2.22 ± 0.08 | 2.4 ± 0.1     | 3.2 ± 0.1   | 2.3 ± 0.1    | 3.6 ± 0.2     | E Africa M   | E Asia M     |
| 24  | 4.5 ± 0.2     | 2.82 ± 0.07 | 3.5 ± 0.1     | 3.52 ± 0.09 | 2.89 ± 0.09  | 3.7 ± 0.2     | 4.3 ± 0.2     | 2.36 ± 0.08 | 2.6 ± 0.1     | 3.1 ± 0.1   | 2.4 ± 0.1    | 3.4 ± 0.2     | E Africa F   | S E Asia M   |
| 25  | 4.6 ± 0.2     | 3.02 ± 0.07 | 3.5 ± 0.1     | 3.48 ± 0.08 | 2.97 ± 0.09  | 4.1 ± 0.2     | 4.5 ± 0.2     | 2.66 ± 0.08 | 2.6 ± 0.1     | 2.96 ± 0.09 | 2.7 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 4.6 ± 0.2     | 3.15 ± 0.06 | 3.52 ± 0.10   | 3.42 ± 0.07 | 3.19 ± 0.08  | 4.3 ± 0.2     | 4.1 ± 0.2     | 2.73 ± 0.07 | 2.8 ± 0.1     | 2.95 ± 0.08 | 2.8 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 27  | 4.8 ± 0.2     | 3.29 ± 0.06 | 3.43 ± 0.09   | 3.36 ± 0.06 | 3.30 ± 0.08  | 4.8 ± 0.2     | 4.3 ± 0.2     | 2.97 ± 0.08 | 2.8 ± 0.1     | 2.94 ± 0.07 | 3.0 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 28  | 5.0 ± 0.2     | 3.51 ± 0.06 | 3.34 ± 0.09   | 3.37 ± 0.06 | 3.40 ± 0.09  | 4.9 ± 0.2     | 3.9 ± 0.2     | 3.16 ± 0.09 | 3.0 ± 0.1     | 2.91 ± 0.07 | 3.0 ± 0.1    | 3.5 ± 0.1     | E Africa F   | S Asia M     |
| 29  | 5.0 ± 0.2     | 3.57 ± 0.06 | 3.34 ± 0.09   | 3.36 ± 0.06 | 3.52 ± 0.09  | 5.2 ± 0.2     | 4.1 ± 0.2     | 3.28 ± 0.08 | 3.2 ± 0.1     | 2.88 ± 0.06 | 3.2 ± 0.1    | 3.7 ± 0.1     | W Africa F   | S Asia M     |
| 30  | 5.1 ± 0.2     | 3.71 ± 0.06 | 3.37 ± 0.10   | 3.37 ± 0.06 | 3.62 ± 0.10  | 5.4 ± 0.2     | 4.0 ± 0.2     | 3.41 ± 0.09 | 3.1 ± 0.1     | 2.93 ± 0.06 | 3.2 ± 0.1    | 3.7 ± 0.1     | W Africa F   | S Asia M     |
| 31  | 5.3 ± 0.2     | 3.77 ± 0.07 | 3.3 ± 0.1     | 3.38 ± 0.07 | 3.7 ± 0.1    | 5.7 ± 0.2     | 4.1 ± 0.2     | 3.57 ± 0.09 | 3.4 ± 0.1     | 2.98 ± 0.06 | 3.5 ± 0.2    | 3.8 ± 0.1     | W Africa F   | S Asia M     |
| 32  | 5.4 ± 0.2     | 3.87 ± 0.08 | 3.3 ± 0.1     | 3.44 ± 0.07 | 3.8 ± 0.1    | 5.7 ± 0.2     | 4.1 ± 0.2     | 3.70 ± 0.09 | 3.5 ± 0.1     | 3.11 ± 0.06 | 3.6 ± 0.2    | 3.9 ± 0.2     | W Africa F   | S Asia M     |
| 33  | 5.7 ± 0.2     | 3.89 ± 0.08 | 3.3 ± 0.1     | 3.51 ± 0.08 | 3.9 ± 0.1    | 5.9 ± 0.2     | 4.2 ± 0.2     | 3.9 ± 0.1   | 3.5 ± 0.1     | 3.24 ± 0.06 | 3.6 ± 0.2    | 4.1 ± 0.2     | W Africa F   | S Asia M     |
| 34  | 5.7 ± 0.3     | 3.94 ± 0.08 | 3.4 ± 0.1     | 3.63 ± 0.08 | 3.9 ± 0.1    | 6.0 ± 0.2     | 4.2 ± 0.2     | 4.0 ± 0.1   | 3.6 ± 0.2     | 3.39 ± 0.07 | 3.9 ± 0.2    | 4.1 ± 0.1     | W Africa F   | E Europe F   |
| 35  | 6.0 ± 0.3     | 4.07 ± 0.09 | 3.5 ± 0.1     | 3.77 ± 0.08 | 3.9 ± 0.1    | 6.1 ± 0.3     | 4.3 ± 0.2     | 4.2 ± 0.1   | 3.8 ± 0.2     | 3.64 ± 0.08 | 4.2 ± 0.2    | 4.5 ± 0.2     | W Africa F   | E Europe F   |
| 36  | 6.0 ± 0.3     | 4.14 ± 0.09 | 3.5 ± 0.1     | 3.98 ± 0.09 | 4.2 ± 0.1    | 6.3 ± 0.2     | 4.5 ± 0.2     | 4.3 ± 0.1   | 4.0 ± 0.2     | 3.85 ± 0.08 | 4.3 ± 0.2    | 4.4 ± 0.2     | W Africa F   | E Europe F   |
| 37  | 6.2 ± 0.3     | 4.26 ± 0.09 | 3.7 ± 0.1     | 4.28 ± 0.10 | 4.2 ± 0.1    | 6.6 ± 0.3     | 4.4 ± 0.2     | 4.5 ± 0.1   | 4.2 ± 0.2     | 4.14 ± 0.08 | 4.3 ± 0.2    | 4.7 ± 0.2     | W Africa F   | E Europe F   |
| 38  | 6.1 ± 0.3     | 4.48 ± 0.09 | 3.7 ± 0.1     | 4.41 ± 0.10 | 4.3 ± 0.1    | 6.8 ± 0.3     | 4.6 ± 0.2     | 4.7 ± 0.1   | 4.2 ± 0.2     | 4.40 ± 0.08 | 4.4 ± 0.2    | 4.8 ± 0.2     | W Africa F   | E Europe F   |
| 39  | 6.2 ± 0.3     | 4.53 ± 0.09 | 3.9 ± 0.1     | 4.6 ± 0.1   | 4.3 ± 0.1    | 6.5 ± 0.3     | 4.6 ± 0.2     | 4.8 ± 0.1   | 4.3 ± 0.2     | 4.46 ± 0.08 | 4.5 ± 0.2    | 5.1 ± 0.2     | W Africa F   | E Europe F   |
| 40  | 6.0 ± 0.3     | 4.70 ± 0.09 | 3.8 ± 0.1     | 4.7 ± 0.1   | 4.5 ± 0.2    | 6.8 ± 0.3     | 4.6 ± 0.2     | 4.9 ± 0.1   | 4.5 ± 0.2     | 4.72 ± 0.09 | 4.5 ± 0.2    | 5.2 ± 0.2     | W Africa F   | E Europe F   |
| 41  | 6.5 ± 0.4     | 4.69 ± 0.10 | 3.8 ± 0.1     | 5.0 ± 0.1   | 4.5 ± 0.2    | 6.9 ± 0.3     | 4.9 ± 0.2     | 4.9 ± 0.1   | 4.6 ± 0.2     | 4.7 ± 0.1   | 4.6 ± 0.2    | 5.4 ± 0.2     | W Africa F   | E Europe F   |
| 42  | 5.9 ± 0.4     | 4.73 ± 0.09 | 3.7 ± 0.2     | 4.8 ± 0.1   | 4.5 ± 0.2    | 6.8 ± 0.3     | 4.7 ± 0.3     | 5.0 ± 0.1   | 4.2 ± 0.2     | 4.8 ± 0.1   | 4.5 ± 0.2    | 5.2 ± 0.2     | W Africa F   | E Europe F   |
| 43  | 5.9 ± 0.4     | 4.86 ± 0.10 | 4.0 ± 0.2     | 5.1 ± 0.2   | 4.7 ± 0.2    | 6.8 ± 0.3     | 4.6 ± 0.3     | 4.8 ± 0.1   | 4.5 ± 0.2     | 4.9 ± 0.1   | 4.2 ± 0.2    | 5.5 ± 0.3     | W Africa F   | E Europe F   |
| 44  | 6.1 ± 0.4     | 5.0 ± 0.1   | 4.0 ± 0.2     | 5.0 ± 0.2   | 4.5 ± 0.2    | 6.9 ± 0.4     | 4.3 ± 0.3     | 4.7 ± 0.1   | 4.1 ± 0.2     | 4.8 ± 0.1   | 4.2 ± 0.2    | 5.2 ± 0.3     | W Africa F   | E Europe F   |
| 45  | 6.0 ± 0.4     | 4.9 ± 0.1   | 3.8 ± 0.2     | 5.1 ± 0.2   | 4.4 ± 0.2    | 7.0 ± 0.4     | 4.5 ± 0.3     | 4.7 ± 0.1   | 4.2 ± 0.2     | 4.8 ± 0.1   | 4.1 ± 0.2    | 5.3 ± 0.3     | W Africa F   | E Europe F   |
| 46  | 5.7 ± 0.4     | 5.1 ± 0.1   | 3.9 ± 0.2     | 4.9 ± 0.2   | 4.7 ± 0.2    | 6.5 ± 0.4     | 4.3 ± 0.3     | 4.5 ± 0.1   | 4.1 ± 0.2     | 4.5 ± 0.1   | 3.9 ± 0.2    | 5.6 ± 0.3     | W Africa F   | E Europe F   |
| 47  | 5.7 ± 0.5     | 5.0 ± 0.1   | 3.8 ± 0.2     | 4.8 ± 0.2   | 4.4 ± 0.2    | 6.5 ± 0.4     | 4.4 ± 0.3     | 4.4 ± 0.1   | 3.7 ± 0.2     | 4.7 ± 0.2   | 3.7 ± 0.2    | 5.3 ± 0.3     | W Africa F   | E Europe M   |
| 48  | 5.6 ± 0.5     | 5.0 ± 0.1   | 3.8 ± 0.2     | 5.0 ± 0.2   | 4.5 ± 0.2    | 6.6 ± 0.4     | 4.5 ± 0.3     | 4.3 ± 0.1   | 3.9 ± 0.3     | 4.6 ± 0.2   | 3.9 ± 0.2    | 5.5 ± 0.3     | W Africa F   | E Europe F   |
| 49  | 5.7 ± 0.5     | 5.1 ± 0.1   | 3.8 ± 0.2     | 4.9 ± 0.2   | 4.3 ± 0.2    | 6.9 ± 0.4     | 4.7 ± 0.4     | 4.3 ± 0.1   | 3.8 ± 0.2     | 4.5 ± 0.2   | 3.8 ± 0.2    | 5.4 ± 0.3     | W Africa F   | E Europe F   |
| 50  | 5.9 ± 0.5     | 5.1 ± 0.1   | 3.7 ± 0.2     | 4.8 ± 0.2   | 4.6 ± 0.2    | 6.7 ± 0.4     | 4.3 ± 0.4     | 4.2 ± 0.1   | 3.9 ± 0.2     | 4.6 ± 0.2   | 3.8 ± 0.2    | 5.3 ± 0.3     | W Africa F   | E Europe F   |
| 51  | 5.2 ± 0.4     | 5.3 ± 0.1   | 3.6 ± 0.2     | 4.9 ± 0.2   | 4.5 ± 0.2    | 6.7 ± 0.4     | 4.7 ± 0.4     | 4.2 ± 0.1   | 3.6 ± 0.2     | 4.6 ± 0.2   | 3.7 ± 0.2    | 5.7 ± 0.3     | W Africa F   | E Europe M   |
| 52  | 5.6 ± 0.5     | 5.2 ± 0.1   | 3.9 ± 0.2     | 5.1 ± 0.2   | 4.4 ± 0.2    | 7.0 ± 0.5     | 4.7 ± 0.4     | 4.1 ± 0.1   | 3.9 ± 0.2     | 4.6 ± 0.2   | 3.9 ± 0.2    | 5.8 ± 0.4     | W Africa F   | E Europe F   |

**Table 81.** Mean absolute error for Application-type images for algorithm dermalog-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.8 ± 0.3     | 5.5 ± 0.2   | 5.2 ± 0.3     | 4.8 ± 0.2   | 5.4 ± 0.3    | 3.7 ± 0.3     | 3.1 ± 0.2     | 3.0 ± 0.1   | 2.6 ± 0.2     | 3.1 ± 0.1   | 3.1 ± 0.2    | 3.2 ± 0.3     | E Asia F     | E Europe M   |
| 15  | 3.6 ± 0.3     | 5.8 ± 0.2   | 5.9 ± 0.4     | 5.2 ± 0.2   | 6.0 ± 0.4    | 4.0 ± 0.4     | 3.0 ± 0.2     | 3.2 ± 0.1   | 2.7 ± 0.2     | 3.3 ± 0.1   | 3.3 ± 0.2    | 3.3 ± 0.3     | S E Asia F   | E Europe M   |
| 16  | 4.1 ± 0.3     | 5.9 ± 0.2   | 5.8 ± 0.3     | 5.3 ± 0.2   | 5.9 ± 0.3    | 4.0 ± 0.4     | 3.1 ± 0.3     | 3.3 ± 0.1   | 2.9 ± 0.3     | 3.3 ± 0.1   | 3.6 ± 0.2    | 3.3 ± 0.3     | S E Asia F   | E Europe M   |
| 17  | 4.3 ± 0.3     | 5.6 ± 0.1   | 5.7 ± 0.3     | 5.3 ± 0.2   | 5.8 ± 0.3    | 4.1 ± 0.3     | 2.9 ± 0.3     | 3.1 ± 0.1   | 3.0 ± 0.2     | 3.4 ± 0.2   | 3.4 ± 0.2    | 3.1 ± 0.3     | S E Asia F   | E Africa M   |
| 18  | 4.3 ± 0.2     | 5.3 ± 0.1   | 5.7 ± 0.3     | 4.9 ± 0.1   | 5.2 ± 0.2    | 3.9 ± 0.3     | 3.1 ± 0.2     | 2.95 ± 0.09 | 3.1 ± 0.2     | 3.5 ± 0.1   | 3.2 ± 0.2    | 3.1 ± 0.2     | E Europe F   | E Asia M     |
| 19  | 4.3 ± 0.3     | 4.8 ± 0.1   | 5.2 ± 0.3     | 4.7 ± 0.1   | 4.9 ± 0.2    | 3.8 ± 0.2     | 3.1 ± 0.2     | 2.9 ± 0.1   | 3.0 ± 0.2     | 3.5 ± 0.1   | 3.0 ± 0.2    | 3.1 ± 0.3     | E Europe F   | E Asia M     |
| 20  | 4.5 ± 0.2     | 4.6 ± 0.1   | 4.9 ± 0.2     | 4.3 ± 0.2   | 4.7 ± 0.2    | 3.8 ± 0.3     | 3.3 ± 0.3     | 2.7 ± 0.1   | 3.0 ± 0.2     | 3.4 ± 0.1   | 3.0 ± 0.2    | 2.9 ± 0.3     | E Europe F   | E Asia M     |
| 21  | 4.4 ± 0.2     | 4.1 ± 0.1   | 4.5 ± 0.2     | 4.1 ± 0.1   | 4.1 ± 0.1    | 3.8 ± 0.2     | 3.1 ± 0.2     | 2.51 ± 0.09 | 3.1 ± 0.2     | 3.3 ± 0.1   | 2.8 ± 0.2    | 3.1 ± 0.3     | E Europe F   | E Asia M     |
| 22  | 4.5 ± 0.2     | 3.6 ± 0.1   | 4.1 ± 0.2     | 3.7 ± 0.1   | 3.7 ± 0.1    | 3.7 ± 0.2     | 3.5 ± 0.2     | 2.50 ± 0.09 | 3.2 ± 0.2     | 3.0 ± 0.1   | 2.5 ± 0.1    | 3.0 ± 0.2     | E Africa F   | E Asia M     |
| 23  | 4.8 ± 0.2     | 3.22 ± 0.09 | 4.1 ± 0.2     | 3.6 ± 0.1   | 3.4 ± 0.1    | 3.7 ± 0.2     | 3.4 ± 0.2     | 2.64 ± 0.09 | 3.0 ± 0.2     | 3.2 ± 0.1   | 2.6 ± 0.1    | 3.2 ± 0.2     | E Africa F   | S E Asia M   |
| 24  | 4.6 ± 0.2     | 2.98 ± 0.08 | 3.8 ± 0.1     | 3.41 ± 0.09 | 3.05 ± 0.10  | 3.8 ± 0.2     | 3.5 ± 0.2     | 2.72 ± 0.09 | 3.0 ± 0.2     | 3.2 ± 0.1   | 2.7 ± 0.1    | 3.2 ± 0.2     | E Africa F   | S E Asia M   |
| 25  | 5.0 ± 0.2     | 2.72 ± 0.07 | 3.7 ± 0.1     | 3.34 ± 0.08 | 2.92 ± 0.10  | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.82 ± 0.09 | 3.0 ± 0.1     | 3.15 ± 0.09 | 2.9 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Asia F     |
| 26  | 5.1 ± 0.2     | 2.67 ± 0.07 | 3.6 ± 0.1     | 3.38 ± 0.08 | 2.89 ± 0.09  | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.89 ± 0.09 | 3.0 ± 0.1     | 3.24 ± 0.08 | 3.0 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Asia F     |
| 27  | 5.3 ± 0.2     | 2.58 ± 0.06 | 3.5 ± 0.1     | 3.46 ± 0.08 | 2.83 ± 0.09  | 4.5 ± 0.2     | 3.7 ± 0.2     | 3.09 ± 0.09 | 3.2 ± 0.1     | 3.29 ± 0.08 | 3.2 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Asia F     |
| 28  | 5.5 ± 0.2     | 2.72 ± 0.06 | 3.5 ± 0.1     | 3.68 ± 0.08 | 2.95 ± 0.09  | 4.8 ± 0.2     | 3.8 ± 0.1     | 3.38 ± 0.10 | 3.2 ± 0.1     | 3.39 ± 0.08 | 3.3 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Asia F     |
| 29  | 5.7 ± 0.2     | 2.84 ± 0.06 | 3.5 ± 0.1     | 3.91 ± 0.08 | 3.16 ± 0.09  | 4.9 ± 0.2     | 4.0 ± 0.2     | 3.39 ± 0.09 | 3.3 ± 0.1     | 3.51 ± 0.07 | 3.5 ± 0.1    | 4.2 ± 0.1     | E Africa F   | E Asia F     |
| 30  | 5.9 ± 0.2     | 3.04 ± 0.06 | 3.7 ± 0.1     | 4.13 ± 0.08 | 3.24 ± 0.10  | 5.8 ± 0.2     | 4.3 ± 0.2     | 3.60 ± 0.09 | 3.2 ± 0.1     | 3.75 ± 0.07 | 3.6 ± 0.1    | 4.3 ± 0.1     | E Africa F   | E Asia F     |
| 31  | 6.2 ± 0.2     | 3.29 ± 0.06 | 3.8 ± 0.1     | 4.40 ± 0.09 | 3.5 ± 0.1    | 6.0 ± 0.2     | 4.2 ± 0.2     | 3.74 ± 0.09 | 3.4 ± 0.1     | 3.95 ± 0.07 | 3.9 ± 0.2    | 4.6 ± 0.1     | E Africa F   | E Asia F     |
| 32  | 6.5 ± 0.3     | 3.46 ± 0.06 | 3.8 ± 0.1     | 4.67 ± 0.09 | 3.7 ± 0.1    | 6.2 ± 0.2     | 4.6 ± 0.2     | 4.0 ± 0.1   | 3.4 ± 0.1     | 4.17 ± 0.08 | 3.9 ± 0.2    | 4.8 ± 0.2     | E Africa F   | E Europe M   |
| 33  | 6.8 ± 0.3     | 3.67 ± 0.07 | 3.8 ± 0.1     | 4.96 ± 0.10 | 3.9 ± 0.1    | 6.5 ± 0.2     | 4.8 ± 0.2     | 4.1 ± 0.1   | 3.5 ± 0.1     | 4.47 ± 0.08 | 4.0 ± 0.2    | 4.9 ± 0.2     | E Africa F   | E Europe M   |
| 34  | 7.0 ± 0.3     | 3.82 ± 0.07 | 3.8 ± 0.1     | 5.2 ± 0.1   | 4.0 ± 0.1    | 6.7 ± 0.2     | 5.0 ± 0.2     | 4.3 ± 0.1   | 3.4 ± 0.1     | 4.67 ± 0.09 | 4.1 ± 0.2    | 5.1 ± 0.2     | E Africa F   | E Europe M   |
| 35  | 7.1 ± 0.3     | 4.12 ± 0.08 | 3.9 ± 0.1     | 5.5 ± 0.1   | 4.2 ± 0.1    | 7.1 ± 0.3     | 5.1 ± 0.2     | 4.4 ± 0.1   | 3.6 ± 0.2     | 4.96 ± 0.10 | 4.5 ± 0.2    | 5.5 ± 0.2     | E Africa F   | E Europe M   |
| 36  | 7.6 ± 0.3     | 4.25 ± 0.08 | 3.9 ± 0.1     | 5.8 ± 0.1   | 4.4 ± 0.1    | 7.4 ± 0.3     | 5.4 ± 0.2     | 4.6 ± 0.1   | 3.5 ± 0.2     | 5.2 ± 0.1   | 4.6 ± 0.2    | 5.7 ± 0.2     | E Africa F   | E Europe M   |
| 37  | 7.8 ± 0.4     | 4.46 ± 0.09 | 4.1 ± 0.1     | 6.0 ± 0.1   | 4.5 ± 0.1    | 7.7 ± 0.3     | 5.4 ± 0.2     | 4.7 ± 0.1   | 3.7 ± 0.2     | 5.6 ± 0.1   | 4.6 ± 0.2    | 5.7 ± 0.2     | E Africa F   | E Europe M   |
| 38  | 7.4 ± 0.4     | 4.68 ± 0.09 | 4.0 ± 0.2     | 6.3 ± 0.1   | 4.6 ± 0.2    | 7.9 ± 0.3     | 5.5 ± 0.3     | 4.9 ± 0.1   | 3.5 ± 0.2     | 5.7 ± 0.1   | 4.6 ± 0.2    | 6.0 ± 0.2     | W Africa F   | E Europe M   |
| 39  | 7.8 ± 0.4     | 4.77 ± 0.09 | 4.1 ± 0.2     | 6.5 ± 0.1   | 4.8 ± 0.2    | 8.1 ± 0.3     | 5.5 ± 0.3     | 5.0 ± 0.1   | 3.7 ± 0.2     | 5.8 ± 0.1   | 4.8 ± 0.2    | 6.4 ± 0.2     | W Africa F   | E Europe M   |
| 40  | 8.1 ± 0.4     | 4.89 ± 0.09 | 4.1 ± 0.2     | 6.6 ± 0.2   | 5.0 ± 0.2    | 8.5 ± 0.3     | 5.7 ± 0.3     | 5.1 ± 0.1   | 3.9 ± 0.2     | 6.1 ± 0.1   | 4.9 ± 0.2    | 6.6 ± 0.3     | W Africa F   | E Europe M   |
| 41  | 8.4 ± 0.4     | 5.00 ± 0.10 | 4.2 ± 0.2     | 6.8 ± 0.2   | 5.1 ± 0.2    | 8.8 ± 0.4     | 6.1 ± 0.3     | 5.1 ± 0.1   | 4.1 ± 0.2     | 6.3 ± 0.1   | 5.2 ± 0.2    | 6.9 ± 0.3     | W Africa F   | E Europe M   |
| 42  | 7.8 ± 0.4     | 5.08 ± 0.10 | 4.3 ± 0.2     | 6.8 ± 0.2   | 5.0 ± 0.2    | 8.5 ± 0.4     | 6.0 ± 0.3     | 5.3 ± 0.1   | 3.9 ± 0.2     | 6.4 ± 0.1   | 5.0 ± 0.2    | 7.0 ± 0.3     | W Africa F   | E Europe M   |
| 43  | 8.2 ± 0.5     | 5.2 ± 0.1   | 4.4 ± 0.2     | 6.9 ± 0.2   | 5.2 ± 0.2    | 8.8 ± 0.4     | 6.0 ± 0.3     | 5.2 ± 0.1   | 4.1 ± 0.2     | 6.4 ± 0.2   | 5.1 ± 0.2    | 7.4 ± 0.3     | W Africa F   | E Europe M   |
| 44  | 8.4 ± 0.5     | 5.3 ± 0.1   | 4.3 ± 0.2     | 6.8 ± 0.2   | 5.1 ± 0.2    | 9.1 ± 0.4     | 6.4 ± 0.4     | 5.3 ± 0.1   | 4.0 ± 0.2     | 6.6 ± 0.2   | 5.1 ± 0.2    | 7.3 ± 0.3     | W Africa F   | E Europe M   |
| 45  | 8.4 ± 0.5     | 5.2 ± 0.1   | 4.2 ± 0.2     | 6.9 ± 0.2   | 5.1 ± 0.2    | 9.0 ± 0.4     | 6.2 ± 0.4     | 5.5 ± 0.1   | 4.1 ± 0.2     | 6.5 ± 0.2   | 5.2 ± 0.3    | 7.6 ± 0.3     | W Africa F   | E Europe M   |
| 46  | 7.8 ± 0.5     | 5.3 ± 0.1   | 4.3 ± 0.2     | 6.7 ± 0.2   | 5.3 ± 0.2    | 9.2 ± 0.5     | 6.1 ± 0.4     | 5.4 ± 0.1   | 4.3 ± 0.2     | 6.3 ± 0.2   | 5.0 ± 0.3    | 7.9 ± 0.4     | W Africa F   | E Europe M   |
| 47  | 8.0 ± 0.6     | 5.2 ± 0.1   | 4.4 ± 0.2     | 6.6 ± 0.2   | 5.0 ± 0.2    | 8.9 ± 0.5     | 6.2 ± 0.4     | 5.2 ± 0.1   | 4.3 ± 0.2     | 6.4 ± 0.2   | 5.0 ± 0.2    | 7.8 ± 0.4     | W Africa F   | E Europe M   |
| 48  | 8.1 ± 0.6     | 5.1 ± 0.1   | 4.2 ± 0.2     | 6.7 ± 0.3   | 5.2 ± 0.2    | 8.8 ± 0.5     | 6.8 ± 0.5     | 5.4 ± 0.1   | 4.2 ± 0.2     | 6.5 ± 0.2   | 5.0 ± 0.2    | 8.0 ± 0.4     | W Africa F   | E Europe F   |
| 49  | 8.1 ± 0.7     | 5.3 ± 0.1   | 4.3 ± 0.2     | 6.4 ± 0.2   | 5.0 ± 0.2    | 9.2 ± 0.5     | 6.6 ± 0.5     | 5.3 ± 0.1   | 4.6 ± 0.3     | 6.5 ± 0.2   | 4.9 ± 0.3    | 8.1 ± 0.4     | W Africa F   | E Europe F   |
| 50  | 7.6 ± 0.7     | 5.1 ± 0.1   | 4.3 ± 0.2     | 6.3 ± 0.2   | 4.9 ± 0.2    | 8.5 ± 0.5     | 7.2 ± 0.5     | 5.3 ± 0.2   | 4.6 ± 0.2     | 6.8 ± 0.2   | 4.8 ± 0.3    | 8.1 ± 0.4     | W Africa F   | E Europe F   |
| 51  | 7.6 ± 0.6     | 5.2 ± 0.1   | 4.2 ± 0.2     | 6.4 ± 0.2   | 5.0 ± 0.2    | 9.2 ± 0.6     | 7.2 ± 0.5     | 5.2 ± 0.2   | 4.3 ± 0.2     | 6.9 ± 0.2   | 4.8 ± 0.3    | 7.9 ± 0.4     | W Africa F   | E Europe F   |
| 52  | 7.8 ± 0.7     | 5.1 ± 0.1   | 4.2 ± 0.2     | 6.5 ± 0.2   | 4.6 ± 0.2    | 9.4 ± 0.6     | 6.9 ± 0.5     | 5.2 ± 0.2   | 4.5 ± 0.3     | 6.8 ± 0.3   | 5.0 ± 0.3    | 8.6 ± 0.5     | W Africa F   | E Europe F   |

**Table 82.** Mean absolute error for Application-type images for algorithm hzailu-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 6.1 ± 0.4     | 6.1 ± 0.2   | 8.4 ± 0.4     | 6.7 ± 0.2   | 6.4 ± 0.4    | 8.4 ± 0.6     | 6.1 ± 0.4     | 5.9 ± 0.2   | 5.5 ± 0.4     | 7.5 ± 0.2   | 6.2 ± 0.4    | 7.6 ± 0.8     | E Europe F   | E Europe M   |
| 15  | 6.3 ± 0.5     | 6.4 ± 0.2   | 8.6 ± 0.4     | 7.3 ± 0.3   | 6.7 ± 0.4    | 8.2 ± 0.8     | 6.4 ± 0.5     | 6.1 ± 0.2   | 6.3 ± 0.4     | 8.2 ± 0.2   | 6.9 ± 0.3    | 8.1 ± 0.8     | E Europe F   | E Asia M     |
| 16  | 6.2 ± 0.4     | 6.4 ± 0.2   | 8.5 ± 0.4     | 7.1 ± 0.3   | 7.2 ± 0.4    | 8.4 ± 0.7     | 8.3 ± 0.5     | 6.0 ± 0.2   | 6.8 ± 0.4     | 8.4 ± 0.2   | 7.3 ± 0.4    | 9.4 ± 0.8     | W Africa M   | E Asia M     |
| 17  | 6.4 ± 0.5     | 6.3 ± 0.2   | 8.5 ± 0.4     | 7.1 ± 0.3   | 6.8 ± 0.4    | 8.7 ± 0.6     | 8.3 ± 0.5     | 5.8 ± 0.1   | 6.6 ± 0.3     | 8.5 ± 0.2   | 7.4 ± 0.3    | 9.5 ± 0.7     | W Africa M   | E Asia M     |
| 18  | 6.6 ± 0.4     | 6.1 ± 0.1   | 8.2 ± 0.3     | 6.6 ± 0.2   | 6.5 ± 0.2    | 8.3 ± 0.5     | 9.0 ± 0.4     | 5.4 ± 0.1   | 6.6 ± 0.3     | 8.4 ± 0.2   | 6.6 ± 0.2    | 10.2 ± 0.5    | W Africa M   | E Asia M     |
| 19  | 6.4 ± 0.3     | 5.6 ± 0.1   | 7.8 ± 0.3     | 6.6 ± 0.2   | 6.3 ± 0.2    | 8.1 ± 0.4     | 8.9 ± 0.4     | 5.1 ± 0.1   | 6.3 ± 0.3     | 8.2 ± 0.2   | 6.2 ± 0.2    | 10.2 ± 0.5    | W Africa M   | E Asia M     |
| 20  | 6.4 ± 0.3     | 5.5 ± 0.1   | 7.5 ± 0.2     | 6.1 ± 0.2   | 5.9 ± 0.2    | 8.0 ± 0.4     | 9.1 ± 0.4     | 4.6 ± 0.1   | 6.3 ± 0.3     | 7.9 ± 0.2   | 6.0 ± 0.2    | 9.9 ± 0.5     | W Africa M   | E Asia M     |
| 21  | 6.2 ± 0.3     | 5.1 ± 0.1   | 7.0 ± 0.2     | 5.8 ± 0.1   | 5.6 ± 0.2    | 8.0 ± 0.4     | 8.7 ± 0.3     | 4.1 ± 0.1   | 6.5 ± 0.3     | 7.7 ± 0.1   | 5.6 ± 0.2    | 10.0 ± 0.4    | W Africa M   | E Asia M     |
| 22  | 6.0 ± 0.3     | 4.6 ± 0.1   | 6.5 ± 0.2     | 5.3 ± 0.1   | 5.1 ± 0.2    | 7.4 ± 0.3     | 8.9 ± 0.3     | 3.7 ± 0.1   | 6.3 ± 0.3     | 7.1 ± 0.1   | 4.9 ± 0.2    | 9.4 ± 0.4     | W Africa M   | E Asia M     |
| 23  | 5.5 ± 0.2     | 4.4 ± 0.1   | 6.1 ± 0.2     | 5.0 ± 0.1   | 4.7 ± 0.1    | 7.2 ± 0.3     | 8.1 ± 0.3     | 3.4 ± 0.1   | 6.0 ± 0.2     | 6.6 ± 0.1   | 4.5 ± 0.2    | 9.2 ± 0.3     | W Africa M   | E Asia M     |
| 24  | 5.3 ± 0.2     | 3.97 ± 0.09 | 5.7 ± 0.1     | 4.56 ± 0.10 | 4.4 ± 0.1    | 6.7 ± 0.3     | 7.8 ± 0.3     | 3.2 ± 0.1   | 5.5 ± 0.2     | 6.4 ± 0.1   | 4.0 ± 0.2    | 8.9 ± 0.3     | W Africa M   | E Asia M     |
| 25  | 5.1 ± 0.2     | 3.68 ± 0.08 | 5.3 ± 0.1     | 4.21 ± 0.09 | 4.0 ± 0.1    | 6.7 ± 0.3     | 7.6 ± 0.3     | 2.87 ± 0.09 | 5.3 ± 0.2     | 5.8 ± 0.1   | 3.7 ± 0.2    | 8.4 ± 0.3     | W Africa M   | E Asia M     |
| 26  | 5.1 ± 0.2     | 3.45 ± 0.08 | 4.8 ± 0.1     | 3.88 ± 0.08 | 3.7 ± 0.1    | 6.0 ± 0.2     | 7.2 ± 0.2     | 2.63 ± 0.09 | 5.0 ± 0.2     | 5.3 ± 0.1   | 3.3 ± 0.2    | 7.9 ± 0.2     | W Africa M   | E Asia M     |
| 27  | 5.1 ± 0.2     | 3.10 ± 0.07 | 4.4 ± 0.1     | 3.60 ± 0.08 | 3.5 ± 0.1    | 5.9 ± 0.2     | 6.8 ± 0.2     | 2.61 ± 0.09 | 4.9 ± 0.2     | 4.73 ± 0.10 | 3.3 ± 0.2    | 7.4 ± 0.2     | W Africa M   | E Asia M     |
| 28  | 4.9 ± 0.2     | 3.06 ± 0.07 | 3.9 ± 0.1     | 3.52 ± 0.08 | 3.22 ± 0.10  | 5.8 ± 0.2     | 6.3 ± 0.2     | 2.66 ± 0.09 | 4.5 ± 0.2     | 4.24 ± 0.09 | 3.1 ± 0.2    | 6.9 ± 0.2     | W Africa M   | E Asia M     |
| 29  | 4.7 ± 0.2     | 2.93 ± 0.07 | 3.6 ± 0.1     | 3.41 ± 0.08 | 3.2 ± 0.1    | 5.4 ± 0.2     | 6.1 ± 0.2     | 2.74 ± 0.09 | 4.2 ± 0.2     | 3.81 ± 0.08 | 3.0 ± 0.1    | 6.5 ± 0.2     | W Africa M   | E Asia M     |
| 30  | 5.0 ± 0.2     | 2.92 ± 0.06 | 3.5 ± 0.1     | 3.55 ± 0.08 | 3.2 ± 0.1    | 5.3 ± 0.2     | 5.5 ± 0.2     | 2.81 ± 0.09 | 3.9 ± 0.2     | 3.43 ± 0.08 | 3.1 ± 0.2    | 6.0 ± 0.2     | W Africa M   | E Asia M     |
| 31  | 4.9 ± 0.2     | 3.04 ± 0.07 | 3.5 ± 0.1     | 3.80 ± 0.09 | 3.3 ± 0.1    | 5.3 ± 0.2     | 5.2 ± 0.2     | 3.18 ± 0.09 | 3.8 ± 0.2     | 3.20 ± 0.07 | 3.3 ± 0.2    | 5.7 ± 0.2     | W Africa M   | E Asia F     |
| 32  | 5.1 ± 0.2     | 3.18 ± 0.07 | 3.4 ± 0.1     | 4.01 ± 0.09 | 3.2 ± 0.1    | 5.2 ± 0.2     | 4.8 ± 0.2     | 3.41 ± 0.09 | 3.9 ± 0.2     | 3.18 ± 0.07 | 3.7 ± 0.2    | 5.2 ± 0.2     | W Africa M   | E Asia F     |
| 33  | 5.4 ± 0.2     | 3.41 ± 0.07 | 3.5 ± 0.1     | 4.34 ± 0.10 | 3.4 ± 0.1    | 5.4 ± 0.2     | 4.7 ± 0.2     | 3.71 ± 0.10 | 3.7 ± 0.2     | 3.24 ± 0.08 | 3.6 ± 0.2    | 4.8 ± 0.2     | W Africa F   | S Asia M     |
| 34  | 5.4 ± 0.3     | 3.68 ± 0.07 | 3.6 ± 0.1     | 4.7 ± 0.1   | 3.6 ± 0.1    | 5.3 ± 0.2     | 4.2 ± 0.2     | 4.08 ± 0.10 | 3.8 ± 0.2     | 3.46 ± 0.08 | 4.1 ± 0.2    | 4.6 ± 0.2     | E Africa F   | S Asia M     |
| 35  | 5.9 ± 0.3     | 4.05 ± 0.08 | 3.9 ± 0.1     | 5.1 ± 0.1   | 3.9 ± 0.1    | 5.4 ± 0.2     | 4.2 ± 0.2     | 4.40 ± 0.10 | 4.0 ± 0.2     | 3.69 ± 0.08 | 4.4 ± 0.2    | 4.5 ± 0.2     | E Africa F   | S Asia M     |
| 36  | 6.2 ± 0.3     | 4.29 ± 0.08 | 4.4 ± 0.2     | 5.5 ± 0.1   | 4.2 ± 0.1    | 5.3 ± 0.2     | 4.4 ± 0.2     | 4.6 ± 0.1   | 4.3 ± 0.2     | 3.87 ± 0.08 | 4.7 ± 0.2    | 4.3 ± 0.2     | E Africa F   | S Asia M     |
| 37  | 6.5 ± 0.3     | 4.60 ± 0.08 | 4.6 ± 0.2     | 5.9 ± 0.1   | 4.7 ± 0.1    | 5.5 ± 0.3     | 4.3 ± 0.2     | 4.8 ± 0.1   | 4.6 ± 0.2     | 4.12 ± 0.08 | 4.7 ± 0.2    | 4.2 ± 0.2     | E Africa F   | S Asia M     |
| 38  | 6.4 ± 0.3     | 4.87 ± 0.09 | 4.8 ± 0.2     | 6.3 ± 0.1   | 4.6 ± 0.1    | 6.0 ± 0.3     | 4.3 ± 0.2     | 4.9 ± 0.1   | 4.6 ± 0.2     | 4.26 ± 0.08 | 4.7 ± 0.2    | 4.2 ± 0.2     | E Africa F   | W Africa M   |
| 39  | 6.5 ± 0.3     | 5.15 ± 0.09 | 5.1 ± 0.2     | 6.7 ± 0.1   | 4.8 ± 0.2    | 5.7 ± 0.3     | 4.4 ± 0.2     | 5.1 ± 0.1   | 4.8 ± 0.2     | 4.48 ± 0.09 | 5.0 ± 0.2    | 4.0 ± 0.2     | S Asia F     | W Africa M   |
| 40  | 6.9 ± 0.4     | 5.36 ± 0.09 | 5.3 ± 0.2     | 6.8 ± 0.2   | 5.3 ± 0.2    | 6.0 ± 0.3     | 4.4 ± 0.2     | 5.1 ± 0.1   | 5.1 ± 0.2     | 4.65 ± 0.10 | 4.7 ± 0.2    | 4.2 ± 0.2     | E Africa F   | W Africa M   |
| 41  | 7.3 ± 0.4     | 5.50 ± 0.10 | 5.5 ± 0.2     | 7.2 ± 0.2   | 5.3 ± 0.2    | 6.2 ± 0.3     | 4.5 ± 0.2     | 5.0 ± 0.1   | 5.1 ± 0.2     | 4.8 ± 0.1   | 4.9 ± 0.2    | 4.4 ± 0.2     | E Africa F   | W Africa M   |
| 42  | 7.0 ± 0.4     | 5.6 ± 0.1   | 5.6 ± 0.2     | 7.3 ± 0.2   | 5.4 ± 0.2    | 6.2 ± 0.3     | 4.5 ± 0.2     | 5.1 ± 0.1   | 5.1 ± 0.2     | 5.0 ± 0.1   | 5.0 ± 0.2    | 4.5 ± 0.2     | S Asia F     | E Africa M   |
| 43  | 7.6 ± 0.5     | 5.8 ± 0.1   | 5.9 ± 0.2     | 7.6 ± 0.2   | 5.7 ± 0.2    | 6.5 ± 0.4     | 4.7 ± 0.3     | 4.9 ± 0.1   | 5.2 ± 0.3     | 5.0 ± 0.1   | 5.0 ± 0.2    | 4.7 ± 0.2     | E Africa F   | E Africa M   |
| 44  | 7.6 ± 0.5     | 6.0 ± 0.1   | 6.0 ± 0.2     | 7.6 ± 0.2   | 5.7 ± 0.2    | 6.5 ± 0.4     | 4.8 ± 0.3     | 4.9 ± 0.1   | 5.2 ± 0.2     | 5.1 ± 0.1   | 5.0 ± 0.2    | 4.8 ± 0.2     | S Asia F     | W Africa M   |
| 45  | 7.9 ± 0.5     | 6.0 ± 0.1   | 5.7 ± 0.2     | 7.8 ± 0.2   | 5.8 ± 0.2    | 6.6 ± 0.4     | 4.9 ± 0.3     | 4.9 ± 0.1   | 4.8 ± 0.2     | 5.1 ± 0.2   | 5.0 ± 0.2    | 5.1 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 7.6 ± 0.5     | 6.1 ± 0.1   | 5.9 ± 0.3     | 7.7 ± 0.3   | 6.2 ± 0.2    | 6.9 ± 0.4     | 4.9 ± 0.3     | 4.7 ± 0.1   | 5.1 ± 0.3     | 5.2 ± 0.2   | 5.1 ± 0.3    | 5.3 ± 0.2     | S Asia F     | E Asia M     |
| 47  | 7.5 ± 0.5     | 6.1 ± 0.1   | 5.8 ± 0.3     | 7.7 ± 0.3   | 5.8 ± 0.2    | 6.8 ± 0.4     | 5.3 ± 0.3     | 4.8 ± 0.1   | 5.1 ± 0.3     | 5.3 ± 0.2   | 4.9 ± 0.2    | 5.8 ± 0.3     | S Asia F     | E Asia M     |
| 48  | 7.8 ± 0.6     | 6.2 ± 0.1   | 5.8 ± 0.3     | 7.7 ± 0.3   | 6.0 ± 0.2    | 7.3 ± 0.5     | 5.4 ± 0.4     | 4.7 ± 0.1   | 5.3 ± 0.3     | 5.4 ± 0.2   | 5.1 ± 0.3    | 5.8 ± 0.3     | E Africa F   | E Asia M     |
| 49  | 7.9 ± 0.6     | 6.2 ± 0.1   | 5.7 ± 0.3     | 7.7 ± 0.3   | 5.9 ± 0.3    | 7.6 ± 0.5     | 5.9 ± 0.4     | 4.7 ± 0.1   | 5.3 ± 0.3     | 5.5 ± 0.2   | 4.9 ± 0.3    | 6.0 ± 0.3     | E Africa F   | E Asia M     |
| 50  | 8.0 ± 0.6     | 6.3 ± 0.1   | 5.5 ± 0.3     | 7.7 ± 0.3   | 6.1 ± 0.3    | 7.1 ± 0.5     | 5.9 ± 0.4     | 4.7 ± 0.1   | 5.1 ± 0.3     | 5.7 ± 0.2   | 4.8 ± 0.3    | 6.2 ± 0.3     | E Africa F   | E Asia M     |
| 51  | 7.4 ± 0.5     | 6.4 ± 0.2   | 5.7 ± 0.3     | 7.6 ± 0.3   | 6.0 ± 0.2    | 8.0 ± 0.5     | 6.4 ± 0.4     | 4.6 ± 0.1   | 5.1 ± 0.3     | 5.6 ± 0.2   | 4.7 ± 0.3    | 6.5 ± 0.3     | W Africa F   | E Asia M     |
| 52  | 8.2 ± 0.7     | 6.4 ± 0.2   | 5.4 ± 0.3     | 7.5 ± 0.3   | 5.9 ± 0.3    | 8.2 ± 0.5     | 6.1 ± 0.5     | 4.6 ± 0.1   | 4.9 ± 0.3     | 5.6 ± 0.2   | 4.7 ± 0.3    | 7.0 ± 0.4     | W Africa F   | E Asia M     |



**Table 83.** Mean absolute error for Application-type images for algorithm idemia-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.7 ± 0.2     | 2.34 ± 0.09 | 1.6 ± 0.1     | 2.2 ± 0.1   | 2.5 ± 0.2    | 2.9 ± 0.3     | 2.5 ± 0.2     | 1.89 ± 0.08 | 1.6 ± 0.1     | 1.89 ± 0.08 | 2.0 ± 0.1    | 2.0 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 2.7 ± 0.2     | 2.4 ± 0.1   | 1.7 ± 0.1     | 2.4 ± 0.1   | 2.6 ± 0.2    | 3.2 ± 0.3     | 2.3 ± 0.2     | 1.92 ± 0.08 | 1.5 ± 0.1     | 1.81 ± 0.09 | 1.9 ± 0.1    | 2.2 ± 0.2     | W Africa F   | E Europe M   |
| 16  | 2.9 ± 0.2     | 2.5 ± 0.1   | 1.8 ± 0.1     | 2.4 ± 0.1   | 2.8 ± 0.3    | 3.5 ± 0.4     | 2.4 ± 0.2     | 1.94 ± 0.08 | 1.6 ± 0.2     | 1.85 ± 0.10 | 2.0 ± 0.2    | 2.3 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 2.9 ± 0.2     | 2.5 ± 0.1   | 2.1 ± 0.2     | 2.5 ± 0.1   | 2.8 ± 0.2    | 3.5 ± 0.3     | 2.4 ± 0.2     | 2.03 ± 0.09 | 1.7 ± 0.1     | 2.1 ± 0.1   | 2.0 ± 0.2    | 2.6 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.0 ± 0.2     | 2.43 ± 0.08 | 2.4 ± 0.2     | 2.46 ± 0.09 | 2.8 ± 0.1    | 3.6 ± 0.3     | 2.7 ± 0.2     | 2.13 ± 0.07 | 2.1 ± 0.1     | 2.41 ± 0.10 | 2.3 ± 0.2    | 2.6 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 3.2 ± 0.2     | 2.35 ± 0.08 | 2.4 ± 0.2     | 2.41 ± 0.10 | 2.7 ± 0.1    | 3.2 ± 0.2     | 2.9 ± 0.2     | 2.23 ± 0.08 | 2.1 ± 0.2     | 2.41 ± 0.10 | 2.4 ± 0.1    | 3.0 ± 0.3     | E Africa F   | E Europe M   |
| 20  | 3.3 ± 0.2     | 2.39 ± 0.09 | 2.4 ± 0.1     | 2.6 ± 0.1   | 2.6 ± 0.1    | 3.3 ± 0.2     | 3.1 ± 0.2     | 2.15 ± 0.08 | 2.0 ± 0.1     | 2.54 ± 0.10 | 2.4 ± 0.1    | 3.1 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 3.2 ± 0.2     | 2.30 ± 0.08 | 2.6 ± 0.1     | 2.53 ± 0.08 | 2.7 ± 0.1    | 3.2 ± 0.2     | 3.2 ± 0.2     | 2.17 ± 0.07 | 2.0 ± 0.1     | 2.7 ± 0.1   | 2.5 ± 0.1    | 3.2 ± 0.2     | W Africa F   | E Europe M   |
| 22  | 3.3 ± 0.2     | 2.27 ± 0.07 | 2.9 ± 0.1     | 2.62 ± 0.08 | 2.7 ± 0.1    | 3.1 ± 0.2     | 3.7 ± 0.2     | 2.18 ± 0.08 | 2.1 ± 0.1     | 2.7 ± 0.1   | 2.3 ± 0.1    | 3.0 ± 0.2     | E Africa M   | E Europe M   |
| 23  | 3.3 ± 0.2     | 2.32 ± 0.06 | 3.1 ± 0.1     | 2.69 ± 0.07 | 2.53 ± 0.09  | 3.2 ± 0.2     | 3.7 ± 0.2     | 2.17 ± 0.08 | 2.1 ± 0.1     | 2.9 ± 0.1   | 2.5 ± 0.1    | 3.3 ± 0.2     | E Africa M   | E Europe M   |
| 24  | 3.4 ± 0.2     | 2.34 ± 0.06 | 3.24 ± 0.09   | 2.78 ± 0.07 | 2.58 ± 0.08  | 3.3 ± 0.2     | 3.7 ± 0.2     | 2.26 ± 0.08 | 2.2 ± 0.1     | 2.99 ± 0.09 | 2.3 ± 0.1    | 3.3 ± 0.2     | E Africa M   | E Europe M   |
| 25  | 3.5 ± 0.2     | 2.43 ± 0.06 | 3.43 ± 0.09   | 2.80 ± 0.06 | 2.61 ± 0.08  | 3.4 ± 0.2     | 3.9 ± 0.2     | 2.35 ± 0.07 | 2.2 ± 0.1     | 2.95 ± 0.08 | 2.5 ± 0.1    | 3.4 ± 0.2     | E Africa M   | E Europe M   |
| 26  | 3.5 ± 0.1     | 2.49 ± 0.06 | 3.69 ± 0.09   | 2.94 ± 0.06 | 2.64 ± 0.07  | 3.2 ± 0.1     | 3.8 ± 0.2     | 2.31 ± 0.07 | 2.3 ± 0.1     | 3.06 ± 0.08 | 2.5 ± 0.1    | 3.2 ± 0.2     | E Africa M   | E Asia M     |
| 27  | 3.7 ± 0.2     | 2.53 ± 0.05 | 3.87 ± 0.09   | 3.08 ± 0.06 | 2.58 ± 0.07  | 3.4 ± 0.1     | 3.7 ± 0.2     | 2.35 ± 0.07 | 2.31 ± 0.09   | 3.00 ± 0.07 | 2.6 ± 0.1    | 3.3 ± 0.1     | E Europe F   | E Europe M   |
| 28  | 3.7 ± 0.2     | 2.66 ± 0.05 | 3.99 ± 0.09   | 3.23 ± 0.06 | 2.69 ± 0.07  | 3.3 ± 0.1     | 3.6 ± 0.2     | 2.46 ± 0.08 | 2.38 ± 0.09   | 3.00 ± 0.07 | 2.6 ± 0.1    | 3.3 ± 0.1     | E Europe F   | E Europe M   |
| 29  | 3.6 ± 0.1     | 2.62 ± 0.05 | 4.08 ± 0.10   | 3.31 ± 0.06 | 2.67 ± 0.08  | 3.5 ± 0.1     | 3.6 ± 0.2     | 2.34 ± 0.07 | 2.45 ± 0.09   | 2.99 ± 0.07 | 2.6 ± 0.1    | 3.4 ± 0.1     | E Europe F   | E Asia M     |
| 30  | 3.8 ± 0.1     | 2.68 ± 0.05 | 4.2 ± 0.1     | 3.44 ± 0.06 | 2.78 ± 0.08  | 3.7 ± 0.1     | 3.6 ± 0.2     | 2.36 ± 0.06 | 2.38 ± 0.09   | 2.96 ± 0.06 | 2.6 ± 0.1    | 3.4 ± 0.1     | E Europe F   | E Asia M     |
| 31  | 3.8 ± 0.2     | 2.76 ± 0.05 | 4.2 ± 0.1     | 3.57 ± 0.07 | 2.81 ± 0.08  | 3.6 ± 0.1     | 3.6 ± 0.2     | 2.39 ± 0.07 | 2.44 ± 0.10   | 2.94 ± 0.06 | 2.7 ± 0.1    | 3.4 ± 0.1     | E Europe F   | E Asia M     |
| 32  | 3.9 ± 0.2     | 2.83 ± 0.05 | 4.2 ± 0.1     | 3.72 ± 0.07 | 2.90 ± 0.09  | 3.8 ± 0.1     | 3.5 ± 0.2     | 2.37 ± 0.06 | 2.5 ± 0.1     | 2.92 ± 0.06 | 2.6 ± 0.1    | 3.5 ± 0.1     | E Europe F   | E Asia M     |
| 33  | 4.0 ± 0.2     | 2.86 ± 0.06 | 4.2 ± 0.1     | 3.76 ± 0.07 | 2.85 ± 0.09  | 3.7 ± 0.2     | 3.6 ± 0.2     | 2.40 ± 0.07 | 2.5 ± 0.1     | 2.89 ± 0.06 | 2.6 ± 0.1    | 3.3 ± 0.1     | E Europe F   | E Asia M     |
| 34  | 4.0 ± 0.2     | 2.87 ± 0.06 | 4.3 ± 0.1     | 3.78 ± 0.07 | 2.88 ± 0.09  | 3.7 ± 0.2     | 3.5 ± 0.2     | 2.39 ± 0.06 | 2.5 ± 0.1     | 2.93 ± 0.06 | 2.6 ± 0.1    | 3.3 ± 0.1     | E Europe F   | E Asia M     |
| 35  | 4.2 ± 0.2     | 2.94 ± 0.06 | 4.2 ± 0.1     | 3.90 ± 0.08 | 2.93 ± 0.10  | 3.8 ± 0.2     | 3.5 ± 0.2     | 2.38 ± 0.07 | 2.5 ± 0.1     | 2.90 ± 0.06 | 2.8 ± 0.1    | 3.4 ± 0.1     | E Europe F   | E Asia M     |
| 36  | 4.3 ± 0.2     | 2.96 ± 0.06 | 4.1 ± 0.1     | 3.98 ± 0.08 | 3.0 ± 0.1    | 3.8 ± 0.2     | 3.5 ± 0.2     | 2.36 ± 0.07 | 2.5 ± 0.1     | 2.83 ± 0.06 | 2.7 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 37  | 4.3 ± 0.2     | 2.92 ± 0.06 | 4.1 ± 0.1     | 4.01 ± 0.09 | 3.0 ± 0.1    | 4.2 ± 0.2     | 3.5 ± 0.2     | 2.39 ± 0.07 | 2.5 ± 0.1     | 2.87 ± 0.06 | 2.7 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Asia M     |
| 38  | 4.0 ± 0.2     | 2.99 ± 0.06 | 4.0 ± 0.1     | 4.04 ± 0.09 | 3.0 ± 0.1    | 4.2 ± 0.2     | 3.5 ± 0.2     | 2.41 ± 0.07 | 2.5 ± 0.1     | 2.75 ± 0.06 | 2.9 ± 0.1    | 3.2 ± 0.1     | W Africa F   | E Asia M     |
| 39  | 4.3 ± 0.2     | 3.05 ± 0.06 | 4.0 ± 0.1     | 4.09 ± 0.10 | 3.0 ± 0.1    | 3.8 ± 0.2     | 3.4 ± 0.2     | 2.39 ± 0.07 | 2.5 ± 0.1     | 2.77 ± 0.06 | 3.0 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 40  | 4.2 ± 0.2     | 3.03 ± 0.06 | 3.6 ± 0.1     | 4.0 ± 0.1   | 3.2 ± 0.1    | 4.0 ± 0.2     | 3.4 ± 0.2     | 2.41 ± 0.06 | 2.6 ± 0.1     | 2.82 ± 0.06 | 2.7 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Asia M     |
| 41  | 4.4 ± 0.2     | 3.06 ± 0.06 | 3.7 ± 0.1     | 4.0 ± 0.1   | 3.0 ± 0.1    | 4.1 ± 0.2     | 3.5 ± 0.2     | 2.45 ± 0.06 | 2.5 ± 0.1     | 2.79 ± 0.07 | 2.8 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 42  | 4.1 ± 0.3     | 3.07 ± 0.06 | 3.6 ± 0.2     | 3.9 ± 0.1   | 2.9 ± 0.1    | 3.9 ± 0.2     | 3.3 ± 0.2     | 2.47 ± 0.07 | 2.6 ± 0.1     | 2.76 ± 0.07 | 2.9 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Asia M     |
| 43  | 4.1 ± 0.2     | 3.17 ± 0.07 | 3.6 ± 0.2     | 3.9 ± 0.1   | 3.1 ± 0.1    | 4.0 ± 0.2     | 3.5 ± 0.2     | 2.52 ± 0.07 | 2.5 ± 0.1     | 2.73 ± 0.08 | 2.9 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 44  | 4.4 ± 0.3     | 3.27 ± 0.07 | 3.7 ± 0.2     | 4.0 ± 0.1   | 3.1 ± 0.1    | 4.1 ± 0.2     | 3.3 ± 0.2     | 2.59 ± 0.07 | 2.6 ± 0.1     | 2.80 ± 0.08 | 3.0 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.2 ± 0.3     | 3.27 ± 0.07 | 3.4 ± 0.2     | 3.9 ± 0.1   | 3.0 ± 0.1    | 4.0 ± 0.2     | 3.4 ± 0.2     | 2.64 ± 0.07 | 2.5 ± 0.1     | 2.81 ± 0.09 | 2.9 ± 0.1    | 3.3 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.4 ± 0.3     | 3.41 ± 0.07 | 3.4 ± 0.2     | 3.7 ± 0.1   | 3.2 ± 0.1    | 4.0 ± 0.2     | 3.3 ± 0.2     | 2.66 ± 0.07 | 2.8 ± 0.1     | 2.85 ± 0.09 | 3.0 ± 0.2    | 3.3 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 4.3 ± 0.3     | 3.38 ± 0.08 | 3.6 ± 0.2     | 3.8 ± 0.1   | 3.0 ± 0.1    | 4.0 ± 0.2     | 3.5 ± 0.2     | 2.66 ± 0.08 | 2.6 ± 0.1     | 3.1 ± 0.1   | 3.1 ± 0.2    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.3 ± 0.3     | 3.44 ± 0.08 | 3.3 ± 0.2     | 3.9 ± 0.1   | 3.2 ± 0.1    | 4.2 ± 0.3     | 3.6 ± 0.3     | 2.73 ± 0.08 | 2.7 ± 0.2     | 2.9 ± 0.1   | 3.0 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia M     |
| 49  | 4.3 ± 0.3     | 3.59 ± 0.08 | 3.4 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.1    | 4.6 ± 0.3     | 3.5 ± 0.3     | 2.82 ± 0.08 | 2.7 ± 0.1     | 3.0 ± 0.1   | 3.1 ± 0.2    | 3.4 ± 0.2     | W Africa F   | E Europe M   |
| 50  | 4.4 ± 0.4     | 3.65 ± 0.08 | 3.2 ± 0.2     | 3.6 ± 0.1   | 3.1 ± 0.1    | 4.2 ± 0.3     | 3.2 ± 0.2     | 2.72 ± 0.08 | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.2 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia M     |
| 51  | 4.1 ± 0.3     | 3.66 ± 0.09 | 3.3 ± 0.2     | 3.6 ± 0.1   | 3.2 ± 0.1    | 4.4 ± 0.3     | 3.7 ± 0.3     | 2.84 ± 0.08 | 2.7 ± 0.1     | 3.2 ± 0.1   | 3.1 ± 0.2    | 3.7 ± 0.2     | W Africa F   | E Europe M   |
| 52  | 4.3 ± 0.4     | 3.76 ± 0.09 | 3.2 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.1    | 4.5 ± 0.3     | 3.5 ± 0.3     | 2.82 ± 0.08 | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.0 ± 0.2    | 3.8 ± 0.3     | W Africa F   | E Europe M   |



**Table 84.** Mean absolute error for Application-type images for algorithm idemia-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.0 ± 0.2     | 1.40 ± 0.06 | 1.6 ± 0.1     | 1.57 ± 0.09 | 1.6 ± 0.1    | 1.8 ± 0.2     | 2.1 ± 0.1     | 1.64 ± 0.07 | 1.5 ± 0.1     | 1.48 ± 0.07 | 1.7 ± 0.1    | 1.8 ± 0.2     | E Africa M   | E Asia F     |
| 15  | 2.2 ± 0.2     | 1.55 ± 0.08 | 1.9 ± 0.1     | 1.8 ± 0.1   | 1.8 ± 0.2    | 2.4 ± 0.3     | 2.0 ± 0.2     | 1.63 ± 0.07 | 1.4 ± 0.1     | 1.47 ± 0.08 | 1.7 ± 0.1    | 1.8 ± 0.2     | W Africa F   | E Europe M   |
| 16  | 2.6 ± 0.2     | 1.83 ± 0.08 | 2.3 ± 0.2     | 2.1 ± 0.1   | 2.1 ± 0.2    | 2.8 ± 0.4     | 2.0 ± 0.2     | 1.61 ± 0.06 | 1.6 ± 0.2     | 1.53 ± 0.08 | 1.7 ± 0.1    | 1.8 ± 0.2     | W Africa F   | S Asia M     |
| 17  | 2.7 ± 0.2     | 2.10 ± 0.09 | 2.6 ± 0.2     | 2.3 ± 0.1   | 2.4 ± 0.2    | 3.1 ± 0.3     | 2.0 ± 0.2     | 1.69 ± 0.08 | 1.5 ± 0.1     | 1.71 ± 0.10 | 1.7 ± 0.1    | 2.1 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.2 ± 0.2     | 2.26 ± 0.07 | 2.9 ± 0.2     | 2.57 ± 0.09 | 2.5 ± 0.1    | 3.3 ± 0.3     | 2.2 ± 0.2     | 1.81 ± 0.07 | 1.9 ± 0.1     | 1.99 ± 0.08 | 1.9 ± 0.2    | 2.0 ± 0.2     | W Africa F   | E Asia M     |
| 19  | 3.4 ± 0.2     | 2.38 ± 0.08 | 2.7 ± 0.2     | 2.7 ± 0.1   | 2.6 ± 0.1    | 3.1 ± 0.2     | 2.3 ± 0.2     | 1.91 ± 0.08 | 1.9 ± 0.1     | 1.97 ± 0.08 | 2.1 ± 0.1    | 2.3 ± 0.2     | E Africa F   | E Asia M     |
| 20  | 3.7 ± 0.2     | 2.53 ± 0.10 | 2.8 ± 0.2     | 3.0 ± 0.1   | 2.6 ± 0.1    | 3.4 ± 0.2     | 2.5 ± 0.2     | 1.89 ± 0.07 | 1.9 ± 0.1     | 2.05 ± 0.09 | 2.1 ± 0.1    | 2.4 ± 0.3     | E Africa F   | E Europe M   |
| 21  | 3.6 ± 0.2     | 2.55 ± 0.08 | 2.9 ± 0.1     | 3.01 ± 0.09 | 2.7 ± 0.1    | 3.5 ± 0.2     | 2.7 ± 0.2     | 2.02 ± 0.07 | 1.8 ± 0.1     | 2.27 ± 0.09 | 2.2 ± 0.1    | 2.7 ± 0.2     | E Africa F   | E Europe M   |
| 22  | 3.8 ± 0.2     | 2.61 ± 0.08 | 3.1 ± 0.1     | 3.16 ± 0.09 | 2.8 ± 0.1    | 3.4 ± 0.2     | 3.2 ± 0.2     | 2.10 ± 0.07 | 1.9 ± 0.1     | 2.32 ± 0.10 | 2.2 ± 0.1    | 2.6 ± 0.2     | E Africa F   | E Europe M   |
| 23  | 3.9 ± 0.2     | 2.70 ± 0.07 | 3.1 ± 0.1     | 3.25 ± 0.09 | 2.69 ± 0.09  | 3.6 ± 0.2     | 3.4 ± 0.2     | 2.19 ± 0.07 | 1.9 ± 0.1     | 2.53 ± 0.09 | 2.4 ± 0.1    | 3.0 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 4.0 ± 0.2     | 2.81 ± 0.07 | 3.10 ± 0.10   | 3.35 ± 0.08 | 2.82 ± 0.08  | 3.8 ± 0.2     | 3.5 ± 0.2     | 2.33 ± 0.07 | 2.03 ± 0.10   | 2.64 ± 0.08 | 2.3 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 4.2 ± 0.2     | 2.92 ± 0.07 | 3.08 ± 0.09   | 3.37 ± 0.07 | 2.93 ± 0.08  | 4.0 ± 0.2     | 3.6 ± 0.2     | 2.43 ± 0.07 | 2.1 ± 0.1     | 2.65 ± 0.07 | 2.5 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 4.2 ± 0.2     | 3.02 ± 0.06 | 3.14 ± 0.09   | 3.48 ± 0.07 | 2.96 ± 0.08  | 3.8 ± 0.2     | 3.6 ± 0.2     | 2.40 ± 0.07 | 2.2 ± 0.1     | 2.80 ± 0.07 | 2.5 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 27  | 4.3 ± 0.2     | 3.05 ± 0.06 | 3.19 ± 0.09   | 3.59 ± 0.07 | 2.94 ± 0.08  | 3.9 ± 0.2     | 3.7 ± 0.2     | 2.46 ± 0.07 | 2.17 ± 0.09   | 2.80 ± 0.07 | 2.6 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 28  | 4.3 ± 0.2     | 3.16 ± 0.06 | 3.10 ± 0.09   | 3.69 ± 0.07 | 3.06 ± 0.08  | 3.9 ± 0.2     | 3.5 ± 0.1     | 2.58 ± 0.08 | 2.25 ± 0.09   | 2.82 ± 0.07 | 2.6 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 4.2 ± 0.2     | 3.10 ± 0.06 | 3.08 ± 0.09   | 3.70 ± 0.07 | 3.06 ± 0.09  | 4.1 ± 0.2     | 3.6 ± 0.2     | 2.46 ± 0.07 | 2.31 ± 0.09   | 2.86 ± 0.06 | 2.7 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 4.4 ± 0.2     | 3.14 ± 0.06 | 3.07 ± 0.09   | 3.77 ± 0.07 | 3.16 ± 0.09  | 4.2 ± 0.2     | 3.5 ± 0.2     | 2.50 ± 0.07 | 2.27 ± 0.09   | 2.85 ± 0.06 | 2.7 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.3 ± 0.2     | 3.19 ± 0.06 | 3.09 ± 0.10   | 3.86 ± 0.07 | 3.20 ± 0.09  | 4.2 ± 0.2     | 3.6 ± 0.2     | 2.53 ± 0.07 | 2.30 ± 0.09   | 2.90 ± 0.06 | 2.8 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 4.4 ± 0.2     | 3.22 ± 0.06 | 3.02 ± 0.10   | 3.89 ± 0.07 | 3.3 ± 0.1    | 4.3 ± 0.2     | 3.6 ± 0.2     | 2.53 ± 0.07 | 2.37 ± 0.10   | 2.89 ± 0.06 | 2.7 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 4.5 ± 0.2     | 3.22 ± 0.06 | 3.1 ± 0.1     | 3.88 ± 0.07 | 3.2 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.56 ± 0.07 | 2.36 ± 0.10   | 2.87 ± 0.06 | 2.7 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 4.5 ± 0.2     | 3.20 ± 0.06 | 3.1 ± 0.1     | 3.82 ± 0.07 | 3.2 ± 0.1    | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.54 ± 0.07 | 2.41 ± 0.10   | 2.93 ± 0.06 | 2.7 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.6 ± 0.2     | 3.24 ± 0.06 | 3.2 ± 0.1     | 3.77 ± 0.08 | 3.3 ± 0.1    | 4.3 ± 0.2     | 3.7 ± 0.2     | 2.51 ± 0.07 | 2.32 ± 0.10   | 2.92 ± 0.06 | 2.9 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 4.7 ± 0.2     | 3.22 ± 0.07 | 3.2 ± 0.1     | 3.69 ± 0.08 | 3.3 ± 0.1    | 4.3 ± 0.2     | 3.7 ± 0.2     | 2.49 ± 0.07 | 2.4 ± 0.1     | 2.89 ± 0.06 | 2.9 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 37  | 4.6 ± 0.2     | 3.13 ± 0.06 | 3.1 ± 0.1     | 3.68 ± 0.08 | 3.3 ± 0.1    | 4.6 ± 0.2     | 3.7 ± 0.2     | 2.49 ± 0.07 | 2.35 ± 0.10   | 2.96 ± 0.06 | 2.8 ± 0.1    | 3.6 ± 0.1     | W Africa F   | E Europe M   |
| 38  | 4.5 ± 0.2     | 3.16 ± 0.07 | 3.1 ± 0.1     | 3.55 ± 0.09 | 3.3 ± 0.1    | 4.6 ± 0.2     | 3.7 ± 0.2     | 2.48 ± 0.07 | 2.31 ± 0.10   | 2.85 ± 0.06 | 2.9 ± 0.1    | 3.6 ± 0.2     | W Africa F   | E Europe M   |
| 39  | 4.5 ± 0.2     | 3.17 ± 0.06 | 3.1 ± 0.1     | 3.44 ± 0.09 | 3.3 ± 0.1    | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.44 ± 0.07 | 2.34 ± 0.10   | 2.88 ± 0.06 | 3.0 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 40  | 4.5 ± 0.2     | 3.10 ± 0.07 | 3.0 ± 0.1     | 3.33 ± 0.10 | 3.4 ± 0.1    | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.44 ± 0.07 | 2.4 ± 0.1     | 2.94 ± 0.07 | 2.8 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 41  | 4.6 ± 0.3     | 3.10 ± 0.07 | 3.1 ± 0.1     | 3.30 ± 0.10 | 3.2 ± 0.1    | 4.2 ± 0.2     | 3.8 ± 0.2     | 2.47 ± 0.07 | 2.4 ± 0.1     | 2.92 ± 0.07 | 2.8 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.4 ± 0.3     | 3.05 ± 0.06 | 3.1 ± 0.1     | 3.3 ± 0.1   | 3.1 ± 0.1    | 4.0 ± 0.2     | 3.6 ± 0.2     | 2.47 ± 0.07 | 2.4 ± 0.1     | 2.89 ± 0.08 | 2.9 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 4.3 ± 0.3     | 3.07 ± 0.07 | 3.0 ± 0.1     | 3.3 ± 0.1   | 3.3 ± 0.1    | 3.9 ± 0.2     | 3.8 ± 0.2     | 2.53 ± 0.07 | 2.4 ± 0.1     | 2.83 ± 0.08 | 2.9 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.6 ± 0.3     | 3.12 ± 0.07 | 3.0 ± 0.1     | 3.4 ± 0.1   | 3.3 ± 0.1    | 3.9 ± 0.2     | 3.7 ± 0.2     | 2.59 ± 0.07 | 2.5 ± 0.1     | 2.88 ± 0.09 | 3.0 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.3 ± 0.3     | 3.01 ± 0.06 | 2.8 ± 0.1     | 3.3 ± 0.1   | 3.1 ± 0.1    | 3.8 ± 0.2     | 3.7 ± 0.2     | 2.64 ± 0.07 | 2.4 ± 0.1     | 2.85 ± 0.09 | 2.9 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.5 ± 0.3     | 3.10 ± 0.07 | 2.9 ± 0.1     | 3.2 ± 0.1   | 3.2 ± 0.1    | 3.7 ± 0.2     | 3.7 ± 0.3     | 2.66 ± 0.07 | 2.8 ± 0.1     | 2.87 ± 0.10 | 3.0 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 4.5 ± 0.3     | 2.99 ± 0.07 | 3.1 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.1    | 3.7 ± 0.2     | 3.8 ± 0.2     | 2.66 ± 0.08 | 2.6 ± 0.1     | 3.1 ± 0.1   | 3.2 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.4 ± 0.3     | 2.93 ± 0.07 | 2.9 ± 0.1     | 3.4 ± 0.1   | 3.1 ± 0.1    | 3.8 ± 0.2     | 4.0 ± 0.3     | 2.72 ± 0.07 | 2.7 ± 0.2     | 2.9 ± 0.1   | 3.0 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 49  | 4.4 ± 0.3     | 2.97 ± 0.08 | 2.9 ± 0.1     | 3.3 ± 0.1   | 3.0 ± 0.1    | 4.1 ± 0.3     | 3.8 ± 0.3     | 2.82 ± 0.08 | 2.7 ± 0.1     | 3.0 ± 0.1   | 3.1 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 50  | 4.3 ± 0.4     | 2.97 ± 0.08 | 2.9 ± 0.1     | 3.2 ± 0.1   | 3.0 ± 0.1    | 3.7 ± 0.2     | 3.5 ± 0.3     | 2.72 ± 0.08 | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.2 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Asia M     |
| 51  | 4.3 ± 0.3     | 2.96 ± 0.08 | 2.9 ± 0.2     | 3.2 ± 0.1   | 2.9 ± 0.1    | 3.9 ± 0.3     | 3.9 ± 0.4     | 2.84 ± 0.08 | 2.6 ± 0.1     | 3.1 ± 0.1   | 3.1 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 52  | 4.4 ± 0.4     | 2.99 ± 0.08 | 2.9 ± 0.1     | 3.2 ± 0.1   | 2.9 ± 0.1    | 3.8 ± 0.3     | 3.8 ± 0.3     | 2.82 ± 0.08 | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.0 ± 0.2    | 3.9 ± 0.3     | E Africa F   | E Europe M   |

**Table 85.** Mean absolute error for Application-type images for algorithm inbut-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 6.9 ± 0.4     | 4.0 ± 0.1   | 5.2 ± 0.4     | 4.6 ± 0.2   | 4.1 ± 0.2    | 7.7 ± 0.5     | 6.8 ± 0.4     | 3.5 ± 0.1   | 3.5 ± 0.3     | 4.1 ± 0.2   | 3.7 ± 0.3    | 6.0 ± 0.5     | W Africa F   | E Europe M   |
| 15  | 7.2 ± 0.4     | 3.7 ± 0.1   | 5.5 ± 0.4     | 4.6 ± 0.2   | 4.0 ± 0.3    | 7.3 ± 0.5     | 6.4 ± 0.4     | 3.5 ± 0.2   | 3.9 ± 0.3     | 4.0 ± 0.2   | 3.7 ± 0.3    | 6.3 ± 0.5     | W Africa F   | E Asia M     |
| 16  | 6.8 ± 0.4     | 3.4 ± 0.1   | 5.7 ± 0.4     | 4.2 ± 0.2   | 3.7 ± 0.3    | 7.8 ± 0.6     | 6.4 ± 0.4     | 3.3 ± 0.2   | 4.3 ± 0.3     | 4.0 ± 0.2   | 3.8 ± 0.4    | 6.1 ± 0.5     | W Africa F   | E Asia M     |
| 17  | 7.1 ± 0.4     | 3.1 ± 0.1   | 5.3 ± 0.4     | 4.2 ± 0.2   | 3.5 ± 0.3    | 7.1 ± 0.4     | 6.1 ± 0.4     | 3.2 ± 0.2   | 3.9 ± 0.3     | 3.8 ± 0.2   | 3.5 ± 0.4    | 6.0 ± 0.5     | W Africa F   | E Asia F     |
| 18  | 6.3 ± 0.3     | 2.8 ± 0.1   | 4.8 ± 0.3     | 3.7 ± 0.2   | 2.7 ± 0.2    | 6.8 ± 0.4     | 5.4 ± 0.3     | 3.0 ± 0.1   | 4.0 ± 0.3     | 3.5 ± 0.2   | 2.6 ± 0.2    | 4.9 ± 0.3     | W Africa F   | S E Asia M   |
| 19  | 6.4 ± 0.3     | 2.7 ± 0.1   | 4.5 ± 0.3     | 3.8 ± 0.2   | 3.0 ± 0.2    | 6.6 ± 0.4     | 5.7 ± 0.4     | 3.1 ± 0.1   | 4.1 ± 0.3     | 3.8 ± 0.2   | 2.8 ± 0.2    | 5.5 ± 0.4     | W Africa F   | E Asia F     |
| 20  | 6.1 ± 0.3     | 2.9 ± 0.1   | 4.2 ± 0.2     | 3.8 ± 0.2   | 2.9 ± 0.1    | 6.7 ± 0.4     | 5.5 ± 0.4     | 3.0 ± 0.1   | 3.9 ± 0.3     | 3.7 ± 0.2   | 2.8 ± 0.2    | 5.2 ± 0.3     | W Africa F   | S E Asia M   |
| 21  | 5.9 ± 0.3     | 2.88 ± 0.10 | 4.0 ± 0.2     | 3.8 ± 0.1   | 3.0 ± 0.1    | 6.2 ± 0.3     | 5.8 ± 0.3     | 2.96 ± 0.10 | 3.7 ± 0.2     | 3.8 ± 0.2   | 3.1 ± 0.2    | 5.1 ± 0.3     | W Africa F   | E Asia F     |
| 22  | 6.0 ± 0.3     | 2.70 ± 0.08 | 3.8 ± 0.2     | 3.7 ± 0.1   | 2.9 ± 0.1    | 5.9 ± 0.3     | 6.5 ± 0.3     | 2.79 ± 0.09 | 3.8 ± 0.2     | 4.0 ± 0.1   | 2.9 ± 0.2    | 4.8 ± 0.3     | E Africa M   | E Asia F     |
| 23  | 5.6 ± 0.2     | 2.67 ± 0.07 | 3.6 ± 0.1     | 3.66 ± 0.10 | 2.72 ± 0.10  | 5.9 ± 0.3     | 6.1 ± 0.3     | 2.59 ± 0.09 | 3.6 ± 0.2     | 4.0 ± 0.1   | 2.8 ± 0.2    | 5.2 ± 0.3     | E Africa M   | E Asia M     |
| 24  | 5.4 ± 0.2     | 2.50 ± 0.06 | 3.3 ± 0.1     | 3.69 ± 0.09 | 2.51 ± 0.08  | 5.6 ± 0.3     | 6.0 ± 0.3     | 2.44 ± 0.08 | 3.5 ± 0.2     | 4.0 ± 0.1   | 2.7 ± 0.1    | 4.9 ± 0.3     | E Africa M   | E Asia M     |
| 25  | 4.8 ± 0.2     | 2.29 ± 0.06 | 3.0 ± 0.1     | 3.48 ± 0.08 | 2.41 ± 0.08  | 5.2 ± 0.2     | 5.8 ± 0.3     | 2.36 ± 0.08 | 3.6 ± 0.2     | 3.8 ± 0.1   | 2.7 ± 0.1    | 4.8 ± 0.2     | E Africa M   | E Asia F     |
| 26  | 4.5 ± 0.2     | 2.17 ± 0.05 | 2.86 ± 0.09   | 3.30 ± 0.07 | 2.40 ± 0.08  | 5.1 ± 0.2     | 5.6 ± 0.2     | 2.20 ± 0.08 | 3.2 ± 0.1     | 3.8 ± 0.1   | 2.7 ± 0.1    | 4.4 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 4.5 ± 0.2     | 2.08 ± 0.05 | 2.74 ± 0.09   | 3.19 ± 0.07 | 2.31 ± 0.07  | 4.9 ± 0.2     | 5.5 ± 0.2     | 2.18 ± 0.08 | 3.3 ± 0.1     | 3.69 ± 0.10 | 2.8 ± 0.1    | 4.6 ± 0.2     | E Africa M   | E Asia F     |
| 28  | 4.5 ± 0.2     | 2.17 ± 0.05 | 2.52 ± 0.08   | 3.08 ± 0.07 | 2.32 ± 0.07  | 4.7 ± 0.2     | 5.2 ± 0.2     | 2.24 ± 0.08 | 3.2 ± 0.1     | 3.67 ± 0.09 | 2.8 ± 0.1    | 4.4 ± 0.2     | E Africa M   | E Asia F     |
| 29  | 4.2 ± 0.2     | 2.22 ± 0.05 | 2.47 ± 0.08   | 2.93 ± 0.06 | 2.42 ± 0.07  | 4.5 ± 0.2     | 5.1 ± 0.2     | 2.37 ± 0.07 | 3.3 ± 0.1     | 3.59 ± 0.09 | 2.8 ± 0.1    | 4.5 ± 0.2     | E Africa M   | E Asia F     |
| 30  | 4.1 ± 0.2     | 2.44 ± 0.05 | 2.52 ± 0.08   | 2.78 ± 0.06 | 2.49 ± 0.08  | 4.3 ± 0.2     | 5.4 ± 0.2     | 2.43 ± 0.07 | 3.1 ± 0.1     | 3.59 ± 0.08 | 2.8 ± 0.1    | 4.5 ± 0.2     | E Africa M   | E Asia M     |
| 31  | 4.0 ± 0.2     | 2.61 ± 0.05 | 2.56 ± 0.09   | 2.85 ± 0.06 | 2.64 ± 0.08  | 4.3 ± 0.2     | 5.2 ± 0.2     | 2.63 ± 0.08 | 3.1 ± 0.1     | 3.61 ± 0.08 | 3.0 ± 0.1    | 4.3 ± 0.2     | E Africa M   | E Europe F   |
| 32  | 4.0 ± 0.2     | 2.83 ± 0.05 | 2.51 ± 0.08   | 2.81 ± 0.06 | 2.80 ± 0.09  | 4.3 ± 0.2     | 5.0 ± 0.2     | 2.73 ± 0.07 | 3.2 ± 0.1     | 3.57 ± 0.08 | 2.8 ± 0.1    | 4.4 ± 0.2     | E Africa M   | E Europe F   |
| 33  | 3.9 ± 0.2     | 2.98 ± 0.05 | 2.58 ± 0.09   | 2.92 ± 0.06 | 2.90 ± 0.09  | 4.1 ± 0.2     | 5.3 ± 0.2     | 2.81 ± 0.08 | 3.2 ± 0.1     | 3.52 ± 0.08 | 2.8 ± 0.1    | 4.4 ± 0.2     | E Africa M   | E Europe F   |
| 34  | 3.9 ± 0.2     | 3.08 ± 0.05 | 2.68 ± 0.10   | 2.95 ± 0.06 | 2.91 ± 0.09  | 3.8 ± 0.2     | 5.0 ± 0.2     | 2.92 ± 0.07 | 3.0 ± 0.1     | 3.55 ± 0.08 | 3.0 ± 0.1    | 4.5 ± 0.2     | E Africa M   | E Europe F   |
| 35  | 4.0 ± 0.2     | 3.28 ± 0.06 | 2.72 ± 0.10   | 3.13 ± 0.06 | 3.14 ± 0.10  | 3.7 ± 0.2     | 5.1 ± 0.2     | 2.88 ± 0.08 | 2.9 ± 0.1     | 3.47 ± 0.08 | 2.9 ± 0.1    | 4.5 ± 0.2     | E Africa M   | E Europe F   |
| 36  | 4.0 ± 0.2     | 3.30 ± 0.06 | 2.67 ± 0.10   | 3.25 ± 0.07 | 3.2 ± 0.1    | 3.8 ± 0.2     | 5.1 ± 0.2     | 2.78 ± 0.08 | 3.0 ± 0.1     | 3.39 ± 0.08 | 2.8 ± 0.1    | 4.5 ± 0.2     | E Africa M   | E Europe F   |
| 37  | 3.9 ± 0.2     | 3.29 ± 0.06 | 2.8 ± 0.1     | 3.47 ± 0.07 | 3.3 ± 0.1    | 3.8 ± 0.2     | 5.1 ± 0.3     | 2.72 ± 0.07 | 2.8 ± 0.1     | 3.26 ± 0.07 | 2.7 ± 0.1    | 4.3 ± 0.2     | E Africa M   | E Asia M     |
| 38  | 4.1 ± 0.2     | 3.41 ± 0.07 | 2.9 ± 0.1     | 3.61 ± 0.08 | 3.4 ± 0.1    | 3.8 ± 0.2     | 4.8 ± 0.2     | 2.57 ± 0.07 | 2.8 ± 0.1     | 3.27 ± 0.08 | 2.7 ± 0.1    | 4.2 ± 0.2     | E Africa M   | E Asia M     |
| 39  | 4.0 ± 0.2     | 3.38 ± 0.07 | 2.9 ± 0.1     | 3.73 ± 0.09 | 3.2 ± 0.1    | 3.7 ± 0.2     | 4.8 ± 0.2     | 2.53 ± 0.08 | 3.1 ± 0.2     | 3.21 ± 0.08 | 2.6 ± 0.1    | 4.3 ± 0.2     | E Africa M   | E Asia M     |
| 40  | 4.2 ± 0.2     | 3.43 ± 0.07 | 3.0 ± 0.1     | 3.79 ± 0.09 | 3.3 ± 0.1    | 3.5 ± 0.2     | 5.0 ± 0.3     | 2.44 ± 0.07 | 3.1 ± 0.2     | 3.30 ± 0.08 | 2.5 ± 0.1    | 4.2 ± 0.2     | E Africa M   | E Asia M     |
| 41  | 4.3 ± 0.2     | 3.45 ± 0.07 | 3.1 ± 0.1     | 3.88 ± 0.10 | 3.3 ± 0.1    | 3.6 ± 0.2     | 4.9 ± 0.3     | 2.47 ± 0.07 | 3.2 ± 0.2     | 3.26 ± 0.08 | 2.6 ± 0.1    | 4.5 ± 0.2     | E Africa M   | E Asia M     |
| 42  | 4.2 ± 0.2     | 3.44 ± 0.07 | 3.2 ± 0.1     | 3.8 ± 0.1   | 3.3 ± 0.1    | 3.6 ± 0.2     | 5.0 ± 0.3     | 2.44 ± 0.07 | 3.3 ± 0.2     | 3.32 ± 0.09 | 2.6 ± 0.1    | 4.2 ± 0.2     | E Africa M   | E Asia M     |
| 43  | 4.2 ± 0.2     | 3.57 ± 0.07 | 3.2 ± 0.1     | 3.9 ± 0.1   | 3.4 ± 0.1    | 3.7 ± 0.2     | 5.2 ± 0.3     | 2.55 ± 0.07 | 3.2 ± 0.2     | 3.32 ± 0.09 | 2.7 ± 0.1    | 4.2 ± 0.2     | E Africa M   | E Asia M     |
| 44  | 4.6 ± 0.3     | 3.70 ± 0.08 | 3.3 ± 0.1     | 3.9 ± 0.1   | 3.4 ± 0.1    | 3.8 ± 0.2     | 4.8 ± 0.3     | 2.66 ± 0.07 | 3.2 ± 0.2     | 3.35 ± 0.10 | 2.8 ± 0.1    | 4.1 ± 0.2     | E Africa M   | E Asia M     |
| 45  | 4.3 ± 0.3     | 3.76 ± 0.08 | 3.1 ± 0.1     | 3.8 ± 0.1   | 3.4 ± 0.1    | 3.9 ± 0.2     | 4.9 ± 0.3     | 2.84 ± 0.08 | 3.3 ± 0.2     | 3.26 ± 0.09 | 2.8 ± 0.1    | 4.3 ± 0.2     | E Africa M   | E Asia M     |
| 46  | 4.7 ± 0.3     | 3.95 ± 0.08 | 3.2 ± 0.2     | 3.6 ± 0.1   | 3.4 ± 0.1    | 3.7 ± 0.2     | 5.0 ± 0.3     | 2.85 ± 0.07 | 3.4 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 4.0 ± 0.2     | E Africa M   | E Asia M     |
| 47  | 4.4 ± 0.3     | 3.88 ± 0.08 | 3.3 ± 0.2     | 3.6 ± 0.1   | 3.3 ± 0.1    | 4.0 ± 0.2     | 5.1 ± 0.3     | 2.98 ± 0.08 | 3.1 ± 0.2     | 3.4 ± 0.1   | 3.0 ± 0.2    | 4.2 ± 0.2     | E Africa M   | E Asia M     |
| 48  | 4.7 ± 0.3     | 4.04 ± 0.08 | 3.2 ± 0.2     | 3.6 ± 0.1   | 3.5 ± 0.1    | 3.9 ± 0.2     | 5.1 ± 0.3     | 2.99 ± 0.08 | 3.2 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 4.2 ± 0.2     | E Africa M   | E Asia M     |
| 49  | 4.8 ± 0.3     | 4.15 ± 0.09 | 3.4 ± 0.2     | 3.4 ± 0.1   | 3.4 ± 0.1    | 4.2 ± 0.2     | 4.9 ± 0.3     | 3.08 ± 0.08 | 3.2 ± 0.2     | 3.2 ± 0.1   | 2.9 ± 0.2    | 3.8 ± 0.2     | E Africa M   | S E Asia M   |
| 50  | 4.5 ± 0.4     | 4.18 ± 0.09 | 3.1 ± 0.2     | 3.4 ± 0.1   | 3.6 ± 0.1    | 4.0 ± 0.3     | 4.4 ± 0.3     | 3.04 ± 0.08 | 3.0 ± 0.2     | 3.1 ± 0.1   | 3.0 ± 0.2    | 3.8 ± 0.2     | E Africa F   | S E Asia M   |
| 51  | 4.2 ± 0.3     | 4.20 ± 0.09 | 3.2 ± 0.2     | 3.4 ± 0.1   | 3.6 ± 0.2    | 4.2 ± 0.3     | 4.6 ± 0.4     | 3.12 ± 0.09 | 2.8 ± 0.2     | 3.2 ± 0.1   | 2.9 ± 0.1    | 4.1 ± 0.2     | E Africa M   | E Europe M   |
| 52  | 4.4 ± 0.3     | 4.2 ± 0.1   | 3.4 ± 0.2     | 3.4 ± 0.1   | 3.4 ± 0.1    | 4.1 ± 0.3     | 4.4 ± 0.3     | 3.09 ± 0.09 | 3.1 ± 0.2     | 3.1 ± 0.1   | 2.9 ± 0.2    | 4.1 ± 0.2     | E Africa M   | S E Asia M   |

**Table 86.** Mean absolute error for Application-type images for algorithm incode-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.7 ± 0.2     | 2.39 ± 0.09 | 2.4 ± 0.2     | 2.3 ± 0.1   | 2.6 ± 0.2    | 2.9 ± 0.3     | 2.6 ± 0.2     | 1.67 ± 0.07 | 1.6 ± 0.1     | 1.81 ± 0.08 | 1.8 ± 0.1    | 2.2 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 2.8 ± 0.3     | 2.5 ± 0.1   | 2.2 ± 0.2     | 2.6 ± 0.2   | 2.9 ± 0.3    | 2.7 ± 0.3     | 2.5 ± 0.2     | 1.74 ± 0.06 | 1.4 ± 0.1     | 1.81 ± 0.08 | 2.0 ± 0.1    | 2.2 ± 0.2     | S E Asia F   | E Europe M   |
| 16  | 2.9 ± 0.3     | 2.5 ± 0.1   | 2.6 ± 0.3     | 2.5 ± 0.2   | 2.8 ± 0.3    | 3.0 ± 0.5     | 2.5 ± 0.2     | 1.60 ± 0.06 | 1.5 ± 0.2     | 1.71 ± 0.08 | 1.8 ± 0.1    | 2.0 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.4 ± 0.4     | 2.6 ± 0.1   | 2.5 ± 0.3     | 2.6 ± 0.2   | 2.8 ± 0.2    | 2.9 ± 0.3     | 2.1 ± 0.2     | 1.50 ± 0.08 | 1.3 ± 0.1     | 1.7 ± 0.1   | 1.6 ± 0.2    | 2.0 ± 0.3     | E Africa F   | E Europe M   |
| 18  | 3.4 ± 0.3     | 2.7 ± 0.1   | 2.7 ± 0.3     | 2.5 ± 0.1   | 2.9 ± 0.2    | 3.0 ± 0.3     | 2.1 ± 0.2     | 1.67 ± 0.08 | 1.6 ± 0.2     | 1.80 ± 0.10 | 1.7 ± 0.2    | 2.0 ± 0.2     | E Africa F   | E Europe M   |
| 19  | 4.1 ± 0.3     | 2.9 ± 0.1   | 3.1 ± 0.2     | 2.9 ± 0.1   | 3.0 ± 0.2    | 3.3 ± 0.3     | 2.8 ± 0.2     | 2.01 ± 0.08 | 1.9 ± 0.2     | 2.07 ± 0.10 | 2.0 ± 0.1    | 2.6 ± 0.3     | E Africa F   | E Europe M   |
| 20  | 4.6 ± 0.3     | 3.1 ± 0.1   | 3.4 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.1    | 3.7 ± 0.2     | 3.2 ± 0.2     | 2.28 ± 0.08 | 2.2 ± 0.1     | 2.54 ± 0.09 | 2.2 ± 0.1    | 3.1 ± 0.2     | E Africa F   | E Europe M   |
| 21  | 4.6 ± 0.2     | 3.18 ± 0.09 | 3.6 ± 0.2     | 3.32 ± 0.10 | 3.3 ± 0.1    | 4.3 ± 0.3     | 3.5 ± 0.2     | 2.47 ± 0.07 | 2.4 ± 0.1     | 2.64 ± 0.09 | 2.5 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 22  | 5.0 ± 0.2     | 3.10 ± 0.08 | 3.5 ± 0.1     | 3.36 ± 0.09 | 3.2 ± 0.1    | 4.3 ± 0.2     | 4.0 ± 0.2     | 2.58 ± 0.08 | 2.5 ± 0.1     | 2.75 ± 0.10 | 2.5 ± 0.1    | 3.4 ± 0.2     | E Africa F   | S E Asia M   |
| 23  | 4.9 ± 0.2     | 2.94 ± 0.08 | 3.5 ± 0.1     | 3.36 ± 0.08 | 3.0 ± 0.1    | 4.3 ± 0.2     | 4.0 ± 0.2     | 2.58 ± 0.08 | 2.3 ± 0.1     | 2.80 ± 0.10 | 2.4 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 5.0 ± 0.2     | 2.87 ± 0.07 | 3.3 ± 0.1     | 3.32 ± 0.08 | 2.84 ± 0.09  | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.48 ± 0.08 | 2.3 ± 0.1     | 2.79 ± 0.09 | 2.3 ± 0.1    | 3.4 ± 0.2     | E Africa F   | S E Asia M   |
| 25  | 4.9 ± 0.2     | 2.74 ± 0.07 | 3.1 ± 0.1     | 3.31 ± 0.07 | 2.80 ± 0.09  | 4.3 ± 0.2     | 3.9 ± 0.2     | 2.52 ± 0.08 | 2.3 ± 0.1     | 2.74 ± 0.08 | 2.3 ± 0.1    | 3.6 ± 0.2     | E Africa F   | S E Asia M   |
| 26  | 4.8 ± 0.2     | 2.75 ± 0.07 | 3.1 ± 0.1     | 3.17 ± 0.07 | 2.80 ± 0.09  | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.43 ± 0.08 | 2.3 ± 0.1     | 2.75 ± 0.08 | 2.3 ± 0.1    | 3.4 ± 0.2     | E Africa F   | S E Asia M   |
| 27  | 4.7 ± 0.2     | 2.74 ± 0.06 | 3.1 ± 0.1     | 3.19 ± 0.07 | 2.77 ± 0.09  | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.52 ± 0.08 | 2.3 ± 0.1     | 2.77 ± 0.07 | 2.4 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 28  | 4.6 ± 0.2     | 2.81 ± 0.06 | 2.97 ± 0.09   | 3.25 ± 0.07 | 2.77 ± 0.09  | 4.2 ± 0.2     | 3.6 ± 0.2     | 2.57 ± 0.08 | 2.4 ± 0.1     | 2.83 ± 0.07 | 2.5 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 29  | 4.6 ± 0.2     | 2.83 ± 0.06 | 3.01 ± 0.10   | 3.32 ± 0.07 | 2.85 ± 0.09  | 4.3 ± 0.2     | 3.6 ± 0.2     | 2.67 ± 0.08 | 2.5 ± 0.1     | 2.89 ± 0.07 | 2.6 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 30  | 4.6 ± 0.2     | 3.02 ± 0.06 | 3.11 ± 0.10   | 3.38 ± 0.07 | 2.92 ± 0.09  | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.80 ± 0.08 | 2.4 ± 0.1     | 2.96 ± 0.06 | 2.8 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.7 ± 0.2     | 3.15 ± 0.06 | 3.2 ± 0.1     | 3.46 ± 0.07 | 3.02 ± 0.09  | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.89 ± 0.08 | 2.5 ± 0.1     | 3.11 ± 0.06 | 2.9 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 4.8 ± 0.2     | 3.29 ± 0.06 | 3.2 ± 0.1     | 3.69 ± 0.07 | 3.19 ± 0.10  | 4.4 ± 0.2     | 3.8 ± 0.2     | 3.05 ± 0.08 | 2.6 ± 0.1     | 3.24 ± 0.06 | 2.9 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 4.7 ± 0.2     | 3.44 ± 0.07 | 3.4 ± 0.1     | 3.70 ± 0.07 | 3.29 ± 0.10  | 4.5 ± 0.2     | 4.1 ± 0.2     | 3.24 ± 0.08 | 2.7 ± 0.1     | 3.26 ± 0.06 | 3.0 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 4.6 ± 0.2     | 3.52 ± 0.07 | 3.4 ± 0.1     | 3.70 ± 0.07 | 3.25 ± 0.10  | 4.4 ± 0.2     | 3.8 ± 0.2     | 3.33 ± 0.08 | 2.8 ± 0.1     | 3.40 ± 0.06 | 3.0 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.9 ± 0.2     | 3.65 ± 0.07 | 3.6 ± 0.1     | 3.79 ± 0.07 | 3.4 ± 0.1    | 4.4 ± 0.2     | 4.0 ± 0.2     | 3.40 ± 0.08 | 2.8 ± 0.1     | 3.46 ± 0.07 | 3.2 ± 0.1    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 36  | 4.7 ± 0.2     | 3.66 ± 0.07 | 3.4 ± 0.1     | 3.78 ± 0.08 | 3.4 ± 0.1    | 4.4 ± 0.2     | 4.1 ± 0.2     | 3.32 ± 0.09 | 2.9 ± 0.1     | 3.39 ± 0.07 | 3.2 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 37  | 4.7 ± 0.2     | 3.56 ± 0.07 | 3.6 ± 0.1     | 3.74 ± 0.08 | 3.3 ± 0.1    | 4.7 ± 0.2     | 4.0 ± 0.2     | 3.27 ± 0.09 | 2.9 ± 0.1     | 3.44 ± 0.07 | 3.1 ± 0.1    | 3.9 ± 0.1     | W Africa F   | E Europe M   |
| 38  | 4.5 ± 0.2     | 3.55 ± 0.07 | 3.4 ± 0.1     | 3.63 ± 0.09 | 3.3 ± 0.1    | 4.5 ± 0.2     | 3.7 ± 0.2     | 3.32 ± 0.09 | 2.7 ± 0.1     | 3.28 ± 0.07 | 3.0 ± 0.1    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 39  | 4.4 ± 0.2     | 3.48 ± 0.08 | 3.2 ± 0.1     | 3.56 ± 0.09 | 3.1 ± 0.1    | 4.3 ± 0.2     | 3.6 ± 0.2     | 3.14 ± 0.09 | 2.7 ± 0.1     | 3.15 ± 0.07 | 2.9 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 40  | 4.2 ± 0.2     | 3.33 ± 0.08 | 3.2 ± 0.1     | 3.36 ± 0.10 | 3.0 ± 0.1    | 4.4 ± 0.2     | 3.5 ± 0.2     | 2.96 ± 0.08 | 2.7 ± 0.1     | 3.12 ± 0.08 | 2.7 ± 0.1    | 3.7 ± 0.2     | W Africa F   | E Europe M   |
| 41  | 4.1 ± 0.3     | 3.17 ± 0.08 | 3.0 ± 0.1     | 3.3 ± 0.1   | 2.9 ± 0.1    | 4.2 ± 0.2     | 3.4 ± 0.2     | 2.88 ± 0.09 | 2.6 ± 0.1     | 2.92 ± 0.08 | 2.6 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Europe M   |
| 42  | 4.3 ± 0.3     | 3.00 ± 0.07 | 3.1 ± 0.1     | 3.1 ± 0.1   | 2.7 ± 0.1    | 4.2 ± 0.2     | 3.3 ± 0.2     | 2.76 ± 0.08 | 2.5 ± 0.1     | 2.79 ± 0.08 | 2.5 ± 0.1    | 3.6 ± 0.2     | E Africa F   | S E Asia M   |
| 43  | 4.2 ± 0.3     | 3.02 ± 0.07 | 3.0 ± 0.1     | 3.1 ± 0.1   | 2.8 ± 0.1    | 4.1 ± 0.2     | 3.3 ± 0.2     | 2.72 ± 0.08 | 2.4 ± 0.1     | 2.84 ± 0.09 | 2.7 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.5 ± 0.3     | 3.05 ± 0.07 | 2.8 ± 0.1     | 3.1 ± 0.1   | 2.9 ± 0.1    | 4.4 ± 0.3     | 3.4 ± 0.2     | 2.66 ± 0.08 | 2.6 ± 0.1     | 2.86 ± 0.09 | 2.7 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.3 ± 0.3     | 3.01 ± 0.07 | 2.7 ± 0.1     | 3.1 ± 0.1   | 2.8 ± 0.1    | 3.9 ± 0.2     | 3.5 ± 0.2     | 2.78 ± 0.08 | 2.7 ± 0.1     | 2.92 ± 0.09 | 2.7 ± 0.1    | 3.9 ± 0.2     | E Africa F   | S E Asia M   |
| 46  | 4.7 ± 0.4     | 3.12 ± 0.08 | 2.9 ± 0.1     | 3.1 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.3     | 3.8 ± 0.3     | 2.83 ± 0.08 | 2.8 ± 0.1     | 2.90 ± 0.10 | 2.8 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 4.4 ± 0.4     | 3.04 ± 0.08 | 2.9 ± 0.1     | 3.1 ± 0.1   | 2.9 ± 0.1    | 4.2 ± 0.3     | 4.0 ± 0.3     | 2.82 ± 0.08 | 2.7 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.5 ± 0.4     | 3.04 ± 0.08 | 3.0 ± 0.1     | 3.2 ± 0.1   | 2.9 ± 0.1    | 4.3 ± 0.3     | 4.3 ± 0.3     | 2.85 ± 0.08 | 3.0 ± 0.2     | 3.1 ± 0.1   | 2.9 ± 0.2    | 4.3 ± 0.3     | E Africa F   | E Asia M     |
| 49  | 4.8 ± 0.4     | 3.14 ± 0.09 | 2.8 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.1    | 4.3 ± 0.3     | 4.0 ± 0.3     | 2.93 ± 0.09 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Europe F   |
| 50  | 4.8 ± 0.4     | 3.15 ± 0.09 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.1    | 4.5 ± 0.3     | 3.9 ± 0.3     | 3.02 ± 0.09 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.2 ± 0.2    | 4.3 ± 0.2     | E Africa F   | E Europe F   |
| 51  | 5.0 ± 0.4     | 3.17 ± 0.09 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.2 ± 0.1    | 4.6 ± 0.3     | 4.3 ± 0.4     | 3.14 ± 0.10 | 3.1 ± 0.2     | 3.6 ± 0.2   | 3.3 ± 0.2    | 4.4 ± 0.3     | E Africa F   | E Europe F   |
| 52  | 5.1 ± 0.4     | 3.24 ± 0.09 | 3.3 ± 0.2     | 3.5 ± 0.1   | 3.4 ± 0.2    | 4.6 ± 0.3     | 4.2 ± 0.3     | 3.14 ± 0.10 | 3.4 ± 0.2     | 3.7 ± 0.1   | 3.5 ± 0.2    | 4.7 ± 0.3     | E Africa F   | E Asia M     |

**Table 87.** Mean absolute error for Application-type images for algorithm incode-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.5 ± 0.2     | 2.6 ± 0.1   | 2.6 ± 0.2     | 2.8 ± 0.1   | 2.9 ± 0.2    | 3.6 ± 0.3     | 2.5 ± 0.2     | 1.72 ± 0.08 | 1.5 ± 0.1     | 1.76 ± 0.08 | 1.7 ± 0.1    | 2.1 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 3.5 ± 0.3     | 2.7 ± 0.1   | 2.6 ± 0.2     | 3.0 ± 0.2   | 3.3 ± 0.3    | 3.4 ± 0.3     | 2.4 ± 0.2     | 1.88 ± 0.07 | 1.50 ± 0.10   | 1.76 ± 0.08 | 1.9 ± 0.1    | 2.3 ± 0.2     | E Africa F   | E Europe M   |
| 16  | 3.5 ± 0.3     | 2.8 ± 0.1   | 2.7 ± 0.2     | 2.9 ± 0.1   | 3.4 ± 0.3    | 3.8 ± 0.5     | 2.4 ± 0.2     | 1.76 ± 0.07 | 1.5 ± 0.2     | 1.68 ± 0.08 | 1.7 ± 0.1    | 1.9 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.6 ± 0.3     | 3.0 ± 0.1   | 2.7 ± 0.3     | 2.9 ± 0.2   | 3.2 ± 0.2    | 3.9 ± 0.4     | 2.1 ± 0.2     | 1.70 ± 0.09 | 1.2 ± 0.1     | 1.5 ± 0.1   | 1.6 ± 0.2    | 2.0 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.4 ± 0.3     | 3.1 ± 0.1   | 2.6 ± 0.3     | 2.8 ± 0.1   | 3.4 ± 0.2    | 3.7 ± 0.3     | 1.9 ± 0.2     | 1.82 ± 0.08 | 1.3 ± 0.1     | 1.66 ± 0.09 | 1.7 ± 0.2    | 1.9 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 4.0 ± 0.3     | 3.2 ± 0.1   | 2.9 ± 0.2     | 2.9 ± 0.1   | 3.4 ± 0.2    | 3.9 ± 0.3     | 2.5 ± 0.2     | 2.16 ± 0.09 | 1.6 ± 0.1     | 1.87 ± 0.10 | 2.0 ± 0.1    | 2.6 ± 0.3     | E Africa F   | E Europe M   |
| 20  | 4.5 ± 0.3     | 3.4 ± 0.1   | 2.9 ± 0.2     | 3.1 ± 0.1   | 3.4 ± 0.2    | 4.2 ± 0.3     | 3.1 ± 0.3     | 2.29 ± 0.09 | 1.9 ± 0.1     | 2.42 ± 0.10 | 2.3 ± 0.1    | 3.0 ± 0.3     | E Africa F   | E Europe M   |
| 21  | 4.5 ± 0.3     | 3.4 ± 0.1   | 3.1 ± 0.1     | 3.2 ± 0.1   | 3.6 ± 0.1    | 4.5 ± 0.3     | 3.6 ± 0.3     | 2.59 ± 0.09 | 2.1 ± 0.1     | 2.67 ± 0.10 | 2.7 ± 0.1    | 3.4 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.9 ± 0.3     | 3.41 ± 0.09 | 3.1 ± 0.1     | 3.26 ± 0.09 | 3.5 ± 0.1    | 4.4 ± 0.3     | 4.4 ± 0.3     | 2.85 ± 0.09 | 2.2 ± 0.1     | 2.8 ± 0.1   | 2.7 ± 0.1    | 3.5 ± 0.3     | E Africa F   | E Europe M   |
| 23  | 5.0 ± 0.2     | 3.27 ± 0.08 | 3.1 ± 0.1     | 3.28 ± 0.09 | 3.5 ± 0.1    | 4.7 ± 0.2     | 4.5 ± 0.3     | 2.87 ± 0.09 | 2.2 ± 0.1     | 3.0 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 5.2 ± 0.3     | 3.21 ± 0.07 | 3.0 ± 0.1     | 3.28 ± 0.08 | 3.49 ± 0.10  | 4.6 ± 0.2     | 4.4 ± 0.2     | 2.92 ± 0.09 | 2.2 ± 0.1     | 3.03 ± 0.09 | 2.8 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 5.1 ± 0.2     | 3.16 ± 0.07 | 2.88 ± 0.09   | 3.22 ± 0.07 | 3.43 ± 0.10  | 4.7 ± 0.2     | 4.6 ± 0.2     | 2.95 ± 0.09 | 2.2 ± 0.1     | 3.02 ± 0.09 | 3.0 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 5.1 ± 0.2     | 3.12 ± 0.07 | 2.81 ± 0.08   | 3.18 ± 0.06 | 3.41 ± 0.09  | 4.6 ± 0.2     | 4.5 ± 0.2     | 2.88 ± 0.08 | 2.2 ± 0.1     | 3.05 ± 0.08 | 2.9 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 27  | 5.1 ± 0.2     | 2.99 ± 0.06 | 2.90 ± 0.09   | 3.19 ± 0.06 | 3.26 ± 0.09  | 4.9 ± 0.2     | 4.4 ± 0.2     | 2.95 ± 0.08 | 2.3 ± 0.1     | 3.00 ± 0.07 | 3.0 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 28  | 5.1 ± 0.2     | 2.98 ± 0.06 | 2.68 ± 0.08   | 3.24 ± 0.06 | 3.27 ± 0.09  | 4.7 ± 0.2     | 4.3 ± 0.2     | 2.93 ± 0.09 | 2.34 ± 0.10   | 3.00 ± 0.07 | 2.9 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 29  | 5.3 ± 0.2     | 2.83 ± 0.06 | 2.80 ± 0.09   | 3.16 ± 0.06 | 3.32 ± 0.09  | 4.8 ± 0.2     | 4.3 ± 0.2     | 2.79 ± 0.08 | 2.30 ± 0.09   | 2.97 ± 0.07 | 2.8 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 30  | 5.3 ± 0.2     | 2.91 ± 0.06 | 2.74 ± 0.08   | 3.16 ± 0.06 | 3.19 ± 0.09  | 4.6 ± 0.2     | 4.2 ± 0.2     | 2.84 ± 0.08 | 2.29 ± 0.09   | 2.97 ± 0.06 | 2.9 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 5.2 ± 0.2     | 2.91 ± 0.06 | 2.78 ± 0.09   | 3.14 ± 0.07 | 3.24 ± 0.09  | 4.5 ± 0.2     | 4.3 ± 0.2     | 2.76 ± 0.08 | 2.33 ± 0.09   | 3.03 ± 0.06 | 2.9 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 5.3 ± 0.2     | 2.90 ± 0.06 | 2.75 ± 0.09   | 3.23 ± 0.07 | 3.22 ± 0.10  | 4.8 ± 0.2     | 4.1 ± 0.2     | 2.79 ± 0.07 | 2.4 ± 0.1     | 3.05 ± 0.06 | 2.9 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 5.3 ± 0.2     | 2.89 ± 0.06 | 2.82 ± 0.09   | 3.21 ± 0.07 | 3.3 ± 0.1    | 4.6 ± 0.2     | 4.3 ± 0.2     | 2.78 ± 0.08 | 2.5 ± 0.1     | 3.00 ± 0.06 | 2.7 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 5.1 ± 0.2     | 2.94 ± 0.06 | 2.85 ± 0.09   | 3.13 ± 0.07 | 3.2 ± 0.1    | 4.6 ± 0.2     | 4.0 ± 0.2     | 2.78 ± 0.07 | 2.5 ± 0.1     | 3.10 ± 0.06 | 2.9 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 5.4 ± 0.2     | 2.96 ± 0.06 | 2.95 ± 0.10   | 3.16 ± 0.07 | 3.3 ± 0.1    | 4.6 ± 0.2     | 4.2 ± 0.2     | 2.86 ± 0.08 | 2.5 ± 0.1     | 3.08 ± 0.06 | 2.9 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 5.3 ± 0.2     | 2.94 ± 0.06 | 2.86 ± 0.10   | 3.15 ± 0.07 | 3.2 ± 0.1    | 4.5 ± 0.2     | 4.1 ± 0.2     | 2.82 ± 0.08 | 2.5 ± 0.1     | 3.04 ± 0.06 | 3.0 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 37  | 5.2 ± 0.2     | 2.90 ± 0.06 | 3.0 ± 0.1     | 3.20 ± 0.08 | 3.1 ± 0.1    | 4.8 ± 0.2     | 4.1 ± 0.2     | 2.82 ± 0.08 | 2.5 ± 0.1     | 3.07 ± 0.06 | 2.8 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 38  | 5.4 ± 0.3     | 2.94 ± 0.06 | 2.9 ± 0.1     | 3.15 ± 0.08 | 3.2 ± 0.1    | 4.5 ± 0.2     | 4.0 ± 0.2     | 2.91 ± 0.08 | 2.5 ± 0.1     | 2.99 ± 0.06 | 2.8 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 39  | 5.1 ± 0.3     | 2.98 ± 0.06 | 2.9 ± 0.1     | 3.17 ± 0.09 | 3.1 ± 0.1    | 4.5 ± 0.2     | 4.1 ± 0.2     | 2.88 ± 0.08 | 2.6 ± 0.1     | 2.93 ± 0.06 | 2.9 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 40  | 5.1 ± 0.3     | 2.95 ± 0.06 | 3.0 ± 0.1     | 3.06 ± 0.09 | 3.1 ± 0.1    | 4.4 ± 0.2     | 4.0 ± 0.2     | 2.83 ± 0.08 | 2.5 ± 0.1     | 2.98 ± 0.07 | 2.8 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 41  | 4.9 ± 0.3     | 2.86 ± 0.06 | 3.0 ± 0.1     | 3.09 ± 0.09 | 3.0 ± 0.1    | 4.3 ± 0.2     | 4.1 ± 0.2     | 2.80 ± 0.08 | 2.5 ± 0.1     | 2.86 ± 0.07 | 2.7 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 5.2 ± 0.3     | 2.93 ± 0.06 | 3.1 ± 0.1     | 3.1 ± 0.1   | 2.9 ± 0.1    | 4.4 ± 0.2     | 4.0 ± 0.2     | 2.87 ± 0.08 | 2.6 ± 0.1     | 2.83 ± 0.08 | 2.8 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 5.4 ± 0.3     | 2.95 ± 0.06 | 3.1 ± 0.1     | 3.0 ± 0.1   | 3.0 ± 0.1    | 4.2 ± 0.2     | 4.2 ± 0.3     | 2.91 ± 0.08 | 2.5 ± 0.1     | 2.86 ± 0.08 | 3.0 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 5.4 ± 0.3     | 2.94 ± 0.06 | 2.8 ± 0.1     | 3.1 ± 0.1   | 3.1 ± 0.1    | 4.4 ± 0.2     | 4.1 ± 0.3     | 2.94 ± 0.08 | 2.7 ± 0.1     | 2.89 ± 0.09 | 3.1 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 5.3 ± 0.3     | 2.91 ± 0.07 | 2.9 ± 0.1     | 3.1 ± 0.1   | 3.0 ± 0.1    | 3.9 ± 0.2     | 4.2 ± 0.3     | 3.02 ± 0.08 | 2.8 ± 0.2     | 2.97 ± 0.09 | 3.1 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 5.7 ± 0.4     | 2.91 ± 0.07 | 3.0 ± 0.1     | 3.0 ± 0.1   | 2.9 ± 0.1    | 4.1 ± 0.2     | 4.4 ± 0.3     | 2.94 ± 0.08 | 2.9 ± 0.2     | 3.0 ± 0.1   | 3.1 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 5.5 ± 0.4     | 2.85 ± 0.07 | 3.0 ± 0.2     | 3.1 ± 0.1   | 2.9 ± 0.1    | 4.1 ± 0.2     | 4.7 ± 0.3     | 2.98 ± 0.09 | 2.8 ± 0.2     | 3.3 ± 0.1   | 3.3 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 5.4 ± 0.4     | 2.73 ± 0.07 | 3.0 ± 0.1     | 3.1 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.3     | 4.6 ± 0.3     | 2.88 ± 0.08 | 3.0 ± 0.2     | 3.2 ± 0.1   | 3.2 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Asia F     |
| 49  | 5.8 ± 0.4     | 2.83 ± 0.08 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.1    | 4.2 ± 0.3     | 4.5 ± 0.3     | 3.04 ± 0.09 | 3.0 ± 0.2     | 3.2 ± 0.1   | 3.3 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia F     |
| 50  | 5.2 ± 0.4     | 2.84 ± 0.07 | 3.0 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.1    | 4.1 ± 0.3     | 4.4 ± 0.3     | 3.02 ± 0.09 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.6 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Asia F     |
| 51  | 5.5 ± 0.4     | 2.83 ± 0.08 | 3.1 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.1    | 4.3 ± 0.3     | 4.4 ± 0.4     | 3.04 ± 0.09 | 3.2 ± 0.2     | 3.6 ± 0.1   | 3.5 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Asia F     |
| 52  | 5.4 ± 0.4     | 2.78 ± 0.07 | 3.2 ± 0.2     | 3.3 ± 0.1   | 3.3 ± 0.1    | 4.0 ± 0.3     | 4.6 ± 0.3     | 3.06 ± 0.09 | 3.4 ± 0.2     | 3.5 ± 0.1   | 3.5 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Asia F     |

**Table 88.** Mean absolute error for Application-type images for algorithm innovatrics-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.3 ± 0.2     | 3.20 ± 0.08 | 3.0 ± 0.2     | 3.9 ± 0.1   | 3.0 ± 0.1    | 5.8 ± 0.3     | 3.7 ± 0.2     | 2.36 ± 0.07 | 2.1 ± 0.1     | 2.69 ± 0.08 | 2.2 ± 0.1    | 3.5 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 5.3 ± 0.2     | 3.05 ± 0.09 | 3.0 ± 0.2     | 3.8 ± 0.2   | 2.8 ± 0.2    | 5.7 ± 0.3     | 3.5 ± 0.2     | 2.27 ± 0.07 | 1.8 ± 0.1     | 2.59 ± 0.09 | 2.1 ± 0.1    | 3.6 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 5.0 ± 0.3     | 2.9 ± 0.1   | 2.9 ± 0.2     | 3.5 ± 0.1   | 2.7 ± 0.2    | 5.8 ± 0.4     | 3.8 ± 0.2     | 2.12 ± 0.07 | 2.2 ± 0.2     | 2.50 ± 0.10 | 1.8 ± 0.1    | 3.5 ± 0.2     | W Africa F   | S E Asia M   |
| 17  | 5.3 ± 0.3     | 2.6 ± 0.1   | 2.8 ± 0.2     | 3.4 ± 0.2   | 2.4 ± 0.2    | 5.3 ± 0.3     | 3.4 ± 0.2     | 2.00 ± 0.09 | 1.9 ± 0.1     | 2.5 ± 0.1   | 1.7 ± 0.1    | 3.9 ± 0.3     | E Africa F   | S E Asia M   |
| 18  | 4.7 ± 0.2     | 2.37 ± 0.08 | 2.6 ± 0.2     | 3.1 ± 0.1   | 2.1 ± 0.1    | 5.0 ± 0.3     | 3.5 ± 0.2     | 1.93 ± 0.07 | 2.1 ± 0.1     | 2.62 ± 0.09 | 1.6 ± 0.1    | 3.6 ± 0.2     | W Africa F   | S E Asia M   |
| 19  | 4.6 ± 0.2     | 2.28 ± 0.08 | 2.4 ± 0.2     | 3.1 ± 0.1   | 2.1 ± 0.1    | 4.9 ± 0.2     | 3.5 ± 0.2     | 1.95 ± 0.07 | 2.1 ± 0.1     | 2.68 ± 0.10 | 1.6 ± 0.1    | 3.6 ± 0.3     | W Africa F   | S E Asia M   |
| 20  | 4.4 ± 0.2     | 2.22 ± 0.09 | 2.1 ± 0.1     | 2.9 ± 0.1   | 2.02 ± 0.10  | 4.7 ± 0.2     | 3.4 ± 0.2     | 1.85 ± 0.07 | 2.0 ± 0.2     | 2.73 ± 0.10 | 1.6 ± 0.1    | 3.6 ± 0.3     | W Africa F   | S E Asia M   |
| 21  | 4.1 ± 0.2     | 2.22 ± 0.07 | 2.1 ± 0.1     | 2.83 ± 0.08 | 2.03 ± 0.09  | 4.2 ± 0.2     | 3.4 ± 0.2     | 1.81 ± 0.06 | 2.0 ± 0.1     | 2.69 ± 0.10 | 1.80 ± 0.10  | 3.6 ± 0.3     | W Africa F   | S E Asia M   |
| 22  | 4.0 ± 0.2     | 2.13 ± 0.06 | 2.07 ± 0.09   | 2.56 ± 0.07 | 2.05 ± 0.08  | 4.0 ± 0.2     | 3.8 ± 0.2     | 1.82 ± 0.06 | 2.1 ± 0.1     | 2.70 ± 0.10 | 1.79 ± 0.08  | 3.3 ± 0.2     | E Africa F   | S E Asia M   |
| 23  | 3.7 ± 0.2     | 2.04 ± 0.05 | 2.04 ± 0.08   | 2.48 ± 0.07 | 2.04 ± 0.07  | 3.8 ± 0.2     | 3.6 ± 0.2     | 1.86 ± 0.06 | 1.9 ± 0.1     | 2.76 ± 0.09 | 1.84 ± 0.09  | 3.5 ± 0.2     | W Africa F   | S E Asia M   |
| 24  | 3.6 ± 0.2     | 2.05 ± 0.05 | 2.05 ± 0.07   | 2.30 ± 0.06 | 2.08 ± 0.06  | 3.5 ± 0.2     | 3.6 ± 0.2     | 1.89 ± 0.06 | 1.88 ± 0.09   | 2.65 ± 0.07 | 1.84 ± 0.09  | 3.3 ± 0.2     | E Africa F   | S E Asia M   |
| 25  | 3.4 ± 0.2     | 2.07 ± 0.05 | 2.06 ± 0.06   | 2.18 ± 0.05 | 2.16 ± 0.06  | 3.5 ± 0.2     | 3.5 ± 0.2     | 1.96 ± 0.06 | 1.86 ± 0.09   | 2.53 ± 0.07 | 2.01 ± 0.09  | 3.3 ± 0.2     | E Africa M   | E Europe M   |
| 26  | 3.2 ± 0.1     | 2.16 ± 0.05 | 2.17 ± 0.06   | 2.18 ± 0.05 | 2.26 ± 0.06  | 3.3 ± 0.2     | 3.4 ± 0.2     | 1.93 ± 0.06 | 1.91 ± 0.08   | 2.42 ± 0.07 | 2.15 ± 0.09  | 3.1 ± 0.1     | E Africa M   | E Europe M   |
| 27  | 3.3 ± 0.2     | 2.21 ± 0.05 | 2.26 ± 0.07   | 2.28 ± 0.05 | 2.43 ± 0.07  | 3.3 ± 0.2     | 3.3 ± 0.2     | 2.08 ± 0.06 | 1.93 ± 0.08   | 2.34 ± 0.06 | 2.25 ± 0.09  | 3.1 ± 0.1     | W Africa F   | E Europe M   |
| 28  | 3.3 ± 0.2     | 2.40 ± 0.05 | 2.28 ± 0.06   | 2.44 ± 0.05 | 2.64 ± 0.07  | 3.2 ± 0.1     | 3.2 ± 0.1     | 2.19 ± 0.07 | 1.92 ± 0.08   | 2.25 ± 0.06 | 2.31 ± 0.10  | 3.0 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 3.3 ± 0.1     | 2.63 ± 0.05 | 2.38 ± 0.07   | 2.59 ± 0.05 | 2.88 ± 0.08  | 3.2 ± 0.1     | 3.2 ± 0.1     | 2.30 ± 0.06 | 1.97 ± 0.08   | 2.23 ± 0.05 | 2.5 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 3.3 ± 0.1     | 2.86 ± 0.05 | 2.50 ± 0.07   | 2.80 ± 0.05 | 2.98 ± 0.08  | 3.2 ± 0.1     | 3.3 ± 0.1     | 2.54 ± 0.06 | 1.99 ± 0.08   | 2.31 ± 0.05 | 2.7 ± 0.1    | 3.1 ± 0.1     | E Africa M   | E Europe M   |
| 31  | 3.3 ± 0.1     | 3.11 ± 0.05 | 2.61 ± 0.08   | 3.10 ± 0.06 | 3.21 ± 0.08  | 3.2 ± 0.1     | 3.4 ± 0.2     | 2.81 ± 0.07 | 2.15 ± 0.08   | 2.40 ± 0.05 | 3.0 ± 0.1    | 3.1 ± 0.1     | E Africa M   | E Europe M   |
| 32  | 3.4 ± 0.1     | 3.27 ± 0.06 | 2.66 ± 0.08   | 3.30 ± 0.06 | 3.44 ± 0.09  | 3.3 ± 0.1     | 3.3 ± 0.1     | 2.90 ± 0.07 | 2.30 ± 0.09   | 2.54 ± 0.05 | 3.0 ± 0.1    | 3.0 ± 0.1     | S E Asia F   | E Europe M   |
| 33  | 3.5 ± 0.1     | 3.47 ± 0.06 | 2.84 ± 0.09   | 3.47 ± 0.06 | 3.52 ± 0.10  | 3.1 ± 0.1     | 3.5 ± 0.2     | 3.11 ± 0.07 | 2.28 ± 0.09   | 2.64 ± 0.05 | 3.1 ± 0.1    | 3.0 ± 0.1     | S E Asia F   | E Europe M   |
| 34  | 3.6 ± 0.2     | 3.57 ± 0.06 | 2.97 ± 0.09   | 3.63 ± 0.07 | 3.6 ± 0.1    | 3.2 ± 0.1     | 3.4 ± 0.2     | 3.18 ± 0.08 | 2.38 ± 0.09   | 2.80 ± 0.05 | 3.2 ± 0.1    | 3.0 ± 0.1     | S E Asia F   | E Europe M   |
| 35  | 3.7 ± 0.2     | 3.83 ± 0.07 | 3.1 ± 0.1     | 3.84 ± 0.07 | 3.8 ± 0.1    | 3.2 ± 0.1     | 3.5 ± 0.2     | 3.26 ± 0.08 | 2.37 ± 0.10   | 2.89 ± 0.06 | 3.4 ± 0.2    | 3.2 ± 0.1     | S Asia F     | E Europe M   |
| 36  | 3.8 ± 0.2     | 3.89 ± 0.07 | 3.0 ± 0.1     | 4.04 ± 0.08 | 3.8 ± 0.1    | 3.2 ± 0.1     | 3.6 ± 0.2     | 3.27 ± 0.08 | 2.4 ± 0.1     | 3.02 ± 0.06 | 3.2 ± 0.2    | 3.1 ± 0.1     | S Asia F     | E Europe M   |
| 37  | 3.8 ± 0.2     | 3.94 ± 0.07 | 3.2 ± 0.1     | 4.12 ± 0.08 | 3.9 ± 0.1    | 3.6 ± 0.2     | 3.6 ± 0.2     | 3.27 ± 0.08 | 2.5 ± 0.1     | 3.19 ± 0.06 | 3.3 ± 0.2    | 3.1 ± 0.1     | S Asia F     | E Europe M   |
| 38  | 3.8 ± 0.2     | 4.13 ± 0.07 | 3.1 ± 0.1     | 4.25 ± 0.09 | 3.9 ± 0.1    | 3.7 ± 0.2     | 3.6 ± 0.2     | 3.31 ± 0.09 | 2.4 ± 0.1     | 3.17 ± 0.07 | 3.3 ± 0.2    | 3.1 ± 0.1     | S Asia F     | E Europe M   |
| 39  | 3.8 ± 0.2     | 4.14 ± 0.08 | 3.1 ± 0.1     | 4.33 ± 0.10 | 3.9 ± 0.1    | 3.5 ± 0.2     | 3.6 ± 0.2     | 3.32 ± 0.09 | 2.6 ± 0.1     | 3.23 ± 0.07 | 3.2 ± 0.1    | 3.2 ± 0.1     | S Asia F     | E Europe M   |
| 40  | 3.9 ± 0.2     | 4.27 ± 0.08 | 3.1 ± 0.1     | 4.4 ± 0.1   | 4.1 ± 0.1    | 3.7 ± 0.2     | 3.7 ± 0.2     | 3.38 ± 0.09 | 2.7 ± 0.1     | 3.29 ± 0.07 | 3.2 ± 0.1    | 3.6 ± 0.2     | S Asia F     | E Europe M   |
| 41  | 4.2 ± 0.2     | 4.32 ± 0.08 | 3.2 ± 0.1     | 4.5 ± 0.1   | 4.1 ± 0.1    | 3.9 ± 0.2     | 3.8 ± 0.2     | 3.38 ± 0.09 | 2.7 ± 0.1     | 3.34 ± 0.08 | 3.4 ± 0.2    | 3.8 ± 0.2     | S Asia F     | E Europe M   |
| 42  | 4.0 ± 0.2     | 4.32 ± 0.08 | 3.3 ± 0.1     | 4.3 ± 0.1   | 4.1 ± 0.1    | 4.0 ± 0.2     | 3.8 ± 0.2     | 3.32 ± 0.09 | 2.6 ± 0.1     | 3.32 ± 0.08 | 3.2 ± 0.1    | 3.6 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 4.0 ± 0.2     | 4.51 ± 0.08 | 3.3 ± 0.1     | 4.5 ± 0.1   | 4.2 ± 0.1    | 4.0 ± 0.2     | 3.9 ± 0.2     | 3.41 ± 0.09 | 2.6 ± 0.1     | 3.26 ± 0.09 | 3.3 ± 0.2    | 3.9 ± 0.2     | E Asia F     | E Europe M   |
| 44  | 4.2 ± 0.3     | 4.68 ± 0.08 | 3.4 ± 0.1     | 4.4 ± 0.1   | 4.2 ± 0.1    | 4.4 ± 0.2     | 3.7 ± 0.2     | 3.41 ± 0.09 | 2.6 ± 0.1     | 3.31 ± 0.10 | 3.3 ± 0.2    | 3.8 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 4.1 ± 0.3     | 4.70 ± 0.08 | 3.4 ± 0.1     | 4.4 ± 0.1   | 4.2 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 3.55 ± 0.09 | 2.6 ± 0.1     | 3.24 ± 0.10 | 3.3 ± 0.2    | 4.0 ± 0.2     | E Asia F     | E Europe M   |
| 46  | 4.2 ± 0.3     | 4.92 ± 0.09 | 3.4 ± 0.2     | 4.1 ± 0.1   | 4.4 ± 0.2    | 4.3 ± 0.2     | 3.8 ± 0.2     | 3.52 ± 0.09 | 2.9 ± 0.1     | 3.1 ± 0.1   | 3.4 ± 0.2    | 4.0 ± 0.2     | E Asia F     | E Europe M   |
| 47  | 4.1 ± 0.3     | 4.79 ± 0.09 | 3.5 ± 0.2     | 4.1 ± 0.1   | 4.2 ± 0.2    | 4.4 ± 0.2     | 3.7 ± 0.3     | 3.52 ± 0.10 | 2.8 ± 0.1     | 3.3 ± 0.1   | 3.2 ± 0.2    | 4.1 ± 0.2     | E Asia F     | E Europe M   |
| 48  | 4.1 ± 0.3     | 4.85 ± 0.09 | 3.4 ± 0.2     | 4.1 ± 0.2   | 4.3 ± 0.2    | 4.4 ± 0.3     | 3.7 ± 0.3     | 3.51 ± 0.09 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.2 ± 0.2    | 4.1 ± 0.2     | E Asia F     | E Europe M   |
| 49  | 3.9 ± 0.3     | 5.0 ± 0.1   | 3.7 ± 0.2     | 3.8 ± 0.1   | 4.1 ± 0.2    | 4.7 ± 0.3     | 3.6 ± 0.3     | 3.52 ± 0.10 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 3.6 ± 0.2     | E Asia F     | E Europe M   |
| 50  | 3.9 ± 0.3     | 4.97 ± 0.10 | 3.5 ± 0.2     | 3.8 ± 0.1   | 4.2 ± 0.2    | 4.5 ± 0.3     | 3.4 ± 0.3     | 3.45 ± 0.09 | 2.8 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 51  | 3.7 ± 0.3     | 5.0 ± 0.1   | 3.7 ± 0.2     | 3.8 ± 0.1   | 4.1 ± 0.2    | 4.5 ± 0.3     | 3.5 ± 0.3     | 3.45 ± 0.10 | 2.8 ± 0.1     | 3.3 ± 0.1   | 3.0 ± 0.2    | 3.9 ± 0.2     | E Asia F     | E Europe M   |
| 52  | 3.9 ± 0.3     | 5.0 ± 0.1   | 3.6 ± 0.2     | 3.8 ± 0.1   | 3.8 ± 0.2    | 4.6 ± 0.3     | 3.2 ± 0.3     | 3.5 ± 0.1   | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.2 ± 0.2    | 3.8 ± 0.3     | E Asia F     | E Europe M   |

**Table 89.** Mean absolute error for Application-type images for algorithm innovatrics-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.2 ± 0.2     | 3.05 ± 0.07 | 3.0 ± 0.2     | 3.7 ± 0.1   | 3.0 ± 0.1    | 5.9 ± 0.3     | 3.7 ± 0.2     | 2.34 ± 0.07 | 2.0 ± 0.1     | 2.62 ± 0.07 | 2.2 ± 0.1    | 3.5 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 5.2 ± 0.2     | 2.83 ± 0.09 | 3.0 ± 0.2     | 3.6 ± 0.1   | 2.7 ± 0.2    | 5.8 ± 0.3     | 3.4 ± 0.2     | 2.20 ± 0.07 | 1.9 ± 0.1     | 2.51 ± 0.09 | 2.03 ± 0.10  | 3.7 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.9 ± 0.3     | 2.6 ± 0.1   | 2.9 ± 0.2     | 3.2 ± 0.1   | 2.5 ± 0.2    | 5.8 ± 0.4     | 3.8 ± 0.2     | 2.00 ± 0.07 | 2.1 ± 0.2     | 2.42 ± 0.10 | 1.8 ± 0.1    | 3.5 ± 0.2     | W Africa F   | S E Asia M   |
| 17  | 5.1 ± 0.3     | 2.4 ± 0.1   | 2.8 ± 0.2     | 3.1 ± 0.1   | 2.2 ± 0.2    | 5.5 ± 0.3     | 3.5 ± 0.2     | 1.82 ± 0.08 | 1.7 ± 0.1     | 2.4 ± 0.1   | 1.6 ± 0.1    | 3.8 ± 0.4     | W Africa F   | S E Asia M   |
| 18  | 4.5 ± 0.2     | 2.19 ± 0.08 | 2.5 ± 0.2     | 2.84 ± 0.10 | 2.0 ± 0.1    | 5.0 ± 0.3     | 3.4 ± 0.2     | 1.71 ± 0.06 | 1.9 ± 0.1     | 2.41 ± 0.09 | 1.5 ± 0.2    | 3.6 ± 0.2     | W Africa F   | S E Asia M   |
| 19  | 4.4 ± 0.2     | 2.13 ± 0.08 | 2.2 ± 0.2     | 2.7 ± 0.1   | 1.97 ± 0.10  | 4.9 ± 0.2     | 3.5 ± 0.2     | 1.69 ± 0.07 | 1.8 ± 0.1     | 2.39 ± 0.09 | 1.4 ± 0.1    | 3.7 ± 0.3     | W Africa F   | S E Asia M   |
| 20  | 4.2 ± 0.2     | 2.05 ± 0.08 | 2.0 ± 0.1     | 2.6 ± 0.1   | 1.91 ± 0.09  | 4.7 ± 0.2     | 3.3 ± 0.2     | 1.61 ± 0.07 | 1.8 ± 0.1     | 2.45 ± 0.09 | 1.50 ± 0.09  | 3.5 ± 0.3     | W Africa F   | S E Asia M   |
| 21  | 3.8 ± 0.2     | 2.04 ± 0.07 | 1.87 ± 0.10   | 2.51 ± 0.08 | 1.88 ± 0.08  | 4.2 ± 0.2     | 3.3 ± 0.2     | 1.62 ± 0.05 | 1.8 ± 0.1     | 2.42 ± 0.09 | 1.69 ± 0.09  | 3.5 ± 0.3     | W Africa F   | E Asia M     |
| 22  | 3.8 ± 0.2     | 1.94 ± 0.06 | 1.93 ± 0.08   | 2.30 ± 0.07 | 1.93 ± 0.08  | 4.0 ± 0.2     | 3.6 ± 0.2     | 1.67 ± 0.06 | 1.8 ± 0.1     | 2.43 ± 0.09 | 1.69 ± 0.08  | 3.2 ± 0.2     | W Africa F   | E Asia M     |
| 23  | 3.4 ± 0.2     | 1.91 ± 0.05 | 1.84 ± 0.08   | 2.27 ± 0.06 | 1.93 ± 0.07  | 3.7 ± 0.2     | 3.4 ± 0.2     | 1.80 ± 0.06 | 1.71 ± 0.10   | 2.54 ± 0.08 | 1.82 ± 0.08  | 3.3 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 3.4 ± 0.2     | 1.97 ± 0.05 | 1.81 ± 0.06   | 2.17 ± 0.06 | 2.04 ± 0.06  | 3.4 ± 0.2     | 3.4 ± 0.2     | 1.84 ± 0.06 | 1.72 ± 0.08   | 2.46 ± 0.07 | 1.81 ± 0.08  | 3.2 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 3.2 ± 0.2     | 2.06 ± 0.05 | 1.94 ± 0.06   | 2.12 ± 0.05 | 2.16 ± 0.06  | 3.3 ± 0.2     | 3.3 ± 0.2     | 1.99 ± 0.06 | 1.74 ± 0.08   | 2.39 ± 0.07 | 2.01 ± 0.08  | 3.2 ± 0.2     | E Africa M   | E Europe M   |
| 26  | 3.1 ± 0.1     | 2.25 ± 0.05 | 2.06 ± 0.06   | 2.19 ± 0.05 | 2.32 ± 0.06  | 3.1 ± 0.2     | 3.2 ± 0.2     | 1.97 ± 0.06 | 1.78 ± 0.08   | 2.31 ± 0.06 | 2.15 ± 0.09  | 2.9 ± 0.1     | E Africa M   | E Europe M   |
| 27  | 3.1 ± 0.1     | 2.34 ± 0.05 | 2.23 ± 0.07   | 2.37 ± 0.05 | 2.55 ± 0.07  | 3.2 ± 0.1     | 3.1 ± 0.2     | 2.13 ± 0.06 | 1.80 ± 0.08   | 2.25 ± 0.05 | 2.25 ± 0.09  | 3.0 ± 0.1     | W Africa F   | E Europe M   |
| 28  | 3.1 ± 0.1     | 2.55 ± 0.05 | 2.32 ± 0.06   | 2.55 ± 0.05 | 2.73 ± 0.07  | 3.0 ± 0.1     | 3.0 ± 0.1     | 2.22 ± 0.07 | 1.79 ± 0.08   | 2.18 ± 0.05 | 2.35 ± 0.10  | 2.9 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 3.2 ± 0.1     | 2.79 ± 0.05 | 2.40 ± 0.07   | 2.71 ± 0.05 | 2.96 ± 0.08  | 3.0 ± 0.1     | 3.1 ± 0.1     | 2.35 ± 0.06 | 1.92 ± 0.08   | 2.15 ± 0.05 | 2.6 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 3.1 ± 0.1     | 3.06 ± 0.05 | 2.62 ± 0.07   | 2.98 ± 0.05 | 3.15 ± 0.08  | 3.1 ± 0.1     | 3.2 ± 0.1     | 2.67 ± 0.06 | 2.02 ± 0.08   | 2.23 ± 0.05 | 2.8 ± 0.1    | 3.0 ± 0.1     | E Africa M   | E Europe M   |
| 31  | 3.2 ± 0.1     | 3.38 ± 0.05 | 2.73 ± 0.08   | 3.30 ± 0.06 | 3.44 ± 0.09  | 3.0 ± 0.1     | 3.2 ± 0.1     | 2.96 ± 0.07 | 2.17 ± 0.08   | 2.32 ± 0.05 | 3.1 ± 0.1    | 3.0 ± 0.1     | S E Asia F   | E Europe M   |
| 32  | 3.3 ± 0.1     | 3.63 ± 0.06 | 2.85 ± 0.08   | 3.51 ± 0.06 | 3.68 ± 0.09  | 3.1 ± 0.1     | 3.2 ± 0.1     | 3.18 ± 0.07 | 2.29 ± 0.09   | 2.47 ± 0.05 | 3.2 ± 0.1    | 3.0 ± 0.1     | S E Asia F   | E Europe M   |
| 33  | 3.4 ± 0.1     | 3.92 ± 0.06 | 3.06 ± 0.09   | 3.72 ± 0.07 | 3.8 ± 0.1    | 3.0 ± 0.1     | 3.4 ± 0.2     | 3.45 ± 0.07 | 2.34 ± 0.09   | 2.60 ± 0.05 | 3.4 ± 0.1    | 3.0 ± 0.1     | E Asia F     | E Europe M   |
| 34  | 3.5 ± 0.1     | 4.09 ± 0.06 | 3.18 ± 0.09   | 3.87 ± 0.07 | 3.9 ± 0.1    | 3.0 ± 0.1     | 3.4 ± 0.2     | 3.65 ± 0.08 | 2.52 ± 0.10   | 2.81 ± 0.05 | 3.5 ± 0.2    | 3.0 ± 0.1     | E Asia F     | E Europe M   |
| 35  | 3.7 ± 0.2     | 4.38 ± 0.07 | 3.3 ± 0.1     | 4.09 ± 0.07 | 4.1 ± 0.1    | 3.1 ± 0.1     | 3.4 ± 0.2     | 3.73 ± 0.08 | 2.5 ± 0.1     | 2.96 ± 0.06 | 3.7 ± 0.2    | 3.2 ± 0.1     | E Asia F     | E Europe M   |
| 36  | 3.8 ± 0.2     | 4.45 ± 0.07 | 3.3 ± 0.1     | 4.31 ± 0.08 | 4.1 ± 0.1    | 3.0 ± 0.1     | 3.6 ± 0.2     | 3.78 ± 0.09 | 2.6 ± 0.1     | 3.12 ± 0.06 | 3.6 ± 0.2    | 3.1 ± 0.1     | E Asia F     | E Europe M   |
| 37  | 3.8 ± 0.2     | 4.54 ± 0.08 | 3.4 ± 0.1     | 4.46 ± 0.08 | 4.2 ± 0.1    | 3.4 ± 0.2     | 3.6 ± 0.2     | 3.86 ± 0.09 | 2.6 ± 0.1     | 3.30 ± 0.06 | 3.6 ± 0.2    | 3.1 ± 0.1     | E Asia F     | E Europe M   |
| 38  | 3.7 ± 0.2     | 4.71 ± 0.08 | 3.3 ± 0.1     | 4.59 ± 0.09 | 4.1 ± 0.1    | 3.6 ± 0.2     | 3.6 ± 0.2     | 3.88 ± 0.09 | 2.6 ± 0.1     | 3.32 ± 0.07 | 3.6 ± 0.2    | 3.1 ± 0.1     | E Asia F     | E Europe M   |
| 39  | 3.8 ± 0.2     | 4.76 ± 0.08 | 3.4 ± 0.1     | 4.82 ± 0.10 | 4.2 ± 0.1    | 3.4 ± 0.2     | 3.7 ± 0.2     | 3.96 ± 0.10 | 2.7 ± 0.1     | 3.41 ± 0.07 | 3.6 ± 0.2    | 3.2 ± 0.1     | S Asia F     | E Europe M   |
| 40  | 4.0 ± 0.2     | 4.90 ± 0.08 | 3.3 ± 0.1     | 4.8 ± 0.1   | 4.5 ± 0.1    | 3.8 ± 0.2     | 3.7 ± 0.2     | 4.00 ± 0.09 | 2.9 ± 0.1     | 3.49 ± 0.07 | 3.5 ± 0.2    | 3.4 ± 0.2     | E Asia F     | E Europe M   |
| 41  | 4.4 ± 0.2     | 5.02 ± 0.08 | 3.5 ± 0.1     | 5.0 ± 0.1   | 4.5 ± 0.1    | 4.1 ± 0.2     | 3.9 ± 0.2     | 4.02 ± 0.09 | 2.8 ± 0.1     | 3.55 ± 0.08 | 3.8 ± 0.2    | 3.7 ± 0.2     | S Asia F     | E Europe M   |
| 42  | 4.3 ± 0.3     | 5.06 ± 0.08 | 3.5 ± 0.1     | 4.9 ± 0.1   | 4.6 ± 0.1    | 4.2 ± 0.2     | 4.0 ± 0.2     | 4.01 ± 0.09 | 2.7 ± 0.1     | 3.61 ± 0.09 | 3.6 ± 0.2    | 3.6 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 4.3 ± 0.2     | 5.34 ± 0.08 | 3.6 ± 0.2     | 5.1 ± 0.1   | 4.8 ± 0.1    | 4.2 ± 0.2     | 4.0 ± 0.2     | 4.09 ± 0.09 | 2.8 ± 0.1     | 3.56 ± 0.09 | 3.7 ± 0.2    | 3.8 ± 0.2     | E Asia F     | E Europe M   |
| 44  | 4.8 ± 0.3     | 5.58 ± 0.08 | 3.8 ± 0.2     | 5.1 ± 0.1   | 4.7 ± 0.1    | 4.6 ± 0.2     | 3.9 ± 0.2     | 4.16 ± 0.09 | 2.7 ± 0.1     | 3.6 ± 0.1   | 3.7 ± 0.2    | 3.8 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 4.7 ± 0.3     | 5.70 ± 0.09 | 3.8 ± 0.2     | 5.1 ± 0.2   | 4.9 ± 0.1    | 4.6 ± 0.2     | 4.2 ± 0.2     | 4.27 ± 0.09 | 2.7 ± 0.1     | 3.6 ± 0.1   | 3.7 ± 0.2    | 4.1 ± 0.2     | E Asia F     | E Europe M   |
| 46  | 5.1 ± 0.4     | 5.96 ± 0.09 | 3.7 ± 0.2     | 4.9 ± 0.1   | 5.1 ± 0.2    | 4.7 ± 0.2     | 4.3 ± 0.3     | 4.32 ± 0.10 | 3.0 ± 0.2     | 3.5 ± 0.1   | 3.9 ± 0.2    | 4.2 ± 0.2     | E Asia F     | E Europe M   |
| 47  | 5.0 ± 0.4     | 5.88 ± 0.09 | 3.9 ± 0.2     | 4.9 ± 0.2   | 5.0 ± 0.2    | 4.7 ± 0.2     | 4.5 ± 0.3     | 4.3 ± 0.1   | 2.9 ± 0.1     | 3.8 ± 0.1   | 3.8 ± 0.2    | 4.3 ± 0.2     | E Asia F     | E Europe M   |
| 48  | 4.9 ± 0.4     | 6.04 ± 0.10 | 3.9 ± 0.2     | 4.9 ± 0.2   | 5.2 ± 0.2    | 5.0 ± 0.3     | 4.4 ± 0.3     | 4.4 ± 0.1   | 3.0 ± 0.2     | 3.6 ± 0.1   | 3.8 ± 0.2    | 4.4 ± 0.2     | E Asia F     | E Europe M   |
| 49  | 4.9 ± 0.4     | 6.2 ± 0.1   | 4.1 ± 0.2     | 4.7 ± 0.2   | 5.1 ± 0.2    | 5.2 ± 0.3     | 4.5 ± 0.3     | 4.4 ± 0.1   | 3.2 ± 0.2     | 3.7 ± 0.1   | 3.7 ± 0.2    | 4.0 ± 0.2     | E Asia F     | E Europe M   |
| 50  | 5.1 ± 0.4     | 6.2 ± 0.1   | 3.9 ± 0.2     | 4.7 ± 0.2   | 5.2 ± 0.2    | 5.2 ± 0.3     | 4.2 ± 0.3     | 4.3 ± 0.1   | 3.1 ± 0.2     | 3.8 ± 0.1   | 3.8 ± 0.2    | 4.3 ± 0.2     | E Asia F     | E Europe M   |
| 51  | 5.2 ± 0.4     | 6.1 ± 0.1   | 4.2 ± 0.2     | 4.7 ± 0.2   | 5.2 ± 0.2    | 5.3 ± 0.3     | 4.4 ± 0.4     | 4.3 ± 0.1   | 3.1 ± 0.2     | 3.8 ± 0.1   | 3.8 ± 0.2    | 4.4 ± 0.3     | E Asia F     | E Europe M   |
| 52  | 5.4 ± 0.5     | 6.1 ± 0.1   | 4.1 ± 0.2     | 4.7 ± 0.2   | 4.9 ± 0.2    | 5.2 ± 0.3     | 4.3 ± 0.4     | 4.3 ± 0.1   | 3.4 ± 0.2     | 3.7 ± 0.1   | 3.9 ± 0.2    | 4.5 ± 0.3     | E Asia F     | E Europe M   |

**Table 90.** Mean absolute error for Application-type images for algorithm innovatrics-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.2 ± 0.3     | 2.11 ± 0.07 | 2.0 ± 0.1     | 2.5 ± 0.1   | 1.9 ± 0.1    | 5.4 ± 0.4     | 2.4 ± 0.2     | 1.48 ± 0.07 | 1.06 ± 0.07   | 1.56 ± 0.06 | 1.29 ± 0.10  | 2.5 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 4.4 ± 0.3     | 2.0 ± 0.1   | 2.0 ± 0.2     | 2.5 ± 0.2   | 1.7 ± 0.2    | 5.5 ± 0.4     | 2.3 ± 0.2     | 1.34 ± 0.07 | 0.87 ± 0.08   | 1.39 ± 0.09 | 1.07 ± 0.09  | 2.8 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.4 ± 0.3     | 2.0 ± 0.1   | 2.2 ± 0.2     | 2.4 ± 0.2   | 1.8 ± 0.2    | 5.8 ± 0.5     | 3.0 ± 0.3     | 1.31 ± 0.07 | 1.2 ± 0.1     | 1.5 ± 0.1   | 1.0 ± 0.1    | 2.9 ± 0.3     | W Africa F   | S E Asia M   |
| 17  | 5.0 ± 0.3     | 2.2 ± 0.1   | 2.3 ± 0.2     | 2.6 ± 0.2   | 1.9 ± 0.2    | 5.8 ± 0.3     | 3.1 ± 0.2     | 1.55 ± 0.09 | 1.3 ± 0.1     | 1.9 ± 0.1   | 1.4 ± 0.1    | 3.6 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.6 ± 0.2     | 2.27 ± 0.08 | 2.4 ± 0.2     | 2.7 ± 0.1   | 2.1 ± 0.1    | 5.3 ± 0.3     | 3.4 ± 0.2     | 1.87 ± 0.07 | 1.8 ± 0.1     | 2.46 ± 0.10 | 1.8 ± 0.1    | 3.6 ± 0.3     | W Africa F   | S E Asia M   |
| 19  | 4.7 ± 0.3     | 2.44 ± 0.08 | 2.5 ± 0.2     | 2.8 ± 0.1   | 2.4 ± 0.1    | 5.6 ± 0.3     | 3.7 ± 0.2     | 2.16 ± 0.07 | 2.0 ± 0.1     | 2.72 ± 0.10 | 2.0 ± 0.1    | 4.0 ± 0.3     | W Africa F   | S E Asia M   |
| 20  | 4.6 ± 0.2     | 2.56 ± 0.09 | 2.5 ± 0.1     | 2.9 ± 0.1   | 2.40 ± 0.10  | 5.4 ± 0.3     | 3.8 ± 0.2     | 2.18 ± 0.07 | 2.1 ± 0.1     | 2.9 ± 0.1   | 2.2 ± 0.1    | 3.9 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 4.3 ± 0.2     | 2.62 ± 0.07 | 2.5 ± 0.1     | 2.88 ± 0.08 | 2.49 ± 0.09  | 5.0 ± 0.3     | 3.8 ± 0.2     | 2.25 ± 0.07 | 2.2 ± 0.1     | 2.91 ± 0.10 | 2.4 ± 0.1    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.3 ± 0.2     | 2.52 ± 0.07 | 2.56 ± 0.10   | 2.68 ± 0.07 | 2.52 ± 0.09  | 4.8 ± 0.2     | 4.3 ± 0.2     | 2.25 ± 0.07 | 2.3 ± 0.1     | 2.91 ± 0.09 | 2.2 ± 0.1    | 4.0 ± 0.2     | W Africa F   | S E Asia M   |
| 23  | 3.9 ± 0.2     | 2.39 ± 0.06 | 2.44 ± 0.10   | 2.60 ± 0.07 | 2.46 ± 0.08  | 4.6 ± 0.2     | 4.1 ± 0.2     | 2.24 ± 0.07 | 2.1 ± 0.1     | 2.97 ± 0.09 | 2.3 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 3.9 ± 0.2     | 2.35 ± 0.06 | 2.31 ± 0.08   | 2.44 ± 0.06 | 2.45 ± 0.07  | 4.2 ± 0.2     | 4.1 ± 0.2     | 2.19 ± 0.08 | 2.07 ± 0.10   | 2.83 ± 0.08 | 2.2 ± 0.1    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 3.7 ± 0.2     | 2.32 ± 0.06 | 2.41 ± 0.08   | 2.38 ± 0.06 | 2.52 ± 0.08  | 4.1 ± 0.2     | 4.0 ± 0.2     | 2.24 ± 0.07 | 2.0 ± 0.1     | 2.62 ± 0.08 | 2.3 ± 0.1    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 3.5 ± 0.2     | 2.40 ± 0.06 | 2.42 ± 0.08   | 2.36 ± 0.05 | 2.59 ± 0.08  | 3.9 ± 0.2     | 4.0 ± 0.2     | 2.14 ± 0.07 | 2.12 ± 0.10   | 2.53 ± 0.07 | 2.4 ± 0.1    | 3.6 ± 0.2     | E Africa M   | E Europe M   |
| 27  | 3.5 ± 0.2     | 2.43 ± 0.05 | 2.47 ± 0.08   | 2.60 ± 0.05 | 2.74 ± 0.08  | 3.8 ± 0.2     | 3.9 ± 0.2     | 2.28 ± 0.07 | 2.14 ± 0.09   | 2.52 ± 0.06 | 2.5 ± 0.1    | 3.6 ± 0.2     | E Africa M   | E Europe M   |
| 28  | 3.5 ± 0.2     | 2.64 ± 0.05 | 2.58 ± 0.07   | 2.83 ± 0.05 | 2.90 ± 0.08  | 3.6 ± 0.2     | 3.7 ± 0.2     | 2.37 ± 0.07 | 2.17 ± 0.09   | 2.43 ± 0.06 | 2.5 ± 0.1    | 3.5 ± 0.1     | E Africa M   | E Europe M   |
| 29  | 3.4 ± 0.1     | 2.78 ± 0.05 | 2.60 ± 0.07   | 3.03 ± 0.06 | 3.07 ± 0.08  | 3.4 ± 0.1     | 3.7 ± 0.2     | 2.43 ± 0.07 | 2.16 ± 0.09   | 2.41 ± 0.06 | 2.6 ± 0.1    | 3.5 ± 0.1     | E Africa M   | E Europe M   |
| 30  | 3.3 ± 0.1     | 2.99 ± 0.06 | 2.76 ± 0.08   | 3.27 ± 0.06 | 3.14 ± 0.09  | 3.3 ± 0.1     | 3.7 ± 0.2     | 2.53 ± 0.07 | 2.17 ± 0.09   | 2.44 ± 0.05 | 2.7 ± 0.1    | 3.3 ± 0.1     | E Africa M   | E Europe M   |
| 31  | 3.4 ± 0.1     | 3.19 ± 0.06 | 2.87 ± 0.09   | 3.55 ± 0.06 | 3.35 ± 0.09  | 3.2 ± 0.1     | 3.7 ± 0.2     | 2.64 ± 0.07 | 2.24 ± 0.09   | 2.52 ± 0.05 | 2.9 ± 0.1    | 3.3 ± 0.1     | E Africa M   | E Europe M   |
| 32  | 3.4 ± 0.2     | 3.35 ± 0.06 | 2.91 ± 0.09   | 3.76 ± 0.07 | 3.5 ± 0.1    | 3.2 ± 0.1     | 3.5 ± 0.2     | 2.74 ± 0.07 | 2.37 ± 0.10   | 2.65 ± 0.05 | 2.9 ± 0.1    | 3.2 ± 0.1     | S Asia F     | E Europe M   |
| 33  | 3.5 ± 0.1     | 3.54 ± 0.07 | 3.02 ± 0.10   | 4.03 ± 0.07 | 3.6 ± 0.1    | 3.1 ± 0.1     | 3.7 ± 0.2     | 2.89 ± 0.08 | 2.40 ± 0.10   | 2.74 ± 0.05 | 3.0 ± 0.1    | 3.1 ± 0.1     | S Asia F     | E Europe M   |
| 34  | 3.6 ± 0.2     | 3.68 ± 0.07 | 3.2 ± 0.1     | 4.20 ± 0.08 | 3.7 ± 0.1    | 3.1 ± 0.1     | 3.6 ± 0.2     | 3.00 ± 0.08 | 2.5 ± 0.1     | 2.94 ± 0.06 | 3.1 ± 0.1    | 3.1 ± 0.1     | S Asia F     | E Europe M   |
| 35  | 3.8 ± 0.2     | 4.01 ± 0.07 | 3.3 ± 0.1     | 4.45 ± 0.08 | 3.9 ± 0.1    | 3.2 ± 0.1     | 3.8 ± 0.2     | 3.08 ± 0.08 | 2.6 ± 0.1     | 3.06 ± 0.06 | 3.3 ± 0.1    | 3.5 ± 0.1     | S Asia F     | E Europe M   |
| 36  | 4.0 ± 0.2     | 4.09 ± 0.07 | 3.4 ± 0.1     | 4.62 ± 0.08 | 4.0 ± 0.1    | 3.1 ± 0.1     | 3.9 ± 0.2     | 3.18 ± 0.08 | 2.7 ± 0.1     | 3.24 ± 0.06 | 3.3 ± 0.2    | 3.4 ± 0.1     | S Asia F     | E Europe M   |
| 37  | 3.9 ± 0.2     | 4.26 ± 0.07 | 3.4 ± 0.1     | 4.80 ± 0.09 | 4.2 ± 0.1    | 3.6 ± 0.2     | 3.9 ± 0.2     | 3.26 ± 0.08 | 2.8 ± 0.1     | 3.41 ± 0.07 | 3.2 ± 0.1    | 3.4 ± 0.1     | S Asia F     | E Europe M   |
| 38  | 3.9 ± 0.2     | 4.50 ± 0.08 | 3.4 ± 0.1     | 4.9 ± 0.1   | 4.1 ± 0.1    | 3.7 ± 0.2     | 4.0 ± 0.2     | 3.29 ± 0.08 | 2.8 ± 0.1     | 3.41 ± 0.07 | 3.3 ± 0.2    | 3.4 ± 0.1     | S Asia F     | E Europe M   |
| 39  | 3.9 ± 0.2     | 4.55 ± 0.08 | 3.3 ± 0.1     | 5.0 ± 0.1   | 4.2 ± 0.1    | 3.5 ± 0.2     | 3.9 ± 0.2     | 3.37 ± 0.09 | 2.9 ± 0.1     | 3.44 ± 0.07 | 3.3 ± 0.2    | 3.5 ± 0.1     | S Asia F     | E Europe M   |
| 40  | 4.1 ± 0.2     | 4.69 ± 0.08 | 3.3 ± 0.1     | 5.0 ± 0.1   | 4.4 ± 0.1    | 3.9 ± 0.2     | 4.2 ± 0.2     | 3.34 ± 0.09 | 3.0 ± 0.1     | 3.48 ± 0.08 | 3.3 ± 0.2    | 3.7 ± 0.2     | S Asia F     | E Europe M   |
| 41  | 4.3 ± 0.2     | 4.78 ± 0.08 | 3.3 ± 0.1     | 5.1 ± 0.1   | 4.4 ± 0.1    | 4.0 ± 0.2     | 4.2 ± 0.2     | 3.35 ± 0.09 | 2.9 ± 0.1     | 3.58 ± 0.08 | 3.5 ± 0.2    | 3.9 ± 0.2     | S Asia F     | E Europe M   |
| 42  | 4.3 ± 0.3     | 4.82 ± 0.09 | 3.3 ± 0.2     | 4.8 ± 0.1   | 4.3 ± 0.2    | 4.1 ± 0.2     | 4.3 ± 0.2     | 3.47 ± 0.09 | 2.9 ± 0.1     | 3.55 ± 0.09 | 3.5 ± 0.2    | 3.9 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 4.3 ± 0.3     | 4.96 ± 0.09 | 3.3 ± 0.1     | 5.0 ± 0.1   | 4.5 ± 0.2    | 4.1 ± 0.2     | 4.3 ± 0.3     | 3.60 ± 0.09 | 2.8 ± 0.1     | 3.54 ± 0.10 | 3.6 ± 0.2    | 4.0 ± 0.2     | S Asia F     | E Europe M   |
| 44  | 4.9 ± 0.3     | 5.14 ± 0.09 | 3.3 ± 0.2     | 4.9 ± 0.2   | 4.4 ± 0.2    | 4.4 ± 0.2     | 4.2 ± 0.3     | 3.63 ± 0.09 | 2.8 ± 0.1     | 3.7 ± 0.1   | 3.7 ± 0.2    | 4.0 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 4.5 ± 0.3     | 5.19 ± 0.09 | 3.4 ± 0.2     | 4.9 ± 0.2   | 4.4 ± 0.2    | 4.4 ± 0.2     | 4.5 ± 0.3     | 3.76 ± 0.09 | 2.9 ± 0.2     | 3.6 ± 0.1   | 3.8 ± 0.2    | 4.4 ± 0.2     | E Asia F     | E Europe M   |
| 46  | 5.1 ± 0.4     | 5.40 ± 0.10 | 3.4 ± 0.2     | 4.6 ± 0.2   | 4.8 ± 0.2    | 4.3 ± 0.2     | 4.6 ± 0.3     | 3.76 ± 0.09 | 3.2 ± 0.2     | 3.5 ± 0.1   | 3.8 ± 0.2    | 4.3 ± 0.2     | E Asia F     | E Europe M   |
| 47  | 5.0 ± 0.4     | 5.3 ± 0.1   | 3.5 ± 0.2     | 4.7 ± 0.2   | 4.5 ± 0.2    | 4.6 ± 0.2     | 5.0 ± 0.4     | 3.8 ± 0.1   | 3.0 ± 0.2     | 3.7 ± 0.1   | 3.8 ± 0.2    | 4.5 ± 0.2     | E Asia F     | E Europe M   |
| 48  | 4.9 ± 0.4     | 5.4 ± 0.1   | 3.6 ± 0.2     | 4.8 ± 0.2   | 4.7 ± 0.2    | 4.7 ± 0.3     | 4.9 ± 0.4     | 3.81 ± 0.10 | 3.3 ± 0.2     | 3.7 ± 0.1   | 3.9 ± 0.2    | 4.6 ± 0.3     | E Asia F     | E Europe M   |
| 49  | 4.9 ± 0.4     | 5.5 ± 0.1   | 4.0 ± 0.2     | 4.6 ± 0.2   | 4.6 ± 0.2    | 5.2 ± 0.3     | 4.9 ± 0.4     | 3.9 ± 0.1   | 3.4 ± 0.2     | 3.8 ± 0.1   | 3.6 ± 0.2    | 4.3 ± 0.2     | E Asia F     | E Europe M   |
| 50  | 5.3 ± 0.4     | 5.5 ± 0.1   | 3.9 ± 0.2     | 4.6 ± 0.2   | 4.7 ± 0.2    | 5.0 ± 0.3     | 4.8 ± 0.4     | 4.0 ± 0.1   | 3.5 ± 0.2     | 4.0 ± 0.1   | 3.9 ± 0.2    | 4.5 ± 0.2     | E Asia F     | E Europe M   |
| 51  | 5.5 ± 0.4     | 5.6 ± 0.1   | 4.2 ± 0.2     | 4.6 ± 0.2   | 4.9 ± 0.2    | 5.2 ± 0.3     | 5.1 ± 0.4     | 4.0 ± 0.1   | 3.4 ± 0.2     | 4.2 ± 0.2   | 3.8 ± 0.2    | 4.8 ± 0.3     | E Asia F     | E Europe M   |
| 52  | 5.7 ± 0.4     | 5.6 ± 0.1   | 4.2 ± 0.2     | 4.9 ± 0.2   | 4.7 ± 0.2    | 5.1 ± 0.3     | 5.0 ± 0.4     | 3.9 ± 0.1   | 3.8 ± 0.2     | 4.1 ± 0.1   | 4.0 ± 0.2    | 5.0 ± 0.3     | E Africa F   | E Europe M   |



**Table 91.** Mean absolute error for Application-type images for algorithm jumio-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.6 ± 0.3     | 4.6 ± 0.1   | 4.7 ± 0.3     | 4.8 ± 0.1   | 4.9 ± 0.3    | 6.0 ± 0.3     | 4.3 ± 0.2     | 3.23 ± 0.10 | 4.0 ± 0.4     | 3.1 ± 0.1   | 3.2 ± 0.2    | 3.8 ± 0.3     | W Africa F   | S Asia M     |
| 15  | 5.9 ± 0.3     | 4.5 ± 0.1   | 4.7 ± 0.3     | 4.9 ± 0.2   | 4.9 ± 0.3    | 6.0 ± 0.4     | 3.4 ± 0.2     | 3.09 ± 0.08 | 4.0 ± 0.5     | 2.7 ± 0.1   | 3.1 ± 0.1    | 3.5 ± 0.3     | W Africa F   | S Asia M     |
| 16  | 5.4 ± 0.4     | 4.2 ± 0.1   | 4.7 ± 0.3     | 4.7 ± 0.2   | 4.9 ± 0.3    | 6.2 ± 0.5     | 3.1 ± 0.2     | 2.73 ± 0.08 | 4.0 ± 0.5     | 2.39 ± 0.10 | 2.6 ± 0.2    | 3.1 ± 0.3     | W Africa F   | S Asia M     |
| 17  | 5.5 ± 0.4     | 4.1 ± 0.1   | 4.6 ± 0.3     | 4.6 ± 0.2   | 4.7 ± 0.3    | 5.9 ± 0.4     | 2.6 ± 0.2     | 2.5 ± 0.1   | 3.6 ± 0.5     | 2.3 ± 0.1   | 2.5 ± 0.2    | 3.2 ± 0.4     | W Africa F   | S Asia M     |
| 18  | 5.0 ± 0.3     | 3.8 ± 0.1   | 4.1 ± 0.3     | 4.1 ± 0.1   | 4.3 ± 0.2    | 5.8 ± 0.4     | 2.6 ± 0.2     | 2.30 ± 0.08 | 3.3 ± 0.3     | 2.13 ± 0.10 | 2.2 ± 0.2    | 2.9 ± 0.2     | W Africa F   | S Asia M     |
| 19  | 5.3 ± 0.3     | 3.7 ± 0.1   | 3.9 ± 0.3     | 4.1 ± 0.1   | 4.2 ± 0.2    | 6.0 ± 0.4     | 2.9 ± 0.2     | 2.36 ± 0.09 | 2.8 ± 0.3     | 2.3 ± 0.1   | 2.4 ± 0.1    | 3.3 ± 0.3     | W Africa F   | S Asia M     |
| 20  | 5.5 ± 0.3     | 3.6 ± 0.1   | 3.7 ± 0.2     | 3.9 ± 0.1   | 4.1 ± 0.2    | 5.7 ± 0.3     | 3.3 ± 0.2     | 2.39 ± 0.09 | 2.9 ± 0.3     | 2.5 ± 0.1   | 2.5 ± 0.1    | 3.4 ± 0.3     | W Africa F   | E Asia M     |
| 21  | 5.3 ± 0.3     | 3.5 ± 0.1   | 3.5 ± 0.2     | 3.7 ± 0.1   | 4.0 ± 0.2    | 5.9 ± 0.3     | 3.4 ± 0.2     | 2.53 ± 0.08 | 2.7 ± 0.2     | 2.63 ± 0.10 | 2.6 ± 0.1    | 3.7 ± 0.3     | W Africa F   | E Asia M     |
| 22  | 5.3 ± 0.3     | 3.38 ± 0.10 | 3.3 ± 0.1     | 3.6 ± 0.1   | 4.0 ± 0.1    | 5.7 ± 0.3     | 4.1 ± 0.2     | 2.60 ± 0.09 | 2.6 ± 0.2     | 2.7 ± 0.1   | 2.6 ± 0.1    | 3.5 ± 0.2     | W Africa F   | S E Asia M   |
| 23  | 5.3 ± 0.2     | 3.26 ± 0.09 | 3.3 ± 0.1     | 3.54 ± 0.10 | 3.8 ± 0.1    | 5.9 ± 0.3     | 3.8 ± 0.2     | 2.55 ± 0.08 | 2.4 ± 0.1     | 2.72 ± 0.10 | 2.6 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 5.4 ± 0.3     | 3.22 ± 0.08 | 3.2 ± 0.1     | 3.39 ± 0.09 | 3.7 ± 0.1    | 5.4 ± 0.3     | 3.8 ± 0.2     | 2.57 ± 0.08 | 2.4 ± 0.1     | 2.76 ± 0.09 | 2.6 ± 0.1    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 5.3 ± 0.2     | 3.16 ± 0.07 | 3.08 ± 0.09   | 3.30 ± 0.08 | 3.7 ± 0.1    | 5.4 ± 0.2     | 3.9 ± 0.2     | 2.58 ± 0.08 | 2.4 ± 0.1     | 2.66 ± 0.08 | 2.6 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 5.2 ± 0.2     | 3.14 ± 0.07 | 2.95 ± 0.09   | 3.16 ± 0.07 | 3.66 ± 0.10  | 5.2 ± 0.2     | 3.8 ± 0.2     | 2.45 ± 0.07 | 2.3 ± 0.1     | 2.70 ± 0.07 | 2.6 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 27  | 5.3 ± 0.2     | 3.08 ± 0.06 | 2.97 ± 0.09   | 3.10 ± 0.07 | 3.56 ± 0.10  | 5.4 ± 0.2     | 3.8 ± 0.2     | 2.50 ± 0.07 | 2.14 ± 0.09   | 2.68 ± 0.07 | 2.8 ± 0.1    | 3.6 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 5.0 ± 0.2     | 3.04 ± 0.06 | 2.82 ± 0.08   | 3.11 ± 0.06 | 3.46 ± 0.09  | 5.1 ± 0.2     | 3.5 ± 0.1     | 2.46 ± 0.08 | 2.32 ± 0.09   | 2.65 ± 0.06 | 2.7 ± 0.1    | 3.5 ± 0.1     | W Africa F   | E Europe M   |
| 29  | 5.2 ± 0.2     | 2.95 ± 0.06 | 2.84 ± 0.08   | 3.13 ± 0.06 | 3.49 ± 0.09  | 5.0 ± 0.2     | 3.6 ± 0.2     | 2.37 ± 0.07 | 2.17 ± 0.09   | 2.64 ± 0.06 | 2.5 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 4.9 ± 0.2     | 2.89 ± 0.06 | 2.81 ± 0.09   | 3.15 ± 0.06 | 3.42 ± 0.10  | 4.8 ± 0.2     | 3.5 ± 0.2     | 2.39 ± 0.06 | 2.10 ± 0.09   | 2.70 ± 0.06 | 2.7 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.9 ± 0.2     | 2.96 ± 0.06 | 2.68 ± 0.09   | 3.11 ± 0.06 | 3.32 ± 0.09  | 4.6 ± 0.2     | 3.5 ± 0.2     | 2.26 ± 0.07 | 2.15 ± 0.09   | 2.75 ± 0.06 | 2.6 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 5.1 ± 0.2     | 2.91 ± 0.06 | 2.72 ± 0.09   | 3.17 ± 0.07 | 3.34 ± 0.10  | 4.8 ± 0.2     | 3.4 ± 0.2     | 2.33 ± 0.06 | 2.15 ± 0.10   | 2.79 ± 0.05 | 2.6 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 4.8 ± 0.2     | 2.88 ± 0.06 | 2.78 ± 0.09   | 3.23 ± 0.07 | 3.25 ± 0.10  | 4.5 ± 0.2     | 3.6 ± 0.2     | 2.35 ± 0.07 | 2.19 ± 0.09   | 2.82 ± 0.06 | 2.5 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 4.8 ± 0.2     | 2.90 ± 0.06 | 2.69 ± 0.09   | 3.19 ± 0.07 | 3.07 ± 0.09  | 4.6 ± 0.2     | 3.5 ± 0.2     | 2.36 ± 0.06 | 2.25 ± 0.09   | 2.91 ± 0.06 | 2.5 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.8 ± 0.2     | 2.93 ± 0.06 | 2.76 ± 0.10   | 3.28 ± 0.07 | 3.02 ± 0.10  | 4.4 ± 0.2     | 3.5 ± 0.2     | 2.45 ± 0.07 | 2.2 ± 0.1     | 2.97 ± 0.06 | 2.6 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 36  | 4.7 ± 0.2     | 2.86 ± 0.06 | 2.76 ± 0.10   | 3.26 ± 0.07 | 3.00 ± 0.10  | 4.4 ± 0.2     | 3.7 ± 0.2     | 2.42 ± 0.07 | 2.3 ± 0.1     | 2.94 ± 0.06 | 2.6 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 37  | 4.7 ± 0.2     | 2.75 ± 0.06 | 2.8 ± 0.1     | 3.41 ± 0.08 | 2.9 ± 0.1    | 4.4 ± 0.2     | 3.6 ± 0.2     | 2.47 ± 0.07 | 2.33 ± 0.10   | 3.03 ± 0.07 | 2.5 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 38  | 4.9 ± 0.2     | 2.81 ± 0.06 | 2.8 ± 0.1     | 3.42 ± 0.08 | 2.9 ± 0.1    | 4.3 ± 0.2     | 3.7 ± 0.2     | 2.56 ± 0.07 | 2.4 ± 0.1     | 3.08 ± 0.07 | 2.6 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 39  | 4.5 ± 0.2     | 2.85 ± 0.06 | 2.8 ± 0.1     | 3.43 ± 0.09 | 2.8 ± 0.1    | 4.2 ± 0.2     | 3.9 ± 0.2     | 2.58 ± 0.07 | 2.5 ± 0.1     | 3.11 ± 0.07 | 2.6 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 40  | 4.5 ± 0.2     | 2.89 ± 0.06 | 2.9 ± 0.1     | 3.38 ± 0.09 | 2.8 ± 0.1    | 4.2 ± 0.2     | 3.9 ± 0.2     | 2.58 ± 0.07 | 2.6 ± 0.1     | 3.22 ± 0.07 | 2.5 ± 0.1    | 4.1 ± 0.2     | E Africa F   | S E Asia M   |
| 41  | 4.6 ± 0.3     | 2.88 ± 0.06 | 3.0 ± 0.1     | 3.46 ± 0.10 | 2.8 ± 0.1    | 4.1 ± 0.2     | 4.0 ± 0.2     | 2.65 ± 0.07 | 2.6 ± 0.1     | 3.13 ± 0.08 | 2.7 ± 0.1    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.8 ± 0.3     | 2.90 ± 0.06 | 3.0 ± 0.1     | 3.5 ± 0.1   | 2.7 ± 0.1    | 4.1 ± 0.2     | 4.0 ± 0.2     | 2.74 ± 0.08 | 2.7 ± 0.1     | 3.22 ± 0.08 | 2.7 ± 0.1    | 4.1 ± 0.2     | E Africa F   | S E Asia F   |
| 43  | 4.8 ± 0.3     | 2.96 ± 0.06 | 3.2 ± 0.1     | 3.5 ± 0.1   | 2.9 ± 0.1    | 4.1 ± 0.2     | 4.1 ± 0.2     | 2.79 ± 0.07 | 2.7 ± 0.1     | 3.22 ± 0.09 | 2.9 ± 0.1    | 4.3 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 5.0 ± 0.3     | 3.02 ± 0.07 | 3.1 ± 0.1     | 3.5 ± 0.1   | 2.9 ± 0.1    | 4.2 ± 0.2     | 4.1 ± 0.2     | 2.89 ± 0.08 | 2.8 ± 0.1     | 3.24 ± 0.09 | 2.9 ± 0.1    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.7 ± 0.3     | 3.04 ± 0.07 | 3.0 ± 0.1     | 3.4 ± 0.1   | 2.8 ± 0.1    | 4.0 ± 0.2     | 4.2 ± 0.2     | 2.95 ± 0.08 | 2.7 ± 0.1     | 3.25 ± 0.10 | 3.0 ± 0.1    | 4.4 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.9 ± 0.4     | 3.16 ± 0.07 | 3.1 ± 0.1     | 3.4 ± 0.1   | 3.0 ± 0.1    | 4.0 ± 0.2     | 4.1 ± 0.3     | 2.95 ± 0.08 | 3.0 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.2    | 4.3 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 4.7 ± 0.3     | 3.14 ± 0.07 | 3.1 ± 0.2     | 3.3 ± 0.1   | 2.9 ± 0.1    | 4.2 ± 0.2     | 4.2 ± 0.3     | 2.98 ± 0.09 | 2.7 ± 0.2     | 3.4 ± 0.1   | 3.1 ± 0.2    | 4.5 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.6 ± 0.3     | 3.10 ± 0.07 | 3.1 ± 0.1     | 3.4 ± 0.1   | 3.0 ± 0.1    | 3.8 ± 0.2     | 4.4 ± 0.3     | 2.99 ± 0.08 | 2.8 ± 0.2     | 3.3 ± 0.1   | 3.0 ± 0.2    | 4.4 ± 0.2     | E Africa F   | E Europe M   |
| 49  | 4.7 ± 0.3     | 3.21 ± 0.08 | 3.0 ± 0.1     | 3.2 ± 0.1   | 3.0 ± 0.1    | 4.0 ± 0.3     | 4.3 ± 0.3     | 3.09 ± 0.09 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.0 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 50  | 4.8 ± 0.4     | 3.20 ± 0.08 | 2.9 ± 0.1     | 3.2 ± 0.1   | 3.0 ± 0.1    | 3.9 ± 0.2     | 3.9 ± 0.3     | 2.99 ± 0.08 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.2    | 4.3 ± 0.2     | E Africa F   | E Europe M   |
| 51  | 4.4 ± 0.3     | 3.22 ± 0.08 | 2.9 ± 0.2     | 3.1 ± 0.1   | 3.0 ± 0.1    | 4.0 ± 0.3     | 4.2 ± 0.4     | 3.07 ± 0.09 | 2.7 ± 0.1     | 3.5 ± 0.1   | 3.0 ± 0.2    | 4.5 ± 0.3     | W Africa M   | E Europe M   |
| 52  | 4.4 ± 0.4     | 3.15 ± 0.08 | 2.9 ± 0.1     | 3.1 ± 0.1   | 3.0 ± 0.1    | 3.8 ± 0.3     | 4.1 ± 0.3     | 3.05 ± 0.09 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.1 ± 0.2    | 4.4 ± 0.3     | W Africa M   | E Europe M   |



**Table 92.** Mean absolute error for Application-type images for algorithm neurotechnology-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 6.3 ± 0.4     | 6.0 ± 0.2   | 4.3 ± 0.2     | 4.5 ± 0.2   | 5.1 ± 0.3    | 9.5 ± 0.5     | 8.9 ± 0.4     | 6.7 ± 0.1   | 4.0 ± 0.2     | 5.3 ± 0.1   | 5.8 ± 0.3    | 11.1 ± 0.4    | W Africa M   | E Europe M   |
| 15  | 5.7 ± 0.4     | 6.2 ± 0.2   | 3.8 ± 0.2     | 4.5 ± 0.2   | 5.2 ± 0.3    | 8.6 ± 0.5     | 9.0 ± 0.4     | 6.8 ± 0.1   | 3.8 ± 0.2     | 5.4 ± 0.2   | 6.2 ± 0.3    | 10.4 ± 0.4    | W Africa M   | E Europe F   |
| 16  | 6.1 ± 0.4     | 5.5 ± 0.2   | 3.9 ± 0.3     | 4.2 ± 0.2   | 5.2 ± 0.3    | 8.6 ± 0.5     | 9.3 ± 0.3     | 6.5 ± 0.1   | 3.9 ± 0.2     | 5.3 ± 0.2   | 6.0 ± 0.3    | 9.9 ± 0.4     | W Africa M   | E Europe F   |
| 17  | 6.1 ± 0.4     | 5.2 ± 0.1   | 3.7 ± 0.3     | 3.9 ± 0.2   | 4.8 ± 0.3    | 8.4 ± 0.4     | 9.2 ± 0.4     | 5.9 ± 0.1   | 3.3 ± 0.2     | 5.1 ± 0.2   | 5.6 ± 0.2    | 9.8 ± 0.4     | W Africa M   | E Europe M   |
| 18  | 5.3 ± 0.3     | 4.5 ± 0.1   | 3.2 ± 0.2     | 3.4 ± 0.1   | 4.2 ± 0.2    | 7.4 ± 0.3     | 8.6 ± 0.3     | 5.34 ± 0.10 | 3.1 ± 0.2     | 4.7 ± 0.1   | 5.1 ± 0.2    | 9.0 ± 0.3     | W Africa M   | E Europe M   |
| 19  | 5.2 ± 0.3     | 3.93 ± 0.10 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.7 ± 0.2    | 7.1 ± 0.3     | 8.4 ± 0.3     | 4.69 ± 0.10 | 2.5 ± 0.2     | 4.4 ± 0.1   | 4.4 ± 0.2    | 8.5 ± 0.3     | W Africa M   | E Europe M   |
| 20  | 4.8 ± 0.2     | 3.4 ± 0.1   | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.3 ± 0.1    | 6.4 ± 0.3     | 7.9 ± 0.3     | 4.06 ± 0.09 | 2.4 ± 0.2     | 4.1 ± 0.1   | 3.7 ± 0.1    | 8.2 ± 0.3     | W Africa M   | E Europe M   |
| 21  | 4.6 ± 0.2     | 3.11 ± 0.09 | 2.7 ± 0.1     | 2.90 ± 0.09 | 3.0 ± 0.1    | 6.1 ± 0.3     | 7.4 ± 0.3     | 3.33 ± 0.08 | 2.5 ± 0.2     | 3.6 ± 0.1   | 3.3 ± 0.1    | 7.6 ± 0.3     | W Africa M   | E Europe M   |
| 22  | 4.6 ± 0.2     | 2.86 ± 0.09 | 2.8 ± 0.1     | 2.83 ± 0.09 | 2.7 ± 0.1    | 5.5 ± 0.3     | 7.2 ± 0.2     | 2.75 ± 0.08 | 2.5 ± 0.2     | 3.2 ± 0.1   | 2.7 ± 0.1    | 7.0 ± 0.2     | E Africa M   | E Europe M   |
| 23  | 4.1 ± 0.2     | 2.72 ± 0.08 | 3.0 ± 0.1     | 2.95 ± 0.08 | 2.7 ± 0.1    | 4.8 ± 0.2     | 6.4 ± 0.2     | 2.43 ± 0.07 | 2.3 ± 0.1     | 3.0 ± 0.1   | 2.3 ± 0.1    | 6.3 ± 0.2     | E Africa M   | S E Asia M   |
| 24  | 3.9 ± 0.2     | 2.82 ± 0.08 | 3.0 ± 0.1     | 3.05 ± 0.08 | 2.61 ± 0.09  | 4.5 ± 0.2     | 6.1 ± 0.2     | 2.09 ± 0.07 | 2.4 ± 0.1     | 2.80 ± 0.09 | 1.94 ± 0.10  | 5.7 ± 0.2     | E Africa M   | S E Asia M   |
| 25  | 3.9 ± 0.2     | 2.98 ± 0.08 | 3.27 ± 0.10   | 3.32 ± 0.07 | 3.0 ± 0.1    | 4.1 ± 0.2     | 5.5 ± 0.2     | 1.90 ± 0.07 | 2.5 ± 0.1     | 2.65 ± 0.08 | 1.99 ± 0.09  | 5.2 ± 0.2     | E Africa M   | E Asia M     |
| 26  | 3.9 ± 0.2     | 3.33 ± 0.09 | 3.44 ± 0.09   | 3.70 ± 0.07 | 3.18 ± 0.10  | 3.7 ± 0.2     | 5.0 ± 0.2     | 1.86 ± 0.07 | 2.8 ± 0.1     | 2.64 ± 0.07 | 2.1 ± 0.1    | 4.6 ± 0.2     | E Africa M   | E Asia M     |
| 27  | 4.0 ± 0.2     | 3.76 ± 0.08 | 3.62 ± 0.10   | 4.11 ± 0.07 | 3.5 ± 0.1    | 3.4 ± 0.2     | 4.3 ± 0.2     | 2.08 ± 0.07 | 2.9 ± 0.1     | 2.68 ± 0.06 | 2.3 ± 0.1    | 4.0 ± 0.1     | E Africa M   | E Asia M     |
| 28  | 4.0 ± 0.2     | 4.16 ± 0.08 | 3.87 ± 0.10   | 4.61 ± 0.08 | 4.0 ± 0.1    | 3.4 ± 0.2     | 3.9 ± 0.2     | 2.46 ± 0.07 | 3.0 ± 0.1     | 2.86 ± 0.06 | 2.7 ± 0.1    | 3.5 ± 0.1     | S Asia F     | E Asia M     |
| 29  | 4.0 ± 0.2     | 4.66 ± 0.08 | 4.1 ± 0.1     | 4.98 ± 0.08 | 4.6 ± 0.1    | 3.4 ± 0.2     | 3.6 ± 0.2     | 2.82 ± 0.07 | 3.3 ± 0.1     | 3.09 ± 0.06 | 3.1 ± 0.1    | 3.2 ± 0.1     | S Asia F     | E Asia M     |
| 30  | 4.4 ± 0.2     | 5.18 ± 0.08 | 4.4 ± 0.1     | 5.50 ± 0.08 | 4.9 ± 0.1    | 3.5 ± 0.1     | 3.5 ± 0.2     | 3.35 ± 0.07 | 3.3 ± 0.1     | 3.37 ± 0.06 | 3.5 ± 0.1    | 2.9 ± 0.1     | S Asia F     | W Africa M   |
| 31  | 4.6 ± 0.2     | 5.72 ± 0.08 | 4.7 ± 0.1     | 6.01 ± 0.09 | 5.4 ± 0.1    | 3.7 ± 0.2     | 3.2 ± 0.1     | 3.89 ± 0.08 | 3.6 ± 0.1     | 3.68 ± 0.07 | 4.1 ± 0.1    | 2.7 ± 0.1     | S Asia F     | W Africa M   |
| 32  | 4.7 ± 0.2     | 6.29 ± 0.09 | 4.8 ± 0.1     | 6.49 ± 0.10 | 5.9 ± 0.1    | 4.1 ± 0.2     | 3.1 ± 0.1     | 4.40 ± 0.08 | 3.7 ± 0.1     | 4.01 ± 0.07 | 4.4 ± 0.1    | 2.6 ± 0.1     | S Asia F     | W Africa M   |
| 33  | 5.1 ± 0.2     | 6.81 ± 0.09 | 5.1 ± 0.1     | 7.0 ± 0.1   | 6.4 ± 0.1    | 4.2 ± 0.2     | 3.3 ± 0.2     | 5.03 ± 0.09 | 3.6 ± 0.1     | 4.39 ± 0.07 | 4.7 ± 0.2    | 2.7 ± 0.1     | S Asia F     | W Africa M   |
| 34  | 5.2 ± 0.2     | 7.27 ± 0.09 | 5.3 ± 0.2     | 7.4 ± 0.1   | 6.8 ± 0.2    | 4.7 ± 0.2     | 3.2 ± 0.2     | 5.47 ± 0.09 | 3.8 ± 0.2     | 4.69 ± 0.08 | 5.3 ± 0.2    | 2.8 ± 0.1     | S Asia F     | W Africa M   |
| 35  | 5.5 ± 0.2     | 7.79 ± 0.10 | 5.5 ± 0.2     | 7.9 ± 0.1   | 7.3 ± 0.2    | 4.8 ± 0.2     | 3.4 ± 0.2     | 5.96 ± 0.10 | 3.9 ± 0.2     | 5.10 ± 0.09 | 5.6 ± 0.2    | 3.2 ± 0.1     | S Asia F     | W Africa M   |
| 36  | 6.1 ± 0.3     | 8.19 ± 0.10 | 5.6 ± 0.2     | 8.3 ± 0.1   | 7.4 ± 0.2    | 5.2 ± 0.2     | 3.8 ± 0.2     | 6.3 ± 0.1   | 4.1 ± 0.2     | 5.41 ± 0.09 | 5.9 ± 0.2    | 3.6 ± 0.1     | S Asia F     | W Africa M   |
| 37  | 5.9 ± 0.3     | 8.6 ± 0.1   | 5.8 ± 0.2     | 8.5 ± 0.1   | 8.0 ± 0.2    | 5.7 ± 0.2     | 3.9 ± 0.2     | 6.6 ± 0.1   | 4.3 ± 0.2     | 5.76 ± 0.10 | 6.2 ± 0.2    | 3.8 ± 0.1     | E Asia F     | W Africa M   |
| 38  | 6.2 ± 0.3     | 9.0 ± 0.1   | 5.7 ± 0.2     | 8.8 ± 0.1   | 8.2 ± 0.2    | 6.3 ± 0.2     | 4.2 ± 0.2     | 7.0 ± 0.1   | 4.3 ± 0.2     | 6.1 ± 0.1   | 6.2 ± 0.2    | 4.2 ± 0.1     | E Asia F     | E Africa M   |
| 39  | 6.3 ± 0.3     | 9.5 ± 0.1   | 5.9 ± 0.2     | 9.3 ± 0.2   | 8.2 ± 0.2    | 6.1 ± 0.2     | 4.4 ± 0.2     | 7.4 ± 0.1   | 4.4 ± 0.2     | 6.2 ± 0.1   | 6.5 ± 0.2    | 4.7 ± 0.2     | E Asia F     | E Africa M   |
| 40  | 6.6 ± 0.3     | 9.7 ± 0.1   | 5.9 ± 0.2     | 9.4 ± 0.2   | 8.7 ± 0.2    | 6.5 ± 0.2     | 4.6 ± 0.2     | 7.6 ± 0.1   | 4.6 ± 0.2     | 6.4 ± 0.1   | 6.5 ± 0.2    | 5.2 ± 0.2     | E Asia F     | E Europe M   |
| 41  | 7.2 ± 0.3     | 9.9 ± 0.1   | 6.1 ± 0.2     | 9.6 ± 0.2   | 9.0 ± 0.2    | 7.3 ± 0.3     | 5.0 ± 0.2     | 7.7 ± 0.1   | 4.7 ± 0.2     | 6.6 ± 0.1   | 6.9 ± 0.2    | 5.8 ± 0.2     | E Asia F     | E Europe M   |
| 42  | 6.7 ± 0.3     | 10.2 ± 0.1  | 6.1 ± 0.3     | 9.5 ± 0.2   | 8.9 ± 0.2    | 7.1 ± 0.3     | 5.1 ± 0.3     | 7.9 ± 0.1   | 4.5 ± 0.2     | 6.7 ± 0.1   | 6.8 ± 0.2    | 5.8 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 7.6 ± 0.4     | 10.3 ± 0.1  | 6.0 ± 0.3     | 9.8 ± 0.2   | 9.2 ± 0.2    | 7.5 ± 0.3     | 5.2 ± 0.3     | 7.9 ± 0.2   | 4.7 ± 0.2     | 6.8 ± 0.2   | 6.9 ± 0.3    | 6.5 ± 0.2     | E Asia F     | E Europe M   |
| 44  | 7.6 ± 0.4     | 10.7 ± 0.1  | 6.1 ± 0.3     | 9.8 ± 0.3   | 8.9 ± 0.3    | 7.7 ± 0.3     | 5.4 ± 0.3     | 8.0 ± 0.2   | 4.6 ± 0.2     | 6.8 ± 0.2   | 6.8 ± 0.3    | 6.4 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 7.9 ± 0.4     | 10.8 ± 0.2  | 6.1 ± 0.3     | 9.7 ± 0.3   | 9.1 ± 0.3    | 7.6 ± 0.4     | 5.5 ± 0.3     | 8.2 ± 0.2   | 4.6 ± 0.2     | 6.8 ± 0.2   | 6.7 ± 0.3    | 6.9 ± 0.3     | E Asia F     | E Europe M   |
| 46  | 7.4 ± 0.5     | 11.1 ± 0.2  | 5.9 ± 0.3     | 9.5 ± 0.3   | 9.3 ± 0.3    | 8.0 ± 0.4     | 5.5 ± 0.3     | 8.1 ± 0.2   | 4.9 ± 0.3     | 6.6 ± 0.2   | 6.3 ± 0.3    | 7.1 ± 0.3     | E Asia F     | E Europe M   |
| 47  | 7.2 ± 0.5     | 10.8 ± 0.2  | 5.9 ± 0.3     | 9.6 ± 0.3   | 8.9 ± 0.3    | 7.9 ± 0.4     | 5.6 ± 0.3     | 8.0 ± 0.2   | 5.0 ± 0.3     | 6.5 ± 0.2   | 6.4 ± 0.3    | 7.2 ± 0.3     | E Asia F     | E Europe M   |
| 48  | 7.5 ± 0.5     | 10.7 ± 0.2  | 5.7 ± 0.3     | 9.5 ± 0.3   | 9.0 ± 0.3    | 8.0 ± 0.4     | 6.2 ± 0.4     | 8.0 ± 0.2   | 5.0 ± 0.3     | 6.6 ± 0.2   | 6.6 ± 0.3    | 7.9 ± 0.3     | E Asia F     | E Europe M   |
| 49  | 7.3 ± 0.5     | 10.9 ± 0.2  | 6.0 ± 0.3     | 9.3 ± 0.3   | 8.5 ± 0.3    | 8.4 ± 0.5     | 6.1 ± 0.4     | 7.9 ± 0.2   | 5.2 ± 0.3     | 6.4 ± 0.2   | 6.0 ± 0.3    | 7.8 ± 0.3     | E Asia F     | E Europe M   |
| 50  | 7.8 ± 0.7     | 10.6 ± 0.2  | 5.5 ± 0.3     | 9.1 ± 0.3   | 8.7 ± 0.3    | 7.9 ± 0.5     | 6.3 ± 0.4     | 7.7 ± 0.2   | 5.0 ± 0.3     | 6.6 ± 0.2   | 6.1 ± 0.3    | 7.6 ± 0.3     | E Asia F     | E Europe M   |
| 51  | 7.1 ± 0.5     | 10.6 ± 0.2  | 5.7 ± 0.3     | 8.9 ± 0.3   | 8.5 ± 0.3    | 8.2 ± 0.5     | 6.6 ± 0.5     | 7.6 ± 0.2   | 4.8 ± 0.3     | 6.6 ± 0.2   | 6.0 ± 0.3    | 8.2 ± 0.4     | E Asia F     | E Europe M   |
| 52  | 7.2 ± 0.6     | 10.5 ± 0.2  | 5.4 ± 0.3     | 9.1 ± 0.3   | 7.7 ± 0.3    | 8.1 ± 0.5     | 6.0 ± 0.4     | 7.3 ± 0.2   | 4.9 ± 0.3     | 6.4 ± 0.2   | 5.9 ± 0.3    | 8.0 ± 0.4     | E Asia F     | E Europe M   |

**Table 93.** Mean absolute error for Application-type images for algorithm neurotechnology-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.5 ± 0.2     | 6.1 ± 0.2   | 4.3 ± 0.2     | 4.5 ± 0.2   | 5.2 ± 0.3    | 5.7 ± 0.3     | 4.8 ± 0.2     | 6.8 ± 0.1   | 4.0 ± 0.2     | 5.3 ± 0.1   | 5.8 ± 0.3    | 4.7 ± 0.3     | E Asia M     | E Europe M   |
| 15  | 4.3 ± 0.3     | 6.2 ± 0.2   | 3.8 ± 0.2     | 4.6 ± 0.2   | 5.3 ± 0.3    | 5.0 ± 0.3     | 4.5 ± 0.3     | 6.9 ± 0.1   | 3.8 ± 0.2     | 5.4 ± 0.2   | 6.2 ± 0.3    | 4.9 ± 0.4     | E Asia M     | E Europe M   |
| 16  | 4.1 ± 0.3     | 5.6 ± 0.2   | 4.0 ± 0.3     | 4.2 ± 0.2   | 5.3 ± 0.3    | 5.0 ± 0.4     | 4.4 ± 0.2     | 6.5 ± 0.1   | 4.0 ± 0.2     | 5.3 ± 0.2   | 6.1 ± 0.3    | 4.6 ± 0.3     | E Asia M     | E Europe M   |
| 17  | 3.7 ± 0.3     | 5.2 ± 0.1   | 3.7 ± 0.3     | 3.9 ± 0.2   | 4.8 ± 0.3    | 4.4 ± 0.3     | 4.6 ± 0.3     | 6.0 ± 0.1   | 3.3 ± 0.2     | 5.0 ± 0.2   | 5.6 ± 0.2    | 4.9 ± 0.4     | E Asia M     | E Europe M   |
| 18  | 3.3 ± 0.2     | 4.5 ± 0.1   | 3.3 ± 0.2     | 3.4 ± 0.1   | 4.3 ± 0.2    | 3.7 ± 0.3     | 4.4 ± 0.3     | 5.36 ± 0.10 | 3.1 ± 0.2     | 4.6 ± 0.1   | 5.1 ± 0.2    | 4.1 ± 0.3     | E Asia M     | E Europe M   |
| 19  | 3.1 ± 0.2     | 3.96 ± 0.10 | 3.1 ± 0.2     | 3.2 ± 0.1   | 3.7 ± 0.1    | 3.2 ± 0.2     | 4.1 ± 0.3     | 4.73 ± 0.10 | 2.5 ± 0.2     | 4.2 ± 0.1   | 4.4 ± 0.2    | 3.7 ± 0.3     | E Asia M     | E Europe M   |
| 20  | 2.9 ± 0.2     | 3.4 ± 0.1   | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.3 ± 0.1    | 2.8 ± 0.2     | 4.1 ± 0.3     | 4.10 ± 0.09 | 2.5 ± 0.2     | 3.9 ± 0.1   | 3.7 ± 0.1    | 3.6 ± 0.3     | E Asia M     | E Europe M   |
| 21  | 2.7 ± 0.2     | 3.12 ± 0.09 | 2.8 ± 0.2     | 2.85 ± 0.09 | 3.0 ± 0.1    | 2.6 ± 0.3     | 3.7 ± 0.3     | 3.38 ± 0.08 | 2.5 ± 0.2     | 3.5 ± 0.1   | 3.3 ± 0.1    | 3.3 ± 0.3     | E Africa M   | E Europe M   |
| 22  | 2.8 ± 0.2     | 2.85 ± 0.09 | 2.8 ± 0.1     | 2.82 ± 0.09 | 2.7 ± 0.1    | 2.6 ± 0.3     | 3.8 ± 0.3     | 2.79 ± 0.08 | 2.5 ± 0.2     | 3.1 ± 0.1   | 2.7 ± 0.1    | 3.1 ± 0.3     | E Africa M   | E Europe M   |
| 23  | 2.7 ± 0.2     | 2.70 ± 0.08 | 3.1 ± 0.1     | 2.94 ± 0.08 | 2.6 ± 0.1    | 2.5 ± 0.2     | 3.4 ± 0.3     | 2.44 ± 0.08 | 2.3 ± 0.1     | 2.8 ± 0.1   | 2.3 ± 0.1    | 3.0 ± 0.2     | E Africa M   | S E Asia M   |
| 24  | 2.9 ± 0.2     | 2.78 ± 0.08 | 3.1 ± 0.1     | 3.03 ± 0.08 | 2.61 ± 0.09  | 2.9 ± 0.2     | 3.4 ± 0.3     | 2.10 ± 0.07 | 2.4 ± 0.1     | 2.73 ± 0.09 | 1.94 ± 0.10  | 2.6 ± 0.2     | E Africa M   | S E Asia M   |
| 25  | 3.4 ± 0.2     | 2.94 ± 0.08 | 3.30 ± 0.10   | 3.31 ± 0.07 | 2.9 ± 0.1    | 3.3 ± 0.2     | 3.4 ± 0.2     | 1.92 ± 0.07 | 2.5 ± 0.1     | 2.59 ± 0.08 | 1.97 ± 0.09  | 2.8 ± 0.2     | E Africa M   | E Asia M     |
| 26  | 3.8 ± 0.2     | 3.27 ± 0.08 | 3.45 ± 0.09   | 3.67 ± 0.07 | 3.16 ± 0.10  | 3.5 ± 0.2     | 3.5 ± 0.2     | 1.86 ± 0.07 | 2.8 ± 0.1     | 2.62 ± 0.07 | 2.2 ± 0.1    | 2.9 ± 0.2     | E Africa F   | E Asia M     |
| 27  | 4.3 ± 0.2     | 3.72 ± 0.08 | 3.63 ± 0.10   | 4.09 ± 0.07 | 3.5 ± 0.1    | 4.3 ± 0.1     | 3.6 ± 0.2     | 2.06 ± 0.06 | 2.9 ± 0.1     | 2.70 ± 0.06 | 2.3 ± 0.1    | 3.3 ± 0.2     | E Africa F   | E Asia M     |
| 28  | 4.9 ± 0.2     | 4.10 ± 0.08 | 3.86 ± 0.10   | 4.59 ± 0.08 | 4.0 ± 0.1    | 4.9 ± 0.1     | 4.0 ± 0.2     | 2.44 ± 0.07 | 3.0 ± 0.1     | 2.89 ± 0.06 | 2.7 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Asia M     |
| 29  | 5.3 ± 0.1     | 4.60 ± 0.08 | 4.1 ± 0.1     | 4.94 ± 0.08 | 4.6 ± 0.1    | 5.4 ± 0.1     | 4.4 ± 0.1     | 2.78 ± 0.07 | 3.3 ± 0.1     | 3.14 ± 0.06 | 3.1 ± 0.1    | 4.2 ± 0.1     | W Africa F   | E Asia M     |
| 30  | 5.8 ± 0.1     | 5.14 ± 0.08 | 4.4 ± 0.1     | 5.48 ± 0.08 | 4.9 ± 0.1    | 5.9 ± 0.1     | 4.8 ± 0.1     | 3.31 ± 0.07 | 3.3 ± 0.1     | 3.47 ± 0.06 | 3.5 ± 0.1    | 4.61 ± 0.10   | W Africa F   | E Asia M     |
| 31  | 6.3 ± 0.2     | 5.66 ± 0.08 | 4.7 ± 0.1     | 5.98 ± 0.09 | 5.4 ± 0.1    | 6.3 ± 0.1     | 4.8 ± 0.1     | 3.85 ± 0.08 | 3.6 ± 0.1     | 3.79 ± 0.07 | 4.1 ± 0.1    | 4.91 ± 0.09   | W Africa F   | E Europe M   |
| 32  | 6.6 ± 0.2     | 6.24 ± 0.09 | 4.8 ± 0.1     | 6.47 ± 0.09 | 5.9 ± 0.1    | 6.6 ± 0.1     | 5.0 ± 0.1     | 4.35 ± 0.08 | 3.6 ± 0.1     | 4.15 ± 0.07 | 4.4 ± 0.1    | 5.1 ± 0.1     | W Africa F   | E Europe M   |
| 33  | 6.7 ± 0.2     | 6.75 ± 0.09 | 5.1 ± 0.1     | 7.0 ± 0.1   | 6.4 ± 0.1    | 6.9 ± 0.2     | 5.0 ± 0.2     | 4.98 ± 0.09 | 3.6 ± 0.1     | 4.52 ± 0.07 | 4.7 ± 0.2    | 5.0 ± 0.1     | S Asia F     | E Europe M   |
| 34  | 6.8 ± 0.2     | 7.20 ± 0.09 | 5.3 ± 0.2     | 7.4 ± 0.1   | 6.7 ± 0.1    | 7.2 ± 0.2     | 4.9 ± 0.2     | 5.43 ± 0.09 | 3.8 ± 0.1     | 4.84 ± 0.08 | 5.3 ± 0.2    | 4.9 ± 0.2     | S Asia F     | E Europe M   |
| 35  | 6.9 ± 0.3     | 7.70 ± 0.10 | 5.4 ± 0.2     | 7.9 ± 0.1   | 7.2 ± 0.2    | 7.2 ± 0.2     | 4.8 ± 0.2     | 5.90 ± 0.10 | 3.8 ± 0.2     | 5.22 ± 0.09 | 5.6 ± 0.2    | 5.1 ± 0.2     | S Asia F     | E Europe M   |
| 36  | 7.5 ± 0.3     | 8.11 ± 0.10 | 5.5 ± 0.2     | 8.3 ± 0.1   | 7.3 ± 0.2    | 7.3 ± 0.3     | 5.2 ± 0.2     | 6.2 ± 0.1   | 4.1 ± 0.2     | 5.53 ± 0.09 | 5.8 ± 0.2    | 5.4 ± 0.2     | S Asia F     | E Europe M   |
| 37  | 7.3 ± 0.3     | 8.5 ± 0.1   | 5.7 ± 0.2     | 8.5 ± 0.1   | 8.0 ± 0.2    | 7.7 ± 0.3     | 5.0 ± 0.2     | 6.6 ± 0.1   | 4.3 ± 0.2     | 5.88 ± 0.10 | 6.2 ± 0.2    | 5.3 ± 0.2     | E Asia F     | E Europe M   |
| 38  | 7.4 ± 0.3     | 8.9 ± 0.1   | 5.7 ± 0.2     | 8.8 ± 0.1   | 8.0 ± 0.2    | 8.0 ± 0.3     | 5.1 ± 0.3     | 6.9 ± 0.1   | 4.2 ± 0.2     | 6.2 ± 0.1   | 6.1 ± 0.2    | 5.4 ± 0.2     | E Asia F     | E Europe M   |
| 39  | 7.2 ± 0.3     | 9.4 ± 0.1   | 5.8 ± 0.2     | 9.2 ± 0.2   | 8.2 ± 0.2    | 7.8 ± 0.3     | 5.2 ± 0.3     | 7.3 ± 0.1   | 4.4 ± 0.2     | 6.3 ± 0.1   | 6.4 ± 0.2    | 5.9 ± 0.2     | E Asia F     | E Europe M   |
| 40  | 7.4 ± 0.4     | 9.7 ± 0.1   | 5.9 ± 0.2     | 9.3 ± 0.2   | 8.6 ± 0.2    | 8.0 ± 0.3     | 5.3 ± 0.3     | 7.6 ± 0.1   | 4.6 ± 0.2     | 6.5 ± 0.1   | 6.5 ± 0.2    | 6.3 ± 0.3     | E Asia F     | E Europe M   |
| 41  | 8.1 ± 0.4     | 9.8 ± 0.1   | 6.1 ± 0.2     | 9.6 ± 0.2   | 8.9 ± 0.2    | 8.8 ± 0.4     | 5.5 ± 0.3     | 7.6 ± 0.1   | 4.6 ± 0.2     | 6.6 ± 0.1   | 6.9 ± 0.2    | 6.8 ± 0.3     | E Asia F     | E Europe M   |
| 42  | 7.4 ± 0.4     | 10.1 ± 0.1  | 6.0 ± 0.3     | 9.4 ± 0.2   | 8.8 ± 0.2    | 8.2 ± 0.4     | 5.6 ± 0.3     | 7.8 ± 0.1   | 4.4 ± 0.2     | 6.8 ± 0.1   | 6.7 ± 0.2    | 6.5 ± 0.3     | E Asia F     | E Europe M   |
| 43  | 8.1 ± 0.5     | 10.3 ± 0.1  | 5.9 ± 0.3     | 9.7 ± 0.2   | 9.0 ± 0.3    | 8.5 ± 0.4     | 5.4 ± 0.3     | 7.8 ± 0.1   | 4.7 ± 0.2     | 6.9 ± 0.2   | 6.9 ± 0.3    | 7.1 ± 0.3     | E Asia F     | E Europe M   |
| 44  | 8.3 ± 0.5     | 10.6 ± 0.1  | 6.0 ± 0.3     | 9.7 ± 0.3   | 8.8 ± 0.3    | 8.4 ± 0.4     | 5.6 ± 0.3     | 8.0 ± 0.2   | 4.7 ± 0.2     | 6.8 ± 0.2   | 6.8 ± 0.3    | 6.9 ± 0.3     | E Asia F     | E Europe M   |
| 45  | 8.4 ± 0.5     | 10.6 ± 0.1  | 6.0 ± 0.3     | 9.7 ± 0.3   | 9.0 ± 0.3    | 8.1 ± 0.4     | 5.7 ± 0.3     | 8.1 ± 0.2   | 4.6 ± 0.2     | 6.8 ± 0.2   | 6.7 ± 0.3    | 7.4 ± 0.3     | E Asia F     | E Europe M   |
| 46  | 7.8 ± 0.5     | 10.9 ± 0.2  | 5.8 ± 0.3     | 9.4 ± 0.3   | 9.2 ± 0.3    | 8.6 ± 0.4     | 5.7 ± 0.4     | 8.1 ± 0.2   | 4.9 ± 0.3     | 6.6 ± 0.2   | 6.3 ± 0.3    | 7.5 ± 0.3     | E Asia F     | E Europe M   |
| 47  | 7.3 ± 0.5     | 10.7 ± 0.2  | 5.8 ± 0.3     | 9.5 ± 0.3   | 8.7 ± 0.3    | 8.3 ± 0.5     | 5.6 ± 0.3     | 7.9 ± 0.2   | 5.0 ± 0.3     | 6.6 ± 0.2   | 6.4 ± 0.3    | 7.5 ± 0.3     | E Asia F     | E Europe M   |
| 48  | 7.8 ± 0.6     | 10.6 ± 0.2  | 5.7 ± 0.3     | 9.4 ± 0.3   | 8.8 ± 0.3    | 8.3 ± 0.5     | 6.3 ± 0.4     | 7.8 ± 0.2   | 4.9 ± 0.3     | 6.6 ± 0.2   | 6.5 ± 0.3    | 8.2 ± 0.4     | E Asia F     | E Europe M   |
| 49  | 7.7 ± 0.6     | 10.8 ± 0.2  | 5.9 ± 0.3     | 9.2 ± 0.3   | 8.4 ± 0.3    | 8.8 ± 0.5     | 6.3 ± 0.4     | 7.8 ± 0.2   | 5.1 ± 0.3     | 6.4 ± 0.2   | 6.1 ± 0.3    | 8.0 ± 0.3     | E Asia F     | E Europe M   |
| 50  | 7.9 ± 0.7     | 10.5 ± 0.2  | 5.4 ± 0.3     | 9.1 ± 0.3   | 8.6 ± 0.3    | 8.2 ± 0.5     | 6.4 ± 0.4     | 7.7 ± 0.2   | 5.0 ± 0.3     | 6.6 ± 0.2   | 6.0 ± 0.3    | 7.8 ± 0.4     | E Asia F     | E Europe M   |
| 51  | 7.2 ± 0.6     | 10.5 ± 0.2  | 5.7 ± 0.3     | 8.9 ± 0.3   | 8.3 ± 0.3    | 8.5 ± 0.5     | 6.7 ± 0.5     | 7.5 ± 0.2   | 4.8 ± 0.3     | 6.5 ± 0.2   | 5.8 ± 0.3    | 8.4 ± 0.4     | E Asia F     | E Europe M   |
| 52  | 7.3 ± 0.6     | 10.4 ± 0.2  | 5.4 ± 0.3     | 9.0 ± 0.3   | 7.5 ± 0.3    | 8.3 ± 0.5     | 5.9 ± 0.5     | 7.2 ± 0.2   | 4.8 ± 0.3     | 6.4 ± 0.2   | 5.8 ± 0.4    | 8.1 ± 0.4     | E Asia F     | E Europe M   |

**Table 94.** Mean absolute error for Application-type images for algorithm nominder-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.9 ± 0.3     | 3.7 ± 0.1   | 2.8 ± 0.2     | 2.5 ± 0.1   | 3.2 ± 0.3    | 4.6 ± 0.5     | 2.2 ± 0.2     | 2.5 ± 0.1   | 1.8 ± 0.2     | 1.8 ± 0.1   | 2.0 ± 0.2    | 2.5 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 4.1 ± 0.3     | 4.0 ± 0.1   | 2.8 ± 0.2     | 3.1 ± 0.2   | 3.6 ± 0.3    | 4.8 ± 0.5     | 2.3 ± 0.2     | 2.9 ± 0.1   | 1.8 ± 0.2     | 2.3 ± 0.1   | 2.7 ± 0.2    | 2.9 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.0 ± 0.3     | 4.1 ± 0.1   | 2.9 ± 0.2     | 3.2 ± 0.2   | 3.8 ± 0.3    | 5.3 ± 0.5     | 2.8 ± 0.3     | 3.2 ± 0.1   | 2.0 ± 0.2     | 2.7 ± 0.1   | 3.0 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 4.6 ± 0.3     | 4.0 ± 0.1   | 3.0 ± 0.2     | 3.4 ± 0.1   | 3.7 ± 0.3    | 5.3 ± 0.4     | 3.1 ± 0.3     | 3.4 ± 0.1   | 1.8 ± 0.2     | 3.2 ± 0.2   | 3.0 ± 0.2    | 3.7 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.2 ± 0.2     | 3.71 ± 0.10 | 2.8 ± 0.2     | 3.1 ± 0.1   | 3.3 ± 0.1    | 5.2 ± 0.3     | 3.3 ± 0.2     | 3.36 ± 0.09 | 2.1 ± 0.2     | 3.4 ± 0.1   | 3.2 ± 0.2    | 3.7 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 4.1 ± 0.3     | 3.43 ± 0.09 | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.2 ± 0.1    | 5.5 ± 0.3     | 3.4 ± 0.2     | 3.33 ± 0.10 | 2.3 ± 0.2     | 3.5 ± 0.1   | 3.1 ± 0.2    | 3.9 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.0 ± 0.2     | 3.2 ± 0.1   | 2.6 ± 0.2     | 3.0 ± 0.1   | 3.0 ± 0.1    | 5.3 ± 0.3     | 3.6 ± 0.2     | 3.13 ± 0.10 | 2.3 ± 0.2     | 3.5 ± 0.1   | 3.0 ± 0.2    | 4.1 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 3.9 ± 0.2     | 2.78 ± 0.09 | 2.5 ± 0.1     | 2.84 ± 0.10 | 2.7 ± 0.1    | 5.3 ± 0.3     | 3.6 ± 0.2     | 2.88 ± 0.09 | 2.4 ± 0.2     | 3.4 ± 0.1   | 2.8 ± 0.2    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 3.9 ± 0.2     | 2.49 ± 0.08 | 2.5 ± 0.1     | 2.72 ± 0.09 | 2.6 ± 0.1    | 4.9 ± 0.3     | 4.0 ± 0.2     | 2.69 ± 0.09 | 2.6 ± 0.2     | 3.4 ± 0.1   | 2.5 ± 0.1    | 3.8 ± 0.3     | W Africa F   | E Europe F   |
| 23  | 3.6 ± 0.2     | 2.35 ± 0.07 | 2.5 ± 0.1     | 2.70 ± 0.08 | 2.39 ± 0.10  | 5.0 ± 0.3     | 3.8 ± 0.2     | 2.58 ± 0.09 | 2.6 ± 0.2     | 3.4 ± 0.1   | 2.6 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Asia F     |
| 24  | 3.8 ± 0.2     | 2.32 ± 0.07 | 2.45 ± 0.09   | 2.75 ± 0.07 | 2.42 ± 0.08  | 4.6 ± 0.2     | 3.8 ± 0.2     | 2.54 ± 0.09 | 2.6 ± 0.1     | 3.5 ± 0.1   | 2.4 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Asia F     |
| 25  | 3.8 ± 0.2     | 2.44 ± 0.06 | 2.62 ± 0.08   | 2.97 ± 0.06 | 2.67 ± 0.08  | 4.8 ± 0.2     | 4.0 ± 0.2     | 2.83 ± 0.09 | 2.7 ± 0.1     | 3.49 ± 0.10 | 2.8 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Asia F     |
| 26  | 3.8 ± 0.2     | 2.60 ± 0.05 | 2.74 ± 0.08   | 3.12 ± 0.06 | 2.78 ± 0.07  | 4.5 ± 0.2     | 3.9 ± 0.2     | 2.72 ± 0.08 | 2.9 ± 0.1     | 3.58 ± 0.09 | 2.9 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Asia F     |
| 27  | 3.9 ± 0.2     | 2.69 ± 0.05 | 2.87 ± 0.09   | 3.30 ± 0.06 | 2.86 ± 0.08  | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.87 ± 0.08 | 3.0 ± 0.1     | 3.43 ± 0.08 | 3.0 ± 0.1    | 3.7 ± 0.2     | W Africa F   | E Asia F     |
| 28  | 3.8 ± 0.2     | 2.83 ± 0.05 | 2.81 ± 0.08   | 3.52 ± 0.06 | 3.05 ± 0.08  | 4.2 ± 0.2     | 3.9 ± 0.2     | 2.96 ± 0.08 | 3.0 ± 0.1     | 3.32 ± 0.07 | 3.0 ± 0.1    | 3.7 ± 0.1     | W Africa F   | E Europe F   |
| 29  | 3.9 ± 0.2     | 2.89 ± 0.05 | 2.94 ± 0.09   | 3.64 ± 0.07 | 3.19 ± 0.08  | 3.9 ± 0.2     | 3.8 ± 0.2     | 2.82 ± 0.07 | 2.9 ± 0.1     | 3.19 ± 0.06 | 3.1 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Asia M     |
| 30  | 3.8 ± 0.2     | 3.06 ± 0.06 | 3.00 ± 0.09   | 3.90 ± 0.07 | 3.20 ± 0.09  | 3.9 ± 0.2     | 3.8 ± 0.2     | 2.90 ± 0.07 | 2.8 ± 0.1     | 3.09 ± 0.06 | 3.0 ± 0.1    | 3.7 ± 0.1     | S Asia F     | E Europe M   |
| 31  | 3.8 ± 0.2     | 3.20 ± 0.06 | 3.14 ± 0.10   | 4.10 ± 0.08 | 3.33 ± 0.09  | 3.7 ± 0.1     | 3.8 ± 0.2     | 2.91 ± 0.08 | 2.7 ± 0.1     | 3.02 ± 0.06 | 3.1 ± 0.1    | 3.6 ± 0.1     | S Asia F     | E Europe M   |
| 32  | 3.9 ± 0.2     | 3.33 ± 0.06 | 3.13 ± 0.10   | 4.26 ± 0.08 | 3.4 ± 0.1    | 3.7 ± 0.2     | 3.6 ± 0.2     | 2.89 ± 0.08 | 2.6 ± 0.1     | 2.92 ± 0.06 | 3.0 ± 0.1    | 3.5 ± 0.1     | S Asia F     | E Europe M   |
| 33  | 4.0 ± 0.2     | 3.49 ± 0.07 | 3.3 ± 0.1     | 4.42 ± 0.08 | 3.6 ± 0.1    | 3.5 ± 0.2     | 3.7 ± 0.2     | 2.85 ± 0.08 | 2.4 ± 0.1     | 2.93 ± 0.06 | 3.0 ± 0.1    | 3.4 ± 0.1     | S Asia F     | E Europe M   |
| 34  | 4.0 ± 0.2     | 3.63 ± 0.07 | 3.4 ± 0.1     | 4.66 ± 0.09 | 3.7 ± 0.1    | 3.5 ± 0.2     | 3.6 ± 0.2     | 2.97 ± 0.08 | 2.6 ± 0.1     | 3.06 ± 0.06 | 3.1 ± 0.2    | 3.6 ± 0.1     | S Asia F     | E Europe M   |
| 35  | 4.2 ± 0.2     | 3.92 ± 0.07 | 3.5 ± 0.1     | 4.85 ± 0.09 | 3.9 ± 0.1    | 3.6 ± 0.2     | 3.7 ± 0.2     | 2.96 ± 0.08 | 2.5 ± 0.1     | 3.16 ± 0.07 | 3.2 ± 0.2    | 3.7 ± 0.1     | S Asia F     | E Europe M   |
| 36  | 4.4 ± 0.2     | 4.06 ± 0.08 | 3.5 ± 0.1     | 5.11 ± 0.10 | 3.9 ± 0.1    | 3.5 ± 0.2     | 3.9 ± 0.2     | 3.06 ± 0.08 | 2.5 ± 0.1     | 3.25 ± 0.06 | 3.2 ± 0.2    | 3.6 ± 0.1     | S Asia F     | E Europe M   |
| 37  | 4.4 ± 0.2     | 4.23 ± 0.08 | 3.5 ± 0.1     | 5.2 ± 0.1   | 4.2 ± 0.1    | 3.8 ± 0.2     | 3.8 ± 0.2     | 3.14 ± 0.08 | 2.6 ± 0.1     | 3.33 ± 0.07 | 3.3 ± 0.2    | 3.5 ± 0.1     | S Asia F     | E Europe M   |
| 38  | 4.5 ± 0.2     | 4.38 ± 0.08 | 3.6 ± 0.1     | 5.3 ± 0.1   | 4.3 ± 0.1    | 4.1 ± 0.2     | 3.8 ± 0.2     | 3.20 ± 0.09 | 2.6 ± 0.1     | 3.31 ± 0.07 | 3.4 ± 0.2    | 3.8 ± 0.2     | S Asia F     | E Europe M   |
| 39  | 4.4 ± 0.2     | 4.57 ± 0.08 | 3.4 ± 0.1     | 5.5 ± 0.1   | 4.3 ± 0.1    | 4.0 ± 0.2     | 3.8 ± 0.2     | 3.20 ± 0.09 | 2.7 ± 0.1     | 3.42 ± 0.07 | 3.5 ± 0.2    | 3.9 ± 0.2     | S Asia F     | E Europe M   |
| 40  | 4.8 ± 0.3     | 4.67 ± 0.09 | 3.4 ± 0.1     | 5.5 ± 0.1   | 4.5 ± 0.1    | 4.3 ± 0.2     | 4.0 ± 0.2     | 3.23 ± 0.09 | 2.8 ± 0.1     | 3.42 ± 0.08 | 3.4 ± 0.2    | 3.9 ± 0.2     | S Asia F     | E Europe M   |
| 41  | 5.0 ± 0.3     | 4.78 ± 0.09 | 3.4 ± 0.1     | 5.7 ± 0.1   | 4.5 ± 0.2    | 4.6 ± 0.2     | 4.1 ± 0.2     | 3.35 ± 0.09 | 2.8 ± 0.1     | 3.53 ± 0.09 | 3.6 ± 0.2    | 4.2 ± 0.2     | S Asia F     | E Europe M   |
| 42  | 4.9 ± 0.3     | 4.93 ± 0.09 | 3.5 ± 0.2     | 5.4 ± 0.1   | 4.6 ± 0.2    | 4.6 ± 0.2     | 3.9 ± 0.2     | 3.34 ± 0.09 | 3.0 ± 0.1     | 3.52 ± 0.09 | 3.5 ± 0.2    | 4.2 ± 0.2     | S Asia F     | E Europe M   |
| 43  | 5.0 ± 0.3     | 5.19 ± 0.09 | 3.4 ± 0.2     | 5.7 ± 0.2   | 4.9 ± 0.2    | 4.9 ± 0.2     | 4.1 ± 0.2     | 3.41 ± 0.09 | 3.0 ± 0.2     | 3.52 ± 0.10 | 3.7 ± 0.2    | 4.5 ± 0.2     | S Asia F     | E Europe M   |
| 44  | 5.5 ± 0.3     | 5.41 ± 0.09 | 3.4 ± 0.2     | 5.7 ± 0.2   | 4.9 ± 0.2    | 5.5 ± 0.3     | 3.9 ± 0.2     | 3.53 ± 0.09 | 3.0 ± 0.2     | 3.6 ± 0.1   | 3.7 ± 0.2    | 4.3 ± 0.2     | S Asia F     | E Europe M   |
| 45  | 5.5 ± 0.3     | 5.58 ± 0.10 | 3.4 ± 0.2     | 5.8 ± 0.2   | 5.0 ± 0.2    | 5.3 ± 0.3     | 4.1 ± 0.2     | 3.73 ± 0.10 | 2.9 ± 0.2     | 3.6 ± 0.1   | 3.8 ± 0.2    | 4.7 ± 0.2     | S Asia F     | E Europe M   |
| 46  | 5.6 ± 0.3     | 5.84 ± 0.10 | 3.4 ± 0.2     | 5.6 ± 0.2   | 5.3 ± 0.2    | 5.5 ± 0.3     | 4.3 ± 0.3     | 3.81 ± 0.10 | 3.3 ± 0.2     | 3.6 ± 0.1   | 3.9 ± 0.2    | 4.6 ± 0.2     | E Asia F     | E Europe M   |
| 47  | 5.2 ± 0.4     | 5.8 ± 0.1   | 3.5 ± 0.2     | 5.6 ± 0.2   | 5.3 ± 0.2    | 5.6 ± 0.3     | 4.0 ± 0.2     | 3.9 ± 0.1   | 3.1 ± 0.2     | 3.8 ± 0.1   | 3.8 ± 0.2    | 4.7 ± 0.2     | E Asia F     | E Europe M   |
| 48  | 5.6 ± 0.4     | 5.9 ± 0.1   | 3.4 ± 0.2     | 5.6 ± 0.2   | 5.3 ± 0.2    | 5.8 ± 0.3     | 4.2 ± 0.3     | 3.85 ± 0.10 | 3.2 ± 0.2     | 3.6 ± 0.1   | 3.8 ± 0.2    | 5.0 ± 0.3     | E Asia F     | E Europe M   |
| 49  | 5.4 ± 0.4     | 6.1 ± 0.1   | 3.4 ± 0.2     | 5.5 ± 0.2   | 5.1 ± 0.2    | 6.2 ± 0.3     | 3.9 ± 0.3     | 3.8 ± 0.1   | 3.2 ± 0.2     | 3.6 ± 0.1   | 3.6 ± 0.2    | 4.9 ± 0.3     | W Africa F   | E Europe M   |
| 50  | 5.9 ± 0.5     | 6.1 ± 0.1   | 3.3 ± 0.2     | 5.3 ± 0.2   | 5.5 ± 0.2    | 6.2 ± 0.3     | 3.9 ± 0.3     | 3.8 ± 0.1   | 3.1 ± 0.2     | 3.5 ± 0.1   | 3.6 ± 0.2    | 5.1 ± 0.3     | W Africa F   | E Europe M   |
| 51  | 5.2 ± 0.4     | 6.1 ± 0.1   | 3.3 ± 0.2     | 5.3 ± 0.2   | 5.5 ± 0.2    | 6.3 ± 0.4     | 4.2 ± 0.4     | 3.8 ± 0.1   | 2.8 ± 0.2     | 3.5 ± 0.2   | 3.6 ± 0.2    | 5.3 ± 0.3     | W Africa F   | E Europe M   |
| 52  | 5.4 ± 0.4     | 6.2 ± 0.1   | 3.4 ± 0.2     | 5.5 ± 0.2   | 5.3 ± 0.2    | 6.7 ± 0.4     | 3.8 ± 0.4     | 3.8 ± 0.1   | 3.1 ± 0.2     | 3.4 ± 0.1   | 3.5 ± 0.2    | 5.1 ± 0.3     | W Africa F   | E Europe M   |

**Table 95.** Mean absolute error for Application-type images for algorithm nominder-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.2 ± 0.2     | 4.1 ± 0.1   | 3.8 ± 0.2     | 3.7 ± 0.1   | 3.8 ± 0.2    | 5.4 ± 0.3     | 3.0 ± 0.2     | 3.2 ± 0.1   | 2.5 ± 0.2     | 2.47 ± 0.09 | 2.8 ± 0.2    | 3.2 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 4.0 ± 0.2     | 4.1 ± 0.1   | 3.4 ± 0.2     | 3.8 ± 0.2   | 4.0 ± 0.2    | 5.3 ± 0.3     | 2.7 ± 0.2     | 3.5 ± 0.1   | 2.3 ± 0.2     | 2.49 ± 0.10 | 3.2 ± 0.2    | 3.1 ± 0.2     | W Africa F   | E Europe M   |
| 16  | 3.8 ± 0.2     | 4.0 ± 0.1   | 3.4 ± 0.2     | 3.6 ± 0.1   | 3.9 ± 0.2    | 5.3 ± 0.4     | 2.7 ± 0.2     | 3.28 ± 0.09 | 2.1 ± 0.2     | 2.5 ± 0.1   | 3.0 ± 0.2    | 2.6 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.9 ± 0.3     | 3.6 ± 0.1   | 2.9 ± 0.2     | 3.4 ± 0.1   | 3.4 ± 0.2    | 5.0 ± 0.3     | 2.3 ± 0.2     | 3.1 ± 0.1   | 1.7 ± 0.2     | 2.4 ± 0.1   | 2.7 ± 0.2    | 2.8 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.3 ± 0.2     | 3.14 ± 0.08 | 2.5 ± 0.2     | 2.85 ± 0.09 | 3.1 ± 0.1    | 4.7 ± 0.3     | 2.3 ± 0.2     | 2.82 ± 0.08 | 1.6 ± 0.1     | 2.21 ± 0.09 | 2.5 ± 0.2    | 2.5 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 3.4 ± 0.2     | 2.71 ± 0.08 | 2.3 ± 0.2     | 2.6 ± 0.1   | 2.7 ± 0.1    | 4.7 ± 0.3     | 2.4 ± 0.2     | 2.61 ± 0.09 | 1.6 ± 0.1     | 2.17 ± 0.09 | 2.2 ± 0.1    | 2.7 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 3.2 ± 0.2     | 2.46 ± 0.09 | 2.0 ± 0.1     | 2.3 ± 0.1   | 2.4 ± 0.1    | 4.3 ± 0.3     | 2.6 ± 0.2     | 2.32 ± 0.08 | 1.7 ± 0.2     | 2.24 ± 0.10 | 2.2 ± 0.1    | 2.9 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 3.0 ± 0.2     | 2.08 ± 0.07 | 2.0 ± 0.1     | 2.17 ± 0.09 | 2.14 ± 0.10  | 4.1 ± 0.3     | 2.7 ± 0.2     | 2.17 ± 0.07 | 1.8 ± 0.1     | 2.23 ± 0.09 | 2.1 ± 0.1    | 3.0 ± 0.2     | W Africa F   | E Europe M   |
| 22  | 3.1 ± 0.2     | 1.80 ± 0.06 | 2.05 ± 0.10   | 2.06 ± 0.07 | 2.07 ± 0.09  | 3.8 ± 0.2     | 3.3 ± 0.2     | 2.05 ± 0.07 | 1.9 ± 0.1     | 2.35 ± 0.10 | 1.8 ± 0.1    | 2.9 ± 0.2     | W Africa F   | E Asia F     |
| 23  | 2.9 ± 0.2     | 1.78 ± 0.06 | 2.15 ± 0.09   | 2.12 ± 0.07 | 1.98 ± 0.08  | 3.9 ± 0.2     | 3.4 ± 0.2     | 2.00 ± 0.07 | 1.9 ± 0.1     | 2.46 ± 0.09 | 2.0 ± 0.1    | 3.2 ± 0.2     | W Africa F   | E Asia F     |
| 24  | 3.0 ± 0.2     | 1.88 ± 0.05 | 2.20 ± 0.07   | 2.28 ± 0.06 | 2.07 ± 0.07  | 3.7 ± 0.2     | 3.4 ± 0.2     | 2.01 ± 0.07 | 2.1 ± 0.1     | 2.60 ± 0.09 | 1.92 ± 0.09  | 3.2 ± 0.2     | W Africa F   | E Asia F     |
| 25  | 3.1 ± 0.1     | 2.06 ± 0.05 | 2.39 ± 0.07   | 2.47 ± 0.05 | 2.31 ± 0.07  | 3.7 ± 0.2     | 3.5 ± 0.2     | 2.16 ± 0.07 | 2.1 ± 0.1     | 2.63 ± 0.08 | 2.08 ± 0.09  | 3.3 ± 0.2     | W Africa F   | E Asia F     |
| 26  | 3.1 ± 0.1     | 2.31 ± 0.05 | 2.54 ± 0.07   | 2.70 ± 0.05 | 2.50 ± 0.07  | 3.6 ± 0.2     | 3.4 ± 0.2     | 2.07 ± 0.06 | 2.17 ± 0.10   | 2.76 ± 0.07 | 2.19 ± 0.09  | 3.3 ± 0.2     | W Africa F   | E Asia M     |
| 27  | 3.2 ± 0.1     | 2.51 ± 0.05 | 2.67 ± 0.08   | 2.94 ± 0.05 | 2.71 ± 0.07  | 3.8 ± 0.2     | 3.4 ± 0.2     | 2.27 ± 0.06 | 2.31 ± 0.10   | 2.73 ± 0.07 | 2.34 ± 0.10  | 3.3 ± 0.2     | W Africa F   | E Asia M     |
| 28  | 3.2 ± 0.1     | 2.79 ± 0.05 | 2.78 ± 0.07   | 3.21 ± 0.05 | 2.84 ± 0.07  | 3.6 ± 0.1     | 3.4 ± 0.1     | 2.35 ± 0.07 | 2.40 ± 0.10   | 2.72 ± 0.06 | 2.39 ± 0.10  | 3.3 ± 0.1     | W Africa F   | E Asia M     |
| 29  | 3.4 ± 0.1     | 3.02 ± 0.05 | 2.83 ± 0.08   | 3.38 ± 0.06 | 3.07 ± 0.08  | 3.6 ± 0.1     | 3.5 ± 0.1     | 2.37 ± 0.06 | 2.44 ± 0.10   | 2.72 ± 0.06 | 2.5 ± 0.1    | 3.4 ± 0.1     | W Africa F   | E Asia M     |
| 30  | 3.3 ± 0.1     | 3.29 ± 0.05 | 2.91 ± 0.08   | 3.59 ± 0.06 | 3.19 ± 0.09  | 3.6 ± 0.1     | 3.4 ± 0.1     | 2.48 ± 0.06 | 2.40 ± 0.09   | 2.77 ± 0.05 | 2.6 ± 0.1    | 3.4 ± 0.1     | W Africa F   | E Europe M   |
| 31  | 3.6 ± 0.1     | 3.54 ± 0.06 | 3.05 ± 0.09   | 3.83 ± 0.07 | 3.30 ± 0.09  | 3.5 ± 0.1     | 3.4 ± 0.1     | 2.61 ± 0.07 | 2.46 ± 0.09   | 2.78 ± 0.05 | 2.7 ± 0.1    | 3.4 ± 0.1     | S Asia F     | E Europe M   |
| 32  | 3.7 ± 0.1     | 3.74 ± 0.06 | 3.11 ± 0.09   | 4.00 ± 0.07 | 3.48 ± 0.10  | 3.6 ± 0.1     | 3.3 ± 0.2     | 2.72 ± 0.07 | 2.5 ± 0.1     | 2.74 ± 0.05 | 2.7 ± 0.1    | 3.3 ± 0.1     | S Asia F     | E Europe M   |
| 33  | 3.7 ± 0.1     | 4.04 ± 0.07 | 3.18 ± 0.10   | 4.04 ± 0.07 | 3.6 ± 0.1    | 3.4 ± 0.1     | 3.6 ± 0.2     | 2.84 ± 0.07 | 2.31 ± 0.10   | 2.74 ± 0.06 | 2.8 ± 0.1    | 3.3 ± 0.1     | S Asia F     | E Europe M   |
| 34  | 3.7 ± 0.2     | 4.19 ± 0.07 | 3.3 ± 0.1     | 4.16 ± 0.08 | 3.6 ± 0.1    | 3.4 ± 0.2     | 3.4 ± 0.2     | 3.07 ± 0.08 | 2.5 ± 0.1     | 2.86 ± 0.06 | 2.9 ± 0.1    | 3.3 ± 0.1     | E Asia F     | E Europe M   |
| 35  | 4.0 ± 0.2     | 4.50 ± 0.07 | 3.4 ± 0.1     | 4.28 ± 0.08 | 3.9 ± 0.1    | 3.5 ± 0.2     | 3.5 ± 0.2     | 3.09 ± 0.08 | 2.4 ± 0.1     | 2.93 ± 0.06 | 3.2 ± 0.1    | 3.5 ± 0.1     | E Asia F     | E Europe M   |
| 36  | 4.0 ± 0.2     | 4.58 ± 0.08 | 3.4 ± 0.1     | 4.42 ± 0.09 | 3.9 ± 0.1    | 3.6 ± 0.2     | 3.6 ± 0.2     | 3.21 ± 0.08 | 2.4 ± 0.1     | 3.00 ± 0.06 | 3.1 ± 0.1    | 3.6 ± 0.1     | E Asia F     | E Europe M   |
| 37  | 4.0 ± 0.2     | 4.72 ± 0.08 | 3.4 ± 0.1     | 4.47 ± 0.09 | 3.9 ± 0.1    | 3.8 ± 0.2     | 3.7 ± 0.2     | 3.41 ± 0.08 | 2.5 ± 0.1     | 3.06 ± 0.06 | 3.3 ± 0.2    | 3.4 ± 0.1     | E Asia F     | E Europe M   |
| 38  | 3.9 ± 0.2     | 4.96 ± 0.08 | 3.4 ± 0.1     | 4.54 ± 0.10 | 4.0 ± 0.1    | 3.8 ± 0.2     | 3.5 ± 0.2     | 3.50 ± 0.09 | 2.4 ± 0.1     | 3.13 ± 0.07 | 3.3 ± 0.2    | 3.5 ± 0.2     | E Asia F     | E Europe M   |
| 39  | 4.0 ± 0.2     | 5.01 ± 0.09 | 3.5 ± 0.1     | 4.7 ± 0.1   | 4.1 ± 0.1    | 3.9 ± 0.2     | 3.6 ± 0.2     | 3.68 ± 0.09 | 2.6 ± 0.1     | 3.13 ± 0.07 | 3.3 ± 0.1    | 3.5 ± 0.2     | E Asia F     | E Europe M   |
| 40  | 4.1 ± 0.2     | 5.11 ± 0.09 | 3.4 ± 0.1     | 4.6 ± 0.1   | 4.1 ± 0.1    | 3.9 ± 0.2     | 3.7 ± 0.2     | 3.71 ± 0.09 | 2.7 ± 0.1     | 3.28 ± 0.07 | 3.3 ± 0.1    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 41  | 4.3 ± 0.2     | 5.18 ± 0.09 | 3.5 ± 0.1     | 4.9 ± 0.1   | 4.1 ± 0.1    | 4.2 ± 0.2     | 3.8 ± 0.2     | 3.81 ± 0.09 | 2.7 ± 0.1     | 3.30 ± 0.08 | 3.6 ± 0.2    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 42  | 4.1 ± 0.2     | 5.17 ± 0.09 | 3.5 ± 0.1     | 4.5 ± 0.1   | 4.0 ± 0.1    | 4.1 ± 0.2     | 3.7 ± 0.2     | 3.82 ± 0.09 | 2.7 ± 0.1     | 3.36 ± 0.09 | 3.5 ± 0.2    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 4.3 ± 0.3     | 5.38 ± 0.09 | 3.6 ± 0.2     | 4.8 ± 0.1   | 4.3 ± 0.2    | 4.2 ± 0.2     | 3.8 ± 0.2     | 3.98 ± 0.09 | 2.7 ± 0.1     | 3.37 ± 0.09 | 3.6 ± 0.2    | 3.9 ± 0.2     | E Asia F     | E Europe M   |
| 44  | 4.5 ± 0.3     | 5.45 ± 0.10 | 3.7 ± 0.2     | 4.8 ± 0.1   | 4.3 ± 0.2    | 4.5 ± 0.3     | 3.7 ± 0.2     | 4.04 ± 0.10 | 2.8 ± 0.1     | 3.47 ± 0.10 | 3.7 ± 0.2    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 4.5 ± 0.3     | 5.47 ± 0.10 | 3.5 ± 0.2     | 4.8 ± 0.1   | 4.3 ± 0.2    | 4.3 ± 0.2     | 3.8 ± 0.2     | 4.16 ± 0.10 | 2.9 ± 0.1     | 3.51 ± 0.10 | 3.7 ± 0.2    | 4.0 ± 0.2     | E Asia F     | E Europe M   |
| 46  | 4.7 ± 0.3     | 5.7 ± 0.1   | 3.4 ± 0.2     | 4.7 ± 0.1   | 4.5 ± 0.2    | 4.4 ± 0.2     | 4.0 ± 0.3     | 4.24 ± 0.10 | 3.0 ± 0.1     | 3.5 ± 0.1   | 3.8 ± 0.2    | 3.8 ± 0.2     | E Asia F     | E Europe M   |
| 47  | 4.4 ± 0.3     | 5.5 ± 0.1   | 3.7 ± 0.2     | 4.7 ± 0.2   | 4.4 ± 0.2    | 4.4 ± 0.2     | 3.9 ± 0.2     | 4.2 ± 0.1   | 3.0 ± 0.2     | 3.7 ± 0.1   | 3.7 ± 0.2    | 4.1 ± 0.2     | E Asia F     | E Europe M   |
| 48  | 4.7 ± 0.3     | 5.5 ± 0.1   | 3.5 ± 0.2     | 4.8 ± 0.2   | 4.4 ± 0.2    | 4.5 ± 0.2     | 4.1 ± 0.3     | 4.3 ± 0.1   | 3.0 ± 0.2     | 3.6 ± 0.1   | 3.7 ± 0.2    | 4.0 ± 0.2     | E Asia F     | E Europe M   |
| 49  | 4.8 ± 0.4     | 5.6 ± 0.1   | 3.5 ± 0.2     | 4.6 ± 0.2   | 4.2 ± 0.2    | 4.9 ± 0.3     | 3.7 ± 0.3     | 4.3 ± 0.1   | 3.1 ± 0.2     | 3.6 ± 0.1   | 3.6 ± 0.2    | 3.8 ± 0.2     | E Asia F     | E Europe M   |
| 50  | 4.8 ± 0.4     | 5.5 ± 0.1   | 3.3 ± 0.2     | 4.6 ± 0.2   | 4.5 ± 0.2    | 4.7 ± 0.3     | 3.6 ± 0.3     | 4.2 ± 0.1   | 3.0 ± 0.2     | 3.5 ± 0.1   | 3.4 ± 0.2    | 3.8 ± 0.2     | E Asia F     | E Europe M   |
| 51  | 4.3 ± 0.4     | 5.4 ± 0.1   | 3.4 ± 0.2     | 4.5 ± 0.2   | 4.4 ± 0.2    | 4.8 ± 0.3     | 3.8 ± 0.3     | 4.3 ± 0.1   | 2.7 ± 0.2     | 3.6 ± 0.1   | 3.5 ± 0.2    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 52  | 5.0 ± 0.4     | 5.4 ± 0.1   | 3.3 ± 0.2     | 4.7 ± 0.2   | 4.3 ± 0.2    | 4.7 ± 0.3     | 3.6 ± 0.3     | 4.3 ± 0.1   | 3.0 ± 0.2     | 3.5 ± 0.1   | 3.6 ± 0.2    | 3.8 ± 0.3     | E Asia F     | E Europe M   |

**Table 96.** Mean absolute error for Application-type images for algorithm ondato-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.5 ± 0.3     | 5.0 ± 0.1   | 3.5 ± 0.2     | 4.6 ± 0.1   | 4.9 ± 0.2    | 5.8 ± 0.4     | 4.1 ± 0.3     | 3.3 ± 0.1   | 2.3 ± 0.2     | 2.8 ± 0.1   | 3.0 ± 0.2    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 5.4 ± 0.3     | 4.9 ± 0.1   | 3.4 ± 0.2     | 4.4 ± 0.2   | 4.8 ± 0.3    | 5.9 ± 0.5     | 3.4 ± 0.3     | 3.1 ± 0.1   | 1.9 ± 0.2     | 2.5 ± 0.1   | 3.0 ± 0.2    | 3.6 ± 0.4     | W Africa F   | E Europe M   |
| 16  | 4.9 ± 0.3     | 4.6 ± 0.1   | 3.6 ± 0.3     | 4.1 ± 0.2   | 4.7 ± 0.3    | 6.0 ± 0.5     | 3.2 ± 0.3     | 2.9 ± 0.1   | 1.9 ± 0.2     | 2.5 ± 0.1   | 2.8 ± 0.2    | 3.4 ± 0.4     | W Africa F   | E Europe M   |
| 17  | 5.1 ± 0.4     | 4.2 ± 0.1   | 3.3 ± 0.3     | 3.9 ± 0.2   | 4.3 ± 0.3    | 5.8 ± 0.4     | 3.1 ± 0.3     | 3.0 ± 0.1   | 1.8 ± 0.2     | 2.8 ± 0.2   | 2.9 ± 0.2    | 3.7 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.6 ± 0.3     | 3.8 ± 0.1   | 2.9 ± 0.2     | 3.5 ± 0.1   | 3.8 ± 0.2    | 5.6 ± 0.4     | 3.3 ± 0.2     | 2.82 ± 0.09 | 2.0 ± 0.1     | 2.7 ± 0.1   | 2.9 ± 0.2    | 3.7 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.6 ± 0.3     | 3.50 ± 0.10 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.4 ± 0.2    | 5.5 ± 0.4     | 3.8 ± 0.3     | 2.83 ± 0.09 | 2.1 ± 0.1     | 2.9 ± 0.1   | 2.9 ± 0.2    | 4.0 ± 0.4     | W Africa F   | E Europe M   |
| 20  | 4.5 ± 0.3     | 3.2 ± 0.1   | 2.7 ± 0.2     | 3.1 ± 0.1   | 3.2 ± 0.1    | 5.5 ± 0.4     | 4.0 ± 0.3     | 2.68 ± 0.09 | 2.1 ± 0.2     | 3.0 ± 0.1   | 2.9 ± 0.1    | 4.3 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 4.2 ± 0.2     | 2.96 ± 0.10 | 2.5 ± 0.1     | 2.92 ± 0.10 | 3.0 ± 0.1    | 5.5 ± 0.4     | 4.3 ± 0.3     | 2.59 ± 0.08 | 2.0 ± 0.1     | 3.1 ± 0.1   | 2.9 ± 0.1    | 4.5 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.5 ± 0.3     | 2.66 ± 0.08 | 2.6 ± 0.1     | 2.73 ± 0.09 | 2.8 ± 0.1    | 5.3 ± 0.3     | 4.8 ± 0.3     | 2.51 ± 0.09 | 2.1 ± 0.1     | 3.1 ± 0.1   | 2.6 ± 0.1    | 4.3 ± 0.3     | W Africa F   | E Europe M   |
| 23  | 4.4 ± 0.2     | 2.43 ± 0.07 | 2.6 ± 0.1     | 2.74 ± 0.08 | 2.66 ± 0.10  | 5.4 ± 0.3     | 4.7 ± 0.3     | 2.43 ± 0.08 | 2.0 ± 0.1     | 3.1 ± 0.1   | 2.6 ± 0.1    | 4.5 ± 0.3     | W Africa F   | E Europe M   |
| 24  | 4.5 ± 0.2     | 2.26 ± 0.06 | 2.51 ± 0.09   | 2.67 ± 0.07 | 2.50 ± 0.08  | 5.0 ± 0.3     | 4.4 ± 0.2     | 2.34 ± 0.08 | 2.1 ± 0.1     | 3.1 ± 0.1   | 2.5 ± 0.1    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 4.5 ± 0.2     | 2.25 ± 0.05 | 2.48 ± 0.08   | 2.71 ± 0.06 | 2.50 ± 0.08  | 5.2 ± 0.3     | 4.7 ± 0.2     | 2.34 ± 0.08 | 2.2 ± 0.1     | 3.07 ± 0.09 | 2.4 ± 0.1    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 4.5 ± 0.2     | 2.23 ± 0.05 | 2.50 ± 0.08   | 2.77 ± 0.06 | 2.53 ± 0.08  | 5.0 ± 0.2     | 4.5 ± 0.2     | 2.26 ± 0.07 | 2.2 ± 0.1     | 3.00 ± 0.08 | 2.5 ± 0.1    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 27  | 4.6 ± 0.2     | 2.28 ± 0.05 | 2.65 ± 0.08   | 2.90 ± 0.06 | 2.55 ± 0.08  | 5.1 ± 0.2     | 4.5 ± 0.2     | 2.35 ± 0.07 | 2.26 ± 0.10   | 2.99 ± 0.08 | 2.6 ± 0.1    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 4.3 ± 0.2     | 2.37 ± 0.05 | 2.65 ± 0.08   | 3.14 ± 0.06 | 2.70 ± 0.08  | 4.9 ± 0.2     | 4.4 ± 0.2     | 2.45 ± 0.08 | 2.35 ± 0.10   | 2.97 ± 0.07 | 2.7 ± 0.1    | 4.2 ± 0.2     | W Africa F   | E Europe M   |
| 29  | 4.6 ± 0.2     | 2.60 ± 0.05 | 2.80 ± 0.08   | 3.28 ± 0.06 | 2.87 ± 0.08  | 4.9 ± 0.2     | 4.4 ± 0.2     | 2.53 ± 0.07 | 2.42 ± 0.09   | 3.00 ± 0.07 | 2.7 ± 0.1    | 4.7 ± 0.2     | W Africa F   | E Europe M   |
| 30  | 4.7 ± 0.2     | 2.78 ± 0.05 | 2.83 ± 0.08   | 3.47 ± 0.06 | 3.02 ± 0.08  | 4.9 ± 0.2     | 4.4 ± 0.2     | 2.63 ± 0.07 | 2.5 ± 0.1     | 3.06 ± 0.06 | 2.8 ± 0.1    | 4.5 ± 0.2     | W Africa F   | E Europe M   |
| 31  | 4.7 ± 0.2     | 3.00 ± 0.05 | 2.96 ± 0.09   | 3.67 ± 0.07 | 3.15 ± 0.09  | 5.1 ± 0.2     | 4.5 ± 0.2     | 2.76 ± 0.08 | 2.6 ± 0.1     | 3.16 ± 0.06 | 3.0 ± 0.1    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 32  | 4.9 ± 0.2     | 3.27 ± 0.06 | 3.07 ± 0.09   | 3.88 ± 0.07 | 3.33 ± 0.10  | 5.2 ± 0.2     | 4.2 ± 0.2     | 2.82 ± 0.07 | 2.6 ± 0.1     | 3.21 ± 0.06 | 3.0 ± 0.1    | 4.6 ± 0.2     | W Africa F   | E Europe M   |
| 33  | 5.1 ± 0.2     | 3.47 ± 0.06 | 3.2 ± 0.1     | 4.07 ± 0.08 | 3.34 ± 0.10  | 5.1 ± 0.2     | 4.7 ± 0.2     | 2.93 ± 0.08 | 2.6 ± 0.1     | 3.21 ± 0.06 | 3.0 ± 0.1    | 4.6 ± 0.2     | W Africa F   | E Europe M   |
| 34  | 4.9 ± 0.2     | 3.68 ± 0.07 | 3.3 ± 0.1     | 4.15 ± 0.08 | 3.4 ± 0.1    | 5.2 ± 0.2     | 4.5 ± 0.2     | 3.02 ± 0.08 | 2.7 ± 0.1     | 3.34 ± 0.07 | 3.0 ± 0.1    | 4.6 ± 0.2     | W Africa F   | E Europe M   |
| 35  | 5.2 ± 0.2     | 3.90 ± 0.07 | 3.4 ± 0.1     | 4.34 ± 0.08 | 3.7 ± 0.1    | 4.9 ± 0.2     | 4.7 ± 0.2     | 3.11 ± 0.08 | 2.6 ± 0.1     | 3.41 ± 0.07 | 3.3 ± 0.2    | 4.6 ± 0.2     | E Africa F   | E Europe M   |
| 36  | 5.0 ± 0.2     | 4.02 ± 0.07 | 3.5 ± 0.1     | 4.49 ± 0.09 | 3.6 ± 0.1    | 5.2 ± 0.2     | 4.7 ± 0.2     | 3.09 ± 0.08 | 2.8 ± 0.1     | 3.36 ± 0.07 | 3.2 ± 0.2    | 4.7 ± 0.2     | W Africa F   | E Europe M   |
| 37  | 5.1 ± 0.2     | 4.12 ± 0.08 | 3.5 ± 0.1     | 4.49 ± 0.10 | 3.5 ± 0.1    | 5.2 ± 0.2     | 4.6 ± 0.2     | 3.07 ± 0.09 | 2.8 ± 0.1     | 3.46 ± 0.07 | 3.2 ± 0.2    | 4.5 ± 0.2     | W Africa F   | E Europe M   |
| 38  | 5.1 ± 0.3     | 4.33 ± 0.08 | 3.5 ± 0.1     | 4.6 ± 0.1   | 3.8 ± 0.1    | 5.0 ± 0.2     | 4.5 ± 0.2     | 3.23 ± 0.09 | 2.6 ± 0.1     | 3.40 ± 0.07 | 3.3 ± 0.2    | 4.5 ± 0.2     | E Africa F   | E Europe M   |
| 39  | 5.0 ± 0.2     | 4.34 ± 0.08 | 3.5 ± 0.1     | 4.6 ± 0.1   | 3.8 ± 0.1    | 5.2 ± 0.2     | 4.5 ± 0.2     | 3.24 ± 0.09 | 2.9 ± 0.1     | 3.39 ± 0.08 | 3.4 ± 0.2    | 4.4 ± 0.2     | W Africa F   | E Europe M   |
| 40  | 5.1 ± 0.3     | 4.49 ± 0.09 | 3.6 ± 0.1     | 4.6 ± 0.1   | 3.9 ± 0.1    | 5.1 ± 0.2     | 4.4 ± 0.2     | 3.30 ± 0.09 | 2.8 ± 0.1     | 3.59 ± 0.08 | 3.2 ± 0.2    | 4.4 ± 0.2     | E Africa F   | E Europe M   |
| 41  | 5.0 ± 0.3     | 4.54 ± 0.09 | 3.6 ± 0.1     | 4.6 ± 0.1   | 4.0 ± 0.1    | 4.9 ± 0.2     | 4.5 ± 0.2     | 3.34 ± 0.09 | 3.0 ± 0.1     | 3.54 ± 0.08 | 3.4 ± 0.2    | 4.5 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 5.0 ± 0.3     | 4.44 ± 0.09 | 3.7 ± 0.2     | 4.5 ± 0.1   | 3.7 ± 0.1    | 4.8 ± 0.2     | 4.5 ± 0.2     | 3.35 ± 0.09 | 3.1 ± 0.1     | 3.54 ± 0.09 | 3.4 ± 0.2    | 4.5 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 4.9 ± 0.3     | 4.56 ± 0.09 | 3.7 ± 0.2     | 4.3 ± 0.1   | 4.1 ± 0.2    | 4.8 ± 0.2     | 4.6 ± 0.2     | 3.38 ± 0.09 | 3.0 ± 0.1     | 3.52 ± 0.09 | 3.5 ± 0.2    | 4.5 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 5.1 ± 0.3     | 4.62 ± 0.09 | 3.5 ± 0.2     | 4.4 ± 0.1   | 3.9 ± 0.1    | 4.8 ± 0.3     | 4.2 ± 0.2     | 3.41 ± 0.09 | 3.0 ± 0.2     | 3.50 ± 0.10 | 3.5 ± 0.2    | 4.3 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.8 ± 0.3     | 4.49 ± 0.09 | 3.5 ± 0.2     | 4.4 ± 0.2   | 3.8 ± 0.2    | 4.4 ± 0.2     | 4.3 ± 0.2     | 3.44 ± 0.09 | 3.0 ± 0.2     | 3.5 ± 0.1   | 3.5 ± 0.2    | 4.3 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 5.0 ± 0.3     | 4.5 ± 0.1   | 3.3 ± 0.2     | 4.1 ± 0.1   | 3.8 ± 0.2    | 4.2 ± 0.2     | 4.4 ± 0.3     | 3.36 ± 0.09 | 3.1 ± 0.2     | 3.4 ± 0.1   | 3.5 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 4.5 ± 0.3     | 4.2 ± 0.1   | 3.5 ± 0.2     | 3.9 ± 0.2   | 3.8 ± 0.2    | 4.4 ± 0.3     | 4.3 ± 0.3     | 3.25 ± 0.09 | 3.0 ± 0.2     | 3.5 ± 0.1   | 3.4 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.5 ± 0.4     | 4.1 ± 0.1   | 3.3 ± 0.2     | 3.9 ± 0.2   | 3.6 ± 0.2    | 4.1 ± 0.2     | 4.4 ± 0.3     | 3.27 ± 0.09 | 3.1 ± 0.2     | 3.4 ± 0.1   | 3.3 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 49  | 4.2 ± 0.3     | 4.1 ± 0.1   | 3.2 ± 0.2     | 3.6 ± 0.2   | 3.6 ± 0.2    | 4.2 ± 0.3     | 4.1 ± 0.3     | 3.23 ± 0.09 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.4 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 50  | 4.5 ± 0.4     | 3.9 ± 0.1   | 3.1 ± 0.2     | 3.6 ± 0.1   | 3.4 ± 0.2    | 3.9 ± 0.3     | 3.6 ± 0.3     | 3.00 ± 0.09 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.2 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 51  | 4.2 ± 0.3     | 3.8 ± 0.1   | 3.0 ± 0.2     | 3.5 ± 0.1   | 3.3 ± 0.2    | 4.1 ± 0.3     | 4.0 ± 0.4     | 3.03 ± 0.09 | 2.9 ± 0.2     | 3.4 ± 0.1   | 3.2 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 52  | 4.4 ± 0.4     | 3.7 ± 0.1   | 3.1 ± 0.2     | 3.6 ± 0.1   | 3.4 ± 0.2    | 3.6 ± 0.3     | 3.7 ± 0.3     | 2.98 ± 0.09 | 2.9 ± 0.2     | 3.1 ± 0.1   | 3.1 ± 0.2    | 3.7 ± 0.3     | E Africa F   | E Europe M   |

**Table 97.** Mean absolute error for Application-type images for algorithm ondato-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.0 ± 0.3     | 4.98 ± 0.10 | 3.8 ± 0.2     | 4.4 ± 0.1   | 4.7 ± 0.2    | 5.8 ± 0.3     | 4.0 ± 0.2     | 3.61 ± 0.09 | 2.4 ± 0.2     | 2.70 ± 0.10 | 3.1 ± 0.2    | 3.7 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 5.2 ± 0.3     | 4.9 ± 0.1   | 3.7 ± 0.2     | 4.5 ± 0.2   | 4.8 ± 0.3    | 5.3 ± 0.4     | 3.4 ± 0.2     | 3.41 ± 0.09 | 2.1 ± 0.2     | 2.7 ± 0.1   | 3.1 ± 0.2    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 16  | 4.8 ± 0.3     | 4.6 ± 0.1   | 3.6 ± 0.2     | 4.2 ± 0.2   | 4.5 ± 0.3    | 5.4 ± 0.4     | 3.3 ± 0.3     | 3.12 ± 0.09 | 2.1 ± 0.2     | 2.6 ± 0.1   | 2.9 ± 0.2    | 3.2 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 4.8 ± 0.3     | 4.2 ± 0.1   | 3.3 ± 0.2     | 4.0 ± 0.2   | 4.3 ± 0.3    | 5.2 ± 0.3     | 3.2 ± 0.3     | 2.90 ± 0.10 | 1.8 ± 0.1     | 2.7 ± 0.1   | 2.9 ± 0.2    | 3.4 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.3 ± 0.2     | 3.79 ± 0.09 | 3.0 ± 0.2     | 3.6 ± 0.1   | 4.0 ± 0.2    | 4.9 ± 0.3     | 3.1 ± 0.2     | 2.56 ± 0.07 | 1.9 ± 0.1     | 2.65 ± 0.10 | 2.6 ± 0.2    | 3.2 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 4.3 ± 0.3     | 3.47 ± 0.10 | 2.9 ± 0.2     | 3.4 ± 0.1   | 3.6 ± 0.1    | 4.8 ± 0.3     | 3.3 ± 0.2     | 2.39 ± 0.08 | 1.9 ± 0.1     | 2.6 ± 0.1   | 2.6 ± 0.2    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.2 ± 0.3     | 3.2 ± 0.1   | 2.7 ± 0.2     | 3.3 ± 0.1   | 3.4 ± 0.1    | 4.5 ± 0.3     | 3.5 ± 0.2     | 2.27 ± 0.08 | 1.9 ± 0.2     | 2.8 ± 0.1   | 2.5 ± 0.1    | 3.7 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 3.9 ± 0.2     | 3.08 ± 0.10 | 2.5 ± 0.1     | 2.92 ± 0.10 | 3.3 ± 0.1    | 4.4 ± 0.3     | 3.6 ± 0.2     | 2.16 ± 0.07 | 1.9 ± 0.1     | 2.70 ± 0.10 | 2.5 ± 0.1    | 3.6 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.0 ± 0.2     | 2.80 ± 0.08 | 2.5 ± 0.1     | 2.71 ± 0.08 | 3.0 ± 0.1    | 4.0 ± 0.2     | 4.1 ± 0.2     | 2.10 ± 0.08 | 2.0 ± 0.1     | 2.7 ± 0.1   | 2.2 ± 0.1    | 3.4 ± 0.2     | E Africa M   | E Europe M   |
| 23  | 3.9 ± 0.2     | 2.62 ± 0.07 | 2.53 ± 0.10   | 2.68 ± 0.08 | 2.82 ± 0.09  | 4.1 ± 0.2     | 3.8 ± 0.2     | 2.10 ± 0.07 | 1.9 ± 0.1     | 2.67 ± 0.09 | 2.3 ± 0.1    | 3.6 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 3.9 ± 0.2     | 2.46 ± 0.06 | 2.46 ± 0.08   | 2.58 ± 0.07 | 2.65 ± 0.08  | 3.8 ± 0.2     | 3.6 ± 0.2     | 2.09 ± 0.07 | 1.97 ± 0.10   | 2.70 ± 0.09 | 2.3 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 3.8 ± 0.2     | 2.37 ± 0.06 | 2.43 ± 0.08   | 2.54 ± 0.06 | 2.68 ± 0.08  | 3.8 ± 0.2     | 3.8 ± 0.2     | 2.12 ± 0.07 | 1.99 ± 0.10   | 2.60 ± 0.08 | 2.2 ± 0.1    | 3.4 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 3.9 ± 0.2     | 2.38 ± 0.06 | 2.42 ± 0.07   | 2.56 ± 0.05 | 2.62 ± 0.08  | 3.7 ± 0.2     | 3.5 ± 0.2     | 2.06 ± 0.07 | 2.03 ± 0.10   | 2.58 ± 0.07 | 2.2 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Europe M   |
| 27  | 4.0 ± 0.2     | 2.32 ± 0.05 | 2.50 ± 0.08   | 2.70 ± 0.05 | 2.55 ± 0.08  | 3.8 ± 0.2     | 3.6 ± 0.2     | 2.18 ± 0.07 | 2.10 ± 0.09   | 2.57 ± 0.07 | 2.4 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Europe M   |
| 28  | 3.9 ± 0.2     | 2.35 ± 0.05 | 2.45 ± 0.07   | 2.88 ± 0.06 | 2.63 ± 0.07  | 3.8 ± 0.2     | 3.4 ± 0.2     | 2.26 ± 0.07 | 2.21 ± 0.09   | 2.60 ± 0.06 | 2.5 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 4.1 ± 0.2     | 2.45 ± 0.05 | 2.56 ± 0.07   | 3.02 ± 0.05 | 2.74 ± 0.08  | 3.9 ± 0.2     | 3.5 ± 0.2     | 2.34 ± 0.06 | 2.24 ± 0.09   | 2.67 ± 0.06 | 2.4 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 4.1 ± 0.2     | 2.60 ± 0.05 | 2.56 ± 0.08   | 3.16 ± 0.06 | 2.82 ± 0.08  | 3.9 ± 0.2     | 3.5 ± 0.2     | 2.40 ± 0.06 | 2.30 ± 0.09   | 2.73 ± 0.05 | 2.6 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.1 ± 0.2     | 2.72 ± 0.05 | 2.64 ± 0.08   | 3.30 ± 0.06 | 2.86 ± 0.08  | 3.9 ± 0.1     | 3.5 ± 0.2     | 2.48 ± 0.07 | 2.39 ± 0.09   | 2.79 ± 0.05 | 2.7 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 4.3 ± 0.2     | 2.88 ± 0.05 | 2.68 ± 0.08   | 3.52 ± 0.07 | 2.96 ± 0.09  | 4.1 ± 0.2     | 3.4 ± 0.1     | 2.53 ± 0.07 | 2.35 ± 0.09   | 2.91 ± 0.05 | 2.7 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 4.4 ± 0.2     | 2.94 ± 0.06 | 2.74 ± 0.09   | 3.65 ± 0.07 | 2.96 ± 0.09  | 4.1 ± 0.2     | 3.7 ± 0.2     | 2.64 ± 0.07 | 2.40 ± 0.10   | 2.89 ± 0.06 | 2.7 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 4.3 ± 0.2     | 3.14 ± 0.06 | 2.82 ± 0.10   | 3.77 ± 0.07 | 2.92 ± 0.09  | 4.1 ± 0.2     | 3.6 ± 0.2     | 2.72 ± 0.07 | 2.33 ± 0.10   | 3.04 ± 0.06 | 2.6 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.4 ± 0.2     | 3.30 ± 0.07 | 2.9 ± 0.1     | 3.89 ± 0.08 | 3.0 ± 0.1    | 4.1 ± 0.2     | 3.7 ± 0.2     | 2.78 ± 0.08 | 2.3 ± 0.1     | 3.08 ± 0.06 | 2.9 ± 0.1    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 4.4 ± 0.2     | 3.39 ± 0.07 | 2.8 ± 0.1     | 4.01 ± 0.08 | 3.1 ± 0.1    | 4.1 ± 0.2     | 3.8 ± 0.2     | 2.82 ± 0.08 | 2.4 ± 0.1     | 3.05 ± 0.06 | 2.8 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 37  | 4.4 ± 0.2     | 3.43 ± 0.07 | 2.9 ± 0.1     | 4.05 ± 0.09 | 3.0 ± 0.1    | 4.4 ± 0.2     | 3.8 ± 0.2     | 2.78 ± 0.08 | 2.5 ± 0.1     | 3.24 ± 0.07 | 2.8 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 38  | 4.4 ± 0.2     | 3.62 ± 0.07 | 2.9 ± 0.1     | 4.16 ± 0.10 | 3.2 ± 0.1    | 4.3 ± 0.2     | 3.7 ± 0.2     | 2.94 ± 0.08 | 2.3 ± 0.1     | 3.19 ± 0.07 | 2.8 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 39  | 4.4 ± 0.2     | 3.67 ± 0.07 | 2.9 ± 0.1     | 4.3 ± 0.1   | 3.2 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.98 ± 0.08 | 2.5 ± 0.1     | 3.17 ± 0.07 | 2.9 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 40  | 4.6 ± 0.2     | 3.80 ± 0.08 | 3.1 ± 0.1     | 4.2 ± 0.1   | 3.4 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.96 ± 0.08 | 2.5 ± 0.1     | 3.40 ± 0.08 | 2.8 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 41  | 4.6 ± 0.3     | 3.89 ± 0.08 | 3.2 ± 0.1     | 4.4 ± 0.1   | 3.4 ± 0.1    | 4.4 ± 0.2     | 3.8 ± 0.2     | 3.05 ± 0.08 | 2.7 ± 0.1     | 3.46 ± 0.08 | 3.1 ± 0.1    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.5 ± 0.3     | 3.91 ± 0.08 | 3.3 ± 0.1     | 4.3 ± 0.1   | 3.4 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 3.17 ± 0.08 | 2.8 ± 0.1     | 3.46 ± 0.09 | 3.1 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 4.6 ± 0.3     | 4.11 ± 0.08 | 3.4 ± 0.1     | 4.4 ± 0.1   | 3.8 ± 0.1    | 4.4 ± 0.2     | 3.9 ± 0.2     | 3.24 ± 0.08 | 2.8 ± 0.1     | 3.54 ± 0.09 | 3.4 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.8 ± 0.3     | 4.25 ± 0.09 | 3.4 ± 0.2     | 4.4 ± 0.1   | 3.8 ± 0.1    | 4.8 ± 0.3     | 3.9 ± 0.2     | 3.28 ± 0.09 | 2.8 ± 0.1     | 3.59 ± 0.10 | 3.3 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.6 ± 0.3     | 4.25 ± 0.09 | 3.3 ± 0.2     | 4.5 ± 0.1   | 3.7 ± 0.1    | 4.4 ± 0.2     | 3.9 ± 0.2     | 3.45 ± 0.09 | 2.9 ± 0.1     | 3.6 ± 0.1   | 3.3 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.9 ± 0.3     | 4.41 ± 0.09 | 3.3 ± 0.1     | 4.2 ± 0.1   | 3.8 ± 0.2    | 4.2 ± 0.2     | 4.2 ± 0.3     | 3.45 ± 0.09 | 3.2 ± 0.2     | 3.5 ± 0.1   | 3.6 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 4.7 ± 0.3     | 4.24 ± 0.10 | 3.6 ± 0.2     | 4.2 ± 0.2   | 3.8 ± 0.2    | 4.4 ± 0.3     | 4.1 ± 0.3     | 3.39 ± 0.10 | 2.9 ± 0.2     | 3.7 ± 0.1   | 3.5 ± 0.2    | 4.4 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.6 ± 0.3     | 4.20 ± 0.10 | 3.4 ± 0.2     | 4.4 ± 0.2   | 3.8 ± 0.2    | 4.4 ± 0.3     | 4.3 ± 0.3     | 3.45 ± 0.09 | 3.2 ± 0.2     | 3.5 ± 0.1   | 3.4 ± 0.2    | 4.4 ± 0.2     | E Africa F   | E Europe M   |
| 49  | 4.5 ± 0.3     | 4.3 ± 0.1   | 3.3 ± 0.2     | 3.9 ± 0.2   | 3.9 ± 0.2    | 4.6 ± 0.3     | 4.1 ± 0.3     | 3.48 ± 0.10 | 3.2 ± 0.2     | 3.5 ± 0.1   | 3.5 ± 0.2    | 4.3 ± 0.2     | W Africa F   | E Europe M   |
| 50  | 4.7 ± 0.4     | 4.2 ± 0.1   | 3.3 ± 0.2     | 4.0 ± 0.2   | 3.8 ± 0.2    | 4.5 ± 0.3     | 3.8 ± 0.3     | 3.35 ± 0.10 | 3.2 ± 0.2     | 3.5 ± 0.1   | 3.4 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 51  | 4.5 ± 0.3     | 4.2 ± 0.1   | 3.4 ± 0.2     | 3.9 ± 0.2   | 3.7 ± 0.2    | 4.6 ± 0.3     | 4.0 ± 0.4     | 3.37 ± 0.10 | 3.1 ± 0.2     | 3.6 ± 0.1   | 3.4 ± 0.2    | 4.5 ± 0.3     | W Africa F   | E Europe M   |
| 52  | 4.5 ± 0.4     | 4.1 ± 0.1   | 3.3 ± 0.2     | 4.0 ± 0.2   | 3.7 ± 0.2    | 4.5 ± 0.3     | 4.0 ± 0.3     | 3.4 ± 0.1   | 3.1 ± 0.2     | 3.4 ± 0.1   | 3.3 ± 0.2    | 4.3 ± 0.3     | W Africa F   | E Europe M   |

**Table 98.** Mean absolute error for Application-type images for algorithm privately-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.2 ± 0.2     | 4.0 ± 0.1   | 1.9 ± 0.2     | 2.8 ± 0.1   | 3.9 ± 0.3    | 3.2 ± 0.3     | 2.9 ± 0.2     | 3.6 ± 0.1   | 1.5 ± 0.1     | 2.5 ± 0.1   | 3.0 ± 0.2    | 2.6 ± 0.2     | E Asia F     | E Europe M   |
| 15  | 3.6 ± 0.3     | 4.3 ± 0.2   | 2.6 ± 0.2     | 3.3 ± 0.2   | 4.3 ± 0.3    | 3.3 ± 0.4     | 2.7 ± 0.2     | 3.6 ± 0.1   | 1.7 ± 0.1     | 2.5 ± 0.1   | 3.0 ± 0.2    | 2.7 ± 0.2     | S E Asia F   | E Europe M   |
| 16  | 3.7 ± 0.3     | 4.5 ± 0.2   | 3.1 ± 0.2     | 3.4 ± 0.2   | 4.4 ± 0.3    | 3.8 ± 0.4     | 2.6 ± 0.2     | 3.4 ± 0.1   | 2.0 ± 0.2     | 2.4 ± 0.1   | 3.1 ± 0.2    | 2.3 ± 0.2     | E Asia F     | E Europe M   |
| 17  | 3.9 ± 0.3     | 4.3 ± 0.1   | 3.7 ± 0.2     | 3.6 ± 0.2   | 4.3 ± 0.3    | 3.6 ± 0.3     | 2.4 ± 0.2     | 3.2 ± 0.1   | 2.2 ± 0.1     | 2.3 ± 0.1   | 2.7 ± 0.2    | 2.1 ± 0.3     | E Asia F     | W Africa M   |
| 18  | 3.7 ± 0.2     | 4.2 ± 0.1   | 3.6 ± 0.2     | 3.4 ± 0.1   | 4.1 ± 0.2    | 3.6 ± 0.2     | 2.4 ± 0.2     | 2.72 ± 0.08 | 2.5 ± 0.2     | 2.20 ± 0.09 | 2.4 ± 0.2    | 2.2 ± 0.2     | E Asia F     | W Africa M   |
| 19  | 3.7 ± 0.2     | 3.8 ± 0.1   | 3.5 ± 0.2     | 3.4 ± 0.1   | 3.8 ± 0.2    | 3.6 ± 0.2     | 2.5 ± 0.2     | 2.46 ± 0.09 | 2.5 ± 0.2     | 2.19 ± 0.09 | 2.1 ± 0.1    | 2.1 ± 0.2     | E Asia F     | W Africa M   |
| 20  | 3.8 ± 0.2     | 3.7 ± 0.1   | 3.4 ± 0.2     | 3.4 ± 0.1   | 3.6 ± 0.2    | 3.7 ± 0.2     | 2.7 ± 0.2     | 2.20 ± 0.09 | 2.5 ± 0.2     | 2.08 ± 0.08 | 2.0 ± 0.1    | 2.3 ± 0.2     | E Africa F   | S E Asia M   |
| 21  | 3.5 ± 0.2     | 3.4 ± 0.1   | 3.2 ± 0.2     | 3.3 ± 0.1   | 3.5 ± 0.1    | 3.9 ± 0.2     | 2.7 ± 0.2     | 1.91 ± 0.08 | 2.3 ± 0.2     | 1.99 ± 0.08 | 1.9 ± 0.1    | 2.6 ± 0.2     | W Africa F   | S E Asia M   |
| 22  | 3.6 ± 0.2     | 3.24 ± 0.10 | 3.0 ± 0.1     | 3.2 ± 0.1   | 3.3 ± 0.1    | 3.8 ± 0.2     | 3.0 ± 0.2     | 1.89 ± 0.08 | 2.1 ± 0.1     | 2.03 ± 0.09 | 1.9 ± 0.1    | 2.7 ± 0.2     | W Africa F   | E Asia M     |
| 23  | 3.5 ± 0.2     | 3.01 ± 0.08 | 2.9 ± 0.1     | 3.29 ± 0.09 | 3.1 ± 0.1    | 3.7 ± 0.2     | 3.1 ± 0.2     | 2.03 ± 0.07 | 2.2 ± 0.1     | 2.21 ± 0.08 | 2.2 ± 0.1    | 2.9 ± 0.1     | W Africa F   | E Asia M     |
| 24  | 3.7 ± 0.2     | 2.90 ± 0.07 | 2.9 ± 0.1     | 3.34 ± 0.08 | 2.98 ± 0.09  | 3.8 ± 0.2     | 3.1 ± 0.2     | 2.32 ± 0.07 | 2.4 ± 0.1     | 2.37 ± 0.07 | 2.39 ± 0.09  | 3.1 ± 0.2     | W Africa F   | E Asia M     |
| 25  | 3.6 ± 0.2     | 2.84 ± 0.07 | 3.1 ± 0.1     | 3.46 ± 0.08 | 2.92 ± 0.09  | 3.8 ± 0.2     | 3.3 ± 0.2     | 2.63 ± 0.07 | 2.6 ± 0.1     | 2.62 ± 0.07 | 2.8 ± 0.1    | 3.4 ± 0.1     | W Africa F   | E Europe M   |
| 26  | 3.7 ± 0.2     | 2.73 ± 0.06 | 3.2 ± 0.1     | 3.70 ± 0.08 | 3.03 ± 0.09  | 4.0 ± 0.2     | 3.3 ± 0.1     | 2.90 ± 0.07 | 3.1 ± 0.1     | 2.89 ± 0.07 | 3.1 ± 0.1    | 3.5 ± 0.1     | W Africa F   | E Asia F     |
| 27  | 3.9 ± 0.2     | 2.73 ± 0.06 | 3.31 ± 0.10   | 3.88 ± 0.08 | 3.02 ± 0.09  | 4.1 ± 0.2     | 3.5 ± 0.1     | 3.14 ± 0.08 | 3.1 ± 0.1     | 3.15 ± 0.07 | 3.3 ± 0.1    | 3.7 ± 0.1     | W Africa F   | E Asia F     |
| 28  | 4.2 ± 0.2     | 2.77 ± 0.06 | 3.46 ± 0.10   | 4.25 ± 0.08 | 3.06 ± 0.09  | 4.3 ± 0.2     | 3.6 ± 0.1     | 3.36 ± 0.08 | 3.2 ± 0.1     | 3.46 ± 0.07 | 3.5 ± 0.1    | 4.1 ± 0.1     | W Africa F   | E Asia F     |
| 29  | 4.2 ± 0.2     | 2.80 ± 0.06 | 3.61 ± 0.10   | 4.36 ± 0.08 | 3.15 ± 0.09  | 4.7 ± 0.2     | 3.7 ± 0.1     | 3.46 ± 0.08 | 3.6 ± 0.1     | 3.78 ± 0.07 | 3.7 ± 0.1    | 4.3 ± 0.1     | W Africa F   | E Asia F     |
| 30  | 4.5 ± 0.2     | 2.94 ± 0.06 | 3.8 ± 0.1     | 4.69 ± 0.09 | 3.22 ± 0.09  | 5.1 ± 0.2     | 4.1 ± 0.2     | 3.68 ± 0.09 | 3.6 ± 0.1     | 4.12 ± 0.07 | 3.8 ± 0.1    | 4.6 ± 0.1     | W Africa F   | E Asia F     |
| 31  | 4.8 ± 0.2     | 3.12 ± 0.06 | 4.0 ± 0.1     | 5.04 ± 0.09 | 3.34 ± 0.10  | 5.2 ± 0.2     | 4.3 ± 0.2     | 3.83 ± 0.09 | 3.9 ± 0.1     | 4.43 ± 0.07 | 4.2 ± 0.2    | 5.0 ± 0.1     | W Africa F   | E Asia F     |
| 32  | 5.1 ± 0.2     | 3.23 ± 0.06 | 4.3 ± 0.1     | 5.31 ± 0.10 | 3.6 ± 0.1    | 5.7 ± 0.2     | 4.4 ± 0.2     | 3.87 ± 0.10 | 3.9 ± 0.1     | 4.64 ± 0.07 | 4.1 ± 0.2    | 5.4 ± 0.2     | W Africa F   | E Asia F     |
| 33  | 5.4 ± 0.2     | 3.45 ± 0.07 | 4.5 ± 0.1     | 5.7 ± 0.1   | 3.6 ± 0.1    | 6.1 ± 0.2     | 4.8 ± 0.2     | 4.1 ± 0.1   | 3.9 ± 0.1     | 5.02 ± 0.08 | 4.3 ± 0.2    | 5.5 ± 0.2     | W Africa F   | E Asia F     |
| 34  | 5.6 ± 0.2     | 3.66 ± 0.07 | 4.8 ± 0.1     | 5.9 ± 0.1   | 3.8 ± 0.1    | 6.3 ± 0.2     | 5.2 ± 0.2     | 4.3 ± 0.1   | 4.1 ± 0.1     | 5.31 ± 0.09 | 4.5 ± 0.2    | 5.9 ± 0.2     | W Africa F   | E Asia F     |
| 35  | 5.7 ± 0.2     | 4.04 ± 0.08 | 5.2 ± 0.1     | 6.3 ± 0.1   | 4.1 ± 0.1    | 6.8 ± 0.2     | 5.2 ± 0.2     | 4.4 ± 0.1   | 4.4 ± 0.2     | 5.71 ± 0.09 | 4.8 ± 0.2    | 6.2 ± 0.2     | W Africa F   | E Asia F     |
| 36  | 6.3 ± 0.2     | 4.25 ± 0.08 | 5.3 ± 0.1     | 6.7 ± 0.1   | 4.4 ± 0.1    | 7.0 ± 0.2     | 5.8 ± 0.2     | 4.7 ± 0.1   | 4.4 ± 0.2     | 5.95 ± 0.09 | 5.0 ± 0.2    | 6.6 ± 0.2     | W Africa F   | E Asia F     |
| 37  | 6.8 ± 0.3     | 4.63 ± 0.08 | 5.7 ± 0.2     | 6.8 ± 0.1   | 4.8 ± 0.1    | 7.5 ± 0.3     | 5.7 ± 0.2     | 4.8 ± 0.1   | 4.7 ± 0.2     | 6.33 ± 0.10 | 5.2 ± 0.2    | 6.9 ± 0.2     | W Africa F   | E Asia F     |
| 38  | 6.7 ± 0.3     | 4.93 ± 0.08 | 5.8 ± 0.2     | 7.2 ± 0.1   | 4.9 ± 0.1    | 8.1 ± 0.3     | 5.9 ± 0.2     | 5.2 ± 0.1   | 4.9 ± 0.2     | 6.6 ± 0.1   | 5.1 ± 0.2    | 7.1 ± 0.2     | W Africa F   | E Europe M   |
| 39  | 7.0 ± 0.3     | 5.24 ± 0.08 | 6.1 ± 0.2     | 7.5 ± 0.1   | 5.2 ± 0.1    | 8.1 ± 0.3     | 6.1 ± 0.2     | 5.5 ± 0.1   | 5.1 ± 0.2     | 6.7 ± 0.1   | 5.2 ± 0.2    | 7.4 ± 0.2     | W Africa F   | E Europe M   |
| 40  | 7.4 ± 0.3     | 5.56 ± 0.09 | 6.1 ± 0.2     | 7.6 ± 0.1   | 5.8 ± 0.2    | 8.6 ± 0.3     | 6.2 ± 0.3     | 5.7 ± 0.1   | 5.4 ± 0.2     | 7.0 ± 0.1   | 5.7 ± 0.2    | 7.8 ± 0.2     | W Africa F   | E Europe M   |
| 41  | 7.7 ± 0.3     | 5.82 ± 0.09 | 6.5 ± 0.2     | 8.0 ± 0.2   | 5.9 ± 0.2    | 9.4 ± 0.3     | 6.4 ± 0.3     | 5.8 ± 0.1   | 5.3 ± 0.2     | 7.2 ± 0.1   | 5.8 ± 0.2    | 8.2 ± 0.3     | W Africa F   | E Europe M   |
| 42  | 7.5 ± 0.3     | 5.93 ± 0.09 | 6.5 ± 0.2     | 7.9 ± 0.2   | 6.1 ± 0.2    | 9.1 ± 0.3     | 6.3 ± 0.3     | 6.0 ± 0.1   | 5.2 ± 0.2     | 7.2 ± 0.1   | 5.6 ± 0.2    | 8.1 ± 0.3     | W Africa F   | E Europe M   |
| 43  | 7.9 ± 0.4     | 6.13 ± 0.10 | 6.8 ± 0.2     | 8.3 ± 0.2   | 6.5 ± 0.2    | 9.6 ± 0.3     | 6.2 ± 0.3     | 6.0 ± 0.1   | 5.4 ± 0.2     | 7.2 ± 0.1   | 5.7 ± 0.2    | 8.5 ± 0.3     | W Africa F   | E Europe M   |
| 44  | 7.9 ± 0.4     | 6.4 ± 0.1   | 6.7 ± 0.2     | 8.2 ± 0.2   | 6.3 ± 0.2    | 9.7 ± 0.4     | 6.4 ± 0.3     | 6.0 ± 0.1   | 5.4 ± 0.2     | 7.3 ± 0.2   | 5.6 ± 0.2    | 8.0 ± 0.3     | W Africa F   | E Europe M   |
| 45  | 8.2 ± 0.4     | 6.4 ± 0.1   | 6.8 ± 0.3     | 8.3 ± 0.2   | 6.5 ± 0.2    | 9.8 ± 0.4     | 6.5 ± 0.4     | 6.2 ± 0.1   | 5.5 ± 0.2     | 7.1 ± 0.2   | 5.6 ± 0.2    | 8.3 ± 0.3     | W Africa F   | E Europe M   |
| 46  | 7.9 ± 0.5     | 6.5 ± 0.1   | 6.8 ± 0.3     | 8.0 ± 0.2   | 6.6 ± 0.2    | 10.0 ± 0.4    | 6.2 ± 0.4     | 6.0 ± 0.1   | 5.4 ± 0.2     | 6.9 ± 0.2   | 5.4 ± 0.3    | 8.5 ± 0.3     | W Africa F   | E Europe M   |
| 47  | 7.5 ± 0.5     | 6.2 ± 0.1   | 6.8 ± 0.3     | 8.0 ± 0.2   | 6.4 ± 0.2    | 10.0 ± 0.4    | 6.1 ± 0.4     | 5.9 ± 0.2   | 5.3 ± 0.3     | 6.9 ± 0.2   | 5.1 ± 0.3    | 8.2 ± 0.4     | W Africa F   | S E Asia M   |
| 48  | 8.0 ± 0.6     | 6.2 ± 0.1   | 6.4 ± 0.3     | 8.1 ± 0.3   | 6.5 ± 0.3    | 9.9 ± 0.5     | 6.2 ± 0.4     | 5.8 ± 0.2   | 5.0 ± 0.3     | 6.7 ± 0.2   | 5.0 ± 0.3    | 8.4 ± 0.4     | W Africa F   | E Europe M   |
| 49  | 8.1 ± 0.6     | 6.3 ± 0.2   | 6.7 ± 0.3     | 7.7 ± 0.3   | 6.2 ± 0.3    | 10.2 ± 0.5    | 6.2 ± 0.5     | 5.6 ± 0.2   | 5.4 ± 0.3     | 6.6 ± 0.2   | 4.8 ± 0.3    | 8.4 ± 0.4     | W Africa F   | S E Asia M   |
| 50  | 8.0 ± 0.6     | 6.3 ± 0.2   | 6.4 ± 0.3     | 7.6 ± 0.3   | 6.4 ± 0.3    | 10.0 ± 0.5    | 5.8 ± 0.5     | 5.6 ± 0.2   | 5.3 ± 0.3     | 6.7 ± 0.2   | 4.6 ± 0.3    | 8.1 ± 0.4     | W Africa F   | S E Asia M   |
| 51  | 7.6 ± 0.5     | 6.3 ± 0.2   | 6.4 ± 0.3     | 7.8 ± 0.3   | 6.6 ± 0.3    | 10.5 ± 0.5    | 6.6 ± 0.5     | 5.6 ± 0.2   | 5.1 ± 0.3     | 6.6 ± 0.3   | 4.6 ± 0.3    | 8.6 ± 0.5     | W Africa F   | S E Asia M   |
| 52  | 7.6 ± 0.6     | 6.4 ± 0.2   | 6.4 ± 0.3     | 7.7 ± 0.3   | 6.2 ± 0.3    | 10.9 ± 0.6    | 6.2 ± 0.5     | 5.5 ± 0.2   | 5.3 ± 0.3     | 6.4 ± 0.2   | 5.0 ± 0.3    | 8.4 ± 0.5     | W Africa F   | S E Asia M   |



**Table 99.** Mean absolute error for Application-type images for algorithm regula-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.8 ± 0.2     | 2.26 ± 0.07 | 1.8 ± 0.1     | 2.7 ± 0.1   | 2.4 ± 0.2    | 2.8 ± 0.3     | 1.8 ± 0.1     | 1.76 ± 0.07 | 1.17 ± 0.09   | 1.69 ± 0.07 | 1.6 ± 0.1    | 1.4 ± 0.2     | E Africa F   | E Europe M   |
| 15  | 3.1 ± 0.2     | 1.97 ± 0.07 | 1.8 ± 0.1     | 2.8 ± 0.1   | 2.5 ± 0.2    | 2.9 ± 0.3     | 1.6 ± 0.1     | 1.53 ± 0.05 | 1.01 ± 0.08   | 1.53 ± 0.07 | 1.50 ± 0.10  | 1.7 ± 0.2     | E Africa F   | E Europe M   |
| 16  | 2.9 ± 0.2     | 1.78 ± 0.08 | 1.9 ± 0.2     | 2.6 ± 0.1   | 2.4 ± 0.2    | 3.5 ± 0.4     | 1.7 ± 0.2     | 1.31 ± 0.05 | 1.2 ± 0.1     | 1.46 ± 0.08 | 1.3 ± 0.1    | 1.7 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.2 ± 0.3     | 1.65 ± 0.09 | 2.0 ± 0.2     | 2.5 ± 0.1   | 2.3 ± 0.2    | 3.5 ± 0.3     | 1.7 ± 0.2     | 1.19 ± 0.07 | 1.09 ± 0.10   | 1.38 ± 0.09 | 1.3 ± 0.1    | 2.2 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 3.0 ± 0.2     | 1.67 ± 0.07 | 2.0 ± 0.2     | 2.37 ± 0.10 | 2.2 ± 0.1    | 3.6 ± 0.3     | 1.9 ± 0.2     | 1.21 ± 0.06 | 1.3 ± 0.1     | 1.54 ± 0.08 | 1.4 ± 0.2    | 2.3 ± 0.2     | W Africa F   | E Asia M     |
| 19  | 3.3 ± 0.2     | 1.76 ± 0.07 | 2.1 ± 0.2     | 2.3 ± 0.1   | 2.3 ± 0.1    | 3.7 ± 0.3     | 2.2 ± 0.2     | 1.37 ± 0.06 | 1.4 ± 0.1     | 1.60 ± 0.08 | 1.6 ± 0.1    | 2.5 ± 0.2     | W Africa F   | E Asia M     |
| 20  | 3.3 ± 0.2     | 1.98 ± 0.09 | 2.1 ± 0.1     | 2.4 ± 0.1   | 2.4 ± 0.1    | 3.6 ± 0.2     | 2.4 ± 0.2     | 1.49 ± 0.06 | 1.6 ± 0.1     | 1.85 ± 0.08 | 1.7 ± 0.1    | 2.8 ± 0.2     | W Africa F   | E Asia M     |
| 21  | 3.1 ± 0.2     | 2.16 ± 0.07 | 2.2 ± 0.1     | 2.29 ± 0.08 | 2.5 ± 0.1    | 3.7 ± 0.2     | 2.5 ± 0.2     | 1.61 ± 0.05 | 1.7 ± 0.1     | 1.98 ± 0.08 | 1.9 ± 0.1    | 2.9 ± 0.2     | W Africa F   | E Asia M     |
| 22  | 3.3 ± 0.2     | 2.22 ± 0.07 | 2.28 ± 0.09   | 2.29 ± 0.07 | 2.5 ± 0.1    | 3.4 ± 0.2     | 3.0 ± 0.2     | 1.80 ± 0.06 | 1.9 ± 0.1     | 2.08 ± 0.08 | 1.96 ± 0.09  | 2.8 ± 0.2     | W Africa F   | E Asia M     |
| 23  | 3.2 ± 0.2     | 2.37 ± 0.06 | 2.36 ± 0.09   | 2.31 ± 0.07 | 2.46 ± 0.09  | 3.5 ± 0.2     | 3.0 ± 0.2     | 2.00 ± 0.06 | 1.82 ± 0.10   | 2.09 ± 0.08 | 2.14 ± 0.10  | 3.0 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 3.3 ± 0.2     | 2.44 ± 0.06 | 2.24 ± 0.07   | 2.35 ± 0.06 | 2.42 ± 0.07  | 3.4 ± 0.2     | 2.9 ± 0.2     | 2.06 ± 0.07 | 1.88 ± 0.09   | 2.18 ± 0.07 | 2.14 ± 0.10  | 3.0 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 3.3 ± 0.2     | 2.58 ± 0.06 | 2.21 ± 0.07   | 2.39 ± 0.06 | 2.51 ± 0.08  | 3.3 ± 0.2     | 3.0 ± 0.2     | 2.22 ± 0.07 | 1.82 ± 0.09   | 2.13 ± 0.06 | 2.19 ± 0.09  | 3.0 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 3.2 ± 0.1     | 2.63 ± 0.06 | 2.25 ± 0.07   | 2.50 ± 0.05 | 2.58 ± 0.07  | 3.1 ± 0.1     | 3.0 ± 0.1     | 2.22 ± 0.06 | 1.88 ± 0.09   | 2.25 ± 0.06 | 2.19 ± 0.10  | 2.9 ± 0.1     | E Africa F   | E Europe M   |
| 27  | 3.3 ± 0.2     | 2.60 ± 0.05 | 2.26 ± 0.07   | 2.66 ± 0.05 | 2.54 ± 0.07  | 3.3 ± 0.1     | 3.1 ± 0.1     | 2.37 ± 0.07 | 1.82 ± 0.08   | 2.30 ± 0.06 | 2.28 ± 0.10  | 2.9 ± 0.1     | E Africa F   | E Europe M   |
| 28  | 3.3 ± 0.1     | 2.64 ± 0.05 | 2.24 ± 0.07   | 2.83 ± 0.06 | 2.57 ± 0.07  | 3.1 ± 0.1     | 2.9 ± 0.1     | 2.39 ± 0.07 | 1.92 ± 0.08   | 2.39 ± 0.06 | 2.23 ± 0.10  | 2.9 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 3.5 ± 0.1     | 2.65 ± 0.05 | 2.27 ± 0.07   | 2.98 ± 0.06 | 2.65 ± 0.08  | 3.1 ± 0.1     | 3.0 ± 0.1     | 2.44 ± 0.07 | 2.03 ± 0.08   | 2.50 ± 0.05 | 2.4 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 3.4 ± 0.1     | 2.76 ± 0.05 | 2.36 ± 0.07   | 3.14 ± 0.06 | 2.68 ± 0.08  | 3.2 ± 0.1     | 3.1 ± 0.1     | 2.53 ± 0.07 | 2.09 ± 0.08   | 2.64 ± 0.05 | 2.4 ± 0.1    | 2.9 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 3.5 ± 0.1     | 2.84 ± 0.05 | 2.47 ± 0.07   | 3.26 ± 0.06 | 2.77 ± 0.08  | 3.1 ± 0.1     | 3.1 ± 0.1     | 2.61 ± 0.07 | 2.22 ± 0.08   | 2.77 ± 0.05 | 2.6 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 3.5 ± 0.2     | 2.88 ± 0.05 | 2.49 ± 0.08   | 3.44 ± 0.06 | 2.85 ± 0.09  | 3.4 ± 0.1     | 3.2 ± 0.1     | 2.62 ± 0.07 | 2.32 ± 0.09   | 2.86 ± 0.05 | 2.6 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 3.7 ± 0.2     | 2.90 ± 0.06 | 2.54 ± 0.08   | 3.46 ± 0.07 | 2.90 ± 0.09  | 3.3 ± 0.1     | 3.4 ± 0.1     | 2.68 ± 0.07 | 2.28 ± 0.09   | 2.90 ± 0.06 | 2.6 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 3.7 ± 0.2     | 2.93 ± 0.06 | 2.60 ± 0.08   | 3.55 ± 0.07 | 2.85 ± 0.09  | 3.2 ± 0.1     | 3.3 ± 0.1     | 2.75 ± 0.07 | 2.39 ± 0.10   | 3.04 ± 0.06 | 2.7 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 3.9 ± 0.2     | 2.95 ± 0.06 | 2.66 ± 0.09   | 3.58 ± 0.07 | 2.94 ± 0.10  | 3.4 ± 0.1     | 3.4 ± 0.2     | 2.74 ± 0.07 | 2.4 ± 0.1     | 2.97 ± 0.06 | 2.8 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 3.9 ± 0.2     | 2.90 ± 0.06 | 2.58 ± 0.09   | 3.57 ± 0.08 | 2.94 ± 0.10  | 3.3 ± 0.2     | 3.6 ± 0.2     | 2.78 ± 0.08 | 2.4 ± 0.1     | 2.98 ± 0.06 | 2.8 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 37  | 4.0 ± 0.2     | 2.85 ± 0.06 | 2.63 ± 0.10   | 3.63 ± 0.08 | 2.9 ± 0.1    | 3.7 ± 0.2     | 3.5 ± 0.2     | 2.79 ± 0.07 | 2.5 ± 0.1     | 3.04 ± 0.07 | 2.8 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Europe M   |
| 38  | 3.9 ± 0.2     | 2.92 ± 0.06 | 2.6 ± 0.1     | 3.59 ± 0.09 | 2.9 ± 0.1    | 3.6 ± 0.2     | 3.5 ± 0.2     | 2.83 ± 0.08 | 2.5 ± 0.1     | 2.96 ± 0.07 | 2.9 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 39  | 4.0 ± 0.2     | 2.86 ± 0.06 | 2.5 ± 0.1     | 3.56 ± 0.09 | 2.9 ± 0.1    | 3.4 ± 0.2     | 3.6 ± 0.2     | 2.86 ± 0.08 | 2.5 ± 0.1     | 2.92 ± 0.07 | 2.9 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 40  | 3.9 ± 0.2     | 2.86 ± 0.06 | 2.6 ± 0.1     | 3.48 ± 0.10 | 3.1 ± 0.1    | 3.6 ± 0.2     | 3.6 ± 0.2     | 2.85 ± 0.07 | 2.7 ± 0.1     | 3.04 ± 0.07 | 2.8 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Europe F   |
| 41  | 4.1 ± 0.2     | 2.90 ± 0.06 | 2.7 ± 0.1     | 3.5 ± 0.1   | 3.0 ± 0.1    | 3.8 ± 0.2     | 3.6 ± 0.2     | 2.83 ± 0.07 | 2.6 ± 0.1     | 2.99 ± 0.07 | 3.0 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.2 ± 0.2     | 2.88 ± 0.06 | 2.8 ± 0.1     | 3.4 ± 0.1   | 2.9 ± 0.1    | 3.8 ± 0.2     | 3.6 ± 0.2     | 2.86 ± 0.08 | 2.5 ± 0.1     | 2.94 ± 0.08 | 2.9 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 4.2 ± 0.3     | 2.93 ± 0.06 | 2.7 ± 0.1     | 3.4 ± 0.1   | 3.1 ± 0.1    | 4.0 ± 0.2     | 3.6 ± 0.2     | 2.99 ± 0.08 | 2.6 ± 0.1     | 2.99 ± 0.08 | 3.1 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.4 ± 0.3     | 3.04 ± 0.06 | 2.7 ± 0.1     | 3.6 ± 0.1   | 3.2 ± 0.1    | 4.1 ± 0.2     | 3.6 ± 0.2     | 2.95 ± 0.08 | 2.5 ± 0.1     | 3.06 ± 0.09 | 3.2 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.2 ± 0.3     | 2.99 ± 0.06 | 2.6 ± 0.1     | 3.5 ± 0.1   | 3.0 ± 0.1    | 3.9 ± 0.2     | 3.8 ± 0.2     | 3.02 ± 0.08 | 2.6 ± 0.1     | 3.03 ± 0.09 | 3.2 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe F   |
| 46  | 4.4 ± 0.3     | 3.06 ± 0.07 | 2.7 ± 0.1     | 3.2 ± 0.1   | 3.1 ± 0.1    | 4.0 ± 0.2     | 3.7 ± 0.3     | 2.93 ± 0.08 | 2.6 ± 0.1     | 3.0 ± 0.1   | 3.2 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 4.4 ± 0.3     | 3.01 ± 0.07 | 2.8 ± 0.1     | 3.4 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.2     | 4.0 ± 0.3     | 2.98 ± 0.09 | 2.7 ± 0.1     | 3.2 ± 0.1   | 3.2 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.3 ± 0.3     | 2.96 ± 0.07 | 2.7 ± 0.1     | 3.5 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.2     | 4.2 ± 0.3     | 2.90 ± 0.08 | 2.8 ± 0.2     | 3.2 ± 0.1   | 3.3 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe F   |
| 49  | 4.3 ± 0.3     | 3.08 ± 0.08 | 2.8 ± 0.1     | 3.2 ± 0.1   | 3.1 ± 0.1    | 4.3 ± 0.3     | 4.2 ± 0.3     | 2.92 ± 0.09 | 2.8 ± 0.1     | 3.2 ± 0.1   | 3.2 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe F   |
| 50  | 4.3 ± 0.3     | 3.03 ± 0.07 | 2.7 ± 0.1     | 3.3 ± 0.1   | 3.1 ± 0.1    | 4.0 ± 0.3     | 3.8 ± 0.3     | 2.89 ± 0.08 | 2.8 ± 0.2     | 3.2 ± 0.1   | 3.3 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe F   |
| 51  | 4.3 ± 0.3     | 2.95 ± 0.08 | 2.8 ± 0.1     | 3.3 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.3     | 4.3 ± 0.4     | 2.87 ± 0.09 | 2.7 ± 0.2     | 3.4 ± 0.1   | 3.3 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 52  | 4.3 ± 0.4     | 2.94 ± 0.08 | 2.8 ± 0.1     | 3.3 ± 0.1   | 2.9 ± 0.1    | 3.9 ± 0.3     | 4.0 ± 0.3     | 2.92 ± 0.08 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.3 ± 0.2    | 4.2 ± 0.3     | E Africa F   | E Europe F   |



**Table 100.** Mean absolute error for Application-type images for algorithm roc-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.3 ± 0.2     | 5.5 ± 0.1   | 3.8 ± 0.1     | 4.0 ± 0.1   | 4.8 ± 0.3    | 5.7 ± 0.3     | 4.4 ± 0.2     | 5.5 ± 0.1   | 3.0 ± 0.1     | 3.62 ± 0.09 | 4.8 ± 0.2    | 4.7 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 4.9 ± 0.3     | 5.5 ± 0.1   | 3.3 ± 0.1     | 3.9 ± 0.2   | 5.0 ± 0.3    | 5.5 ± 0.3     | 4.3 ± 0.2     | 5.4 ± 0.1   | 2.74 ± 0.10   | 3.4 ± 0.1   | 4.7 ± 0.2    | 4.7 ± 0.3     | E Asia F     | E Europe M   |
| 16  | 4.8 ± 0.3     | 5.4 ± 0.1   | 2.8 ± 0.1     | 3.5 ± 0.1   | 4.9 ± 0.3    | 5.7 ± 0.4     | 4.2 ± 0.2     | 4.9 ± 0.1   | 2.4 ± 0.1     | 3.1 ± 0.1   | 4.4 ± 0.2    | 4.3 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 5.0 ± 0.3     | 5.1 ± 0.1   | 2.3 ± 0.2     | 3.2 ± 0.2   | 4.6 ± 0.3    | 5.2 ± 0.3     | 3.7 ± 0.2     | 4.7 ± 0.1   | 1.59 ± 0.09   | 2.9 ± 0.1   | 4.2 ± 0.2    | 4.5 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 4.2 ± 0.2     | 4.6 ± 0.1   | 1.9 ± 0.2     | 2.5 ± 0.1   | 4.4 ± 0.2    | 4.8 ± 0.3     | 3.6 ± 0.2     | 4.30 ± 0.10 | 1.3 ± 0.1     | 2.4 ± 0.1   | 4.0 ± 0.2    | 4.0 ± 0.2     | W Africa F   | E Europe M   |
| 19  | 4.2 ± 0.3     | 4.3 ± 0.1   | 1.8 ± 0.2     | 2.4 ± 0.1   | 4.2 ± 0.2    | 4.8 ± 0.3     | 3.8 ± 0.2     | 4.2 ± 0.1   | 1.1 ± 0.1     | 2.4 ± 0.1   | 3.8 ± 0.2    | 4.2 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.1 ± 0.3     | 4.1 ± 0.1   | 1.9 ± 0.1     | 2.5 ± 0.1   | 4.0 ± 0.2    | 4.5 ± 0.3     | 3.8 ± 0.3     | 4.0 ± 0.1   | 1.4 ± 0.1     | 2.7 ± 0.1   | 3.8 ± 0.2    | 4.2 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 3.9 ± 0.2     | 4.0 ± 0.1   | 2.3 ± 0.1     | 2.6 ± 0.1   | 3.8 ± 0.1    | 4.5 ± 0.3     | 4.0 ± 0.3     | 3.9 ± 0.1   | 1.7 ± 0.1     | 2.9 ± 0.1   | 3.9 ± 0.2    | 4.3 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.2 ± 0.3     | 3.7 ± 0.1   | 2.6 ± 0.1     | 2.79 ± 0.09 | 3.9 ± 0.1    | 4.2 ± 0.3     | 4.6 ± 0.3     | 3.7 ± 0.1   | 2.2 ± 0.1     | 3.1 ± 0.1   | 3.5 ± 0.1    | 4.1 ± 0.3     | E Africa M   | E Europe M   |
| 23  | 4.2 ± 0.2     | 3.67 ± 0.09 | 3.1 ± 0.1     | 3.08 ± 0.08 | 3.8 ± 0.1    | 4.4 ± 0.2     | 4.5 ± 0.3     | 3.6 ± 0.1   | 2.4 ± 0.1     | 3.2 ± 0.1   | 3.6 ± 0.1    | 4.3 ± 0.2     | E Africa M   | E Europe M   |
| 24  | 4.3 ± 0.2     | 3.48 ± 0.08 | 3.12 ± 0.09   | 3.21 ± 0.07 | 3.7 ± 0.1    | 4.3 ± 0.2     | 4.4 ± 0.2     | 3.41 ± 0.10 | 2.7 ± 0.1     | 3.3 ± 0.1   | 3.4 ± 0.1    | 4.2 ± 0.2     | E Africa M   | E Europe M   |
| 25  | 4.4 ± 0.2     | 3.46 ± 0.08 | 3.41 ± 0.09   | 3.35 ± 0.07 | 3.6 ± 0.1    | 4.2 ± 0.2     | 4.5 ± 0.2     | 3.34 ± 0.09 | 2.8 ± 0.1     | 3.25 ± 0.09 | 3.4 ± 0.1    | 4.2 ± 0.2     | E Africa M   | E Europe M   |
| 26  | 4.4 ± 0.2     | 3.33 ± 0.07 | 3.56 ± 0.09   | 3.49 ± 0.07 | 3.57 ± 0.10  | 4.2 ± 0.2     | 4.6 ± 0.2     | 3.15 ± 0.09 | 2.9 ± 0.1     | 3.29 ± 0.08 | 3.3 ± 0.1    | 3.9 ± 0.2     | E Africa M   | E Europe M   |
| 27  | 4.8 ± 0.2     | 3.13 ± 0.06 | 3.53 ± 0.10   | 3.66 ± 0.07 | 3.42 ± 0.09  | 4.3 ± 0.2     | 4.3 ± 0.2     | 3.09 ± 0.09 | 2.9 ± 0.1     | 3.21 ± 0.08 | 3.3 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 28  | 4.5 ± 0.2     | 3.07 ± 0.06 | 3.61 ± 0.10   | 3.87 ± 0.07 | 3.38 ± 0.09  | 4.1 ± 0.2     | 4.3 ± 0.2     | 2.96 ± 0.09 | 2.8 ± 0.1     | 3.13 ± 0.07 | 3.1 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 29  | 4.7 ± 0.2     | 2.99 ± 0.06 | 3.6 ± 0.1     | 3.94 ± 0.07 | 3.34 ± 0.10  | 4.1 ± 0.2     | 4.2 ± 0.2     | 2.87 ± 0.08 | 2.9 ± 0.1     | 3.09 ± 0.07 | 3.0 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 30  | 4.5 ± 0.2     | 2.96 ± 0.06 | 3.7 ± 0.1     | 4.04 ± 0.07 | 3.26 ± 0.10  | 4.1 ± 0.2     | 4.3 ± 0.2     | 2.74 ± 0.07 | 2.8 ± 0.1     | 3.09 ± 0.06 | 3.0 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 31  | 4.6 ± 0.2     | 2.96 ± 0.06 | 3.7 ± 0.1     | 4.15 ± 0.08 | 3.21 ± 0.09  | 4.1 ± 0.2     | 4.3 ± 0.2     | 2.65 ± 0.08 | 2.9 ± 0.1     | 3.03 ± 0.06 | 2.9 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Asia M     |
| 32  | 4.8 ± 0.2     | 2.97 ± 0.06 | 3.7 ± 0.1     | 4.29 ± 0.08 | 3.3 ± 0.1    | 4.3 ± 0.2     | 4.1 ± 0.2     | 2.69 ± 0.07 | 2.9 ± 0.1     | 3.07 ± 0.06 | 2.9 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 33  | 4.8 ± 0.2     | 2.93 ± 0.06 | 3.8 ± 0.1     | 4.31 ± 0.08 | 3.25 ± 0.10  | 4.1 ± 0.2     | 4.3 ± 0.2     | 2.59 ± 0.07 | 2.9 ± 0.1     | 3.05 ± 0.06 | 2.7 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 34  | 4.8 ± 0.2     | 2.98 ± 0.06 | 3.8 ± 0.1     | 4.34 ± 0.09 | 3.2 ± 0.1    | 4.0 ± 0.2     | 4.0 ± 0.2     | 2.60 ± 0.07 | 2.9 ± 0.1     | 3.08 ± 0.06 | 3.0 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 35  | 5.1 ± 0.2     | 3.01 ± 0.06 | 4.0 ± 0.1     | 4.41 ± 0.09 | 3.2 ± 0.1    | 4.2 ± 0.2     | 4.1 ± 0.2     | 2.66 ± 0.07 | 2.8 ± 0.1     | 3.07 ± 0.06 | 2.9 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Asia M     |
| 36  | 5.0 ± 0.2     | 3.10 ± 0.06 | 3.9 ± 0.1     | 4.36 ± 0.09 | 3.3 ± 0.1    | 4.1 ± 0.2     | 4.2 ± 0.2     | 2.66 ± 0.07 | 3.0 ± 0.1     | 3.10 ± 0.07 | 3.1 ± 0.2    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 37  | 5.0 ± 0.2     | 3.08 ± 0.07 | 4.0 ± 0.1     | 4.5 ± 0.1   | 3.3 ± 0.1    | 4.4 ± 0.2     | 4.0 ± 0.2     | 2.70 ± 0.07 | 2.9 ± 0.1     | 3.13 ± 0.07 | 2.9 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 38  | 4.9 ± 0.3     | 3.20 ± 0.07 | 4.1 ± 0.2     | 4.4 ± 0.1   | 3.4 ± 0.1    | 4.4 ± 0.2     | 3.9 ± 0.2     | 2.71 ± 0.08 | 2.7 ± 0.1     | 3.08 ± 0.07 | 3.2 ± 0.2    | 3.4 ± 0.1     | E Africa F   | E Asia M     |
| 39  | 4.8 ± 0.3     | 3.22 ± 0.07 | 3.9 ± 0.2     | 4.5 ± 0.1   | 3.4 ± 0.1    | 4.1 ± 0.2     | 3.8 ± 0.2     | 2.76 ± 0.08 | 2.9 ± 0.1     | 3.10 ± 0.07 | 3.3 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Asia M     |
| 40  | 4.9 ± 0.3     | 3.30 ± 0.07 | 4.1 ± 0.2     | 4.4 ± 0.1   | 3.5 ± 0.1    | 4.3 ± 0.2     | 3.9 ± 0.2     | 2.74 ± 0.08 | 2.9 ± 0.1     | 3.20 ± 0.07 | 3.1 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Asia M     |
| 41  | 4.8 ± 0.3     | 3.34 ± 0.07 | 4.2 ± 0.2     | 4.4 ± 0.1   | 3.5 ± 0.1    | 4.3 ± 0.2     | 4.0 ± 0.2     | 2.89 ± 0.08 | 2.9 ± 0.1     | 3.20 ± 0.08 | 3.2 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Asia M     |
| 42  | 5.2 ± 0.3     | 3.44 ± 0.07 | 4.3 ± 0.2     | 4.3 ± 0.1   | 3.5 ± 0.1    | 4.6 ± 0.2     | 3.8 ± 0.2     | 2.96 ± 0.08 | 2.9 ± 0.1     | 3.20 ± 0.08 | 3.1 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 5.1 ± 0.3     | 3.55 ± 0.07 | 4.4 ± 0.2     | 4.3 ± 0.1   | 3.7 ± 0.1    | 4.4 ± 0.2     | 4.0 ± 0.2     | 3.00 ± 0.08 | 2.9 ± 0.2     | 3.21 ± 0.09 | 3.4 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 5.3 ± 0.3     | 3.58 ± 0.08 | 4.4 ± 0.2     | 4.5 ± 0.2   | 3.8 ± 0.1    | 4.5 ± 0.3     | 3.7 ± 0.2     | 3.11 ± 0.09 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.4 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 5.0 ± 0.3     | 3.62 ± 0.08 | 4.3 ± 0.2     | 4.4 ± 0.2   | 3.6 ± 0.1    | 4.4 ± 0.2     | 3.8 ± 0.2     | 3.10 ± 0.08 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.4 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 5.5 ± 0.4     | 3.61 ± 0.08 | 4.6 ± 0.2     | 4.1 ± 0.2   | 3.7 ± 0.1    | 4.4 ± 0.3     | 3.8 ± 0.2     | 3.11 ± 0.09 | 3.4 ± 0.2     | 3.2 ± 0.1   | 3.5 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 5.2 ± 0.4     | 3.76 ± 0.09 | 4.7 ± 0.2     | 4.2 ± 0.2   | 3.7 ± 0.2    | 4.4 ± 0.3     | 4.0 ± 0.3     | 3.21 ± 0.09 | 3.2 ± 0.2     | 3.5 ± 0.1   | 3.5 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 5.1 ± 0.4     | 3.80 ± 0.09 | 4.7 ± 0.2     | 4.3 ± 0.2   | 3.7 ± 0.2    | 4.5 ± 0.3     | 3.9 ± 0.3     | 3.26 ± 0.09 | 3.5 ± 0.2     | 3.5 ± 0.1   | 3.7 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 49  | 5.2 ± 0.4     | 3.82 ± 0.09 | 4.7 ± 0.2     | 4.1 ± 0.2   | 3.8 ± 0.2    | 4.8 ± 0.3     | 3.8 ± 0.3     | 3.34 ± 0.10 | 3.3 ± 0.2     | 3.4 ± 0.1   | 3.6 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 50  | 4.9 ± 0.4     | 3.91 ± 0.10 | 5.0 ± 0.2     | 4.1 ± 0.2   | 3.7 ± 0.2    | 4.4 ± 0.3     | 3.6 ± 0.3     | 3.32 ± 0.10 | 3.6 ± 0.2     | 3.6 ± 0.1   | 3.7 ± 0.2    | 3.9 ± 0.2     | E Europe F   | E Asia M     |
| 51  | 5.1 ± 0.4     | 3.94 ± 0.10 | 5.0 ± 0.2     | 4.2 ± 0.2   | 3.8 ± 0.2    | 4.4 ± 0.3     | 3.8 ± 0.3     | 3.5 ± 0.1   | 3.5 ± 0.2     | 3.7 ± 0.1   | 3.8 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Asia M     |
| 52  | 5.1 ± 0.4     | 4.0 ± 0.1   | 5.0 ± 0.2     | 4.1 ± 0.2   | 3.9 ± 0.2    | 4.5 ± 0.3     | 3.6 ± 0.3     | 3.5 ± 0.1   | 3.5 ± 0.2     | 3.6 ± 0.1   | 3.7 ± 0.2    | 4.1 ± 0.3     | E Africa F   | E Asia M     |

**Table 101.** Mean absolute error for Application-type images for algorithm roc-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.8 ± 0.2     | 4.99 ± 0.09 | 3.4 ± 0.1     | 4.27 ± 0.10 | 4.8 ± 0.2    | 4.9 ± 0.2     | 3.7 ± 0.1     | 4.73 ± 0.08 | 2.82 ± 0.08   | 3.60 ± 0.07 | 4.3 ± 0.1    | 3.7 ± 0.2     | E Asia F     | E Europe M   |
| 15  | 4.8 ± 0.2     | 4.9 ± 0.1   | 2.8 ± 0.1     | 4.1 ± 0.1   | 4.7 ± 0.2    | 4.9 ± 0.3     | 3.4 ± 0.2     | 4.44 ± 0.07 | 2.33 ± 0.09   | 3.24 ± 0.08 | 4.0 ± 0.1    | 3.4 ± 0.2     | W Africa F   | E Europe M   |
| 16  | 4.6 ± 0.3     | 4.8 ± 0.1   | 2.4 ± 0.1     | 3.8 ± 0.1   | 4.5 ± 0.3    | 4.7 ± 0.4     | 3.4 ± 0.2     | 4.05 ± 0.08 | 1.9 ± 0.1     | 2.84 ± 0.09 | 3.7 ± 0.2    | 3.1 ± 0.2     | E Asia F     | E Europe M   |
| 17  | 4.9 ± 0.3     | 4.5 ± 0.1   | 2.0 ± 0.1     | 3.8 ± 0.2   | 4.4 ± 0.2    | 4.6 ± 0.3     | 3.2 ± 0.2     | 3.9 ± 0.1   | 1.30 ± 0.10   | 2.5 ± 0.1   | 3.5 ± 0.2    | 3.6 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 4.1 ± 0.2     | 4.16 ± 0.10 | 1.8 ± 0.2     | 3.3 ± 0.1   | 4.1 ± 0.1    | 4.3 ± 0.3     | 3.0 ± 0.2     | 3.66 ± 0.08 | 1.13 ± 0.10   | 2.23 ± 0.09 | 3.3 ± 0.2    | 3.3 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.4 ± 0.3     | 3.95 ± 0.10 | 1.7 ± 0.2     | 3.2 ± 0.1   | 3.9 ± 0.1    | 4.5 ± 0.3     | 3.4 ± 0.2     | 3.65 ± 0.09 | 1.1 ± 0.1     | 2.3 ± 0.1   | 3.1 ± 0.1    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.6 ± 0.3     | 3.8 ± 0.1   | 1.9 ± 0.1     | 3.2 ± 0.1   | 3.8 ± 0.1    | 4.6 ± 0.3     | 3.5 ± 0.3     | 3.59 ± 0.09 | 1.5 ± 0.1     | 2.6 ± 0.1   | 3.2 ± 0.1    | 3.8 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 4.5 ± 0.2     | 3.65 ± 0.10 | 2.1 ± 0.1     | 3.1 ± 0.1   | 3.6 ± 0.1    | 4.3 ± 0.3     | 3.7 ± 0.2     | 3.49 ± 0.08 | 1.8 ± 0.1     | 2.7 ± 0.1   | 3.2 ± 0.1    | 3.9 ± 0.3     | E Africa F   | E Europe M   |
| 22  | 4.6 ± 0.2     | 3.43 ± 0.09 | 2.24 ± 0.09   | 3.10 ± 0.09 | 3.5 ± 0.1    | 4.1 ± 0.2     | 4.4 ± 0.2     | 3.38 ± 0.09 | 1.8 ± 0.1     | 2.8 ± 0.1   | 3.0 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 23  | 4.6 ± 0.2     | 3.19 ± 0.08 | 2.39 ± 0.10   | 3.09 ± 0.08 | 3.26 ± 0.10  | 4.1 ± 0.2     | 4.2 ± 0.2     | 3.20 ± 0.08 | 2.0 ± 0.1     | 2.9 ± 0.1   | 3.0 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 4.7 ± 0.2     | 3.02 ± 0.07 | 2.44 ± 0.08   | 2.95 ± 0.08 | 3.08 ± 0.09  | 4.0 ± 0.2     | 4.2 ± 0.2     | 3.02 ± 0.08 | 2.01 ± 0.09   | 2.80 ± 0.09 | 2.8 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 4.5 ± 0.2     | 2.86 ± 0.06 | 2.48 ± 0.07   | 2.88 ± 0.07 | 2.92 ± 0.09  | 4.0 ± 0.2     | 4.2 ± 0.2     | 2.86 ± 0.08 | 2.22 ± 0.10   | 2.73 ± 0.08 | 2.7 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 4.2 ± 0.2     | 2.73 ± 0.06 | 2.58 ± 0.07   | 2.82 ± 0.06 | 2.80 ± 0.08  | 3.7 ± 0.2     | 4.2 ± 0.2     | 2.63 ± 0.08 | 2.2 ± 0.1     | 2.78 ± 0.08 | 2.6 ± 0.1    | 3.5 ± 0.2     | E Africa M   | E Europe M   |
| 27  | 4.6 ± 0.2     | 2.62 ± 0.05 | 2.61 ± 0.08   | 2.82 ± 0.06 | 2.67 ± 0.08  | 3.7 ± 0.2     | 4.1 ± 0.2     | 2.54 ± 0.07 | 2.21 ± 0.09   | 2.63 ± 0.07 | 2.5 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 28  | 4.4 ± 0.2     | 2.53 ± 0.05 | 2.63 ± 0.07   | 2.90 ± 0.06 | 2.68 ± 0.08  | 3.6 ± 0.2     | 3.9 ± 0.2     | 2.42 ± 0.08 | 2.34 ± 0.09   | 2.63 ± 0.07 | 2.5 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 4.6 ± 0.2     | 2.47 ± 0.05 | 2.63 ± 0.08   | 2.87 ± 0.06 | 2.63 ± 0.08  | 3.6 ± 0.1     | 3.9 ± 0.2     | 2.34 ± 0.07 | 2.35 ± 0.09   | 2.61 ± 0.06 | 2.4 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Asia M     |
| 30  | 4.3 ± 0.2     | 2.47 ± 0.05 | 2.68 ± 0.08   | 2.88 ± 0.06 | 2.63 ± 0.08  | 3.6 ± 0.1     | 3.9 ± 0.2     | 2.25 ± 0.06 | 2.24 ± 0.09   | 2.57 ± 0.05 | 2.4 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.3 ± 0.2     | 2.46 ± 0.05 | 2.62 ± 0.08   | 2.93 ± 0.06 | 2.66 ± 0.08  | 3.5 ± 0.1     | 3.9 ± 0.2     | 2.24 ± 0.07 | 2.42 ± 0.09   | 2.56 ± 0.05 | 2.4 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 32  | 4.5 ± 0.2     | 2.47 ± 0.05 | 2.75 ± 0.08   | 2.97 ± 0.06 | 2.72 ± 0.09  | 3.7 ± 0.2     | 3.9 ± 0.2     | 2.31 ± 0.06 | 2.40 ± 0.10   | 2.59 ± 0.05 | 2.4 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 33  | 4.3 ± 0.2     | 2.54 ± 0.05 | 2.73 ± 0.09   | 2.99 ± 0.06 | 2.67 ± 0.09  | 3.7 ± 0.2     | 4.0 ± 0.2     | 2.28 ± 0.06 | 2.4 ± 0.1     | 2.64 ± 0.05 | 2.4 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Asia M     |
| 34  | 4.4 ± 0.2     | 2.58 ± 0.05 | 2.78 ± 0.09   | 3.01 ± 0.06 | 2.78 ± 0.09  | 3.7 ± 0.2     | 3.8 ± 0.2     | 2.35 ± 0.06 | 2.5 ± 0.1     | 2.72 ± 0.06 | 2.5 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Asia M     |
| 35  | 4.7 ± 0.2     | 2.54 ± 0.05 | 2.85 ± 0.10   | 3.04 ± 0.07 | 2.65 ± 0.09  | 3.6 ± 0.2     | 3.9 ± 0.2     | 2.36 ± 0.07 | 2.4 ± 0.1     | 2.76 ± 0.06 | 2.7 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 36  | 4.5 ± 0.2     | 2.58 ± 0.05 | 2.82 ± 0.10   | 3.04 ± 0.07 | 2.77 ± 0.09  | 3.6 ± 0.2     | 4.0 ± 0.2     | 2.37 ± 0.07 | 2.6 ± 0.1     | 2.73 ± 0.06 | 2.7 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 37  | 4.6 ± 0.2     | 2.55 ± 0.05 | 2.9 ± 0.1     | 3.25 ± 0.07 | 2.76 ± 0.09  | 3.8 ± 0.2     | 4.0 ± 0.2     | 2.44 ± 0.07 | 2.5 ± 0.1     | 2.83 ± 0.06 | 2.7 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Asia M     |
| 38  | 4.8 ± 0.2     | 2.62 ± 0.05 | 3.1 ± 0.1     | 3.23 ± 0.08 | 2.85 ± 0.10  | 3.7 ± 0.2     | 3.9 ± 0.2     | 2.44 ± 0.07 | 2.8 ± 0.1     | 2.87 ± 0.06 | 2.9 ± 0.2    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 39  | 4.6 ± 0.2     | 2.67 ± 0.06 | 3.0 ± 0.1     | 3.21 ± 0.08 | 2.9 ± 0.1    | 3.6 ± 0.2     | 3.9 ± 0.2     | 2.55 ± 0.07 | 2.8 ± 0.1     | 2.93 ± 0.07 | 2.9 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 40  | 4.6 ± 0.3     | 2.66 ± 0.06 | 3.2 ± 0.1     | 3.24 ± 0.09 | 3.0 ± 0.1    | 3.6 ± 0.2     | 3.9 ± 0.2     | 2.53 ± 0.07 | 2.9 ± 0.1     | 3.03 ± 0.07 | 2.9 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Asia M     |
| 41  | 4.6 ± 0.3     | 2.72 ± 0.06 | 3.3 ± 0.1     | 3.40 ± 0.10 | 3.0 ± 0.1    | 3.6 ± 0.2     | 4.0 ± 0.2     | 2.60 ± 0.07 | 2.9 ± 0.1     | 3.03 ± 0.07 | 2.9 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 42  | 4.8 ± 0.3     | 2.77 ± 0.06 | 3.5 ± 0.2     | 3.4 ± 0.1   | 2.9 ± 0.1    | 3.7 ± 0.2     | 4.0 ± 0.2     | 2.68 ± 0.08 | 3.2 ± 0.2     | 3.07 ± 0.08 | 3.1 ± 0.1    | 3.3 ± 0.1     | E Africa F   | E Asia M     |
| 43  | 4.9 ± 0.3     | 2.80 ± 0.06 | 3.4 ± 0.2     | 3.4 ± 0.1   | 3.1 ± 0.1    | 3.6 ± 0.2     | 4.2 ± 0.3     | 2.79 ± 0.08 | 3.2 ± 0.2     | 3.16 ± 0.09 | 3.3 ± 0.2    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 44  | 5.2 ± 0.3     | 2.88 ± 0.06 | 3.5 ± 0.2     | 3.7 ± 0.1   | 3.3 ± 0.1    | 3.7 ± 0.2     | 4.0 ± 0.2     | 2.90 ± 0.08 | 3.4 ± 0.2     | 3.26 ± 0.09 | 3.4 ± 0.2    | 3.3 ± 0.2     | E Africa F   | E Asia F     |
| 45  | 4.9 ± 0.3     | 2.91 ± 0.06 | 3.4 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.1    | 3.6 ± 0.2     | 4.0 ± 0.2     | 2.98 ± 0.08 | 3.4 ± 0.2     | 3.20 ± 0.10 | 3.4 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Asia F     |
| 46  | 5.5 ± 0.4     | 2.93 ± 0.06 | 3.7 ± 0.2     | 3.5 ± 0.1   | 3.3 ± 0.1    | 3.6 ± 0.2     | 4.0 ± 0.3     | 2.95 ± 0.08 | 3.7 ± 0.2     | 3.2 ± 0.1   | 3.4 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia F     |
| 47  | 5.2 ± 0.4     | 3.02 ± 0.07 | 3.8 ± 0.2     | 3.6 ± 0.1   | 3.1 ± 0.1    | 3.8 ± 0.2     | 4.2 ± 0.3     | 3.05 ± 0.08 | 3.6 ± 0.2     | 3.4 ± 0.1   | 3.6 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Asia F     |
| 48  | 5.1 ± 0.4     | 2.98 ± 0.07 | 3.7 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.1    | 3.8 ± 0.2     | 3.8 ± 0.3     | 3.03 ± 0.08 | 4.1 ± 0.2     | 3.3 ± 0.1   | 3.5 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia F     |
| 49  | 5.3 ± 0.4     | 3.03 ± 0.07 | 3.6 ± 0.2     | 3.7 ± 0.1   | 3.3 ± 0.1    | 3.9 ± 0.3     | 4.2 ± 0.3     | 3.17 ± 0.09 | 3.8 ± 0.2     | 3.3 ± 0.1   | 3.5 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia F     |
| 50  | 5.0 ± 0.4     | 3.10 ± 0.08 | 3.7 ± 0.2     | 3.7 ± 0.1   | 3.2 ± 0.1    | 3.7 ± 0.2     | 3.8 ± 0.3     | 3.12 ± 0.09 | 3.8 ± 0.2     | 3.3 ± 0.1   | 3.7 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia F     |
| 51  | 5.4 ± 0.4     | 3.16 ± 0.08 | 3.7 ± 0.2     | 3.8 ± 0.1   | 3.2 ± 0.1    | 3.8 ± 0.3     | 4.0 ± 0.3     | 3.14 ± 0.09 | 3.8 ± 0.2     | 3.5 ± 0.1   | 3.6 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia M     |
| 52  | 5.4 ± 0.4     | 3.08 ± 0.08 | 3.7 ± 0.2     | 3.7 ± 0.1   | 3.3 ± 0.1    | 3.4 ± 0.2     | 4.2 ± 0.3     | 3.11 ± 0.09 | 3.7 ± 0.2     | 3.5 ± 0.1   | 3.6 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia F     |

**Table 102.** Mean absolute error for Application-type images for algorithm roc-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 2.8 ± 0.2     | 3.49 ± 0.09 | 2.6 ± 0.1     | 2.60 ± 0.09 | 3.0 ± 0.2    | 3.0 ± 0.2     | 1.9 ± 0.1     | 2.86 ± 0.09 | 1.8 ± 0.1     | 2.13 ± 0.07 | 2.1 ± 0.1    | 1.9 ± 0.2     | E Asia F     | E Europe M   |
| 15  | 2.9 ± 0.2     | 3.50 ± 0.10 | 2.1 ± 0.1     | 2.5 ± 0.1   | 2.9 ± 0.2    | 2.8 ± 0.2     | 1.8 ± 0.1     | 2.91 ± 0.08 | 1.6 ± 0.1     | 1.96 ± 0.08 | 2.3 ± 0.1    | 1.9 ± 0.2     | E Asia F     | E Europe M   |
| 16  | 2.6 ± 0.2     | 3.4 ± 0.1   | 1.78 ± 0.10   | 2.0 ± 0.1   | 3.0 ± 0.2    | 2.8 ± 0.4     | 1.8 ± 0.2     | 2.82 ± 0.08 | 1.5 ± 0.1     | 1.77 ± 0.08 | 2.1 ± 0.2    | 1.8 ± 0.2     | E Asia F     | E Europe M   |
| 17  | 2.7 ± 0.3     | 3.1 ± 0.1   | 1.4 ± 0.1     | 1.9 ± 0.1   | 2.9 ± 0.2    | 2.5 ± 0.2     | 1.7 ± 0.2     | 2.70 ± 0.10 | 1.09 ± 0.09   | 1.52 ± 0.10 | 2.0 ± 0.2    | 2.1 ± 0.3     | E Asia F     | E Europe M   |
| 18  | 2.4 ± 0.2     | 2.91 ± 0.09 | 1.3 ± 0.2     | 1.72 ± 0.09 | 2.6 ± 0.1    | 2.6 ± 0.2     | 1.8 ± 0.1     | 2.57 ± 0.08 | 1.1 ± 0.1     | 1.48 ± 0.08 | 2.0 ± 0.2    | 2.2 ± 0.2     | E Asia F     | E Europe M   |
| 19  | 2.9 ± 0.2     | 2.80 ± 0.10 | 1.4 ± 0.1     | 1.8 ± 0.1   | 2.6 ± 0.1    | 2.7 ± 0.2     | 2.3 ± 0.2     | 2.50 ± 0.09 | 1.3 ± 0.1     | 1.59 ± 0.08 | 1.9 ± 0.1    | 2.4 ± 0.3     | E Africa F   | E Europe M   |
| 20  | 3.2 ± 0.2     | 2.9 ± 0.1   | 1.6 ± 0.1     | 2.0 ± 0.1   | 2.7 ± 0.1    | 2.9 ± 0.2     | 2.6 ± 0.2     | 2.48 ± 0.09 | 1.4 ± 0.1     | 1.93 ± 0.09 | 2.1 ± 0.1    | 2.6 ± 0.2     | E Africa F   | E Europe M   |
| 21  | 3.1 ± 0.2     | 2.86 ± 0.09 | 2.03 ± 0.09   | 2.31 ± 0.08 | 2.7 ± 0.1    | 3.0 ± 0.2     | 2.9 ± 0.2     | 2.45 ± 0.08 | 1.8 ± 0.1     | 2.16 ± 0.09 | 2.2 ± 0.1    | 2.8 ± 0.2     | E Africa F   | E Europe M   |
| 22  | 3.4 ± 0.2     | 2.85 ± 0.08 | 2.30 ± 0.08   | 2.48 ± 0.07 | 2.8 ± 0.1    | 3.0 ± 0.2     | 3.4 ± 0.2     | 2.44 ± 0.08 | 1.9 ± 0.1     | 2.37 ± 0.09 | 2.2 ± 0.1    | 2.7 ± 0.2     | E Africa M   | E Europe M   |
| 23  | 3.5 ± 0.2     | 2.71 ± 0.07 | 2.52 ± 0.09   | 2.68 ± 0.07 | 2.72 ± 0.09  | 3.0 ± 0.2     | 3.4 ± 0.2     | 2.39 ± 0.08 | 2.1 ± 0.1     | 2.59 ± 0.08 | 2.3 ± 0.1    | 3.0 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 3.5 ± 0.2     | 2.69 ± 0.06 | 2.48 ± 0.07   | 2.70 ± 0.06 | 2.65 ± 0.08  | 3.0 ± 0.2     | 3.3 ± 0.2     | 2.29 ± 0.07 | 2.11 ± 0.09   | 2.59 ± 0.08 | 2.27 ± 0.10  | 3.0 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 3.5 ± 0.2     | 2.59 ± 0.06 | 2.55 ± 0.07   | 2.80 ± 0.06 | 2.67 ± 0.08  | 3.1 ± 0.2     | 3.4 ± 0.2     | 2.35 ± 0.07 | 2.15 ± 0.10   | 2.55 ± 0.07 | 2.30 ± 0.09  | 2.9 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 3.5 ± 0.2     | 2.49 ± 0.05 | 2.59 ± 0.07   | 2.84 ± 0.06 | 2.61 ± 0.07  | 3.0 ± 0.1     | 3.4 ± 0.2     | 2.23 ± 0.07 | 2.09 ± 0.09   | 2.53 ± 0.07 | 2.4 ± 0.1    | 2.7 ± 0.1     | E Africa F   | E Europe M   |
| 27  | 3.7 ± 0.2     | 2.40 ± 0.05 | 2.54 ± 0.08   | 2.91 ± 0.06 | 2.52 ± 0.07  | 3.2 ± 0.1     | 3.3 ± 0.2     | 2.21 ± 0.07 | 2.11 ± 0.09   | 2.40 ± 0.06 | 2.4 ± 0.1    | 2.8 ± 0.1     | E Africa F   | E Europe M   |
| 28  | 3.6 ± 0.2     | 2.37 ± 0.05 | 2.55 ± 0.07   | 3.03 ± 0.06 | 2.59 ± 0.07  | 3.1 ± 0.1     | 3.2 ± 0.1     | 2.20 ± 0.07 | 2.05 ± 0.09   | 2.36 ± 0.06 | 2.4 ± 0.1    | 2.7 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 3.8 ± 0.2     | 2.29 ± 0.05 | 2.48 ± 0.07   | 3.08 ± 0.06 | 2.56 ± 0.08  | 3.1 ± 0.1     | 3.1 ± 0.1     | 2.08 ± 0.06 | 2.04 ± 0.08   | 2.31 ± 0.05 | 2.3 ± 0.1    | 2.8 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 3.7 ± 0.2     | 2.34 ± 0.05 | 2.55 ± 0.08   | 3.13 ± 0.06 | 2.57 ± 0.07  | 3.3 ± 0.1     | 3.2 ± 0.1     | 2.19 ± 0.06 | 2.05 ± 0.08   | 2.35 ± 0.05 | 2.3 ± 0.1    | 2.8 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 3.9 ± 0.2     | 2.40 ± 0.05 | 2.61 ± 0.08   | 3.25 ± 0.06 | 2.57 ± 0.07  | 3.4 ± 0.1     | 3.3 ± 0.1     | 2.17 ± 0.06 | 2.21 ± 0.09   | 2.40 ± 0.05 | 2.5 ± 0.1    | 2.9 ± 0.1     | E Africa F   | E Asia M     |
| 32  | 4.0 ± 0.2     | 2.44 ± 0.05 | 2.67 ± 0.08   | 3.30 ± 0.07 | 2.62 ± 0.08  | 3.5 ± 0.1     | 3.3 ± 0.1     | 2.22 ± 0.06 | 2.24 ± 0.09   | 2.50 ± 0.05 | 2.4 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Asia M     |
| 33  | 4.0 ± 0.2     | 2.51 ± 0.05 | 2.67 ± 0.08   | 3.33 ± 0.07 | 2.69 ± 0.08  | 3.5 ± 0.1     | 3.4 ± 0.1     | 2.22 ± 0.06 | 2.22 ± 0.09   | 2.56 ± 0.05 | 2.4 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Asia M     |
| 34  | 4.0 ± 0.2     | 2.54 ± 0.05 | 2.68 ± 0.09   | 3.29 ± 0.07 | 2.61 ± 0.08  | 3.5 ± 0.1     | 3.3 ± 0.1     | 2.25 ± 0.06 | 2.29 ± 0.09   | 2.64 ± 0.05 | 2.5 ± 0.1    | 3.0 ± 0.1     | E Africa F   | E Asia M     |
| 35  | 4.3 ± 0.2     | 2.54 ± 0.05 | 2.74 ± 0.09   | 3.35 ± 0.07 | 2.65 ± 0.09  | 3.7 ± 0.2     | 3.3 ± 0.2     | 2.29 ± 0.06 | 2.3 ± 0.1     | 2.73 ± 0.06 | 2.5 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 36  | 4.4 ± 0.2     | 2.60 ± 0.05 | 2.76 ± 0.10   | 3.36 ± 0.07 | 2.68 ± 0.09  | 3.5 ± 0.2     | 3.4 ± 0.2     | 2.32 ± 0.06 | 2.3 ± 0.1     | 2.78 ± 0.06 | 2.4 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 37  | 4.3 ± 0.2     | 2.61 ± 0.05 | 2.66 ± 0.10   | 3.47 ± 0.08 | 2.62 ± 0.09  | 3.9 ± 0.2     | 3.4 ± 0.2     | 2.34 ± 0.06 | 2.39 ± 0.10   | 2.89 ± 0.06 | 2.4 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Asia M     |
| 38  | 4.3 ± 0.2     | 2.64 ± 0.05 | 2.7 ± 0.1     | 3.40 ± 0.08 | 2.68 ± 0.09  | 3.8 ± 0.2     | 3.2 ± 0.2     | 2.38 ± 0.07 | 2.31 ± 0.10   | 2.91 ± 0.06 | 2.7 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 39  | 4.3 ± 0.2     | 2.67 ± 0.06 | 2.6 ± 0.1     | 3.45 ± 0.09 | 2.64 ± 0.09  | 3.6 ± 0.2     | 3.4 ± 0.2     | 2.41 ± 0.07 | 2.3 ± 0.1     | 2.92 ± 0.06 | 2.6 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Europe M   |
| 40  | 4.3 ± 0.2     | 2.68 ± 0.06 | 2.7 ± 0.1     | 3.35 ± 0.09 | 2.8 ± 0.1    | 3.9 ± 0.2     | 3.3 ± 0.2     | 2.38 ± 0.06 | 2.4 ± 0.1     | 3.03 ± 0.07 | 2.4 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Asia M     |
| 41  | 4.4 ± 0.3     | 2.72 ± 0.06 | 2.8 ± 0.1     | 3.44 ± 0.10 | 2.78 ± 0.10  | 4.0 ± 0.2     | 3.4 ± 0.2     | 2.48 ± 0.07 | 2.4 ± 0.1     | 2.99 ± 0.07 | 2.7 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.4 ± 0.3     | 2.77 ± 0.06 | 2.8 ± 0.1     | 3.4 ± 0.1   | 2.7 ± 0.1    | 3.8 ± 0.2     | 3.4 ± 0.2     | 2.50 ± 0.07 | 2.6 ± 0.1     | 3.02 ± 0.08 | 2.7 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 43  | 4.4 ± 0.3     | 2.81 ± 0.06 | 3.0 ± 0.1     | 3.3 ± 0.1   | 2.9 ± 0.1    | 3.8 ± 0.2     | 3.4 ± 0.2     | 2.58 ± 0.07 | 2.6 ± 0.1     | 3.15 ± 0.08 | 2.8 ± 0.1    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 44  | 4.6 ± 0.3     | 2.82 ± 0.06 | 2.9 ± 0.1     | 3.6 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.2     | 3.5 ± 0.2     | 2.66 ± 0.07 | 2.6 ± 0.1     | 3.22 ± 0.09 | 2.8 ± 0.1    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.6 ± 0.3     | 2.87 ± 0.06 | 2.7 ± 0.1     | 3.5 ± 0.1   | 2.9 ± 0.1    | 3.9 ± 0.2     | 3.5 ± 0.2     | 2.75 ± 0.07 | 2.6 ± 0.1     | 3.19 ± 0.10 | 2.9 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.8 ± 0.3     | 2.94 ± 0.06 | 3.0 ± 0.1     | 3.4 ± 0.1   | 3.0 ± 0.1    | 4.0 ± 0.2     | 3.5 ± 0.2     | 2.78 ± 0.08 | 2.8 ± 0.1     | 3.15 ± 0.10 | 3.0 ± 0.1    | 4.0 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 4.6 ± 0.4     | 2.93 ± 0.07 | 3.2 ± 0.2     | 3.5 ± 0.1   | 2.9 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.76 ± 0.08 | 2.8 ± 0.1     | 3.4 ± 0.1   | 2.9 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.7 ± 0.4     | 2.88 ± 0.07 | 3.2 ± 0.2     | 3.5 ± 0.1   | 2.9 ± 0.1    | 4.1 ± 0.3     | 3.7 ± 0.3     | 2.77 ± 0.07 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.0 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Asia M     |
| 49  | 4.6 ± 0.4     | 2.94 ± 0.07 | 3.3 ± 0.2     | 3.4 ± 0.1   | 3.0 ± 0.1    | 4.4 ± 0.3     | 3.6 ± 0.3     | 2.94 ± 0.08 | 3.0 ± 0.2     | 3.3 ± 0.1   | 2.9 ± 0.2    | 4.1 ± 0.2     | E Africa F   | S E Asia M   |
| 50  | 4.3 ± 0.4     | 3.02 ± 0.08 | 3.3 ± 0.1     | 3.3 ± 0.1   | 3.0 ± 0.1    | 4.2 ± 0.3     | 3.4 ± 0.3     | 2.93 ± 0.08 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.1 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Europe M   |
| 51  | 4.7 ± 0.4     | 3.08 ± 0.08 | 3.2 ± 0.2     | 3.4 ± 0.1   | 2.9 ± 0.1    | 4.1 ± 0.3     | 3.9 ± 0.3     | 3.00 ± 0.09 | 2.7 ± 0.1     | 3.4 ± 0.1   | 3.1 ± 0.2    | 4.6 ± 0.3     | E Africa F   | E Europe M   |
| 52  | 4.6 ± 0.4     | 3.06 ± 0.08 | 3.3 ± 0.1     | 3.4 ± 0.1   | 3.0 ± 0.1    | 4.1 ± 0.3     | 3.6 ± 0.3     | 2.97 ± 0.09 | 2.9 ± 0.2     | 3.3 ± 0.1   | 3.0 ± 0.2    | 4.5 ± 0.3     | E Africa F   | E Europe M   |

**Table 103.** Mean absolute error for Application-type images for algorithm shufti-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 5.2 ± 0.3     | 3.9 ± 0.1   | 3.6 ± 0.3     | 4.2 ± 0.1   | 3.8 ± 0.2    | 4.8 ± 0.3     | 3.7 ± 0.2     | 3.0 ± 0.1   | 2.5 ± 0.1     | 2.8 ± 0.1   | 2.7 ± 0.2    | 3.6 ± 0.3     | E Africa F   | E Europe M   |
| 15  | 5.1 ± 0.3     | 3.9 ± 0.1   | 4.0 ± 0.3     | 4.4 ± 0.2   | 3.9 ± 0.3    | 4.7 ± 0.4     | 3.5 ± 0.3     | 3.0 ± 0.1   | 2.5 ± 0.2     | 2.9 ± 0.1   | 2.8 ± 0.2    | 3.5 ± 0.3     | E Africa F   | E Europe M   |
| 16  | 4.9 ± 0.3     | 3.6 ± 0.1   | 4.2 ± 0.3     | 4.1 ± 0.2   | 3.9 ± 0.3    | 4.7 ± 0.4     | 3.9 ± 0.3     | 2.9 ± 0.1   | 2.7 ± 0.2     | 3.1 ± 0.1   | 2.8 ± 0.2    | 3.5 ± 0.3     | E Africa F   | E Europe M   |
| 17  | 5.1 ± 0.4     | 3.4 ± 0.1   | 4.2 ± 0.3     | 4.2 ± 0.2   | 3.4 ± 0.3    | 4.6 ± 0.3     | 3.6 ± 0.3     | 2.9 ± 0.1   | 2.2 ± 0.2     | 3.0 ± 0.2   | 2.8 ± 0.2    | 3.6 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 4.7 ± 0.3     | 3.3 ± 0.1   | 3.9 ± 0.3     | 3.9 ± 0.1   | 3.4 ± 0.2    | 4.3 ± 0.3     | 3.7 ± 0.3     | 3.0 ± 0.1   | 2.4 ± 0.2     | 3.4 ± 0.1   | 3.0 ± 0.2    | 3.1 ± 0.3     | E Africa F   | E Europe M   |
| 19  | 5.2 ± 0.3     | 3.3 ± 0.1   | 4.2 ± 0.3     | 4.0 ± 0.2   | 3.6 ± 0.2    | 4.4 ± 0.3     | 3.9 ± 0.3     | 3.3 ± 0.1   | 2.9 ± 0.3     | 3.7 ± 0.1   | 3.3 ± 0.2    | 3.9 ± 0.4     | E Africa F   | E Europe M   |
| 20  | 5.1 ± 0.3     | 3.4 ± 0.1   | 4.4 ± 0.3     | 3.9 ± 0.2   | 3.8 ± 0.2    | 4.2 ± 0.3     | 4.1 ± 0.3     | 3.5 ± 0.1   | 3.2 ± 0.3     | 4.0 ± 0.2   | 3.5 ± 0.2    | 3.9 ± 0.3     | E Africa F   | E Europe M   |
| 21  | 5.0 ± 0.3     | 3.5 ± 0.1   | 4.4 ± 0.2     | 4.1 ± 0.1   | 3.8 ± 0.2    | 4.6 ± 0.3     | 4.8 ± 0.3     | 3.7 ± 0.1   | 3.6 ± 0.3     | 4.3 ± 0.2   | 3.9 ± 0.2    | 4.2 ± 0.3     | E Africa F   | E Asia F     |
| 22  | 5.2 ± 0.3     | 3.6 ± 0.1   | 4.4 ± 0.2     | 4.1 ± 0.1   | 3.8 ± 0.2    | 4.2 ± 0.3     | 5.6 ± 0.3     | 4.0 ± 0.1   | 3.9 ± 0.3     | 4.5 ± 0.2   | 3.8 ± 0.2    | 4.0 ± 0.3     | E Africa M   | E Asia F     |
| 23  | 5.3 ± 0.3     | 3.56 ± 0.10 | 4.5 ± 0.2     | 4.2 ± 0.1   | 3.9 ± 0.1    | 4.7 ± 0.3     | 5.5 ± 0.3     | 4.0 ± 0.1   | 4.3 ± 0.3     | 4.7 ± 0.1   | 3.9 ± 0.2    | 4.8 ± 0.3     | E Africa M   | E Asia F     |
| 24  | 5.4 ± 0.3     | 3.78 ± 0.09 | 4.4 ± 0.1     | 4.2 ± 0.1   | 3.9 ± 0.1    | 4.6 ± 0.2     | 5.6 ± 0.3     | 4.1 ± 0.1   | 4.3 ± 0.2     | 4.8 ± 0.1   | 3.9 ± 0.2    | 4.8 ± 0.3     | E Africa M   | E Asia F     |
| 25  | 5.3 ± 0.2     | 3.64 ± 0.09 | 4.3 ± 0.1     | 4.17 ± 0.09 | 4.0 ± 0.1    | 4.4 ± 0.2     | 5.6 ± 0.2     | 4.0 ± 0.1   | 4.3 ± 0.2     | 5.0 ± 0.1   | 4.0 ± 0.2    | 4.7 ± 0.2     | E Africa M   | E Asia F     |
| 26  | 5.3 ± 0.2     | 3.75 ± 0.08 | 4.4 ± 0.1     | 4.13 ± 0.08 | 4.1 ± 0.1    | 4.4 ± 0.2     | 5.4 ± 0.2     | 3.9 ± 0.1   | 4.3 ± 0.2     | 5.0 ± 0.1   | 3.9 ± 0.2    | 4.9 ± 0.2     | E Africa M   | E Asia F     |
| 27  | 5.5 ± 0.2     | 3.80 ± 0.08 | 4.5 ± 0.1     | 4.13 ± 0.08 | 3.9 ± 0.1    | 4.6 ± 0.2     | 5.4 ± 0.2     | 4.0 ± 0.1   | 4.3 ± 0.2     | 5.0 ± 0.1   | 3.9 ± 0.2    | 4.7 ± 0.2     | E Africa F   | E Asia F     |
| 28  | 5.1 ± 0.2     | 3.63 ± 0.07 | 4.3 ± 0.1     | 4.14 ± 0.08 | 4.0 ± 0.1    | 4.3 ± 0.2     | 5.2 ± 0.2     | 3.9 ± 0.1   | 4.6 ± 0.2     | 4.98 ± 0.10 | 3.9 ± 0.2    | 4.6 ± 0.2     | E Africa F   | E Asia F     |
| 29  | 5.3 ± 0.2     | 3.60 ± 0.07 | 4.1 ± 0.1     | 4.07 ± 0.08 | 4.0 ± 0.1    | 4.3 ± 0.2     | 5.2 ± 0.2     | 3.8 ± 0.1   | 4.3 ± 0.2     | 4.96 ± 0.09 | 3.8 ± 0.2    | 4.8 ± 0.2     | E Africa F   | E Asia F     |
| 30  | 5.3 ± 0.2     | 3.70 ± 0.07 | 4.0 ± 0.1     | 4.09 ± 0.08 | 4.0 ± 0.1    | 4.5 ± 0.2     | 5.1 ± 0.2     | 3.8 ± 0.1   | 4.2 ± 0.2     | 4.98 ± 0.08 | 3.9 ± 0.2    | 4.5 ± 0.2     | E Africa F   | E Asia F     |
| 31  | 5.1 ± 0.2     | 3.64 ± 0.07 | 4.0 ± 0.1     | 3.99 ± 0.08 | 3.9 ± 0.1    | 4.4 ± 0.2     | 4.9 ± 0.2     | 3.65 ± 0.10 | 4.1 ± 0.2     | 4.90 ± 0.08 | 3.9 ± 0.2    | 4.4 ± 0.2     | E Africa F   | E Asia F     |
| 32  | 5.4 ± 0.2     | 3.60 ± 0.07 | 3.9 ± 0.1     | 3.99 ± 0.08 | 4.0 ± 0.1    | 4.4 ± 0.2     | 4.9 ± 0.2     | 3.67 ± 0.09 | 4.0 ± 0.1     | 4.78 ± 0.08 | 3.9 ± 0.2    | 4.4 ± 0.2     | E Africa F   | E Asia F     |
| 33  | 5.2 ± 0.2     | 3.53 ± 0.07 | 3.9 ± 0.1     | 3.96 ± 0.08 | 3.9 ± 0.1    | 4.2 ± 0.2     | 5.0 ± 0.2     | 3.70 ± 0.10 | 3.9 ± 0.1     | 4.66 ± 0.08 | 3.6 ± 0.2    | 4.3 ± 0.2     | E Africa F   | E Asia F     |
| 34  | 4.9 ± 0.2     | 3.59 ± 0.08 | 3.7 ± 0.1     | 3.89 ± 0.08 | 3.7 ± 0.1    | 4.2 ± 0.2     | 4.6 ± 0.2     | 3.59 ± 0.09 | 3.9 ± 0.1     | 4.57 ± 0.08 | 3.8 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Asia M     |
| 35  | 5.1 ± 0.2     | 3.57 ± 0.08 | 3.8 ± 0.1     | 3.80 ± 0.08 | 3.7 ± 0.1    | 4.3 ± 0.2     | 4.7 ± 0.2     | 3.59 ± 0.09 | 3.7 ± 0.2     | 4.45 ± 0.08 | 3.6 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Asia F     |
| 36  | 5.0 ± 0.2     | 3.46 ± 0.07 | 3.7 ± 0.1     | 3.84 ± 0.08 | 3.8 ± 0.1    | 4.4 ± 0.2     | 4.7 ± 0.2     | 3.42 ± 0.09 | 3.9 ± 0.1     | 4.28 ± 0.08 | 3.7 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Asia M     |
| 37  | 5.0 ± 0.2     | 3.49 ± 0.07 | 3.7 ± 0.1     | 3.89 ± 0.09 | 3.5 ± 0.1    | 4.6 ± 0.2     | 4.6 ± 0.2     | 3.42 ± 0.09 | 3.6 ± 0.1     | 4.14 ± 0.08 | 3.5 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia M     |
| 38  | 5.0 ± 0.3     | 3.47 ± 0.08 | 3.6 ± 0.1     | 3.81 ± 0.09 | 3.5 ± 0.1    | 4.4 ± 0.2     | 4.3 ± 0.2     | 3.43 ± 0.09 | 3.6 ± 0.2     | 4.05 ± 0.08 | 3.7 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 39  | 4.7 ± 0.2     | 3.46 ± 0.07 | 3.7 ± 0.2     | 3.79 ± 0.10 | 3.5 ± 0.1    | 4.4 ± 0.2     | 4.4 ± 0.2     | 3.32 ± 0.09 | 3.6 ± 0.1     | 3.97 ± 0.08 | 3.6 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 40  | 4.8 ± 0.3     | 3.44 ± 0.08 | 3.7 ± 0.2     | 3.7 ± 0.1   | 3.6 ± 0.1    | 4.2 ± 0.2     | 4.6 ± 0.2     | 3.24 ± 0.09 | 3.5 ± 0.2     | 3.87 ± 0.08 | 3.4 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 41  | 4.7 ± 0.3     | 3.41 ± 0.08 | 3.8 ± 0.2     | 3.7 ± 0.1   | 3.3 ± 0.1    | 4.3 ± 0.2     | 4.5 ± 0.3     | 3.22 ± 0.09 | 3.6 ± 0.2     | 3.78 ± 0.08 | 3.4 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia M     |
| 42  | 4.8 ± 0.3     | 3.37 ± 0.08 | 3.7 ± 0.2     | 3.7 ± 0.1   | 3.3 ± 0.1    | 4.4 ± 0.2     | 4.6 ± 0.3     | 3.13 ± 0.09 | 3.6 ± 0.2     | 3.72 ± 0.10 | 3.3 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 43  | 4.8 ± 0.3     | 3.33 ± 0.08 | 3.8 ± 0.2     | 3.6 ± 0.1   | 3.3 ± 0.1    | 4.5 ± 0.3     | 4.7 ± 0.3     | 3.11 ± 0.08 | 3.4 ± 0.2     | 3.71 ± 0.10 | 3.4 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia M     |
| 44  | 5.0 ± 0.3     | 3.27 ± 0.08 | 3.7 ± 0.2     | 3.7 ± 0.1   | 3.4 ± 0.1    | 4.5 ± 0.3     | 4.4 ± 0.3     | 3.11 ± 0.09 | 3.7 ± 0.2     | 3.54 ± 0.10 | 3.4 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 45  | 4.9 ± 0.3     | 3.24 ± 0.08 | 3.5 ± 0.2     | 3.7 ± 0.1   | 3.4 ± 0.1    | 4.2 ± 0.3     | 4.5 ± 0.3     | 3.03 ± 0.09 | 3.6 ± 0.2     | 3.6 ± 0.1   | 3.3 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia M     |
| 46  | 5.1 ± 0.4     | 3.24 ± 0.08 | 3.9 ± 0.2     | 3.6 ± 0.1   | 3.3 ± 0.1    | 4.5 ± 0.3     | 4.8 ± 0.3     | 2.88 ± 0.09 | 3.7 ± 0.2     | 3.5 ± 0.1   | 3.4 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 4.9 ± 0.4     | 3.29 ± 0.08 | 4.0 ± 0.3     | 3.6 ± 0.1   | 3.5 ± 0.2    | 4.4 ± 0.3     | 4.7 ± 0.3     | 2.97 ± 0.09 | 3.6 ± 0.2     | 3.8 ± 0.1   | 3.5 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Asia M     |
| 48  | 5.0 ± 0.4     | 3.26 ± 0.09 | 3.9 ± 0.2     | 3.8 ± 0.1   | 3.4 ± 0.1    | 4.2 ± 0.3     | 4.5 ± 0.3     | 2.87 ± 0.08 | 3.6 ± 0.2     | 3.7 ± 0.1   | 3.4 ± 0.2    | 4.2 ± 0.2     | E Africa F   | E Asia M     |
| 49  | 5.4 ± 0.4     | 3.24 ± 0.09 | 3.9 ± 0.2     | 3.8 ± 0.2   | 3.5 ± 0.2    | 4.4 ± 0.3     | 4.4 ± 0.3     | 2.95 ± 0.09 | 3.7 ± 0.2     | 3.6 ± 0.1   | 3.3 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Asia M     |
| 50  | 5.0 ± 0.4     | 3.28 ± 0.09 | 3.9 ± 0.2     | 3.8 ± 0.1   | 3.5 ± 0.2    | 4.3 ± 0.3     | 4.4 ± 0.3     | 2.90 ± 0.09 | 3.6 ± 0.2     | 3.7 ± 0.1   | 3.5 ± 0.2    | 4.1 ± 0.3     | E Africa F   | E Asia M     |
| 51  | 4.9 ± 0.3     | 3.25 ± 0.09 | 3.8 ± 0.2     | 3.7 ± 0.1   | 3.4 ± 0.1    | 4.5 ± 0.3     | 4.6 ± 0.4     | 2.99 ± 0.09 | 3.5 ± 0.2     | 3.7 ± 0.1   | 3.6 ± 0.2    | 4.4 ± 0.2     | E Africa F   | E Asia M     |
| 52  | 4.6 ± 0.4     | 3.14 ± 0.09 | 4.1 ± 0.2     | 3.6 ± 0.1   | 3.7 ± 0.2    | 4.4 ± 0.3     | 4.5 ± 0.4     | 2.88 ± 0.09 | 3.6 ± 0.2     | 3.5 ± 0.1   | 3.4 ± 0.2    | 4.3 ± 0.3     | E Africa F   | E Asia M     |

**Table 104.** Mean absolute error for Application-type images for algorithm unissey-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.1 ± 0.3     | 4.1 ± 0.2   | 2.8 ± 0.2     | 2.8 ± 0.1   | 3.4 ± 0.3    | 4.4 ± 0.5     | 3.0 ± 0.2     | 3.7 ± 0.1   | 2.1 ± 0.2     | 2.6 ± 0.1   | 3.2 ± 0.3    | 3.1 ± 0.3     | W Africa F   | E Europe M   |
| 15  | 3.2 ± 0.3     | 4.3 ± 0.2   | 3.0 ± 0.3     | 3.1 ± 0.2   | 3.6 ± 0.3    | 4.3 ± 0.5     | 3.0 ± 0.3     | 3.9 ± 0.2   | 2.2 ± 0.2     | 3.0 ± 0.2   | 3.6 ± 0.3    | 3.1 ± 0.3     | E Asia F     | E Europe M   |
| 16  | 3.3 ± 0.3     | 4.4 ± 0.2   | 3.1 ± 0.3     | 3.2 ± 0.2   | 4.0 ± 0.3    | 4.5 ± 0.5     | 3.6 ± 0.3     | 4.2 ± 0.2   | 2.7 ± 0.3     | 3.3 ± 0.2   | 4.1 ± 0.3    | 3.0 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 3.6 ± 0.4     | 4.3 ± 0.2   | 3.7 ± 0.4     | 3.3 ± 0.2   | 3.7 ± 0.3    | 4.9 ± 0.5     | 3.7 ± 0.3     | 4.3 ± 0.2   | 2.2 ± 0.2     | 3.5 ± 0.2   | 3.8 ± 0.3    | 3.5 ± 0.4     | W Africa F   | E Europe M   |
| 18  | 3.7 ± 0.3     | 4.2 ± 0.1   | 3.5 ± 0.3     | 3.2 ± 0.1   | 3.6 ± 0.2    | 4.8 ± 0.4     | 4.1 ± 0.3     | 4.3 ± 0.1   | 2.7 ± 0.2     | 3.8 ± 0.1   | 3.9 ± 0.3    | 3.6 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.0 ± 0.3     | 4.2 ± 0.1   | 3.4 ± 0.3     | 3.5 ± 0.1   | 3.7 ± 0.2    | 4.9 ± 0.4     | 4.4 ± 0.3     | 4.3 ± 0.1   | 3.0 ± 0.2     | 3.9 ± 0.2   | 3.9 ± 0.2    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.4 ± 0.3     | 4.2 ± 0.1   | 4.0 ± 0.2     | 3.6 ± 0.1   | 3.8 ± 0.2    | 5.4 ± 0.4     | 4.7 ± 0.3     | 4.3 ± 0.1   | 3.1 ± 0.2     | 4.1 ± 0.1   | 4.2 ± 0.2    | 3.9 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 4.4 ± 0.2     | 4.1 ± 0.1   | 4.0 ± 0.2     | 3.8 ± 0.1   | 3.8 ± 0.1    | 5.5 ± 0.3     | 4.7 ± 0.3     | 4.3 ± 0.1   | 3.5 ± 0.2     | 4.0 ± 0.1   | 4.4 ± 0.2    | 4.0 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.7 ± 0.2     | 3.83 ± 0.09 | 4.5 ± 0.2     | 3.82 ± 0.10 | 3.7 ± 0.1    | 5.7 ± 0.3     | 4.9 ± 0.2     | 4.2 ± 0.1   | 3.9 ± 0.2     | 4.0 ± 0.1   | 4.1 ± 0.2    | 3.8 ± 0.2     | W Africa F   | S E Asia F   |
| 23  | 4.6 ± 0.2     | 3.72 ± 0.08 | 4.5 ± 0.2     | 3.93 ± 0.09 | 3.7 ± 0.1    | 5.4 ± 0.3     | 4.7 ± 0.2     | 4.1 ± 0.1   | 3.7 ± 0.2     | 4.0 ± 0.1   | 4.1 ± 0.2    | 3.7 ± 0.2     | W Africa F   | S E Asia F   |
| 24  | 4.7 ± 0.2     | 3.48 ± 0.08 | 4.5 ± 0.1     | 3.86 ± 0.09 | 3.54 ± 0.09  | 5.5 ± 0.3     | 4.7 ± 0.2     | 3.9 ± 0.1   | 3.8 ± 0.2     | 3.7 ± 0.1   | 3.8 ± 0.1    | 3.6 ± 0.2     | W Africa F   | E Asia F     |
| 25  | 4.8 ± 0.2     | 3.23 ± 0.07 | 4.7 ± 0.1     | 3.97 ± 0.08 | 3.36 ± 0.09  | 5.5 ± 0.2     | 4.6 ± 0.2     | 3.8 ± 0.1   | 3.7 ± 0.2     | 3.57 ± 0.10 | 3.8 ± 0.1    | 3.7 ± 0.2     | W Africa F   | E Asia F     |
| 26  | 4.7 ± 0.2     | 3.08 ± 0.07 | 4.6 ± 0.1     | 3.92 ± 0.08 | 3.33 ± 0.09  | 5.4 ± 0.2     | 4.6 ± 0.2     | 3.48 ± 0.10 | 3.8 ± 0.2     | 3.55 ± 0.09 | 3.6 ± 0.2    | 3.4 ± 0.2     | W Africa F   | E Asia F     |
| 27  | 4.9 ± 0.2     | 2.86 ± 0.06 | 4.6 ± 0.1     | 3.99 ± 0.08 | 3.17 ± 0.09  | 5.3 ± 0.2     | 4.3 ± 0.2     | 3.28 ± 0.09 | 3.8 ± 0.2     | 3.35 ± 0.09 | 3.6 ± 0.2    | 3.5 ± 0.2     | W Africa F   | E Asia F     |
| 28  | 4.8 ± 0.2     | 2.72 ± 0.06 | 4.6 ± 0.1     | 4.13 ± 0.08 | 3.12 ± 0.10  | 5.3 ± 0.2     | 4.1 ± 0.2     | 3.2 ± 0.1   | 3.7 ± 0.2     | 3.17 ± 0.08 | 3.4 ± 0.2    | 3.4 ± 0.1     | W Africa F   | E Asia F     |
| 29  | 4.7 ± 0.2     | 2.68 ± 0.06 | 4.6 ± 0.1     | 4.18 ± 0.09 | 3.2 ± 0.1    | 5.0 ± 0.2     | 4.1 ± 0.2     | 2.97 ± 0.10 | 3.7 ± 0.2     | 3.19 ± 0.08 | 3.2 ± 0.2    | 3.5 ± 0.1     | W Africa F   | E Asia F     |
| 30  | 4.8 ± 0.2     | 2.83 ± 0.06 | 4.8 ± 0.1     | 4.40 ± 0.09 | 3.2 ± 0.1    | 5.2 ± 0.2     | 4.2 ± 0.2     | 2.89 ± 0.09 | 3.6 ± 0.2     | 3.26 ± 0.07 | 3.2 ± 0.2    | 3.7 ± 0.1     | W Africa F   | E Asia F     |
| 31  | 4.9 ± 0.2     | 3.06 ± 0.07 | 4.9 ± 0.1     | 4.56 ± 0.09 | 3.5 ± 0.1    | 5.4 ± 0.2     | 4.2 ± 0.2     | 2.99 ± 0.09 | 3.7 ± 0.1     | 3.43 ± 0.07 | 3.3 ± 0.2    | 3.9 ± 0.1     | W Africa F   | E Asia M     |
| 32  | 5.2 ± 0.2     | 3.31 ± 0.06 | 5.0 ± 0.1     | 4.80 ± 0.09 | 3.8 ± 0.1    | 5.3 ± 0.2     | 4.2 ± 0.2     | 3.14 ± 0.08 | 3.8 ± 0.1     | 3.58 ± 0.07 | 3.4 ± 0.2    | 4.2 ± 0.1     | W Africa F   | E Asia M     |
| 33  | 5.3 ± 0.2     | 3.66 ± 0.06 | 5.3 ± 0.1     | 5.01 ± 0.09 | 4.0 ± 0.1    | 5.4 ± 0.2     | 4.5 ± 0.2     | 3.26 ± 0.08 | 4.0 ± 0.1     | 3.79 ± 0.06 | 3.5 ± 0.1    | 4.4 ± 0.1     | W Africa F   | E Asia M     |
| 34  | 5.3 ± 0.2     | 3.93 ± 0.06 | 5.3 ± 0.2     | 5.22 ± 0.09 | 4.2 ± 0.1    | 5.4 ± 0.2     | 4.3 ± 0.2     | 3.44 ± 0.07 | 3.9 ± 0.1     | 3.99 ± 0.06 | 3.8 ± 0.2    | 4.8 ± 0.1     | W Africa F   | E Asia M     |
| 35  | 5.5 ± 0.2     | 4.25 ± 0.07 | 5.2 ± 0.2     | 5.31 ± 0.10 | 4.5 ± 0.1    | 5.4 ± 0.2     | 4.4 ± 0.2     | 3.43 ± 0.08 | 3.8 ± 0.1     | 4.10 ± 0.07 | 3.8 ± 0.2    | 5.0 ± 0.2     | E Africa F   | E Asia M     |
| 36  | 5.6 ± 0.2     | 4.34 ± 0.08 | 5.1 ± 0.2     | 5.5 ± 0.1   | 4.5 ± 0.1    | 5.5 ± 0.2     | 4.5 ± 0.2     | 3.40 ± 0.08 | 3.8 ± 0.1     | 4.14 ± 0.07 | 3.8 ± 0.2    | 5.2 ± 0.2     | E Africa F   | E Asia M     |
| 37  | 5.2 ± 0.3     | 4.35 ± 0.08 | 5.0 ± 0.2     | 5.4 ± 0.1   | 4.5 ± 0.1    | 5.4 ± 0.2     | 4.1 ± 0.2     | 3.32 ± 0.09 | 3.3 ± 0.1     | 4.03 ± 0.09 | 3.6 ± 0.2    | 5.2 ± 0.2     | S Asia F     | E Europe M   |
| 38  | 5.0 ± 0.3     | 4.51 ± 0.09 | 5.0 ± 0.2     | 5.5 ± 0.1   | 4.5 ± 0.2    | 5.6 ± 0.3     | 4.3 ± 0.2     | 3.3 ± 0.1   | 3.1 ± 0.2     | 4.07 ± 0.09 | 3.5 ± 0.2    | 5.5 ± 0.2     | W Africa F   | E Europe M   |
| 39  | 5.3 ± 0.3     | 4.68 ± 0.10 | 4.8 ± 0.2     | 5.6 ± 0.1   | 4.6 ± 0.2    | 5.3 ± 0.3     | 4.1 ± 0.2     | 3.4 ± 0.1   | 3.1 ± 0.2     | 4.09 ± 0.10 | 3.6 ± 0.2    | 5.9 ± 0.2     | W Africa M   | E Europe M   |
| 40  | 5.2 ± 0.3     | 4.88 ± 0.10 | 5.0 ± 0.2     | 5.5 ± 0.1   | 4.9 ± 0.2    | 5.5 ± 0.3     | 4.4 ± 0.2     | 3.5 ± 0.1   | 3.3 ± 0.2     | 4.3 ± 0.1   | 3.5 ± 0.2    | 6.2 ± 0.2     | W Africa M   | E Europe M   |
| 41  | 5.9 ± 0.3     | 5.03 ± 0.10 | 5.0 ± 0.2     | 5.8 ± 0.2   | 5.2 ± 0.2    | 5.6 ± 0.3     | 4.6 ± 0.3     | 3.65 ± 0.10 | 3.3 ± 0.2     | 4.4 ± 0.1   | 3.9 ± 0.2    | 6.4 ± 0.3     | W Africa M   | E Europe M   |
| 42  | 5.5 ± 0.3     | 5.17 ± 0.10 | 5.1 ± 0.2     | 5.7 ± 0.2   | 5.1 ± 0.2    | 5.4 ± 0.3     | 4.7 ± 0.3     | 3.92 ± 0.10 | 3.3 ± 0.2     | 4.6 ± 0.1   | 3.9 ± 0.2    | 6.6 ± 0.3     | W Africa M   | E Europe M   |
| 43  | 5.7 ± 0.3     | 5.4 ± 0.1   | 4.9 ± 0.2     | 5.8 ± 0.2   | 5.4 ± 0.2    | 5.9 ± 0.3     | 4.8 ± 0.3     | 4.0 ± 0.1   | 3.5 ± 0.2     | 4.7 ± 0.1   | 4.1 ± 0.2    | 6.9 ± 0.3     | W Africa M   | E Europe M   |
| 44  | 5.9 ± 0.3     | 5.7 ± 0.1   | 4.9 ± 0.2     | 5.9 ± 0.2   | 5.4 ± 0.2    | 6.1 ± 0.3     | 4.8 ± 0.3     | 4.3 ± 0.1   | 3.5 ± 0.2     | 4.8 ± 0.1   | 4.4 ± 0.2    | 7.0 ± 0.3     | W Africa M   | E Europe M   |
| 45  | 5.9 ± 0.4     | 5.7 ± 0.1   | 4.6 ± 0.2     | 6.0 ± 0.2   | 5.4 ± 0.2    | 6.1 ± 0.3     | 5.0 ± 0.3     | 4.6 ± 0.1   | 3.4 ± 0.2     | 4.7 ± 0.1   | 4.4 ± 0.2    | 7.3 ± 0.3     | W Africa M   | E Europe M   |
| 46  | 6.0 ± 0.4     | 5.9 ± 0.1   | 4.6 ± 0.2     | 5.6 ± 0.2   | 5.8 ± 0.2    | 5.9 ± 0.3     | 5.0 ± 0.3     | 4.6 ± 0.1   | 3.5 ± 0.2     | 4.7 ± 0.1   | 4.3 ± 0.2    | 7.7 ± 0.3     | W Africa M   | E Europe M   |
| 47  | 5.8 ± 0.4     | 5.8 ± 0.1   | 4.6 ± 0.2     | 5.6 ± 0.2   | 5.5 ± 0.2    | 5.9 ± 0.4     | 5.0 ± 0.3     | 4.6 ± 0.1   | 3.4 ± 0.2     | 4.8 ± 0.2   | 4.2 ± 0.2    | 7.3 ± 0.3     | W Africa M   | E Europe M   |
| 48  | 5.9 ± 0.5     | 5.9 ± 0.1   | 4.4 ± 0.2     | 5.7 ± 0.2   | 5.5 ± 0.2    | 5.6 ± 0.4     | 5.6 ± 0.4     | 4.6 ± 0.1   | 3.3 ± 0.2     | 4.7 ± 0.2   | 4.4 ± 0.2    | 7.7 ± 0.4     | W Africa M   | E Europe M   |
| 49  | 5.8 ± 0.4     | 6.0 ± 0.1   | 4.2 ± 0.2     | 5.6 ± 0.2   | 5.3 ± 0.2    | 6.1 ± 0.4     | 5.3 ± 0.4     | 4.8 ± 0.1   | 3.4 ± 0.2     | 4.7 ± 0.2   | 4.2 ± 0.2    | 7.8 ± 0.4     | W Africa M   | E Europe M   |
| 50  | 6.1 ± 0.6     | 6.0 ± 0.1   | 4.3 ± 0.2     | 5.3 ± 0.2   | 5.6 ± 0.2    | 5.8 ± 0.4     | 5.4 ± 0.4     | 4.8 ± 0.1   | 3.3 ± 0.2     | 4.8 ± 0.2   | 4.2 ± 0.2    | 8.0 ± 0.4     | W Africa M   | E Europe M   |
| 51  | 5.4 ± 0.5     | 6.0 ± 0.1   | 4.1 ± 0.2     | 5.4 ± 0.2   | 5.6 ± 0.2    | 6.2 ± 0.4     | 5.6 ± 0.4     | 4.9 ± 0.1   | 3.3 ± 0.2     | 4.7 ± 0.2   | 4.4 ± 0.2    | 8.2 ± 0.4     | W Africa M   | E Europe M   |
| 52  | 6.2 ± 0.5     | 6.0 ± 0.2   | 4.1 ± 0.2     | 5.5 ± 0.2   | 5.2 ± 0.2    | 6.4 ± 0.4     | 5.1 ± 0.4     | 5.0 ± 0.1   | 3.6 ± 0.2     | 4.8 ± 0.2   | 4.3 ± 0.2    | 8.2 ± 0.4     | W Africa M   | E Europe M   |

**Table 105.** Mean absolute error for Application-type images for algorithm viante-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 6.5 ± 0.4     | 5.4 ± 0.1   | 2.7 ± 0.3     | 6.2 ± 0.2   | 5.9 ± 0.3    | 7.3 ± 0.4     | 4.8 ± 0.3     | 4.0 ± 0.1   | 1.1 ± 0.1     | 3.9 ± 0.2   | 4.3 ± 0.3    | 5.4 ± 0.4     | W Africa F   | E Europe M   |
| 15  | 7.6 ± 0.4     | 6.0 ± 0.1   | 3.4 ± 0.3     | 7.1 ± 0.2   | 6.8 ± 0.3    | 7.6 ± 0.4     | 5.2 ± 0.3     | 4.9 ± 0.1   | 1.4 ± 0.2     | 5.1 ± 0.2   | 5.4 ± 0.3    | 6.5 ± 0.4     | W Africa F   | E Europe M   |
| 16  | 7.5 ± 0.3     | 6.1 ± 0.1   | 4.2 ± 0.3     | 7.3 ± 0.2   | 7.2 ± 0.3    | 7.6 ± 0.4     | 6.5 ± 0.3     | 5.4 ± 0.1   | 2.3 ± 0.2     | 5.9 ± 0.2   | 6.0 ± 0.3    | 6.7 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 7.8 ± 0.3     | 6.0 ± 0.1   | 4.5 ± 0.2     | 7.2 ± 0.2   | 6.8 ± 0.3    | 7.4 ± 0.3     | 6.7 ± 0.3     | 5.5 ± 0.1   | 2.8 ± 0.2     | 6.3 ± 0.2   | 6.2 ± 0.3    | 7.2 ± 0.4     | E Africa F   | E Europe M   |
| 18  | 7.3 ± 0.2     | 5.54 ± 0.09 | 4.5 ± 0.2     | 6.7 ± 0.1   | 6.3 ± 0.1    | 7.1 ± 0.3     | 6.6 ± 0.2     | 5.36 ± 0.09 | 3.2 ± 0.2     | 6.2 ± 0.1   | 6.0 ± 0.2    | 6.8 ± 0.3     | E Africa F   | E Europe M   |
| 19  | 7.2 ± 0.2     | 5.13 ± 0.08 | 4.1 ± 0.2     | 6.5 ± 0.1   | 6.0 ± 0.1    | 7.0 ± 0.2     | 6.8 ± 0.2     | 5.30 ± 0.09 | 3.2 ± 0.1     | 6.4 ± 0.1   | 5.9 ± 0.2    | 6.7 ± 0.3     | E Africa F   | E Europe M   |
| 20  | 6.8 ± 0.2     | 4.83 ± 0.09 | 3.8 ± 0.1     | 6.1 ± 0.1   | 5.6 ± 0.1    | 6.5 ± 0.2     | 6.7 ± 0.2     | 5.02 ± 0.09 | 3.0 ± 0.1     | 6.3 ± 0.1   | 5.8 ± 0.2    | 6.6 ± 0.2     | E Africa F   | E Europe M   |
| 21  | 6.4 ± 0.2     | 4.35 ± 0.08 | 3.1 ± 0.1     | 5.8 ± 0.1   | 5.1 ± 0.1    | 6.0 ± 0.2     | 6.6 ± 0.2     | 4.73 ± 0.08 | 2.8 ± 0.1     | 6.1 ± 0.1   | 5.5 ± 0.2    | 6.3 ± 0.2     | E Africa M   | E Europe M   |
| 22  | 6.1 ± 0.2     | 3.77 ± 0.07 | 2.70 ± 0.08   | 5.4 ± 0.1   | 4.7 ± 0.1    | 5.5 ± 0.2     | 6.9 ± 0.2     | 4.41 ± 0.09 | 2.6 ± 0.1     | 5.9 ± 0.1   | 5.0 ± 0.1    | 5.9 ± 0.2     | E Africa M   | E Europe M   |
| 23  | 5.6 ± 0.2     | 3.36 ± 0.06 | 2.24 ± 0.08   | 5.17 ± 0.10 | 4.26 ± 0.09  | 5.2 ± 0.2     | 6.5 ± 0.2     | 4.02 ± 0.08 | 2.1 ± 0.1     | 5.7 ± 0.1   | 4.9 ± 0.1    | 5.8 ± 0.2     | E Africa M   | E Europe M   |
| 24  | 5.3 ± 0.2     | 2.83 ± 0.06 | 1.80 ± 0.06   | 4.78 ± 0.09 | 3.74 ± 0.08  | 4.8 ± 0.2     | 6.2 ± 0.2     | 3.65 ± 0.08 | 1.89 ± 0.09   | 5.6 ± 0.1   | 4.5 ± 0.1    | 5.4 ± 0.2     | E Africa M   | E Europe F   |
| 25  | 4.7 ± 0.2     | 2.37 ± 0.05 | 1.45 ± 0.05   | 4.35 ± 0.08 | 3.29 ± 0.08  | 4.4 ± 0.2     | 5.9 ± 0.2     | 3.38 ± 0.08 | 1.75 ± 0.10   | 5.23 ± 0.10 | 4.3 ± 0.1    | 5.1 ± 0.2     | E Africa M   | E Europe F   |
| 26  | 4.3 ± 0.2     | 2.08 ± 0.05 | 1.37 ± 0.06   | 3.93 ± 0.07 | 3.00 ± 0.08  | 4.0 ± 0.2     | 5.6 ± 0.2     | 3.04 ± 0.07 | 1.65 ± 0.10   | 5.03 ± 0.09 | 4.0 ± 0.1    | 4.5 ± 0.1     | E Africa M   | E Europe F   |
| 27  | 4.2 ± 0.2     | 1.85 ± 0.04 | 1.52 ± 0.06   | 3.60 ± 0.07 | 2.65 ± 0.07  | 3.7 ± 0.2     | 5.2 ± 0.2     | 2.77 ± 0.07 | 1.82 ± 0.10   | 4.76 ± 0.09 | 3.9 ± 0.1    | 4.1 ± 0.2     | E Africa M   | E Europe F   |
| 28  | 3.8 ± 0.2     | 1.79 ± 0.04 | 1.84 ± 0.06   | 3.37 ± 0.07 | 2.47 ± 0.07  | 3.4 ± 0.2     | 4.8 ± 0.2     | 2.57 ± 0.08 | 2.11 ± 0.09   | 4.45 ± 0.08 | 3.5 ± 0.1    | 3.6 ± 0.1     | E Africa M   | E Asia F     |
| 29  | 3.7 ± 0.2     | 1.79 ± 0.04 | 2.29 ± 0.06   | 3.11 ± 0.06 | 2.31 ± 0.07  | 3.2 ± 0.1     | 4.6 ± 0.2     | 2.41 ± 0.07 | 2.40 ± 0.09   | 4.22 ± 0.08 | 3.3 ± 0.1    | 3.5 ± 0.1     | E Africa M   | E Asia F     |
| 30  | 3.4 ± 0.2     | 1.88 ± 0.04 | 2.59 ± 0.06   | 2.94 ± 0.06 | 2.28 ± 0.07  | 3.0 ± 0.2     | 4.5 ± 0.2     | 2.19 ± 0.07 | 2.57 ± 0.09   | 4.04 ± 0.08 | 3.3 ± 0.1    | 3.1 ± 0.1     | E Africa M   | E Asia F     |
| 31  | 3.5 ± 0.2     | 2.05 ± 0.04 | 2.79 ± 0.06   | 2.84 ± 0.06 | 2.24 ± 0.07  | 2.9 ± 0.1     | 4.5 ± 0.2     | 2.23 ± 0.07 | 2.73 ± 0.09   | 3.89 ± 0.08 | 3.1 ± 0.2    | 3.0 ± 0.1     | E Africa M   | E Asia F     |
| 32  | 3.7 ± 0.2     | 2.27 ± 0.04 | 2.80 ± 0.07   | 2.82 ± 0.07 | 2.40 ± 0.08  | 3.3 ± 0.2     | 4.3 ± 0.2     | 2.32 ± 0.07 | 2.72 ± 0.09   | 3.86 ± 0.08 | 3.2 ± 0.2    | 3.0 ± 0.1     | E Africa M   | E Asia F     |
| 33  | 4.0 ± 0.2     | 2.46 ± 0.05 | 2.67 ± 0.08   | 2.89 ± 0.07 | 2.41 ± 0.08  | 3.4 ± 0.2     | 4.5 ± 0.2     | 2.39 ± 0.07 | 2.54 ± 0.09   | 3.76 ± 0.08 | 3.2 ± 0.2    | 3.1 ± 0.1     | E Africa M   | E Asia M     |
| 34  | 4.0 ± 0.2     | 2.65 ± 0.05 | 2.47 ± 0.09   | 2.95 ± 0.07 | 2.52 ± 0.08  | 3.5 ± 0.2     | 4.3 ± 0.2     | 2.45 ± 0.07 | 2.4 ± 0.1     | 3.88 ± 0.09 | 3.4 ± 0.2    | 3.3 ± 0.1     | E Africa M   | E Europe M   |
| 35  | 4.4 ± 0.2     | 2.92 ± 0.05 | 2.43 ± 0.10   | 3.16 ± 0.07 | 2.70 ± 0.09  | 3.8 ± 0.2     | 4.6 ± 0.2     | 2.56 ± 0.07 | 2.1 ± 0.1     | 3.89 ± 0.09 | 3.5 ± 0.2    | 3.6 ± 0.1     | E Africa M   | E Europe M   |
| 36  | 4.6 ± 0.2     | 3.05 ± 0.06 | 2.3 ± 0.1     | 3.38 ± 0.07 | 2.88 ± 0.10  | 4.0 ± 0.2     | 4.6 ± 0.2     | 2.61 ± 0.07 | 2.1 ± 0.1     | 3.90 ± 0.08 | 3.6 ± 0.2    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 37  | 4.7 ± 0.2     | 3.22 ± 0.06 | 2.4 ± 0.1     | 3.68 ± 0.07 | 3.08 ± 0.10  | 4.3 ± 0.2     | 4.6 ± 0.2     | 2.67 ± 0.08 | 2.1 ± 0.1     | 3.99 ± 0.08 | 3.5 ± 0.2    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 38  | 4.9 ± 0.2     | 3.46 ± 0.06 | 2.7 ± 0.1     | 3.87 ± 0.07 | 3.3 ± 0.1    | 4.6 ± 0.2     | 4.4 ± 0.2     | 2.71 ± 0.08 | 2.3 ± 0.1     | 4.00 ± 0.08 | 3.8 ± 0.2    | 3.7 ± 0.1     | E Africa F   | E Europe M   |
| 39  | 4.8 ± 0.2     | 3.63 ± 0.07 | 2.9 ± 0.1     | 3.94 ± 0.08 | 3.4 ± 0.1    | 4.3 ± 0.2     | 4.4 ± 0.2     | 2.85 ± 0.08 | 2.8 ± 0.1     | 3.99 ± 0.07 | 4.0 ± 0.2    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 40  | 4.7 ± 0.2     | 3.85 ± 0.07 | 3.3 ± 0.1     | 3.93 ± 0.09 | 3.6 ± 0.1    | 4.3 ± 0.2     | 4.4 ± 0.2     | 2.91 ± 0.08 | 3.1 ± 0.1     | 4.01 ± 0.08 | 3.8 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 41  | 4.7 ± 0.2     | 3.96 ± 0.07 | 3.6 ± 0.1     | 3.81 ± 0.09 | 3.6 ± 0.1    | 4.4 ± 0.2     | 4.3 ± 0.2     | 3.02 ± 0.08 | 3.2 ± 0.1     | 3.84 ± 0.08 | 3.9 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Asia M     |
| 42  | 4.7 ± 0.3     | 4.00 ± 0.07 | 3.6 ± 0.1     | 3.6 ± 0.1   | 3.6 ± 0.1    | 4.4 ± 0.2     | 4.0 ± 0.2     | 3.10 ± 0.08 | 3.3 ± 0.1     | 3.68 ± 0.08 | 3.9 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |
| 43  | 4.6 ± 0.3     | 4.07 ± 0.07 | 3.6 ± 0.1     | 3.5 ± 0.1   | 3.6 ± 0.1    | 4.3 ± 0.2     | 4.2 ± 0.2     | 3.08 ± 0.08 | 3.1 ± 0.1     | 3.47 ± 0.08 | 3.9 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.8 ± 0.3     | 4.07 ± 0.08 | 3.3 ± 0.1     | 3.3 ± 0.1   | 3.6 ± 0.1    | 4.5 ± 0.3     | 3.7 ± 0.2     | 3.06 ± 0.09 | 2.9 ± 0.1     | 3.21 ± 0.09 | 3.9 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.3 ± 0.3     | 3.93 ± 0.08 | 2.9 ± 0.1     | 3.2 ± 0.1   | 3.5 ± 0.1    | 4.1 ± 0.2     | 3.6 ± 0.2     | 3.07 ± 0.08 | 2.6 ± 0.1     | 2.97 ± 0.09 | 3.9 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.9 ± 0.4     | 3.91 ± 0.08 | 2.8 ± 0.1     | 3.0 ± 0.1   | 3.5 ± 0.1    | 3.9 ± 0.2     | 3.6 ± 0.2     | 3.00 ± 0.08 | 2.5 ± 0.1     | 2.89 ± 0.09 | 3.9 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 4.5 ± 0.4     | 3.71 ± 0.08 | 2.8 ± 0.2     | 2.9 ± 0.1   | 3.3 ± 0.1    | 4.1 ± 0.2     | 3.5 ± 0.2     | 3.01 ± 0.08 | 2.3 ± 0.1     | 2.9 ± 0.1   | 3.8 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.3 ± 0.4     | 3.51 ± 0.09 | 2.4 ± 0.1     | 3.1 ± 0.1   | 3.2 ± 0.1    | 4.0 ± 0.3     | 3.5 ± 0.3     | 2.98 ± 0.08 | 2.2 ± 0.1     | 2.63 ± 0.10 | 3.6 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 49  | 4.4 ± 0.4     | 3.36 ± 0.09 | 2.4 ± 0.1     | 3.0 ± 0.1   | 3.0 ± 0.1    | 4.1 ± 0.3     | 3.2 ± 0.3     | 2.94 ± 0.08 | 2.2 ± 0.1     | 2.7 ± 0.1   | 3.4 ± 0.2    | 3.3 ± 0.2     | E Africa F   | E Europe M   |
| 50  | 4.3 ± 0.4     | 3.24 ± 0.09 | 2.3 ± 0.1     | 3.1 ± 0.1   | 3.1 ± 0.1    | 3.9 ± 0.3     | 3.0 ± 0.2     | 2.74 ± 0.08 | 2.2 ± 0.1     | 2.6 ± 0.1   | 3.4 ± 0.2    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 51  | 4.4 ± 0.3     | 3.17 ± 0.09 | 2.4 ± 0.1     | 3.4 ± 0.1   | 2.9 ± 0.1    | 4.1 ± 0.3     | 3.3 ± 0.3     | 2.73 ± 0.08 | 2.2 ± 0.1     | 2.7 ± 0.1   | 3.3 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 52  | 4.6 ± 0.4     | 3.08 ± 0.08 | 2.8 ± 0.1     | 3.7 ± 0.1   | 3.0 ± 0.1    | 4.0 ± 0.3     | 3.2 ± 0.3     | 2.64 ± 0.07 | 2.6 ± 0.2     | 2.7 ± 0.1   | 3.1 ± 0.2    | 3.7 ± 0.3     | E Africa F   | E Europe M   |

**Table 106.** Mean absolute error for Application-type images for algorithm yoti-001. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.0 ± 0.2     | 2.4 ± 0.1   | 2.6 ± 0.2     | 2.9 ± 0.1   | 2.5 ± 0.3    | 4.4 ± 0.4     | 2.4 ± 0.2     | 1.22 ± 0.08 | 1.47 ± 0.10   | 1.56 ± 0.07 | 1.0 ± 0.1    | 2.3 ± 0.2     | W Africa F   | S E Asia M   |
| 15  | 4.1 ± 0.3     | 2.7 ± 0.2   | 3.2 ± 0.3     | 3.4 ± 0.2   | 3.0 ± 0.3    | 4.6 ± 0.4     | 2.4 ± 0.2     | 1.40 ± 0.08 | 1.4 ± 0.1     | 1.7 ± 0.1   | 1.3 ± 0.1    | 2.4 ± 0.3     | W Africa F   | S E Asia M   |
| 16  | 4.3 ± 0.3     | 3.1 ± 0.2   | 3.4 ± 0.4     | 3.9 ± 0.2   | 3.9 ± 0.4    | 5.1 ± 0.5     | 2.5 ± 0.2     | 1.65 ± 0.08 | 1.5 ± 0.2     | 1.8 ± 0.1   | 1.6 ± 0.2    | 2.6 ± 0.3     | W Africa F   | E Europe M   |
| 17  | 4.5 ± 0.4     | 3.6 ± 0.2   | 4.1 ± 0.4     | 4.2 ± 0.2   | 4.2 ± 0.4    | 4.9 ± 0.4     | 2.4 ± 0.2     | 2.2 ± 0.1   | 1.4 ± 0.1     | 2.0 ± 0.1   | 2.0 ± 0.2    | 2.9 ± 0.3     | W Africa F   | E Europe M   |
| 18  | 4.3 ± 0.3     | 3.8 ± 0.1   | 4.3 ± 0.3     | 4.4 ± 0.2   | 4.6 ± 0.2    | 5.5 ± 0.4     | 2.7 ± 0.2     | 2.50 ± 0.09 | 2.0 ± 0.2     | 2.4 ± 0.1   | 2.6 ± 0.2    | 2.9 ± 0.3     | W Africa F   | E Europe M   |
| 19  | 4.7 ± 0.3     | 4.3 ± 0.1   | 4.7 ± 0.3     | 4.7 ± 0.2   | 5.0 ± 0.2    | 5.5 ± 0.4     | 2.8 ± 0.2     | 2.97 ± 0.10 | 2.3 ± 0.2     | 2.8 ± 0.1   | 2.9 ± 0.2    | 3.3 ± 0.3     | W Africa F   | E Europe M   |
| 20  | 4.9 ± 0.3     | 4.6 ± 0.1   | 4.7 ± 0.2     | 4.9 ± 0.2   | 5.2 ± 0.2    | 5.4 ± 0.3     | 3.2 ± 0.2     | 3.22 ± 0.10 | 2.6 ± 0.2     | 3.1 ± 0.1   | 3.3 ± 0.2    | 3.6 ± 0.3     | W Africa F   | E Europe M   |
| 21  | 5.1 ± 0.3     | 4.7 ± 0.1   | 4.6 ± 0.2     | 5.1 ± 0.2   | 5.4 ± 0.2    | 5.8 ± 0.3     | 3.2 ± 0.2     | 3.34 ± 0.09 | 2.7 ± 0.2     | 3.3 ± 0.1   | 3.5 ± 0.2    | 3.7 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 5.2 ± 0.3     | 4.8 ± 0.1   | 4.5 ± 0.2     | 5.0 ± 0.1   | 5.5 ± 0.2    | 5.5 ± 0.3     | 3.9 ± 0.2     | 3.44 ± 0.10 | 2.6 ± 0.1     | 3.3 ± 0.1   | 3.4 ± 0.1    | 3.7 ± 0.2     | S E Asia F   | E Europe M   |
| 23  | 5.2 ± 0.2     | 4.8 ± 0.1   | 4.2 ± 0.1     | 5.1 ± 0.1   | 5.4 ± 0.2    | 5.6 ± 0.3     | 3.9 ± 0.2     | 3.38 ± 0.10 | 2.4 ± 0.1     | 3.4 ± 0.1   | 3.5 ± 0.1    | 4.1 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 5.5 ± 0.3     | 4.7 ± 0.1   | 4.0 ± 0.1     | 5.1 ± 0.1   | 5.4 ± 0.1    | 5.5 ± 0.3     | 3.7 ± 0.2     | 3.27 ± 0.09 | 2.3 ± 0.1     | 3.4 ± 0.1   | 3.5 ± 0.1    | 3.7 ± 0.2     | W Africa F   | E Europe M   |
| 25  | 5.3 ± 0.2     | 4.7 ± 0.1   | 3.8 ± 0.1     | 4.9 ± 0.1   | 5.4 ± 0.1    | 5.3 ± 0.3     | 3.8 ± 0.2     | 3.17 ± 0.09 | 2.1 ± 0.1     | 3.29 ± 0.10 | 3.5 ± 0.2    | 3.9 ± 0.2     | S E Asia F   | E Europe M   |
| 26  | 5.4 ± 0.2     | 4.68 ± 0.10 | 3.5 ± 0.1     | 4.78 ± 0.10 | 5.4 ± 0.1    | 5.3 ± 0.2     | 3.8 ± 0.2     | 2.93 ± 0.09 | 1.9 ± 0.1     | 3.29 ± 0.09 | 3.4 ± 0.2    | 3.8 ± 0.2     | S E Asia F   | E Europe M   |
| 27  | 5.7 ± 0.2     | 4.67 ± 0.09 | 3.4 ± 0.1     | 4.62 ± 0.09 | 5.1 ± 0.1    | 5.5 ± 0.2     | 3.7 ± 0.2     | 2.94 ± 0.09 | 1.93 ± 0.09   | 3.09 ± 0.08 | 3.5 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 28  | 5.4 ± 0.2     | 4.54 ± 0.09 | 3.2 ± 0.1     | 4.63 ± 0.09 | 5.2 ± 0.1    | 5.2 ± 0.2     | 3.5 ± 0.2     | 3.0 ± 0.1   | 1.99 ± 0.09   | 3.14 ± 0.08 | 3.4 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 29  | 5.7 ± 0.2     | 4.57 ± 0.09 | 3.2 ± 0.1     | 4.50 ± 0.09 | 5.2 ± 0.1    | 5.4 ± 0.2     | 3.7 ± 0.2     | 2.90 ± 0.10 | 2.00 ± 0.09   | 3.09 ± 0.07 | 3.3 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 30  | 5.5 ± 0.2     | 4.59 ± 0.08 | 3.2 ± 0.1     | 4.46 ± 0.09 | 5.2 ± 0.1    | 5.3 ± 0.2     | 3.8 ± 0.2     | 3.06 ± 0.09 | 2.03 ± 0.09   | 3.18 ± 0.07 | 3.5 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 31  | 5.5 ± 0.2     | 4.55 ± 0.08 | 3.2 ± 0.1     | 4.34 ± 0.09 | 5.2 ± 0.1    | 5.2 ± 0.2     | 3.8 ± 0.2     | 3.02 ± 0.09 | 2.20 ± 0.09   | 3.16 ± 0.07 | 3.5 ± 0.2    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 5.7 ± 0.2     | 4.46 ± 0.08 | 3.2 ± 0.1     | 4.26 ± 0.08 | 4.9 ± 0.1    | 5.5 ± 0.2     | 3.9 ± 0.2     | 3.25 ± 0.09 | 2.32 ± 0.10   | 3.21 ± 0.06 | 3.7 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 33  | 5.5 ± 0.2     | 4.40 ± 0.08 | 3.2 ± 0.1     | 4.21 ± 0.08 | 4.9 ± 0.1    | 5.2 ± 0.2     | 4.0 ± 0.2     | 3.33 ± 0.09 | 2.41 ± 0.10   | 3.19 ± 0.07 | 3.4 ± 0.2    | 4.0 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 5.5 ± 0.2     | 4.28 ± 0.08 | 3.2 ± 0.1     | 4.00 ± 0.08 | 4.8 ± 0.1    | 5.3 ± 0.2     | 3.9 ± 0.2     | 3.38 ± 0.09 | 2.40 ± 0.10   | 3.26 ± 0.07 | 3.6 ± 0.2    | 4.0 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 5.7 ± 0.2     | 4.12 ± 0.08 | 3.2 ± 0.1     | 3.84 ± 0.08 | 4.5 ± 0.1    | 5.0 ± 0.2     | 4.1 ± 0.2     | 3.44 ± 0.09 | 2.4 ± 0.1     | 3.25 ± 0.07 | 3.7 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 36  | 5.3 ± 0.2     | 4.00 ± 0.08 | 3.1 ± 0.1     | 3.75 ± 0.08 | 4.4 ± 0.1    | 5.1 ± 0.2     | 4.1 ± 0.2     | 3.43 ± 0.09 | 2.5 ± 0.1     | 3.21 ± 0.07 | 3.8 ± 0.2    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 37  | 5.3 ± 0.2     | 3.84 ± 0.07 | 3.2 ± 0.1     | 3.75 ± 0.08 | 4.2 ± 0.1    | 5.1 ± 0.2     | 4.1 ± 0.2     | 3.36 ± 0.08 | 2.5 ± 0.1     | 3.23 ± 0.07 | 3.5 ± 0.2    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 38  | 5.0 ± 0.2     | 3.71 ± 0.07 | 3.0 ± 0.1     | 3.62 ± 0.09 | 4.1 ± 0.1    | 4.9 ± 0.2     | 3.9 ± 0.2     | 3.30 ± 0.09 | 2.5 ± 0.1     | 3.10 ± 0.07 | 3.6 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 39  | 4.7 ± 0.2     | 3.54 ± 0.07 | 3.0 ± 0.1     | 3.43 ± 0.09 | 3.9 ± 0.1    | 4.8 ± 0.2     | 3.9 ± 0.2     | 3.24 ± 0.09 | 2.6 ± 0.1     | 3.06 ± 0.07 | 3.6 ± 0.2    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 40  | 4.5 ± 0.2     | 3.42 ± 0.07 | 3.0 ± 0.1     | 3.30 ± 0.09 | 3.8 ± 0.1    | 4.6 ± 0.2     | 3.8 ± 0.2     | 3.11 ± 0.08 | 2.6 ± 0.1     | 3.09 ± 0.07 | 3.4 ± 0.2    | 3.8 ± 0.2     | W Africa F   | E Europe M   |
| 41  | 4.5 ± 0.3     | 3.26 ± 0.07 | 3.0 ± 0.1     | 3.22 ± 0.10 | 3.6 ± 0.1    | 4.5 ± 0.2     | 3.8 ± 0.2     | 3.06 ± 0.08 | 2.5 ± 0.1     | 2.97 ± 0.07 | 3.4 ± 0.2    | 4.0 ± 0.2     | W Africa F   | E Europe M   |
| 42  | 4.8 ± 0.3     | 3.24 ± 0.07 | 3.1 ± 0.1     | 3.4 ± 0.1   | 3.3 ± 0.1    | 4.4 ± 0.2     | 3.7 ± 0.2     | 2.97 ± 0.08 | 2.7 ± 0.1     | 3.04 ± 0.08 | 3.3 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 4.5 ± 0.3     | 3.13 ± 0.07 | 3.0 ± 0.1     | 3.2 ± 0.1   | 3.4 ± 0.1    | 4.4 ± 0.2     | 3.8 ± 0.2     | 2.99 ± 0.08 | 2.7 ± 0.1     | 3.03 ± 0.09 | 3.4 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.6 ± 0.3     | 2.99 ± 0.06 | 3.0 ± 0.2     | 3.3 ± 0.1   | 3.3 ± 0.1    | 4.4 ± 0.3     | 3.6 ± 0.2     | 2.95 ± 0.08 | 2.9 ± 0.1     | 3.05 ± 0.09 | 3.4 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.2 ± 0.3     | 2.97 ± 0.06 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.2 ± 0.1    | 4.3 ± 0.2     | 3.7 ± 0.2     | 2.84 ± 0.08 | 2.9 ± 0.2     | 3.08 ± 0.09 | 3.2 ± 0.1    | 3.9 ± 0.2     | W Africa F   | E Asia M     |
| 46  | 4.3 ± 0.3     | 2.91 ± 0.07 | 3.2 ± 0.2     | 3.2 ± 0.1   | 3.2 ± 0.1    | 4.2 ± 0.3     | 3.6 ± 0.3     | 2.72 ± 0.07 | 3.0 ± 0.2     | 3.05 ± 0.10 | 3.3 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 47  | 4.1 ± 0.3     | 2.83 ± 0.07 | 3.3 ± 0.2     | 3.1 ± 0.1   | 3.1 ± 0.1    | 4.3 ± 0.3     | 3.6 ± 0.2     | 2.65 ± 0.08 | 2.9 ± 0.2     | 3.2 ± 0.1   | 3.3 ± 0.2    | 3.9 ± 0.2     | W Africa F   | E Asia M     |
| 48  | 4.3 ± 0.4     | 2.69 ± 0.07 | 3.1 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.1    | 4.2 ± 0.3     | 3.9 ± 0.3     | 2.54 ± 0.07 | 3.1 ± 0.2     | 3.0 ± 0.1   | 3.0 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia M     |
| 49  | 4.1 ± 0.3     | 2.69 ± 0.07 | 3.1 ± 0.2     | 3.1 ± 0.1   | 3.0 ± 0.1    | 4.1 ± 0.3     | 3.7 ± 0.3     | 2.59 ± 0.08 | 3.0 ± 0.2     | 3.0 ± 0.1   | 3.0 ± 0.2    | 3.5 ± 0.2     | W Africa F   | E Asia M     |
| 50  | 4.4 ± 0.4     | 2.68 ± 0.07 | 3.1 ± 0.2     | 3.1 ± 0.1   | 3.0 ± 0.1    | 4.2 ± 0.3     | 3.3 ± 0.3     | 2.52 ± 0.08 | 3.0 ± 0.2     | 3.0 ± 0.1   | 3.1 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Asia M     |
| 51  | 4.2 ± 0.3     | 2.65 ± 0.08 | 3.2 ± 0.2     | 3.1 ± 0.1   | 3.0 ± 0.1    | 4.6 ± 0.3     | 3.7 ± 0.3     | 2.43 ± 0.08 | 3.0 ± 0.2     | 3.1 ± 0.1   | 3.0 ± 0.2    | 3.9 ± 0.2     | W Africa F   | E Asia M     |
| 52  | 4.4 ± 0.4     | 2.59 ± 0.07 | 3.2 ± 0.2     | 3.3 ± 0.1   | 3.1 ± 0.1    | 4.5 ± 0.3     | 3.5 ± 0.3     | 2.46 ± 0.08 | 3.2 ± 0.2     | 3.0 ± 0.1   | 3.0 ± 0.2    | 3.9 ± 0.2     | W Africa F   | E Asia M     |



**Table 107.** Mean absolute error for Application-type images for algorithm yoti-002. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.2 ± 0.2     | 2.7 ± 0.1   | 1.7 ± 0.1     | 2.8 ± 0.1   | 2.8 ± 0.3    | 3.4 ± 0.3     | 2.3 ± 0.1     | 1.22 ± 0.08 | 1.14 ± 0.08   | 1.48 ± 0.07 | 1.1 ± 0.1    | 2.1 ± 0.2     | W Africa F   | S E Asia M   |
| 15  | 3.3 ± 0.2     | 3.0 ± 0.1   | 1.8 ± 0.2     | 3.3 ± 0.2   | 3.2 ± 0.3    | 3.3 ± 0.3     | 2.3 ± 0.1     | 1.45 ± 0.07 | 0.98 ± 0.09   | 1.55 ± 0.07 | 1.4 ± 0.1    | 2.2 ± 0.2     | S Asia F     | E Europe M   |
| 16  | 3.2 ± 0.3     | 3.5 ± 0.2   | 2.0 ± 0.2     | 3.5 ± 0.2   | 4.0 ± 0.4    | 3.9 ± 0.5     | 2.2 ± 0.2     | 1.65 ± 0.08 | 1.1 ± 0.2     | 1.74 ± 0.10 | 1.7 ± 0.2    | 1.9 ± 0.2     | S E Asia F   | E Europe M   |
| 17  | 3.4 ± 0.3     | 3.7 ± 0.2   | 2.3 ± 0.3     | 3.6 ± 0.2   | 4.0 ± 0.3    | 3.6 ± 0.4     | 2.2 ± 0.2     | 2.1 ± 0.1   | 1.3 ± 0.1     | 2.0 ± 0.1   | 2.1 ± 0.2    | 2.3 ± 0.3     | S E Asia F   | E Europe M   |
| 18  | 3.0 ± 0.2     | 3.7 ± 0.1   | 2.7 ± 0.2     | 3.7 ± 0.1   | 4.1 ± 0.2    | 3.9 ± 0.3     | 2.3 ± 0.2     | 2.48 ± 0.08 | 1.9 ± 0.1     | 2.31 ± 0.10 | 2.5 ± 0.2    | 2.3 ± 0.2     | S E Asia F   | E Europe M   |
| 19  | 3.7 ± 0.3     | 3.9 ± 0.1   | 3.0 ± 0.2     | 3.9 ± 0.1   | 4.4 ± 0.2    | 4.2 ± 0.3     | 2.7 ± 0.2     | 2.86 ± 0.09 | 2.2 ± 0.1     | 2.6 ± 0.1   | 2.9 ± 0.2    | 2.9 ± 0.3     | S E Asia F   | E Europe M   |
| 20  | 3.7 ± 0.3     | 4.0 ± 0.1   | 3.2 ± 0.2     | 4.1 ± 0.1   | 4.5 ± 0.2    | 4.2 ± 0.3     | 3.0 ± 0.2     | 3.02 ± 0.09 | 2.4 ± 0.2     | 2.9 ± 0.1   | 3.1 ± 0.2    | 3.1 ± 0.3     | S E Asia F   | E Europe M   |
| 21  | 3.9 ± 0.2     | 3.9 ± 0.1   | 3.3 ± 0.1     | 4.2 ± 0.1   | 4.4 ± 0.1    | 4.6 ± 0.3     | 3.3 ± 0.2     | 3.16 ± 0.09 | 2.5 ± 0.1     | 3.1 ± 0.1   | 3.3 ± 0.1    | 3.5 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.3 ± 0.2     | 3.74 ± 0.10 | 3.2 ± 0.1     | 4.1 ± 0.1   | 4.3 ± 0.1    | 4.4 ± 0.2     | 4.0 ± 0.2     | 3.08 ± 0.09 | 2.5 ± 0.1     | 3.2 ± 0.1   | 3.2 ± 0.1    | 3.4 ± 0.2     | W Africa F   | E Europe M   |
| 23  | 4.4 ± 0.2     | 3.59 ± 0.09 | 3.0 ± 0.1     | 4.07 ± 0.09 | 4.2 ± 0.1    | 4.7 ± 0.2     | 3.9 ± 0.2     | 2.94 ± 0.09 | 2.3 ± 0.1     | 3.17 ± 0.09 | 3.0 ± 0.1    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 4.6 ± 0.2     | 3.41 ± 0.08 | 2.65 ± 0.08   | 3.80 ± 0.08 | 4.0 ± 0.1    | 4.5 ± 0.2     | 3.7 ± 0.2     | 2.77 ± 0.09 | 2.08 ± 0.09   | 3.05 ± 0.09 | 2.9 ± 0.1    | 3.9 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 4.6 ± 0.2     | 3.33 ± 0.08 | 2.40 ± 0.08   | 3.58 ± 0.08 | 3.9 ± 0.1    | 4.5 ± 0.2     | 3.9 ± 0.2     | 2.64 ± 0.08 | 1.84 ± 0.09   | 2.88 ± 0.08 | 2.8 ± 0.1    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 4.4 ± 0.2     | 3.21 ± 0.08 | 2.14 ± 0.08   | 3.32 ± 0.07 | 3.8 ± 0.1    | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.35 ± 0.08 | 1.68 ± 0.09   | 2.75 ± 0.07 | 2.6 ± 0.1    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 27  | 4.4 ± 0.2     | 3.18 ± 0.07 | 2.09 ± 0.08   | 3.12 ± 0.07 | 3.6 ± 0.1    | 4.6 ± 0.2     | 3.7 ± 0.2     | 2.30 ± 0.08 | 1.61 ± 0.08   | 2.57 ± 0.07 | 2.7 ± 0.1    | 3.9 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 4.5 ± 0.2     | 3.10 ± 0.07 | 2.03 ± 0.07   | 3.11 ± 0.07 | 3.6 ± 0.1    | 4.2 ± 0.2     | 3.6 ± 0.2     | 2.37 ± 0.09 | 1.67 ± 0.08   | 2.57 ± 0.07 | 2.6 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 29  | 4.5 ± 0.2     | 3.12 ± 0.07 | 2.11 ± 0.07   | 3.00 ± 0.07 | 3.8 ± 0.1    | 4.3 ± 0.2     | 3.6 ± 0.2     | 2.37 ± 0.08 | 1.79 ± 0.08   | 2.51 ± 0.06 | 2.6 ± 0.1    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 30  | 4.6 ± 0.2     | 3.25 ± 0.06 | 2.29 ± 0.07   | 2.98 ± 0.07 | 3.7 ± 0.1    | 4.2 ± 0.2     | 3.8 ± 0.2     | 2.50 ± 0.08 | 1.87 ± 0.07   | 2.59 ± 0.06 | 2.8 ± 0.1    | 4.1 ± 0.2     | E Africa F   | E Europe M   |
| 31  | 4.4 ± 0.2     | 3.27 ± 0.06 | 2.49 ± 0.07   | 2.96 ± 0.07 | 3.69 ± 0.10  | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.60 ± 0.08 | 2.11 ± 0.08   | 2.62 ± 0.06 | 2.9 ± 0.1    | 4.0 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 4.5 ± 0.2     | 3.27 ± 0.06 | 2.58 ± 0.08   | 3.03 ± 0.07 | 3.7 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.77 ± 0.08 | 2.28 ± 0.09   | 2.70 ± 0.05 | 3.0 ± 0.1    | 4.1 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 4.4 ± 0.2     | 3.27 ± 0.06 | 2.69 ± 0.09   | 3.05 ± 0.07 | 3.6 ± 0.1    | 4.1 ± 0.2     | 3.9 ± 0.2     | 2.89 ± 0.08 | 2.30 ± 0.09   | 2.68 ± 0.05 | 2.9 ± 0.1    | 4.0 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 4.3 ± 0.2     | 3.29 ± 0.06 | 2.78 ± 0.09   | 2.98 ± 0.07 | 3.45 ± 0.10  | 4.0 ± 0.2     | 3.7 ± 0.2     | 3.02 ± 0.08 | 2.33 ± 0.09   | 2.79 ± 0.06 | 3.1 ± 0.1    | 4.0 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.5 ± 0.2     | 3.20 ± 0.06 | 2.86 ± 0.09   | 2.98 ± 0.06 | 3.4 ± 0.1    | 3.8 ± 0.2     | 3.8 ± 0.2     | 3.10 ± 0.08 | 2.4 ± 0.1     | 2.75 ± 0.06 | 3.1 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 4.3 ± 0.2     | 3.04 ± 0.06 | 2.74 ± 0.09   | 2.96 ± 0.07 | 3.3 ± 0.1    | 3.9 ± 0.2     | 3.7 ± 0.2     | 3.11 ± 0.08 | 2.5 ± 0.1     | 2.76 ± 0.06 | 3.3 ± 0.2    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 37  | 4.3 ± 0.2     | 2.96 ± 0.06 | 2.8 ± 0.1     | 3.05 ± 0.07 | 3.2 ± 0.1    | 4.0 ± 0.2     | 3.7 ± 0.2     | 3.05 ± 0.08 | 2.4 ± 0.1     | 2.76 ± 0.06 | 3.0 ± 0.1    | 3.6 ± 0.1     | E Africa F   | E Europe M   |
| 38  | 4.1 ± 0.2     | 2.85 ± 0.06 | 2.7 ± 0.1     | 3.05 ± 0.07 | 3.1 ± 0.1    | 3.7 ± 0.2     | 3.5 ± 0.2     | 3.04 ± 0.08 | 2.4 ± 0.1     | 2.71 ± 0.06 | 3.2 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 39  | 4.0 ± 0.2     | 2.71 ± 0.06 | 2.7 ± 0.1     | 3.05 ± 0.08 | 2.9 ± 0.1    | 3.7 ± 0.2     | 3.5 ± 0.2     | 3.01 ± 0.08 | 2.4 ± 0.1     | 2.69 ± 0.06 | 3.1 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 40  | 4.0 ± 0.2     | 2.62 ± 0.06 | 2.6 ± 0.1     | 3.05 ± 0.08 | 3.0 ± 0.1    | 3.7 ± 0.2     | 3.4 ± 0.2     | 2.95 ± 0.08 | 2.4 ± 0.1     | 2.73 ± 0.06 | 2.9 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 41  | 4.1 ± 0.3     | 2.56 ± 0.06 | 2.7 ± 0.1     | 3.15 ± 0.09 | 2.8 ± 0.1    | 3.8 ± 0.2     | 3.3 ± 0.2     | 2.86 ± 0.08 | 2.4 ± 0.1     | 2.66 ± 0.07 | 2.9 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.4 ± 0.3     | 2.52 ± 0.06 | 2.7 ± 0.1     | 3.2 ± 0.1   | 2.7 ± 0.1    | 3.8 ± 0.2     | 3.3 ± 0.2     | 2.84 ± 0.08 | 2.4 ± 0.1     | 2.71 ± 0.07 | 3.0 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 4.1 ± 0.3     | 2.51 ± 0.06 | 2.8 ± 0.1     | 3.2 ± 0.1   | 2.9 ± 0.1    | 3.8 ± 0.2     | 3.4 ± 0.2     | 2.88 ± 0.08 | 2.5 ± 0.1     | 2.70 ± 0.08 | 3.0 ± 0.1    | 3.3 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.4 ± 0.3     | 2.53 ± 0.06 | 2.7 ± 0.1     | 3.3 ± 0.1   | 2.9 ± 0.1    | 3.9 ± 0.2     | 3.2 ± 0.2     | 2.81 ± 0.08 | 2.6 ± 0.1     | 2.77 ± 0.09 | 3.2 ± 0.2    | 3.2 ± 0.2     | E Africa F   | E Asia F     |
| 45  | 4.4 ± 0.3     | 2.52 ± 0.06 | 2.7 ± 0.1     | 3.3 ± 0.1   | 2.8 ± 0.1    | 3.7 ± 0.2     | 3.4 ± 0.2     | 2.83 ± 0.07 | 2.7 ± 0.1     | 2.85 ± 0.09 | 3.2 ± 0.2    | 3.3 ± 0.2     | E Africa F   | E Asia F     |
| 46  | 4.5 ± 0.3     | 2.59 ± 0.06 | 2.8 ± 0.1     | 3.2 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.3     | 3.4 ± 0.2     | 2.74 ± 0.07 | 2.8 ± 0.1     | 2.79 ± 0.09 | 3.1 ± 0.2    | 3.4 ± 0.2     | E Africa F   | E Asia F     |
| 47  | 4.0 ± 0.3     | 2.60 ± 0.06 | 2.9 ± 0.2     | 3.4 ± 0.1   | 2.9 ± 0.1    | 4.0 ± 0.2     | 3.5 ± 0.2     | 2.68 ± 0.08 | 2.7 ± 0.2     | 3.1 ± 0.1   | 3.3 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia F     |
| 48  | 4.6 ± 0.4     | 2.56 ± 0.06 | 2.8 ± 0.1     | 3.5 ± 0.1   | 2.8 ± 0.1    | 4.2 ± 0.3     | 3.9 ± 0.3     | 2.54 ± 0.07 | 2.8 ± 0.2     | 2.9 ± 0.1   | 3.1 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Asia M     |
| 49  | 4.5 ± 0.4     | 2.61 ± 0.07 | 2.8 ± 0.1     | 3.5 ± 0.1   | 2.9 ± 0.1    | 4.4 ± 0.3     | 3.6 ± 0.3     | 2.53 ± 0.08 | 2.7 ± 0.2     | 2.9 ± 0.1   | 3.0 ± 0.2    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 50  | 4.5 ± 0.4     | 2.62 ± 0.07 | 2.7 ± 0.1     | 3.3 ± 0.1   | 2.9 ± 0.1    | 4.5 ± 0.3     | 3.3 ± 0.3     | 2.47 ± 0.07 | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.2 ± 0.2    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 51  | 4.6 ± 0.4     | 2.63 ± 0.07 | 2.9 ± 0.2     | 3.5 ± 0.1   | 2.9 ± 0.1    | 4.6 ± 0.3     | 3.9 ± 0.3     | 2.34 ± 0.08 | 2.8 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 3.5 ± 0.2     | W Africa F   | E Asia M     |
| 52  | 4.7 ± 0.4     | 2.62 ± 0.07 | 3.0 ± 0.1     | 3.6 ± 0.2   | 2.9 ± 0.1    | 4.6 ± 0.3     | 3.7 ± 0.3     | 2.27 ± 0.07 | 2.9 ± 0.2     | 3.2 ± 0.1   | 2.8 ± 0.2    | 3.8 ± 0.2     | E Africa F   | E Asia M     |



**Table 108.** Mean absolute error for Application-type images for algorithm yoti-003. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.3 ± 0.2     | 2.7 ± 0.1   | 1.6 ± 0.1     | 2.9 ± 0.1   | 3.2 ± 0.3    | 3.6 ± 0.3     | 2.3 ± 0.1     | 1.53 ± 0.09 | 0.96 ± 0.08   | 1.56 ± 0.07 | 1.4 ± 0.1    | 2.1 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 3.3 ± 0.3     | 3.0 ± 0.2   | 1.5 ± 0.2     | 3.2 ± 0.2   | 3.4 ± 0.3    | 3.2 ± 0.3     | 1.9 ± 0.2     | 1.62 ± 0.09 | 0.84 ± 0.07   | 1.35 ± 0.08 | 1.5 ± 0.1    | 2.0 ± 0.2     | S E Asia F   | E Europe M   |
| 16  | 3.2 ± 0.3     | 3.3 ± 0.2   | 1.7 ± 0.2     | 3.1 ± 0.2   | 3.7 ± 0.3    | 3.7 ± 0.5     | 1.7 ± 0.2     | 1.66 ± 0.09 | 1.0 ± 0.2     | 1.42 ± 0.09 | 1.6 ± 0.2    | 1.6 ± 0.2     | S E Asia F   | E Europe M   |
| 17  | 3.4 ± 0.3     | 3.5 ± 0.2   | 1.9 ± 0.2     | 3.3 ± 0.2   | 3.9 ± 0.3    | 3.4 ± 0.3     | 1.5 ± 0.2     | 1.9 ± 0.1   | 1.1 ± 0.1     | 1.6 ± 0.1   | 2.0 ± 0.2    | 1.8 ± 0.3     | S E Asia F   | E Europe M   |
| 18  | 3.0 ± 0.2     | 3.5 ± 0.1   | 2.4 ± 0.2     | 3.3 ± 0.1   | 3.9 ± 0.2    | 3.4 ± 0.3     | 1.8 ± 0.2     | 2.11 ± 0.08 | 1.6 ± 0.1     | 1.93 ± 0.10 | 2.1 ± 0.2    | 2.0 ± 0.2     | S E Asia F   | E Europe M   |
| 19  | 3.9 ± 0.3     | 3.7 ± 0.1   | 2.8 ± 0.2     | 3.6 ± 0.1   | 4.1 ± 0.2    | 3.8 ± 0.3     | 2.3 ± 0.2     | 2.48 ± 0.10 | 1.9 ± 0.1     | 2.25 ± 0.10 | 2.4 ± 0.1    | 2.7 ± 0.3     | S E Asia F   | E Europe M   |
| 20  | 4.1 ± 0.3     | 3.9 ± 0.1   | 2.8 ± 0.2     | 3.7 ± 0.1   | 4.3 ± 0.2    | 4.0 ± 0.3     | 2.8 ± 0.2     | 2.66 ± 0.09 | 2.2 ± 0.1     | 2.6 ± 0.1   | 2.7 ± 0.1    | 2.9 ± 0.3     | S E Asia F   | E Europe M   |
| 21  | 4.3 ± 0.2     | 3.9 ± 0.1   | 3.0 ± 0.1     | 3.9 ± 0.1   | 4.3 ± 0.1    | 4.4 ± 0.3     | 3.0 ± 0.2     | 2.77 ± 0.09 | 2.3 ± 0.1     | 2.87 ± 0.10 | 2.9 ± 0.1    | 3.3 ± 0.3     | W Africa F   | E Europe M   |
| 22  | 4.5 ± 0.3     | 3.72 ± 0.10 | 3.0 ± 0.1     | 3.84 ± 0.10 | 4.1 ± 0.1    | 4.1 ± 0.2     | 3.8 ± 0.2     | 2.78 ± 0.09 | 2.3 ± 0.1     | 3.0 ± 0.1   | 2.8 ± 0.1    | 3.3 ± 0.2     | E Africa F   | E Europe M   |
| 23  | 4.7 ± 0.2     | 3.61 ± 0.09 | 2.80 ± 0.10   | 3.83 ± 0.09 | 4.1 ± 0.1    | 4.3 ± 0.2     | 3.8 ± 0.2     | 2.73 ± 0.08 | 2.1 ± 0.1     | 3.01 ± 0.09 | 2.9 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 24  | 4.8 ± 0.2     | 3.45 ± 0.08 | 2.53 ± 0.09   | 3.65 ± 0.08 | 3.8 ± 0.1    | 4.4 ± 0.2     | 3.7 ± 0.2     | 2.69 ± 0.08 | 1.99 ± 0.09   | 3.04 ± 0.08 | 2.9 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 4.6 ± 0.2     | 3.37 ± 0.07 | 2.27 ± 0.08   | 3.44 ± 0.07 | 3.7 ± 0.1    | 4.4 ± 0.2     | 3.9 ± 0.2     | 2.60 ± 0.08 | 1.80 ± 0.09   | 2.95 ± 0.08 | 2.8 ± 0.1    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 26  | 4.5 ± 0.2     | 3.29 ± 0.07 | 2.04 ± 0.08   | 3.26 ± 0.07 | 3.7 ± 0.1    | 4.1 ± 0.2     | 3.8 ± 0.2     | 2.43 ± 0.08 | 1.63 ± 0.09   | 2.91 ± 0.07 | 2.7 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 27  | 4.4 ± 0.2     | 3.13 ± 0.07 | 2.08 ± 0.08   | 3.10 ± 0.06 | 3.50 ± 0.10  | 4.4 ± 0.2     | 3.7 ± 0.2     | 2.40 ± 0.08 | 1.56 ± 0.08   | 2.74 ± 0.07 | 2.8 ± 0.1    | 3.8 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 4.3 ± 0.2     | 3.03 ± 0.06 | 2.02 ± 0.07   | 3.07 ± 0.07 | 3.42 ± 0.10  | 4.2 ± 0.2     | 3.5 ± 0.2     | 2.46 ± 0.08 | 1.70 ± 0.08   | 2.75 ± 0.06 | 2.8 ± 0.1    | 3.8 ± 0.2     | E Africa F   | E Europe M   |
| 29  | 4.4 ± 0.2     | 2.97 ± 0.06 | 2.15 ± 0.07   | 3.01 ± 0.06 | 3.46 ± 0.10  | 4.2 ± 0.2     | 3.6 ± 0.2     | 2.39 ± 0.07 | 1.77 ± 0.07   | 2.69 ± 0.06 | 2.6 ± 0.1    | 4.0 ± 0.2     | E Africa F   | E Europe M   |
| 30  | 4.3 ± 0.2     | 3.00 ± 0.06 | 2.32 ± 0.07   | 2.99 ± 0.06 | 3.37 ± 0.10  | 4.1 ± 0.2     | 3.8 ± 0.2     | 2.44 ± 0.07 | 1.91 ± 0.08   | 2.72 ± 0.06 | 2.9 ± 0.1    | 4.0 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 4.2 ± 0.2     | 2.99 ± 0.06 | 2.51 ± 0.08   | 2.96 ± 0.06 | 3.36 ± 0.09  | 4.1 ± 0.2     | 3.7 ± 0.1     | 2.48 ± 0.07 | 2.06 ± 0.08   | 2.73 ± 0.06 | 2.8 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 32  | 4.4 ± 0.2     | 2.93 ± 0.06 | 2.60 ± 0.07   | 3.02 ± 0.06 | 3.39 ± 0.10  | 4.2 ± 0.2     | 3.7 ± 0.2     | 2.61 ± 0.07 | 2.23 ± 0.09   | 2.74 ± 0.06 | 2.9 ± 0.1    | 3.9 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 4.2 ± 0.2     | 2.90 ± 0.06 | 2.71 ± 0.08   | 2.95 ± 0.06 | 3.31 ± 0.10  | 4.2 ± 0.2     | 3.8 ± 0.2     | 2.67 ± 0.07 | 2.23 ± 0.09   | 2.71 ± 0.06 | 2.8 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 4.2 ± 0.2     | 2.97 ± 0.06 | 2.71 ± 0.09   | 2.91 ± 0.06 | 3.20 ± 0.09  | 4.1 ± 0.2     | 3.6 ± 0.2     | 2.77 ± 0.07 | 2.29 ± 0.09   | 2.74 ± 0.06 | 3.0 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.3 ± 0.2     | 2.85 ± 0.06 | 2.75 ± 0.09   | 2.80 ± 0.06 | 3.20 ± 0.10  | 4.0 ± 0.2     | 3.6 ± 0.2     | 2.87 ± 0.08 | 2.3 ± 0.1     | 2.73 ± 0.06 | 3.1 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 4.1 ± 0.2     | 2.77 ± 0.06 | 2.67 ± 0.09   | 2.75 ± 0.06 | 3.09 ± 0.10  | 4.0 ± 0.2     | 3.6 ± 0.2     | 2.85 ± 0.07 | 2.40 ± 0.10   | 2.67 ± 0.06 | 3.2 ± 0.1    | 3.8 ± 0.1     | E Africa F   | E Europe M   |
| 37  | 4.0 ± 0.2     | 2.71 ± 0.06 | 2.7 ± 0.1     | 2.91 ± 0.07 | 2.93 ± 0.10  | 4.0 ± 0.2     | 3.6 ± 0.2     | 2.87 ± 0.08 | 2.3 ± 0.1     | 2.64 ± 0.06 | 3.0 ± 0.1    | 3.5 ± 0.1     | W Africa F   | E Europe M   |
| 38  | 4.0 ± 0.2     | 2.70 ± 0.06 | 2.60 ± 0.10   | 2.80 ± 0.07 | 2.9 ± 0.1    | 4.0 ± 0.2     | 3.4 ± 0.2     | 2.91 ± 0.08 | 2.3 ± 0.1     | 2.58 ± 0.06 | 3.2 ± 0.2    | 3.4 ± 0.1     | W Africa F   | E Europe M   |
| 39  | 3.9 ± 0.2     | 2.68 ± 0.06 | 2.5 ± 0.1     | 2.76 ± 0.08 | 2.83 ± 0.10  | 3.8 ± 0.2     | 3.5 ± 0.2     | 2.92 ± 0.08 | 2.3 ± 0.1     | 2.54 ± 0.06 | 3.2 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 40  | 3.9 ± 0.2     | 2.62 ± 0.06 | 2.5 ± 0.1     | 2.79 ± 0.08 | 2.9 ± 0.1    | 3.7 ± 0.2     | 3.3 ± 0.2     | 2.92 ± 0.08 | 2.3 ± 0.1     | 2.63 ± 0.06 | 3.1 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 41  | 3.9 ± 0.2     | 2.62 ± 0.06 | 2.5 ± 0.1     | 2.87 ± 0.09 | 2.8 ± 0.1    | 3.7 ± 0.2     | 3.4 ± 0.2     | 2.89 ± 0.07 | 2.3 ± 0.1     | 2.54 ± 0.06 | 3.0 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.1 ± 0.2     | 2.63 ± 0.06 | 2.6 ± 0.1     | 2.91 ± 0.10 | 2.6 ± 0.1    | 3.6 ± 0.2     | 3.3 ± 0.2     | 2.90 ± 0.08 | 2.3 ± 0.1     | 2.59 ± 0.07 | 3.1 ± 0.1    | 3.3 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 3.9 ± 0.3     | 2.65 ± 0.06 | 2.6 ± 0.1     | 2.89 ± 0.10 | 2.9 ± 0.1    | 3.7 ± 0.2     | 3.3 ± 0.2     | 2.94 ± 0.08 | 2.5 ± 0.1     | 2.58 ± 0.07 | 3.2 ± 0.1    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.1 ± 0.3     | 2.66 ± 0.06 | 2.7 ± 0.1     | 3.0 ± 0.1   | 2.9 ± 0.1    | 3.8 ± 0.2     | 3.4 ± 0.2     | 2.91 ± 0.08 | 2.5 ± 0.1     | 2.63 ± 0.08 | 3.3 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.0 ± 0.3     | 2.63 ± 0.06 | 2.6 ± 0.1     | 3.0 ± 0.1   | 2.8 ± 0.1    | 3.5 ± 0.2     | 3.6 ± 0.2     | 2.91 ± 0.08 | 2.5 ± 0.1     | 2.70 ± 0.09 | 3.1 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.1 ± 0.3     | 2.69 ± 0.06 | 2.7 ± 0.1     | 2.9 ± 0.1   | 2.8 ± 0.1    | 3.8 ± 0.2     | 3.5 ± 0.2     | 2.77 ± 0.07 | 2.7 ± 0.1     | 2.70 ± 0.09 | 3.2 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Europe M   |
| 47  | 4.1 ± 0.3     | 2.64 ± 0.06 | 2.9 ± 0.2     | 3.0 ± 0.1   | 2.8 ± 0.1    | 3.8 ± 0.2     | 3.5 ± 0.2     | 2.73 ± 0.08 | 2.6 ± 0.1     | 2.9 ± 0.1   | 3.3 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.3 ± 0.3     | 2.50 ± 0.06 | 2.8 ± 0.1     | 3.2 ± 0.1   | 2.7 ± 0.1    | 3.6 ± 0.2     | 3.7 ± 0.3     | 2.55 ± 0.07 | 2.7 ± 0.2     | 2.9 ± 0.1   | 3.1 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia F     |
| 49  | 4.2 ± 0.3     | 2.52 ± 0.07 | 2.8 ± 0.1     | 3.1 ± 0.1   | 2.8 ± 0.1    | 3.9 ± 0.3     | 3.5 ± 0.3     | 2.61 ± 0.08 | 2.7 ± 0.2     | 2.9 ± 0.1   | 3.0 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia F     |
| 50  | 4.4 ± 0.4     | 2.50 ± 0.06 | 2.7 ± 0.1     | 3.0 ± 0.1   | 2.8 ± 0.1    | 3.9 ± 0.2     | 3.4 ± 0.3     | 2.48 ± 0.07 | 2.8 ± 0.2     | 3.0 ± 0.1   | 3.2 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 51  | 4.2 ± 0.3     | 2.49 ± 0.07 | 2.7 ± 0.1     | 3.1 ± 0.1   | 2.7 ± 0.1    | 4.1 ± 0.3     | 3.8 ± 0.3     | 2.40 ± 0.07 | 2.8 ± 0.2     | 3.2 ± 0.1   | 3.0 ± 0.2    | 3.7 ± 0.2     | E Africa F   | E Asia M     |
| 52  | 4.2 ± 0.3     | 2.47 ± 0.07 | 3.0 ± 0.1     | 3.2 ± 0.1   | 2.8 ± 0.1    | 4.1 ± 0.3     | 3.5 ± 0.3     | 2.31 ± 0.07 | 2.9 ± 0.2     | 3.2 ± 0.1   | 2.9 ± 0.2    | 3.9 ± 0.2     | E Africa F   | E Asia M     |

**Table 109.** Mean absolute error for Application-type images for algorithm yoti-004. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 3.5 ± 0.2     | 2.8 ± 0.1   | 1.8 ± 0.1     | 3.0 ± 0.1   | 3.0 ± 0.2    | 3.7 ± 0.3     | 2.3 ± 0.1     | 1.26 ± 0.07 | 1.05 ± 0.09   | 1.50 ± 0.07 | 1.3 ± 0.1    | 2.3 ± 0.2     | W Africa F   | E Europe M   |
| 15  | 3.5 ± 0.2     | 3.1 ± 0.1   | 1.8 ± 0.2     | 3.5 ± 0.2   | 3.5 ± 0.3    | 3.5 ± 0.3     | 2.2 ± 0.2     | 1.48 ± 0.07 | 0.93 ± 0.09   | 1.45 ± 0.08 | 1.6 ± 0.1    | 2.4 ± 0.2     | W Africa F   | E Europe M   |
| 16  | 3.4 ± 0.2     | 3.5 ± 0.1   | 2.0 ± 0.2     | 3.4 ± 0.2   | 3.8 ± 0.3    | 3.8 ± 0.4     | 2.2 ± 0.2     | 1.70 ± 0.07 | 1.1 ± 0.2     | 1.69 ± 0.09 | 1.8 ± 0.2    | 2.2 ± 0.2     | W Africa F   | E Europe M   |
| 17  | 3.5 ± 0.2     | 3.7 ± 0.1   | 2.0 ± 0.2     | 3.6 ± 0.2   | 3.9 ± 0.3    | 3.6 ± 0.3     | 2.0 ± 0.2     | 2.01 ± 0.09 | 1.2 ± 0.1     | 1.8 ± 0.1   | 2.1 ± 0.2    | 2.4 ± 0.3     | S E Asia F   | E Europe M   |
| 18  | 3.1 ± 0.2     | 3.55 ± 0.09 | 2.3 ± 0.2     | 3.4 ± 0.1   | 3.9 ± 0.2    | 3.5 ± 0.3     | 2.1 ± 0.2     | 2.14 ± 0.07 | 1.5 ± 0.1     | 2.04 ± 0.09 | 2.2 ± 0.2    | 2.2 ± 0.2     | S E Asia F   | E Europe M   |
| 19  | 3.4 ± 0.2     | 3.51 ± 0.09 | 2.4 ± 0.2     | 3.3 ± 0.1   | 3.8 ± 0.1    | 3.5 ± 0.2     | 2.1 ± 0.2     | 2.31 ± 0.08 | 1.8 ± 0.1     | 2.17 ± 0.08 | 2.3 ± 0.1    | 2.6 ± 0.3     | S E Asia F   | E Europe M   |
| 20  | 3.2 ± 0.2     | 3.4 ± 0.1   | 2.3 ± 0.1     | 3.2 ± 0.1   | 3.7 ± 0.1    | 3.5 ± 0.2     | 2.3 ± 0.2     | 2.27 ± 0.08 | 1.8 ± 0.1     | 2.24 ± 0.09 | 2.4 ± 0.1    | 2.5 ± 0.3     | S E Asia F   | E Europe M   |
| 21  | 3.1 ± 0.2     | 3.13 ± 0.09 | 2.23 ± 0.10   | 3.09 ± 0.09 | 3.4 ± 0.1    | 3.5 ± 0.2     | 2.5 ± 0.2     | 2.17 ± 0.07 | 1.8 ± 0.1     | 2.29 ± 0.08 | 2.3 ± 0.1    | 2.7 ± 0.2     | W Africa F   | E Europe M   |
| 22  | 3.4 ± 0.2     | 2.84 ± 0.08 | 2.14 ± 0.08   | 2.90 ± 0.08 | 3.2 ± 0.1    | 3.3 ± 0.2     | 2.9 ± 0.2     | 2.05 ± 0.07 | 1.7 ± 0.1     | 2.21 ± 0.09 | 2.1 ± 0.1    | 2.5 ± 0.2     | E Africa F   | E Europe M   |
| 23  | 3.3 ± 0.2     | 2.57 ± 0.07 | 1.96 ± 0.08   | 2.78 ± 0.08 | 2.98 ± 0.10  | 3.4 ± 0.2     | 2.9 ± 0.2     | 1.89 ± 0.07 | 1.51 ± 0.09   | 2.13 ± 0.08 | 2.1 ± 0.1    | 3.0 ± 0.2     | W Africa F   | E Europe M   |
| 24  | 3.4 ± 0.2     | 2.38 ± 0.06 | 1.76 ± 0.07   | 2.64 ± 0.07 | 2.74 ± 0.08  | 3.4 ± 0.2     | 2.8 ± 0.2     | 1.84 ± 0.07 | 1.38 ± 0.07   | 2.14 ± 0.07 | 1.99 ± 0.10  | 2.9 ± 0.2     | E Africa F   | E Europe M   |
| 25  | 3.3 ± 0.2     | 2.30 ± 0.06 | 1.63 ± 0.06   | 2.50 ± 0.06 | 2.74 ± 0.09  | 3.4 ± 0.2     | 3.0 ± 0.2     | 1.83 ± 0.07 | 1.36 ± 0.08   | 2.07 ± 0.07 | 1.94 ± 0.09  | 3.1 ± 0.2     | W Africa F   | E Europe M   |
| 26  | 3.4 ± 0.2     | 2.30 ± 0.06 | 1.65 ± 0.06   | 2.48 ± 0.06 | 2.63 ± 0.08  | 3.2 ± 0.2     | 3.1 ± 0.2     | 1.77 ± 0.06 | 1.39 ± 0.08   | 2.13 ± 0.06 | 2.0 ± 0.1    | 3.1 ± 0.2     | E Africa F   | E Europe M   |
| 27  | 3.5 ± 0.2     | 2.32 ± 0.06 | 1.85 ± 0.06   | 2.50 ± 0.05 | 2.61 ± 0.08  | 3.6 ± 0.2     | 3.2 ± 0.2     | 1.89 ± 0.07 | 1.51 ± 0.07   | 2.11 ± 0.06 | 2.1 ± 0.1    | 3.1 ± 0.2     | W Africa F   | E Europe M   |
| 28  | 3.5 ± 0.2     | 2.38 ± 0.05 | 2.01 ± 0.06   | 2.67 ± 0.06 | 2.62 ± 0.08  | 3.5 ± 0.2     | 3.0 ± 0.1     | 2.06 ± 0.07 | 1.71 ± 0.07   | 2.23 ± 0.06 | 2.2 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 29  | 3.6 ± 0.2     | 2.43 ± 0.05 | 2.25 ± 0.06   | 2.72 ± 0.06 | 2.78 ± 0.08  | 3.6 ± 0.2     | 3.2 ± 0.1     | 2.13 ± 0.06 | 1.96 ± 0.07   | 2.34 ± 0.05 | 2.2 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 30  | 3.7 ± 0.1     | 2.59 ± 0.05 | 2.48 ± 0.07   | 2.80 ± 0.06 | 2.80 ± 0.09  | 3.6 ± 0.2     | 3.3 ± 0.1     | 2.31 ± 0.06 | 2.11 ± 0.08   | 2.48 ± 0.05 | 2.5 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 31  | 3.6 ± 0.1     | 2.66 ± 0.05 | 2.71 ± 0.07   | 2.89 ± 0.06 | 2.86 ± 0.09  | 3.7 ± 0.1     | 3.3 ± 0.1     | 2.39 ± 0.07 | 2.34 ± 0.08   | 2.60 ± 0.05 | 2.6 ± 0.1    | 3.4 ± 0.1     | W Africa F   | E Europe M   |
| 32  | 3.9 ± 0.2     | 2.70 ± 0.05 | 2.86 ± 0.08   | 3.02 ± 0.06 | 3.04 ± 0.09  | 3.8 ± 0.1     | 3.4 ± 0.1     | 2.53 ± 0.06 | 2.52 ± 0.09   | 2.69 ± 0.05 | 2.7 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 33  | 3.8 ± 0.2     | 2.77 ± 0.05 | 3.02 ± 0.09   | 3.01 ± 0.06 | 3.04 ± 0.09  | 3.8 ± 0.2     | 3.6 ± 0.1     | 2.62 ± 0.07 | 2.50 ± 0.10   | 2.72 ± 0.05 | 2.7 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 34  | 3.9 ± 0.2     | 2.87 ± 0.06 | 3.03 ± 0.09   | 3.02 ± 0.06 | 2.92 ± 0.09  | 3.9 ± 0.2     | 3.4 ± 0.1     | 2.69 ± 0.07 | 2.55 ± 0.10   | 2.80 ± 0.06 | 2.9 ± 0.1    | 3.4 ± 0.1     | E Africa F   | E Europe M   |
| 35  | 4.1 ± 0.2     | 2.88 ± 0.06 | 3.13 ± 0.10   | 2.94 ± 0.06 | 3.03 ± 0.09  | 3.7 ± 0.2     | 3.5 ± 0.2     | 2.79 ± 0.07 | 2.5 ± 0.1     | 2.77 ± 0.06 | 2.9 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 36  | 4.0 ± 0.2     | 2.82 ± 0.06 | 2.95 ± 0.10   | 2.89 ± 0.07 | 3.04 ± 0.10  | 3.7 ± 0.2     | 3.4 ± 0.2     | 2.74 ± 0.07 | 2.5 ± 0.1     | 2.74 ± 0.06 | 3.0 ± 0.1    | 3.5 ± 0.1     | E Africa F   | E Europe M   |
| 37  | 3.9 ± 0.2     | 2.75 ± 0.06 | 2.9 ± 0.1     | 3.03 ± 0.07 | 2.87 ± 0.10  | 3.9 ± 0.2     | 3.5 ± 0.2     | 2.74 ± 0.07 | 2.4 ± 0.1     | 2.71 ± 0.06 | 2.9 ± 0.1    | 3.2 ± 0.1     | W Africa F   | E Europe M   |
| 38  | 4.0 ± 0.2     | 2.78 ± 0.06 | 2.8 ± 0.1     | 2.91 ± 0.07 | 2.8 ± 0.1    | 3.7 ± 0.2     | 3.2 ± 0.2     | 2.76 ± 0.08 | 2.4 ± 0.1     | 2.61 ± 0.06 | 3.0 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 39  | 4.0 ± 0.2     | 2.75 ± 0.06 | 2.7 ± 0.1     | 2.91 ± 0.08 | 2.81 ± 0.10  | 3.6 ± 0.2     | 3.3 ± 0.2     | 2.72 ± 0.08 | 2.4 ± 0.1     | 2.57 ± 0.06 | 2.9 ± 0.1    | 3.1 ± 0.1     | E Africa F   | E Europe M   |
| 40  | 4.0 ± 0.2     | 2.71 ± 0.06 | 2.6 ± 0.1     | 2.79 ± 0.08 | 2.9 ± 0.1    | 3.6 ± 0.2     | 3.3 ± 0.2     | 2.71 ± 0.07 | 2.3 ± 0.1     | 2.65 ± 0.06 | 2.8 ± 0.1    | 3.2 ± 0.1     | E Africa F   | E Europe M   |
| 41  | 4.0 ± 0.2     | 2.67 ± 0.06 | 2.6 ± 0.1     | 2.97 ± 0.09 | 2.7 ± 0.1    | 3.8 ± 0.2     | 3.3 ± 0.2     | 2.66 ± 0.07 | 2.3 ± 0.1     | 2.52 ± 0.06 | 2.9 ± 0.1    | 3.3 ± 0.2     | E Africa F   | E Europe M   |
| 42  | 4.3 ± 0.3     | 2.61 ± 0.06 | 2.5 ± 0.1     | 2.91 ± 0.10 | 2.6 ± 0.1    | 3.7 ± 0.2     | 3.3 ± 0.2     | 2.70 ± 0.07 | 2.3 ± 0.1     | 2.57 ± 0.07 | 2.8 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Europe M   |
| 43  | 4.2 ± 0.3     | 2.60 ± 0.06 | 2.6 ± 0.1     | 2.9 ± 0.1   | 2.7 ± 0.1    | 4.1 ± 0.2     | 3.1 ± 0.2     | 2.73 ± 0.07 | 2.3 ± 0.1     | 2.60 ± 0.08 | 3.0 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 44  | 4.4 ± 0.3     | 2.54 ± 0.06 | 2.6 ± 0.1     | 3.0 ± 0.1   | 2.8 ± 0.1    | 4.0 ± 0.2     | 3.3 ± 0.2     | 2.72 ± 0.07 | 2.4 ± 0.1     | 2.61 ± 0.08 | 3.1 ± 0.1    | 3.2 ± 0.2     | E Africa F   | E Europe M   |
| 45  | 4.2 ± 0.3     | 2.53 ± 0.06 | 2.5 ± 0.1     | 3.0 ± 0.1   | 2.7 ± 0.1    | 4.0 ± 0.2     | 3.3 ± 0.2     | 2.71 ± 0.07 | 2.5 ± 0.1     | 2.71 ± 0.09 | 3.0 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Europe M   |
| 46  | 4.5 ± 0.3     | 2.50 ± 0.06 | 2.5 ± 0.1     | 2.9 ± 0.1   | 2.6 ± 0.1    | 4.2 ± 0.3     | 3.4 ± 0.2     | 2.53 ± 0.07 | 2.5 ± 0.1     | 2.61 ± 0.09 | 3.0 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia F     |
| 47  | 4.5 ± 0.4     | 2.47 ± 0.06 | 2.6 ± 0.1     | 3.0 ± 0.1   | 2.6 ± 0.1    | 4.3 ± 0.2     | 3.4 ± 0.2     | 2.48 ± 0.07 | 2.5 ± 0.1     | 2.9 ± 0.1   | 3.1 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Europe M   |
| 48  | 4.7 ± 0.4     | 2.37 ± 0.06 | 2.6 ± 0.1     | 3.1 ± 0.1   | 2.6 ± 0.1    | 4.4 ± 0.3     | 3.6 ± 0.3     | 2.37 ± 0.07 | 2.5 ± 0.2     | 2.8 ± 0.1   | 2.9 ± 0.2    | 3.6 ± 0.2     | E Africa F   | E Asia F     |
| 49  | 4.6 ± 0.4     | 2.45 ± 0.07 | 2.6 ± 0.1     | 3.1 ± 0.1   | 2.6 ± 0.1    | 4.4 ± 0.3     | 3.5 ± 0.3     | 2.41 ± 0.07 | 2.5 ± 0.1     | 2.9 ± 0.1   | 2.8 ± 0.1    | 3.4 ± 0.2     | E Africa F   | E Asia M     |
| 50  | 4.7 ± 0.4     | 2.43 ± 0.06 | 2.5 ± 0.1     | 3.1 ± 0.1   | 2.7 ± 0.1    | 4.6 ± 0.3     | 3.3 ± 0.3     | 2.31 ± 0.07 | 2.5 ± 0.1     | 2.9 ± 0.1   | 3.0 ± 0.2    | 3.5 ± 0.2     | E Africa F   | E Asia M     |
| 51  | 4.6 ± 0.4     | 2.45 ± 0.07 | 2.6 ± 0.1     | 3.1 ± 0.1   | 2.7 ± 0.1    | 4.9 ± 0.3     | 3.7 ± 0.3     | 2.26 ± 0.07 | 2.5 ± 0.1     | 3.1 ± 0.1   | 2.8 ± 0.2    | 3.6 ± 0.2     | W Africa F   | E Asia M     |
| 52  | 4.8 ± 0.4     | 2.39 ± 0.07 | 2.8 ± 0.1     | 3.2 ± 0.1   | 2.7 ± 0.1    | 5.0 ± 0.3     | 3.4 ± 0.3     | 2.22 ± 0.07 | 2.7 ± 0.2     | 3.1 ± 0.1   | 2.7 ± 0.2    | 3.7 ± 0.2     | W Africa F   | E Asia M     |

**Table 110.** Mean absolute error for Application-type images for algorithm youverse-000. Values closer to zero are better.

| Age | F<br>E Africa | F<br>E Asia | F<br>E Europe | F<br>S Asia | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia | M<br>E Europe | M<br>S Asia | M<br>SE Asia | M<br>W Africa | max<br>Group | min<br>Group |
|-----|---------------|-------------|---------------|-------------|--------------|---------------|---------------|-------------|---------------|-------------|--------------|---------------|--------------|--------------|
| 14  | 4.0 ± 0.3     | 2.87 ± 0.09 | 4.1 ± 0.2     | 4.0 ± 0.1   | 2.7 ± 0.2    | 4.5 ± 0.5     | 3.2 ± 0.2     | 2.53 ± 0.07 | 2.6 ± 0.1     | 3.0 ± 0.1   | 2.5 ± 0.1    | 3.1 ± 0.3     | W Africa F   | S E Asia M   |
| 15  | 4.1 ± 0.3     | 2.7 ± 0.1   | 4.1 ± 0.3     | 4.0 ± 0.2   | 2.6 ± 0.3    | 4.8 ± 0.5     | 2.9 ± 0.3     | 2.13 ± 0.08 | 2.4 ± 0.2     | 2.8 ± 0.1   | 2.3 ± 0.2    | 3.6 ± 0.4     | W Africa F   | E Asia M     |
| 16  | 4.3 ± 0.4     | 2.4 ± 0.1   | 4.2 ± 0.3     | 3.6 ± 0.2   | 2.5 ± 0.3    | 4.4 ± 0.5     | 3.3 ± 0.3     | 1.88 ± 0.09 | 2.6 ± 0.2     | 2.8 ± 0.2   | 2.5 ± 0.2    | 3.6 ± 0.4     | W Africa F   | E Asia M     |
| 17  | 4.5 ± 0.4     | 2.3 ± 0.1   | 4.0 ± 0.3     | 3.2 ± 0.2   | 2.3 ± 0.2    | 4.8 ± 0.4     | 3.6 ± 0.3     | 1.83 ± 0.09 | 2.4 ± 0.2     | 3.1 ± 0.2   | 2.2 ± 0.2    | 3.9 ± 0.4     | W Africa F   | E Asia M     |
| 18  | 4.0 ± 0.3     | 2.38 ± 0.09 | 3.8 ± 0.2     | 3.3 ± 0.1   | 2.4 ± 0.1    | 4.5 ± 0.3     | 4.3 ± 0.3     | 1.90 ± 0.08 | 2.6 ± 0.2     | 3.1 ± 0.1   | 2.1 ± 0.2    | 4.4 ± 0.4     | W Africa F   | E Asia M     |
| 19  | 4.1 ± 0.3     | 2.39 ± 0.08 | 3.3 ± 0.2     | 3.1 ± 0.1   | 2.4 ± 0.1    | 4.5 ± 0.3     | 4.1 ± 0.3     | 2.05 ± 0.08 | 2.6 ± 0.2     | 3.4 ± 0.1   | 2.2 ± 0.2    | 4.2 ± 0.3     | W Africa F   | E Asia M     |
| 20  | 4.2 ± 0.3     | 2.49 ± 0.09 | 3.4 ± 0.2     | 3.0 ± 0.1   | 2.4 ± 0.1    | 4.3 ± 0.3     | 4.4 ± 0.3     | 2.28 ± 0.08 | 2.7 ± 0.2     | 3.5 ± 0.1   | 2.5 ± 0.1    | 4.6 ± 0.3     | W Africa M   | E Asia M     |
| 21  | 3.9 ± 0.2     | 2.65 ± 0.08 | 3.3 ± 0.2     | 3.01 ± 0.10 | 2.64 ± 0.10  | 4.4 ± 0.3     | 4.6 ± 0.3     | 2.48 ± 0.08 | 3.0 ± 0.2     | 3.3 ± 0.1   | 2.6 ± 0.1    | 4.9 ± 0.3     | W Africa M   | E Asia M     |
| 22  | 3.9 ± 0.2     | 2.81 ± 0.08 | 3.2 ± 0.1     | 3.02 ± 0.09 | 2.84 ± 0.09  | 4.0 ± 0.2     | 4.7 ± 0.3     | 2.71 ± 0.08 | 3.0 ± 0.2     | 3.4 ± 0.1   | 2.6 ± 0.1    | 4.4 ± 0.3     | E Africa M   | S E Asia M   |
| 23  | 3.8 ± 0.2     | 3.04 ± 0.07 | 3.0 ± 0.1     | 3.09 ± 0.08 | 3.07 ± 0.09  | 4.2 ± 0.2     | 4.7 ± 0.2     | 2.80 ± 0.08 | 3.1 ± 0.2     | 3.4 ± 0.1   | 2.8 ± 0.1    | 4.3 ± 0.2     | E Africa M   | S E Asia M   |
| 24  | 4.0 ± 0.2     | 3.37 ± 0.08 | 3.0 ± 0.1     | 3.14 ± 0.07 | 3.33 ± 0.09  | 3.9 ± 0.2     | 4.2 ± 0.2     | 3.04 ± 0.09 | 3.0 ± 0.1     | 3.2 ± 0.1   | 2.9 ± 0.1    | 4.2 ± 0.2     | W Africa M   | S E Asia M   |
| 25  | 3.9 ± 0.2     | 3.65 ± 0.08 | 2.87 ± 0.10   | 3.18 ± 0.07 | 3.68 ± 0.10  | 3.9 ± 0.2     | 4.2 ± 0.2     | 3.15 ± 0.09 | 3.0 ± 0.1     | 3.15 ± 0.09 | 3.1 ± 0.1    | 4.2 ± 0.2     | W Africa M   | E Europe F   |
| 26  | 4.0 ± 0.2     | 4.07 ± 0.08 | 2.78 ± 0.09   | 3.18 ± 0.07 | 4.00 ± 0.10  | 3.9 ± 0.2     | 4.1 ± 0.2     | 3.25 ± 0.09 | 3.1 ± 0.1     | 3.04 ± 0.08 | 3.2 ± 0.1    | 4.0 ± 0.2     | E Africa M   | E Europe F   |
| 27  | 4.0 ± 0.2     | 4.48 ± 0.08 | 2.92 ± 0.09   | 3.35 ± 0.07 | 4.3 ± 0.1    | 3.8 ± 0.2     | 3.9 ± 0.2     | 3.38 ± 0.10 | 3.1 ± 0.1     | 3.00 ± 0.08 | 3.4 ± 0.1    | 4.0 ± 0.2     | E Asia F     | E Europe F   |
| 28  | 4.2 ± 0.2     | 4.91 ± 0.08 | 3.03 ± 0.09   | 3.55 ± 0.07 | 4.7 ± 0.1    | 4.0 ± 0.2     | 3.7 ± 0.2     | 3.5 ± 0.1   | 3.0 ± 0.1     | 2.95 ± 0.07 | 3.4 ± 0.1    | 3.8 ± 0.2     | E Asia F     | S Asia M     |
| 29  | 4.3 ± 0.2     | 5.24 ± 0.08 | 3.20 ± 0.09   | 3.79 ± 0.07 | 5.2 ± 0.1    | 4.2 ± 0.2     | 3.6 ± 0.2     | 3.7 ± 0.1   | 3.1 ± 0.1     | 2.98 ± 0.06 | 3.6 ± 0.2    | 3.8 ± 0.1     | E Asia F     | S Asia M     |
| 30  | 4.4 ± 0.2     | 5.74 ± 0.08 | 3.28 ± 0.10   | 4.04 ± 0.07 | 5.4 ± 0.1    | 4.3 ± 0.2     | 3.7 ± 0.2     | 3.82 ± 0.10 | 3.1 ± 0.1     | 3.02 ± 0.06 | 3.6 ± 0.2    | 3.8 ± 0.1     | E Asia F     | S Asia M     |
| 31  | 4.7 ± 0.2     | 6.06 ± 0.09 | 3.6 ± 0.1     | 4.37 ± 0.08 | 6.0 ± 0.1    | 4.6 ± 0.2     | 3.5 ± 0.2     | 4.1 ± 0.1   | 3.2 ± 0.1     | 3.17 ± 0.06 | 4.0 ± 0.2    | 3.7 ± 0.1     | E Asia F     | E Europe M   |
| 32  | 4.9 ± 0.2     | 6.47 ± 0.09 | 3.7 ± 0.1     | 4.65 ± 0.08 | 6.3 ± 0.1    | 5.1 ± 0.2     | 3.6 ± 0.2     | 4.3 ± 0.1   | 3.3 ± 0.1     | 3.37 ± 0.06 | 4.0 ± 0.2    | 3.7 ± 0.1     | E Asia F     | E Europe M   |
| 33  | 5.3 ± 0.2     | 6.79 ± 0.10 | 4.0 ± 0.1     | 4.94 ± 0.08 | 6.7 ± 0.1    | 5.1 ± 0.2     | 3.8 ± 0.2     | 4.5 ± 0.1   | 3.2 ± 0.1     | 3.45 ± 0.06 | 4.4 ± 0.2    | 3.8 ± 0.1     | E Asia F     | E Europe M   |
| 34  | 5.3 ± 0.2     | 7.10 ± 0.10 | 4.1 ± 0.1     | 5.09 ± 0.09 | 7.1 ± 0.2    | 5.4 ± 0.2     | 3.7 ± 0.2     | 4.9 ± 0.1   | 3.3 ± 0.1     | 3.70 ± 0.07 | 4.5 ± 0.2    | 3.8 ± 0.1     | E Asia F     | E Europe M   |
| 35  | 5.6 ± 0.2     | 7.6 ± 0.1   | 4.1 ± 0.1     | 5.46 ± 0.09 | 7.5 ± 0.2    | 5.8 ± 0.2     | 3.8 ± 0.2     | 5.0 ± 0.1   | 3.5 ± 0.1     | 3.86 ± 0.07 | 4.6 ± 0.2    | 4.1 ± 0.1     | E Asia F     | E Europe M   |
| 36  | 5.9 ± 0.2     | 7.9 ± 0.1   | 4.3 ± 0.1     | 5.71 ± 0.10 | 7.6 ± 0.2    | 6.1 ± 0.2     | 4.1 ± 0.2     | 5.2 ± 0.1   | 3.4 ± 0.1     | 4.05 ± 0.08 | 4.8 ± 0.2    | 4.2 ± 0.2     | E Asia F     | E Europe M   |
| 37  | 6.1 ± 0.2     | 8.2 ± 0.1   | 4.3 ± 0.1     | 6.0 ± 0.1   | 8.1 ± 0.2    | 6.5 ± 0.2     | 3.9 ± 0.2     | 5.3 ± 0.1   | 3.5 ± 0.1     | 4.14 ± 0.08 | 4.9 ± 0.2    | 4.4 ± 0.2     | E Asia F     | E Europe M   |
| 38  | 6.2 ± 0.3     | 8.6 ± 0.1   | 4.5 ± 0.2     | 6.1 ± 0.1   | 8.2 ± 0.2    | 6.8 ± 0.3     | 4.2 ± 0.2     | 5.4 ± 0.1   | 3.6 ± 0.2     | 4.36 ± 0.08 | 5.0 ± 0.2    | 4.4 ± 0.2     | E Asia F     | E Europe M   |
| 39  | 6.5 ± 0.3     | 8.8 ± 0.1   | 4.4 ± 0.2     | 6.5 ± 0.1   | 8.5 ± 0.2    | 6.9 ± 0.3     | 4.2 ± 0.2     | 5.6 ± 0.1   | 3.5 ± 0.2     | 4.51 ± 0.09 | 5.1 ± 0.2    | 4.7 ± 0.2     | E Asia F     | E Europe M   |
| 40  | 6.7 ± 0.3     | 9.1 ± 0.1   | 4.5 ± 0.2     | 6.6 ± 0.1   | 8.5 ± 0.2    | 7.4 ± 0.3     | 4.0 ± 0.2     | 5.8 ± 0.1   | 3.7 ± 0.2     | 4.69 ± 0.10 | 5.0 ± 0.2    | 4.9 ± 0.2     | E Asia F     | E Europe M   |
| 41  | 7.2 ± 0.4     | 9.4 ± 0.1   | 4.6 ± 0.2     | 6.8 ± 0.2   | 8.9 ± 0.2    | 8.2 ± 0.3     | 4.5 ± 0.2     | 5.9 ± 0.1   | 3.8 ± 0.2     | 4.9 ± 0.1   | 5.3 ± 0.2    | 5.1 ± 0.2     | E Asia F     | E Europe M   |
| 42  | 7.0 ± 0.4     | 9.5 ± 0.1   | 4.7 ± 0.2     | 6.9 ± 0.2   | 8.9 ± 0.2    | 8.4 ± 0.3     | 4.7 ± 0.2     | 6.1 ± 0.1   | 3.8 ± 0.2     | 5.2 ± 0.1   | 5.1 ± 0.2    | 5.3 ± 0.2     | E Asia F     | E Europe M   |
| 43  | 7.4 ± 0.4     | 9.9 ± 0.1   | 4.8 ± 0.2     | 7.1 ± 0.2   | 9.4 ± 0.2    | 9.0 ± 0.4     | 4.7 ± 0.3     | 6.2 ± 0.1   | 4.0 ± 0.2     | 5.3 ± 0.1   | 5.3 ± 0.2    | 5.9 ± 0.2     | E Asia F     | E Europe M   |
| 44  | 7.6 ± 0.4     | 10.3 ± 0.1  | 5.1 ± 0.2     | 7.4 ± 0.2   | 9.5 ± 0.3    | 8.9 ± 0.4     | 5.3 ± 0.3     | 6.3 ± 0.2   | 3.9 ± 0.2     | 5.4 ± 0.1   | 5.3 ± 0.2    | 6.0 ± 0.2     | E Asia F     | E Europe M   |
| 45  | 7.7 ± 0.4     | 10.4 ± 0.1  | 5.1 ± 0.2     | 7.4 ± 0.2   | 9.8 ± 0.3    | 9.2 ± 0.4     | 5.7 ± 0.3     | 6.5 ± 0.1   | 4.2 ± 0.2     | 5.5 ± 0.1   | 5.5 ± 0.2    | 6.6 ± 0.3     | E Asia F     | E Europe M   |
| 46  | 7.7 ± 0.5     | 10.8 ± 0.1  | 5.2 ± 0.2     | 7.4 ± 0.2   | 10.0 ± 0.3   | 9.8 ± 0.4     | 5.5 ± 0.3     | 6.8 ± 0.2   | 4.2 ± 0.2     | 5.6 ± 0.2   | 5.6 ± 0.3    | 6.6 ± 0.3     | E Asia F     | E Europe M   |
| 47  | 8.1 ± 0.5     | 10.9 ± 0.2  | 5.6 ± 0.3     | 7.7 ± 0.2   | 10.3 ± 0.3   | 10.0 ± 0.4    | 5.8 ± 0.3     | 7.0 ± 0.2   | 4.5 ± 0.2     | 6.0 ± 0.2   | 5.4 ± 0.3    | 7.0 ± 0.3     | E Asia F     | E Europe M   |
| 48  | 8.6 ± 0.5     | 11.2 ± 0.2  | 5.6 ± 0.3     | 7.8 ± 0.2   | 10.3 ± 0.3   | 10.5 ± 0.4    | 6.3 ± 0.4     | 7.2 ± 0.2   | 4.6 ± 0.2     | 6.0 ± 0.2   | 5.8 ± 0.3    | 7.4 ± 0.3     | E Asia F     | E Europe M   |
| 49  | 8.4 ± 0.6     | 11.6 ± 0.2  | 5.8 ± 0.3     | 8.0 ± 0.2   | 10.5 ± 0.3   | 10.8 ± 0.4    | 6.7 ± 0.4     | 7.4 ± 0.2   | 4.7 ± 0.2     | 6.2 ± 0.2   | 5.8 ± 0.3    | 7.9 ± 0.3     | E Asia F     | E Europe M   |
| 50  | 9.0 ± 0.6     | 11.7 ± 0.2  | 5.9 ± 0.3     | 8.0 ± 0.2   | 10.8 ± 0.3   | 11.1 ± 0.5    | 7.0 ± 0.4     | 7.5 ± 0.2   | 4.8 ± 0.2     | 6.5 ± 0.2   | 5.9 ± 0.3    | 8.1 ± 0.3     | E Asia F     | E Europe M   |
| 51  | 9.5 ± 0.6     | 12.0 ± 0.2  | 6.0 ± 0.3     | 8.4 ± 0.2   | 11.1 ± 0.3   | 11.6 ± 0.5    | 7.4 ± 0.5     | 7.7 ± 0.2   | 4.7 ± 0.2     | 6.5 ± 0.2   | 5.8 ± 0.3    | 9.0 ± 0.3     | E Asia F     | E Europe M   |
| 52  | 9.3 ± 0.6     | 12.2 ± 0.2  | 6.2 ± 0.3     | 8.7 ± 0.3   | 11.0 ± 0.3   | 11.9 ± 0.5    | 7.0 ± 0.4     | 8.1 ± 0.2   | 4.9 ± 0.3     | 6.9 ± 0.2   | 6.0 ± 0.3    | 8.9 ± 0.4     | E Asia F     | E Europe M   |

### Appendix A.3. Challenge Age

The following pages include two tables for each algorithm, one for *Visa* images and another for lower quality *Border* images. The values are Challenge-25 accept rates - the proportion of people of the given age who are judged to have age 25 or over. For an application with a legal age (LA) of 18, these rates would be properly termed false positive rates (FPR) for everyone below 18, and true positive rates for those 18 or over. The vocabulary would change depending on the LA. For this reason we use the generic term “accept rates”.

**Table 111.** For algorithm *cognitec-000* with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.007 ± 0.007 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.008 ± 0.008 | 0.004 ± 0.004 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.004 ± 0.004 | W Africa F    | E Europe F    | 0.46 |
| 15  | 0.04 ± 0.02   | 0.005 ± 0.004 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.005 ± 0.005 | 0.03 ± 0.02   | 0.005 ± 0.005 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.009 ± 0.009 | E Africa F    | E Europe M    | 0.67 |
| 16  | 0.05 ± 0.02   | 0.008 ± 0.005 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.01 ± 0.01   | 0.06 ± 0.03   | 0.02 ± 0.01   | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.01 ± 0.01   | W Africa F    | E Europe F    | 0.66 |
| 17  | 0.08 ± 0.03   | 0.015 ± 0.006 | 0.006 ± 0.006 | 0.03 ± 0.01   | 0.02 ± 0.01   | 0.09 ± 0.03   | 0.006 ± 0.006 | 0.010 ± 0.004 | 0.000 ± 0.000 | 0.014 ± 0.007 | 0.02 ± 0.01   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.55 |
| 18  | 0.11 ± 0.03   | 0.027 ± 0.007 | 0.02 ± 0.01   | 0.030 ± 0.007 | 0.04 ± 0.01   | 0.14 ± 0.03   | 0.04 ± 0.01   | 0.018 ± 0.005 | 0.004 ± 0.004 | 0.029 ± 0.008 | 0.03 ± 0.01   | 0.05 ± 0.02   | W Africa F    | E Europe M    | 0.45 |
| 19  | 0.16 ± 0.03   | 0.045 ± 0.008 | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.08 ± 0.02   | 0.21 ± 0.04   | 0.11 ± 0.02   | 0.040 ± 0.008 | 0.009 ± 0.008 | 0.06 ± 0.01   | 0.05 ± 0.02   | 0.08 ± 0.02   | W Africa F    | E Europe M    | 0.40 |
| 20  | 0.26 ± 0.03   | 0.10 ± 0.01   | 0.04 ± 0.01   | 0.09 ± 0.01   | 0.12 ± 0.02   | 0.24 ± 0.03   | 0.18 ± 0.03   | 0.07 ± 0.01   | 0.03 ± 0.01   | 0.12 ± 0.01   | 0.11 ± 0.02   | 0.12 ± 0.03   | E Africa F    | E Europe M    | 0.32 |
| 21  | 0.27 ± 0.03   | 0.17 ± 0.02   | 0.06 ± 0.02   | 0.16 ± 0.02   | 0.17 ± 0.02   | 0.36 ± 0.03   | 0.23 ± 0.03   | 0.13 ± 0.01   | 0.06 ± 0.02   | 0.19 ± 0.02   | 0.17 ± 0.02   | 0.19 ± 0.04   | W Africa F    | E Europe M    | 0.26 |
| 22  | 0.35 ± 0.03   | 0.24 ± 0.02   | 0.09 ± 0.02   | 0.22 ± 0.02   | 0.27 ± 0.02   | 0.37 ± 0.03   | 0.36 ± 0.03   | 0.22 ± 0.02   | 0.14 ± 0.03   | 0.29 ± 0.02   | 0.25 ± 0.03   | 0.23 ± 0.03   | W Africa F    | E Europe F    | 0.19 |
| 23  | 0.41 ± 0.03   | 0.33 ± 0.02   | 0.19 ± 0.02   | 0.31 ± 0.02   | 0.37 ± 0.02   | 0.44 ± 0.03   | 0.42 ± 0.03   | 0.31 ± 0.02   | 0.23 ± 0.03   | 0.39 ± 0.02   | 0.37 ± 0.03   | 0.37 ± 0.03   | W Africa F    | E Europe F    | 0.13 |
| 24  | 0.47 ± 0.03   | 0.42 ± 0.02   | 0.28 ± 0.02   | 0.39 ± 0.02   | 0.45 ± 0.02   | 0.50 ± 0.03   | 0.55 ± 0.03   | 0.41 ± 0.02   | 0.38 ± 0.03   | 0.54 ± 0.02   | 0.49 ± 0.03   | 0.44 ± 0.03   | E Africa M    | E Europe F    | 0.10 |
| 25  | 0.50 ± 0.03   | 0.53 ± 0.01   | 0.37 ± 0.02   | 0.45 ± 0.01   | 0.55 ± 0.02   | 0.58 ± 0.03   | 0.62 ± 0.03   | 0.53 ± 0.02   | 0.51 ± 0.03   | 0.66 ± 0.02   | 0.60 ± 0.03   | 0.49 ± 0.03   | S Asia M      | E Europe F    | 0.08 |
| 26  | 0.58 ± 0.03   | 0.62 ± 0.01   | 0.47 ± 0.02   | 0.54 ± 0.01   | 0.65 ± 0.02   | 0.64 ± 0.03   | 0.69 ± 0.03   | 0.67 ± 0.02   | 0.64 ± 0.03   | 0.75 ± 0.01   | 0.70 ± 0.02   | 0.60 ± 0.03   | S Asia M      | E Europe F    | 0.07 |
| 27  | 0.64 ± 0.02   | 0.70 ± 0.01   | 0.59 ± 0.02   | 0.61 ± 0.01   | 0.71 ± 0.02   | 0.71 ± 0.03   | 0.76 ± 0.02   | 0.75 ± 0.02   | 0.76 ± 0.02   | 0.82 ± 0.01   | 0.82 ± 0.02   | 0.69 ± 0.02   | S Asia M      | E Europe F    | 0.06 |
| 28  | 0.70 ± 0.02   | 0.79 ± 0.01   | 0.71 ± 0.02   | 0.69 ± 0.01   | 0.78 ± 0.02   | 0.77 ± 0.02   | 0.82 ± 0.02   | 0.83 ± 0.01   | 0.86 ± 0.02   | 0.880 ± 0.009 | 0.87 ± 0.02   | 0.77 ± 0.02   | S Asia M      | S Asia F      | 0.05 |
| 29  | 0.77 ± 0.02   | 0.84 ± 0.01   | 0.80 ± 0.02   | 0.76 ± 0.01   | 0.83 ± 0.01   | 0.80 ± 0.02   | 0.87 ± 0.02   | 0.90 ± 0.01   | 0.92 ± 0.01   | 0.925 ± 0.007 | 0.90 ± 0.02   | 0.83 ± 0.02   | S Asia M      | S Asia F      | 0.04 |
| 30  | 0.83 ± 0.02   | 0.896 ± 0.007 | 0.87 ± 0.01   | 0.81 ± 0.01   | 0.89 ± 0.01   | 0.84 ± 0.02   | 0.92 ± 0.01   | 0.933 ± 0.010 | 0.95 ± 0.01   | 0.948 ± 0.005 | 0.93 ± 0.02   | 0.88 ± 0.02   | S Asia M      | S Asia F      | 0.03 |
| 31  | 0.87 ± 0.02   | 0.933 ± 0.007 | 0.92 ± 0.01   | 0.848 ± 0.009 | 0.92 ± 0.01   | 0.89 ± 0.02   | 0.94 ± 0.01   | 0.960 ± 0.007 | 0.971 ± 0.008 | 0.968 ± 0.005 | 0.96 ± 0.01   | 0.90 ± 0.02   | E Europe M    | S Asia F      | 0.03 |
| 32  | 0.90 ± 0.02   | 0.952 ± 0.006 | 0.957 ± 0.008 | 0.892 ± 0.008 | 0.947 ± 0.009 | 0.92 ± 0.01   | 0.95 ± 0.01   | 0.973 ± 0.005 | 0.983 ± 0.008 | 0.983 ± 0.003 | 0.980 ± 0.009 | 0.93 ± 0.01   | E Europe M    | S Asia F      | 0.02 |
| 33  | 0.91 ± 0.01   | 0.972 ± 0.004 | 0.968 ± 0.008 | 0.932 ± 0.007 | 0.965 ± 0.008 | 0.94 ± 0.01   | 0.972 ± 0.008 | 0.987 ± 0.004 | 0.996 ± 0.004 | 0.989 ± 0.003 | 0.987 ± 0.007 | 0.958 ± 0.009 | E Europe M    | E Africa F    | 0.01 |
| 34  | 0.93 ± 0.01   | 0.982 ± 0.004 | 0.985 ± 0.006 | 0.949 ± 0.006 | 0.974 ± 0.007 | 0.96 ± 0.01   | 0.981 ± 0.008 | 0.993 ± 0.003 | 0.995 ± 0.004 | 0.995 ± 0.002 | 0.993 ± 0.006 | 0.971 ± 0.008 | S Asia M      | E Africa F    | 0.01 |

**Table 112.** For algorithm *cognitec-000* with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.11 ± 0.05   | 0.012 ± 0.004 | 0.011 ± 0.009 | 0.04 ± 0.01   | 0.014 ± 0.009 | 0.08 ± 0.04   | 0.07 ± 0.04   | 0.011 ± 0.004 | 0.015 ± 0.010 | 0.022 ± 0.007 | 0.013 ± 0.008 | 0.11 ± 0.05   | E Africa F    | E Europe F    | 0.51 |
| 15  | 0.17 ± 0.05   | 0.017 ± 0.005 | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.011 ± 0.007 | 0.18 ± 0.05   | 0.15 ± 0.05   | 0.021 ± 0.005 | 0.004 ± 0.004 | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.13 ± 0.05   | W Africa F    | E Europe M    | 0.55 |
| 16  | 0.28 ± 0.06   | 0.020 ± 0.005 | 0.02 ± 0.01   | 0.08 ± 0.02   | 0.03 ± 0.01   | 0.22 ± 0.05   | 0.16 ± 0.05   | 0.032 ± 0.006 | 0.020 ± 0.010 | 0.07 ± 0.01   | 0.06 ± 0.02   | 0.16 ± 0.04   | E Africa F    | E Europe M    | 0.51 |
| 17  | 0.27 ± 0.05   | 0.041 ± 0.006 | 0.04 ± 0.01   | 0.10 ± 0.01   | 0.05 ± 0.01   | 0.25 ± 0.05   | 0.28 ± 0.06   | 0.065 ± 0.008 | 0.04 ± 0.01   | 0.12 ± 0.02   | 0.07 ± 0.02   | 0.24 ± 0.05   | E Africa M    | E Europe M    | 0.43 |
| 18  | 0.40 ± 0.05   | 0.054 ± 0.006 | 0.08 ± 0.02   | 0.15 ± 0.01   | 0.07 ± 0.01   | 0.29 ± 0.05   | 0.36 ± 0.05   | 0.097 ± 0.008 | 0.08 ± 0.02   | 0.18 ± 0.02   | 0.13 ± 0.02   | 0.26 ± 0.05   | E Africa F    | E Asia F      | 0.38 |
| 19  | 0.46 ± 0.05   | 0.090 ± 0.006 | 0.10 ± 0.01   | 0.20 ± 0.02   | 0.09 ± 0.01   | 0.40 ± 0.05   | 0.44 ± 0.05   | 0.153 ± 0.009 | 0.10 ± 0.02   | 0.28 ± 0.02   | 0.19 ± 0.02   | 0.31 ± 0.04   | E Africa F    | E Asia F      | 0.35 |
| 20  | 0.50 ± 0.04   | 0.141 ± 0.008 | 0.13 ± 0.01   | 0.24 ± 0.02   | 0.16 ± 0.01   | 0.48 ± 0.04   | 0.51 ± 0.05   | 0.21 ± 0.01   | 0.18 ± 0.02   | 0.37 ± 0.02   | 0.27 ± 0.02   | 0.42 ± 0.04   | E Africa M    | E Europe F    | 0.28 |
| 21  | 0.57 ± 0.04   | 0.185 ± 0.008 | 0.18 ± 0.01   | 0.31 ± 0.02   | 0.20 ± 0.02   | 0.52 ± 0.04   | 0.60 ± 0.04   | 0.31 ± 0.01   | 0.31 ± 0.03   | 0.48 ± 0.02   | 0.34 ± 0.02   | 0.44 ± 0.04   | E Africa M    | E Europe F    | 0.24 |
| 22  | 0.62 ± 0.04   | 0.255 ± 0.008 | 0.26 ± 0.02   | 0.38 ± 0.01   | 0.27 ± 0.02   | 0.59 ± 0.03   | 0.66 ± 0.04   | 0.41 ± 0.01   | 0.38 ± 0.02   | 0.59 ± 0.02   | 0.45 ± 0.03   | 0.53 ± 0.04   | E Africa M    | E Asia F      | 0.19 |
| 23  | 0.67 ± 0.03   | 0.314 ± 0.008 | 0.33 ± 0.01   | 0.43 ± 0.01   | 0.36 ± 0.02   | 0.62 ± 0.03   | 0.78 ± 0.03   | 0.51 ± 0.01   | 0.54 ± 0.02   | 0.69 ± 0.01   | 0.53 ± 0.03   | 0.59 ± 0.03   | E Africa M    | E Asia F      | 0.17 |
| 24  | 0.72 ± 0.03   | 0.394 ± 0.009 | 0.42 ± 0.01   | 0.51 ± 0.01   | 0.44 ± 0.02   | 0.70 ± 0.02   | 0.79 ± 0.03   | 0.61 ± 0.01   | 0.64 ± 0.02   | 0.76 ± 0.01   | 0.63 ± 0.02   | 0.66 ± 0.03   | E Africa M    | E Asia F      | 0.13 |
| 25  | 0.76 ± 0.02   | 0.483 ± 0.008 | 0.51 ± 0.01   | 0.556 ± 0.010 | 0.52 ± 0.02   | 0.76 ± 0.02   | 0.84 ± 0.03   | 0.70 ± 0.01   | 0.74 ± 0.02   | 0.832 ± 0.009 | 0.75 ± 0.02   | 0.72 ± 0.02   | E Africa M    | E Asia F      | 0.11 |
| 26  | 0.81 ± 0.02   | 0.569 ± 0.008 | 0.57 ± 0.01   | 0.622 ± 0.009 | 0.61 ± 0.02   | 0.79 ± 0.02   | 0.89 ± 0.02   | 0.771 ± 0.009 | 0.82 ± 0.01   | 0.876 ± 0.007 | 0.80 ± 0.02   | 0.78 ± 0.02   | E Africa M    | E Asia F      | 0.09 |
| 27  | 0.81 ± 0.02   | 0.638 ± 0.007 | 0.68 ± 0.01   | 0.676 ± 0.008 | 0.69 ± 0.01   | 0.83 ± 0.02   | 0.91 ± 0.02   | 0.826 ± 0.008 | 0.88 ± 0.01   | 0.910 ± 0.005 | 0.86 ± 0.02   | 0.84 ± 0.02   | E Africa M    | E Asia F      | 0.07 |
| 28  | 0.88 ± 0.02   | 0.711 ± 0.006 | 0.750 ± 0.009 | 0.725 ± 0.008 | 0.75 ± 0.01   | 0.85 ± 0.02   | 0.93 ± 0.02   | 0.888 ± 0.007 | 0.930 ± 0.009 | 0.943 ± 0.004 | 0.92 ± 0.01   | 0.87 ± 0.01   | S Asia M      | E Asia F      | 0.06 |
| 29  | 0.89 ± 0.02   | 0.776 ± 0.006 | 0.825 ± 0.009 | 0.764 ± 0.008 | 0.80 ± 0.01   | 0.89 ± 0.01   | 0.96 ± 0.01   | 0.923 ± 0.005 | 0.955 ± 0.007 | 0.962 ± 0.003 | 0.94 ± 0.01   | 0.91 ± 0.01   | S Asia M      | S Asia F      | 0.05 |
| 30  | 0.92 ± 0.02   | 0.836 ± 0.005 | 0.870 ± 0.008 | 0.819 ± 0.006 | 0.85 ± 0.01   | 0.91 ± 0.01   | 0.973 ± 0.010 | 0.948 ± 0.004 | 0.974 ± 0.005 | 0.973 ± 0.003 | 0.956 ± 0.010 | 0.93 ± 0.01   | E Europe M    | S Asia F      | 0.04 |
| 31  | 0.93 ± 0.01   | 0.884 ± 0.005 | 0.908 ± 0.007 | 0.850 ± 0.006 | 0.884 ± 0.009 | 0.93 ± 0.01   | 0.979 ± 0.008 | 0.969 ± 0.003 | 0.986 ± 0.004 | 0.983 ± 0.002 | 0.969 ± 0.008 | 0.947 ± 0.009 | E Europe M    | S Asia F      | 0.03 |
| 32  | 0.94 ± 0.01   | 0.916 ± 0.004 | 0.946 ± 0.006 | 0.886 ± 0.006 | 0.917 ± 0.008 | 0.95 ± 0.01   | 0.989 ± 0.006 | 0.979 ± 0.003 | 0.992 ± 0.003 | 0.991 ± 0.001 | 0.981 ± 0.007 | 0.961 ± 0.008 | E Europe M    | S Asia F      | 0.02 |
| 33  | 0.97 ± 0.01   | 0.942 ± 0.003 | 0.964 ± 0.005 | 0.920 ± 0.004 | 0.937 ± 0.008 | 0.96 ± 0.01   | 0.990 ± 0.006 | 0.989 ± 0.002 | 0.995 ± 0.002 | 0.994 ± 0.001 | 0.983 ± 0.005 | 0.974 ± 0.007 | E Europe M    | S Asia F      | 0.01 |
| 34  | 0.980 ± 0.009 | 0.960 ± 0.003 | 0.977 ± 0.004 | 0.939 ± 0.004 | 0.962 ± 0.006 | 0.971 ± 0.009 | 0.996 ± 0.003 | 0.992 ± 0.002 | 0.999 ± 0.001 | 0.996 ± 0.001 | 0.995 ± 0.003 | 0.983 ± 0.005 | E Europe M    | S Asia F      | 0.01 |

**Table 113.** For algorithm cvut-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.23 ± 0.04   | 0.028 ± 0.008 | 0.002 ± 0.002 | 0.015 ± 0.006 | 0.02 ± 0.01  | 0.21 ± 0.05   | 0.06 ± 0.02   | 0.010 ± 0.004 | 0.000 ± 0.000 | 0.005 ± 0.003 | 0.009 ± 0.009 | 0.07 ± 0.03   | E Africa F    | E Europe M    | 0.71 |
| 15  | 0.30 ± 0.05   | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.06 ± 0.02   | 0.04 ± 0.02  | 0.21 ± 0.05   | 0.08 ± 0.03   | 0.016 ± 0.006 | 0.006 ± 0.006 | 0.016 ± 0.007 | 0.010 ± 0.007 | 0.10 ± 0.04   | E Africa F    | E Europe M    | 0.61 |
| 16  | 0.38 ± 0.05   | 0.08 ± 0.01   | 0.03 ± 0.02   | 0.07 ± 0.02   | 0.06 ± 0.02  | 0.29 ± 0.05   | 0.11 ± 0.03   | 0.024 ± 0.007 | 0.009 ± 0.009 | 0.03 ± 0.01   | 0.01 ± 0.01   | 0.16 ± 0.05   | E Africa F    | E Europe M    | 0.58 |
| 17  | 0.46 ± 0.05   | 0.12 ± 0.01   | 0.04 ± 0.02   | 0.11 ± 0.02   | 0.07 ± 0.02  | 0.34 ± 0.05   | 0.17 ± 0.04   | 0.05 ± 0.01   | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.05 ± 0.02   | 0.27 ± 0.05   | E Africa F    | E Europe M    | 0.50 |
| 18  | 0.50 ± 0.04   | 0.13 ± 0.01   | 0.05 ± 0.02   | 0.14 ± 0.02   | 0.09 ± 0.02  | 0.37 ± 0.05   | 0.25 ± 0.03   | 0.056 ± 0.009 | 0.06 ± 0.02   | 0.10 ± 0.01   | 0.03 ± 0.01   | 0.28 ± 0.04   | E Africa F    | S E Asia M    | 0.47 |
| 19  | 0.59 ± 0.04   | 0.18 ± 0.02   | 0.08 ± 0.02   | 0.21 ± 0.02   | 0.12 ± 0.02  | 0.49 ± 0.04   | 0.35 ± 0.04   | 0.08 ± 0.01   | 0.11 ± 0.03   | 0.19 ± 0.02   | 0.06 ± 0.02   | 0.43 ± 0.04   | E Africa F    | S E Asia M    | 0.42 |
| 20  | 0.66 ± 0.03   | 0.23 ± 0.02   | 0.12 ± 0.02   | 0.24 ± 0.02   | 0.18 ± 0.02  | 0.54 ± 0.04   | 0.48 ± 0.04   | 0.12 ± 0.01   | 0.20 ± 0.03   | 0.29 ± 0.02   | 0.09 ± 0.02   | 0.50 ± 0.05   | E Africa F    | S E Asia M    | 0.36 |
| 21  | 0.71 ± 0.03   | 0.29 ± 0.02   | 0.17 ± 0.02   | 0.33 ± 0.02   | 0.21 ± 0.02  | 0.59 ± 0.04   | 0.59 ± 0.04   | 0.16 ± 0.02   | 0.29 ± 0.04   | 0.38 ± 0.02   | 0.14 ± 0.02   | 0.58 ± 0.04   | E Africa F    | S E Asia M    | 0.31 |
| 22  | 0.75 ± 0.03   | 0.33 ± 0.02   | 0.23 ± 0.02   | 0.40 ± 0.02   | 0.25 ± 0.02  | 0.66 ± 0.04   | 0.71 ± 0.03   | 0.21 ± 0.02   | 0.43 ± 0.04   | 0.49 ± 0.02   | 0.19 ± 0.02   | 0.69 ± 0.04   | E Africa F    | S E Asia M    | 0.28 |
| 23  | 0.80 ± 0.03   | 0.42 ± 0.02   | 0.27 ± 0.02   | 0.49 ± 0.02   | 0.30 ± 0.02  | 0.68 ± 0.03   | 0.77 ± 0.03   | 0.29 ± 0.02   | 0.51 ± 0.03   | 0.61 ± 0.02   | 0.25 ± 0.03   | 0.76 ± 0.03   | E Africa F    | S E Asia M    | 0.24 |
| 24  | 0.83 ± 0.02   | 0.46 ± 0.02   | 0.34 ± 0.02   | 0.56 ± 0.02   | 0.35 ± 0.02  | 0.76 ± 0.03   | 0.83 ± 0.02   | 0.37 ± 0.02   | 0.62 ± 0.03   | 0.70 ± 0.02   | 0.30 ± 0.03   | 0.83 ± 0.03   | E Africa M    | S E Asia M    | 0.21 |
| 25  | 0.84 ± 0.02   | 0.54 ± 0.01   | 0.40 ± 0.02   | 0.60 ± 0.01   | 0.39 ± 0.02  | 0.79 ± 0.02   | 0.86 ± 0.02   | 0.45 ± 0.02   | 0.72 ± 0.03   | 0.78 ± 0.01   | 0.41 ± 0.03   | 0.87 ± 0.02   | W Africa M    | S E Asia F    | 0.18 |
| 26  | 0.87 ± 0.02   | 0.60 ± 0.01   | 0.47 ± 0.02   | 0.66 ± 0.01   | 0.46 ± 0.02  | 0.81 ± 0.02   | 0.91 ± 0.02   | 0.55 ± 0.02   | 0.79 ± 0.02   | 0.83 ± 0.01   | 0.49 ± 0.03   | 0.89 ± 0.02   | E Africa M    | S E Asia F    | 0.15 |
| 27  | 0.90 ± 0.02   | 0.67 ± 0.01   | 0.55 ± 0.02   | 0.71 ± 0.01   | 0.52 ± 0.02  | 0.85 ± 0.02   | 0.94 ± 0.01   | 0.61 ± 0.02   | 0.87 ± 0.02   | 0.88 ± 0.01   | 0.58 ± 0.03   | 0.91 ± 0.01   | E Africa M    | S E Asia F    | 0.12 |
| 28  | 0.90 ± 0.02   | 0.72 ± 0.01   | 0.61 ± 0.02   | 0.76 ± 0.01   | 0.58 ± 0.02  | 0.87 ± 0.02   | 0.96 ± 0.01   | 0.69 ± 0.02   | 0.92 ± 0.01   | 0.908 ± 0.009 | 0.66 ± 0.03   | 0.94 ± 0.01   | E Africa M    | S E Asia F    | 0.10 |
| 29  | 0.91 ± 0.02   | 0.77 ± 0.01   | 0.70 ± 0.02   | 0.80 ± 0.01   | 0.64 ± 0.02  | 0.90 ± 0.02   | 0.966 ± 0.009 | 0.77 ± 0.02   | 0.94 ± 0.01   | 0.939 ± 0.006 | 0.72 ± 0.03   | 0.96 ± 0.01   | E Africa M    | S E Asia F    | 0.08 |
| 30  | 0.93 ± 0.01   | 0.823 ± 0.009 | 0.76 ± 0.02   | 0.835 ± 0.010 | 0.70 ± 0.02  | 0.92 ± 0.01   | 0.978 ± 0.008 | 0.82 ± 0.01   | 0.96 ± 0.01   | 0.945 ± 0.006 | 0.77 ± 0.02   | 0.963 ± 0.009 | E Africa M    | S E Asia F    | 0.06 |
| 31  | 0.95 ± 0.01   | 0.857 ± 0.008 | 0.84 ± 0.01   | 0.850 ± 0.009 | 0.74 ± 0.02  | 0.93 ± 0.01   | 0.980 ± 0.008 | 0.86 ± 0.01   | 0.971 ± 0.009 | 0.965 ± 0.005 | 0.82 ± 0.02   | 0.975 ± 0.008 | E Africa M    | S E Asia F    | 0.05 |
| 32  | 0.966 ± 0.010 | 0.885 ± 0.008 | 0.87 ± 0.01   | 0.894 ± 0.008 | 0.78 ± 0.02  | 0.94 ± 0.01   | 0.989 ± 0.006 | 0.90 ± 0.01   | 0.978 ± 0.009 | 0.971 ± 0.004 | 0.87 ± 0.02   | 0.985 ± 0.006 | E Africa M    | S E Asia F    | 0.04 |
| 33  | 0.97 ± 0.01   | 0.916 ± 0.007 | 0.89 ± 0.01   | 0.903 ± 0.009 | 0.81 ± 0.02  | 0.964 ± 0.010 | 0.992 ± 0.005 | 0.924 ± 0.010 | 0.990 ± 0.006 | 0.981 ± 0.003 | 0.90 ± 0.02   | 0.987 ± 0.006 | E Africa M    | S E Asia F    | 0.03 |
| 34  | 0.97 ± 0.01   | 0.931 ± 0.007 | 0.92 ± 0.01   | 0.930 ± 0.007 | 0.86 ± 0.02  | 0.966 ± 0.010 | 0.994 ± 0.004 | 0.935 ± 0.009 | 0.990 ± 0.006 | 0.984 ± 0.003 | 0.93 ± 0.02   | 0.987 ± 0.006 | E Africa M    | S E Asia F    | 0.02 |

**Table 114.** For algorithm cvut-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.34 ± 0.07   | 0.023 ± 0.006 | 0.04 ± 0.02   | 0.07 ± 0.01   | 0.02 ± 0.01  | 0.41 ± 0.07   | 0.14 ± 0.05   | 0.003 ± 0.002 | 0.010 ± 0.007 | 0.007 ± 0.004 | 0.02 ± 0.01   | 0.10 ± 0.04   | W Africa F    | E Asia M      | 0.69 |
| 15  | 0.42 ± 0.07   | 0.045 ± 0.007 | 0.05 ± 0.02   | 0.12 ± 0.02   | 0.03 ± 0.01  | 0.46 ± 0.07   | 0.20 ± 0.06   | 0.005 ± 0.002 | 0.02 ± 0.01   | 0.025 ± 0.008 | 0.014 ± 0.009 | 0.17 ± 0.06   | W Africa F    | E Asia M      | 0.64 |
| 16  | 0.58 ± 0.06   | 0.072 ± 0.009 | 0.07 ± 0.02   | 0.16 ± 0.02   | 0.05 ± 0.01  | 0.49 ± 0.07   | 0.23 ± 0.06   | 0.021 ± 0.005 | 0.03 ± 0.01   | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.25 ± 0.05   | E Africa F    | S E Asia M    | 0.60 |
| 17  | 0.55 ± 0.06   | 0.10 ± 0.01   | 0.13 ± 0.02   | 0.21 ± 0.02   | 0.07 ± 0.01  | 0.57 ± 0.06   | 0.34 ± 0.07   | 0.029 ± 0.005 | 0.07 ± 0.02   | 0.12 ± 0.02   | 0.04 ± 0.01   | 0.38 ± 0.05   | W Africa F    | E Asia M      | 0.50 |
| 18  | 0.66 ± 0.05   | 0.141 ± 0.009 | 0.19 ± 0.02   | 0.28 ± 0.02   | 0.09 ± 0.01  | 0.66 ± 0.05   | 0.50 ± 0.06   | 0.042 ± 0.005 | 0.10 ± 0.02   | 0.19 ± 0.02   | 0.06 ± 0.01   | 0.47 ± 0.05   | W Africa F    | E Asia M      | 0.47 |
| 19  | 0.74 ± 0.04   | 0.196 ± 0.009 | 0.25 ± 0.02   | 0.34 ± 0.02   | 0.12 ± 0.01  | 0.69 ± 0.04   | 0.60 ± 0.05   | 0.070 ± 0.007 | 0.18 ± 0.02   | 0.29 ± 0.02   | 0.08 ± 0.02   | 0.59 ± 0.05   | E Africa F    | E Asia M      | 0.41 |
| 20  | 0.78 ± 0.04   | 0.233 ± 0.009 | 0.31 ± 0.02   | 0.43 ± 0.02   | 0.16 ± 0.01  | 0.73 ± 0.03   | 0.69 ± 0.04   | 0.107 ± 0.008 | 0.27 ± 0.03   | 0.42 ± 0.02   | 0.13 ± 0.02   | 0.66 ± 0.04   | E Africa F    | E Asia M      | 0.36 |
| 21  | 0.80 ± 0.03   | 0.278 ± 0.010 | 0.38 ± 0.02   | 0.50 ± 0.02   | 0.20 ± 0.02  | 0.79 ± 0.03   | 0.76 ± 0.03   | 0.153 ± 0.009 | 0.42 ± 0.02   | 0.53 ± 0.02   | 0.17 ± 0.02   | 0.75 ± 0.03   | E Africa F    | E Asia M      | 0.31 |
| 22  | 0.84 ± 0.03   | 0.315 ± 0.009 | 0.45 ± 0.02   | 0.55 ± 0.01   | 0.25 ± 0.02  | 0.83 ± 0.03   | 0.83 ± 0.03   | 0.21 ± 0.01   | 0.49 ± 0.02   | 0.63 ± 0.02   | 0.25 ± 0.02   | 0.81 ± 0.03   | E Africa F    | E Asia M      | 0.27 |
| 23  | 0.88 ± 0.02   | 0.364 ± 0.009 | 0.55 ± 0.01   | 0.60 ± 0.01   | 0.31 ± 0.02  | 0.85 ± 0.02   | 0.86 ± 0.03   | 0.26 ± 0.01   | 0.60 ± 0.02   | 0.70 ± 0.01   | 0.31 ± 0.02   | 0.85 ± 0.02   | E Africa F    | E Asia M      | 0.23 |
| 24  | 0.88 ± 0.02   | 0.427 ± 0.009 | 0.59 ± 0.01   | 0.65 ± 0.01   | 0.37 ± 0.02  | 0.89 ± 0.02   | 0.91 ± 0.02   | 0.33 ± 0.01   | 0.70 ± 0.02   | 0.77 ± 0.01   | 0.37 ± 0.02   | 0.87 ± 0.02   | E Africa M    | E Asia M      | 0.20 |
| 25  | 0.91 ± 0.02   | 0.481 ± 0.008 | 0.67 ± 0.01   | 0.68 ± 0.01   | 0.44 ± 0.02  | 0.89 ± 0.02   | 0.92 ± 0.02   | 0.41 ± 0.01   | 0.76 ± 0.02   | 0.816 ± 0.009 | 0.46 ± 0.02   | 0.90 ± 0.02   | E Africa M    | E Asia M      | 0.17 |
| 26  | 0.90 ± 0.02   | 0.527 ± 0.008 | 0.71 ± 0.01   | 0.717 ± 0.009 | 0.48 ± 0.01  | 0.89 ± 0.01   | 0.93 ± 0.02   | 0.46 ± 0.01   | 0.82 ± 0.01   | 0.846 ± 0.008 | 0.51 ± 0.02   | 0.92 ± 0.01   | E Africa M    | E Asia M      | 0.15 |
| 27  | 0.92 ± 0.02   | 0.570 ± 0.008 | 0.765 ± 0.010 | 0.742 ± 0.007 | 0.55 ± 0.02  | 0.92 ± 0.01   | 0.93 ± 0.02   | 0.52 ± 0.01   | 0.85 ± 0.01   | 0.877 ± 0.007 | 0.57 ± 0.02   | 0.94 ± 0.01   | W Africa M    | E Asia M      | 0.13 |
| 28  | 0.94 ± 0.01   | 0.620 ± 0.007 | 0.805 ± 0.009 | 0.771 ± 0.007 | 0.59 ± 0.02  | 0.92 ± 0.01   | 0.95 ± 0.01   | 0.60 ± 0.01   | 0.89 ± 0.01   | 0.899 ± 0.006 | 0.65 ± 0.02   | 0.94 ± 0.01   | E Africa M    | S E Asia F    | 0.10 |
| 29  | 0.94 ± 0.01   | 0.669 ± 0.007 | 0.841 ± 0.009 | 0.796 ± 0.006 | 0.63 ± 0.02  | 0.93 ± 0.01   | 0.96 ± 0.01   | 0.65 ± 0.01   | 0.90 ± 0.01   | 0.914 ± 0.005 | 0.70 ± 0.02   | 0.94 ± 0.01   | E Africa M    | S E Asia F    | 0.09 |
| 30  | 0.93 ± 0.01   | 0.705 ± 0.006 | 0.870 ± 0.008 | 0.826 ± 0.006 | 0.69 ± 0.01  | 0.93 ± 0.01   | 0.95 ± 0.01   | 0.700 ± 0.009 | 0.931 ± 0.009 | 0.928 ± 0.004 | 0.76 ± 0.02   | 0.951 ± 0.009 | E Africa M    | S E Asia F    | 0.07 |
| 31  | 0.95 ± 0.01   | 0.739 ± 0.006 | 0.901 ± 0.007 | 0.855 ± 0.006 | 0.71 ± 0.01  | 0.947 ± 0.009 | 0.965 ± 0.010 | 0.756 ± 0.009 | 0.939 ± 0.007 | 0.939 ± 0.004 | 0.79 ± 0.02   | 0.952 ± 0.009 | E Africa M    | S E Asia F    | 0.06 |
| 32  | 0.95 ± 0.01   | 0.771 ± 0.006 | 0.919 ± 0.007 | 0.871 ± 0.006 | 0.74 ± 0.01  | 0.95 ± 0.01   | 0.968 ± 0.010 | 0.795 ± 0.008 | 0.952 ± 0.007 | 0.952 ± 0.003 | 0.82 ± 0.02   | 0.953 ± 0.008 | E Africa M    | S E Asia F    | 0.05 |
| 33  | 0.96 ± 0.01   | 0.804 ± 0.006 | 0.924 ± 0.006 | 0.890 ± 0.005 | 0.79 ± 0.01  | 0.95 ± 0.01   | 0.95 ± 0.01   | 0.829 ± 0.007 | 0.950 ± 0.007 | 0.954 ± 0.003 | 0.85 ± 0.02   | 0.966 ± 0.008 | W Africa M    | S E Asia F    | 0.04 |
| 34  | 0.96 ± 0.01   | 0.829 ± 0.006 | 0.936 ± 0.006 | 0.903 ± 0.005 | 0.81 ± 0.01  | 0.95 ± 0.01   | 0.976 ± 0.008 | 0.865 ± 0.007 | 0.960 ± 0.006 | 0.956 ± 0.003 | 0.88 ± 0.01   | 0.965 ± 0.008 | E Africa M    | S E Asia F    | 0.04 |

**Table 115.** For algorithm cvut-002 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.05 ± 0.02   | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.011 ± 0.009 | 0.06 ± 0.03   | 0.013 ± 0.010 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.005 ± 0.003 | 0.000 ± 0.000 | 0.009 ± 0.009 | W Africa F    | E Europe F    | 0.74 |
| 15  | 0.05 ± 0.02   | 0.007 ± 0.004 | 0.008 ± 0.008 | 0.021 ± 0.009 | 0.02 ± 0.01   | 0.07 ± 0.03   | 0.02 ± 0.01   | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.02 ± 0.01   | W Africa F    | E Europe M    | 0.63 |
| 16  | 0.12 ± 0.03   | 0.015 ± 0.006 | 0.02 ± 0.01   | 0.019 ± 0.009 | 0.02 ± 0.01   | 0.14 ± 0.04   | 0.04 ± 0.02   | 0.003 ± 0.002 | 0.009 ± 0.009 | 0.013 ± 0.007 | 0.003 ± 0.003 | 0.03 ± 0.02   | W Africa F    | E Asia M      | 0.62 |
| 17  | 0.17 ± 0.04   | 0.024 ± 0.007 | 0.03 ± 0.02   | 0.05 ± 0.01   | 0.04 ± 0.02   | 0.22 ± 0.04   | 0.05 ± 0.02   | 0.006 ± 0.003 | 0.008 ± 0.008 | 0.04 ± 0.01   | 0.01 ± 0.01   | 0.11 ± 0.03   | W Africa F    | E Asia M      | 0.55 |
| 18  | 0.19 ± 0.03   | 0.028 ± 0.007 | 0.05 ± 0.02   | 0.08 ± 0.01   | 0.04 ± 0.01   | 0.25 ± 0.04   | 0.09 ± 0.02   | 0.014 ± 0.004 | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.008 ± 0.006 | 0.15 ± 0.04   | W Africa F    | S E Asia M    | 0.52 |
| 19  | 0.27 ± 0.03   | 0.046 ± 0.009 | 0.10 ± 0.03   | 0.12 ± 0.02   | 0.08 ± 0.01   | 0.34 ± 0.04   | 0.18 ± 0.03   | 0.020 ± 0.006 | 0.07 ± 0.02   | 0.14 ± 0.02   | 0.03 ± 0.01   | 0.23 ± 0.04   | W Africa F    | E Asia M      | 0.43 |
| 20  | 0.34 ± 0.04   | 0.08 ± 0.01   | 0.15 ± 0.02   | 0.15 ± 0.02   | 0.10 ± 0.02   | 0.41 ± 0.04   | 0.29 ± 0.04   | 0.040 ± 0.008 | 0.15 ± 0.03   | 0.23 ± 0.02   | 0.06 ± 0.02   | 0.36 ± 0.04   | W Africa F    | E Asia M      | 0.38 |
| 21  | 0.41 ± 0.04   | 0.11 ± 0.01   | 0.18 ± 0.02   | 0.24 ± 0.02   | 0.13 ± 0.02   | 0.49 ± 0.04   | 0.42 ± 0.04   | 0.07 ± 0.01   | 0.26 ± 0.03   | 0.34 ± 0.02   | 0.10 ± 0.02   | 0.43 ± 0.04   | W Africa F    | E Asia M      | 0.33 |
| 22  | 0.49 ± 0.03   | 0.12 ± 0.01   | 0.25 ± 0.02   | 0.30 ± 0.02   | 0.15 ± 0.02   | 0.56 ± 0.03   | 0.52 ± 0.03   | 0.10 ± 0.01   | 0.38 ± 0.03   | 0.44 ± 0.02   | 0.14 ± 0.02   | 0.57 ± 0.04   | W Africa M    | E Asia M      | 0.32 |
| 23  | 0.55 ± 0.03   | 0.19 ± 0.01   | 0.34 ± 0.02   | 0.39 ± 0.02   | 0.21 ± 0.02   | 0.65 ± 0.03   | 0.61 ± 0.03   | 0.17 ± 0.01   | 0.47 ± 0.03   | 0.54 ± 0.02   | 0.21 ± 0.02   | 0.69 ± 0.03   | W Africa M    | E Asia M      | 0.27 |
| 24  | 0.62 ± 0.03   | 0.24 ± 0.01   | 0.41 ± 0.02   | 0.46 ± 0.02   | 0.26 ± 0.02   | 0.68 ± 0.03   | 0.70 ± 0.03   | 0.23 ± 0.02   | 0.60 ± 0.03   | 0.68 ± 0.02   | 0.29 ± 0.03   | 0.76 ± 0.03   | W Africa M    | E Asia M      | 0.24 |
| 25  | 0.62 ± 0.03   | 0.30 ± 0.01   | 0.50 ± 0.02   | 0.51 ± 0.02   | 0.32 ± 0.02   | 0.72 ± 0.03   | 0.77 ± 0.03   | 0.33 ± 0.02   | 0.73 ± 0.03   | 0.75 ± 0.02   | 0.40 ± 0.03   | 0.80 ± 0.02   | W Africa M    | E Asia F      | 0.20 |
| 26  | 0.69 ± 0.03   | 0.36 ± 0.01   | 0.58 ± 0.02   | 0.57 ± 0.01   | 0.40 ± 0.02   | 0.77 ± 0.02   | 0.84 ± 0.02   | 0.42 ± 0.02   | 0.78 ± 0.02   | 0.81 ± 0.01   | 0.48 ± 0.03   | 0.87 ± 0.02   | W Africa M    | E Asia F      | 0.17 |
| 27  | 0.73 ± 0.02   | 0.44 ± 0.01   | 0.67 ± 0.02   | 0.63 ± 0.01   | 0.46 ± 0.02   | 0.82 ± 0.02   | 0.87 ± 0.02   | 0.51 ± 0.02   | 0.86 ± 0.02   | 0.86 ± 0.01   | 0.59 ± 0.03   | 0.89 ± 0.02   | W Africa M    | E Asia F      | 0.14 |
| 28  | 0.78 ± 0.02   | 0.52 ± 0.01   | 0.75 ± 0.02   | 0.69 ± 0.01   | 0.55 ± 0.02   | 0.86 ± 0.02   | 0.91 ± 0.01   | 0.61 ± 0.02   | 0.93 ± 0.01   | 0.891 ± 0.009 | 0.67 ± 0.03   | 0.92 ± 0.01   | E Europe M    | E Asia F      | 0.12 |
| 29  | 0.81 ± 0.02   | 0.60 ± 0.01   | 0.82 ± 0.01   | 0.74 ± 0.01   | 0.61 ± 0.02   | 0.89 ± 0.02   | 0.93 ± 0.01   | 0.72 ± 0.02   | 0.95 ± 0.01   | 0.921 ± 0.007 | 0.72 ± 0.02   | 0.95 ± 0.01   | E Europe M    | E Asia F      | 0.09 |
| 30  | 0.86 ± 0.02   | 0.67 ± 0.01   | 0.87 ± 0.01   | 0.79 ± 0.01   | 0.69 ± 0.02   | 0.90 ± 0.02   | 0.95 ± 0.01   | 0.77 ± 0.01   | 0.973 ± 0.009 | 0.937 ± 0.007 | 0.80 ± 0.02   | 0.965 ± 0.009 | E Europe M    | E Asia F      | 0.07 |
| 31  | 0.89 ± 0.02   | 0.73 ± 0.01   | 0.92 ± 0.01   | 0.818 ± 0.010 | 0.73 ± 0.02   | 0.93 ± 0.01   | 0.96 ± 0.01   | 0.83 ± 0.01   | 0.979 ± 0.008 | 0.955 ± 0.005 | 0.84 ± 0.02   | 0.975 ± 0.007 | E Europe M    | S E Asia F    | 0.06 |
| 32  | 0.92 ± 0.01   | 0.78 ± 0.01   | 0.938 ± 0.010 | 0.861 ± 0.008 | 0.79 ± 0.02   | 0.94 ± 0.01   | 0.976 ± 0.009 | 0.87 ± 0.01   | 0.983 ± 0.007 | 0.965 ± 0.005 | 0.90 ± 0.02   | 0.978 ± 0.007 | E Europe M    | E Asia F      | 0.04 |
| 33  | 0.92 ± 0.01   | 0.843 ± 0.009 | 0.950 ± 0.009 | 0.894 ± 0.008 | 0.83 ± 0.02   | 0.967 ± 0.010 | 0.980 ± 0.008 | 0.91 ± 0.01   | 0.996 ± 0.004 | 0.975 ± 0.004 | 0.92 ± 0.02   | 0.990 ± 0.005 | E Europe M    | S E Asia F    | 0.04 |
| 34  | 0.94 ± 0.01   | 0.877 ± 0.009 | 0.969 ± 0.008 | 0.918 ± 0.008 | 0.88 ± 0.01   | 0.96 ± 0.01   | 0.988 ± 0.007 | 0.934 ± 0.009 | 0.996 ± 0.004 | 0.980 ± 0.004 | 0.93 ± 0.02   | 0.992 ± 0.005 | E Europe M    | E Asia F      | 0.03 |

**Table 116.** For algorithm cvut-002 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.21 ± 0.06   | 0.037 ± 0.008 | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.06 ± 0.02   | 0.25 ± 0.07   | 0.13 ± 0.05   | 0.010 ± 0.004 | 0.002 ± 0.002 | 0.017 ± 0.007 | 0.02 ± 0.01   | 0.13 ± 0.06   | W Africa F    | E Europe M    | 0.57 |
| 15  | 0.27 ± 0.06   | 0.056 ± 0.008 | 0.04 ± 0.02   | 0.12 ± 0.02   | 0.08 ± 0.02   | 0.41 ± 0.07   | 0.27 ± 0.07   | 0.024 ± 0.005 | 0.011 ± 0.009 | 0.05 ± 0.01   | 0.04 ± 0.01   | 0.24 ± 0.06   | W Africa F    | E Europe M    | 0.54 |
| 16  | 0.46 ± 0.06   | 0.10 ± 0.01   | 0.07 ± 0.02   | 0.17 ± 0.02   | 0.12 ± 0.02   | 0.45 ± 0.06   | 0.34 ± 0.06   | 0.047 ± 0.007 | 0.02 ± 0.01   | 0.10 ± 0.01   | 0.06 ± 0.02   | 0.27 ± 0.05   | E Africa F    | E Europe M    | 0.48 |
| 17  | 0.46 ± 0.06   | 0.15 ± 0.01   | 0.12 ± 0.02   | 0.25 ± 0.02   | 0.17 ± 0.02   | 0.51 ± 0.06   | 0.51 ± 0.07   | 0.084 ± 0.009 | 0.06 ± 0.02   | 0.23 ± 0.02   | 0.12 ± 0.02   | 0.46 ± 0.06   | E Africa M    | E Europe M    | 0.38 |
| 18  | 0.58 ± 0.05   | 0.22 ± 0.01   | 0.18 ± 0.02   | 0.31 ± 0.02   | 0.25 ± 0.02   | 0.53 ± 0.05   | 0.64 ± 0.06   | 0.144 ± 0.010 | 0.14 ± 0.03   | 0.34 ± 0.02   | 0.19 ± 0.02   | 0.57 ± 0.04   | E Africa M    | E Europe M    | 0.32 |
| 19  | 0.63 ± 0.04   | 0.28 ± 0.01   | 0.24 ± 0.02   | 0.38 ± 0.02   | 0.32 ± 0.02   | 0.64 ± 0.04   | 0.71 ± 0.05   | 0.21 ± 0.01   | 0.22 ± 0.03   | 0.49 ± 0.02   | 0.30 ± 0.02   | 0.67 ± 0.05   | E Africa M    | E Asia M      | 0.26 |
| 20  | 0.73 ± 0.04   | 0.35 ± 0.01   | 0.33 ± 0.02   | 0.46 ± 0.02   | 0.41 ± 0.02   | 0.74 ± 0.04   | 0.82 ± 0.04   | 0.30 ± 0.01   | 0.33 ± 0.03   | 0.62 ± 0.02   | 0.41 ± 0.03   | 0.77 ± 0.03   | E Africa M    | E Asia M      | 0.22 |
| 21  | 0.77 ± 0.03   | 0.41 ± 0.01   | 0.37 ± 0.02   | 0.53 ± 0.02   | 0.47 ± 0.02   | 0.79 ± 0.03   | 0.88 ± 0.03   | 0.41 ± 0.01   | 0.53 ± 0.03   | 0.74 ± 0.02   | 0.52 ± 0.03   | 0.87 ± 0.03   | E Africa M    | E Europe F    | 0.18 |
| 22  | 0.82 ± 0.03   | 0.485 ± 0.009 | 0.48 ± 0.02   | 0.61 ± 0.01   | 0.56 ± 0.02   | 0.83 ± 0.03   | 0.94 ± 0.02   | 0.51 ± 0.01   | 0.63 ± 0.02   | 0.83 ± 0.01   | 0.60 ± 0.03   | 0.90 ± 0.02   | E Africa M    | E Europe F    | 0.15 |
| 23  | 0.86 ± 0.03   | 0.535 ± 0.009 | 0.55 ± 0.02   | 0.66 ± 0.01   | 0.63 ± 0.02   | 0.86 ± 0.02   | 0.97 ± 0.01   | 0.60 ± 0.01   | 0.73 ± 0.02   | 0.87 ± 0.01   | 0.68 ± 0.02   | 0.93 ± 0.02   | E Africa M    | E Asia F      | 0.12 |
| 24  | 0.89 ± 0.02   | 0.607 ± 0.009 | 0.62 ± 0.01   | 0.72 ± 0.01   | 0.70 ± 0.02   | 0.91 ± 0.02   | 0.96 ± 0.01   | 0.69 ± 0.01   | 0.83 ± 0.02   | 0.918 ± 0.008 | 0.77 ± 0.02   | 0.95 ± 0.01   | E Africa M    | E Asia F      | 0.10 |
| 25  | 0.91 ± 0.02   | 0.672 ± 0.008 | 0.70 ± 0.01   | 0.750 ± 0.009 | 0.75 ± 0.01   | 0.93 ± 0.01   | 0.989 ± 0.007 | 0.772 ± 0.010 | 0.87 ± 0.01   | 0.943 ± 0.005 | 0.84 ± 0.02   | 0.977 ± 0.008 | E Africa M    | E Asia F      | 0.08 |
| 26  | 0.92 ± 0.02   | 0.722 ± 0.008 | 0.758 ± 0.010 | 0.792 ± 0.007 | 0.79 ± 0.01   | 0.92 ± 0.01   | 0.987 ± 0.008 | 0.828 ± 0.008 | 0.92 ± 0.01   | 0.960 ± 0.005 | 0.89 ± 0.01   | 0.980 ± 0.007 | E Africa M    | E Asia F      | 0.06 |
| 27  | 0.94 ± 0.01   | 0.765 ± 0.006 | 0.822 ± 0.010 | 0.822 ± 0.007 | 0.84 ± 0.01   | 0.960 ± 0.010 | 0.991 ± 0.007 | 0.871 ± 0.007 | 0.953 ± 0.008 | 0.972 ± 0.003 | 0.92 ± 0.01   | 0.990 ± 0.005 | E Africa M    | E Asia F      | 0.05 |
| 28  | 0.96 ± 0.01   | 0.814 ± 0.005 | 0.860 ± 0.008 | 0.858 ± 0.006 | 0.88 ± 0.01   | 0.964 ± 0.008 | 0.995 ± 0.005 | 0.912 ± 0.006 | 0.968 ± 0.006 | 0.983 ± 0.002 | 0.94 ± 0.01   | 0.992 ± 0.004 | E Africa M    | E Asia F      | 0.04 |
| 29  | 0.96 ± 0.01   | 0.854 ± 0.005 | 0.906 ± 0.007 | 0.882 ± 0.006 | 0.89 ± 0.01   | 0.973 ± 0.008 | 0.997 ± 0.004 | 0.942 ± 0.005 | 0.977 ± 0.005 | 0.988 ± 0.002 | 0.956 ± 0.010 | 0.993 ± 0.004 | E Africa M    | E Asia F      | 0.03 |
| 30  | 0.96 ± 0.01   | 0.888 ± 0.004 | 0.932 ± 0.006 | 0.905 ± 0.005 | 0.921 ± 0.009 | 0.970 ± 0.008 | 0.998 ± 0.003 | 0.952 ± 0.005 | 0.990 ± 0.003 | 0.992 ± 0.001 | 0.976 ± 0.007 | 0.997 ± 0.002 | E Africa M    | E Asia F      | 0.02 |
| 31  | 0.983 ± 0.008 | 0.919 ± 0.004 | 0.949 ± 0.005 | 0.933 ± 0.005 | 0.938 ± 0.007 | 0.980 ± 0.007 | 0.999 ± 0.002 | 0.972 ± 0.003 | 0.993 ± 0.003 | 0.996 ± 0.001 | 0.977 ± 0.007 | 0.997 ± 0.002 | E Africa M    | E Asia F      | 0.02 |
| 32  | 0.986 ± 0.006 | 0.938 ± 0.003 | 0.970 ± 0.004 | 0.942 ± 0.004 | 0.952 ± 0.006 | 0.989 ± 0.005 | 1.000 ± 0.000 | 0.983 ± 0.002 | 0.994 ± 0.002 | 0.995 ± 0.001 | 0.984 ± 0.005 | 1.000 ± 0.000 | E Africa M    | E Asia F      | 0.01 |
| 33  | 0.988 ± 0.007 | 0.954 ± 0.003 | 0.978 ± 0.003 | 0.959 ± 0.004 | 0.964 ± 0.006 | 0.992 ± 0.004 | 1.000 ± 0.000 | 0.987 ± 0.002 | 0.997 ± 0.002 | 0.998 ± 0.001 | 0.991 ± 0.004 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.01 |
| 34  | 0.996 ± 0.004 | 0.968 ± 0.002 | 0.983 ± 0.003 | 0.969 ± 0.003 | 0.972 ± 0.005 | 0.991 ± 0.005 | 0.999 ± 0.002 | 0.992 ± 0.002 | 0.999 ± 0.001 | 0.998 ± 0.001 | 0.994 ± 0.004 | 0.999 ± 0.002 | E Africa M    | E Asia F      | 0.01 |

**Table 117.** For algorithm cvut-003 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.03 ± 0.02   | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.005 ± 0.003 | 0.011 ± 0.009 | 0.07 ± 0.03   | 0.012 ± 0.008 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.009 ± 0.009 | W Africa F    | E Europe F    | 0.75 |
| 15  | 0.05 ± 0.02   | 0.005 ± 0.003 | 0.010 ± 0.008 | 0.015 ± 0.008 | 0.02 ± 0.01   | 0.09 ± 0.04   | 0.010 ± 0.008 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.01 ± 0.01   | W Africa F    | E Asia M      | 0.69 |
| 16  | 0.10 ± 0.03   | 0.011 ± 0.005 | 0.02 ± 0.01   | 0.011 ± 0.008 | 0.02 ± 0.01   | 0.15 ± 0.04   | 0.04 ± 0.02   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.02 ± 0.01   | W Africa F    | S E Asia M    | 0.70 |
| 17  | 0.15 ± 0.04   | 0.022 ± 0.007 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.02   | 0.20 ± 0.05   | 0.03 ± 0.02   | 0.004 ± 0.003 | 0.003 ± 0.003 | 0.029 ± 0.009 | 0.006 ± 0.006 | 0.08 ± 0.03   | W Africa F    | E Europe M    | 0.60 |
| 18  | 0.14 ± 0.03   | 0.026 ± 0.006 | 0.03 ± 0.01   | 0.053 ± 0.010 | 0.04 ± 0.01   | 0.25 ± 0.04   | 0.10 ± 0.02   | 0.011 ± 0.004 | 0.015 ± 0.009 | 0.030 ± 0.008 | 0.006 ± 0.004 | 0.12 ± 0.03   | W Africa F    | S E Asia M    | 0.56 |
| 19  | 0.23 ± 0.03   | 0.043 ± 0.008 | 0.06 ± 0.02   | 0.10 ± 0.01   | 0.08 ± 0.02   | 0.36 ± 0.04   | 0.13 ± 0.03   | 0.015 ± 0.005 | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.022 ± 0.010 | 0.18 ± 0.03   | W Africa F    | E Asia M      | 0.50 |
| 20  | 0.27 ± 0.04   | 0.069 ± 0.010 | 0.11 ± 0.02   | 0.12 ± 0.02   | 0.11 ± 0.02   | 0.43 ± 0.04   | 0.26 ± 0.04   | 0.029 ± 0.007 | 0.07 ± 0.02   | 0.14 ± 0.02   | 0.05 ± 0.01   | 0.29 ± 0.04   | W Africa F    | E Asia M      | 0.42 |
| 21  | 0.35 ± 0.04   | 0.11 ± 0.01   | 0.15 ± 0.02   | 0.14 ± 0.02   | 0.14 ± 0.02   | 0.47 ± 0.04   | 0.35 ± 0.04   | 0.049 ± 0.009 | 0.12 ± 0.03   | 0.21 ± 0.02   | 0.08 ± 0.02   | 0.37 ± 0.04   | W Africa F    | E Asia M      | 0.36 |
| 22  | 0.45 ± 0.03   | 0.13 ± 0.01   | 0.22 ± 0.02   | 0.25 ± 0.02   | 0.17 ± 0.02   | 0.56 ± 0.04   | 0.48 ± 0.03   | 0.10 ± 0.01   | 0.22 ± 0.03   | 0.29 ± 0.02   | 0.11 ± 0.02   | 0.49 ± 0.04   | W Africa F    | E Asia M      | 0.33 |
| 23  | 0.48 ± 0.03   | 0.21 ± 0.01   | 0.29 ± 0.02   | 0.33 ± 0.02   | 0.23 ± 0.02   | 0.65 ± 0.03   | 0.55 ± 0.03   | 0.16 ± 0.02   | 0.28 ± 0.03   | 0.42 ± 0.02   | 0.18 ± 0.03   | 0.60 ± 0.03   | W Africa F    | E Asia M      | 0.28 |
| 24  | 0.57 ± 0.03   | 0.26 ± 0.01   | 0.35 ± 0.02   | 0.40 ± 0.02   | 0.31 ± 0.02   | 0.68 ± 0.03   | 0.66 ± 0.03   | 0.23 ± 0.02   | 0.44 ± 0.03   | 0.53 ± 0.02   | 0.25 ± 0.03   | 0.70 ± 0.03   | W Africa M    | E Asia M      | 0.23 |
| 25  | 0.57 ± 0.03   | 0.32 ± 0.01   | 0.45 ± 0.02   | 0.46 ± 0.01   | 0.36 ± 0.02   | 0.73 ± 0.02   | 0.73 ± 0.03   | 0.33 ± 0.02   | 0.55 ± 0.03   | 0.63 ± 0.02   | 0.37 ± 0.03   | 0.74 ± 0.03   | W Africa M    | E Asia F      | 0.18 |
| 26  | 0.64 ± 0.03   | 0.39 ± 0.01   | 0.51 ± 0.02   | 0.52 ± 0.01   | 0.46 ± 0.02   | 0.78 ± 0.02   | 0.80 ± 0.02   | 0.43 ± 0.02   | 0.63 ± 0.03   | 0.70 ± 0.02   | 0.45 ± 0.03   | 0.83 ± 0.02   | W Africa M    | E Asia F      | 0.15 |
| 27  | 0.69 ± 0.03   | 0.48 ± 0.01   | 0.62 ± 0.02   | 0.58 ± 0.01   | 0.53 ± 0.02   | 0.82 ± 0.02   | 0.83 ± 0.02   | 0.53 ± 0.02   | 0.77 ± 0.02   | 0.77 ± 0.01   | 0.58 ± 0.03   | 0.87 ± 0.02   | W Africa M    | E Asia F      | 0.12 |
| 28  | 0.74 ± 0.02   | 0.56 ± 0.01   | 0.70 ± 0.02   | 0.64 ± 0.01   | 0.60 ± 0.02   | 0.86 ± 0.02   | 0.89 ± 0.02   | 0.64 ± 0.02   | 0.84 ± 0.02   | 0.83 ± 0.01   | 0.67 ± 0.02   | 0.90 ± 0.02   | W Africa M    | E Asia F      | 0.10 |
| 29  | 0.78 ± 0.02   | 0.65 ± 0.01   | 0.78 ± 0.02   | 0.71 ± 0.01   | 0.67 ± 0.02   | 0.89 ± 0.02   | 0.91 ± 0.01   | 0.75 ± 0.02   | 0.89 ± 0.02   | 0.874 ± 0.009 | 0.70 ± 0.03   | 0.94 ± 0.01   | W Africa M    | E Asia F      | 0.07 |
| 30  | 0.82 ± 0.02   | 0.72 ± 0.01   | 0.83 ± 0.01   | 0.76 ± 0.01   | 0.74 ± 0.02   | 0.91 ± 0.02   | 0.94 ± 0.01   | 0.81 ± 0.01   | 0.94 ± 0.01   | 0.901 ± 0.008 | 0.80 ± 0.02   | 0.957 ± 0.009 | W Africa M    | E Asia F      | 0.06 |
| 31  | 0.88 ± 0.02   | 0.78 ± 0.01   | 0.90 ± 0.01   | 0.794 ± 0.010 | 0.79 ± 0.02   | 0.93 ± 0.01   | 0.95 ± 0.01   | 0.87 ± 0.01   | 0.966 ± 0.009 | 0.928 ± 0.007 | 0.85 ± 0.02   | 0.974 ± 0.008 | W Africa M    | E Asia F      | 0.05 |
| 32  | 0.91 ± 0.02   | 0.83 ± 0.01   | 0.93 ± 0.01   | 0.839 ± 0.010 | 0.83 ± 0.02   | 0.95 ± 0.01   | 0.976 ± 0.008 | 0.903 ± 0.009 | 0.973 ± 0.009 | 0.950 ± 0.005 | 0.90 ± 0.02   | 0.976 ± 0.008 | W Africa M    | E Asia F      | 0.04 |
| 33  | 0.92 ± 0.02   | 0.880 ± 0.009 | 0.946 ± 0.009 | 0.875 ± 0.010 | 0.87 ± 0.02   | 0.970 ± 0.010 | 0.983 ± 0.007 | 0.939 ± 0.008 | 0.988 ± 0.006 | 0.968 ± 0.005 | 0.92 ± 0.02   | 0.985 ± 0.006 | E Europe M    | S E Asia F    | 0.03 |
| 34  | 0.93 ± 0.02   | 0.913 ± 0.007 | 0.962 ± 0.010 | 0.901 ± 0.008 | 0.92 ± 0.01   | 0.970 ± 0.009 | 0.988 ± 0.007 | 0.963 ± 0.007 | 0.992 ± 0.005 | 0.974 ± 0.004 | 0.95 ± 0.01   | 0.992 ± 0.004 | W Africa M    | S Asia F      | 0.02 |

**Table 118.** For algorithm cvut-003 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.25 ± 0.06   | 0.020 ± 0.006 | 0.013 ± 0.009 | 0.07 ± 0.01   | 0.05 ± 0.02   | 0.33 ± 0.08   | 0.12 ± 0.05   | 0.005 ± 0.003 | 0.000 ± 0.000 | 0.011 ± 0.005 | 0.01 ± 0.01   | 0.10 ± 0.04   | W Africa F    | E Europe M    | 0.67 |
| 15  | 0.27 ± 0.06   | 0.032 ± 0.006 | 0.03 ± 0.02   | 0.12 ± 0.02   | 0.05 ± 0.02   | 0.47 ± 0.07   | 0.21 ± 0.06   | 0.008 ± 0.003 | 0.009 ± 0.007 | 0.032 ± 0.009 | 0.012 ± 0.009 | 0.19 ± 0.06   | W Africa F    | E Asia M      | 0.62 |
| 16  | 0.45 ± 0.07   | 0.061 ± 0.008 | 0.06 ± 0.02   | 0.18 ± 0.02   | 0.10 ± 0.02   | 0.43 ± 0.06   | 0.30 ± 0.07   | 0.016 ± 0.004 | 0.005 ± 0.005 | 0.06 ± 0.01   | 0.04 ± 0.01   | 0.22 ± 0.05   | E Africa F    | E Europe M    | 0.55 |
| 17  | 0.45 ± 0.05   | 0.112 ± 0.010 | 0.10 ± 0.02   | 0.27 ± 0.02   | 0.14 ± 0.02   | 0.51 ± 0.06   | 0.41 ± 0.07   | 0.033 ± 0.006 | 0.04 ± 0.02   | 0.17 ± 0.02   | 0.07 ± 0.02   | 0.40 ± 0.06   | W Africa F    | E Asia M      | 0.45 |
| 18  | 0.55 ± 0.05   | 0.164 ± 0.010 | 0.17 ± 0.02   | 0.34 ± 0.02   | 0.22 ± 0.02   | 0.56 ± 0.05   | 0.55 ± 0.06   | 0.070 ± 0.007 | 0.09 ± 0.02   | 0.25 ± 0.02   | 0.12 ± 0.02   | 0.50 ± 0.05   | W Africa F    | E Asia M      | 0.38 |
| 19  | 0.61 ± 0.05   | 0.227 ± 0.010 | 0.24 ± 0.02   | 0.42 ± 0.02   | 0.29 ± 0.02   | 0.67 ± 0.04   | 0.65 ± 0.05   | 0.106 ± 0.008 | 0.14 ± 0.02   | 0.39 ± 0.02   | 0.20 ± 0.02   | 0.60 ± 0.05   | W Africa F    | E Asia M      | 0.32 |
| 20  | 0.74 ± 0.04   | 0.30 ± 0.01   | 0.32 ± 0.02   | 0.50 ± 0.02   | 0.39 ± 0.02   | 0.75 ± 0.03   | 0.77 ± 0.04   | 0.171 ± 0.009 | 0.22 ± 0.02   | 0.52 ± 0.02   | 0.28 ± 0.03   | 0.71 ± 0.04   | E Africa M    | E Asia M      | 0.28 |
| 21  | 0.77 ± 0.03   | 0.37 ± 0.01   | 0.38 ± 0.02   | 0.57 ± 0.02   | 0.46 ± 0.02   | 0.79 ± 0.03   | 0.81 ± 0.03   | 0.27 ± 0.01   | 0.38 ± 0.03   | 0.65 ± 0.02   | 0.39 ± 0.03   | 0.81 ± 0.03   | E Africa M    | E Asia M      | 0.21 |
| 22  | 0.82 ± 0.03   | 0.450 ± 0.010 | 0.47 ± 0.02   | 0.65 ± 0.01   | 0.54 ± 0.02   | 0.85 ± 0.02   | 0.91 ± 0.02   | 0.37 ± 0.01   | 0.49 ± 0.03   | 0.76 ± 0.01   | 0.49 ± 0.03   | 0.85 ± 0.03   | E Africa M    | E Asia M      | 0.18 |
| 23  | 0.86 ± 0.02   | 0.511 ± 0.009 | 0.55 ± 0.01   | 0.69 ± 0.01   | 0.64 ± 0.02   | 0.88 ± 0.02   | 0.93 ± 0.02   | 0.47 ± 0.01   | 0.63 ± 0.02   | 0.82 ± 0.01   | 0.58 ± 0.02   | 0.90 ± 0.02   | E Africa M    | E Asia M      | 0.14 |
| 24  | 0.89 ± 0.02   | 0.587 ± 0.009 | 0.64 ± 0.01   | 0.753 ± 0.010 | 0.71 ± 0.02   | 0.91 ± 0.01   | 0.95 ± 0.02   | 0.58 ± 0.01   | 0.74 ± 0.02   | 0.878 ± 0.009 | 0.68 ± 0.02   | 0.93 ± 0.01   | E Africa M    | E Asia M      | 0.10 |
| 25  | 0.92 ± 0.02   | 0.663 ± 0.008 | 0.72 ± 0.01   | 0.786 ± 0.008 | 0.77 ± 0.01   | 0.93 ± 0.01   | 0.98 ± 0.01   | 0.67 ± 0.01   | 0.80 ± 0.02   | 0.918 ± 0.007 | 0.78 ± 0.02   | 0.96 ± 0.01   | E Africa M    | E Asia F      | 0.08 |
| 26  | 0.92 ± 0.02   | 0.718 ± 0.007 | 0.77 ± 0.01   | 0.826 ± 0.007 | 0.81 ± 0.01   | 0.93 ± 0.01   | 0.984 ± 0.009 | 0.756 ± 0.010 | 0.87 ± 0.01   | 0.943 ± 0.005 | 0.84 ± 0.02   | 0.977 ± 0.007 | E Africa M    | E Asia F      | 0.06 |
| 27  | 0.94 ± 0.01   | 0.765 ± 0.006 | 0.837 ± 0.009 | 0.849 ± 0.006 | 0.85 ± 0.01   | 0.967 ± 0.009 | 0.988 ± 0.007 | 0.813 ± 0.009 | 0.91 ± 0.01   | 0.958 ± 0.004 | 0.88 ± 0.02   | 0.983 ± 0.006 | E Africa M    | E Asia F      | 0.05 |
| 28  | 0.96 ± 0.01   | 0.820 ± 0.006 | 0.872 ± 0.008 | 0.885 ± 0.006 | 0.894 ± 0.010 | 0.968 ± 0.009 | 0.991 ± 0.006 | 0.868 ± 0.007 | 0.950 ± 0.009 | 0.975 ± 0.003 | 0.92 ± 0.01   | 0.990 ± 0.005 | E Africa M    | E Asia F      | 0.04 |
| 29  | 0.96 ± 0.01   | 0.860 ± 0.005 | 0.915 ± 0.007 | 0.903 ± 0.005 | 0.908 ± 0.009 | 0.976 ± 0.007 | 0.996 ± 0.004 | 0.911 ± 0.006 | 0.963 ± 0.006 | 0.984 ± 0.002 | 0.94 ± 0.01   | 0.991 ± 0.004 | E Africa M    | E Asia F      | 0.03 |
| 30  | 0.969 ± 0.009 | 0.895 ± 0.004 | 0.940 ± 0.006 | 0.921 ± 0.004 | 0.938 ± 0.008 | 0.976 ± 0.007 | 0.997 ± 0.003 | 0.937 ± 0.005 | 0.983 ± 0.004 | 0.987 ± 0.002 | 0.963 ± 0.009 | 0.995 ± 0.003 | E Africa M    | E Asia F      | 0.02 |
| 31  | 0.985 ± 0.007 | 0.925 ± 0.004 | 0.955 ± 0.005 | 0.948 ± 0.004 | 0.950 ± 0.007 | 0.985 ± 0.006 | 0.999 ± 0.002 | 0.959 ± 0.004 | 0.990 ± 0.003 | 0.993 ± 0.001 | 0.970 ± 0.008 | 0.996 ± 0.002 | E Africa M    | E Asia F      | 0.01 |
| 32  | 0.984 ± 0.007 | 0.944 ± 0.003 | 0.976 ± 0.004 | 0.954 ± 0.004 | 0.960 ± 0.006 | 0.991 ± 0.005 | 1.000 ± 0.000 | 0.974 ± 0.003 | 0.993 ± 0.003 | 0.994 ± 0.001 | 0.981 ± 0.007 | 0.999 ± 0.002 | E Africa M    | E Asia F      | 0.01 |
| 33  | 0.992 ± 0.005 | 0.960 ± 0.003 | 0.982 ± 0.003 | 0.968 ± 0.003 | 0.968 ± 0.005 | 0.994 ± 0.004 | 1.000 ± 0.000 | 0.983 ± 0.003 | 0.996 ± 0.002 | 0.996 ± 0.001 | 0.988 ± 0.005 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.01 |
| 34  | 0.995 ± 0.004 | 0.973 ± 0.002 | 0.988 ± 0.003 | 0.976 ± 0.003 | 0.977 ± 0.005 | 0.993 ± 0.005 | 0.999 ± 0.002 | 0.988 ± 0.002 | 0.998 ± 0.001 | 0.997 ± 0.001 | 0.993 ± 0.004 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.01 |



**Table 119.** For algorithm cyberlink-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.10 ± 0.03   | 0.026 ± 0.008 | 0.002 ± 0.002 | 0.023 ± 0.008 | 0.06 ± 0.02   | 0.11 ± 0.04   | 0.06 ± 0.02   | 0.013 ± 0.006 | 0.002 ± 0.002 | 0.021 ± 0.007 | 0.03 ± 0.02   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.49 |
| 15  | 0.10 ± 0.03   | 0.06 ± 0.01   | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.10 ± 0.03   | 0.15 ± 0.04   | 0.07 ± 0.02   | 0.022 ± 0.007 | 0.009 ± 0.009 | 0.028 ± 0.010 | 0.05 ± 0.02   | 0.06 ± 0.03   | W Africa F    | E Europe M    | 0.40 |
| 16  | 0.18 ± 0.04   | 0.10 ± 0.02   | 0.05 ± 0.02   | 0.08 ± 0.02   | 0.10 ± 0.03   | 0.22 ± 0.05   | 0.14 ± 0.04   | 0.033 ± 0.009 | 0.01 ± 0.01   | 0.05 ± 0.02   | 0.08 ± 0.03   | 0.09 ± 0.04   | W Africa F    | E Europe M    | 0.36 |
| 17  | 0.28 ± 0.04   | 0.14 ± 0.02   | 0.06 ± 0.02   | 0.12 ± 0.02   | 0.17 ± 0.04   | 0.28 ± 0.05   | 0.20 ± 0.04   | 0.09 ± 0.01   | 0.04 ± 0.02   | 0.12 ± 0.02   | 0.11 ± 0.03   | 0.19 ± 0.05   | W Africa F    | E Europe M    | 0.30 |
| 18  | 0.26 ± 0.04   | 0.20 ± 0.02   | 0.11 ± 0.03   | 0.15 ± 0.02   | 0.22 ± 0.03   | 0.32 ± 0.04   | 0.31 ± 0.04   | 0.13 ± 0.01   | 0.06 ± 0.02   | 0.18 ± 0.02   | 0.21 ± 0.03   | 0.24 ± 0.04   | W Africa F    | E Europe M    | 0.23 |
| 19  | 0.34 ± 0.04   | 0.26 ± 0.02   | 0.14 ± 0.03   | 0.19 ± 0.02   | 0.30 ± 0.03   | 0.44 ± 0.04   | 0.39 ± 0.04   | 0.22 ± 0.02   | 0.12 ± 0.03   | 0.30 ± 0.02   | 0.29 ± 0.03   | 0.38 ± 0.05   | W Africa F    | E Europe M    | 0.21 |
| 20  | 0.44 ± 0.04   | 0.34 ± 0.02   | 0.19 ± 0.03   | 0.25 ± 0.02   | 0.38 ± 0.03   | 0.48 ± 0.04   | 0.52 ± 0.04   | 0.35 ± 0.02   | 0.19 ± 0.03   | 0.43 ± 0.02   | 0.43 ± 0.04   | 0.49 ± 0.05   | E Africa M    | E Europe F    | 0.18 |
| 21  | 0.51 ± 0.04   | 0.42 ± 0.02   | 0.27 ± 0.03   | 0.29 ± 0.02   | 0.43 ± 0.03   | 0.52 ± 0.04   | 0.65 ± 0.04   | 0.48 ± 0.02   | 0.29 ± 0.03   | 0.54 ± 0.02   | 0.54 ± 0.03   | 0.59 ± 0.04   | E Africa M    | E Europe F    | 0.16 |
| 22  | 0.57 ± 0.03   | 0.50 ± 0.02   | 0.34 ± 0.02   | 0.34 ± 0.02   | 0.53 ± 0.03   | 0.61 ± 0.03   | 0.76 ± 0.03   | 0.59 ± 0.02   | 0.47 ± 0.04   | 0.66 ± 0.02   | 0.64 ± 0.03   | 0.69 ± 0.04   | E Africa M    | S Asia F      | 0.13 |
| 23  | 0.60 ± 0.03   | 0.60 ± 0.02   | 0.43 ± 0.02   | 0.40 ± 0.02   | 0.59 ± 0.02   | 0.66 ± 0.03   | 0.79 ± 0.02   | 0.70 ± 0.02   | 0.57 ± 0.04   | 0.76 ± 0.02   | 0.72 ± 0.03   | 0.77 ± 0.03   | E Africa M    | S Asia F      | 0.12 |
| 24  | 0.66 ± 0.03   | 0.66 ± 0.02   | 0.53 ± 0.02   | 0.46 ± 0.01   | 0.69 ± 0.02   | 0.70 ± 0.03   | 0.89 ± 0.02   | 0.79 ± 0.02   | 0.71 ± 0.03   | 0.83 ± 0.01   | 0.81 ± 0.02   | 0.83 ± 0.02   | E Africa M    | S Asia F      | 0.10 |
| 25  | 0.70 ± 0.02   | 0.73 ± 0.01   | 0.59 ± 0.02   | 0.51 ± 0.01   | 0.74 ± 0.02   | 0.75 ± 0.03   | 0.91 ± 0.02   | 0.85 ± 0.01   | 0.81 ± 0.02   | 0.89 ± 0.01   | 0.87 ± 0.02   | 0.90 ± 0.02   | E Africa M    | S Asia F      | 0.10 |
| 26  | 0.75 ± 0.02   | 0.78 ± 0.01   | 0.68 ± 0.02   | 0.54 ± 0.01   | 0.79 ± 0.01   | 0.79 ± 0.02   | 0.94 ± 0.01   | 0.91 ± 0.01   | 0.86 ± 0.02   | 0.924 ± 0.009 | 0.91 ± 0.02   | 0.94 ± 0.01   | E Africa M    | S Asia F      | 0.08 |
| 27  | 0.76 ± 0.02   | 0.82 ± 0.01   | 0.76 ± 0.02   | 0.60 ± 0.01   | 0.83 ± 0.01   | 0.83 ± 0.02   | 0.97 ± 0.01   | 0.948 ± 0.008 | 0.92 ± 0.01   | 0.946 ± 0.007 | 0.95 ± 0.01   | 0.96 ± 0.01   | E Africa M    | S Asia F      | 0.07 |
| 28  | 0.81 ± 0.02   | 0.869 ± 0.008 | 0.81 ± 0.01   | 0.66 ± 0.01   | 0.87 ± 0.01   | 0.87 ± 0.02   | 0.971 ± 0.010 | 0.966 ± 0.006 | 0.95 ± 0.01   | 0.965 ± 0.005 | 0.972 ± 0.009 | 0.96 ± 0.01   | S E Asia M    | S Asia F      | 0.06 |
| 29  | 0.86 ± 0.02   | 0.911 ± 0.008 | 0.88 ± 0.01   | 0.72 ± 0.01   | 0.90 ± 0.01   | 0.89 ± 0.02   | 0.987 ± 0.006 | 0.982 ± 0.005 | 0.971 ± 0.010 | 0.978 ± 0.004 | 0.97 ± 0.01   | 0.985 ± 0.006 | E Africa M    | S Asia F      | 0.05 |
| 30  | 0.88 ± 0.02   | 0.930 ± 0.006 | 0.91 ± 0.01   | 0.76 ± 0.01   | 0.932 ± 0.010 | 0.90 ± 0.02   | 0.989 ± 0.006 | 0.990 ± 0.004 | 0.983 ± 0.007 | 0.987 ± 0.003 | 0.988 ± 0.007 | 0.984 ± 0.006 | E Asia M      | S Asia F      | 0.04 |
| 31  | 0.91 ± 0.02   | 0.955 ± 0.006 | 0.939 ± 0.010 | 0.81 ± 0.01   | 0.951 ± 0.009 | 0.92 ± 0.01   | 0.993 ± 0.004 | 0.993 ± 0.003 | 0.988 ± 0.006 | 0.993 ± 0.003 | 0.990 ± 0.007 | 0.992 ± 0.005 | E Africa M    | S Asia F      | 0.03 |
| 32  | 0.93 ± 0.01   | 0.964 ± 0.005 | 0.958 ± 0.008 | 0.851 ± 0.009 | 0.957 ± 0.008 | 0.94 ± 0.01   | 0.998 ± 0.003 | 0.997 ± 0.002 | 0.996 ± 0.004 | 0.994 ± 0.002 | 0.991 ± 0.005 | 0.994 ± 0.004 | E Africa M    | S Asia F      | 0.02 |
| 33  | 0.95 ± 0.01   | 0.976 ± 0.004 | 0.969 ± 0.008 | 0.879 ± 0.009 | 0.978 ± 0.006 | 0.95 ± 0.01   | 0.999 ± 0.002 | 0.998 ± 0.002 | 0.999 ± 0.002 | 0.997 ± 0.001 | 0.997 ± 0.004 | 0.997 ± 0.003 | E Africa M    | S Asia F      | 0.02 |
| 34  | 0.96 ± 0.01   | 0.985 ± 0.003 | 0.976 ± 0.007 | 0.908 ± 0.008 | 0.984 ± 0.006 | 0.96 ± 0.01   | 0.999 ± 0.002 | 0.998 ± 0.001 | 0.997 ± 0.003 | 0.998 ± 0.001 | 1.000 ± 0.000 | 0.996 ± 0.003 | S E Asia M    | S Asia F      | 0.01 |

**Table 120.** For algorithm cyberlink-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.36 ± 0.07   | 0.033 ± 0.008 | 0.03 ± 0.02   | 0.09 ± 0.02   | 0.05 ± 0.02   | 0.29 ± 0.07   | 0.36 ± 0.08   | 0.017 ± 0.005 | 0.02 ± 0.01   | 0.034 ± 0.009 | 0.04 ± 0.02   | 0.23 ± 0.06   | E Africa M    | E Asia M      | 0.57 |
| 15  | 0.34 ± 0.07   | 0.042 ± 0.007 | 0.06 ± 0.02   | 0.12 ± 0.02   | 0.05 ± 0.02   | 0.31 ± 0.06   | 0.46 ± 0.07   | 0.027 ± 0.005 | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.03 ± 0.01   | 0.26 ± 0.06   | E Africa M    | E Europe M    | 0.55 |
| 16  | 0.43 ± 0.06   | 0.064 ± 0.009 | 0.04 ± 0.01   | 0.14 ± 0.02   | 0.07 ± 0.02   | 0.34 ± 0.07   | 0.49 ± 0.07   | 0.040 ± 0.007 | 0.07 ± 0.02   | 0.11 ± 0.02   | 0.06 ± 0.02   | 0.30 ± 0.06   | E Africa M    | E Asia M      | 0.50 |
| 17  | 0.47 ± 0.05   | 0.10 ± 0.01   | 0.10 ± 0.02   | 0.17 ± 0.02   | 0.10 ± 0.02   | 0.40 ± 0.06   | 0.48 ± 0.07   | 0.067 ± 0.007 | 0.08 ± 0.02   | 0.18 ± 0.02   | 0.09 ± 0.02   | 0.44 ± 0.05   | E Africa M    | E Asia M      | 0.42 |
| 18  | 0.46 ± 0.05   | 0.125 ± 0.009 | 0.12 ± 0.02   | 0.22 ± 0.02   | 0.14 ± 0.02   | 0.45 ± 0.05   | 0.56 ± 0.06   | 0.100 ± 0.008 | 0.19 ± 0.03   | 0.27 ± 0.02   | 0.14 ± 0.02   | 0.48 ± 0.05   | E Africa M    | E Asia M      | 0.35 |
| 19  | 0.47 ± 0.05   | 0.164 ± 0.009 | 0.17 ± 0.02   | 0.25 ± 0.02   | 0.18 ± 0.02   | 0.48 ± 0.05   | 0.66 ± 0.05   | 0.146 ± 0.009 | 0.23 ± 0.03   | 0.36 ± 0.02   | 0.21 ± 0.02   | 0.55 ± 0.04   | E Africa M    | E Asia M      | 0.31 |
| 20  | 0.57 ± 0.04   | 0.206 ± 0.010 | 0.22 ± 0.02   | 0.27 ± 0.01   | 0.23 ± 0.02   | 0.54 ± 0.04   | 0.74 ± 0.04   | 0.21 ± 0.01   | 0.30 ± 0.03   | 0.49 ± 0.02   | 0.28 ± 0.02   | 0.62 ± 0.04   | E Africa M    | E Asia F      | 0.28 |
| 21  | 0.59 ± 0.04   | 0.264 ± 0.009 | 0.28 ± 0.02   | 0.31 ± 0.01   | 0.29 ± 0.02   | 0.58 ± 0.04   | 0.78 ± 0.04   | 0.30 ± 0.01   | 0.44 ± 0.03   | 0.58 ± 0.02   | 0.35 ± 0.02   | 0.72 ± 0.03   | E Africa M    | E Asia F      | 0.23 |
| 22  | 0.65 ± 0.04   | 0.313 ± 0.009 | 0.34 ± 0.02   | 0.36 ± 0.01   | 0.35 ± 0.02   | 0.68 ± 0.03   | 0.87 ± 0.03   | 0.38 ± 0.01   | 0.54 ± 0.03   | 0.67 ± 0.01   | 0.45 ± 0.03   | 0.76 ± 0.03   | E Africa M    | E Asia F      | 0.21 |
| 23  | 0.66 ± 0.04   | 0.368 ± 0.008 | 0.42 ± 0.02   | 0.39 ± 0.01   | 0.43 ± 0.02   | 0.71 ± 0.03   | 0.88 ± 0.03   | 0.48 ± 0.01   | 0.65 ± 0.02   | 0.75 ± 0.01   | 0.55 ± 0.03   | 0.84 ± 0.03   | E Africa M    | E Asia F      | 0.18 |
| 24  | 0.72 ± 0.03   | 0.439 ± 0.009 | 0.49 ± 0.01   | 0.46 ± 0.01   | 0.49 ± 0.02   | 0.73 ± 0.03   | 0.91 ± 0.02   | 0.57 ± 0.01   | 0.76 ± 0.02   | 0.831 ± 0.010 | 0.64 ± 0.03   | 0.85 ± 0.02   | E Africa M    | E Asia F      | 0.15 |
| 25  | 0.74 ± 0.03   | 0.513 ± 0.009 | 0.57 ± 0.01   | 0.49 ± 0.01   | 0.56 ± 0.02   | 0.78 ± 0.02   | 0.94 ± 0.02   | 0.65 ± 0.01   | 0.80 ± 0.02   | 0.879 ± 0.008 | 0.71 ± 0.02   | 0.89 ± 0.02   | E Africa M    | S Asia F      | 0.13 |
| 26  | 0.80 ± 0.02   | 0.575 ± 0.008 | 0.63 ± 0.01   | 0.538 ± 0.009 | 0.61 ± 0.02   | 0.79 ± 0.02   | 0.95 ± 0.01   | 0.72 ± 0.01   | 0.86 ± 0.01   | 0.899 ± 0.007 | 0.76 ± 0.02   | 0.92 ± 0.01   | E Africa M    | S Asia F      | 0.11 |
| 27  | 0.79 ± 0.02   | 0.628 ± 0.008 | 0.71 ± 0.01   | 0.578 ± 0.009 | 0.68 ± 0.01   | 0.83 ± 0.02   | 0.96 ± 0.01   | 0.775 ± 0.009 | 0.91 ± 0.01   | 0.930 ± 0.005 | 0.82 ± 0.02   | 0.94 ± 0.01   | E Africa M    | S Asia F      | 0.10 |
| 28  | 0.84 ± 0.02   | 0.685 ± 0.007 | 0.767 ± 0.010 | 0.627 ± 0.008 | 0.73 ± 0.01   | 0.84 ± 0.01   | 0.97 ± 0.01   | 0.834 ± 0.008 | 0.942 ± 0.008 | 0.949 ± 0.004 | 0.87 ± 0.02   | 0.958 ± 0.009 | E Africa M    | S Asia F      | 0.08 |
| 29  | 0.87 ± 0.02   | 0.736 ± 0.006 | 0.821 ± 0.009 | 0.668 ± 0.008 | 0.76 ± 0.01   | 0.88 ± 0.01   | 0.985 ± 0.007 | 0.873 ± 0.007 | 0.962 ± 0.006 | 0.963 ± 0.003 | 0.91 ± 0.01   | 0.965 ± 0.008 | E Africa M    | S Asia F      | 0.07 |
| 30  | 0.89 ± 0.02   | 0.787 ± 0.006 | 0.860 ± 0.008 | 0.709 ± 0.007 | 0.81 ± 0.01   | 0.90 ± 0.01   | 0.989 ± 0.007 | 0.911 ± 0.006 | 0.975 ± 0.005 | 0.973 ± 0.003 | 0.94 ± 0.01   | 0.973 ± 0.007 | E Africa M    | S Asia F      | 0.06 |
| 31  | 0.90 ± 0.02   | 0.825 ± 0.005 | 0.899 ± 0.007 | 0.758 ± 0.007 | 0.85 ± 0.01   | 0.91 ± 0.01   | 0.993 ± 0.005 | 0.932 ± 0.005 | 0.985 ± 0.004 | 0.982 ± 0.002 | 0.95 ± 0.01   | 0.982 ± 0.005 | E Africa M    | S Asia F      | 0.05 |
| 32  | 0.94 ± 0.01   | 0.861 ± 0.005 | 0.926 ± 0.007 | 0.799 ± 0.007 | 0.88 ± 0.01   | 0.93 ± 0.01   | 0.993 ± 0.005 | 0.955 ± 0.004 | 0.992 ± 0.003 | 0.987 ± 0.002 | 0.958 ± 0.009 | 0.981 ± 0.005 | E Africa M    | S Asia F      | 0.04 |
| 33  | 0.93 ± 0.02   | 0.889 ± 0.005 | 0.940 ± 0.006 | 0.833 ± 0.006 | 0.899 ± 0.010 | 0.94 ± 0.01   | 0.993 ± 0.005 | 0.964 ± 0.004 | 0.991 ± 0.003 | 0.990 ± 0.001 | 0.970 ± 0.008 | 0.991 ± 0.004 | E Africa M    | S Asia F      | 0.03 |
| 34  | 0.94 ± 0.01   | 0.910 ± 0.004 | 0.955 ± 0.006 | 0.867 ± 0.006 | 0.924 ± 0.009 | 0.95 ± 0.01   | 0.998 ± 0.003 | 0.975 ± 0.003 | 0.996 ± 0.002 | 0.993 ± 0.001 | 0.975 ± 0.007 | 0.994 ± 0.003 | E Africa M    | S Asia F      | 0.02 |

**Table 121.** For algorithm cyberlink-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.02 ± 0.01   | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.002 ± 0.002 | 0.03 ± 0.02   | 0.010 ± 0.008 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.005 ± 0.005 | 0.000 ± 0.000 | W Africa F    | E Europe F    | 0.72 |
| 15  | 0.02 ± 0.01   | 0.005 ± 0.003 | 0.010 ± 0.008 | 0.015 ± 0.008 | 0.02 ± 0.01   | 0.04 ± 0.02   | 0.02 ± 0.01   | 0.003 ± 0.002 | 0.006 ± 0.006 | 0.005 ± 0.004 | 0.002 ± 0.002 | 0.01 ± 0.01   | W Africa F    | S E Asia M    | 0.46 |
| 16  | 0.05 ± 0.02   | 0.021 ± 0.007 | 0.02 ± 0.01   | 0.02 ± 0.01   | 0.03 ± 0.02   | 0.08 ± 0.03   | 0.04 ± 0.02   | 0.010 ± 0.005 | 0.009 ± 0.009 | 0.015 ± 0.008 | 0.011 ± 0.008 | 0.02 ± 0.02   | W Africa F    | E Europe M    | 0.39 |
| 17  | 0.10 ± 0.03   | 0.034 ± 0.009 | 0.03 ± 0.02   | 0.03 ± 0.01   | 0.05 ± 0.02   | 0.12 ± 0.03   | 0.04 ± 0.02   | 0.020 ± 0.006 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.02   | 0.05 ± 0.02   | W Africa F    | E Europe M    | 0.34 |
| 18  | 0.12 ± 0.03   | 0.054 ± 0.009 | 0.04 ± 0.01   | 0.06 ± 0.01   | 0.06 ± 0.02   | 0.14 ± 0.03   | 0.09 ± 0.02   | 0.041 ± 0.008 | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.04 ± 0.01   | 0.09 ± 0.03   | W Africa F    | E Europe M    | 0.30 |
| 19  | 0.18 ± 0.03   | 0.09 ± 0.01   | 0.08 ± 0.02   | 0.09 ± 0.01   | 0.11 ± 0.02   | 0.22 ± 0.03   | 0.15 ± 0.03   | 0.08 ± 0.01   | 0.06 ± 0.02   | 0.12 ± 0.02   | 0.11 ± 0.02   | 0.17 ± 0.03   | W Africa F    | E Europe M    | 0.23 |
| 20  | 0.23 ± 0.03   | 0.14 ± 0.01   | 0.10 ± 0.02   | 0.14 ± 0.02   | 0.17 ± 0.02   | 0.29 ± 0.04   | 0.24 ± 0.03   | 0.14 ± 0.01   | 0.09 ± 0.02   | 0.23 ± 0.02   | 0.17 ± 0.03   | 0.25 ± 0.04   | W Africa F    | E Europe M    | 0.21 |
| 21  | 0.29 ± 0.03   | 0.20 ± 0.02   | 0.16 ± 0.02   | 0.16 ± 0.02   | 0.21 ± 0.02   | 0.35 ± 0.04   | 0.35 ± 0.04   | 0.22 ± 0.02   | 0.17 ± 0.03   | 0.32 ± 0.02   | 0.26 ± 0.03   | 0.29 ± 0.04   | E Africa M    | S Asia F      | 0.17 |
| 22  | 0.36 ± 0.03   | 0.26 ± 0.02   | 0.24 ± 0.02   | 0.22 ± 0.02   | 0.28 ± 0.02   | 0.39 ± 0.03   | 0.49 ± 0.03   | 0.35 ± 0.02   | 0.32 ± 0.03   | 0.44 ± 0.02   | 0.37 ± 0.03   | 0.39 ± 0.04   | E Africa M    | S Asia F      | 0.14 |
| 23  | 0.39 ± 0.03   | 0.38 ± 0.02   | 0.31 ± 0.02   | 0.29 ± 0.02   | 0.37 ± 0.02   | 0.48 ± 0.03   | 0.56 ± 0.03   | 0.46 ± 0.02   | 0.41 ± 0.04   | 0.59 ± 0.02   | 0.47 ± 0.03   | 0.53 ± 0.03   | S Asia M      | S Asia F      | 0.13 |
| 24  | 0.45 ± 0.03   | 0.42 ± 0.02   | 0.42 ± 0.02   | 0.36 ± 0.02   | 0.45 ± 0.02   | 0.52 ± 0.03   | 0.68 ± 0.03   | 0.58 ± 0.02   | 0.58 ± 0.03   | 0.71 ± 0.02   | 0.60 ± 0.03   | 0.63 ± 0.03   | S Asia M      | S Asia F      | 0.13 |
| 25  | 0.52 ± 0.03   | 0.54 ± 0.01   | 0.52 ± 0.02   | 0.41 ± 0.01   | 0.52 ± 0.02   | 0.58 ± 0.03   | 0.76 ± 0.02   | 0.70 ± 0.02   | 0.71 ± 0.03   | 0.79 ± 0.01   | 0.70 ± 0.03   | 0.69 ± 0.03   | S Asia M      | S Asia F      | 0.11 |
| 26  | 0.57 ± 0.02   | 0.61 ± 0.01   | 0.62 ± 0.02   | 0.45 ± 0.01   | 0.63 ± 0.02   | 0.63 ± 0.03   | 0.82 ± 0.02   | 0.79 ± 0.02   | 0.77 ± 0.03   | 0.85 ± 0.01   | 0.79 ± 0.03   | 0.77 ± 0.02   | S Asia M      | S Asia F      | 0.10 |
| 27  | 0.62 ± 0.03   | 0.68 ± 0.01   | 0.72 ± 0.02   | 0.52 ± 0.01   | 0.68 ± 0.02   | 0.67 ± 0.03   | 0.86 ± 0.02   | 0.86 ± 0.01   | 0.88 ± 0.02   | 0.89 ± 0.009  | 0.86 ± 0.02   | 0.83 ± 0.02   | S Asia M      | S Asia F      | 0.09 |
| 28  | 0.68 ± 0.02   | 0.75 ± 0.01   | 0.79 ± 0.02   | 0.58 ± 0.01   | 0.75 ± 0.02   | 0.71 ± 0.03   | 0.89 ± 0.02   | 0.91 ± 0.01   | 0.93 ± 0.01   | 0.929 ± 0.008 | 0.91 ± 0.02   | 0.87 ± 0.02   | S Asia M      | S Asia F      | 0.08 |
| 29  | 0.75 ± 0.02   | 0.81 ± 0.01   | 0.86 ± 0.01   | 0.66 ± 0.01   | 0.81 ± 0.01   | 0.77 ± 0.02   | 0.94 ± 0.01   | 0.949 ± 0.008 | 0.96 ± 0.01   | 0.956 ± 0.006 | 0.93 ± 0.02   | 0.91 ± 0.01   | S Asia M      | S Asia F      | 0.07 |
| 30  | 0.78 ± 0.02   | 0.861 ± 0.008 | 0.91 ± 0.01   | 0.71 ± 0.01   | 0.86 ± 0.01   | 0.79 ± 0.02   | 0.96 ± 0.01   | 0.970 ± 0.006 | 0.973 ± 0.008 | 0.972 ± 0.004 | 0.96 ± 0.01   | 0.93 ± 0.01   | E Europe M    | S Asia F      | 0.06 |
| 31  | 0.80 ± 0.02   | 0.901 ± 0.008 | 0.93 ± 0.01   | 0.75 ± 0.01   | 0.89 ± 0.01   | 0.84 ± 0.02   | 0.969 ± 0.010 | 0.982 ± 0.005 | 0.979 ± 0.008 | 0.981 ± 0.004 | 0.97 ± 0.01   | 0.949 ± 0.010 | E Asia M      | S Asia F      | 0.05 |
| 32  | 0.85 ± 0.02   | 0.929 ± 0.007 | 0.958 ± 0.008 | 0.80 ± 0.01   | 0.92 ± 0.01   | 0.86 ± 0.02   | 0.985 ± 0.007 | 0.989 ± 0.003 | 0.992 ± 0.004 | 0.988 ± 0.003 | 0.985 ± 0.007 | 0.965 ± 0.009 | E Europe M    | S Asia F      | 0.04 |
| 33  | 0.87 ± 0.02   | 0.951 ± 0.006 | 0.968 ± 0.008 | 0.84 ± 0.01   | 0.932 ± 0.010 | 0.89 ± 0.02   | 0.987 ± 0.006 | 0.991 ± 0.004 | 0.996 ± 0.003 | 0.994 ± 0.002 | 0.988 ± 0.006 | 0.976 ± 0.009 | E Europe M    | S Asia F      | 0.03 |
| 34  | 0.90 ± 0.02   | 0.968 ± 0.005 | 0.973 ± 0.007 | 0.874 ± 0.009 | 0.962 ± 0.008 | 0.91 ± 0.02   | 0.988 ± 0.007 | 0.994 ± 0.003 | 0.998 ± 0.003 | 0.995 ± 0.002 | 0.997 ± 0.004 | 0.984 ± 0.006 | E Europe M    | S Asia F      | 0.02 |

**Table 122.** For algorithm cyberlink-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.06 ± 0.03   | 0.005 ± 0.003 | 0.006 ± 0.006 | 0.012 ± 0.007 | 0.009 ± 0.007 | 0.05 ± 0.03   | 0.06 ± 0.04   | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.008 ± 0.005 | 0.004 ± 0.004 | 0.06 ± 0.04   | W Africa M    | E Europe M    | 0.58 |
| 15  | 0.10 ± 0.04   | 0.007 ± 0.003 | 0.02 ± 0.01   | 0.024 ± 0.008 | 0.016 ± 0.009 | 0.12 ± 0.04   | 0.21 ± 0.06   | 0.006 ± 0.003 | 0.004 ± 0.004 | 0.019 ± 0.007 | 0.011 ± 0.007 | 0.13 ± 0.05   | E Africa M    | E Europe M    | 0.64 |
| 16  | 0.15 ± 0.05   | 0.016 ± 0.004 | 0.009 ± 0.006 | 0.04 ± 0.01   | 0.014 ± 0.008 | 0.12 ± 0.04   | 0.18 ± 0.05   | 0.016 ± 0.004 | 0.015 ± 0.008 | 0.031 ± 0.009 | 0.017 ± 0.009 | 0.15 ± 0.04   | E Africa M    | E Europe F    | 0.56 |
| 17  | 0.15 ± 0.04   | 0.023 ± 0.005 | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.03 ± 0.01   | 0.20 ± 0.04   | 0.26 ± 0.06   | 0.025 ± 0.005 | 0.03 ± 0.01   | 0.08 ± 0.01   | 0.04 ± 0.01   | 0.25 ± 0.05   | E Africa M    | E Asia F      | 0.51 |
| 18  | 0.23 ± 0.04   | 0.043 ± 0.006 | 0.05 ± 0.01   | 0.10 ± 0.01   | 0.044 ± 0.010 | 0.22 ± 0.04   | 0.33 ± 0.05   | 0.042 ± 0.005 | 0.08 ± 0.02   | 0.14 ± 0.01   | 0.05 ± 0.01   | 0.30 ± 0.05   | E Africa M    | E Asia M      | 0.44 |
| 19  | 0.26 ± 0.04   | 0.060 ± 0.005 | 0.06 ± 0.01   | 0.11 ± 0.01   | 0.06 ± 0.01   | 0.31 ± 0.04   | 0.45 ± 0.05   | 0.070 ± 0.007 | 0.12 ± 0.02   | 0.21 ± 0.02   | 0.10 ± 0.02   | 0.34 ± 0.04   | E Africa M    | E Asia F      | 0.42 |
| 20  | 0.32 ± 0.04   | 0.081 ± 0.006 | 0.10 ± 0.01   | 0.14 ± 0.01   | 0.10 ± 0.01   | 0.33 ± 0.04   | 0.56 ± 0.05   | 0.109 ± 0.008 | 0.20 ± 0.02   | 0.34 ± 0.02   | 0.15 ± 0.02   | 0.43 ± 0.04   | E Africa M    | E Asia F      | 0.37 |
| 21  | 0.39 ± 0.04   | 0.116 ± 0.007 | 0.13 ± 0.01   | 0.17 ± 0.01   | 0.14 ± 0.01   | 0.41 ± 0.04   | 0.63 ± 0.04   | 0.172 ± 0.009 | 0.32 ± 0.03   | 0.45 ± 0.02   | 0.23 ± 0.02   | 0.57 ± 0.03   | E Africa M    | E Asia F      | 0.34 |
| 22  | 0.45 ± 0.04   | 0.154 ± 0.007 | 0.18 ± 0.01   | 0.21 ± 0.01   | 0.18 ± 0.01   | 0.49 ± 0.03   | 0.73 ± 0.04   | 0.25 ± 0.01   | 0.42 ± 0.03   | 0.55 ± 0.02   | 0.32 ± 0.02   | 0.62 ± 0.03   | E Africa M    | E Asia F      | 0.30 |
| 23  | 0.46 ± 0.04   | 0.198 ± 0.007 | 0.24 ± 0.01   | 0.25 ± 0.01   | 0.25 ± 0.02   | 0.53 ± 0.03   | 0.76 ± 0.03   | 0.33 ± 0.01   | 0.56 ± 0.02   | 0.65 ± 0.01   | 0.39 ± 0.03   | 0.71 ± 0.03   | E Africa M    | E Asia F      | 0.26 |
| 24  | 0.56 ± 0.03   | 0.245 ± 0.008 | 0.29 ± 0.01   | 0.31 ± 0.01   | 0.29 ± 0.02   | 0.58 ± 0.03   | 0.85 ± 0.02   | 0.43 ± 0.01   | 0.66 ± 0.02   | 0.74 ± 0.01   | 0.50 ± 0.03   | 0.77 ± 0.03   | E Africa M    | E Asia F      | 0.24 |
| 25  | 0.59 ± 0.03   | 0.312 ± 0.008 | 0.39 ± 0.01   | 0.346 ± 0.009 | 0.35 ± 0.02   | 0.64 ± 0.02   | 0.87 ± 0.02   | 0.51 ± 0.01   | 0.74 ± 0.02   | 0.801 ± 0.009 | 0.58 ± 0.02   | 0.83 ± 0.02   | E Africa M    | E Asia F      | 0.21 |
| 26  | 0.66 ± 0.03   | 0.375 ± 0.008 | 0.45 ± 0.01   | 0.397 ± 0.010 | 0.43 ± 0.02   | 0.66 ± 0.02   | 0.92 ± 0.02   | 0.603 ± 0.010 | 0.80 ± 0.01   | 0.844 ± 0.008 | 0.65 ± 0.02   | 0.85 ± 0.02   | E Africa M    | E Asia F      | 0.18 |
| 27  | 0.66 ± 0.03   | 0.437 ± 0.007 | 0.54 ± 0.01   | 0.443 ± 0.010 | 0.48 ± 0.01   | 0.71 ± 0.02   | 0.92 ± 0.02   | 0.672 ± 0.010 | 0.86 ± 0.01   | 0.885 ± 0.006 | 0.71 ± 0.02   | 0.90 ± 0.01   | E Africa M    | E Asia F      | 0.16 |
| 28  | 0.72 ± 0.02   | 0.508 ± 0.007 | 0.61 ± 0.01   | 0.493 ± 0.009 | 0.54 ± 0.02   | 0.72 ± 0.02   | 0.94 ± 0.01   | 0.747 ± 0.009 | 0.901 ± 0.010 | 0.911 ± 0.006 | 0.78 ± 0.02   | 0.92 ± 0.01   | E Africa M    | S Asia F      | 0.13 |
| 29  | 0.76 ± 0.02   | 0.568 ± 0.007 | 0.69 ± 0.01   | 0.546 ± 0.008 | 0.59 ± 0.02   | 0.77 ± 0.02   | 0.96 ± 0.01   | 0.800 ± 0.008 | 0.925 ± 0.009 | 0.929 ± 0.005 | 0.83 ± 0.02   | 0.93 ± 0.01   | E Africa M    | S Asia F      | 0.11 |
| 30  | 0.79 ± 0.02   | 0.630 ± 0.007 | 0.75 ± 0.01   | 0.591 ± 0.008 | 0.65 ± 0.01   | 0.80 ± 0.02   | 0.977 ± 0.008 | 0.848 ± 0.007 | 0.955 ± 0.008 | 0.945 ± 0.004 | 0.87 ± 0.01   | 0.948 ± 0.009 | E Africa M    | S Asia F      | 0.10 |
| 31  | 0.81 ± 0.02   | 0.689 ± 0.007 | 0.79 ± 0.01   | 0.645 ± 0.008 | 0.71 ± 0.01   | 0.83 ± 0.02   | 0.984 ± 0.007 | 0.892 ± 0.006 | 0.968 ± 0.005 | 0.961 ± 0.003 | 0.89 ± 0.02   | 0.958 ± 0.008 | E Africa M    | S Asia F      | 0.08 |
| 32  | 0.86 ± 0.02   | 0.745 ± 0.006 | 0.845 ± 0.008 | 0.695 ± 0.008 | 0.76 ± 0.01   | 0.84 ± 0.02   | 0.984 ± 0.008 | 0.919 ± 0.005 | 0.978 ± 0.005 | 0.969 ± 0.003 | 0.92 ± 0.01   | 0.963 ± 0.008 | E Africa M    | S Asia F      | 0.07 |
| 33  | 0.87 ± 0.02   | 0.793 ± 0.006 | 0.871 ± 0.008 | 0.737 ± 0.008 | 0.79 ± 0.01   | 0.88 ± 0.02   | 0.988 ± 0.006 | 0.941 ± 0.005 | 0.984 ± 0.004 | 0.974 ± 0.002 | 0.94 ± 0.01   | 0.978 ± 0.006 | E Africa M    | S Asia F      | 0.06 |
| 34  | 0.87 ± 0.02   | 0.832 ± 0.006 | 0.901 ± 0.008 | 0.781 ± 0.007 | 0.83 ± 0.01   | 0.89 ± 0.02   | 0.994 ± 0.005 | 0.955 ± 0.004 | 0.989 ± 0.004 | 0.980 ± 0.002 | 0.958 ± 0.009 | 0.981 ± 0.006 | E Africa M    | S Asia F      | 0.05 |

**Table 123.** For algorithm daon-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.05 ± 0.02   | 0.025 ± 0.007 | 0.000 ± 0.000 | 0.026 ± 0.008 | 0.03 ± 0.01  | 0.05 ± 0.03   | 0.05 ± 0.02   | 0.06 ± 0.01   | 0.000 ± 0.000 | 0.040 ± 0.010 | 0.04 ± 0.02   | 0.01 ± 0.01   | E Asia M      | E Europe F    | 0.37 |
| 15  | 0.07 ± 0.03   | 0.05 ± 0.01   | 0.000 ± 0.000 | 0.06 ± 0.01   | 0.06 ± 0.02  | 0.09 ± 0.04   | 0.08 ± 0.03   | 0.10 ± 0.01   | 0.009 ± 0.009 | 0.06 ± 0.01   | 0.10 ± 0.03   | 0.04 ± 0.02   | S E Asia M    | E Europe F    | 0.31 |
| 16  | 0.12 ± 0.03   | 0.08 ± 0.01   | 0.01 ± 0.01   | 0.06 ± 0.02   | 0.08 ± 0.03  | 0.16 ± 0.04   | 0.15 ± 0.04   | 0.12 ± 0.02   | 0.03 ± 0.01   | 0.12 ± 0.02   | 0.13 ± 0.03   | 0.07 ± 0.03   | W Africa F    | E Europe F    | 0.29 |
| 17  | 0.21 ± 0.04   | 0.10 ± 0.02   | 0.03 ± 0.02   | 0.09 ± 0.02   | 0.13 ± 0.03  | 0.25 ± 0.05   | 0.23 ± 0.04   | 0.17 ± 0.02   | 0.03 ± 0.01   | 0.19 ± 0.02   | 0.16 ± 0.04   | 0.13 ± 0.04   | W Africa F    | E Europe M    | 0.30 |
| 18  | 0.18 ± 0.03   | 0.11 ± 0.01   | 0.04 ± 0.01   | 0.09 ± 0.01   | 0.11 ± 0.02  | 0.27 ± 0.04   | 0.31 ± 0.04   | 0.19 ± 0.02   | 0.05 ± 0.02   | 0.26 ± 0.02   | 0.19 ± 0.03   | 0.23 ± 0.04   | E Africa M    | E Europe F    | 0.31 |
| 19  | 0.32 ± 0.04   | 0.15 ± 0.01   | 0.05 ± 0.02   | 0.14 ± 0.02   | 0.15 ± 0.02  | 0.33 ± 0.04   | 0.43 ± 0.04   | 0.25 ± 0.02   | 0.08 ± 0.02   | 0.38 ± 0.02   | 0.26 ± 0.03   | 0.28 ± 0.04   | E Africa M    | E Europe F    | 0.30 |
| 20  | 0.36 ± 0.04   | 0.19 ± 0.02   | 0.09 ± 0.02   | 0.19 ± 0.02   | 0.20 ± 0.02  | 0.41 ± 0.04   | 0.52 ± 0.04   | 0.30 ± 0.02   | 0.14 ± 0.03   | 0.49 ± 0.02   | 0.30 ± 0.03   | 0.39 ± 0.04   | E Africa M    | E Europe F    | 0.27 |
| 21  | 0.41 ± 0.03   | 0.24 ± 0.02   | 0.14 ± 0.02   | 0.22 ± 0.02   | 0.24 ± 0.02  | 0.49 ± 0.04   | 0.65 ± 0.03   | 0.35 ± 0.02   | 0.20 ± 0.03   | 0.55 ± 0.02   | 0.38 ± 0.03   | 0.45 ± 0.04   | E Africa M    | E Europe F    | 0.26 |
| 22  | 0.50 ± 0.04   | 0.31 ± 0.02   | 0.19 ± 0.02   | 0.27 ± 0.02   | 0.32 ± 0.02  | 0.53 ± 0.04   | 0.75 ± 0.03   | 0.43 ± 0.02   | 0.31 ± 0.04   | 0.66 ± 0.02   | 0.41 ± 0.03   | 0.56 ± 0.04   | E Africa M    | E Europe F    | 0.23 |
| 23  | 0.53 ± 0.03   | 0.36 ± 0.02   | 0.28 ± 0.02   | 0.32 ± 0.02   | 0.38 ± 0.02  | 0.58 ± 0.03   | 0.78 ± 0.03   | 0.51 ± 0.02   | 0.41 ± 0.03   | 0.73 ± 0.02   | 0.51 ± 0.03   | 0.59 ± 0.03   | E Africa M    | E Europe F    | 0.18 |
| 24  | 0.59 ± 0.03   | 0.43 ± 0.02   | 0.37 ± 0.02   | 0.37 ± 0.02   | 0.44 ± 0.02  | 0.62 ± 0.03   | 0.82 ± 0.03   | 0.56 ± 0.02   | 0.56 ± 0.03   | 0.80 ± 0.02   | 0.59 ± 0.03   | 0.68 ± 0.03   | E Africa M    | E Europe F    | 0.15 |
| 25  | 0.62 ± 0.02   | 0.50 ± 0.01   | 0.45 ± 0.02   | 0.43 ± 0.01   | 0.52 ± 0.02  | 0.64 ± 0.03   | 0.87 ± 0.02   | 0.64 ± 0.02   | 0.65 ± 0.03   | 0.85 ± 0.01   | 0.66 ± 0.03   | 0.74 ± 0.03   | E Africa M    | S Asia F      | 0.13 |
| 26  | 0.71 ± 0.03   | 0.55 ± 0.01   | 0.56 ± 0.02   | 0.46 ± 0.01   | 0.59 ± 0.02  | 0.71 ± 0.02   | 0.90 ± 0.02   | 0.72 ± 0.02   | 0.73 ± 0.02   | 0.88 ± 0.01   | 0.71 ± 0.02   | 0.77 ± 0.02   | E Africa M    | S Asia F      | 0.11 |
| 27  | 0.74 ± 0.02   | 0.62 ± 0.01   | 0.65 ± 0.02   | 0.52 ± 0.01   | 0.64 ± 0.02  | 0.75 ± 0.02   | 0.92 ± 0.02   | 0.78 ± 0.02   | 0.84 ± 0.02   | 0.915 ± 0.008 | 0.79 ± 0.02   | 0.83 ± 0.02   | E Africa M    | S Asia F      | 0.10 |
| 28  | 0.75 ± 0.02   | 0.67 ± 0.01   | 0.72 ± 0.02   | 0.57 ± 0.01   | 0.69 ± 0.02  | 0.76 ± 0.02   | 0.93 ± 0.01   | 0.83 ± 0.01   | 0.88 ± 0.02   | 0.934 ± 0.007 | 0.85 ± 0.02   | 0.85 ± 0.02   | S Asia M      | S Asia F      | 0.08 |
| 29  | 0.80 ± 0.02   | 0.74 ± 0.01   | 0.80 ± 0.01   | 0.63 ± 0.01   | 0.77 ± 0.01  | 0.78 ± 0.02   | 0.96 ± 0.01   | 0.87 ± 0.01   | 0.92 ± 0.01   | 0.951 ± 0.006 | 0.88 ± 0.02   | 0.89 ± 0.01   | E Africa M    | S Asia F      | 0.07 |
| 30  | 0.83 ± 0.02   | 0.78 ± 0.01   | 0.85 ± 0.01   | 0.68 ± 0.01   | 0.81 ± 0.02  | 0.81 ± 0.02   | 0.966 ± 0.010 | 0.91 ± 0.01   | 0.95 ± 0.01   | 0.963 ± 0.005 | 0.92 ± 0.01   | 0.92 ± 0.01   | E Africa M    | S Asia F      | 0.06 |
| 31  | 0.85 ± 0.02   | 0.817 ± 0.010 | 0.90 ± 0.01   | 0.71 ± 0.01   | 0.83 ± 0.01  | 0.84 ± 0.02   | 0.975 ± 0.009 | 0.943 ± 0.009 | 0.972 ± 0.009 | 0.975 ± 0.004 | 0.93 ± 0.02   | 0.93 ± 0.01   | S Asia M      | S Asia F      | 0.05 |
| 32  | 0.88 ± 0.02   | 0.861 ± 0.008 | 0.92 ± 0.01   | 0.76 ± 0.01   | 0.87 ± 0.01  | 0.85 ± 0.02   | 0.975 ± 0.009 | 0.961 ± 0.007 | 0.983 ± 0.007 | 0.978 ± 0.004 | 0.96 ± 0.01   | 0.95 ± 0.01   | E Europe M    | S Asia F      | 0.04 |
| 33  | 0.88 ± 0.02   | 0.892 ± 0.008 | 0.94 ± 0.01   | 0.80 ± 0.01   | 0.90 ± 0.01  | 0.89 ± 0.02   | 0.980 ± 0.008 | 0.970 ± 0.006 | 0.993 ± 0.005 | 0.988 ± 0.003 | 0.97 ± 0.01   | 0.96 ± 0.01   | E Europe M    | S Asia F      | 0.04 |
| 34  | 0.91 ± 0.02   | 0.913 ± 0.007 | 0.954 ± 0.009 | 0.84 ± 0.01   | 0.92 ± 0.01  | 0.88 ± 0.02   | 0.989 ± 0.007 | 0.981 ± 0.005 | 0.997 ± 0.004 | 0.990 ± 0.003 | 0.980 ± 0.009 | 0.97 ± 0.01   | E Europe M    | S Asia F      | 0.03 |

**Table 124.** For algorithm daon-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.21 ± 0.06   | 0.035 ± 0.007 | 0.03 ± 0.01   | 0.12 ± 0.02   | 0.04 ± 0.02  | 0.14 ± 0.05   | 0.20 ± 0.06   | 0.10 ± 0.01   | 0.02 ± 0.01   | 0.10 ± 0.02   | 0.10 ± 0.03   | 0.17 ± 0.06   | E Africa F    | E Europe M    | 0.37 |
| 15  | 0.19 ± 0.06   | 0.062 ± 0.010 | 0.05 ± 0.02   | 0.13 ± 0.02   | 0.06 ± 0.02  | 0.24 ± 0.06   | 0.41 ± 0.07   | 0.15 ± 0.01   | 0.06 ± 0.02   | 0.18 ± 0.02   | 0.13 ± 0.03   | 0.27 ± 0.07   | E Africa M    | E Europe F    | 0.37 |
| 16  | 0.30 ± 0.06   | 0.083 ± 0.009 | 0.07 ± 0.02   | 0.18 ± 0.02   | 0.09 ± 0.02  | 0.29 ± 0.06   | 0.42 ± 0.07   | 0.21 ± 0.01   | 0.07 ± 0.02   | 0.24 ± 0.02   | 0.18 ± 0.03   | 0.27 ± 0.06   | E Africa M    | E Europe F    | 0.32 |
| 17  | 0.35 ± 0.05   | 0.10 ± 0.01   | 0.10 ± 0.02   | 0.21 ± 0.02   | 0.11 ± 0.02  | 0.40 ± 0.06   | 0.55 ± 0.07   | 0.28 ± 0.01   | 0.11 ± 0.02   | 0.37 ± 0.02   | 0.24 ± 0.03   | 0.47 ± 0.06   | E Africa M    | E Europe F    | 0.34 |
| 18  | 0.45 ± 0.05   | 0.129 ± 0.009 | 0.15 ± 0.02   | 0.22 ± 0.02   | 0.13 ± 0.02  | 0.39 ± 0.05   | 0.66 ± 0.06   | 0.31 ± 0.01   | 0.19 ± 0.03   | 0.44 ± 0.02   | 0.28 ± 0.03   | 0.50 ± 0.05   | E Africa M    | S E Asia F    | 0.31 |
| 19  | 0.44 ± 0.05   | 0.155 ± 0.009 | 0.18 ± 0.02   | 0.26 ± 0.02   | 0.15 ± 0.02  | 0.50 ± 0.05   | 0.69 ± 0.05   | 0.35 ± 0.01   | 0.24 ± 0.03   | 0.53 ± 0.02   | 0.35 ± 0.03   | 0.57 ± 0.05   | E Africa M    | S E Asia F    | 0.29 |
| 20  | 0.54 ± 0.04   | 0.187 ± 0.008 | 0.23 ± 0.02   | 0.28 ± 0.02   | 0.18 ± 0.02  | 0.56 ± 0.04   | 0.74 ± 0.04   | 0.42 ± 0.01   | 0.36 ± 0.03   | 0.62 ± 0.02   | 0.41 ± 0.03   | 0.60 ± 0.04   | E Africa M    | S E Asia F    | 0.26 |
| 21  | 0.58 ± 0.04   | 0.226 ± 0.008 | 0.28 ± 0.02   | 0.34 ± 0.02   | 0.21 ± 0.02  | 0.59 ± 0.04   | 0.78 ± 0.04   | 0.48 ± 0.01   | 0.47 ± 0.03   | 0.69 ± 0.02   | 0.48 ± 0.03   | 0.67 ± 0.04   | E Africa M    | S E Asia F    | 0.23 |
| 22  | 0.65 ± 0.03   | 0.256 ± 0.008 | 0.33 ± 0.02   | 0.37 ± 0.01   | 0.24 ± 0.02  | 0.64 ± 0.03   | 0.87 ± 0.03   | 0.54 ± 0.01   | 0.52 ± 0.03   | 0.75 ± 0.01   | 0.54 ± 0.03   | 0.70 ± 0.03   | E Africa M    | S E Asia F    | 0.22 |
| 23  | 0.67 ± 0.03   | 0.294 ± 0.009 | 0.42 ± 0.02   | 0.41 ± 0.01   | 0.32 ± 0.02  | 0.67 ± 0.03   | 0.88 ± 0.03   | 0.60 ± 0.01   | 0.65 ± 0.02   | 0.80 ± 0.01   | 0.60 ± 0.02   | 0.74 ± 0.03   | E Africa M    | E Asia F      | 0.19 |
| 24  | 0.70 ± 0.03   | 0.341 ± 0.008 | 0.47 ± 0.01   | 0.47 ± 0.01   | 0.35 ± 0.02  | 0.71 ± 0.03   | 0.90 ± 0.02   | 0.64 ± 0.01   | 0.73 ± 0.02   | 0.853 ± 0.010 | 0.66 ± 0.02   | 0.78 ± 0.03   | E Africa M    | E Asia F      | 0.17 |
| 25  | 0.75 ± 0.02   | 0.385 ± 0.009 | 0.57 ± 0.01   | 0.482 ± 0.010 | 0.42 ± 0.02  | 0.75 ± 0.02   | 0.92 ± 0.02   | 0.70 ± 0.01   | 0.80 ± 0.02   | 0.880 ± 0.007 | 0.73 ± 0.02   | 0.80 ± 0.02   | E Africa M    | E Asia F      | 0.15 |
| 26  | 0.75 ± 0.02   | 0.438 ± 0.008 | 0.62 ± 0.01   | 0.531 ± 0.010 | 0.48 ± 0.01  | 0.75 ± 0.02   | 0.95 ± 0.02   | 0.75 ± 0.01   | 0.85 ± 0.01   | 0.904 ± 0.007 | 0.77 ± 0.02   | 0.85 ± 0.02   | E Africa M    | E Asia F      | 0.14 |
| 27  | 0.79 ± 0.02   | 0.484 ± 0.008 | 0.69 ± 0.01   | 0.567 ± 0.009 | 0.52 ± 0.02  | 0.77 ± 0.02   | 0.95 ± 0.01   | 0.786 ± 0.009 | 0.90 ± 0.01   | 0.923 ± 0.005 | 0.81 ± 0.02   | 0.88 ± 0.02   | E Africa M    | E Asia F      | 0.12 |
| 28  | 0.81 ± 0.02   | 0.535 ± 0.007 | 0.74 ± 0.01   | 0.604 ± 0.009 | 0.57 ± 0.01  | 0.81 ± 0.02   | 0.95 ± 0.01   | 0.830 ± 0.008 | 0.935 ± 0.008 | 0.940 ± 0.004 | 0.85 ± 0.02   | 0.90 ± 0.01   | E Africa M    | E Asia F      | 0.11 |
| 29  | 0.83 ± 0.02   | 0.580 ± 0.007 | 0.801 ± 0.009 | 0.639 ± 0.008 | 0.61 ± 0.02  | 0.82 ± 0.02   | 0.95 ± 0.01   | 0.858 ± 0.007 | 0.949 ± 0.008 | 0.949 ± 0.004 | 0.90 ± 0.01   | 0.90 ± 0.01   | E Africa M    | E Asia F      | 0.09 |
| 30  | 0.86 ± 0.02   | 0.632 ± 0.008 | 0.838 ± 0.008 | 0.674 ± 0.008 | 0.66 ± 0.01  | 0.81 ± 0.02   | 0.96 ± 0.01   | 0.890 ± 0.007 | 0.968 ± 0.006 | 0.965 ± 0.003 | 0.91 ± 0.01   | 0.92 ± 0.01   | E Europe M    | E Asia F      | 0.08 |
| 31  | 0.87 ± 0.02   | 0.674 ± 0.007 | 0.873 ± 0.008 | 0.712 ± 0.007 | 0.71 ± 0.01  | 0.85 ± 0.02   | 0.977 ± 0.009 | 0.912 ± 0.006 | 0.982 ± 0.004 | 0.969 ± 0.003 | 0.91 ± 0.01   | 0.939 ± 0.009 | E Europe M    | E Asia F      | 0.07 |
| 32  | 0.89 ± 0.02   | 0.713 ± 0.007 | 0.904 ± 0.008 | 0.745 ± 0.007 | 0.74 ± 0.01  | 0.85 ± 0.02   | 0.977 ± 0.008 | 0.930 ± 0.005 | 0.987 ± 0.004 | 0.977 ± 0.002 | 0.92 ± 0.01   | 0.936 ± 0.010 | E Europe M    | E Asia F      | 0.06 |
| 33  | 0.89 ± 0.02   | 0.748 ± 0.006 | 0.924 ± 0.007 | 0.776 ± 0.007 | 0.77 ± 0.01  | 0.87 ± 0.02   | 0.985 ± 0.007 | 0.945 ± 0.004 | 0.992 ± 0.003 | 0.982 ± 0.002 | 0.93 ± 0.01   | 0.948 ± 0.009 | E Europe M    | E Asia F      | 0.06 |
| 34  | 0.89 ± 0.02   | 0.783 ± 0.006 | 0.942 ± 0.006 | 0.806 ± 0.007 | 0.79 ± 0.01  | 0.89 ± 0.02   | 0.986 ± 0.007 | 0.955 ± 0.004 | 0.995 ± 0.002 | 0.986 ± 0.002 | 0.958 ± 0.009 | 0.963 ± 0.007 | E Europe M    | E Asia F      | 0.05 |

**Table 125.** For algorithm daon-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.02 ± 0.01   | 0.010 ± 0.005 | 0.000 ± 0.000 | 0.005 ± 0.003 | 0.009 ± 0.007 | 0.03 ± 0.02   | 0.03 ± 0.01   | 0.024 ± 0.007 | 0.000 ± 0.000 | 0.008 ± 0.005 | 0.02 ± 0.01   | 0.004 ± 0.004 | E Africa M    | E Europe F    | 0.48 |
| 15  | 0.03 ± 0.02   | 0.025 ± 0.008 | 0.000 ± 0.000 | 0.018 ± 0.008 | 0.02 ± 0.01   | 0.06 ± 0.03   | 0.03 ± 0.02   | 0.033 ± 0.009 | 0.006 ± 0.006 | 0.018 ± 0.008 | 0.03 ± 0.01   | 0.02 ± 0.02   | W Africa F    | E Europe F    | 0.34 |
| 16  | 0.07 ± 0.02   | 0.03 ± 0.01   | 0.008 ± 0.008 | 0.023 ± 0.009 | 0.03 ± 0.02   | 0.11 ± 0.04   | 0.07 ± 0.02   | 0.07 ± 0.01   | 0.003 ± 0.003 | 0.04 ± 0.01   | 0.04 ± 0.02   | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.39 |
| 17  | 0.14 ± 0.04   | 0.06 ± 0.01   | 0.02 ± 0.01   | 0.04 ± 0.01   | 0.06 ± 0.02   | 0.18 ± 0.04   | 0.13 ± 0.04   | 0.10 ± 0.01   | 0.008 ± 0.008 | 0.09 ± 0.02   | 0.10 ± 0.03   | 0.11 ± 0.03   | W Africa F    | E Europe M    | 0.35 |
| 18  | 0.12 ± 0.03   | 0.08 ± 0.01   | 0.03 ± 0.01   | 0.05 ± 0.01   | 0.06 ± 0.02   | 0.23 ± 0.04   | 0.21 ± 0.03   | 0.12 ± 0.01   | 0.03 ± 0.01   | 0.17 ± 0.02   | 0.11 ± 0.02   | 0.15 ± 0.04   | W Africa F    | E Europe F    | 0.34 |
| 19  | 0.24 ± 0.03   | 0.12 ± 0.01   | 0.06 ± 0.02   | 0.09 ± 0.01   | 0.10 ± 0.02   | 0.29 ± 0.04   | 0.33 ± 0.04   | 0.18 ± 0.02   | 0.06 ± 0.02   | 0.28 ± 0.02   | 0.15 ± 0.02   | 0.24 ± 0.04   | E Africa M    | E Europe M    | 0.32 |
| 20  | 0.31 ± 0.03   | 0.16 ± 0.02   | 0.08 ± 0.02   | 0.13 ± 0.01   | 0.14 ± 0.02   | 0.38 ± 0.04   | 0.49 ± 0.04   | 0.25 ± 0.02   | 0.12 ± 0.03   | 0.41 ± 0.02   | 0.24 ± 0.03   | 0.37 ± 0.04   | E Africa M    | E Europe F    | 0.31 |
| 21  | 0.39 ± 0.03   | 0.22 ± 0.02   | 0.14 ± 0.02   | 0.17 ± 0.01   | 0.19 ± 0.02   | 0.45 ± 0.04   | 0.61 ± 0.04   | 0.33 ± 0.02   | 0.17 ± 0.03   | 0.51 ± 0.02   | 0.33 ± 0.03   | 0.44 ± 0.04   | E Africa M    | E Europe F    | 0.28 |
| 22  | 0.48 ± 0.03   | 0.28 ± 0.02   | 0.21 ± 0.02   | 0.22 ± 0.02   | 0.27 ± 0.02   | 0.50 ± 0.04   | 0.73 ± 0.03   | 0.41 ± 0.02   | 0.30 ± 0.03   | 0.63 ± 0.02   | 0.38 ± 0.03   | 0.58 ± 0.04   | E Africa M    | E Europe F    | 0.24 |
| 23  | 0.52 ± 0.03   | 0.35 ± 0.02   | 0.30 ± 0.02   | 0.29 ± 0.02   | 0.34 ± 0.02   | 0.59 ± 0.03   | 0.78 ± 0.03   | 0.53 ± 0.02   | 0.41 ± 0.03   | 0.74 ± 0.02   | 0.50 ± 0.03   | 0.63 ± 0.03   | E Africa M    | S Asia F      | 0.20 |
| 24  | 0.58 ± 0.03   | 0.42 ± 0.02   | 0.38 ± 0.02   | 0.35 ± 0.01   | 0.41 ± 0.02   | 0.64 ± 0.03   | 0.85 ± 0.02   | 0.61 ± 0.02   | 0.57 ± 0.03   | 0.82 ± 0.02   | 0.60 ± 0.03   | 0.73 ± 0.03   | E Africa M    | S Asia F      | 0.17 |
| 25  | 0.64 ± 0.03   | 0.51 ± 0.02   | 0.49 ± 0.02   | 0.41 ± 0.01   | 0.49 ± 0.02   | 0.68 ± 0.03   | 0.90 ± 0.02   | 0.69 ± 0.02   | 0.69 ± 0.03   | 0.88 ± 0.01   | 0.70 ± 0.03   | 0.79 ± 0.02   | E Africa M    | S Asia F      | 0.14 |
| 26  | 0.71 ± 0.03   | 0.58 ± 0.01   | 0.61 ± 0.02   | 0.47 ± 0.01   | 0.59 ± 0.02   | 0.75 ± 0.02   | 0.93 ± 0.01   | 0.78 ± 0.02   | 0.78 ± 0.02   | 0.916 ± 0.009 | 0.78 ± 0.02   | 0.83 ± 0.02   | E Africa M    | S Asia F      | 0.11 |
| 27  | 0.76 ± 0.02   | 0.66 ± 0.01   | 0.71 ± 0.02   | 0.54 ± 0.01   | 0.66 ± 0.02   | 0.78 ± 0.02   | 0.95 ± 0.01   | 0.84 ± 0.01   | 0.88 ± 0.02   | 0.939 ± 0.007 | 0.84 ± 0.02   | 0.88 ± 0.02   | E Africa M    | S Asia F      | 0.10 |
| 28  | 0.80 ± 0.02   | 0.71 ± 0.01   | 0.78 ± 0.02   | 0.60 ± 0.01   | 0.73 ± 0.02   | 0.81 ± 0.02   | 0.96 ± 0.01   | 0.89 ± 0.01   | 0.92 ± 0.02   | 0.958 ± 0.006 | 0.91 ± 0.02   | 0.92 ± 0.02   | E Africa M    | S Asia F      | 0.08 |
| 29  | 0.85 ± 0.02   | 0.78 ± 0.01   | 0.86 ± 0.01   | 0.67 ± 0.01   | 0.80 ± 0.02   | 0.84 ± 0.02   | 0.979 ± 0.008 | 0.92 ± 0.01   | 0.95 ± 0.01   | 0.973 ± 0.005 | 0.93 ± 0.02   | 0.94 ± 0.01   | E Africa M    | S Asia F      | 0.06 |
| 30  | 0.88 ± 0.02   | 0.829 ± 0.010 | 0.90 ± 0.01   | 0.72 ± 0.01   | 0.84 ± 0.01   | 0.84 ± 0.02   | 0.984 ± 0.007 | 0.953 ± 0.008 | 0.978 ± 0.008 | 0.983 ± 0.003 | 0.96 ± 0.01   | 0.957 ± 0.010 | E Africa M    | S Asia F      | 0.05 |
| 31  | 0.89 ± 0.02   | 0.870 ± 0.009 | 0.936 ± 0.009 | 0.77 ± 0.01   | 0.88 ± 0.01   | 0.90 ± 0.02   | 0.985 ± 0.007 | 0.976 ± 0.005 | 0.985 ± 0.006 | 0.989 ± 0.003 | 0.97 ± 0.01   | 0.969 ± 0.008 | S Asia M      | S Asia F      | 0.04 |
| 32  | 0.92 ± 0.01   | 0.901 ± 0.007 | 0.956 ± 0.008 | 0.823 ± 0.010 | 0.91 ± 0.01   | 0.92 ± 0.01   | 0.991 ± 0.005 | 0.987 ± 0.004 | 0.991 ± 0.005 | 0.991 ± 0.002 | 0.984 ± 0.007 | 0.982 ± 0.007 | E Europe M    | S Asia F      | 0.03 |
| 33  | 0.93 ± 0.01   | 0.932 ± 0.007 | 0.970 ± 0.007 | 0.859 ± 0.009 | 0.93 ± 0.01   | 0.93 ± 0.01   | 0.995 ± 0.004 | 0.991 ± 0.004 | 0.998 ± 0.003 | 0.995 ± 0.002 | 0.990 ± 0.006 | 0.984 ± 0.007 | E Europe M    | S Asia F      | 0.02 |
| 34  | 0.95 ± 0.01   | 0.953 ± 0.005 | 0.974 ± 0.007 | 0.895 ± 0.008 | 0.951 ± 0.009 | 0.95 ± 0.01   | 0.998 ± 0.002 | 0.994 ± 0.003 | 0.999 ± 0.002 | 0.997 ± 0.001 | 0.997 ± 0.004 | 0.990 ± 0.005 | E Europe M    | S Asia F      | 0.02 |

**Table 126.** For algorithm daon-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.14 ± 0.05   | 0.020 ± 0.006 | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.02 ± 0.01  | 0.11 ± 0.05   | 0.12 ± 0.05   | 0.036 ± 0.007 | 0.002 ± 0.002 | 0.031 ± 0.009 | 0.06 ± 0.02   | 0.09 ± 0.04   | E Africa F    | E Europe M    | 0.45 |
| 15  | 0.16 ± 0.05   | 0.036 ± 0.007 | 0.03 ± 0.01   | 0.08 ± 0.02   | 0.04 ± 0.01  | 0.20 ± 0.06   | 0.29 ± 0.07   | 0.075 ± 0.009 | 0.02 ± 0.01   | 0.08 ± 0.01   | 0.06 ± 0.02   | 0.15 ± 0.05   | E Africa M    | E Europe M    | 0.45 |
| 16  | 0.27 ± 0.06   | 0.053 ± 0.007 | 0.05 ± 0.02   | 0.12 ± 0.02   | 0.07 ± 0.02  | 0.27 ± 0.06   | 0.32 ± 0.06   | 0.11 ± 0.01   | 0.04 ± 0.01   | 0.14 ± 0.02   | 0.09 ± 0.02   | 0.23 ± 0.06   | E Africa M    | E Europe M    | 0.39 |
| 17  | 0.30 ± 0.05   | 0.076 ± 0.008 | 0.07 ± 0.02   | 0.15 ± 0.02   | 0.08 ± 0.02  | 0.36 ± 0.06   | 0.49 ± 0.07   | 0.17 ± 0.01   | 0.06 ± 0.02   | 0.25 ± 0.02   | 0.14 ± 0.02   | 0.38 ± 0.06   | E Africa M    | E Europe M    | 0.39 |
| 18  | 0.40 ± 0.05   | 0.108 ± 0.009 | 0.11 ± 0.02   | 0.17 ± 0.02   | 0.10 ± 0.02  | 0.36 ± 0.05   | 0.56 ± 0.06   | 0.21 ± 0.01   | 0.11 ± 0.02   | 0.33 ± 0.02   | 0.19 ± 0.02   | 0.43 ± 0.05   | E Africa M    | S E Asia F    | 0.34 |
| 19  | 0.45 ± 0.05   | 0.146 ± 0.008 | 0.15 ± 0.02   | 0.21 ± 0.02   | 0.14 ± 0.02  | 0.46 ± 0.04   | 0.63 ± 0.05   | 0.27 ± 0.01   | 0.15 ± 0.02   | 0.42 ± 0.02   | 0.24 ± 0.02   | 0.52 ± 0.05   | E Africa M    | S E Asia F    | 0.31 |
| 20  | 0.55 ± 0.04   | 0.189 ± 0.009 | 0.20 ± 0.02   | 0.26 ± 0.02   | 0.16 ± 0.02  | 0.57 ± 0.04   | 0.72 ± 0.04   | 0.33 ± 0.01   | 0.24 ± 0.02   | 0.55 ± 0.02   | 0.33 ± 0.03   | 0.56 ± 0.04   | E Africa M    | S E Asia F    | 0.28 |
| 21  | 0.59 ± 0.04   | 0.227 ± 0.008 | 0.26 ± 0.02   | 0.30 ± 0.02   | 0.20 ± 0.02  | 0.61 ± 0.04   | 0.77 ± 0.03   | 0.41 ± 0.01   | 0.36 ± 0.03   | 0.65 ± 0.02   | 0.41 ± 0.03   | 0.64 ± 0.04   | E Africa M    | S E Asia F    | 0.25 |
| 22  | 0.66 ± 0.04   | 0.265 ± 0.009 | 0.32 ± 0.02   | 0.35 ± 0.01   | 0.24 ± 0.02  | 0.66 ± 0.03   | 0.87 ± 0.03   | 0.48 ± 0.01   | 0.48 ± 0.03   | 0.71 ± 0.01   | 0.49 ± 0.02   | 0.71 ± 0.04   | E Africa M    | S E Asia F    | 0.23 |
| 23  | 0.68 ± 0.04   | 0.311 ± 0.008 | 0.40 ± 0.02   | 0.39 ± 0.01   | 0.33 ± 0.02  | 0.69 ± 0.03   | 0.89 ± 0.02   | 0.55 ± 0.01   | 0.60 ± 0.02   | 0.79 ± 0.01   | 0.55 ± 0.03   | 0.77 ± 0.03   | E Africa M    | E Asia F      | 0.20 |
| 24  | 0.73 ± 0.03   | 0.373 ± 0.008 | 0.48 ± 0.01   | 0.46 ± 0.01   | 0.39 ± 0.02  | 0.72 ± 0.03   | 0.93 ± 0.02   | 0.62 ± 0.01   | 0.72 ± 0.02   | 0.844 ± 0.009 | 0.64 ± 0.02   | 0.80 ± 0.02   | E Africa M    | E Asia F      | 0.17 |
| 25  | 0.78 ± 0.02   | 0.424 ± 0.008 | 0.59 ± 0.01   | 0.50 ± 0.01   | 0.45 ± 0.02  | 0.78 ± 0.02   | 0.94 ± 0.01   | 0.68 ± 0.01   | 0.79 ± 0.02   | 0.889 ± 0.008 | 0.71 ± 0.02   | 0.83 ± 0.02   | E Africa M    | E Asia F      | 0.15 |
| 26  | 0.80 ± 0.02   | 0.489 ± 0.008 | 0.64 ± 0.01   | 0.551 ± 0.009 | 0.51 ± 0.02  | 0.78 ± 0.02   | 0.96 ± 0.01   | 0.751 ± 0.010 | 0.86 ± 0.01   | 0.915 ± 0.007 | 0.77 ± 0.02   | 0.87 ± 0.02   | E Africa M    | E Asia F      | 0.13 |
| 27  | 0.83 ± 0.02   | 0.532 ± 0.007 | 0.72 ± 0.01   | 0.595 ± 0.009 | 0.56 ± 0.02  | 0.82 ± 0.02   | 0.96 ± 0.01   | 0.792 ± 0.009 | 0.90 ± 0.01   | 0.934 ± 0.005 | 0.83 ± 0.02   | 0.92 ± 0.01   | E Africa M    | E Asia F      | 0.11 |
| 28  | 0.85 ± 0.02   | 0.597 ± 0.007 | 0.78 ± 0.01   | 0.639 ± 0.008 | 0.63 ± 0.01  | 0.84 ± 0.02   | 0.97 ± 0.01   | 0.845 ± 0.008 | 0.946 ± 0.008 | 0.953 ± 0.004 | 0.86 ± 0.02   | 0.93 ± 0.01   | E Africa M    | E Asia F      | 0.09 |
| 29  | 0.87 ± 0.02   | 0.652 ± 0.007 | 0.842 ± 0.009 | 0.682 ± 0.008 | 0.67 ± 0.01  | 0.87 ± 0.02   | 0.97 ± 0.01   | 0.874 ± 0.006 | 0.958 ± 0.007 | 0.964 ± 0.003 | 0.91 ± 0.01   | 0.94 ± 0.01   | E Africa M    | E Asia F      | 0.08 |
| 30  | 0.90 ± 0.02   | 0.704 ± 0.006 | 0.882 ± 0.008 | 0.730 ± 0.007 | 0.72 ± 0.01  | 0.86 ± 0.02   | 0.979 ± 0.009 | 0.909 ± 0.006 | 0.976 ± 0.005 | 0.977 ± 0.002 | 0.92 ± 0.01   | 0.953 ± 0.009 | E Africa M    | E Asia F      | 0.07 |
| 31  | 0.91 ± 0.02   | 0.747 ± 0.006 | 0.914 ± 0.007 | 0.769 ± 0.007 | 0.77 ± 0.01  | 0.90 ± 0.01   | 0.989 ± 0.007 | 0.931 ± 0.005 | 0.990 ± 0.003 | 0.981 ± 0.002 | 0.93 ± 0.01   | 0.964 ± 0.007 | E Europe M    | E Asia F      | 0.06 |
| 32  | 0.94 ± 0.01   | 0.785 ± 0.006 | 0.944 ± 0.006 | 0.803 ± 0.007 | 0.81 ± 0.01  | 0.92 ± 0.01   | 0.990 ± 0.005 | 0.949 ± 0.005 | 0.991 ± 0.003 | 0.987 ± 0.002 | 0.95 ± 0.01   | 0.967 ± 0.007 | E Europe M    | E Asia F      | 0.04 |
| 33  | 0.94 ± 0.01   | 0.820 ± 0.005 | 0.959 ± 0.005 | 0.832 ± 0.007 | 0.84 ± 0.01  | 0.92 ± 0.01   | 0.997 ± 0.004 | 0.964 ± 0.004 | 0.996 ± 0.002 | 0.991 ± 0.001 | 0.963 ± 0.009 | 0.980 ± 0.005 | E Africa M    | E Asia F      | 0.04 |
| 34  | 0.94 ± 0.01   | 0.847 ± 0.005 | 0.969 ± 0.005 | 0.863 ± 0.006 | 0.86 ± 0.01  | 0.93 ± 0.01   | 0.995 ± 0.004 | 0.969 ± 0.003 | 0.998 ± 0.001 | 0.993 ± 0.001 | 0.976 ± 0.007 | 0.989 ± 0.004 | E Europe M    | E Asia F      | 0.03 |

**Table 127.** For algorithm dermalog-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.08 ± 0.03   | 0.06 ± 0.01   | 0.09 ± 0.02   | 0.02 ± 0.01   | 0.02 ± 0.01   | 0.017 ± 0.006 | 0.016 ± 0.009 | 0.04 ± 0.01   | 0.02 ± 0.01  | 0.02 ± 0.02   | S E Asia F    | E Europe M    | 0.38 |
| 15  | 0.08 ± 0.03   | 0.07 ± 0.01   | 0.14 ± 0.03   | 0.13 ± 0.02   | 0.11 ± 0.03   | 0.02 ± 0.02   | 0.01 ± 0.01   | 0.032 ± 0.009 | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.03 ± 0.01  | 0.01 ± 0.01   | E Europe F    | E Africa M    | 0.43 |
| 16  | 0.10 ± 0.03   | 0.10 ± 0.02   | 0.21 ± 0.04   | 0.16 ± 0.03   | 0.16 ± 0.04   | 0.06 ± 0.03   | 0.05 ± 0.02   | 0.05 ± 0.01   | 0.04 ± 0.02   | 0.10 ± 0.02   | 0.08 ± 0.03  | 0.04 ± 0.02   | E Europe F    | W Africa M    | 0.32 |
| 17  | 0.13 ± 0.03   | 0.14 ± 0.02   | 0.31 ± 0.05   | 0.24 ± 0.03   | 0.20 ± 0.04   | 0.07 ± 0.03   | 0.10 ± 0.03   | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.12 ± 0.02   | 0.10 ± 0.03  | 0.09 ± 0.03   | E Europe F    | E Europe M    | 0.33 |
| 18  | 0.13 ± 0.03   | 0.16 ± 0.02   | 0.27 ± 0.04   | 0.25 ± 0.02   | 0.20 ± 0.03   | 0.12 ± 0.03   | 0.13 ± 0.03   | 0.07 ± 0.01   | 0.05 ± 0.02   | 0.18 ± 0.02   | 0.10 ± 0.02  | 0.14 ± 0.03   | E Europe F    | E Europe M    | 0.26 |
| 19  | 0.23 ± 0.03   | 0.20 ± 0.02   | 0.41 ± 0.04   | 0.34 ± 0.02   | 0.24 ± 0.03   | 0.17 ± 0.03   | 0.26 ± 0.04   | 0.11 ± 0.01   | 0.11 ± 0.02   | 0.27 ± 0.02   | 0.15 ± 0.02  | 0.24 ± 0.04   | E Europe F    | E Asia M      | 0.23 |
| 20  | 0.28 ± 0.03   | 0.26 ± 0.02   | 0.43 ± 0.04   | 0.39 ± 0.02   | 0.32 ± 0.03   | 0.18 ± 0.03   | 0.34 ± 0.04   | 0.15 ± 0.02   | 0.17 ± 0.03   | 0.38 ± 0.02   | 0.22 ± 0.03  | 0.28 ± 0.04   | E Europe F    | E Asia M      | 0.19 |
| 21  | 0.31 ± 0.03   | 0.30 ± 0.02   | 0.51 ± 0.03   | 0.49 ± 0.02   | 0.38 ± 0.02   | 0.22 ± 0.03   | 0.40 ± 0.04   | 0.19 ± 0.02   | 0.20 ± 0.03   | 0.46 ± 0.02   | 0.29 ± 0.03  | 0.39 ± 0.04   | E Europe F    | E Asia M      | 0.19 |
| 22  | 0.38 ± 0.03   | 0.35 ± 0.02   | 0.59 ± 0.03   | 0.56 ± 0.02   | 0.42 ± 0.02   | 0.27 ± 0.03   | 0.55 ± 0.03   | 0.26 ± 0.02   | 0.33 ± 0.03   | 0.58 ± 0.02   | 0.35 ± 0.03  | 0.49 ± 0.04   | E Europe F    | E Asia M      | 0.17 |
| 23  | 0.42 ± 0.03   | 0.42 ± 0.02   | 0.67 ± 0.02   | 0.63 ± 0.02   | 0.50 ± 0.02   | 0.30 ± 0.03   | 0.61 ± 0.03   | 0.32 ± 0.02   | 0.40 ± 0.03   | 0.66 ± 0.02   | 0.42 ± 0.03  | 0.59 ± 0.03   | E Europe F    | W Africa F    | 0.15 |
| 24  | 0.48 ± 0.03   | 0.48 ± 0.02   | 0.73 ± 0.02   | 0.69 ± 0.01   | 0.57 ± 0.02   | 0.33 ± 0.03   | 0.70 ± 0.03   | 0.42 ± 0.02   | 0.50 ± 0.03   | 0.74 ± 0.02   | 0.49 ± 0.03  | 0.66 ± 0.03   | S Asia M      | W Africa F    | 0.14 |
| 25  | 0.49 ± 0.03   | 0.55 ± 0.01   | 0.77 ± 0.02   | 0.75 ± 0.01   | 0.61 ± 0.02   | 0.38 ± 0.03   | 0.77 ± 0.02   | 0.46 ± 0.02   | 0.58 ± 0.03   | 0.80 ± 0.01   | 0.62 ± 0.03  | 0.69 ± 0.03   | S Asia M      | W Africa F    | 0.13 |
| 26  | 0.57 ± 0.03   | 0.62 ± 0.01   | 0.81 ± 0.01   | 0.78 ± 0.01   | 0.69 ± 0.02   | 0.45 ± 0.03   | 0.81 ± 0.02   | 0.58 ± 0.02   | 0.67 ± 0.03   | 0.85 ± 0.01   | 0.67 ± 0.03  | 0.75 ± 0.02   | S Asia M      | W Africa F    | 0.10 |
| 27  | 0.60 ± 0.03   | 0.69 ± 0.01   | 0.87 ± 0.01   | 0.82 ± 0.01   | 0.73 ± 0.02   | 0.49 ± 0.03   | 0.85 ± 0.02   | 0.66 ± 0.02   | 0.80 ± 0.02   | 0.88 ± 0.01   | 0.75 ± 0.02  | 0.80 ± 0.02   | S Asia M      | W Africa F    | 0.09 |
| 28  | 0.67 ± 0.02   | 0.72 ± 0.01   | 0.90 ± 0.01   | 0.855 ± 0.009 | 0.78 ± 0.02   | 0.54 ± 0.03   | 0.89 ± 0.02   | 0.73 ± 0.02   | 0.84 ± 0.02   | 0.919 ± 0.009 | 0.82 ± 0.02  | 0.85 ± 0.02   | S Asia M      | W Africa F    | 0.08 |
| 29  | 0.70 ± 0.03   | 0.790 ± 0.010 | 0.92 ± 0.01   | 0.884 ± 0.008 | 0.83 ± 0.01   | 0.61 ± 0.02   | 0.91 ± 0.02   | 0.80 ± 0.01   | 0.85 ± 0.02   | 0.944 ± 0.006 | 0.85 ± 0.02  | 0.87 ± 0.02   | S Asia M      | W Africa F    | 0.07 |
| 30  | 0.74 ± 0.02   | 0.833 ± 0.010 | 0.93 ± 0.01   | 0.912 ± 0.007 | 0.86 ± 0.01   | 0.65 ± 0.03   | 0.95 ± 0.01   | 0.84 ± 0.01   | 0.92 ± 0.02   | 0.955 ± 0.005 | 0.89 ± 0.02  | 0.90 ± 0.01   | S Asia M      | W Africa F    | 0.06 |
| 31  | 0.76 ± 0.02   | 0.873 ± 0.009 | 0.950 ± 0.008 | 0.923 ± 0.007 | 0.89 ± 0.01   | 0.69 ± 0.03   | 0.95 ± 0.01   | 0.90 ± 0.01   | 0.93 ± 0.01   | 0.967 ± 0.004 | 0.92 ± 0.02  | 0.93 ± 0.01   | S Asia M      | W Africa F    | 0.05 |
| 32  | 0.80 ± 0.02   | 0.903 ± 0.008 | 0.970 ± 0.007 | 0.941 ± 0.006 | 0.92 ± 0.01   | 0.74 ± 0.02   | 0.95 ± 0.01   | 0.917 ± 0.009 | 0.96 ± 0.01   | 0.977 ± 0.004 | 0.94 ± 0.01  | 0.94 ± 0.01   | S Asia M      | W Africa F    | 0.04 |
| 33  | 0.81 ± 0.02   | 0.928 ± 0.007 | 0.978 ± 0.006 | 0.957 ± 0.005 | 0.93 ± 0.01   | 0.78 ± 0.02   | 0.968 ± 0.010 | 0.943 ± 0.008 | 0.978 ± 0.009 | 0.982 ± 0.003 | 0.96 ± 0.01  | 0.961 ± 0.010 | S Asia M      | W Africa F    | 0.04 |
| 34  | 0.86 ± 0.02   | 0.949 ± 0.006 | 0.981 ± 0.006 | 0.967 ± 0.005 | 0.952 ± 0.009 | 0.81 ± 0.02   | 0.979 ± 0.008 | 0.955 ± 0.007 | 0.981 ± 0.006 | 0.986 ± 0.003 | 0.97 ± 0.01  | 0.969 ± 0.010 | S Asia M      | W Africa F    | 0.03 |

**Table 128.** For algorithm dermalog-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.21 ± 0.06   | 0.16 ± 0.02   | 0.15 ± 0.03   | 0.21 ± 0.03   | 0.19 ± 0.03   | 0.13 ± 0.05   | 0.15 ± 0.06   | 0.064 ± 0.010 | 0.06 ± 0.02   | 0.10 ± 0.02   | 0.07 ± 0.02   | 0.15 ± 0.06   | E Africa F    | E Europe M    | 0.24 |
| 15  | 0.24 ± 0.06   | 0.20 ± 0.02   | 0.22 ± 0.04   | 0.32 ± 0.03   | 0.22 ± 0.04   | 0.24 ± 0.06   | 0.26 ± 0.07   | 0.10 ± 0.01   | 0.08 ± 0.02   | 0.16 ± 0.02   | 0.10 ± 0.02   | 0.21 ± 0.06   | S Asia F      | E Europe M    | 0.21 |
| 16  | 0.35 ± 0.06   | 0.27 ± 0.02   | 0.27 ± 0.03   | 0.38 ± 0.03   | 0.34 ± 0.04   | 0.23 ± 0.06   | 0.30 ± 0.06   | 0.15 ± 0.01   | 0.10 ± 0.02   | 0.22 ± 0.02   | 0.15 ± 0.03   | 0.31 ± 0.06   | S Asia F      | E Europe M    | 0.20 |
| 17  | 0.37 ± 0.06   | 0.34 ± 0.01   | 0.34 ± 0.03   | 0.44 ± 0.02   | 0.37 ± 0.03   | 0.31 ± 0.05   | 0.43 ± 0.07   | 0.22 ± 0.01   | 0.12 ± 0.03   | 0.35 ± 0.02   | 0.22 ± 0.03   | 0.46 ± 0.06   | W Africa M    | E Europe M    | 0.17 |
| 18  | 0.43 ± 0.05   | 0.42 ± 0.01   | 0.44 ± 0.03   | 0.52 ± 0.02   | 0.45 ± 0.02   | 0.35 ± 0.05   | 0.54 ± 0.06   | 0.30 ± 0.01   | 0.23 ± 0.03   | 0.45 ± 0.02   | 0.32 ± 0.03   | 0.55 ± 0.05   | W Africa M    | E Europe M    | 0.14 |
| 19  | 0.50 ± 0.05   | 0.49 ± 0.01   | 0.51 ± 0.02   | 0.60 ± 0.02   | 0.50 ± 0.03   | 0.43 ± 0.04   | 0.63 ± 0.05   | 0.38 ± 0.01   | 0.30 ± 0.03   | 0.56 ± 0.02   | 0.38 ± 0.03   | 0.60 ± 0.04   | E Africa M    | E Europe M    | 0.12 |
| 20  | 0.58 ± 0.04   | 0.57 ± 0.01   | 0.55 ± 0.02   | 0.66 ± 0.02   | 0.58 ± 0.02   | 0.50 ± 0.04   | 0.73 ± 0.04   | 0.47 ± 0.01   | 0.36 ± 0.03   | 0.68 ± 0.02   | 0.49 ± 0.02   | 0.71 ± 0.04   | E Africa M    | E Europe M    | 0.11 |
| 21  | 0.66 ± 0.04   | 0.63 ± 0.01   | 0.61 ± 0.02   | 0.71 ± 0.01   | 0.62 ± 0.02   | 0.57 ± 0.04   | 0.80 ± 0.03   | 0.56 ± 0.01   | 0.47 ± 0.03   | 0.76 ± 0.01   | 0.60 ± 0.02   | 0.76 ± 0.03   | E Africa M    | E Europe M    | 0.09 |
| 22  | 0.66 ± 0.03   | 0.689 ± 0.010 | 0.66 ± 0.02   | 0.77 ± 0.01   | 0.69 ± 0.02   | 0.63 ± 0.03   | 0.86 ± 0.03   | 0.64 ± 0.01   | 0.56 ± 0.03   | 0.82 ± 0.01   | 0.67 ± 0.03   | 0.82 ± 0.03   | E Africa M    | E Europe M    | 0.07 |
| 23  | 0.73 ± 0.03   | 0.738 ± 0.008 | 0.72 ± 0.01   | 0.811 ± 0.010 | 0.75 ± 0.01   | 0.65 ± 0.03   | 0.91 ± 0.02   | 0.72 ± 0.01   | 0.66 ± 0.02   | 0.866 ± 0.010 | 0.72 ± 0.02   | 0.88 ± 0.02   | E Africa M    | W Africa F    | 0.06 |
| 24  | 0.78 ± 0.03   | 0.780 ± 0.007 | 0.76 ± 0.01   | 0.845 ± 0.008 | 0.80 ± 0.01   | 0.71 ± 0.03   | 0.91 ± 0.02   | 0.773 ± 0.009 | 0.74 ± 0.02   | 0.909 ± 0.008 | 0.78 ± 0.02   | 0.90 ± 0.02   | E Africa M    | W Africa F    | 0.05 |
| 25  | 0.79 ± 0.02   | 0.829 ± 0.006 | 0.81 ± 0.01   | 0.863 ± 0.007 | 0.84 ± 0.01   | 0.77 ± 0.02   | 0.94 ± 0.02   | 0.842 ± 0.009 | 0.80 ± 0.02   | 0.934 ± 0.006 | 0.84 ± 0.02   | 0.95 ± 0.01   | W Africa M    | W Africa F    | 0.04 |
| 26  | 0.83 ± 0.02   | 0.864 ± 0.005 | 0.834 ± 0.009 | 0.883 ± 0.006 | 0.87 ± 0.01   | 0.80 ± 0.02   | 0.96 ± 0.01   | 0.874 ± 0.007 | 0.85 ± 0.01   | 0.947 ± 0.004 | 0.88 ± 0.02   | 0.95 ± 0.01   | E Africa M    | W Africa F    | 0.03 |
| 27  | 0.85 ± 0.02   | 0.888 ± 0.004 | 0.868 ± 0.008 | 0.904 ± 0.005 | 0.897 ± 0.010 | 0.83 ± 0.02   | 0.974 ± 0.010 | 0.901 ± 0.006 | 0.90 ± 0.01   | 0.963 ± 0.004 | 0.91 ± 0.01   | 0.966 ± 0.009 | E Africa M    | W Africa F    | 0.03 |
| 28  | 0.88 ± 0.02   | 0.915 ± 0.004 | 0.907 ± 0.007 | 0.921 ± 0.004 | 0.914 ± 0.009 | 0.83 ± 0.02   | 0.984 ± 0.008 | 0.928 ± 0.006 | 0.935 ± 0.008 | 0.973 ± 0.003 | 0.94 ± 0.01   | 0.974 ± 0.007 | E Africa M    | W Africa F    | 0.03 |
| 29  | 0.91 ± 0.02   | 0.932 ± 0.004 | 0.925 ± 0.006 | 0.932 ± 0.004 | 0.932 ± 0.008 | 0.88 ± 0.01   | 0.990 ± 0.006 | 0.951 ± 0.005 | 0.946 ± 0.007 | 0.979 ± 0.003 | 0.94 ± 0.01   | 0.981 ± 0.005 | E Africa M    | W Africa F    | 0.02 |
| 30  | 0.91 ± 0.02   | 0.951 ± 0.003 | 0.939 ± 0.005 | 0.946 ± 0.004 | 0.949 ± 0.007 | 0.90 ± 0.01   | 0.993 ± 0.005 | 0.963 ± 0.006 | 0.963 ± 0.006 | 0.983 ± 0.002 | 0.962 ± 0.009 | 0.984 ± 0.005 | E Africa M    | W Africa F    | 0.02 |
| 31  | 0.94 ± 0.01   | 0.963 ± 0.003 | 0.959 ± 0.005 | 0.959 ± 0.003 | 0.959 ± 0.006 | 0.92 ± 0.01   | 0.996 ± 0.004 | 0.975 ± 0.003 | 0.975 ± 0.005 | 0.987 ± 0.002 | 0.974 ± 0.008 | 0.991 ± 0.004 | E Africa M    | W Africa F    | 0.01 |
| 32  | 0.94 ± 0.01   | 0.971 ± 0.002 | 0.975 ± 0.004 | 0.966 ± 0.003 | 0.963 ± 0.006 | 0.93 ± 0.01   | 0.992 ± 0.005 | 0.982 ± 0.003 | 0.985 ± 0.004 | 0.990 ± 0.002 | 0.980 ± 0.006 | 0.992 ± 0.004 | W Africa M    | W Africa F    | 0.01 |
| 33  | 0.95 ± 0.01   | 0.979 ± 0.002 | 0.978 ± 0.004 | 0.972 ± 0.003 | 0.969 ± 0.005 | 0.96 ± 0.01   | 0.998 ± 0.003 | 0.986 ± 0.002 | 0.989 ± 0.003 | 0.992 ± 0.001 | 0.984 ± 0.006 | 0.995 ± 0.003 | E Africa M    | E Africa F    | 0.01 |
| 34  | 0.96 ± 0.01   | 0.984 ± 0.002 | 0.982 ± 0.004 | 0.981 ± 0.002 | 0.982 ± 0.004 | 0.968 ± 0.008 | 0.998 ± 0.003 | 0.990 ± 0.002 | 0.993 ± 0.002 | 0.994 ± 0.001 | 0.991 ± 0.004 | 0.994 ± 0.003 | E Africa M    | E Africa F    | 0.01 |

**Table 129.** For algorithm dermalog-002 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.04 ± 0.02   | 0.10 ± 0.01   | 0.10 ± 0.03   | 0.08 ± 0.01   | 0.12 ± 0.03   | 0.04 ± 0.02   | 0.010 ± 0.008 | 0.014 ± 0.005 | 0.011 ± 0.009 | 0.021 ± 0.007 | 0.005 ± 0.005 | 0.02 ± 0.01   | S E Asia F    | S E Asia M    | 0.52 |
| 15  | 0.03 ± 0.02   | 0.17 ± 0.02   | 0.20 ± 0.04   | 0.14 ± 0.02   | 0.18 ± 0.04   | 0.05 ± 0.03   | 0.02 ± 0.01   | 0.028 ± 0.008 | 0.02 ± 0.02   | 0.025 ± 0.009 | 0.012 ± 0.010 | 0.009 ± 0.009 | E Europe F    | W Africa M    | 0.54 |
| 16  | 0.07 ± 0.03   | 0.23 ± 0.02   | 0.24 ± 0.04   | 0.19 ± 0.03   | 0.21 ± 0.04   | 0.10 ± 0.04   | 0.04 ± 0.02   | 0.038 ± 0.009 | 0.04 ± 0.02   | 0.03 ± 0.01   | 0.04 ± 0.02   | 0.03 ± 0.02   | E Europe F    | W Africa M    | 0.45 |
| 17  | 0.11 ± 0.03   | 0.29 ± 0.02   | 0.31 ± 0.05   | 0.26 ± 0.03   | 0.30 ± 0.05   | 0.14 ± 0.03   | 0.07 ± 0.03   | 0.054 ± 0.010 | 0.06 ± 0.02   | 0.09 ± 0.02   | 0.08 ± 0.03   | 0.05 ± 0.02   | E Europe F    | W Africa M    | 0.39 |
| 18  | 0.13 ± 0.03   | 0.35 ± 0.02   | 0.39 ± 0.04   | 0.29 ± 0.02   | 0.32 ± 0.03   | 0.14 ± 0.03   | 0.12 ± 0.03   | 0.08 ± 0.01   | 0.11 ± 0.03   | 0.13 ± 0.02   | 0.11 ± 0.02   | 0.11 ± 0.03   | E Europe F    | E Asia M      | 0.33 |
| 19  | 0.17 ± 0.03   | 0.40 ± 0.02   | 0.45 ± 0.04   | 0.37 ± 0.02   | 0.43 ± 0.03   | 0.18 ± 0.03   | 0.19 ± 0.03   | 0.13 ± 0.01   | 0.15 ± 0.03   | 0.22 ± 0.02   | 0.15 ± 0.03   | 0.17 ± 0.04   | E Europe F    | E Asia M      | 0.27 |
| 20  | 0.22 ± 0.03   | 0.52 ± 0.02   | 0.53 ± 0.04   | 0.40 ± 0.02   | 0.54 ± 0.03   | 0.26 ± 0.04   | 0.26 ± 0.04   | 0.17 ± 0.02   | 0.23 ± 0.03   | 0.29 ± 0.02   | 0.24 ± 0.03   | 0.20 ± 0.04   | S E Asia F    | E Asia M      | 0.24 |
| 21  | 0.24 ± 0.03   | 0.57 ± 0.02   | 0.59 ± 0.03   | 0.48 ± 0.02   | 0.60 ± 0.02   | 0.33 ± 0.04   | 0.32 ± 0.04   | 0.24 ± 0.02   | 0.32 ± 0.04   | 0.38 ± 0.02   | 0.30 ± 0.03   | 0.31 ± 0.04   | S E Asia F    | E Africa F    | 0.20 |
| 22  | 0.29 ± 0.03   | 0.64 ± 0.02   | 0.66 ± 0.02   | 0.51 ± 0.02   | 0.64 ± 0.02   | 0.32 ± 0.04   | 0.44 ± 0.03   | 0.29 ± 0.02   | 0.46 ± 0.04   | 0.45 ± 0.02   | 0.34 ± 0.03   | 0.35 ± 0.03   | E Europe F    | E Africa F    | 0.18 |
| 23  | 0.34 ± 0.03   | 0.71 ± 0.02   | 0.74 ± 0.02   | 0.60 ± 0.02   | 0.71 ± 0.02   | 0.38 ± 0.03   | 0.47 ± 0.03   | 0.40 ± 0.02   | 0.52 ± 0.03   | 0.55 ± 0.02   | 0.41 ± 0.03   | 0.46 ± 0.04   | E Europe F    | E Africa F    | 0.15 |
| 24  | 0.39 ± 0.03   | 0.76 ± 0.01   | 0.81 ± 0.02   | 0.63 ± 0.02   | 0.77 ± 0.02   | 0.42 ± 0.03   | 0.56 ± 0.03   | 0.46 ± 0.02   | 0.60 ± 0.03   | 0.62 ± 0.02   | 0.50 ± 0.03   | 0.53 ± 0.03   | E Europe F    | E Africa F    | 0.14 |
| 25  | 0.42 ± 0.03   | 0.81 ± 0.01   | 0.83 ± 0.01   | 0.67 ± 0.01   | 0.79 ± 0.02   | 0.44 ± 0.03   | 0.63 ± 0.03   | 0.55 ± 0.02   | 0.71 ± 0.03   | 0.70 ± 0.02   | 0.57 ± 0.03   | 0.57 ± 0.03   | E Europe F    | E Africa F    | 0.13 |
| 26  | 0.46 ± 0.03   | 0.850 ± 0.010 | 0.86 ± 0.01   | 0.70 ± 0.01   | 0.83 ± 0.01   | 0.49 ± 0.03   | 0.67 ± 0.03   | 0.61 ± 0.02   | 0.75 ± 0.02   | 0.73 ± 0.01   | 0.62 ± 0.03   | 0.63 ± 0.03   | E Europe F    | E Africa F    | 0.11 |
| 27  | 0.51 ± 0.03   | 0.886 ± 0.009 | 0.90 ± 0.01   | 0.73 ± 0.01   | 0.86 ± 0.01   | 0.54 ± 0.02   | 0.74 ± 0.02   | 0.70 ± 0.02   | 0.84 ± 0.02   | 0.77 ± 0.01   | 0.73 ± 0.03   | 0.68 ± 0.02   | E Europe F    | E Africa F    | 0.10 |
| 28  | 0.56 ± 0.03   | 0.914 ± 0.007 | 0.924 ± 0.009 | 0.76 ± 0.01   | 0.89 ± 0.01   | 0.58 ± 0.03   | 0.78 ± 0.02   | 0.76 ± 0.02   | 0.88 ± 0.02   | 0.80 ± 0.01   | 0.79 ± 0.02   | 0.72 ± 0.02   | E Europe F    | E Africa F    | 0.09 |
| 29  | 0.61 ± 0.03   | 0.937 ± 0.007 | 0.949 ± 0.008 | 0.78 ± 0.01   | 0.913 ± 0.010 | 0.65 ± 0.03   | 0.80 ± 0.02   | 0.82 ± 0.01   | 0.90 ± 0.02   | 0.83 ± 0.01   | 0.82 ± 0.02   | 0.77 ± 0.02   | E Europe F    | E Africa F    | 0.07 |
| 30  | 0.64 ± 0.02   | 0.953 ± 0.005 | 0.953 ± 0.008 | 0.821 ± 0.010 | 0.936 ± 0.010 | 0.65 ± 0.02   | 0.82 ± 0.02   | 0.86 ± 0.01   | 0.93 ± 0.01   | 0.850 ± 0.010 | 0.86 ± 0.02   | 0.81 ± 0.02   | E Asia F      | E Africa F    | 0.07 |
| 31  | 0.68 ± 0.02   | 0.955 ± 0.005 | 0.964 ± 0.007 | 0.844 ± 0.009 | 0.950 ± 0.008 | 0.68 ± 0.02   | 0.86 ± 0.02   | 0.90 ± 0.01   | 0.94 ± 0.01   | 0.876 ± 0.008 | 0.88 ± 0.02   | 0.83 ± 0.02   | E Europe F    | E Africa F    | 0.06 |
| 32  | 0.71 ± 0.02   | 0.971 ± 0.004 | 0.975 ± 0.007 | 0.869 ± 0.010 | 0.961 ± 0.008 | 0.75 ± 0.02   | 0.88 ± 0.02   | 0.912 ± 0.009 | 0.971 ± 0.009 | 0.890 ± 0.008 | 0.92 ± 0.02   | 0.85 ± 0.02   | E Europe F    | E Africa F    | 0.05 |
| 33  | 0.73 ± 0.02   | 0.979 ± 0.004 | 0.980 ± 0.006 | 0.893 ± 0.008 | 0.964 ± 0.008 | 0.76 ± 0.02   | 0.90 ± 0.02   | 0.937 ± 0.009 | 0.978 ± 0.008 | 0.909 ± 0.007 | 0.95 ± 0.01   | 0.89 ± 0.02   | E Europe F    | E Africa F    | 0.05 |
| 34  | 0.77 ± 0.02   | 0.985 ± 0.003 | 0.985 ± 0.006 | 0.911 ± 0.008 | 0.979 ± 0.006 | 0.80 ± 0.02   | 0.92 ± 0.02   | 0.951 ± 0.008 | 0.985 ± 0.007 | 0.921 ± 0.008 | 0.95 ± 0.01   | 0.89 ± 0.02   | E Asia F      | E Africa F    | 0.04 |

**Table 130.** For algorithm dermalog-002 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.18 ± 0.06   | 0.22 ± 0.02   | 0.27 ± 0.04   | 0.24 ± 0.03   | 0.23 ± 0.04   | 0.17 ± 0.06   | 0.15 ± 0.06   | 0.09 ± 0.01   | 0.10 ± 0.03   | 0.13 ± 0.02   | 0.10 ± 0.03   | 0.16 ± 0.06   | E Europe F    | E Asia M      | 0.21 |
| 15  | 0.18 ± 0.05   | 0.31 ± 0.02   | 0.38 ± 0.04   | 0.26 ± 0.03   | 0.26 ± 0.04   | 0.22 ± 0.06   | 0.29 ± 0.07   | 0.13 ± 0.01   | 0.14 ± 0.03   | 0.15 ± 0.02   | 0.13 ± 0.03   | 0.28 ± 0.07   | E Europe F    | S E Asia M    | 0.21 |
| 16  | 0.33 ± 0.06   | 0.37 ± 0.02   | 0.43 ± 0.04   | 0.33 ± 0.03   | 0.38 ± 0.03   | 0.23 ± 0.05   | 0.32 ± 0.06   | 0.19 ± 0.01   | 0.20 ± 0.03   | 0.23 ± 0.02   | 0.16 ± 0.03   | 0.28 ± 0.05   | E Europe F    | S E Asia M    | 0.18 |
| 17  | 0.30 ± 0.05   | 0.45 ± 0.02   | 0.51 ± 0.03   | 0.41 ± 0.02   | 0.44 ± 0.03   | 0.34 ± 0.06   | 0.46 ± 0.07   | 0.24 ± 0.01   | 0.27 ± 0.04   | 0.32 ± 0.02   | 0.22 ± 0.03   | 0.43 ± 0.06   | E Europe F    | S E Asia M    | 0.15 |
| 18  | 0.36 ± 0.05   | 0.52 ± 0.01   | 0.58 ± 0.03   | 0.45 ± 0.02   | 0.49 ± 0.02   | 0.33 ± 0.05   | 0.50 ± 0.06   | 0.31 ± 0.01   | 0.39 ± 0.04   | 0.40 ± 0.02   | 0.31 ± 0.03   | 0.52 ± 0.05   | E Europe F    | E Asia M      | 0.13 |
| 19  | 0.43 ± 0.05   | 0.59 ± 0.01   | 0.64 ± 0.02   | 0.50 ± 0.02   | 0.57 ± 0.02   | 0.39 ± 0.04   | 0.52 ± 0.05   | 0.36 ± 0.01   | 0.50 ± 0.04   | 0.50 ± 0.02   | 0.36 ± 0.03   | 0.56 ± 0.05   | E Europe F    | E Asia M      | 0.11 |
| 20  | 0.46 ± 0.04   | 0.65 ± 0.01   | 0.70 ± 0.02   | 0.54 ± 0.02   | 0.62 ± 0.02   | 0.41 ± 0.04   | 0.65 ± 0.04   | 0.44 ± 0.01   | 0.60 ± 0.03   | 0.58 ± 0.02   | 0.46 ± 0.03   | 0.61 ± 0.04   | E Europe F    | W Africa F    | 0.10 |
| 21  | 0.49 ± 0.04   | 0.70 ± 0.01   | 0.74 ± 0.02   | 0.60 ± 0.02   | 0.66 ± 0.02   | 0.48 ± 0.04   | 0.72 ± 0.04   | 0.52 ± 0.01   | 0.69 ± 0.03   | 0.66 ± 0.02   | 0.52 ± 0.03   | 0.71 ± 0.03   | E Europe F    | W Africa F    | 0.09 |
| 22  | 0.55 ± 0.04   | 0.751 ± 0.008 | 0.78 ± 0.01   | 0.64 ± 0.01   | 0.71 ± 0.02   | 0.53 ± 0.03   | 0.76 ± 0.04   | 0.58 ± 0.01   | 0.74 ± 0.02   | 0.72 ± 0.01   | 0.59 ± 0.03   | 0.73 ± 0.03   | E Europe F    | W Africa F    | 0.08 |
| 23  | 0.58 ± 0.04   | 0.789 ± 0.007 | 0.82 ± 0.01   | 0.67 ± 0.01   | 0.78 ± 0.01   | 0.59 ± 0.03   | 0.80 ± 0.03   | 0.65 ± 0.01   | 0.81 ± 0.02   | 0.78 ± 0.01   | 0.65 ± 0.03   | 0.79 ± 0.03   | E Europe F    | E Africa F    | 0.07 |
| 24  | 0.65 ± 0.03   | 0.822 ± 0.007 | 0.85 ± 0.01   | 0.72 ± 0.01   | 0.81 ± 0.01   | 0.60 ± 0.03   | 0.85 ± 0.03   | 0.70 ± 0.01   | 0.87 ± 0.01   | 0.83 ± 0.01   | 0.71 ± 0.02   | 0.83 ± 0.02   | E Europe M    | W Africa F    | 0.07 |
| 25  | 0.71 ± 0.02   | 0.850 ± 0.006 | 0.880 ± 0.008 | 0.740 ± 0.009 | 0.84 ± 0.01   | 0.65 ± 0.02   | 0.89 ± 0.02   | 0.766 ± 0.010 | 0.89 ± 0.01   | 0.865 ± 0.008 | 0.76 ± 0.02   | 0.87 ± 0.02   | E Europe M    | W Africa F    | 0.06 |
| 26  | 0.71 ± 0.03   | 0.884 ± 0.005 | 0.900 ± 0.007 | 0.770 ± 0.008 | 0.868 ± 0.010 | 0.67 ± 0.02   | 0.93 ± 0.02   | 0.803 ± 0.010 | 0.92 ± 0.01   | 0.884 ± 0.007 | 0.80 ± 0.02   | 0.88 ± 0.02   | E Africa M    | W Africa F    | 0.06 |
| 27  | 0.74 ± 0.02   | 0.900 ± 0.004 | 0.923 ± 0.006 | 0.795 ± 0.007 | 0.900 ± 0.009 | 0.70 ± 0.02   | 0.93 ± 0.02   | 0.836 ± 0.008 | 0.941 ± 0.009 | 0.903 ± 0.006 | 0.86 ± 0.02   | 0.91 ± 0.01   | E Europe M    | W Africa F    | 0.05 |
| 28  | 0.80 ± 0.02   | 0.920 ± 0.004 | 0.942 ± 0.006 | 0.826 ± 0.007 | 0.915 ± 0.009 | 0.73 ± 0.02   | 0.94 ± 0.01   | 0.871 ± 0.007 | 0.960 ± 0.006 | 0.921 ± 0.005 | 0.89 ± 0.01   | 0.92 ± 0.01   | E Europe M    | W Africa F    | 0.04 |
| 29  | 0.83 ± 0.02   | 0.935 ± 0.004 | 0.958 ± 0.005 | 0.847 ± 0.006 | 0.923 ± 0.009 | 0.76 ± 0.02   | 0.96 ± 0.01   | 0.898 ± 0.006 | 0.960 ± 0.007 | 0.929 ± 0.005 | 0.91 ± 0.01   | 0.930 ± 0.010 | E Europe M    | W Africa F    | 0.04 |
| 30  | 0.83 ± 0.02   | 0.948 ± 0.003 | 0.964 ± 0.004 | 0.870 ± 0.005 | 0.941 ± 0.007 | 0.80 ± 0.02   | 0.96 ± 0.01   | 0.916 ± 0.006 | 0.976 ± 0.006 | 0.940 ± 0.004 | 0.92 ± 0.01   | 0.954 ± 0.009 | E Europe M    | W Africa F    | 0.03 |
| 31  | 0.85 ± 0.02   | 0.959 ± 0.003 | 0.970 ± 0.004 | 0.887 ± 0.005 | 0.948 ± 0.007 | 0.82 ± 0.02   | 0.975 ± 0.009 | 0.932 ± 0.005 | 0.982 ± 0.004 | 0.952 ± 0.003 | 0.94 ± 0.01   | 0.962 ± 0.007 | E Europe M    | W Africa F    | 0.03 |
| 32  | 0.86 ± 0.02   | 0.965 ± 0.003 | 0.977 ± 0.004 | 0.907 ± 0.004 | 0.959 ± 0.006 | 0.84 ± 0.02   | 0.971 ± 0.009 | 0.952 ± 0.004 | 0.983 ± 0.004 | 0.962 ± 0.003 | 0.95 ± 0.01   | 0.963 ± 0.007 | E Europe M    | W Africa F    | 0.03 |
| 33  | 0.89 ± 0.02   | 0.973 ± 0.002 | 0.982 ± 0.003 | 0.920 ± 0.005 | 0.962 ± 0.007 | 0.87 ± 0.02   | 0.978 ± 0.009 | 0.957 ± 0.004 | 0.988 ± 0.004 | 0.968 ± 0.003 | 0.954 ± 0.010 | 0.967 ± 0.007 | E Europe M    | W Africa F    | 0.02 |
| 34  | 0.90 ± 0.02   | 0.977 ± 0.002 | 0.982 ± 0.004 | 0.931 ± 0.004 | 0.974 ± 0.005 | 0.87 ± 0.02   | 0.979 ± 0.008 | 0.965 ± 0.004 | 0.992 ± 0.003 | 0.972 ± 0.002 | 0.967 ± 0.008 | 0.979 ± 0.006 | E Europe M    | W Africa F    | 0.02 |

**Table 131.** For algorithm hzailu-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.24 ± 0.04   | 0.19 ± 0.02   | 0.41 ± 0.04   | 0.25 ± 0.02   | 0.21 ± 0.04   | 0.38 ± 0.06   | 0.21 ± 0.03   | 0.12 ± 0.01   | 0.17 ± 0.03   | 0.29 ± 0.02   | 0.18 ± 0.03   | 0.30 ± 0.06   | E Europe F    | E Asia M      | 0.20 |
| 15  | 0.29 ± 0.05   | 0.27 ± 0.02   | 0.51 ± 0.05   | 0.38 ± 0.03   | 0.29 ± 0.04   | 0.41 ± 0.06   | 0.26 ± 0.04   | 0.21 ± 0.02   | 0.27 ± 0.05   | 0.44 ± 0.03   | 0.28 ± 0.04   | 0.42 ± 0.07   | E Europe F    | E Asia M      | 0.16 |
| 16  | 0.33 ± 0.05   | 0.33 ± 0.02   | 0.56 ± 0.05   | 0.43 ± 0.03   | 0.43 ± 0.05   | 0.52 ± 0.06   | 0.51 ± 0.05   | 0.25 ± 0.02   | 0.38 ± 0.05   | 0.58 ± 0.03   | 0.46 ± 0.05   | 0.56 ± 0.06   | S Asia M      | E Asia M      | 0.14 |
| 17  | 0.39 ± 0.05   | 0.40 ± 0.02   | 0.66 ± 0.05   | 0.51 ± 0.03   | 0.48 ± 0.06   | 0.61 ± 0.05   | 0.55 ± 0.06   | 0.34 ± 0.02   | 0.44 ± 0.05   | 0.68 ± 0.03   | 0.53 ± 0.05   | 0.67 ± 0.05   | S Asia M      | E Asia M      | 0.13 |
| 18  | 0.46 ± 0.04   | 0.47 ± 0.02   | 0.73 ± 0.04   | 0.55 ± 0.02   | 0.51 ± 0.03   | 0.63 ± 0.04   | 0.70 ± 0.04   | 0.40 ± 0.02   | 0.55 ± 0.04   | 0.77 ± 0.02   | 0.57 ± 0.03   | 0.74 ± 0.04   | S Asia M      | E Asia M      | 0.12 |
| 19  | 0.53 ± 0.04   | 0.52 ± 0.02   | 0.78 ± 0.03   | 0.64 ± 0.02   | 0.61 ± 0.03   | 0.66 ± 0.04   | 0.78 ± 0.03   | 0.48 ± 0.02   | 0.66 ± 0.04   | 0.82 ± 0.02   | 0.64 ± 0.03   | 0.84 ± 0.04   | W Africa M    | E Asia M      | 0.11 |
| 20  | 0.56 ± 0.04   | 0.60 ± 0.02   | 0.83 ± 0.02   | 0.67 ± 0.02   | 0.66 ± 0.03   | 0.71 ± 0.04   | 0.85 ± 0.03   | 0.54 ± 0.02   | 0.71 ± 0.04   | 0.88 ± 0.01   | 0.74 ± 0.03   | 0.86 ± 0.03   | S Asia M      | E Asia M      | 0.10 |
| 21  | 0.60 ± 0.04   | 0.66 ± 0.02   | 0.87 ± 0.02   | 0.71 ± 0.02   | 0.72 ± 0.02   | 0.74 ± 0.04   | 0.88 ± 0.03   | 0.62 ± 0.02   | 0.83 ± 0.03   | 0.93 ± 0.01   | 0.81 ± 0.03   | 0.90 ± 0.03   | S Asia M      | E Africa F    | 0.09 |
| 22  | 0.62 ± 0.03   | 0.71 ± 0.02   | 0.89 ± 0.02   | 0.74 ± 0.02   | 0.76 ± 0.02   | 0.78 ± 0.03   | 0.92 ± 0.02   | 0.67 ± 0.02   | 0.86 ± 0.03   | 0.94 ± 0.01   | 0.83 ± 0.02   | 0.93 ± 0.02   | S Asia M      | E Africa F    | 0.08 |
| 23  | 0.65 ± 0.03   | 0.77 ± 0.02   | 0.92 ± 0.01   | 0.78 ± 0.02   | 0.82 ± 0.02   | 0.78 ± 0.03   | 0.92 ± 0.02   | 0.72 ± 0.02   | 0.88 ± 0.03   | 0.951 ± 0.010 | 0.86 ± 0.02   | 0.93 ± 0.02   | S Asia M      | E Africa F    | 0.07 |
| 24  | 0.69 ± 0.03   | 0.79 ± 0.01   | 0.93 ± 0.01   | 0.82 ± 0.01   | 0.85 ± 0.02   | 0.80 ± 0.03   | 0.93 ± 0.02   | 0.78 ± 0.02   | 0.93 ± 0.02   | 0.961 ± 0.008 | 0.88 ± 0.02   | 0.96 ± 0.01   | S Asia M      | E Africa F    | 0.06 |
| 25  | 0.73 ± 0.03   | 0.84 ± 0.01   | 0.94 ± 0.01   | 0.83 ± 0.01   | 0.87 ± 0.01   | 0.81 ± 0.02   | 0.95 ± 0.01   | 0.82 ± 0.02   | 0.96 ± 0.01   | 0.971 ± 0.006 | 0.91 ± 0.02   | 0.95 ± 0.01   | S Asia M      | E Africa F    | 0.05 |
| 26  | 0.74 ± 0.02   | 0.859 ± 0.010 | 0.960 ± 0.008 | 0.851 ± 0.010 | 0.89 ± 0.01   | 0.85 ± 0.02   | 0.96 ± 0.01   | 0.87 ± 0.01   | 0.965 ± 0.010 | 0.978 ± 0.005 | 0.93 ± 0.01   | 0.96 ± 0.01   | S Asia M      | E Africa F    | 0.04 |
| 27  | 0.78 ± 0.02   | 0.895 ± 0.008 | 0.969 ± 0.007 | 0.862 ± 0.009 | 0.91 ± 0.01   | 0.86 ± 0.02   | 0.964 ± 0.009 | 0.89 ± 0.01   | 0.980 ± 0.007 | 0.981 ± 0.004 | 0.95 ± 0.01   | 0.967 ± 0.010 | S Asia M      | E Africa F    | 0.04 |
| 28  | 0.79 ± 0.02   | 0.913 ± 0.007 | 0.976 ± 0.006 | 0.868 ± 0.010 | 0.933 ± 0.009 | 0.86 ± 0.02   | 0.96 ± 0.01   | 0.92 ± 0.01   | 0.987 ± 0.006 | 0.983 ± 0.004 | 0.96 ± 0.01   | 0.966 ± 0.009 | E Europe M    | E Africa F    | 0.04 |
| 29  | 0.84 ± 0.02   | 0.932 ± 0.007 | 0.979 ± 0.005 | 0.893 ± 0.008 | 0.942 ± 0.008 | 0.89 ± 0.02   | 0.971 ± 0.009 | 0.953 ± 0.008 | 0.990 ± 0.006 | 0.988 ± 0.003 | 0.973 ± 0.009 | 0.979 ± 0.007 | E Europe M    | E Africa F    | 0.03 |
| 30  | 0.84 ± 0.02   | 0.946 ± 0.006 | 0.980 ± 0.006 | 0.896 ± 0.008 | 0.952 ± 0.009 | 0.90 ± 0.01   | 0.981 ± 0.007 | 0.957 ± 0.007 | 0.996 ± 0.004 | 0.991 ± 0.002 | 0.984 ± 0.008 | 0.981 ± 0.006 | E Europe M    | E Africa F    | 0.03 |
| 31  | 0.86 ± 0.02   | 0.960 ± 0.005 | 0.988 ± 0.004 | 0.900 ± 0.008 | 0.961 ± 0.007 | 0.90 ± 0.01   | 0.988 ± 0.006 | 0.969 ± 0.006 | 0.992 ± 0.005 | 0.993 ± 0.002 | 0.983 ± 0.008 | 0.984 ± 0.006 | S Asia M      | E Africa F    | 0.03 |
| 32  | 0.87 ± 0.02   | 0.966 ± 0.005 | 0.985 ± 0.006 | 0.914 ± 0.007 | 0.968 ± 0.007 | 0.91 ± 0.01   | 0.986 ± 0.007 | 0.975 ± 0.006 | 0.998 ± 0.002 | 0.993 ± 0.002 | 0.982 ± 0.009 | 0.990 ± 0.005 | E Europe M    | E Africa F    | 0.02 |
| 33  | 0.87 ± 0.02   | 0.975 ± 0.004 | 0.988 ± 0.005 | 0.921 ± 0.007 | 0.978 ± 0.006 | 0.92 ± 0.01   | 0.987 ± 0.006 | 0.981 ± 0.005 | 0.997 ± 0.003 | 0.995 ± 0.002 | 0.991 ± 0.006 | 0.989 ± 0.005 | E Europe M    | E Africa F    | 0.02 |
| 34  | 0.90 ± 0.02   | 0.984 ± 0.003 | 0.991 ± 0.004 | 0.931 ± 0.007 | 0.979 ± 0.006 | 0.94 ± 0.01   | 0.989 ± 0.007 | 0.988 ± 0.004 | 0.999 ± 0.002 | 0.996 ± 0.002 | 0.991 ± 0.006 | 0.989 ± 0.005 | E Europe M    | E Africa F    | 0.02 |

**Table 132.** For algorithm hzailu-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.32 ± 0.07   | 0.20 ± 0.02   | 0.46 ± 0.05   | 0.36 ± 0.03   | 0.23 ± 0.04   | 0.48 ± 0.08   | 0.41 ± 0.07   | 0.17 ± 0.01   | 0.26 ± 0.04   | 0.42 ± 0.03   | 0.24 ± 0.04   | 0.49 ± 0.09   | W Africa M    | E Asia M      | 0.20 |
| 15  | 0.39 ± 0.07   | 0.28 ± 0.02   | 0.56 ± 0.04   | 0.43 ± 0.03   | 0.31 ± 0.04   | 0.62 ± 0.07   | 0.62 ± 0.07   | 0.25 ± 0.01   | 0.43 ± 0.04   | 0.58 ± 0.03   | 0.30 ± 0.04   | 0.66 ± 0.08   | W Africa M    | E Asia M      | 0.20 |
| 16  | 0.51 ± 0.06   | 0.32 ± 0.02   | 0.61 ± 0.04   | 0.50 ± 0.03   | 0.38 ± 0.03   | 0.58 ± 0.07   | 0.70 ± 0.07   | 0.33 ± 0.02   | 0.52 ± 0.04   | 0.65 ± 0.02   | 0.40 ± 0.04   | 0.73 ± 0.06   | W Africa M    | E Asia F      | 0.16 |
| 17  | 0.50 ± 0.07   | 0.39 ± 0.02   | 0.70 ± 0.03   | 0.57 ± 0.03   | 0.39 ± 0.03   | 0.65 ± 0.06   | 0.82 ± 0.05   | 0.41 ± 0.02   | 0.66 ± 0.04   | 0.76 ± 0.02   | 0.49 ± 0.03   | 0.86 ± 0.04   | W Africa M    | E Asia F      | 0.17 |
| 18  | 0.56 ± 0.05   | 0.43 ± 0.01   | 0.74 ± 0.03   | 0.63 ± 0.02   | 0.46 ± 0.03   | 0.63 ± 0.05   | 0.88 ± 0.04   | 0.45 ± 0.01   | 0.73 ± 0.03   | 0.83 ± 0.02   | 0.57 ± 0.03   | 0.90 ± 0.03   | W Africa M    | E Asia F      | 0.15 |
| 19  | 0.64 ± 0.04   | 0.50 ± 0.01   | 0.81 ± 0.02   | 0.67 ± 0.02   | 0.49 ± 0.02   | 0.68 ± 0.04   | 0.87 ± 0.03   | 0.53 ± 0.01   | 0.80 ± 0.03   | 0.87 ± 0.01   | 0.63 ± 0.03   | 0.93 ± 0.03   | W Africa M    | S E Asia F    | 0.13 |
| 20  | 0.61 ± 0.04   | 0.57 ± 0.01   | 0.83 ± 0.02   | 0.72 ± 0.02   | 0.58 ± 0.02   | 0.74 ± 0.03   | 0.93 ± 0.02   | 0.60 ± 0.01   | 0.85 ± 0.02   | 0.90 ± 0.01   | 0.71 ± 0.02   | 0.94 ± 0.02   | W Africa M    | E Asia F      | 0.11 |
| 21  | 0.71 ± 0.04   | 0.613 ± 0.010 | 0.85 ± 0.01   | 0.75 ± 0.01   | 0.63 ± 0.02   | 0.76 ± 0.03   | 0.94 ± 0.02   | 0.66 ± 0.01   | 0.90 ± 0.02   | 0.934 ± 0.009 | 0.78 ± 0.02   | 0.96 ± 0.02   | W Africa M    | E Asia F      | 0.09 |
| 22  | 0.74 ± 0.03   | 0.660 ± 0.009 | 0.88 ± 0.01   | 0.79 ± 0.01   | 0.66 ± 0.02   | 0.79 ± 0.03   | 0.95 ± 0.02   | 0.72 ± 0.01   | 0.93 ± 0.01   | 0.950 ± 0.006 | 0.80 ± 0.02   | 0.97 ± 0.01   | W Africa M    | E Asia F      | 0.08 |
| 23  | 0.75 ± 0.03   | 0.702 ± 0.008 | 0.89 ± 0.01   | 0.82 ± 0.01   | 0.74 ± 0.02   | 0.79 ± 0.02   | 0.96 ± 0.01   | 0.76 ± 0.01   | 0.95 ± 0.01   | 0.959 ± 0.005 | 0.83 ± 0.02   | 0.97 ± 0.01   | W Africa M    | E Asia F      | 0.07 |
| 24  | 0.76 ± 0.03   | 0.735 ± 0.008 | 0.915 ± 0.008 | 0.832 ± 0.009 | 0.77 ± 0.01   | 0.82 ± 0.02   | 0.97 ± 0.01   | 0.800 ± 0.010 | 0.972 ± 0.007 | 0.974 ± 0.004 | 0.88 ± 0.02   | 0.976 ± 0.010 | W Africa M    | E Asia F      | 0.06 |
| 25  | 0.80 ± 0.03   | 0.779 ± 0.007 | 0.936 ± 0.007 | 0.842 ± 0.008 | 0.81 ± 0.01   | 0.83 ± 0.02   | 0.98 ± 0.01   | 0.841 ± 0.009 | 0.973 ± 0.007 | 0.978 ± 0.003 | 0.89 ± 0.02   | 0.989 ± 0.006 | W Africa M    | E Asia F      | 0.05 |
| 26  | 0.83 ± 0.02   | 0.812 ± 0.006 | 0.943 ± 0.006 | 0.859 ± 0.007 | 0.82 ± 0.01   | 0.85 ± 0.02   | 0.983 ± 0.009 | 0.868 ± 0.007 | 0.979 ± 0.006 | 0.982 ± 0.003 | 0.91 ± 0.01   | 0.985 ± 0.006 | W Africa M    | E Asia F      | 0.04 |
| 27  | 0.83 ± 0.02   | 0.838 ± 0.006 | 0.950 ± 0.005 | 0.868 ± 0.006 | 0.87 ± 0.01   | 0.86 ± 0.02   | 0.980 ± 0.009 | 0.887 ± 0.007 | 0.984 ± 0.005 | 0.984 ± 0.003 | 0.93 ± 0.01   | 0.986 ± 0.006 | W Africa M    | E Africa F    | 0.04 |
| 28  | 0.85 ± 0.02   | 0.863 ± 0.005 | 0.954 ± 0.005 | 0.877 ± 0.006 | 0.88 ± 0.01   | 0.87 ± 0.01   | 0.988 ± 0.007 | 0.913 ± 0.006 | 0.992 ± 0.003 | 0.986 ± 0.002 | 0.948 ± 0.010 | 0.987 ± 0.006 | E Europe M    | E Africa F    | 0.04 |
| 29  | 0.89 ± 0.02   | 0.887 ± 0.005 | 0.965 ± 0.005 | 0.886 ± 0.005 | 0.896 ± 0.010 | 0.86 ± 0.02   | 0.982 ± 0.008 | 0.929 ± 0.005 | 0.990 ± 0.003 | 0.991 ± 0.002 | 0.958 ± 0.010 | 0.987 ± 0.005 | S Asia M      | W Africa F    | 0.03 |
| 30  | 0.87 ± 0.02   | 0.909 ± 0.004 | 0.969 ± 0.004 | 0.897 ± 0.005 | 0.910 ± 0.009 | 0.88 ± 0.02   | 0.989 ± 0.007 | 0.941 ± 0.005 | 0.992 ± 0.003 | 0.990 ± 0.002 | 0.957 ± 0.009 | 0.993 ± 0.003 | W Africa M    | E Africa F    | 0.03 |
| 31  | 0.88 ± 0.02   | 0.926 ± 0.004 | 0.971 ± 0.004 | 0.905 ± 0.005 | 0.922 ± 0.009 | 0.89 ± 0.01   | 0.993 ± 0.005 | 0.955 ± 0.004 | 0.994 ± 0.003 | 0.992 ± 0.001 | 0.968 ± 0.009 | 0.995 ± 0.003 | W Africa M    | E Africa F    | 0.03 |
| 32  | 0.89 ± 0.02   | 0.938 ± 0.003 | 0.980 ± 0.003 | 0.915 ± 0.005 | 0.936 ± 0.007 | 0.90 ± 0.01   | 0.993 ± 0.005 | 0.968 ± 0.003 | 0.996 ± 0.002 | 0.993 ± 0.001 | 0.973 ± 0.007 | 0.991 ± 0.004 | E Europe M    | E Africa F    | 0.02 |
| 33  | 0.92 ± 0.02   | 0.952 ± 0.003 | 0.976 ± 0.004 | 0.926 ± 0.004 | 0.946 ± 0.007 | 0.91 ± 0.01   | 0.993 ± 0.005 | 0.974 ± 0.003 | 0.995 ± 0.002 | 0.994 ± 0.001 | 0.979 ± 0.006 | 0.996 ± 0.002 | W Africa M    | W Africa F    | 0.02 |
| 34  | 0.92 ± 0.01   | 0.957 ± 0.003 | 0.981 ± 0.004 | 0.932 ± 0.004 | 0.951 ± 0.007 | 0.92 ± 0.01   | 0.992 ± 0.005 | 0.978 ± 0.003 | 0.997 ± 0.002 | 0.995 ± 0.001 | 0.985 ± 0.006 | 0.993 ± 0.004 | E Europe M    | E Africa F    | 0.02 |

**Table 133.** For algorithm idemia-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.004 ± 0.004 | 0.000 ± 0.000 | 0.004 ± 0.004 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.000 ± 0.000 | S E Asia F    | E Africa F    | 0.77 |
| 15  | 0.009 ± 0.009 | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.006 ± 0.004 | 0.01 ± 0.01   | 0.04 ± 0.02   | 0.005 ± 0.005 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.005 | W Africa F    | E Europe F    | 0.70 |
| 16  | 0.02 ± 0.01   | 0.007 ± 0.004 | 0.000 ± 0.000 | 0.004 ± 0.004 | 0.03 ± 0.02   | 0.05 ± 0.03   | 0.02 ± 0.01   | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.005 ± 0.005 | W Africa F    | E Europe F    | 0.67 |
| 17  | 0.04 ± 0.02   | 0.023 ± 0.007 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.05 ± 0.02   | 0.09 ± 0.03   | 0.01 ± 0.01   | 0.007 ± 0.004 | 0.000 ± 0.000 | 0.009 ± 0.005 | 0.006 ± 0.006 | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.55 |
| 18  | 0.08 ± 0.02   | 0.030 ± 0.007 | 0.014 ± 0.009 | 0.031 ± 0.009 | 0.04 ± 0.01   | 0.11 ± 0.03   | 0.04 ± 0.02   | 0.019 ± 0.005 | 0.000 ± 0.000 | 0.031 ± 0.008 | 0.02 ± 0.01   | 0.06 ± 0.02   | W Africa F    | E Europe M    | 0.42 |
| 19  | 0.12 ± 0.03   | 0.055 ± 0.010 | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.11 ± 0.02   | 0.17 ± 0.03   | 0.12 ± 0.03   | 0.048 ± 0.009 | 0.009 ± 0.008 | 0.07 ± 0.01   | 0.04 ± 0.01   | 0.12 ± 0.03   | W Africa F    | E Europe M    | 0.37 |
| 20  | 0.19 ± 0.03   | 0.10 ± 0.01   | 0.03 ± 0.01   | 0.11 ± 0.01   | 0.18 ± 0.02   | 0.22 ± 0.03   | 0.22 ± 0.03   | 0.09 ± 0.01   | 0.03 ± 0.01   | 0.13 ± 0.02   | 0.10 ± 0.02   | 0.18 ± 0.04   | W Africa F    | E Europe M    | 0.30 |
| 21  | 0.26 ± 0.03   | 0.16 ± 0.01   | 0.04 ± 0.01   | 0.16 ± 0.02   | 0.23 ± 0.02   | 0.31 ± 0.04   | 0.31 ± 0.04   | 0.17 ± 0.02   | 0.04 ± 0.01   | 0.25 ± 0.02   | 0.21 ± 0.03   | 0.26 ± 0.04   | W Africa F    | E Europe F    | 0.26 |
| 22  | 0.33 ± 0.03   | 0.23 ± 0.02   | 0.07 ± 0.01   | 0.22 ± 0.02   | 0.33 ± 0.02   | 0.34 ± 0.04   | 0.47 ± 0.04   | 0.26 ± 0.02   | 0.13 ± 0.02   | 0.36 ± 0.02   | 0.30 ± 0.03   | 0.32 ± 0.04   | E Africa M    | E Europe F    | 0.22 |
| 23  | 0.40 ± 0.03   | 0.34 ± 0.02   | 0.11 ± 0.01   | 0.31 ± 0.02   | 0.45 ± 0.02   | 0.44 ± 0.03   | 0.53 ± 0.03   | 0.37 ± 0.02   | 0.20 ± 0.03   | 0.52 ± 0.02   | 0.44 ± 0.03   | 0.47 ± 0.03   | E Africa M    | E Europe F    | 0.19 |
| 24  | 0.49 ± 0.03   | 0.43 ± 0.01   | 0.15 ± 0.02   | 0.40 ± 0.01   | 0.55 ± 0.02   | 0.51 ± 0.03   | 0.64 ± 0.03   | 0.51 ± 0.02   | 0.35 ± 0.03   | 0.66 ± 0.02   | 0.56 ± 0.03   | 0.56 ± 0.03   | S Asia M      | E Europe F    | 0.16 |
| 25  | 0.53 ± 0.03   | 0.55 ± 0.01   | 0.22 ± 0.02   | 0.48 ± 0.01   | 0.63 ± 0.02   | 0.57 ± 0.03   | 0.76 ± 0.02   | 0.66 ± 0.02   | 0.50 ± 0.03   | 0.78 ± 0.02   | 0.71 ± 0.03   | 0.68 ± 0.03   | S Asia M      | E Europe F    | 0.15 |
| 26  | 0.62 ± 0.03   | 0.63 ± 0.01   | 0.29 ± 0.02   | 0.55 ± 0.01   | 0.74 ± 0.02   | 0.67 ± 0.03   | 0.83 ± 0.02   | 0.77 ± 0.02   | 0.65 ± 0.03   | 0.86 ± 0.01   | 0.81 ± 0.02   | 0.76 ± 0.03   | S Asia M      | E Europe F    | 0.13 |
| 27  | 0.69 ± 0.02   | 0.73 ± 0.01   | 0.39 ± 0.02   | 0.63 ± 0.01   | 0.81 ± 0.01   | 0.74 ± 0.02   | 0.87 ± 0.02   | 0.85 ± 0.01   | 0.78 ± 0.02   | 0.916 ± 0.009 | 0.90 ± 0.02   | 0.81 ± 0.02   | S Asia M      | E Europe F    | 0.10 |
| 28  | 0.76 ± 0.02   | 0.80 ± 0.01   | 0.49 ± 0.02   | 0.71 ± 0.01   | 0.85 ± 0.01   | 0.81 ± 0.02   | 0.93 ± 0.01   | 0.92 ± 0.01   | 0.86 ± 0.02   | 0.953 ± 0.006 | 0.94 ± 0.01   | 0.87 ± 0.02   | S Asia M      | E Europe F    | 0.08 |
| 29  | 0.83 ± 0.02   | 0.869 ± 0.009 | 0.61 ± 0.02   | 0.779 ± 0.010 | 0.91 ± 0.01   | 0.85 ± 0.02   | 0.96 ± 0.01   | 0.953 ± 0.008 | 0.92 ± 0.01   | 0.975 ± 0.005 | 0.95 ± 0.01   | 0.93 ± 0.01   | S Asia M      | E Europe F    | 0.06 |
| 30  | 0.87 ± 0.02   | 0.926 ± 0.007 | 0.71 ± 0.02   | 0.838 ± 0.009 | 0.943 ± 0.009 | 0.86 ± 0.02   | 0.978 ± 0.008 | 0.975 ± 0.005 | 0.964 ± 0.009 | 0.986 ± 0.003 | 0.982 ± 0.008 | 0.95 ± 0.01   | S Asia M      | E Europe F    | 0.05 |
| 31  | 0.90 ± 0.02   | 0.953 ± 0.006 | 0.80 ± 0.02   | 0.877 ± 0.008 | 0.968 ± 0.007 | 0.93 ± 0.01   | 0.986 ± 0.007 | 0.987 ± 0.004 | 0.979 ± 0.008 | 0.991 ± 0.002 | 0.989 ± 0.007 | 0.974 ± 0.008 | S Asia M      | E Europe F    | 0.03 |
| 32  | 0.94 ± 0.01   | 0.970 ± 0.004 | 0.87 ± 0.01   | 0.917 ± 0.007 | 0.968 ± 0.007 | 0.95 ± 0.01   | 0.987 ± 0.007 | 0.991 ± 0.003 | 0.989 ± 0.006 | 0.997 ± 0.001 | 0.994 ± 0.005 | 0.983 ± 0.006 | S Asia M      | E Europe F    | 0.02 |
| 33  | 0.95 ± 0.01   | 0.984 ± 0.003 | 0.92 ± 0.01   | 0.952 ± 0.006 | 0.985 ± 0.005 | 0.96 ± 0.01   | 0.996 ± 0.004 | 0.995 ± 0.003 | 0.996 ± 0.004 | 0.998 ± 0.001 | 0.997 ± 0.004 | 0.987 ± 0.007 | S Asia M      | E Europe F    | 0.01 |
| 34  | 0.96 ± 0.01   | 0.991 ± 0.002 | 0.94 ± 0.01   | 0.968 ± 0.005 | 0.991 ± 0.004 | 0.978 ± 0.007 | 0.996 ± 0.004 | 0.998 ± 0.002 | 0.997 ± 0.003 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.994 ± 0.004 | S E Asia M    | E Europe F    | 0.01 |

**Table 134.** For algorithm idemia-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.03 ± 0.02   | 0.002 ± 0.001 | 0.000 ± 0.000 | 0.007 ± 0.005 | 0.000 ± 0.000 | 0.007 ± 0.007 | 0.03 ± 0.02   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.01 ± 0.01   | E Africa M    | E Europe F    | 0.76 |
| 15  | 0.03 ± 0.02   | 0.002 ± 0.001 | 0.000 ± 0.000 | 0.008 ± 0.005 | 0.000 ± 0.000 | 0.09 ± 0.04   | 0.04 ± 0.02   | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.05 ± 0.03   | W Africa F    | E Europe F    | 0.78 |
| 16  | 0.06 ± 0.03   | 0.006 ± 0.003 | 0.003 ± 0.003 | 0.021 ± 0.008 | 0.010 ± 0.007 | 0.06 ± 0.03   | 0.04 ± 0.03   | 0.004 ± 0.002 | 0.002 ± 0.002 | 0.001 ± 0.001 | 0.005 ± 0.005 | 0.06 ± 0.03   | E Africa F    | S Asia M      | 0.61 |
| 17  | 0.10 ± 0.04   | 0.017 ± 0.004 | 0.010 ± 0.006 | 0.035 ± 0.009 | 0.022 ± 0.008 | 0.11 ± 0.04   | 0.11 ± 0.04   | 0.008 ± 0.003 | 0.003 ± 0.003 | 0.022 ± 0.007 | 0.009 ± 0.006 | 0.07 ± 0.03   | W Africa F    | E Europe M    | 0.55 |
| 18  | 0.17 ± 0.04   | 0.020 ± 0.004 | 0.023 ± 0.009 | 0.057 ± 0.010 | 0.028 ± 0.008 | 0.13 ± 0.03   | 0.16 ± 0.04   | 0.013 ± 0.003 | 0.008 ± 0.006 | 0.039 ± 0.008 | 0.016 ± 0.007 | 0.10 ± 0.03   | E Africa F    | E Europe M    | 0.52 |
| 19  | 0.24 ± 0.04   | 0.041 ± 0.005 | 0.033 ± 0.009 | 0.10 ± 0.01   | 0.07 ± 0.01   | 0.22 ± 0.04   | 0.26 ± 0.04   | 0.033 ± 0.005 | 0.010 ± 0.006 | 0.10 ± 0.01   | 0.05 ± 0.01   | 0.14 ± 0.03   | E Africa M    | E Europe M    | 0.47 |
| 20  | 0.30 ± 0.04   | 0.074 ± 0.006 | 0.047 ± 0.008 | 0.12 ± 0.01   | 0.12 ± 0.01   | 0.29 ± 0.04   | 0.34 ± 0.04   | 0.066 ± 0.007 | 0.028 ± 0.010 | 0.17 ± 0.01   | 0.10 ± 0.02   | 0.23 ± 0.03   | E Africa M    | E Europe M    | 0.40 |
| 21  | 0.36 ± 0.04   | 0.106 ± 0.006 | 0.071 ± 0.009 | 0.18 ± 0.01   | 0.16 ± 0.01   | 0.35 ± 0.04   | 0.39 ± 0.04   | 0.122 ± 0.009 | 0.06 ± 0.01   | 0.26 ± 0.01   | 0.18 ± 0.02   | 0.30 ± 0.03   | E Africa M    | E Europe M    | 0.33 |
| 22  | 0.46 ± 0.04   | 0.170 ± 0.007 | 0.11 ± 0.01   | 0.24 ± 0.01   | 0.24 ± 0.02   | 0.43 ± 0.04   | 0.56 ± 0.04   | 0.210 ± 0.009 | 0.11 ± 0.02   | 0.41 ± 0.02   | 0.27 ± 0.02   | 0.40 ± 0.03   | E Africa M    | E Europe M    | 0.29 |
| 23  | 0.51 ± 0.04   | 0.232 ± 0.008 | 0.15 ± 0.01   | 0.31 ± 0.01   | 0.35 ± 0.02   | 0.50 ± 0.03   | 0.63 ± 0.04   | 0.31 ± 0.01   | 0.21 ± 0.02   | 0.53 ± 0.01   | 0.36 ± 0.02   | 0.48 ± 0.03   | E Africa M    | E Europe F    | 0.23 |
| 24  | 0.57 ± 0.03   | 0.309 ± 0.008 | 0.22 ± 0.01   | 0.39 ± 0.01   | 0.45 ± 0.02   | 0.54 ± 0.03   | 0.71 ± 0.03   | 0.44 ± 0.01   | 0.33 ± 0.02   | 0.64 ± 0.01   | 0.48 ± 0.02   | 0.58 ± 0.03   | E Africa M    | E Europe F    | 0.18 |
| 25  | 0.63 ± 0.03   | 0.411 ± 0.008 | 0.30 ± 0.01   | 0.45 ± 0.01   | 0.53 ± 0.02   | 0.65 ± 0.02   | 0.81 ± 0.03   | 0.56 ± 0.01   | 0.46 ± 0.02   | 0.753 ± 0.010 | 0.61 ± 0.02   | 0.69 ± 0.02   | E Africa M    | E Europe F    | 0.15 |
| 26  | 0.72 ± 0.02   | 0.508 ± 0.008 | 0.37 ± 0.01   | 0.530 ± 0.009 | 0.63 ± 0.01   | 0.68 ± 0.02   | 0.88 ± 0.02   | 0.67 ± 0.01   | 0.60 ± 0.02   | 0.824 ± 0.008 | 0.72 ± 0.02   | 0.77 ± 0.02   | E Africa M    | E Europe F    | 0.13 |
| 27  | 0.75 ± 0.02   | 0.599 ± 0.007 | 0.46 ± 0.01   | 0.589 ± 0.009 | 0.71 ± 0.01   | 0.75 ± 0.02   | 0.91 ± 0.02   | 0.758 ± 0.009 | 0.70 ± 0.02   | 0.882 ± 0.007 | 0.79 ± 0.02   | 0.85 ± 0.02   | E Africa M    | E Europe F    | 0.10 |
| 28  | 0.81 ± 0.02   | 0.689 ± 0.007 | 0.55 ± 0.01   | 0.664 ± 0.008 | 0.78 ± 0.01   | 0.79 ± 0.02   | 0.94 ± 0.01   | 0.829 ± 0.008 | 0.80 ± 0.01   | 0.928 ± 0.005 | 0.87 ± 0.01   | 0.88 ± 0.02   | E Africa M    | E Europe F    | 0.08 |
| 29  | 0.83 ± 0.02   | 0.760 ± 0.007 | 0.65 ± 0.01   | 0.718 ± 0.008 | 0.83 ± 0.01   | 0.85 ± 0.02   | 0.95 ± 0.01   | 0.887 ± 0.007 | 0.86 ± 0.01   | 0.957 ± 0.004 | 0.91 ± 0.01   | 0.93 ± 0.01   | S Asia M      | E Europe F    | 0.07 |
| 30  | 0.87 ± 0.02   | 0.832 ± 0.005 | 0.72 ± 0.01   | 0.785 ± 0.007 | 0.89 ± 0.01   | 0.88 ± 0.01   | 0.979 ± 0.008 | 0.930 ± 0.005 | 0.922 ± 0.009 | 0.972 ± 0.003 | 0.956 ± 0.009 | 0.954 ± 0.008 | E Africa M    | E Europe F    | 0.05 |
| 31  | 0.91 ± 0.02   | 0.886 ± 0.005 | 0.799 ± 0.010 | 0.839 ± 0.006 | 0.915 ± 0.009 | 0.91 ± 0.01   | 0.985 ± 0.007 | 0.958 ± 0.004 | 0.955 ± 0.007 | 0.986 ± 0.002 | 0.963 ± 0.009 | 0.963 ± 0.007 | S Asia M      | E Europe F    | 0.04 |
| 32  | 0.94 ± 0.01   | 0.926 ± 0.004 | 0.864 ± 0.008 | 0.880 ± 0.006 | 0.949 ± 0.007 | 0.92 ± 0.01   | 0.992 ± 0.004 | 0.976 ± 0.003 | 0.977 ± 0.005 | 0.991 ± 0.001 | 0.982 ± 0.006 | 0.983 ± 0.005 | E Africa M    | E Europe F    | 0.03 |
| 33  | 0.94 ± 0.01   | 0.951 ± 0.003 | 0.906 ± 0.008 | 0.917 ± 0.005 | 0.961 ± 0.006 | 0.96 ± 0.01   | 0.994 ± 0.005 | 0.986 ± 0.002 | 0.989 ± 0.003 | 0.995 ± 0.001 | 0.989 ± 0.004 | 0.986 ± 0.005 | S Asia M      | E Europe F    | 0.02 |
| 34  | 0.96 ± 0.01   | 0.968 ± 0.003 | 0.936 ± 0.007 | 0.942 ± 0.004 | 0.975 ± 0.005 | 0.969 ± 0.009 | 0.995 ± 0.004 | 0.991 ± 0.002 | 0.995 ± 0.002 | 0.997 ± 0.001 | 0.994 ± 0.004 | 0.993 ± 0.004 | S Asia M      | E Europe F    | 0.01 |



**Table 135.** For algorithm idemia-002 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.000 ± 0.000 | S Asia F      | E Africa F    | 0.78 |
| 15  | 0.009 ± 0.009 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.006 ± 0.004 | 0.010 ± 0.007 | 0.03 ± 0.02   | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.000 ± 0.000 | W Africa F    | E Europe F    | 0.76 |
| 16  | 0.03 ± 0.02   | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.01 ± 0.01   | 0.03 ± 0.02   | 0.01 ± 0.01   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | W Africa F    | E Europe F    | 0.71 |
| 17  | 0.03 ± 0.02   | 0.013 ± 0.005 | 0.03 ± 0.02   | 0.018 ± 0.008 | 0.02 ± 0.02   | 0.07 ± 0.02   | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.006 ± 0.004 | 0.006 ± 0.006 | 0.03 ± 0.02   | W Africa F    | E Africa M    | 0.55 |
| 18  | 0.07 ± 0.02   | 0.015 ± 0.005 | 0.04 ± 0.01   | 0.025 ± 0.007 | 0.026 ± 0.009 | 0.09 ± 0.03   | 0.02 ± 0.01   | 0.012 ± 0.004 | 0.000 ± 0.000 | 0.012 ± 0.005 | 0.011 ± 0.007 | 0.03 ± 0.01   | W Africa F    | E Europe M    | 0.48 |
| 19  | 0.11 ± 0.02   | 0.031 ± 0.007 | 0.04 ± 0.02   | 0.051 ± 0.010 | 0.06 ± 0.01   | 0.13 ± 0.03   | 0.08 ± 0.02   | 0.027 ± 0.007 | 0.013 ± 0.009 | 0.040 ± 0.010 | 0.022 ± 0.010 | 0.08 ± 0.02   | W Africa F    | E Europe M    | 0.37 |
| 20  | 0.18 ± 0.03   | 0.060 ± 0.010 | 0.09 ± 0.02   | 0.11 ± 0.01   | 0.11 ± 0.02   | 0.19 ± 0.03   | 0.16 ± 0.03   | 0.052 ± 0.009 | 0.04 ± 0.01   | 0.08 ± 0.01   | 0.06 ± 0.02   | 0.10 ± 0.03   | W Africa F    | E Europe M    | 0.28 |
| 21  | 0.22 ± 0.03   | 0.10 ± 0.01   | 0.13 ± 0.02   | 0.15 ± 0.01   | 0.15 ± 0.02   | 0.24 ± 0.03   | 0.22 ± 0.03   | 0.10 ± 0.01   | 0.07 ± 0.02   | 0.15 ± 0.01   | 0.14 ± 0.02   | 0.15 ± 0.03   | W Africa F    | E Europe M    | 0.19 |
| 22  | 0.29 ± 0.03   | 0.15 ± 0.01   | 0.21 ± 0.02   | 0.22 ± 0.02   | 0.23 ± 0.02   | 0.30 ± 0.03   | 0.33 ± 0.03   | 0.17 ± 0.01   | 0.16 ± 0.03   | 0.23 ± 0.02   | 0.21 ± 0.03   | 0.19 ± 0.03   | E Africa M    | E Asia F      | 0.15 |
| 23  | 0.37 ± 0.03   | 0.22 ± 0.02   | 0.28 ± 0.02   | 0.31 ± 0.02   | 0.33 ± 0.02   | 0.37 ± 0.03   | 0.40 ± 0.03   | 0.28 ± 0.02   | 0.27 ± 0.03   | 0.34 ± 0.02   | 0.34 ± 0.03   | 0.30 ± 0.03   | E Africa M    | E Asia F      | 0.10 |
| 24  | 0.44 ± 0.03   | 0.31 ± 0.01   | 0.35 ± 0.02   | 0.41 ± 0.02   | 0.43 ± 0.02   | 0.46 ± 0.03   | 0.50 ± 0.03   | 0.40 ± 0.02   | 0.45 ± 0.03   | 0.48 ± 0.02   | 0.46 ± 0.03   | 0.38 ± 0.03   | E Africa M    | E Asia F      | 0.07 |
| 25  | 0.50 ± 0.03   | 0.41 ± 0.01   | 0.47 ± 0.02   | 0.49 ± 0.01   | 0.52 ± 0.02   | 0.52 ± 0.03   | 0.63 ± 0.03   | 0.55 ± 0.02   | 0.60 ± 0.03   | 0.60 ± 0.02   | 0.61 ± 0.03   | 0.48 ± 0.03   | E Africa M    | E Asia F      | 0.07 |
| 26  | 0.59 ± 0.03   | 0.51 ± 0.01   | 0.55 ± 0.02   | 0.56 ± 0.01   | 0.64 ± 0.02   | 0.61 ± 0.03   | 0.70 ± 0.02   | 0.69 ± 0.02   | 0.73 ± 0.03   | 0.71 ± 0.02   | 0.73 ± 0.03   | 0.58 ± 0.03   | E Europe M    | E Asia F      | 0.07 |
| 27  | 0.67 ± 0.03   | 0.62 ± 0.01   | 0.66 ± 0.02   | 0.64 ± 0.01   | 0.72 ± 0.02   | 0.68 ± 0.02   | 0.77 ± 0.02   | 0.78 ± 0.02   | 0.85 ± 0.02   | 0.81 ± 0.01   | 0.85 ± 0.02   | 0.67 ± 0.03   | E Europe M    | E Asia F      | 0.07 |
| 28  | 0.72 ± 0.02   | 0.70 ± 0.01   | 0.74 ± 0.02   | 0.72 ± 0.01   | 0.78 ± 0.02   | 0.75 ± 0.02   | 0.84 ± 0.02   | 0.86 ± 0.01   | 0.90 ± 0.02   | 0.874 ± 0.009 | 0.89 ± 0.02   | 0.75 ± 0.02   | E Europe M    | E Asia F      | 0.05 |
| 29  | 0.81 ± 0.02   | 0.80 ± 0.01   | 0.83 ± 0.01   | 0.79 ± 0.01   | 0.86 ± 0.01   | 0.80 ± 0.02   | 0.90 ± 0.02   | 0.92 ± 0.01   | 0.95 ± 0.01   | 0.921 ± 0.008 | 0.93 ± 0.02   | 0.84 ± 0.02   | E Europe M    | S Africa F    | 0.04 |
| 30  | 0.85 ± 0.02   | 0.875 ± 0.008 | 0.88 ± 0.01   | 0.849 ± 0.010 | 0.91 ± 0.01   | 0.83 ± 0.02   | 0.94 ± 0.01   | 0.958 ± 0.008 | 0.982 ± 0.006 | 0.950 ± 0.005 | 0.97 ± 0.01   | 0.88 ± 0.02   | E Europe M    | W Africa F    | 0.03 |
| 31  | 0.88 ± 0.02   | 0.911 ± 0.007 | 0.924 ± 0.009 | 0.882 ± 0.008 | 0.938 ± 0.010 | 0.89 ± 0.02   | 0.96 ± 0.01   | 0.977 ± 0.005 | 0.985 ± 0.007 | 0.970 ± 0.004 | 0.979 ± 0.008 | 0.92 ± 0.01   | E Europe M    | E Africa F    | 0.03 |
| 32  | 0.92 ± 0.01   | 0.944 ± 0.006 | 0.958 ± 0.008 | 0.925 ± 0.007 | 0.953 ± 0.009 | 0.92 ± 0.01   | 0.974 ± 0.008 | 0.986 ± 0.004 | 0.995 ± 0.004 | 0.988 ± 0.003 | 0.988 ± 0.007 | 0.95 ± 0.01   | E Europe M    | W Africa F    | 0.02 |
| 33  | 0.94 ± 0.01   | 0.966 ± 0.005 | 0.974 ± 0.007 | 0.956 ± 0.005 | 0.975 ± 0.007 | 0.94 ± 0.01   | 0.984 ± 0.007 | 0.992 ± 0.003 | 0.998 ± 0.003 | 0.993 ± 0.002 | 0.996 ± 0.005 | 0.962 ± 0.009 | E Europe M    | W Africa F    | 0.01 |
| 34  | 0.95 ± 0.01   | 0.983 ± 0.003 | 0.986 ± 0.005 | 0.972 ± 0.005 | 0.985 ± 0.005 | 0.966 ± 0.010 | 0.987 ± 0.007 | 0.997 ± 0.002 | 0.999 ± 0.002 | 0.997 ± 0.002 | 0.998 ± 0.003 | 0.987 ± 0.006 | E Europe M    | E Africa F    | 0.01 |

**Table 136.** For algorithm idemia-002 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.03 ± 0.02   | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.007 ± 0.004 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.01 ± 0.01   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.007 ± 0.007 | E Africa F    | E Europe F    | 0.82 |
| 15  | 0.04 ± 0.02   | 0.000 ± 0.000 | 0.004 ± 0.004 | 0.011 ± 0.005 | 0.000 ± 0.000 | 0.09 ± 0.04   | 0.02 ± 0.02   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.02 ± 0.02   | W Africa F    | S E Asia F    | 0.81 |
| 16  | 0.06 ± 0.03   | 0.004 ± 0.002 | 0.011 ± 0.008 | 0.024 ± 0.008 | 0.003 ± 0.003 | 0.05 ± 0.03   | 0.02 ± 0.02   | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.04 ± 0.02   | E Africa F    | S Asia M      | 0.62 |
| 17  | 0.09 ± 0.03   | 0.010 ± 0.003 | 0.03 ± 0.01   | 0.034 ± 0.009 | 0.014 ± 0.006 | 0.10 ± 0.03   | 0.08 ± 0.03   | 0.005 ± 0.002 | 0.003 ± 0.003 | 0.016 ± 0.006 | 0.005 ± 0.004 | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.55 |
| 18  | 0.15 ± 0.04   | 0.011 ± 0.002 | 0.07 ± 0.01   | 0.058 ± 0.009 | 0.018 ± 0.007 | 0.10 ± 0.03   | 0.11 ± 0.04   | 0.007 ± 0.002 | 0.014 ± 0.009 | 0.021 ± 0.006 | 0.009 ± 0.006 | 0.05 ± 0.02   | E Africa F    | E Asia M      | 0.52 |
| 19  | 0.24 ± 0.04   | 0.022 ± 0.003 | 0.11 ± 0.01   | 0.10 ± 0.01   | 0.041 ± 0.009 | 0.19 ± 0.04   | 0.18 ± 0.04   | 0.021 ± 0.004 | 0.017 ± 0.008 | 0.057 ± 0.009 | 0.033 ± 0.009 | 0.10 ± 0.03   | E Africa F    | E Europe M    | 0.46 |
| 20  | 0.29 ± 0.04   | 0.043 ± 0.005 | 0.13 ± 0.01   | 0.13 ± 0.01   | 0.07 ± 0.01   | 0.24 ± 0.03   | 0.27 ± 0.04   | 0.041 ± 0.005 | 0.04 ± 0.01   | 0.10 ± 0.01   | 0.07 ± 0.01   | 0.14 ± 0.03   | E Africa F    | E Asia M      | 0.39 |
| 21  | 0.35 ± 0.04   | 0.064 ± 0.005 | 0.19 ± 0.01   | 0.19 ± 0.01   | 0.09 ± 0.01   | 0.30 ± 0.03   | 0.33 ± 0.04   | 0.078 ± 0.006 | 0.09 ± 0.02   | 0.15 ± 0.01   | 0.12 ± 0.02   | 0.18 ± 0.03   | E Africa F    | E Asia F      | 0.32 |
| 22  | 0.45 ± 0.04   | 0.105 ± 0.006 | 0.26 ± 0.02   | 0.26 ± 0.01   | 0.16 ± 0.01   | 0.38 ± 0.03   | 0.46 ± 0.04   | 0.142 ± 0.008 | 0.16 ± 0.02   | 0.24 ± 0.01   | 0.20 ± 0.02   | 0.28 ± 0.03   | E Africa M    | E Asia F      | 0.27 |
| 23  | 0.48 ± 0.04   | 0.149 ± 0.006 | 0.34 ± 0.01   | 0.32 ± 0.01   | 0.26 ± 0.02   | 0.44 ± 0.03   | 0.48 ± 0.04   | 0.225 ± 0.010 | 0.29 ± 0.02   | 0.34 ± 0.01   | 0.28 ± 0.02   | 0.33 ± 0.03   | E Africa M    | E Asia F      | 0.18 |
| 24  | 0.57 ± 0.03   | 0.216 ± 0.007 | 0.43 ± 0.01   | 0.41 ± 0.01   | 0.34 ± 0.02   | 0.48 ± 0.03   | 0.59 ± 0.03   | 0.33 ± 0.01   | 0.41 ± 0.02   | 0.45 ± 0.01   | 0.40 ± 0.02   | 0.41 ± 0.03   | E Africa M    | E Asia F      | 0.14 |
| 25  | 0.62 ± 0.03   | 0.300 ± 0.008 | 0.54 ± 0.01   | 0.46 ± 0.01   | 0.42 ± 0.02   | 0.59 ± 0.02   | 0.68 ± 0.03   | 0.46 ± 0.01   | 0.55 ± 0.02   | 0.56 ± 0.01   | 0.51 ± 0.03   | 0.52 ± 0.03   | E Africa M    | E Asia F      | 0.11 |
| 26  | 0.69 ± 0.03   | 0.393 ± 0.008 | 0.60 ± 0.01   | 0.548 ± 0.010 | 0.52 ± 0.01   | 0.62 ± 0.02   | 0.77 ± 0.03   | 0.56 ± 0.01   | 0.67 ± 0.02   | 0.66 ± 0.01   | 0.62 ± 0.03   | 0.61 ± 0.02   | E Africa M    | E Asia F      | 0.09 |
| 27  | 0.72 ± 0.02   | 0.482 ± 0.007 | 0.69 ± 0.01   | 0.608 ± 0.009 | 0.62 ± 0.02   | 0.71 ± 0.02   | 0.82 ± 0.02   | 0.667 ± 0.010 | 0.76 ± 0.01   | 0.750 ± 0.009 | 0.72 ± 0.02   | 0.71 ± 0.02   | E Africa M    | E Asia F      | 0.07 |
| 28  | 0.80 ± 0.03   | 0.585 ± 0.007 | 0.77 ± 0.01   | 0.680 ± 0.008 | 0.70 ± 0.01   | 0.75 ± 0.02   | 0.87 ± 0.02   | 0.756 ± 0.009 | 0.85 ± 0.01   | 0.825 ± 0.007 | 0.81 ± 0.02   | 0.77 ± 0.02   | E Africa M    | E Asia F      | 0.06 |
| 29  | 0.82 ± 0.02   | 0.666 ± 0.007 | 0.835 ± 0.009 | 0.735 ± 0.008 | 0.76 ± 0.01   | 0.81 ± 0.02   | 0.90 ± 0.02   | 0.837 ± 0.008 | 0.90 ± 0.01   | 0.882 ± 0.006 | 0.86 ± 0.02   | 0.84 ± 0.01   | E Africa M    | E Asia F      | 0.05 |
| 30  | 0.86 ± 0.02   | 0.751 ± 0.006 | 0.882 ± 0.007 | 0.798 ± 0.006 | 0.83 ± 0.01   | 0.86 ± 0.02   | 0.95 ± 0.01   | 0.891 ± 0.007 | 0.949 ± 0.007 | 0.921 ± 0.005 | 0.92 ± 0.01   | 0.89 ± 0.01   | E Africa M    | E Asia F      | 0.04 |
| 31  | 0.89 ± 0.02   | 0.823 ± 0.005 | 0.916 ± 0.006 | 0.851 ± 0.006 | 0.874 ± 0.009 | 0.89 ± 0.01   | 0.96 ± 0.01   | 0.935 ± 0.005 | 0.970 ± 0.005 | 0.952 ± 0.004 | 0.95 ± 0.01   | 0.91 ± 0.01   | E Europe M    | E Asia F      | 0.03 |
| 32  | 0.93 ± 0.01   | 0.880 ± 0.005 | 0.949 ± 0.005 | 0.891 ± 0.005 | 0.915 ± 0.009 | 0.90 ± 0.01   | 0.980 ± 0.007 | 0.961 ± 0.004 | 0.984 ± 0.004 | 0.969 ± 0.003 | 0.968 ± 0.009 | 0.941 ± 0.009 | E Europe M    | E Asia F      | 0.02 |
| 33  | 0.93 ± 0.02   | 0.918 ± 0.004 | 0.964 ± 0.005 | 0.925 ± 0.004 | 0.937 ± 0.008 | 0.94 ± 0.01   | 0.978 ± 0.009 | 0.977 ± 0.003 | 0.991 ± 0.003 | 0.983 ± 0.002 | 0.978 ± 0.007 | 0.962 ± 0.008 | E Europe M    | E Asia F      | 0.01 |
| 34  | 0.96 ± 0.01   | 0.947 ± 0.003 | 0.979 ± 0.004 | 0.947 ± 0.004 | 0.960 ± 0.006 | 0.96 ± 0.01   | 0.987 ± 0.006 | 0.984 ± 0.003 | 0.997 ± 0.002 | 0.990 ± 0.001 | 0.989 ± 0.005 | 0.974 ± 0.006 | E Europe M    | E Asia F      | 0.01 |

**Table 137.** For algorithm inbut-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.17 ± 0.03   | 0.019 ± 0.007 | 0.10 ± 0.03   | 0.05 ± 0.01   | 0.03 ± 0.02   | 0.22 ± 0.05   | 0.20 ± 0.03   | 0.034 ± 0.008 | 0.04 ± 0.02   | 0.07 ± 0.01   | 0.06 ± 0.02   | 0.10 ± 0.04   | W Africa F    | E Asia F      | 0.43 |
| 15  | 0.26 ± 0.04   | 0.04 ± 0.01   | 0.15 ± 0.04   | 0.09 ± 0.02   | 0.06 ± 0.02   | 0.24 ± 0.05   | 0.19 ± 0.04   | 0.06 ± 0.01   | 0.06 ± 0.02   | 0.08 ± 0.02   | 0.07 ± 0.02   | 0.18 ± 0.05   | E Africa F    | E Asia F      | 0.36 |
| 16  | 0.26 ± 0.04   | 0.06 ± 0.01   | 0.19 ± 0.04   | 0.09 ± 0.02   | 0.09 ± 0.03   | 0.35 ± 0.06   | 0.23 ± 0.04   | 0.07 ± 0.01   | 0.12 ± 0.03   | 0.11 ± 0.02   | 0.12 ± 0.03   | 0.22 ± 0.05   | W Africa F    | E Asia F      | 0.33 |
| 17  | 0.39 ± 0.05   | 0.08 ± 0.01   | 0.22 ± 0.04   | 0.16 ± 0.02   | 0.11 ± 0.03   | 0.40 ± 0.05   | 0.30 ± 0.05   | 0.10 ± 0.02   | 0.12 ± 0.03   | 0.13 ± 0.02   | 0.13 ± 0.03   | 0.27 ± 0.05   | W Africa F    | E Asia F      | 0.32 |
| 18  | 0.40 ± 0.04   | 0.10 ± 0.01   | 0.23 ± 0.04   | 0.15 ± 0.02   | 0.11 ± 0.02   | 0.45 ± 0.05   | 0.27 ± 0.04   | 0.10 ± 0.01   | 0.15 ± 0.03   | 0.14 ± 0.02   | 0.08 ± 0.02   | 0.24 ± 0.04   | W Africa F    | S E Asia M    | 0.33 |
| 19  | 0.53 ± 0.04   | 0.15 ± 0.01   | 0.29 ± 0.04   | 0.25 ± 0.02   | 0.16 ± 0.02   | 0.54 ± 0.04   | 0.39 ± 0.04   | 0.16 ± 0.02   | 0.24 ± 0.04   | 0.22 ± 0.02   | 0.14 ± 0.02   | 0.38 ± 0.05   | W Africa F    | S E Asia M    | 0.28 |
| 20  | 0.60 ± 0.04   | 0.22 ± 0.02   | 0.37 ± 0.03   | 0.34 ± 0.02   | 0.23 ± 0.02   | 0.66 ± 0.04   | 0.51 ± 0.04   | 0.22 ± 0.02   | 0.33 ± 0.04   | 0.30 ± 0.02   | 0.18 ± 0.03   | 0.49 ± 0.04   | W Africa F    | S E Asia M    | 0.25 |
| 21  | 0.68 ± 0.03   | 0.31 ± 0.02   | 0.50 ± 0.03   | 0.44 ± 0.02   | 0.31 ± 0.02   | 0.70 ± 0.04   | 0.64 ± 0.04   | 0.33 ± 0.02   | 0.43 ± 0.04   | 0.41 ± 0.02   | 0.29 ± 0.03   | 0.61 ± 0.04   | W Africa F    | S E Asia M    | 0.19 |
| 22  | 0.78 ± 0.03   | 0.39 ± 0.02   | 0.58 ± 0.03   | 0.54 ± 0.02   | 0.40 ± 0.02   | 0.77 ± 0.03   | 0.79 ± 0.03   | 0.44 ± 0.02   | 0.61 ± 0.03   | 0.57 ± 0.02   | 0.42 ± 0.03   | 0.68 ± 0.04   | E Africa M    | E Asia F      | 0.15 |
| 23  | 0.83 ± 0.02   | 0.52 ± 0.02   | 0.67 ± 0.02   | 0.65 ± 0.02   | 0.52 ± 0.02   | 0.87 ± 0.02   | 0.84 ± 0.02   | 0.56 ± 0.02   | 0.73 ± 0.03   | 0.70 ± 0.02   | 0.52 ± 0.03   | 0.81 ± 0.03   | W Africa F    | S E Asia M    | 0.12 |
| 24  | 0.87 ± 0.02   | 0.60 ± 0.02   | 0.77 ± 0.02   | 0.73 ± 0.01   | 0.62 ± 0.02   | 0.87 ± 0.02   | 0.91 ± 0.02   | 0.69 ± 0.02   | 0.83 ± 0.02   | 0.81 ± 0.02   | 0.67 ± 0.03   | 0.87 ± 0.02   | E Africa M    | E Asia F      | 0.08 |
| 25  | 0.90 ± 0.02   | 0.71 ± 0.01   | 0.83 ± 0.02   | 0.81 ± 0.01   | 0.69 ± 0.02   | 0.91 ± 0.02   | 0.93 ± 0.02   | 0.79 ± 0.02   | 0.89 ± 0.02   | 0.89 ± 0.01   | 0.77 ± 0.02   | 0.91 ± 0.02   | E Africa M    | S E Asia F    | 0.06 |
| 26  | 0.93 ± 0.01   | 0.78 ± 0.01   | 0.90 ± 0.01   | 0.850 ± 0.009 | 0.79 ± 0.01   | 0.94 ± 0.01   | 0.965 ± 0.010 | 0.87 ± 0.01   | 0.93 ± 0.01   | 0.928 ± 0.009 | 0.86 ± 0.02   | 0.94 ± 0.01   | E Africa M    | E Asia F      | 0.04 |
| 27  | 0.96 ± 0.01   | 0.855 ± 0.009 | 0.93 ± 0.01   | 0.890 ± 0.007 | 0.85 ± 0.01   | 0.96 ± 0.01   | 0.974 ± 0.009 | 0.923 ± 0.010 | 0.965 ± 0.010 | 0.955 ± 0.006 | 0.93 ± 0.02   | 0.96 ± 0.01   | E Africa M    | E Asia F      | 0.03 |
| 28  | 0.967 ± 0.009 | 0.893 ± 0.008 | 0.963 ± 0.007 | 0.921 ± 0.007 | 0.90 ± 0.01   | 0.975 ± 0.008 | 0.985 ± 0.007 | 0.961 ± 0.008 | 0.988 ± 0.006 | 0.972 ± 0.004 | 0.95 ± 0.01   | 0.980 ± 0.008 | E Europe M    | E Asia F      | 0.02 |
| 29  | 0.985 ± 0.007 | 0.937 ± 0.006 | 0.980 ± 0.005 | 0.946 ± 0.006 | 0.925 ± 0.010 | 0.980 ± 0.007 | 0.995 ± 0.004 | 0.980 ± 0.005 | 0.995 ± 0.004 | 0.984 ± 0.003 | 0.97 ± 0.01   | 0.994 ± 0.004 | E Africa M    | S E Asia F    | 0.01 |
| 30  | 0.981 ± 0.007 | 0.957 ± 0.005 | 0.983 ± 0.005 | 0.962 ± 0.005 | 0.947 ± 0.008 | 0.984 ± 0.006 | 0.995 ± 0.004 | 0.988 ± 0.004 | 0.997 ± 0.003 | 0.984 ± 0.003 | 0.984 ± 0.008 | 0.994 ± 0.004 | E Europe M    | S E Asia F    | 0.01 |
| 31  | 0.990 ± 0.006 | 0.975 ± 0.004 | 0.994 ± 0.003 | 0.972 ± 0.004 | 0.970 ± 0.007 | 0.991 ± 0.005 | 0.999 ± 0.002 | 0.997 ± 0.002 | 0.999 ± 0.002 | 0.995 ± 0.002 | 0.991 ± 0.006 | 0.995 ± 0.004 | E Europe M    | S E Asia F    | 0.01 |
| 32  | 0.995 ± 0.004 | 0.984 ± 0.003 | 0.997 ± 0.002 | 0.981 ± 0.004 | 0.984 ± 0.005 | 0.994 ± 0.004 | 0.999 ± 0.001 | 0.997 ± 0.002 | 0.999 ± 0.002 | 0.997 ± 0.001 | 0.998 ± 0.003 | 0.999 ± 0.002 | E Africa M    | S Asia F      | 0.00 |
| 33  | 0.995 ± 0.005 | 0.994 ± 0.002 | 0.995 ± 0.003 | 0.989 ± 0.003 | 0.988 ± 0.005 | 0.994 ± 0.004 | 1.000 ± 0.000 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.997 ± 0.001 | 0.997 ± 0.004 | 0.999 ± 0.002 | E Africa M    | S Asia F      | 0.00 |
| 34  | 0.995 ± 0.004 | 0.996 ± 0.002 | 0.998 ± 0.002 | 0.992 ± 0.003 | 0.993 ± 0.004 | 0.995 ± 0.004 | 1.000 ± 0.000 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.999 ± 0.001 | 1.000 ± 0.000 | 0.998 ± 0.003 | E Africa M    | S Asia F      | 0.00 |

**Table 138.** For algorithm inbut-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.31 ± 0.06   | 0.011 ± 0.004 | 0.006 ± 0.006 | 0.04 ± 0.01   | 0.011 ± 0.009 | 0.37 ± 0.07   | 0.23 ± 0.06   | 0.007 ± 0.003 | 0.000 ± 0.000 | 0.008 ± 0.005 | 0.02 ± 0.01   | 0.15 ± 0.06   | W Africa F    | E Europe M    | 0.72 |
| 15  | 0.39 ± 0.07   | 0.015 ± 0.004 | 0.05 ± 0.02   | 0.08 ± 0.02   | 0.03 ± 0.01   | 0.51 ± 0.07   | 0.28 ± 0.07   | 0.015 ± 0.004 | 0.004 ± 0.004 | 0.023 ± 0.007 | 0.02 ± 0.01   | 0.20 ± 0.06   | W Africa F    | E Europe M    | 0.67 |
| 16  | 0.48 ± 0.06   | 0.039 ± 0.007 | 0.06 ± 0.02   | 0.12 ± 0.02   | 0.04 ± 0.01   | 0.52 ± 0.07   | 0.40 ± 0.07   | 0.040 ± 0.007 | 0.03 ± 0.01   | 0.05 ± 0.01   | 0.04 ± 0.01   | 0.26 ± 0.06   | W Africa F    | E Europe M    | 0.58 |
| 17  | 0.53 ± 0.06   | 0.071 ± 0.008 | 0.09 ± 0.02   | 0.17 ± 0.02   | 0.06 ± 0.01   | 0.64 ± 0.06   | 0.47 ± 0.07   | 0.083 ± 0.008 | 0.04 ± 0.02   | 0.14 ± 0.02   | 0.08 ± 0.02   | 0.44 ± 0.05   | W Africa F    | E Europe M    | 0.51 |
| 18  | 0.62 ± 0.05   | 0.096 ± 0.008 | 0.15 ± 0.02   | 0.22 ± 0.02   | 0.10 ± 0.02   | 0.63 ± 0.05   | 0.57 ± 0.06   | 0.138 ± 0.010 | 0.13 ± 0.02   | 0.22 ± 0.02   | 0.13 ± 0.02   | 0.52 ± 0.05   | W Africa F    | E Asia F      | 0.41 |
| 19  | 0.66 ± 0.04   | 0.148 ± 0.009 | 0.20 ± 0.02   | 0.30 ± 0.02   | 0.14 ± 0.02   | 0.72 ± 0.04   | 0.69 ± 0.05   | 0.21 ± 0.01   | 0.19 ± 0.03   | 0.35 ± 0.02   | 0.18 ± 0.02   | 0.59 ± 0.05   | W Africa F    | S E Asia F    | 0.36 |
| 20  | 0.70 ± 0.04   | 0.210 ± 0.010 | 0.27 ± 0.02   | 0.36 ± 0.02   | 0.23 ± 0.02   | 0.80 ± 0.03   | 0.76 ± 0.04   | 0.29 ± 0.01   | 0.31 ± 0.03   | 0.48 ± 0.02   | 0.29 ± 0.02   | 0.70 ± 0.04   | W Africa F    | E Asia F      | 0.29 |
| 21  | 0.76 ± 0.03   | 0.264 ± 0.009 | 0.33 ± 0.02   | 0.42 ± 0.01   | 0.30 ± 0.02   | 0.82 ± 0.03   | 0.78 ± 0.04   | 0.39 ± 0.01   | 0.50 ± 0.03   | 0.61 ± 0.02   | 0.39 ± 0.02   | 0.83 ± 0.03   | W Africa M    | E Asia F      | 0.24 |
| 22  | 0.85 ± 0.03   | 0.334 ± 0.009 | 0.42 ± 0.02   | 0.50 ± 0.01   | 0.36 ± 0.02   | 0.87 ± 0.02   | 0.89 ± 0.03   | 0.49 ± 0.01   | 0.59 ± 0.03   | 0.72 ± 0.01   | 0.49 ± 0.02   | 0.86 ± 0.03   | E Africa M    | E Asia F      | 0.20 |
| 23  | 0.86 ± 0.02   | 0.399 ± 0.009 | 0.49 ± 0.02   | 0.56 ± 0.01   | 0.49 ± 0.02   | 0.90 ± 0.02   | 0.89 ± 0.03   | 0.57 ± 0.01   | 0.69 ± 0.02   | 0.79 ± 0.01   | 0.58 ± 0.03   | 0.90 ± 0.02   | W Africa M    | E Asia F      | 0.16 |
| 24  | 0.87 ± 0.02   | 0.482 ± 0.009 | 0.57 ± 0.01   | 0.64 ± 0.01   | 0.55 ± 0.02   | 0.91 ± 0.01   | 0.95 ± 0.01   | 0.66 ± 0.01   | 0.80 ± 0.02   | 0.863 ± 0.009 | 0.66 ± 0.02   | 0.92 ± 0.02   | E Africa M    | E Asia F      | 0.13 |
| 25  | 0.91 ± 0.02   | 0.556 ± 0.008 | 0.66 ± 0.01   | 0.691 ± 0.009 | 0.63 ± 0.02   | 0.94 ± 0.01   | 0.96 ± 0.01   | 0.73 ± 0.01   | 0.86 ± 0.01   | 0.894 ± 0.007 | 0.75 ± 0.02   | 0.958 ± 0.010 | E Africa M    | E Asia F      | 0.11 |
| 26  | 0.93 ± 0.01   | 0.632 ± 0.008 | 0.72 ± 0.01   | 0.736 ± 0.009 | 0.69 ± 0.01   | 0.94 ± 0.01   | 0.97 ± 0.01   | 0.794 ± 0.009 | 0.90 ± 0.01   | 0.926 ± 0.006 | 0.81 ± 0.02   | 0.959 ± 0.009 | E Africa M    | E Asia F      | 0.08 |
| 27  | 0.94 ± 0.01   | 0.684 ± 0.007 | 0.790 ± 0.009 | 0.776 ± 0.008 | 0.76 ± 0.01   | 0.962 ± 0.009 | 0.980 ± 0.009 | 0.837 ± 0.008 | 0.940 ± 0.008 | 0.947 ± 0.005 | 0.86 ± 0.02   | 0.984 ± 0.006 | W Africa M    | E Asia F      | 0.07 |
| 28  | 0.95 ± 0.01   | 0.750 ± 0.006 | 0.848 ± 0.008 | 0.816 ± 0.007 | 0.81 ± 0.01   | 0.965 ± 0.008 | 0.985 ± 0.007 | 0.878 ± 0.007 | 0.965 ± 0.006 | 0.965 ± 0.004 | 0.90 ± 0.01   | 0.983 ± 0.006 | E Africa M    | E Asia F      | 0.05 |
| 29  | 0.969 ± 0.010 | 0.804 ± 0.006 | 0.894 ± 0.007 | 0.849 ± 0.006 | 0.84 ± 0.01   | 0.975 ± 0.007 | 0.991 ± 0.006 | 0.908 ± 0.006 | 0.974 ± 0.005 | 0.970 ± 0.003 | 0.92 ± 0.01   | 0.989 ± 0.005 | E Africa M    | E Asia F      | 0.04 |
| 30  | 0.967 ± 0.009 | 0.841 ± 0.005 | 0.924 ± 0.006 | 0.883 ± 0.005 | 0.88 ± 0.01   | 0.979 ± 0.007 | 0.997 ± 0.003 | 0.929 ± 0.005 | 0.984 ± 0.004 | 0.982 ± 0.002 | 0.94 ± 0.01   | 0.993 ± 0.003 | E Africa M    | E Asia F      | 0.03 |
| 31  | 0.985 ± 0.007 | 0.873 ± 0.004 | 0.948 ± 0.005 | 0.911 ± 0.005 | 0.904 ± 0.009 | 0.986 ± 0.005 | 0.999 ± 0.002 | 0.949 ± 0.005 | 0.995 ± 0.002 | 0.986 ± 0.002 | 0.95 ± 0.01   | 0.994 ± 0.003 | E Africa M    | E Asia F      | 0.03 |
| 32  | 0.980 ± 0.008 | 0.901 ± 0.004 | 0.969 ± 0.004 | 0.929 ± 0.004 | 0.921 ± 0.008 | 0.987 ± 0.006 | 0.998 ± 0.002 | 0.961 ± 0.003 | 0.995 ± 0.002 | 0.991 ± 0.001 | 0.965 ± 0.009 | 0.995 ± 0.003 | E Africa M    | E Asia F      | 0.02 |
| 33  | 0.985 ± 0.007 | 0.926 ± 0.004 | 0.975 ± 0.004 | 0.945 ± 0.004 | 0.939 ± 0.008 | 0.991 ± 0.006 | 0.999 ± 0.002 | 0.970 ± 0.003 | 0.997 ± 0.002 | 0.992 ± 0.001 | 0.971 ± 0.007 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.01 |
| 34  | 0.985 ± 0.007 | 0.943 ± 0.003 | 0.981 ± 0.003 | 0.960 ± 0.004 | 0.950 ± 0.006 | 0.993 ± 0.004 | 0.998 ± 0.003 | 0.977 ± 0.003 | 0.999 ± 0.001 | 0.995 ± 0.001 | 0.977 ± 0.007 | 0.998 ± 0.002 | E Europe M    | E Asia F      | 0.01 |

**Table 139.** For algorithm incode-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.02 ± 0.01   | 0.007 ± 0.004 | 0.02 ± 0.01   | 0.008 ± 0.004 | 0.013 ± 0.009 | 0.008 ± 0.008 | 0.010 ± 0.008 | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.004 ± 0.004 | E Africa F    | S E Asia M    | 0.47 |
| 15  | 0.03 ± 0.01   | 0.010 ± 0.005 | 0.02 ± 0.01   | 0.018 ± 0.008 | 0.02 ± 0.01   | 0.02 ± 0.02   | 0.008 ± 0.008 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.01 ± 0.01   | E Africa F    | E Europe M    | 0.47 |
| 16  | 0.08 ± 0.02   | 0.026 ± 0.009 | 0.04 ± 0.02   | 0.03 ± 0.01   | 0.05 ± 0.02   | 0.09 ± 0.03   | 0.03 ± 0.02   | 0.002 ± 0.002 | 0.006 ± 0.006 | 0.004 ± 0.003 | 0.006 ± 0.006 | 0.01 ± 0.01   | W Africa F    | E Asia M      | 0.52 |
| 17  | 0.11 ± 0.03   | 0.05 ± 0.01   | 0.07 ± 0.03   | 0.06 ± 0.01   | 0.06 ± 0.02   | 0.11 ± 0.03   | 0.02 ± 0.01   | 0.006 ± 0.004 | 0.003 ± 0.003 | 0.017 ± 0.007 | 0.01 ± 0.01   | 0.05 ± 0.02   | E Africa F    | E Europe M    | 0.46 |
| 18  | 0.15 ± 0.03   | 0.07 ± 0.01   | 0.08 ± 0.02   | 0.08 ± 0.01   | 0.09 ± 0.02   | 0.13 ± 0.03   | 0.04 ± 0.01   | 0.013 ± 0.004 | 0.013 ± 0.009 | 0.023 ± 0.007 | 0.015 ± 0.008 | 0.06 ± 0.02   | E Africa F    | E Europe M    | 0.42 |
| 19  | 0.24 ± 0.03   | 0.11 ± 0.01   | 0.14 ± 0.03   | 0.13 ± 0.02   | 0.14 ± 0.02   | 0.19 ± 0.03   | 0.12 ± 0.03   | 0.032 ± 0.008 | 0.03 ± 0.01   | 0.06 ± 0.01   | 0.03 ± 0.01   | 0.11 ± 0.03   | E Africa F    | E Europe M    | 0.34 |
| 20  | 0.30 ± 0.04   | 0.17 ± 0.02   | 0.22 ± 0.03   | 0.20 ± 0.02   | 0.22 ± 0.02   | 0.26 ± 0.04   | 0.20 ± 0.03   | 0.044 ± 0.008 | 0.07 ± 0.02   | 0.13 ± 0.02   | 0.09 ± 0.02   | 0.17 ± 0.03   | E Africa F    | E Asia M      | 0.27 |
| 21  | 0.39 ± 0.03   | 0.24 ± 0.02   | 0.29 ± 0.03   | 0.26 ± 0.02   | 0.31 ± 0.02   | 0.36 ± 0.04   | 0.30 ± 0.03   | 0.10 ± 0.01   | 0.12 ± 0.02   | 0.20 ± 0.02   | 0.16 ± 0.02   | 0.24 ± 0.04   | E Africa F    | E Asia M      | 0.22 |
| 22  | 0.47 ± 0.03   | 0.32 ± 0.02   | 0.40 ± 0.03   | 0.34 ± 0.02   | 0.38 ± 0.02   | 0.39 ± 0.04   | 0.46 ± 0.04   | 0.18 ± 0.02   | 0.23 ± 0.03   | 0.31 ± 0.02   | 0.21 ± 0.03   | 0.31 ± 0.04   | E Africa F    | E Asia M      | 0.17 |
| 23  | 0.56 ± 0.03   | 0.41 ± 0.02   | 0.50 ± 0.02   | 0.45 ± 0.02   | 0.49 ± 0.02   | 0.50 ± 0.03   | 0.51 ± 0.03   | 0.28 ± 0.02   | 0.34 ± 0.04   | 0.45 ± 0.02   | 0.35 ± 0.03   | 0.46 ± 0.04   | E Africa F    | E Asia M      | 0.11 |
| 24  | 0.63 ± 0.03   | 0.49 ± 0.02   | 0.61 ± 0.02   | 0.54 ± 0.02   | 0.57 ± 0.02   | 0.54 ± 0.03   | 0.64 ± 0.03   | 0.38 ± 0.02   | 0.50 ± 0.03   | 0.59 ± 0.02   | 0.46 ± 0.03   | 0.53 ± 0.03   | E Africa M    | E Asia M      | 0.08 |
| 25  | 0.69 ± 0.03   | 0.60 ± 0.01   | 0.70 ± 0.02   | 0.61 ± 0.01   | 0.67 ± 0.02   | 0.64 ± 0.03   | 0.74 ± 0.03   | 0.52 ± 0.02   | 0.65 ± 0.03   | 0.71 ± 0.02   | 0.61 ± 0.03   | 0.62 ± 0.03   | E Africa M    | E Asia M      | 0.05 |
| 26  | 0.77 ± 0.02   | 0.68 ± 0.01   | 0.79 ± 0.02   | 0.69 ± 0.01   | 0.75 ± 0.02   | 0.71 ± 0.02   | 0.80 ± 0.02   | 0.64 ± 0.02   | 0.75 ± 0.03   | 0.79 ± 0.01   | 0.72 ± 0.03   | 0.70 ± 0.03   | E Africa M    | E Asia M      | 0.04 |
| 27  | 0.81 ± 0.02   | 0.75 ± 0.01   | 0.85 ± 0.01   | 0.75 ± 0.01   | 0.82 ± 0.01   | 0.77 ± 0.02   | 0.86 ± 0.02   | 0.74 ± 0.02   | 0.86 ± 0.02   | 0.85 ± 0.01   | 0.81 ± 0.02   | 0.76 ± 0.02   | E Africa M    | E Asia M      | 0.04 |
| 28  | 0.85 ± 0.02   | 0.812 ± 0.010 | 0.912 ± 0.010 | 0.81 ± 0.01   | 0.86 ± 0.01   | 0.83 ± 0.02   | 0.90 ± 0.01   | 0.82 ± 0.01   | 0.91 ± 0.02   | 0.902 ± 0.008 | 0.87 ± 0.02   | 0.83 ± 0.02   | E Europe F    | S Asia F      | 0.03 |
| 29  | 0.90 ± 0.02   | 0.866 ± 0.008 | 0.946 ± 0.009 | 0.853 ± 0.009 | 0.91 ± 0.01   | 0.85 ± 0.02   | 0.94 ± 0.01   | 0.88 ± 0.01   | 0.95 ± 0.01   | 0.934 ± 0.007 | 0.91 ± 0.02   | 0.89 ± 0.01   | E Europe M    | S Asia F      | 0.02 |
| 30  | 0.92 ± 0.01   | 0.911 ± 0.007 | 0.963 ± 0.008 | 0.895 ± 0.007 | 0.942 ± 0.010 | 0.87 ± 0.02   | 0.95 ± 0.01   | 0.92 ± 0.01   | 0.973 ± 0.009 | 0.955 ± 0.006 | 0.96 ± 0.01   | 0.92 ± 0.01   | E Europe M    | W Africa F    | 0.02 |
| 31  | 0.93 ± 0.01   | 0.940 ± 0.006 | 0.984 ± 0.005 | 0.926 ± 0.007 | 0.962 ± 0.007 | 0.92 ± 0.01   | 0.968 ± 0.010 | 0.955 ± 0.008 | 0.984 ± 0.007 | 0.966 ± 0.005 | 0.96 ± 0.01   | 0.95 ± 0.01   | E Europe F    | W Africa F    | 0.01 |
| 32  | 0.96 ± 0.01   | 0.958 ± 0.005 | 0.990 ± 0.004 | 0.939 ± 0.006 | 0.971 ± 0.007 | 0.94 ± 0.01   | 0.976 ± 0.008 | 0.968 ± 0.006 | 0.994 ± 0.004 | 0.978 ± 0.004 | 0.986 ± 0.007 | 0.958 ± 0.010 | E Europe M    | W Africa F    | 0.01 |
| 33  | 0.974 ± 0.009 | 0.971 ± 0.005 | 0.993 ± 0.004 | 0.966 ± 0.005 | 0.984 ± 0.006 | 0.95 ± 0.01   | 0.989 ± 0.006 | 0.981 ± 0.005 | 0.989 ± 0.003 | 0.989 ± 0.003 | 0.991 ± 0.006 | 0.973 ± 0.009 | E Europe M    | W Africa F    | 0.01 |
| 34  | 0.97 ± 0.01   | 0.985 ± 0.003 | 0.996 ± 0.003 | 0.980 ± 0.004 | 0.990 ± 0.005 | 0.96 ± 0.01   | 0.993 ± 0.005 | 0.988 ± 0.004 | 0.997 ± 0.003 | 0.992 ± 0.002 | 0.992 ± 0.006 | 0.987 ± 0.006 | E Europe M    | W Africa F    | 0.01 |

**Table 140.** For algorithm incode-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.33 ± 0.07   | 0.06 ± 0.01   | 0.22 ± 0.04   | 0.20 ± 0.02   | 0.06 ± 0.02   | 0.20 ± 0.06   | 0.27 ± 0.07   | 0.012 ± 0.004 | 0.06 ± 0.02   | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.22 ± 0.06   | E Africa F    | E Asia M      | 0.44 |
| 15  | 0.33 ± 0.07   | 0.063 ± 0.009 | 0.27 ± 0.04   | 0.25 ± 0.03   | 0.10 ± 0.02   | 0.34 ± 0.06   | 0.37 ± 0.07   | 0.017 ± 0.005 | 0.06 ± 0.02   | 0.06 ± 0.01   | 0.02 ± 0.01   | 0.23 ± 0.06   | E Africa M    | E Asia M      | 0.45 |
| 16  | 0.43 ± 0.06   | 0.080 ± 0.009 | 0.32 ± 0.04   | 0.30 ± 0.03   | 0.13 ± 0.02   | 0.35 ± 0.06   | 0.34 ± 0.07   | 0.024 ± 0.005 | 0.11 ± 0.02   | 0.08 ± 0.01   | 0.03 ± 0.01   | 0.20 ± 0.05   | E Africa F    | E Asia M      | 0.42 |
| 17  | 0.48 ± 0.06   | 0.12 ± 0.01   | 0.40 ± 0.03   | 0.33 ± 0.02   | 0.14 ± 0.02   | 0.42 ± 0.06   | 0.43 ± 0.07   | 0.038 ± 0.006 | 0.12 ± 0.03   | 0.15 ± 0.02   | 0.05 ± 0.02   | 0.36 ± 0.06   | E Africa F    | E Asia M      | 0.38 |
| 18  | 0.55 ± 0.05   | 0.139 ± 0.009 | 0.45 ± 0.03   | 0.36 ± 0.02   | 0.20 ± 0.02   | 0.42 ± 0.05   | 0.46 ± 0.06   | 0.053 ± 0.006 | 0.19 ± 0.03   | 0.20 ± 0.02   | 0.07 ± 0.02   | 0.36 ± 0.05   | E Africa F    | E Asia M      | 0.34 |
| 19  | 0.58 ± 0.05   | 0.179 ± 0.009 | 0.50 ± 0.02   | 0.42 ± 0.02   | 0.23 ± 0.02   | 0.53 ± 0.05   | 0.52 ± 0.05   | 0.077 ± 0.007 | 0.24 ± 0.03   | 0.28 ± 0.02   | 0.13 ± 0.02   | 0.45 ± 0.05   | E Africa F    | E Asia M      | 0.30 |
| 20  | 0.65 ± 0.04   | 0.224 ± 0.009 | 0.56 ± 0.02   | 0.49 ± 0.02   | 0.29 ± 0.02   | 0.57 ± 0.04   | 0.60 ± 0.05   | 0.111 ± 0.008 | 0.35 ± 0.03   | 0.40 ± 0.02   | 0.16 ± 0.02   | 0.50 ± 0.04   | E Africa F    | E Asia M      | 0.26 |
| 21  | 0.69 ± 0.04   | 0.262 ± 0.009 | 0.62 ± 0.02   | 0.55 ± 0.02   | 0.37 ± 0.02   | 0.64 ± 0.03   | 0.69 ± 0.04   | 0.160 ± 0.009 | 0.44 ± 0.03   | 0.50 ± 0.02   | 0.23 ± 0.02   | 0.60 ± 0.04   | E Africa M    | E Asia M      | 0.23 |
| 22  | 0.78 ± 0.03   | 0.319 ± 0.009 | 0.68 ± 0.02   | 0.60 ± 0.01   | 0.41 ± 0.02   | 0.70 ± 0.03   | 0.76 ± 0.04   | 0.222 ± 0.009 | 0.56 ± 0.03   | 0.59 ± 0.02   | 0.34 ± 0.02   | 0.65 ± 0.03   | E Africa F    | E Asia M      | 0.20 |
| 23  | 0.79 ± 0.03   | 0.370 ± 0.009 | 0.74 ± 0.01   | 0.66 ± 0.01   | 0.52 ± 0.02   | 0.73 ± 0.03   | 0.82 ± 0.03   | 0.30 ± 0.01   | 0.67 ± 0.02   | 0.67 ± 0.01   | 0.42 ± 0.03   | 0.71 ± 0.03   | E Africa M    | E Asia M      | 0.16 |
| 24  | 0.84 ± 0.02   | 0.436 ± 0.009 | 0.80 ± 0.01   | 0.71 ± 0.01   | 0.56 ± 0.02   | 0.77 ± 0.02   | 0.86 ± 0.03   | 0.38 ± 0.01   | 0.75 ± 0.02   | 0.76 ± 0.01   | 0.49 ± 0.02   | 0.76 ± 0.03   | E Africa M    | E Asia M      | 0.14 |
| 25  | 0.83 ± 0.02   | 0.502 ± 0.009 | 0.849 ± 0.009 | 0.747 ± 0.008 | 0.64 ± 0.02   | 0.82 ± 0.02   | 0.89 ± 0.02   | 0.82 ± 0.01   | 0.82 ± 0.02   | 0.815 ± 0.010 | 0.59 ± 0.02   | 0.83 ± 0.02   | E Africa M    | E Asia M      | 0.11 |
| 26  | 0.89 ± 0.02   | 0.561 ± 0.007 | 0.874 ± 0.008 | 0.795 ± 0.008 | 0.69 ± 0.01   | 0.84 ± 0.02   | 0.92 ± 0.02   | 0.55 ± 0.01   | 0.88 ± 0.01   | 0.852 ± 0.008 | 0.67 ± 0.02   | 0.86 ± 0.02   | E Africa M    | E Asia M      | 0.09 |
| 27  | 0.90 ± 0.02   | 0.622 ± 0.007 | 0.906 ± 0.007 | 0.828 ± 0.007 | 0.75 ± 0.01   | 0.87 ± 0.02   | 0.95 ± 0.01   | 0.61 ± 0.01   | 0.91 ± 0.01   | 0.887 ± 0.007 | 0.76 ± 0.02   | 0.89 ± 0.01   | E Africa M    | E Asia M      | 0.08 |
| 28  | 0.93 ± 0.01   | 0.688 ± 0.007 | 0.935 ± 0.006 | 0.858 ± 0.006 | 0.80 ± 0.01   | 0.88 ± 0.02   | 0.96 ± 0.01   | 0.698 ± 0.010 | 0.947 ± 0.008 | 0.920 ± 0.005 | 0.81 ± 0.02   | 0.92 ± 0.01   | E Africa M    | E Asia F      | 0.06 |
| 29  | 0.94 ± 0.01   | 0.737 ± 0.006 | 0.954 ± 0.005 | 0.886 ± 0.005 | 0.83 ± 0.01   | 0.91 ± 0.01   | 0.979 ± 0.008 | 0.767 ± 0.008 | 0.953 ± 0.007 | 0.940 ± 0.004 | 0.87 ± 0.02   | 0.94 ± 0.01   | E Africa M    | E Asia F      | 0.05 |
| 30  | 0.94 ± 0.01   | 0.787 ± 0.006 | 0.970 ± 0.004 | 0.907 ± 0.004 | 0.88 ± 0.01   | 0.93 ± 0.01   | 0.979 ± 0.009 | 0.821 ± 0.008 | 0.975 ± 0.005 | 0.957 ± 0.003 | 0.89 ± 0.01   | 0.957 ± 0.008 | E Africa M    | E Asia F      | 0.04 |
| 31  | 0.97 ± 0.01   | 0.829 ± 0.006 | 0.978 ± 0.004 | 0.929 ± 0.004 | 0.905 ± 0.009 | 0.95 ± 0.01   | 0.983 ± 0.007 | 0.867 ± 0.007 | 0.987 ± 0.004 | 0.969 ± 0.003 | 0.91 ± 0.01   | 0.960 ± 0.008 | E Europe M    | E Asia F      | 0.03 |
| 32  | 0.976 ± 0.009 | 0.867 ± 0.005 | 0.986 ± 0.003 | 0.947 ± 0.004 | 0.929 ± 0.007 | 0.957 ± 0.010 | 0.991 ± 0.005 | 0.901 ± 0.006 | 0.992 ± 0.003 | 0.977 ± 0.002 | 0.94 ± 0.01   | 0.970 ± 0.006 | E Europe M    | E Asia F      | 0.02 |
| 33  | 0.978 ± 0.009 | 0.896 ± 0.004 | 0.989 ± 0.003 | 0.956 ± 0.004 | 0.934 ± 0.008 | 0.966 ± 0.009 | 0.991 ± 0.005 | 0.926 ± 0.005 | 0.992 ± 0.002 | 0.983 ± 0.002 | 0.95 ± 0.01   | 0.980 ± 0.006 | E Europe M    | E Asia F      | 0.02 |
| 34  | 0.980 ± 0.009 | 0.918 ± 0.004 | 0.991 ± 0.002 | 0.973 ± 0.003 | 0.958 ± 0.007 | 0.979 ± 0.008 | 0.993 ± 0.005 | 0.943 ± 0.005 | 0.997 ± 0.002 | 0.988 ± 0.002 | 0.971 ± 0.007 | 0.983 ± 0.005 | E Europe M    | E Asia F      | 0.01 |

**Table 141.** For algorithm incode-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.02 ± 0.01   | 0.003 ± 0.002 | 0.005 ± 0.005 | 0.005 ± 0.004 | 0.007 ± 0.007 | 0.02 ± 0.01   | 0.004 ± 0.004 | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.004 ± 0.004 | E Africa F    | E Europe M    | 0.53 |
| 15  | 0.03 ± 0.02   | 0.011 ± 0.005 | 0.01 ± 0.01   | 0.015 ± 0.008 | 0.02 ± 0.01   | 0.03 ± 0.02   | 0.01 ± 0.01   | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.005 | E Africa F    | E Europe M    | 0.53 |
| 16  | 0.07 ± 0.02   | 0.023 ± 0.008 | 0.04 ± 0.02   | 0.03 ± 0.01   | 0.06 ± 0.02   | 0.08 ± 0.03   | 0.02 ± 0.01   | 0.005 ± 0.003 | 0.003 ± 0.003 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.005 | W Africa F    | S E Asia M    | 0.57 |
| 17  | 0.11 ± 0.03   | 0.07 ± 0.01   | 0.06 ± 0.03   | 0.06 ± 0.02   | 0.06 ± 0.02   | 0.16 ± 0.04   | 0.02 ± 0.01   | 0.011 ± 0.004 | 0.000 ± 0.000 | 0.011 ± 0.005 | 0.02 ± 0.01   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.49 |
| 18  | 0.13 ± 0.03   | 0.10 ± 0.01   | 0.08 ± 0.02   | 0.10 ± 0.01   | 0.15 ± 0.02   | 0.19 ± 0.03   | 0.04 ± 0.01   | 0.022 ± 0.007 | 0.009 ± 0.007 | 0.025 ± 0.007 | 0.03 ± 0.01   | 0.06 ± 0.02   | W Africa F    | E Europe M    | 0.43 |
| 19  | 0.26 ± 0.03   | 0.17 ± 0.02   | 0.13 ± 0.03   | 0.16 ± 0.02   | 0.21 ± 0.02   | 0.27 ± 0.04   | 0.14 ± 0.03   | 0.07 ± 0.01   | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.05 ± 0.02   | 0.14 ± 0.03   | W Africa F    | E Europe M    | 0.34 |
| 20  | 0.36 ± 0.04   | 0.26 ± 0.02   | 0.20 ± 0.03   | 0.24 ± 0.02   | 0.30 ± 0.03   | 0.35 ± 0.04   | 0.21 ± 0.03   | 0.13 ± 0.01   | 0.06 ± 0.02   | 0.13 ± 0.02   | 0.13 ± 0.02   | 0.20 ± 0.03   | E Africa F    | E Europe M    | 0.26 |
| 21  | 0.47 ± 0.04   | 0.37 ± 0.02   | 0.28 ± 0.03   | 0.32 ± 0.02   | 0.42 ± 0.03   | 0.46 ± 0.04   | 0.34 ± 0.04   | 0.24 ± 0.02   | 0.10 ± 0.02   | 0.23 ± 0.02   | 0.26 ± 0.03   | 0.29 ± 0.04   | E Africa F    | E Europe M    | 0.20 |
| 22  | 0.53 ± 0.04   | 0.50 ± 0.02   | 0.39 ± 0.03   | 0.41 ± 0.02   | 0.51 ± 0.02   | 0.54 ± 0.04   | 0.53 ± 0.04   | 0.40 ± 0.02   | 0.22 ± 0.03   | 0.35 ± 0.02   | 0.36 ± 0.03   | 0.37 ± 0.03   | W Africa F    | E Europe M    | 0.13 |
| 23  | 0.65 ± 0.03   | 0.60 ± 0.02   | 0.51 ± 0.02   | 0.52 ± 0.02   | 0.64 ± 0.02   | 0.67 ± 0.03   | 0.62 ± 0.03   | 0.55 ± 0.02   | 0.36 ± 0.03   | 0.52 ± 0.02   | 0.54 ± 0.03   | 0.54 ± 0.03   | W Africa F    | E Europe M    | 0.09 |
| 24  | 0.73 ± 0.03   | 0.69 ± 0.01   | 0.62 ± 0.02   | 0.60 ± 0.02   | 0.72 ± 0.02   | 0.71 ± 0.03   | 0.75 ± 0.03   | 0.68 ± 0.02   | 0.53 ± 0.03   | 0.65 ± 0.02   | 0.67 ± 0.03   | 0.64 ± 0.03   | E Africa M    | E Europe M    | 0.05 |
| 25  | 0.77 ± 0.02   | 0.78 ± 0.01   | 0.72 ± 0.02   | 0.67 ± 0.01   | 0.80 ± 0.02   | 0.78 ± 0.03   | 0.83 ± 0.02   | 0.80 ± 0.02   | 0.70 ± 0.03   | 0.78 ± 0.01   | 0.80 ± 0.02   | 0.76 ± 0.03   | E Africa M    | S Asia F      | 0.04 |
| 26  | 0.85 ± 0.02   | 0.84 ± 0.01   | 0.82 ± 0.01   | 0.74 ± 0.01   | 0.86 ± 0.01   | 0.86 ± 0.02   | 0.87 ± 0.02   | 0.88 ± 0.01   | 0.81 ± 0.02   | 0.86 ± 0.01   | 0.87 ± 0.02   | 0.82 ± 0.02   | E Asia M      | S Asia F      | 0.03 |
| 27  | 0.89 ± 0.02   | 0.894 ± 0.009 | 0.89 ± 0.01   | 0.804 ± 0.009 | 0.91 ± 0.01   | 0.88 ± 0.02   | 0.92 ± 0.01   | 0.926 ± 0.010 | 0.90 ± 0.02   | 0.90 ± 0.01   | 0.93 ± 0.02   | 0.88 ± 0.02   | S E Asia M    | S Asia F      | 0.02 |
| 28  | 0.93 ± 0.01   | 0.932 ± 0.006 | 0.947 ± 0.009 | 0.852 ± 0.009 | 0.938 ± 0.009 | 0.92 ± 0.01   | 0.95 ± 0.01   | 0.957 ± 0.008 | 0.94 ± 0.01   | 0.944 ± 0.007 | 0.96 ± 0.01   | 0.93 ± 0.01   | S E Asia M    | S Asia F      | 0.01 |
| 29  | 0.95 ± 0.01   | 0.954 ± 0.005 | 0.962 ± 0.007 | 0.893 ± 0.008 | 0.965 ± 0.007 | 0.94 ± 0.01   | 0.971 ± 0.010 | 0.977 ± 0.006 | 0.974 ± 0.009 | 0.965 ± 0.005 | 0.97 ± 0.01   | 0.960 ± 0.010 | E Asia M      | S Asia F      | 0.01 |
| 30  | 0.969 ± 0.009 | 0.974 ± 0.004 | 0.979 ± 0.006 | 0.928 ± 0.006 | 0.980 ± 0.005 | 0.96 ± 0.01   | 0.984 ± 0.007 | 0.989 ± 0.004 | 0.989 ± 0.006 | 0.976 ± 0.004 | 0.985 ± 0.008 | 0.974 ± 0.008 | E Europe M    | S Asia F      | 0.01 |
| 31  | 0.976 ± 0.009 | 0.981 ± 0.003 | 0.992 ± 0.004 | 0.952 ± 0.006 | 0.986 ± 0.004 | 0.971 ± 0.009 | 0.993 ± 0.004 | 0.992 ± 0.003 | 0.996 ± 0.003 | 0.987 ± 0.003 | 0.989 ± 0.007 | 0.984 ± 0.006 | E Europe M    | S Asia F      | 0.01 |
| 32  | 0.984 ± 0.007 | 0.991 ± 0.003 | 0.998 ± 0.002 | 0.968 ± 0.004 | 0.987 ± 0.005 | 0.986 ± 0.006 | 0.996 ± 0.004 | 0.996 ± 0.002 | 0.997 ± 0.003 | 0.993 ± 0.002 | 0.991 ± 0.006 | 0.988 ± 0.005 | E Europe F    | S Asia F      | 0.00 |
| 33  | 0.989 ± 0.006 | 0.992 ± 0.002 | 0.995 ± 0.003 | 0.978 ± 0.004 | 0.994 ± 0.004 | 0.990 ± 0.006 | 0.998 ± 0.002 | 0.997 ± 0.002 | 0.998 ± 0.003 | 0.995 ± 0.002 | 0.997 ± 0.004 | 0.992 ± 0.005 | E Africa M    | S Asia F      | 0.00 |
| 34  | 0.987 ± 0.007 | 0.996 ± 0.002 | 0.997 ± 0.003 | 0.989 ± 0.003 | 0.995 ± 0.003 | 0.991 ± 0.005 | 0.998 ± 0.002 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.998 ± 0.001 | 0.995 ± 0.005 | 0.997 ± 0.003 | E Asia M      | E Africa F    | 0.00 |

**Table 142.** For algorithm incode-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.28 ± 0.06   | 0.047 ± 0.008 | 0.12 ± 0.03   | 0.09 ± 0.02   | 0.05 ± 0.02   | 0.22 ± 0.07   | 0.17 ± 0.06   | 0.011 ± 0.004 | 0.02 ± 0.01   | 0.018 ± 0.007 | 0.02 ± 0.01   | 0.14 ± 0.05   | E Africa F    | E Asia M      | 0.51 |
| 15  | 0.29 ± 0.06   | 0.058 ± 0.009 | 0.17 ± 0.03   | 0.14 ± 0.02   | 0.07 ± 0.02   | 0.36 ± 0.06   | 0.30 ± 0.07   | 0.016 ± 0.005 | 0.02 ± 0.01   | 0.018 ± 0.007 | 0.007 ± 0.005 | 0.18 ± 0.06   | W Africa F    | S E Asia M    | 0.53 |
| 16  | 0.38 ± 0.07   | 0.075 ± 0.009 | 0.22 ± 0.03   | 0.16 ± 0.02   | 0.11 ± 0.02   | 0.37 ± 0.06   | 0.28 ± 0.06   | 0.026 ± 0.005 | 0.05 ± 0.02   | 0.029 ± 0.008 | 0.011 ± 0.006 | 0.19 ± 0.05   | E Africa F    | S E Asia M    | 0.48 |
| 17  | 0.48 ± 0.06   | 0.11 ± 0.01   | 0.29 ± 0.03   | 0.21 ± 0.02   | 0.13 ± 0.02   | 0.46 ± 0.06   | 0.38 ± 0.06   | 0.041 ± 0.006 | 0.07 ± 0.02   | 0.09 ± 0.01   | 0.04 ± 0.01   | 0.29 ± 0.05   | E Africa F    | S E Asia M    | 0.44 |
| 18  | 0.50 ± 0.05   | 0.148 ± 0.008 | 0.36 ± 0.03   | 0.24 ± 0.02   | 0.17 ± 0.02   | 0.46 ± 0.05   | 0.44 ± 0.06   | 0.063 ± 0.006 | 0.14 ± 0.03   | 0.13 ± 0.01   | 0.05 ± 0.01   | 0.30 ± 0.05   | E Africa F    | S E Asia M    | 0.38 |
| 19  | 0.55 ± 0.04   | 0.20 ± 0.01   | 0.39 ± 0.02   | 0.30 ± 0.02   | 0.23 ± 0.02   | 0.62 ± 0.04   | 0.55 ± 0.05   | 0.115 ± 0.008 | 0.20 ± 0.03   | 0.21 ± 0.01   | 0.09 ± 0.02   | 0.44 ± 0.05   | W Africa F    | S E Asia M    | 0.33 |
| 20  | 0.65 ± 0.04   | 0.262 ± 0.010 | 0.47 ± 0.02   | 0.36 ± 0.02   | 0.29 ± 0.02   | 0.68 ± 0.04   | 0.61 ± 0.05   | 0.176 ± 0.010 | 0.31 ± 0.03   | 0.33 ± 0.02   | 0.17 ± 0.02   | 0.53 ± 0.04   | W Africa F    | S E Asia M    | 0.26 |
| 21  | 0.68 ± 0.04   | 0.31 ± 0.01   | 0.56 ± 0.02   | 0.41 ± 0.02   | 0.35 ± 0.02   | 0.73 ± 0.03   | 0.70 ± 0.04   | 0.26 ± 0.01   | 0.44 ± 0.03   | 0.46 ± 0.02   | 0.26 ± 0.02   | 0.63 ± 0.03   | W Africa F    | E Asia M      | 0.21 |
| 22  | 0.76 ± 0.03   | 0.370 ± 0.009 | 0.63 ± 0.02   | 0.47 ± 0.02   | 0.42 ± 0.02   | 0.78 ± 0.03   | 0.80 ± 0.04   | 0.350 ± 0.010 | 0.56 ± 0.03   | 0.56 ± 0.01   | 0.36 ± 0.02   | 0.72 ± 0.03   | E Africa M    | E Asia M      | 0.18 |
| 23  | 0.76 ± 0.03   | 0.434 ± 0.009 | 0.69 ± 0.02   | 0.52 ± 0.01   | 0.50 ± 0.02   | 0.79 ± 0.03   | 0.86 ± 0.03   | 0.45 ± 0.01   | 0.70 ± 0.02   | 0.65 ± 0.01   | 0.47 ± 0.03   | 0.79 ± 0.02   | E Africa M    | E Asia F      | 0.14 |
| 24  | 0.81 ± 0.02   | 0.500 ± 0.009 | 0.76 ± 0.01   | 0.59 ± 0.01   | 0.56 ± 0.02   | 0.84 ± 0.02   | 0.89 ± 0.02   | 0.55 ± 0.01   | 0.80 ± 0.02   | 0.75 ± 0.01   | 0.55 ± 0.02   | 0.84 ± 0.02   | E Africa M    | E Asia F      | 0.12 |
| 25  | 0.83 ± 0.02   | 0.570 ± 0.008 | 0.82 ± 0.01   | 0.64 ± 0.01   | 0.62 ± 0.02   | 0.88 ± 0.02   | 0.93 ± 0.02   | 0.65 ± 0.01   | 0.86 ± 0.01   | 0.812 ± 0.010 | 0.65 ± 0.02   | 0.90 ± 0.02   | E Africa M    | E Asia F      | 0.10 |
| 26  | 0.89 ± 0.02   | 0.634 ± 0.008 | 0.851 ± 0.009 | 0.691 ± 0.008 | 0.69 ± 0.01   | 0.91 ± 0.01   | 0.95 ± 0.02   | 0.735 ± 0.009 | 0.91 ± 0.01   | 0.859 ± 0.008 | 0.75 ± 0.02   | 0.91 ± 0.01   | E Africa M    | E Asia F      | 0.08 |
| 27  | 0.90 ± 0.02   | 0.689 ± 0.006 | 0.895 ± 0.008 | 0.737 ± 0.008 | 0.73 ± 0.01   | 0.93 ± 0.01   | 0.97 ± 0.01   | 0.795 ± 0.009 | 0.942 ± 0.008 | 0.896 ± 0.006 | 0.82 ± 0.02   | 0.95 ± 0.01   | E Africa M    | E Asia F      | 0.07 |
| 28  | 0.91 ± 0.02   | 0.748 ± 0.007 | 0.929 ± 0.006 | 0.778 ± 0.007 | 0.79 ± 0.01   | 0.93 ± 0.01   | 0.982 ± 0.008 | 0.850 ± 0.008 | 0.970 ± 0.006 | 0.928 ± 0.005 | 0.87 ± 0.02   | 0.963 ± 0.009 | E Africa M    | E Asia F      | 0.05 |
| 29  | 0.94 ± 0.01   | 0.798 ± 0.006 | 0.952 ± 0.005 | 0.814 ± 0.007 | 0.83 ± 0.01   | 0.95 ± 0.01   | 0.985 ± 0.007 | 0.891 ± 0.006 | 0.979 ± 0.005 | 0.950 ± 0.004 | 0.90 ± 0.01   | 0.974 ± 0.007 | E Africa M    | E Asia F      | 0.04 |
| 30  | 0.95 ± 0.01   | 0.840 ± 0.006 | 0.964 ± 0.006 | 0.855 ± 0.006 | 0.870 ± 0.010 | 0.963 ± 0.009 | 0.992 ± 0.005 | 0.923 ± 0.005 | 0.987 ± 0.004 | 0.966 ± 0.003 | 0.93 ± 0.01   | 0.978 ± 0.006 | E Africa M    | E Asia F      | 0.03 |
| 31  | 0.96 ± 0.01   | 0.873 ± 0.005 | 0.975 ± 0.004 | 0.881 ± 0.005 | 0.894 ± 0.009 | 0.972 ± 0.008 | 0.992 ± 0.006 | 0.948 ± 0.004 | 0.994 ± 0.003 | 0.976 ± 0.003 | 0.95 ± 0.01   | 0.986 ± 0.005 | E Europe M    | E Asia F      | 0.03 |
| 32  | 0.979 ± 0.009 | 0.900 ± 0.004 | 0.983 ± 0.003 | 0.906 ± 0.005 | 0.923 ± 0.008 | 0.978 ± 0.008 | 0.996 ± 0.004 | 0.964 ± 0.004 | 0.995 ± 0.002 | 0.985 ± 0.002 | 0.967 ± 0.008 | 0.990 ± 0.004 | E Africa M    | E Asia F      | 0.02 |
| 33  | 0.978 ± 0.009 | 0.927 ± 0.004 | 0.988 ± 0.003 | 0.925 ± 0.005 | 0.938 ± 0.007 | 0.983 ± 0.007 | 0.995 ± 0.004 | 0.974 ± 0.003 | 0.996 ± 0.002 | 0.987 ± 0.002 | 0.981 ± 0.006 | 0.997 ± 0.002 | W Africa M    | S Asia F      | 0.01 |
| 34  | 0.987 ± 0.007 | 0.941 ± 0.003 | 0.990 ± 0.003 | 0.943 ± 0.004 | 0.954 ± 0.007 | 0.988 ± 0.006 | 0.998 ± 0.002 | 0.981 ± 0.003 | 0.999 ± 0.001 | 0.992 ± 0.001 | 0.987 ± 0.005 | 0.995 ± 0.003 | E Europe M    | E Asia F      | 0.01 |

**Table 143.** For algorithm *innovatrics-000* with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.03 ± 0.02   | 0.001 ± 0.001 | 0.002 ± 0.002 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.04 ± 0.02   | 0.012 ± 0.008 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.000 ± 0.000 | W Africa F    | S E Asia F    | 0.77 |
| 15  | 0.05 ± 0.02   | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.018 ± 0.009 | 0.003 ± 0.003 | 0.08 ± 0.03   | 0.008 ± 0.008 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.005 | W Africa F    | E Europe F    | 0.79 |
| 16  | 0.11 ± 0.03   | 0.015 ± 0.006 | 0.005 ± 0.005 | 0.015 ± 0.008 | 0.02 ± 0.01   | 0.12 ± 0.04   | 0.03 ± 0.02   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.02 ± 0.01   | W Africa F    | S E Asia M    | 0.70 |
| 17  | 0.20 ± 0.04   | 0.020 ± 0.007 | 0.01 ± 0.01   | 0.07 ± 0.02   | 0.01 ± 0.01   | 0.22 ± 0.04   | 0.03 ± 0.01   | 0.006 ± 0.003 | 0.000 ± 0.000 | 0.020 ± 0.007 | 0.003 ± 0.003 | 0.07 ± 0.03   | W Africa F    | E Europe M    | 0.68 |
| 18  | 0.23 ± 0.03   | 0.029 ± 0.007 | 0.02 ± 0.01   | 0.08 ± 0.01   | 0.027 ± 0.009 | 0.25 ± 0.04   | 0.07 ± 0.02   | 0.015 ± 0.005 | 0.011 ± 0.007 | 0.04 ± 0.01   | 0.010 ± 0.007 | 0.10 ± 0.03   | W Africa F    | S E Asia M    | 0.58 |
| 19  | 0.34 ± 0.04   | 0.07 ± 0.01   | 0.06 ± 0.02   | 0.14 ± 0.02   | 0.05 ± 0.01   | 0.37 ± 0.04   | 0.17 ± 0.03   | 0.030 ± 0.007 | 0.03 ± 0.01   | 0.10 ± 0.01   | 0.022 ± 0.010 | 0.20 ± 0.03   | W Africa F    | S E Asia M    | 0.50 |
| 20  | 0.45 ± 0.04   | 0.12 ± 0.01   | 0.09 ± 0.02   | 0.20 ± 0.02   | 0.08 ± 0.01   | 0.48 ± 0.04   | 0.28 ± 0.03   | 0.057 ± 0.009 | 0.07 ± 0.02   | 0.19 ± 0.02   | 0.04 ± 0.01   | 0.27 ± 0.04   | W Africa F    | S E Asia M    | 0.44 |
| 21  | 0.53 ± 0.04   | 0.18 ± 0.02   | 0.15 ± 0.02   | 0.31 ± 0.02   | 0.15 ± 0.02   | 0.53 ± 0.04   | 0.37 ± 0.04   | 0.12 ± 0.01   | 0.15 ± 0.03   | 0.30 ± 0.02   | 0.09 ± 0.02   | 0.40 ± 0.04   | W Africa F    | S E Asia M    | 0.34 |
| 22  | 0.63 ± 0.03   | 0.25 ± 0.02   | 0.24 ± 0.02   | 0.39 ± 0.02   | 0.21 ± 0.02   | 0.63 ± 0.04   | 0.56 ± 0.03   | 0.19 ± 0.02   | 0.28 ± 0.03   | 0.43 ± 0.02   | 0.14 ± 0.02   | 0.51 ± 0.04   | W Africa F    | S E Asia M    | 0.28 |
| 23  | 0.71 ± 0.03   | 0.35 ± 0.02   | 0.32 ± 0.02   | 0.51 ± 0.02   | 0.29 ± 0.02   | 0.72 ± 0.03   | 0.65 ± 0.03   | 0.30 ± 0.02   | 0.40 ± 0.03   | 0.58 ± 0.02   | 0.23 ± 0.02   | 0.66 ± 0.03   | W Africa F    | S E Asia M    | 0.22 |
| 24  | 0.76 ± 0.02   | 0.43 ± 0.02   | 0.47 ± 0.02   | 0.59 ± 0.02   | 0.39 ± 0.02   | 0.75 ± 0.03   | 0.78 ± 0.03   | 0.40 ± 0.02   | 0.57 ± 0.03   | 0.72 ± 0.02   | 0.35 ± 0.03   | 0.74 ± 0.03   | E Africa M    | S E Asia M    | 0.17 |
| 25  | 0.81 ± 0.02   | 0.55 ± 0.01   | 0.56 ± 0.02   | 0.68 ± 0.01   | 0.49 ± 0.02   | 0.82 ± 0.02   | 0.84 ± 0.02   | 0.54 ± 0.02   | 0.71 ± 0.03   | 0.82 ± 0.01   | 0.47 ± 0.03   | 0.82 ± 0.03   | E Africa M    | S E Asia M    | 0.12 |
| 26  | 0.85 ± 0.02   | 0.63 ± 0.01   | 0.68 ± 0.02   | 0.74 ± 0.01   | 0.60 ± 0.02   | 0.90 ± 0.02   | 0.89 ± 0.02   | 0.66 ± 0.02   | 0.79 ± 0.02   | 0.88 ± 0.01   | 0.61 ± 0.03   | 0.89 ± 0.02   | W Africa F    | S E Asia F    | 0.09 |
| 27  | 0.89 ± 0.02   | 0.72 ± 0.01   | 0.76 ± 0.02   | 0.796 ± 0.010 | 0.67 ± 0.02   | 0.91 ± 0.01   | 0.92 ± 0.01   | 0.75 ± 0.02   | 0.89 ± 0.02   | 0.923 ± 0.008 | 0.72 ± 0.03   | 0.92 ± 0.01   | S Asia M      | S E Asia F    | 0.07 |
| 28  | 0.92 ± 0.01   | 0.79 ± 0.01   | 0.85 ± 0.01   | 0.848 ± 0.010 | 0.76 ± 0.02   | 0.94 ± 0.01   | 0.96 ± 0.01   | 0.84 ± 0.01   | 0.94 ± 0.01   | 0.949 ± 0.007 | 0.82 ± 0.02   | 0.96 ± 0.01   | W Africa M    | S E Asia F    | 0.05 |
| 29  | 0.95 ± 0.01   | 0.852 ± 0.009 | 0.91 ± 0.01   | 0.887 ± 0.008 | 0.82 ± 0.01   | 0.96 ± 0.01   | 0.972 ± 0.009 | 0.90 ± 0.01   | 0.968 ± 0.010 | 0.971 ± 0.004 | 0.87 ± 0.02   | 0.976 ± 0.008 | W Africa M    | S E Asia F    | 0.03 |
| 30  | 0.96 ± 0.01   | 0.901 ± 0.007 | 0.935 ± 0.009 | 0.919 ± 0.007 | 0.88 ± 0.01   | 0.96 ± 0.01   | 0.986 ± 0.007 | 0.939 ± 0.008 | 0.987 ± 0.006 | 0.981 ± 0.003 | 0.92 ± 0.02   | 0.985 ± 0.006 | E Europe M    | S E Asia F    | 0.02 |
| 31  | 0.980 ± 0.007 | 0.929 ± 0.007 | 0.958 ± 0.008 | 0.936 ± 0.006 | 0.91 ± 0.01   | 0.985 ± 0.006 | 0.987 ± 0.006 | 0.966 ± 0.007 | 0.994 ± 0.004 | 0.989 ± 0.003 | 0.95 ± 0.01   | 0.992 ± 0.004 | E Europe M    | S E Asia F    | 0.02 |
| 32  | 0.984 ± 0.007 | 0.957 ± 0.005 | 0.980 ± 0.007 | 0.961 ± 0.004 | 0.944 ± 0.010 | 0.986 ± 0.007 | 0.993 ± 0.004 | 0.977 ± 0.006 | 0.994 ± 0.004 | 0.995 ± 0.002 | 0.97 ± 0.01   | 0.994 ± 0.004 | S Asia M      | S E Asia F    | 0.01 |
| 33  | 0.987 ± 0.007 | 0.976 ± 0.004 | 0.987 ± 0.005 | 0.975 ± 0.004 | 0.961 ± 0.008 | 0.996 ± 0.004 | 0.999 ± 0.002 | 0.988 ± 0.004 | 0.997 ± 0.004 | 0.998 ± 0.001 | 0.984 ± 0.008 | 0.998 ± 0.002 | E Africa M    | S E Asia F    | 0.01 |
| 34  | 0.987 ± 0.007 | 0.982 ± 0.004 | 0.991 ± 0.004 | 0.984 ± 0.003 | 0.974 ± 0.007 | 0.993 ± 0.004 | 0.998 ± 0.002 | 0.993 ± 0.003 | 0.999 ± 0.002 | 0.998 ± 0.001 | 0.990 ± 0.007 | 0.999 ± 0.002 | W Africa M    | S E Asia F    | 0.00 |

**Table 144.** For algorithm *innovatrics-000* with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.12 ± 0.05   | 0.002 ± 0.001 | 0.004 ± 0.004 | 0.024 ± 0.009 | 0.000 ± 0.000 | 0.11 ± 0.05   | 0.07 ± 0.04   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.004 ± 0.004 | 0.04 ± 0.03   | E Africa F    | S E Asia F    | 0.75 |
| 15  | 0.16 ± 0.05   | 0.002 ± 0.001 | 0.004 ± 0.004 | 0.04 ± 0.01   | 0.000 ± 0.000 | 0.22 ± 0.06   | 0.11 ± 0.04   | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.004 ± 0.004 | 0.12 ± 0.05   | W Africa F    | S E Asia F    | 0.73 |
| 16  | 0.30 ± 0.06   | 0.007 ± 0.003 | 0.012 ± 0.008 | 0.07 ± 0.01   | 0.007 ± 0.005 | 0.25 ± 0.06   | 0.16 ± 0.05   | 0.004 ± 0.002 | 0.007 ± 0.005 | 0.009 ± 0.004 | 0.000 ± 0.000 | 0.11 ± 0.04   | E Africa F    | S E Asia M    | 0.71 |
| 17  | 0.35 ± 0.06   | 0.019 ± 0.004 | 0.03 ± 0.01   | 0.10 ± 0.02   | 0.008 ± 0.005 | 0.38 ± 0.06   | 0.20 ± 0.06   | 0.011 ± 0.003 | 0.005 ± 0.005 | 0.051 ± 0.010 | 0.005 ± 0.004 | 0.21 ± 0.05   | W Africa F    | S E Asia M    | 0.65 |
| 18  | 0.43 ± 0.05   | 0.028 ± 0.004 | 0.07 ± 0.01   | 0.15 ± 0.02   | 0.014 ± 0.006 | 0.37 ± 0.05   | 0.29 ± 0.05   | 0.019 ± 0.004 | 0.03 ± 0.01   | 0.09 ± 0.01   | 0.011 ± 0.006 | 0.27 ± 0.04   | E Africa F    | S E Asia M    | 0.58 |
| 19  | 0.48 ± 0.05   | 0.045 ± 0.005 | 0.10 ± 0.01   | 0.22 ± 0.02   | 0.033 ± 0.007 | 0.52 ± 0.04   | 0.46 ± 0.05   | 0.041 ± 0.005 | 0.06 ± 0.02   | 0.20 ± 0.01   | 0.025 ± 0.009 | 0.36 ± 0.04   | W Africa F    | S E Asia M    | 0.52 |
| 20  | 0.61 ± 0.04   | 0.080 ± 0.006 | 0.14 ± 0.01   | 0.29 ± 0.02   | 0.06 ± 0.01   | 0.62 ± 0.04   | 0.53 ± 0.05   | 0.077 ± 0.007 | 0.12 ± 0.02   | 0.32 ± 0.02   | 0.06 ± 0.01   | 0.48 ± 0.04   | W Africa F    | S E Asia M    | 0.46 |
| 21  | 0.67 ± 0.04   | 0.112 ± 0.007 | 0.19 ± 0.02   | 0.35 ± 0.02   | 0.08 ± 0.01   | 0.69 ± 0.03   | 0.62 ± 0.04   | 0.125 ± 0.008 | 0.23 ± 0.02   | 0.45 ± 0.02   | 0.10 ± 0.02   | 0.64 ± 0.04   | W Africa F    | S E Asia F    | 0.40 |
| 22  | 0.77 ± 0.03   | 0.157 ± 0.007 | 0.28 ± 0.02   | 0.44 ± 0.01   | 0.14 ± 0.01   | 0.77 ± 0.03   | 0.75 ± 0.04   | 0.21 ± 0.01   | 0.35 ± 0.03   | 0.58 ± 0.02   | 0.16 ± 0.02   | 0.73 ± 0.03   | E Africa F    | S E Asia F    | 0.35 |
| 23  | 0.81 ± 0.03   | 0.217 ± 0.007 | 0.36 ± 0.02   | 0.51 ± 0.01   | 0.22 ± 0.01   | 0.81 ± 0.02   | 0.81 ± 0.03   | 0.30 ± 0.01   | 0.49 ± 0.02   | 0.68 ± 0.01   | 0.24 ± 0.02   | 0.81 ± 0.03   | W Africa F    | E Asia F      | 0.28 |
| 24  | 0.85 ± 0.02   | 0.286 ± 0.008 | 0.45 ± 0.01   | 0.60 ± 0.01   | 0.28 ± 0.02   | 0.85 ± 0.02   | 0.88 ± 0.02   | 0.40 ± 0.01   | 0.63 ± 0.02   | 0.78 ± 0.01   | 0.34 ± 0.02   | 0.86 ± 0.02   | E Africa M    | S E Asia F    | 0.24 |
| 25  | 0.89 ± 0.02   | 0.365 ± 0.008 | 0.56 ± 0.01   | 0.66 ± 0.01   | 0.37 ± 0.02   | 0.89 ± 0.02   | 0.92 ± 0.02   | 0.49 ± 0.01   | 0.73 ± 0.02   | 0.850 ± 0.008 | 0.44 ± 0.02   | 0.91 ± 0.02   | E Africa M    | E Asia F      | 0.19 |
| 26  | 0.91 ± 0.02   | 0.454 ± 0.008 | 0.64 ± 0.01   | 0.714 ± 0.009 | 0.46 ± 0.02   | 0.91 ± 0.01   | 0.96 ± 0.01   | 0.59 ± 0.01   | 0.81 ± 0.02   | 0.896 ± 0.007 | 0.55 ± 0.02   | 0.93 ± 0.01   | E Africa M    | E Asia F      | 0.15 |
| 27  | 0.92 ± 0.02   | 0.525 ± 0.008 | 0.728 ± 0.009 | 0.766 ± 0.008 | 0.54 ± 0.01   | 0.94 ± 0.01   | 0.96 ± 0.01   | 0.67 ± 0.01   | 0.89 ± 0.01   | 0.932 ± 0.005 | 0.66 ± 0.02   | 0.967 ± 0.009 | W Africa M    | E Asia F      | 0.12 |
| 28  | 0.95 ± 0.01   | 0.608 ± 0.007 | 0.798 ± 0.009 | 0.817 ± 0.007 | 0.62 ± 0.01   | 0.95 ± 0.01   | 0.977 ± 0.009 | 0.748 ± 0.009 | 0.934 ± 0.010 | 0.956 ± 0.004 | 0.73 ± 0.02   | 0.977 ± 0.007 | W Africa M    | E Asia F      | 0.09 |
| 29  | 0.966 ± 0.010 | 0.682 ± 0.007 | 0.860 ± 0.008 | 0.848 ± 0.006 | 0.69 ± 0.01   | 0.969 ± 0.008 | 0.991 ± 0.005 | 0.817 ± 0.007 | 0.951 ± 0.007 | 0.966 ± 0.003 | 0.80 ± 0.02   | 0.988 ± 0.005 | E Africa M    | E Asia F      | 0.07 |
| 30  | 0.970 ± 0.009 | 0.743 ± 0.006 | 0.905 ± 0.007 | 0.886 ± 0.005 | 0.75 ± 0.01   | 0.975 ± 0.008 | 0.994 ± 0.005 | 0.867 ± 0.007 | 0.980 ± 0.005 | 0.980 ± 0.002 | 0.86 ± 0.02   | 0.990 ± 0.004 | E Africa M    | E Asia F      | 0.06 |
| 31  | 0.979 ± 0.008 | 0.802 ± 0.006 | 0.936 ± 0.006 | 0.915 ± 0.004 | 0.80 ± 0.01   | 0.980 ± 0.007 | 0.997 ± 0.003 | 0.908 ± 0.005 | 0.988 ± 0.004 | 0.988 ± 0.002 | 0.89 ± 0.01   | 0.993 ± 0.003 | E Africa M    | S E Asia F    | 0.04 |
| 32  | 0.987 ± 0.006 | 0.852 ± 0.005 | 0.962 ± 0.005 | 0.934 ± 0.004 | 0.85 ± 0.01   | 0.989 ± 0.005 | 0.997 ± 0.003 | 0.940 ± 0.004 | 0.992 ± 0.003 | 0.992 ± 0.002 | 0.92 ± 0.01   | 0.995 ± 0.003 | E Africa M    | S E Asia F    | 0.03 |
| 33  | 0.987 ± 0.008 | 0.887 ± 0.004 | 0.974 ± 0.004 | 0.953 ± 0.004 | 0.895 ± 0.010 | 0.993 ± 0.004 | 0.999 ± 0.002 | 0.956 ± 0.004 | 0.998 ± 0.001 | 0.995 ± 0.001 | 0.956 ± 0.009 | 0.997 ± 0.002 | E Africa M    | E Asia F      | 0.02 |
| 34  | 0.990 ± 0.006 | 0.920 ± 0.004 | 0.982 ± 0.003 | 0.967 ± 0.003 | 0.920 ± 0.009 | 0.994 ± 0.004 | 0.999 ± 0.002 | 0.969 ± 0.003 | 0.998 ± 0.001 | 0.997 ± 0.001 | 0.963 ± 0.009 | 0.998 ± 0.002 | E Africa M    | S E Asia F    | 0.02 |

**Table 145.** For algorithm *innovatrics-001* with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.02 ± 0.01   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.005 ± 0.003 | 0.000 ± 0.000 | 0.04 ± 0.02   | 0.010 ± 0.008 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.000 ± 0.000 | W Africa F    | E Asia F      | 0.84 |
| 15  | 0.04 ± 0.02   | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.011 ± 0.006 | 0.003 ± 0.003 | 0.07 ± 0.03   | 0.005 ± 0.005 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.005 | W Africa F    | E Europe F    | 0.82 |
| 16  | 0.09 ± 0.03   | 0.008 ± 0.004 | 0.005 ± 0.005 | 0.011 ± 0.006 | 0.005 ± 0.005 | 0.13 ± 0.04   | 0.03 ± 0.02   | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.01 ± 0.01   | W Africa F    | E Asia M      | 0.75 |
| 17  | 0.17 ± 0.04   | 0.014 ± 0.005 | 0.02 ± 0.01   | 0.04 ± 0.01   | 0.01 ± 0.01   | 0.22 ± 0.04   | 0.02 ± 0.01   | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.012 ± 0.006 | 0.003 ± 0.003 | 0.06 ± 0.02   | W Africa F    | E Europe M    | 0.71 |
| 18  | 0.20 ± 0.03   | 0.024 ± 0.006 | 0.016 ± 0.009 | 0.051 ± 0.010 | 0.020 ± 0.008 | 0.25 ± 0.04   | 0.06 ± 0.02   | 0.011 ± 0.004 | 0.007 ± 0.005 | 0.026 ± 0.007 | 0.006 ± 0.004 | 0.11 ± 0.03   | W Africa F    | S E Asia M    | 0.63 |
| 19  | 0.29 ± 0.04   | 0.05 ± 0.01   | 0.05 ± 0.02   | 0.10 ± 0.01   | 0.04 ± 0.01   | 0.37 ± 0.04   | 0.17 ± 0.03   | 0.019 ± 0.006 | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.011 ± 0.007 | 0.19 ± 0.04   | W Africa F    | S E Asia M    | 0.56 |
| 20  | 0.40 ± 0.04   | 0.08 ± 0.01   | 0.06 ± 0.02   | 0.15 ± 0.02   | 0.06 ± 0.01   | 0.49 ± 0.04   | 0.26 ± 0.04   | 0.034 ± 0.008 | 0.05 ± 0.02   | 0.15 ± 0.02   | 0.022 ± 0.010 | 0.26 ± 0.04   | W Africa F    | S E Asia M    | 0.51 |
| 21  | 0.49 ± 0.04   | 0.14 ± 0.01   | 0.11 ± 0.02   | 0.24 ± 0.02   | 0.10 ± 0.02   | 0.55 ± 0.04   | 0.36 ± 0.04   | 0.08 ± 0.01   | 0.11 ± 0.02   | 0.25 ± 0.02   | 0.07 ± 0.02   | 0.37 ± 0.04   | W Africa F    | S E Asia M    | 0.40 |
| 22  | 0.59 ± 0.03   | 0.19 ± 0.01   | 0.19 ± 0.02   | 0.32 ± 0.02   | 0.16 ± 0.02   | 0.65 ± 0.04   | 0.53 ± 0.04   | 0.14 ± 0.01   | 0.22 ± 0.03   | 0.36 ± 0.02   | 0.10 ± 0.02   | 0.50 ± 0.04   | W Africa F    | S E Asia M    | 0.34 |
| 23  | 0.66 ± 0.03   | 0.28 ± 0.02   | 0.27 ± 0.02   | 0.44 ± 0.02   | 0.24 ± 0.02   | 0.75 ± 0.03   | 0.63 ± 0.03   | 0.22 ± 0.02   | 0.34 ± 0.03   | 0.53 ± 0.02   | 0.20 ± 0.02   | 0.63 ± 0.03   | W Africa F    | S E Asia M    | 0.27 |
| 24  | 0.72 ± 0.03   | 0.35 ± 0.01   | 0.40 ± 0.02   | 0.52 ± 0.02   | 0.34 ± 0.02   | 0.77 ± 0.03   | 0.73 ± 0.03   | 0.34 ± 0.02   | 0.49 ± 0.03   | 0.67 ± 0.02   | 0.30 ± 0.03   | 0.73 ± 0.03   | W Africa F    | S E Asia M    | 0.20 |
| 25  | 0.77 ± 0.02   | 0.48 ± 0.02   | 0.51 ± 0.02   | 0.61 ± 0.01   | 0.44 ± 0.02   | 0.84 ± 0.02   | 0.82 ± 0.02   | 0.47 ± 0.02   | 0.66 ± 0.03   | 0.77 ± 0.02   | 0.41 ± 0.03   | 0.80 ± 0.02   | W Africa F    | S E Asia M    | 0.15 |
| 26  | 0.82 ± 0.02   | 0.56 ± 0.01   | 0.63 ± 0.02   | 0.68 ± 0.01   | 0.55 ± 0.02   | 0.90 ± 0.02   | 0.88 ± 0.02   | 0.59 ± 0.02   | 0.76 ± 0.02   | 0.85 ± 0.01   | 0.56 ± 0.03   | 0.88 ± 0.02   | W Africa F    | S E Asia F    | 0.11 |
| 27  | 0.88 ± 0.02   | 0.66 ± 0.01   | 0.73 ± 0.02   | 0.74 ± 0.01   | 0.62 ± 0.02   | 0.92 ± 0.01   | 0.91 ± 0.02   | 0.70 ± 0.02   | 0.87 ± 0.02   | 0.898 ± 0.009 | 0.68 ± 0.03   | 0.91 ± 0.01   | W Africa F    | S E Asia F    | 0.08 |
| 28  | 0.91 ± 0.01   | 0.75 ± 0.01   | 0.82 ± 0.01   | 0.80 ± 0.01   | 0.72 ± 0.02   | 0.95 ± 0.01   | 0.95 ± 0.01   | 0.80 ± 0.02   | 0.93 ± 0.01   | 0.938 ± 0.007 | 0.79 ± 0.02   | 0.95 ± 0.01   | E Africa M    | S E Asia F    | 0.06 |
| 29  | 0.94 ± 0.01   | 0.810 ± 0.010 | 0.90 ± 0.01   | 0.855 ± 0.009 | 0.79 ± 0.01   | 0.96 ± 0.01   | 0.96 ± 0.01   | 0.89 ± 0.01   | 0.96 ± 0.01   | 0.962 ± 0.005 | 0.85 ± 0.02   | 0.973 ± 0.008 | W Africa M    | S E Asia F    | 0.04 |
| 30  | 0.95 ± 0.01   | 0.869 ± 0.009 | 0.922 ± 0.010 | 0.891 ± 0.008 | 0.85 ± 0.01   | 0.969 ± 0.009 | 0.982 ± 0.007 | 0.923 ± 0.009 | 0.982 ± 0.008 | 0.975 ± 0.004 | 0.90 ± 0.02   | 0.983 ± 0.007 | W Africa M    | S E Asia F    | 0.03 |
| 31  | 0.971 ± 0.009 | 0.911 ± 0.008 | 0.953 ± 0.009 | 0.919 ± 0.007 | 0.90 ± 0.01   | 0.984 ± 0.007 | 0.987 ± 0.006 | 0.957 ± 0.007 | 0.989 ± 0.006 | 0.984 ± 0.003 | 0.94 ± 0.01   | 0.991 ± 0.005 | W Africa M    | S E Asia F    | 0.02 |
| 32  | 0.981 ± 0.007 | 0.943 ± 0.006 | 0.976 ± 0.006 | 0.949 ± 0.006 | 0.93 ± 0.01   | 0.982 ± 0.007 | 0.993 ± 0.005 | 0.974 ± 0.006 | 0.994 ± 0.004 | 0.992 ± 0.002 | 0.97 ± 0.01   | 0.994 ± 0.004 | E Europe M    | S E Asia F    | 0.01 |
| 33  | 0.988 ± 0.006 | 0.963 ± 0.005 | 0.985 ± 0.005 | 0.965 ± 0.005 | 0.951 ± 0.009 | 0.995 ± 0.004 | 0.997 ± 0.003 | 0.985 ± 0.005 | 0.998 ± 0.003 | 0.996 ± 0.002 | 0.986 ± 0.007 | 0.996 ± 0.004 | E Europe M    | S E Asia F    | 0.01 |
| 34  | 0.985 ± 0.008 | 0.976 ± 0.004 | 0.990 ± 0.004 | 0.979 ± 0.004 | 0.971 ± 0.008 | 0.993 ± 0.004 | 0.999 ± 0.002 | 0.993 ± 0.003 | 0.999 ± 0.002 | 0.997 ± 0.002 | 0.986 ± 0.007 | 0.996 ± 0.004 | E Africa M    | S E Asia F    | 0.01 |

**Table 146.** For algorithm *innovatrics-001* with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.12 ± 0.05   | 0.001 ± 0.001 | 0.004 ± 0.004 | 0.021 ± 0.008 | 0.000 ± 0.000 | 0.09 ± 0.05   | 0.06 ± 0.04   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.03 ± 0.02   | E Africa F    | S E Asia F    | 0.76 |
| 15  | 0.12 ± 0.05   | 0.001 ± 0.001 | 0.004 ± 0.004 | 0.03 ± 0.01   | 0.002 ± 0.002 | 0.19 ± 0.05   | 0.07 ± 0.04   | 0.002 ± 0.001 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.10 ± 0.04   | W Africa F    | E Europe M    | 0.75 |
| 16  | 0.23 ± 0.06   | 0.004 ± 0.002 | 0.008 ± 0.006 | 0.06 ± 0.01   | 0.003 ± 0.003 | 0.21 ± 0.05   | 0.12 ± 0.04   | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.007 ± 0.004 | 0.000 ± 0.000 | 0.10 ± 0.04   | E Africa F    | S E Asia M    | 0.72 |
| 17  | 0.29 ± 0.05   | 0.012 ± 0.003 | 0.03 ± 0.01   | 0.09 ± 0.01   | 0.007 ± 0.005 | 0.33 ± 0.06   | 0.18 ± 0.05   | 0.005 ± 0.002 | 0.003 ± 0.003 | 0.040 ± 0.009 | 0.003 ± 0.003 | 0.20 ± 0.04   | W Africa F    | S E Asia M    | 0.66 |
| 18  | 0.40 ± 0.05   | 0.021 ± 0.004 | 0.06 ± 0.01   | 0.12 ± 0.02   | 0.011 ± 0.005 | 0.33 ± 0.05   | 0.30 ± 0.05   | 0.011 ± 0.003 | 0.019 ± 0.009 | 0.07 ± 0.01   | 0.006 ± 0.005 | 0.26 ± 0.04   | E Africa F    | S E Asia M    | 0.61 |
| 19  | 0.45 ± 0.05   | 0.028 ± 0.004 | 0.08 ± 0.01   | 0.18 ± 0.01   | 0.022 ± 0.007 | 0.51 ± 0.05   | 0.44 ± 0.05   | 0.023 ± 0.004 | 0.05 ± 0.01   | 0.15 ± 0.01   | 0.015 ± 0.006 | 0.34 ± 0.04   | W Africa F    | S E Asia M    | 0.56 |
| 20  | 0.57 ± 0.04   | 0.054 ± 0.005 | 0.13 ± 0.01   | 0.23 ± 0.02   | 0.041 ± 0.008 | 0.60 ± 0.04   | 0.54 ± 0.05   | 0.044 ± 0.005 | 0.10 ± 0.02   | 0.26 ± 0.02   | 0.040 ± 0.010 | 0.47 ± 0.04   | W Africa F    | S E Asia M    | 0.50 |
| 21  | 0.65 ± 0.04   | 0.078 ± 0.005 | 0.17 ± 0.01   | 0.30 ± 0.02   | 0.06 ± 0.01   | 0.67 ± 0.04   | 0.61 ± 0.04   | 0.081 ± 0.007 | 0.21 ± 0.02   | 0.38 ± 0.02   | 0.07 ± 0.01   | 0.61 ± 0.04   | W Africa F    | S E Asia F    | 0.44 |
| 22  | 0.72 ± 0.03   | 0.110 ± 0.006 | 0.26 ± 0.02   | 0.38 ± 0.01   | 0.11 ± 0.01   | 0.76 ± 0.03   | 0.73 ± 0.04   | 0.141 ± 0.009 | 0.33 ± 0.02   | 0.51 ± 0.02   | 0.12 ± 0.02   | 0.73 ± 0.03   | W Africa F    | S E Asia F    | 0.39 |
| 23  | 0.77 ± 0.03   | 0.156 ± 0.007 | 0.33 ± 0.01   | 0.44 ± 0.01   | 0.16 ± 0.01   | 0.79 ± 0.02   | 0.80 ± 0.03   | 0.219 ± 0.009 | 0.48 ± 0.02   | 0.62 ± 0.01   | 0.19 ± 0.02   | 0.80 ± 0.03   | E Africa M    | E Asia F      | 0.33 |
| 24  | 0.81 ± 0.02   | 0.214 ± 0.007 | 0.43 ± 0.01   | 0.53 ± 0.01   | 0.22 ± 0.01   | 0.84 ± 0.02   | 0.87 ± 0.03   | 0.31 ± 0.01   | 0.61 ± 0.02   | 0.73 ± 0.01   | 0.27 ± 0.02   | 0.86 ± 0.02   | E Africa M    | E Asia F      | 0.28 |
| 25  | 0.86 ± 0.02   | 0.284 ± 0.008 | 0.53 ± 0.01   | 0.60 ± 0.01   | 0.30 ± 0.02   | 0.89 ± 0.02   | 0.92 ± 0.02   | 0.40 ± 0.01   | 0.72 ± 0.02   | 0.809 ± 0.009 | 0.37 ± 0.02   | 0.91 ± 0.02   | E Africa M    | E Asia F      | 0.23 |
| 26  | 0.90 ± 0.02   | 0.364 ± 0.007 | 0.62 ± 0.01   | 0.658 ± 0.009 | 0.38 ± 0.01   | 0.91 ± 0.01   | 0.95 ± 0.01   | 0.51 ± 0.01   | 0.81 ± 0.02   | 0.859 ± 0.008 | 0.48 ± 0.02   | 0.93 ± 0.01   | E Africa M    | E Asia F      | 0.19 |
| 27  | 0.90 ± 0.02   | 0.433 ± 0.008 | 0.71 ± 0.01   | 0.713 ± 0.007 | 0.47 ± 0.02   | 0.94 ± 0.01   | 0.96 ± 0.01   | 0.58 ± 0.01   | 0.88 ± 0.01   | 0.906 ± 0.006 | 0.59 ± 0.02   | 0.969 ± 0.008 | W Africa M    | E Asia F      | 0.15 |
| 28  | 0.93 ± 0.01   | 0.522 ± 0.008 | 0.782 ± 0.010 | 0.766 ± 0.008 | 0.55 ± 0.01   | 0.95 ± 0.01   | 0.974 ± 0.010 | 0.68 ± 0.01   | 0.936 ± 0.008 | 0.935 ± 0.005 | 0.66 ± 0.02   | 0.975 ± 0.007 | W Africa M    | E Asia F      | 0.12 |
| 29  | 0.95 ± 0.01   | 0.601 ± 0.007 | 0.852 ± 0.008 | 0.803 ± 0.007 | 0.62 ± 0.01   | 0.967 ± 0.008 | 0.989 ± 0.006 | 0.761 ± 0.009 | 0.951 ± 0.007 | 0.951 ± 0.004 | 0.75 ± 0.02   | 0.986 ± 0.005 | E Africa M    | E Asia F      | 0.10 |
| 30  | 0.96 ± 0.01   | 0.671 ± 0.007 | 0.897 ± 0.008 | 0.847 ± 0.006 | 0.69 ± 0.01   | 0.972 ± 0.007 | 0.990 ± 0.006 | 0.823 ± 0.008 | 0.976 ± 0.005 | 0.969 ± 0.003 | 0.82 ± 0.02   | 0.987 ± 0.005 | E Africa M    | E Asia F      | 0.07 |
| 31  | 0.97 ± 0.01   | 0.741 ± 0.006 | 0.933 ± 0.006 | 0.885 ± 0.005 | 0.75 ± 0.01   | 0.982 ± 0.006 | 0.995 ± 0.004 | 0.878 ± 0.006 | 0.983 ± 0.004 | 0.979 ± 0.002 | 0.87 ± 0.02   | 0.993 ± 0.004 | E Africa M    | E Asia F      | 0.06 |
| 32  | 0.987 ± 0.006 | 0.805 ± 0.006 | 0.959 ± 0.005 | 0.910 ± 0.005 | 0.81 ± 0.01   | 0.990 ± 0.005 | 0.997 ± 0.003 | 0.916 ± 0.005 | 0.991 ± 0.003 | 0.987 ± 0.002 | 0.91 ± 0.01   | 0.995 ± 0.003 | E Africa M    | E Asia F      | 0.04 |
| 33  | 0.990 ± 0.006 | 0.849 ± 0.005 | 0.973 ± 0.004 | 0.936 ± 0.004 | 0.85 ± 0.01   | 0.990 ± 0.005 | 0.999 ± 0.002 | 0.940 ± 0.005 | 0.997 ± 0.002 | 0.993 ± 0.001 | 0.938 ± 0.010 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.03 |
| 34  | 0.990 ± 0.006 | 0.887 ± 0.005 | 0.982 ± 0.003 | 0.955 ± 0.004 | 0.895 ± 0.010 | 0.993 ± 0.004 | 1.000 ± 0.000 | 0.961 ± 0.004 | 0.998 ± 0.001 | 0.995 ± 0.001 | 0.959 ± 0.009 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.02 |

**Table 147.** For algorithm *innovatrics-002* with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.03 ± 0.02   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.06 ± 0.03   | 0.008 ± 0.006 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.009 ± 0.009 | W Africa F    | E Asia F      | 0.81 |
| 15  | 0.05 ± 0.02   | 0.005 ± 0.004 | 0.003 ± 0.003 | 0.016 ± 0.008 | 0.003 ± 0.003 | 0.10 ± 0.04   | 0.01 ± 0.01   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.009 ± 0.009 | W Africa F    | E Asia M      | 0.79 |
| 16  | 0.15 ± 0.03   | 0.011 ± 0.005 | 0.01 ± 0.01   | 0.018 ± 0.009 | 0.010 ± 0.008 | 0.18 ± 0.04   | 0.03 ± 0.02   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.006 ± 0.003 | 0.003 ± 0.003 | 0.03 ± 0.02   | W Africa F    | E Asia M      | 0.73 |
| 17  | 0.22 ± 0.04   | 0.024 ± 0.008 | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.31 ± 0.05   | 0.05 ± 0.02   | 0.008 ± 0.004 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.003 ± 0.003 | 0.09 ± 0.03   | W Africa F    | S E Asia M    | 0.68 |
| 18  | 0.26 ± 0.04   | 0.032 ± 0.008 | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.04 ± 0.01   | 0.33 ± 0.04   | 0.09 ± 0.02   | 0.018 ± 0.006 | 0.015 ± 0.009 | 0.044 ± 0.009 | 0.010 ± 0.007 | 0.13 ± 0.03   | W Africa F    | S E Asia M    | 0.59 |
| 19  | 0.37 ± 0.04   | 0.07 ± 0.01   | 0.08 ± 0.02   | 0.12 ± 0.02   | 0.07 ± 0.02   | 0.48 ± 0.04   | 0.21 ± 0.03   | 0.035 ± 0.007 | 0.03 ± 0.01   | 0.11 ± 0.02   | 0.03 ± 0.01   | 0.26 ± 0.04   | W Africa F    | S E Asia M    | 0.51 |
| 20  | 0.47 ± 0.04   | 0.12 ± 0.01   | 0.11 ± 0.02   | 0.18 ± 0.02   | 0.10 ± 0.02   | 0.60 ± 0.04   | 0.32 ± 0.04   | 0.07 ± 0.01   | 0.07 ± 0.02   | 0.20 ± 0.02   | 0.06 ± 0.02   | 0.34 ± 0.04   | W Africa F    | S E Asia M    | 0.44 |
| 21  | 0.53 ± 0.04   | 0.20 ± 0.02   | 0.18 ± 0.02   | 0.27 ± 0.02   | 0.15 ± 0.02   | 0.64 ± 0.04   | 0.44 ± 0.04   | 0.14 ± 0.01   | 0.15 ± 0.03   | 0.31 ± 0.02   | 0.11 ± 0.02   | 0.44 ± 0.04   | W Africa F    | S E Asia M    | 0.34 |
| 22  | 0.64 ± 0.03   | 0.26 ± 0.02   | 0.27 ± 0.02   | 0.34 ± 0.02   | 0.23 ± 0.02   | 0.73 ± 0.03   | 0.62 ± 0.04   | 0.22 ± 0.02   | 0.30 ± 0.03   | 0.44 ± 0.02   | 0.16 ± 0.02   | 0.60 ± 0.04   | W Africa F    | S E Asia M    | 0.28 |
| 23  | 0.71 ± 0.03   | 0.36 ± 0.02   | 0.34 ± 0.02   | 0.44 ± 0.02   | 0.32 ± 0.02   | 0.82 ± 0.03   | 0.69 ± 0.04   | 0.34 ± 0.02   | 0.39 ± 0.03   | 0.61 ± 0.02   | 0.26 ± 0.03   | 0.70 ± 0.04   | W Africa F    | S E Asia M    | 0.22 |
| 24  | 0.76 ± 0.03   | 0.43 ± 0.02   | 0.48 ± 0.02   | 0.53 ± 0.02   | 0.40 ± 0.02   | 0.84 ± 0.02   | 0.79 ± 0.03   | 0.44 ± 0.02   | 0.56 ± 0.03   | 0.71 ± 0.02   | 0.39 ± 0.03   | 0.78 ± 0.03   | W Africa F    | S E Asia M    | 0.17 |
| 25  | 0.81 ± 0.02   | 0.54 ± 0.01   | 0.60 ± 0.02   | 0.60 ± 0.01   | 0.50 ± 0.02   | 0.88 ± 0.02   | 0.85 ± 0.02   | 0.57 ± 0.02   | 0.72 ± 0.03   | 0.81 ± 0.01   | 0.51 ± 0.03   | 0.85 ± 0.02   | W Africa F    | S E Asia F    | 0.13 |
| 26  | 0.84 ± 0.02   | 0.61 ± 0.01   | 0.69 ± 0.02   | 0.67 ± 0.01   | 0.59 ± 0.02   | 0.93 ± 0.01   | 0.91 ± 0.02   | 0.69 ± 0.02   | 0.78 ± 0.02   | 0.88 ± 0.01   | 0.64 ± 0.03   | 0.92 ± 0.02   | W Africa F    | S E Asia F    | 0.10 |
| 27  | 0.90 ± 0.02   | 0.71 ± 0.01   | 0.76 ± 0.02   | 0.72 ± 0.01   | 0.66 ± 0.02   | 0.95 ± 0.01   | 0.93 ± 0.02   | 0.78 ± 0.02   | 0.88 ± 0.02   | 0.916 ± 0.008 | 0.75 ± 0.02   | 0.94 ± 0.01   | W Africa F    | S E Asia F    | 0.07 |
| 28  | 0.92 ± 0.01   | 0.78 ± 0.01   | 0.84 ± 0.01   | 0.78 ± 0.01   | 0.74 ± 0.02   | 0.96 ± 0.01   | 0.96 ± 0.01   | 0.86 ± 0.01   | 0.94 ± 0.01   | 0.946 ± 0.006 | 0.83 ± 0.02   | 0.96 ± 0.01   | W Africa F    | S E Asia F    | 0.05 |
| 29  | 0.95 ± 0.01   | 0.835 ± 0.010 | 0.90 ± 0.01   | 0.838 ± 0.009 | 0.80 ± 0.02   | 0.967 ± 0.010 | 0.969 ± 0.009 | 0.91 ± 0.01   | 0.96 ± 0.01   | 0.966 ± 0.005 | 0.87 ± 0.02   | 0.982 ± 0.006 | W Africa M    | S E Asia F    | 0.04 |
| 30  | 0.96 ± 0.01   | 0.883 ± 0.007 | 0.92 ± 0.01   | 0.871 ± 0.008 | 0.86 ± 0.01   | 0.978 ± 0.008 | 0.987 ± 0.006 | 0.943 ± 0.008 | 0.982 ± 0.007 | 0.976 ± 0.004 | 0.92 ± 0.02   | 0.988 ± 0.005 | W Africa M    | S E Asia F    | 0.03 |
| 31  | 0.970 ± 0.009 | 0.919 ± 0.007 | 0.953 ± 0.009 | 0.896 ± 0.008 | 0.89 ± 0.01   | 0.991 ± 0.005 | 0.988 ± 0.006 | 0.968 ± 0.006 | 0.988 ± 0.006 | 0.987 ± 0.003 | 0.95 ± 0.01   | 0.995 ± 0.003 | W Africa M    | S E Asia F    | 0.02 |
| 32  | 0.981 ± 0.007 | 0.941 ± 0.006 | 0.975 ± 0.006 | 0.932 ± 0.007 | 0.92 ± 0.01   | 0.986 ± 0.006 | 0.996 ± 0.004 | 0.979 ± 0.005 | 0.994 ± 0.004 | 0.992 ± 0.002 | 0.97 ± 0.01   | 0.996 ± 0.003 | W Africa M    | S E Asia F    | 0.01 |
| 33  | 0.988 ± 0.006 | 0.962 ± 0.005 | 0.979 ± 0.005 | 0.947 ± 0.006 | 0.952 ± 0.008 | 0.997 ± 0.003 | 0.997 ± 0.003 | 0.988 ± 0.004 | 0.999 ± 0.002 | 0.996 ± 0.002 | 0.986 ± 0.007 | 0.998 ± 0.002 | E Europe M    | S Asia F      | 0.01 |
| 34  | 0.988 ± 0.007 | 0.975 ± 0.004 | 0.988 ± 0.005 | 0.965 ± 0.005 | 0.970 ± 0.007 | 0.994 ± 0.004 | 0.998 ± 0.002 | 0.995 ± 0.002 | 0.998 ± 0.003 | 0.996 ± 0.002 | 0.989 ± 0.007 | 0.999 ± 0.001 | W Africa M    | S Asia F      | 0.01 |

**Table 148.** For algorithm *innovatrics-002* with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.07 ± 0.03   | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.010 ± 0.005 | 0.000 ± 0.000 | 0.05 ± 0.03   | 0.03 ± 0.02   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.04 ± 0.03   | E Africa F    | E Europe F    | 0.77 |
| 15  | 0.10 ± 0.04   | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.009 ± 0.005 | 0.002 ± 0.002 | 0.16 ± 0.05   | 0.03 ± 0.02   | 0.001 ± 0.001 | 0.002 ± 0.002 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.07 ± 0.04   | W Africa F    | E Asia F      | 0.81 |
| 16  | 0.18 ± 0.05   | 0.002 ± 0.001 | 0.005 ± 0.005 | 0.026 ± 0.009 | 0.000 ± 0.000 | 0.17 ± 0.05   | 0.07 ± 0.04   | 0.001 ± 0.001 | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.09 ± 0.04   | E Africa F    | S E Asia F    | 0.75 |
| 17  | 0.19 ± 0.04   | 0.007 ± 0.003 | 0.018 ± 0.009 | 0.031 ± 0.009 | 0.003 ± 0.003 | 0.24 ± 0.05   | 0.15 ± 0.05   | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.020 ± 0.006 | 0.003 ± 0.003 | 0.19 ± 0.05   | W Africa F    | E Europe M    | 0.68 |
| 18  | 0.32 ± 0.05   | 0.010 ± 0.003 | 0.031 ± 0.010 | 0.061 ± 0.010 | 0.003 ± 0.002 | 0.26 ± 0.04   | 0.22 ± 0.04   | 0.005 ± 0.002 | 0.008 ± 0.006 | 0.030 ± 0.007 | 0.003 ± 0.003 | 0.20 ± 0.04   | E Africa F    | S E Asia F    | 0.67 |
| 19  | 0.38 ± 0.05   | 0.018 ± 0.003 | 0.043 ± 0.009 | 0.10 ± 0.01   | 0.011 ± 0.004 | 0.40 ± 0.04   | 0.35 ± 0.05   | 0.014 ± 0.003 | 0.019 ± 0.008 | 0.09 ± 0.01   | 0.009 ± 0.004 | 0.29 ± 0.04   | W Africa F    | S E Asia M    | 0.61 |
| 20  | 0.49 ± 0.04   | 0.031 ± 0.004 | 0.08 ± 0.01   | 0.13 ± 0.01   | 0.022 ± 0.006 | 0.49 ± 0.04   | 0.48 ± 0.04   | 0.027 ± 0.004 | 0.05 ± 0.01   | 0.17 ± 0.01   | 0.025 ± 0.008 | 0.39 ± 0.04   | W Africa F    | S E Asia F    | 0.56 |
| 21  | 0.57 ± 0.04   | 0.047 ± 0.004 | 0.11 ± 0.01   | 0.18 ± 0.01   | 0.038 ± 0.008 | 0.57 ± 0.04   | 0.53 ± 0.04   | 0.052 ± 0.006 | 0.12 ± 0.02   | 0.25 ± 0.02   | 0.05 ± 0.01   | 0.52 ± 0.04   | W Africa F    | S E Asia F    | 0.50 |
| 22  | 0.65 ± 0.04   | 0.074 ± 0.005 | 0.17 ± 0.01   | 0.24 ± 0.01   | 0.068 ± 0.009 | 0.66 ± 0.03   | 0.68 ± 0.04   | 0.098 ± 0.007 | 0.19 ± 0.02   | 0.36 ± 0.02   | 0.09 ± 0.01   | 0.64 ± 0.04   | E Africa M    | S E Asia F    | 0.45 |
| 23  | 0.69 ± 0.03   | 0.110 ± 0.006 | 0.24 ± 0.01   | 0.29 ± 0.01   | 0.12 ± 0.01   | 0.72 ± 0.03   | 0.75 ± 0.03   | 0.164 ± 0.009 | 0.32 ± 0.02   | 0.47 ± 0.02   | 0.15 ± 0.02   | 0.73 ± 0.03   | E Africa M    | E Asia F      | 0.38 |
| 24  | 0.76 ± 0.03   | 0.154 ± 0.006 | 0.30 ± 0.01   | 0.37 ± 0.01   | 0.15 ± 0.01   | 0.77 ± 0.02   | 0.82 ± 0.03   | 0.24 ± 0.01   | 0.46 ± 0.02   | 0.59 ± 0.01   | 0.21 ± 0.02   | 0.81 ± 0.02   | E Africa M    | S E Asia F    | 0.34 |
| 25  | 0.81 ± 0.02   | 0.213 ± 0.007 | 0.41 ± 0.01   | 0.423 ± 0.009 | 0.22 ± 0.02   | 0.82 ± 0.02   | 0.87 ± 0.02   | 0.32 ± 0.01   | 0.57 ± 0.02   | 0.68 ± 0.01   | 0.30 ± 0.02   | 0.86 ± 0.02   | E Africa M    | E Asia F      | 0.28 |
| 26  | 0.84 ± 0.02   | 0.280 ± 0.007 | 0.48 ± 0.01   | 0.491 ± 0.010 | 0.29 ± 0.01   | 0.85 ± 0.02   | 0.92 ± 0.02   | 0.42 ± 0.01   | 0.68 ± 0.02   | 0.757 ± 0.009 | 0.40 ± 0.02   | 0.89 ± 0.02   | E Africa M    | E Asia F      | 0.23 |
| 27  | 0.84 ± 0.02   | 0.345 ± 0.007 | 0.59 ± 0.01   | 0.551 ± 0.009 | 0.36 ± 0.01   | 0.90 ± 0.01   | 0.94 ± 0.02   | 0.50 ± 0.01   | 0.76 ± 0.02   | 0.817 ± 0.008 | 0.51 ± 0.03   | 0.95 ± 0.01   | W Africa M    | E Asia F      | 0.19 |
| 28  | 0.90 ± 0.02   | 0.428 ± 0.007 | 0.67 ± 0.01   | 0.609 ± 0.009 | 0.44 ± 0.02   | 0.91 ± 0.01   | 0.96 ± 0.01   | 0.598 ± 0.010 | 0.84 ± 0.01   | 0.874 ± 0.006 | 0.58 ± 0.02   | 0.958 ± 0.009 | W Africa M    | E Asia F      | 0.16 |
| 29  | 0.92 ± 0.01   | 0.502 ± 0.007 | 0.761 ± 0.010 | 0.666 ± 0.008 | 0.51 ± 0.01   | 0.94 ± 0.01   | 0.975 ± 0.010 | 0.684 ± 0.010 | 0.88 ± 0.01   | 0.904 ± 0.005 | 0.68 ± 0.02   | 0.975 ± 0.007 | E Africa M    | E Asia F      | 0.13 |
| 30  | 0.93 ± 0.01   | 0.576 ± 0.007 | 0.815 ± 0.009 | 0.724 ± 0.007 | 0.58 ± 0.02   | 0.94 ± 0.01   | 0.984 ± 0.007 | 0.756 ± 0.009 | 0.931 ± 0.008 | 0.934 ± 0.004 | 0.77 ± 0.02   | 0.981 ± 0.005 | E Africa M    | E Asia F      | 0.10 |
| 31  | 0.96 ± 0.01   | 0.648 ± 0.007 | 0.865 ± 0.008 | 0.776 ± 0.007 | 0.65 ± 0.01   | 0.970 ± 0.008 | 0.987 ± 0.007 | 0.823 ± 0.007 | 0.954 ± 0.007 | 0.953 ± 0.003 | 0.82 ± 0.02   | 0.990 ± 0.004 | W Africa M    | E Asia F      | 0.08 |
| 32  | 0.975 ± 0.009 | 0.722 ± 0.006 | 0.917 ± 0.007 | 0.817 ± 0.006 | 0.72 ± 0.01   | 0.974 ± 0.008 | 0.995 ± 0.005 | 0.875 ± 0.006 | 0.978 ± 0.005 | 0.967 ± 0.003 | 0.85 ± 0.02   | 0.991 ± 0.004 | E Africa M    | S E Asia F    | 0.06 |
| 33  | 0.973 ± 0.009 | 0.777 ± 0.006 | 0.938 ± 0.006 | 0.862 ± 0.006 | 0.77 ± 0.01   | 0.981 ± 0.007 | 0.998 ± 0.002 | 0.906 ± 0.006 | 0.987 ± 0.004 | 0.979 ± 0.002 | 0.91 ± 0.01   | 0.995 ± 0.003 | E Africa M    | S E Asia F    | 0.05 |
| 34  | 0.983 ± 0.008 | 0.826 ± 0.006 | 0.956 ± 0.005 | 0.896 ± 0.005 | 0.83 ± 0.01   | 0.986 ± 0.006 | 0.996 ± 0.004 | 0.936 ± 0.005 | 0.994 ± 0.003 | 0.987 ± 0.002 | 0.93 ± 0.01   | 0.997 ± 0.002 | W Africa M    | E Asia F      | 0.04 |

**Table 149.** For algorithm jumio-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.07 ± 0.02   | 0.022 ± 0.007 | 0.06 ± 0.02   | 0.029 ± 0.008 | 0.04 ± 0.02   | 0.07 ± 0.03   | 0.04 ± 0.02   | 0.003 ± 0.002 | 0.08 ± 0.02   | 0.012 ± 0.005 | 0.005 ± 0.005 | 0.02 ± 0.01   | E Europe M    | E Asia M      | 0.42 |
| 15  | 0.11 ± 0.03   | 0.041 ± 0.009 | 0.08 ± 0.03   | 0.06 ± 0.02   | 0.07 ± 0.02   | 0.13 ± 0.04   | 0.03 ± 0.01   | 0.002 ± 0.002 | 0.08 ± 0.03   | 0.016 ± 0.007 | 0.005 ± 0.005 | 0.01 ± 0.01   | W Africa F    | E Asia M      | 0.47 |
| 16  | 0.18 ± 0.04   | 0.06 ± 0.01   | 0.11 ± 0.03   | 0.10 ± 0.02   | 0.12 ± 0.03   | 0.22 ± 0.05   | 0.04 ± 0.02   | 0.009 ± 0.005 | 0.08 ± 0.03   | 0.006 ± 0.004 | 0.011 ± 0.008 | 0.02 ± 0.01   | W Africa F    | S Asia M      | 0.50 |
| 17  | 0.25 ± 0.04   | 0.10 ± 0.01   | 0.16 ± 0.04   | 0.16 ± 0.02   | 0.15 ± 0.04   | 0.29 ± 0.04   | 0.04 ± 0.02   | 0.019 ± 0.006 | 0.11 ± 0.03   | 0.017 ± 0.007 | 0.03 ± 0.01   | 0.06 ± 0.03   | W Africa F    | S Asia M      | 0.45 |
| 18  | 0.28 ± 0.04   | 0.15 ± 0.01   | 0.18 ± 0.03   | 0.18 ± 0.02   | 0.20 ± 0.02   | 0.37 ± 0.04   | 0.07 ± 0.02   | 0.039 ± 0.007 | 0.13 ± 0.03   | 0.041 ± 0.009 | 0.03 ± 0.01   | 0.10 ± 0.03   | W Africa F    | S E Asia M    | 0.40 |
| 19  | 0.41 ± 0.04   | 0.23 ± 0.02   | 0.23 ± 0.03   | 0.28 ± 0.02   | 0.30 ± 0.03   | 0.47 ± 0.05   | 0.16 ± 0.03   | 0.09 ± 0.01   | 0.11 ± 0.03   | 0.08 ± 0.01   | 0.06 ± 0.02   | 0.21 ± 0.04   | W Africa F    | S E Asia M    | 0.34 |
| 20  | 0.51 ± 0.04   | 0.31 ± 0.02   | 0.31 ± 0.03   | 0.36 ± 0.02   | 0.41 ± 0.03   | 0.54 ± 0.04   | 0.27 ± 0.04   | 0.14 ± 0.01   | 0.16 ± 0.03   | 0.17 ± 0.02   | 0.17 ± 0.02   | 0.27 ± 0.04   | W Africa F    | E Asia M      | 0.26 |
| 21  | 0.59 ± 0.04   | 0.41 ± 0.02   | 0.39 ± 0.03   | 0.44 ± 0.02   | 0.50 ± 0.03   | 0.65 ± 0.04   | 0.36 ± 0.03   | 0.27 ± 0.02   | 0.23 ± 0.03   | 0.26 ± 0.02   | 0.27 ± 0.03   | 0.41 ± 0.04   | W Africa F    | E Europe M    | 0.20 |
| 22  | 0.65 ± 0.03   | 0.52 ± 0.02   | 0.49 ± 0.03   | 0.54 ± 0.02   | 0.60 ± 0.02   | 0.70 ± 0.03   | 0.54 ± 0.03   | 0.40 ± 0.02   | 0.37 ± 0.03   | 0.38 ± 0.02   | 0.39 ± 0.03   | 0.49 ± 0.04   | W Africa F    | E Europe M    | 0.13 |
| 23  | 0.73 ± 0.03   | 0.61 ± 0.02   | 0.58 ± 0.03   | 0.64 ± 0.02   | 0.68 ± 0.02   | 0.81 ± 0.03   | 0.62 ± 0.03   | 0.52 ± 0.02   | 0.47 ± 0.03   | 0.52 ± 0.02   | 0.54 ± 0.03   | 0.65 ± 0.03   | W Africa F    | E Europe M    | 0.09 |
| 24  | 0.80 ± 0.02   | 0.69 ± 0.02   | 0.70 ± 0.02   | 0.71 ± 0.01   | 0.78 ± 0.02   | 0.80 ± 0.03   | 0.76 ± 0.03   | 0.67 ± 0.02   | 0.63 ± 0.03   | 0.66 ± 0.02   | 0.67 ± 0.03   | 0.76 ± 0.03   | W Africa F    | E Europe M    | 0.05 |
| 25  | 0.83 ± 0.02   | 0.78 ± 0.01   | 0.78 ± 0.02   | 0.77 ± 0.01   | 0.83 ± 0.02   | 0.86 ± 0.02   | 0.83 ± 0.02   | 0.77 ± 0.02   | 0.76 ± 0.03   | 0.78 ± 0.02   | 0.77 ± 0.02   | 0.82 ± 0.02   | W Africa F    | E Europe M    | 0.02 |
| 26  | 0.88 ± 0.02   | 0.836 ± 0.010 | 0.83 ± 0.01   | 0.819 ± 0.010 | 0.88 ± 0.01   | 0.89 ± 0.02   | 0.89 ± 0.02   | 0.85 ± 0.01   | 0.84 ± 0.02   | 0.84 ± 0.01   | 0.86 ± 0.02   | 0.88 ± 0.02   | W Africa F    | S Asia F      | 0.02 |
| 27  | 0.90 ± 0.02   | 0.883 ± 0.009 | 0.90 ± 0.01   | 0.868 ± 0.009 | 0.92 ± 0.01   | 0.93 ± 0.01   | 0.91 ± 0.02   | 0.90 ± 0.01   | 0.93 ± 0.01   | 0.890 ± 0.010 | 0.92 ± 0.02   | 0.92 ± 0.01   | E Europe M    | S Asia F      | 0.01 |
| 28  | 0.93 ± 0.01   | 0.917 ± 0.007 | 0.937 ± 0.010 | 0.909 ± 0.007 | 0.943 ± 0.008 | 0.95 ± 0.01   | 0.95 ± 0.01   | 0.943 ± 0.009 | 0.95 ± 0.01   | 0.928 ± 0.007 | 0.94 ± 0.01   | 0.95 ± 0.01   | E Europe M    | S Asia F      | 0.01 |
| 29  | 0.95 ± 0.01   | 0.946 ± 0.006 | 0.956 ± 0.008 | 0.934 ± 0.006 | 0.963 ± 0.007 | 0.966 ± 0.009 | 0.972 ± 0.009 | 0.966 ± 0.007 | 0.969 ± 0.009 | 0.951 ± 0.006 | 0.97 ± 0.01   | 0.970 ± 0.008 | E Africa M    | S Asia F      | 0.01 |
| 30  | 0.966 ± 0.009 | 0.969 ± 0.005 | 0.972 ± 0.006 | 0.951 ± 0.006 | 0.978 ± 0.006 | 0.973 ± 0.008 | 0.980 ± 0.008 | 0.977 ± 0.006 | 0.985 ± 0.007 | 0.964 ± 0.005 | 0.981 ± 0.008 | 0.976 ± 0.007 | E Europe M    | S Asia F      | 0.01 |
| 31  | 0.97 ± 0.01   | 0.976 ± 0.004 | 0.985 ± 0.005 | 0.966 ± 0.005 | 0.986 ± 0.004 | 0.978 ± 0.008 | 0.990 ± 0.006 | 0.990 ± 0.004 | 0.992 ± 0.005 | 0.980 ± 0.004 | 0.981 ± 0.008 | 0.989 ± 0.005 | E Europe M    | S Asia F      | 0.01 |
| 32  | 0.978 ± 0.008 | 0.987 ± 0.003 | 0.989 ± 0.005 | 0.976 ± 0.004 | 0.988 ± 0.005 | 0.982 ± 0.007 | 0.993 ± 0.004 | 0.995 ± 0.003 | 0.996 ± 0.004 | 0.989 ± 0.003 | 0.989 ± 0.006 | 0.991 ± 0.005 | E Europe M    | S Asia F      | 0.00 |
| 33  | 0.993 ± 0.005 | 0.992 ± 0.003 | 0.994 ± 0.003 | 0.988 ± 0.003 | 0.996 ± 0.003 | 0.994 ± 0.005 | 0.998 ± 0.002 | 0.995 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.004 | 0.995 ± 0.004 | E Europe M    | S Asia F      | 0.00 |
| 34  | 0.990 ± 0.005 | 0.995 ± 0.002 | 0.997 ± 0.002 | 0.993 ± 0.002 | 0.998 ± 0.002 | 0.993 ± 0.004 | 0.995 ± 0.005 | 0.997 ± 0.002 | 0.999 ± 0.002 | 0.995 ± 0.002 | 0.996 ± 0.004 | 0.997 ± 0.003 | E Europe M    | E Africa F    | 0.00 |

**Table 150.** For algorithm jumio-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.58 ± 0.07   | 0.26 ± 0.02   | 0.42 ± 0.04   | 0.41 ± 0.03   | 0.33 ± 0.04   | 0.42 ± 0.08   | 0.52 ± 0.07   | 0.15 ± 0.01   | 0.26 ± 0.04   | 0.29 ± 0.03   | 0.18 ± 0.04   | 0.45 ± 0.08   | E Africa F    | E Asia M      | 0.22 |
| 15  | 0.57 ± 0.07   | 0.28 ± 0.02   | 0.48 ± 0.05   | 0.44 ± 0.03   | 0.34 ± 0.04   | 0.60 ± 0.07   | 0.44 ± 0.08   | 0.15 ± 0.01   | 0.22 ± 0.03   | 0.22 ± 0.02   | 0.17 ± 0.03   | 0.33 ± 0.08   | W Africa F    | E Asia M      | 0.26 |
| 16  | 0.66 ± 0.07   | 0.32 ± 0.02   | 0.55 ± 0.04   | 0.50 ± 0.03   | 0.41 ± 0.04   | 0.65 ± 0.06   | 0.41 ± 0.07   | 0.16 ± 0.01   | 0.22 ± 0.03   | 0.20 ± 0.02   | 0.15 ± 0.03   | 0.35 ± 0.06   | E Africa F    | S E Asia M    | 0.28 |
| 17  | 0.69 ± 0.06   | 0.38 ± 0.02   | 0.65 ± 0.03   | 0.54 ± 0.03   | 0.45 ± 0.03   | 0.70 ± 0.06   | 0.47 ± 0.07   | 0.20 ± 0.01   | 0.22 ± 0.03   | 0.27 ± 0.02   | 0.20 ± 0.03   | 0.40 ± 0.06   | W Africa F    | S E Asia M    | 0.25 |
| 18  | 0.73 ± 0.05   | 0.42 ± 0.01   | 0.65 ± 0.03   | 0.60 ± 0.02   | 0.48 ± 0.03   | 0.66 ± 0.05   | 0.49 ± 0.06   | 0.24 ± 0.01   | 0.25 ± 0.03   | 0.29 ± 0.02   | 0.22 ± 0.03   | 0.43 ± 0.05   | E Africa F    | S E Asia M    | 0.24 |
| 19  | 0.77 ± 0.04   | 0.48 ± 0.01   | 0.71 ± 0.02   | 0.67 ± 0.02   | 0.53 ± 0.02   | 0.79 ± 0.04   | 0.50 ± 0.05   | 0.30 ± 0.01   | 0.32 ± 0.03   | 0.36 ± 0.02   | 0.30 ± 0.03   | 0.51 ± 0.05   | W Africa F    | E Asia M      | 0.20 |
| 20  | 0.80 ± 0.04   | 0.55 ± 0.01   | 0.76 ± 0.02   | 0.72 ± 0.02   | 0.59 ± 0.02   | 0.83 ± 0.03   | 0.65 ± 0.04   | 0.37 ± 0.01   | 0.42 ± 0.03   | 0.45 ± 0.02   | 0.39 ± 0.03   | 0.57 ± 0.04   | W Africa F    | E Asia M      | 0.16 |
| 21  | 0.85 ± 0.03   | 0.59 ± 0.01   | 0.80 ± 0.02   | 0.75 ± 0.02   | 0.65 ± 0.02   | 0.88 ± 0.03   | 0.67 ± 0.04   | 0.46 ± 0.01   | 0.50 ± 0.03   | 0.56 ± 0.02   | 0.46 ± 0.03   | 0.68 ± 0.03   | W Africa F    | S E Asia M    | 0.13 |
| 22  | 0.88 ± 0.03   | 0.661 ± 0.009 | 0.85 ± 0.01   | 0.81 ± 0.01   | 0.72 ± 0.02   | 0.89 ± 0.02   | 0.79 ± 0.04   | 0.56 ± 0.01   | 0.59 ± 0.03   | 0.65 ± 0.02   | 0.56 ± 0.03   | 0.76 ± 0.03   | W Africa F    | S E Asia M    | 0.10 |
| 23  | 0.91 ± 0.02   | 0.706 ± 0.008 | 0.88 ± 0.01   | 0.835 ± 0.010 | 0.78 ± 0.01   | 0.90 ± 0.02   | 0.81 ± 0.03   | 0.65 ± 0.01   | 0.71 ± 0.02   | 0.74 ± 0.01   | 0.63 ± 0.02   | 0.83 ± 0.02   | E Africa F    | S E Asia M    | 0.07 |
| 24  | 0.92 ± 0.02   | 0.750 ± 0.007 | 0.914 ± 0.008 | 0.874 ± 0.007 | 0.81 ± 0.01   | 0.92 ± 0.02   | 0.88 ± 0.02   | 0.73 ± 0.01   | 0.80 ± 0.02   | 0.81 ± 0.01   | 0.73 ± 0.02   | 0.88 ± 0.02   | E Africa F    | E Asia M      | 0.05 |
| 25  | 0.93 ± 0.02   | 0.800 ± 0.007 | 0.939 ± 0.007 | 0.884 ± 0.007 | 0.85 ± 0.01   | 0.93 ± 0.01   | 0.91 ± 0.02   | 0.801 ± 0.009 | 0.86 ± 0.01   | 0.865 ± 0.008 | 0.80 ± 0.02   | 0.89 ± 0.02   | E Europe F    | E Asia F      | 0.04 |
| 26  | 0.96 ± 0.01   | 0.840 ± 0.006 | 0.954 ± 0.005 | 0.910 ± 0.006 | 0.890 ± 0.010 | 0.95 ± 0.01   | 0.93 ± 0.02   | 0.851 ± 0.008 | 0.90 ± 0.01   | 0.898 ± 0.007 | 0.85 ± 0.02   | 0.93 ± 0.01   | E Africa F    | E Asia F      | 0.03 |
| 27  | 0.96 ± 0.01   | 0.870 ± 0.005 | 0.968 ± 0.004 | 0.928 ± 0.005 | 0.910 ± 0.009 | 0.961 ± 0.008 | 0.95 ± 0.01   | 0.901 ± 0.006 | 0.939 ± 0.009 | 0.924 ± 0.005 | 0.89 ± 0.02   | 0.95 ± 0.01   | E Europe F    | E Asia F      | 0.02 |
| 28  | 0.970 ± 0.009 | 0.900 ± 0.004 | 0.981 ± 0.003 | 0.943 ± 0.004 | 0.931 ± 0.007 | 0.964 ± 0.009 | 0.975 ± 0.009 | 0.931 ± 0.006 | 0.971 ± 0.006 | 0.948 ± 0.004 | 0.93 ± 0.01   | 0.963 ± 0.008 | E Europe F    | E Asia F      | 0.01 |
| 29  | 0.976 ± 0.009 | 0.923 ± 0.004 | 0.987 ± 0.003 | 0.957 ± 0.004 | 0.944 ± 0.007 | 0.975 ± 0.007 | 0.988 ± 0.006 | 0.953 ± 0.004 | 0.970 ± 0.006 | 0.963 ± 0.004 | 0.95 ± 0.01   | 0.975 ± 0.006 | E Africa M    | E Asia F      | 0.01 |
| 30  | 0.981 ± 0.007 | 0.942 ± 0.003 | 0.991 ± 0.002 | 0.968 ± 0.003 | 0.961 ± 0.006 | 0.982 ± 0.006 | 0.979 ± 0.009 | 0.967 ± 0.004 | 0.985 ± 0.004 | 0.976 ± 0.003 | 0.969 ± 0.009 | 0.981 ± 0.006 | E Europe F    | E Asia F      | 0.01 |
| 31  | 0.991 ± 0.006 | 0.958 ± 0.003 | 0.994 ± 0.002 | 0.973 ± 0.003 | 0.968 ± 0.006 | 0.981 ± 0.006 | 0.989 ± 0.007 | 0.979 ± 0.003 | 0.992 ± 0.003 | 0.983 ± 0.002 | 0.977 ± 0.008 | 0.984 ± 0.005 | E Europe F    | E Asia F      | 0.01 |
| 32  | 0.987 ± 0.007 | 0.970 ± 0.003 | 0.996 ± 0.002 | 0.981 ± 0.002 | 0.972 ± 0.005 | 0.990 ± 0.005 | 0.991 ± 0.005 | 0.985 ± 0.002 | 0.994 ± 0.003 | 0.988 ± 0.002 | 0.984 ± 0.006 | 0.992 ± 0.004 | E Europe F    | E Asia F      | 0.00 |
| 33  | 0.991 ± 0.006 | 0.977 ± 0.002 | 0.998 ± 0.001 | 0.985 ± 0.002 | 0.978 ± 0.004 | 0.992 ± 0.004 | 0.998 ± 0.003 | 0.989 ± 0.002 | 0.995 ± 0.002 | 0.991 ± 0.001 | 0.991 ± 0.005 | 0.993 ± 0.003 | E Europe F    | E Asia F      | 0.00 |
| 34  | 0.992 ± 0.006 | 0.982 ± 0.002 | 0.999 ± 0.001 | 0.990 ± 0.002 | 0.988 ± 0.003 | 0.995 ± 0.004 | 0.994 ± 0.004 | 0.990 ± 0.002 | 0.996 ± 0.002 | 0.994 ± 0.001 | 0.994 ± 0.003 | 0.993 ± 0.003 | E Europe F    | E Asia F      | 0.00 |



**Table 151.** For algorithm neurotechnology-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.21 ± 0.04   | 0.12 ± 0.02   | 0.03 ± 0.02   | 0.040 ± 0.010 | 0.06 ± 0.02  | 0.53 ± 0.06   | 0.42 ± 0.04   | 0.16 ± 0.02   | 0.013 ± 0.009 | 0.05 ± 0.01   | 0.12 ± 0.03  | 0.65 ± 0.06   | W Africa M    | E Europe M    | 0.57 |
| 15  | 0.20 ± 0.04   | 0.19 ± 0.02   | 0.04 ± 0.02   | 0.07 ± 0.02   | 0.12 ± 0.03  | 0.49 ± 0.06   | 0.49 ± 0.05   | 0.22 ± 0.02   | 0.02 ± 0.01   | 0.10 ± 0.02   | 0.16 ± 0.03  | 0.67 ± 0.07   | W Africa M    | E Europe M    | 0.49 |
| 16  | 0.31 ± 0.05   | 0.20 ± 0.02   | 0.08 ± 0.03   | 0.10 ± 0.02   | 0.16 ± 0.03  | 0.59 ± 0.06   | 0.66 ± 0.05   | 0.29 ± 0.02   | 0.03 ± 0.01   | 0.14 ± 0.02   | 0.19 ± 0.04  | 0.76 ± 0.06   | W Africa M    | E Europe M    | 0.47 |
| 17  | 0.39 ± 0.05   | 0.24 ± 0.02   | 0.12 ± 0.03   | 0.13 ± 0.02   | 0.21 ± 0.04  | 0.64 ± 0.05   | 0.76 ± 0.04   | 0.36 ± 0.02   | 0.06 ± 0.02   | 0.20 ± 0.02   | 0.28 ± 0.04  | 0.84 ± 0.04   | W Africa M    | E Europe M    | 0.41 |
| 18  | 0.40 ± 0.04   | 0.26 ± 0.02   | 0.13 ± 0.03   | 0.14 ± 0.02   | 0.23 ± 0.03  | 0.66 ± 0.05   | 0.77 ± 0.04   | 0.41 ± 0.02   | 0.10 ± 0.03   | 0.27 ± 0.02   | 0.34 ± 0.03  | 0.85 ± 0.03   | W Africa M    | E Europe M    | 0.38 |
| 19  | 0.43 ± 0.04   | 0.28 ± 0.02   | 0.16 ± 0.03   | 0.18 ± 0.02   | 0.23 ± 0.03  | 0.74 ± 0.04   | 0.86 ± 0.03   | 0.46 ± 0.02   | 0.11 ± 0.02   | 0.36 ± 0.02   | 0.37 ± 0.03  | 0.91 ± 0.03   | W Africa M    | E Europe M    | 0.37 |
| 20  | 0.47 ± 0.04   | 0.30 ± 0.02   | 0.20 ± 0.03   | 0.22 ± 0.02   | 0.26 ± 0.02  | 0.75 ± 0.04   | 0.88 ± 0.03   | 0.51 ± 0.02   | 0.19 ± 0.03   | 0.44 ± 0.02   | 0.42 ± 0.03  | 0.95 ± 0.02   | W Africa M    | E Europe M    | 0.33 |
| 21  | 0.51 ± 0.04   | 0.33 ± 0.02   | 0.26 ± 0.03   | 0.26 ± 0.02   | 0.31 ± 0.02  | 0.79 ± 0.03   | 0.91 ± 0.02   | 0.55 ± 0.02   | 0.28 ± 0.04   | 0.51 ± 0.02   | 0.49 ± 0.04  | 0.95 ± 0.02   | W Africa M    | S Asia F      | 0.28 |
| 22  | 0.59 ± 0.03   | 0.35 ± 0.02   | 0.32 ± 0.03   | 0.28 ± 0.02   | 0.31 ± 0.02  | 0.83 ± 0.03   | 0.94 ± 0.02   | 0.59 ± 0.02   | 0.38 ± 0.04   | 0.58 ± 0.02   | 0.54 ± 0.03  | 0.96 ± 0.02   | W Africa M    | S Asia F      | 0.25 |
| 23  | 0.61 ± 0.03   | 0.40 ± 0.02   | 0.38 ± 0.03   | 0.34 ± 0.02   | 0.38 ± 0.02  | 0.84 ± 0.02   | 0.94 ± 0.02   | 0.66 ± 0.02   | 0.41 ± 0.03   | 0.64 ± 0.02   | 0.59 ± 0.03  | 0.98 ± 0.01   | W Africa M    | S Asia F      | 0.22 |
| 24  | 0.63 ± 0.03   | 0.42 ± 0.02   | 0.44 ± 0.02   | 0.38 ± 0.02   | 0.42 ± 0.02  | 0.86 ± 0.02   | 0.96 ± 0.01   | 0.69 ± 0.02   | 0.52 ± 0.03   | 0.71 ± 0.02   | 0.65 ± 0.03  | 0.987 ± 0.009 | W Africa M    | S Asia F      | 0.19 |
| 25  | 0.66 ± 0.03   | 0.47 ± 0.01   | 0.51 ± 0.02   | 0.42 ± 0.01   | 0.45 ± 0.02  | 0.85 ± 0.02   | 0.97 ± 0.01   | 0.73 ± 0.02   | 0.61 ± 0.03   | 0.76 ± 0.02   | 0.68 ± 0.03  | 0.986 ± 0.008 | W Africa M    | S Asia F      | 0.17 |
| 26  | 0.69 ± 0.02   | 0.50 ± 0.01   | 0.57 ± 0.02   | 0.45 ± 0.01   | 0.51 ± 0.02  | 0.89 ± 0.02   | 0.977 ± 0.009 | 0.78 ± 0.02   | 0.68 ± 0.03   | 0.79 ± 0.01   | 0.74 ± 0.03  | 0.991 ± 0.006 | W Africa M    | S Asia F      | 0.15 |
| 27  | 0.75 ± 0.02   | 0.53 ± 0.01   | 0.62 ± 0.02   | 0.48 ± 0.01   | 0.56 ± 0.02  | 0.91 ± 0.02   | 0.984 ± 0.008 | 0.81 ± 0.02   | 0.75 ± 0.02   | 0.82 ± 0.01   | 0.79 ± 0.02  | 0.990 ± 0.006 | W Africa M    | S Asia F      | 0.13 |
| 28  | 0.75 ± 0.02   | 0.59 ± 0.01   | 0.68 ± 0.02   | 0.52 ± 0.01   | 0.60 ± 0.02  | 0.92 ± 0.02   | 0.986 ± 0.007 | 0.85 ± 0.01   | 0.82 ± 0.02   | 0.83 ± 0.01   | 0.82 ± 0.02  | 0.991 ± 0.006 | W Africa M    | S Asia F      | 0.12 |
| 29  | 0.80 ± 0.02   | 0.63 ± 0.01   | 0.73 ± 0.02   | 0.56 ± 0.01   | 0.64 ± 0.02  | 0.93 ± 0.01   | 0.989 ± 0.006 | 0.90 ± 0.01   | 0.84 ± 0.02   | 0.857 ± 0.009 | 0.87 ± 0.02  | 0.997 ± 0.003 | W Africa M    | S Asia F      | 0.10 |
| 30  | 0.82 ± 0.02   | 0.68 ± 0.01   | 0.77 ± 0.02   | 0.59 ± 0.01   | 0.69 ± 0.02  | 0.93 ± 0.01   | 0.994 ± 0.004 | 0.924 ± 0.010 | 0.90 ± 0.02   | 0.878 ± 0.008 | 0.89 ± 0.02  | 0.996 ± 0.003 | W Africa M    | S Asia F      | 0.09 |
| 31  | 0.84 ± 0.02   | 0.71 ± 0.01   | 0.81 ± 0.02   | 0.63 ± 0.01   | 0.73 ± 0.02  | 0.95 ± 0.01   | 0.996 ± 0.004 | 0.937 ± 0.009 | 0.91 ± 0.01   | 0.899 ± 0.008 | 0.91 ± 0.02  | 0.995 ± 0.003 | E Africa M    | S Asia F      | 0.08 |
| 32  | 0.89 ± 0.02   | 0.76 ± 0.01   | 0.85 ± 0.01   | 0.66 ± 0.01   | 0.76 ± 0.02  | 0.94 ± 0.01   | 0.993 ± 0.005 | 0.947 ± 0.008 | 0.93 ± 0.01   | 0.917 ± 0.007 | 0.92 ± 0.01  | 0.999 ± 0.002 | W Africa M    | S Asia F      | 0.07 |
| 33  | 0.88 ± 0.02   | 0.78 ± 0.01   | 0.87 ± 0.01   | 0.69 ± 0.01   | 0.79 ± 0.02  | 0.95 ± 0.01   | 0.993 ± 0.005 | 0.964 ± 0.007 | 0.968 ± 0.010 | 0.936 ± 0.006 | 0.95 ± 0.01  | 0.999 ± 0.001 | W Africa M    | S Asia F      | 0.06 |
| 34  | 0.90 ± 0.02   | 0.827 ± 0.009 | 0.89 ± 0.01   | 0.73 ± 0.01   | 0.83 ± 0.02  | 0.96 ± 0.01   | 0.997 ± 0.003 | 0.970 ± 0.007 | 0.97 ± 0.01   | 0.948 ± 0.006 | 0.96 ± 0.01  | 0.999 ± 0.002 | W Africa M    | S Asia F      | 0.05 |

**Table 152.** For algorithm neurotechnology-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.26 ± 0.06   | 0.061 ± 0.010 | 0.07 ± 0.02   | 0.09 ± 0.02   | 0.05 ± 0.02  | 0.47 ± 0.08   | 0.36 ± 0.07   | 0.10 ± 0.01   | 0.04 ± 0.02   | 0.08 ± 0.01   | 0.08 ± 0.02  | 0.52 ± 0.08   | W Africa M    | E Europe M    | 0.51 |
| 15  | 0.24 ± 0.06   | 0.08 ± 0.01   | 0.10 ± 0.02   | 0.12 ± 0.02   | 0.08 ± 0.02  | 0.49 ± 0.07   | 0.60 ± 0.08   | 0.14 ± 0.01   | 0.05 ± 0.02   | 0.12 ± 0.02   | 0.10 ± 0.02  | 0.65 ± 0.07   | W Africa M    | E Europe M    | 0.49 |
| 16  | 0.37 ± 0.06   | 0.10 ± 0.01   | 0.11 ± 0.02   | 0.14 ± 0.02   | 0.08 ± 0.02  | 0.51 ± 0.06   | 0.60 ± 0.07   | 0.19 ± 0.01   | 0.07 ± 0.02   | 0.14 ± 0.02   | 0.13 ± 0.03  | 0.70 ± 0.06   | W Africa M    | E Europe M    | 0.46 |
| 17  | 0.36 ± 0.05   | 0.122 ± 0.010 | 0.17 ± 0.03   | 0.17 ± 0.02   | 0.09 ± 0.02  | 0.60 ± 0.06   | 0.72 ± 0.07   | 0.24 ± 0.01   | 0.08 ± 0.02   | 0.22 ± 0.02   | 0.16 ± 0.03  | 0.74 ± 0.05   | W Africa M    | E Europe M    | 0.43 |
| 18  | 0.42 ± 0.05   | 0.131 ± 0.009 | 0.19 ± 0.02   | 0.19 ± 0.02   | 0.13 ± 0.02  | 0.63 ± 0.06   | 0.75 ± 0.05   | 0.27 ± 0.01   | 0.16 ± 0.03   | 0.26 ± 0.02   | 0.18 ± 0.02  | 0.82 ± 0.04   | W Africa M    | S E Asia F    | 0.40 |
| 19  | 0.45 ± 0.05   | 0.153 ± 0.008 | 0.25 ± 0.02   | 0.22 ± 0.02   | 0.13 ± 0.02  | 0.66 ± 0.04   | 0.82 ± 0.04   | 0.31 ± 0.01   | 0.18 ± 0.02   | 0.32 ± 0.02   | 0.23 ± 0.02  | 0.86 ± 0.03   | W Africa M    | S E Asia F    | 0.37 |
| 20  | 0.51 ± 0.04   | 0.180 ± 0.009 | 0.30 ± 0.02   | 0.25 ± 0.01   | 0.18 ± 0.02  | 0.67 ± 0.04   | 0.87 ± 0.03   | 0.35 ± 0.01   | 0.21 ± 0.03   | 0.43 ± 0.02   | 0.29 ± 0.03  | 0.84 ± 0.03   | E Africa M    | E Asia F      | 0.33 |
| 21  | 0.56 ± 0.04   | 0.195 ± 0.008 | 0.33 ± 0.02   | 0.28 ± 0.02   | 0.18 ± 0.01  | 0.69 ± 0.04   | 0.85 ± 0.03   | 0.40 ± 0.01   | 0.34 ± 0.03   | 0.49 ± 0.02   | 0.32 ± 0.02  | 0.89 ± 0.02   | W Africa M    | S E Asia F    | 0.30 |
| 22  | 0.60 ± 0.04   | 0.228 ± 0.009 | 0.38 ± 0.02   | 0.32 ± 0.01   | 0.22 ± 0.02  | 0.75 ± 0.03   | 0.88 ± 0.03   | 0.45 ± 0.01   | 0.38 ± 0.03   | 0.56 ± 0.02   | 0.38 ± 0.03  | 0.91 ± 0.02   | W Africa M    | S E Asia F    | 0.27 |
| 23  | 0.60 ± 0.04   | 0.237 ± 0.008 | 0.44 ± 0.02   | 0.34 ± 0.01   | 0.27 ± 0.02  | 0.76 ± 0.03   | 0.92 ± 0.02   | 0.49 ± 0.01   | 0.48 ± 0.02   | 0.61 ± 0.01   | 0.43 ± 0.03  | 0.94 ± 0.02   | W Africa M    | E Asia F      | 0.25 |
| 24  | 0.65 ± 0.03   | 0.276 ± 0.008 | 0.47 ± 0.01   | 0.38 ± 0.01   | 0.29 ± 0.02  | 0.80 ± 0.02   | 0.94 ± 0.02   | 0.55 ± 0.01   | 0.57 ± 0.02   | 0.67 ± 0.01   | 0.49 ± 0.03  | 0.95 ± 0.01   | W Africa M    | E Asia F      | 0.23 |
| 25  | 0.70 ± 0.03   | 0.321 ± 0.008 | 0.55 ± 0.01   | 0.41 ± 0.01   | 0.34 ± 0.02  | 0.84 ± 0.02   | 0.96 ± 0.01   | 0.60 ± 0.01   | 0.62 ± 0.02   | 0.71 ± 0.01   | 0.54 ± 0.02  | 0.96 ± 0.01   | W Africa M    | E Asia F      | 0.20 |
| 26  | 0.73 ± 0.02   | 0.350 ± 0.007 | 0.58 ± 0.01   | 0.44 ± 0.01   | 0.37 ± 0.01  | 0.84 ± 0.02   | 0.97 ± 0.01   | 0.64 ± 0.01   | 0.68 ± 0.02   | 0.738 ± 0.010 | 0.62 ± 0.02  | 0.975 ± 0.008 | W Africa M    | E Asia F      | 0.19 |
| 27  | 0.74 ± 0.02   | 0.379 ± 0.007 | 0.63 ± 0.01   | 0.473 ± 0.009 | 0.43 ± 0.02  | 0.85 ± 0.02   | 0.96 ± 0.01   | 0.682 ± 0.010 | 0.74 ± 0.02   | 0.779 ± 0.008 | 0.65 ± 0.02  | 0.973 ± 0.008 | W Africa M    | E Asia F      | 0.16 |
| 28  | 0.79 ± 0.02   | 0.422 ± 0.007 | 0.67 ± 0.01   | 0.513 ± 0.009 | 0.48 ± 0.02  | 0.87 ± 0.02   | 0.96 ± 0.01   | 0.73 ± 0.01   | 0.80 ± 0.01   | 0.811 ± 0.007 | 0.71 ± 0.02  | 0.982 ± 0.006 | W Africa M    | E Asia F      | 0.15 |
| 29  | 0.80 ± 0.02   | 0.460 ± 0.007 | 0.71 ± 0.01   | 0.537 ± 0.009 | 0.51 ± 0.02  | 0.87 ± 0.01   | 0.969 ± 0.010 | 0.767 ± 0.008 | 0.82 ± 0.01   | 0.834 ± 0.006 | 0.76 ± 0.02  | 0.988 ± 0.005 | W Africa M    | E Asia F      | 0.13 |
| 30  | 0.81 ± 0.02   | 0.506 ± 0.008 | 0.756 ± 0.010 | 0.570 ± 0.008 | 0.56 ± 0.01  | 0.89 ± 0.01   | 0.978 ± 0.009 | 0.798 ± 0.008 | 0.88 ± 0.01   | 0.856 ± 0.006 | 0.78 ± 0.02  | 0.983 ± 0.005 | W Africa M    | E Asia F      | 0.12 |
| 31  | 0.85 ± 0.02   | 0.553 ± 0.007 | 0.793 ± 0.010 | 0.594 ± 0.008 | 0.61 ± 0.02  | 0.90 ± 0.02   | 0.978 ± 0.009 | 0.832 ± 0.007 | 0.899 ± 0.010 | 0.879 ± 0.005 | 0.80 ± 0.02  | 0.990 ± 0.004 | W Africa M    | E Asia F      | 0.10 |
| 32  | 0.87 ± 0.02   | 0.595 ± 0.007 | 0.826 ± 0.010 | 0.642 ± 0.008 | 0.64 ± 0.01  | 0.92 ± 0.01   | 0.986 ± 0.007 | 0.858 ± 0.007 | 0.925 ± 0.008 | 0.902 ± 0.005 | 0.83 ± 0.02  | 0.991 ± 0.004 | W Africa M    | E Asia F      | 0.09 |
| 33  | 0.88 ± 0.02   | 0.631 ± 0.008 | 0.843 ± 0.010 | 0.674 ± 0.008 | 0.68 ± 0.01  | 0.92 ± 0.01   | 0.991 ± 0.005 | 0.890 ± 0.006 | 0.940 ± 0.007 | 0.921 ± 0.004 | 0.87 ± 0.02  | 0.990 ± 0.004 | E Africa M    | E Asia F      | 0.08 |
| 34  | 0.87 ± 0.02   | 0.677 ± 0.006 | 0.867 ± 0.010 | 0.711 ± 0.008 | 0.71 ± 0.01  | 0.94 ± 0.01   | 0.987 ± 0.007 | 0.896 ± 0.006 | 0.964 ± 0.005 | 0.930 ± 0.004 | 0.90 ± 0.01  | 0.990 ± 0.004 | W Africa M    | E Asia F      | 0.07 |

**Table 153.** For algorithm neurotechnology-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.04 ± 0.02   | 0.13 ± 0.02   | 0.03 ± 0.01   | 0.042 ± 0.010 | 0.06 ± 0.02  | 0.02 ± 0.02   | 0.04 ± 0.02   | 0.16 ± 0.02   | 0.011 ± 0.009 | 0.05 ± 0.01   | 0.10 ± 0.03  | 0.01 ± 0.01   | E Asia M      | E Europe M    | 0.45 |
| 15  | 0.04 ± 0.02   | 0.19 ± 0.02   | 0.04 ± 0.02   | 0.07 ± 0.02   | 0.12 ± 0.03  | 0.04 ± 0.02   | 0.04 ± 0.02   | 0.23 ± 0.02   | 0.02 ± 0.01   | 0.09 ± 0.02   | 0.17 ± 0.04  | 0.04 ± 0.02   | E Asia M      | E Europe M    | 0.44 |
| 16  | 0.07 ± 0.02   | 0.21 ± 0.02   | 0.08 ± 0.03   | 0.10 ± 0.02   | 0.17 ± 0.04  | 0.06 ± 0.03   | 0.05 ± 0.02   | 0.30 ± 0.02   | 0.03 ± 0.02   | 0.12 ± 0.02   | 0.21 ± 0.04  | 0.03 ± 0.02   | E Asia M      | E Europe M    | 0.41 |
| 17  | 0.08 ± 0.03   | 0.26 ± 0.02   | 0.14 ± 0.04   | 0.13 ± 0.02   | 0.23 ± 0.04  | 0.09 ± 0.03   | 0.12 ± 0.03   | 0.37 ± 0.02   | 0.06 ± 0.02   | 0.20 ± 0.02   | 0.26 ± 0.05  | 0.10 ± 0.03   | E Asia M      | E Europe M    | 0.32 |
| 18  | 0.12 ± 0.03   | 0.26 ± 0.02   | 0.13 ± 0.03   | 0.14 ± 0.02   | 0.23 ± 0.03  | 0.11 ± 0.03   | 0.16 ± 0.03   | 0.41 ± 0.02   | 0.10 ± 0.02   | 0.25 ± 0.02   | 0.35 ± 0.04  | 0.15 ± 0.03   | E Asia M      | E Europe M    | 0.28 |
| 19  | 0.16 ± 0.03   | 0.29 ± 0.02   | 0.18 ± 0.03   | 0.18 ± 0.02   | 0.23 ± 0.02  | 0.15 ± 0.03   | 0.24 ± 0.03   | 0.46 ± 0.02   | 0.12 ± 0.03   | 0.34 ± 0.02   | 0.37 ± 0.03  | 0.21 ± 0.04   | E Asia M      | E Europe M    | 0.24 |
| 20  | 0.20 ± 0.03   | 0.31 ± 0.02   | 0.21 ± 0.03   | 0.21 ± 0.02   | 0.27 ± 0.03  | 0.20 ± 0.03   | 0.35 ± 0.04   | 0.52 ± 0.02   | 0.20 ± 0.03   | 0.42 ± 0.02   | 0.42 ± 0.04  | 0.31 ± 0.04   | E Asia M      | E Europe M    | 0.20 |
| 21  | 0.20 ± 0.03   | 0.34 ± 0.02   | 0.27 ± 0.03   | 0.26 ± 0.02   | 0.31 ± 0.02  | 0.23 ± 0.03   | 0.47 ± 0.04   | 0.55 ± 0.02   | 0.30 ± 0.04   | 0.49 ± 0.02   | 0.49 ± 0.03  | 0.38 ± 0.04   | E Asia M      | E Africa F    | 0.19 |
| 22  | 0.29 ± 0.03   | 0.35 ± 0.02   | 0.33 ± 0.03   | 0.28 ± 0.02   | 0.32 ± 0.02  | 0.28 ± 0.03   | 0.57 ± 0.03   | 0.60 ± 0.02   | 0.38 ± 0.03   | 0.56 ± 0.02   | 0.54 ± 0.03  | 0.47 ± 0.04   | E Asia M      | S Asia F      | 0.17 |
| 23  | 0.32 ± 0.03   | 0.41 ± 0.02   | 0.39 ± 0.02   | 0.35 ± 0.02   | 0.39 ± 0.02  | 0.34 ± 0.03   | 0.61 ± 0.03   | 0.66 ± 0.02   | 0.42 ± 0.03   | 0.61 ± 0.02   | 0.60 ± 0.03  | 0.57 ± 0.03   | E Asia M      | E Africa F    | 0.15 |
| 24  | 0.39 ± 0.03   | 0.43 ± 0.02   | 0.45 ± 0.02   | 0.39 ± 0.02   | 0.43 ± 0.02  | 0.36 ± 0.03   | 0.71 ± 0.03   | 0.69 ± 0.02   | 0.52 ± 0.03   | 0.69 ± 0.02   | 0.66 ± 0.03  | 0.64 ± 0.03   | E Africa M    | W Africa F    | 0.15 |
| 25  | 0.42 ± 0.02   | 0.48 ± 0.02   | 0.53 ± 0.02   | 0.42 ± 0.01   | 0.45 ± 0.02  | 0.42 ± 0.03   | 0.76 ± 0.03   | 0.74 ± 0.02   | 0.61 ± 0.03   | 0.74 ± 0.02   | 0.70 ± 0.03  | 0.70 ± 0.03   | E Africa M    | W Africa F    | 0.14 |
| 26  | 0.47 ± 0.03   | 0.51 ± 0.01   | 0.58 ± 0.02   | 0.45 ± 0.01   | 0.52 ± 0.02  | 0.48 ± 0.03   | 0.79 ± 0.02   | 0.79 ± 0.02   | 0.68 ± 0.03   | 0.77 ± 0.01   | 0.73 ± 0.03  | 0.76 ± 0.03   | E Asia M      | S Asia F      | 0.13 |
| 27  | 0.53 ± 0.02   | 0.55 ± 0.01   | 0.63 ± 0.02   | 0.48 ± 0.01   | 0.57 ± 0.02  | 0.51 ± 0.02   | 0.84 ± 0.02   | 0.82 ± 0.01   | 0.76 ± 0.02   | 0.81 ± 0.01   | 0.80 ± 0.02  | 0.82 ± 0.02   | E Africa M    | S Asia F      | 0.12 |
| 28  | 0.56 ± 0.02   | 0.60 ± 0.01   | 0.69 ± 0.02   | 0.52 ± 0.01   | 0.60 ± 0.02  | 0.56 ± 0.03   | 0.86 ± 0.02   | 0.86 ± 0.01   | 0.82 ± 0.02   | 0.83 ± 0.01   | 0.82 ± 0.02  | 0.84 ± 0.02   | E Asia M      | S Asia F      | 0.11 |
| 29  | 0.63 ± 0.02   | 0.64 ± 0.01   | 0.75 ± 0.02   | 0.57 ± 0.01   | 0.64 ± 0.02  | 0.62 ± 0.03   | 0.89 ± 0.02   | 0.89 ± 0.01   | 0.84 ± 0.02   | 0.851 ± 0.009 | 0.87 ± 0.02  | 0.88 ± 0.02   | E Asia M      | S Asia F      | 0.10 |
| 30  | 0.66 ± 0.03   | 0.68 ± 0.01   | 0.77 ± 0.02   | 0.60 ± 0.01   | 0.69 ± 0.02  | 0.66 ± 0.02   | 0.93 ± 0.01   | 0.924 ± 0.010 | 0.90 ± 0.02   | 0.868 ± 0.009 | 0.90 ± 0.02  | 0.90 ± 0.01   | E Africa M    | S Asia F      | 0.09 |
| 31  | 0.68 ± 0.02   | 0.72 ± 0.01   | 0.81 ± 0.02   | 0.63 ± 0.01   | 0.74 ± 0.02  | 0.70 ± 0.03   | 0.94 ± 0.01   | 0.942 ± 0.008 | 0.92 ± 0.01   | 0.894 ± 0.008 | 0.90 ± 0.02  | 0.93 ± 0.01   | E Asia M      | S Asia F      | 0.08 |
| 32  | 0.76 ± 0.02   | 0.76 ± 0.01   | 0.86 ± 0.01   | 0.66 ± 0.01   | 0.77 ± 0.02  | 0.75 ± 0.02   | 0.96 ± 0.01   | 0.948 ± 0.008 | 0.93 ± 0.01   | 0.910 ± 0.007 | 0.93 ± 0.02  | 0.93 ± 0.01   | E Africa M    | S Asia F      | 0.07 |
| 33  | 0.77 ± 0.02   | 0.79 ± 0.01   | 0.87 ± 0.02   | 0.69 ± 0.01   | 0.79 ± 0.02  | 0.78 ± 0.02   | 0.963 ± 0.010 | 0.964 ± 0.007 | 0.97 ± 0.01   | 0.927 ± 0.007 | 0.95 ± 0.01  | 0.961 ± 0.010 | E Europe M    | S Asia F      | 0.07 |
| 34  | 0.80 ± 0.02   | 0.831 ± 0.010 | 0.89 ± 0.01   | 0.74 ± 0.01   | 0.83 ± 0.02  | 0.82 ± 0.02   | 0.963 ± 0.010 | 0.972 ± 0.005 | 0.97 ± 0.01   | 0.940 ± 0.006 | 0.95 ± 0.01  | 0.96 ± 0.01   | E Asia M      | S Asia F      | 0.05 |

**Table 154.** For algorithm neurotechnology-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.06 ± 0.03   | 0.061 ± 0.009 | 0.07 ± 0.02   | 0.09 ± 0.02   | 0.05 ± 0.02  | 0.007 ± 0.007 | 0.04 ± 0.02   | 0.10 ± 0.01   | 0.03 ± 0.01   | 0.08 ± 0.01   | 0.08 ± 0.02  | 0.04 ± 0.03   | E Asia M      | W Africa F    | 0.28 |
| 15  | 0.08 ± 0.03   | 0.083 ± 0.010 | 0.10 ± 0.03   | 0.12 ± 0.02   | 0.07 ± 0.02  | 0.06 ± 0.03   | 0.11 ± 0.04   | 0.14 ± 0.01   | 0.05 ± 0.02   | 0.11 ± 0.02   | 0.10 ± 0.02  | 0.06 ± 0.04   | E Asia M      | E Europe M    | 0.19 |
| 16  | 0.11 ± 0.04   | 0.10 ± 0.01   | 0.11 ± 0.02   | 0.14 ± 0.02   | 0.07 ± 0.02  | 0.09 ± 0.04   | 0.12 ± 0.05   | 0.19 ± 0.01   | 0.07 ± 0.02   | 0.14 ± 0.02   | 0.13 ± 0.03  | 0.11 ± 0.04   | E Asia M      | E Europe M    | 0.17 |
| 17  | 0.14 ± 0.04   | 0.12 ± 0.01   | 0.16 ± 0.02   | 0.16 ± 0.02   | 0.09 ± 0.02  | 0.10 ± 0.03   | 0.26 ± 0.06   | 0.24 ± 0.01   | 0.09 ± 0.02   | 0.23 ± 0.02   | 0.16 ± 0.03  | 0.19 ± 0.04   | E Africa M    | E Europe M    | 0.21 |
| 18  | 0.19 ± 0.04   | 0.136 ± 0.009 | 0.19 ± 0.03   | 0.19 ± 0.02   | 0.12 ± 0.02  | 0.12 ± 0.04   | 0.32 ± 0.06   | 0.27 ± 0.01   | 0.14 ± 0.03   | 0.26 ± 0.02   | 0.20 ± 0.02  | 0.25 ± 0.04   | E Africa M    | W Africa F    | 0.19 |
| 19  | 0.23 ± 0.04   | 0.154 ± 0.009 | 0.25 ± 0.02   | 0.21 ± 0.02   | 0.14 ± 0.02  | 0.17 ± 0.04   | 0.38 ± 0.05   | 0.30 ± 0.01   | 0.18 ± 0.03   | 0.32 ± 0.02   | 0.24 ± 0.02  | 0.31 ± 0.04   | E Africa M    | S E Asia F    | 0.18 |
| 20  | 0.24 ± 0.04   | 0.185 ± 0.009 | 0.29 ± 0.02   | 0.25 ± 0.02   | 0.18 ± 0.02  | 0.20 ± 0.03   | 0.47 ± 0.05   | 0.36 ± 0.01   | 0.21 ± 0.02   | 0.43 ± 0.02   | 0.30 ± 0.02  | 0.33 ± 0.04   | E Africa M    | S E Asia F    | 0.19 |
| 21  | 0.32 ± 0.04   | 0.197 ± 0.008 | 0.34 ± 0.02   | 0.28 ± 0.01   | 0.19 ± 0.01  | 0.25 ± 0.03   | 0.53 ± 0.04   | 0.40 ± 0.01   | 0.33 ± 0.03   | 0.48 ± 0.02   | 0.33 ± 0.03  | 0.44 ± 0.04   | E Africa M    | S E Asia F    | 0.18 |
| 22  | 0.38 ± 0.04   | 0.236 ± 0.008 | 0.39 ± 0.02   | 0.32 ± 0.01   | 0.22 ± 0.02  | 0.30 ± 0.03   | 0.58 ± 0.04   | 0.45 ± 0.01   | 0.38 ± 0.03   | 0.55 ± 0.02   | 0.39 ± 0.03  | 0.49 ± 0.04   | E Africa M    | S E Asia F    | 0.17 |
| 23  | 0.36 ± 0.03   | 0.248 ± 0.008 | 0.45 ± 0.02   | 0.34 ± 0.01   | 0.27 ± 0.02  | 0.33 ± 0.03   | 0.64 ± 0.03   | 0.50 ± 0.01   | 0.49 ± 0.02   | 0.60 ± 0.01   | 0.44 ± 0.03  | 0.58 ± 0.03   | E Africa M    | E Asia F      | 0.18 |
| 24  | 0.43 ± 0.03   | 0.281 ± 0.008 | 0.48 ± 0.01   | 0.39 ± 0.01   | 0.30 ± 0.02  | 0.36 ± 0.03   | 0.71 ± 0.03   | 0.55 ± 0.01   | 0.57 ± 0.02   | 0.66 ± 0.01   | 0.50 ± 0.03  | 0.62 ± 0.03   | E Africa M    | E Asia F      | 0.17 |
| 25  | 0.47 ± 0.03   | 0.322 ± 0.008 | 0.55 ± 0.01   | 0.41 ± 0.01   | 0.35 ± 0.02  | 0.40 ± 0.03   | 0.77 ± 0.03   | 0.60 ± 0.01   | 0.63 ± 0.02   | 0.70 ± 0.01   | 0.54 ± 0.02  | 0.67 ± 0.02   | E Africa M    | E Asia F      | 0.16 |
| 26  | 0.49 ± 0.03   | 0.354 ± 0.007 | 0.58 ± 0.01   | 0.446 ± 0.009 | 0.38 ± 0.01  | 0.43 ± 0.02   | 0.80 ± 0.02   | 0.650 ± 0.010 | 0.68 ± 0.02   | 0.727 ± 0.010 | 0.62 ± 0.02  | 0.71 ± 0.02   | E Africa M    | E Asia F      | 0.15 |
| 27  | 0.56 ± 0.03   | 0.386 ± 0.007 | 0.63 ± 0.01   | 0.475 ± 0.009 | 0.43 ± 0.02  | 0.48 ± 0.02   | 0.83 ± 0.02   | 0.68 ± 0.01   | 0.74 ± 0.02   | 0.772 ± 0.008 | 0.65 ± 0.02  | 0.76 ± 0.02   | E Africa M    | E Asia F      | 0.14 |
| 28  | 0.58 ± 0.03   | 0.425 ± 0.007 | 0.67 ± 0.01   | 0.513 ± 0.009 | 0.48 ± 0.02  | 0.50 ± 0.02   | 0.85 ± 0.02   | 0.731 ± 0.009 | 0.80 ± 0.01   | 0.801 ± 0.008 | 0.71 ± 0.02  | 0.81 ± 0.02   | E Africa M    | E Asia F      | 0.14 |
| 29  | 0.62 ± 0.03   | 0.465 ± 0.008 | 0.71 ± 0.01   | 0.541 ± 0.008 | 0.52 ± 0.02  | 0.54 ± 0.02   | 0.86 ± 0.02   | 0.769 ± 0.009 | 0.82 ± 0.01   | 0.824 ± 0.007 | 0.75 ± 0.02  | 0.82 ± 0.02   | E Africa M    | E Asia F      | 0.12 |
| 30  | 0.62 ± 0.03   | 0.511 ± 0.007 | 0.75 ± 0.01   | 0.570 ± 0.009 | 0.56 ± 0.02  | 0.60 ± 0.02   | 0.87 ± 0.02   | 0.800 ± 0.008 | 0.88 ± 0.01   | 0.846 ± 0.006 | 0.78 ± 0.02  | 0.84 ± 0.02   | E Europe M    | E Asia F      | 0.11 |
| 31  | 0.67 ± 0.03   | 0.559 ± 0.007 | 0.794 ± 0.009 | 0.599 ± 0.008 | 0.60 ± 0.01  | 0.64 ± 0.02   | 0.89 ± 0.02   | 0.836 ± 0.007 | 0.902 ± 0.009 | 0.867 ± 0.005 | 0.80 ± 0.02  | 0.85 ± 0.01   | E Europe M    | E Asia F      | 0.10 |
| 32  | 0.69 ± 0.03   | 0.600 ± 0.007 | 0.830 ± 0.009 | 0.644 ± 0.008 | 0.64 ± 0.01  | 0.65 ± 0.02   | 0.92 ± 0.01   | 0.859 ± 0.007 | 0.923 ± 0.009 | 0.892 ± 0.005 | 0.83 ± 0.02  | 0.86 ± 0.01   | E Europe M    | E Asia F      | 0.09 |
| 33  | 0.74 ± 0.03   | 0.637 ± 0.007 | 0.847 ± 0.009 | 0.675 ± 0.008 | 0.68 ± 0.01  | 0.70 ± 0.02   | 0.92 ± 0.01   | 0.890 ± 0.007 | 0.940 ± 0.008 | 0.910 ± 0.004 | 0.88 ± 0.02  | 0.88 ± 0.01   | E Europe M    | E Asia F      | 0.08 |
| 34  | 0.75 ± 0.03   | 0.682 ± 0.006 | 0.871 ± 0.008 | 0.710 ± 0.008 | 0.72 ± 0.01  | 0.75 ± 0.02   | 0.92 ± 0.01   | 0.899 ± 0.006 | 0.963 ± 0.006 | 0.921 ± 0.004 | 0.90 ± 0.01  | 0.90 ± 0.01   | E Europe M    | E Asia F      | 0.07 |

**Table 155.** For algorithm nominder-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.04 ± 0.02   | 0.007 ± 0.004 | 0.005 ± 0.005 | 0.008 ± 0.005 | 0.009 ± 0.007 | 0.09 ± 0.04   | 0.006 ± 0.006 | 0.007 ± 0.004 | 0.004 ± 0.004 | 0.006 ± 0.003 | 0.007 ± 0.007 | 0.01 ± 0.01   | W Africa F    | E Europe M    | 0.61 |
| 15  | 0.05 ± 0.03   | 0.018 ± 0.007 | 0.01 ± 0.01   | 0.019 ± 0.009 | 0.01 ± 0.01   | 0.11 ± 0.04   | 0.02 ± 0.01   | 0.007 ± 0.004 | 0.012 ± 0.009 | 0.006 ± 0.004 | 0.012 ± 0.010 | 0.02 ± 0.01   | W Africa F    | S Asia M      | 0.51 |
| 16  | 0.08 ± 0.03   | 0.028 ± 0.009 | 0.01 ± 0.01   | 0.03 ± 0.01   | 0.05 ± 0.02   | 0.18 ± 0.05   | 0.03 ± 0.02   | 0.026 ± 0.008 | 0.009 ± 0.009 | 0.013 ± 0.007 | 0.01 ± 0.01   | 0.02 ± 0.02   | W Africa F    | E Europe M    | 0.52 |
| 17  | 0.14 ± 0.04   | 0.06 ± 0.01   | 0.04 ± 0.02   | 0.04 ± 0.01   | 0.06 ± 0.02   | 0.27 ± 0.05   | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.008 ± 0.008 | 0.04 ± 0.01   | 0.03 ± 0.02   | 0.10 ± 0.03   | W Africa F    | E Europe M    | 0.47 |
| 18  | 0.17 ± 0.03   | 0.10 ± 0.01   | 0.05 ± 0.02   | 0.05 ± 0.01   | 0.07 ± 0.02   | 0.31 ± 0.04   | 0.09 ± 0.02   | 0.07 ± 0.01   | 0.03 ± 0.01   | 0.08 ± 0.01   | 0.06 ± 0.02   | 0.12 ± 0.03   | W Africa F    | E Europe M    | 0.37 |
| 19  | 0.23 ± 0.03   | 0.12 ± 0.01   | 0.09 ± 0.02   | 0.09 ± 0.01   | 0.11 ± 0.02   | 0.44 ± 0.04   | 0.17 ± 0.03   | 0.13 ± 0.01   | 0.07 ± 0.02   | 0.15 ± 0.02   | 0.10 ± 0.02   | 0.24 ± 0.04   | W Africa F    | E Europe M    | 0.32 |
| 20  | 0.29 ± 0.03   | 0.18 ± 0.02   | 0.12 ± 0.02   | 0.15 ± 0.02   | 0.16 ± 0.02   | 0.49 ± 0.04   | 0.29 ± 0.04   | 0.19 ± 0.02   | 0.11 ± 0.02   | 0.26 ± 0.02   | 0.18 ± 0.03   | 0.32 ± 0.04   | W Africa F    | E Europe M    | 0.26 |
| 21  | 0.37 ± 0.04   | 0.23 ± 0.02   | 0.18 ± 0.02   | 0.19 ± 0.02   | 0.21 ± 0.02   | 0.56 ± 0.04   | 0.38 ± 0.04   | 0.26 ± 0.02   | 0.17 ± 0.03   | 0.34 ± 0.02   | 0.23 ± 0.03   | 0.45 ± 0.04   | W Africa F    | E Europe M    | 0.23 |
| 22  | 0.45 ± 0.03   | 0.30 ± 0.02   | 0.26 ± 0.02   | 0.25 ± 0.02   | 0.27 ± 0.02   | 0.60 ± 0.04   | 0.52 ± 0.04   | 0.35 ± 0.02   | 0.31 ± 0.03   | 0.47 ± 0.02   | 0.29 ± 0.03   | 0.50 ± 0.04   | W Africa F    | S Asia F      | 0.18 |
| 23  | 0.50 ± 0.03   | 0.38 ± 0.02   | 0.35 ± 0.02   | 0.33 ± 0.02   | 0.33 ± 0.02   | 0.70 ± 0.03   | 0.57 ± 0.03   | 0.45 ± 0.02   | 0.40 ± 0.03   | 0.57 ± 0.02   | 0.38 ± 0.03   | 0.61 ± 0.04   | W Africa F    | S Asia F      | 0.16 |
| 24  | 0.58 ± 0.03   | 0.45 ± 0.02   | 0.45 ± 0.02   | 0.37 ± 0.02   | 0.42 ± 0.02   | 0.71 ± 0.03   | 0.69 ± 0.03   | 0.54 ± 0.02   | 0.54 ± 0.03   | 0.69 ± 0.02   | 0.50 ± 0.03   | 0.71 ± 0.03   | W Africa F    | S Asia F      | 0.13 |
| 25  | 0.61 ± 0.03   | 0.54 ± 0.01   | 0.55 ± 0.02   | 0.45 ± 0.01   | 0.49 ± 0.02   | 0.79 ± 0.02   | 0.76 ± 0.02   | 0.63 ± 0.02   | 0.68 ± 0.03   | 0.77 ± 0.01   | 0.60 ± 0.03   | 0.76 ± 0.02   | W Africa F    | S Asia F      | 0.11 |
| 26  | 0.69 ± 0.02   | 0.61 ± 0.01   | 0.65 ± 0.02   | 0.49 ± 0.01   | 0.59 ± 0.02   | 0.83 ± 0.02   | 0.81 ± 0.02   | 0.72 ± 0.02   | 0.76 ± 0.02   | 0.83 ± 0.01   | 0.69 ± 0.03   | 0.83 ± 0.02   | W Africa M    | S Asia F      | 0.09 |
| 27  | 0.73 ± 0.02   | 0.67 ± 0.01   | 0.74 ± 0.02   | 0.55 ± 0.01   | 0.66 ± 0.02   | 0.86 ± 0.02   | 0.87 ± 0.02   | 0.79 ± 0.02   | 0.86 ± 0.02   | 0.876 ± 0.009 | 0.78 ± 0.02   | 0.86 ± 0.02   | S Asia M      | S Asia F      | 0.08 |
| 28  | 0.79 ± 0.02   | 0.739 ± 0.010 | 0.81 ± 0.01   | 0.62 ± 0.01   | 0.72 ± 0.02   | 0.89 ± 0.02   | 0.89 ± 0.02   | 0.84 ± 0.01   | 0.91 ± 0.01   | 0.904 ± 0.009 | 0.82 ± 0.02   | 0.90 ± 0.02   | E Europe M    | S Asia F      | 0.06 |
| 29  | 0.83 ± 0.02   | 0.805 ± 0.010 | 0.87 ± 0.01   | 0.69 ± 0.01   | 0.78 ± 0.02   | 0.92 ± 0.01   | 0.93 ± 0.01   | 0.90 ± 0.01   | 0.94 ± 0.01   | 0.937 ± 0.006 | 0.87 ± 0.02   | 0.94 ± 0.01   | E Europe M    | S Asia F      | 0.05 |
| 30  | 0.86 ± 0.02   | 0.846 ± 0.009 | 0.93 ± 0.01   | 0.73 ± 0.01   | 0.84 ± 0.01   | 0.93 ± 0.01   | 0.95 ± 0.01   | 0.925 ± 0.009 | 0.969 ± 0.009 | 0.949 ± 0.006 | 0.91 ± 0.02   | 0.949 ± 0.010 | E Europe M    | S Asia F      | 0.04 |
| 31  | 0.89 ± 0.02   | 0.886 ± 0.008 | 0.948 ± 0.009 | 0.78 ± 0.01   | 0.88 ± 0.01   | 0.95 ± 0.01   | 0.96 ± 0.01   | 0.946 ± 0.008 | 0.981 ± 0.007 | 0.967 ± 0.005 | 0.94 ± 0.01   | 0.968 ± 0.008 | E Europe M    | S Asia F      | 0.03 |
| 32  | 0.92 ± 0.01   | 0.919 ± 0.007 | 0.965 ± 0.007 | 0.83 ± 0.01   | 0.91 ± 0.01   | 0.96 ± 0.01   | 0.981 ± 0.007 | 0.966 ± 0.006 | 0.991 ± 0.005 | 0.981 ± 0.004 | 0.96 ± 0.01   | 0.979 ± 0.007 | E Europe M    | S Asia F      | 0.03 |
| 33  | 0.93 ± 0.01   | 0.950 ± 0.006 | 0.975 ± 0.007 | 0.869 ± 0.009 | 0.94 ± 0.01   | 0.971 ± 0.010 | 0.984 ± 0.007 | 0.977 ± 0.005 | 0.996 ± 0.004 | 0.987 ± 0.003 | 0.977 ± 0.009 | 0.982 ± 0.007 | E Europe M    | S Asia F      | 0.02 |
| 34  | 0.96 ± 0.01   | 0.963 ± 0.005 | 0.981 ± 0.006 | 0.897 ± 0.009 | 0.948 ± 0.010 | 0.972 ± 0.010 | 0.990 ± 0.006 | 0.984 ± 0.004 | 0.998 ± 0.003 | 0.990 ± 0.003 | 0.979 ± 0.009 | 0.992 ± 0.004 | E Europe M    | S Asia F      | 0.01 |

**Table 156.** For algorithm nominder-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.14 ± 0.05   | 0.015 ± 0.005 | 0.03 ± 0.02   | 0.04 ± 0.01   | 0.03 ± 0.02   | 0.17 ± 0.06   | 0.15 ± 0.06   | 0.011 ± 0.004 | 0.03 ± 0.01   | 0.024 ± 0.008 | 0.03 ± 0.01   | 0.18 ± 0.06   | W Africa M    | E Asia M      | 0.50 |
| 15  | 0.17 ± 0.05   | 0.023 ± 0.006 | 0.04 ± 0.02   | 0.08 ± 0.01   | 0.04 ± 0.02   | 0.26 ± 0.06   | 0.27 ± 0.06   | 0.023 ± 0.006 | 0.04 ± 0.02   | 0.07 ± 0.01   | 0.04 ± 0.01   | 0.21 ± 0.06   | E Africa M    | E Asia M      | 0.50 |
| 16  | 0.33 ± 0.06   | 0.032 ± 0.006 | 0.07 ± 0.02   | 0.11 ± 0.02   | 0.05 ± 0.02   | 0.33 ± 0.07   | 0.28 ± 0.06   | 0.038 ± 0.007 | 0.05 ± 0.02   | 0.10 ± 0.02   | 0.08 ± 0.02   | 0.26 ± 0.06   | W Africa F    | E Asia F      | 0.45 |
| 17  | 0.30 ± 0.05   | 0.063 ± 0.008 | 0.12 ± 0.02   | 0.16 ± 0.02   | 0.08 ± 0.02   | 0.41 ± 0.06   | 0.38 ± 0.07   | 0.068 ± 0.008 | 0.08 ± 0.02   | 0.20 ± 0.02   | 0.08 ± 0.02   | 0.43 ± 0.05   | W Africa M    | E Asia F      | 0.41 |
| 18  | 0.40 ± 0.05   | 0.086 ± 0.007 | 0.16 ± 0.02   | 0.19 ± 0.02   | 0.10 ± 0.01   | 0.44 ± 0.05   | 0.56 ± 0.06   | 0.108 ± 0.008 | 0.15 ± 0.03   | 0.29 ± 0.02   | 0.18 ± 0.02   | 0.48 ± 0.05   | E Africa M    | E Asia F      | 0.36 |
| 19  | 0.48 ± 0.05   | 0.134 ± 0.009 | 0.20 ± 0.02   | 0.25 ± 0.02   | 0.16 ± 0.02   | 0.56 ± 0.05   | 0.58 ± 0.05   | 0.158 ± 0.009 | 0.21 ± 0.03   | 0.40 ± 0.02   | 0.24 ± 0.02   | 0.54 ± 0.05   | E Africa M    | E Asia F      | 0.31 |
| 20  | 0.54 ± 0.04   | 0.169 ± 0.009 | 0.23 ± 0.02   | 0.30 ± 0.02   | 0.22 ± 0.02   | 0.61 ± 0.04   | 0.68 ± 0.04   | 0.23 ± 0.01   | 0.30 ± 0.03   | 0.52 ± 0.02   | 0.32 ± 0.03   | 0.60 ± 0.04   | E Africa M    | E Asia F      | 0.27 |
| 21  | 0.60 ± 0.04   | 0.220 ± 0.008 | 0.29 ± 0.02   | 0.35 ± 0.02   | 0.27 ± 0.02   | 0.65 ± 0.03   | 0.74 ± 0.03   | 0.32 ± 0.01   | 0.43 ± 0.03   | 0.63 ± 0.02   | 0.45 ± 0.02   | 0.69 ± 0.03   | E Africa M    | E Asia F      | 0.23 |
| 22  | 0.66 ± 0.04   | 0.276 ± 0.009 | 0.35 ± 0.02   | 0.40 ± 0.01   | 0.34 ± 0.02   | 0.74 ± 0.03   | 0.79 ± 0.03   | 0.40 ± 0.01   | 0.52 ± 0.03   | 0.72 ± 0.01   | 0.51 ± 0.02   | 0.75 ± 0.03   | E Africa M    | E Asia F      | 0.20 |
| 23  | 0.70 ± 0.03   | 0.325 ± 0.009 | 0.43 ± 0.01   | 0.44 ± 0.01   | 0.44 ± 0.02   | 0.76 ± 0.02   | 0.85 ± 0.03   | 0.48 ± 0.01   | 0.64 ± 0.02   | 0.78 ± 0.01   | 0.57 ± 0.03   | 0.83 ± 0.02   | E Africa M    | E Asia F      | 0.18 |
| 24  | 0.74 ± 0.03   | 0.393 ± 0.009 | 0.50 ± 0.01   | 0.51 ± 0.01   | 0.50 ± 0.02   | 0.78 ± 0.02   | 0.91 ± 0.02   | 0.57 ± 0.01   | 0.72 ± 0.02   | 0.84 ± 0.01   | 0.68 ± 0.02   | 0.84 ± 0.02   | E Africa M    | E Asia F      | 0.15 |
| 25  | 0.80 ± 0.03   | 0.469 ± 0.009 | 0.57 ± 0.01   | 0.551 ± 0.010 | 0.56 ± 0.02   | 0.83 ± 0.02   | 0.94 ± 0.02   | 0.65 ± 0.01   | 0.79 ± 0.02   | 0.875 ± 0.008 | 0.74 ± 0.02   | 0.90 ± 0.02   | E Africa M    | E Asia F      | 0.13 |
| 26  | 0.82 ± 0.02   | 0.531 ± 0.008 | 0.63 ± 0.01   | 0.591 ± 0.009 | 0.63 ± 0.02   | 0.84 ± 0.02   | 0.95 ± 0.01   | 0.721 ± 0.010 | 0.84 ± 0.01   | 0.894 ± 0.007 | 0.80 ± 0.02   | 0.92 ± 0.01   | E Africa M    | E Asia F      | 0.11 |
| 27  | 0.85 ± 0.02   | 0.592 ± 0.007 | 0.70 ± 0.01   | 0.631 ± 0.009 | 0.70 ± 0.01   | 0.88 ± 0.02   | 0.96 ± 0.01   | 0.771 ± 0.009 | 0.89 ± 0.01   | 0.920 ± 0.006 | 0.84 ± 0.02   | 0.94 ± 0.01   | E Africa M    | E Asia F      | 0.09 |
| 28  | 0.87 ± 0.02   | 0.658 ± 0.007 | 0.76 ± 0.01   | 0.671 ± 0.008 | 0.74 ± 0.01   | 0.88 ± 0.01   | 0.969 ± 0.010 | 0.831 ± 0.008 | 0.924 ± 0.009 | 0.942 ± 0.004 | 0.88 ± 0.01   | 0.959 ± 0.010 | E Africa M    | E Asia F      | 0.08 |
| 29  | 0.90 ± 0.02   | 0.707 ± 0.006 | 0.814 ± 0.009 | 0.721 ± 0.007 | 0.78 ± 0.01   | 0.91 ± 0.01   | 0.984 ± 0.007 | 0.864 ± 0.007 | 0.947 ± 0.007 | 0.957 ± 0.004 | 0.91 ± 0.01   | 0.965 ± 0.008 | E Africa M    | E Asia F      | 0.06 |
| 30  | 0.90 ± 0.02   | 0.761 ± 0.006 | 0.865 ± 0.008 | 0.754 ± 0.007 | 0.83 ± 0.01   | 0.93 ± 0.01   | 0.989 ± 0.007 | 0.899 ± 0.006 | 0.964 ± 0.006 | 0.969 ± 0.003 | 0.94 ± 0.01   | 0.976 ± 0.007 | E Africa M    | S Asia F      | 0.05 |
| 31  | 0.93 ± 0.01   | 0.805 ± 0.006 | 0.883 ± 0.008 | 0.793 ± 0.007 | 0.85 ± 0.01   | 0.95 ± 0.01   | 0.989 ± 0.007 | 0.925 ± 0.005 | 0.977 ± 0.005 | 0.978 ± 0.002 | 0.94 ± 0.01   | 0.983 ± 0.006 | E Africa M    | S Asia F      | 0.04 |
| 32  | 0.93 ± 0.02   | 0.847 ± 0.005 | 0.925 ± 0.006 | 0.832 ± 0.007 | 0.88 ± 0.01   | 0.953 ± 0.010 | 0.990 ± 0.005 | 0.946 ± 0.005 | 0.984 ± 0.004 | 0.982 ± 0.002 | 0.95 ± 0.01   | 0.979 ± 0.006 | E Africa M    | S Asia F      | 0.03 |
| 33  | 0.94 ± 0.02   | 0.880 ± 0.004 | 0.934 ± 0.006 | 0.861 ± 0.006 | 0.902 ± 0.010 | 0.965 ± 0.010 | 0.991 ± 0.006 | 0.957 ± 0.004 | 0.989 ± 0.003 | 0.986 ± 0.002 | 0.967 ± 0.008 | 0.986 ± 0.004 | E Africa M    | S Asia F      | 0.03 |
| 34  | 0.96 ± 0.01   | 0.907 ± 0.004 | 0.955 ± 0.006 | 0.884 ± 0.005 | 0.925 ± 0.009 | 0.965 ± 0.009 | 0.993 ± 0.005 | 0.969 ± 0.004 | 0.993 ± 0.003 | 0.991 ± 0.001 | 0.978 ± 0.007 | 0.989 ± 0.004 | E Africa M    | S Asia F      | 0.02 |

**Table 157.** For algorithm nominder-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.007 ± 0.007 | 0.002 ± 0.002 | 0.004 ± 0.004 | 0.02 ± 0.02   | 0.004 ± 0.004 | 0.003 ± 0.002 | 0.004 ± 0.004 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.000 ± 0.000 | W Africa F    | E Asia F      | 0.65 |
| 15  | 0.006 ± 0.006 | 0.003 ± 0.002 | 0.010 ± 0.008 | 0.010 ± 0.005 | 0.005 ± 0.005 | 0.04 ± 0.02   | 0.003 ± 0.003 | 0.004 ± 0.003 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.009 ± 0.009 | W Africa F    | S E Asia M    | 0.56 |
| 16  | 0.04 ± 0.02   | 0.015 ± 0.006 | 0.02 ± 0.01   | 0.010 ± 0.006 | 0.03 ± 0.02   | 0.12 ± 0.04   | 0.01 ± 0.01   | 0.009 ± 0.004 | 0.003 ± 0.003 | 0.004 ± 0.003 | 0.008 ± 0.008 | 0.005 ± 0.005 | W Africa F    | E Europe M    | 0.61 |
| 17  | 0.08 ± 0.03   | 0.024 ± 0.007 | 0.03 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.01   | 0.18 ± 0.04   | 0.01 ± 0.01   | 0.016 ± 0.006 | 0.008 ± 0.008 | 0.016 ± 0.007 | 0.03 ± 0.01   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.51 |
| 18  | 0.10 ± 0.02   | 0.033 ± 0.008 | 0.04 ± 0.02   | 0.033 ± 0.009 | 0.05 ± 0.01   | 0.24 ± 0.04   | 0.04 ± 0.02   | 0.039 ± 0.008 | 0.016 ± 0.009 | 0.031 ± 0.008 | 0.03 ± 0.01   | 0.07 ± 0.02   | W Africa F    | E Europe M    | 0.43 |
| 19  | 0.17 ± 0.03   | 0.050 ± 0.010 | 0.06 ± 0.02   | 0.07 ± 0.01   | 0.07 ± 0.02   | 0.34 ± 0.04   | 0.08 ± 0.02   | 0.07 ± 0.01   | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.06 ± 0.02   | 0.12 ± 0.03   | W Africa F    | E Europe M    | 0.40 |
| 20  | 0.25 ± 0.03   | 0.09 ± 0.01   | 0.08 ± 0.02   | 0.10 ± 0.01   | 0.12 ± 0.02   | 0.40 ± 0.04   | 0.20 ± 0.03   | 0.10 ± 0.01   | 0.04 ± 0.02   | 0.12 ± 0.01   | 0.09 ± 0.02   | 0.20 ± 0.04   | W Africa F    | E Europe M    | 0.34 |
| 21  | 0.31 ± 0.03   | 0.14 ± 0.01   | 0.14 ± 0.02   | 0.17 ± 0.01   | 0.16 ± 0.02   | 0.47 ± 0.04   | 0.26 ± 0.03   | 0.18 ± 0.02   | 0.10 ± 0.02   | 0.19 ± 0.02   | 0.17 ± 0.03   | 0.29 ± 0.04   | W Africa F    | E Europe M    | 0.25 |
| 22  | 0.43 ± 0.04   | 0.18 ± 0.02   | 0.20 ± 0.02   | 0.22 ± 0.02   | 0.23 ± 0.02   | 0.53 ± 0.03   | 0.42 ± 0.03   | 0.27 ± 0.02   | 0.19 ± 0.03   | 0.30 ± 0.02   | 0.21 ± 0.03   | 0.36 ± 0.04   | W Africa F    | E Asia F      | 0.22 |
| 23  | 0.49 ± 0.03   | 0.28 ± 0.02   | 0.28 ± 0.02   | 0.30 ± 0.02   | 0.29 ± 0.02   | 0.64 ± 0.03   | 0.50 ± 0.03   | 0.38 ± 0.02   | 0.27 ± 0.03   | 0.45 ± 0.02   | 0.33 ± 0.03   | 0.51 ± 0.04   | W Africa F    | E Europe M    | 0.18 |
| 24  | 0.58 ± 0.03   | 0.34 ± 0.01   | 0.38 ± 0.02   | 0.38 ± 0.01   | 0.39 ± 0.02   | 0.69 ± 0.03   | 0.64 ± 0.03   | 0.50 ± 0.02   | 0.42 ± 0.03   | 0.57 ± 0.02   | 0.42 ± 0.03   | 0.59 ± 0.03   | W Africa F    | E Asia F      | 0.14 |
| 25  | 0.63 ± 0.02   | 0.45 ± 0.01   | 0.45 ± 0.02   | 0.45 ± 0.01   | 0.46 ± 0.02   | 0.75 ± 0.03   | 0.72 ± 0.03   | 0.59 ± 0.02   | 0.58 ± 0.03   | 0.70 ± 0.02   | 0.57 ± 0.02   | 0.70 ± 0.03   | W Africa F    | E Europe F    | 0.12 |
| 26  | 0.72 ± 0.02   | 0.53 ± 0.01   | 0.57 ± 0.02   | 0.51 ± 0.01   | 0.57 ± 0.02   | 0.84 ± 0.02   | 0.80 ± 0.02   | 0.70 ± 0.02   | 0.70 ± 0.02   | 0.78 ± 0.01   | 0.68 ± 0.03   | 0.79 ± 0.02   | W Africa F    | S Asia F      | 0.10 |
| 27  | 0.77 ± 0.02   | 0.62 ± 0.01   | 0.67 ± 0.02   | 0.59 ± 0.01   | 0.65 ± 0.02   | 0.86 ± 0.02   | 0.86 ± 0.02   | 0.79 ± 0.02   | 0.82 ± 0.02   | 0.85 ± 0.01   | 0.78 ± 0.02   | 0.85 ± 0.02   | E Africa M    | S Asia F      | 0.07 |
| 28  | 0.83 ± 0.02   | 0.70 ± 0.01   | 0.75 ± 0.02   | 0.67 ± 0.01   | 0.73 ± 0.02   | 0.91 ± 0.02   | 0.89 ± 0.02   | 0.85 ± 0.01   | 0.88 ± 0.02   | 0.889 ± 0.009 | 0.85 ± 0.02   | 0.88 ± 0.02   | W Africa F    | S Asia F      | 0.06 |
| 29  | 0.89 ± 0.02   | 0.776 ± 0.010 | 0.84 ± 0.01   | 0.73 ± 0.01   | 0.80 ± 0.02   | 0.93 ± 0.01   | 0.94 ± 0.01   | 0.91 ± 0.01   | 0.93 ± 0.01   | 0.927 ± 0.007 | 0.89 ± 0.02   | 0.94 ± 0.01   | E Africa M    | S Asia F      | 0.05 |
| 30  | 0.92 ± 0.01   | 0.833 ± 0.009 | 0.90 ± 0.01   | 0.79 ± 0.01   | 0.85 ± 0.01   | 0.94 ± 0.01   | 0.969 ± 0.009 | 0.942 ± 0.008 | 0.95 ± 0.01   | 0.946 ± 0.006 | 0.94 ± 0.01   | 0.955 ± 0.010 | E Africa M    | S Asia F      | 0.03 |
| 31  | 0.93 ± 0.01   | 0.878 ± 0.009 | 0.94 ± 0.01   | 0.835 ± 0.010 | 0.89 ± 0.01   | 0.96 ± 0.01   | 0.987 ± 0.007 | 0.969 ± 0.006 | 0.979 ± 0.007 | 0.968 ± 0.004 | 0.97 ± 0.01   | 0.967 ± 0.009 | E Africa M    | S Asia F      | 0.03 |
| 32  | 0.96 ± 0.01   | 0.916 ± 0.007 | 0.960 ± 0.007 | 0.875 ± 0.009 | 0.93 ± 0.01   | 0.965 ± 0.010 | 0.990 ± 0.005 | 0.979 ± 0.005 | 0.989 ± 0.006 | 0.982 ± 0.003 | 0.971 ± 0.010 | 0.982 ± 0.006 | E Africa M    | S Asia F      | 0.02 |
| 33  | 0.967 ± 0.009 | 0.940 ± 0.006 | 0.976 ± 0.007 | 0.918 ± 0.007 | 0.954 ± 0.009 | 0.981 ± 0.007 | 0.992 ± 0.005 | 0.988 ± 0.004 | 0.996 ± 0.004 | 0.992 ± 0.002 | 0.987 ± 0.006 | 0.984 ± 0.006 | E Europe M    | S Asia F      | 0.01 |
| 34  | 0.97 ± 0.01   | 0.960 ± 0.005 | 0.983 ± 0.006 | 0.949 ± 0.006 | 0.974 ± 0.007 | 0.980 ± 0.008 | 0.995 ± 0.004 | 0.993 ± 0.003 | 0.996 ± 0.004 | 0.992 ± 0.003 | 0.983 ± 0.009 | 0.992 ± 0.004 | E Europe M    | S Asia F      | 0.01 |

**Table 158.** For algorithm nominder-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.05 ± 0.03   | 0.007 ± 0.003 | 0.02 ± 0.01   | 0.018 ± 0.008 | 0.02 ± 0.01   | 0.09 ± 0.04   | 0.02 ± 0.02   | 0.005 ± 0.003 | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.013 ± 0.008 | 0.04 ± 0.03   | W Africa F    | E Europe M    | 0.54 |
| 15  | 0.08 ± 0.04   | 0.009 ± 0.004 | 0.02 ± 0.01   | 0.034 ± 0.010 | 0.02 ± 0.01   | 0.13 ± 0.04   | 0.04 ± 0.02   | 0.009 ± 0.003 | 0.007 ± 0.007 | 0.013 ± 0.006 | 0.011 ± 0.007 | 0.06 ± 0.04   | W Africa F    | E Europe M    | 0.53 |
| 16  | 0.12 ± 0.04   | 0.019 ± 0.005 | 0.03 ± 0.01   | 0.05 ± 0.01   | 0.022 ± 0.010 | 0.19 ± 0.05   | 0.07 ± 0.03   | 0.019 ± 0.005 | 0.008 ± 0.007 | 0.025 ± 0.007 | 0.03 ± 0.01   | 0.09 ± 0.04   | W Africa F    | E Europe M    | 0.51 |
| 17  | 0.15 ± 0.04   | 0.031 ± 0.005 | 0.04 ± 0.01   | 0.06 ± 0.01   | 0.03 ± 0.01   | 0.30 ± 0.05   | 0.13 ± 0.04   | 0.038 ± 0.006 | 0.008 ± 0.007 | 0.07 ± 0.01   | 0.05 ± 0.02   | 0.17 ± 0.04   | W Africa F    | E Europe M    | 0.48 |
| 18  | 0.27 ± 0.05   | 0.047 ± 0.005 | 0.08 ± 0.02   | 0.11 ± 0.01   | 0.06 ± 0.01   | 0.31 ± 0.04   | 0.17 ± 0.04   | 0.065 ± 0.007 | 0.03 ± 0.01   | 0.10 ± 0.01   | 0.07 ± 0.02   | 0.23 ± 0.04   | W Africa F    | E Europe M    | 0.40 |
| 19  | 0.31 ± 0.04   | 0.071 ± 0.006 | 0.11 ± 0.01   | 0.13 ± 0.01   | 0.09 ± 0.01   | 0.39 ± 0.04   | 0.23 ± 0.05   | 0.109 ± 0.008 | 0.05 ± 0.01   | 0.18 ± 0.02   | 0.15 ± 0.02   | 0.27 ± 0.04   | W Africa F    | E Europe M    | 0.35 |
| 20  | 0.36 ± 0.04   | 0.103 ± 0.006 | 0.14 ± 0.01   | 0.17 ± 0.01   | 0.14 ± 0.02   | 0.47 ± 0.04   | 0.40 ± 0.04   | 0.160 ± 0.009 | 0.10 ± 0.02   | 0.26 ± 0.02   | 0.22 ± 0.02   | 0.31 ± 0.04   | W Africa F    | E Europe M    | 0.31 |
| 21  | 0.44 ± 0.04   | 0.139 ± 0.008 | 0.19 ± 0.02   | 0.21 ± 0.01   | 0.17 ± 0.01   | 0.55 ± 0.04   | 0.44 ± 0.04   | 0.24 ± 0.01   | 0.15 ± 0.02   | 0.35 ± 0.02   | 0.29 ± 0.02   | 0.39 ± 0.04   | W Africa F    | E Asia F      | 0.27 |
| 22  | 0.49 ± 0.04   | 0.184 ± 0.008 | 0.26 ± 0.01   | 0.25 ± 0.01   | 0.24 ± 0.02   | 0.64 ± 0.03   | 0.51 ± 0.04   | 0.33 ± 0.01   | 0.24 ± 0.02   | 0.45 ± 0.02   | 0.36 ± 0.02   | 0.48 ± 0.04   | W Africa F    | E Asia F      | 0.22 |
| 23  | 0.58 ± 0.03   | 0.229 ± 0.008 | 0.31 ± 0.01   | 0.29 ± 0.01   | 0.32 ± 0.02   | 0.66 ± 0.03   | 0.59 ± 0.04   | 0.42 ± 0.01   | 0.35 ± 0.02   | 0.55 ± 0.01   | 0.46 ± 0.03   | 0.56 ± 0.03   | W Africa F    | E Asia F      | 0.19 |
| 24  | 0.64 ± 0.03   | 0.291 ± 0.007 | 0.38 ± 0.01   | 0.35 ± 0.01   | 0.38 ± 0.02   | 0.72 ± 0.03   | 0.67 ± 0.03   | 0.52 ± 0.01   | 0.47 ± 0.02   | 0.65 ± 0.01   | 0.56 ± 0.02   | 0.62 ± 0.03   | W Africa F    | E Asia F      | 0.16 |
| 25  | 0.70 ± 0.03   | 0.364 ± 0.008 | 0.48 ± 0.01   | 0.395 ± 0.010 | 0.46 ± 0.02   | 0.78 ± 0.02   | 0.74 ± 0.03   | 0.61 ± 0.01   | 0.57 ± 0.02   | 0.72 ± 0.01   | 0.66 ± 0.02   | 0.72 ± 0.02   | W Africa F    | E Asia F      | 0.14 |
| 26  | 0.74 ± 0.03   | 0.430 ± 0.008 | 0.54 ± 0.01   | 0.443 ± 0.010 | 0.53 ± 0.02   | 0.78 ± 0.02   | 0.81 ± 0.03   | 0.68 ± 0.01   | 0.66 ± 0.02   | 0.780 ± 0.009 | 0.72 ± 0.02   | 0.78 ± 0.02   | E Africa M    | E Asia F      | 0.12 |
| 27  | 0.77 ± 0.02   | 0.494 ± 0.007 | 0.63 ± 0.01   | 0.497 ± 0.009 | 0.61 ± 0.01   | 0.84 ± 0.02   | 0.84 ± 0.02   | 0.738 ± 0.009 | 0.76 ± 0.01   | 0.824 ± 0.008 | 0.79 ± 0.02   | 0.84 ± 0.02   | E Africa M    | E Asia F      | 0.10 |
| 28  | 0.82 ± 0.02   | 0.574 ± 0.008 | 0.71 ± 0.01   | 0.553 ± 0.008 | 0.66 ± 0.01   | 0.85 ± 0.02   | 0.89 ± 0.02   | 0.803 ± 0.009 | 0.83 ± 0.01   | 0.874 ± 0.006 | 0.84 ± 0.02   | 0.87 ± 0.02   | E Africa M    | S Asia F      | 0.09 |
| 29  | 0.84 ± 0.02   | 0.631 ± 0.007 | 0.775 ± 0.010 | 0.604 ± 0.008 | 0.71 ± 0.01   | 0.87 ± 0.01   | 0.93 ± 0.02   | 0.854 ± 0.007 | 0.88 ± 0.01   | 0.907 ± 0.005 | 0.87 ± 0.02   | 0.91 ± 0.01   | E Africa M    | S Asia F      | 0.07 |
| 30  | 0.86 ± 0.02   | 0.699 ± 0.006 | 0.831 ± 0.009 | 0.660 ± 0.007 | 0.77 ± 0.01   | 0.91 ± 0.01   | 0.95 ± 0.01   | 0.891 ± 0.007 | 0.919 ± 0.008 | 0.927 ± 0.004 | 0.91 ± 0.01   | 0.94 ± 0.01   | E Africa M    | S Asia F      | 0.06 |
| 31  | 0.89 ± 0.02   | 0.756 ± 0.006 | 0.865 ± 0.008 | 0.707 ± 0.007 | 0.81 ± 0.01   | 0.93 ± 0.01   | 0.96 ± 0.01   | 0.921 ± 0.005 | 0.953 ± 0.007 | 0.949 ± 0.004 | 0.93 ± 0.01   | 0.943 ± 0.009 | E Africa M    | S Asia F      | 0.05 |
| 32  | 0.92 ± 0.01   | 0.805 ± 0.006 | 0.909 ± 0.007 | 0.761 ± 0.007 | 0.84 ± 0.01   | 0.94 ± 0.01   | 0.980 ± 0.008 | 0.949 ± 0.004 | 0.976 ± 0.005 | 0.963 ± 0.003 | 0.94 ± 0.01   | 0.950 ± 0.009 | E Africa M    | S Asia F      | 0.04 |
| 33  | 0.92 ± 0.02   | 0.843 ± 0.005 | 0.936 ± 0.006 | 0.807 ± 0.006 | 0.881 ± 0.010 | 0.962 ± 0.010 | 0.979 ± 0.007 | 0.962 ± 0.004 | 0.982 ± 0.004 | 0.974 ± 0.002 | 0.964 ± 0.008 | 0.968 ± 0.007 | E Europe M    | S Asia F      | 0.03 |
| 34  | 0.94 ± 0.01   | 0.884 ± 0.005 | 0.957 ± 0.006 | 0.848 ± 0.006 | 0.906 ± 0.009 | 0.96 ± 0.01   | 0.983 ± 0.007 | 0.968 ± 0.004 | 0.989 ± 0.003 | 0.983 ± 0.002 | 0.980 ± 0.006 | 0.980 ± 0.006 | E Europe M    | S Asia F      | 0.03 |

**Table 159.** For algorithm ondato-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.08 ± 0.03   | 0.027 ± 0.008 | 0.005 ± 0.005 | 0.023 ± 0.008 | 0.03 ± 0.02   | 0.09 ± 0.04   | 0.03 ± 0.02   | 0.011 ± 0.004 | 0.004 ± 0.004 | 0.007 ± 0.003 | 0.005 ± 0.005 | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.54 |
| 15  | 0.10 ± 0.03   | 0.044 ± 0.010 | 0.02 ± 0.01   | 0.04 ± 0.01   | 0.06 ± 0.02   | 0.15 ± 0.04   | 0.04 ± 0.02   | 0.013 ± 0.006 | 0.009 ± 0.009 | 0.009 ± 0.005 | 0.010 ± 0.007 | 0.04 ± 0.02   | W Africa F    | S Asia M      | 0.50 |
| 16  | 0.14 ± 0.03   | 0.06 ± 0.01   | 0.05 ± 0.02   | 0.05 ± 0.01   | 0.08 ± 0.03   | 0.23 ± 0.05   | 0.07 ± 0.02   | 0.026 ± 0.008 | 0.009 ± 0.009 | 0.017 ± 0.008 | 0.03 ± 0.02   | 0.05 ± 0.03   | W Africa F    | E Europe M    | 0.47 |
| 17  | 0.23 ± 0.04   | 0.11 ± 0.02   | 0.05 ± 0.02   | 0.10 ± 0.02   | 0.10 ± 0.03   | 0.30 ± 0.05   | 0.10 ± 0.03   | 0.04 ± 0.01   | 0.011 ± 0.008 | 0.05 ± 0.01   | 0.05 ± 0.02   | 0.16 ± 0.04   | W Africa F    | E Europe M    | 0.43 |
| 18  | 0.24 ± 0.03   | 0.14 ± 0.01   | 0.07 ± 0.02   | 0.12 ± 0.02   | 0.13 ± 0.02   | 0.34 ± 0.04   | 0.14 ± 0.03   | 0.06 ± 0.01   | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.06 ± 0.02   | 0.16 ± 0.04   | W Africa F    | E Europe M    | 0.38 |
| 19  | 0.33 ± 0.03   | 0.19 ± 0.02   | 0.13 ± 0.03   | 0.17 ± 0.02   | 0.19 ± 0.02   | 0.43 ± 0.04   | 0.26 ± 0.04   | 0.11 ± 0.01   | 0.05 ± 0.02   | 0.13 ± 0.02   | 0.14 ± 0.02   | 0.26 ± 0.04   | W Africa F    | E Europe M    | 0.30 |
| 20  | 0.40 ± 0.04   | 0.26 ± 0.02   | 0.18 ± 0.03   | 0.23 ± 0.02   | 0.26 ± 0.03   | 0.52 ± 0.04   | 0.36 ± 0.04   | 0.17 ± 0.01   | 0.10 ± 0.02   | 0.24 ± 0.02   | 0.20 ± 0.03   | 0.38 ± 0.05   | W Africa F    | E Europe M    | 0.25 |
| 21  | 0.47 ± 0.04   | 0.32 ± 0.02   | 0.24 ± 0.03   | 0.30 ± 0.02   | 0.34 ± 0.02   | 0.58 ± 0.04   | 0.49 ± 0.04   | 0.26 ± 0.02   | 0.13 ± 0.03   | 0.33 ± 0.02   | 0.33 ± 0.03   | 0.48 ± 0.04   | W Africa F    | E Europe M    | 0.21 |
| 22  | 0.56 ± 0.03   | 0.42 ± 0.02   | 0.35 ± 0.03   | 0.37 ± 0.02   | 0.43 ± 0.02   | 0.63 ± 0.03   | 0.62 ± 0.03   | 0.37 ± 0.02   | 0.26 ± 0.03   | 0.46 ± 0.02   | 0.41 ± 0.03   | 0.56 ± 0.04   | W Africa F    | E Europe M    | 0.15 |
| 23  | 0.63 ± 0.03   | 0.51 ± 0.02   | 0.44 ± 0.02   | 0.47 ± 0.02   | 0.50 ± 0.02   | 0.72 ± 0.03   | 0.67 ± 0.03   | 0.49 ± 0.02   | 0.36 ± 0.03   | 0.59 ± 0.02   | 0.53 ± 0.03   | 0.70 ± 0.03   | W Africa F    | E Europe M    | 0.12 |
| 24  | 0.69 ± 0.03   | 0.58 ± 0.02   | 0.55 ± 0.02   | 0.53 ± 0.02   | 0.60 ± 0.02   | 0.74 ± 0.03   | 0.77 ± 0.03   | 0.60 ± 0.02   | 0.51 ± 0.03   | 0.70 ± 0.02   | 0.64 ± 0.03   | 0.77 ± 0.03   | E Africa M    | E Europe M    | 0.09 |
| 25  | 0.74 ± 0.03   | 0.67 ± 0.01   | 0.63 ± 0.02   | 0.60 ± 0.01   | 0.67 ± 0.02   | 0.79 ± 0.02   | 0.84 ± 0.02   | 0.71 ± 0.02   | 0.66 ± 0.03   | 0.80 ± 0.01   | 0.75 ± 0.02   | 0.84 ± 0.02   | W Africa M    | S Asia F      | 0.07 |
| 26  | 0.80 ± 0.02   | 0.74 ± 0.01   | 0.73 ± 0.02   | 0.66 ± 0.01   | 0.75 ± 0.02   | 0.82 ± 0.02   | 0.89 ± 0.02   | 0.81 ± 0.02   | 0.74 ± 0.02   | 0.86 ± 0.01   | 0.82 ± 0.02   | 0.89 ± 0.02   | E Africa M    | S Asia F      | 0.05 |
| 27  | 0.85 ± 0.02   | 0.79 ± 0.01   | 0.79 ± 0.01   | 0.72 ± 0.01   | 0.81 ± 0.01   | 0.88 ± 0.02   | 0.91 ± 0.01   | 0.87 ± 0.01   | 0.86 ± 0.02   | 0.902 ± 0.009 | 0.89 ± 0.02   | 0.92 ± 0.01   | W Africa M    | S Asia F      | 0.04 |
| 28  | 0.88 ± 0.02   | 0.851 ± 0.009 | 0.86 ± 0.01   | 0.77 ± 0.01   | 0.86 ± 0.01   | 0.90 ± 0.02   | 0.96 ± 0.01   | 0.922 ± 0.010 | 0.92 ± 0.01   | 0.936 ± 0.007 | 0.93 ± 0.01   | 0.94 ± 0.01   | E Africa M    | S Asia F      | 0.03 |
| 29  | 0.91 ± 0.01   | 0.885 ± 0.008 | 0.90 ± 0.01   | 0.82 ± 0.01   | 0.89 ± 0.01   | 0.92 ± 0.01   | 0.972 ± 0.009 | 0.949 ± 0.008 | 0.94 ± 0.01   | 0.957 ± 0.006 | 0.96 ± 0.01   | 0.970 ± 0.008 | E Africa M    | S Asia F      | 0.03 |
| 30  | 0.93 ± 0.01   | 0.925 ± 0.007 | 0.938 ± 0.009 | 0.858 ± 0.009 | 0.92 ± 0.01   | 0.94 ± 0.01   | 0.978 ± 0.008 | 0.974 ± 0.006 | 0.972 ± 0.009 | 0.972 ± 0.004 | 0.97 ± 0.01   | 0.975 ± 0.008 | E Africa M    | S Asia F      | 0.02 |
| 31  | 0.95 ± 0.01   | 0.955 ± 0.006 | 0.964 ± 0.007 | 0.895 ± 0.007 | 0.945 ± 0.008 | 0.94 ± 0.01   | 0.985 ± 0.007 | 0.984 ± 0.005 | 0.981 ± 0.007 | 0.981 ± 0.003 | 0.984 ± 0.007 | 0.988 ± 0.005 | W Africa M    | S Asia F      | 0.02 |
| 32  | 0.95 ± 0.01   | 0.964 ± 0.004 | 0.974 ± 0.007 | 0.927 ± 0.007 | 0.965 ± 0.007 | 0.95 ± 0.01   | 0.991 ± 0.005 | 0.990 ± 0.003 | 0.991 ± 0.005 | 0.988 ± 0.003 | 0.989 ± 0.006 | 0.990 ± 0.005 | E Africa M    | S Asia F      | 0.01 |
| 33  | 0.96 ± 0.01   | 0.977 ± 0.004 | 0.983 ± 0.006 | 0.943 ± 0.006 | 0.978 ± 0.006 | 0.969 ± 0.009 | 0.995 ± 0.004 | 0.995 ± 0.003 | 0.994 ± 0.004 | 0.993 ± 0.002 | 0.994 ± 0.005 | 0.991 ± 0.005 | E Africa M    | S Asia F      | 0.01 |
| 34  | 0.978 ± 0.009 | 0.984 ± 0.003 | 0.987 ± 0.005 | 0.965 ± 0.005 | 0.986 ± 0.005 | 0.973 ± 0.009 | 0.997 ± 0.003 | 0.998 ± 0.002 | 0.997 ± 0.004 | 0.997 ± 0.001 | 0.997 ± 0.004 | 0.997 ± 0.003 | E Asia M      | S Asia F      | 0.01 |

**Table 160.** For algorithm ondato-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.19 ± 0.06   | 0.015 ± 0.005 | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.02 ± 0.01  | 0.17 ± 0.06   | 0.16 ± 0.06   | 0.010 ± 0.004 | 0.010 ± 0.007 | 0.031 ± 0.009 | 0.01 ± 0.01   | 0.15 ± 0.06   | E Africa F    | E Asia M      | 0.56 |
| 15  | 0.17 ± 0.05   | 0.018 ± 0.005 | 0.04 ± 0.02   | 0.08 ± 0.01   | 0.02 ± 0.01  | 0.26 ± 0.06   | 0.29 ± 0.07   | 0.014 ± 0.004 | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.016 ± 0.009 | 0.23 ± 0.06   | E Africa M    | E Asia M      | 0.57 |
| 16  | 0.23 ± 0.06   | 0.025 ± 0.005 | 0.03 ± 0.01   | 0.10 ± 0.02   | 0.04 ± 0.01  | 0.25 ± 0.06   | 0.28 ± 0.06   | 0.022 ± 0.005 | 0.04 ± 0.01   | 0.08 ± 0.01   | 0.02 ± 0.01   | 0.22 ± 0.05   | E Africa M    | E Asia M      | 0.51 |
| 17  | 0.25 ± 0.05   | 0.042 ± 0.006 | 0.07 ± 0.02   | 0.12 ± 0.02   | 0.04 ± 0.01  | 0.24 ± 0.05   | 0.41 ± 0.07   | 0.040 ± 0.006 | 0.05 ± 0.02   | 0.16 ± 0.02   | 0.05 ± 0.01   | 0.38 ± 0.06   | E Africa M    | S E Asia F    | 0.49 |
| 18  | 0.33 ± 0.05   | 0.051 ± 0.006 | 0.10 ± 0.02   | 0.15 ± 0.02   | 0.06 ± 0.01  | 0.32 ± 0.05   | 0.43 ± 0.06   | 0.061 ± 0.007 | 0.10 ± 0.02   | 0.22 ± 0.02   | 0.06 ± 0.01   | 0.41 ± 0.05   | E Africa M    | E Asia F      | 0.44 |
| 19  | 0.37 ± 0.05   | 0.070 ± 0.006 | 0.13 ± 0.02   | 0.18 ± 0.01   | 0.06 ± 0.01  | 0.44 ± 0.04   | 0.47 ± 0.05   | 0.088 ± 0.008 | 0.11 ± 0.02   | 0.29 ± 0.02   | 0.14 ± 0.02   | 0.46 ± 0.05   | E Africa M    | S E Asia F    | 0.40 |
| 20  | 0.43 ± 0.04   | 0.093 ± 0.007 | 0.15 ± 0.01   | 0.22 ± 0.01   | 0.10 ± 0.01  | 0.44 ± 0.04   | 0.60 ± 0.04   | 0.126 ± 0.008 | 0.18 ± 0.02   | 0.41 ± 0.02   | 0.19 ± 0.02   | 0.53 ± 0.04   | E Africa M    | E Asia F      | 0.36 |
| 21  | 0.43 ± 0.04   | 0.109 ± 0.006 | 0.19 ± 0.02   | 0.26 ± 0.02   | 0.12 ± 0.01  | 0.53 ± 0.04   | 0.69 ± 0.04   | 0.194 ± 0.009 | 0.27 ± 0.03   | 0.51 ± 0.02   | 0.24 ± 0.02   | 0.62 ± 0.03   | E Africa M    | E Asia F      | 0.34 |
| 22  | 0.57 ± 0.04   | 0.142 ± 0.007 | 0.24 ± 0.01   | 0.31 ± 0.01   | 0.16 ± 0.01  | 0.62 ± 0.03   | 0.75 ± 0.03   | 0.25 ± 0.01   | 0.39 ± 0.03   | 0.59 ± 0.02   | 0.33 ± 0.03   | 0.67 ± 0.04   | E Africa M    | E Asia F      | 0.30 |
| 23  | 0.58 ± 0.03   | 0.178 ± 0.007 | 0.31 ± 0.01   | 0.35 ± 0.01   | 0.21 ± 0.01  | 0.64 ± 0.03   | 0.80 ± 0.03   | 0.32 ± 0.01   | 0.46 ± 0.02   | 0.67 ± 0.01   | 0.39 ± 0.02   | 0.76 ± 0.03   | E Africa M    | E Asia F      | 0.27 |
| 24  | 0.63 ± 0.03   | 0.227 ± 0.007 | 0.36 ± 0.01   | 0.41 ± 0.01   | 0.25 ± 0.01  | 0.65 ± 0.03   | 0.86 ± 0.03   | 0.40 ± 0.01   | 0.58 ± 0.02   | 0.74 ± 0.01   | 0.46 ± 0.02   | 0.79 ± 0.02   | E Africa M    | E Asia F      | 0.24 |
| 25  | 0.64 ± 0.03   | 0.269 ± 0.007 | 0.43 ± 0.01   | 0.449 ± 0.010 | 0.31 ± 0.01  | 0.72 ± 0.02   | 0.89 ± 0.02   | 0.48 ± 0.01   | 0.67 ± 0.02   | 0.799 ± 0.010 | 0.56 ± 0.02   | 0.83 ± 0.02   | E Africa M    | E Asia F      | 0.21 |
| 26  | 0.70 ± 0.03   | 0.322 ± 0.007 | 0.49 ± 0.01   | 0.50 ± 0.01   | 0.36 ± 0.01  | 0.73 ± 0.02   | 0.92 ± 0.02   | 0.57 ± 0.01   | 0.72 ± 0.02   | 0.833 ± 0.008 | 0.63 ± 0.02   | 0.88 ± 0.02   | E Africa M    | E Asia F      | 0.18 |
| 27  | 0.71 ± 0.03   | 0.378 ± 0.007 | 0.57 ± 0.01   | 0.538 ± 0.009 | 0.42 ± 0.02  | 0.78 ± 0.02   | 0.91 ± 0.02   | 0.63 ± 0.01   | 0.79 ± 0.01   | 0.868 ± 0.007 | 0.70 ± 0.02   | 0.90 ± 0.01   | E Africa M    | E Asia F      | 0.16 |
| 28  | 0.77 ± 0.02   | 0.445 ± 0.007 | 0.64 ± 0.01   | 0.582 ± 0.009 | 0.49 ± 0.01  | 0.81 ± 0.02   | 0.94 ± 0.01   | 0.70 ± 0.01   | 0.85 ± 0.01   | 0.900 ± 0.005 | 0.74 ± 0.02   | 0.92 ± 0.01   | E Africa M    | E Asia F      | 0.13 |
| 29  | 0.79 ± 0.02   | 0.498 ± 0.007 | 0.70 ± 0.01   | 0.625 ± 0.009 | 0.53 ± 0.02  | 0.83 ± 0.02   | 0.96 ± 0.01   | 0.754 ± 0.009 | 0.88 ± 0.01   | 0.920 ± 0.005 | 0.79 ± 0.02   | 0.94 ± 0.01   | E Africa M    | E Asia F      | 0.12 |
| 30  | 0.83 ± 0.02   | 0.563 ± 0.007 | 0.758 ± 0.009 | 0.672 ± 0.008 | 0.60 ± 0.01  | 0.86 ± 0.02   | 0.97 ± 0.01   | 0.801 ± 0.008 | 0.916 ± 0.009 | 0.933 ± 0.004 | 0.84 ± 0.02   | 0.944 ± 0.009 | E Africa M    | E Asia F      | 0.10 |
| 31  | 0.84 ± 0.02   | 0.619 ± 0.007 | 0.811 ± 0.009 | 0.722 ± 0.007 | 0.66 ± 0.01  | 0.89 ± 0.01   | 0.971 ± 0.010 | 0.853 ± 0.008 | 0.950 ± 0.007 | 0.952 ± 0.004 | 0.87 ± 0.02   | 0.946 ± 0.008 | E Africa M    | E Asia F      | 0.08 |
| 32  | 0.90 ± 0.02   | 0.671 ± 0.007 | 0.848 ± 0.009 | 0.757 ± 0.007 | 0.70 ± 0.01  | 0.89 ± 0.01   | 0.980 ± 0.008 | 0.879 ± 0.006 | 0.954 ± 0.006 | 0.958 ± 0.003 | 0.89 ± 0.01   | 0.964 ± 0.007 | E Africa M    | E Asia F      | 0.07 |
| 33  | 0.91 ± 0.02   | 0.724 ± 0.006 | 0.877 ± 0.009 | 0.796 ± 0.007 | 0.74 ± 0.01  | 0.91 ± 0.01   | 0.984 ± 0.007 | 0.907 ± 0.005 | 0.968 ± 0.005 | 0.967 ± 0.003 | 0.92 ± 0.01   | 0.973 ± 0.006 | E Africa M    | E Asia F      | 0.06 |
| 34  | 0.91 ± 0.02   | 0.769 ± 0.006 | 0.901 ± 0.008 | 0.834 ± 0.007 | 0.79 ± 0.01  | 0.93 ± 0.01   | 0.989 ± 0.006 | 0.926 ± 0.005 | 0.980 ± 0.004 | 0.973 ± 0.002 | 0.94 ± 0.01   | 0.976 ± 0.006 | E Africa M    | E Asia F      | 0.05 |

**Table 161.** For algorithm ondato-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.04 ± 0.02   | 0.010 ± 0.005 | 0.005 ± 0.005 | 0.011 ± 0.005 | 0.013 ± 0.009 | 0.05 ± 0.02   | 0.02 ± 0.01   | 0.004 ± 0.003 | 0.004 ± 0.004 | 0.004 ± 0.003 | 0.002 ± 0.002 | 0.01 ± 0.01   | W Africa F    | S E Asia M    | 0.52 |
| 15  | 0.08 ± 0.03   | 0.025 ± 0.008 | 0.010 ± 0.008 | 0.028 ± 0.010 | 0.04 ± 0.02   | 0.10 ± 0.04   | 0.03 ± 0.02   | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.007 ± 0.005 | 0.002 ± 0.002 | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.57 |
| 16  | 0.10 ± 0.03   | 0.04 ± 0.01   | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.05 ± 0.02   | 0.16 ± 0.04   | 0.05 ± 0.02   | 0.014 ± 0.006 | 0.003 ± 0.003 | 0.007 ± 0.004 | 0.008 ± 0.008 | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.50 |
| 17  | 0.16 ± 0.04   | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.10 ± 0.02   | 0.11 ± 0.03   | 0.22 ± 0.04   | 0.06 ± 0.02   | 0.018 ± 0.006 | 0.006 ± 0.006 | 0.024 ± 0.009 | 0.04 ± 0.02   | 0.09 ± 0.03   | W Africa F    | E Europe M    | 0.46 |
| 18  | 0.20 ± 0.03   | 0.12 ± 0.01   | 0.07 ± 0.02   | 0.12 ± 0.02   | 0.14 ± 0.02   | 0.26 ± 0.04   | 0.11 ± 0.03   | 0.036 ± 0.008 | 0.013 ± 0.007 | 0.05 ± 0.01   | 0.04 ± 0.01   | 0.12 ± 0.03   | W Africa F    | E Europe M    | 0.39 |
| 19  | 0.29 ± 0.04   | 0.18 ± 0.02   | 0.12 ± 0.03   | 0.18 ± 0.02   | 0.21 ± 0.02   | 0.36 ± 0.04   | 0.20 ± 0.03   | 0.06 ± 0.01   | 0.04 ± 0.02   | 0.09 ± 0.01   | 0.08 ± 0.02   | 0.21 ± 0.04   | W Africa F    | E Europe M    | 0.33 |
| 20  | 0.38 ± 0.04   | 0.27 ± 0.02   | 0.19 ± 0.03   | 0.26 ± 0.02   | 0.30 ± 0.03   | 0.42 ± 0.04   | 0.31 ± 0.04   | 0.12 ± 0.01   | 0.09 ± 0.02   | 0.19 ± 0.02   | 0.15 ± 0.02   | 0.30 ± 0.04   | W Africa F    | E Europe M    | 0.24 |
| 21  | 0.45 ± 0.04   | 0.36 ± 0.02   | 0.24 ± 0.03   | 0.34 ± 0.02   | 0.41 ± 0.02   | 0.51 ± 0.04   | 0.39 ± 0.03   | 0.19 ± 0.02   | 0.13 ± 0.03   | 0.29 ± 0.02   | 0.26 ± 0.03   | 0.36 ± 0.04   | W Africa F    | E Europe M    | 0.20 |
| 22  | 0.52 ± 0.03   | 0.46 ± 0.02   | 0.36 ± 0.02   | 0.40 ± 0.02   | 0.49 ± 0.02   | 0.55 ± 0.04   | 0.57 ± 0.04   | 0.30 ± 0.02   | 0.26 ± 0.03   | 0.41 ± 0.02   | 0.35 ± 0.03   | 0.48 ± 0.04   | E Africa M    | E Europe M    | 0.14 |
| 23  | 0.62 ± 0.03   | 0.56 ± 0.02   | 0.47 ± 0.02   | 0.50 ± 0.02   | 0.58 ± 0.02   | 0.66 ± 0.03   | 0.65 ± 0.03   | 0.43 ± 0.02   | 0.34 ± 0.03   | 0.55 ± 0.02   | 0.48 ± 0.03   | 0.63 ± 0.03   | W Africa F    | E Europe M    | 0.10 |
| 24  | 0.68 ± 0.03   | 0.64 ± 0.02   | 0.58 ± 0.02   | 0.55 ± 0.02   | 0.66 ± 0.02   | 0.70 ± 0.03   | 0.73 ± 0.03   | 0.56 ± 0.02   | 0.53 ± 0.03   | 0.67 ± 0.02   | 0.62 ± 0.03   | 0.73 ± 0.03   | E Africa M    | E Europe M    | 0.07 |
| 25  | 0.73 ± 0.02   | 0.73 ± 0.01   | 0.68 ± 0.02   | 0.64 ± 0.01   | 0.75 ± 0.02   | 0.77 ± 0.03   | 0.83 ± 0.02   | 0.68 ± 0.02   | 0.67 ± 0.03   | 0.79 ± 0.01   | 0.73 ± 0.03   | 0.79 ± 0.02   | E Africa M    | S Asia F      | 0.05 |
| 26  | 0.79 ± 0.02   | 0.79 ± 0.01   | 0.77 ± 0.02   | 0.70 ± 0.01   | 0.82 ± 0.02   | 0.81 ± 0.02   | 0.87 ± 0.02   | 0.78 ± 0.02   | 0.77 ± 0.02   | 0.84 ± 0.01   | 0.83 ± 0.02   | 0.86 ± 0.02   | E Africa M    | S Asia F      | 0.03 |
| 27  | 0.84 ± 0.02   | 0.849 ± 0.009 | 0.84 ± 0.02   | 0.76 ± 0.01   | 0.86 ± 0.01   | 0.85 ± 0.02   | 0.91 ± 0.02   | 0.85 ± 0.01   | 0.88 ± 0.02   | 0.89 ± 0.01   | 0.88 ± 0.02   | 0.90 ± 0.02   | E Africa M    | S Asia F      | 0.03 |
| 28  | 0.88 ± 0.02   | 0.890 ± 0.008 | 0.90 ± 0.01   | 0.81 ± 0.01   | 0.89 ± 0.01   | 0.89 ± 0.02   | 0.94 ± 0.01   | 0.911 ± 0.010 | 0.92 ± 0.01   | 0.931 ± 0.007 | 0.93 ± 0.01   | 0.92 ± 0.01   | E Africa M    | S Asia F      | 0.02 |
| 29  | 0.91 ± 0.02   | 0.921 ± 0.007 | 0.931 ± 0.009 | 0.857 ± 0.008 | 0.927 ± 0.009 | 0.91 ± 0.02   | 0.96 ± 0.01   | 0.945 ± 0.008 | 0.968 ± 0.010 | 0.951 ± 0.006 | 0.95 ± 0.01   | 0.95 ± 0.01   | E Europe M    | S Asia F      | 0.02 |
| 30  | 0.93 ± 0.01   | 0.945 ± 0.006 | 0.955 ± 0.009 | 0.887 ± 0.008 | 0.947 ± 0.009 | 0.93 ± 0.01   | 0.975 ± 0.009 | 0.976 ± 0.005 | 0.968 ± 0.008 | 0.968 ± 0.004 | 0.979 ± 0.009 | 0.965 ± 0.009 | S E Asia M    | S Asia F      | 0.02 |
| 31  | 0.95 ± 0.01   | 0.963 ± 0.005 | 0.974 ± 0.007 | 0.920 ± 0.007 | 0.963 ± 0.007 | 0.95 ± 0.01   | 0.982 ± 0.007 | 0.984 ± 0.005 | 0.987 ± 0.006 | 0.981 ± 0.004 | 0.981 ± 0.009 | 0.980 ± 0.007 | E Europe M    | S Asia F      | 0.01 |
| 32  | 0.963 ± 0.009 | 0.978 ± 0.004 | 0.984 ± 0.005 | 0.942 ± 0.006 | 0.973 ± 0.006 | 0.96 ± 0.01   | 0.991 ± 0.006 | 0.988 ± 0.004 | 0.995 ± 0.004 | 0.989 ± 0.003 | 0.988 ± 0.007 | 0.984 ± 0.007 | E Europe M    | S Asia F      | 0.01 |
| 33  | 0.971 ± 0.009 | 0.985 ± 0.003 | 0.989 ± 0.005 | 0.960 ± 0.006 | 0.983 ± 0.005 | 0.966 ± 0.010 | 0.992 ± 0.005 | 0.994 ± 0.003 | 0.998 ± 0.003 | 0.992 ± 0.002 | 0.993 ± 0.006 | 0.991 ± 0.005 | E Europe M    | S Asia F      | 0.01 |
| 34  | 0.981 ± 0.008 | 0.989 ± 0.003 | 0.988 ± 0.005 | 0.974 ± 0.005 | 0.991 ± 0.004 | 0.978 ± 0.008 | 0.998 ± 0.003 | 0.996 ± 0.002 | 0.996 ± 0.003 | 0.995 ± 0.002 | 0.996 ± 0.004 | 0.994 ± 0.004 | E Africa M    | S Asia F      | 0.00 |

**Table 162.** For algorithm ondato-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.11 ± 0.05   | 0.013 ± 0.005 | 0.03 ± 0.01   | 0.04 ± 0.01   | 0.014 ± 0.009 | 0.09 ± 0.04   | 0.09 ± 0.04   | 0.006 ± 0.003 | 0.007 ± 0.007 | 0.020 ± 0.007 | 0.011 ± 0.008 | 0.08 ± 0.04   | E Africa F    | E Asia M      | 0.51 |
| 15  | 0.12 ± 0.05   | 0.012 ± 0.004 | 0.04 ± 0.02   | 0.07 ± 0.01   | 0.012 ± 0.009 | 0.18 ± 0.05   | 0.20 ± 0.06   | 0.008 ± 0.003 | 0.02 ± 0.01   | 0.031 ± 0.009 | 0.016 ± 0.009 | 0.13 ± 0.05   | E Africa M    | E Asia M      | 0.56 |
| 16  | 0.18 ± 0.05   | 0.018 ± 0.004 | 0.04 ± 0.01   | 0.07 ± 0.01   | 0.03 ± 0.01   | 0.18 ± 0.05   | 0.19 ± 0.05   | 0.016 ± 0.004 | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.03 ± 0.01   | 0.13 ± 0.04   | E Africa M    | E Asia M      | 0.49 |
| 17  | 0.21 ± 0.05   | 0.031 ± 0.006 | 0.09 ± 0.02   | 0.11 ± 0.02   | 0.024 ± 0.010 | 0.21 ± 0.04   | 0.30 ± 0.06   | 0.031 ± 0.005 | 0.05 ± 0.02   | 0.12 ± 0.02   | 0.05 ± 0.02   | 0.29 ± 0.05   | E Africa M    | S E Asia F    | 0.47 |
| 18  | 0.28 ± 0.05   | 0.039 ± 0.005 | 0.11 ± 0.02   | 0.13 ± 0.01   | 0.05 ± 0.01   | 0.29 ± 0.05   | 0.39 ± 0.06   | 0.051 ± 0.006 | 0.09 ± 0.02   | 0.17 ± 0.02   | 0.06 ± 0.01   | 0.33 ± 0.05   | E Africa M    | E Asia F      | 0.43 |
| 19  | 0.34 ± 0.04   | 0.061 ± 0.005 | 0.13 ± 0.02   | 0.18 ± 0.01   | 0.05 ± 0.01   | 0.36 ± 0.04   | 0.46 ± 0.05   | 0.078 ± 0.006 | 0.14 ± 0.02   | 0.25 ± 0.02   | 0.12 ± 0.02   | 0.38 ± 0.05   | E Africa M    | S E Asia F    | 0.38 |
| 20  | 0.40 ± 0.04   | 0.086 ± 0.007 | 0.17 ± 0.02   | 0.21 ± 0.01   | 0.09 ± 0.01   | 0.38 ± 0.04   | 0.55 ± 0.04   | 0.122 ± 0.009 | 0.20 ± 0.03   | 0.37 ± 0.02   | 0.18 ± 0.02   | 0.43 ± 0.04   | E Africa M    | E Asia F      | 0.34 |
| 21  | 0.44 ± 0.04   | 0.115 ± 0.007 | 0.21 ± 0.01   | 0.25 ± 0.01   | 0.11 ± 0.01   | 0.47 ± 0.04   | 0.62 ± 0.04   | 0.188 ± 0.010 | 0.31 ± 0.02   | 0.48 ± 0.02   | 0.24 ± 0.02   | 0.55 ± 0.04   | E Africa M    | S E Asia F    | 0.31 |
| 22  | 0.54 ± 0.04   | 0.147 ± 0.007 | 0.27 ± 0.02   | 0.31 ± 0.01   | 0.15 ± 0.01   | 0.56 ± 0.04   | 0.73 ± 0.04   | 0.26 ± 0.01   | 0.44 ± 0.02   | 0.57 ± 0.02   | 0.34 ± 0.03   | 0.61 ± 0.04   | E Africa M    | E Asia F      | 0.28 |
| 23  | 0.58 ± 0.04   | 0.186 ± 0.007 | 0.34 ± 0.01   | 0.35 ± 0.01   | 0.21 ± 0.01   | 0.59 ± 0.03   | 0.78 ± 0.03   | 0.34 ± 0.01   | 0.50 ± 0.02   | 0.65 ± 0.01   | 0.42 ± 0.02   | 0.70 ± 0.03   | E Africa M    | E Asia F      | 0.24 |
| 24  | 0.59 ± 0.03   | 0.236 ± 0.007 | 0.41 ± 0.02   | 0.41 ± 0.01   | 0.25 ± 0.01   | 0.63 ± 0.03   | 0.84 ± 0.03   | 0.43 ± 0.01   | 0.63 ± 0.02   | 0.73 ± 0.01   | 0.50 ± 0.02   | 0.74 ± 0.03   | E Africa M    | E Asia F      | 0.22 |
| 25  | 0.64 ± 0.03   | 0.294 ± 0.008 | 0.49 ± 0.01   | 0.46 ± 0.01   | 0.32 ± 0.01   | 0.70 ± 0.02   | 0.88 ± 0.02   | 0.51 ± 0.01   | 0.72 ± 0.02   | 0.796 ± 0.010 | 0.58 ± 0.03   | 0.81 ± 0.02   | E Africa M    | E Asia F      | 0.19 |
| 26  | 0.69 ± 0.02   | 0.359 ± 0.007 | 0.55 ± 0.01   | 0.51 ± 0.01   | 0.39 ± 0.02   | 0.71 ± 0.02   | 0.91 ± 0.02   | 0.60 ± 0.01   | 0.78 ± 0.01   | 0.832 ± 0.008 | 0.66 ± 0.02   | 0.85 ± 0.02   | E Africa M    | E Asia F      | 0.16 |
| 27  | 0.72 ± 0.03   | 0.410 ± 0.007 | 0.64 ± 0.01   | 0.552 ± 0.009 | 0.43 ± 0.02   | 0.75 ± 0.02   | 0.92 ± 0.02   | 0.67 ± 0.01   | 0.85 ± 0.01   | 0.867 ± 0.007 | 0.73 ± 0.02   | 0.88 ± 0.02   | E Africa M    | E Asia F      | 0.14 |
| 28  | 0.77 ± 0.02   | 0.482 ± 0.007 | 0.70 ± 0.01   | 0.599 ± 0.008 | 0.51 ± 0.01   | 0.77 ± 0.02   | 0.95 ± 0.01   | 0.744 ± 0.009 | 0.90 ± 0.01   | 0.902 ± 0.006 | 0.78 ± 0.02   | 0.90 ± 0.01   | E Africa M    | E Asia F      | 0.12 |
| 29  | 0.78 ± 0.02   | 0.535 ± 0.007 | 0.758 ± 0.009 | 0.646 ± 0.009 | 0.56 ± 0.01   | 0.82 ± 0.02   | 0.96 ± 0.01   | 0.795 ± 0.008 | 0.918 ± 0.009 | 0.921 ± 0.005 | 0.83 ± 0.02   | 0.93 ± 0.01   | E Africa M    | E Asia F      | 0.10 |
| 30  | 0.82 ± 0.02   | 0.597 ± 0.007 | 0.814 ± 0.010 | 0.694 ± 0.007 | 0.63 ± 0.01   | 0.83 ± 0.02   | 0.968 ± 0.010 | 0.836 ± 0.007 | 0.939 ± 0.008 | 0.935 ± 0.004 | 0.86 ± 0.02   | 0.945 ± 0.010 | E Africa M    | E Asia F      | 0.09 |
| 31  | 0.86 ± 0.02   | 0.657 ± 0.007 | 0.858 ± 0.008 | 0.737 ± 0.007 | 0.68 ± 0.01   | 0.86 ± 0.02   | 0.970 ± 0.010 | 0.882 ± 0.007 | 0.967 ± 0.006 | 0.957 ± 0.003 | 0.90 ± 0.01   | 0.946 ± 0.009 | E Africa M    | E Asia F      | 0.07 |
| 32  | 0.89 ± 0.02   | 0.705 ± 0.007 | 0.898 ± 0.007 | 0.774 ± 0.007 | 0.71 ± 0.01   | 0.89 ± 0.02   | 0.977 ± 0.008 | 0.902 ± 0.006 | 0.972 ± 0.005 | 0.963 ± 0.003 | 0.92 ± 0.01   | 0.954 ± 0.008 | E Africa M    | E Asia F      | 0.06 |
| 33  | 0.91 ± 0.02   | 0.753 ± 0.006 | 0.915 ± 0.007 | 0.809 ± 0.007 | 0.77 ± 0.01   | 0.90 ± 0.01   | 0.985 ± 0.007 | 0.929 ± 0.005 | 0.982 ± 0.004 | 0.971 ± 0.002 | 0.93 ± 0.01   | 0.970 ± 0.007 | E Africa M    | E Asia F      | 0.05 |
| 34  | 0.92 ± 0.01   | 0.791 ± 0.006 | 0.934 ± 0.007 | 0.845 ± 0.006 | 0.81 ± 0.01   | 0.93 ± 0.01   | 0.992 ± 0.005 | 0.940 ± 0.005 | 0.989 ± 0.003 | 0.978 ± 0.002 | 0.955 ± 0.009 | 0.974 ± 0.006 | E Africa M    | E Asia F      | 0.04 |

**Table 163.** For algorithm privately-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.02 ± 0.01   | 0.030 ± 0.008 | 0.005 ± 0.005 | 0.014 ± 0.006 | 0.05 ± 0.02   | 0.02 ± 0.02   | 0.006 ± 0.006 | 0.015 ± 0.006 | 0.002 ± 0.002 | 0.005 ± 0.003 | 0.005 ± 0.005 | 0.000 ± 0.000 | S E Asia F    | W Africa M    | 0.54 |
| 15  | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.02 ± 0.01   | 0.04 ± 0.01   | 0.09 ± 0.03   | 0.04 ± 0.02   | 0.005 ± 0.005 | 0.015 ± 0.006 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.007 ± 0.007 | 0.009 ± 0.009 | S E Asia F    | E Europe M    | 0.57 |
| 16  | 0.06 ± 0.02   | 0.11 ± 0.02   | 0.03 ± 0.02   | 0.06 ± 0.02   | 0.13 ± 0.03   | 0.08 ± 0.03   | 0.02 ± 0.01   | 0.026 ± 0.008 | 0.012 ± 0.009 | 0.006 ± 0.004 | 0.03 ± 0.01   | 0.005 ± 0.005 | S E Asia F    | W Africa M    | 0.49 |
| 17  | 0.08 ± 0.03   | 0.14 ± 0.02   | 0.07 ± 0.03   | 0.08 ± 0.02   | 0.12 ± 0.03   | 0.11 ± 0.03   | 0.01 ± 0.01   | 0.037 ± 0.009 | 0.003 ± 0.003 | 0.015 ± 0.007 | 0.03 ± 0.01   | 0.03 ± 0.02   | E Asia F      | E Europe M    | 0.46 |
| 18  | 0.10 ± 0.02   | 0.19 ± 0.02   | 0.08 ± 0.02   | 0.10 ± 0.01   | 0.18 ± 0.03   | 0.11 ± 0.03   | 0.05 ± 0.02   | 0.043 ± 0.008 | 0.007 ± 0.005 | 0.010 ± 0.004 | 0.03 ± 0.01   | 0.03 ± 0.02   | E Asia F      | E Europe M    | 0.45 |
| 19  | 0.15 ± 0.03   | 0.24 ± 0.02   | 0.11 ± 0.03   | 0.15 ± 0.02   | 0.23 ± 0.03   | 0.15 ± 0.03   | 0.08 ± 0.02   | 0.07 ± 0.01   | 0.02 ± 0.01   | 0.034 ± 0.009 | 0.04 ± 0.01   | 0.04 ± 0.02   | E Asia F      | E Europe M    | 0.40 |
| 20  | 0.22 ± 0.03   | 0.31 ± 0.02   | 0.15 ± 0.02   | 0.18 ± 0.02   | 0.31 ± 0.03   | 0.22 ± 0.03   | 0.16 ± 0.03   | 0.10 ± 0.01   | 0.03 ± 0.01   | 0.08 ± 0.01   | 0.07 ± 0.02   | 0.09 ± 0.02   | S E Asia F    | E Europe M    | 0.34 |
| 21  | 0.24 ± 0.03   | 0.40 ± 0.02   | 0.16 ± 0.02   | 0.23 ± 0.02   | 0.37 ± 0.02   | 0.27 ± 0.03   | 0.19 ± 0.03   | 0.13 ± 0.01   | 0.05 ± 0.02   | 0.11 ± 0.01   | 0.11 ± 0.02   | 0.13 ± 0.03   | E Asia F      | E Europe M    | 0.31 |
| 22  | 0.32 ± 0.03   | 0.46 ± 0.02   | 0.24 ± 0.02   | 0.25 ± 0.02   | 0.45 ± 0.02   | 0.33 ± 0.03   | 0.29 ± 0.03   | 0.18 ± 0.02   | 0.10 ± 0.02   | 0.16 ± 0.02   | 0.14 ± 0.02   | 0.19 ± 0.03   | E Asia F      | E Europe M    | 0.26 |
| 23  | 0.34 ± 0.03   | 0.53 ± 0.02   | 0.31 ± 0.02   | 0.32 ± 0.02   | 0.52 ± 0.02   | 0.36 ± 0.03   | 0.33 ± 0.03   | 0.22 ± 0.02   | 0.16 ± 0.03   | 0.22 ± 0.02   | 0.18 ± 0.02   | 0.24 ± 0.03   | E Asia F      | E Europe M    | 0.22 |
| 24  | 0.42 ± 0.03   | 0.59 ± 0.02   | 0.38 ± 0.02   | 0.37 ± 0.01   | 0.59 ± 0.02   | 0.39 ± 0.03   | 0.41 ± 0.03   | 0.31 ± 0.02   | 0.21 ± 0.02   | 0.32 ± 0.02   | 0.25 ± 0.02   | 0.28 ± 0.03   | E Asia F      | E Europe M    | 0.18 |
| 25  | 0.48 ± 0.03   | 0.68 ± 0.01   | 0.43 ± 0.02   | 0.42 ± 0.01   | 0.64 ± 0.02   | 0.44 ± 0.03   | 0.50 ± 0.03   | 0.39 ± 0.02   | 0.32 ± 0.03   | 0.39 ± 0.02   | 0.32 ± 0.03   | 0.34 ± 0.03   | E Asia F      | S E Asia M    | 0.15 |
| 26  | 0.53 ± 0.03   | 0.73 ± 0.01   | 0.51 ± 0.02   | 0.48 ± 0.01   | 0.71 ± 0.02   | 0.48 ± 0.03   | 0.57 ± 0.03   | 0.45 ± 0.02   | 0.38 ± 0.03   | 0.47 ± 0.02   | 0.41 ± 0.03   | 0.42 ± 0.03   | E Asia F      | E Europe M    | 0.12 |
| 27  | 0.60 ± 0.03   | 0.79 ± 0.01   | 0.59 ± 0.02   | 0.53 ± 0.01   | 0.75 ± 0.02   | 0.54 ± 0.02   | 0.62 ± 0.03   | 0.55 ± 0.02   | 0.51 ± 0.03   | 0.54 ± 0.02   | 0.51 ± 0.03   | 0.48 ± 0.03   | E Asia F      | W Africa M    | 0.09 |
| 28  | 0.63 ± 0.02   | 0.843 ± 0.009 | 0.66 ± 0.02   | 0.58 ± 0.01   | 0.81 ± 0.01   | 0.58 ± 0.03   | 0.68 ± 0.02   | 0.62 ± 0.02   | 0.65 ± 0.03   | 0.62 ± 0.01   | 0.61 ± 0.03   | 0.54 ± 0.02   | E Asia F      | W Africa M    | 0.07 |
| 29  | 0.69 ± 0.02   | 0.884 ± 0.008 | 0.73 ± 0.02   | 0.64 ± 0.01   | 0.85 ± 0.01   | 0.61 ± 0.03   | 0.75 ± 0.02   | 0.72 ± 0.02   | 0.69 ± 0.02   | 0.68 ± 0.01   | 0.68 ± 0.03   | 0.63 ± 0.02   | E Asia F      | W Africa F    | 0.07 |
| 30  | 0.73 ± 0.02   | 0.908 ± 0.007 | 0.78 ± 0.02   | 0.67 ± 0.01   | 0.89 ± 0.01   | 0.64 ± 0.02   | 0.79 ± 0.02   | 0.77 ± 0.02   | 0.78 ± 0.02   | 0.73 ± 0.01   | 0.76 ± 0.02   | 0.68 ± 0.02   | E Asia F      | W Africa F    | 0.06 |
| 31  | 0.76 ± 0.02   | 0.926 ± 0.007 | 0.83 ± 0.02   | 0.72 ± 0.01   | 0.91 ± 0.01   | 0.71 ± 0.02   | 0.84 ± 0.02   | 0.83 ± 0.01   | 0.84 ± 0.02   | 0.78 ± 0.01   | 0.79 ± 0.03   | 0.73 ± 0.02   | E Asia F      | W Africa F    | 0.05 |
| 32  | 0.80 ± 0.02   | 0.946 ± 0.006 | 0.87 ± 0.01   | 0.77 ± 0.01   | 0.923 ± 0.010 | 0.72 ± 0.02   | 0.88 ± 0.02   | 0.88 ± 0.01   | 0.91 ± 0.02   | 0.829 ± 0.009 | 0.86 ± 0.02   | 0.77 ± 0.02   | E Asia F      | W Africa F    | 0.05 |
| 33  | 0.83 ± 0.02   | 0.963 ± 0.005 | 0.90 ± 0.01   | 0.80 ± 0.01   | 0.944 ± 0.010 | 0.77 ± 0.02   | 0.90 ± 0.02   | 0.91 ± 0.01   | 0.95 ± 0.01   | 0.860 ± 0.010 | 0.90 ± 0.02   | 0.82 ± 0.02   | E Asia F      | W Africa F    | 0.04 |
| 34  | 0.86 ± 0.02   | 0.975 ± 0.004 | 0.93 ± 0.01   | 0.846 ± 0.010 | 0.964 ± 0.008 | 0.80 ± 0.02   | 0.91 ± 0.02   | 0.936 ± 0.008 | 0.96 ± 0.01   | 0.894 ± 0.009 | 0.93 ± 0.02   | 0.85 ± 0.02   | E Asia F      | W Africa F    | 0.04 |

**Table 164.** For algorithm privately-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.05 ± 0.03   | 0.08 ± 0.01   | 0.04 ± 0.02   | 0.06 ± 0.01   | 0.09 ± 0.02   | 0.02 ± 0.02   | 0.06 ± 0.03   | 0.040 ± 0.008 | 0.02 ± 0.01   | 0.023 ± 0.008 | 0.05 ± 0.02  | 0.05 ± 0.04   | S E Asia F    | E Europe M    | 0.28 |
| 15  | 0.14 ± 0.04   | 0.11 ± 0.01   | 0.07 ± 0.02   | 0.07 ± 0.02   | 0.13 ± 0.03   | 0.07 ± 0.04   | 0.11 ± 0.05   | 0.073 ± 0.009 | 0.03 ± 0.02   | 0.04 ± 0.01   | 0.04 ± 0.02  | 0.06 ± 0.03   | E Africa F    | E Europe M    | 0.27 |
| 16  | 0.12 ± 0.04   | 0.15 ± 0.01   | 0.07 ± 0.02   | 0.12 ± 0.02   | 0.16 ± 0.03   | 0.11 ± 0.04   | 0.09 ± 0.04   | 0.10 ± 0.01   | 0.04 ± 0.01   | 0.06 ± 0.01   | 0.09 ± 0.02  | 0.07 ± 0.04   | S E Asia F    | E Europe M    | 0.21 |
| 17  | 0.14 ± 0.04   | 0.21 ± 0.01   | 0.11 ± 0.02   | 0.17 ± 0.02   | 0.21 ± 0.03   | 0.16 ± 0.04   | 0.20 ± 0.05   | 0.15 ± 0.01   | 0.06 ± 0.02   | 0.13 ± 0.02   | 0.13 ± 0.02  | 0.16 ± 0.04   | E Asia F      | E Europe M    | 0.17 |
| 18  | 0.23 ± 0.04   | 0.25 ± 0.01   | 0.13 ± 0.02   | 0.19 ± 0.02   | 0.27 ± 0.02   | 0.17 ± 0.04   | 0.27 ± 0.05   | 0.20 ± 0.01   | 0.11 ± 0.03   | 0.16 ± 0.02   | 0.18 ± 0.02  | 0.19 ± 0.04   | E Africa M    | E Europe M    | 0.15 |
| 19  | 0.23 ± 0.04   | 0.30 ± 0.01   | 0.18 ± 0.02   | 0.24 ± 0.02   | 0.31 ± 0.02   | 0.19 ± 0.04   | 0.30 ± 0.05   | 0.25 ± 0.01   | 0.14 ± 0.02   | 0.22 ± 0.02   | 0.22 ± 0.02  | 0.23 ± 0.04   | S E Asia F    | E Europe M    | 0.13 |
| 20  | 0.29 ± 0.04   | 0.36 ± 0.01   | 0.19 ± 0.02   | 0.27 ± 0.01   | 0.38 ± 0.02   | 0.24 ± 0.03   | 0.41 ± 0.05   | 0.32 ± 0.01   | 0.20 ± 0.02   | 0.30 ± 0.02   | 0.28 ± 0.02  | 0.27 ± 0.03   | E Africa M    | E Europe F    | 0.14 |
| 21  | 0.34 ± 0.04   | 0.42 ± 0.01   | 0.24 ± 0.02   | 0.32 ± 0.01   | 0.42 ± 0.02   | 0.30 ± 0.04   | 0.46 ± 0.04   | 0.38 ± 0.01   | 0.28 ± 0.02   | 0.38 ± 0.01   | 0.33 ± 0.03  | 0.34 ± 0.03   | E Africa M    | E Europe F    | 0.11 |
| 22  | 0.43 ± 0.04   | 0.486 ± 0.010 | 0.30 ± 0.02   | 0.36 ± 0.01   | 0.47 ± 0.02   | 0.38 ± 0.03   | 0.53 ± 0.04   | 0.45 ± 0.01   | 0.34 ± 0.02   | 0.47 ± 0.02   | 0.43 ± 0.03  | 0.40 ± 0.04   | E Africa M    | E Europe F    | 0.10 |
| 23  | 0.43 ± 0.03   | 0.538 ± 0.008 | 0.35 ± 0.01   | 0.41 ± 0.01   | 0.53 ± 0.02   | 0.39 ± 0.03   | 0.62 ± 0.04   | 0.51 ± 0.01   | 0.45 ± 0.02   | 0.54 ± 0.01   | 0.49 ± 0.02  | 0.50 ± 0.03   | E Africa M    | E Europe F    | 0.09 |
| 24  | 0.49 ± 0.03   | 0.595 ± 0.009 | 0.40 ± 0.01   | 0.46 ± 0.01   | 0.59 ± 0.02   | 0.43 ± 0.03   | 0.67 ± 0.03   | 0.58 ± 0.01   | 0.57 ± 0.02   | 0.63 ± 0.01   | 0.56 ± 0.02  | 0.54 ± 0.03   | E Africa M    | E Europe F    | 0.09 |
| 25  | 0.53 ± 0.03   | 0.654 ± 0.008 | 0.47 ± 0.01   | 0.51 ± 0.01   | 0.65 ± 0.02   | 0.51 ± 0.03   | 0.73 ± 0.03   | 0.64 ± 0.01   | 0.63 ± 0.02   | 0.68 ± 0.01   | 0.62 ± 0.02  | 0.62 ± 0.03   | E Africa M    | E Europe F    | 0.08 |
| 26  | 0.59 ± 0.03   | 0.705 ± 0.007 | 0.53 ± 0.01   | 0.556 ± 0.009 | 0.69 ± 0.01   | 0.50 ± 0.03   | 0.77 ± 0.03   | 0.700 ± 0.010 | 0.71 ± 0.02   | 0.736 ± 0.009 | 0.67 ± 0.02  | 0.65 ± 0.02   | E Africa M    | W Africa F    | 0.08 |
| 27  | 0.61 ± 0.03   | 0.740 ± 0.006 | 0.60 ± 0.01   | 0.590 ± 0.009 | 0.74 ± 0.01   | 0.56 ± 0.02   | 0.80 ± 0.03   | 0.74 ± 0.01   | 0.77 ± 0.02   | 0.778 ± 0.009 | 0.76 ± 0.02  | 0.73 ± 0.02   | E Africa M    | W Africa F    | 0.07 |
| 28  | 0.68 ± 0.02   | 0.788 ± 0.006 | 0.66 ± 0.01   | 0.635 ± 0.008 | 0.78 ± 0.01   | 0.58 ± 0.02   | 0.84 ± 0.02   | 0.796 ± 0.009 | 0.85 ± 0.01   | 0.824 ± 0.007 | 0.80 ± 0.02  | 0.77 ± 0.02   | E Europe M    | W Africa F    | 0.07 |
| 29  | 0.73 ± 0.03   | 0.825 ± 0.006 | 0.71 ± 0.01   | 0.673 ± 0.007 | 0.81 ± 0.01   | 0.63 ± 0.02   | 0.86 ± 0.02   | 0.830 ± 0.008 | 0.87 ± 0.01   | 0.854 ± 0.006 | 0.83 ± 0.02  | 0.80 ± 0.02   | E Europe M    | W Africa F    | 0.06 |
| 30  | 0.72 ± 0.02   | 0.854 ± 0.005 | 0.764 ± 0.010 | 0.712 ± 0.008 | 0.84 ± 0.01   | 0.66 ± 0.02   | 0.89 ± 0.02   | 0.859 ± 0.007 | 0.91 ± 0.01   | 0.878 ± 0.005 | 0.86 ± 0.02  | 0.84 ± 0.01   | E Europe M    | W Africa F    | 0.06 |
| 31  | 0.77 ± 0.02   | 0.883 ± 0.005 | 0.807 ± 0.009 | 0.750 ± 0.008 | 0.868 ± 0.010 | 0.71 ± 0.02   | 0.91 ± 0.02   | 0.889 ± 0.007 | 0.932 ± 0.008 | 0.909 ± 0.005 | 0.90 ± 0.01  | 0.85 ± 0.01   | E Europe M    | W Africa F    | 0.05 |
| 32  | 0.79 ± 0.02   | 0.903 ± 0.004 | 0.848 ± 0.009 | 0.786 ± 0.007 | 0.898 ± 0.010 | 0.72 ± 0.02   | 0.93 ± 0.01   | 0.918 ± 0.006 | 0.951 ± 0.007 | 0.925 ± 0.004 | 0.91 ± 0.01  | 0.88 ± 0.01   | E Europe M    | W Africa F    | 0.05 |
| 33  | 0.83 ± 0.02   | 0.917 ± 0.004 | 0.877 ± 0.009 | 0.810 ± 0.006 | 0.902 ± 0.010 | 0.74 ± 0.02   | 0.93 ± 0.01   | 0.933 ± 0.005 | 0.963 ± 0.006 | 0.938 ± 0.004 | 0.93 ± 0.01  | 0.90 ± 0.01   | E Europe M    | W Africa F    | 0.04 |
| 34  | 0.81 ± 0.02   | 0.933 ± 0.004 | 0.905 ± 0.008 | 0.840 ± 0.006 | 0.923 ± 0.008 | 0.76 ± 0.02   | 0.95 ± 0.01   | 0.947 ± 0.004 | 0.973 ± 0.005 | 0.950 ± 0.003 | 0.95 ± 0.01  | 0.91 ± 0.01   | E Europe M    | W Africa F    | 0.04 |

**Table 165.** For algorithm regula-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.004 ± 0.004 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.000 ± 0.000 | E Africa F    | E Europe F    | 0.74 |
| 15  | 0.01 ± 0.01   | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.009 ± 0.005 | 0.010 ± 0.007 | 0.009 ± 0.009 | 0.003 ± 0.003 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.005 ± 0.005 | E Africa F    | E Europe F    | 0.62 |
| 16  | 0.03 ± 0.02   | 0.003 ± 0.002 | 0.003 ± 0.003 | 0.004 ± 0.004 | 0.02 ± 0.01   | 0.05 ± 0.02   | 0.02 ± 0.01   | 0.001 ± 0.001 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.003 ± 0.003 | 0.009 ± 0.009 | W Africa F    | E Asia M      | 0.62 |
| 17  | 0.07 ± 0.02   | 0.013 ± 0.005 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.02 ± 0.01   | 0.09 ± 0.03   | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.007 ± 0.005 | 0.011 ± 0.008 | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.59 |
| 18  | 0.10 ± 0.03   | 0.015 ± 0.005 | 0.03 ± 0.01   | 0.034 ± 0.009 | 0.04 ± 0.01   | 0.14 ± 0.03   | 0.015 ± 0.009 | 0.009 ± 0.004 | 0.002 ± 0.002 | 0.014 ± 0.005 | 0.011 ± 0.007 | 0.06 ± 0.02   | W Africa F    | E Europe M    | 0.56 |
| 19  | 0.19 ± 0.03   | 0.040 ± 0.008 | 0.06 ± 0.02   | 0.07 ± 0.01   | 0.10 ± 0.02   | 0.23 ± 0.04   | 0.07 ± 0.02   | 0.012 ± 0.004 | 0.011 ± 0.008 | 0.029 ± 0.008 | 0.016 ± 0.008 | 0.09 ± 0.02   | W Africa F    | E Europe M    | 0.49 |
| 20  | 0.24 ± 0.03   | 0.08 ± 0.01   | 0.08 ± 0.02   | 0.11 ± 0.01   | 0.14 ± 0.02   | 0.33 ± 0.04   | 0.13 ± 0.03   | 0.020 ± 0.005 | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.05 ± 0.01   | 0.17 ± 0.04   | W Africa F    | E Europe M    | 0.43 |
| 21  | 0.31 ± 0.03   | 0.14 ± 0.01   | 0.16 ± 0.02   | 0.19 ± 0.02   | 0.21 ± 0.02   | 0.40 ± 0.04   | 0.22 ± 0.03   | 0.054 ± 0.009 | 0.04 ± 0.02   | 0.13 ± 0.01   | 0.11 ± 0.02   | 0.26 ± 0.04   | W Africa F    | E Europe M    | 0.33 |
| 22  | 0.43 ± 0.03   | 0.19 ± 0.02   | 0.24 ± 0.02   | 0.26 ± 0.02   | 0.31 ± 0.02   | 0.46 ± 0.04   | 0.37 ± 0.03   | 0.11 ± 0.01   | 0.13 ± 0.02   | 0.22 ± 0.02   | 0.16 ± 0.02   | 0.33 ± 0.04   | W Africa F    | E Asia M      | 0.26 |
| 23  | 0.52 ± 0.03   | 0.30 ± 0.02   | 0.35 ± 0.02   | 0.37 ± 0.02   | 0.41 ± 0.02   | 0.60 ± 0.03   | 0.45 ± 0.03   | 0.19 ± 0.02   | 0.23 ± 0.03   | 0.33 ± 0.02   | 0.30 ± 0.03   | 0.49 ± 0.04   | W Africa F    | E Asia M      | 0.19 |
| 24  | 0.62 ± 0.03   | 0.40 ± 0.02   | 0.49 ± 0.02   | 0.48 ± 0.02   | 0.52 ± 0.02   | 0.66 ± 0.03   | 0.57 ± 0.03   | 0.30 ± 0.02   | 0.40 ± 0.03   | 0.50 ± 0.02   | 0.42 ± 0.03   | 0.60 ± 0.03   | W Africa F    | E Asia M      | 0.13 |
| 25  | 0.66 ± 0.03   | 0.54 ± 0.01   | 0.59 ± 0.02   | 0.55 ± 0.01   | 0.64 ± 0.02   | 0.76 ± 0.02   | 0.70 ± 0.03   | 0.44 ± 0.02   | 0.57 ± 0.03   | 0.62 ± 0.02   | 0.57 ± 0.03   | 0.70 ± 0.03   | W Africa F    | E Asia M      | 0.08 |
| 26  | 0.73 ± 0.03   | 0.63 ± 0.01   | 0.71 ± 0.02   | 0.65 ± 0.01   | 0.74 ± 0.02   | 0.83 ± 0.02   | 0.78 ± 0.02   | 0.59 ± 0.02   | 0.70 ± 0.03   | 0.73 ± 0.01   | 0.68 ± 0.03   | 0.80 ± 0.02   | W Africa F    | E Asia M      | 0.06 |
| 27  | 0.82 ± 0.02   | 0.74 ± 0.01   | 0.80 ± 0.02   | 0.71 ± 0.01   | 0.81 ± 0.01   | 0.88 ± 0.02   | 0.84 ± 0.02   | 0.70 ± 0.02   | 0.84 ± 0.02   | 0.82 ± 0.01   | 0.81 ± 0.02   | 0.88 ± 0.02   | W Africa F    | E Asia M      | 0.04 |
| 28  | 0.87 ± 0.02   | 0.81 ± 0.01   | 0.90 ± 0.01   | 0.796 ± 0.009 | 0.88 ± 0.01   | 0.91 ± 0.02   | 0.89 ± 0.02   | 0.81 ± 0.01   | 0.90 ± 0.02   | 0.887 ± 0.009 | 0.88 ± 0.02   | 0.91 ± 0.01   | W Africa F    | S Asia F      | 0.03 |
| 29  | 0.91 ± 0.01   | 0.869 ± 0.009 | 0.940 ± 0.009 | 0.852 ± 0.008 | 0.926 ± 0.010 | 0.94 ± 0.01   | 0.94 ± 0.01   | 0.89 ± 0.01   | 0.96 ± 0.01   | 0.927 ± 0.007 | 0.91 ± 0.02   | 0.95 ± 0.01   | E Europe M    | S Asia F      | 0.02 |
| 30  | 0.93 ± 0.01   | 0.922 ± 0.007 | 0.960 ± 0.008 | 0.889 ± 0.008 | 0.950 ± 0.008 | 0.95 ± 0.01   | 0.966 ± 0.010 | 0.937 ± 0.009 | 0.981 ± 0.008 | 0.958 ± 0.005 | 0.96 ± 0.01   | 0.962 ± 0.010 | E Europe M    | S Asia F      | 0.01 |
| 31  | 0.95 ± 0.01   | 0.950 ± 0.006 | 0.982 ± 0.006 | 0.924 ± 0.007 | 0.970 ± 0.007 | 0.974 ± 0.009 | 0.976 ± 0.008 | 0.965 ± 0.007 | 0.989 ± 0.006 | 0.975 ± 0.004 | 0.981 ± 0.008 | 0.983 ± 0.006 | E Europe M    | S Asia F      | 0.01 |
| 32  | 0.969 ± 0.008 | 0.973 ± 0.004 | 0.993 ± 0.003 | 0.957 ± 0.005 | 0.978 ± 0.006 | 0.982 ± 0.007 | 0.986 ± 0.007 | 0.982 ± 0.005 | 0.995 ± 0.004 | 0.985 ± 0.003 | 0.987 ± 0.006 | 0.988 ± 0.006 | E Europe M    | S Asia F      | 0.01 |
| 33  | 0.983 ± 0.007 | 0.985 ± 0.003 | 0.992 ± 0.004 | 0.972 ± 0.004 | 0.991 ± 0.004 | 0.991 ± 0.005 | 0.993 ± 0.005 | 0.992 ± 0.004 | 0.998 ± 0.003 | 0.993 ± 0.002 | 0.988 ± 0.007 | 0.990 ± 0.005 | E Europe M    | S Asia F      | 0.00 |
| 34  | 0.979 ± 0.009 | 0.991 ± 0.003 | 0.997 ± 0.002 | 0.983 ± 0.004 | 0.993 ± 0.004 | 0.993 ± 0.005 | 0.995 ± 0.004 | 0.995 ± 0.002 | 0.998 ± 0.003 | 0.995 ± 0.002 | 0.996 ± 0.004 | 0.996 ± 0.003 | E Europe M    | E Africa F    | 0.00 |

**Table 166.** For algorithm regula-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.06 ± 0.03   | 0.012 ± 0.005 | 0.02 ± 0.01   | 0.018 ± 0.008 | 0.011 ± 0.009 | 0.03 ± 0.03   | 0.02 ± 0.02   | 0.005 ± 0.003 | 0.002 ± 0.002 | 0.003 ± 0.002 | 0.006 ± 0.006 | 0.04 ± 0.03   | E Africa F    | E Europe M    | 0.50 |
| 15  | 0.09 ± 0.04   | 0.017 ± 0.005 | 0.03 ± 0.01   | 0.030 ± 0.010 | 0.02 ± 0.01   | 0.14 ± 0.05   | 0.04 ± 0.02   | 0.008 ± 0.003 | 0.004 ± 0.004 | 0.005 ± 0.003 | 0.005 ± 0.005 | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.58 |
| 16  | 0.13 ± 0.05   | 0.024 ± 0.006 | 0.03 ± 0.01   | 0.042 ± 0.010 | 0.03 ± 0.01   | 0.10 ± 0.04   | 0.04 ± 0.02   | 0.007 ± 0.002 | 0.003 ± 0.003 | 0.005 ± 0.003 | 0.008 ± 0.006 | 0.04 ± 0.02   | E Africa F    | E Europe M    | 0.55 |
| 17  | 0.17 ± 0.04   | 0.049 ± 0.007 | 0.05 ± 0.02   | 0.07 ± 0.01   | 0.04 ± 0.01   | 0.17 ± 0.04   | 0.11 ± 0.03   | 0.010 ± 0.003 | 0.003 ± 0.003 | 0.013 ± 0.005 | 0.013 ± 0.008 | 0.10 ± 0.04   | W Africa F    | E Europe M    | 0.51 |
| 18  | 0.27 ± 0.05   | 0.062 ± 0.006 | 0.11 ± 0.02   | 0.11 ± 0.01   | 0.07 ± 0.01   | 0.19 ± 0.04   | 0.12 ± 0.03   | 0.023 ± 0.004 | 0.009 ± 0.006 | 0.028 ± 0.007 | 0.021 ± 0.008 | 0.11 ± 0.03   | E Africa F    | E Europe M    | 0.46 |
| 19  | 0.36 ± 0.05   | 0.093 ± 0.006 | 0.13 ± 0.02   | 0.18 ± 0.02   | 0.11 ± 0.01   | 0.32 ± 0.04   | 0.20 ± 0.04   | 0.041 ± 0.005 | 0.04 ± 0.01   | 0.053 ± 0.009 | 0.04 ± 0.01   | 0.15 ± 0.03   | E Africa F    | E Europe M    | 0.42 |
| 20  | 0.43 ± 0.04   | 0.146 ± 0.008 | 0.19 ± 0.02   | 0.22 ± 0.01   | 0.17 ± 0.01   | 0.38 ± 0.04   | 0.27 ± 0.04   | 0.068 ± 0.006 | 0.05 ± 0.01   | 0.09 ± 0.01   | 0.09 ± 0.01   | 0.21 ± 0.03   | E Africa F    | E Europe M    | 0.36 |
| 21  | 0.49 ± 0.04   | 0.205 ± 0.008 | 0.26 ± 0.02   | 0.28 ± 0.01   | 0.22 ± 0.02   | 0.49 ± 0.04   | 0.33 ± 0.04   | 0.121 ± 0.008 | 0.09 ± 0.02   | 0.15 ± 0.01   | 0.12 ± 0.02   | 0.30 ± 0.03   | E Africa F    | E Europe M    | 0.30 |
| 22  | 0.60 ± 0.04   | 0.272 ± 0.008 | 0.35 ± 0.02   | 0.36 ± 0.01   | 0.31 ± 0.02   | 0.57 ± 0.03   | 0.44 ± 0.04   | 0.188 ± 0.010 | 0.17 ± 0.02   | 0.23 ± 0.01   | 0.23 ± 0.02   | 0.40 ± 0.04   | E Africa F    | E Europe M    | 0.24 |
| 23  | 0.63 ± 0.03   | 0.349 ± 0.009 | 0.43 ± 0.02   | 0.43 ± 0.01   | 0.41 ± 0.02   | 0.62 ± 0.03   | 0.50 ± 0.04   | 0.28 ± 0.01   | 0.27 ± 0.02   | 0.31 ± 0.01   | 0.31 ± 0.02   | 0.48 ± 0.03   | E Africa F    | E Europe M    | 0.17 |
| 24  | 0.72 ± 0.03   | 0.436 ± 0.009 | 0.53 ± 0.01   | 0.51 ± 0.01   | 0.50 ± 0.02   | 0.67 ± 0.02   | 0.62 ± 0.04   | 0.39 ± 0.01   | 0.40 ± 0.02   | 0.42 ± 0.01   | 0.42 ± 0.02   | 0.57 ± 0.03   | E Africa F    | E Asia M      | 0.13 |
| 25  | 0.75 ± 0.03   | 0.533 ± 0.008 | 0.63 ± 0.01   | 0.57 ± 0.01   | 0.58 ± 0.01   | 0.77 ± 0.02   | 0.69 ± 0.03   | 0.49 ± 0.01   | 0.51 ± 0.02   | 0.53 ± 0.01   | 0.54 ± 0.02   | 0.67 ± 0.03   | W Africa F    | E Asia M      | 0.09 |
| 26  | 0.81 ± 0.02   | 0.624 ± 0.008 | 0.70 ± 0.01   | 0.63 ± 0.01   | 0.67 ± 0.01   | 0.78 ± 0.02   | 0.78 ± 0.03   | 0.61 ± 0.01   | 0.64 ± 0.02   | 0.61 ± 0.01   | 0.64 ± 0.02   | 0.75 ± 0.02   | E Africa F    | E Asia M      | 0.06 |
| 27  | 0.84 ± 0.02   | 0.690 ± 0.007 | 0.792 ± 0.009 | 0.695 ± 0.008 | 0.74 ± 0.01   | 0.86 ± 0.02   | 0.82 ± 0.02   | 0.687 ± 0.009 | 0.74 ± 0.01   | 0.696 ± 0.009 | 0.74 ± 0.02   | 0.82 ± 0.02   | W Africa F    | E Asia M      | 0.05 |
| 28  | 0.89 ± 0.02   | 0.759 ± 0.006 | 0.855 ± 0.009 | 0.744 ± 0.008 | 0.81 ± 0.01   | 0.88 ± 0.02   | 0.89 ± 0.02   | 0.769 ± 0.009 | 0.84 ± 0.01   | 0.779 ± 0.008 | 0.81 ± 0.02   | 0.88 ± 0.02   | E Africa F    | S Asia F      | 0.04 |
| 29  | 0.91 ± 0.02   | 0.816 ± 0.005 | 0.900 ± 0.007 | 0.793 ± 0.007 | 0.85 ± 0.01   | 0.91 ± 0.01   | 0.91 ± 0.02   | 0.838 ± 0.008 | 0.89 ± 0.01   | 0.836 ± 0.006 | 0.86 ± 0.02   | 0.90 ± 0.01   | E Africa M    | S Asia F      | 0.03 |
| 30  | 0.93 ± 0.01   | 0.868 ± 0.005 | 0.932 ± 0.006 | 0.842 ± 0.006 | 0.900 ± 0.010 | 0.93 ± 0.01   | 0.95 ± 0.01   | 0.886 ± 0.007 | 0.936 ± 0.008 | 0.879 ± 0.006 | 0.90 ± 0.01   | 0.93 ± 0.01   | E Africa M    | S Asia F      | 0.02 |
| 31  | 0.94 ± 0.01   | 0.906 ± 0.004 | 0.953 ± 0.005 | 0.881 ± 0.005 | 0.915 ± 0.009 | 0.960 ± 0.009 | 0.97 ± 0.01   | 0.922 ± 0.005 | 0.961 ± 0.006 | 0.916 ± 0.004 | 0.92 ± 0.01   | 0.954 ± 0.009 | E Africa M    | S Asia F      | 0.02 |
| 32  | 0.97 ± 0.01   | 0.930 ± 0.003 | 0.974 ± 0.004 | 0.909 ± 0.005 | 0.938 ± 0.007 | 0.96 ± 0.01   | 0.974 ± 0.009 | 0.947 ± 0.004 | 0.979 ± 0.004 | 0.936 ± 0.004 | 0.95 ± 0.01   | 0.964 ± 0.007 | E Europe M    | S Asia F      | 0.01 |
| 33  | 0.96 ± 0.01   | 0.952 ± 0.003 | 0.982 ± 0.003 | 0.935 ± 0.004 | 0.961 ± 0.006 | 0.973 ± 0.009 | 0.985 ± 0.007 | 0.963 ± 0.004 | 0.984 ± 0.004 | 0.956 ± 0.003 | 0.966 ± 0.008 | 0.977 ± 0.006 | E Africa M    | S Asia F      | 0.01 |
| 34  | 0.980 ± 0.009 | 0.965 ± 0.003 | 0.988 ± 0.003 | 0.953 ± 0.004 | 0.969 ± 0.006 | 0.984 ± 0.007 | 0.989 ± 0.006 | 0.972 ± 0.003 | 0.992 ± 0.003 | 0.971 ± 0.002 | 0.976 ± 0.007 | 0.987 ± 0.004 | E Europe M    | S Asia F      | 0.01 |



**Table 167.** For algorithm roc-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.04 ± 0.02   | 0.034 ± 0.008 | 0.000 ± 0.000 | 0.006 ± 0.004 | 0.04 ± 0.02   | 0.04 ± 0.02   | 0.013 ± 0.008 | 0.030 ± 0.008 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.03 ± 0.02   | 0.004 ± 0.004 | S E Asia F    | E Europe F    | 0.49 |
| 15  | 0.06 ± 0.02   | 0.07 ± 0.02   | 0.003 ± 0.003 | 0.027 ± 0.010 | 0.06 ± 0.02   | 0.08 ± 0.04   | 0.03 ± 0.02   | 0.05 ± 0.01   | 0.000 ± 0.000 | 0.011 ± 0.006 | 0.03 ± 0.01   | 0.02 ± 0.02   | W Africa F    | E Europe M    | 0.43 |
| 16  | 0.11 ± 0.03   | 0.14 ± 0.02   | 0.005 ± 0.005 | 0.03 ± 0.01   | 0.11 ± 0.03   | 0.14 ± 0.04   | 0.05 ± 0.02   | 0.06 ± 0.01   | 0.003 ± 0.003 | 0.018 ± 0.008 | 0.05 ± 0.02   | 0.05 ± 0.03   | W Africa F    | E Europe M    | 0.45 |
| 17  | 0.18 ± 0.04   | 0.17 ± 0.02   | 0.02 ± 0.01   | 0.07 ± 0.01   | 0.13 ± 0.03   | 0.22 ± 0.04   | 0.07 ± 0.02   | 0.12 ± 0.01   | 0.000 ± 0.000 | 0.04 ± 0.01   | 0.11 ± 0.03   | 0.11 ± 0.03   | W Africa F    | E Europe M    | 0.38 |
| 18  | 0.19 ± 0.03   | 0.22 ± 0.02   | 0.05 ± 0.02   | 0.07 ± 0.01   | 0.20 ± 0.03   | 0.26 ± 0.04   | 0.13 ± 0.03   | 0.17 ± 0.02   | 0.007 ± 0.005 | 0.07 ± 0.01   | 0.14 ± 0.03   | 0.15 ± 0.03   | W Africa F    | E Europe M    | 0.33 |
| 19  | 0.26 ± 0.03   | 0.31 ± 0.02   | 0.08 ± 0.02   | 0.12 ± 0.02   | 0.29 ± 0.03   | 0.38 ± 0.04   | 0.25 ± 0.04   | 0.28 ± 0.02   | 0.02 ± 0.01   | 0.11 ± 0.01   | 0.23 ± 0.03   | 0.28 ± 0.04   | W Africa F    | E Europe M    | 0.29 |
| 20  | 0.37 ± 0.04   | 0.40 ± 0.02   | 0.10 ± 0.02   | 0.18 ± 0.02   | 0.38 ± 0.03   | 0.41 ± 0.04   | 0.32 ± 0.04   | 0.39 ± 0.02   | 0.04 ± 0.02   | 0.21 ± 0.02   | 0.35 ± 0.03   | 0.39 ± 0.05   | W Africa F    | E Europe M    | 0.24 |
| 21  | 0.41 ± 0.04   | 0.52 ± 0.02   | 0.15 ± 0.02   | 0.23 ± 0.02   | 0.48 ± 0.02   | 0.50 ± 0.04   | 0.46 ± 0.04   | 0.51 ± 0.02   | 0.06 ± 0.02   | 0.30 ± 0.02   | 0.50 ± 0.03   | 0.51 ± 0.04   | E Asia F      | E Europe M    | 0.23 |
| 22  | 0.51 ± 0.03   | 0.60 ± 0.02   | 0.22 ± 0.02   | 0.28 ± 0.02   | 0.59 ± 0.02   | 0.57 ± 0.04   | 0.63 ± 0.03   | 0.62 ± 0.02   | 0.18 ± 0.03   | 0.42 ± 0.02   | 0.58 ± 0.03   | 0.61 ± 0.04   | E Africa M    | E Europe M    | 0.19 |
| 23  | 0.57 ± 0.03   | 0.70 ± 0.02   | 0.29 ± 0.02   | 0.35 ± 0.02   | 0.68 ± 0.02   | 0.65 ± 0.03   | 0.66 ± 0.03   | 0.74 ± 0.02   | 0.23 ± 0.03   | 0.55 ± 0.02   | 0.72 ± 0.03   | 0.70 ± 0.03   | E Asia M      | E Europe M    | 0.17 |
| 24  | 0.65 ± 0.03   | 0.76 ± 0.01   | 0.39 ± 0.02   | 0.42 ± 0.02   | 0.76 ± 0.02   | 0.70 ± 0.03   | 0.78 ± 0.03   | 0.82 ± 0.02   | 0.41 ± 0.03   | 0.66 ± 0.02   | 0.81 ± 0.02   | 0.79 ± 0.03   | E Asia M      | E Europe F    | 0.14 |
| 25  | 0.69 ± 0.02   | 0.83 ± 0.01   | 0.47 ± 0.02   | 0.47 ± 0.02   | 0.82 ± 0.02   | 0.78 ± 0.02   | 0.83 ± 0.02   | 0.88 ± 0.01   | 0.55 ± 0.03   | 0.73 ± 0.02   | 0.88 ± 0.02   | 0.86 ± 0.02   | E Asia M      | E Europe F    | 0.12 |
| 26  | 0.74 ± 0.02   | 0.872 ± 0.009 | 0.57 ± 0.02   | 0.55 ± 0.01   | 0.87 ± 0.01   | 0.82 ± 0.02   | 0.88 ± 0.02   | 0.94 ± 0.01   | 0.64 ± 0.02   | 0.81 ± 0.01   | 0.92 ± 0.02   | 0.91 ± 0.02   | E Asia M      | S Asia F      | 0.10 |
| 27  | 0.78 ± 0.02   | 0.915 ± 0.008 | 0.67 ± 0.02   | 0.61 ± 0.01   | 0.90 ± 0.01   | 0.86 ± 0.02   | 0.91 ± 0.02   | 0.965 ± 0.007 | 0.77 ± 0.02   | 0.86 ± 0.01   | 0.95 ± 0.01   | 0.93 ± 0.01   | E Asia M      | S Asia F      | 0.08 |
| 28  | 0.83 ± 0.02   | 0.939 ± 0.006 | 0.75 ± 0.02   | 0.67 ± 0.01   | 0.932 ± 0.009 | 0.90 ± 0.02   | 0.94 ± 0.01   | 0.982 ± 0.005 | 0.85 ± 0.02   | 0.903 ± 0.009 | 0.97 ± 0.01   | 0.956 ± 0.010 | E Asia M      | S Asia F      | 0.06 |
| 29  | 0.88 ± 0.02   | 0.968 ± 0.005 | 0.83 ± 0.01   | 0.73 ± 0.01   | 0.949 ± 0.008 | 0.93 ± 0.01   | 0.97 ± 0.01   | 0.989 ± 0.004 | 0.91 ± 0.02   | 0.943 ± 0.006 | 0.980 ± 0.008 | 0.975 ± 0.008 | E Asia M      | S Asia F      | 0.04 |
| 30  | 0.91 ± 0.02   | 0.981 ± 0.004 | 0.88 ± 0.01   | 0.78 ± 0.01   | 0.976 ± 0.006 | 0.93 ± 0.01   | 0.972 ± 0.008 | 0.992 ± 0.003 | 0.95 ± 0.01   | 0.955 ± 0.005 | 0.987 ± 0.006 | 0.984 ± 0.006 | E Asia M      | S Asia F      | 0.03 |
| 31  | 0.92 ± 0.01   | 0.988 ± 0.003 | 0.91 ± 0.01   | 0.825 ± 0.009 | 0.982 ± 0.005 | 0.96 ± 0.01   | 0.981 ± 0.007 | 0.998 ± 0.002 | 0.970 ± 0.008 | 0.974 ± 0.004 | 0.995 ± 0.005 | 0.989 ± 0.005 | E Asia M      | S Asia F      | 0.03 |
| 32  | 0.95 ± 0.01   | 0.991 ± 0.003 | 0.953 ± 0.009 | 0.875 ± 0.009 | 0.986 ± 0.005 | 0.966 ± 0.010 | 0.990 ± 0.006 | 0.999 ± 0.001 | 0.980 ± 0.008 | 0.985 ± 0.003 | 0.995 ± 0.004 | 0.995 ± 0.004 | E Asia M      | S Asia F      | 0.02 |
| 33  | 0.96 ± 0.01   | 0.995 ± 0.002 | 0.963 ± 0.008 | 0.903 ± 0.007 | 0.992 ± 0.004 | 0.978 ± 0.004 | 0.995 ± 0.004 | 0.999 ± 0.001 | 0.994 ± 0.004 | 0.989 ± 0.003 | 0.996 ± 0.004 | 0.996 ± 0.003 | E Asia M      | S Asia F      | 0.01 |
| 34  | 0.979 ± 0.008 | 0.998 ± 0.001 | 0.980 ± 0.006 | 0.933 ± 0.007 | 0.997 ± 0.002 | 0.989 ± 0.006 | 0.995 ± 0.005 | 0.999 ± 0.001 | 0.996 ± 0.004 | 0.993 ± 0.002 | 1.000 ± 0.000 | 0.996 ± 0.004 | S E Asia M    | S Asia F      | 0.01 |

**Table 168.** For algorithm roc-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.14 ± 0.05   | 0.10 ± 0.01   | 0.009 ± 0.006 | 0.03 ± 0.01   | 0.05 ± 0.02   | 0.13 ± 0.05   | 0.12 ± 0.04   | 0.045 ± 0.008 | 0.002 ± 0.002 | 0.007 ± 0.004 | 0.03 ± 0.01   | 0.09 ± 0.04   | E Africa F    | E Europe M    | 0.47 |
| 15  | 0.18 ± 0.05   | 0.15 ± 0.01   | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.08 ± 0.02   | 0.25 ± 0.06   | 0.19 ± 0.06   | 0.09 ± 0.01   | 0.004 ± 0.004 | 0.010 ± 0.005 | 0.06 ± 0.02   | 0.18 ± 0.06   | W Africa F    | E Europe M    | 0.47 |
| 16  | 0.29 ± 0.06   | 0.19 ± 0.01   | 0.03 ± 0.01   | 0.07 ± 0.01   | 0.14 ± 0.02   | 0.27 ± 0.06   | 0.22 ± 0.06   | 0.13 ± 0.01   | 0.002 ± 0.002 | 0.023 ± 0.007 | 0.09 ± 0.02   | 0.13 ± 0.04   | E Africa F    | E Europe M    | 0.43 |
| 17  | 0.38 ± 0.06   | 0.27 ± 0.01   | 0.05 ± 0.02   | 0.10 ± 0.01   | 0.20 ± 0.03   | 0.37 ± 0.06   | 0.25 ± 0.06   | 0.19 ± 0.01   | 0.014 ± 0.008 | 0.07 ± 0.01   | 0.15 ± 0.02   | 0.25 ± 0.05   | E Africa F    | E Europe M    | 0.37 |
| 18  | 0.45 ± 0.05   | 0.35 ± 0.01   | 0.09 ± 0.02   | 0.14 ± 0.02   | 0.27 ± 0.02   | 0.38 ± 0.05   | 0.32 ± 0.05   | 0.26 ± 0.01   | 0.02 ± 0.01   | 0.10 ± 0.01   | 0.22 ± 0.03   | 0.34 ± 0.05   | E Africa F    | E Europe M    | 0.32 |
| 19  | 0.49 ± 0.04   | 0.42 ± 0.01   | 0.12 ± 0.02   | 0.20 ± 0.02   | 0.32 ± 0.02   | 0.47 ± 0.05   | 0.44 ± 0.05   | 0.36 ± 0.01   | 0.03 ± 0.01   | 0.16 ± 0.02   | 0.31 ± 0.02   | 0.42 ± 0.05   | E Africa F    | E Europe M    | 0.28 |
| 20  | 0.58 ± 0.04   | 0.51 ± 0.01   | 0.18 ± 0.02   | 0.24 ± 0.02   | 0.42 ± 0.02   | 0.58 ± 0.04   | 0.58 ± 0.04   | 0.47 ± 0.01   | 0.07 ± 0.01   | 0.26 ± 0.02   | 0.42 ± 0.02   | 0.52 ± 0.04   | E Africa M    | E Europe M    | 0.25 |
| 21  | 0.61 ± 0.04   | 0.58 ± 0.01   | 0.23 ± 0.02   | 0.32 ± 0.01   | 0.49 ± 0.02   | 0.61 ± 0.04   | 0.61 ± 0.04   | 0.59 ± 0.01   | 0.13 ± 0.02   | 0.34 ± 0.02   | 0.55 ± 0.03   | 0.61 ± 0.04   | E Africa F    | E Europe M    | 0.20 |
| 22  | 0.72 ± 0.04   | 0.663 ± 0.010 | 0.31 ± 0.02   | 0.38 ± 0.01   | 0.58 ± 0.02   | 0.70 ± 0.03   | 0.75 ± 0.04   | 0.69 ± 0.01   | 0.22 ± 0.02   | 0.46 ± 0.02   | 0.67 ± 0.03   | 0.73 ± 0.03   | E Africa M    | E Europe M    | 0.18 |
| 23  | 0.76 ± 0.03   | 0.727 ± 0.008 | 0.37 ± 0.02   | 0.44 ± 0.01   | 0.67 ± 0.02   | 0.76 ± 0.02   | 0.80 ± 0.03   | 0.768 ± 0.009 | 0.32 ± 0.02   | 0.57 ± 0.02   | 0.74 ± 0.02   | 0.80 ± 0.02   | W Africa M    | E Europe M    | 0.15 |
| 24  | 0.79 ± 0.02   | 0.781 ± 0.008 | 0.45 ± 0.01   | 0.51 ± 0.01   | 0.74 ± 0.01   | 0.80 ± 0.02   | 0.85 ± 0.02   | 0.831 ± 0.009 | 0.46 ± 0.02   | 0.66 ± 0.01   | 0.81 ± 0.02   | 0.86 ± 0.02   | W Africa M    | E Europe F    | 0.12 |
| 25  | 0.82 ± 0.02   | 0.836 ± 0.007 | 0.53 ± 0.01   | 0.56 ± 0.01   | 0.79 ± 0.01   | 0.84 ± 0.02   | 0.91 ± 0.02   | 0.893 ± 0.007 | 0.57 ± 0.02   | 0.74 ± 0.01   | 0.89 ± 0.02   | 0.90 ± 0.02   | E Africa M    | E Europe F    | 0.10 |
| 26  | 0.86 ± 0.02   | 0.875 ± 0.005 | 0.60 ± 0.01   | 0.632 ± 0.009 | 0.85 ± 0.01   | 0.86 ± 0.02   | 0.93 ± 0.02   | 0.927 ± 0.006 | 0.67 ± 0.02   | 0.797 ± 0.009 | 0.92 ± 0.01   | 0.93 ± 0.01   | E Africa M    | E Europe F    | 0.08 |
| 27  | 0.85 ± 0.02   | 0.905 ± 0.004 | 0.68 ± 0.01   | 0.677 ± 0.009 | 0.875 ± 0.010 | 0.90 ± 0.01   | 0.94 ± 0.01   | 0.948 ± 0.005 | 0.76 ± 0.02   | 0.855 ± 0.007 | 0.949 ± 0.010 | 0.953 ± 0.010 | W Africa M    | S Asia F      | 0.07 |
| 28  | 0.92 ± 0.02   | 0.927 ± 0.004 | 0.75 ± 0.01   | 0.735 ± 0.008 | 0.912 ± 0.009 | 0.91 ± 0.01   | 0.96 ± 0.01   | 0.966 ± 0.004 | 0.84 ± 0.01   | 0.898 ± 0.006 | 0.958 ± 0.010 | 0.969 ± 0.008 | W Africa M    | S Asia F      | 0.05 |
| 29  | 0.93 ± 0.02   | 0.949 ± 0.003 | 0.815 ± 0.009 | 0.773 ± 0.007 | 0.933 ± 0.008 | 0.94 ± 0.01   | 0.982 ± 0.008 | 0.978 ± 0.003 | 0.86 ± 0.01   | 0.922 ± 0.005 | 0.973 ± 0.007 | 0.980 ± 0.006 | E Africa M    | S Asia F      | 0.04 |
| 30  | 0.93 ± 0.01   | 0.961 ± 0.003 | 0.861 ± 0.009 | 0.824 ± 0.006 | 0.952 ± 0.007 | 0.95 ± 0.01   | 0.986 ± 0.007 | 0.984 ± 0.002 | 0.922 ± 0.009 | 0.943 ± 0.004 | 0.983 ± 0.006 | 0.984 ± 0.005 | E Africa M    | S Asia F      | 0.03 |
| 31  | 0.95 ± 0.01   | 0.972 ± 0.002 | 0.892 ± 0.007 | 0.855 ± 0.006 | 0.962 ± 0.006 | 0.970 ± 0.008 | 0.988 ± 0.007 | 0.986 ± 0.002 | 0.949 ± 0.007 | 0.961 ± 0.003 | 0.982 ± 0.007 | 0.989 ± 0.004 | W Africa M    | S Asia F      | 0.02 |
| 32  | 0.970 ± 0.010 | 0.979 ± 0.002 | 0.931 ± 0.007 | 0.884 ± 0.005 | 0.967 ± 0.006 | 0.974 ± 0.008 | 0.991 ± 0.005 | 0.991 ± 0.002 | 0.964 ± 0.006 | 0.971 ± 0.003 | 0.986 ± 0.005 | 0.992 ± 0.004 | W Africa M    | S Asia F      | 0.02 |
| 33  | 0.981 ± 0.008 | 0.983 ± 0.002 | 0.949 ± 0.005 | 0.914 ± 0.005 | 0.977 ± 0.005 | 0.981 ± 0.007 | 0.996 ± 0.004 | 0.994 ± 0.002 | 0.979 ± 0.004 | 0.980 ± 0.002 | 0.990 ± 0.005 | 0.995 ± 0.003 | E Africa M    | S Asia F      | 0.01 |
| 34  | 0.975 ± 0.010 | 0.986 ± 0.002 | 0.967 ± 0.005 | 0.933 ± 0.004 | 0.984 ± 0.004 | 0.989 ± 0.005 | 0.999 ± 0.002 | 0.994 ± 0.001 | 0.983 ± 0.004 | 0.985 ± 0.002 | 0.995 ± 0.004 | 0.996 ± 0.002 | E Africa M    | S Asia F      | 0.01 |

**Table 169.** For algorithm roc-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.011 ± 0.009 | 0.005 ± 0.003 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.004 ± 0.004 | 0.02 ± 0.01   | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.004 ± 0.004 | W Africa F    | E Europe F    | 0.54 |
| 15  | 0.04 ± 0.02   | 0.016 ± 0.006 | 0.000 ± 0.000 | 0.023 ± 0.010 | 0.03 ± 0.01   | 0.07 ± 0.03   | 0.010 ± 0.008 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.002 ± 0.002 | 0.005 ± 0.005 | W Africa F    | E Europe F    | 0.64 |
| 16  | 0.09 ± 0.03   | 0.04 ± 0.01   | 0.000 ± 0.000 | 0.02 ± 0.01   | 0.05 ± 0.02   | 0.09 ± 0.03   | 0.03 ± 0.01   | 0.014 ± 0.005 | 0.003 ± 0.003 | 0.004 ± 0.003 | 0.01 ± 0.01   | 0.01 ± 0.01   | E Africa F    | E Europe F    | 0.57 |
| 17  | 0.14 ± 0.03   | 0.10 ± 0.01   | 0.003 ± 0.003 | 0.07 ± 0.01   | 0.10 ± 0.03   | 0.18 ± 0.04   | 0.05 ± 0.02   | 0.042 ± 0.009 | 0.003 ± 0.003 | 0.019 ± 0.008 | 0.03 ± 0.02   | 0.08 ± 0.03   | W Africa F    | E Europe F    | 0.47 |
| 18  | 0.19 ± 0.03   | 0.15 ± 0.02   | 0.02 ± 0.01   | 0.10 ± 0.01   | 0.14 ± 0.02   | 0.23 ± 0.04   | 0.09 ± 0.02   | 0.09 ± 0.01   | 0.002 ± 0.002 | 0.04 ± 0.01   | 0.06 ± 0.02   | 0.13 ± 0.03   | W Africa F    | E Europe M    | 0.39 |
| 19  | 0.32 ± 0.04   | 0.25 ± 0.02   | 0.04 ± 0.01   | 0.17 ± 0.02   | 0.25 ± 0.03   | 0.36 ± 0.04   | 0.19 ± 0.03   | 0.18 ± 0.02   | 0.013 ± 0.008 | 0.10 ± 0.01   | 0.13 ± 0.02   | 0.24 ± 0.04   | W Africa F    | E Europe M    | 0.33 |
| 20  | 0.44 ± 0.04   | 0.35 ± 0.02   | 0.08 ± 0.02   | 0.26 ± 0.02   | 0.37 ± 0.03   | 0.43 ± 0.04   | 0.30 ± 0.04   | 0.31 ± 0.02   | 0.03 ± 0.01   | 0.18 ± 0.02   | 0.25 ± 0.03   | 0.32 ± 0.04   | E Africa F    | E Europe M    | 0.26 |
| 21  | 0.54 ± 0.04   | 0.48 ± 0.02   | 0.14 ± 0.02   | 0.35 ± 0.02   | 0.48 ± 0.03   | 0.51 ± 0.04   | 0.45 ± 0.04   | 0.48 ± 0.02   | 0.08 ± 0.02   | 0.29 ± 0.02   | 0.41 ± 0.03   | 0.45 ± 0.04   | E Africa F    | E Europe M    | 0.21 |
| 22  | 0.64 ± 0.03   | 0.61 ± 0.02   | 0.23 ± 0.02   | 0.48 ± 0.02   | 0.60 ± 0.02   | 0.61 ± 0.04   | 0.61 ± 0.03   | 0.62 ± 0.02   | 0.14 ± 0.03   | 0.43 ± 0.02   | 0.56 ± 0.03   | 0.57 ± 0.04   | E Africa F    | E Europe M    | 0.17 |
| 23  | 0.72 ± 0.03   | 0.72 ± 0.02   | 0.35 ± 0.02   | 0.60 ± 0.02   | 0.72 ± 0.02   | 0.69 ± 0.03   | 0.68 ± 0.03   | 0.76 ± 0.02   | 0.29 ± 0.03   | 0.58 ± 0.02   | 0.72 ± 0.03   | 0.72 ± 0.03   | E Asia M      | E Europe M    | 0.12 |
| 24  | 0.78 ± 0.03   | 0.80 ± 0.01   | 0.49 ± 0.02   | 0.68 ± 0.02   | 0.80 ± 0.02   | 0.76 ± 0.03   | 0.81 ± 0.02   | 0.85 ± 0.01   | 0.47 ± 0.03   | 0.70 ± 0.02   | 0.81 ± 0.02   | 0.80 ± 0.03   | E Asia M      | E Europe M    | 0.09 |
| 25  | 0.82 ± 0.02   | 0.87 ± 0.01   | 0.61 ± 0.02   | 0.76 ± 0.01   | 0.87 ± 0.01   | 0.85 ± 0.02   | 0.87 ± 0.02   | 0.92 ± 0.01   | 0.61 ± 0.03   | 0.81 ± 0.01   | 0.88 ± 0.02   | 0.87 ± 0.02   | E Asia M      | E Europe F    | 0.07 |
| 26  | 0.89 ± 0.02   | 0.914 ± 0.008 | 0.74 ± 0.02   | 0.820 ± 0.010 | 0.92 ± 0.01   | 0.89 ± 0.02   | 0.91 ± 0.02   | 0.956 ± 0.009 | 0.77 ± 0.03   | 0.88 ± 0.01   | 0.94 ± 0.01   | 0.92 ± 0.02   | E Asia M      | E Europe F    | 0.04 |
| 27  | 0.92 ± 0.01   | 0.953 ± 0.006 | 0.83 ± 0.01   | 0.872 ± 0.009 | 0.945 ± 0.009 | 0.92 ± 0.01   | 0.95 ± 0.01   | 0.976 ± 0.006 | 0.89 ± 0.02   | 0.927 ± 0.009 | 0.970 ± 0.010 | 0.95 ± 0.01   | E Asia M      | E Europe F    | 0.03 |
| 28  | 0.95 ± 0.01   | 0.972 ± 0.004 | 0.91 ± 0.01   | 0.919 ± 0.007 | 0.971 ± 0.006 | 0.96 ± 0.01   | 0.972 ± 0.009 | 0.989 ± 0.004 | 0.93 ± 0.01   | 0.958 ± 0.006 | 0.986 ± 0.008 | 0.965 ± 0.009 | E Asia M      | E Europe F    | 0.01 |
| 29  | 0.97 ± 0.01   | 0.989 ± 0.003 | 0.947 ± 0.008 | 0.947 ± 0.005 | 0.987 ± 0.004 | 0.96 ± 0.01   | 0.981 ± 0.007 | 0.996 ± 0.002 | 0.97 ± 0.01   | 0.978 ± 0.004 | 0.992 ± 0.005 | 0.985 ± 0.006 | E Asia M      | E Europe F    | 0.01 |
| 30  | 0.978 ± 0.008 | 0.995 ± 0.002 | 0.974 ± 0.006 | 0.969 ± 0.004 | 0.992 ± 0.001 | 0.978 ± 0.008 | 0.990 ± 0.006 | 0.999 ± 0.001 | 0.992 ± 0.005 | 0.988 ± 0.003 | 0.993 ± 0.005 | 0.992 ± 0.005 | E Asia M      | S Asia F      | 0.01 |
| 31  | 0.990 ± 0.005 | 0.997 ± 0.002 | 0.991 ± 0.004 | 0.980 ± 0.004 | 0.994 ± 0.003 | 0.984 ± 0.006 | 0.995 ± 0.004 | 0.999 ± 0.001 | 0.995 ± 0.003 | 0.994 ± 0.002 | 0.997 ± 0.004 | 0.995 ± 0.003 | E Asia M      | S Asia F      | 0.00 |
| 32  | 0.990 ± 0.006 | 0.999 ± 0.001 | 0.996 ± 0.002 | 0.990 ± 0.002 | 0.997 ± 0.002 | 0.988 ± 0.005 | 0.998 ± 0.003 | 0.999 ± 0.001 | 0.997 ± 0.003 | 0.997 ± 0.001 | 0.998 ± 0.003 | 0.998 ± 0.003 | E Asia M      | W Africa F    | 0.00 |
| 33  | 0.997 ± 0.003 | 0.999 ± 0.001 | 0.996 ± 0.003 | 0.993 ± 0.002 | 0.999 ± 0.001 | 0.994 ± 0.004 | 1.000 ± 0.001 | 0.999 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.996 ± 0.003 | E Africa M    | S Asia F      | 0.00 |
| 34  | 0.999 ± 0.002 | 1.000 ± 0.000 | 1.000 ± 0.001 | 0.997 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.999 ± 0.002 | 1.000 ± 0.001 | 0.999 ± 0.002 | 0.999 ± 0.001 | 1.000 ± 0.000 | 1.000 ± 0.000 | E Asia F      | S Asia F      | 0.00 |

**Table 170.** For algorithm roc-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.11 ± 0.05   | 0.009 ± 0.004 | 0.000 ± 0.000 | 0.017 ± 0.007 | 0.011 ± 0.009 | 0.07 ± 0.03   | 0.03 ± 0.02   | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.008 ± 0.006 | 0.04 ± 0.03   | E Africa F    | E Europe F    | 0.68 |
| 15  | 0.16 ± 0.05   | 0.022 ± 0.006 | 0.006 ± 0.006 | 0.028 ± 0.010 | 0.02 ± 0.01   | 0.14 ± 0.04   | 0.000 ± 0.000 | 0.010 ± 0.003 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.009 ± 0.007 | 0.07 ± 0.04   | E Africa F    | E Africa M    | 0.70 |
| 16  | 0.23 ± 0.06   | 0.056 ± 0.007 | 0.006 ± 0.005 | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.22 ± 0.06   | 0.08 ± 0.04   | 0.024 ± 0.005 | 0.000 ± 0.000 | 0.009 ± 0.004 | 0.03 ± 0.01   | 0.09 ± 0.04   | E Africa F    | E Europe M    | 0.57 |
| 17  | 0.32 ± 0.05   | 0.11 ± 0.01   | 0.03 ± 0.01   | 0.10 ± 0.02   | 0.09 ± 0.02   | 0.29 ± 0.05   | 0.15 ± 0.05   | 0.073 ± 0.008 | 0.005 ± 0.005 | 0.034 ± 0.008 | 0.07 ± 0.02   | 0.18 ± 0.04   | E Africa F    | E Europe M    | 0.46 |
| 18  | 0.43 ± 0.05   | 0.17 ± 0.01   | 0.07 ± 0.01   | 0.17 ± 0.02   | 0.15 ± 0.02   | 0.34 ± 0.05   | 0.26 ± 0.05   | 0.137 ± 0.009 | 0.015 ± 0.009 | 0.056 ± 0.009 | 0.10 ± 0.02   | 0.26 ± 0.04   | E Africa F    | E Europe M    | 0.39 |
| 19  | 0.52 ± 0.04   | 0.27 ± 0.01   | 0.10 ± 0.02   | 0.25 ± 0.02   | 0.26 ± 0.02   | 0.47 ± 0.04   | 0.36 ± 0.05   | 0.24 ± 0.01   | 0.02 ± 0.01   | 0.14 ± 0.01   | 0.20 ± 0.02   | 0.34 ± 0.05   | E Africa F    | E Europe M    | 0.32 |
| 20  | 0.63 ± 0.05   | 0.38 ± 0.01   | 0.17 ± 0.02   | 0.34 ± 0.02   | 0.39 ± 0.02   | 0.60 ± 0.04   | 0.48 ± 0.04   | 0.38 ± 0.01   | 0.05 ± 0.01   | 0.21 ± 0.01   | 0.37 ± 0.03   | 0.48 ± 0.04   | E Africa F    | E Europe M    | 0.26 |
| 21  | 0.71 ± 0.04   | 0.49 ± 0.01   | 0.25 ± 0.02   | 0.43 ± 0.02   | 0.49 ± 0.02   | 0.67 ± 0.04   | 0.61 ± 0.04   | 0.52 ± 0.01   | 0.16 ± 0.02   | 0.32 ± 0.02   | 0.50 ± 0.03   | 0.63 ± 0.04   | E Africa F    | E Europe M    | 0.21 |
| 22  | 0.80 ± 0.03   | 0.603 ± 0.010 | 0.36 ± 0.02   | 0.53 ± 0.01   | 0.60 ± 0.02   | 0.77 ± 0.03   | 0.71 ± 0.04   | 0.65 ± 0.01   | 0.27 ± 0.02   | 0.46 ± 0.01   | 0.62 ± 0.03   | 0.73 ± 0.03   | E Africa F    | E Europe M    | 0.16 |
| 23  | 0.83 ± 0.03   | 0.693 ± 0.008 | 0.46 ± 0.02   | 0.61 ± 0.01   | 0.71 ± 0.01   | 0.81 ± 0.02   | 0.78 ± 0.03   | 0.77 ± 0.01   | 0.44 ± 0.02   | 0.59 ± 0.01   | 0.74 ± 0.02   | 0.82 ± 0.02   | E Africa F    | E Europe M    | 0.11 |
| 24  | 0.88 ± 0.02   | 0.775 ± 0.008 | 0.58 ± 0.01   | 0.71 ± 0.01   | 0.79 ± 0.01   | 0.86 ± 0.02   | 0.87 ± 0.03   | 0.847 ± 0.009 | 0.57 ± 0.02   | 0.70 ± 0.01   | 0.82 ± 0.02   | 0.88 ± 0.02   | W Africa M    | E Europe M    | 0.08 |
| 25  | 0.91 ± 0.02   | 0.843 ± 0.007 | 0.69 ± 0.01   | 0.765 ± 0.009 | 0.85 ± 0.01   | 0.91 ± 0.01   | 0.92 ± 0.02   | 0.912 ± 0.007 | 0.71 ± 0.02   | 0.803 ± 0.009 | 0.89 ± 0.02   | 0.92 ± 0.01   | W Africa M    | E Europe F    | 0.06 |
| 26  | 0.94 ± 0.01   | 0.897 ± 0.005 | 0.78 ± 0.01   | 0.826 ± 0.007 | 0.902 ± 0.009 | 0.92 ± 0.01   | 0.95 ± 0.01   | 0.942 ± 0.005 | 0.81 ± 0.02   | 0.866 ± 0.008 | 0.94 ± 0.01   | 0.95 ± 0.01   | E Africa M    | E Europe F    | 0.04 |
| 27  | 0.95 ± 0.01   | 0.932 ± 0.003 | 0.860 ± 0.008 | 0.868 ± 0.006 | 0.939 ± 0.007 | 0.957 ± 0.009 | 0.97 ± 0.01   | 0.971 ± 0.004 | 0.89 ± 0.01   | 0.917 ± 0.006 | 0.964 ± 0.009 | 0.976 ± 0.007 | W Africa M    | E Europe F    | 0.03 |
| 28  | 0.97 ± 0.01   | 0.956 ± 0.003 | 0.912 ± 0.007 | 0.905 ± 0.005 | 0.960 ± 0.006 | 0.963 ± 0.008 | 0.982 ± 0.008 | 0.982 ± 0.003 | 0.945 ± 0.008 | 0.949 ± 0.005 | 0.974 ± 0.008 | 0.980 ± 0.007 | E Asia M      | S Asia F      | 0.01 |
| 29  | 0.977 ± 0.008 | 0.976 ± 0.002 | 0.948 ± 0.005 | 0.936 ± 0.004 | 0.975 ± 0.005 | 0.979 ± 0.007 | 0.992 ± 0.005 | 0.990 ± 0.002 | 0.959 ± 0.007 | 0.970 ± 0.003 | 0.987 ± 0.005 | 0.988 ± 0.005 | E Africa M    | S Asia F      | 0.01 |
| 30  | 0.981 ± 0.007 | 0.983 ± 0.002 | 0.970 ± 0.004 | 0.955 ± 0.003 | 0.987 ± 0.003 | 0.985 ± 0.005 | 0.996 ± 0.004 | 0.994 ± 0.002 | 0.982 ± 0.004 | 0.982 ± 0.002 | 0.992 ± 0.004 | 0.992 ± 0.004 | E Africa M    | S Asia F      | 0.01 |
| 31  | 0.995 ± 0.004 | 0.990 ± 0.001 | 0.982 ± 0.003 | 0.971 ± 0.003 | 0.994 ± 0.002 | 0.988 ± 0.005 | 0.995 ± 0.004 | 0.996 ± 0.001 | 0.992 ± 0.003 | 0.990 ± 0.002 | 0.997 ± 0.003 | 0.995 ± 0.003 | S E Asia M    | S Asia F      | 0.00 |
| 32  | 0.991 ± 0.006 | 0.995 ± 0.001 | 0.993 ± 0.002 | 0.981 ± 0.003 | 0.995 ± 0.002 | 0.994 ± 0.004 | 0.995 ± 0.004 | 0.999 ± 0.001 | 0.995 ± 0.002 | 0.993 ± 0.001 | 0.997 ± 0.003 | 0.998 ± 0.002 | E Asia M      | S Asia F      | 0.00 |
| 33  | 0.997 ± 0.003 | 0.996 ± 0.001 | 0.996 ± 0.002 | 0.988 ± 0.002 | 0.995 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.997 ± 0.002 | 0.996 ± 0.001 | 0.999 ± 0.001 | 0.999 ± 0.001 | S E Asia M    | S Asia F      | 0.00 |
| 34  | 0.998 ± 0.003 | 0.997 ± 0.001 | 0.998 ± 0.001 | 0.993 ± 0.001 | 0.998 ± 0.002 | 0.998 ± 0.003 | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.000 ± 0.001 | 0.998 ± 0.001 | 0.999 ± 0.002 | 1.000 ± 0.000 | E Africa M    | S Asia F      | 0.00 |

**Table 171.** For algorithm roc-002 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.004 ± 0.004 | 0.004 ± 0.004 | 0.002 ± 0.002 | 0.003 ± 0.002 | 0.000 ± 0.000 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.004 ± 0.004 | S E Asia F    | E Africa F    | 0.58 |
| 15  | 0.012 ± 0.009 | 0.008 ± 0.005 | 0.000 ± 0.000 | 0.006 ± 0.004 | 0.01 ± 0.01   | 0.009 ± 0.009 | 0.005 ± 0.005 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.005 ± 0.005 | S E Asia F    | E Europe F    | 0.56 |
| 16  | 0.03 ± 0.02   | 0.019 ± 0.007 | 0.000 ± 0.000 | 0.005 ± 0.004 | 0.04 ± 0.02   | 0.04 ± 0.02   | 0.02 ± 0.01   | 0.004 ± 0.003 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.008 ± 0.008 | 0.005 ± 0.005 | W Africa F    | E Europe F    | 0.57 |
| 17  | 0.05 ± 0.02   | 0.040 ± 0.009 | 0.008 ± 0.008 | 0.025 ± 0.009 | 0.06 ± 0.02   | 0.05 ± 0.02   | 0.012 ± 0.009 | 0.014 ± 0.005 | 0.000 ± 0.000 | 0.010 ± 0.006 | 0.02 ± 0.01   | 0.04 ± 0.02   | S E Asia F    | E Europe M    | 0.42 |
| 18  | 0.08 ± 0.02   | 0.07 ± 0.01   | 0.007 ± 0.005 | 0.030 ± 0.008 | 0.07 ± 0.02   | 0.06 ± 0.02   | 0.03 ± 0.01   | 0.037 ± 0.007 | 0.002 ± 0.002 | 0.023 ± 0.006 | 0.022 ± 0.010 | 0.05 ± 0.02   | E Africa F    | E Europe M    | 0.39 |
| 19  | 0.15 ± 0.03   | 0.12 ± 0.01   | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.10 ± 0.02   | 0.13 ± 0.03   | 0.11 ± 0.03   | 0.08 ± 0.01   | 0.011 ± 0.008 | 0.045 ± 0.010 | 0.06 ± 0.02   | 0.11 ± 0.03   | E Africa F    | E Europe M    | 0.31 |
| 20  | 0.24 ± 0.03   | 0.22 ± 0.02   | 0.03 ± 0.01   | 0.08 ± 0.01   | 0.20 ± 0.02   | 0.21 ± 0.03   | 0.17 ± 0.03   | 0.15 ± 0.02   | 0.03 ± 0.01   | 0.09 ± 0.01   | 0.11 ± 0.02   | 0.15 ± 0.03   | E Africa F    | E Europe F    | 0.30 |
| 21  | 0.31 ± 0.03   | 0.32 ± 0.02   | 0.07 ± 0.02   | 0.16 ± 0.02   | 0.29 ± 0.02   | 0.24 ± 0.04   | 0.28 ± 0.03   | 0.26 ± 0.02   | 0.06 ± 0.02   | 0.16 ± 0.02   | 0.18 ± 0.02   | 0.25 ± 0.04   | E Asia F      | E Europe M    | 0.24 |
| 22  | 0.41 ± 0.03   | 0.44 ± 0.02   | 0.13 ± 0.02   | 0.24 ± 0.02   | 0.39 ± 0.02   | 0.35 ± 0.03   | 0.42 ± 0.03   | 0.37 ± 0.02   | 0.14 ± 0.03   | 0.27 ± 0.02   | 0.28 ± 0.03   | 0.31 ± 0.04   | E Asia F      | E Europe F    | 0.20 |
| 23  | 0.49 ± 0.03   | 0.55 ± 0.02   | 0.24 ± 0.02   | 0.34 ± 0.02   | 0.51 ± 0.02   | 0.45 ± 0.03   | 0.49 ± 0.03   | 0.50 ± 0.02   | 0.23 ± 0.03   | 0.42 ± 0.02   | 0.43 ± 0.03   | 0.46 ± 0.03   | E Asia F      | E Europe M    | 0.14 |
| 24  | 0.59 ± 0.03   | 0.67 ± 0.02   | 0.35 ± 0.02   | 0.44 ± 0.02   | 0.62 ± 0.02   | 0.53 ± 0.03   | 0.61 ± 0.03   | 0.63 ± 0.02   | 0.39 ± 0.03   | 0.56 ± 0.02   | 0.55 ± 0.03   | 0.57 ± 0.03   | E Asia F      | E Europe F    | 0.11 |
| 25  | 0.62 ± 0.03   | 0.76 ± 0.01   | 0.48 ± 0.02   | 0.54 ± 0.01   | 0.70 ± 0.02   | 0.61 ± 0.03   | 0.73 ± 0.03   | 0.76 ± 0.02   | 0.54 ± 0.03   | 0.69 ± 0.02   | 0.71 ± 0.03   | 0.69 ± 0.03   | E Asia F      | E Europe F    | 0.08 |
| 26  | 0.72 ± 0.03   | 0.83 ± 0.01   | 0.60 ± 0.02   | 0.62 ± 0.01   | 0.80 ± 0.02   | 0.70 ± 0.02   | 0.80 ± 0.02   | 0.85 ± 0.01   | 0.69 ± 0.02   | 0.80 ± 0.01   | 0.79 ± 0.02   | 0.77 ± 0.02   | E Asia M      | E Europe F    | 0.06 |
| 27  | 0.78 ± 0.02   | 0.892 ± 0.009 | 0.72 ± 0.02   | 0.70 ± 0.01   | 0.86 ± 0.01   | 0.76 ± 0.02   | 0.85 ± 0.02   | 0.91 ± 0.01   | 0.81 ± 0.02   | 0.86 ± 0.01   | 0.88 ± 0.02   | 0.84 ± 0.02   | E Asia M      | S Asia F      | 0.05 |
| 28  | 0.84 ± 0.02   | 0.930 ± 0.007 | 0.82 ± 0.01   | 0.78 ± 0.01   | 0.90 ± 0.01   | 0.83 ± 0.02   | 0.91 ± 0.02   | 0.954 ± 0.008 | 0.88 ± 0.02   | 0.922 ± 0.008 | 0.94 ± 0.01   | 0.89 ± 0.02   | E Asia M      | S Asia F      | 0.04 |
| 29  | 0.89 ± 0.02   | 0.958 ± 0.005 | 0.89 ± 0.01   | 0.843 ± 0.009 | 0.941 ± 0.008 | 0.89 ± 0.02   | 0.95 ± 0.01   | 0.978 ± 0.005 | 0.95 ± 0.01   | 0.949 ± 0.006 | 0.95 ± 0.01   | 0.94 ± 0.01   | E Asia M      | S Asia F      | 0.02 |
| 30  | 0.93 ± 0.01   | 0.978 ± 0.004 | 0.942 ± 0.008 | 0.890 ± 0.008 | 0.963 ± 0.008 | 0.91 ± 0.01   | 0.970 ± 0.010 | 0.987 ± 0.004 | 0.980 ± 0.007 | 0.970 ± 0.004 | 0.983 ± 0.009 | 0.958 ± 0.009 | E Asia M      | S Asia F      | 0.02 |
| 31  | 0.95 ± 0.01   | 0.985 ± 0.003 | 0.973 ± 0.006 | 0.922 ± 0.007 | 0.984 ± 0.005 | 0.94 ± 0.01   | 0.975 ± 0.009 | 0.996 ± 0.002 | 0.983 ± 0.007 | 0.985 ± 0.003 | 0.984 ± 0.008 | 0.977 ± 0.008 | E Asia M      | S Asia F      | 0.01 |
| 32  | 0.96 ± 0.01   | 0.993 ± 0.002 | 0.988 ± 0.005 | 0.954 ± 0.005 | 0.986 ± 0.005 | 0.96 ± 0.01   | 0.987 ± 0.006 | 0.997 ± 0.002 | 0.993 ± 0.005 | 0.992 ± 0.002 | 0.994 ± 0.005 | 0.983 ± 0.007 | E Asia M      | S Asia F      | 0.01 |
| 33  | 0.977 ± 0.009 | 0.996 ± 0.002 | 0.991 ± 0.004 | 0.972 ± 0.004 | 0.992 ± 0.004 | 0.978 ± 0.008 | 0.995 ± 0.004 | 0.998 ± 0.002 | 0.998 ± 0.003 | 0.996 ± 0.001 | 0.996 ± 0.004 | 0.991 ± 0.005 | E Asia M      | S Asia F      | 0.00 |
| 34  | 0.983 ± 0.008 | 0.999 ± 0.001 | 0.997 ± 0.002 | 0.983 ± 0.004 | 0.998 ± 0.002 | 0.985 ± 0.007 | 0.993 ± 0.005 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.998 ± 0.001 | 0.998 ± 0.003 | 0.994 ± 0.004 | E Asia F      | S Asia F      | 0.00 |

**Table 172.** For algorithm roc-002 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.05 ± 0.03   | 0.010 ± 0.004 | 0.000 ± 0.000 | 0.007 ± 0.005 | 0.005 ± 0.005 | 0.03 ± 0.03   | 0.02 ± 0.02   | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.000 ± 0.000 | 0.01 ± 0.01   | E Africa F    | E Europe F    | 0.70 |
| 15  | 0.06 ± 0.03   | 0.023 ± 0.006 | 0.004 ± 0.004 | 0.011 ± 0.005 | 0.018 ± 0.009 | 0.10 ± 0.04   | 0.006 ± 0.006 | 0.012 ± 0.004 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.66 |
| 16  | 0.13 ± 0.05   | 0.054 ± 0.008 | 0.006 ± 0.005 | 0.04 ± 0.01   | 0.04 ± 0.01   | 0.11 ± 0.04   | 0.03 ± 0.02   | 0.028 ± 0.005 | 0.002 ± 0.002 | 0.005 ± 0.003 | 0.016 ± 0.009 | 0.06 ± 0.03   | E Africa F    | E Europe M    | 0.53 |
| 17  | 0.17 ± 0.04   | 0.108 ± 0.010 | 0.017 ± 0.009 | 0.06 ± 0.01   | 0.08 ± 0.02   | 0.18 ± 0.05   | 0.08 ± 0.03   | 0.067 ± 0.008 | 0.002 ± 0.002 | 0.028 ± 0.008 | 0.04 ± 0.01   | 0.15 ± 0.04   | W Africa F    | E Europe M    | 0.43 |
| 18  | 0.25 ± 0.05   | 0.17 ± 0.01   | 0.06 ± 0.01   | 0.09 ± 0.01   | 0.11 ± 0.02   | 0.22 ± 0.04   | 0.14 ± 0.04   | 0.129 ± 0.010 | 0.014 ± 0.008 | 0.044 ± 0.009 | 0.06 ± 0.01   | 0.17 ± 0.04   | E Africa F    | E Europe M    | 0.36 |
| 19  | 0.32 ± 0.04   | 0.26 ± 0.01   | 0.09 ± 0.01   | 0.16 ± 0.01   | 0.18 ± 0.02   | 0.32 ± 0.04   | 0.22 ± 0.04   | 0.222 ± 0.010 | 0.03 ± 0.01   | 0.10 ± 0.01   | 0.14 ± 0.02   | 0.24 ± 0.04   | E Africa F    | E Europe M    | 0.29 |
| 20  | 0.43 ± 0.04   | 0.37 ± 0.01   | 0.14 ± 0.01   | 0.21 ± 0.01   | 0.29 ± 0.02   | 0.43 ± 0.04   | 0.35 ± 0.04   | 0.36 ± 0.01   | 0.05 ± 0.01   | 0.15 ± 0.01   | 0.27 ± 0.02   | 0.35 ± 0.04   | E Africa F    | E Europe M    | 0.25 |
| 21  | 0.53 ± 0.04   | 0.45 ± 0.01   | 0.20 ± 0.02   | 0.30 ± 0.01   | 0.36 ± 0.02   | 0.52 ± 0.04   | 0.44 ± 0.04   | 0.47 ± 0.01   | 0.12 ± 0.02   | 0.24 ± 0.01   | 0.36 ± 0.02   | 0.47 ± 0.04   | E Africa F    | E Europe M    | 0.21 |
| 22  | 0.61 ± 0.04   | 0.557 ± 0.009 | 0.31 ± 0.02   | 0.38 ± 0.01   | 0.49 ± 0.02   | 0.61 ± 0.04   | 0.55 ± 0.04   | 0.61 ± 0.01   | 0.21 ± 0.02   | 0.37 ± 0.02   | 0.50 ± 0.02   | 0.60 ± 0.04   | E Africa F    | E Europe M    | 0.16 |
| 23  | 0.66 ± 0.03   | 0.643 ± 0.009 | 0.40 ± 0.02   | 0.46 ± 0.01   | 0.59 ± 0.02   | 0.66 ± 0.03   | 0.65 ± 0.04   | 0.72 ± 0.01   | 0.35 ± 0.02   | 0.50 ± 0.01   | 0.59 ± 0.03   | 0.69 ± 0.03   | E Asia M      | E Europe M    | 0.12 |
| 24  | 0.76 ± 0.03   | 0.731 ± 0.008 | 0.52 ± 0.01   | 0.56 ± 0.01   | 0.68 ± 0.01   | 0.77 ± 0.03   | 0.77 ± 0.03   | 0.800 ± 0.009 | 0.48 ± 0.02   | 0.64 ± 0.01   | 0.70 ± 0.02   | 0.79 ± 0.03   | E Asia M      | E Europe M    | 0.09 |
| 25  | 0.78 ± 0.03   | 0.797 ± 0.007 | 0.63 ± 0.01   | 0.623 ± 0.010 | 0.74 ± 0.01   | 0.82 ± 0.02   | 0.82 ± 0.03   | 0.875 ± 0.007 | 0.61 ± 0.02   | 0.74 ± 0.01   | 0.80 ± 0.02   | 0.86 ± 0.02   | E Asia M      | E Europe M    | 0.07 |
| 26  | 0.83 ± 0.02   | 0.860 ± 0.005 | 0.71 ± 0.01   | 0.707 ± 0.009 | 0.82 ± 0.01   | 0.85 ± 0.02   | 0.89 ± 0.02   | 0.923 ± 0.006 | 0.73 ± 0.02   | 0.814 ± 0.009 | 0.88 ± 0.02   | 0.89 ± 0.02   | E Asia M      | S Asia F      | 0.05 |
| 27  | 0.88 ± 0.02   | 0.897 ± 0.005 | 0.801 ± 0.009 | 0.760 ± 0.008 | 0.872 ± 0.010 | 0.91 ± 0.01   | 0.92 ± 0.02   | 0.949 ± 0.005 | 0.83 ± 0.01   | 0.883 ± 0.006 | 0.91 ± 0.01   | 0.94 ± 0.01   | E Asia M      | S Asia F      | 0.04 |
| 28  | 0.93 ± 0.01   | 0.930 ± 0.003 | 0.872 ± 0.008 | 0.824 ± 0.007 | 0.905 ± 0.010 | 0.92 ± 0.01   | 0.96 ± 0.01   | 0.972 ± 0.003 | 0.903 ± 0.010 | 0.929 ± 0.005 | 0.966 ± 0.008 | 0.955 ± 0.010 | E Asia M      | S Asia F      | 0.03 |
| 29  | 0.94 ± 0.01   | 0.957 ± 0.003 | 0.920 ± 0.007 | 0.866 ± 0.006 | 0.935 ± 0.008 | 0.959 ± 0.010 | 0.975 ± 0.010 | 0.986 ± 0.003 | 0.927 ± 0.009 | 0.955 ± 0.004 | 0.973 ± 0.007 | 0.974 ± 0.007 | E Asia M      | S Asia F      | 0.02 |
| 30  | 0.95 ± 0.01   | 0.969 ± 0.003 | 0.949 ± 0.003 | 0.902 ± 0.005 | 0.960 ± 0.006 | 0.965 ± 0.009 | 0.983 ± 0.007 | 0.989 ± 0.002 | 0.966 ± 0.006 | 0.973 ± 0.003 | 0.988 ± 0.005 | 0.981 ± 0.006 | E Asia M      | S Asia F      | 0.01 |
| 31  | 0.966 ± 0.010 | 0.981 ± 0.002 | 0.970 ± 0.004 | 0.930 ± 0.004 | 0.973 ± 0.005 | 0.980 ± 0.007 | 0.990 ± 0.006 | 0.995 ± 0.001 | 0.980 ± 0.004 | 0.984 ± 0.002 | 0.991 ± 0.005 | 0.988 ± 0.005 | E Asia M      | S Asia F      | 0.01 |
| 32  | 0.976 ± 0.008 | 0.988 ± 0.002 | 0.985 ± 0.003 | 0.954 ± 0.004 | 0.985 ± 0.004 | 0.985 ± 0.006 | 0.996 ± 0.004 | 0.998 ± 0.001 | 0.988 ± 0.004 | 0.991 ± 0.001 | 0.994 ± 0.004 | 0.994 ± 0.003 | E Asia M      | S Asia F      | 0.01 |
| 33  | 0.990 ± 0.007 | 0.993 ± 0.001 | 0.990 ± 0.002 | 0.971 ± 0.003 | 0.987 ± 0.004 | 0.993 ± 0.004 | 0.998 ± 0.003 | 0.999 ± 0.000 | 0.995 ± 0.002 | 0.994 ± 0.001 | 0.998 ± 0.002 | 0.998 ± 0.002 | E Asia M      | S Asia F      | 0.00 |
| 34  | 0.992 ± 0.005 | 0.996 ± 0.001 | 0.994 ± 0.002 | 0.980 ± 0.002 | 0.992 ± 0.003 | 0.995 ± 0.004 | 0.998 ± 0.003 | 0.999 ± 0.000 | 0.999 ± 0.001 | 0.997 ± 0.001 | 0.998 ± 0.002 | 0.998 ± 0.002 | E Asia M      | S Asia F      | 0.00 |

**Table 173.** For algorithm shufti-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.06 ± 0.02   | 0.021 ± 0.007 | 0.02 ± 0.01   | 0.016 ± 0.006 | 0.02 ± 0.01   | 0.05 ± 0.02   | 0.04 ± 0.02   | 0.016 ± 0.006 | 0.002 ± 0.002 | 0.007 ± 0.004 | 0.002 ± 0.002 | 0.009 ± 0.009 | E Africa F    | E Europe M    | 0.47 |
| 15  | 0.08 ± 0.03   | 0.04 ± 0.01   | 0.06 ± 0.02   | 0.05 ± 0.01   | 0.03 ± 0.02   | 0.06 ± 0.03   | 0.06 ± 0.02   | 0.018 ± 0.006 | 0.006 ± 0.006 | 0.011 ± 0.006 | 0.012 ± 0.010 | 0.01 ± 0.01   | E Africa F    | E Europe M    | 0.39 |
| 16  | 0.13 ± 0.03   | 0.05 ± 0.01   | 0.11 ± 0.03   | 0.06 ± 0.02   | 0.07 ± 0.03   | 0.12 ± 0.04   | 0.10 ± 0.03   | 0.027 ± 0.007 | 0.02 ± 0.01   | 0.022 ± 0.009 | 0.02 ± 0.01   | 0.03 ± 0.02   | E Africa F    | E Europe M    | 0.38 |
| 17  | 0.21 ± 0.04   | 0.09 ± 0.01   | 0.14 ± 0.04   | 0.14 ± 0.02   | 0.11 ± 0.03   | 0.17 ± 0.04   | 0.11 ± 0.03   | 0.05 ± 0.01   | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.06 ± 0.02   | 0.09 ± 0.03   | E Africa F    | E Europe M    | 0.32 |
| 18  | 0.24 ± 0.03   | 0.14 ± 0.01   | 0.17 ± 0.03   | 0.18 ± 0.02   | 0.15 ± 0.02   | 0.20 ± 0.04   | 0.16 ± 0.03   | 0.09 ± 0.01   | 0.03 ± 0.01   | 0.12 ± 0.01   | 0.09 ± 0.02   | 0.11 ± 0.03   | E Africa F    | E Europe M    | 0.22 |
| 19  | 0.38 ± 0.04   | 0.19 ± 0.02   | 0.27 ± 0.04   | 0.28 ± 0.02   | 0.24 ± 0.02   | 0.31 ± 0.04   | 0.24 ± 0.03   | 0.19 ± 0.02   | 0.14 ± 0.03   | 0.22 ± 0.02   | 0.19 ± 0.03   | 0.24 ± 0.04   | E Africa F    | E Europe M    | 0.15 |
| 20  | 0.45 ± 0.04   | 0.27 ± 0.02   | 0.39 ± 0.04   | 0.35 ± 0.02   | 0.33 ± 0.03   | 0.37 ± 0.04   | 0.33 ± 0.04   | 0.30 ± 0.02   | 0.25 ± 0.03   | 0.36 ± 0.02   | 0.31 ± 0.03   | 0.32 ± 0.04   | E Africa F    | E Europe M    | 0.09 |
| 21  | 0.55 ± 0.04   | 0.37 ± 0.02   | 0.47 ± 0.03   | 0.47 ± 0.02   | 0.44 ± 0.03   | 0.48 ± 0.04   | 0.52 ± 0.04   | 0.44 ± 0.02   | 0.38 ± 0.04   | 0.50 ± 0.02   | 0.46 ± 0.03   | 0.45 ± 0.04   | E Africa F    | E Asia F      | 0.07 |
| 22  | 0.61 ± 0.03   | 0.48 ± 0.02   | 0.56 ± 0.02   | 0.56 ± 0.02   | 0.50 ± 0.02   | 0.51 ± 0.04   | 0.66 ± 0.04   | 0.60 ± 0.02   | 0.53 ± 0.04   | 0.64 ± 0.02   | 0.58 ± 0.03   | 0.53 ± 0.04   | E Africa M    | E Asia F      | 0.06 |
| 23  | 0.70 ± 0.03   | 0.56 ± 0.02   | 0.64 ± 0.02   | 0.66 ± 0.02   | 0.62 ± 0.02   | 0.65 ± 0.03   | 0.73 ± 0.03   | 0.70 ± 0.02   | 0.65 ± 0.03   | 0.76 ± 0.02   | 0.70 ± 0.03   | 0.66 ± 0.03   | S Asia M      | E Asia F      | 0.04 |
| 24  | 0.75 ± 0.03   | 0.67 ± 0.01   | 0.73 ± 0.02   | 0.73 ± 0.01   | 0.71 ± 0.02   | 0.69 ± 0.03   | 0.81 ± 0.03   | 0.82 ± 0.02   | 0.79 ± 0.03   | 0.85 ± 0.01   | 0.80 ± 0.02   | 0.76 ± 0.03   | S Asia M      | E Asia F      | 0.04 |
| 25  | 0.78 ± 0.02   | 0.75 ± 0.01   | 0.81 ± 0.02   | 0.80 ± 0.01   | 0.79 ± 0.02   | 0.73 ± 0.02   | 0.87 ± 0.02   | 0.88 ± 0.01   | 0.89 ± 0.02   | 0.921 ± 0.009 | 0.88 ± 0.02   | 0.83 ± 0.02   | S Asia M      | W Africa F    | 0.04 |
| 26  | 0.85 ± 0.02   | 0.81 ± 0.01   | 0.86 ± 0.01   | 0.841 ± 0.010 | 0.86 ± 0.01   | 0.82 ± 0.02   | 0.92 ± 0.02   | 0.93 ± 0.01   | 0.92 ± 0.01   | 0.953 ± 0.007 | 0.93 ± 0.01   | 0.90 ± 0.02   | S Asia M      | E Asia F      | 0.03 |
| 27  | 0.88 ± 0.02   | 0.854 ± 0.009 | 0.91 ± 0.01   | 0.885 ± 0.008 | 0.89 ± 0.01   | 0.85 ± 0.02   | 0.95 ± 0.01   | 0.953 ± 0.007 | 0.971 ± 0.008 | 0.968 ± 0.006 | 0.95 ± 0.01   | 0.93 ± 0.01   | E Europe M    | W Africa F    | 0.03 |
| 28  | 0.91 ± 0.02   | 0.908 ± 0.007 | 0.944 ± 0.009 | 0.920 ± 0.006 | 0.92 ± 0.01   | 0.89 ± 0.02   | 0.969 ± 0.009 | 0.972 ± 0.006 | 0.976 ± 0.008 | 0.980 ± 0.004 | 0.974 ± 0.009 | 0.95 ± 0.01   | S Asia M      | W Africa F    | 0.02 |
| 29  | 0.94 ± 0.01   | 0.937 ± 0.006 | 0.963 ± 0.007 | 0.942 ± 0.006 | 0.950 ± 0.008 | 0.91 ± 0.01   | 0.987 ± 0.007 | 0.983 ± 0.005 | 0.986 ± 0.006 | 0.990 ± 0.003 | 0.973 ± 0.009 | 0.973 ± 0.008 | S Asia M      | W Africa F    | 0.01 |
| 30  | 0.96 ± 0.01   | 0.956 ± 0.005 | 0.972 ± 0.006 | 0.956 ± 0.005 | 0.966 ± 0.007 | 0.93 ± 0.01   | 0.988 ± 0.006 | 0.993 ± 0.003 | 0.992 ± 0.005 | 0.991 ± 0.002 | 0.986 ± 0.007 | 0.987 ± 0.006 | E Asia M      | W Africa F    | 0.01 |
| 31  | 0.972 ± 0.009 | 0.968 ± 0.005 | 0.981 ± 0.006 | 0.970 ± 0.005 | 0.972 ± 0.006 | 0.95 ± 0.01   | 0.992 ± 0.005 | 0.996 ± 0.002 | 0.997 ± 0.003 | 0.994 ± 0.002 | 0.995 ± 0.005 | 0.988 ± 0.005 | E Europe M    | W Africa F    | 0.01 |
| 32  | 0.97 ± 0.01   | 0.985 ± 0.003 | 0.991 ± 0.004 | 0.979 ± 0.004 | 0.983 ± 0.005 | 0.967 ± 0.010 | 0.993 ± 0.004 | 0.994 ± 0.003 | 0.997 ± 0.004 | 0.999 ± 0.001 | 0.992 ± 0.005 | 0.991 ± 0.005 | S Asia M      | E Africa F    | 0.01 |
| 33  | 0.984 ± 0.007 | 0.987 ± 0.003 | 0.988 ± 0.005 | 0.988 ± 0.003 | 0.990 ± 0.004 | 0.978 ± 0.007 | 0.996 ± 0.004 | 0.998 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.996 ± 0.004 | 0.996 ± 0.003 | S Asia M      | W Africa F    | 0.00 |
| 34  | 0.994 ± 0.004 | 0.992 ± 0.002 | 0.992 ± 0.004 | 0.991 ± 0.003 | 0.995 ± 0.003 | 0.984 ± 0.007 | 0.994 ± 0.004 | 0.998 ± 0.002 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.998 ± 0.003 | 0.997 ± 0.003 | S Asia M      | W Africa F    | 0.00 |

**Table 174.** For algorithm shufti-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.07 ± 0.03   | 0.032 ± 0.007 | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.02 ± 0.01  | 0.08 ± 0.04   | 0.09 ± 0.04   | 0.018 ± 0.005 | 0.005 ± 0.005 | 0.013 ± 0.006 | 0.02 ± 0.01   | 0.06 ± 0.04   | E Africa M    | E Europe M    | 0.40 |
| 15  | 0.13 ± 0.05   | 0.039 ± 0.007 | 0.08 ± 0.02   | 0.08 ± 0.02   | 0.03 ± 0.01  | 0.11 ± 0.04   | 0.16 ± 0.05   | 0.031 ± 0.006 | 0.013 ± 0.009 | 0.028 ± 0.009 | 0.04 ± 0.02   | 0.09 ± 0.04   | E Africa M    | E Europe M    | 0.39 |
| 16  | 0.14 ± 0.05   | 0.062 ± 0.008 | 0.10 ± 0.02   | 0.12 ± 0.02   | 0.06 ± 0.02  | 0.15 ± 0.04   | 0.27 ± 0.06   | 0.055 ± 0.007 | 0.02 ± 0.01   | 0.06 ± 0.01   | 0.04 ± 0.01   | 0.13 ± 0.04   | E Africa M    | E Europe M    | 0.36 |
| 17  | 0.20 ± 0.05   | 0.094 ± 0.009 | 0.13 ± 0.02   | 0.14 ± 0.02   | 0.09 ± 0.02  | 0.19 ± 0.04   | 0.32 ± 0.06   | 0.092 ± 0.009 | 0.05 ± 0.02   | 0.10 ± 0.02   | 0.09 ± 0.02   | 0.21 ± 0.05   | E Africa M    | E Europe M    | 0.29 |
| 18  | 0.30 ± 0.05   | 0.120 ± 0.009 | 0.21 ± 0.02   | 0.20 ± 0.02   | 0.13 ± 0.02  | 0.29 ± 0.04   | 0.40 ± 0.06   | 0.129 ± 0.009 | 0.09 ± 0.02   | 0.16 ± 0.01   | 0.11 ± 0.02   | 0.25 ± 0.04   | E Africa M    | E Europe M    | 0.27 |
| 19  | 0.36 ± 0.04   | 0.158 ± 0.009 | 0.24 ± 0.02   | 0.26 ± 0.02   | 0.16 ± 0.02  | 0.39 ± 0.04   | 0.50 ± 0.05   | 0.19 ± 0.01   | 0.13 ± 0.02   | 0.26 ± 0.02   | 0.18 ± 0.02   | 0.29 ± 0.05   | E Africa M    | E Europe M    | 0.24 |
| 20  | 0.42 ± 0.04   | 0.208 ± 0.009 | 0.31 ± 0.02   | 0.32 ± 0.02   | 0.19 ± 0.02  | 0.43 ± 0.04   | 0.55 ± 0.04   | 0.27 ± 0.01   | 0.19 ± 0.02   | 0.36 ± 0.02   | 0.28 ± 0.03   | 0.40 ± 0.04   | E Africa M    | S E Asia F    | 0.20 |
| 21  | 0.48 ± 0.04   | 0.245 ± 0.009 | 0.36 ± 0.02   | 0.39 ± 0.02   | 0.24 ± 0.02  | 0.52 ± 0.04   | 0.65 ± 0.04   | 0.38 ± 0.01   | 0.31 ± 0.02   | 0.47 ± 0.02   | 0.36 ± 0.02   | 0.53 ± 0.04   | E Africa M    | S E Asia F    | 0.17 |
| 22  | 0.55 ± 0.04   | 0.307 ± 0.009 | 0.42 ± 0.02   | 0.45 ± 0.01   | 0.30 ± 0.02  | 0.55 ± 0.03   | 0.74 ± 0.04   | 0.47 ± 0.01   | 0.40 ± 0.03   | 0.60 ± 0.02   | 0.45 ± 0.03   | 0.57 ± 0.03   | E Africa M    | S E Asia F    | 0.15 |
| 23  | 0.56 ± 0.03   | 0.367 ± 0.009 | 0.51 ± 0.02   | 0.51 ± 0.01   | 0.39 ± 0.02  | 0.57 ± 0.03   | 0.76 ± 0.04   | 0.56 ± 0.01   | 0.52 ± 0.02   | 0.67 ± 0.01   | 0.55 ± 0.03   | 0.65 ± 0.03   | E Africa M    | E Asia F      | 0.12 |
| 24  | 0.60 ± 0.03   | 0.424 ± 0.008 | 0.57 ± 0.01   | 0.57 ± 0.01   | 0.45 ± 0.02  | 0.63 ± 0.03   | 0.80 ± 0.03   | 0.64 ± 0.01   | 0.63 ± 0.02   | 0.77 ± 0.01   | 0.67 ± 0.02   | 0.73 ± 0.03   | E Africa M    | E Asia F      | 0.11 |
| 25  | 0.66 ± 0.03   | 0.492 ± 0.008 | 0.66 ± 0.01   | 0.63 ± 0.01   | 0.51 ± 0.01  | 0.67 ± 0.02   | 0.85 ± 0.02   | 0.724 ± 0.009 | 0.71 ± 0.02   | 0.836 ± 0.009 | 0.74 ± 0.02   | 0.80 ± 0.02   | E Africa M    | E Asia F      | 0.10 |
| 26  | 0.72 ± 0.02   | 0.557 ± 0.008 | 0.71 ± 0.01   | 0.670 ± 0.009 | 0.59 ± 0.02  | 0.74 ± 0.02   | 0.91 ± 0.02   | 0.792 ± 0.010 | 0.80 ± 0.02   | 0.881 ± 0.007 | 0.80 ± 0.02   | 0.85 ± 0.02   | E Africa M    | E Asia F      | 0.09 |
| 27  | 0.74 ± 0.03   | 0.609 ± 0.007 | 0.769 ± 0.010 | 0.711 ± 0.008 | 0.63 ± 0.01  | 0.76 ± 0.02   | 0.91 ± 0.02   | 0.842 ± 0.008 | 0.86 ± 0.01   | 0.914 ± 0.006 | 0.86 ± 0.02   | 0.90 ± 0.01   | S Asia M      | E Asia F      | 0.08 |
| 28  | 0.78 ± 0.02   | 0.677 ± 0.007 | 0.823 ± 0.008 | 0.757 ± 0.008 | 0.68 ± 0.01  | 0.77 ± 0.02   | 0.95 ± 0.01   | 0.884 ± 0.007 | 0.91 ± 0.01   | 0.938 ± 0.004 | 0.89 ± 0.01   | 0.91 ± 0.01   | E Africa M    | E Asia F      | 0.07 |
| 29  | 0.79 ± 0.02   | 0.714 ± 0.007 | 0.851 ± 0.008 | 0.785 ± 0.007 | 0.74 ± 0.01  | 0.81 ± 0.02   | 0.95 ± 0.01   | 0.920 ± 0.005 | 0.930 ± 0.008 | 0.956 ± 0.004 | 0.92 ± 0.01   | 0.93 ± 0.01   | S Asia M      | E Asia F      | 0.06 |
| 30  | 0.82 ± 0.02   | 0.770 ± 0.006 | 0.882 ± 0.007 | 0.826 ± 0.007 | 0.78 ± 0.01  | 0.84 ± 0.02   | 0.96 ± 0.01   | 0.943 ± 0.005 | 0.950 ± 0.007 | 0.966 ± 0.003 | 0.94 ± 0.01   | 0.948 ± 0.009 | S Asia M      | E Asia F      | 0.05 |
| 31  | 0.84 ± 0.02   | 0.800 ± 0.006 | 0.910 ± 0.006 | 0.856 ± 0.006 | 0.82 ± 0.01  | 0.86 ± 0.01   | 0.972 ± 0.009 | 0.954 ± 0.005 | 0.967 ± 0.006 | 0.977 ± 0.003 | 0.957 ± 0.010 | 0.964 ± 0.007 | S Asia M      | E Asia F      | 0.04 |
| 32  | 0.85 ± 0.02   | 0.833 ± 0.005 | 0.930 ± 0.006 | 0.877 ± 0.006 | 0.85 ± 0.01  | 0.88 ± 0.02   | 0.971 ± 0.009 | 0.966 ± 0.004 | 0.975 ± 0.005 | 0.979 ± 0.002 | 0.968 ± 0.009 | 0.971 ± 0.007 | S Asia M      | E Asia F      | 0.04 |
| 33  | 0.86 ± 0.02   | 0.862 ± 0.005 | 0.939 ± 0.006 | 0.897 ± 0.005 | 0.85 ± 0.01  | 0.88 ± 0.02   | 0.971 ± 0.009 | 0.977 ± 0.003 | 0.980 ± 0.004 | 0.982 ± 0.002 | 0.976 ± 0.007 | 0.978 ± 0.006 | S Asia M      | S E Asia F    | 0.03 |
| 34  | 0.88 ± 0.02   | 0.884 ± 0.005 | 0.947 ± 0.006 | 0.915 ± 0.005 | 0.89 ± 0.01  | 0.90 ± 0.02   | 0.981 ± 0.008 | 0.981 ± 0.003 | 0.988 ± 0.003 | 0.987 ± 0.002 | 0.983 ± 0.006 | 0.977 ± 0.006 | E Europe M    | E Africa F    | 0.03 |

**Table 175.** For algorithm unissey-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.04 ± 0.02   | 0.07 ± 0.01   | 0.02 ± 0.01   | 0.024 ± 0.008 | 0.05 ± 0.02   | 0.11 ± 0.04   | 0.04 ± 0.02   | 0.07 ± 0.01   | 0.009 ± 0.007 | 0.022 ± 0.007 | 0.06 ± 0.02   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.35 |
| 15  | 0.09 ± 0.03   | 0.13 ± 0.02   | 0.06 ± 0.02   | 0.06 ± 0.01   | 0.10 ± 0.03   | 0.15 ± 0.04   | 0.06 ± 0.02   | 0.12 ± 0.02   | 0.02 ± 0.01   | 0.05 ± 0.01   | 0.09 ± 0.02   | 0.06 ± 0.03   | W Africa F    | E Europe M    | 0.27 |
| 16  | 0.12 ± 0.03   | 0.19 ± 0.02   | 0.10 ± 0.03   | 0.10 ± 0.02   | 0.16 ± 0.04   | 0.19 ± 0.05   | 0.11 ± 0.03   | 0.17 ± 0.02   | 0.05 ± 0.02   | 0.10 ± 0.02   | 0.17 ± 0.04   | 0.08 ± 0.03   | E Asia F      | E Europe M    | 0.21 |
| 17  | 0.17 ± 0.04   | 0.26 ± 0.02   | 0.19 ± 0.04   | 0.15 ± 0.03   | 0.21 ± 0.04   | 0.29 ± 0.05   | 0.20 ± 0.04   | 0.26 ± 0.02   | 0.04 ± 0.02   | 0.15 ± 0.02   | 0.23 ± 0.04   | 0.14 ± 0.04   | W Africa F    | E Europe M    | 0.20 |
| 18  | 0.22 ± 0.03   | 0.32 ± 0.02   | 0.22 ± 0.03   | 0.18 ± 0.02   | 0.24 ± 0.03   | 0.32 ± 0.04   | 0.27 ± 0.04   | 0.32 ± 0.02   | 0.12 ± 0.03   | 0.26 ± 0.02   | 0.28 ± 0.03   | 0.25 ± 0.04   | E Asia F      | E Europe M    | 0.14 |
| 19  | 0.28 ± 0.04   | 0.41 ± 0.02   | 0.25 ± 0.04   | 0.27 ± 0.02   | 0.32 ± 0.03   | 0.41 ± 0.04   | 0.41 ± 0.04   | 0.40 ± 0.02   | 0.22 ± 0.03   | 0.36 ± 0.02   | 0.36 ± 0.03   | 0.29 ± 0.05   | E Asia F      | E Europe M    | 0.12 |
| 20  | 0.37 ± 0.04   | 0.48 ± 0.02   | 0.36 ± 0.03   | 0.32 ± 0.02   | 0.39 ± 0.03   | 0.50 ± 0.04   | 0.52 ± 0.04   | 0.49 ± 0.02   | 0.30 ± 0.03   | 0.47 ± 0.02   | 0.46 ± 0.03   | 0.42 ± 0.04   | E Africa M    | E Europe M    | 0.10 |
| 21  | 0.42 ± 0.04   | 0.55 ± 0.02   | 0.42 ± 0.03   | 0.38 ± 0.02   | 0.46 ± 0.03   | 0.52 ± 0.04   | 0.57 ± 0.03   | 0.57 ± 0.02   | 0.40 ± 0.04   | 0.54 ± 0.02   | 0.55 ± 0.03   | 0.48 ± 0.04   | E Africa M    | S Asia F      | 0.08 |
| 22  | 0.48 ± 0.03   | 0.61 ± 0.02   | 0.54 ± 0.03   | 0.44 ± 0.02   | 0.53 ± 0.02   | 0.59 ± 0.04   | 0.69 ± 0.03   | 0.64 ± 0.02   | 0.54 ± 0.03   | 0.65 ± 0.02   | 0.60 ± 0.03   | 0.52 ± 0.04   | E Africa M    | S Asia F      | 0.08 |
| 23  | 0.52 ± 0.03   | 0.69 ± 0.02   | 0.60 ± 0.02   | 0.51 ± 0.02   | 0.61 ± 0.02   | 0.65 ± 0.03   | 0.73 ± 0.03   | 0.74 ± 0.02   | 0.60 ± 0.03   | 0.73 ± 0.02   | 0.69 ± 0.03   | 0.60 ± 0.04   | E Asia M      | S Asia F      | 0.07 |
| 24  | 0.58 ± 0.03   | 0.73 ± 0.01   | 0.68 ± 0.02   | 0.57 ± 0.02   | 0.68 ± 0.02   | 0.68 ± 0.03   | 0.77 ± 0.03   | 0.76 ± 0.02   | 0.73 ± 0.03   | 0.80 ± 0.02   | 0.75 ± 0.02   | 0.66 ± 0.03   | S Asia M      | S Asia F      | 0.06 |
| 25  | 0.64 ± 0.02   | 0.78 ± 0.01   | 0.75 ± 0.02   | 0.60 ± 0.02   | 0.72 ± 0.02   | 0.74 ± 0.03   | 0.84 ± 0.02   | 0.83 ± 0.02   | 0.80 ± 0.02   | 0.85 ± 0.01   | 0.81 ± 0.02   | 0.72 ± 0.03   | S Asia M      | S Asia F      | 0.06 |
| 26  | 0.72 ± 0.03   | 0.825 ± 0.010 | 0.82 ± 0.02   | 0.65 ± 0.01   | 0.77 ± 0.02   | 0.77 ± 0.02   | 0.88 ± 0.02   | 0.87 ± 0.01   | 0.85 ± 0.02   | 0.87 ± 0.01   | 0.85 ± 0.02   | 0.77 ± 0.02   | E Africa M    | S Asia F      | 0.05 |
| 27  | 0.73 ± 0.02   | 0.87 ± 0.01   | 0.85 ± 0.01   | 0.69 ± 0.01   | 0.81 ± 0.01   | 0.81 ± 0.02   | 0.90 ± 0.02   | 0.90 ± 0.01   | 0.90 ± 0.02   | 0.906 ± 0.010 | 0.88 ± 0.02   | 0.81 ± 0.02   | S Asia M      | S Asia F      | 0.05 |
| 28  | 0.78 ± 0.02   | 0.894 ± 0.008 | 0.88 ± 0.01   | 0.72 ± 0.01   | 0.85 ± 0.01   | 0.85 ± 0.02   | 0.92 ± 0.02   | 0.929 ± 0.010 | 0.93 ± 0.01   | 0.926 ± 0.008 | 0.91 ± 0.02   | 0.82 ± 0.02   | E Europe M    | S Asia F      | 0.04 |
| 29  | 0.83 ± 0.02   | 0.922 ± 0.007 | 0.92 ± 0.01   | 0.78 ± 0.01   | 0.88 ± 0.01   | 0.88 ± 0.02   | 0.93 ± 0.01   | 0.954 ± 0.008 | 0.95 ± 0.01   | 0.944 ± 0.007 | 0.93 ± 0.01   | 0.89 ± 0.02   | E Asia M      | S Asia F      | 0.03 |
| 30  | 0.85 ± 0.02   | 0.942 ± 0.006 | 0.934 ± 0.009 | 0.81 ± 0.01   | 0.91 ± 0.01   | 0.88 ± 0.02   | 0.94 ± 0.01   | 0.966 ± 0.007 | 0.971 ± 0.008 | 0.954 ± 0.005 | 0.95 ± 0.01   | 0.89 ± 0.01   | E Europe M    | S Asia F      | 0.03 |
| 31  | 0.87 ± 0.02   | 0.951 ± 0.006 | 0.961 ± 0.008 | 0.842 ± 0.009 | 0.925 ± 0.009 | 0.89 ± 0.02   | 0.969 ± 0.010 | 0.977 ± 0.005 | 0.981 ± 0.006 | 0.964 ± 0.005 | 0.97 ± 0.01   | 0.91 ± 0.01   | E Europe M    | S Asia F      | 0.03 |
| 32  | 0.90 ± 0.02   | 0.958 ± 0.005 | 0.969 ± 0.007 | 0.873 ± 0.008 | 0.940 ± 0.010 | 0.91 ± 0.01   | 0.974 ± 0.010 | 0.982 ± 0.004 | 0.989 ± 0.006 | 0.976 ± 0.004 | 0.978 ± 0.009 | 0.92 ± 0.01   | E Europe M    | S Asia F      | 0.02 |
| 33  | 0.91 ± 0.01   | 0.976 ± 0.004 | 0.969 ± 0.008 | 0.895 ± 0.008 | 0.95 ± 0.01   | 0.94 ± 0.01   | 0.967 ± 0.010 | 0.991 ± 0.004 | 0.996 ± 0.004 | 0.981 ± 0.003 | 0.986 ± 0.007 | 0.95 ± 0.01   | E Europe M    | S Asia F      | 0.02 |
| 34  | 0.93 ± 0.02   | 0.982 ± 0.004 | 0.982 ± 0.006 | 0.922 ± 0.007 | 0.971 ± 0.007 | 0.94 ± 0.01   | 0.988 ± 0.007 | 0.993 ± 0.003 | 0.998 ± 0.003 | 0.987 ± 0.003 | 0.987 ± 0.008 | 0.95 ± 0.01   | E Europe M    | S Asia F      | 0.01 |

**Table 176.** For algorithm unissey-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.10 ± 0.05   | 0.050 ± 0.009 | 0.03 ± 0.02   | 0.034 ± 0.010 | 0.03 ± 0.02  | 0.06 ± 0.03   | 0.05 ± 0.03   | 0.033 ± 0.007 | 0.007 ± 0.007 | 0.013 ± 0.006 | 0.04 ± 0.01  | 0.01 ± 0.01   | E Africa F    | E Europe M    | 0.34 |
| 15  | 0.09 ± 0.04   | 0.082 ± 0.010 | 0.07 ± 0.02   | 0.05 ± 0.01   | 0.08 ± 0.02  | 0.13 ± 0.05   | 0.15 ± 0.05   | 0.060 ± 0.009 | 0.03 ± 0.01   | 0.034 ± 0.009 | 0.04 ± 0.01  | 0.06 ± 0.03   | E Africa M    | E Europe M    | 0.29 |
| 16  | 0.14 ± 0.05   | 0.12 ± 0.01   | 0.10 ± 0.02   | 0.08 ± 0.02   | 0.08 ± 0.02  | 0.18 ± 0.05   | 0.12 ± 0.04   | 0.10 ± 0.01   | 0.03 ± 0.01   | 0.06 ± 0.01   | 0.08 ± 0.02  | 0.08 ± 0.04   | W Africa F    | E Europe M    | 0.22 |
| 17  | 0.14 ± 0.04   | 0.16 ± 0.01   | 0.12 ± 0.02   | 0.11 ± 0.01   | 0.11 ± 0.02  | 0.24 ± 0.06   | 0.14 ± 0.04   | 0.14 ± 0.01   | 0.03 ± 0.01   | 0.11 ± 0.01   | 0.12 ± 0.02  | 0.13 ± 0.04   | W Africa F    | E Europe M    | 0.19 |
| 18  | 0.25 ± 0.04   | 0.197 ± 0.010 | 0.20 ± 0.02   | 0.14 ± 0.02   | 0.14 ± 0.02  | 0.24 ± 0.04   | 0.24 ± 0.04   | 0.19 ± 0.01   | 0.10 ± 0.02   | 0.15 ± 0.01   | 0.15 ± 0.02  | 0.19 ± 0.04   | E Africa F    | E Europe M    | 0.15 |
| 19  | 0.29 ± 0.04   | 0.25 ± 0.01   | 0.21 ± 0.02   | 0.18 ± 0.01   | 0.18 ± 0.02  | 0.31 ± 0.04   | 0.32 ± 0.05   | 0.24 ± 0.01   | 0.12 ± 0.02   | 0.22 ± 0.02   | 0.20 ± 0.02  | 0.21 ± 0.04   | E Africa M    | E Europe M    | 0.14 |
| 20  | 0.33 ± 0.04   | 0.308 ± 0.010 | 0.28 ± 0.02   | 0.21 ± 0.02   | 0.23 ± 0.02  | 0.37 ± 0.04   | 0.39 ± 0.04   | 0.30 ± 0.01   | 0.19 ± 0.02   | 0.31 ± 0.02   | 0.28 ± 0.02  | 0.23 ± 0.03   | E Africa M    | E Europe M    | 0.13 |
| 21  | 0.39 ± 0.04   | 0.35 ± 0.01   | 0.32 ± 0.02   | 0.25 ± 0.01   | 0.26 ± 0.02  | 0.43 ± 0.04   | 0.44 ± 0.04   | 0.36 ± 0.01   | 0.29 ± 0.03   | 0.40 ± 0.02   | 0.34 ± 0.02  | 0.29 ± 0.03   | E Africa M    | S Asia F      | 0.11 |
| 22  | 0.45 ± 0.04   | 0.388 ± 0.010 | 0.40 ± 0.02   | 0.30 ± 0.01   | 0.31 ± 0.02  | 0.47 ± 0.03   | 0.52 ± 0.04   | 0.43 ± 0.01   | 0.36 ± 0.02   | 0.47 ± 0.02   | 0.41 ± 0.03  | 0.35 ± 0.03   | E Africa M    | S Asia F      | 0.10 |
| 23  | 0.51 ± 0.04   | 0.438 ± 0.009 | 0.45 ± 0.02   | 0.34 ± 0.01   | 0.38 ± 0.02  | 0.53 ± 0.03   | 0.57 ± 0.04   | 0.50 ± 0.01   | 0.47 ± 0.02   | 0.55 ± 0.01   | 0.51 ± 0.03  | 0.41 ± 0.03   | E Africa M    | S Asia F      | 0.09 |
| 24  | 0.54 ± 0.03   | 0.506 ± 0.008 | 0.51 ± 0.01   | 0.40 ± 0.01   | 0.45 ± 0.02  | 0.56 ± 0.03   | 0.61 ± 0.04   | 0.57 ± 0.01   | 0.57 ± 0.02   | 0.63 ± 0.01   | 0.56 ± 0.03  | 0.44 ± 0.03   | S Asia M      | S Asia F      | 0.08 |
| 25  | 0.61 ± 0.03   | 0.558 ± 0.008 | 0.58 ± 0.01   | 0.41 ± 0.01   | 0.51 ± 0.02  | 0.61 ± 0.02   | 0.70 ± 0.03   | 0.62 ± 0.01   | 0.63 ± 0.02   | 0.68 ± 0.01   | 0.63 ± 0.02  | 0.52 ± 0.03   | E Africa M    | S Asia F      | 0.08 |
| 26  | 0.67 ± 0.03   | 0.615 ± 0.008 | 0.64 ± 0.01   | 0.478 ± 0.009 | 0.54 ± 0.02  | 0.62 ± 0.02   | 0.73 ± 0.03   | 0.69 ± 0.01   | 0.71 ± 0.02   | 0.72 ± 0.01   | 0.68 ± 0.02  | 0.56 ± 0.02   | E Africa M    | S Asia F      | 0.07 |
| 27  | 0.68 ± 0.03   | 0.653 ± 0.007 | 0.69 ± 0.01   | 0.501 ± 0.009 | 0.61 ± 0.01  | 0.68 ± 0.02   | 0.76 ± 0.03   | 0.738 ± 0.010 | 0.78 ± 0.02   | 0.766 ± 0.008 | 0.74 ± 0.02  | 0.63 ± 0.02   | E Europe M    | S Asia F      | 0.07 |
| 28  | 0.74 ± 0.02   | 0.710 ± 0.006 | 0.742 ± 0.009 | 0.550 ± 0.008 | 0.65 ± 0.01  | 0.68 ± 0.02   | 0.82 ± 0.02   | 0.788 ± 0.009 | 0.84 ± 0.01   | 0.810 ± 0.007 | 0.78 ± 0.02  | 0.68 ± 0.02   | E Europe M    | S Asia F      | 0.07 |
| 29  | 0.77 ± 0.03   | 0.745 ± 0.007 | 0.796 ± 0.010 | 0.586 ± 0.007 | 0.69 ± 0.02  | 0.74 ± 0.02   | 0.83 ± 0.02   | 0.825 ± 0.008 | 0.87 ± 0.01   | 0.841 ± 0.007 | 0.83 ± 0.02  | 0.72 ± 0.02   | E Europe M    | S Asia F      | 0.06 |
| 30  | 0.77 ± 0.02   | 0.781 ± 0.006 | 0.834 ± 0.009 | 0.628 ± 0.008 | 0.74 ± 0.01  | 0.75 ± 0.02   | 0.85 ± 0.02   | 0.859 ± 0.007 | 0.90 ± 0.01   | 0.867 ± 0.006 | 0.85 ± 0.02  | 0.76 ± 0.02   | E Europe M    | S Asia F      | 0.05 |
| 31  | 0.81 ± 0.02   | 0.821 ± 0.006 | 0.861 ± 0.008 | 0.665 ± 0.008 | 0.77 ± 0.01  | 0.78 ± 0.02   | 0.89 ± 0.02   | 0.886 ± 0.006 | 0.933 ± 0.008 | 0.891 ± 0.005 | 0.86 ± 0.02  | 0.76 ± 0.02   | E Europe M    | S Asia F      | 0.05 |
| 32  | 0.83 ± 0.02   | 0.845 ± 0.005 | 0.893 ± 0.008 | 0.715 ± 0.007 | 0.80 ± 0.01  | 0.80 ± 0.02   | 0.92 ± 0.02   | 0.912 ± 0.005 | 0.951 ± 0.007 | 0.910 ± 0.004 | 0.88 ± 0.01  | 0.79 ± 0.02   | E Europe M    | S Asia F      | 0.05 |
| 33  | 0.87 ± 0.02   | 0.872 ± 0.005 | 0.914 ± 0.008 | 0.755 ± 0.007 | 0.84 ± 0.01  | 0.84 ± 0.02   | 0.92 ± 0.02   | 0.930 ± 0.005 | 0.961 ± 0.006 | 0.925 ± 0.004 | 0.93 ± 0.01  | 0.82 ± 0.02   | E Europe M    | S Asia F      | 0.04 |
| 34  | 0.87 ± 0.02   | 0.895 ± 0.004 | 0.935 ± 0.007 | 0.791 ± 0.007 | 0.86 ± 0.01  | 0.85 ± 0.02   | 0.94 ± 0.01   | 0.942 ± 0.005 | 0.979 ± 0.005 | 0.938 ± 0.003 | 0.94 ± 0.01  | 0.86 ± 0.01   | E Europe M    | S Asia F      | 0.04 |

**Table 177.** For algorithm viante-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.25 ± 0.04   | 0.06 ± 0.01   | 0.007 ± 0.007 | 0.17 ± 0.02   | 0.14 ± 0.03   | 0.21 ± 0.05   | 0.09 ± 0.02   | 0.022 ± 0.006 | 0.000 ± 0.000 | 0.036 ± 0.009 | 0.05 ± 0.02   | 0.07 ± 0.03   | E Africa F    | E Europe M    | 0.52 |
| 15  | 0.36 ± 0.05   | 0.11 ± 0.02   | 0.02 ± 0.01   | 0.29 ± 0.03   | 0.21 ± 0.04   | 0.29 ± 0.06   | 0.11 ± 0.03   | 0.05 ± 0.01   | 0.003 ± 0.003 | 0.09 ± 0.02   | 0.09 ± 0.03   | 0.15 ± 0.05   | E Africa F    | E Europe M    | 0.45 |
| 16  | 0.49 ± 0.05   | 0.21 ± 0.02   | 0.03 ± 0.02   | 0.41 ± 0.04   | 0.39 ± 0.05   | 0.43 ± 0.06   | 0.28 ± 0.05   | 0.14 ± 0.02   | 0.01 ± 0.01   | 0.21 ± 0.03   | 0.23 ± 0.04   | 0.28 ± 0.06   | E Africa F    | E Europe M    | 0.34 |
| 17  | 0.60 ± 0.05   | 0.33 ± 0.02   | 0.12 ± 0.03   | 0.51 ± 0.03   | 0.46 ± 0.05   | 0.57 ± 0.05   | 0.43 ± 0.05   | 0.25 ± 0.02   | 0.02 ± 0.01   | 0.35 ± 0.03   | 0.37 ± 0.05   | 0.47 ± 0.06   | E Africa F    | E Europe M    | 0.27 |
| 18  | 0.65 ± 0.04   | 0.42 ± 0.02   | 0.18 ± 0.03   | 0.56 ± 0.02   | 0.56 ± 0.03   | 0.61 ± 0.05   | 0.54 ± 0.04   | 0.36 ± 0.02   | 0.06 ± 0.02   | 0.48 ± 0.02   | 0.47 ± 0.04   | 0.55 ± 0.05   | E Africa F    | E Europe M    | 0.21 |
| 19  | 0.74 ± 0.04   | 0.55 ± 0.02   | 0.25 ± 0.04   | 0.70 ± 0.02   | 0.68 ± 0.03   | 0.71 ± 0.04   | 0.70 ± 0.04   | 0.54 ± 0.02   | 0.12 ± 0.03   | 0.67 ± 0.02   | 0.61 ± 0.03   | 0.69 ± 0.04   | E Africa F    | E Europe M    | 0.17 |
| 20  | 0.80 ± 0.03   | 0.66 ± 0.02   | 0.42 ± 0.03   | 0.77 ± 0.02   | 0.78 ± 0.03   | 0.78 ± 0.03   | 0.82 ± 0.03   | 0.68 ± 0.02   | 0.22 ± 0.03   | 0.78 ± 0.02   | 0.78 ± 0.03   | 0.81 ± 0.04   | E Africa M    | E Europe M    | 0.13 |
| 21  | 0.88 ± 0.02   | 0.76 ± 0.02   | 0.51 ± 0.03   | 0.84 ± 0.01   | 0.85 ± 0.02   | 0.83 ± 0.03   | 0.90 ± 0.02   | 0.80 ± 0.02   | 0.38 ± 0.04   | 0.86 ± 0.02   | 0.84 ± 0.02   | 0.86 ± 0.03   | E Africa M    | E Europe M    | 0.10 |
| 22  | 0.91 ± 0.02   | 0.84 ± 0.01   | 0.64 ± 0.03   | 0.90 ± 0.01   | 0.90 ± 0.01   | 0.87 ± 0.02   | 0.95 ± 0.01   | 0.88 ± 0.01   | 0.61 ± 0.04   | 0.92 ± 0.01   | 0.91 ± 0.02   | 0.92 ± 0.02   | E Africa M    | E Europe M    | 0.06 |
| 23  | 0.92 ± 0.02   | 0.90 ± 0.01   | 0.75 ± 0.02   | 0.934 ± 0.008 | 0.93 ± 0.01   | 0.92 ± 0.02   | 0.97 ± 0.01   | 0.94 ± 0.01   | 0.71 ± 0.03   | 0.961 ± 0.009 | 0.95 ± 0.01   | 0.96 ± 0.01   | E Africa M    | E Europe M    | 0.04 |
| 24  | 0.95 ± 0.01   | 0.935 ± 0.008 | 0.87 ± 0.02   | 0.963 ± 0.006 | 0.959 ± 0.008 | 0.94 ± 0.02   | 0.989 ± 0.007 | 0.957 ± 0.008 | 0.86 ± 0.02   | 0.981 ± 0.006 | 0.972 ± 0.009 | 0.97 ± 0.01   | E Africa M    | E Europe M    | 0.02 |
| 25  | 0.96 ± 0.01   | 0.957 ± 0.006 | 0.92 ± 0.01   | 0.967 ± 0.005 | 0.970 ± 0.007 | 0.96 ± 0.01   | 0.994 ± 0.005 | 0.978 ± 0.006 | 0.93 ± 0.01   | 0.992 ± 0.003 | 0.988 ± 0.006 | 0.988 ± 0.007 | E Africa M    | E Europe F    | 0.01 |
| 26  | 0.97 ± 0.01   | 0.971 ± 0.005 | 0.956 ± 0.008 | 0.981 ± 0.003 | 0.987 ± 0.004 | 0.96 ± 0.01   | 0.995 ± 0.005 | 0.989 ± 0.004 | 0.95 ± 0.01   | 0.995 ± 0.002 | 0.995 ± 0.004 | 0.994 ± 0.005 | E Africa M    | E Europe M    | 0.01 |
| 27  | 0.979 ± 0.008 | 0.983 ± 0.004 | 0.975 ± 0.006 | 0.986 ± 0.003 | 0.988 ± 0.004 | 0.981 ± 0.008 | 0.995 ± 0.004 | 0.995 ± 0.003 | 0.988 ± 0.007 | 0.997 ± 0.002 | 0.997 ± 0.003 | 0.996 ± 0.004 | S Asia M      | E Europe F    | 0.00 |
| 28  | 0.991 ± 0.006 | 0.989 ± 0.003 | 0.993 ± 0.003 | 0.989 ± 0.003 | 0.995 ± 0.002 | 0.984 ± 0.007 | 0.999 ± 0.001 | 0.998 ± 0.002 | 0.991 ± 0.006 | 0.998 ± 0.001 | 0.997 ± 0.003 | 0.998 ± 0.003 | E Africa M    | W Africa F    | 0.00 |
| 29  | 0.993 ± 0.005 | 0.996 ± 0.002 | 0.997 ± 0.003 | 0.994 ± 0.002 | 0.997 ± 0.002 | 0.988 ± 0.005 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.996 ± 0.003 | 1.000 ± 0.001 | 0.999 ± 0.002 | 1.000 ± 0.000 | W Africa M    | W Africa F    | 0.00 |
| 30  | 0.992 ± 0.005 | 0.997 ± 0.001 | 0.996 ± 0.003 | 0.997 ± 0.002 | 0.998 ± 0.002 | 0.991 ± 0.005 | 0.999 ± 0.002 | 1.000 ± 0.001 | 0.999 ± 0.002 | 1.000 ± 0.000 | 0.998 ± 0.003 | 1.000 ± 0.000 | W Africa M    | W Africa F    | 0.00 |
| 31  | 0.997 ± 0.003 | 0.998 ± 0.001 | 0.999 ± 0.001 | 0.997 ± 0.002 | 1.000 ± 0.001 | 0.997 ± 0.003 | 0.999 ± 0.001 | 1.000 ± 0.001 | 0.999 ± 0.001 | 1.000 ± 0.000 | 0.999 ± 0.002 | 0.999 ± 0.002 | S Asia M      | W Africa F    | 0.00 |
| 32  | 0.998 ± 0.002 | 0.999 ± 0.001 | 0.999 ± 0.001 | 0.999 ± 0.001 | 1.000 ± 0.001 | 0.997 ± 0.003 | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.998 ± 0.003 | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.000 ± 0.000 | E Africa M    | W Africa F    | 0.00 |
| 33  | 0.995 ± 0.004 | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.999 ± 0.001 | 1.000 ± 0.000 | 0.999 ± 0.002 | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.999 ± 0.002 | 1.000 ± 0.000 | 0.999 ± 0.002 | 1.000 ± 0.000 | E Europe F    | E Africa F    | 0.00 |
| 34  | 0.995 ± 0.004 | 1.000 ± 0.001 | 1.000 ± 0.001 | 1.000 ± 0.001 | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.000 ± 0.001 | 0.999 ± 0.001 | 1.000 ± 0.000 | 1.000 ± 0.000 | 1.000 ± 0.000 | S E Asia F    | E Africa F    | 0.00 |

**Table 178.** For algorithm viante-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.38 ± 0.07   | 0.025 ± 0.006 | 0.009 ± 0.006 | 0.26 ± 0.03   | 0.10 ± 0.03   | 0.44 ± 0.08   | 0.25 ± 0.06   | 0.009 ± 0.004 | 0.000 ± 0.000 | 0.032 ± 0.010 | 0.03 ± 0.01   | 0.25 ± 0.06   | W Africa F    | E Europe M    | 0.60 |
| 15  | 0.58 ± 0.07   | 0.043 ± 0.007 | 0.04 ± 0.02   | 0.37 ± 0.03   | 0.14 ± 0.03   | 0.55 ± 0.07   | 0.40 ± 0.07   | 0.014 ± 0.004 | 0.007 ± 0.007 | 0.09 ± 0.01   | 0.05 ± 0.02   | 0.38 ± 0.07   | E Africa F    | E Europe M    | 0.56 |
| 16  | 0.69 ± 0.06   | 0.074 ± 0.009 | 0.06 ± 0.02   | 0.50 ± 0.03   | 0.23 ± 0.03   | 0.71 ± 0.06   | 0.55 ± 0.07   | 0.035 ± 0.006 | 0.018 ± 0.010 | 0.21 ± 0.02   | 0.12 ± 0.02   | 0.56 ± 0.06   | W Africa F    | E Europe M    | 0.50 |
| 17  | 0.74 ± 0.05   | 0.13 ± 0.01   | 0.11 ± 0.02   | 0.63 ± 0.02   | 0.30 ± 0.03   | 0.76 ± 0.05   | 0.70 ± 0.06   | 0.086 ± 0.009 | 0.03 ± 0.01   | 0.40 ± 0.02   | 0.23 ± 0.03   | 0.66 ± 0.05   | W Africa F    | E Europe M    | 0.42 |
| 18  | 0.81 ± 0.04   | 0.21 ± 0.01   | 0.18 ± 0.02   | 0.72 ± 0.02   | 0.41 ± 0.02   | 0.83 ± 0.04   | 0.83 ± 0.04   | 0.16 ± 0.01   | 0.08 ± 0.02   | 0.55 ± 0.02   | 0.35 ± 0.03   | 0.78 ± 0.04   | W Africa F    | E Europe M    | 0.35 |
| 19  | 0.90 ± 0.03   | 0.29 ± 0.01   | 0.25 ± 0.02   | 0.79 ± 0.02   | 0.53 ± 0.03   | 0.86 ± 0.03   | 0.89 ± 0.03   | 0.23 ± 0.01   | 0.10 ± 0.02   | 0.71 ± 0.02   | 0.49 ± 0.03   | 0.84 ± 0.03   | E Africa F    | E Europe M    | 0.30 |
| 20  | 0.91 ± 0.02   | 0.40 ± 0.01   | 0.33 ± 0.02   | 0.86 ± 0.01   | 0.62 ± 0.02   | 0.90 ± 0.02   | 0.94 ± 0.02   | 0.33 ± 0.01   | 0.20 ± 0.02   | 0.82 ± 0.01   | 0.60 ± 0.03   | 0.89 ± 0.03   | E Africa M    | E Europe M    | 0.24 |
| 21  | 0.96 ± 0.01   | 0.49 ± 0.01   | 0.43 ± 0.02   | 0.897 ± 0.010 | 0.70 ± 0.02   | 0.94 ± 0.02   | 0.98 ± 0.01   | 0.45 ± 0.01   | 0.33 ± 0.03   | 0.90 ± 0.01   | 0.70 ± 0.02   | 0.97 ± 0.02   | E Africa M    | E Europe M    | 0.19 |
| 22  | 0.97 ± 0.01   | 0.57 ± 0.01   | 0.54 ± 0.02   | 0.926 ± 0.008 | 0.78 ± 0.02   | 0.95 ± 0.01   | 0.99 ± 0.01   | 0.57 ± 0.01   | 0.45 ± 0.03   | 0.940 ± 0.007 | 0.79 ± 0.02   | 0.97 ± 0.01   | E Africa M    | E Europe M    | 0.15 |
| 23  | 0.97 ± 0.01   | 0.651 ± 0.008 | 0.64 ± 0.02   | 0.951 ± 0.006 | 0.84 ± 0.01   | 0.97 ± 0.01   | 0.990 ± 0.009 | 0.66 ± 0.01   | 0.59 ± 0.02   | 0.966 ± 0.006 | 0.85 ± 0.02   | 0.98 ± 0.01   | E Africa M    | E Europe M    | 0.11 |
| 24  | 0.977 ± 0.010 | 0.726 ± 0.008 | 0.73 ± 0.01   | 0.970 ± 0.004 | 0.88 ± 0.01   | 0.981 ± 0.007 | 0.997 ± 0.004 | 0.75 ± 0.01   | 0.73 ± 0.02   | 0.985 ± 0.003 | 0.89 ± 0.02   | 0.991 ± 0.007 | E Africa M    | E Asia F      | 0.07 |
| 25  | 0.987 ± 0.007 | 0.796 ± 0.007 | 0.824 ± 0.010 | 0.972 ± 0.004 | 0.912 ± 0.009 | 0.983 ± 0.007 | 1.000 ± 0.000 | 0.830 ± 0.009 | 0.82 ± 0.01   | 0.989 ± 0.003 | 0.93 ± 0.01   | 0.995 ± 0.004 | E Africa M    | E Asia F      | 0.05 |
| 26  | 0.990 ± 0.006 | 0.848 ± 0.006 | 0.869 ± 0.009 | 0.982 ± 0.003 | 0.943 ± 0.008 | 0.982 ± 0.007 | 0.999 ± 0.002 | 0.883 ± 0.007 | 0.89 ± 0.01   | 0.994 ± 0.002 | 0.95 ± 0.01   | 0.995 ± 0.004 | E Africa M    | E Asia F      | 0.03 |
| 27  | 0.996 ± 0.003 | 0.886 ± 0.004 | 0.923 ± 0.006 | 0.987 ± 0.002 | 0.958 ± 0.007 | 0.993 ± 0.004 | 0.999 ± 0.002 | 0.918 ± 0.006 | 0.935 ± 0.009 | 0.996 ± 0.001 | 0.973 ± 0.007 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.02 |
| 28  | 0.998 ± 0.003 | 0.920 ± 0.004 | 0.952 ± 0.005 | 0.989 ± 0.002 | 0.971 ± 0.005 | 0.992 ± 0.004 | 0.999 ± 0.002 | 0.949 ± 0.005 | 0.966 ± 0.006 | 0.998 ± 0.001 | 0.977 ± 0.007 | 0.998 ± 0.002 | E Africa M    | E Asia F      | 0.01 |
| 29  | 0.996 ± 0.004 | 0.941 ± 0.003 | 0.972 ± 0.004 | 0.993 ± 0.002 | 0.978 ± 0.005 | 0.996 ± 0.003 | 1.000 ± 0.000 | 0.964 ± 0.004 | 0.978 ± 0.005 | 0.998 ± 0.001 | 0.982 ± 0.007 | 0.999 ± 0.001 | E Africa M    | E Asia F      | 0.01 |
| 30  | 0.995 ± 0.004 | 0.960 ± 0.003 | 0.987 ± 0.003 | 0.995 ± 0.001 | 0.984 ± 0.004 | 0.995 ± 0.004 | 1.000 ± 0.000 | 0.976 ± 0.003 | 0.992 ± 0.003 | 1.000 ± 0.000 | 0.990 ± 0.004 | 1.000 ± 0.001 | E Africa M    | E Asia F      | 0.01 |
| 31  | 0.998 ± 0.003 | 0.974 ± 0.002 | 0.992 ± 0.002 | 0.996 ± 0.001 | 0.990 ± 0.003 | 0.997 ± 0.003 | 1.000 ± 0.000 | 0.986 ± 0.002 | 0.996 ± 0.002 | 1.000 ± 0.000 | 0.990 ± 0.005 | 0.999 ± 0.001 | E Africa M    | E Asia F      | 0.00 |
| 32  | 0.998 ± 0.003 | 0.981 ± 0.002 | 0.997 ± 0.001 | 0.997 ± 0.001 | 0.992 ± 0.003 | 0.999 ± 0.002 | 1.000 ± 0.000 | 0.992 ± 0.002 | 0.998 ± 0.002 | 1.000 ± 0.000 | 0.993 ± 0.004 | 1.000 ± 0.001 | E Africa M    | E Asia F      | 0.00 |
| 33  | 0.998 ± 0.003 | 0.988 ± 0.002 | 0.997 ± 0.001 | 0.998 ± 0.001 | 0.994 ± 0.003 | 0.999 ± 0.001 | 1.000 ± 0.000 | 0.994 ± 0.002 | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.998 ± 0.002 | 1.000 ± 0.000 | E Africa M    | E Asia F      | 0.00 |
| 34  | 1.000 ± 0.000 | 0.990 ± 0.001 | 0.998 ± 0.001 | 0.998 ± 0.001 | 0.994 ± 0.002 | 0.998 ± 0.003 | 1.000 ± 0.000 | 0.995 ± 0.001 | 1.000 ± 0.000 | 1.000 ± 0.000 | 0.999 ± 0.001 | 1.000 ± 0.000 | E Africa F    | E Asia F      | 0.00 |

**Table 179.** For algorithm yoti-001 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.03 ± 0.02   | 0.014 ± 0.005 | 0.03 ± 0.01   | 0.027 ± 0.008 | 0.04 ± 0.02   | 0.06 ± 0.03   | 0.006 ± 0.006 | 0.005 ± 0.003 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.005 ± 0.005 | 0.000 ± 0.000 | W Africa F    | E Europe M    | 0.60 |
| 15  | 0.05 ± 0.02   | 0.04 ± 0.01   | 0.07 ± 0.02   | 0.07 ± 0.02   | 0.04 ± 0.02   | 0.09 ± 0.04   | 0.008 ± 0.008 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.007 ± 0.005 | 0.002 ± 0.002 | 0.01 ± 0.01   | W Africa F    | E Europe M    | 0.54 |
| 16  | 0.13 ± 0.04   | 0.08 ± 0.01   | 0.11 ± 0.03   | 0.12 ± 0.02   | 0.15 ± 0.03   | 0.17 ± 0.05   | 0.02 ± 0.02   | 0.010 ± 0.005 | 0.003 ± 0.003 | 0.006 ± 0.004 | 0.003 ± 0.003 | 0.02 ± 0.02   | W Africa F    | S E Asia M    | 0.54 |
| 17  | 0.18 ± 0.04   | 0.14 ± 0.02   | 0.19 ± 0.04   | 0.21 ± 0.03   | 0.18 ± 0.03   | 0.24 ± 0.04   | 0.02 ± 0.01   | 0.036 ± 0.009 | 0.003 ± 0.003 | 0.021 ± 0.009 | 0.02 ± 0.01   | 0.08 ± 0.03   | W Africa F    | E Europe M    | 0.46 |
| 18  | 0.23 ± 0.04   | 0.19 ± 0.02   | 0.26 ± 0.03   | 0.25 ± 0.02   | 0.26 ± 0.03   | 0.32 ± 0.04   | 0.07 ± 0.02   | 0.059 ± 0.009 | 0.03 ± 0.01   | 0.043 ± 0.009 | 0.04 ± 0.02   | 0.10 ± 0.03   | W Africa F    | E Europe M    | 0.40 |
| 19  | 0.34 ± 0.04   | 0.29 ± 0.02   | 0.36 ± 0.04   | 0.35 ± 0.02   | 0.38 ± 0.03   | 0.44 ± 0.04   | 0.13 ± 0.03   | 0.12 ± 0.01   | 0.05 ± 0.02   | 0.09 ± 0.01   | 0.11 ± 0.02   | 0.18 ± 0.04   | W Africa F    | E Europe M    | 0.34 |
| 20  | 0.39 ± 0.04   | 0.40 ± 0.02   | 0.45 ± 0.04   | 0.43 ± 0.02   | 0.48 ± 0.03   | 0.49 ± 0.04   | 0.24 ± 0.03   | 0.22 ± 0.02   | 0.13 ± 0.03   | 0.18 ± 0.02   | 0.22 ± 0.03   | 0.28 ± 0.04   | W Africa F    | E Europe M    | 0.23 |
| 21  | 0.50 ± 0.04   | 0.51 ± 0.02   | 0.56 ± 0.04   | 0.56 ± 0.02   | 0.62 ± 0.02   | 0.61 ± 0.04   | 0.32 ± 0.04   | 0.35 ± 0.02   | 0.23 ± 0.03   | 0.30 ± 0.02   | 0.35 ± 0.03   | 0.38 ± 0.04   | S E Asia F    | E Europe M    | 0.18 |
| 22  | 0.56 ± 0.03   | 0.63 ± 0.02   | 0.68 ± 0.03   | 0.63 ± 0.02   | 0.70 ± 0.02   | 0.65 ± 0.03   | 0.50 ± 0.04   | 0.51 ± 0.02   | 0.40 ± 0.04   | 0.43 ± 0.02   | 0.49 ± 0.03   | 0.50 ± 0.04   | S E Asia F    | E Europe M    | 0.11 |
| 23  | 0.67 ± 0.03   | 0.73 ± 0.02   | 0.76 ± 0.02   | 0.72 ± 0.02   | 0.78 ± 0.02   | 0.77 ± 0.03   | 0.59 ± 0.03   | 0.62 ± 0.02   | 0.55 ± 0.03   | 0.60 ± 0.02   | 0.64 ± 0.03   | 0.65 ± 0.03   | S E Asia F    | E Europe M    | 0.07 |
| 24  | 0.72 ± 0.03   | 0.80 ± 0.01   | 0.86 ± 0.01   | 0.79 ± 0.01   | 0.86 ± 0.01   | 0.79 ± 0.03   | 0.73 ± 0.03   | 0.75 ± 0.02   | 0.73 ± 0.03   | 0.72 ± 0.02   | 0.77 ± 0.02   | 0.73 ± 0.03   | S E Asia F    | E Africa F    | 0.04 |
| 25  | 0.77 ± 0.02   | 0.870 ± 0.010 | 0.90 ± 0.01   | 0.84 ± 0.01   | 0.90 ± 0.01   | 0.84 ± 0.02   | 0.78 ± 0.02   | 0.84 ± 0.01   | 0.85 ± 0.02   | 0.83 ± 0.01   | 0.85 ± 0.02   | 0.81 ± 0.02   | E Europe F    | E Africa F    | 0.03 |
| 26  | 0.82 ± 0.02   | 0.913 ± 0.008 | 0.949 ± 0.008 | 0.883 ± 0.008 | 0.939 ± 0.009 | 0.88 ± 0.02   | 0.87 ± 0.02   | 0.92 ± 0.01   | 0.91 ± 0.02   | 0.89 ± 0.01   | 0.92 ± 0.02   | 0.88 ± 0.02   | E Europe F    | E Africa F    | 0.02 |
| 27  | 0.86 ± 0.02   | 0.950 ± 0.006 | 0.967 ± 0.007 | 0.912 ± 0.007 | 0.955 ± 0.008 | 0.91 ± 0.02   | 0.89 ± 0.02   | 0.949 ± 0.008 | 0.95 ± 0.01   | 0.941 ± 0.008 | 0.95 ± 0.01   | 0.91 ± 0.02   | E Europe F    | E Africa F    | 0.02 |
| 28  | 0.90 ± 0.02   | 0.967 ± 0.005 | 0.983 ± 0.005 | 0.931 ± 0.007 | 0.965 ± 0.007 | 0.94 ± 0.01   | 0.95 ± 0.01   | 0.971 ± 0.007 | 0.977 ± 0.008 | 0.963 ± 0.006 | 0.97 ± 0.01   | 0.95 ± 0.01   | E Europe F    | E Africa F    | 0.01 |
| 29  | 0.93 ± 0.01   | 0.980 ± 0.004 | 0.989 ± 0.004 | 0.954 ± 0.005 | 0.985 ± 0.005 | 0.94 ± 0.01   | 0.96 ± 0.01   | 0.986 ± 0.004 | 0.990 ± 0.006 | 0.978 ± 0.004 | 0.978 ± 0.008 | 0.972 ± 0.008 | E Europe M    | E Africa F    | 0.01 |
| 30  | 0.95 ± 0.01   | 0.987 ± 0.003 | 0.995 ± 0.003 | 0.965 ± 0.005 | 0.990 ± 0.004 | 0.96 ± 0.01   | 0.972 ± 0.010 | 0.991 ± 0.004 | 0.995 ± 0.004 | 0.986 ± 0.003 | 0.987 ± 0.007 | 0.978 ± 0.006 | E Europe M    | E Africa F    | 0.01 |
| 31  | 0.96 ± 0.01   | 0.991 ± 0.003 | 0.996 ± 0.003 | 0.975 ± 0.004 | 0.991 ± 0.004 | 0.96 ± 0.01   | 0.985 ± 0.007 | 0.996 ± 0.002 | 0.997 ± 0.003 | 0.992 ± 0.002 | 0.991 ± 0.007 | 0.989 ± 0.005 | E Europe M    | E Africa F    | 0.01 |
| 32  | 0.964 ± 0.010 | 0.995 ± 0.002 | 0.999 ± 0.002 | 0.982 ± 0.004 | 0.993 ± 0.003 | 0.965 ± 0.010 | 0.987 ± 0.007 | 0.996 ± 0.003 | 0.998 ± 0.003 | 0.997 ± 0.002 | 0.993 ± 0.005 | 0.994 ± 0.004 | E Europe F    | E Africa F    | 0.01 |
| 33  | 0.980 ± 0.008 | 0.996 ± 0.002 | 0.997 ± 0.002 | 0.987 ± 0.003 | 0.996 ± 0.003 | 0.978 ± 0.008 | 0.995 ± 0.005 | 0.998 ± 0.002 | 0.999 ± 0.002 | 0.997 ± 0.001 | 0.996 ± 0.004 | 0.994 ± 0.004 | E Europe M    | W Africa F    | 0.00 |
| 34  | 0.980 ± 0.008 | 0.998 ± 0.001 | 0.997 ± 0.002 | 0.993 ± 0.002 | 0.997 ± 0.002 | 0.983 ± 0.007 | 0.997 ± 0.003 | 0.998 ± 0.002 | 0.999 ± 0.002 | 0.998 ± 0.001 | 0.997 ± 0.003 | 0.997 ± 0.003 | E Europe M    | E Africa F    | 0.00 |

**Table 180.** For algorithm yoti-001 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.15 ± 0.05   | 0.11 ± 0.01   | 0.08 ± 0.02   | 0.07 ± 0.02   | 0.13 ± 0.03   | 0.17 ± 0.05   | 0.06 ± 0.03   | 0.033 ± 0.007 | 0.012 ± 0.010 | 0.020 ± 0.007 | 0.05 ± 0.02   | 0.07 ± 0.04   | W Africa F    | E Europe M    | 0.37 |
| 15  | 0.16 ± 0.05   | 0.14 ± 0.01   | 0.12 ± 0.03   | 0.10 ± 0.02   | 0.19 ± 0.03   | 0.20 ± 0.05   | 0.09 ± 0.04   | 0.072 ± 0.009 | 0.02 ± 0.01   | 0.026 ± 0.008 | 0.07 ± 0.02   | 0.14 ± 0.05   | W Africa F    | E Europe M    | 0.31 |
| 16  | 0.21 ± 0.05   | 0.20 ± 0.01   | 0.18 ± 0.03   | 0.15 ± 0.02   | 0.28 ± 0.03   | 0.25 ± 0.05   | 0.16 ± 0.05   | 0.11 ± 0.01   | 0.04 ± 0.01   | 0.05 ± 0.01   | 0.12 ± 0.03   | 0.18 ± 0.04   | S E Asia F    | E Europe M    | 0.27 |
| 17  | 0.23 ± 0.05   | 0.28 ± 0.01   | 0.26 ± 0.03   | 0.21 ± 0.02   | 0.34 ± 0.03   | 0.29 ± 0.06   | 0.22 ± 0.06   | 0.19 ± 0.01   | 0.06 ± 0.02   | 0.12 ± 0.01   | 0.20 ± 0.03   | 0.28 ± 0.05   | S E Asia F    | E Europe M    | 0.19 |
| 18  | 0.36 ± 0.05   | 0.34 ± 0.01   | 0.33 ± 0.03   | 0.27 ± 0.02   | 0.44 ± 0.02   | 0.38 ± 0.05   | 0.33 ± 0.06   | 0.28 ± 0.01   | 0.13 ± 0.02   | 0.15 ± 0.02   | 0.32 ± 0.03   | 0.33 ± 0.04   | S E Asia F    | E Europe M    | 0.16 |
| 19  | 0.40 ± 0.04   | 0.42 ± 0.01   | 0.40 ± 0.02   | 0.35 ± 0.02   | 0.48 ± 0.02   | 0.44 ± 0.05   | 0.39 ± 0.05   | 0.37 ± 0.01   | 0.20 ± 0.03   | 0.26 ± 0.02   | 0.41 ± 0.03   | 0.43 ± 0.04   | S E Asia F    | E Europe M    | 0.11 |
| 20  | 0.49 ± 0.04   | 0.51 ± 0.01   | 0.47 ± 0.02   | 0.43 ± 0.02   | 0.56 ± 0.02   | 0.50 ± 0.04   | 0.52 ± 0.04   | 0.49 ± 0.01   | 0.31 ± 0.03   | 0.36 ± 0.02   | 0.54 ± 0.03   | 0.53 ± 0.04   | S E Asia F    | E Europe M    | 0.09 |
| 21  | 0.54 ± 0.04   | 0.58 ± 0.01   | 0.55 ± 0.02   | 0.49 ± 0.02   | 0.63 ± 0.02   | 0.60 ± 0.04   | 0.61 ± 0.04   | 0.61 ± 0.01   | 0.47 ± 0.03   | 0.45 ± 0.02   | 0.63 ± 0.02   | 0.66 ± 0.04   | W Africa M    | S Asia M      | 0.07 |
| 22  | 0.62 ± 0.04   | 0.643 ± 0.009 | 0.61 ± 0.02   | 0.56 ± 0.01   | 0.68 ± 0.02   | 0.66 ± 0.03   | 0.73 ± 0.04   | 0.72 ± 0.01   | 0.58 ± 0.03   | 0.58 ± 0.02   | 0.74 ± 0.02   | 0.73 ± 0.03   | S E Asia M    | S Asia F      | 0.06 |
| 23  | 0.62 ± 0.03   | 0.719 ± 0.008 | 0.69 ± 0.02   | 0.62 ± 0.01   | 0.76 ± 0.02   | 0.67 ± 0.03   | 0.76 ± 0.03   | 0.79 ± 0.01   | 0.71 ± 0.02   | 0.67 ± 0.01   | 0.82 ± 0.02   | 0.80 ± 0.03   | S E Asia M    | S Asia F      | 0.06 |
| 24  | 0.70 ± 0.03   | 0.759 ± 0.007 | 0.75 ± 0.01   | 0.67 ± 0.01   | 0.79 ± 0.01   | 0.71 ± 0.03   | 0.82 ± 0.03   | 0.845 ± 0.009 | 0.80 ± 0.02   | 0.76 ± 0.01   | 0.86 ± 0.02   | 0.86 ± 0.02   | S E Asia M    | S Asia F      | 0.05 |
| 25  | 0.72 ± 0.03   | 0.810 ± 0.007 | 0.81 ± 0.01   | 0.706 ± 0.010 | 0.83 ± 0.01   | 0.75 ± 0.02   | 0.90 ± 0.02   | 0.902 ± 0.006 | 0.87 ± 0.01   | 0.835 ± 0.009 | 0.91 ± 0.01   | 0.90 ± 0.02   | S E Asia M    | S Asia F      | 0.05 |
| 26  | 0.77 ± 0.03   | 0.850 ± 0.006 | 0.840 ± 0.009 | 0.748 ± 0.008 | 0.87 ± 0.01   | 0.80 ± 0.02   | 0.91 ± 0.02   | 0.932 ± 0.005 | 0.90 ± 0.01   | 0.877 ± 0.007 | 0.94 ± 0.01   | 0.93 ± 0.01   | S E Asia M    | S Asia F      | 0.04 |
| 27  | 0.81 ± 0.02   | 0.881 ± 0.005 | 0.885 ± 0.008 | 0.783 ± 0.008 | 0.893 ± 0.009 | 0.83 ± 0.02   | 0.94 ± 0.01   | 0.951 ± 0.005 | 0.937 ± 0.009 | 0.908 ± 0.006 | 0.959 ± 0.009 | 0.94 ± 0.01   | S E Asia M    | S Asia F      | 0.04 |
| 28  | 0.83 ± 0.02   | 0.908 ± 0.004 | 0.914 ± 0.007 | 0.815 ± 0.007 | 0.913 ± 0.008 | 0.83 ± 0.02   | 0.96 ± 0.01   | 0.964 ± 0.004 | 0.965 ± 0.006 | 0.936 ± 0.005 | 0.968 ± 0.008 | 0.959 ± 0.010 | S E Asia M    | S Asia F      | 0.04 |
| 29  | 0.88 ± 0.02   | 0.929 ± 0.004 | 0.938 ± 0.006 | 0.833 ± 0.006 | 0.928 ± 0.009 | 0.87 ± 0.01   | 0.978 ± 0.009 | 0.978 ± 0.003 | 0.968 ± 0.006 | 0.950 ± 0.004 | 0.978 ± 0.007 | 0.974 ± 0.007 | S E Asia M    | S Asia F      | 0.03 |
| 30  | 0.88 ± 0.02   | 0.944 ± 0.003 | 0.954 ± 0.005 | 0.858 ± 0.006 | 0.944 ± 0.007 | 0.88 ± 0.02   | 0.978 ± 0.010 | 0.983 ± 0.003 | 0.977 ± 0.005 | 0.967 ± 0.003 | 0.986 ± 0.006 | 0.977 ± 0.006 | S E Asia M    | S Asia F      | 0.03 |
| 31  | 0.90 ± 0.02   | 0.957 ± 0.003 | 0.966 ± 0.004 | 0.888 ± 0.005 | 0.946 ± 0.007 | 0.90 ± 0.01   | 0.984 ± 0.007 | 0.990 ± 0.002 | 0.989 ± 0.003 | 0.975 ± 0.002 | 0.991 ± 0.005 | 0.987 ± 0.005 | S E Asia M    | S Asia F      | 0.02 |
| 32  | 0.92 ± 0.02   | 0.965 ± 0.003 | 0.976 ± 0.004 | 0.904 ± 0.005 | 0.965 ± 0.005 | 0.91 ± 0.01   | 0.986 ± 0.007 | 0.990 ± 0.002 | 0.990 ± 0.003 | 0.982 ± 0.002 | 0.984 ± 0.006 | 0.987 ± 0.004 | E Asia M      | S Asia F      | 0.02 |
| 33  | 0.93 ± 0.02   | 0.972 ± 0.002 | 0.979 ± 0.004 | 0.912 ± 0.004 | 0.968 ± 0.006 | 0.93 ± 0.01   | 0.993 ± 0.005 | 0.993 ± 0.002 | 0.996 ± 0.002 | 0.987 ± 0.002 | 0.992 ± 0.004 | 0.994 ± 0.003 | E Europe M    | S Asia F      | 0.02 |
| 34  | 0.93 ± 0.01   | 0.977 ± 0.002 | 0.984 ± 0.003 | 0.930 ± 0.004 | 0.975 ± 0.005 | 0.95 ± 0.01   | 0.993 ± 0.005 | 0.994 ± 0.002 | 0.997 ± 0.002 | 0.990 ± 0.001 | 0.995 ± 0.003 | 0.993 ± 0.004 | E Europe M    | S Asia F      | 0.01 |

**Table 181.** For algorithm yoti-002 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.013 ± 0.009 | 0.021 ± 0.007 | 0.007 ± 0.007 | 0.014 ± 0.006 | 0.04 ± 0.02   | 0.03 ± 0.02   | 0.006 ± 0.006 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.003 ± 0.002 | 0.002 ± 0.002 | 0.000 ± 0.000 | S E Asia F    | E Europe M    | 0.60 |
| 15  | 0.03 ± 0.02   | 0.044 ± 0.010 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.04 ± 0.02   | 0.05 ± 0.03   | 0.008 ± 0.008 | 0.006 ± 0.004 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.005 ± 0.005 | 0.005 ± 0.005 | W Africa F    | E Europe M    | 0.52 |
| 16  | 0.07 ± 0.03   | 0.07 ± 0.01   | 0.04 ± 0.02   | 0.07 ± 0.02   | 0.13 ± 0.03   | 0.11 ± 0.04   | 0.02 ± 0.01   | 0.011 ± 0.005 | 0.003 ± 0.003 | 0.008 ± 0.006 | 0.011 ± 0.008 | 0.009 ± 0.009 | S E Asia F    | E Europe M    | 0.54 |
| 17  | 0.10 ± 0.03   | 0.12 ± 0.02   | 0.06 ± 0.03   | 0.14 ± 0.02   | 0.16 ± 0.03   | 0.18 ± 0.04   | 0.012 ± 0.009 | 0.028 ± 0.008 | 0.006 ± 0.006 | 0.018 ± 0.007 | 0.011 ± 0.008 | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.51 |
| 18  | 0.12 ± 0.03   | 0.16 ± 0.02   | 0.09 ± 0.02   | 0.18 ± 0.02   | 0.22 ± 0.03   | 0.19 ± 0.04   | 0.04 ± 0.02   | 0.043 ± 0.008 | 0.005 ± 0.005 | 0.043 ± 0.009 | 0.03 ± 0.01   | 0.07 ± 0.02   | S E Asia F    | E Europe M    | 0.43 |
| 19  | 0.22 ± 0.03   | 0.24 ± 0.02   | 0.15 ± 0.03   | 0.29 ± 0.02   | 0.33 ± 0.03   | 0.29 ± 0.04   | 0.11 ± 0.03   | 0.10 ± 0.01   | 0.02 ± 0.01   | 0.08 ± 0.01   | 0.08 ± 0.02   | 0.15 ± 0.03   | S E Asia F    | E Europe M    | 0.34 |
| 20  | 0.29 ± 0.04   | 0.37 ± 0.02   | 0.19 ± 0.03   | 0.39 ± 0.02   | 0.45 ± 0.03   | 0.38 ± 0.04   | 0.16 ± 0.03   | 0.16 ± 0.01   | 0.06 ± 0.02   | 0.16 ± 0.02   | 0.16 ± 0.02   | 0.19 ± 0.03   | S E Asia F    | E Europe M    | 0.28 |
| 21  | 0.36 ± 0.03   | 0.49 ± 0.02   | 0.32 ± 0.03   | 0.51 ± 0.02   | 0.57 ± 0.03   | 0.46 ± 0.04   | 0.27 ± 0.03   | 0.29 ± 0.02   | 0.14 ± 0.03   | 0.28 ± 0.02   | 0.29 ± 0.03   | 0.30 ± 0.04   | S E Asia F    | E Europe M    | 0.20 |
| 22  | 0.47 ± 0.03   | 0.60 ± 0.02   | 0.43 ± 0.03   | 0.62 ± 0.02   | 0.67 ± 0.02   | 0.55 ± 0.04   | 0.46 ± 0.03   | 0.44 ± 0.02   | 0.29 ± 0.03   | 0.43 ± 0.02   | 0.40 ± 0.03   | 0.36 ± 0.04   | S E Asia F    | E Europe M    | 0.14 |
| 23  | 0.56 ± 0.03   | 0.71 ± 0.02   | 0.53 ± 0.02   | 0.71 ± 0.02   | 0.77 ± 0.02   | 0.66 ± 0.03   | 0.54 ± 0.03   | 0.58 ± 0.02   | 0.43 ± 0.03   | 0.61 ± 0.02   | 0.57 ± 0.03   | 0.55 ± 0.04   | S E Asia F    | E Europe M    | 0.09 |
| 24  | 0.64 ± 0.03   | 0.80 ± 0.01   | 0.66 ± 0.02   | 0.79 ± 0.01   | 0.84 ± 0.02   | 0.70 ± 0.03   | 0.66 ± 0.03   | 0.72 ± 0.02   | 0.63 ± 0.03   | 0.73 ± 0.02   | 0.71 ± 0.02   | 0.66 ± 0.03   | S E Asia F    | E Europe M    | 0.06 |
| 25  | 0.70 ± 0.02   | 0.86 ± 0.01   | 0.77 ± 0.02   | 0.84 ± 0.01   | 0.89 ± 0.01   | 0.78 ± 0.03   | 0.75 ± 0.03   | 0.82 ± 0.01   | 0.78 ± 0.02   | 0.85 ± 0.01   | 0.82 ± 0.02   | 0.76 ± 0.02   | S E Asia F    | E Africa F    | 0.04 |
| 26  | 0.77 ± 0.02   | 0.915 ± 0.007 | 0.85 ± 0.01   | 0.881 ± 0.009 | 0.936 ± 0.009 | 0.83 ± 0.02   | 0.84 ± 0.02   | 0.91 ± 0.01   | 0.88 ± 0.02   | 0.908 ± 0.009 | 0.89 ± 0.02   | 0.83 ± 0.02   | S E Asia F    | E Africa F    | 0.03 |
| 27  | 0.82 ± 0.02   | 0.950 ± 0.006 | 0.91 ± 0.01   | 0.912 ± 0.008 | 0.953 ± 0.007 | 0.85 ± 0.02   | 0.89 ± 0.02   | 0.947 ± 0.009 | 0.94 ± 0.01   | 0.948 ± 0.007 | 0.92 ± 0.02   | 0.89 ± 0.02   | S E Asia F    | E Africa F    | 0.03 |
| 28  | 0.85 ± 0.02   | 0.971 ± 0.004 | 0.954 ± 0.008 | 0.938 ± 0.006 | 0.969 ± 0.006 | 0.91 ± 0.02   | 0.94 ± 0.01   | 0.969 ± 0.006 | 0.968 ± 0.009 | 0.965 ± 0.006 | 0.97 ± 0.01   | 0.93 ± 0.01   | E Asia F      | E Africa F    | 0.02 |
| 29  | 0.92 ± 0.02   | 0.983 ± 0.003 | 0.977 ± 0.006 | 0.951 ± 0.006 | 0.981 ± 0.005 | 0.93 ± 0.01   | 0.96 ± 0.01   | 0.983 ± 0.005 | 0.986 ± 0.007 | 0.982 ± 0.004 | 0.981 ± 0.008 | 0.958 ± 0.010 | E Europe M    | E Africa F    | 0.01 |
| 30  | 0.92 ± 0.01   | 0.989 ± 0.003 | 0.983 ± 0.003 | 0.967 ± 0.005 | 0.987 ± 0.004 | 0.94 ± 0.01   | 0.97 ± 0.01   | 0.992 ± 0.003 | 0.994 ± 0.004 | 0.987 ± 0.003 | 0.987 ± 0.007 | 0.973 ± 0.008 | E Europe M    | E Africa F    | 0.01 |
| 31  | 0.93 ± 0.01   | 0.995 ± 0.002 | 0.991 ± 0.004 | 0.977 ± 0.003 | 0.988 ± 0.004 | 0.95 ± 0.01   | 0.990 ± 0.006 | 0.994 ± 0.003 | 0.994 ± 0.004 | 0.993 ± 0.002 | 0.989 ± 0.007 | 0.987 ± 0.005 | E Asia F      | E Africa F    | 0.01 |
| 32  | 0.96 ± 0.01   | 0.997 ± 0.002 | 0.997 ± 0.002 | 0.987 ± 0.003 | 0.991 ± 0.004 | 0.967 ± 0.010 | 0.988 ± 0.006 | 0.998 ± 0.002 | 0.998 ± 0.003 | 0.996 ± 0.002 | 0.992 ± 0.005 | 0.990 ± 0.005 | E Asia M      | E Africa F    | 0.01 |
| 33  | 0.973 ± 0.009 | 0.998 ± 0.001 | 0.994 ± 0.003 | 0.988 ± 0.003 | 0.998 ± 0.002 | 0.984 ± 0.007 | 0.996 ± 0.003 | 0.998 ± 0.002 | 0.999 ± 0.002 | 0.997 ± 0.001 | 0.997 ± 0.004 | 0.991 ± 0.004 | E Europe M    | E Africa F    | 0.00 |
| 34  | 0.97 ± 0.01   | 0.998 ± 0.001 | 0.997 ± 0.002 | 0.992 ± 0.002 | 0.999 ± 0.002 | 0.988 ± 0.006 | 0.998 ± 0.002 | 0.998 ± 0.002 | 0.999 ± 0.001 | 0.997 ± 0.001 | 0.998 ± 0.003 | 0.997 ± 0.003 | E Europe M    | E Africa F    | 0.00 |

**Table 182.** For algorithm yoti-002 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.08 ± 0.04   | 0.12 ± 0.01   | 0.04 ± 0.02   | 0.05 ± 0.01   | 0.19 ± 0.04   | 0.07 ± 0.04   | 0.02 ± 0.02   | 0.021 ± 0.005 | 0.007 ± 0.007 | 0.012 ± 0.006 | 0.03 ± 0.01   | 0.06 ± 0.04   | S E Asia F    | E Europe M    | 0.49 |
| 15  | 0.09 ± 0.04   | 0.18 ± 0.01   | 0.07 ± 0.02   | 0.07 ± 0.02   | 0.23 ± 0.03   | 0.17 ± 0.05   | 0.08 ± 0.04   | 0.051 ± 0.008 | 0.02 ± 0.01   | 0.015 ± 0.006 | 0.04 ± 0.02   | 0.08 ± 0.04   | S E Asia F    | S Asia M      | 0.41 |
| 16  | 0.20 ± 0.05   | 0.25 ± 0.01   | 0.11 ± 0.02   | 0.12 ± 0.02   | 0.34 ± 0.04   | 0.17 ± 0.05   | 0.07 ± 0.03   | 0.081 ± 0.010 | 0.018 ± 0.010 | 0.027 ± 0.007 | 0.07 ± 0.02   | 0.08 ± 0.03   | S E Asia F    | E Europe M    | 0.42 |
| 17  | 0.21 ± 0.05   | 0.34 ± 0.01   | 0.16 ± 0.02   | 0.15 ± 0.02   | 0.39 ± 0.03   | 0.26 ± 0.05   | 0.17 ± 0.05   | 0.15 ± 0.01   | 0.03 ± 0.01   | 0.08 ± 0.01   | 0.15 ± 0.02   | 0.20 ± 0.05   | S E Asia F    | E Europe M    | 0.30 |
| 18  | 0.30 ± 0.05   | 0.41 ± 0.01   | 0.24 ± 0.02   | 0.23 ± 0.02   | 0.50 ± 0.03   | 0.29 ± 0.05   | 0.27 ± 0.06   | 0.22 ± 0.01   | 0.08 ± 0.02   | 0.10 ± 0.01   | 0.22 ± 0.03   | 0.23 ± 0.04   | S E Asia F    | E Europe M    | 0.24 |
| 19  | 0.35 ± 0.04   | 0.50 ± 0.01   | 0.29 ± 0.02   | 0.30 ± 0.02   | 0.54 ± 0.02   | 0.39 ± 0.04   | 0.34 ± 0.05   | 0.32 ± 0.01   | 0.15 ± 0.02   | 0.19 ± 0.02   | 0.31 ± 0.03   | 0.34 ± 0.04   | S E Asia F    | E Europe M    | 0.19 |
| 20  | 0.47 ± 0.04   | 0.60 ± 0.01   | 0.37 ± 0.02   | 0.36 ± 0.02   | 0.63 ± 0.02   | 0.50 ± 0.04   | 0.45 ± 0.04   | 0.43 ± 0.01   | 0.26 ± 0.02   | 0.28 ± 0.02   | 0.44 ± 0.03   | 0.45 ± 0.04   | S E Asia F    | E Europe M    | 0.15 |
| 21  | 0.51 ± 0.04   | 0.669 ± 0.010 | 0.45 ± 0.02   | 0.43 ± 0.02   | 0.69 ± 0.02   | 0.55 ± 0.04   | 0.55 ± 0.04   | 0.56 ± 0.01   | 0.43 ± 0.03   | 0.38 ± 0.02   | 0.56 ± 0.02   | 0.58 ± 0.04   | S E Asia F    | S Asia M      | 0.10 |
| 22  | 0.63 ± 0.04   | 0.733 ± 0.008 | 0.53 ± 0.02   | 0.51 ± 0.02   | 0.74 ± 0.02   | 0.64 ± 0.03   | 0.68 ± 0.04   | 0.68 ± 0.01   | 0.56 ± 0.03   | 0.51 ± 0.02   | 0.68 ± 0.02   | 0.65 ± 0.04   | S E Asia F    | S Asia M      | 0.08 |
| 23  | 0.62 ± 0.04   | 0.798 ± 0.008 | 0.62 ± 0.02   | 0.56 ± 0.01   | 0.82 ± 0.01   | 0.66 ± 0.03   | 0.72 ± 0.04   | 0.77 ± 0.01   | 0.70 ± 0.02   | 0.61 ± 0.01   | 0.76 ± 0.02   | 0.76 ± 0.03   | S E Asia F    | S Asia F      | 0.07 |
| 24  | 0.69 ± 0.03   | 0.837 ± 0.006 | 0.70 ± 0.01   | 0.63 ± 0.01   | 0.85 ± 0.01   | 0.71 ± 0.03   | 0.81 ± 0.03   | 0.833 ± 0.008 | 0.81 ± 0.02   | 0.72 ± 0.01   | 0.84 ± 0.02   | 0.82 ± 0.02   | S E Asia F    | S Asia F      | 0.06 |
| 25  | 0.72 ± 0.03   | 0.880 ± 0.005 | 0.77 ± 0.01   | 0.668 ± 0.010 | 0.87 ± 0.01   | 0.75 ± 0.02   | 0.86 ± 0.02   | 0.899 ± 0.007 | 0.87 ± 0.01   | 0.803 ± 0.010 | 0.89 ± 0.02   | 0.88 ± 0.02   | E Asia M      | S Asia F      | 0.05 |
| 26  | 0.78 ± 0.02   | 0.906 ± 0.005 | 0.81 ± 0.01   | 0.712 ± 0.008 | 0.909 ± 0.009 | 0.80 ± 0.02   | 0.91 ± 0.02   | 0.936 ± 0.005 | 0.92 ± 0.01   | 0.856 ± 0.009 | 0.92 ± 0.01   | 0.91 ± 0.01   | E Asia M      | S Asia F      | 0.05 |
| 27  | 0.80 ± 0.02   | 0.933 ± 0.004 | 0.869 ± 0.008 | 0.753 ± 0.008 | 0.918 ± 0.008 | 0.83 ± 0.02   | 0.93 ± 0.02   | 0.960 ± 0.004 | 0.948 ± 0.008 | 0.900 ± 0.006 | 0.95 ± 0.01   | 0.93 ± 0.01   | E Asia M      | S Asia F      | 0.04 |
| 28  | 0.85 ± 0.02   | 0.951 ± 0.003 | 0.909 ± 0.007 | 0.784 ± 0.007 | 0.940 ± 0.008 | 0.85 ± 0.02   | 0.95 ± 0.01   | 0.972 ± 0.003 | 0.972 ± 0.006 | 0.929 ± 0.005 | 0.963 ± 0.009 | 0.96 ± 0.01   | E Europe M    | S Asia F      | 0.04 |
| 29  | 0.87 ± 0.02   | 0.966 ± 0.003 | 0.935 ± 0.006 | 0.813 ± 0.007 | 0.955 ± 0.006 | 0.88 ± 0.01   | 0.97 ± 0.01   | 0.984 ± 0.003 | 0.978 ± 0.005 | 0.947 ± 0.004 | 0.982 ± 0.007 | 0.971 ± 0.006 | E Asia M      | S Asia F      | 0.03 |
| 30  | 0.91 ± 0.02   | 0.975 ± 0.002 | 0.951 ± 0.006 | 0.847 ± 0.006 | 0.963 ± 0.006 | 0.91 ± 0.01   | 0.980 ± 0.009 | 0.989 ± 0.002 | 0.987 ± 0.004 | 0.962 ± 0.003 | 0.986 ± 0.006 | 0.974 ± 0.006 | E Asia M      | S Asia F      | 0.02 |
| 31  | 0.90 ± 0.02   | 0.982 ± 0.002 | 0.968 ± 0.004 | 0.866 ± 0.006 | 0.970 ± 0.005 | 0.93 ± 0.01   | 0.984 ± 0.007 | 0.994 ± 0.002 | 0.994 ± 0.003 | 0.974 ± 0.002 | 0.992 ± 0.005 | 0.985 ± 0.005 | E Europe M    | S Asia F      | 0.02 |
| 32  | 0.91 ± 0.02   | 0.988 ± 0.002 | 0.979 ± 0.004 | 0.888 ± 0.005 | 0.978 ± 0.005 | 0.93 ± 0.01   | 0.990 ± 0.006 | 0.996 ± 0.001 | 0.994 ± 0.003 | 0.981 ± 0.002 | 0.992 ± 0.004 | 0.986 ± 0.005 | E Asia M      | S Asia F      | 0.02 |
| 33  | 0.93 ± 0.01   | 0.991 ± 0.001 | 0.983 ± 0.003 | 0.903 ± 0.005 | 0.985 ± 0.004 | 0.94 ± 0.01   | 0.993 ± 0.005 | 0.997 ± 0.001 | 0.996 ± 0.002 | 0.986 ± 0.002 | 0.993 ± 0.004 | 0.994 ± 0.003 | E Asia M      | S Asia F      | 0.02 |
| 34  | 0.94 ± 0.01   | 0.992 ± 0.001 | 0.988 ± 0.003 | 0.924 ± 0.005 | 0.986 ± 0.004 | 0.96 ± 0.01   | 0.992 ± 0.005 | 0.997 ± 0.001 | 0.999 ± 0.001 | 0.990 ± 0.001 | 0.997 ± 0.003 | 0.994 ± 0.003 | E Europe M    | S Asia F      | 0.01 |



**Table 183.** For algorithm yoti-003 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.013 ± 0.009 | 0.018 ± 0.006 | 0.005 ± 0.005 | 0.008 ± 0.005 | 0.04 ± 0.02   | 0.02 ± 0.02   | 0.002 ± 0.002 | 0.008 ± 0.003 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.004 ± 0.004 | S E Asia F    | E Europe M    | 0.61 |
| 15  | 0.03 ± 0.02   | 0.04 ± 0.01   | 0.02 ± 0.01   | 0.026 ± 0.010 | 0.05 ± 0.02   | 0.01 ± 0.01   | 0.005 ± 0.005 | 0.004 ± 0.003 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.002 ± 0.002 | 0.005 ± 0.005 | S E Asia F    | E Europe M    | 0.56 |
| 16  | 0.07 ± 0.02   | 0.06 ± 0.01   | 0.04 ± 0.02   | 0.04 ± 0.01   | 0.10 ± 0.03   | 0.10 ± 0.03   | 0.02 ± 0.01   | 0.010 ± 0.005 | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.008 ± 0.008 | 0.009 ± 0.009 | W Africa F    | E Europe M    | 0.54 |
| 17  | 0.11 ± 0.03   | 0.12 ± 0.02   | 0.05 ± 0.02   | 0.12 ± 0.02   | 0.14 ± 0.03   | 0.14 ± 0.04   | 0.012 ± 0.009 | 0.020 ± 0.006 | 0.006 ± 0.006 | 0.016 ± 0.007 | 0.03 ± 0.02   | 0.04 ± 0.02   | W Africa F    | E Europe M    | 0.46 |
| 18  | 0.14 ± 0.03   | 0.15 ± 0.01   | 0.05 ± 0.02   | 0.14 ± 0.02   | 0.19 ± 0.03   | 0.16 ± 0.03   | 0.03 ± 0.01   | 0.041 ± 0.009 | 0.005 ± 0.005 | 0.037 ± 0.009 | 0.03 ± 0.01   | 0.06 ± 0.02   | S E Asia F    | E Europe M    | 0.42 |
| 19  | 0.27 ± 0.03   | 0.24 ± 0.02   | 0.10 ± 0.02   | 0.25 ± 0.02   | 0.30 ± 0.03   | 0.25 ± 0.04   | 0.10 ± 0.02   | 0.09 ± 0.01   | 0.02 ± 0.01   | 0.08 ± 0.01   | 0.08 ± 0.02   | 0.15 ± 0.03   | S E Asia F    | E Europe M    | 0.34 |
| 20  | 0.34 ± 0.04   | 0.37 ± 0.02   | 0.15 ± 0.02   | 0.35 ± 0.02   | 0.44 ± 0.03   | 0.34 ± 0.04   | 0.17 ± 0.03   | 0.16 ± 0.02   | 0.07 ± 0.02   | 0.17 ± 0.02   | 0.18 ± 0.03   | 0.17 ± 0.03   | S E Asia F    | E Europe M    | 0.28 |
| 21  | 0.44 ± 0.04   | 0.49 ± 0.02   | 0.26 ± 0.03   | 0.48 ± 0.02   | 0.56 ± 0.03   | 0.45 ± 0.04   | 0.26 ± 0.03   | 0.30 ± 0.02   | 0.12 ± 0.03   | 0.29 ± 0.02   | 0.31 ± 0.03   | 0.26 ± 0.04   | S E Asia F    | E Europe M    | 0.21 |
| 22  | 0.53 ± 0.03   | 0.60 ± 0.02   | 0.37 ± 0.03   | 0.59 ± 0.02   | 0.66 ± 0.02   | 0.52 ± 0.04   | 0.43 ± 0.04   | 0.44 ± 0.02   | 0.28 ± 0.03   | 0.43 ± 0.02   | 0.43 ± 0.03   | 0.34 ± 0.04   | S E Asia F    | E Europe M    | 0.14 |
| 23  | 0.62 ± 0.03   | 0.71 ± 0.02   | 0.48 ± 0.03   | 0.70 ± 0.02   | 0.77 ± 0.02   | 0.63 ± 0.03   | 0.51 ± 0.03   | 0.59 ± 0.02   | 0.45 ± 0.04   | 0.61 ± 0.02   | 0.62 ± 0.03   | 0.53 ± 0.04   | S E Asia F    | E Europe M    | 0.10 |
| 24  | 0.72 ± 0.03   | 0.79 ± 0.01   | 0.65 ± 0.02   | 0.79 ± 0.01   | 0.85 ± 0.01   | 0.69 ± 0.03   | 0.64 ± 0.03   | 0.73 ± 0.02   | 0.66 ± 0.03   | 0.75 ± 0.02   | 0.78 ± 0.02   | 0.64 ± 0.03   | S E Asia F    | E Africa M    | 0.06 |
| 25  | 0.77 ± 0.02   | 0.877 ± 0.009 | 0.75 ± 0.02   | 0.84 ± 0.01   | 0.89 ± 0.01   | 0.77 ± 0.03   | 0.75 ± 0.03   | 0.85 ± 0.02   | 0.79 ± 0.02   | 0.86 ± 0.01   | 0.86 ± 0.02   | 0.74 ± 0.03   | S E Asia F    | W Africa M    | 0.04 |
| 26  | 0.81 ± 0.02   | 0.919 ± 0.008 | 0.85 ± 0.01   | 0.883 ± 0.008 | 0.942 ± 0.009 | 0.83 ± 0.02   | 0.84 ± 0.02   | 0.92 ± 0.01   | 0.88 ± 0.02   | 0.920 ± 0.009 | 0.93 ± 0.02   | 0.83 ± 0.02   | S E Asia F    | E Africa F    | 0.03 |
| 27  | 0.88 ± 0.02   | 0.955 ± 0.006 | 0.90 ± 0.01   | 0.924 ± 0.007 | 0.963 ± 0.007 | 0.88 ± 0.02   | 0.88 ± 0.02   | 0.954 ± 0.008 | 0.95 ± 0.01   | 0.960 ± 0.007 | 0.96 ± 0.01   | 0.89 ± 0.02   | S E Asia M    | W Africa F    | 0.02 |
| 28  | 0.91 ± 0.02   | 0.976 ± 0.004 | 0.951 ± 0.008 | 0.946 ± 0.005 | 0.977 ± 0.006 | 0.91 ± 0.01   | 0.93 ± 0.01   | 0.980 ± 0.005 | 0.973 ± 0.008 | 0.977 ± 0.005 | 0.987 ± 0.007 | 0.93 ± 0.02   | S E Asia M    | E Africa F    | 0.02 |
| 29  | 0.94 ± 0.01   | 0.987 ± 0.003 | 0.975 ± 0.006 | 0.960 ± 0.005 | 0.986 ± 0.005 | 0.93 ± 0.01   | 0.95 ± 0.01   | 0.988 ± 0.004 | 0.990 ± 0.006 | 0.991 ± 0.003 | 0.993 ± 0.005 | 0.961 ± 0.010 | S E Asia M    | W Africa F    | 0.01 |
| 30  | 0.96 ± 0.01   | 0.993 ± 0.002 | 0.985 ± 0.005 | 0.977 ± 0.004 | 0.992 ± 0.003 | 0.96 ± 0.01   | 0.973 ± 0.009 | 0.996 ± 0.002 | 0.993 ± 0.002 | 0.994 ± 0.005 | 0.997 ± 0.008 | 0.973 ± 0.008 | E Europe M    | E Africa F    | 0.01 |
| 31  | 0.96 ± 0.01   | 0.996 ± 0.002 | 0.993 ± 0.004 | 0.983 ± 0.003 | 0.996 ± 0.003 | 0.96 ± 0.01   | 0.988 ± 0.006 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.996 ± 0.002 | 0.994 ± 0.005 | 0.987 ± 0.006 | E Asia M      | W Africa F    | 0.01 |
| 32  | 0.972 ± 0.009 | 0.998 ± 0.001 | 0.998 ± 0.002 | 0.990 ± 0.003 | 0.997 ± 0.002 | 0.976 ± 0.008 | 0.988 ± 0.006 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.998 ± 0.001 | 0.995 ± 0.004 | 0.991 ± 0.005 | E Asia M      | E Africa F    | 0.00 |
| 33  | 0.984 ± 0.007 | 0.999 ± 0.001 | 0.996 ± 0.003 | 0.992 ± 0.002 | 1.000 ± 0.000 | 0.989 ± 0.006 | 0.998 ± 0.003 | 1.000 ± 0.001 | 0.999 ± 0.002 | 0.998 ± 0.001 | 0.998 ± 0.003 | 0.993 ± 0.004 | S E Asia F    | E Africa F    | 0.00 |
| 34  | 0.986 ± 0.007 | 0.999 ± 0.001 | 0.998 ± 0.002 | 0.996 ± 0.002 | 1.000 ± 0.001 | 0.987 ± 0.007 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.999 ± 0.001 | 0.999 ± 0.001 | 1.000 ± 0.000 | 0.998 ± 0.003 | S E Asia M    | E Africa F    | 0.00 |

**Table 184.** For algorithm yoti-003 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.09 ± 0.04   | 0.13 ± 0.01   | 0.03 ± 0.02   | 0.05 ± 0.01   | 0.19 ± 0.04   | 0.07 ± 0.04   | 0.06 ± 0.03   | 0.041 ± 0.008 | 0.010 ± 0.007 | 0.008 ± 0.004 | 0.03 ± 0.01   | 0.04 ± 0.03   | S E Asia F    | S Asia M      | 0.46 |
| 15  | 0.11 ± 0.04   | 0.19 ± 0.01   | 0.08 ± 0.02   | 0.08 ± 0.02   | 0.24 ± 0.04   | 0.16 ± 0.05   | 0.07 ± 0.04   | 0.067 ± 0.009 | 0.007 ± 0.007 | 0.022 ± 0.007 | 0.05 ± 0.02   | 0.04 ± 0.03   | S E Asia F    | E Europe M    | 0.43 |
| 16  | 0.21 ± 0.05   | 0.24 ± 0.01   | 0.10 ± 0.02   | 0.12 ± 0.02   | 0.33 ± 0.03   | 0.17 ± 0.04   | 0.05 ± 0.03   | 0.103 ± 0.010 | 0.018 ± 0.010 | 0.034 ± 0.008 | 0.10 ± 0.02   | 0.06 ± 0.03   | S E Asia F    | E Europe M    | 0.42 |
| 17  | 0.22 ± 0.04   | 0.32 ± 0.02   | 0.15 ± 0.02   | 0.18 ± 0.02   | 0.40 ± 0.03   | 0.19 ± 0.05   | 0.15 ± 0.05   | 0.16 ± 0.01   | 0.02 ± 0.01   | 0.08 ± 0.01   | 0.15 ± 0.02   | 0.15 ± 0.04   | S E Asia F    | E Europe M    | 0.30 |
| 18  | 0.30 ± 0.05   | 0.40 ± 0.01   | 0.24 ± 0.03   | 0.25 ± 0.02   | 0.48 ± 0.03   | 0.26 ± 0.04   | 0.15 ± 0.04   | 0.23 ± 0.01   | 0.06 ± 0.02   | 0.10 ± 0.01   | 0.24 ± 0.03   | 0.16 ± 0.03   | S E Asia F    | E Europe M    | 0.28 |
| 19  | 0.35 ± 0.05   | 0.49 ± 0.01   | 0.31 ± 0.02   | 0.34 ± 0.02   | 0.50 ± 0.02   | 0.36 ± 0.04   | 0.24 ± 0.05   | 0.35 ± 0.01   | 0.10 ± 0.02   | 0.18 ± 0.01   | 0.36 ± 0.03   | 0.27 ± 0.04   | S E Asia F    | E Europe M    | 0.20 |
| 20  | 0.46 ± 0.04   | 0.58 ± 0.01   | 0.39 ± 0.02   | 0.40 ± 0.02   | 0.62 ± 0.02   | 0.44 ± 0.04   | 0.37 ± 0.04   | 0.46 ± 0.01   | 0.22 ± 0.02   | 0.27 ± 0.02   | 0.49 ± 0.03   | 0.36 ± 0.04   | S E Asia F    | E Europe M    | 0.16 |
| 21  | 0.51 ± 0.04   | 0.652 ± 0.010 | 0.48 ± 0.02   | 0.49 ± 0.02   | 0.66 ± 0.02   | 0.51 ± 0.04   | 0.45 ± 0.04   | 0.59 ± 0.01   | 0.35 ± 0.03   | 0.38 ± 0.02   | 0.58 ± 0.02   | 0.48 ± 0.04   | S E Asia F    | E Europe M    | 0.11 |
| 22  | 0.64 ± 0.04   | 0.724 ± 0.009 | 0.56 ± 0.02   | 0.56 ± 0.01   | 0.73 ± 0.02   | 0.60 ± 0.04   | 0.56 ± 0.04   | 0.70 ± 0.01   | 0.50 ± 0.03   | 0.52 ± 0.02   | 0.70 ± 0.02   | 0.58 ± 0.04   | S E Asia F    | E Europe M    | 0.08 |
| 23  | 0.61 ± 0.04   | 0.789 ± 0.007 | 0.64 ± 0.01   | 0.61 ± 0.01   | 0.79 ± 0.01   | 0.60 ± 0.03   | 0.64 ± 0.04   | 0.795 ± 0.010 | 0.64 ± 0.02   | 0.62 ± 0.02   | 0.79 ± 0.02   | 0.65 ± 0.03   | E Asia M      | W Africa F    | 0.07 |
| 24  | 0.69 ± 0.03   | 0.829 ± 0.007 | 0.73 ± 0.01   | 0.682 ± 0.010 | 0.84 ± 0.01   | 0.64 ± 0.03   | 0.72 ± 0.03   | 0.852 ± 0.009 | 0.77 ± 0.02   | 0.72 ± 0.01   | 0.87 ± 0.02   | 0.76 ± 0.03   | S E Asia M    | W Africa F    | 0.06 |
| 25  | 0.74 ± 0.03   | 0.869 ± 0.006 | 0.80 ± 0.01   | 0.732 ± 0.010 | 0.86 ± 0.01   | 0.72 ± 0.02   | 0.79 ± 0.03   | 0.916 ± 0.006 | 0.86 ± 0.01   | 0.809 ± 0.010 | 0.91 ± 0.01   | 0.82 ± 0.02   | E Asia M      | W Africa F    | 0.05 |
| 26  | 0.80 ± 0.02   | 0.906 ± 0.005 | 0.851 ± 0.008 | 0.776 ± 0.008 | 0.908 ± 0.008 | 0.75 ± 0.02   | 0.87 ± 0.02   | 0.948 ± 0.005 | 0.90 ± 0.01   | 0.870 ± 0.008 | 0.94 ± 0.01   | 0.87 ± 0.02   | E Asia M      | W Africa F    | 0.04 |
| 27  | 0.82 ± 0.02   | 0.934 ± 0.004 | 0.897 ± 0.007 | 0.807 ± 0.007 | 0.931 ± 0.008 | 0.81 ± 0.02   | 0.89 ± 0.02   | 0.969 ± 0.004 | 0.944 ± 0.008 | 0.907 ± 0.006 | 0.964 ± 0.009 | 0.92 ± 0.01   | E Asia M      | W Africa F    | 0.04 |
| 28  | 0.86 ± 0.02   | 0.952 ± 0.003 | 0.929 ± 0.006 | 0.841 ± 0.006 | 0.948 ± 0.007 | 0.84 ± 0.02   | 0.94 ± 0.01   | 0.981 ± 0.003 | 0.970 ± 0.006 | 0.939 ± 0.004 | 0.974 ± 0.008 | 0.93 ± 0.01   | E Asia M      | W Africa F    | 0.03 |
| 29  | 0.89 ± 0.02   | 0.967 ± 0.003 | 0.951 ± 0.005 | 0.864 ± 0.005 | 0.954 ± 0.007 | 0.85 ± 0.02   | 0.96 ± 0.01   | 0.990 ± 0.002 | 0.977 ± 0.005 | 0.960 ± 0.003 | 0.983 ± 0.006 | 0.959 ± 0.007 | E Asia M      | W Africa F    | 0.03 |
| 30  | 0.91 ± 0.02   | 0.979 ± 0.002 | 0.967 ± 0.004 | 0.896 ± 0.005 | 0.973 ± 0.005 | 0.89 ± 0.01   | 0.96 ± 0.01   | 0.994 ± 0.002 | 0.988 ± 0.004 | 0.973 ± 0.003 | 0.992 ± 0.004 | 0.970 ± 0.007 | E Asia M      | W Africa F    | 0.02 |
| 31  | 0.92 ± 0.01   | 0.985 ± 0.002 | 0.977 ± 0.003 | 0.915 ± 0.005 | 0.977 ± 0.005 | 0.91 ± 0.01   | 0.980 ± 0.008 | 0.995 ± 0.001 | 0.993 ± 0.003 | 0.984 ± 0.002 | 0.996 ± 0.003 | 0.976 ± 0.006 | S E Asia M    | W Africa F    | 0.02 |
| 32  | 0.95 ± 0.01   | 0.991 ± 0.001 | 0.987 ± 0.003 | 0.930 ± 0.004 | 0.981 ± 0.004 | 0.91 ± 0.01   | 0.987 ± 0.006 | 0.998 ± 0.001 | 0.995 ± 0.002 | 0.988 ± 0.002 | 0.995 ± 0.004 | 0.987 ± 0.004 | E Asia M      | W Africa F    | 0.01 |
| 33  | 0.94 ± 0.01   | 0.994 ± 0.001 | 0.991 ± 0.002 | 0.946 ± 0.004 | 0.987 ± 0.004 | 0.93 ± 0.01   | 0.993 ± 0.005 | 0.998 ± 0.001 | 0.998 ± 0.001 | 0.991 ± 0.001 | 0.999 ± 0.001 | 0.992 ± 0.004 | S E Asia M    | W Africa F    | 0.01 |
| 34  | 0.97 ± 0.01   | 0.994 ± 0.001 | 0.993 ± 0.002 | 0.958 ± 0.004 | 0.991 ± 0.003 | 0.962 ± 0.010 | 0.997 ± 0.003 | 0.998 ± 0.001 | 0.999 ± 0.001 | 0.995 ± 0.001 | 1.000 ± 0.000 | 0.997 ± 0.002 | S E Asia M    | S Asia F      | 0.01 |

**Table 185.** For algorithm yoti-004 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.007 ± 0.007 | 0.008 ± 0.005 | 0.000 ± 0.000 | 0.004 ± 0.003 | 0.009 ± 0.007 | 0.004 ± 0.004 | 0.002 ± 0.002 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.000 ± 0.000 | S E Asia F    | E Europe F    | 0.59 |
| 15  | 0.01 ± 0.01   | 0.013 ± 0.006 | 0.000 ± 0.000 | 0.015 ± 0.008 | 0.03 ± 0.01   | 0.009 ± 0.009 | 0.010 ± 0.008 | 0.001 ± 0.001 | 0.000 ± 0.000 | 0.003 ± 0.003 | 0.002 ± 0.002 | 0.005 ± 0.005 | S E Asia F    | E Europe F    | 0.55 |
| 16  | 0.04 ± 0.02   | 0.031 ± 0.009 | 0.011 ± 0.008 | 0.02 ± 0.01   | 0.05 ± 0.02   | 0.03 ± 0.02   | 0.01 ± 0.01   | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.003 ± 0.003 | 0.000 ± 0.000 | S E Asia F    | W Africa M    | 0.57 |
| 17  | 0.06 ± 0.02   | 0.05 ± 0.01   | 0.011 ± 0.008 | 0.06 ± 0.01   | 0.09 ± 0.03   | 0.09 ± 0.03   | 0.012 ± 0.009 | 0.011 ± 0.005 | 0.003 ± 0.003 | 0.011 ± 0.006 | 0.011 ± 0.008 | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.50 |
| 18  | 0.08 ± 0.02   | 0.09 ± 0.01   | 0.02 ± 0.01   | 0.08 ± 0.01   | 0.12 ± 0.02   | 0.12 ± 0.03   | 0.02 ± 0.01   | 0.014 ± 0.005 | 0.000 ± 0.000 | 0.022 ± 0.006 | 0.013 ± 0.008 | 0.04 ± 0.02   | S E Asia F    | E Europe M    | 0.48 |
| 19  | 0.17 ± 0.03   | 0.15 ± 0.01   | 0.06 ± 0.02   | 0.15 ± 0.02   | 0.21 ± 0.03   | 0.21 ± 0.04   | 0.08 ± 0.02   | 0.042 ± 0.008 | 0.006 ± 0.006 | 0.040 ± 0.010 | 0.05 ± 0.01   | 0.10 ± 0.03   | S E Asia F    | E Europe M    | 0.39 |
| 20  | 0.22 ± 0.03   | 0.25 ± 0.02   | 0.07 ± 0.02   | 0.23 ± 0.02   | 0.34 ± 0.03   | 0.27 ± 0.04   | 0.13 ± 0.03   | 0.07 ± 0.01   | 0.03 ± 0.01   | 0.11 ± 0.01   | 0.10 ± 0.02   | 0.14 ± 0.03   | S E Asia F    | E Europe M    | 0.34 |
| 21  | 0.32 ± 0.03   | 0.36 ± 0.02   | 0.13 ± 0.02   | 0.35 ± 0.02   | 0.43 ± 0.03   | 0.38 ± 0.04   | 0.21 ± 0.03   | 0.16 ± 0.02   | 0.05 ± 0.02   | 0.19 ± 0.02   | 0.20 ± 0.03   | 0.23 ± 0.04   | S E Asia F    | E Europe M    | 0.27 |
| 22  | 0.41 ± 0.03   | 0.48 ± 0.02   | 0.24 ± 0.02   | 0.46 ± 0.02   | 0.56 ± 0.02   | 0.45 ± 0.04   | 0.35 ± 0.03   | 0.29 ± 0.02   | 0.13 ± 0.02   | 0.31 ± 0.02   | 0.31 ± 0.03   | 0.29 ± 0.03   | S E Asia F    | E Europe M    | 0.20 |
| 23  | 0.51 ± 0.03   | 0.59 ± 0.02   | 0.33 ± 0.02   | 0.59 ± 0.02   | 0.67 ± 0.02   | 0.58 ± 0.03   | 0.42 ± 0.03   | 0.42 ± 0.02   | 0.27 ± 0.03   | 0.47 ± 0.02   | 0.48 ± 0.03   | 0.48 ± 0.04   | S E Asia F    | E Europe M    | 0.14 |
| 24  | 0.62 ± 0.03   | 0.71 ± 0.02   | 0.50 ± 0.02   | 0.67 ± 0.02   | 0.78 ± 0.02   | 0.61 ± 0.03   | 0.56 ± 0.03   | 0.58 ± 0.02   | 0.48 ± 0.03   | 0.63 ± 0.02   | 0.65 ± 0.03   | 0.57 ± 0.03   | S E Asia F    | E Europe M    | 0.08 |
| 25  | 0.67 ± 0.02   | 0.81 ± 0.01   | 0.63 ± 0.02   | 0.75 ± 0.01   | 0.83 ± 0.02   | 0.73 ± 0.03   | 0.67 ± 0.03   | 0.73 ± 0.02   | 0.64 ± 0.03   | 0.77 ± 0.01   | 0.78 ± 0.02   | 0.67 ± 0.03   | S E Asia F    | E Europe F    | 0.06 |
| 26  | 0.75 ± 0.02   | 0.870 ± 0.009 | 0.75 ± 0.02   | 0.817 ± 0.010 | 0.90 ± 0.01   | 0.80 ± 0.02   | 0.77 ± 0.02   | 0.84 ± 0.01   | 0.78 ± 0.02   | 0.86 ± 0.01   | 0.86 ± 0.02   | 0.78 ± 0.02   | S E Asia F    | E Europe F    | 0.04 |
| 27  | 0.82 ± 0.02   | 0.925 ± 0.007 | 0.84 ± 0.01   | 0.872 ± 0.008 | 0.933 ± 0.009 | 0.85 ± 0.02   | 0.83 ± 0.02   | 0.91 ± 0.01   | 0.91 ± 0.02   | 0.919 ± 0.008 | 0.93 ± 0.01   | 0.86 ± 0.02   | S E Asia F    | E Africa F    | 0.03 |
| 28  | 0.86 ± 0.02   | 0.956 ± 0.005 | 0.919 ± 0.010 | 0.907 ± 0.007 | 0.961 ± 0.007 | 0.91 ± 0.02   | 0.89 ± 0.02   | 0.956 ± 0.006 | 0.94 ± 0.01   | 0.958 ± 0.006 | 0.971 ± 0.009 | 0.90 ± 0.02   | S E Asia M    | E Africa F    | 0.02 |
| 29  | 0.92 ± 0.01   | 0.979 ± 0.004 | 0.954 ± 0.007 | 0.933 ± 0.007 | 0.978 ± 0.006 | 0.92 ± 0.01   | 0.94 ± 0.01   | 0.980 ± 0.005 | 0.975 ± 0.009 | 0.977 ± 0.004 | 0.982 ± 0.008 | 0.95 ± 0.01   | S E Asia M    | W Africa F    | 0.01 |
| 30  | 0.94 ± 0.01   | 0.988 ± 0.003 | 0.968 ± 0.006 | 0.957 ± 0.005 | 0.990 ± 0.004 | 0.94 ± 0.01   | 0.96 ± 0.01   | 0.989 ± 0.004 | 0.993 ± 0.004 | 0.984 ± 0.003 | 0.991 ± 0.006 | 0.961 ± 0.009 | E Europe M    | E Africa F    | 0.01 |
| 31  | 0.95 ± 0.01   | 0.994 ± 0.002 | 0.985 ± 0.005 | 0.971 ± 0.004 | 0.991 ± 0.004 | 0.963 ± 0.009 | 0.980 ± 0.008 | 0.995 ± 0.003 | 0.993 ± 0.005 | 0.992 ± 0.002 | 0.990 ± 0.007 | 0.981 ± 0.006 | E Asia M      | E Africa F    | 0.01 |
| 32  | 0.968 ± 0.010 | 0.997 ± 0.001 | 0.995 ± 0.003 | 0.981 ± 0.004 | 0.995 ± 0.003 | 0.969 ± 0.010 | 0.984 ± 0.007 | 0.998 ± 0.002 | 0.998 ± 0.003 | 0.996 ± 0.002 | 0.995 ± 0.004 | 0.988 ± 0.005 | E Europe M    | E Africa F    | 0.01 |
| 33  | 0.973 ± 0.009 | 0.998 ± 0.001 | 0.994 ± 0.003 | 0.990 ± 0.003 | 0.999 ± 0.001 | 0.987 ± 0.007 | 0.992 ± 0.005 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.997 ± 0.001 | 0.998 ± 0.003 | 0.993 ± 0.004 | S E Asia F    | E Africa F    | 0.00 |
| 34  | 0.977 ± 0.009 | 0.999 ± 0.001 | 0.997 ± 0.002 | 0.992 ± 0.003 | 1.000 ± 0.001 | 0.987 ± 0.006 | 0.995 ± 0.004 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.999 ± 0.001 | 0.999 ± 0.002 | 0.997 ± 0.003 | S E Asia F    | E Africa F    | 0.00 |

**Table 186.** For algorithm yoti-004 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

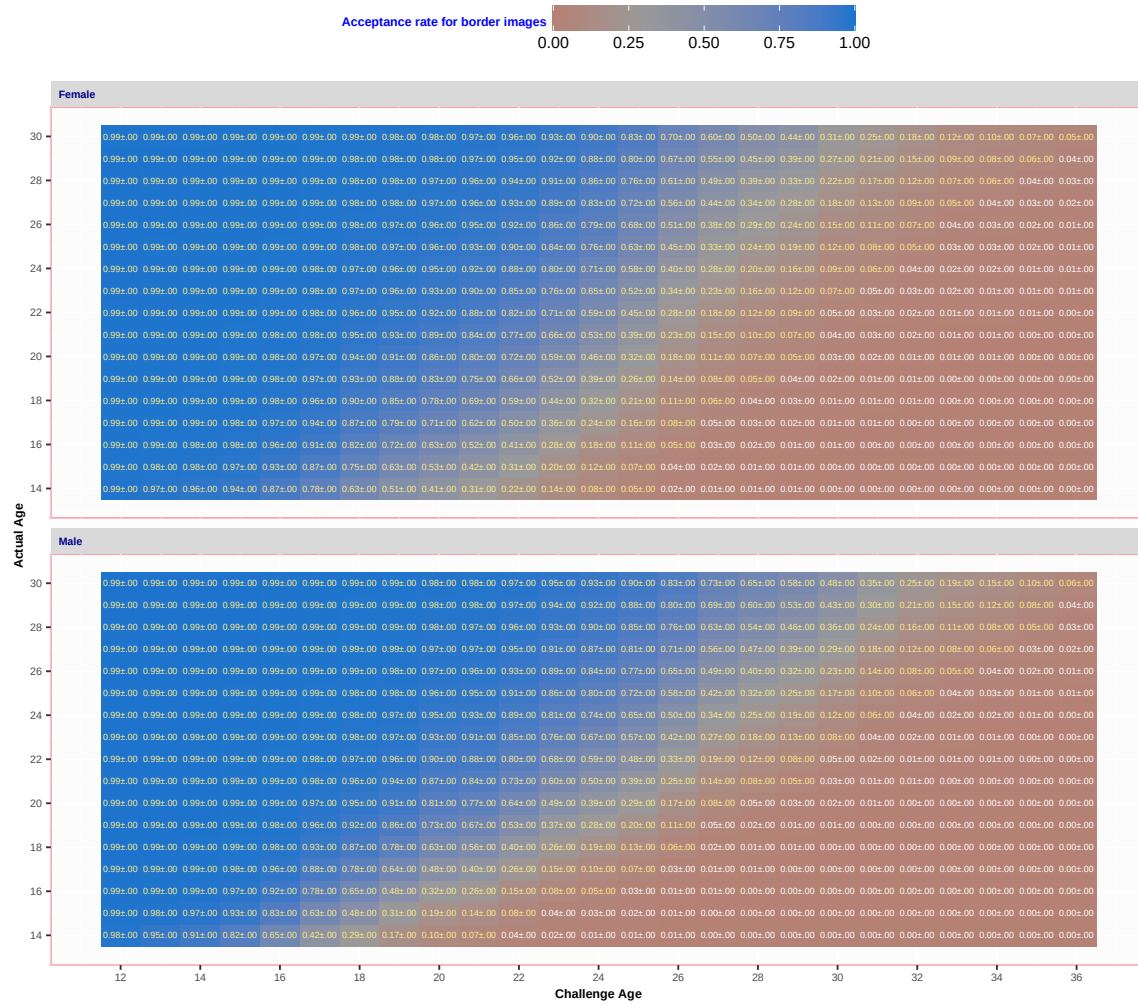
| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.08 ± 0.04   | 0.11 ± 0.01   | 0.03 ± 0.02   | 0.04 ± 0.01   | 0.14 ± 0.03   | 0.06 ± 0.03   | 0.04 ± 0.02   | 0.028 ± 0.006 | 0.005 ± 0.005 | 0.007 ± 0.004 | 0.02 ± 0.01   | 0.04 ± 0.03   | S E Asia F    | E Europe M    | 0.45 |
| 15  | 0.08 ± 0.04   | 0.16 ± 0.01   | 0.05 ± 0.02   | 0.06 ± 0.01   | 0.21 ± 0.03   | 0.12 ± 0.05   | 0.06 ± 0.03   | 0.043 ± 0.007 | 0.000 ± 0.000 | 0.017 ± 0.007 | 0.02 ± 0.01   | 0.05 ± 0.03   | S E Asia F    | E Europe M    | 0.47 |
| 16  | 0.15 ± 0.05   | 0.22 ± 0.01   | 0.08 ± 0.02   | 0.10 ± 0.02   | 0.28 ± 0.03   | 0.13 ± 0.04   | 0.10 ± 0.04   | 0.068 ± 0.008 | 0.020 ± 0.010 | 0.018 ± 0.007 | 0.05 ± 0.02   | 0.07 ± 0.03   | S E Asia F    | S Asia M      | 0.41 |
| 17  | 0.17 ± 0.04   | 0.28 ± 0.02   | 0.11 ± 0.02   | 0.13 ± 0.02   | 0.35 ± 0.03   | 0.17 ± 0.04   | 0.14 ± 0.05   | 0.109 ± 0.010 | 0.015 ± 0.008 | 0.05 ± 0.01   | 0.08 ± 0.02   | 0.13 ± 0.04   | S E Asia F    | E Europe M    | 0.36 |
| 18  | 0.27 ± 0.05   | 0.36 ± 0.01   | 0.19 ± 0.02   | 0.18 ± 0.02   | 0.42 ± 0.03   | 0.20 ± 0.04   | 0.16 ± 0.04   | 0.16 ± 0.01   | 0.04 ± 0.01   | 0.07 ± 0.01   | 0.15 ± 0.02   | 0.16 ± 0.04   | S E Asia F    | E Europe M    | 0.30 |
| 19  | 0.31 ± 0.04   | 0.45 ± 0.01   | 0.23 ± 0.02   | 0.26 ± 0.02   | 0.47 ± 0.02   | 0.28 ± 0.04   | 0.23 ± 0.04   | 0.25 ± 0.01   | 0.08 ± 0.02   | 0.14 ± 0.01   | 0.23 ± 0.02   | 0.23 ± 0.04   | S E Asia F    | E Europe M    | 0.23 |
| 20  | 0.43 ± 0.04   | 0.54 ± 0.01   | 0.32 ± 0.02   | 0.33 ± 0.02   | 0.59 ± 0.02   | 0.39 ± 0.04   | 0.33 ± 0.04   | 0.36 ± 0.01   | 0.17 ± 0.02   | 0.21 ± 0.02   | 0.36 ± 0.02   | 0.31 ± 0.04   | S E Asia F    | E Europe M    | 0.19 |
| 21  | 0.46 ± 0.04   | 0.618 ± 0.010 | 0.39 ± 0.02   | 0.40 ± 0.02   | 0.63 ± 0.02   | 0.47 ± 0.04   | 0.42 ± 0.04   | 0.48 ± 0.01   | 0.30 ± 0.03   | 0.30 ± 0.02   | 0.48 ± 0.03   | 0.41 ± 0.04   | S E Asia F    | E Europe M    | 0.13 |
| 22  | 0.54 ± 0.04   | 0.691 ± 0.009 | 0.48 ± 0.02   | 0.49 ± 0.02   | 0.70 ± 0.02   | 0.54 ± 0.04   | 0.53 ± 0.04   | 0.62 ± 0.01   | 0.43 ± 0.02   | 0.44 ± 0.02   | 0.60 ± 0.03   | 0.52 ± 0.04   | S E Asia F    | E Europe M    | 0.09 |
| 23  | 0.58 ± 0.04   | 0.760 ± 0.008 | 0.57 ± 0.02   | 0.56 ± 0.01   | 0.78 ± 0.02   | 0.57 ± 0.03   | 0.61 ± 0.04   | 0.72 ± 0.01   | 0.58 ± 0.02   | 0.55 ± 0.01   | 0.71 ± 0.02   | 0.63 ± 0.03   | S E Asia F    | S Asia M      | 0.08 |
| 24  | 0.64 ± 0.03   | 0.810 ± 0.007 | 0.66 ± 0.01   | 0.64 ± 0.01   | 0.82 ± 0.01   | 0.65 ± 0.03   | 0.69 ± 0.03   | 0.805 ± 0.010 | 0.71 ± 0.02   | 0.65 ± 0.01   | 0.81 ± 0.02   | 0.70 ± 0.03   | S E Asia F    | E Africa F    | 0.06 |
| 25  | 0.71 ± 0.03   | 0.858 ± 0.006 | 0.74 ± 0.01   | 0.691 ± 0.010 | 0.85 ± 0.01   | 0.71 ± 0.02   | 0.74 ± 0.03   | 0.881 ± 0.008 | 0.82 ± 0.02   | 0.75 ± 0.01   | 0.88 ± 0.02   | 0.79 ± 0.02   | E Asia M      | S Asia F      | 0.05 |
| 26  | 0.75 ± 0.02   | 0.901 ± 0.005 | 0.804 ± 0.010 | 0.737 ± 0.009 | 0.900 ± 0.010 | 0.75 ± 0.02   | 0.84 ± 0.02   | 0.923 ± 0.006 | 0.88 ± 0.01   | 0.828 ± 0.008 | 0.92 ± 0.01   | 0.84 ± 0.02   | E Asia M      | S Asia F      | 0.05 |
| 27  | 0.80 ± 0.02   | 0.929 ± 0.004 | 0.865 ± 0.008 | 0.783 ± 0.008 | 0.920 ± 0.009 | 0.80 ± 0.02   | 0.90 ± 0.02   | 0.951 ± 0.004 | 0.932 ± 0.009 | 0.875 ± 0.007 | 0.95 ± 0.01   | 0.90 ± 0.02   | E Asia M      | S Asia F      | 0.04 |
| 28  | 0.85 ± 0.02   | 0.952 ± 0.003 | 0.910 ± 0.007 | 0.813 ± 0.007 | 0.945 ± 0.007 | 0.83 ± 0.02   | 0.93 ± 0.01   | 0.972 ± 0.004 | 0.959 ± 0.007 | 0.915 ± 0.005 | 0.971 ± 0.008 | 0.92 ± 0.01   | E Asia M      | S Asia F      | 0.03 |
| 29  | 0.88 ± 0.02   | 0.968 ± 0.003 | 0.939 ± 0.006 | 0.850 ± 0.006 | 0.952 ± 0.006 | 0.87 ± 0.02   | 0.95 ± 0.01   | 0.984 ± 0.003 | 0.972 ± 0.006 | 0.944 ± 0.004 | 0.981 ± 0.006 | 0.952 ± 0.010 | E Asia M      | S Asia F      | 0.03 |
| 30  | 0.89 ± 0.02   | 0.977 ± 0.002 | 0.958 ± 0.005 | 0.880 ± 0.005 | 0.969 ± 0.006 | 0.89 ± 0.01   | 0.97 ± 0.01   | 0.992 ± 0.002 | 0.986 ± 0.004 | 0.964 ± 0.003 | 0.985 ± 0.006 | 0.963 ± 0.008 | E Asia M      | S Asia F      | 0.02 |
| 31  | 0.92 ± 0.02   | 0.984 ± 0.002 | 0.970 ± 0.004 | 0.904 ± 0.005 | 0.978 ± 0.004 | 0.92 ± 0.01   | 0.967 ± 0.010 | 0.994 ± 0.002 | 0.992 ± 0.003 | 0.975 ± 0.003 | 0.989 ± 0.005 | 0.974 ± 0.007 | E Asia M      | S Asia F      | 0.02 |
| 32  | 0.93 ± 0.01   | 0.990 ± 0.001 | 0.984 ± 0.003 | 0.920 ± 0.004 | 0.981 ± 0.004 | 0.93 ± 0.01   | 0.984 ± 0.007 | 0.998 ± 0.001 | 0.995 ± 0.002 | 0.983 ± 0.002 | 0.992 ± 0.004 | 0.980 ± 0.005 | E Asia M      | S Asia F      | 0.01 |
| 33  | 0.94 ± 0.01   | 0.994 ± 0.001 | 0.988 ± 0.003 | 0.937 ± 0.004 | 0.987 ± 0.004 | 0.95 ± 0.01   | 0.989 ± 0.006 | 0.997 ± 0.001 | 0.998 ± 0.002 | 0.988 ± 0.002 | 0.997 ± 0.002 | 0.991 ± 0.004 | E Europe M    | S Asia F      | 0.01 |
| 34  | 0.96 ± 0.01   | 0.995 ± 0.001 | 0.992 ± 0.003 | 0.951 ± 0.004 | 0.990 ± 0.003 | 0.967 ± 0.009 | 0.991 ± 0.005 | 0.999 ± 0.001 | 0.999 ± 0.001 | 0.993 ± 0.001 | 0.996 ± 0.003 | 0.993 ± 0.003 | E Europe M    | S Asia F      | 0.01 |

**Table 187.** For algorithm youverse-000 with Application images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

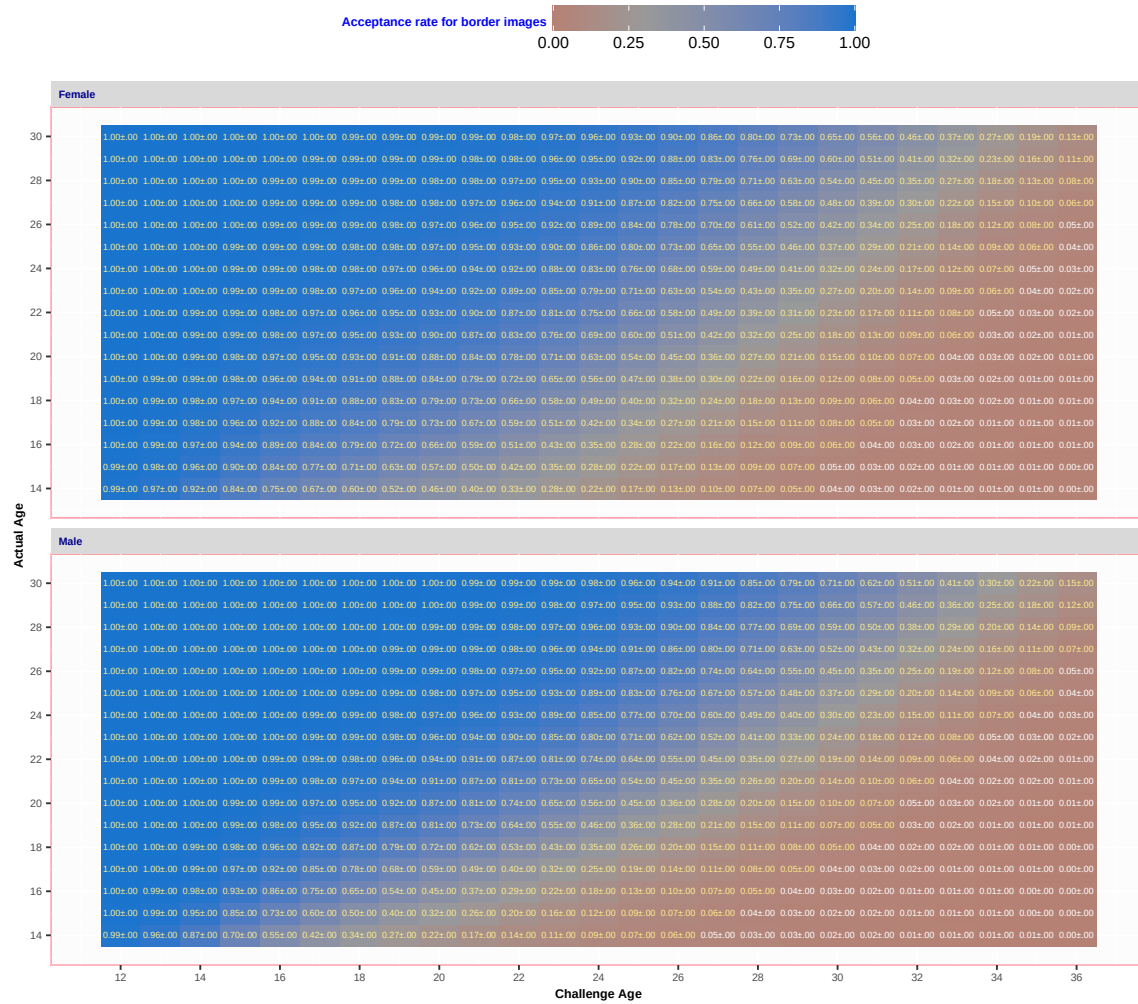
| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia  | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia  | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------|
| 14  | 0.06 ± 0.02   | 0.005 ± 0.003 | 0.016 ± 0.009 | 0.027 ± 0.008 | 0.009 ± 0.007 | 0.07 ± 0.03   | 0.03 ± 0.01   | 0.002 ± 0.002 | 0.000 ± 0.000 | 0.013 ± 0.005 | 0.005 ± 0.005 | 0.03 ± 0.02   | W Africa F    | E Europe M    | 0.56 |
| 15  | 0.07 ± 0.03   | 0.015 ± 0.006 | 0.05 ± 0.02   | 0.06 ± 0.02   | 0.03 ± 0.01   | 0.12 ± 0.04   | 0.04 ± 0.02   | 0.004 ± 0.003 | 0.012 ± 0.009 | 0.022 ± 0.009 | 0.002 ± 0.002 | 0.06 ± 0.03   | W Africa F    | S E Asia M    | 0.49 |
| 16  | 0.13 ± 0.03   | 0.015 ± 0.006 | 0.07 ± 0.02   | 0.06 ± 0.02   | 0.02 ± 0.01   | 0.13 ± 0.04   | 0.07 ± 0.02   | 0.009 ± 0.005 | 0.02 ± 0.01   | 0.03 ± 0.01   | 0.03 ± 0.01   | 0.09 ± 0.04   | E Africa F    | E Asia M      | 0.44 |
| 17  | 0.18 ± 0.04   | 0.037 ± 0.009 | 0.13 ± 0.03   | 0.09 ± 0.02   | 0.04 ± 0.02   | 0.24 ± 0.05   | 0.12 ± 0.03   | 0.021 ± 0.007 | 0.03 ± 0.02   | 0.07 ± 0.02   | 0.02 ± 0.01   | 0.17 ± 0.04   | W Africa F    | S E Asia M    | 0.44 |
| 18  | 0.20 ± 0.03   | 0.046 ± 0.009 | 0.16 ± 0.03   | 0.13 ± 0.02   | 0.05 ± 0.01   | 0.24 ± 0.04   | 0.22 ± 0.03   | 0.031 ± 0.007 | 0.08 ± 0.02   | 0.11 ± 0.02   | 0.04 ± 0.01   | 0.25 ± 0.04   | W Africa M    | E Asia M      | 0.37 |
| 19  | 0.25 ± 0.04   | 0.05 ± 0.01   | 0.21 ± 0.03   | 0.17 ± 0.02   | 0.07 ± 0.02   | 0.29 ± 0.04   | 0.29 ± 0.04   | 0.048 ± 0.009 | 0.11 ± 0.03   | 0.21 ± 0.02   | 0.09 ± 0.02   | 0.28 ± 0.04   | W Africa F    | E Asia M      | 0.32 |
| 20  | 0.31 ± 0.03   | 0.08 ± 0.01   | 0.30 ± 0.03   | 0.22 ± 0.02   | 0.06 ± 0.01   | 0.35 ± 0.04   | 0.40 ± 0.04   | 0.07 ± 0.01   | 0.18 ± 0.03   | 0.29 ± 0.02   | 0.12 ± 0.02   | 0.38 ± 0.04   | E Africa M    | S E Asia F    | 0.32 |
| 21  | 0.36 ± 0.03   | 0.11 ± 0.01   | 0.35 ± 0.03   | 0.27 ± 0.02   | 0.11 ± 0.02   | 0.38 ± 0.04   | 0.48 ± 0.04   | 0.13 ± 0.01   | 0.26 ± 0.04   | 0.36 ± 0.02   | 0.18 ± 0.03   | 0.50 ± 0.04   | W Africa M    | S E Asia F    | 0.28 |
| 22  | 0.42 ± 0.03   | 0.15 ± 0.01   | 0.43 ± 0.03   | 0.34 ± 0.02   | 0.12 ± 0.02   | 0.44 ± 0.04   | 0.58 ± 0.03   | 0.19 ± 0.02   | 0.38 ± 0.04   | 0.50 ± 0.02   | 0.22 ± 0.03   | 0.55 ± 0.04   | E Africa M    | S E Asia F    | 0.26 |
| 23  | 0.43 ± 0.03   | 0.18 ± 0.01   | 0.52 ± 0.02   | 0.41 ± 0.02   | 0.16 ± 0.02   | 0.53 ± 0.03   | 0.66 ± 0.03   | 0.25 ± 0.02   | 0.51 ± 0.04   | 0.60 ± 0.02   | 0.30 ± 0.03   | 0.63 ± 0.03   | E Africa M    | S E Asia F    | 0.24 |
| 24  | 0.52 ± 0.03   | 0.21 ± 0.01   | 0.63 ± 0.02   | 0.46 ± 0.02   | 0.22 ± 0.02   | 0.54 ± 0.03   | 0.71 ± 0.03   | 0.32 ± 0.02   | 0.60 ± 0.03   | 0.69 ± 0.02   | 0.37 ± 0.03   | 0.72 ± 0.03   | W Africa M    | E Asia F      | 0.22 |
| 25  | 0.57 ± 0.03   | 0.25 ± 0.01   | 0.68 ± 0.02   | 0.53 ± 0.01   | 0.25 ± 0.02   | 0.59 ± 0.03   | 0.77 ± 0.03   | 0.41 ± 0.02   | 0.70 ± 0.03   | 0.76 ± 0.01   | 0.47 ± 0.03   | 0.75 ± 0.02   | E Africa M    | S E Asia F    | 0.19 |
| 26  | 0.59 ± 0.03   | 0.30 ± 0.01   | 0.75 ± 0.02   | 0.58 ± 0.01   | 0.31 ± 0.02   | 0.60 ± 0.03   | 0.84 ± 0.02   | 0.48 ± 0.02   | 0.73 ± 0.03   | 0.82 ± 0.01   | 0.56 ± 0.03   | 0.82 ± 0.02   | E Africa M    | E Asia F      | 0.18 |
| 27  | 0.64 ± 0.02   | 0.35 ± 0.01   | 0.80 ± 0.01   | 0.62 ± 0.01   | 0.38 ± 0.02   | 0.66 ± 0.02   | 0.85 ± 0.02   | 0.56 ± 0.02   | 0.83 ± 0.02   | 0.85 ± 0.01   | 0.62 ± 0.03   | 0.85 ± 0.02   | S Asia M      | E Asia F      | 0.15 |
| 28  | 0.69 ± 0.02   | 0.41 ± 0.01   | 0.85 ± 0.01   | 0.67 ± 0.01   | 0.43 ± 0.02   | 0.69 ± 0.03   | 0.89 ± 0.02   | 0.64 ± 0.02   | 0.89 ± 0.02   | 0.879 ± 0.009 | 0.68 ± 0.03   | 0.86 ± 0.02   | E Europe M    | E Asia F      | 0.13 |
| 29  | 0.73 ± 0.02   | 0.48 ± 0.01   | 0.89 ± 0.01   | 0.72 ± 0.01   | 0.49 ± 0.02   | 0.73 ± 0.02   | 0.93 ± 0.01   | 0.71 ± 0.02   | 0.92 ± 0.01   | 0.914 ± 0.008 | 0.76 ± 0.03   | 0.91 ± 0.01   | E Africa M    | E Asia F      | 0.12 |
| 30  | 0.77 ± 0.02   | 0.52 ± 0.01   | 0.93 ± 0.01   | 0.77 ± 0.01   | 0.57 ± 0.02   | 0.76 ± 0.02   | 0.93 ± 0.01   | 0.78 ± 0.02   | 0.94 ± 0.01   | 0.932 ± 0.007 | 0.82 ± 0.02   | 0.92 ± 0.01   | E Europe M    | E Asia F      | 0.10 |
| 31  | 0.79 ± 0.02   | 0.59 ± 0.01   | 0.945 ± 0.009 | 0.80 ± 0.01   | 0.60 ± 0.02   | 0.79 ± 0.02   | 0.96 ± 0.01   | 0.82 ± 0.01   | 0.96 ± 0.01   | 0.950 ± 0.005 | 0.85 ± 0.02   | 0.94 ± 0.01   | E Europe M    | E Asia F      | 0.09 |
| 32  | 0.84 ± 0.02   | 0.64 ± 0.01   | 0.957 ± 0.008 | 0.84 ± 0.01   | 0.67 ± 0.02   | 0.79 ± 0.02   | 0.963 ± 0.010 | 0.86 ± 0.01   | 0.976 ± 0.009 | 0.963 ± 0.005 | 0.89 ± 0.02   | 0.95 ± 0.01   | E Europe M    | E Asia F      | 0.07 |
| 33  | 0.83 ± 0.02   | 0.70 ± 0.01   | 0.969 ± 0.008 | 0.874 ± 0.009 | 0.71 ± 0.02   | 0.85 ± 0.02   | 0.976 ± 0.009 | 0.892 ± 0.010 | 0.984 ± 0.007 | 0.973 ± 0.004 | 0.91 ± 0.02   | 0.965 ± 0.009 | E Europe M    | E Asia F      | 0.06 |
| 34  | 0.88 ± 0.02   | 0.75 ± 0.01   | 0.978 ± 0.007 | 0.909 ± 0.007 | 0.76 ± 0.02   | 0.86 ± 0.02   | 0.982 ± 0.007 | 0.918 ± 0.010 | 0.992 ± 0.005 | 0.979 ± 0.004 | 0.94 ± 0.02   | 0.973 ± 0.008 | E Europe M    | E Asia F      | 0.05 |

**Table 188.** For algorithm youverse-000 with Border images, Challenge-25 accept rate by age, sex and region of birth. The final columns give groups that give the highest and lowest accept rate and the Gini summary of variability. Gini is computed over the 12 accept-rate values to its left; lower values are better.

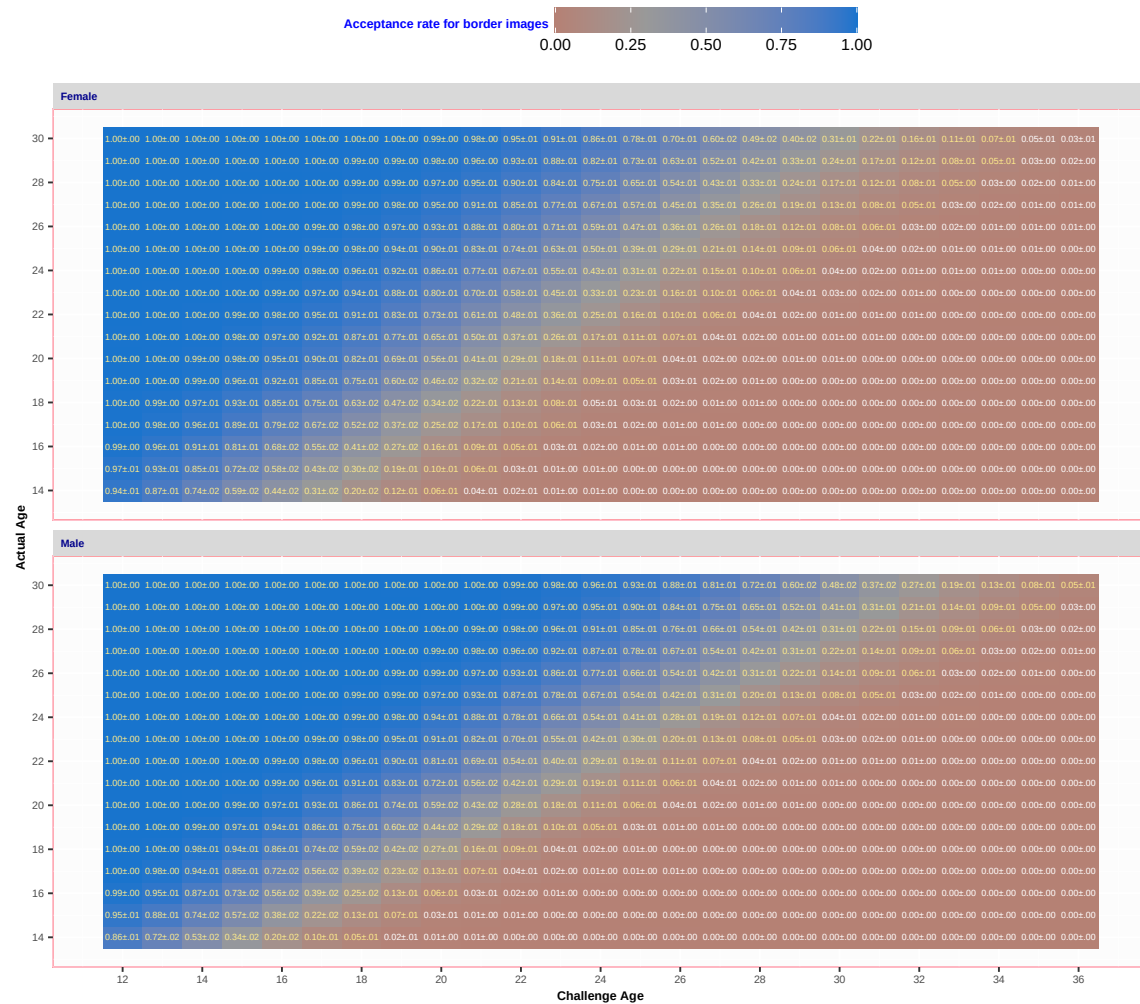
| Age | F<br>E Africa | F<br>E Asia   | F<br>E Europe | F<br>S Asia   | F<br>SE Asia | F<br>W Africa | M<br>E Africa | M<br>E Asia   | M<br>E Europe | M<br>S Asia   | M<br>SE Asia | M<br>W Africa | max<br>Accept | min<br>Accept | Gini |
|-----|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|------|
| 14  | 0.31 ± 0.06   | 0.046 ± 0.009 | 0.06 ± 0.02   | 0.20 ± 0.02   | 0.04 ± 0.02  | 0.28 ± 0.07   | 0.48 ± 0.08   | 0.09 ± 0.01   | 0.10 ± 0.03   | 0.25 ± 0.03   | 0.11 ± 0.03  | 0.50 ± 0.08   | W Africa M    | S E Asia F    | 0.45 |
| 15  | 0.37 ± 0.07   | 0.065 ± 0.009 | 0.11 ± 0.03   | 0.24 ± 0.02   | 0.07 ± 0.02  | 0.35 ± 0.07   | 0.65 ± 0.07   | 0.13 ± 0.01   | 0.13 ± 0.03   | 0.35 ± 0.02   | 0.12 ± 0.03  | 0.63 ± 0.07   | E Africa M    | E Asia F      | 0.43 |
| 16  | 0.44 ± 0.06   | 0.073 ± 0.009 | 0.15 ± 0.03   | 0.27 ± 0.03   | 0.08 ± 0.02  | 0.41 ± 0.06   | 0.72 ± 0.07   | 0.16 ± 0.01   | 0.19 ± 0.03   | 0.43 ± 0.03   | 0.18 ± 0.03  | 0.70 ± 0.05   | E Africa M    | E Asia F      | 0.41 |
| 17  | 0.44 ± 0.05   | 0.082 ± 0.009 | 0.22 ± 0.03   | 0.30 ± 0.02   | 0.06 ± 0.01  | 0.45 ± 0.06   | 0.83 ± 0.05   | 0.21 ± 0.01   | 0.25 ± 0.03   | 0.50 ± 0.02   | 0.22 ± 0.03  | 0.77 ± 0.05   | E Africa M    | S E Asia F    | 0.39 |
| 18  | 0.53 ± 0.06   | 0.102 ± 0.008 | 0.22 ± 0.02   | 0.33 ± 0.02   | 0.07 ± 0.01  | 0.50 ± 0.05   | 0.87 ± 0.04   | 0.25 ± 0.01   | 0.34 ± 0.03   | 0.56 ± 0.02   | 0.27 ± 0.03  | 0.84 ± 0.04   | E Africa M    | S E Asia F    | 0.37 |
| 19  | 0.58 ± 0.05   | 0.117 ± 0.008 | 0.25 ± 0.02   | 0.32 ± 0.02   | 0.10 ± 0.01  | 0.56 ± 0.05   | 0.84 ± 0.04   | 0.28 ± 0.01   | 0.38 ± 0.03   | 0.64 ± 0.02   | 0.33 ± 0.03  | 0.88 ± 0.03   | W Africa M    | S E Asia F    | 0.35 |
| 20  | 0.59 ± 0.04   | 0.143 ± 0.008 | 0.30 ± 0.02   | 0.36 ± 0.02   | 0.11 ± 0.01  | 0.59 ± 0.04   | 0.91 ± 0.03   | 0.33 ± 0.01   | 0.46 ± 0.03   | 0.71 ± 0.02   | 0.37 ± 0.02  | 0.89 ± 0.02   | E Africa M    | S E Asia F    | 0.32 |
| 21  | 0.62 ± 0.04   | 0.156 ± 0.007 | 0.35 ± 0.02   | 0.40 ± 0.02   | 0.14 ± 0.01  | 0.64 ± 0.04   | 0.92 ± 0.02   | 0.38 ± 0.01   | 0.55 ± 0.03   | 0.74 ± 0.01   | 0.40 ± 0.03  | 0.93 ± 0.02   | W Africa M    | S E Asia F    | 0.30 |
| 22  | 0.64 ± 0.04   | 0.176 ± 0.007 | 0.40 ± 0.02   | 0.43 ± 0.02   | 0.14 ± 0.01  | 0.65 ± 0.03   | 0.95 ± 0.02   | 0.42 ± 0.01   | 0.62 ± 0.03   | 0.80 ± 0.01   | 0.48 ± 0.03  | 0.92 ± 0.02   | E Africa M    | S E Asia F    | 0.28 |
| 23  | 0.68 ± 0.03   | 0.199 ± 0.007 | 0.47 ± 0.02   | 0.47 ± 0.01   | 0.19 ± 0.01  | 0.67 ± 0.03   | 0.96 ± 0.01   | 0.46 ± 0.01   | 0.70 ± 0.02   | 0.83 ± 0.01   | 0.54 ± 0.03  | 0.94 ± 0.02   | E Africa M    | S E Asia F    | 0.25 |
| 24  | 0.70 ± 0.03   | 0.223 ± 0.007 | 0.52 ± 0.01   | 0.51 ± 0.01   | 0.21 ± 0.01  | 0.71 ± 0.02   | 0.96 ± 0.02   | 0.51 ± 0.01   | 0.73 ± 0.02   | 0.862 ± 0.010 | 0.56 ± 0.03  | 0.95 ± 0.01   | E Africa M    | S E Asia F    | 0.23 |
| 25  | 0.73 ± 0.03   | 0.246 ± 0.007 | 0.59 ± 0.01   | 0.56 ± 0.01   | 0.24 ± 0.01  | 0.74 ± 0.02   | 0.98 ± 0.01   | 0.55 ± 0.01   | 0.79 ± 0.02   | 0.881 ± 0.008 | 0.62 ± 0.03  | 0.96 ± 0.01   | E Africa M    | S E Asia F    | 0.21 |
| 26  | 0.76 ± 0.02   | 0.278 ± 0.007 | 0.63 ± 0.01   | 0.586 ± 0.009 | 0.28 ± 0.01  | 0.78 ± 0.02   | 0.98 ± 0.01   | 0.60 ± 0.01   | 0.83 ± 0.01   | 0.906 ± 0.007 | 0.65 ± 0.02  | 0.971 ± 0.008 | E Africa M    | S E Asia F    | 0.20 |
| 27  | 0.77 ± 0.02   | 0.305 ± 0.006 | 0.69 ± 0.01   | 0.623 ± 0.009 | 0.32 ± 0.02  | 0.78 ± 0.02   | 0.975 ± 0.010 | 0.64 ± 0.01   | 0.87 ± 0.01   | 0.921 ± 0.005 | 0.72 ± 0.02  | 0.977 ± 0.008 | W Africa M    | E Asia F      | 0.18 |
| 28  | 0.78 ± 0.02   | 0.341 ± 0.007 | 0.72 ± 0.01   | 0.645 ± 0.009 | 0.35 ± 0.01  | 0.81 ± 0.02   | 0.982 ± 0.009 | 0.671 ± 0.010 | 0.89 ± 0.01   | 0.935 ± 0.005 | 0.73 ± 0.02  | 0.976 ± 0.007 | E Africa M    | E Asia F      | 0.17 |
| 29  | 0.82 ± 0.02   | 0.374 ± 0.007 | 0.772 ± 0.009 | 0.685 ± 0.008 | 0.38 ± 0.02  | 0.83 ± 0.02   | 0.982 ± 0.008 | 0.707 ± 0.010 | 0.907 ± 0.010 | 0.943 ± 0.004 | 0.78 ± 0.02  | 0.981 ± 0.006 | E Africa M    | E Asia F      | 0.15 |
| 30  | 0.83 ± 0.02   | 0.416 ± 0.007 | 0.793 ± 0.010 | 0.708 ± 0.008 | 0.43 ± 0.02  | 0.83 ± 0.02   | 0.986 ± 0.007 | 0.74 ± 0.01   | 0.931 ± 0.008 | 0.954 ± 0.003 | 0.81 ± 0.02  | 0.991 ± 0.004 | W Africa M    | E Asia F      | 0.14 |
| 31  | 0.84 ± 0.02   | 0.451 ± 0.008 | 0.837 ± 0.009 | 0.751 ± 0.007 | 0.47 ± 0.02  | 0.86 ± 0.02   | 0.986 ± 0.007 | 0.782 ± 0.008 | 0.946 ± 0.007 | 0.963 ± 0.003 | 0.83 ± 0.02  | 0.990 ± 0.004 | W Africa M    | E Asia F      | 0.12 |
| 32  | 0.86 ± 0.02   | 0.490 ± 0.006 | 0.871 ± 0.008 | 0.784 ± 0.006 | 0.51 ± 0.02  | 0.85 ± 0.02   | 0.986 ± 0.007 | 0.809 ± 0.008 | 0.962 ± 0.006 | 0.968 ± 0.003 | 0.84 ± 0.02  | 0.988 ± 0.004 | W Africa M    | E Asia F      | 0.11 |
| 33  | 0.88 ± 0.02   | 0.530 ± 0.007 | 0.881 ± 0.009 | 0.804 ± 0.007 | 0.54 ± 0.01  | 0.89 ± 0.01   | 0.991 ± 0.006 | 0.842 ± 0.007 | 0.969 ± 0.005 | 0.977 ± 0.002 | 0.88 ± 0.01  | 0.991 ± 0.004 | W Africa M    | E Asia F      | 0.10 |
| 34  | 0.89 ± 0.02   | 0.566 ± 0.007 | 0.907 ± 0.008 | 0.827 ± 0.007 | 0.57 ± 0.02  | 0.91 ± 0.02   | 0.991 ± 0.005 | 0.868 ± 0.007 | 0.979 ± 0.004 | 0.980 ± 0.002 | 0.89 ± 0.01  | 0.989 ± 0.004 | E Africa M    | E Asia F      | 0.09 |



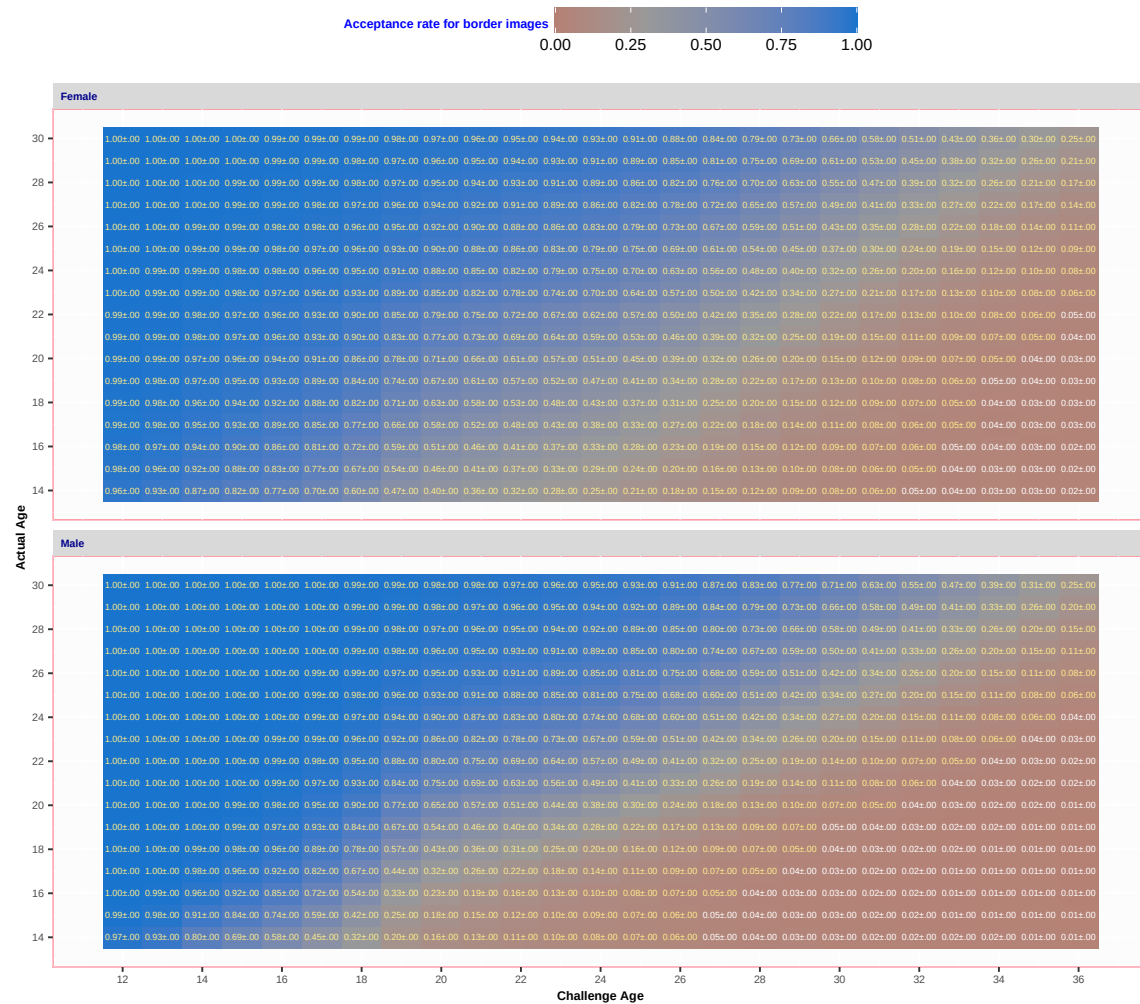
**Fig. 107.** For algorithm CVUT 001 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.



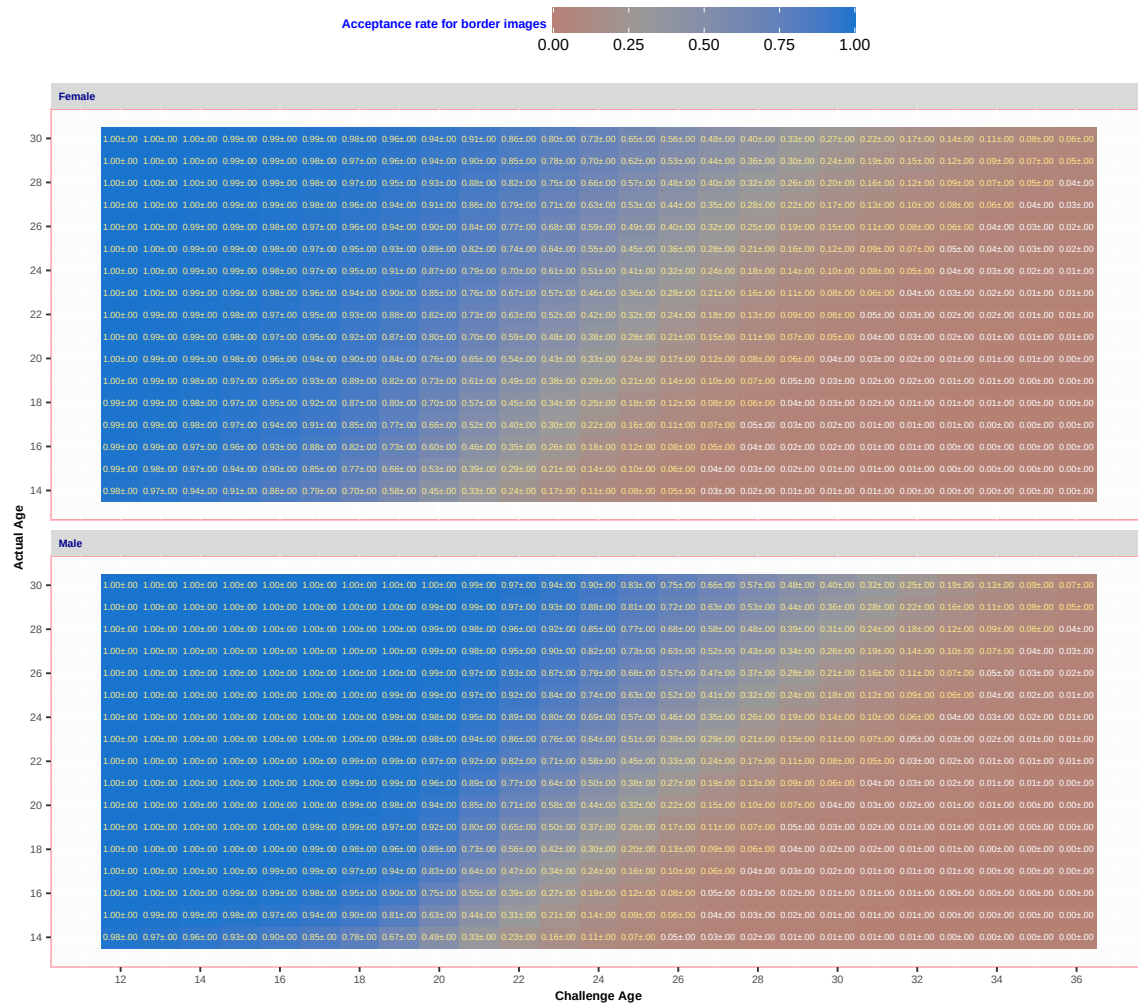
**Fig. 108.** For algorithm Dermalog 001 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.



**Fig. 109.** For algorithm Idemia 001 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.

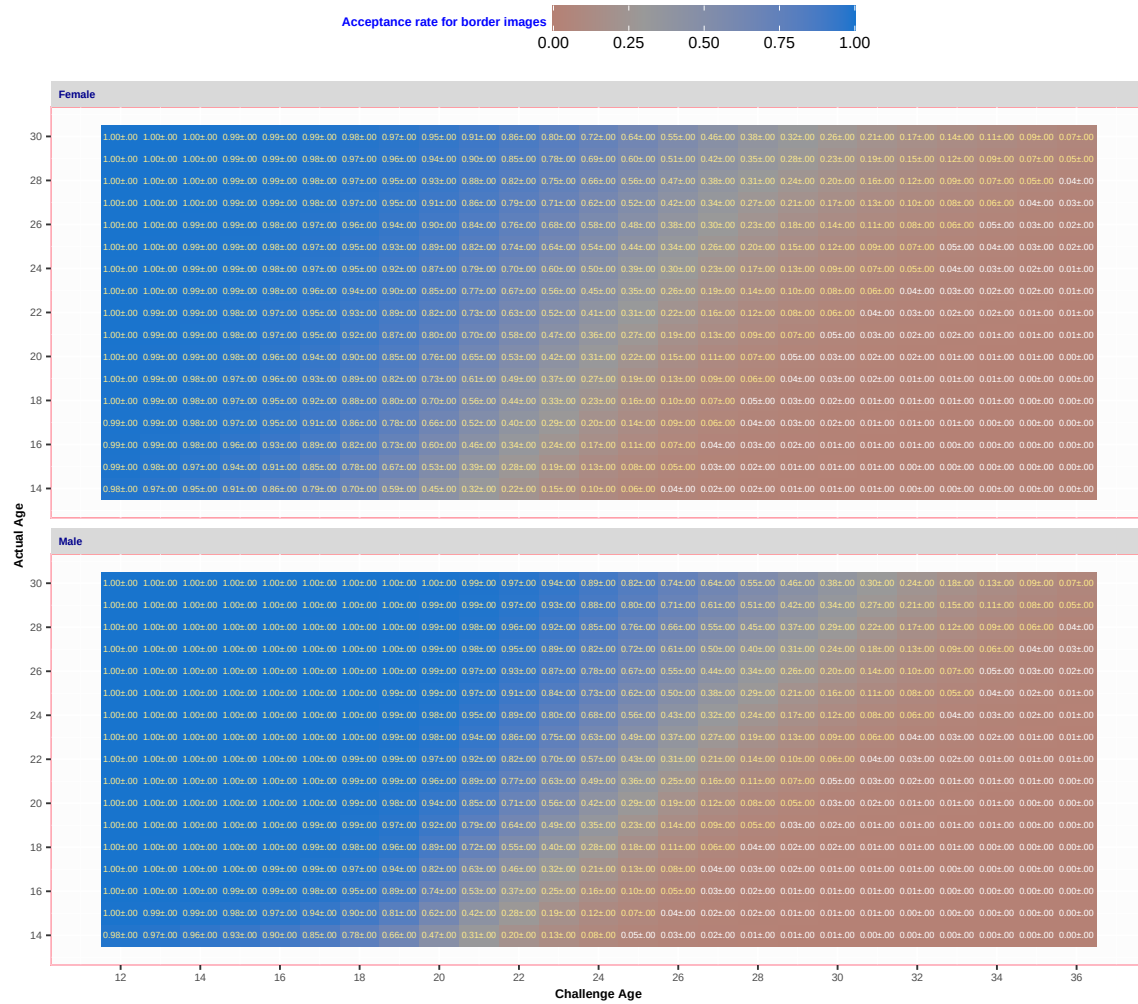


**Fig. 110.** For algorithm Incode 000 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.

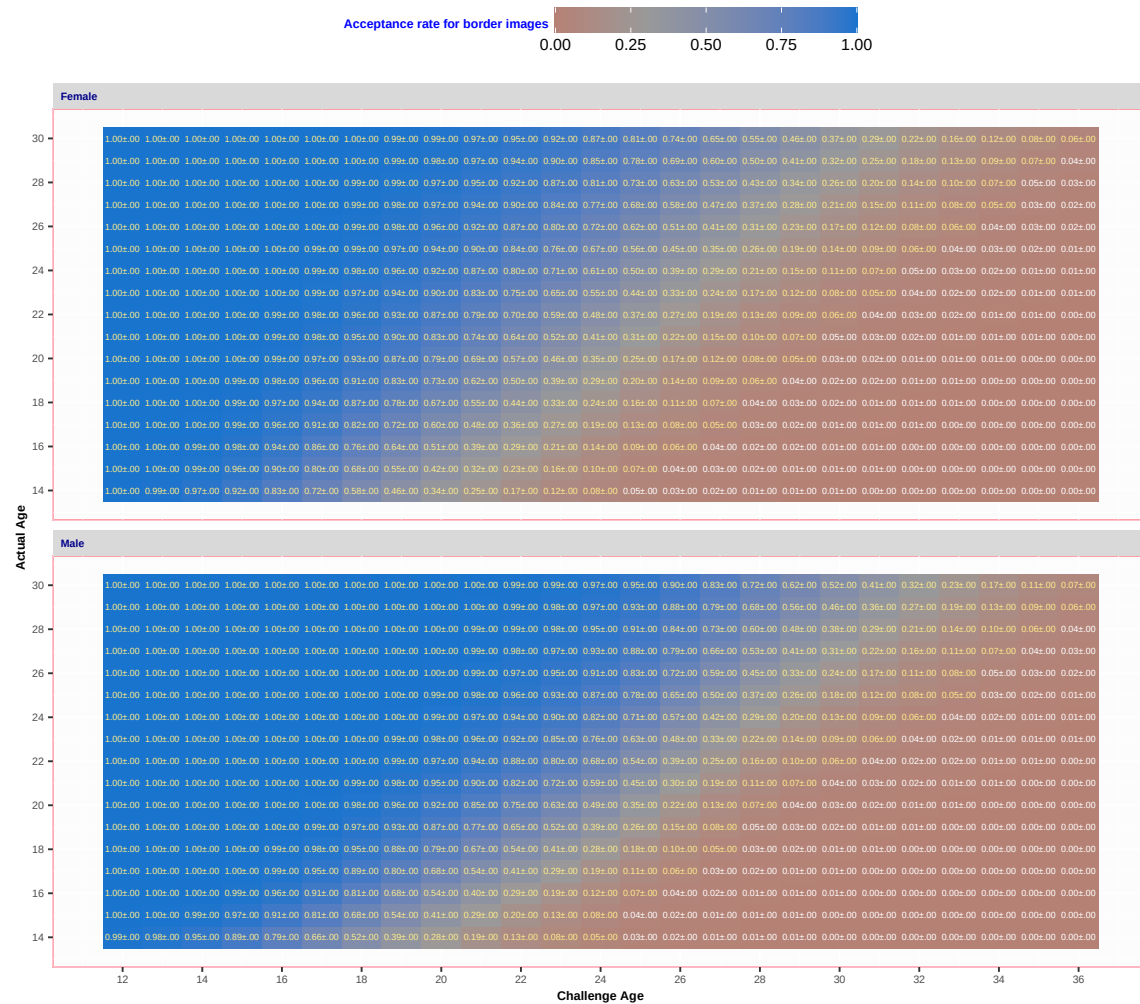


**Fig. 111.** For algorithm Neurotechnology 000 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.

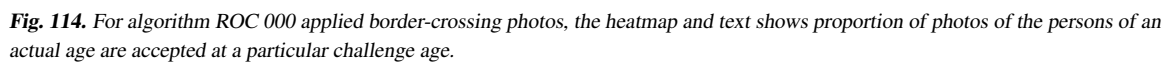


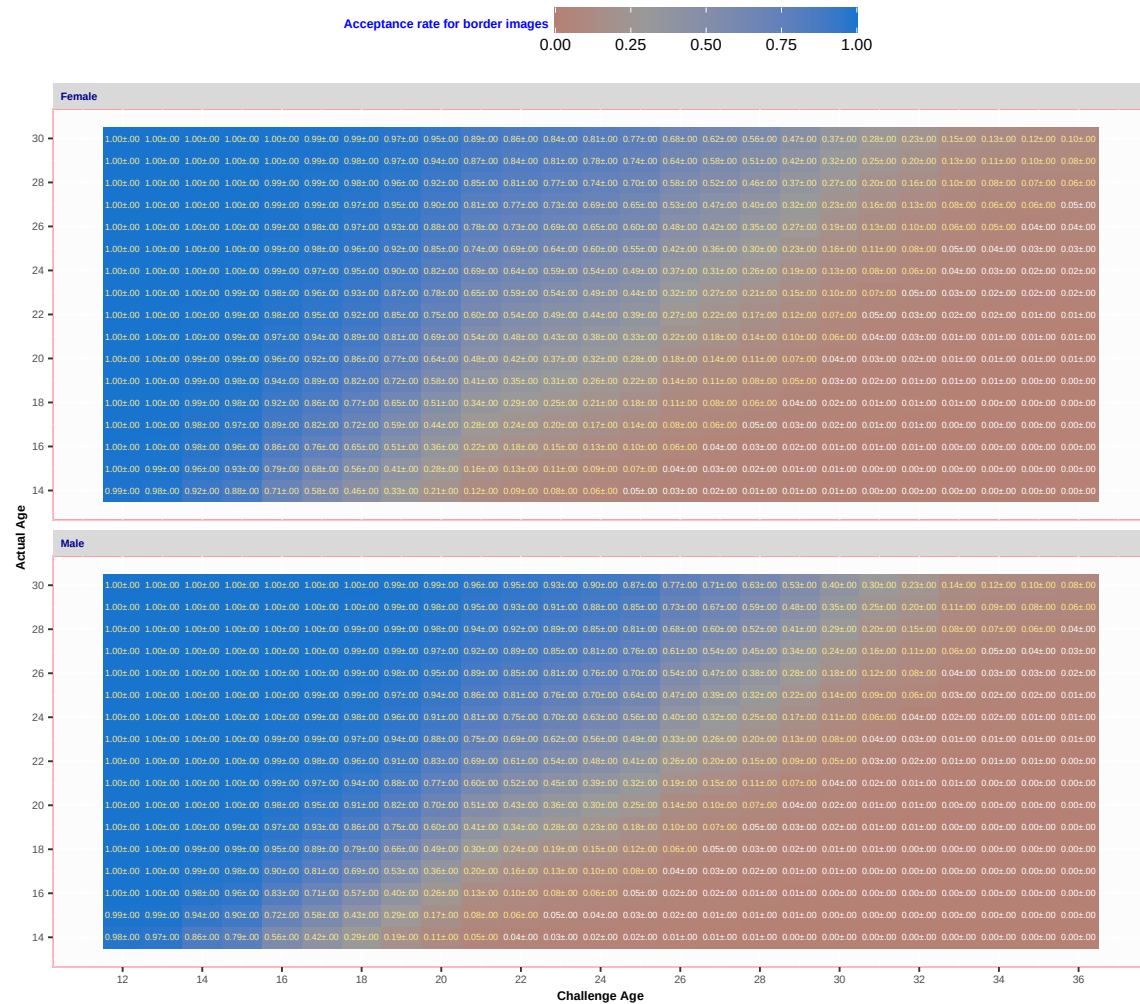


**Fig. 112.** For algorithm Neurotechnology 001 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.

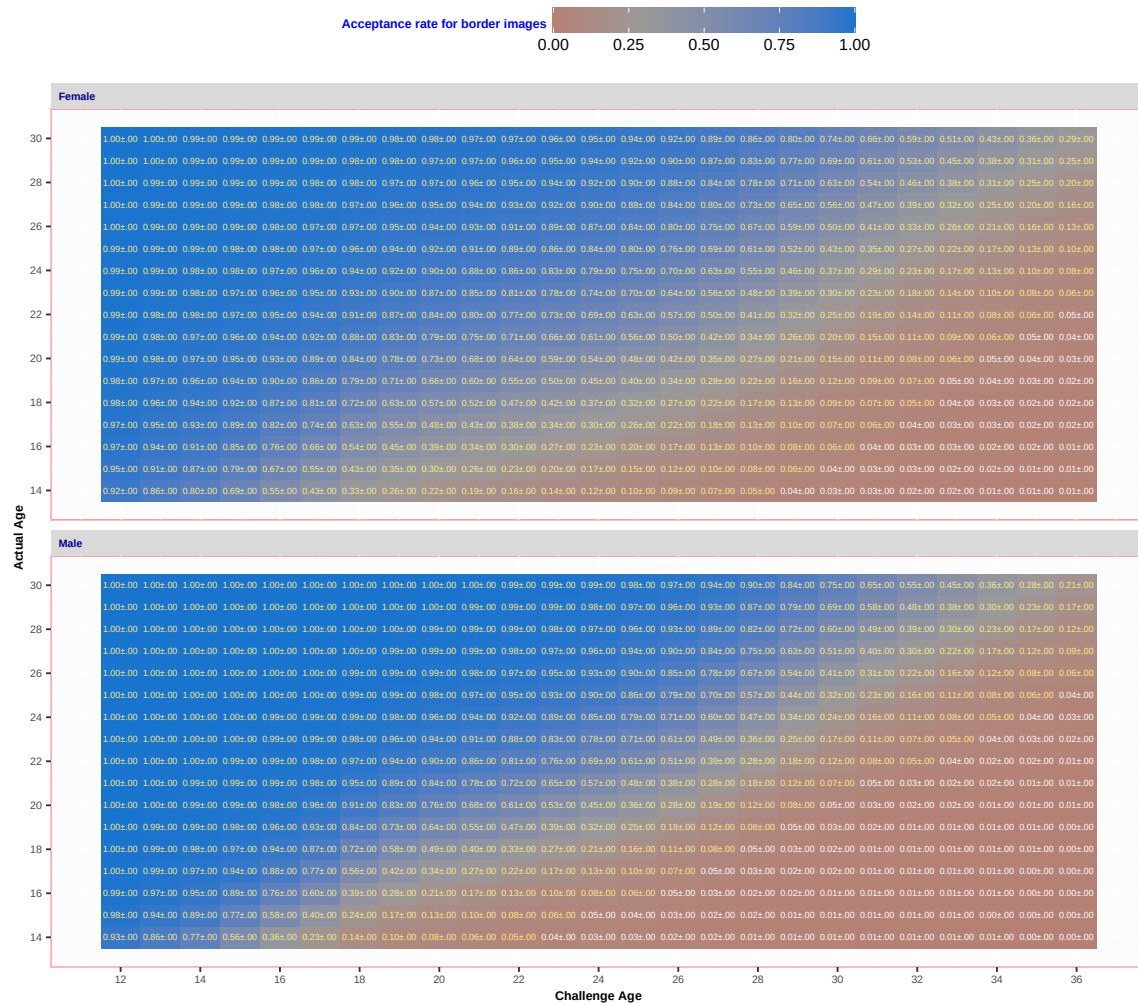


**Fig. 113.** For algorithm Nominder 000 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.





**Fig. 115.** For algorithm Unissey 001 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.



**Fig. 116.** For algorithm Yoti 001 applied border-crossing photos, the heatmap and text shows proportion of photos of the persons of an actual age are accepted at a particular challenge age.

**Table 189.** Mean absolute error for Application-type images for all algorithms. Values closer to zero are better.

| Algorithm           | F E Africa | M E Africa | F E Asia    | M E Asia    | F E Europe  | M E Europe  | F S Asia    | M S Asia    | F SE Asia   | M SE Asia | F W Africa | M W Africa |
|---------------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----------|------------|------------|
| cognitec-000        | 3.5 ± 0.2  | 2.9 ± 0.2  | 2.52 ± 0.08 | 2.26 ± 0.08 | 2.4 ± 0.1   | 1.9 ± 0.1   | 2.77 ± 0.09 | 2.37 ± 0.08 | 2.6 ± 0.1   | 2.3 ± 0.1 | 3.5 ± 0.2  | 2.7 ± 0.2  |
| cvut-001            | 5.3 ± 0.3  | 4.0 ± 0.2  | 3.01 ± 0.09 | 2.92 ± 0.10 | 3.1 ± 0.1   | 2.7 ± 0.2   | 3.3 ± 0.1   | 3.0 ± 0.1   | 3.2 ± 0.1   | 3.0 ± 0.2 | 5.2 ± 0.3  | 3.9 ± 0.3  |
| cvut-002            | 3.8 ± 0.2  | 3.3 ± 0.2  | 3.06 ± 0.08 | 2.60 ± 0.08 | 2.7 ± 0.1   | 2.3 ± 0.1   | 2.97 ± 0.10 | 2.68 ± 0.10 | 3.1 ± 0.1   | 2.5 ± 0.1 | 3.9 ± 0.2  | 3.3 ± 0.2  |
| cvut-003            | 3.9 ± 0.2  | 3.3 ± 0.2  | 2.81 ± 0.07 | 2.32 ± 0.07 | 2.7 ± 0.1   | 2.1 ± 0.1   | 2.90 ± 0.09 | 2.58 ± 0.09 | 2.9 ± 0.1   | 2.4 ± 0.1 | 4.2 ± 0.2  | 3.2 ± 0.2  |
| cyberlink-000       | 4.6 ± 0.2  | 4.5 ± 0.2  | 3.36 ± 0.08 | 3.46 ± 0.10 | 3.2 ± 0.1   | 2.8 ± 0.2   | 3.6 ± 0.1   | 3.6 ± 0.1   | 3.6 ± 0.1   | 3.7 ± 0.2 | 4.8 ± 0.3  | 4.2 ± 0.2  |
| cyberlink-001       | 3.6 ± 0.2  | 3.2 ± 0.2  | 2.84 ± 0.07 | 2.68 ± 0.08 | 2.6 ± 0.1   | 2.3 ± 0.1   | 3.12 ± 0.09 | 2.88 ± 0.09 | 2.9 ± 0.1   | 2.8 ± 0.1 | 4.0 ± 0.2  | 3.1 ± 0.2  |
| daon-000            | 4.7 ± 0.3  | 5.1 ± 0.3  | 4.2 ± 0.1   | 3.8 ± 0.1   | 2.9 ± 0.1   | 2.6 ± 0.2   | 3.8 ± 0.1   | 4.2 ± 0.1   | 4.2 ± 0.2   | 3.9 ± 0.2 | 5.8 ± 0.3  | 4.2 ± 0.3  |
| daon-001            | 4.2 ± 0.2  | 4.7 ± 0.2  | 3.7 ± 0.1   | 3.2 ± 0.1   | 2.7 ± 0.1   | 2.3 ± 0.1   | 3.4 ± 0.1   | 3.7 ± 0.1   | 3.6 ± 0.2   | 3.2 ± 0.2 | 5.0 ± 0.3  | 3.9 ± 0.3  |
| dermalog-001        | 4.4 ± 0.2  | 3.7 ± 0.2  | 3.49 ± 0.10 | 2.91 ± 0.09 | 4.2 ± 0.2   | 2.8 ± 0.2   | 4.0 ± 0.1   | 3.5 ± 0.1   | 3.8 ± 0.2   | 3.1 ± 0.2 | 3.8 ± 0.2  | 3.4 ± 0.2  |
| dermalog-002        | 4.7 ± 0.2  | 3.4 ± 0.2  | 4.0 ± 0.1   | 2.98 ± 0.10 | 4.5 ± 0.2   | 3.0 ± 0.2   | 4.2 ± 0.1   | 3.3 ± 0.1   | 4.1 ± 0.2   | 3.1 ± 0.2 | 4.2 ± 0.3  | 3.4 ± 0.2  |
| hzailu-001          | 5.7 ± 0.3  | 7.6 ± 0.3  | 4.7 ± 0.1   | 4.1 ± 0.1   | 6.4 ± 0.2   | 5.7 ± 0.3   | 5.3 ± 0.1   | 6.6 ± 0.1   | 5.1 ± 0.2   | 5.0 ± 0.2 | 7.2 ± 0.4  | 8.6 ± 0.4  |
| idemia-001          | 3.3 ± 0.2  | 3.2 ± 0.2  | 2.45 ± 0.07 | 2.19 ± 0.08 | 2.9 ± 0.1   | 2.1 ± 0.1   | 2.73 ± 0.09 | 2.60 ± 0.09 | 2.7 ± 0.1   | 2.3 ± 0.1 | 3.3 ± 0.2  | 3.0 ± 0.2  |
| idemia-002          | 3.6 ± 0.2  | 2.9 ± 0.2  | 2.54 ± 0.07 | 2.10 ± 0.07 | 2.8 ± 0.1   | 1.9 ± 0.1   | 2.96 ± 0.09 | 2.28 ± 0.08 | 2.6 ± 0.1   | 2.2 ± 0.1 | 3.4 ± 0.2  | 2.7 ± 0.2  |
| inbut-000           | 5.7 ± 0.3  | 5.8 ± 0.3  | 2.75 ± 0.09 | 2.8 ± 0.1   | 3.9 ± 0.2   | 3.6 ± 0.2   | 3.7 ± 0.1   | 3.8 ± 0.1   | 2.9 ± 0.2   | 3.0 ± 0.2 | 6.0 ± 0.3  | 5.1 ± 0.3  |
| incode-000          | 4.2 ± 0.2  | 3.3 ± 0.2  | 2.80 ± 0.09 | 2.24 ± 0.07 | 3.0 ± 0.2   | 2.1 ± 0.1   | 3.0 ± 0.1   | 2.43 ± 0.09 | 2.9 ± 0.1   | 2.2 ± 0.1 | 3.8 ± 0.2  | 3.0 ± 0.2  |
| incode-001          | 4.5 ± 0.3  | 3.5 ± 0.2  | 3.07 ± 0.09 | 2.46 ± 0.08 | 2.8 ± 0.1   | 1.9 ± 0.1   | 3.1 ± 0.1   | 2.48 ± 0.09 | 3.4 ± 0.1   | 2.5 ± 0.1 | 4.3 ± 0.3  | 3.1 ± 0.2  |
| innovatrics-000     | 4.1 ± 0.2  | 3.5 ± 0.2  | 2.43 ± 0.07 | 2.05 ± 0.07 | 2.4 ± 0.1   | 2.0 ± 0.1   | 2.84 ± 0.09 | 2.54 ± 0.08 | 2.4 ± 0.1   | 2.0 ± 0.1 | 4.3 ± 0.2  | 3.4 ± 0.2  |
| innovatrics-001     | 3.9 ± 0.2  | 3.4 ± 0.2  | 2.36 ± 0.07 | 1.98 ± 0.06 | 2.3 ± 0.1   | 1.8 ± 0.1   | 2.70 ± 0.08 | 2.38 ± 0.08 | 2.35 ± 0.10 | 2.0 ± 0.1 | 4.2 ± 0.2  | 3.3 ± 0.2  |
| innovatrics-002     | 4.1 ± 0.2  | 3.6 ± 0.2  | 2.41 ± 0.07 | 2.05 ± 0.07 | 2.4 ± 0.1   | 1.9 ± 0.1   | 2.68 ± 0.09 | 2.41 ± 0.08 | 2.4 ± 0.1   | 2.0 ± 0.1 | 4.7 ± 0.3  | 3.6 ± 0.2  |
| jumio-000           | 5.3 ± 0.3  | 3.5 ± 0.2  | 3.53 ± 0.10 | 2.56 ± 0.08 | 3.6 ± 0.2   | 2.8 ± 0.2   | 3.8 ± 0.1   | 2.60 ± 0.09 | 4.0 ± 0.2   | 2.6 ± 0.1 | 5.6 ± 0.3  | 3.5 ± 0.2  |
| neurotechnology-000 | 4.8 ± 0.2  | 6.7 ± 0.3  | 4.1 ± 0.1   | 3.83 ± 0.09 | 3.5 ± 0.2   | 3.0 ± 0.2   | 3.8 ± 0.1   | 3.8 ± 0.1   | 3.9 ± 0.2   | 3.7 ± 0.2 | 5.8 ± 0.3  | 6.9 ± 0.2  |
| neurotechnology-001 | 3.8 ± 0.2  | 4.0 ± 0.2  | 4.1 ± 0.1   | 3.85 ± 0.09 | 3.5 ± 0.2   | 3.0 ± 0.2   | 3.8 ± 0.1   | 3.7 ± 0.1   | 3.9 ± 0.2   | 3.7 ± 0.2 | 4.0 ± 0.2  | 3.8 ± 0.2  |
| nominder-000        | 4.0 ± 0.2  | 3.5 ± 0.2  | 3.09 ± 0.09 | 2.92 ± 0.10 | 2.7 ± 0.1   | 2.5 ± 0.2   | 3.1 ± 0.1   | 3.2 ± 0.1   | 3.0 ± 0.1   | 2.8 ± 0.2 | 4.8 ± 0.3  | 3.6 ± 0.2  |
| nominder-001        | 3.4 ± 0.2  | 3.0 ± 0.2  | 2.80 ± 0.07 | 2.52 ± 0.08 | 2.6 ± 0.1   | 2.1 ± 0.1   | 2.89 ± 0.08 | 2.49 ± 0.08 | 2.8 ± 0.1   | 2.4 ± 0.1 | 4.2 ± 0.2  | 3.1 ± 0.2  |
| ondato-000          | 4.7 ± 0.3  | 4.1 ± 0.3  | 3.18 ± 0.08 | 2.65 ± 0.09 | 2.9 ± 0.1   | 2.1 ± 0.1   | 3.3 ± 0.1   | 2.9 ± 0.1   | 3.3 ± 0.1   | 2.8 ± 0.2 | 5.4 ± 0.3  | 4.2 ± 0.3  |
| ondato-001          | 4.2 ± 0.2  | 3.5 ± 0.2  | 3.21 ± 0.08 | 2.48 ± 0.08 | 2.8 ± 0.1   | 2.0 ± 0.1   | 3.24 ± 0.10 | 2.66 ± 0.09 | 3.4 ± 0.1   | 2.6 ± 0.1 | 4.4 ± 0.2  | 3.4 ± 0.2  |
| privately-000       | 3.8 ± 0.2  | 3.0 ± 0.2  | 3.42 ± 0.10 | 2.85 ± 0.09 | 3.2 ± 0.2   | 2.5 ± 0.1   | 3.6 ± 0.1   | 2.64 ± 0.08 | 3.5 ± 0.2   | 2.8 ± 0.1 | 3.9 ± 0.2  | 3.0 ± 0.2  |
| regula-000          | 3.2 ± 0.2  | 2.5 ± 0.2  | 2.24 ± 0.07 | 1.85 ± 0.06 | 2.1 ± 0.1   | 1.62 ± 0.10 | 2.56 ± 0.08 | 1.98 ± 0.07 | 2.5 ± 0.1   | 1.9 ± 0.1 | 3.3 ± 0.2  | 2.6 ± 0.2  |
| roc-000             | 4.5 ± 0.2  | 4.2 ± 0.2  | 4.0 ± 0.1   | 3.87 ± 0.10 | 3.0 ± 0.1   | 2.3 ± 0.1   | 3.3 ± 0.1   | 3.06 ± 0.10 | 4.0 ± 0.2   | 3.7 ± 0.2 | 4.6 ± 0.3  | 4.1 ± 0.2  |
| roc-001             | 4.5 ± 0.2  | 3.8 ± 0.2  | 3.53 ± 0.08 | 3.30 ± 0.08 | 2.4 ± 0.1   | 1.95 ± 0.10 | 3.24 ± 0.10 | 2.73 ± 0.08 | 3.5 ± 0.1   | 3.1 ± 0.1 | 4.2 ± 0.2  | 3.5 ± 0.2  |
| roc-002             | 3.2 ± 0.2  | 2.7 ± 0.2  | 2.81 ± 0.08 | 2.45 ± 0.08 | 2.17 ± 0.10 | 1.8 ± 0.1   | 2.50 ± 0.08 | 2.15 ± 0.07 | 2.7 ± 0.1   | 2.2 ± 0.1 | 2.9 ± 0.2  | 2.5 ± 0.2  |
| shufti-000          | 5.2 ± 0.3  | 4.7 ± 0.3  | 3.6 ± 0.1   | 3.6 ± 0.1   | 4.2 ± 0.2   | 3.6 ± 0.2   | 4.1 ± 0.1   | 4.2 ± 0.1   | 3.8 ± 0.2   | 3.5 ± 0.2 | 4.5 ± 0.3  | 4.2 ± 0.3  |
| unisseey-001        | 4.3 ± 0.3  | 4.2 ± 0.2  | 3.7 ± 0.1   | 3.8 ± 0.1   | 4.1 ± 0.2   | 3.2 ± 0.2   | 3.7 ± 0.1   | 3.5 ± 0.1   | 3.5 ± 0.2   | 3.8 ± 0.2 | 5.1 ± 0.3  | 3.5 ± 0.2  |
| viant-000           | 5.8 ± 0.2  | 5.9 ± 0.2  | 3.83 ± 0.08 | 4.03 ± 0.09 | 2.8 ± 0.1   | 2.3 ± 0.1   | 5.3 ± 0.1   | 5.3 ± 0.1   | 4.6 ± 0.1   | 4.8 ± 0.2 | 5.5 ± 0.2  | 5.5 ± 0.2  |
| yoti-001            | 5.0 ± 0.3  | 3.3 ± 0.2  | 4.2 ± 0.1   | 2.74 ± 0.09 | 3.8 ± 0.2   | 2.0 ± 0.1   | 4.5 ± 0.1   | 2.8 ± 0.1   | 4.8 ± 0.2   | 2.9 ± 0.2 | 5.3 ± 0.3  | 3.4 ± 0.2  |
| yoti-002            | 4.0 ± 0.2  | 3.2 ± 0.2  | 3.44 ± 0.10 | 2.43 ± 0.08 | 2.5 ± 0.1   | 1.8 ± 0.1   | 3.5 ± 0.1   | 2.52 ± 0.09 | 3.9 ± 0.2   | 2.5 ± 0.1 | 4.2 ± 0.3  | 3.3 ± 0.2  |
| yoti-003            | 4.1 ± 0.2  | 3.0 ± 0.2  | 3.4 ± 0.1   | 2.33 ± 0.08 | 2.3 ± 0.1   | 1.7 ± 0.1   | 3.4 ± 0.1   | 2.43 ± 0.09 | 3.8 ± 0.2   | 2.5 ± 0.1 | 4.0 ± 0.3  | 3.1 ± 0.2  |
| yoti-004            | 3.4 ± 0.2  | 2.7 ± 0.2  | 2.87 ± 0.08 | 1.95 ± 0.07 | 2.0 ± 0.1   | 1.52 ± 0.10 | 2.97 ± 0.10 | 2.06 ± 0.08 | 3.2 ± 0.1   | 2.1 ± 0.1 | 3.5 ± 0.2  | 2.8 ± 0.2  |
| youverse-000        | 4.1 ± 0.2  | 3.9 ± 0.2  | 3.38 ± 0.09 | 2.73 ± 0.09 | 3.4 ± 0.2   | 2.8 ± 0.2   | 3.4 ± 0.1   | 3.1 ± 0.1   | 3.3 ± 0.1   | 2.8 ± 0.2 | 4.3 ± 0.3  | 4.1 ± 0.3  |

**Table 190.** Mean absolute error for Border-type images for all algorithms. Values closer to zero are better.

| Algorithm     | F E Africa | M E Africa | F E Asia    | M E Asia    | F E Europe  | M E Europe | F S Asia    | M S Asia    | F SE Asia | M SE Asia | F W Africa | M W Africa |
|---------------|------------|------------|-------------|-------------|-------------|------------|-------------|-------------|-----------|-----------|------------|------------|
| cognitec-000  | 4.8 ± 0.3  | 4.7 ± 0.3  | 2.85 ± 0.05 | 3.07 ± 0.06 | 2.83 ± 0.10 | 2.8 ± 0.1  | 3.38 ± 0.09 | 3.74 ± 0.09 | 2.9 ± 0.1 | 3.2 ± 0.1 | 4.6 ± 0.3  | 4.1 ± 0.3  |
| cvut-001      | 6.6 ± 0.4  | 5.2 ± 0.4  | 3.82 ± 0.09 | 3.47 ± 0.10 | 4.0 ± 0.2   | 3.5 ± 0.2  | 4.3 ± 0.1   | 3.7 ± 0.1   | 3.8 ± 0.2 | 3.4 ± 0.2 | 6.7 ± 0.4  | 4.9 ± 0.3  |
| cvut-002      | 5.9 ± 0.3  | 6.1 ± 0.3  | 3.51 ± 0.06 | 3.13 ± 0.06 | 3.4 ± 0.1   | 2.9 ± 0.1  | 3.92 ± 0.09 | 3.83 ± 0.09 | 3.8 ± 0.1 | 3.5 ± 0.1 | 5.8 ± 0.3  | 5.4 ± 0.3  |
| cvut-003      | 5.9 ± 0.3  | 5.7 ± 0.3  | 3.25 ± 0.05 | 2.76 ± 0.05 | 3.3 ± 0.1   | 2.6 ± 0.1  | 3.97 ± 0.09 | 3.61 ± 0.08 | 3.6 ± 0.1 | 3.2 ± 0.1 | 6.1 ± 0.3  | 5.2 ± 0.3  |
| cyberlink-000 | 5.1 ± 0.3  | 5.5 ± 0.3  | 3.21 ± 0.06 | 3.06 ± 0.06 | 3.2 ± 0.1   | 3.2 ± 0.1  | 3.70 ± 0.09 | 3.68 ± 0.09 | 3.3 ± 0.1 | 3.4 ± 0.1 | 5.3 ± 0.3  | 4.9 ± 0.3  |

|                     |           |           |             |             |             |             |             |             |             |             |           |            |
|---------------------|-----------|-----------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----------|------------|
| cyberlink-001       | 4.0 ± 0.3 | 4.4 ± 0.3 | 3.05 ± 0.05 | 2.65 ± 0.05 | 2.95 ± 0.09 | 2.8 ± 0.1   | 3.46 ± 0.08 | 3.24 ± 0.08 | 3.1 ± 0.1   | 2.8 ± 0.1   | 4.2 ± 0.2 | 4.0 ± 0.2  |
| daon-000            | 5.6 ± 0.4 | 6.8 ± 0.4 | 5.23 ± 0.08 | 4.62 ± 0.10 | 3.8 ± 0.1   | 3.6 ± 0.2   | 4.2 ± 0.1   | 5.0 ± 0.1   | 4.8 ± 0.2   | 4.4 ± 0.2   | 6.1 ± 0.4 | 6.0 ± 0.4  |
| daon-001            | 5.1 ± 0.3 | 6.1 ± 0.4 | 4.35 ± 0.07 | 3.71 ± 0.08 | 3.3 ± 0.1   | 3.0 ± 0.1   | 3.69 ± 0.10 | 4.24 ± 0.10 | 4.0 ± 0.1   | 3.6 ± 0.2   | 5.5 ± 0.3 | 5.1 ± 0.3  |
| dermalog-001        | 5.9 ± 0.4 | 6.7 ± 0.4 | 4.75 ± 0.08 | 3.96 ± 0.08 | 4.8 ± 0.2   | 3.8 ± 0.2   | 5.2 ± 0.1   | 4.9 ± 0.1   | 4.9 ± 0.2   | 4.1 ± 0.2   | 5.4 ± 0.3 | 6.7 ± 0.4  |
| dermalog-002        | 5.4 ± 0.4 | 6.2 ± 0.4 | 5.26 ± 0.09 | 4.40 ± 0.09 | 5.9 ± 0.2   | 5.0 ± 0.2   | 4.9 ± 0.1   | 4.8 ± 0.1   | 5.1 ± 0.2   | 4.3 ± 0.2   | 5.2 ± 0.3 | 6.5 ± 0.4  |
| hzailu-001          | 6.4 ± 0.4 | 9.2 ± 0.5 | 4.65 ± 0.08 | 4.51 ± 0.08 | 6.6 ± 0.2   | 7.1 ± 0.2   | 5.6 ± 0.1   | 7.3 ± 0.1   | 4.8 ± 0.2   | 5.0 ± 0.2   | 7.6 ± 0.4 | 10.1 ± 0.4 |
| idemia-001          | 3.7 ± 0.2 | 3.8 ± 0.3 | 2.52 ± 0.04 | 2.16 ± 0.04 | 2.93 ± 0.08 | 2.23 ± 0.09 | 2.97 ± 0.08 | 2.69 ± 0.07 | 2.60 ± 0.09 | 2.3 ± 0.1   | 3.6 ± 0.2 | 3.2 ± 0.2  |
| idemia-002          | 4.0 ± 0.3 | 3.4 ± 0.3 | 2.85 ± 0.04 | 2.15 ± 0.04 | 3.05 ± 0.10 | 2.09 ± 0.09 | 3.22 ± 0.08 | 2.37 ± 0.06 | 2.74 ± 0.09 | 2.27 ± 0.10 | 3.7 ± 0.2 | 2.9 ± 0.2  |
| inbut-000           | 5.7 ± 0.3 | 5.6 ± 0.3 | 2.93 ± 0.05 | 2.77 ± 0.06 | 3.2 ± 0.1   | 3.0 ± 0.1   | 3.49 ± 0.09 | 3.34 ± 0.08 | 2.9 ± 0.1   | 2.9 ± 0.1   | 6.1 ± 0.3 | 5.3 ± 0.3  |
| incode-000          | 7.3 ± 0.5 | 6.8 ± 0.5 | 4.12 ± 0.09 | 3.17 ± 0.07 | 5.8 ± 0.2   | 4.1 ± 0.2   | 5.2 ± 0.2   | 4.2 ± 0.1   | 4.2 ± 0.2   | 3.1 ± 0.2   | 6.9 ± 0.5 | 6.0 ± 0.5  |
| incode-001          | 7.5 ± 0.5 | 7.1 ± 0.5 | 3.78 ± 0.07 | 3.17 ± 0.07 | 5.2 ± 0.2   | 3.9 ± 0.2   | 4.3 ± 0.1   | 3.94 ± 0.10 | 4.0 ± 0.1   | 3.2 ± 0.1   | 7.6 ± 0.4 | 6.3 ± 0.4  |
| innovatrics-000     | 4.9 ± 0.3 | 4.5 ± 0.3 | 2.67 ± 0.04 | 2.25 ± 0.04 | 2.62 ± 0.09 | 2.20 ± 0.09 | 3.17 ± 0.08 | 2.84 ± 0.07 | 2.57 ± 0.08 | 2.20 ± 0.09 | 5.0 ± 0.2 | 4.3 ± 0.2  |
| innovatrics-001     | 4.7 ± 0.2 | 4.3 ± 0.3 | 2.70 ± 0.04 | 2.22 ± 0.04 | 2.56 ± 0.08 | 2.10 ± 0.09 | 3.06 ± 0.07 | 2.73 ± 0.06 | 2.61 ± 0.08 | 2.21 ± 0.09 | 4.8 ± 0.2 | 4.2 ± 0.2  |
| innovatrics-002     | 4.4 ± 0.3 | 4.0 ± 0.3 | 2.96 ± 0.04 | 2.35 ± 0.04 | 2.62 ± 0.08 | 2.00 ± 0.08 | 2.86 ± 0.07 | 2.32 ± 0.06 | 2.95 ± 0.08 | 2.39 ± 0.09 | 4.4 ± 0.3 | 3.9 ± 0.2  |
| jumio-000           | 8.2 ± 0.4 | 6.5 ± 0.4 | 5.6 ± 0.1   | 4.5 ± 0.1   | 7.0 ± 0.2   | 4.6 ± 0.2   | 6.6 ± 0.2   | 5.0 ± 0.2   | 6.1 ± 0.2   | 4.6 ± 0.2   | 7.8 ± 0.4 | 5.9 ± 0.3  |
| neurotechnology-000 | 5.0 ± 0.3 | 6.8 ± 0.4 | 4.18 ± 0.07 | 3.70 ± 0.06 | 4.0 ± 0.1   | 3.4 ± 0.1   | 4.05 ± 0.10 | 4.05 ± 0.09 | 4.0 ± 0.1   | 3.5 ± 0.1   | 5.6 ± 0.3 | 6.8 ± 0.3  |
| neurotechnology-001 | 4.2 ± 0.3 | 4.7 ± 0.3 | 4.18 ± 0.07 | 3.70 ± 0.06 | 4.0 ± 0.1   | 3.4 ± 0.1   | 4.04 ± 0.10 | 4.01 ± 0.09 | 4.0 ± 0.1   | 3.5 ± 0.1   | 4.1 ± 0.2 | 4.4 ± 0.3  |
| nominder-000        | 5.1 ± 0.3 | 5.4 ± 0.3 | 3.13 ± 0.05 | 3.03 ± 0.05 | 3.4 ± 0.1   | 3.3 ± 0.1   | 3.73 ± 0.09 | 4.00 ± 0.09 | 3.3 ± 0.1   | 3.4 ± 0.1   | 5.6 ± 0.3 | 5.2 ± 0.3  |
| nominder-001        | 4.2 ± 0.3 | 4.0 ± 0.3 | 2.98 ± 0.05 | 2.83 ± 0.05 | 3.02 ± 0.10 | 2.5 ± 0.1   | 3.35 ± 0.08 | 3.13 ± 0.08 | 3.1 ± 0.1   | 3.0 ± 0.1   | 4.8 ± 0.3 | 4.0 ± 0.3  |
| ondato-000          | 4.5 ± 0.3 | 5.2 ± 0.3 | 3.27 ± 0.05 | 2.82 ± 0.06 | 3.3 ± 0.1   | 2.9 ± 0.1   | 3.58 ± 0.09 | 3.60 ± 0.09 | 3.3 ± 0.1   | 2.9 ± 0.1   | 4.8 ± 0.3 | 4.9 ± 0.3  |
| ondato-001          | 4.3 ± 0.3 | 4.6 ± 0.3 | 3.16 ± 0.05 | 2.68 ± 0.05 | 3.2 ± 0.1   | 2.8 ± 0.1   | 3.49 ± 0.09 | 3.29 ± 0.09 | 3.2 ± 0.1   | 2.8 ± 0.1   | 4.3 ± 0.3 | 4.1 ± 0.3  |
| privately-000       | 4.7 ± 0.3 | 4.5 ± 0.3 | 3.82 ± 0.06 | 3.59 ± 0.07 | 4.0 ± 0.1   | 3.3 ± 0.1   | 4.3 ± 0.1   | 3.42 ± 0.09 | 4.0 ± 0.1   | 3.5 ± 0.2   | 4.5 ± 0.3 | 4.0 ± 0.3  |
| regula-000          | 4.3 ± 0.3 | 3.2 ± 0.3 | 2.79 ± 0.05 | 2.25 ± 0.05 | 2.9 ± 0.1   | 1.96 ± 0.08 | 3.11 ± 0.09 | 2.29 ± 0.06 | 2.9 ± 0.1   | 2.3 ± 0.1   | 4.0 ± 0.3 | 3.0 ± 0.2  |
| roc-000             | 6.0 ± 0.4 | 5.6 ± 0.3 | 4.61 ± 0.07 | 4.33 ± 0.07 | 3.4 ± 0.1   | 2.77 ± 0.10 | 3.66 ± 0.09 | 3.46 ± 0.08 | 4.3 ± 0.1   | 4.3 ± 0.2   | 5.6 ± 0.3 | 5.1 ± 0.3  |
| roc-001             | 5.9 ± 0.3 | 4.7 ± 0.3 | 3.59 ± 0.05 | 3.67 ± 0.05 | 2.96 ± 0.09 | 2.45 ± 0.08 | 3.57 ± 0.08 | 2.95 ± 0.07 | 3.7 ± 0.1   | 3.6 ± 0.1   | 5.1 ± 0.3 | 4.5 ± 0.2  |
| roc-002             | 4.4 ± 0.3 | 3.6 ± 0.2 | 3.39 ± 0.05 | 3.32 ± 0.05 | 2.64 ± 0.08 | 2.22 ± 0.08 | 2.94 ± 0.08 | 2.48 ± 0.06 | 3.1 ± 0.1   | 2.9 ± 0.1   | 4.0 ± 0.2 | 3.5 ± 0.2  |
| shufti-000          | 6.3 ± 0.4 | 6.0 ± 0.4 | 4.16 ± 0.08 | 3.53 ± 0.08 | 4.6 ± 0.2   | 3.4 ± 0.2   | 4.6 ± 0.1   | 4.2 ± 0.1   | 4.1 ± 0.1   | 3.5 ± 0.2   | 5.8 ± 0.3 | 4.8 ± 0.3  |
| unissey-001         | 4.1 ± 0.3 | 3.6 ± 0.3 | 3.54 ± 0.06 | 3.35 ± 0.07 | 3.8 ± 0.1   | 2.9 ± 0.1   | 3.87 ± 0.09 | 3.04 ± 0.08 | 3.6 ± 0.1   | 3.2 ± 0.2   | 4.3 ± 0.3 | 3.3 ± 0.2  |
| viant-000           | 6.7 ± 0.3 | 6.7 ± 0.3 | 2.93 ± 0.05 | 2.78 ± 0.05 | 2.73 ± 0.10 | 2.17 ± 0.10 | 5.39 ± 0.10 | 4.91 ± 0.09 | 3.9 ± 0.1   | 3.9 ± 0.1   | 6.2 ± 0.2 | 6.1 ± 0.2  |
| yoti-001            | 5.4 ± 0.3 | 4.7 ± 0.3 | 5.15 ± 0.09 | 4.02 ± 0.08 | 5.3 ± 0.2   | 3.3 ± 0.2   | 4.6 ± 0.1   | 3.61 ± 0.10 | 5.4 ± 0.2   | 4.2 ± 0.2   | 5.0 ± 0.3 | 4.6 ± 0.3  |
| yoti-002            | 4.8 ± 0.3 | 4.3 ± 0.3 | 4.68 ± 0.08 | 3.58 ± 0.08 | 3.6 ± 0.1   | 2.6 ± 0.1   | 4.0 ± 0.1   | 3.04 ± 0.09 | 4.9 ± 0.2   | 3.6 ± 0.2   | 4.6 ± 0.3 | 4.2 ± 0.3  |
| yoti-003            | 4.8 ± 0.3 | 4.0 ± 0.3 | 4.64 ± 0.08 | 3.67 ± 0.08 | 3.5 ± 0.1   | 2.3 ± 0.1   | 3.9 ± 0.1   | 3.00 ± 0.09 | 4.9 ± 0.2   | 3.6 ± 0.2   | 4.4 ± 0.3 | 4.0 ± 0.3  |
| yoti-004            | 4.3 ± 0.3 | 3.6 ± 0.3 | 4.24 ± 0.07 | 3.28 ± 0.07 | 3.0 ± 0.1   | 2.1 ± 0.1   | 3.58 ± 0.10 | 2.65 ± 0.08 | 4.4 ± 0.1   | 3.2 ± 0.1   | 4.1 ± 0.3 | 3.6 ± 0.2  |
| youverse-000        | 6.0 ± 0.4 | 8.6 ± 0.4 | 4.80 ± 0.08 | 4.38 ± 0.09 | 4.3 ± 0.2   | 4.5 ± 0.2   | 4.8 ± 0.1   | 5.8 ± 0.1   | 4.7 ± 0.1   | 4.3 ± 0.2   | 5.8 ± 0.3 | 8.2 ± 0.4  |